# Finite group actions, loop structures, and profinite extensions

A mathematical development of *La Longue Marche*: LM4

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**Working manuscript: INCOMPLETE.** The numbered statements distinguish proved assertions, complete counterexamples, source theorems with their reductions, and the exact obligations still under investigation. This working record makes no assertion of completion of the task.

**Notation and its manuscript concordance.** Local generators, base points, coefficients and comparison maps are defined at their use. The table fixes the global symbols and the meaning of each variant.

| Our notation | Meaning and Grothendieck's notation | Introducing passage / scans |
|---|---|---|
| $X, S, U = X - S$ | Compact surface, marked set, punctured surface; $X, S, U$ | §12: 91–92; §19: 204 |
| $(g, \nu), I$ | Genus, number and index set of punctures; $(g, \nu), I$ | §13 bis: 115–116 |
| $\pi, \pi_0, P$ | Discrete surface group, chosen discretization, its completion; $\pi, \pi_0, \pi$ | §12: 91; §28: 352 |
| $L_i, \kappa_i$ | Indexed peripheral subgroup and exterior source map; $L_i, \kappa_i$ | §12: 91–99 |
| $T = T_\pi$ | Covariant orientation line; dual of the earlier cohomological line | 91; 116; 130 |
| $A, A^0$ | Surface homeomorphism group and identity component; $A_{g,\nu}, A^0$ | §19: 204–205 |
| $M, \Gamma_{g,\nu}$ | Geometric mapping class group; $\mathcal{T}_{g,\nu}, \Gamma_{g,\nu}$ | 193–203; 204–205 |
| $!, +$ | Pointwise marked-point preservation; orientation preservation | 204–205 |
| $G, \Gamma, E$ | Finite action, operator group, lift extension; his $\Gamma, G, E$ as locally specified | §15: 137–139; 307–312 |
| $\Phi, \mu$ | Maximal partial-section data and torsion multiplier; $\Phi, \mu$ | 150–160; 197–203 |
| $SE, S\Gamma$ | Special actual and exterior automorphism groups; $SE, S\Gamma$ | §25: 296–300 |
| $B, \overline{M}$ | $\mathrm{Out}_{\mathrm{lac}}(P)$ and closure of a specified geometric group; $\widehat{\widehat{\mathcal{T}}}, \widehat{\mathcal{T}}$ where identified | 318–320; 352–358 |
| Discr, Prediscr | Actual, outer, or closed-stabilizer cosets as explicitly specified | §28: 352–358 |
| $GT_0, \lambda, f$ | The normalized centralizer of $S_3$, multiplier and commutator parameter | Two-group problem: 312, 316–317 |

## Reading the proof record

We work in the order A–F of the supplied task. A group with loops retains its indexed exterior peripheral maps and its orientation line; this includes the exceptional cases where the image subgroups alone lose that data. All homomorphisms between profinite or pro-$p$ groups are continuous. A section retains its prescribed quotient labels, and its equivalence is conjugacy by the named kernel. Every passage citation gives the manuscript section, folio, and scan; a folio followed by "verso" names its reverse side. A transcription alternative is retained when mathematical evidence bears on it. Our corrections are assertions about the displayed mathematics, not declarations that a doubtful reading has been settled.

The results register contains the full numbered assertions and dependencies. The readings register records the editorial consequences. The report gives the obligation-by-obligation state, source checks and computation provenance. An "Open" label records an unfinished obligation in this working manuscript; it does not replace a requested proof. A "Known" label includes the derivation from the precise source theorem. The code uses exact finite arithmetic, and the proofs identify what the computations certify and what requires a separate infinite-group argument.

## Contents


**Reading the proof record** **2**

**1 Groups with loops and their rigidity** **4**

**2 Finite actions, sections, and rotation data** **19**

2.1 The plane lemma and the section correspondence . . . . . 20

2.2 Finite automorphisms and the withdrawn assertions . . . . . 22

2.3 The multiplier and the abelian surfaces . . . . . 25

2.4 Anti-involutions and the later tube construction . . . . . 29

2.5 Computations and the limits of the requested table . . . . . 34

2.6 The full tube action and the strip subgroups . . . . . 51

**3 C. Equivariant comparison, drilling, and the topology of special groups** **74**

3.1 The categories and their exact fidelity conditions . . . . . 75

3.2 Finite actions on hyperbolic surfaces . . . . . 78

3.3 The exceptional types and infinite operator groups . . . . . 81

3.4 Configuration invariants and drilling . . . . . 91

3.5 Special homeomorphisms: the topology and the extension class . . . . . 95

3.6 The integral sphere comparison . . . . . 98

**4 D. Profinite extensions and the limits of the realization problem** **305**

4.1 Completion of geometric mapping class groups and full loop automorphisms . . . 310

4.2 Descent along a finite covering . . . . . 316

4.3 Torsion-free extensions and recognition . . . . . 319

4.4 A full profinite lifting criterion and nonnilpotent finite tests . . . . . 324

**5 E. Special groups and two operator groups** **480**

5.1 E(a). Special groups from conjugators and from framed diagrams . . . . . 480

5.2 E(b). Two commuting operator groups . . . . . 482

**6 F. Statements usable in the later sections 486**

# 1 Groups with loops and their rigidity

We use $[x,y]=xyx^{-1}y^{-1}$. A peripheral subgroup is indexed even when two indices give the same subgroup. The notation $\mathrm{Out}_{\mathrm{lac}}$ means preservation of the indexed peripheral conjugacy classes, with their permitted permutations; in hyperbolic type these permutations are determined by the automorphism. We specify the additional data needed in the exceptional types below. All profinite homomorphisms and automorphisms in this section are continuous.

**A.1. [Proved] The presentation, ranks, and exceptional types.** Let $X$ be a connected closed oriented topological surface of genus $g$, let $S$ consist of $\nu$ distinct points, and put $U=X\setminus S$. A discrete group with loops of type $(g,\nu)$ consists of a group $\pi$, a finite index set $I$ of cardinality $\nu$, a free rank-one integral orientation module $T$, and exterior homomorphisms $\kappa_i : T\to\pi$. Here an exterior homomorphism is a conjugacy class, under conjugation in the target, of actual homomorphisms. After choosing a generator of $T$, representatives $c_i=\kappa_i(1)$, an ordering of $I$, and the handle generators, these data admit the presentation

$$\pi_{g,\nu}=\left\langle a_1,b_1,\dots,a_g,b_g,c_1,\dots,c_\nu \;\middle|\; \prod_{j=1}^{g}[a_j,b_j]c_1\cdots c_\nu=1\right\rangle,\qquad L_i=\langle c_i\rangle.$$

The profinite version uses a free rank-one $\widehat{\mathbb{Z}}$-module $T$, continuous exterior homomorphisms, and the corresponding presentation in profinite groups, namely the quotient of the free profinite group by the closed normal subgroup generated by this relator. When $I$ is empty one additionally identifies $T$ with the dual of $H^2(\pi;\mathbb{Z})$, or its continuous profinite counterpart. This is the covariant orientation convention of scan 116. More precisely, an earlier cohomological marking $\kappa : H^2(\pi;\mathbb{Z})\xrightarrow{\sim} T_{\mathrm{coh}}$ on scan 91 gives, by dualization, $\kappa^\vee : T_{\mathrm{coh}}^\vee\xrightarrow{\sim} H_2(\pi;\mathbb{Z})$; the covariant module used here is $T=T_{\mathrm{coh}}^\vee$. The same typed dual map specifies the profinite convention. For an isomorphism of groups with loops the data are an isomorphism $f:\pi\to\pi_1$, a bijection $s:I\to I_1$, and an isomorphism $t:T\to T_1$ such that $f\kappa_i$ is conjugate to $\kappa_{s(i)}t$ for every $i$, with the corresponding dual-cohomology compatibility when $I$ is empty. An exterior group with loops regards two such $f$ as equivalent when they differ by target inner conjugation, with the same $s,t$. Thus its isomorphism sets are explicitly the quotient of the preceding isomorphism sets by $\mathrm{Inn}(\pi_1)$. In all nontrivial admitted types $L_i=\kappa_i(T)$ recovers its source by A.2; the one-puncture sphere requires the original source map. For $\nu>0$ the group is free of rank $2g+\nu-1$, and its abelianization has that rank. For $\nu=0$ the abelianization has rank $2g$. The abelian types are exactly

$$(0,0),\quad (0,1),\quad (0,2),\quad (1,0),$$

with groups $1,1,\mathbb{Z},\mathbb{Z}^2$, respectively. The same list holds for their profinite completions. The peripheral maps from the source orientation group $\mathbb{Z}$ to $\pi$ are injective except at $(0,1)$, among the types admitted in scans 91–92. The closed sphere is explicitly excluded there.

*Manuscript:* 12; 44–48; 91–100; 13 bis; 56–57; 115–118.

*Proof.* Eliminate $c_\nu$ when $\nu>0$; the remaining generators have no relation. Abelianizing the displayed presentation leaves only $c_1+\cdots+c_\nu=0$, giving both ranks. A free group of rank at least two is nonabelian, as witnessed by its quotient $S_3$, generated by (12) and (123). A closed genus-$g$ group for $g\geq 2$ maps onto the free group on $a_1,\dots,a_g$ by killing all the $b_j$. The same finite quotient witnesses noncommutativity after completion. For $\nu\geq 2$, each peripheral generator belongs to a free basis obtained by eliminating a different peripheral generator. For $\nu=1$, $g\geq 1$, the nonempty cyclically reduced word $\prod[a_j,b_j]$ is nontrivial and has infinite order. Its image has infinite order in every pro-$p$ completion: map the first pair into the upper unitriangular $3\times 3$ matrices over $\mathbb{Z}_p$, and all other pairs to 1; the commutator is $1+E_{13}$. This also proves that its procyclic closure in the full completion is $\widehat{\mathbb{Z}}$, since every prime component

injects. At $(0,1)$ the peripheral word is 1, so the source map $\mathbb{Z} \to 1$ is not injective. These calculations prove the correction $\nu \leq 2$ to the bracket $\nu \leq 3$ on scan 116. This is mathematical evidence of an error, not a changed reading of the visible numeral. □

*Manuscript:* §12, folios 44–48, scans 91–100; §13 bis, folios 56–57, scans 115–118.

**A.2. [Proved] Canonical comparison of the orientation modules.** Assume $\nu > 0$ and $\pi \neq 1$. There is a canonical exact sequence

$$0 \longrightarrow T_\pi \longrightarrow \bigoplus_{i \in S} L_i \longrightarrow \pi^{\mathrm{ab}} \longrightarrow H_1(X;\mathbb{Z}) \longrightarrow 0, \qquad T_\pi = \ker\left(\bigoplus_i L_i \longrightarrow \pi^{\mathrm{ab}}\right).$$

Each projection $T_\pi \to L_i$ is an isomorphism of rank-one integral modules. If $M \subset U$ is obtained by removing small open discs at the punctures, the boundary morphism and evaluation give canonical isomorphisms

$$H_2(M, \partial M; \mathbb{Z}) \xrightarrow{\sim} T_\pi, \qquad T_\pi \xrightarrow{\sim} \mathrm{Hom}\big(H_c^2(U;\mathbb{Z}), \mathbb{Z}\big).$$

For $\nu = 0$, $g \geq 1$, the corresponding formula is $T_\pi = \mathrm{Hom}(H^2(\pi;\mathbb{Z}),\mathbb{Z}) = H_2(X;\mathbb{Z})$. Filling the discs identifies both orientation lines with $H_2(X;\mathbb{Z})$, naturally for homeomorphisms of the marked surface. All these comparisons persist with $\widehat{\mathbb{Z}}$ and continuous relative group cohomology. At $(0,1)$ the kernel formed from the image subgroup $L_1 = 1$ is zero, although $H_c^2(\mathbb{R}^2;\mathbb{Z}) = \mathbb{Z}$; the source peripheral morphism is therefore essential. At $(0,0)$ group cohomology $H^2(1;\mathbb{Z}) = 0$ cannot supply the surface orientation line.

*Manuscript:* 13 bis; 56–57; 116–118; 14; 64 verso; 130.

*Proof.* The relevant portion of the integral homology sequence of the compact pair is

$$0 \to H_2(M, \partial M) \xrightarrow{\partial} H_1(\partial M) \to H_1(M) \to H_1(M, \partial M).$$

The initial zero follows from $H_2(M) = 0$. A relative fundamental class has boundary the sum of the consistently oriented boundary circles. Identifying each boundary circle with its peripheral subgroup, the boundary morphism identifies $H_2(M,\partial M)$ with the stated kernel and has degree 1 on each factor. The cokernel is $H_1(X)$ by attaching the discs and using the homology sequence. Universal coefficients for the pair, whose homology is free, identify $H^2(M,\partial M)$ with the dual of $H_2(M,\partial M)$; excision identifies the former with $H_c^2(U)$. Let $D \subset X$ be the union of the filled discs. The canonical isomorphisms $H_2(X) \xrightarrow{\sim} H_2(X,D) \xrightarrow{\sim} H_2(M,\partial M)$ are, respectively, the pair map and excision. Their inverse is the filling comparison, sending a relative fundamental class to the closed fundamental class and proving compatibility without a chosen sign. Conjugating a peripheral subgroup changes its identification by an inner normalizer action, which is trivial by A.3; hence the sequence is independent of paths to the boundary. For the annulus the same conclusion follows directly because its group is abelian.

For clarity, the profinite compact-support notation is the relative continuous cochain complex

$$C^*_{\mathrm{rel}}(\widehat{\pi}; (\widehat{L}_i); \mathbb{Z}/n) = \mathrm{Cone}\left(C^*_{\mathrm{cont}}(\widehat{\pi};\mathbb{Z}/n) \longrightarrow \bigoplus_i C^*_{\mathrm{cont}}(\widehat{L}_i;\mathbb{Z}/n)\right)[-1].$$

For $\nu > 0$, $\widehat{\pi}$ has cohomological dimension one and each $\widehat{L}_i \simeq \widehat{\mathbb{Z}}$; thus its degree-two group is the cokernel of restriction on $H^1$. In the displayed presentation the restriction matrix is the transpose of the integral peripheral inclusion matrix. Its cokernel is canonically $\mathrm{Hom}(T_\pi, \mathbb{Z}/n)$. Taking the inverse limit over $n$ gives the claimed duality. For a closed surface of genus at least two, the continuous duality pairing identifies $H^2(\widehat{\pi};\widehat{\mathbb{Z}})$ with the completed surface orientation cohomology; this is the surface Poincare duality statement in [53, Proposition 1.3]. At genus

one, for each prime $p$ the Koszul resolution of the trivial module over $\mathbb{Z}_p[[\mathbb{Z}_p^2]]$ has differentials given by the two generators minus 1. Applying trivial coefficients $\mathbb{Z}/p^r$ makes both differentials zero, so the cup product identifies degree two with the exterior square of degree one, exactly as for the discrete torus. The other prime factors have zero positive-degree cohomology with these coefficients. The Chinese remainder theorem gives the comparison for every finite coefficient $\mathbb{Z}/n$. The transition maps on degree one are surjective, so their first derived inverse limit vanishes; consequently passing to inverse limits introduces no extra degree-two term. The canonical map from the discrete presentation to its completion induces the identity on the surface relation and on every peripheral source; consequently the finite-coefficient comparisons commute with filling. The two trivial-group exceptions follow by the explicit cohomology calculations in the statement. □

*Manuscript:* §13 bis, folios 56–57, scans 116–118; §14, folio 64 verso, scan 130. We read $H^2_!$ on scan 130 as the relative complex just specified, rather than as ordinary cohomology of the underlying abstract profinite group.

**A.3. [Proved] Discrete peripheral rigidity in every hyperbolic type.** For $2g+\nu>2$, the group $\pi=\pi_{g,\nu}$ has trivial centre and $C_\pi(H)=1$ for every finite-index subgroup $H$. For every peripheral subgroup and every integer $n\geq 1$,

$$N_\pi(L_i)=C_\pi(L_i)=C_\pi(L_i^n)=L_i.$$

Distinct peripheral indices give distinct, noncommensurable conjugacy classes.

*Manuscript:* 12; 47 verso–48; 98–99; 13 bis; 57; 117–118.

*Proof.* For $\nu>0$, the free action of a free group on its Cayley tree shows that a nonidentity element has an axis and that its centralizer is the infinite cyclic group of translations generated by its primitive root. An element normalizing this cyclic group preserves the axis. A reversal of the axis fixes a vertex or an edge midpoint, which is incompatible with the free, inversion-free action; the normalizer is the same cyclic group. For $\nu\geq 2$ each peripheral element is primitive in a free basis. For $\nu=1$, the cyclic word $\prod[a_j,b_j]$ contains the positive letter $a_1$ exactly once. A proper cyclic power has the number of occurrences of every signed letter divisible by its exponent, so this word is not a proper power. This proves the peripheral formulas including the marginal case of scan 117.

For the closed case, the hyperbolic polygon realization of [22, Theorem 1.2] embeds $\pi_g$ as a torsion-free cocompact group of orientation-preserving isometries of the hyperbolic plane. Every nonidentity element is hyperbolic. Its centralizer preserves its axis and acts discretely by translations, hence is cyclic. An axis reversal would be an order-two elliptic element and is absent. A finite-index subgroup is a surface group of genus $1+[\pi:H](g-1)\geq 2$. In the punctured case the covering has free rank $1+[\pi:H](2g+\nu-2)\geq 2$. Thus a finite-index subgroup cannot lie in a cyclic centralizer. This proves $C_\pi(H)=1$, including $H=\pi$.

For distinct punctures with $\nu\geq 3$, their primitive abelianization vectors span distinct lines; for $\nu=2$, $g\geq 1$, choose the free basis containing $c_1$. The cyclically reduced lengths of $c_1$ and $c_2=(\prod[a_j,b_j]c_1)^{-1}$ are 1 and $4g+1$, and their abelianizations force any commensurability to match equal absolute exponents. They cannot be conjugate. This proves the final assertion; at $\nu=1$ it is vacuous. □

*Manuscript:* §12, folios 47 verso–48, scans 98–99; §13 bis, folio 57, scans 117–118.

**A.4. [Proved] Exact exceptional-case table.** The discrete and profinite exceptional groups have the following properties; in the profinite column replace $\mathbb{Z}$ by $\widehat{\mathbb{Z}}$.

| $(g,\nu)$ | $\pi$ | $Z(\pi)$ | $C_\pi(H)$ ($H$ finite index/open) | peripherals |
|---|---|---|---|---|
| $(0,0)$ | 1 | 1 | 1 | none |
| $(0,1)$ | 1 | 1 | 1 | $L_1 = 1$ |
| $(0,2)$ | $\mathbb{Z}$ | $\mathbb{Z}$ | $\mathbb{Z}$ | $L_1 = L_2 = \pi$ |
| $(1,0)$ | $\mathbb{Z}^2$ | $\mathbb{Z}^2$ | $\mathbb{Z}^2$ | none |

Consequently the peripheral normalizer and power-centralizer equalities hold in the plane and annulus, while distinctness of the two peripheral classes fails on the annulus. The nonzero vector 1 on the annulus and $(1,0)$ on the torus witness failure of both centre and finite-index-centralizer assertions. The plane's failure is the noninjective source map, not a nontrivial centre. For the standard base-element formula of scan 501, the annulus satisfies the equality since its basis element generates its whole group. On the torus $C(\langle(1,0)\rangle^n) = \mathbb{Z}^2$ for every $n \geq 1$, and $(0,1)$ witnesses its failure to equal $\langle(1,0)\rangle$. The same vector works after completion. The sphere and plane have no nontrivial base elements to test.

*Manuscript:* 20; 111–115 verso; 222–230; 12; 47 verso; 98; 13 bis; 56 verso; 116.

*Proof.* A.1 identifies the groups and peripheral images. Every subgroup of an abelian group is centralized by the entire group. The assertions now follow by direct substitution. For the profinite version the listed elements remain nonzero, as their reductions modulo 2 show. □

*Manuscript:* §20, folios 111–115 verso, scans 222–230; omissions on scans 98 and 116.

**A.5. [Proved] Peripheral decomposition groups and transporters.** Let $1 \to \pi \to E \xrightarrow{p} \Gamma \to 1$ arise from an action on a hyperbolic marked surface. Choose one peripheral subgroup $L_s$ and the corresponding peripheral lift group $E_s \subset E$; thus $E_s \cap \pi = L_s$ and $p(E_s) = \Gamma_s$, the stabilizer of that puncture. Define

$$\mathrm{Trans}_E(L_s^d, L_s) = \{e \in E : eL_s^d e^{-1} \subseteq L_s\} \qquad (d \geq 1).$$

Then $N_E(L_s) = \mathrm{Trans}_E(L_s^d, L_s) = E_s$. The same assertions hold for the corresponding profinite extension and closed peripheral groups. On the annulus these assertions fail for $z \mapsto 1/z$: with $E = \mathbb{Z} \rtimes C_2$ and the involution acting by inversion, $E_s = \mathbb{Z}$ but $N_E(L_s) = E$ and every transporter is $E$. More generally for every annulus action both normalizer and transporter equal $E$, while $E_s$ is the inverse image of the end stabilizer. Thus their equality with $E_s$ holds exactly when the action fixes both ends. On the one-puncture sphere the peripheral mixed group maps onto $E = \Gamma$ with kernel $\mathbb{Z}$; using its image, both equalities hold, whereas identifying that mixed group itself with a subgroup is invalid. The closed sphere and torus have no peripheral statement.

*Manuscript:* 9; 33; 69; 12; 48; 99.

*Proof.* A normalizer element maps to $\Gamma_s$, since the peripheral conjugacy classes are distinct. Choose $e_s \in E_s$ with the same image. Then $e_s^{-1}e$ belongs to $N_\pi(L_s) = L_s$, proving the normalizer equality. For the transporter assertion, $eL_s^d e^{-1} \subset L_s$ implies $L_s^d \subset e^{-1}L_s e$. The latter is abelian, so it lies in $C_\pi(L_s^d) = L_s$, by A.3 or A.7. In the discrete case a peripheral generator cannot be a proper power, so the inclusion of peripheral groups $e^{-1}L_s e \subset L_s$ is an equality. In the profinite case use compactness of $E$: a closed subgroup $L$ of a compact group cannot be properly contained in an inner conjugate. Indeed $aLa^{-1} \subset L$ gives $a^k La^{-k} \subset L$ for all $k \geq 1$, and a net of positive powers of $a$ converges to $a^{-1}$ inside its procyclic closure. Passing to the limit gives the reverse conjugation inclusion and hence equality. Thus the transporter normalizes $L_s$ in both cases. This argument only uses the exact power-centralizer statement and avoids replacing

the one-sided edge statement in [53, Proposition 1.2(i)] by a stronger unstated assertion about chosen subgroup representatives. The peripheral lift groups for the action on a punctured disc map onto the puncture stabilizer by construction, so the preceding choice of $e_s$ is legitimate. For the annulus the involution swaps the two punctures, whose inertia subgroups both equal $\pi$. Its lift is the explicit element in the normalizer outside $E_s$. Replacing $\mathbb{Z}$ with $\widehat{\mathbb{Z}}$ gives the completed counterexample. □

*Manuscript:* §9, folio 33, scan 69; §12, folio 48, scan 99. In the plane, the map from the peripheral mixed group to $E$ has a kernel $\mathbb{Z}$; writing that mixed group as a subgroup would already discard essential data.

**A.6. [Proved] Centre-free extension reconstruction and its abelian correction.** For a centre-free discrete group $\pi$ and a homomorphism $\rho : \Gamma \to \mathrm{Out}(\pi)$, the extension inducing $\rho$ is, up to a unique isomorphism inducing the identity on its prescribed kernel and quotient,

$$E_\rho = \{(a, \gamma) \in \mathrm{Aut}(\pi) \times \Gamma : [a] = \rho(\gamma)\}.$$

The same formula holds for a topologically finitely generated centre-free profinite $\pi$ and continuous $\rho$, with the profinite topology. For $A = \mathbb{Z}$ or $\mathbb{Z}^2$ and a prescribed action $\rho : \Gamma \to \mathrm{Aut}(A)$, extension classes with their kernel and quotient identified are instead classified by $H^2(\Gamma; A_\rho)$. Their automorphisms inducing the identity on both ends form $Z^1(\Gamma; A_\rho)$. For finite $\Gamma$ this statement holds with $A = \widehat{\mathbb{Z}}$ or $\widehat{\mathbb{Z}}^2$ and continuous cochains. Thus both the annulus and the torus require the abelian correction.

*Manuscript:* 12; 47 verso; 98; 17; 77–80; 155–160.

*Proof.* An extension $E$ maps to $E_\rho$ by $e \mapsto (\mathrm{conj}_e, p(e))$. Its kernel is $Z(\pi) = 1$. Given $(a, \gamma)$, choose $e_0$ above $\gamma$. The automorphism $a\,\mathrm{conj}_{e_0}^{-1}$ is inner, implemented by a unique element of $\pi$, so the map is onto. An isomorphism inducing the identity on both ends fixes both components of this formula and is unique. In the profinite case this continuous bijection between compact Hausdorff groups is an isomorphism.

For abelian $A$, choose a set-theoretic section $t$ of the quotient with $t(1) = 1$ and write $t(g)t(h) = c(g, h)t(gh)$. Associativity is exactly $c(g, h) + c(gh, k) = \rho(g)c(h, k) + c(g, hk)$. Changing $t$ by a function $b : \Gamma \to A$ adds its coboundary. Conversely this identity makes $(a, g)(b, h) = (a + \rho(g)b + c(g, h), gh)$ associative. An automorphism fixing both ends has the form $(a, g) \mapsto (a + b(g), g)$, where the homomorphism equation says $b$ is a 1-cocycle. For finite $\Gamma$ all choices of section are continuous. For example $\Gamma = C_2$, $A = \mathbb{Z}$ with trivial action has the two extension classes $\mathbb{Z} \times C_2$ and $\mathbb{Z}$, with kernel $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ in the latter. They induce the same outer action and are not isomorphic as extensions; reduction modulo 2 detects the different value of $t(\sigma)^2$. □

*Manuscript:* §12, folio 47 verso, scan 98; §17, folios 77–80, scans 155–160.

**A.7. [Known] The full profinite rigidity statements, with reductions.** For the full profinite completion $\Pi = \widehat{\pi}_{g,\nu}$ of a hyperbolic type, $\Pi$ is torsion-free, $C_\Pi(H) = 1$ for every open subgroup $H$, and

$$N_\Pi(\widehat{L}_i) = C_\Pi(\widehat{L}_i^n) = \widehat{L}_i \quad (n \in \mathbb{Z}_{>0}).$$

Distinct peripheral classes are noncommensurable. The same conclusions hold for the pro-$p$ completions, for every prime $p$, including 2 and 3. The four exceptional types have exactly the completed table of A.4, and A.5 supplies the extension and transporter assertions.

*Manuscript:* 13 bis; 57 verso; 118; 35; 253–254; 499–501.

*Proof.* Take the full formation of finite groups, respectively the formation of finite $p$-groups, in [54, Remark 1.2.2 and Proposition 1.4]. A hyperbolic complex curve with the prescribed topological presentation identifies its geometric group with $\Pi$, respectively the pro-$p$ completion. Proposition 1.4 asserts precisely the vanishing of centralizers of open subgroups, with no properness assumption, so taking $H = \Pi$ gives centre-freeness. Remark 1.2.2 gives finite $\ell$-cohomological dimension for every relevant prime, hence excludes an element of order $\ell$: such an element would give a closed $C_\ell$ subgroup, whose $\ell$-cohomological dimension is infinite. Every nontrivial finite-order element has a prime-order power; this proves torsion-freeness.

For peripherals, [52, Lemma 1.3.7] gives $\mathrm{Comm}_\Pi(\widehat{L}_i) = \widehat{L}_i$. A normalizer is contained in that commensurator. An element centralizing $\widehat{L}_i^n$ conjugates the finite-index subgroup $n\widehat{L}_i$ to itself, so it too belongs to the commensurator. Conversely the procyclic group $\widehat{L}_i$ centralizes every power. This proves both equalities. In the semi-graph of the smooth pointed curve, peripherals are the cusp edge groups. Proposition 1.2(i) of [53] says that an intersection open in one edge group identifies the underlying edge; hence distinct punctures cannot have commensurable inertia groups. This gives the needed distinction of peripheral indices; A.5 proves its transporter assertion separately using power-centralizers and compactness. No claim about centralizers of arbitrary non-peripheral elements is needed here.

For $\nu \geq 2$ there is also a direct source for the power-centralizer assertion: each peripheral generator can be made a free basis element, and [56, Lemma (2.1.2)] gives the centralizer of every nonidentity element of its procyclic closure. For $\nu = 1$ the commensurator argument covers precisely the case that this basis argument does not cover. For every $\nu \geq 1$, each handle generator $a_j, b_j$ belongs to the free basis obtained by eliminating $c_\nu$. Applying that same lemma proves equation (4) of scan 501 for all these base elements, including the once-punctured case. The basis condition is material if the phrase "systeme de generateurs on scan 500 is read literally without a cardinality restriction: in a free profinite group on $a, b$, the redundant generating list $a, b, a^2$ has $C(a^2) = \overline{\langle a \rangle} \supsetneq \overline{\langle a^2 \rangle}$. The last index is 2, detected in abelianization modulo 2. For the pro-$p$ version use $a, b, a^p$ and index $p$. The stated source theorem applies to a member of a free basis; it does not supply the assertion for every redundant generating list. □

*Manuscript:* §13 bis, folio 57 verso, scan 118; §35, folios 253–254, scans 499–501.

**A.8. [Proved] Lemme 5 from open-subgroup centralizers alone.** Let $P$ be a discrete or profinite group for which the centralizer of every finite-index, respectively open, subgroup is trivial. Let $H \subset P$ be such a subgroup and let $u \in \mathrm{Aut}(P)$ preserve $H$. An automorphism restricting to the identity on $H$ is the identity on $P$. An automorphism whose restriction equals $\mathrm{conj}_h$ for some $h \in H$ equals $\mathrm{conj}_h$ on $P$.

*Manuscript:* 35; 252 verso–254; 498–501.

*Proof.* Take the finite-index normal core $N = \bigcap_{x \in P} xHx^{-1} \subset H$. If $u|_H = \mathrm{id}$, then for $x \in P$ and $n \in N$,

$$u(x)nu(x)^{-1} = u(xnx^{-1}) = xnx^{-1}.$$

Thus $x^{-1}u(x) \in C_P(N) = 1$ for every $x$. For the second assertion apply the first to $\mathrm{conj}_{h^{-1}} u$. In the profinite case the core is an intersection of finitely many open subgroups and is open. This proof uses neither peripheral generators nor their power-centralizers. □

*Manuscript:* §35, folio 252 verso, scan 498, Lemme 5; folios 253–254, scans 500–501, proposed proof.

**A.9. [Proved] Faithfulness ascends along a finite etale cover.** Let $k$ be a characteristic-zero field, $U/k$ a hyperbolic geometrically connected smooth curve, and $V \to U$ a finite etale

geometrically connected covering defined over $k$. Put $\Gamma = \mathrm{Gal}(\overline{k}/k)$ and choose compatible geometric base points. Faithfulness of $\Gamma \to \mathrm{Out}(\pi_1(U_{\overline{k}}))$ implies faithfulness of $\Gamma \to \mathrm{Out}(\pi_1(V_{\overline{k}}))$.

*Manuscript:* 35; 253; 499–500.

*Proof.* The covering gives an open embedding $\Pi_V \hookrightarrow \Pi_U$ of arithmetic fundamental groups, compatible with their surjections onto the same $\Gamma$. Their geometric kernels are an open subgroup $H \subset P$. Let $\gamma \in \Gamma$ act trivially in $\mathrm{Out}(H)$ and choose a lift $e \in \Pi_V$. Conjugation by $e$ preserves $H$ and restricts there to conjugation by some $h \in H$. By A.7 and A.8 its action on $P$ is also conjugation by $h$. Thus $\gamma$ acts trivially in $\mathrm{Out}(P)$ and equals 1. A change of compatible base points conjugates these homomorphisms and does not change their kernels. The proof does not use the structure of finite subgroups of $\Gamma$. □

*Manuscript:* §35, folio 253, scans 499–500, Corollaire 2.

**A.10. [Known] Closed-surface base elements: two separate source reductions.** Let $\Pi = \widehat{\pi}_g$ with $g \geq 2$, let $x$ be any of the standard surface generators $a_i, b_i$, and let $\Lambda = \overline{\langle x \rangle}$. Then $\Lambda \simeq \widehat{\mathbb{Z}}$ and, for every positive integer $n$,

$$C_\Pi(\Lambda^n) = N_\Pi(\Lambda^n) = \Lambda.$$

*Manuscript:* 35; 254; 501.

*Proof.* The standard generator maps to a primitive vector of $\Pi^{\mathrm{ab}} = \widehat{\mathbb{Z}}^{2g}$; hence its procyclic closure injects there and is $\widehat{\mathbb{Z}}$. For the first source reduction, $x$ is represented by a nonseparating simple closed curve, and is therefore algebraically simple in the terminology of [10, Definition 1.3]. Proposition 3.6 of that paper gives $N_\Pi(\Lambda^n) = \Lambda$. Centralizers lie in normalizers and contain $\Lambda$, proving the centralizer equality.

For the second reduction, pinch that curve on the genus-$g$ surface. The resulting stable one-node curve has a normalization of genus $g-1$ with two marked inverse branches; at $g = 2$ this is a twice-marked genus-one component, still hyperbolic. Its associated semi-graph of anabelioids has one edge group whose image in the smoothing's profinite group is exactly $\Lambda$, up to the chosen base-path conjugation. Proposition 1.2(ii) of [53] makes this edge group commensurably terminal. An element normalizing $\Lambda^n$ commensurates $\Lambda$, proving the same normalizer equality. These are independent applications of the two stated source results, and neither replaces a closed surface group by a free group. □

*Manuscript:* §35, folio 254, scan 501, equation (4).

**A.11. [Proved] What the abstract profinite presentation specifies.** The full profinite presentation required by scan 318, equation (4), already determines a profinite group with loops isomorphic to $\widehat{\pi}_{g,\nu}$ with its standard loops. Hence A.7 and A.10 apply to every abstract object satisfying that presentation axiom. Merely requiring generators to satisfy the displayed equation would not suffice: $P = \widehat{\mathbb{Z}}^2$, $c_1 = (1, 0)$, $c_2 = (0, 1)$, $c_3 = (-1, -1)$ satisfies the $(0, 3)$ relation, has three injective procyclic peripheral subgroups and the rank-one orientation kernel, but has centre $P$.

*Manuscript:* 26; 161; 318–319.

*Proof.* The universal property gives a continuous homomorphism from the quotient of the free profinite group by the closed normal closure of the relator to $P$. The assertion that this is a defining relation makes this homomorphism an isomorphism; its inverse sends the specified base to the standard base. This gives exactly the comparison map in scan 319, equation (5). For the

weaker condition the explicit abelian example has $c_1 + c_2 + c_3 = 0$ and generates $P$, but has the additional relation $[c_1, c_2] = 1$. Its peripheral inclusion matrix has kernel generated by $(1, 1, 1)$, and its centre contains $(1, 0) \neq 0$. The finite reductions and the rank calculation are checked by the accompanying code. □

*Manuscript:* §26, folio 161, scans 318–319, equations (4)–(5). The words "relation generatrice" are indispensable; weakening them changes the objects by the explicit quotient above.

**A.12. [Proved] The discrete filling isomorphism and all compatibility maps.** Let $U = X \setminus S$ be hyperbolic, choose $b \in U$, and put $U' = U \setminus \{b\}$, $\pi' = \pi_1(U')$, and $\pi = \pi'/\langle\langle L_b \rangle\rangle$. Choose the peripheral representative $L_b \subset \pi'$. The normalization of representatives defines an isomorphism of discrete groups

$$d_b : \mathrm{Out}_{\mathrm{lac}}(\pi'; b) \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{lac}}(\pi).$$

On both sides old punctures may be permuted and orientation may be reversed. It identifies the pure orientation-preserving subgroups, whose common index is $2\nu!$, where $\nu = |S|$. It commutes with forgetting the distinguished puncture, with the maps to $\mathrm{Out}_{\mathrm{lac}}(\pi)$, with orientation characters, with the permutation maps to $\mathfrak{S}_\nu$, and with point-pushing and inner conjugation on their kernels.

*Manuscript:* 18; 93–94; 186–189.

*Proof.* An outer class fixing $[L_b]$ has a representative $\alpha$ preserving $L_b$ itself: conjugate any representative by an element taking its peripheral image back to $L_b$. Two normalized representatives differ by conjugation by an element of $N_{\pi'}(L_b) = L_b$, by A.3. Such conjugation becomes the identity after killing $L_b$. Thus $\alpha$ induces a well-defined actual automorphism $d_b([\alpha])$ of the specified quotient $\pi$. Products of normalized representatives are normalized, so $d_b$ is a homomorphism. Its definition also proves compatibility with the permutation of the remaining loops and with the orientation module, using A.2's filling comparison.

We give the group-theoretic identification of this homomorphism with the topological one. The Dehn–Nielsen–Baer theorem identifies the extended mapping class group of a hyperbolic punctured surface with the outer automorphisms preserving its peripheral conjugacy classes, and in the closed case with $\mathrm{Out}(\pi)$; the precise versions are [22, Theorems 8.1 and 8.8]. The evaluation map of the homeomorphism group of $(X, S)$ at $b$ is a locally trivial fibration over $U$: over a sufficiently small disc, move $b$ to a nearby point by a homeomorphism supported in that disc, which constructs a local section. Its connecting map pushes $b$ round a loop. The identity component of this homeomorphism group is contractible in hyperbolic type, by the surface homeomorphism-group theorem in [74, Lemma 2.1 and Theorem 1.1]; hence the homotopy sequence gives

$$1 \to \pi \xrightarrow{\mathrm{push}} \mathrm{Out}_{\mathrm{lac}}(\pi'; b) \xrightarrow{\mathrm{fill}} \mathrm{Out}_{\mathrm{lac}}(\pi) \to 1.$$

One may alternatively use the extended Birman exact sequence, [22, Theorem 4.6 and the immediately following extended-group remark]. Surjectivity follows directly by moving the image of $b$ back to $b$ along an arc avoiding $S$.

Choose the convention that the point-push associated to a based loop $\gamma$ moves the base-point track in the direction for which the induced based map is $\eta \mapsto \gamma\eta\gamma^{-1}$. This can also be checked on the boundary of the homotopy square traversed by $\eta$ under the isotopy. It proves $d_b \circ \mathrm{push} = \mathrm{conj}$. The filled mapping class gives the same outer automorphism as $d_b$ by construction. Thus the last sequence maps to

$$1 \to \pi \xrightarrow{\mathrm{conj}} \mathrm{Aut}_{\mathrm{lac}}(\pi) \to \mathrm{Out}_{\mathrm{lac}}(\pi) \to 1$$

by the identity on the end groups. Injectivity and surjectivity of $d_b$ follow by the elementary exact-sequence argument: a kernel element belongs to point-pushing and maps trivially by the

injective inner map, and any target element can first be matched in the outer quotient and then in the kernel.

Finally every permutation of old punctures is realized by half-twists supported in small discs along arcs, and an orientation-reversing homeomorphism can be chosen to fix the finite marked set pointwise. Hence the map to $\{\pm 1\} \times \mathfrak{S}_\nu$ is onto; its kernel is precisely the pure orientation-preserving group. Its index is $2\nu!$. This also identifies the subgroups denoted $\mathcal{T}^{00}$ on scan 188 and proves the compatibility checks left on scan 189. □

*Manuscript:* §18, folios 93–94, scans 186–189. On scan 187 the inequality must be $2g+\nu' \geq 4$ when $\nu'$ counts punctures before filling; the meaningful alternative $\leq 4$ contradicts both examples and the stated hyperbolicity requirement. The mathematical argument supports the first alternative without settling the handwriting.

**A.13. [Proved] Filling into the two abelian targets.** Exactly one of the drilled and filled groups is abelian in the two cases $(0,3) \to (0,2)$ and $(1,1) \to (1,0)$. In the first, the homomorphism on automorphisms of the underlying groups with peripheral subgroup classes is

$$C_2 \times C_2 = \mathrm{Out}_{\mathrm{lac}}(F_2; [\langle xy \rangle]) \longrightarrow \mathrm{Aut}(\mathbb{Z}) = C_2,$$

with kernel of order two. Its nontrivial element has representative $x \mapsto y^{-1}$, $y \mapsto x^{-1}$. Its restriction to the positive-orientation subgroup is an isomorphism. Retaining the two indexed peripheral source maps and their orientation line on the annulus enlarges the target automorphism group to $C_2 \times C_2$ and restores the full isomorphism. In the second case the filling map is the isomorphism $\mathrm{GL}_2(\mathbb{Z}) \to \mathrm{GL}_2(\mathbb{Z})$, with positive subgroup $\mathrm{SL}_2(\mathbb{Z})$.

*Manuscript:* 18; 94–97; 189–194.

*Proof.* A.1 lists the two cases. For a three-punctured sphere the extended mapping class group is $\mathfrak{S}_3 \times C_2$: a homeomorphism fixing the three punctures is isotopic, in each orientation class, to the identity or a fixed reflection, by cutting along three joining arcs and applying the disc isotopy. The stabilizer of the distinguished puncture is $C_2 \times C_2$. On $F_2 = \langle x, y \rangle$, the old-puncture transposition $x \leftrightarrow y$ and the reflection $x \mapsto x^{-1}, y \mapsto y^{-1}$ both induce inversion after imposing $xy = 1$. Their product is the displayed nontrivial kernel element. It takes the distinguished loop to its inverse and reverses orientation. Direct free-word and finite-group calculations appear in the attached program.

For an annulus with ordered end sources $\kappa_1(1) = t$ and $\kappa_2(1) = t^{-1}$, an automorphism includes a sign $\epsilon \in \{\pm 1\}$ on $T$ and a permutation $s \in \mathfrak{S}_2$. Compatibility forces the underlying map $t \mapsto t^{\epsilon \operatorname{sgn}(s)}$. Thus there are four automorphisms of this full data, and the map forgetting $s, \epsilon$ is exactly the preceding quotient. This calculation specifies the information lost by retaining only subgroup images.

For the torus, every matrix in $\mathrm{GL}_2(\mathbb{Z})$ is realized by a linear torus homeomorphism fixing the origin. A based torus homeomorphism inducing the identity matrix is isotopic relative to that point to the identity. Indeed the restriction of evaluation to the translation subgroup $\mathbb{R}^2/\mathbb{Z}^2$ is the identity of the torus, so evaluation is surjective on fundamental groups and the point-pushing kernel is zero. The unbased kernel is the identity component: isotope the images of two primitive curves intersecting once back to those curves, use the identity matrix to remove their remaining twists, and cut along them to a disc, where the Alexander isotopy finishes the argument. Removing the fixed point identifies this mapping class group with the peripheral-preserving outer group of the once-punctured torus by Dehn–Nielsen–Baer. Filling induces abelianization, hence the asserted identity map on matrices. Its orientation sign is the determinant. □

*Manuscript:* §18, folios 94–97, scans 189–194. The abelian cases in the margin of scan 194 refer to the filled types; the corresponding drilled groups are both $F_2$.

**A.14. [Known] Primitive loops in finite-step solvable quotients.** Let $\Sigma$ be a nonempty set of primes, $F$ a free pro-$\Sigma$ group of finite rank at least two, and $F^{[j]}$ its closed derived series. Put $F_m = F/F^{[m]}$, $m \geq 2$. For any element $x$ of a free basis and any positive integer $n$,

$$C_{F_m}(x^n) = N_{F_m}(\overline{\langle x \rangle}) = \overline{\langle x \rangle} \simeq \mathbb{Z}_\Sigma, \qquad \mathbb{Z}_\Sigma = \prod_{p \in \Sigma} \mathbb{Z}_p.$$

Every open subgroup of $F_m$ has trivial centralizer. For $\nu \geq 2$, all peripheral loops of the pro-$\Sigma$ group of type $(g, \nu)$ satisfy these formulas in every $F_m$. Their indexed conjugacy classes are pairwise distinct in all hyperbolic types.

*Manuscript:* 13 bis; 57–57 verso; 117–118; strengthened PartA(c).

*Proof.* The power-centralizer formula is [77, Theorem 1.5, finite-rank case]; the source explicitly treats every nonzero integer power, including powers divisible by 2 or 3. We checked its finite-rank proof: Proposition 1.3 uses the finite semidirect product $(\mathbb{Z}/N)[G/\langle \bar{x} \rangle] \rtimes (G \times \langle \bar{x} \rangle)$ to force the centralizer into $\langle x \rangle F_m^{[m-1]}$; the coefficient at the distinguished coset forces the first $G$ coordinate into $\langle \bar{x} \rangle$. At $m = 2$, the Blanchfield–Lyndon injection of $F'/F''$ into $\mathbb{Z}_\Sigma[[F^{\mathrm{ab}}]]^r$ and injectivity of multiplication by $\bar{x}^n - 1$ remove the intersection with $F_m'$. That injectivity is proved by passing from $C_M$ to $C_{kM}$: invariant coefficients downstairs are divisible by every $\Sigma$-integer $k$, so are zero. The finite-rank induction passes to finite abelian covers where a positive power of $x$ is a free basis element. These are exactly the finite-rank hypotheses used here; no infinite-rank inverse-limit assertion is used.

If $u$ normalizes $\langle x \rangle$, write $uxu^{-1} = x^\lambda$, $\lambda \in \mathbb{Z}_\Sigma^\times$. The primitive abelianization coordinate gives $\lambda = 1$, so the normalizer lies in the already computed centralizer. An open subgroup contains positive powers of two distinct free generators; their cyclic centralizers intersect trivially because their abelianization coordinates are independent. This proves the open-centralizer assertion directly.

Each peripheral generator for $\nu \geq 2$ belongs to a free basis after eliminating another boundary generator. For $\nu \geq 3$ distinct peripheral abelianization lines separate the conjugacy classes. When $\nu = 2$ and $g \geq 1$, kill $c_1$ and all but one handle pair, mapping onto a free metabelian group on $a, b$. The image of $c_1$ is 1 and that of $c_2$ is $[a, b]^{-1} \neq 1$; the latter has infinite order in the upper unitriangular $3 \times 3$ group over $\mathbb{Z}_\Sigma$. Hence they cannot be conjugate, or commensurable with finite indices. These maps factor through every $F_m$, $m \geq 2$. □

*Manuscript:* §13 bis, scans 117–118, and the strengthened finite-step task in Part A(c).

**A.15. [Proved] The compact intersection calculation used in descent.** Let $(H_\alpha)$ be a decreasing directed family of closed subgroups of a profinite group $P$, and put $K = \bigcap_\alpha H_\alpha$. Then

$$K^{[j]} = \bigcap_\alpha H_\alpha^{[j]} \quad (j \geq 0).$$

For a closed subgroup $L \subset P$ and decreasing closed normal subgroups $N_\alpha$ of $P$, one also has $\bigcap_\alpha LN_\alpha = L(\bigcap_\alpha N_\alpha)$.

*Manuscript:* Auxiliary compact-group lemma for finite-step PartA(c).

*Proof.* For any open normal $V \subset P$, the images of $H_\alpha$ in the finite group $P/V$ eventually stabilize. Their eventual image equals that of $K$: each compatible coset fibre is a decreasing family of nonempty compact sets. Taking a fixed number of derived subgroups commutes with homomorphic images, and taking the inverse limit of all finite images recovers the closed derived subgroup. This proves the first equality. For the second equality, an element $z$ of every $LN_\alpha$ gives nonempty closed sets $L \cap zN_\alpha$; compactness makes their intersection nonempty. A member of that intersection gives the required single factorization of $z$. □

**A.16. [Proved] The one-puncture solvable threshold, with a finite certificate.** Let $F$ be the pro-$\Sigma$ completion of $\pi_{g,1}$, $g \geq 1$, let $c = \prod_{j=1}^{g}[a_j, b_j]$ (the inverse peripheral generator), and write $c$ for its image in $F_m = F/F^{[m]}$. For every positive integer $n$,

$$\begin{aligned} m = 2: &\quad C_{F_2}(c^n) = N_{F_2}(\overline{\langle c\rangle}) = F_2', \\ m \geq 3: &\quad C_{F_m}(c^n) = N_{F_m}(\overline{\langle c\rangle}) = \overline{\langle c\rangle}. \end{aligned}$$

In the first line the inclusion $\overline{\langle c\rangle} \subset F_2'$ is strict. Thus peripheral rigidity fails at the smallest solvable depth $m = 2$ and is restored at every $m \geq 3$, for the full profinite and every pro-$p$ version.

*Manuscript:* 13 bis; 57 verso; 118; strengthened PartA(c).

*Proof.* Let $H \subset F$ run over open normal subgroups containing $F'$, with $|F/H| \geq 3$. They form a family whose intersection is $F'$. The corresponding finite abelian covering of the once-punctured surface has exactly $d = |F/H|$ punctures, because $c \in F'$ has trivial image in the deck group. Choose the cusp above the original based peripheral loop. Its loop is $c$ itself; the other loops are its conjugates by deck representatives. The submodule of $H^{\mathrm{ab}}$ generated by these cusps is

$$\mathbb{Z}_\Sigma^{F/H}/\mathbb{Z}_\Sigma\left(\sum_{q\in F/H} e_q\right),$$

with the deck group acting by left translation. This follows either from the relative surface homology sequence of A.2 or directly from the free-group peripheral presentation. For $d \geq 3$ the lines generated by distinct $e_q$ in this module are distinct; moreover $ne_q = ae_1$, for $0 \neq n \in \mathbb{Z}$ and $a \in \mathbb{Z}_\Sigma$, forces $q = 1$ and $a = n$. To check this last assertion, lift the equality to the free module and compare a coordinate different from both 1 and $q$; its coefficient forces the possible multiple of $\sum e_q$ to be zero, after which the coordinates at $q$ and 1 give the result.

For $m \geq 2$, $F^{[m]} \subset H'$, so the natural map $H \to H^{\mathrm{ab}}$ factors through the image of $H$ in $F_m$. Let $u \in F$ represent an element of $C_{F_m}(c^n)$. The relation $uc^nu^{-1} = c^n$ modulo $F^{[m]}$ implies $ne_{\bar{u}} = ne_1$ in the cusp submodule of $H^{\mathrm{ab}}$. Hence $u \in H$ for every such $H$, so its image belongs to $F_m'$. The same argument for a normalizer, using $ucu^{-1} = c^\lambda$ modulo $F^{[m]}$, $\lambda \in \mathbb{Z}_\Sigma^\times$, gives $u \in F'$ and $\lambda = 1$. The subgroup $\overline{\langle c\rangle}$ is indeed $\mathbb{Z}_\Sigma$: the unitriangular $3 \times 3$ quotient used in A.1 proves injectivity of all its prime components. Thus its normalizer equals its centralizer.

When $m = 2$, the derived subgroup is abelian, contains $c$, and centralizes it. This gives equality with $F_2'$. Now let $m \geq 3$ and let $uF^{[m]}$ centralize $c^n$. We have proved $u \in F' \subset H$. Since $F' \subset H$, $F^{[m]} = (F')^{[m-1]} \subset H^{[m-1]}$; therefore there is a well-defined continuous epimorphism

$$H/F^{[m]} \longrightarrow H/H^{[m-1]}.$$

The group $H$ is free pro-$\Sigma$, and $c$ is a member of a free basis because this cover has at least two punctures. By A.14, applied at depth $m - 1 \geq 2$, the image of $u$ in this quotient belongs to $\overline{\langle c\rangle}$. Consequently

$$u \in \bigcap_H \overline{\langle c\rangle} H^{[m-1]} = \overline{\langle c\rangle}\left(\bigcap_H H^{[m-1]}\right) = \overline{\langle c\rangle}(F')^{[m-1]} = \overline{\langle c\rangle}F^{[m]},$$

where both interchanges are proved in A.15. This proves the second line.

For strictness in the first line, set $a = a_1$, $b = b_1$, and kill the remaining handle pairs. In $\mathrm{UT}_4(\mathbb{F}_p)$ take

$$x = 1 + E_{12} + E_{23}, \qquad y = 1 + E_{34}.$$

Their group is metabelian; direct multiplication gives $[x, y] = 1+E_{24}$ and $x[x, y]x^{-1} = 1+E_{24}+E_{14}$. The second matrix is not a power of the first. Thus $aca^{-1} \in F_2'$ centralizes $c$ but does not belong

to its procyclic closure. For every $p \in \Sigma$ this quotient factors through $F_2$ and proves strictness; the attached program independently enumerates and verifies the cases $p = 2, 3, 5$. In particular there is no excluded residue-characteristic caveat. □

*Manuscript:* §13 bis, scan 118, in the strengthened finite-step question of Part A(c). This assertion does not assert a new literature priority; the searched sources and the proof audit are listed in the report.

**A.17. [Proved] Centre and torsion in all finite-step surface quotients.** Let $P$ be the full profinite or pro-$\Sigma$ completion of a hyperbolic surface group, open or closed, and let $P_m = P/P^{[m]}$. For every $m \geq 2$, $P_m$ is torsion-free and centre-free. This verifies the form of [77, Theorem 2.9 and Corollary 2.10] needed here by the following argument from finite covers and transfer.

*Manuscript:* Strengthened PartA(c); Yamaguchi2026 Theorem2.9 and Corollary2.10.

*Proof.* Every open subgroup of $P$ is the pro-$\Sigma$ group of a finite covering surface, so its abelianization is a free $\mathbb{Z}_\Sigma$-module. A closed subgroup $K$ is the intersection of its open neighbourhood subgroups; by A.15 its abelianization is the inverse limit of their abelianizations. More explicitly, a compatible system in those abelianizations lifts to a compatible coset system in the compact group, and the kernel is the intersection of the closed derived subgroups. Thus $K^{\mathrm{ab}}$ is torsion-free. Applying this to each $P^{[j]}$ and inducting in

$$1 \to (P^{[m-1]})^{\mathrm{ab}} \to P_m \to P_{m-1} \to 1$$

proves torsion-freeness, starting with the torsion-free abelianization.

We also need the faithful action of a finite deck group $Q = H/N$ on $N^{\mathrm{ab}}$, for open $N \trianglelefteq H \subset P$. Choose $\ell \in \Sigma$. If a subgroup $Q_0$ acts trivially on $H^1$ of the covering surface with $\mathbb{Q}_\ell$ coefficients, transfer gives an isomorphism of that $H^1$ with the $H^1$ of the quotient by $Q_0$. The two surfaces are both proper or both nonproper, and their Euler characteristics are consequently equal. Multiplicativity of Euler characteristic gives $\chi(Y) = |Q_0|\chi(Y/Q_0)$. Both are negative, so $Q_0 = 1$. The comparison with $N^{\mathrm{ab}} \otimes_{\mathbb{Z}_\Sigma} \mathbb{Q}_\ell$ is the canonical duality between first homology and first cohomology. Thus the deck action on $N^{\mathrm{ab}}$ is faithful.

Put $K = P^{[m-1]}$, $A = K/K'$, and $Q = P/K'$. For an open normal subgroup $N \subset Q$ containing $A$, its preimage $\widetilde{N} \subset P$ contains $K$, hence $K' \subset \widetilde{N}'$. Therefore $N^{\mathrm{ab}} = \widetilde{N}^{\mathrm{ab}}$, and the preceding deck faithfulness implies that an element of $Z(Q)$ belongs to every such $N$. Their intersection is $A$.

We finish by proving $A^{P/K} = 0$. For any open normal $V \subset P$ containing $K$, write $j : V^{\mathrm{ab}} \to P^{\mathrm{ab}}$ for inclusion and $t : P^{\mathrm{ab}} \to V^{\mathrm{ab}}$ for transfer. The composite $tj$ on the $P/V$-invariants is multiplication by $[P : V]$, which is injective because $V^{\mathrm{ab}}$ is torsion-free. Hence inclusion is injective on those invariants. Passing to the inverse limit over $V$ shows that $K^{\mathrm{ab},P/K} \to P^{\mathrm{ab}}$ is injective. But $K \subset P'$, so this map is zero. Therefore $A^{P/K} = 0$. Since $Z(Q) \subset A$, the centre is zero, as claimed. This proof checks precisely the transfer and finite-cover steps used in the cited unrefereed result. □

**A.18. [Proved] The remaining affine solvable rigidity list and a closed bound.** For all hyperbolic types with $\nu > 0$, and all $m \geq 2$, the maximal $m$-step-solvable full profinite or pro-$\Sigma$ quotient has trivial centre and trivial centralizers of open subgroups, and is torsion-free. Peripheral conjugacy classes remain distinct. For $\nu \geq 2$ all peripheral normalizer and integer-power-centralizer equalities hold at every $m \geq 2$; for $\nu = 1$ their exact values are those of A.16. In an extension obtained by quotienting a geometric action extension by the $m$th closed derived subgroup of its kernel, the equality $N_E(L_s) = E_s$ holds in the former cases and at $m \geq 3$ in the latter. At $\nu = 1, m = 2$ its correct value is $N_E(L_s) = F_2' E_s$, with the intersection $F_2' \cap E_s = L_s$.

In all these cases every transporter $\mathrm{Trans}_E(L_s^d, L_s)$, $d \geq 1$, equals the same displayed normalizer. Since the kernels are centre-free by A.17, the extension reconstruction of A.6 also applies to all these solvable quotients. Standard free-basis generators satisfy the power-centralizer formula at every $m \geq 2$ by A.14. For a closed hyperbolic group $P$ and an open subgroup $H \subset P_m$, the proven universal bound is

$$C_{P_m}(H) \subseteq P_m^{[m-1]}.$$

*Manuscript:* 13 bis; 57 verso; 118; 35; 252 verso–254; 498–501; strengthened PartA(c).

*Proof.* The first assertions are A.14, A.16, and A.17. For the extension calculation, the quotient map is injective on the peripheral group, since the peripheral generator has infinite order in $P_2$ by A.1. The image of a geometric peripheral lift group still maps onto the puncture stabilizer. The kernel of the normalizer map to that stabilizer is the normalizer calculated in A.14 or A.16, so the same lifting argument as A.5 proves the equalities. In the metabelian one-puncture case there is just one puncture, so the stabilizer is the whole acting group and multiplication by $E_s$ gives the asserted $F_2' E_s$.

It remains to justify the one-sided transporter equality in these quotients. Suppose $eL_i^d e^{-1} \subset L_i$. The action sends $L_i$ to a conjugate of some $L_j$ and its chosen generator to a conjugate of $c_j^{\lambda}$, with $\lambda \in \mathbb{Z}_\Sigma^\times$. For $\nu \geq 3$, the primitive abelianization lines imply $j = i$. For $\nu = 2$, $g \geq 1$, the quotient killing $c_i$ and all but one handle pair sends $c_j$ to a nonzero commutator of infinite order in $\mathrm{UT}_3(\mathbb{Z}_\Sigma)$; thus $j \neq i$ would contradict the given containment. Choose an element of $E_i$ with the same image as $e$ in the acting group and divide it out. The resulting element $u$ lies in $P_m$ and satisfies $uc_i^d u^{-1} = c_i^a$ for some $a \in \mathbb{Z}_\Sigma$. The primitive abelianization coordinate forces $a = d$, so A.14 puts $u$ in $L_i$. This proves the equality for $\nu \geq 2$. For $\nu = 1$, the acting group already fixes the only puncture. After the same division by $E_i$, apply the finite abelian cover calculation in A.16 to $uc^d u^{-1} = c^a$. It forces $a = d$ and reduces the transporter to the integer-power centralizer calculated there. This gives $E_i$ at $m \geq 3$ and $F_2' E_i$ at $m = 2$.

The closed bound recorded here is strengthened to triviality in A.19 below. For this bound, put $A = P_m^{[m-1]}$ and choose any open normal $N \subset P_m$ containing $A$. The group $N^{\mathrm{ab}}$ is the torsion-free abelianization of its preimage in $P$, as in A.17. If $z$ centralizes $H$, its conjugation action fixes the image of $H \cap N$ in $N^{\mathrm{ab}}$. This image has finite index, so some nonzero integer multiple of every element is fixed. Torsion-freeness then makes the action trivial on all of $N^{\mathrm{ab}}$. Deck faithfulness from A.17 implies $z \in N$. Intersecting all these $N$ yields $z \in A$. □

*Manuscript:* §13 bis, scan 118; §35, scans 498–501; additional finite-step task A(c).

**A.19. [Proved] Closed finite-step power centralizers and slimness.** Let $P$ be the pro-$\Sigma$ completion of a closed oriented surface group of genus $g \geq 2$, let $m \geq 2$, and put $P_m = P/P^{[m]}$. Let $x$ be represented by a nonseparating simple closed curve, in particular by any standard handle generator. Then for every positive integer $n$,

$$C_{P_m}(x^n) = N_{P_m}(\overline{\langle x^n \rangle}) = \overline{\langle x \rangle} \simeq \mathbb{Z}_\Sigma.$$

Consequently $C_{P_m}(H) = 1$ for every open subgroup $H$. This includes the full profinite and every pro-$p$ quotient, without a restriction at $p = 2$ or 3.

*Manuscript:* 35; 254; 501; strengthened PartA(c).

*Proof.* We first give the completed module construction required for the metabelian case. Let $F$ be a free pro-$\Sigma$ group on $z_1, \ldots, z_r$, let $A = F/F'$, and put $R = \mathbb{Z}_\Sigma[[A]]$. The continuous homomorphism

$$\mu : F \longrightarrow R^r \rtimes A, \qquad z_i \longmapsto (e_i, \bar{z}_i),$$

exists by freeness; $A$ acts by multiplication on the coefficient module, so the product in the target is $(v,a)(w,b) = (v + aw, ab)$. Its kernel is exactly $F''$. The inclusion $F'' \subset \ker \mu$ follows because the target is metabelian. To prove the reverse inclusion without assuming any exactness result about completed chain complexes, take an arbitrary finite metabelian quotient $G$ of $F$. Write $K = G'$, $Q = G/K$, and choose a section $t : Q \to G$ with $t(1) = 1$. There is an injective homomorphism

$$G \longrightarrow K^Q \rtimes Q, \qquad g \longmapsto (f_g, \bar{g}), \qquad f_g(q) = t(q)\, g\, t(q\bar{g})^{-1}.$$

Indeed $f_{gh}(q) = f_g(q) f_h(q\bar{g})$, and, when $\bar{g} = 1$, evaluation at $q = 1$ recovers $g$. Use additive notation for the finite abelian group $K$. The finite module $K^Q$ is an $R$-module through $A \to Q$ and right translation on $Q$. Sending $e_i$ to the function coordinate of the image of $z_i$ gives a continuous $R$-linear map $R^r \to K^Q$. The resulting homomorphism of semidirect products makes the homomorphism from $F$ factor through $\mu$, since it agrees on every free generator. Hence $\ker \mu$ maps trivially to every finite metabelian quotient of $F$. These quotients separate $F/F''$, proving the assertion. In particular the function coordinate gives an injective $R$-linear morphism

$$F'/F'' \longhookrightarrow R^r.$$

Its image is a closed $R$-submodule: conjugation gives the $A$-action, integral powers give the scalar action, and continuity extends these to the completed group ring.

Apply this construction with $r = 2g$ and

$$P = F/\langle\langle r_0 \rangle\rangle, \qquad r_0 = \prod_{i=1}^{g} [a_i, b_i], \qquad A = P^{\mathrm{ab}} = \mathbb{Z}_\Sigma^{2g}.$$

The function coordinate of $r_0$ is

$$v = (1 - \bar{b}_1, \bar{a}_1 - 1, \ldots, 1 - \bar{b}_g, \bar{a}_g - 1) \in R^{2g}.$$

This is obtained by multiplying the four factors of each commutator in $R^{2g} \rtimes A$. Since $r_0 \in F'$, its closed normal closure maps in $F'/F''$ onto exactly $Rv$: its conjugates give the $A$-multiples of $v$, their products give their integral sums, and their closure is $Rv$. The last module is closed because it is the image of the compact ring $R$. Thus the natural quotient maps induce the injective comparison

$$B := P'/P'' \longhookrightarrow R^{2g}/Rv. \tag{A.19.1}$$

Indeed $P''$ is the image of $F''$ and the kernel in $F'/F''$ is precisely the just calculated module. This establishes the needed completed Fox-module comparison directly.

Choose a geometric basis with $x = a_1$, and set $h = \bar{a}_1^n - 1$. We use two elementary facts about these completed group rings. First, multiplication by $h$ on $R$ is injective. Second, multiplication by $\bar{a}_2 - 1$ on $R/hR$ is injective. Here

$$R/hR \xrightarrow{\sim} \mathbb{Z}_\Sigma[[A/\overline{\langle \bar{a}_1^n \rangle}]] \tag{A.19.2}$$

is the map induced by the group quotient. Its kernel is the closed ideal generated by $h$; that ideal equals $hR$ since $hR$ is compact. To prove the injectivity assertions, one can use the following coefficient calculation. In a completed group ring of $\mathbb{Z}_\Sigma \times D$, consider an element invariant under the chosen infinite cyclic coordinate, or under its $n$th power. At finite coefficient level $\mathbb{Z}/k$ and finite coordinate level $C_M$, enlarge the coordinate level to $C_{kM}$, taking $M$ divisible by the $\Sigma$-part of $n$ in the second case. Invariance makes the coefficients constant on each orbit. The projection down to $C_M$ sums $k$ equal coefficients on each orbit fibre and therefore is zero modulo $k$. This works with every finite quotient of the remaining factor $D$. All finite images of the original element vanish, so the element is zero. Apply this argument first to the $a_1$ coordinate of $R$, and

then to the independent $a_2$ coordinate in the group of (A.19.2). It applies to arbitrary $\Sigma$, because $k$ and the chosen coordinate orders have prime factors in $\Sigma$.

These ring facts show that multiplication by $h$ is injective on $R^{2g}/Rv$. If $hz = vt$, inspect the $b_2$ coordinate, which is $hz_{b_2} = (\bar{a}_2 - 1)t$. Reducing modulo $h$ and using the second injectivity gives $t = ht'$ for some $t' \in R$. The first injectivity then gives $z = vt'$ in every coordinate. Consequently (A.19.1) implies

$$B^{\langle x^n \rangle} = 0. \tag{A.19.3}$$

For completeness, we next constrain the abelianization of an arbitrary centralizer element, at every depth $m \geq 2$. Put $V = \mathbb{Z}_\Sigma^{2g}$ with its ordered handle basis and put $W = \bigwedge^2 V/\mathbb{Z}_\Sigma\omega$, where $\omega = \sum_i a_i \wedge b_i$. For every $\Sigma$-integer $k$, form the finite class-two group with underlying set $(V/kV) \times (W/kW)$ and product

$$(q, w)(q', w') = (q + q', w + w' + \beta(q, q')), \qquad \beta(q, q') = \sum_{i<j} q_i q'_j \, z_i \wedge z_j.$$

Its commutator is $(0, q \wedge q')$. The standard surface relation maps to $\omega = 0$, so sending the handle generators to their coordinate vectors defines a quotient homomorphism from $P$. The finite group is metabelian and a $\Sigma$-group; the homomorphism therefore factors through every $P_m$, $m \geq 2$. If $u \in P_m$ centralizes $a_1^n$ and has abelianization $q$, these homomorphisms give $n(q \wedge a_1) = 0$ in $W$. The module $W$ is free because $\omega$ is primitive, so multiplication by the nonzero integer $n$ is injective. Write $q \wedge a_1 = t\omega$. The coefficient of $a_2 \wedge b_2$ on the left is zero; on the right it is $t$. Hence $t = 0$, and the remaining coordinates give $q = \lambda a_1$ for some $\lambda \in \mathbb{Z}_\Sigma$. The procyclic group generated by $a_1$ injects into $V$, so dividing $u$ by $a_1^\lambda$ puts it in $P'_m$ without changing its centralization property. At $m = 2$, (A.19.3) makes this remaining element trivial, proving the centralizer formula.

We now induct on $m$, simultaneously for every genus at least two and every nonseparating simple closed curve. Suppose $m \geq 3$, and divide a centralizer element as just explained so that it is represented by $u \in P'$. Let $H$ run over all open normal subgroups of $P$ containing $P'$, and let $d$ be the order of $x$ in $P/H$. The corresponding finite abelian covering of the closed surface has genus at least two. The based lift of the simple curve $x$ is represented by $x^d$, traversed once. This lift is a simple nonseparating curve: its integral homology class pushes forward to $d[x] \neq 0$, whereas a separating curve has zero homology class. It is therefore a standard handle generator for some geometric basis of the covering surface group $H$.

Since $P' \subset H$, we have $P^{[m]} \subset H^{[m-1]}$. The quotient morphism

$$H/P^{[m]} \longrightarrow H/H^{[m-1]}$$

is consequently defined. The image of $u$ centralizes $(x^d)^n$ in its target. By the induction hypothesis at depth $m - 1$, that image belongs to $\overline{\langle x^d \rangle}$. Hence

$$u \in \bigcap_H \overline{\langle x \rangle} H^{[m-1]} = \overline{\langle x \rangle} \left( \bigcap_H H \right)^{[m-1]} = \overline{\langle x \rangle} P^{[m]},$$

by A.15 and $\bigcap_H H = P'$. The intersection of $\overline{\langle x \rangle}$ with $P'$ is trivial, again by its primitive abelianization coordinate. Since $u \in P'$, its image in $P_m$ is trivial. This completes the induction.

If an element normalizes $\overline{\langle x^n \rangle}$, its conjugation sends $x^n$ to $x^{n\lambda}$ for a unit $\lambda \in \mathbb{Z}_\Sigma^\times$. Abelianization gives $n\lambda = n$, whence $\lambda = 1$, so the normalizer equals the calculated centralizer. Finally an open subgroup contains positive integer powers of $a_1$ and $b_1$. Their calculated centralizers have trivial intersection because they inject into distinct independent abelianization coordinates. This proves slimness. □

*Manuscript:* §35, scan 501, equation (4), and the finite-step strengthening in Part A(c). The proof establishes the exact module comparison and all closed-group descent maps used; it does not rely on the unknown general slimness statement mentioned before Proposition 2.6 in [77], nor claim a literature priority.

**A.20. [Proved] The typed completion comparison for filling.** For the two hyperbolic groups in A.12, put $\Pi' = \widehat{\pi'}$ and $\Pi = \widehat{\pi}$. There is a commutative diagram of profinite groups

$$\begin{array}{ccc} \widehat{\mathrm{Out}_{\mathrm{lac}}(\pi'; b)} & \xrightarrow{\widehat{d_b}} & \widehat{\mathrm{Aut}_{\mathrm{lac}}(\pi)} \\ \downarrow c_{\mathrm{out}} & & \downarrow c_{\mathrm{aut}} \\ \mathrm{Out}_{\mathrm{lac}}(\Pi'; b) & \xrightarrow{d_b^{\mathrm{cont}}} & \mathrm{Aut}_{\mathrm{lac}}(\Pi). \end{array}$$

The upper horizontal map is an isomorphism. The lower horizontal map is the normalized filling homomorphism on all continuous loop automorphisms. The vertical maps are induced by extending discrete automorphisms to group completions. The same construction with pro-$p$ completions of the surface groups still has the full profinite completions of the discrete automorphism groups in its upper row. These assertions specify the precise consequence of the discrete filling theorem; no isomorphism assertion for the lower horizontal or vertical arrows is used.

*Manuscript:* 18; 93–94; 186–189; comparison with profinite PartA(c).

*Proof.* The presentation gives $\Pi'/\overline{\langle\langle L_b \rangle\rangle} = \Pi$. By A.7, $N_{\Pi'}(L_b) = L_b$. Normalize an outer representative to preserve $L_b$, then pass to this quotient. Two normalized representatives differ by conjugation by an element of $L_b$, which dies in the quotient. This proves that $d_b^{\mathrm{cont}}$ is a well-defined homomorphism. It is continuous: the closed subgroup of $\mathrm{Aut}(\Pi')$ preserving $L_b$ is compact, its quotient onto the indicated outer group has kernel the inner automorphisms by $L_b$, and the continuous quotient action factors through this quotient. The same proof works for the pro-$p$ groups using A.7.

Every discrete automorphism extends uniquely to an automorphism of its completion; extensions of inner automorphisms remain inner. Since the resulting target automorphism groups are profinite, the extension maps themselves extend to the full profinite completions of the discrete automorphism groups. Normalizing a discrete representative and then completing gives the same quotient automorphism as completing and then normalizing. Thus the square commutes on the dense discrete subgroup and therefore everywhere. A.12 implies that its upper arrow is an isomorphism. The quotient maps $\widehat{\pi} \twoheadrightarrow \pi^{(p)}$ and $\widehat{\pi'} \twoheadrightarrow (\pi')^{(p)}$ have characteristic kernels and commute with filling, giving the further comparison from the full square to its pro-$p$ version.

For odd $p$, replacing the upper groups by their pro-$p$ completions would in general fail to define these same maps: an orientation-reversing involution is killed by every homomorphism to a pro-$p$ group, whereas its action on the pro-$p$ orientation line is multiplication by $-1 \neq 1$. This is an explicit obstruction to that proposed factorization, rather than an identification of two differently typed completion operations. □

## 2 Finite actions, sections, and rotation data

In this section $U = X \setminus S$ is an orientable connected surface of finite type, unless another hypothesis is explicitly stated. A universal covering $p : \widetilde{U} \to U$ is fixed, and $\pi = \mathrm{Deck}(p)$ acts on the left. For a faithful action of a finite group $G$ on $U$, we write $E$ for the group of all lifts of its elements. Thus $1 \to \pi \to E \to G \to 1$ is an extension of *discrete* groups. A partial section means a finite subgroup of $E$ projecting isomorphically onto a subgroup of $G$; its equivalence relation throughout this section is conjugacy by $\pi$. We use $G^+$ for the orientation-preserving subgroup. The claims below also specify the corrections required when $G^+ = 1$.

## 2.1 The plane lemma and the section correspondence

**B.1. [Proved] Finite plane actions and simultaneous local models.** Every faithful action of a finite group $H$ on $\mathbb{R}^2$ by homeomorphisms is conjugate to an orthogonal action. When $H^+ \neq 1$, $H^+$ is cyclic, $H$ is cyclic or dihedral, and $(\mathbb{R}^2)^H$ is a single point. A nontrivial orientation-reversing involution has a fixed line. Every stabilizer in a finite action on a surface with universal cover $\mathbb{R}^2$ has an invariant disc on which the whole stabilizer acts by an orthogonal action. These conclusions supply the two local assertions used in §15, folios 70–74, scans 141–149, and §33, scan 440.

*Manuscript:* 15; 70-74; 141-149; 33; 222; 440.

*Proof.* We use precisely Kolev's Theorem 1.2: a compact subgroup of $\mathrm{Homeo}(S^2)$ is conjugate to a closed subgroup of $O(3)$ [44]. Each homeomorphism of $\mathbb{R}^2$ extends to its one-point compactification, and the finite group fixes the added point $\infty$. After the stated conjugacy the group fixes a point $q$ of the round sphere. Its action on $q^\perp$ gives an injective homomorphism into $O(2)$. Stereographic projection from $q$ is equivariant and identifies the action on $S^2 \setminus \{q\}$ with that linear action. Finite subgroups of $SO(2)$ are the cyclic groups of roots of unity: an element of least positive angle generates the group, by division with remainder of angles. An orientation-reversing orthogonal matrix in dimension two is a reflection and conjugates every rotation to its inverse. This proves the cyclic/dihedral assertion and all assertions about fixed sets.

For the local statement, let $H$ stabilize $x \in U$ and choose $\widetilde{x} \in p^{-1}(x)$. For each $h \in H$ there is a unique lift fixing $\widetilde{x}$. Uniqueness makes the assignment a homomorphism, and it is injective because its projection to $U$ is the original action. The preceding plane argument linearizes these lifts simultaneously. Pull back a sufficiently small round disc about the common fixed point; choose it inside an evenly covered neighbourhood of $\widetilde{x}$. Its projection is the required $H$-invariant disc. The same assertion for a sphere follows directly from Kolev's theorem. In particular the fixed locus of an anti-involution is locally a diameter, with the involution interchanging its two sides. No differentiability of the original action has entered the argument. The single-element versions are Constantin–Kolev, Lemma 2.5, Theorems 3.1 and 4.1, and Corollary 4.2 [18]. □

**B.2. [Refuted] The omitted orientation hypothesis in scans 146–147.** The displayed lemma on scan 147, taken without its preceding orientation hypothesis, is false. Its correct form is: an orientation-preserving finite-order homeomorphism $u$ of $\mathbb{C}$ commuting with $T(z) = z + 1$ and satisfying $u(0) = 0$ is the identity. The quotient $\mathbb{C}/\mathbb{Z}$ provides a complete proof along the route proposed in the manuscript.

*Manuscript:* 15; 72-73; 146-147.

*Proof.* Complex conjugation is a counterexample to the unqualified sentence: it has order two, fixes zero, and commutes with $T$. For the corrected statement, $u(k) = k$ for every integer $k$. B.1 gives at most one fixed point for a nonidentity orientation-preserving periodic plane homeomorphism, so $u = 1$.

Here is the quotient argument. Commutation with $T$ gives a finite-order homeomorphism $\bar{u}$ of the annulus $A = \mathbb{C}/\mathbb{Z}$, inducing $+1$ on $\pi_1(A) = \mathbb{Z}$. It preserves the orientation of $A$. The two end loops, oriented as boundary loops of a compact subannulus, represent opposite homology classes. Consequently an orientation-preserving map inducing $+1$ on $H_1(A;\mathbb{Z})$ fixes each end. Filling the ends extends $\bar{u}$ to an orientation-preserving periodic homeomorphism of $S^2$ fixing both added points and [0]. By Constantin–Kolev, Theorem 4.1, a nonidentity such map has exactly two fixed points. Hence $\bar{u} = 1$. Its lift $u$ is a deck transformation $T^k$; $u(0) = 0$ forces $k = 0$. On scan 146 the equation $G = G^\circ$ is exactly what makes the lifted action orientation

preserving. The mathematical repair therefore restores an inherited hypothesis; it does not justify a spectral-sequence argument. □

**B.3. [Proved] Sections and isolated fixed points.** Let $\widetilde{U} \cong \mathbb{R}^2$. A faithful finite action on $U$ with $G^+ \neq 1$ has a bijection

$$U^G \longrightarrow \mathrm{Sec}(E \to G)/\pi.$$

For every section subgroup $K$, $C_\pi(K) = 1$. Existence of a section forces $G$ to be cyclic or dihedral. More generally, every nontrivial finite subgroup of $E$ whose orientation-preserving part is nontrivial is contained in exactly one maximal finite subgroup of $E$. For an orientation-preserving action the map from points with nontrivial stabilizer to maximal partial-section classes is a $G$-equivariant bijection. These are the theorem and corollary of §15, folios 73–75, scans 148–151, with their stated orientation hypotheses.

*Manuscript:* 15; 73-75; 148-151.

*Proof.* For $x \in U^G$, the lifts of $G$ fixing a chosen point over $x$ form a section. Changing that point conjugates the section by the corresponding element of $\pi$. Conversely a section $K$ is a finite faithful group of plane homeomorphisms. B.1 supplies a unique fixed point $\widetilde{x}$. Its image is fixed by $G$, and the section fixing $\widetilde{x}$ is $K$. These constructions are inverse: conjugate sections have deck-translated fixed points and hence the same image in $U$. An element of $C_\pi(K)$ preserves the unique fixed point. A deck transformation with a fixed point is the identity. This proves the centralizer assertion. B.1 gives the possible finite groups.

Let $K \leq E$ have nontrivial positive part and fix $\widetilde{x}$. The stabilizer $E_{\widetilde{x}}$ is finite, since it intersects $\pi$ trivially and injects into the finite group $G$. Any finite subgroup containing $K$ has a fixed point by B.1; that point must be $\widetilde{x}$. It is therefore contained in $E_{\widetilde{x}}$. This proves uniqueness as an actual subgroup, before any conjugacy quotient. In the orientation-preserving case every nontrivial partial section has this property. Applying the same construction to each stabilizer $G_x$ gives the final bijection and its equivariance. The argument also applies to a nonorientable $U$ whenever the finite lifted group has nontrivial positive part; in particular the cardinality condition $|G| \geq 3$ on scan 149 ensures it. □

**B.4. [Proved] Reflection sections, ovals, and punctured ovals.** Let $\sigma$ be an orientation-reversing involution of $U$ with $\widetilde{U} \cong \mathbb{R}^2$, and let $E \to C_2$ be its lift extension. There is a bijection

$$\pi_0(U^\sigma) \longrightarrow \mathrm{Sec}(E \to C_2)/\pi.$$

For the section $s$ associated with a component $Z$, inclusion induces an isomorphism $\pi_1(Z) \cong C_\pi(s)$. Thus the centralizer is 1 for a line and $\mathbb{Z}$ for an oval. This includes punctures fixed or permuted by $\sigma$ and proves §33, folios 222–225, scans 440–445, in the required noncompact cases.

*Manuscript:* 33; 222-225; 440-445.

*Proof.* Each order-two lift $s$ is a reflection of the plane by B.1. Its fixed set $L$ is a line. The restriction $p : L \to U^\sigma$ is locally a homeomorphism onto the real locus, by the local model in B.1. Its image is the entire component $Z$ containing any of its points. To verify this without a properness assumption, lift a path in $Z$ starting at $p(\widetilde{x})$: applying $s$ gives another lift with the same initial point, so uniqueness of path lifting fixes the lifted path. The same argument for a homotopy shows that $L \to Z$ is its universal covering. Every section has such a fixed line, so every section class arises. Two fixed lines projecting to the same component contain lifts of the same point; a deck transformation taking one lift to the other conjugates the associated involutions. The two lifts fixing the resulting point agree, because lifts of a specified base homeomorphism are unique once the value at one point is fixed.

A deck transformation preserving $L$ normalizes $\langle s\rangle$ and hence centralizes $s$. Conversely a deck transformation centralizing $s$ preserves $L$. Therefore the deck group of $L \to Z$ is exactly $C_\pi(s)$. Since $L \cong \mathbb{R}$, this proves both the asserted isomorphism and injectivity of $\pi_1(Z) \to \pi$. A finite-type real locus is a finite union of circles and open intervals after its real punctures are removed. Each interval is homeomorphic to $\mathbb{R}$. This argument is also the explicit reduction of Mochizuki's Theorem 3.6 and Corollary 3.7 [51] to Grothendieck's formulation: his full centralizer is $\pi_1(Z) \times C_2$, and intersecting it with $\pi$ gives the centralizer used here. □

For $G = 1$ the section set has one element and its centralizer is $\pi$; the comparison is the bijection $\pi_0(U) \to \{*\}$. The map from individual points is not injective. The hypothesis on the universal cover has content: on $S^2$ a nontrivial rotation has two fixed points while its lift extension, with kernel 1, has one section; the antipodal involution has no fixed point and still has one section. These examples explain exactly why the same formula does not extend to that excluded surface.

**B.5. [Proved] Recovering the incidence and cyclic order of real punctures.** For a hyperbolic $U = X \setminus S$ with anti-involution $\sigma$, its discrete extension together with its indexed peripheral subgroups determines the following finite graph. Its vertices are the section classes of B.4, its edges are the real punctures, and each edge is attached to its two adjacent real half-branches. Every punctured oval gives one cycle, allowing a loop or two parallel edges. The graph recovers which punctures lie on each oval and their cyclic order up to reversal. Unpunctured ovals are the vertices with infinite cyclic centralizer. The claim uses the extension and peripheral incidence maps; a naked set of section classes alone has no such incidence information.

*Manuscript:* 33; 221-225; 437,440-445.

*Proof.* A puncture is real exactly when its $C_2$-stabilizer is all of $C_2$. For such a puncture its peripheral extension is

$$E_i = \langle t, s \mid s^2 = 1, ; sts^{-1} = t^{-1}\rangle, \qquad L_i = \langle t\rangle.$$

Indeed the local anti-involution of the filled disc is reflection. On the logarithmic covering $z = \exp(2\pi i w)$ of the punctured disc its two types of order-two lifts are $w \mapsto -\overline{w}$ and $w \mapsto 1 - \overline{w}$. Algebraically the reflections $t^k s$ have two $L_i$-conjugacy classes, distinguished by the parity of $k$, since conjugation by $t^m$ adds $2m$. The two inclusions of these classes into the section set of $E$ are precisely the two real half-branches. They can have the same image. Attach the edge $i$ using these two maps. Walking through the intervals of a punctured circle gives a cycle whose edge labels are its punctures. Conversely these cycles determine the partition of the punctures and their circular order, with the orientation of each circle unchosen. B.4 identifies every remaining vertex with an unpunctured oval. A nonreal pair of punctures has trivial stabilizer and supplies no edge. Finally any two configurations on circles with the same circular order are related by a homeomorphism: map successive closed arcs monotonically, and extend over collars. Thus this recovery specifies the topological position of the punctures, including the single-puncture and single-interval case, without specifying their analytic positions. □

## 2.2 Finite automorphisms and the withdrawn assertions

**B.6. [Proved] The fundamental loop lemma of §18.** Let $\pi = \pi_1(U)$ be hyperbolic and discrete, with its indexed peripheral structure and orientation module. A finite-order orientation-preserving automorphism of $\pi$ fixing a peripheral index and normalizing its subgroup $L_i$ is the identity. Consequently the peripheral extension attached to a nontrivial finite-order positive outer automorphism fixing $i$ cannot split. This proves the lemma and corollary of scans 193–194 without adding realizability to the hypotheses.

*Manuscript:* 18; 96-98; 193-196.

*Proof.* The based Dehn–Nielsen–Baer correspondence identifies the given finite cyclic group with a finite subgroup of the mapping class group with one additional marked base point. Kerckhoff's realization theorem realizes this group by homeomorphisms fixing that point and preserving the marked punctures. The original paper explicitly treats cusps in §IV, pp. 255–256, in addition to the orientation-reversing convention on p. 235 [41]; hence the punctured case used here is included. The section determined by the fixed base point identifies the induced genuine action on $\pi$ with the given one.

An orientation-preserving automorphism fixing $i$ acts as $+1$ on $L_i \cong \mathbb{Z}$, so it fixes a generator $\ell$ of $L_i$. A nontrivial finite positive cyclic group of plane lifts has one fixed point by B.1. Commutation with the nontrivial deck transformation $\ell$ would give the second fixed point $\ell\widetilde{x}$, a contradiction. Thus the automorphism is the identity. A splitting of the peripheral extension gives a finite-order lift normalizing $L_i$; the result just proved makes its outer class trivial.

The margin on scan 194 writes the *filled* type as $(g, \nu)$ and the acting group as type $(g, \nu+1)$. For $(g, \nu) = (0, 2)$ or $(1, 0)$ the acting group is still a hyperbolic free group of rank two, so the same proof applies. For a literal annulus, a positive automorphism fixing one of its two indexed ends acts trivially on $\pi = \mathbb{Z}$. Forgetting the indices permits the positive end interchange, which induces $-1$ on $\mathbb{Z}$ while normalizing its common peripheral subgroup. For a literal unpunctured torus there is no peripheral index, so the displayed loop lemma has no instance. These are the precise comparisons required by the marginal remark, also reflected in A.13. □

**B.7. [Refuted] The unrestricted fixed-subgroup question, with its replacement.** The assertion $\pi^G = 1$ for every finite $G \leq \mathrm{Aut}_{\mathrm{lac}}(\pi)$ is false. For a hyperbolic discrete $\pi$, a nontrivial finite genuine loop action has fixed subgroup 1 when its positive part is nontrivial. Its remaining possibility is a reflection action of $C_2$; its fixed subgroup is 1 or $\mathbb{Z}$, as in B.4. For any finite-type faithful geometric action there are finitely many classes of full and partial sections. This answers questions (c) and (e), scans 197–198.

*Manuscript:* 18; 98-99; 197-198.

*Proof.* A finite subgroup of $\mathrm{Aut}_{\mathrm{lac}}(\pi)$ intersects $\mathrm{Inn}(\pi)$ trivially: $\pi$ is torsion-free and centre-free. Its outer image is therefore finite and is realized by Kerckhoff. The canonical centre-free extension identifies the given genuine action with a section of its geometric lift extension. B.3 and B.4 now give the complete list of fixed subgroups. An explicit counterexample is the double of an orientable one-holed torus, a genus-two surface, with the reflection interchanging the two halves. Its fixed oval is essential and a base point on it gives a genuine order-two automorphism with fixed subgroup $\mathbb{Z}$.

For finiteness, compactify $U$ by its finitely many ends. Every nonidentity positive periodic homeomorphism has isolated fixed points by B.1. The resulting closed discrete subset of the compact surface is finite. A reflection has a compact one-dimensional fixed submanifold, hence finitely many components; removing finitely many punctures preserves finiteness. Apply B.3 or B.4 to each of the finitely many subgroups of $G$. This proves the asserted finiteness. For the abelian groups the genuine actions can be read directly: a nonidentity positive finite-order element of $SL_2(\mathbb{Z})$ fixes no nonzero lattice vector, whereas a determinant $-1$ involution fixes a primitive rank-one sublattice. For $\mathbb{Z}$, the nonidentity automorphism is $-1$ and its fixed subgroup is zero. The trivial group of operators fixes all of $\pi$ in every type. □

**B.8. [Refuted] Unique maximal sections fail for reflections, already in type** $(0, 3)$**.** The orientation-free extension of question (d), scan 198, and of the expectation on scan 155 is false.

Put

$$E = \langle s_0, s_1, s_2 \mid s_j^2 = 1, (s_0 s_1)^2 = 1, (s_1 s_2)^3 = 1 \rangle.$$

There is a geometric extension $1 \to \pi_{0,3} \to E \to S_3 \times C_2 \to 1$. The reflection subgroup $\langle s_1 \rangle$ is contained in two distinct maximal finite subgroups $\langle s_0, s_1 \rangle \cong C_2^2$ and $\langle s_1, s_2 \rangle \cong S_3$, which are not $\pi_{0,3}$-conjugate. The valid uniqueness statement is B.3, requiring a nontrivial positive part.

*Manuscript:* 15; 77; 155; 18; 99-101; 198,201-203.

*Proof.* Take a hyperbolic triangle with angles $\pi/2, \pi/3, 0$ and reflect in its three sides. Reflected triangles tile the plane; the local tilings at the two finite vertices give exactly the displayed relations. Equivalently its group is the amalgam $C_2^2 *_{C_2} S_3$, with common subgroup generated by $s_1$. Write $S_3 = \langle r, a \mid r^3 = a^2 = 1, ara = r^{-1} \rangle$, and let $\tau$ generate the central $C_2$. Define the surjection

$$s_0 \mapsto \tau, \qquad s_1 \mapsto a\tau, \qquad s_2 \mapsto ra\tau.$$

It is injective on both vertex groups. A finite subgroup of this amalgam fixes a vertex of its Bass–Serre tree: the finite orbit spans a finite tree, whose centre is fixed (there are no edge inversions). Thus the kernel is torsion-free. The character sending each $s_j$ to $-1$ factors through the exponent of $\tau$, so the kernel preserves orientation. The group Euler characteristic is $1/4 + 1/6 - 1/2 = -1/12$, and the index is 12; hence the kernel has Euler characteristic $-1$. The stabilizer of the cusp is $\langle s_0, s_2 \rangle$; its image has order four. The kernel therefore has $12/4 = 3$ cusps, and its quotient surface has genus zero. This constructs the asserted geometric extension.

Each vertex group is maximal finite, again by the tree action: a finite overgroup would fix a vertex, and a vertex group of order greater than the edge group fixes only its vertex. Their intersection is $\langle s_1 \rangle$ by the normal form for the amalgam. Their orders differ, and their generated group is all of the infinite amalgam. This proves every claimed failure. The finite quotient, its vertex and cusp images, and the Euler-characteristic calculation are checked in `code/check_part_b.py`. Grothendieck's proof on scans 201–203 applies when the finite overgroups are positive cyclic groups: they centralize their intersection, and B.3 makes the kernel of that centralizer map trivial. A reflection has a line of fixed points, so that argument supplies neither of these requirements in the generalization just refuted. □

**B.9. [Proved] The valid normalizer assertion.** For a hyperbolic discrete group with loops, a nontrivial finite subgroup $K \leq \mathrm{Aut}^+_{\mathrm{lac}}(\pi)$ is cyclic and

$$N_{\mathrm{Aut}^+_{\mathrm{lac}}(\pi)}(K) = C_{\mathrm{Aut}^+_{\mathrm{lac}}(\pi)}(K).$$

An orientation-reversing element of its full normalizer acts on $K$ by inversion. Every finite positive overgroup of $K$ is cyclic. These are the valid assertions within the passage labelled "Ultra fort!" on scans 198–199.

*Manuscript:* 18; 99-100; 198-200.

*Proof.* Realize the outer image of $K$ and use its given section. B.3 makes $K$ cyclic. The boundary rotation character of B.12 is injective on $K$ and is natural for loop automorphisms with their action on the orientation module. A positive normalizer therefore preserves every value of that character, so fixes every element of $K$. A negative normalizer changes its sign, so inverts every element. The same realization argument proves cyclicity of each finite positive overgroup. This proof compares a normalizing automorphism on the canonical oriented boundary; it makes no assertion that an infinite normalizer acts as a finite isometry group. □

**B.10. [Refuted] The withdrawn uniqueness of topological roots.** The lemma marked "faux!" on scan 200 is false even for the punctured torus: two different finite-order homeomorphisms fixing the base point can have the same nontrivial square and induce the same automorphism of $\pi_1$.

*Manuscript:* 18; 100; 200.

*Proof.* On $U = \mathbb{R}^2/\mathbb{Z}^2 \setminus \{0\}$ choose $p = (1/2, 1/2)$ and let $r(x, y) = (-y, x)$, $h = r^2$. Choose a small disc $D$ whose four $r$-translates are disjoint and avoid both 0 and $p$. Choose a nonidentity homeomorphism $f_0$ supported in $D$ and isotopic to the identity there relative to its boundary; for example use a radial bump rotation in disc coordinates. Put $f = f_0(hf_0h^{-1})$. Its two disjoint supports show that $fh = hf$; the same supports show that $fr \neq rf$, since $r$ moves their union to the disjoint pair $rD, r^3D$. Hence $v = frf^{-1} \neq r$, but $v^2 = h$ and $v(p) = p$. The isotopy of $f$ fixes 0 and $p$, so $f_* = 1$ on $\pi_1(U, p)$ and therefore $v_* = r_*$. This verifies the exact data of the withdrawn statement. What survives is uniqueness up to conjugacy by the identity component in the realization comparison, not equality of the homeomorphisms. □

**B.11. [Refuted] Torsion in the centralizer need not be a subgroup.** The assertion following the normalizer assertion on scan 199, that the torsion elements of the centralizer form one largest finite cyclic overgroup, is false for the hyperbolic type $(1, 1)$.

*Manuscript:* 18; 99-100; 198-200.

*Proof.* Keep $U, p, r, h$ as in B.10. The essential simple curve $c = \{x + y = 1/2 \pmod 1\}$ avoids $0, p$ and is invariant under $h$. The based mapping class $T_c$ commutes with $h$, since conjugation of a positive Dehn twist by an orientation-preserving map is the positive twist about the image curve. Thus the two genuine automorphisms induced by $r$ and $T_c r T_c^{-1}$ have order four and common square $h_*$; both centralize $h_*$. On $H_1(U; \mathbb{Z})$, with a consistent choice of twist sign, their matrices are

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad TST^{-1} = \begin{pmatrix} -2 & -5 \\ 1 & 2 \end{pmatrix}, \qquad T = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}.$$

Both square to $-I$, while their product has trace $-6$ and determinant one. Its eigenvalues are $-3 \pm \sqrt{8}$, so it has infinite order. The corresponding product of automorphisms has infinite order as well. These integral equalities are checked by the supplied program. Consequently the centralizer's torsion set is not closed under multiplication, and there is no finite subgroup containing all its finite overgroups of $\langle h_* \rangle$. This coexists with the exact equality of normalizer and centralizer in B.9. □

## 2.3 The multiplier and the abelian surfaces

**B.12. [Proved] An intrinsic discrete formula for the multiplier.** For a geometric finite extension of a hyperbolic discrete surface group, $E$ with its peripheral structure and orientation module $T$ determines $\mu_E$ on all torsion elements. On $E^+$ it is the rotation number of its canonical oriented boundary action. More explicitly, let $\widetilde{\mathrm{Homeo}}^+(S^1) \to \mathrm{Homeo}^+(S^1)$ be the central extension by the positively oriented deck translation $z$. For $u \in E^+$ of order $n$ and any lift $\widetilde{u}$, write $\widetilde{u}^n = z^k$. Then

$$\mu_E(u) = \frac{k}{n} \pmod{\mathbb{Z}} \quad \text{in } T \otimes \mathbb{Q}/\mathbb{Z}.$$

For a nonidentity orientation-reversing torsion element we put $\mu_E(u)$ equal to the unique nonzero element of order two. With this convention the formula is natural under isomorphisms preserving

the loops and transporting $T$, satisfies $\mu(u^m) = m\mu(u)$, and has the same order as $u$. This answers the discrete intrinsicity question of §15, folios 77–79, scans 155–160.

*Manuscript:* 15; 77-79; 155-160.

*Proof.* Choose an oriented finite-area hyperbolic uniformization of $U$. Its boundary at infinity is a circle. We explain the ambiguity in this choice. A marking isomorphism of surface groups preserving their peripheral subgroups and the orientation is induced by an orientation-preserving homeomorphism, by the Dehn–Nielsen–Baer correspondence. We can choose that marking quasiconformal: on a compact core smooth it, and on each cusp use standard cylindrical coordinates and extend with bounded dilatation. Its lift extends to an oriented boundary homeomorphism. This extension is uniquely determined by its $\pi$-equivariance. Indeed it must carry the attracting and repelling endpoints of every hyperbolic element to those of the corresponding element. These endpoints are dense in the boundary circle of a finite-area surface group. For completeness, density follows from its full limit set: a boundary arc meets translates of a fundamental ideal polygon arbitrarily near infinity; conjugating either of two hyperbolic elements with disjoint endpoints by the corresponding translates gives endpoints in that arc. Thus two equivariant boundary maps agree on a dense set and hence everywhere.

For each automorphism of $\pi$ preserving the peripheral structure, Dehn–Nielsen–Baer provides the boundary map conjugating its deck action. The preceding uniqueness makes composition exact and makes this circle, together with its $E$-action, independent of the uniformization up to a unique $\pi$-equivariant boundary isomorphism. The boundary orientation is recorded by $T$, so changing orientation acts by $-1$. Realize the finite action and choose an invariant conformal structure, using B.1's cyclic and dihedral local charts on the quotient. Its hyperbolic metric makes a finite-order positive lift an elliptic isometry. In the disc model centred at its fixed point it is the same rotation on the tangent circle and on the circle at infinity. The displayed formula therefore gives its local multiplier.

Replacing $\widetilde{u}$ by $z^j\widetilde{u}$ replaces $k$ by $k + nj$, so the value is independent of the lift. An orientation-preserving finite-order circle homeomorphism of order $n$ is conjugate to a rotation by $k/n$ with $\gcd(k, n) = 1$; this also follows by ordering the finite orbit on the circle and extending a homeomorphism of one fundamental interval around that orbit. This proves the order and power statements. A negative finite plane isometry is a reflection, of order two, so its assigned value has exactly the required power property and is unchanged by sign reversal.

The normalizer formula suggested in the margin of scan 159 is recovered as follows. Drill the orbit of the fixed point of $u$. Above the new puncture the positive peripheral normalizer is infinite cyclic, generated by a positively oriented element $v$, with peripheral subgroup generated by $v^n$. A lift of $u$ to this normalizer is $v^a$ modulo $\langle v^n\rangle$, and $\mu(u) = a/n$. In a local coordinate this is the equality between the rotation $z \mapsto e^{2\pi ia/n}z$ and monodromy in $z \mapsto z^n$. Harvey's Theorem 7 [32] computes exactly this inverse-residue relation from the elliptic generators of the orbifold epimorphism. His convention uses an inverse deck transformation, accounting for the sign visible in his proof on pp. 393–394. Our local-coordinate calculation also applies to cusps and to the positive cyclic part of a dihedral stabilizer; B.1 supplies the reflection part. □

**B.13. [Refuted] Forgetting loops can change the multiplier.** An abstract extension, with its orientation line regarded as an additional rank-one module, need not determine the multiplier after the peripheral structure is forgotten. There are isomorphic extensions with kernel a free group of rank two and quotient $C_3$ whose geometric multipliers disagree, even after allowing either isomorphism between their orientation lines.

*Manuscript:* 15; 77-79; 155-160.

*Proof.* Let $E_0 = \langle u, v \mid u^3 = v^3 = 1 \rangle = C_3 * C_3$. Realize it as the positive orbifold group of a sphere with one cusp and two cone points of order three; choose $u, v$ as its positive local generators. The epimorphisms

$$q_+(u) = q_+(v) = 1, \qquad q_-(u) = 1, \quad q_-(v) = -1 \quad \text{(values in } \mathbb{Z}/3)$$

are injective on every finite subgroup, because finite subgroups of a free product are conjugate into its factors (the tree argument in B.8 applies). Their kernels are torsion-free of index three and Euler characteristic $-1$. The cusp generator is $(uv)^{-1}$. Its image under $q_+$ is nonzero, so there is one cusp and the kernel is a once-punctured torus group. Its image under $q_-$ is zero, so there are three cusps and the kernel is a pair-of-pants group. Both have free rank two.

The automorphism $\psi(u) = u,\ \psi(v) = v^{-1}$ satisfies $q_-\psi = q_+$ and is an isomorphism of the two abstract extensions. The positive local generator values in each realization are $\mu(u) = \mu(v) = 1/3$. Thus $\psi$ preserves the value at $u$ but changes the value at $v$ to $-1/3$. Reversing the isomorphism of orientation lines would instead break the equality at $u$. No orientation-line identification repairs both. The isomorphism sends the cusp word $uv$ to $uv^{-1}$, displaying precisely which peripheral comparison has been discarded. The finite monodromy calculation is included in the computational record. □

**B.14. [Proved] Finite annulus and torus actions, and their multipliers.** Every finite topological action on the torus is topologically conjugate to an affine action on $V/\Lambda$, where $\Lambda \cong \mathbb{Z}^2$ and $V = \Lambda \otimes \mathbb{R}$. For a fixed finite operator group $G$ such an action is specified by a homomorphism $A : G \to GL(\Lambda)$ and a cocycle $b \in Z^1(G, V/\Lambda)$, subject to faithfulness of $g \mapsto (x \mapsto A_g x + b_g)$. Translation conjugacy corresponds exactly to a coboundary, and

$$H^1(G, V/\Lambda) \xrightarrow{\sim} H^2(G, \Lambda)$$

is the connecting homomorphism. Linear changes of the lattice give the further conjugacies. Every finite annulus action has a simultaneous normal form

$$(x, t) \longmapsto (\epsilon_g x + a_g, \eta_g t), \qquad x \in \mathbb{R}/\mathbb{Z}, \quad t \in \mathbb{R}, \quad \epsilon_g, \eta_g \in \{1, -1\}.$$

These descriptions determine the multiplier of every finite-order element in the respective lift extensions.

*Manuscript:* 15; 78-79; 157-159; 17; 89-90; 179-181.

*Proof.* We first justify simultaneous conformalization. B.1 gives genuine cyclic or dihedral charts for every finite stabilizer. The quotient is a compact surface orbifold after the ends are added; its interior special points have cyclic charts, and its mirror edges and corners have reflection and dihedral charts. Choose a smooth metric on the quotient away from these charts, choose round cone and mirror metrics in them, and patch in collars using an invariant partition of unity on their finite local covers. The local charts make the patched lifts smooth. Pullback is a smooth invariant conformal structure on the original surface. This construction proves the required passage from the original topological action; it does not average a merely topological metric and call the result Riemannian.

Uniformization of the torus gives $\mathbb{C}/\Lambda$. Every holomorphic automorphism lifts to an entire map $f$ with $f(z + \lambda) - f(z) \in \Lambda$. Its derivative is periodic and hence bounded on $\mathbb{C}$, so it is constant; $f(z) = az + c$. An antiholomorphic automorphism is obtained by composing such a map with complex conjugation on its source. Hence every element is real affine on the same lattice. The action law is precisely $b_{gh} = b_g + A_g b_h$. Averaging cochains over the finite group gives $H^q(G, V) = 0$ for $q > 0$. The exact sequence $0 \to \Lambda \to V \to V/\Lambda \to 0$ therefore gives the asserted isomorphism. In concrete terms, lifts $v_g \in V$ of $b_g$ give the extension cocycle $v_g + A_g v_h - v_{gh} \in \Lambda$.

Conjugating by translation adds a coboundary. Moreover any topological conjugacy between two affine actions can be replaced by an affine conjugacy. A lift of the conjugacy has form $Bx + f(x)$, with $B \in GL_2(\mathbb{Z})$ and $f$ periodic. Average its equivariance equation over the torus with Haar measure. Affine lattice automorphisms preserve that measure, so $Bx + \int f$ satisfies the same equation modulo $\Lambda$.

For the annulus, extend the action to its two-point sphere compactification and apply Kolev. In the resulting conformal sphere choose a Möbius coordinate sending the two ends to zero and infinity. The maps are then of the forms $z \mapsto az$, $a/z$, $a\overline{z}$, or $a/\overline{z}$. On the logarithmic radial coordinate they act affinely with slopes $\eta = \pm 1$. A finite group of affine transformations of the real line fixes the average of any orbit; translation to this common fixed point removes every radial constant. The angular coordinate gives the claimed $\epsilon x + a$. This proves the displayed simultaneous normal form. The action on $\pi_1$ is $\epsilon_g$ and its orientation sign is $\epsilon_g \eta_g$.

For the promised computation on the torus, an element of the lift extension is $x \mapsto Ax + v$. It has finite order exactly when $A$ has finite order, say $n$, and $(1 + A + \cdots + A^{n-1})v = 0$. Averaging its orbit gives a fixed point, and translation to that point conjugates it to $A$. A nonidentity positive finite-order matrix has trace $-2, -1, 0, 1$, giving orders $2, 3, 4, 6$, respectively. For order two its multiplier is $1/2$. For the other orders it is $+1/n$ or $-1/n$ according to the sign of $\det(w, Aw)$ for any nonzero $w$; this sign is independent of $w$, as is seen after averaging an inner product and writing $A$ as a rotation. A negative finite-order matrix is a reflection and receives $1/2$. A lift with $A = I$ has finite order only when it is the identity. For the annulus a lift is $(x, t) \mapsto (\epsilon x + a + k, \eta t)$. With $\epsilon = -1$ its square is the identity; with $\epsilon = 1$ finite order requires $a + k = 0$, leaving the identity or $t \mapsto -t$. Thus every nonidentity torsion lift on an annulus has order two and multiplier $1/2$. □

**B.15. [Refuted] The "only counterexample" of scan 179.** The assertion that a faithful homotopically trivial finite annulus action failing to be a translation action must be the single anti-involution $z \mapsto 1/\overline{z}$ is false. For the two non-simply-connected abelian types, the exact replacement is: homotopically trivial torus actions are translation actions; homotopically trivial annulus actions are the finite subgroups of $SO(2) \times C_2$ in the normal form of B.14. Those contained in $SO(2)$ are translations, while those with nontrivial second projection include two different types of anti-involution and groups of arbitrarily large order.

*Manuscript:* 17; 89-90; 179-181.

*Proof.* On the torus, homotopic triviality is $A_g = I$ for every $g$, so B.14 proves the translation statement and the implication doubted on scans 180–181. On the annulus it is $\epsilon_g = 1$, which gives the asserted subgroup. For each $n \geq 1$ the maps

$$r(z) = e^{2\pi i/n} z, \qquad s(z) = 1/\overline{z}$$

generate a faithful $C_n \times C_2$ action, and every element induces the identity on $\pi_1(\mathbb{C}^*)$. The fixed circle of $s$ excludes a translation realization: a nonidentity translation in any group is fixed-point free. In addition $s_-(z) = -1/\overline{z}$ is an involution inducing $+1$ on $\pi_1$, but it has no fixed point, since its fixed-point equation is $|z|^2 = -1$. It cannot be conjugate to $s$, and it cannot be a translation of a connected topological group: every such translation is isotopic to the identity along a path in that group and hence preserves orientation. These formulas give the required full counterexamples, not merely a gap in the argument.

More explicitly let the rotation subgroup have order $n$ and generator $(x, t) \mapsto (x + 1/n, t)$. A second-coset generator can be chosen as $(x, t) \mapsto (x + m/(2n), -t)$, with $m = 0$ or $1$ after multiplying by a rotation. Its square is the $m$th rotation. This exhausts the finite subgroups projecting onto $C_2$, and includes both the reflection and glide cases. For the omitted simply connected surfaces, finite plane actions are the orthogonal actions of B.1 and are all homotopically

trivial; positive sphere actions are likewise homotopic to the identity. Nontrivial rotations fixing a point already refute any extension of the translation formulation to those types. □

## 2.4 Anti-involutions and the later tube construction

**B.16. [Proved] Complete topological classification of anti-involutions.** Conjugacy classes of anti-involutions of a closed oriented genus-$g$ surface are classified by the orientability of the quotient and its number $b$ of boundary components. The possibilities are

$$\begin{aligned} &\text{nonorientable quotient:} && 0 \le b \le g, \quad h = g + 1 - b \ge 1 \text{ crosscaps;} \\ &\text{orientable quotient:} && b \ge 1, \quad g = 2h + b - 1, \quad h \ge 0. \end{aligned}$$

There are $(g+1) + \lfloor (g+2)/2 \rfloor = \lfloor (3g+4)/2 \rfloor$ classes. For $\nu$ unlabelled invariant punctures the additional data are an integer $r \ge 0$ and an unordered list $(q_1, \dots, q_b)$ of nonnegative integers with $2r + \sum q_i = \nu$: $r$ is the number of interior punctures of the quotient, and $q_i$ counts punctures on its $i$th boundary circle.

*Manuscript:* 33; 221-222; 437-440.

*Proof.* By B.1 the quotient $Y = X/\sigma$ is a compact surface with boundary $X^\sigma$. The complement of the real locus covers its interior with degree two; the Euler characteristic of the boundary is zero. Hence $\chi(X) = 2\chi(Y)$. If $Y$ is nonorientable with $h$ crosscaps, $\chi(Y) = 2 - h - b$, yielding $g = h + b - 1$; if orientable of genus $h$ it yields $g = 2h + b - 1$. In the latter case $b$ must be positive, since the oriented double of a closed orientable quotient would be disconnected. Conversely form the orientation double over the interior of each allowed $Y$ and join the two local sides at its boundary. The result is a connected closed orientable surface, with the deck interchange extended to reflection along the boundary; its genus is given by the displayed formula. The quotient recovers $Y$, so the classification of compact surfaces proves both existence and uniqueness of these conjugacy classes. This also proves the complete list underlying Gross–Harris, Proposition 3.1 [28], including the two corrections in the margins of scan 439. Grothendieck's nonorientable parameter $\gamma$ is $h - 1$; thus his formula $g = \gamma + b$ agrees exactly with ours.

An anti-involution of a punctured finite-type surface extends to its end compactification. Punctures in its quotient lie either in the interior, with two-point inverse image, or on the boundary, with one-point inverse image. Any two finite interior configurations of the same cardinality are related by an isotopy supported away from the boundary. Boundary components may be permuted, and any two unlabelled configurations of the same cardinality on one circle are related by a circle homeomorphism, extended over a collar. These facts prove completeness of the punctured invariants. With individually labelled punctures one must instead retain their assignments and their circular orders; B.5 provides the corresponding incidence reconstruction. □

**B.17. [Proved] Real moduli components, with the group and marking convention fixed.** Let $\mathcal{M}_{g,\nu}$ mean the real moduli stack of smooth proper geometrically connected genus-$g$ curves with an *unlabelled* degree-$\nu$ reduced puncture divisor. We retain the stack and its real forms also in the four unstable types. Its complex mapping class group is $\mathcal{T}^+_{g,\nu}$, allowing permutations of the punctures. Write $p_b(N)$ for the number of partitions of $N$ into at most $b$ parts, with $p_0(0) = 1$ and $p_0(N) = 0$ for $N > 0$. Then

$$\#\pi_0(\mathcal{M}_{g,\nu}(\mathbb{R})) = \sum_{b=0}^{g} \sum_{r=0}^{\lfloor \nu/2 \rfloor} p_b(\nu - 2r) + \sum_{\substack{1 \le b \le g+1 \\ b \equiv g+1 \ (2)}} \sum_{r=0}^{\lfloor \nu/2 \rfloor} p_b(\nu - 2r).$$

For closed hyperbolic surfaces this is $\lfloor (3g+4)/2 \rfloor$. In the four unstable types $(0,0)$, $(0,1)$, $(0,2)$ and $(1,0)$ the counts are, respectively, $2, 1, 3, 3$. In the last case no origin is prescribed. Distinct

topological anti-involution classes remain distinct in the extended mapping class group in these hyperbolic types. The last assertion fails for $(0,0)$, $(0,2)$, and $(1,0)$.

*Manuscript:* 33; 222; 439-440.

*Proof.* C.5 and C.6, finite-action realization and uniqueness up to the identity component, identify conjugacy classes of anti-involutions on the fixed punctured surface with conjugacy classes of orientation-reversing involutions in $\mathcal{T}_{g,\nu}$. One can use positive conjugacy here: an orientation-reversing conjugator can be multiplied by the target involution to become positive. Mochizuki, Corollary 3.8, identifies these conjugacy classes with the connected components of the real moduli *stack* [51]. His Example 3.3 uses the full modular group of a punctured surface, which is the unlabelled convention just specified. B.16 supplies the list. For fixed $b, r$, unordered nonnegative lists of length $b$ summing to $\nu - 2r$ are exactly the partitions counted by $p_b(\nu - 2r)$. Summing over the two quotient families proves the formula. The supplied code enumerates these numbers for $2g + \nu \leq 6$ and checks the closed formula for $0 \leq g \leq 20$; the proof holds for all $g$.

We verify the unstable component counts without using mapping-class injectivity. A real genus-zero curve is a split or nonsplit real conic; analytically these are the sphere with $z \mapsto \overline{z}$ or with $z \mapsto -1/\overline{z}$. To see these are the two possibilities, conformalize its anti-involution as in B.14 and apply the two quotient types in B.16; uniformization identifies the resulting conformal double with the sphere. A real point exists exactly in the first type. There is no complex moduli parameter in genus zero. With one puncture the point is real, and $PGL_2(\mathbb{R})$ acts transitively on the real projective line, giving one component. With two unlabelled punctures the possibilities are two real points on the split conic, one conjugate pair on the split conic, or one conjugate pair on the nonsplit conic. The first is transitive under $PGL_2(\mathbb{R})$, the second under its action on the upper half-plane, and the third under the rotations commuting with the antipodal map. These give three components. With no puncture there are the two conic components.

For an unpointed genus-one curve, B.14 and B.16 give the three anti-involution types with zero, one, or two real ovals. Each type has a connected parameter space of complex structures. Explicit representatives are

| $b$ | $\Lambda_t$ | $\sigma(z)$ |
|---|---|---|
| 0 | $\mathbb{Z} + it\mathbb{Z}$ | $\overline{z} + \frac{1}{2}$ |
| 1 | $\mathbb{Z} + (\frac{1}{2} + it)\mathbb{Z}$ | $\overline{z}$ |
| 2 | $\mathbb{Z} + it\mathbb{Z}$ | $\overline{z}$ |

$(t > 0)$.

To obtain this list from an invariant lattice, rotate a primitive real period to 1. Conjugation of a second period forces its real part modulo $\mathbb{Z}$ to be 0 or $1/2$. The translation cocycle of B.14 is then removed by a change of origin, except for the nonzero half-period class in the rectangular case. That class gives the first row. Thus every real genus-one curve occurs in one of these connected families. The number of real ovals is locally constant in a smooth proper real family: local trivialization can be chosen invariant under the involution by an invariant horizontal metric and its path lifting. Distinct rows consequently belong to different components. This proves the number three, with no origin added to the moduli problem. Direct substitution of the four unstable types into the displayed partition formula gives exactly these counts.

The mapping-class exceptions can be checked directly. On $S^2$ reflection and the antipodal map have different fixed sets, but both represent the orientation-reversing mapping class. On $\mathbb{R}^2/\mathbb{Z}^2$ the maps $(x, y) \mapsto (x, -y)$ and $(x, y) \mapsto (x + 1/2, -y)$ have respectively two fixed circles and no fixed points, but both induce $\mathrm{diag}(1, -1)$ and hence the same mapping class. On the annulus, the reflection and glide anti-involutions of B.15 also have the same mapping class: the path $(x, t) \mapsto (x + s/2, -t)$ joins them through homeomorphisms preserving the two ends as a set. These examples establish the exact failure of the unrestricted injectivity hoped for on scan 440. For labelled moduli spaces or a coarse moduli space, the quotient and its real locus have to

be specified separately; the displayed formula is for the stack and puncture convention stated above. □

**B.18. [Proved] The two centralizer corollaries in both quotient families.** Let $g \geq 2$, and let an anti-involution of $X_g$ have $b > 0$ ovals, with outer class $\sigma$. There are exactly $b$ classes of order-two genuine loop automorphisms lifting $\sigma$, under conjugacy by $\pi_g$. For each lift $u$, $\pi_g^u \cong \mathbb{Z}$. The positive centralizer of $\sigma$ acts transitively on those $b$ classes. More precisely, writing $A = \mathrm{Aut}_{\mathrm{lac}}(\pi_g)$ and $O = \mathrm{Out}_{\mathrm{lac}}(\pi_g)$, there is an exact sequence

$$1 \longrightarrow \pi_g^u \longrightarrow C_{A^+}(u) \longrightarrow \mathrm{Stab}_{C_{O^+}(\sigma)}([u]) \longrightarrow 1,$$

and the last group has index $b$ in $C_{O^+}(\sigma)$. The corresponding full centralizer image also has index $b$. This proves Corollaries 1 and 2, folios 225–226, scans 445–448, for both orientable and nonorientable quotients.

*Manuscript:* 33; 225-226; 445-448.

*Proof.* B.4 proves the count and the fixed subgroup. Any permutation of the boundary components of a compact connected quotient surface is realized by a homeomorphism: join two components by an embedded arc, interchange them in a neighbourhood of that arc and the two collars, and use transpositions. Each quotient homeomorphism lifts to the oriented double; among its two lifts choose the orientation-preserving one. It commutes with the reflection. Consequently the positive centralizer is transitive on the ovals, and hence on the section classes.

For exactness, the centre of $\pi_g$ is trivial, so $\mathrm{Inn}(\pi_g)$ is identified with $\pi_g$. Its intersection with $C_A(u)$ is exactly $\pi_g^u$. An outer class $\alpha$ centralizing $\sigma$ preserves $[u]$ precisely when one, hence every, lift $\widetilde{\alpha}$ satisfies $\widetilde{\alpha} u \widetilde{\alpha}^{-1} = vuv^{-1}$ for an inner automorphism $v$. Then $v^{-1}\widetilde{\alpha}$ centralizes $u$, proving surjectivity onto the displayed stabilizer. Orbit–stabilizer gives index $b$ in both versions. For $b = 0$ there are no order-two genuine lifts and these assertions about a chosen lift have no instance. With punctures the same exact sequence holds, but the orbit size records the boundary decorations from B.16; it need not be the total number of ovals when different ovals carry different puncture data. □

**B.19. [Proved] The tube twists, their maximality property, and their normalizer.** For $g \geq 2$, the elements $\rho_j$ of §33, scans 453–470, are full Dehn twists about a cut system $c_1, \ldots, c_g$: disjoint nonseparating curves with connected complement. They generate $H \cong \mathbb{Z}^g$. This is not a maximal free abelian subgroup of $\mathcal{T}_g$: it is contained in a multitwist subgroup of rank $3g - 3$. It has the precise maximality property of being a full multitwist group of largest rank whose supporting multicurve has connected complement. Furthermore

$$N_{\mathcal{T}_g}(H) = \mathrm{Stab}_{\mathcal{T}_g}(\{c_1, \ldots, c_g\}), \qquad N_{\mathcal{T}_g}(H)/C_{\mathcal{T}_g}(H) \cong S_g \times C_2.$$

The action on $H$ permutes its basis and multiplies every basis vector by the same orientation sign. Cutting and capping gives an exact sequence

$$1 \longrightarrow H \longrightarrow N_{\mathcal{T}_g}(H) \longrightarrow \mathcal{T}_{0,2g}^{\mathrm{pairs}} \longrightarrow 1,$$

where the last extended sphere mapping class group preserves the partition of its $2g$ punctures into the $g$ unordered gluing pairs.

*Manuscript:* 33; 229-237; 453-470.

*Proof.* In the coordinates of scan 461 the map is $\rho_j(z, t) = (-ze^{\pi i t}, t)$ on $S^1 \times [-1, 1]$ and is the identity outside the tube. Its angular displacement increases from 0 to $2\pi$, so its relative

boundary mapping class is one full Dehn twist, with sign fixed by that coordinate orientation. The complement of the tubes in his construction is a sphere with $2g$ boundary circles, establishing the cut-system assertion.

Twists in disjoint collars commute. They are independent: cut the surface along the $c_j$, choose transverse curves dual to these cuts, and measure the integer winding of the resulting arcs when the collars are glued back. A product $\prod T_{c_j}^{n_j}$ changes the $j$th such winding by $n_j$; an isotopy cannot change these relative-end winding differences. This proves $H \cong \mathbb{Z}^g$. Complete the $2g$-holed sphere to a pants decomposition by adding $2g-3$ essential disjoint curves. The same winding argument gives a rank-$3g-3$ multitwist group containing $H$. In contrast, cutting $k$ disjoint curves while keeping the complement connected reduces the genus to $g-k$, so $k \le g$. This proves the stated maximality among connected-complement multitwist systems.

For the normalizer, conjugation satisfies $fT_cf^{-1} = T_{f(c)}^{\epsilon(f)}$. A single twist can equal a product of twists about the $c_j$ only by being one of these twists to the corresponding power. Here is the needed support argument. Such a twist commutes with every $T_{c_j}$. The elementary intersection formula $i(T_d(c), c) = i(c, d)^2$ for a single twist forces $d$ to be disjoint from each $c_j$; otherwise its conjugation changes $T_{c_j}$. Add $d$ to the disjoint system if it is not already there and apply the preceding independent-winding argument. The resulting nontrivial relation is impossible. Thus a normalizer permutes precisely the $c_j$, proving the first equality. Conversely every such permutation of curves normalizes $H$. Every permutation of the handles is realized positively, and a reflection of the standard cut-system model reverses the global orientation while preserving each curve. This gives the full image $S_g \times C_2$. Its kernel is exactly the centralizer.

A representative preserving the multicurve can be isotoped to preserve its collars. Its restriction to the cut surface, after capping each boundary to a puncture, preserves the gluing pairs. Conversely a homeomorphism of the capped sphere preserving those pairs can be made radial in small puncture discs and then extended over the paired collars. Two such extensions differ by integer collar twists. An isotopy on the capped sphere lifts after allowing exactly these twists, by the isotopy extension theorem on collars. This proves both surjectivity and the stated kernel, and types the comparison with the complement suggested on scan 470. □

**B.20. [Refuted] Presentation (52) and its exact correction.** For $g \ge 2$, presentation (52) on scan 469 does not present the stated $\mathbb{Z}^g$-by-finite group. Put $r = \varepsilon_0\varepsilon_1$ and $\rho_j = r^j\tau'\tau r^{-j}$, $0 \le j < g$. The printed presentation gives

$$F_g \rtimes (D_{4g} \times \langle\tau\rangle),$$

where $D_{4g} = \langle r, \varepsilon_1 \mid r^{2g} = \varepsilon_1^2 = 1, \varepsilon_1 r \varepsilon_1 = r^{-1}\rangle$ permutes the free generators through its action on $\mathbb{Z}/g$, and $\tau$ inverts each free generator. Adding the relations $[\rho_i, \rho_j] = 1$ for $i < j$ gives exactly $\mathbb{Z}^g \rtimes (D_{4g} \times C_2)$ with the manuscript's specified action.

*Manuscript:* 33; 237; 469.

*Proof.* The relations say that $\tau$ commutes with $D_{4g}$, that $\tau'$ commutes with $\varepsilon_1$ and $r^g$, and that both $\tau, \tau'$ are involutions. Set $\rho_0 = \tau'\tau$. Then $\tau\rho_0\tau = \rho_0^{-1}$, $\varepsilon_1\rho_0\varepsilon_1 = \rho_0$, and $r^g\rho_0 r^{-g} = \rho_0$. Its conjugates are indexed by the $g$ cosets of $\langle\varepsilon_1, r^g\rangle$ in $D_{4g}$, with no further relation among them. To check the last assertion directly, form the displayed semidirect product with a free basis $\rho_0, \dots, \rho_{g-1}$ and define $\tau' = \rho_0\tau$ in it. All printed relations hold. The reverse map sends these free generators to $r^j\tau'\tau r^{-j}$; the two maps agree inversely on all generators. This is a proof by explicit mutually inverse homomorphisms. Abelianizing only the free factor adds exactly the stated commutators and gives the corrected group.

There is also a finite certificate. For $g = 2$ set $\varepsilon_0 = (13)$, $\varepsilon_1 = \tau = 1$, $\tau' = (12)$ in $S_3$. All nine displayed relators evaluate to 1, but $\rho_0 = (12)$ and $\rho_1 = (23)$ do not commute. The program

checks every one of these evaluations. Relations (33), scan 454, already assert commutation for the geometric twists; they are exactly the missing input to (52). □

**B.21. [Proved] The one-holed-torus subgroups and the proved part of the analogy.** For $g \geq 2$, choose a one-holed torus $W_j$ containing $c_j$ and a transverse curve $d_j$ with $i(c_j, d_j) = 1$. Its group fixing its boundary pointwise embeds by extension by the identity into $\mathcal{T}_g^+$. It is

$$B_3 = \langle A_j, B_j \mid A_j B_j A_j = B_j A_j B_j \rangle,$$

and capping the boundary gives

$$1 \longrightarrow \langle T_{\partial W_j} \rangle \cong \mathbb{Z} \longrightarrow B_3 \longrightarrow SL_2(\mathbb{Z}) \longrightarrow 1, \qquad (A_j B_j)^6 = T_{\partial W_j}^{\pm 1}.$$

This extension has no section. For disjoint chosen $W_j$, their supported subgroups commute and, together with $H$, preserve their boundary multicurve; they do not generate $\mathcal{T}_g^+$. On explicit generators the tube subgroup and these two twists act by

$$\begin{array}{ll} \rho_j : & a_j \mapsto a_j, \quad b_j \mapsto b_j a_j, \\ B_j : & a_j \mapsto a_j b_j^{-1}, \quad b_j \mapsto b_j, \end{array}$$

fixing the other handle generators of $\pi_g = \langle a_j, b_j \mid \prod_j [a_j, b_j] = 1 \rangle$.

*Manuscript:* 33; 236-237; 467-470.

*Proof.* Choose the based handle loops so that their commutator is the boundary loop. The displayed substitutions preserve $[a_j, b_j]$ exactly, and are the effect of crossing the appropriate twist collar. Twists in curves meeting once obey the braid relation: in these generators substitution gives $ABA = BAB$. The sixfold product $(AB)^6$ conjugates both generators by $[a, b]^{-1}$, which is the boundary twist with this collar convention. Both word equalities are checked by free reduction in `code/free_words.py`; the stated substitutions also give their direct verification. Filling the boundary gives the standard two elementary matrices generating $SL_2(\mathbb{Z})$. Their action on primitive slopes, reduced by the Euclidean algorithm, shows that the filled mapping class group is exactly $SL_2(\mathbb{Z})$. The kernel of filling is the boundary winding: cut along two based generating arcs, isotope a kernel representative to fix them, and read its remaining integer twist in the boundary collar. The kernel is therefore the infinite cyclic group generated by the boundary twist. The braid group on three strands has the same central extension: with $x = ABA, y = AB$ it is $\langle x, y \mid x^2 = y^3 \rangle$, and its quotient by $y^6$ is $SL_2(\mathbb{Z})$. The comparison sends the kernel generator to the boundary twist, so it is an isomorphism of these extensions.

Extension by the identity is injective because an isotopy to the identity on the closed surface would fix the essential boundary and, after collar adjustment, give an isotopy on the one-holed torus; a nonzero boundary twist remains nontrivial by the winding argument in B.19. The braid group is torsion-free: $\langle x, y \mid x^2 = y^3 \rangle$ is the amalgam of two infinite cyclic groups along their subgroups of indices two and three; a finite subgroup fixes a vertex in its tree and is conjugate into an infinite cyclic group. A section from $SL_2(\mathbb{Z})$ would inject its order-four element, contradicting this fact.

Disjoint supports commute. Each supported homeomorphism fixes the chosen boundary curves, and so does every generator of $H$. An essential curve crossing one of those boundaries has a twist which does not preserve it, by the intersection formula in B.19. Hence the generated subgroup is proper. The exact analogy proved here is the homomorphism to the usual rank-two integral representation and its central boundary-winding extension, with braid and disjoint-support relations. For overlapping choices of the strips one must specify their intersections before computing the remaining relations; no unspecified choice of strips supplies additional relations by itself. □

## 2.5 Computations and the limits of the requested table

**B.22. [Refuted] A finite subgroup alone does not index the requested abelian table.** For all types in Part B(f), a table indexed only by a finite subgroup of the mapping class group cannot assign unique splitting numbers or $\Phi$. The type $(1,0)$ and the single subgroup $\langle \mathrm{diag}(1,-1)\rangle$ already have two different answers. The additional extension datum is precisely the class in $H^2(G,\pi)$ in the abelian cases of B.14.

*Manuscript:* 15; 78-79; 157-159; prompt B.f.

*Proof.* On the torus compare $\sigma_0(x,y)=(x,-y)$ and $\sigma_1(x,y)=(x+1/2,-y)$. Both are faithful involutions with the same homomorphism to $GL_2(\mathbb{Z})$, but the first fixes two circles and the second fixes nothing. The corresponding extensions, with commuting lattice generators $a,b$, have relations

$$\widetilde{\sigma}a\widetilde{\sigma}^{-1}=a, \qquad \widetilde{\sigma}b\widetilde{\sigma}^{-1}=b^{-1}, \qquad \widetilde{\sigma}^2=a^{\epsilon}, \quad \epsilon=0 \text{ or } 1.$$

The square of $a^m b^n\widetilde{\sigma}$ is $a^{2m+\epsilon}$. For $\epsilon=1$ there is no order-two lift. For $\epsilon=0$ such lifts require $m=0$, and conjugation by $b$ changes $n$ by two. Thus there are exactly two section classes in the first case and zero in the second. These are the two parity classes of the extension cocycle. B.14 proves that retaining the extension class gives the correct affine classification, with all translation conjugacies accounted for. Thus the counterexample identifies the missing index and its exact replacement. □

**B.23. [Proved] Complete finite-subgroup table in type $(0,3)$.** The ten conjugacy classes in the extended group $\mathcal{T}_{0,3}\cong S_3\times C_2$, and their discrete section invariants, are the following. In the table $F$ counts full-section classes, $P$ counts all nontrivial partial-section classes with their subgroup images distinguished, $M=|\Phi|$ counts maximal partial-section classes, and $O$ counts the group orbits on $\Phi$.

| $G$ | $\lvert G\rvert$ | $F$ | $P$ | $M$ | $O$ |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | 0 |
| $C_2$ (positive) | 2 | 1 | 1 | 1 | 1 |
| $C_2$ (horizontal mirror) | 2 | 3 | 3 | 3 | 3 |
| $C_2$ (vertical mirror) | 2 | 1 | 1 | 1 | 1 |
| $C_3$ | 3 | 2 | 2 | 2 | 2 |
| $C_2^2$ | 4 | 1 | 6 | 3 | 2 |
| $C_6$ | 6 | 0 | 5 | 5 | 2 |
| $S_3$ (positive) | 6 | 0 | 5 | 5 | 2 |
| $S_3$ (mixed) | 6 | 2 | 7 | 2 | 2 |
| $S_3\times C_2$ | 12 | 0 | 16 | 5 | 2 |

The attached CSV gives every maximal stabilizer; the JSON table gives every partial section and all of its multiplier values. In particular it checks the reflection failure of maximal-section uniqueness from B.8 inside this whole list.

*Manuscript:* 15; 75-77; 150-155; prompt B.f.

*Proof.* Place the three punctures at the vertices of an equatorial equilateral triangle. The positive rotation group of that triangle is $S_3$: it is generated by the 120-degree vertical rotation $r$ and a half-turn $a$ around an axis through a puncture. Reflection $\tau$ in the equator commutes with both. The mapping class group map to permutations and orientation is an isomorphism. For its kernel, fix three arcs forming a tree through the punctures by isotopy; its complement is a disc, and Alexander's disc isotopy makes the remaining map isotopic to the identity. This proves completeness of the ambient finite group and its stated realization.

The program represents this group exactly by pairs $(p, e) \in S_3 \times C_2$. Starting with the trivial subgroup it repeatedly adjoins every group element and takes closure, until no new subgroup occurs. Every subgroup is generated by its elements, so this terminating procedure is exhaustive. It finds sixteen subgroups and ten conjugacy classes, represented in the displayed order by

$$1,\ \langle a\rangle,\ \langle\tau\rangle,\ \langle a\tau\rangle,\ \langle r\rangle,\ \langle a,\tau\rangle,\ \langle r,\tau\rangle,\ \langle r,a\rangle,\ \langle r,a\tau\rangle,\ \langle r,a,\tau\rangle.$$

Here is the independent geometric input to its section calculation. Denote the poles by $N, S$, and the three surviving opposite equatorial axis points by $Q_i$; the other three axis points are the removed punctures. The fixed set of $\langle r\rangle$ is $\{N, S\}$, and that of a positive half-turn is its single surviving $Q_i$. The horizontal mirror has three open intervals, each containing one $Q_i$. A vertical mirror has one real line after its single puncture is removed, and this line contains $N, S, Q_i$. Intersect these fixed sets to find the fixed sets of every subgroup. B.3 and B.4 turn their components into exactly its section classes. A partial section on a mirror interval is contained in a point stabilizer precisely when that point lies on the interval; this determines all inclusions and hence all maximal classes. The code records and checks these inclusions and the conjugation action of every element.

Every reflection and positive half-turn has multiplier $1/2$. For $r$, the values at $N, S$ are respectively $1/3, -1/3$, and its powers give all remaining values. This computes all entries in the attached JSON. The geometry also verifies that all the centralizers occurring for nontrivial sections here are trivial: every mirror component is a line, and every positive section fixes an isolated point. The absence of a common finite overgroup for the two vertex stabilizers along a vertical mirror is retained in the inclusion data. □

**B.24. [Proved] Completed finite-subgroup catalogue and exact indexing.** Every finite-subgroup conjugacy class in the requested range $2g + \nu \le 6$ has now been enumerated. For the hyperbolic types, the numbers of classes in the extended mapping class group are

| $(g, \nu)$ | # classes | proofs and executed tables |
|---|---|---|
| $(0,3), (0,4), (0,5), (0,6)$ | $10, 30, 20, 57$ | $B.23, B.25, B.26$ |
| $(1,1), (1,2), (1,3), (1,4)$ | $13, 40, 38, 115$ | $B.25, B.27$ |
| $(2,0), (2,1), (2,2)$ | $92, 32, 121$ | $B.28, B.29$ |
| $(3,0)$ | $292$ | $B.33, B.37$–$B.41$. |

(B.24.1)

Each table includes the full and partial section classes, multipliers, centralizers, extension incidences, maximal section classes $\Phi$, and their group orbits and stabilizers. In the four exceptional abelian types, the finite mapping-class subgroup counts are respectively $2, 2, 5, 13$ for sphere, plane, annulus and torus. Their injectively realized finite-action counts are $3, 2, 7, 17$; the exact covering extension is retained as part of the section index in B.36, as required by the explicit failures of extension-independent indexing in B.22. The families with a nontrivial kernel on passage to mapping classes are described in B.14. The exact tube words, stabilizer action, strip relations and arbitrary transversal obstruction formerly required here are proved and executed in B.30–B.32.

*Manuscript:* §18; folios 96–101; scans 193–203; §33; folios 222–237; scans 440–470; Part B(e) and B(f) of the task.

*Proof.* The first three rows of (B.24.1) are precisely the complete catalogues cited there. We spell out the closed genus-three reconciliation to record the coverage and the absence of double counting. The positive portion has 55 classes by B.33. For a mixed finite action let $H$ be its positive subgroup. The positive quotient has genus zero, one, two or three, by the positive classification. The genus-zero case gives either a mirror disc (145 classes, B.37) or a projective plane (11 classes, B.39). The genus-one case gives an annulus or Möbius band (57 classes, B.40), or a Klein bottle (6 classes, B.41). The genus-two case is the free positive $C_2$ action and gives 12

mixed classes in B.38. The genus-three case has trivial positive part and gives its six negative involution classes in B.38. Thus the number of mixed classes is

$$145 + 11 + 57 + 6 + 12 + 6 = 237, \qquad 55 + 237 = 292. \tag{B.24.2}$$

Every conjugacy descends to homeomorphisms of both the positive quotient and the full quotient orbifold. The positive quotient genus, and the full quotient's boundary and orientability, separate the listed portions. Within each portion the cited proofs calculate exactly the geometric kernel-normalizer orbits, so these counts are irreducible counts of surface-action conjugacy classes.

B.34 supplies an independent hyperelliptic construction and is counted through its explicit overlaps, not added a second time. All 79 of its rows are identified by monodromy isomorphisms: 18 with positive B.33 rows, 57 with B.37 rows, and four with B.39 rows. The executed program `reconcile_genus3_catalogue.py` records these 79 maps and checks the corresponding section counts. Its unified `type_3_0_complete_catalogue.csv` has 292 different class identifiers, and `type_3_0_complete_sections.json` contains the full finite group and section model for every row. The exceptional replacement in B.36 is complete with its stated realization index; it retains, in particular, the sphere antipodal action's algebraic splitting without identifying it with a geometric fixed point. This proves the claimed coverage with exactly the comparisons and indices used by the computations. □

**B.25. [Proved] Complete executed catalogues in types $(1,1)$ and $(0,4)$.** The extended mapping class groups of types $(1,1)$ and $(0,4)$ have, respectively, 13 and 30 conjugacy classes of finite subgroups, including the trivial subgroup. The tables below give every class and the numbers of full splittings, nontrivial partial splittings, maximal partial section classes $\Phi$, and group orbits on $\Phi$. The accompanying exact-arithmetic files give every section, its image subgroup, its multiplier on every element, its centralizer, all inclusion relations, the group stabilizers, and the puncture stabilizers. These are exhaustive catalogues for the two specified surface types.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; Part B(f) of the task; no further manuscript assertion.

*Proof.* Put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For the punctured torus use their linear actions on $\mathbb{R}^2/\mathbb{Z}^2 \setminus \{0\}$. The homology map identifies its extended mapping class group with $\mathrm{GL}_2(\mathbb{Z})$: surjectivity is supplied by linear maps, and a map acting trivially on homology can be isotoped to fix two transverse basis curves, after which the complementary punctured disc is handled by the Alexander isotopy. This is the punctured version of [22, Theorem 2.5 and pp. 54–55].

For the four-punctured sphere use the pillowcase

$$(\mathbb{R}^2/\mathbb{Z}^2 \setminus (\tfrac{1}{2}\mathbb{Z}^2/\mathbb{Z}^2))/\{p \sim -p\}.$$

Write $U(p) = p + (1/2, 0)$ and $V(p) = p + (0, 1/2)$. The extended mapping class group is $V_4 \rtimes \mathrm{PGL}_2(\mathbb{Z})$ with these affine representatives. The map to $\mathrm{PGL}_2(\mathbb{Z})$ is obtained by lifting through the double branched torus cover and acting on torus homology. Its kernel consists of the four half translations: a kernel element fixes all slope classes; after a half translation it fixes the four punctures and two filling curves, so the Alexander isotopy on their complementary discs kills it. Linear maps split the homomorphism and act on half translations by reduction modulo two. This is the explicit morphism of [22, Proposition 2.7 and the extended statement on p. 57].

Here are finite amalgams that make enumeration complete. The matrices $S, R$ generate $\mathrm{SL}_2(\mathbb{Z})$ by the Euclidean algorithm, since $S^{-1}R = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Modulo $\{\pm I\}$ their images have orders

two and three, with no further relation: on the real projective line, $\bar{S}$ sends the negative half-line to the positive half-line, and both nonidentity powers of $\bar{R}$ send the positive half-line into the negative half-line. The free-product ping-pong argument gives $\mathrm{PSL}_2(\mathbb{Z}) = C_2 * C_3$. Conjugation by $F$ fixes $\bar{S}$ and inverts $\bar{R}$. Consequently

$$\mathrm{PGL}_2(\mathbb{Z}) = V_4 *_{C_2} S_3, \qquad \mathrm{GL}_2(\mathbb{Z}) = \langle S, F\rangle *_{\langle -I, F\rangle} \langle R, F\rangle = D_8 *_{V_4} D_{12},$$

where $D_k$ has order $k$. The second identity follows also by lifting the first through their common central subgroup $\langle -I\rangle$; the normal form for the amalgam leaves no additional kernel. Adding the normal group of half translations gives the pillowcase amalgam

$$(V_4 \rtimes V_4) *_{V_4 \rtimes C_2} (V_4 \rtimes S_3),$$

whose vertex groups have orders 16 and 24, and whose edge group has order 8. The two vertex groups are $D_8 \times C_2$ and $S_4$.

A finite subgroup of an amalgam fixes a vertex in its Bass–Serre tree. Indeed the convex hull of a finite orbit is a finite tree; its centre is fixed, and the action preserves the two vertex types, so an edge centre also gives fixed endpoints. Two vertex subgroups are conjugate in the amalgam precisely through the equivalence generated by conjugacy in each vertex group and identification of edge subgroups. To prove the nontrivial direction, a conjugacy makes the subgroup fix both endpoints of the intervening path; it fixes every edge on that path, which decomposes the conjugacy into those successive vertex and edge steps. Thus finite enumeration in the two vertices, followed by these exact fusion steps, is a complete conjugacy calculation in the infinite group. The program finds 10 and 16 actual subgroups in the torus vertices, and 35 and 30 in the pillowcase vertices. Fusion leaves exactly 13 and 30 classes. All subgroup lists are saturated under adjoining an arbitrary vertex-group element, so the enumeration has no search bound that could omit a subgroup.

We next justify the computation of sections. Let $Q = (\frac{1}{12}\mathbb{Z}^2)/\mathbb{Z}^2$. Every isolated fixed point needed for either model lies in $Q$. For the torus it solves $(A - I)p = 0$; for the pillowcase it solves

$$(A - \epsilon I)p = -t/2 \pmod{\mathbb{Z}^2}, \qquad \epsilon \in \{1, -1\}, \quad t \in (\mathbb{Z}/2)^2.$$

For every positive matrix in the finite vertices, the inverse matrices and the particular solutions of the nonsingular equations have denominators dividing 12. The program certifies this by exact fractions for every matrix and translation, rather than inferring it from a grid count. The singular case for a nonidentity positive element is a nontrivial translation and contributes no solutions to that sign.

For a reflection, the same equations are affine circle equations. The greatest common divisor of the nonzero matrix entries is one or two, so every component meets $Q$. On each such circle the primitive integral kernel direction, divided by 12, joins consecutive sampled points. In the pillowcase the plus and minus circles meet only at two-torsion points, all of which are included in $Q$. Quotienting the sampled graph by $p \sim -p$ therefore gives a subdivision of the true fixed circles. Deleting the prescribed punctures gives exactly their connected components. A component adjacent to a deleted point is a line; the others are ovals. All rotations and all punctures are retained as vertices, so no incidence needed for restriction of sections is lost.

B.3 and B.4 identify these fixed components with section classes. For a partial section with a rotation use its fixed point; for a reflection use its fixed line or oval. Inclusion of image subgroups and containment of the fixed point in the fixed component determine exactly when one section extends another. Applying each affine element gives the exact action on this finite section poset. This computes $\Phi$, its orbits and its stabilizers without replacing pointwise section data by conjugacy of image subgroups in the quotient. Centralizers are 1 for isolated points and lines and $\mathbb{Z}$ for ovals, by B.3–B.4.

At a torus fixed point the local positive derivative is $A$. At a nonbranch pillowcase fixed point it is $\epsilon A$, with the unique sign specified by $Ap + t/2 = \epsilon p$. For these matrices the traces $-2, -1, 0, 1$ give orders $2, 3, 4, 6$; the half-turn has multiplier $1/2$, and otherwise the sign of $\det(e_1, Ae_1)$ selects $+1/n$ or $-1/n$. Reflections have multiplier $1/2$. At a removed pillowcase branch point the quotient coordinate squares a torus coordinate, so the positive rotation angle doubles. The program records these puncture data separately. It checks the order and power identities for every element of every section and checks that every section containing a rotation has exactly one maximal overgroup. The reflection classes with several maximal extensions are retained explicitly.

Finally, C.5–C.6 show that the chosen finite actions compute the action attached to each mapping class subgroup up to conjugacy, so these fixed-component computations depend on the listed subgroup class, not on an unrecorded realization choice. □

In the tables $f$ is the number of full splitting classes, $p$ the number of nontrivial partial splitting classes with image subgroup retained, $\phi = |\Phi|$, and $o = |\Phi/\Gamma|$. Juxtaposition in the generator column means composition of the matrices or affine maps defined above. For $(0, 4)$ the letters $S, R, F$ mean their pillowcase classes. The complete output files are `type_1_1_finite_subgroups.csv`, `type_0_4_finite_subgroups.csv`, and the corresponding `partial_sections.json` files, produced by `catalogue_low_complexity.py`.

Type $(1, 1)$

| Class | Group | Generators | $\lvert\Gamma^+\rvert$ | $f$ | $p$ | $\phi$ | $o$ |
|---|---|---|---|---|---|---|---|
| T01 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| T02 | $C_2$ | $SF$ | 1 | 2 | 2 | 2 | 2 |
| T03 | $C_2$ | $SSF$ | 1 | 1 | 1 | 1 | 1 |
| T04 | $C_2$ | $SS$ | 2 | 3 | 3 | 3 | 3 |
| T05 | $C_3$ | $RR$ | 3 | 2 | 2 | 2 | 2 |
| T06 | $C_2^2$ | $SS, SF$ | 2 | 3 | 10 | 3 | 3 |
| T07 | $C_2^2$ | $SS, SSF$ | 2 | 1 | 6 | 3 | 2 |
| T08 | $C_4$ | $S$ | 4 | 1 | 4 | 3 | 2 |
| T09 | $S_3$ | $RR, RRF$ | 3 | 2 | 7 | 2 | 2 |
| T10 | $S_3$ | $RR, RF$ | 3 | 0 | 5 | 5 | 2 |
| T11 | $C_6$ | $R$ | 6 | 0 | 5 | 5 | 2 |
| T12 | $D_8$ | $S, SF$ | 4 | 1 | 15 | 3 | 2 |
| T13 | $D_{12}$ | $R, RRF$ | 6 | 0 | 16 | 5 | 2 |

Type $(0, 4)$

| Class | Group | Generators | $\lvert\Gamma^+\rvert$ | $f$ | $p$ | $\phi$ | $o$ |
|---|---|---|---|---|---|---|---|
| S01 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| S02 | $C_2$ | $SF$ | 1 | 4 | 4 | 4 | 4 |
| S03 | $C_2$ | $SFV$ | 1 | 1 | 1 | 1 | 1 |
| S04 | $C_2$ | $SFUV$ | 1 | 0 | 0 | 0 | 0 |
| S05 | $C_2$ | $F$ | 1 | 2 | 2 | 2 | 2 |
| S06 | $C_2$ | $V$ | 2 | 2 | 2 | 2 | 2 |
| S07 | $C_2$ | $S$ | 2 | 0 | 0 | 0 | 0 |
| S08 | $C_3$ | $RR$ | 3 | 1 | 1 | 1 | 1 |

| Class | Group | Generators | $\lvert\Gamma^+\rvert$ | $f$ | $p$ | $\phi$ | $o$ |
|---|---|---|---|---|---|---|---|
| S09 | $C_2^2$ | $V, SF$ | 2 | 2 | 9 | 4 | 3 |
| S10 | $C_2^2$ | $V, SFU$ | 2 | 0 | 3 | 3 | 2 |
| S11 | $C_2^2$ | $UV, SF$ | 2 | 0 | 6 | 6 | 3 |
| S12 | $C_2^2$ | $UV, SFV$ | 2 | 2 | 6 | 2 | 2 |
| S13 | $C_2^2$ | $UV, F$ | 2 | 2 | 8 | 2 | 2 |
| S14 | $C_2^2$ | $SF, F$ | 2 | 0 | 6 | 6 | 3 |
| S15 | $C_2^2$ | $SFUV, F$ | 2 | 0 | 2 | 2 | 1 |
| S16 | $C_4$ | $FU$ | 2 | 0 | 2 | 2 | 1 |
| S17 | $C_2^2$ | $V, U$ | 4 | 0 | 6 | 6 | 3 |
| S18 | $C_2^2$ | $UV, S$ | 4 | 0 | 2 | 2 | 1 |
| S19 | $C_4$ | $SU$ | 4 | 2 | 4 | 2 | 2 |
| S20 | $S_3$ | $RR, FR$ | 3 | 1 | 8 | 4 | 2 |
| S21 | $C_2^3$ | $V, U, SF$ | 4 | 0 | 18 | 6 | 3 |
| S22 | $C_2^3$ | $UV, SF, F$ | 4 | 0 | 12 | 6 | 2 |
| S23 | $C_4 \times C_2$ | $FU, SU$ | 4 | 0 | 8 | 6 | 2 |
| S24 | $D_8$ | $FU, V$ | 4 | 0 | 12 | 6 | 2 |
| S25 | $D_8$ | $SU, SFV$ | 4 | 2 | 16 | 2 | 2 |
| S26 | $D_8$ | $FU, SFV$ | 4 | 0 | 6 | 2 | 1 |
| S27 | $D_8$ | $SU, V$ | 8 | 0 | 8 | 6 | 2 |
| S28 | $A_4$ | $RR, RUR$ | 12 | 0 | 10 | 10 | 2 |
| S29 | $D_8 \times C_2$ | $FU, SU, V$ | 8 | 0 | 28 | 6 | 2 |
| S30 | $S_4$ | $FUR, URF$ | 12 | 0 | 32 | 10 | 2 |

**B.26. [Proved] Complete executed spherical catalogues through six punctures.** For $\nu = 3, 4, 5, 6$, the extended mapping class group of the sphere with $\nu$ unlabelled punctures has respectively

$$10, \quad 30, \quad 20, \quad 57$$

conjugacy classes of finite subgroups, including the trivial subgroup. The exact finite parameterization below gives every class once. The executed files `type_0_ν_orbifold_catalogue.csv` and `type_0_ν_orbifold_sections.json`, for each of these four values of $\nu$, give all partial-section classes, all their inclusion relations, $\Phi$, its group action and stabilizers, the centralizers, the multiplier on each section element, and the puncture stabilizers. Thus these files settle the whole computation requested in B(f) for the hyperbolic spherical types with $2g + \nu \leq 6$.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

*Proof.* Extend an action over the punctures. By B.1 and the sphere theorem of [44, Theorem 1.2], a finite topological action on $S^2$ is conjugate to an orthogonal action. Write $Q = S^2/G$ with its local orbifold groups. A point has cyclic positive stabilizer, or a dihedral stabilizer with mirrors; the mirror-only stabilizer has order two. The covering formula is

$$\chi_{\mathrm{orb}}(Q) = \chi(|Q|) - \sum_{j=1}^{r}(1 - 1/m_j) - \frac{1}{2}\sum_{j=1}^{k}(1 - 1/n_j) = \frac{2}{N}, \qquad N = |G|. \tag{B.26.1}$$

Here the $m_j$ are interior cone orders and the $n_j$ are corner orders. One proves this formula directly by a triangulation whose vertices include all cone and corner points: count each cell in the covering with weight equal to the reciprocal of its stabilizer order. Mirrors are boundary strata; they introduce no additional term after the boundary cells and their endpoints are counted together.

Since (B.26.1) is positive, the underlying compact surface is $S^2$, $\mathbb{RP}^2$, or a disc. The sphere has at most three cones. A single cone is impossible, since its local generator would be trivial in the orbifold presentation. Two cones have equal orders, since the two local generators are inverses and both local cyclic groups inject. For three cones the inequality $1/m_1 + 1/m_2 + 1/m_3 > 1$ gives precisely $(2,2,n), (2,3,3), (2,3,4), (2,3,5)$. The projective plane has at most one cone. For a disc, positivity gives $2r + k \leq 3$. A single corner without a cone is impossible; two corners without a cone have equal orders, by the same local-injectivity argument. Three corners give the same four triples. One cone and one corner give $2/m + 1/n > 1$, hence $(m,n) = (2,n)$ or $(3,2)$. Consequently the following table is exhaustive. In its notation $S(m_1, \ldots, m_r)$ is a sphere with cones, $P(m)$ is a projective plane with one cone, $D(m_1, \ldots, m_r; n_1, \ldots, n_k)$ is a mirror disc with the specified interior cones and cyclically ordered corners, and an empty list means no such singularities.

| $Q$ | $N$ | $Q$ | $N$ |
|---|---|---|---|
| $S()$ | 1 | $P()$ | 2 |
| $S(n,n)$ | $n$ | $P(n)$ | $2n$ |
| $S(2,2,n)$ | $2n$ | $D(;)$ | 2 |
| $S(2,3,3)$ | 12 | $D(;n,n)$ | $2n$ |
| $S(2,3,4)$ | 24 | $D(n;)$ | $2n$ |
| $S(2,3,5)$ | 60 | $D(;2,2,n)$ | $4n$ |
| | | $D(2;n)$ | $4n$ |
| | | $D(;2,3,3)$ | 24 |
| | | $D(;2,3,4)$ | 48 |
| | | $D(;2,3,5)$ | 120 |
| | | $D(3;2)$ | 24 |

(B.26.2)

Every integer $n$ in this table is at least two. For $3 \leq \nu \leq 6$ only $2 \leq n \leq 6$ can occur. Indeed, in every family depending on $n$, the union of all point orbits of size strictly smaller than $n$ has at most two points; an invariant set of at least three punctures therefore contains an orbit of size at least $n$. This assertion also follows immediately by dividing $N$ by each local group order in (B.26.2). In the two icosahedral rows every point orbit has at least twelve points. Those rows cannot occur in the present range. These are proved finite bounds, rather than bounds inferred from the output.

We now give the exact decoration rule used in the program. A cone orbit may be removed or retained; put $c_j \in \{0,1\}$ accordingly. A corner orbit similarly has $d_j \in \{0,1\}$. For each open mirror edge let $e_i \geq 0$ be the number of removed orbits on that edge; a cornerless mirror circle is one edge for this purpose. Finally let $a \geq 0$ be the number of regular interior orbits removed. The single equation is

$$\sum_j c_j \frac{N}{m_j} + \sum_j d_j \frac{N}{2n_j} + \sum_i e_i \frac{N}{2} + aN = \nu. \tag{B.26.3}$$

On a sphere, permutations of cones of equal order identify decorations. On a mirror disc, identify them under precisely the rotations and reflections of the cyclic corner list that preserve its order labels. The corresponding edge counts move with their edges. For two corners this rule includes the independent exchanges of the two corners and the two edges. For one or zero corners it introduces no further identification of the single edge count. There is at most one interior cone in every remaining disc or projective-plane row.

For clarity, this rule also proves completeness of the conjugacy classification. A homeomorphism of an interval moves any two unlabelled finite subsets of the same size into one another, and the same assertion holds in the connected regular interior of each quotient surface. Such moves extend to the quotient surface by disjoint small discs and strips. The only remaining data are therefore the selected singular strata and the cyclic mirror-edge counts in (B.26.3),

modulo the explicitly stated symmetries. Conversely every quotient homeomorphism preserves those strata, their orders, and their cyclic incidence. Since $S^2$ is the universal orbifold cover, a homeomorphism between the decorated quotient orbifolds lifts to a conjugacy between their covering actions; this follows by lifting paths and their local cone or mirror charts. A conjugacy of actions descends to such a quotient homeomorphism. Thus no extra equivalence of decorations is possible. Finally C.5 and C.6 identify these topological action classes with conjugacy classes of finite subgroups in the extended mapping class group of the punctured sphere. The induced action there is faithful: a nonidentity positive periodic map isotopic to the identity would fix every puncture individually, whereas a nonidentity sphere rotation has exactly two fixed points; a negative map cannot be isotopic to the identity. Here $\nu \geq 3$ is essential.

Here are certificates for the finite group calculations. The source `catalogue_spheres.py` constructs the cyclic and dihedral families as regular permutation groups, the tetrahedral and octahedral positive groups as $A_4, S_4$, their needed extensions by an explicitly given central involution, and the full octahedral group as signed permutations of three axes. All generators, products, orders and signs are recorded as finite permutations. The program verifies each orbifold relation, injection of every local cyclic or dihedral group, and $N\chi_{\mathrm{orb}}(Q) = 2$ using rational arithmetic. These checks prove that the permutation group is the entire orbifold group, as follows. The kernel of the displayed quotient of the orbifold presentation produces a covering with no cone, corner or mirror singularity because all local groups inject. Its orientation character is trivial, and its Euler characteristic is two. It is therefore a closed orientable sphere; its fundamental group is trivial. This proves injectivity of the quotient presentation as well as its already checked surjectivity. In particular these certificates do not merely exhibit a possibly proper finite quotient of an unexamined presentation.

We describe how the section data are recovered without numerical sampling. Cone and corner lifts are cosets of their cyclic and dihedral stabilizers. In a disc presentation choose reflections $r_0, \ldots, r_k$ with

$$(r_j r_{j+1})^{n_j} = 1, \qquad r_k = e r_0 e^{-1}, \qquad x_1 \cdots x_r e = 1. \tag{B.26.4}$$

Indices of the corner orders are shifted here to start at zero. The code stores $r_0, \ldots, r_{k-1}$ and derives the final reflection. An edge lift is a coset of $\langle r_i \rangle$ and meets the adjacent corner cosets; the closing incidence at edge zero is multiplied by $e^{-1}$. With no corners it instead joins the corresponding two cosets of $\langle r_0 \rangle$ by that same rule. Insert the prescribed removed vertices along each edge, retaining two auxiliary vertices in each complementary segment. These auxiliary vertices prevent a one-edge loop from collapsing when represented as an ordinary finite adjacency graph. This is exactly the lifted mirror locus with a finite subdivision.

For any reflection $r \in G$, keep the vertices whose stabilizers contain $r$. Before punctures are deleted the program verifies degree two at every such vertex. Its components are consequently the full fixed circles of $r$. Deleting the selected puncture vertices gives all the fixed lines and ovals, with their exact incidence. For a subgroup whose positive part is nontrivial, the fixed points are precisely the retained cone and corner lifts whose stabilizers contain that subgroup. A negative involution without a mirror has an empty fixed set. Thus B.3 and B.4 supply every section class, with centralizer 1 for a point or a line and $\mathbb{Z}$ for an oval. A section over $H$ extends a section over $K < H$ exactly when its fixed component is contained in the latter fixed component. Hence the same finite incidence calculation gives the entire partial-section poset and all its maximal elements, including the reflection cases with several maximal extensions. Left multiplication on cosets gives the group action and every stabilizer without a choice of orbit representatives.

The multiplier is exact as well. Orient a local lift of each cone, and at a corner use the positive generator $r_j r_{j+1}$ determined by the oriented boundary order. Write this generator as $u$, of order $m$. At the point represented by a coset $gD$, an element $h$ of its positive stabilizer satisfies $g^{-1}hg = u^b$, and

$$\mu(h) = \operatorname{sgn}(g)\frac{b}{m} \pmod{\mathbb{Z}}. \tag{B.26.5}$$

Replacing $g$ by another coset representative leaves this value unchanged: a positive local element commutes with $u$, while a negative local element both reverses the sign and inverts $u$. For a reflection the value is $1/2$, and for the identity it is zero. Formula (B.26.5) is the local instance of B.12, and applies to all sections in the files. The code checks its order and every power identity, as well as uniqueness of the maximal extension for every section with nontrivial positive part. All nontrivial subgroups of each finite permutation group are enumerated by saturation under adjoining each group element; this stops only when the subgroup set is closed under every possible enlargement.

The following compact census is part of the executed output; each entry is the number of conjugacy classes of that group order. A dash denotes zero. Complete individual rows, rather than just this census, are in the CSV files specified in the statement.

| $\nu\backslash N$ | 1 | 2 | 3 | 4 | 5 | 6 | 8 | 10 | 12 | 16 | 20 | 24 | 48 | total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 3 | 1 | 1 | – | 3 | – | – | 1 | – | – | – | – | 10 |
| 4 | 1 | 6 | 1 | 11 | – | 1 | 7 | – | 1 | 1 | – | 1 | – | 30 |
| 5 | 1 | 4 | 1 | 4 | 1 | 3 | 1 | 3 | 1 | – | 1 | – | – | 20 |
| 6 | 1 | 7 | 1 | 14 | 1 | 9 | 8 | 1 | 9 | 1 | – | 4 | 1 | 57 |

(B.26.6)

As an independent check, the program compares the multisets of all six columns consisting of group order, positive order, full-section count, nontrivial partial-section count, maximal-section count, and group-orbit count, for $\nu = 3, 4$, against the previous permutation and amalgam calculations in B.23 and B.25. Both multisets agree, with multiplicities. The check and its inputs are attached; the completeness proof above does not depend on that agreement. □

**B.27. [Proved] Complete executed punctured-torus catalogues through four punctures.** The extended mapping class groups of a torus with $\nu = 1, 2, 3, 4$ unlabelled punctures have, respectively,

$$13, \quad 40, \quad 38, \quad 115$$

conjugacy classes of finite subgroups. The files `type_1_`$\nu$`_affine_catalogue.csv`, together with their `affine_sections.json` companions and `torus_affine_normal_forms.json`, list every class once. They specify each group by its complete finite set of rational affine transformations, its punctures, every full and nontrivial partial section class, all extension incidences, $\Phi$ and its group orbits and stabilizers, the centralizers, and $\mu$ on every section and puncture-stabilizer element. Thus the torus portion of B(f) is complete for every hyperbolic type in the requested range.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; Part B(f) of the task; no further manuscript assertion.

*Proof.* We give the finite classification underlying the program `catalogue_tori.py`; this also specifies the tables without relying on abstract finite-group names. By B.14 extend over the punctures and put the action in affine form on $V/\Lambda = \mathbb{R}^2/\mathbb{Z}^2$. Write

$$1 \longrightarrow T \longrightarrow G \longrightarrow H \longrightarrow 1, \qquad T \subset V/\Lambda, \quad H \subset \mathrm{GL}_2(\mathbb{Z}). \tag{B.27.1}$$

Here $T$ consists exactly of the translations in $G$. It acts freely on $V/\Lambda$, so each $T$-orbit on the puncture set has $|T|$ elements. Consequently $|T| \mid \nu$ and in particular $|T| \leq 4$. The complete list of subgroups of $V/\Lambda$ of order at most four consists of the trivial subgroup, the three cyclic subgroups of order two, the four cyclic subgroups of order three, the six cyclic subgroups of order four, and $(\frac{1}{2}\Lambda)/\Lambda$. Indeed a group of order at most four is cyclic or $C_2^2$; a cyclic subgroup is generated by a point of that exact order, and division by the number of generators gives precisely these counts. All fifteen groups lie in $(\frac{1}{12}\Lambda)/\Lambda$. The thirteen possibilities for $H$, up to integral conjugacy, are exactly those proved in B.25. For each we retain only the $H$-invariant $T$.

We now enumerate the affine extensions, including the actions with no common fixed point. If $H$ contains a nonidentity positive element, choose a generator $r$ of its cyclic positive subgroup. Every affine map with linear part $r$ has a fixed point, since $I-r$ is invertible on $V$ and is therefore onto on $V/\Lambda$. Translation of that point to the origin makes this generator $(r,0)$. A positive cyclic $H$ then gives exactly $G=\langle T,(r,0)\rangle$. For a dihedral $H$ choose a negative generator $f$ and write its lift as $(f,b)$. The two relations in $H$ give exactly

$$(I-r)b\in T,\qquad (I+f)b\in T. \tag{B.27.2}$$

Conversely these equations make the subgroup generated by $T,(r,0),(f,b)$ a finite group with kernel $T$ and quotient $H$: every word reduces to a translation followed by $r^j$ or $r^jf$, and its linear part distinguishes these cosets. Values of $b$ modulo $T$ are enough. The finite set $(I-r)^{-1}T$ therefore supplies every candidate, with no bound on words or finite quotients involved. Translations of the origin preserving this normal form are exactly $a$ with $(I-r)a\in T$; they change $b$ by $(I-f)a$ modulo $T$.

The remaining nontrivial linear groups have $H=\langle f\rangle$ of order two. Put $W=(V/\Lambda)/T$. Their extensions are represented by $b\in W$ with $(I+f)b=0$, modulo $(I-f)W$. Every such class has a representative in $W[2]$. To verify this assertion directly, choose a lift $v\in V$ of $b$ for the enlarged lattice $\Lambda_T=\{v\in V : v \bmod \Lambda\in T\}$. Then $(I+f)v\in\Lambda_T$. Write $v=v_+ + v_-$ in the real $+1$ and $-1$ eigenspaces of $f$. Subtracting $(I-f)(v_-/2)$ removes $v_-$, and the remaining $v_+$ satisfies $2v_+\in\Lambda_T$. Thus the program uses precisely

$$2b\in T,\qquad (I+f)b\in T. \tag{B.27.3}$$

For two candidates $b,b'$ the translation-conjugacy test is also exact. Let $w$ be a primitive integral row vector satisfying $wf=w$. The image of $I-f$ on $V/\Lambda$ is the connected circle $\ker w$: its real image is that eigenspace, and multiplication by any nonzero integer is onto on a circle. Hence the two candidates are equivalent precisely when $w(b-b')\in w(T)$ in $\mathbb{R}/\mathbb{Z}$. This includes both reflections and fixed-point-free glide reflections.

Integral changes of basis are handled without truncating an infinite matrix group. For $H=1$ or $\{\pm I\}$, use the generators $S,R,F$ of B.25 on the complete fifteen-element list of possible $T$'s. Their orbit calculation is the full $\mathrm{GL}_2(\mathbb{Z})$-conjugacy calculation. For a fixed $T$, a spanning tree of this finite orbit and Schreier's formula give generators for its entire integral stabilizer. Explicitly, with a chosen representative $B_U$ carrying $T$ to $U$, the generator associated to a matrix $A\in\{S,R,F,S^{-1},R^{-1}\}$ and the edge $U\mapsto AU$ is

$$B_{AU}^{-1}AB_U. \tag{B.27.4}$$

For every other $H$ its integral normalizer is finite. Here is an exhaustiveness check for the small matrix enumeration in the code. For a reflection conjugate to $\mathrm{diag}(-1,1)$, a commuting integral unimodular matrix is diagonal with entries $\pm1$. For a reflection conjugate to $\left(\begin{smallmatrix}0&-1\\-1&0\end{smallmatrix}\right)$, the equation is $A=\left(\begin{smallmatrix}a&b\\b&a\end{smallmatrix}\right)$, with $(a-b)(a+b)=\pm1$, giving four solutions. A normalizer of the four-group generated by this reflection and $-I$ may also exchange the two eigenspaces; the same calculation gives eight solutions. A group with a rotation of order four has its normalizer in the eight signed permutation matrices, since a commuting matrix has determinant $a^2+b^2=1$. A group with rotation of order three or six has its normalizer in the twelve matrices for the hexagonal lattice: the commuting equation gives the norm $a^2+ab+b^2=1$, and one reflection supplies the other coset. Testing these finite sets against $H$ gives the full normalizer. Every resulting matrix has entries of absolute value at most one, so the code's box of entries from $-2$ to 2 contains every solution and introduces none outside the tested normalizer equations. Applying these matrices and the origin changes above fuses all and only affine-conjugate candidates. The result is 93 affine action classes with $|T|\le 4$ before choosing punctures; some admit no invariant set of a given required size.

We explain why decorating these affine actions gives topological conjugacy, including all permutations of the singular strata. B.14 proves that a topological conjugacy between finite affine actions can be replaced by an affine conjugacy. The replacement also induces the same permutation on the strata used here. In detail write a lift of the conjugacy as $h(x) = Ax + \delta(x)$ with $\delta$ periodic, and put $\bar{\delta} = \int_{V/\Lambda} \delta$. The equivariance equations and invariance of Haar measure show that $x \mapsto Ax + \bar{\delta}$ has the same equivariance. The straight interpolation of these two lifts is an equivariant homotopy. An isolated fixed point of a nontrivial positive subgroup stays constant under this homotopy. Each reflection circle stays in the same fixed-set component. The degree on every such circle is fixed by the homology map $A$; these circles are essential, since they lift to parallel lines. Together with the images of the isolated vertices this determines the permutation of the intervening open mirror intervals. Thus the affine normalizer gives exactly the possible stratum permutations.

To compute them, let $\mathcal{V}$ be the union of the fixed points of all positive nontranslation elements. It is a finite set. The fixed circles of the reflections, cut at $\mathcal{V}$, are open mirror intervals or uncut mirror circles. Take the $G$-orbits of the vertices and of these mirror components; call their numbers $v$ and $e$. There is one regular interior stratum in the quotient, since its interior after removal of finitely many cone points is connected. A vertex orbit may be removed once, a mirror-component orbit may contain any nonnegative number of puncture orbits, and the regular interior may contain any nonnegative number. Thus the decoration is

$$(d_1, \dots, d_v; c_1, \dots, c_e; a), \qquad d_i \in \{0, 1\},\ c_j, a \geq 0,$$

subject to the exact equation

$$\sum_i d_i |V_i| + \sum_j c_j \frac{|G|}{2} + a|G| = \nu. \tag{B.27.5}$$

Divide these finitely many decorations by the stratum permutations of the affine normalizer computed above. Counts are sufficient on an open mirror interval or circle: an increasing interval map, or a circle map of the prescribed degree, moves any two unlabelled finite sets of the same size into each other. The moves extend through small strips in the quotient. In the regular interior use disjoint moving discs; its configuration space of finitely many unlabelled points is connected. Such quotient isotopies lift through the finite orbifold covering. They preserve all cone and corner charts and therefore give equivariant homeomorphisms upstairs. Conversely a conjugacy preserves the displayed orbit sizes and the incidence of these strata. This proves completeness and injectivity of the decorated classification.

All computations use exact integers on $Q = (\frac{1}{144}\mathbb{Z}^2)/\mathbb{Z}^2$. We record why this finite mesh loses no component or conjugacy. Coordinates of $T$ have denominator dividing its order $k \leq 4$. For the rotations of orders $2, 3, 4, 6$, the inverses of $I - r$ have denominators $2, 3, 2, 1$, respectively. Thus all solutions needed for (B.27.2), all normalized origin changes, and all isolated fixed points have denominators dividing $2k$ or $3k$. The representatives (B.27.3) have denominator dividing $2k$. For a reflection normalizer, the origin equation is a rank-one integral equation with nonzero Smith invariant 1 or 2; every connected solution component has a representative with denominator dividing $4k$. The integer 144 is divisible by each of these integers for $1 \leq k \leq 4$. The code separately checks the inverse matrix and particular-solution denominators for every positive affine element. On each reflection circle, moving by its primitive integral direction divided by 144 connects consecutive sampled points. Every isolated intersection and every selected puncture is retained as a vertex. This graph is a subdivision of the actual fixed-circle arrangement, so its components after removing punctures are exactly the ovals and lines of the action. The mesh has enough unused representatives on each eligible stratum to place the at most four punctures; the executed program explicitly verifies the prescribed orbit counts and total cardinality for every decoration.

For each unremoved vertex $p$, enumerate all subgroups of its finite cyclic or dihedral stabilizer that contain a positive nonidentity element. Each gives the section class with fixed point $p$. For each reflection enumerate the components of its fixed circles after deleting the punctures. B.3–B.4 identify exactly these data with all nontrivial partial-section classes, and identify their centralizers as 1 at points and lines and $\mathbb{Z}$ at ovals. Containment of image subgroups and reverse containment of their fixed components give all extension relations. Applying every affine matrix computes the group action on this poset and hence $\Phi$, its orbit decomposition and every stabilizer. At a positive fixed point the local matrix is its actual linear part. Its trace $-2, -1, 0, 1$ gives order $2, 3, 4, 6$, and the sign of $\det(e_1, Ae_1)$ gives the sign of the angle except for a half-turn. Reflections have multiplier $1/2$. The program checks the order and power identities on every element of every section. It retains rather than deletes the reflection section classes with several maximal extensions.

Finally a nonidentity finite affine map acting trivially in the punctured-torus mapping class group would have trivial homology matrix and hence be a nonidentity translation. It would also fix each puncture, contradicting freeness of translations. Thus each action gives a faithful finite mapping class subgroup. The realization and uniqueness statements C.5–C.6 identify their conjugacy classes with the topological action classes just enumerated. The computation returns the four counts in the statement. For $\nu = 1$, the multiset of group orders, positive orders, full splitting counts and partial splitting counts agrees, class for class, with the independent amalgam calculation of B.25. A first complete run with denominator 72 and the final run with denominator 144 also give the same four counts; the proof of completeness uses the latter mesh. □

**B.28. [Proved] Complete executed extended genus-two catalogue.** The extended mapping class group of the closed orientable genus-two surface has exactly 92 conjugacy classes of finite subgroups. Of these, 57 contain the hyperelliptic involution and 35 do not; 21 classes preserve orientation. The files `type_2_0_extended_catalogue.csv` and `type_2_0_extended_sections.json` specify every group by an exact finite permutation representation and give every full and partial splitting, its multiplier and centralizer, all extension incidences, $\Phi$, and its group orbits and stabilizers.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

*Proof.* Let $\iota$ be the hyperelliptic mapping class. The genus-two Birman–Hilden homomorphism, with orientation reversal allowed, is

$$1 \longrightarrow \langle \iota \rangle \longrightarrow \mathcal{T}_{2,0} \longrightarrow \mathcal{T}_{0,6} \longrightarrow 1. \tag{B.28.1}$$

We recall its geometric content, to specify every comparison used here. The involution is the deck map of the double cover of $S^2$ branched at six points. The covering is determined by the homomorphism from the fundamental group of the six-times-punctured sphere to $C_2$ that sends each peripheral generator to the nonidentity element; the product relation is respected because six is even. Every homeomorphism preserving the branch set preserves this homomorphism, and hence has exactly two lifts, differing by $\iota$. The genus-two hyperelliptic mapping class is central: the usual chain of five nonseparating curves is invariant under the involution, their Dehn twists generate the oriented genus-two mapping class group, and a reflection of the standard branched-cover model commutes with $\iota$ and supplies the negative coset. The Birman–Hilden isotopy property identifies the kernel in (B.28.1) with these two deck classes; it also makes the lifting map onto. This is the hyperelliptic Birman–Hilden description used in B.25, in genus two instead of the four-point pillowcase; see [22, §9.4.2, p. 257, with the proof in §9.4.3].

For a finite $K \subset \mathcal{T}_{2,0}$ put $E = K\langle \iota \rangle$ and let $H$ be its image in $\mathcal{T}_{0,6}$. The group $E$ is exactly the full inverse image of $H$ in (B.28.1), and has order $2|H|$. Thus B.26 supplies its complete list

of 57 conjugacy classes. A subgroup $K$ with this fixed $E$ either equals $E$, or is a complement to $\langle\iota\rangle$. Every complement is normal in $E$ and is uniquely the kernel of a homomorphism

$$\chi : E \longrightarrow \mathbf{F}_2, \qquad \chi(\iota) = 1. \tag{B.28.2}$$

Indeed $E = K \times \langle\iota\rangle$ for a complement, giving this homomorphism; conversely the kernel of (B.28.2) has index two and maps isomorphically onto $H$. Because $\iota$ is central in the whole mapping class group, conjugate $K$'s have conjugate associated $E$'s. For a fixed $E$, their further fusion is therefore exactly the action of its normalizer on the characters (B.28.2).

Here are explicit presentations of all the finite groups $E$. They are also used to calculate the sections, so the statement is independent of the names assigned to abstract finite groups. Let $Q = S^2/H$ be one of the spherical orbifolds in B.26, with its six-branch-point decoration. Write $z = \iota$, $z^2 = 1$, and require $z$ to commute with all generators. An interior cone of order $m_j$ has a generator $x_j$ with

$$x_j^{m_j} = z^{d_j}, \tag{B.28.3}$$

where $d_j = 1$ exactly when its orbit is a branch orbit. Let $a$ be the number of regular interior branch orbits. For a spherical underlying quotient the remaining relation is

$$\prod_j x_j = z^a. \tag{B.28.4}$$

For a projective-plane quotient, with crosscap generator $u$, it is $u^2 \prod_j x_j = z^a$.

For a mirror disc, expand the cyclic corner list by inserting each branch orbit on an open mirror edge as an extra entry $(m, d) = (1, 1)$. An original corner contributes $(m, d)$ with its original order $m$ and its branch flag $d$. The distinction is retained as a color. Let the resulting cyclic list be $(m_j, d_j)_{j=0}^{L-1}$. There are reflections $R_0, \ldots, R_{L-1}$ and a boundary element $e$, with

$$\begin{aligned} R_j^2 &= 1, \\ \prod_i x_i\, e &= z^a, \\ (R_j R_{j+1})^{m_j} &= z^{d_j}, \\ R_L &= e R_0 e^{-1}. \end{aligned} \tag{B.28.5}$$

When $L = 0$ there is one reflection $R_0$ and the last relation is $eR_0e^{-1} = R_0$. A newly inserted mirror branch entry says exactly $R_{j+1} = R_j z$.

These are presentations, rather than only relations satisfied by some finite images. To verify this, use the branched cover described above. Its full group of lifts of $H$ is a group of order $2|H|$. A small positive circle about a branch point has deck monodromy $z$; about an unbranched point it has trivial deck monodromy. This gives (B.28.3). The product of the boundary loops gives (B.28.4), or its crosscap or disc version. At a branch point on a mirror, the local map is $w \mapsto w^2$: the two reflections on the adjacent lifted half-branches differ by $w \mapsto -w$. This gives precisely the inserted entry in (B.28.5). At a corner the positive local cyclic generator has its order doubled exactly when it is branched, giving the other entries. These local monodromies and $z$ generate the full lift group. Conversely quotienting the displayed presentation by $z = 1$ gives the original spherical orbifold presentation, which is $H$ by the local-injectivity and Euler-characteristic certificate in B.26. Since $z$ is central of order at most two, the presented group has at most $2|H|$ elements. Its surjection onto the geometric lift group proves equality and the presentation assertion.

We determine the entire normalizer action needed in (B.28.2), not just conjugation inside the finite group. A normalizing mapping class descends through (B.28.1) and through the $H$-action to an orbifold homeomorphism of $Q$ preserving its branch decoration. Conversely any such

orbifold homeomorphism lifts to $S^2$, because $S^2$ is the universal orbifold cover of $Q$, and then to the double branched cover, because the six peripheral mod-two monodromies are preserved. Thus its lifts give the whole normalizer action, with inner automorphisms and the deck involution as the harmless lifting ambiguities. Both act trivially on characters to $\mathbf{F}_2$.

On a sphere, an orbifold homeomorphism sends each local cone generator to a conjugate of a local generator with the same original order and branch flag, or to its inverse when it reverses orientation. Characters ignore conjugation and inversion. All permutations among cones of the same pair $(m_j, d_j)$ are realized by homeomorphisms of the sphere. Point-pushing and pure braids conjugate the local generators and therefore act trivially on these characters. The ordinary branch generators already equal the central element $z$. Consequently the exact action on (B.28.2) is the permutation action on these colored cone coordinates. The character equations are

$$m_j\chi(x_j) = d_j, \qquad \sum_j \chi(x_j) = a \pmod 2. \tag{B.28.6}$$

This includes, for example, the two elliptic complements in a four-group lying over a sphere involution with no fixed branch points: exchanging the two cone points exchanges those complements. They must be fused, and the program does so.

On a mirror disc there is at most one interior cone. The normalizer preserves the cyclic order on its boundary, up to reversal. It therefore induces exactly the rotations and reversals of the expanded colored corner list that preserve its colors. Each such symmetry is realized by a disc homeomorphism preserving the interior decoration. To see its action explicitly, extend the reflection sequence by $R_{j+L} = eR_je^{-1}$. A cyclic shift by $s$ sends $R_j$ to $R_{j+s}$ and preserves $e$; a reversal sends $R_j$ to $R_{s-j}$ and $e$ to $e^{-1}$. The interior cone generator is sent to a conjugate of itself or its inverse. Modulo two these formulas are literal permutations of the reflection coordinates and leave the interior coordinate fixed. They respect every relation in (B.28.5). Every remaining disc mapping class comes from moving the interior marked points or from a boundary twist; its effect on local generators is conjugation and hence is trivial on the characters. The character equations are therefore

$$\begin{aligned} m_i\chi(x_i) &= d_i, \\ \chi(e) &= a + \sum_i \chi(x_i), \\ m_j(\chi(R_j) + \chi(R_{j+1})) &= d_j \pmod 2. \end{aligned} \tag{B.28.7}$$

The program enumerates these bit vectors and divides by exactly the colored dihedral permutations just proved. For a projective-plane quotient, the six-branch-point equation allows only the three cases $P()$ with $a = 3$, $P(2)$ with $(d, a) = (1, 1)$, and $P(3)$ with $(d, a) = (0, 1)$. Their full lifts are respectively $C_4, C_8, C_{12}$, with $z$ the involution. None has a character taking $z$ to one. Thus no further complement fusion is needed in these cases.

Executing (B.28.6)–(B.28.7) gives 35 character orbits, in addition to the 57 full groups. The following order distribution is a compact view of the complete 92-row table:

| $\lvert K\rvert$ | 1 | 2 | 3 | 4 | 5 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|---|---|
| # | 1 | 7 | 1 | 13 | 1 | 11 | 14 | 3 |
| $\lvert K\rvert$ | 12 | 16 | 20 | 24 | 32 | 48 | 96 | |
| # | 17 | 8 | 1 | 9 | 1 | 4 | 1 | |

(B.28.8)

In particular its seven involution classes consist of the hyperelliptic class, the elliptic positive class, and the five negative classes predicted independently by B.16.

We finish by specifying the verification of all section data. For each of the 57 presentations, the program performs an exact Todd–Coxeter enumeration and obtains the regular action on exactly

$2|H|$ cosets. It evaluates every relator in these permutations and checks that $z$ is nonidentity of order two. In the quotient orbifold for the genus-two action, an originally branched cone of order $m$ now has order $2m$, a regular branch orbit contributes a cone of order two, and a branched corner has its order doubled. Every local cyclic or dihedral group injects into the computed finite group. The orbifold Euler characteristic multiplied by $2|H|$ is $-2$, and the covering subgroup is the kernel of a monodromy homomorphism compatible with orientation. Its torsion-free covering surface is therefore orientable of genus two. The program also checks that $z$ has exactly six section classes.

The cone and corner cosets, together with the lifted mirror coset graph of B.26, give every fixed point and reflection oval. Enumerating subgroups of these local stabilizers and conjugating them by the finite group is enough: every finite subgroup with a nonempty fixed set is contained in such a local stabilizer. The section multiplier is read from the power of the corresponding positive local generator; conjugating by a negative element reverses that angle. For each complement, retain exactly the section classes whose image subgroup lies in its character kernel. This restriction is exact because the lift extension for the complement is the inverse image of that subgroup in the full lift extension. Recompute maximality in the retained poset, intersect the section stabilizers with the complement, and compute its actual orbits. Thus no maximality or orbit count is inferred merely by restricting the full-group summary. All order and power identities for $\mu$ and all unique-maximal-extension claims for sections containing a rotation are checked. Finally the 21 positive rows agree with the independent primary-source oriented genus-two catalogue used in B.33 on the complete multiset of five invariants: group order, full sections, nontrivial partial sections, $|\Phi|$, and $|\Phi/K|$. □

**B.29. [Proved] Complete executed punctured genus-two catalogues.** The extended mapping class groups of types $(2,1)$ and $(2,2)$ have respectively 32 and 121 conjugacy classes of finite subgroups. Their positive subgroups form 10 and 23 of these classes. The files `type_2_`$\nu$`_extended_catalogue.csv` and `type_2_`$\nu$`_extended_sections.json`, for $\nu = 1, 2$, list every class with its exact group, puncture orbits and stabilizers, all section classes and their incidences, $\Phi$, its group action and stabilizers, centralizers, and all section and puncture multipliers.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

*Proof.* Fill the punctures of a finite action on the genus-two surface. The resulting finite subgroup in the closed mapping class group is faithful: a finite homeomorphism isotopic to the identity on a closed hyperbolic surface is the identity, by B.6. Thus B.28 supplies the complete list of closed actions that must be decorated. For such an action $K$, keep the notation $E = K\langle\iota\rangle$, $H = E/\langle\iota\rangle$, and $Q = S^2/H$ from that proof.

The strata relevant to punctures are intrinsic to $K$. Let $\mathcal{V}_K$ be the set of points whose stabilizer in $K$ contains a positive nonidentity element. Take its $K$-orbits. Cut the fixed circles of the reflections in $K$ at $\mathcal{V}_K$ and take the $K$-orbits of the resulting open intervals or uncut circles. These are the mirror strata. There is one connected regular interior in the quotient $S_2/K$. A vertex orbit is selected zero or one time; a mirror stratum receives any nonnegative number of puncture orbits; so does the regular interior. Their weights are, respectively,

$$|Kp|, \qquad |K|/2, \qquad |K|. \tag{B.29.1}$$

The weighted sum must equal $\nu$. Points fixed by $\iota$ but having no positive stabilizer in $K$ are ordinary points of these $K$-strata. They introduce no extra puncture label. In particular punctures need not be preserved by $\iota$ when $K$ is a complement in $E$.

We prove the normalizer permutations used to identify these decorations. Every mapping class normalizing $K$ normalizes $E$, since $\iota$ is central in $\mathcal{T}_{2,0}$. At the homeomorphism level it may be represented by a normalizer of the chosen finite action $E$ without changing its action on the intrinsic $K$-strata. Indeed compare the two commuting involutions $\iota$ and $h\iota h^{-1}$ after conjugating $K$ by $h$. The finite actions of the abstract extension $K\langle\iota\rangle$ have the same mapping classes. Uniqueness of realization in C.6 gives a conjugating homeomorphism isotopic to the identity and commuting with $K$. C.7 gives an isotopy through such commuting homeomorphisms. It fixes the isolated $K$-fixed points individually and preserves every reflection component. Composing with it makes $h$ normalize the chosen $E$ and leaves the required stratum permutation unchanged. Thus it suffices, and is necessary, to compute the action of the normalizer of $E$ that preserves $K$.

The full-group case $K = E$ is direct. A cone in $Q$ has the color consisting of its original order and its branch flag. An ordinary branch orbit has its own color $(1, 1)$. The normalizer permits all permutations of identical interior cone colors and exactly the dihedral symmetries of the expanded colored boundary list of B.28. Each is realized by a quotient orbifold homeomorphism and hence lifts. Pure interior braids conjugate local generators, so they leave the corresponding $E$-orbits unchanged. These observations prove that this finite permutation group is the whole normalizer action on the strata.

Suppose now $K = \ker\chi$ with $\chi(z) = 1$. For a local stabilizer $L \subset E$, its point orbit is $E/L$. It is a single $K$-orbit when $\chi(L) = \mathbf{F}_2$. When $\chi(L) = 0$, it consists of two $K$-orbits, labelled

$$b = \chi(g), \qquad gL \in E/L. \tag{B.29.2}$$

This is well defined precisely in the latter case. It gives the exact orbit comparison, including the ambiguity in choosing a lift of a quotient homeomorphism: composing with $z$ simultaneously changes every available sheet label $b$ to $b + 1$. The code includes this deck permutation.

An interior cone with generator $x$ contributes an intrinsic $K$-vertex only when $K \cap \langle x\rangle$ is nontrivial. For cone permutations, retain exactly the equal-color permutations that also preserve the character values on their local generators. On a sphere quotient, every split cone pair has an additional independent interchange. Here is the explicit lift proving that assertion. Choose a branch local generator $y$ with $\chi(y) = 1$; one exists because the branch decoration is nonempty. An ordinary branch generator is $z$. A branched old cone has $y^m = z$ with $m$ odd whenever a character with $\chi(z) = 1$ exists, and therefore also has $\chi(y) = 1$. For a split cone generator $x$, we have $\chi(x) = 0$. A pure braid about these two quotient points acts by

$$x \longmapsto (xy)x(xy)^{-1}, \qquad y \longmapsto xyx^{-1}. \tag{B.29.3}$$

The other local generators are unchanged. The character is preserved. The conjugator for $x$ has character one, so (B.29.2) interchanges its two $K$-orbits; the conjugator for $y$ has character zero. Thus it leaves the other split pairs unchanged. This proves the claimed independent interchanges. Together with cone permutations they generate the whole stratum action: pure braids generate the remaining mapping classes of a punctured sphere, and their action on the two-element fibers can only be these interchanges.

For a disc quotient there is at most one old interior cone. An ordinary interior branch point permits exactly the same interchange (B.29.3) of that split cone pair, supported away from the boundary. When there is no such interior branch point, the disc has at most one interior marked cone; its pure braid group introduces no additional permutation. Boundary permutations are computed as follows. Let $L$ be the length of the expanded corner list, write $R_{j+L} = eR_je^{-1}$, and put $\epsilon = \chi(e)$. A colored cyclic shift sends a corner index $j$ to $s + j$ and an edge index $i$ to $s + i$. A reversal sends them to $s - j - 1$ and $s - i$, respectively. For either image write the resulting integer as $qL + t$, $0 \le t < L$. Its local generator is conjugated by $e^q$; hence the sheet label changes by

$$b \longmapsto b + q\epsilon. \tag{B.29.4}$$

Retain precisely the colored symmetries that preserve the reflection character values, and include the global deck interchange. These formulas give both vertex and mirror-component permutations. For a cornerless boundary the same rule is its circular covering with monodromy $\chi(e)$; the lifted mirror graph records whether the two local sheet labels belong to one or two components.

There are no omitted boundary permutations. An orbifold homeomorphism of the disc preserves its cyclic boundary order up to reversal. After its colored dihedral permutation has been removed, its remaining mapping class is an interior braid or a boundary twist. An interior braid has the local conjugators already described and fixes the boundary. A boundary twist has the conjugator $e$, already represented by (B.29.4) and the deck permutation. Inner automorphisms coming from $K$ leave $K$-orbits unchanged. The projective-plane cases in B.28 have no complements; their full-group decorations require only permutations of identical interior cone colors. Thus every case has an explicit complete finite normalizer permutation group.

We spell out how this calculation handles smooth points of the $K$-mirror locus that were corners for $E$. The code starts with the exact finite mirror coset graph of the full action $E$. It finds the vertices with positive stabilizer in $K$, removes only these vertices to form the open mirror strata, and computes the fixed circles of each actual reflection in $K$. Thus an $E$-corner that has only a reflection stabilizer in $K$ lies inside a single $K$-mirror stratum. The sheet and boundary formulas above are applied to an ordinary edge sample in each such component. They determine its whole component by connectedness. This is also a direct check that a puncture can move through a noncritical Weierstrass point whenever the $K$-stratum allows it.

Enumerate all vectors with the weights (B.29.1) summing to $\nu = 1$ or 2, and divide by the proved finite permutation group. These are precisely the topological action classes with punctures. Indeed unlabelled finite subsets of an interval or circle with the same cardinality are related by a homeomorphism of prescribed degree, and the regular interior configuration space is connected. Extend such moves by small discs and strips in the quotient and lift the isotopies through the finite orbifold cover. Conversely an equivariant homeomorphism preserves the intrinsic strata, their weights, and a stratum permutation in the normalizer just computed. C.5–C.6 then identify these action classes with the finite-subgroup conjugacy classes in the punctured mapping class group.

The executed program `catalogue_genus2_punctures.py` returns 32 and 121 classes. It places the punctures on the exact coset graph, with one midpoint inserted in every edge before any vertex is removed. This subdivision guarantees that even two punctures at formerly adjacent samples leave the intervening open interval represented. The program checks that an element stabilizing an edge fixes both its endpoints, so the inserted midpoint has exactly the edge stabilizer and creates no artificial critical point. It records regular puncture orbits by the free $K$-set $K$, with trivial stabilizer.

All positive section classes from B.28 are retained exactly when their fixed point has not been removed and their image subgroup lies in $K$. For every reflection, remove the punctures from its actual fixed-circle graph and recompute the resulting ovals and lines. These operations give all partial-section classes by B.3–B.4. Their inclusion poset, maximal classes, $K$-action, stabilizers and centralizers are then recomputed. Local multipliers at punctures are retained from the positive local generators or assigned $1/2$ to a reflection. The positive portions contain 10 and 23 classes and agree with the independent primary-source calculation in B.33 on the full multiset of group order, full sections, partial sections, $|\Phi|$, and $|\Phi/K|$. The negative involutions provide another independent check: there are 4 classes with one puncture and 11 with two, exactly the decorations of the five closed real types in B.16. □

## 2.6 The full tube action and the strip subgroups

**B.30. [Proved] Explicit outer words for the antipodally paired tube model.** For $g \geq 2$, put $n = 2g$ and label the equatorial holes by $j \in \mathbb{Z}/n$, with $j$ glued by a tube to $j + g$. The following presentation and substitutions compute the entire exterior action requested on scan 467:

$$\pi_g = \langle x_0, \dots, x_{g-1}, p_0, \dots, p_{g-1} \mid R = 1 \rangle, \qquad R = x_0 \cdots x_{g-1} \prod_{j=0}^{g-1} (p_j^{-1} x_j^{-1} p_j).$$

Extend the notation, with all subscripts read modulo $n$, by

$$x_{j+g} = p_j^{-1} x_j^{-1} p_j, \qquad p_{j+g} = p_j^{-1},$$
$$u_j = (x_0 \cdots x_j)^{-1}, \quad u_{n-1} = 1, \qquad q_j = u_j^{-1} p_j u_{j+g-1}.$$

Let $r = \varepsilon_0 \varepsilon_1$ and $s = \varepsilon_1$ as in B.20. Representatives of their outer automorphisms, and of $\tau$ and $\rho_k$, are specified on the $2g$ displayed generators by

| | $x_j$ | $p_j$ |
|---|---|---|
| $r$ | $x_{j+1}$ | $p_{j+1}$ |
| $s$ | $u_{-j}^{-1} x_{-j} u_{-j}$ | $q_{-j}$ |
| $\tau$ | $u_{j+g-1}^{-1} x_{j+g}^{-1} u_{j+g-1}$ | $q_j^{-1}$ |
| $\rho_k$ | $x_j$ | $x_j^{\delta_{jk}} p_j$. |

The map to $\mathrm{Out}(\pi_g)$ defined by these substitutions is the geometric action of $\mathbb{Z}^g \rtimes (D_{4g} \times C_2)$ of B.20. The finite factor acts by

$$r \rho_j r^{-1} = \rho_{j+1}, \qquad s \rho_j s^{-1} = \rho_{-j}, \qquad \tau \rho_j \tau^{-1} = \rho_j^{-1},$$

with indices in these three formulas taken modulo $g$.

*Manuscript:* §33; folios 234–236; scans 463–467.

*Proof.* We specify the base paths so that the formulas can be used as words, rather than only as unbased pictures. Let $N, S$ be the north and south poles in the sphere with the $n$ discs removed. Draw disjoint meridian arcs $d_j : N \to S$ through the gap immediately following hole $j$. The loop about that hole, based at $N$, is $x_j = d_{j-1} d_j^{-1}$. Collapse the maximal-tree edge $d_{n-1}$ and identify $N, S$ by that edge. The remaining arcs become $u_j = (x_0 \cdots x_j)^{-1}$, because the product telescopes to $d_{n-1} d_j^{-1}$. In particular $\prod_{j=0}^{n-1} x_j = 1$.

Let $p_j$ run from $N$ to the northern side of hole $j$, through the tube to hole $j + g$, and back to $N$ on its northern side. The opposite traversal is $p_{j+g} = p_j^{-1}$. Its boundary identification reads $x_{j+g} = p_j^{-1} x_j^{-1} p_j$. Van Kampen therefore gives exactly the presentation in the claim. The parallel path based at $S$, on the southern side of the tube, is $q_j = d_j^{-1} p_j d_{j+g-1}$ after inserting the northern paths, which becomes the displayed formula on collapsing the tree edge. These identities can also be taken as a two-vertex cellular description: the sphere faces give the telescoping boundary relation and each tube face gives the indicated boundary conjugation.

Rotation about the polar axis sends hole $j$ to $j + 1$, preserves $N$, and sends $p_j$ to $p_{j+1}$. This proves the $r$ row. The half-turn $s$ about the equatorial axis through hole 0 exchanges $N, S$ and sends hole $j$ to $-j$. The southern boundary loop at hole $k$ is $d_k^{-1} x_k d_k$, and the southern tube path is $q_k$, giving the $s$ row. The antipodal map exchanges $N, S$, sends $j$ to $j + g$, and reverses orientation. It therefore inverts the southern boundary loop and reverses the corresponding tube path. The boundary-loop formula written with $u_{j+g-1}$ equals that written with $u_{j+g}$, since $u_{k-1} = x_k u_k$. This proves the $\tau$ row. Finally a positive full twist of tube $k$ inserts $x_k$ into the

crossing $p_k$ and leaves all other chosen paths unchanged. This proves the $\rho_k$ row. Each row comes from an actual homeomorphism, so it is an automorphism of the stated group and has the prescribed orientation sign.

For $s$ and $\tau$ the chosen path from $N$ to the image of $N$ is $d_{n-1}$. We consequently get actual automorphisms only after this base-path choice. For example their squares are

$$s^2 = \mathrm{Inn}(x_0), \qquad \tau^2 = \mathrm{Inn}(x_0 \cdots x_{g-1}),$$

whereas $r^{2g} = 1$ as an actual automorphism. These formulas follow by concatenating the base path with its image: $s(d_{n-1}) = d_0^{-1}$ and $\tau(d_{n-1}) = d_{g-1}^{-1}$. Thus the displayed group relation $s^2 = \tau^2 = 1$ holds in the outer group with precisely the stated inner ambiguity. The geometric symmetries satisfy $srs = r^{-1}$ and commute with the antipodal involution on the surface. They preserve and permute the tube collars; a positive symmetry preserves twist sign and the antipodal map reverses it. This proves every relation of the asserted semidirect product in the outer group. Changing the maximal-tree edge conjugates each based comparison by its corresponding base-path word and leaves this exterior action unchanged.

The accompanying programs check all relator images for $2 \le g \le 8$; $r, s$ carry $R$ to a conjugate of $R$, and $\tau$ carries it to a conjugate of $R^{-1}$. They additionally check all the finite-factor and twist-conjugacy relations in the surface group, with the actual inner conjugators above. For instance $srsr = 1$ and $\tau r \tau r^{-1} = \mathrm{Inn}(x_0 \cdots x_g)$. The word checker uses the symmetrized relator: its largest common piece has length one, so the length-$4g$ presentation satisfies $C'(1/6)$ for $g \ge 2$, and the Dehn reduction is exact. These finite certificates supplement the cellular proof for arbitrary $g$. □

**B.31. [Proved] The eight symmetries of one tube.** The stabilizer of tube $T_0$ in $D_{4g} \times \langle \tau \rangle$ is $\langle r^g, s, \tau \rangle \cong C_2^3$. There are coordinates $(\theta, t) \in (\mathbb{R}/2\pi\mathbb{Z}) \times [-1, 1]$ on that tube in which its three generators act as

$$r^g(\theta, t) = (-\theta, -t), \qquad s(\theta, t) = (\theta + \pi, t), \qquad \tau(\theta, t) = (\theta + \pi, -t).$$

These formulas agree with the outer action in B.30 and with the manuscript's free antipodal extension, after the explicit coordinate change $\theta_{\mathrm{new}} = \theta_{\mathrm{old}} + \pi t/2$.

*Manuscript:* §33; folios 229–230; scans 453–454; §33; folios 233–236; scans 461–467.

*Proof.* The stabilizer of the unordered pair $\{0, g\}$ under the polygon's dihedral group is $\{1, r^g, s, r^g s\}$: a stabilizing rotation is either 1 or $r^g$, and every stabilizing reflection is one of these followed by $s$. The independent central antipodal factor doubles it. In a symmetric tube, $r^g$ interchanges its ends and reverses its angular direction, while $s$ preserves both ends and rotates the cross section by $\pi$. The displayed representatives commute, are involutions, and have eight distinct combinations; the antipodal representative is free. They extend the same boundary maps as the corresponding sphere symmetries and hence give the desired tube extension.

The formula of scan 453, retained through scans 461–462, is $\tau_{\mathrm{old}}(\theta, t) = (\theta + \pi + \pi t, -t)$. Conjugating it by the stated change gives exactly $(\theta + \pi, -t)$; this also transports both boundary identifications. Thus it is an explicit isomorphism of the two actions. The full twist is $(\theta, t) \mapsto (\theta + \pi + \pi t, t)$ in either coordinate. Conjugation by the two positive generators preserves its class, while conjugation by the negative generator inverts its class, agreeing with B.30. All eight maps and their products are checked symbolically in `code/verify_tube_local.py`. □

**B.32. [Proved] Relations for arbitrary strips and a uniform obstruction to generation.** Choose for each $c_j$ one simple transversal $d_j$, using a strip in the complement of the other tubes, so that $i(c_i, d_j) = \delta_{ij}$. Let $W_j$ be a regular neighbourhood of $c_j \cup d_j$, a one-holed torus, and put

$A_j = T_{c_j} = \rho_j$, $B_j = T_{d_j}$. Its supported subgroup is the central extension $\langle A_j, B_j\rangle \cong B_3$ of B.21, with quotient $SL_2(\mathbb{Z})$. The exact pairwise relations are

$$[A_i, A_j] = 1, \quad [A_i, B_j] = 1 \ (i \neq j), \qquad A_j B_j A_j = B_j A_j B_j.$$

For distinct $i, j$, the two supported subgroups commute precisely when $i(d_i, d_j) = 0$. For intersection one, $B_i, B_j$ obey the braid relation and do not commute; for intersection at least two they freely generate a free group of rank two. For every such choice of one strip per tube and every $g \geq 2$, these $g$ supported subgroups, together with $H$, generate a proper subgroup of $\mathcal{T}_g^+$.

*Manuscript:* §33; folios 236–237; scans 467–470.

*Proof.* Each $W_j$ has genus one and one boundary component because the two core curves meet transversely once. The extension-by-identity map and its faithful boundary-twist kernel are the explicit maps of B.21. The disjointness conditions give the first two commutation relations, and the calculation on its two free generators in B.21 proves the braid relation. For any two essential simple curves $a, b$, their twists commute exactly when they are disjoint: conjugation sends $T_b$ to $T_{T_a(b)}$, and $i(T_a(b), b) = i(a, b)^2$. Hence intersecting curves cannot commute. If $i(a, b) = 1$, their regular neighbourhood is a one-holed torus, so the same braid calculation applies.

For $k = i(a, b) \geq 2$ the freedom assertion has an elementary ping-pong proof. On isotopy classes of essential curves set

$$X_a = \{x : i(x, a) < i(x, b)\}, \qquad X_b = \{x : i(x, b) < i(x, a)\}.$$

Both are nonempty and disjoint. Counting crossings in a twist collar and cancelling only crossings already present outside it gives

$$i(T_a^m(x), b) \geq |m|\, i(x, a)k - i(x, b).$$

For $m \neq 0$ and $x \in X_b$ this is strictly greater than $i(x, a) = i(T_a^m(x), a)$, so $T_a^m(X_b) \subset X_a$. The symmetric inclusion holds for powers of $T_b$. The ping-pong argument on a reduced alternating word proves that it is nontrivial: apply it to the appropriate one of $X_a, X_b$, inserting one further nonzero power from the opposite factor when the first and last factors agree. Thus $\langle T_a, T_b\rangle \cong \mathbb{Z} * \mathbb{Z}$. These statements compute the required pairwise relations for arbitrary intersections of the selected strips; no disjoint-strip convention is being used to conclude the general generation assertion.

For that assertion work on $V = H_1(X_g; \mathbb{F}_2)$ with its nonsingular intersection pairing. The $2g$ classes $[c_j], [d_j]$ form a basis: pairing a linear relation first with the $[c_j]$ kills all coefficients of the $[d_j]$, and pairing next with the $[d_j]$ kills the rest. There is a unique quadratic refinement $q : V \to \mathbb{F}_2$ with $q([c_j]) = q([d_j]) = 1$. Explicitly, for this ordered basis $(v_i)$,

$$q\Big(\sum_i a_i v_i\Big) = \sum_i a_i + \sum_{i<j} a_i a_j \langle v_i, v_j\rangle.$$

It obeys $q(v + w) = q(v) + q(w) + \langle v, w\rangle$ by expansion. A twist about a curve of class $v$ acts by $x \mapsto x + \langle x, v\rangle v$; substitution gives

$$q(T_v x) - q(x) = \langle x, v\rangle (q(v) + 1).$$

Every generator $A_j, B_j$ therefore preserves $q$, without any restriction on the intersections among the $d_j$.

Join $c_1$ and $c_2$ by an embedded arc in their connected complement and form a band sum $e$. This is a simple curve of nonzero class $[e] = [c_1] + [c_2]$, so it is nonseparating. Since the two cores are disjoint, $q([e]) = 0$. Taking $x = [d_1]$ gives $\langle x, [e]\rangle = 1$, so $T_e$ fails to preserve $q$. Consequently $T_e$ belongs to $\mathcal{T}_g^+$ and lies outside the group generated by the strip subgroups. This proves

properness for all choices under discussion. The quadratic calculation has been exhaustively checked for every possible matrix of transversal intersections modulo two in genera two, three and four; the general identity is the displayed algebraic proof.

The rank-one comparison is thus the specified central quotient $B_3 \to SL_2(\mathbb{Z})$ and its action on the homology of $W_j$, together with the above braid and intersection relations. The subgroup they generate fixes the explicit quadratic refinement. This supplies the precise algebraic maps and the explicit obstruction governing the proposed generation in scan 470. □

**B.33. [Proved] Executed positive catalogues in genera two and three.** For $(g, \nu) = (2,0), (2,1), (2,2), (3,0)$, the numbers of conjugacy classes of finite orientation-preserving subgroups, allowing conjugacy in the full extended mapping class group, are respectively

$$21, \qquad 10, \qquad 23, \qquad 55.$$

The trivial subgroup is included. The files `oriented_higher_genus_catalogue.csv` and `oriented_higher_genus_sections.json` give each class, its actual finite permutation group, every subgroup's number of sections, all nontrivial partial-section classes and their inclusion relations, the $G$-set $\Phi$, its stabilizers, and every value of $\mu$ on every section. Punctures are unlabelled. These assertions concern positive subgroups; they do not enumerate subgroups containing orientation-reversing elements.

*Manuscript:* §15; folios73–75; scans148–151; §17; folios77–79verso; scans155–160; §18; folios96–101; scans193–203; Part B(f) of the task; positive higher-genus cases.

*Proof.* We specify the classification input and the computation using it. Karabáš–Nedela–Skyvová [40, Theorem 2.2, Corollary 2.3, Theorem 4.1 and §8] identify the positive topological classes with the orbits of smooth epimorphisms from the Fuchsian group

$$\Delta = \langle x_1, \dots, x_r, a_1, b_1, \dots, a_h, b_h \mid x_i^{m_i} = 1,\ [a_1, b_1] \cdots [a_h, b_h] x_1 \cdots x_r = 1 \rangle$$

under target automorphisms and orientation-preserving orbifold automorphisms. Here smooth means that every image of $x_i$ has order exactly $m_i$. Their §8 gives a complete genus-two list of 21 classes and a genus-three list of 55 classes in 49 group/signature pairs, including the trivial actions. We opened the accompanying data [39], downloaded the two original JSON files, and retained their attribution and SHA-256 digests. The attached program reads those exact files. It verifies that each generating vector generates the group of the stated order, each elliptic generator has its specified order, every presentation relation holds, and

$$|G| \left( 2 - 2h - \sum_i (1 - 1/m_i) \right) = 2 - 2g.$$

The conversion from the source's right-acting permutations to the left-acting permutations in our code is the isomorphism sending each permutation to its inverse. For a handle word the two commutator conventions are compared explicitly; replacing both handle generators by their inverses converts $a^{-1}b^{-1}ab$ to the convention $aba^{-1}b^{-1}$. The files retain the actual generating vector, so this convention does not affect a recorded elliptic rotation.

The local-order checks imply that the kernel of each epimorphism is torsion-free: every finite-order element of a Fuchsian group is conjugate to a power of an elliptic generator. The covering formula then identifies its quotient surface as genus $g$. Conversely a positive finite action determines this epimorphism by lifting paths in the quotient. The two constructions are inverse modulo the stated source and target automorphisms. Finally C.5–C.6 identify these action classes with conjugacy classes of finite mapping-class subgroups. This supplies the exact reduction from the classification input to the groups asked for here.

We also recompute the equivalence relation whenever $h = 0$, rather than using only the published class count. For an ordered generating vector $(u_1, \dots, u_r)$ with product one, the braid generator acts by

$$\sigma_i : (u_i, u_{i+1}) \longmapsto (u_i u_{i+1} u_i^{-1}, u_i).$$

Use adjacent such generators when $m_i = m_{i+1}$ and all pure braid generators, obtained by moving the $j$th entry next to the $i$th, applying $\sigma_i^2$, and reversing the moves. Theorem 4.1 of the cited paper proves that these transformations induce the full orientation-preserving orbifold automorphism action after inner automorphisms have been divided out. A generating vector is compared modulo $\mathrm{Aut}(G)$ by its rooted, edge-coloured Cayley graph: breadth-first enumeration with the ordered generators gives a canonical transition table. Equality of tables gives the explicit bijection of vertices preserving every labelled multiplication edge; induction on words proves that it is the unique target-group isomorphism carrying the generating vectors to one another. Thus the canonicalization does not approximate group automorphisms.

To allow conjugacy by negative homeomorphisms, we add the reflection of the marked sphere. Its effect on the tuple is

$$(u_i)_{i=1}^r \longmapsto \left(p_{i-1} u_i^{-1} p_{i-1}^{-1}\right)_{i=1}^r, \qquad p_{i-1} = u_1 \cdots u_{i-1}.$$

The product of the new tuple is $(u_1 \cdots u_r)^{-1} = 1$, as successive cancellation shows. The program computes the orbits both before and after adding this transformation; it recovers the published planar counts and finds no fusion. All nonplanar group/signature pairs in these two genera have only one positive class in the input classification. Reversing orientation preserves the group and signature and hence fixes that class. Consequently the positive counts remain 21 and 55 under conjugacy in the extended group.

For genus two we add all puncture configurations with $\nu \le 2$. An invariant finite puncture set consists of selected elliptic orbits and a number $a$ of ordinary orbits. Write $c_i \in \{0, 1\}$ for selection of the $i$th elliptic orbit. The exact equation is

$$\sum_i c_i \frac{|G|}{m_i} + a|G| = \nu. \tag{B.33.1}$$

The program enumerates every solution. It carries the bits $c_i$ along under each braid move, exchanging the two bits when the corresponding two elliptic entries are exchanged. It then takes orbits of these decorated generating vectors under the same coloured braids, target automorphisms, and reflection. These are exactly the decorated topological classes: an equivariant homeomorphism descends to the quotient orbifold, and its action on the covering is the compatibility of its induced automorphism with the epimorphism just described. Conversely that compatibility lifts the quotient homeomorphism to an equivariant homeomorphism. Ordinary unlabelled punctures can be moved to any other such configuration through paths in the regular part of the quotient; point-pushing the paths is isotopic to the identity after these additional points are forgotten and so preserves the original covering kernel. Thus $a$ records all their remaining data.

There are just two nonplanar genus-two entries. For the trivial group, there is one configuration for each $\nu$. The other has group $C_2$ and signature $(1; 2, 2)$. Its two elliptic images are the unique involution, and exchanging their small discs preserves the monodromy after an adjustment of the two handle paths. The four possible pairs of handle images in $C_2^2$ are connected by point-pushing either branch point around a handle: crossing the branch cut adds its nonzero monodromy to the transverse handle image. These two pushes change either handle coordinate independently. Hence the exchange lifts and only the number of selected branch orbits matters. These elementary moves account for all four vectors in the supplied data and prove the nonplanar decoration step used by the code. Equation (B.33.1) and the decorated orbits give 10 and 23 classes at one and two punctures.

We now derive every section entry from the checked finite groups. A retained elliptic orbit is the left $G$-set $G/\langle u_i\rangle$. Its point at the coset $v\langle u_i\rangle$ has stabilizer $v\langle u_i\rangle v^{-1}$ and positive generator $vu_iv^{-1}$. B.3 gives exactly one kernel-conjugacy class of sections for every point fixed by a specified nontrivial subgroup $K \leq G$. Such a subgroup is cyclic because it is contained in the cyclic point stabilizer. Conversely every such containment gives a section. Thus the exact count is

$$\#\operatorname{Sec}(E_K \to K)/\pi = \sum_{i:c_i=0} \#\{v\langle u_i\rangle : K \leq v\langle u_i\rangle v^{-1}\}. \tag{B.33.2}$$

The formula is used for every subgroup, which the program enumerates by saturation under adjoining each remaining group element. It checks that no noncyclic subgroup has a section. The trivial subgroup has one section, and it is not included in the nontrivial partial-section totals.

The maximal partial sections are precisely these individual elliptic points, with $G$ acting by left multiplication on the cosets. A partial section lies below another exactly when they have the same point and its subgroup is contained in the other subgroup. This determines the whole inclusion relation and proves uniqueness of a maximal extension in the positive case. Each nontrivial section has centralizer one by B.3. The total number of nontrivial partial-section classes is independently checked against

$$\sum_{i:c_i=0} \frac{|G|}{m_i}(\tau(m_i) - 1),$$

where $\tau(m)$ is the number of positive divisors of $m$. Finally, for $h = vu_i^b v^{-1}$ the value of $\mu$ at this section is $b/m_i \pmod{\mathbb{Z}}$. This lists every power of every local generator and is exactly the intrinsic multiplier identified in B.12. Puncture stabilizers are the removed cyclic groups or the trivial group for ordinary orbits. All these data are computed with integer permutations and rational fractions. The output has 109 individual action rows and 59 independent planar braid-orbit checks. □

**B.34. [Proved] The complete genus-three subcatalogue containing a hyperelliptic involution.** There are exactly 79 conjugacy classes of finite subgroups of the extended mapping class group of a closed orientable genus-three surface that contain a hyperelliptic involution. Exactly 18 preserve orientation and 61 contain orientation-reversing elements. The latter 61 classes are listed below and are specified, together with every section and its extension incidences, in `type_3_0_hyperelliptic_mixed_catalogue.csv` and `type_3_0_hyperelliptic_sections.json`.

*Manuscript:* §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; the genus-three hyperelliptic subcatalogue is an added result, not a further manuscript assertion.

*Proof.* Fix a hyperelliptic involution $z$ and its branched quotient $p : S_3 \longrightarrow S^2$, with branch set $P$ of size eight. On the unbranched complements this is the connected double cover classified by the homomorphism

$$\pi_1(S^2 \setminus P) \longrightarrow C_2 \quad \text{sending each of the eight meridians to 1.} \tag{B.34.1}$$

Their sum is zero modulo two. Every homeomorphism of the sphere preserving $P$ preserves (B.34.1), and therefore has exactly two lifts. The Birman–Hilden isotopy property gives the exact sequence

$$1 \longrightarrow \langle z\rangle \longrightarrow C_{\mathcal{T}_{3,0}}(z) \longrightarrow \mathcal{T}_{0,8} \longrightarrow 1. \tag{B.34.2}$$

Here the left group is the centralizer of this specified involution; the map is induced by the quotient $p$. The lifting and isotopy argument is precisely the fully ramified hyperelliptic case of [22,

§9.4]. Allowing orientation reversal adds the negative lifts of the same sphere homeomorphisms, with the same kernel.

The involution $z$ is the unique hyperelliptic involution inside any finite subgroup $K$ containing it. To verify the assertion directly, realize $K$ by isometries and use the resulting invariant complex structure. A hyperelliptic involution acts as $-1$ on $H^1(S_3;\mathbb{R})$: its invariant subspace is the pullback of $H^1(S^2;\mathbb{R}) = 0$ under its quotient, as follows by averaging cochains, or by the transfer outside the branch points and filling their meridians. If $z'$ were another, the finite-order holomorphic map $zz'$ would act trivially on $H^1$. A nonidentity such map has isolated fixed points, each of positive local Lefschetz index; a fixed-point-free map has Lefschetz number zero. But its Lefschetz number from homology is $2-6=-4$. Thus $zz'=1$, proving uniqueness. Consequently every element of $K$ centralizes $z$, and any conjugacy between two such finite groups maps their hyperelliptic involutions to one another. Once $z$ is fixed, (B.34.2) therefore gives a bijection

$$\left\{\begin{array}{c}\text{finite subgroup conjugacy classes}\\ \text{in } \mathcal{T}_{3,0} \text{ containing } z\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{finite subgroup conjugacy classes}\\ \text{in } \mathcal{T}_{0,8}\end{array}\right\}. \quad \text{(B.34.3)}$$

The inverse takes the full preimage, of order twice the sphere group. The inclusion of every kernel element follows from the assumption $z \in K$. This also proves that no additional fusion occurs when the chosen hyperelliptic involution is forgotten.

For completeness, B.26's decorated spherical-orbifold enumeration applies verbatim to eight marked points: a selected cone orbit has size $|H|/m$, a selected corner orbit has size $|H|/(2m)$, an orbit on an open mirror edge has size $|H|/2$, and a regular orbit has size $|H|$. Their sum is eight. The cyclic and dihedral rotation orders are at most eight, since every non-polar point orbit has at least that many points. The tetrahedral and octahedral cases are included. An icosahedral orbit has at least twelve points, so cannot occur. The spherical classification and the homeomorphism classification of these decorations in B.26 prove exhaustiveness and irredundancy. Executing the integer weight equation gives 79 decorated quotients, 18 spherical and 61 with a negative element.

For each decorated quotient, the full lift is given by the exact central extension presentation (B.28.3)–(B.28.5), now with eight branch points. Its presentation proof uses (B.34.1) in exactly the same way: the displayed group surjects onto the full geometric lift group of order $2|H|$, while its quotient by the central element $z$ is $H$, so it has at most $2|H|$ elements. The two groups are therefore isomorphic by that surjection. For a projective-plane quotient its generators include both $z$ and the crosscap element; their relation is $u^2\prod x_i = z^a$. In particular no cyclicity assumption from the six-branch-point case is imposed.

The program `catalogue_genus3_hyperelliptic.py` was executed. For every presentation it computes an exact regular permutation representation of degree $2|H|$, verifies every defining relator, checks that $z$ has order two, and checks injectivity on every local cyclic or dihedral stabilizer. The resulting orbifold Euler characteristic is $-4/|K|$, so its oriented surface kernel has genus three. Exactly eight point sections have image $\langle z\rangle$, providing an independent branch-number check. The finite coset incidence graph of B.26 then computes every point and oval fixed component and every partial section. Each point section has trivial centralizer in the surface group, each oval section has infinite-cyclic centralizer, and the rotation multipliers, extensions, maximal sections $\Phi$, group orbits and stabilizers are recorded explicitly in the JSON file. The calculation retains all intersections of reflection ovals at dihedral stabilizers. □

In the following table, $b$ is both the G3H class suffix and the decorated sphere-eight row number in `type_3_0_hyperelliptic_catalogue.csv`; $N$ is the group order; $p$ counts all nontrivial partial splitting classes; and $m, o$ are respectively $|\Phi|$ and $|K\backslash\Phi|$. The CSV gives the cone, corner, mirror and ordinary branch decorations of every row, and the JSON gives its finite group generators and all relators.

| b | N | p | m | o |
|---|---|---|---|---|
| 2 | 4 | 8 | 8 | 4 |
| 3 | 4 | 10 | 10 | 5 |
| 4 | 4 | 12 | 8 | 5 |
| 5 | 4 | 16 | 8 | 6 |
| 6 | 4 | 20 | 8 | 7 |
| 7 | 4 | 24 | 8 | 8 |
| 11 | 8 | 12 | 12 | 3 |
| 12 | 8 | 14 | 14 | 4 |
| 13 | 8 | 20 | 12 | 4 |
| 14 | 8 | 28 | 12 | 5 |
| 15 | 8 | 14 | 8 | 3 |
| 16 | 8 | 22 | 8 | 4 |
| 17 | 8 | 20 | 12 | 4 |
| 18 | 8 | 26 | 12 | 5 |
| 19 | 8 | 34 | 12 | 6 |
| 20 | 8 | 24 | 12 | 5 |
| 21 | 8 | 32 | 12 | 6 |
| 22 | 8 | 32 | 12 | 6 |
| 23 | 8 | 24 | 8 | 4 |
| 24 | 8 | 32 | 8 | 5 |
| 25 | 8 | 32 | 8 | 5 |
| 29 | 12 | 12 | 8 | 2 |
| 30 | 12 | 14 | 10 | 3 |
| 31 | 12 | 24 | 8 | 3 |
| 32 | 12 | 30 | 8 | 3 |
| 33 | 12 | 42 | 8 | 4 |
| 34 | 12 | 42 | 8 | 4 |
| 39 | 16 | 16 | 12 | 2 |
| 40 | 16 | 28 | 20 | 3 |
| 41 | 16 | 40 | 20 | 4 |
| 42 | 16 | 28 | 12 | 3 |
| 43 | 16 | 18 | 14 | 3 |
| 44 | 16 | 32 | 12 | 3 |
| 45 | 16 | 38 | 20 | 4 |
| 46 | 16 | 52 | 20 | 5 |
| 47 | 16 | 50 | 20 | 5 |
| 48 | 16 | 48 | 12 | 4 |
| 49 | 16 | 48 | 12 | 4 |
| 50 | 16 | 36 | 12 | 3 |
| 51 | 16 | 48 | 12 | 4 |
| 52 | 16 | 44 | 12 | 4 |
| 56 | 24 | 54 | 20 | 3 |
| 57 | 24 | 36 | 8 | 2 |
| 58 | 24 | 28 | 8 | 2 |
| 59 | 24 | 68 | 20 | 4 |
| 60 | 24 | 66 | 20 | 4 |
| 61 | 24 | 66 | 8 | 3 |
| 62 | 24 | 72 | 8 | 3 |
| 65 | 28 | 45 | 10 | 3 |
| 66 | 32 | 60 | 28 | 3 |

| b | N | p | m | o |
|---|---|---|---|---|
| 67 | 32 | 44 | 12 | 2 |
| 68 | 32 | 36 | 12 | 2 |
| 69 | 32 | 82 | 28 | 4 |
| 70 | 32 | 84 | 28 | 4 |
| 71 | 32 | 76 | 12 | 3 |
| 72 | 32 | 72 | 12 | 3 |
| 74 | 48 | 54 | 20 | 2 |
| 75 | 48 | 120 | 20 | 3 |
| 76 | 48 | 120 | 20 | 3 |
| 78 | 64 | 136 | 28 | 3 |
| 79 | 96 | 210 | 44 | 3 |

The mixed-class order distribution is

| $\lvert K\rvert$ | 4 | 8 | 12 | 16 | 24 | 28 | 32 | 48 | 64 | 96 |
|---|---|---|---|---|---|---|---|---|---|---|
| # | 6 | 15 | 6 | 14 | 7 | 1 | 7 | 3 | 1 | 1. |

(B.34.4)

The scope of (B.34.3) is the condition that the subgroup itself contain a hyperelliptic involution. A finite subgroup that only centralizes an involution outside itself is represented by a complement in a lift group; its conjugacy after forgetting the external involution requires an additional calculation. No such complement is counted among these 79 classes.

**B.36. [Proved] Complete exceptional catalogues and their realization fibers.** For the four exceptional types, the numbers of finite-subgroup conjugacy classes in the extended mapping class group, and the numbers of finite action classes inducing these subgroups isomorphically, are as follows:

| $(g,\nu)$ | $\mathcal{T}_{g,\nu}$ | finite subgroup classes | realized action classes |
|---|---|---|---|
| $(0,0)$ | $C_2$ | 2 | 3 |
| $(0,1)$ | $C_2$ | 2 | 2 |
| $(0,2)$ | $C_2^2$ | 5 | 7 |
| $(1,0)$ | $\mathrm{GL}_2(\mathbb{Z})$ | 13 | 17 |

(B.36.1)

The executed files `exceptional_finite_subgroups_and_lifts.csv` and `exceptional_sections.json` give all 29 action rows over these 22 mapping class subgroup rows. They retain the extension, all full and partial section classes, $\Phi$ and its stabilizers, centralizers, and every defined local multiplier. The request in B(f) for one set of section data attached only to a finite mapping class subgroup is false in these types: the displayed realization fibers are required. For the antipodal action on $S^2$, the algebraic section exists and the fixed-point locus is empty; there is no fixed-point rotation multiplier for that section.

*Manuscript:* §15; folios 68–75; scans 137–151; §17; folios 88–90; scans 179–181; §19; folios 111–115; scans 222–230; Part B(f) of the task; no further manuscript assertion.

*Proof.* We first identify the groups in (B.36.1) and the comparison being classified. Orientation is the component invariant for $\mathrm{Homeo}(S^2)$ and for $\mathrm{Homeo}(S^2,\{\infty\})$. An orientation-preserving homeomorphism is isotopic to the identity: fix a disc and its complementary disc by the Alexander isotopy, with the marked point kept fixed in the second case. For the annulus, the two component invariants are the sign $\epsilon$ on its fundamental group and the sign $\eta$ of its permutation of the two ends. Their four representatives are

$$(x,t)\longmapsto(\epsilon x,\eta t),\qquad x\in\mathbb{R}/\mathbb{Z},\quad t\in\mathbb{R}. \tag{B.36.2}$$

The kernel of these invariants is connected: compactify the two ends, isotope an arc joining them to a fixed arc, and apply the Alexander isotopy on the complementary disc. Winding of the arc is absorbed by rotating a puncture neighborhood, since its end is a puncture and has no prescribed boundary parametrization. For the torus, homology gives $\pi_0 \operatorname{Homeo}(T^2) = \mathrm{GL}_2(\mathbb{Z})$. Surjectivity is given by linear maps. A homologically trivial homeomorphism is first isotoped to fix the origin, then the two basis curves, and then their complementary disc; translations lie in the identity component. B.25 gives its thirteen finite subgroup classes.

In this statement a realized action over a finite subgroup $H \subset \pi_0 \operatorname{Homeo}(U)$ means a finite subgroup $G \subset \operatorname{Homeo}(U)$ for which the projection $G \to \pi_0 \operatorname{Homeo}(U)$ is an isomorphism onto a conjugate of $H$. Actions are identified by homeomorphism conjugacy. This specifies the exact fiber of the comparison map. The additional finite actions having a nontrivial kernel in that projection are the full affine families already classified in B.14; their kernels are retained in that result and are not silently identified with an injective realization.

On $S^2$, a negative involution is conjugate either to a reflection or to the antipodal map. Indeed the sphere theorem in B.1 conjugates the finite action into $O(3)$. An orthogonal involution has eigenvalues $\pm 1$; negative determinant leaves precisely the patterns $(1, 1, -1)$ and $(-1, -1, -1)$. Their fixed sets are a circle and the empty set. Together with the trivial action this gives three rows over the two mapping class subgroups. On the plane, a negative involution is conjugate to a reflection by B.1, giving two rows including the trivial action. For both surfaces $\pi = 1$, so the lift extension is simply $E = G$. For the negative $C_2$ there is exactly one full section, exactly one nontrivial partial section, and one maximal class, with centralizer 1 in $\pi$ and stabilizer $C_2$. For the trivial group there is one full section and no nontrivial partial section. These are algebraic assertions about the extension and include the antipodal action. In that action $p = -p$ implies $p = 0$, which has no solution on the unit sphere. The program verifies $A^2 = I$, $\det A = -1$, and $\det(I - A) = 8$ for $A = -I_3$. It therefore records zero fixed components next to the one algebraic full section.

A reflection has the local multiplier $1/2$. For the antipodal section no fixed-point germ exists from which to read such a multiplier; the table marks that entry as undefined. It separately records the formal value $1/2$ obtained by extending the orientation-sign convention to every negative involution. This convention has a stated source and is not substituted for a nonexistent fixed-point comparison. In particular the circle fixed by a sphere reflection has trivial image in $\pi_1(S^2)$, so the hyperbolic oval-centralizer formula $\mathbb{Z}$ does not transfer to this case.

For the annulus use the simultaneous normal form of B.14. Let

$$a(x,t) = (-x,t), \qquad b(x,t) = (x,-t), \qquad b'(x,t) = (x+\tfrac{1}{2},-t), \qquad c(x,t) = (-x,-t). \tag{B.36.3}$$

The three nontrivial mapping class elements are represented by $a, b, c$. A lift of the class of $a$ or $c$ has an arbitrary angular shift, removed by translation conjugacy. A lift of the class of $b$ has angular shift 0 or $1/2$, because its square is the identity. These give $b$ and $b'$; their fixed sets distinguish them. For the whole four-group, normalize its angular reflection to $a$. Commutation forces the radial generator's angular shift to be 0 or $1/2$, giving exactly $\langle a, b\rangle$ and $\langle a, b'\rangle$. Conjugation cannot exchange the three nontrivial mapping class elements, since their component group is abelian. The fixed oval or its absence again distinguishes these two actions. This proves the seven-row classification.

Here are all its section counts; $f, p, \phi, o$ mean full sections, nontrivial partial sections, $|\Phi|$,

and $|\Phi/G|$:

| action | $f$ | $p$ | $\phi$ | $o$ |
|---|---|---|---|---|
| $1$ | 1 | 0 | 0 | 0 |
| $\langle a\rangle$ | 2 | 2 | 2 | 2 |
| $\langle b\rangle$ | 1 | 1 | 1 | 1 |
| $\langle b'\rangle$ | 0 | 0 | 0 | 0 |
| $\langle c\rangle$ | 2 | 2 | 2 | 2 |
| $\langle a,b\rangle$ | 2 | 7 | 2 | 2 |
| $\langle a,b'\rangle$ | 0 | 4 | 4 | 2 |

(B.36.4)

To verify them, the fixed sets of $a, b, b', c$ are respectively two lines, one oval, the empty set, and two points. In the split four-group, each of its two global fixed points extends the corresponding point section and the incident reflection sections; the oval section has both maximal extensions. In the glide four-group, the two angular-reflection lines and the two fixed points of $ab'$ are disjoint. They are all maximal, and the group interchanges each pair. Thus (B.36.4) also gives the required partial-section distinction. Every nonidentity torsion element in these sections has multiplier $1/2$. The centralizer in $\pi = \mathbb{Z}$ is $\mathbb{Z}$ for the radial oval section and 1 for every point or line section.

These calculations can also be read directly from the extension. For an affine element write its chosen angular shift as $a_g \in [0, 1)$ and let $\epsilon_g$ be its action on $\pi = \mathbb{Z}$. The program records the integer cocycle

$$c(g,h) = a_g + \epsilon_g a_h - a_{gh} \quad \text{and} \quad (n,g)(m,h) = (n + \epsilon_g m + c(g,h), gh). \tag{B.36.5}$$

It checks the cocycle identity on every triple. In the universal cover $\mathbb{R}^2$, a finite subgroup of the lift extension fixes a point; its fixed set is affine and connected. Passing to the deck quotient identifies its conjugacy class with the corresponding fixed component just listed. This proves the section calculation without using a center-free hypothesis.

For the torus apply the exact affine classification in B.27 and retain precisely the normal forms whose translation kernel has order one. The linear projection of such a finite group is faithful, which is exactly the comparison stipulated above. Conversely every injective realized mapping class subgroup has this property and is included by B.14. There are seventeen affine forms over the thirteen integral linear classes. In the order of the linear classes in B.27, the fiber sizes are

$$1, 2, 1, 1, 1, 3, 1, 1, 1, 1, 1, 2, 1. \tag{B.36.6}$$

The proof of B.27 enumerates all cocycles and all affine conjugacies without a word bound. Its averaging comparison in B.14 proves that no further topological conjugacy is missing. Run the same exact fixed-point and reflection-circle computation with no punctures removed. Every isolated point section has centralizer 1 in $\mathbb{Z}^2$; every reflection oval has centralizer the primitive invariant copy of $\mathbb{Z}$. The trivial section has centralizer $\mathbb{Z}^2$. Local rotation matrices give the multipliers, and the complete section poset gives the full and partial counts, $\Phi$, and every stabilizer. The resulting seventeen rows and their exact rational matrices are included in the JSON.

The executed program `catalogue_exceptional.py` carries out all these finite calculations, records the two annulus-end stabilizers and the plane's puncture stabilizer, and verifies the four mapping class counts and the four realization-fiber totals. Consequently it supplies a complete corrected computational answer for the exceptional types, including the precise fibers where B.22 prevents attaching one extension or one section table to a bare mapping class subgroup. □

**B.37. [Proved] Complete genus-three mirror-disc quotient catalogue.** There are exactly 145 conjugacy classes of finite subgroups $K$ of $\mathcal{T}_{3,0}$ for which $H = K \cap \mathcal{T}_{3,0}^{+}$ has spherical quotient and the induced involution on $S_3/H$ has a fixed circle. Exactly 57 contain a hyperelliptic

involution, and 88 do not. The files `type_3_0_reflection_quotient_catalogue.csv`, `type_3_0_reflection_quotient_certificates.json`, and `type_3_0_reflection_quotient_sections.json` give every class, its exact finite group, and all its section data.

*Manuscript:* §18; folios 96–101; scans 193–203; Part B(f) of the task; genus-three mirror-disc quotient subcatalogue.

*Proof.* The quotient by the induced involution is a mirror disc $Q$. Let $(m_1, \dots, m_k)$ be its cyclic list of corner orders and $(n_1, \dots, n_l)$ its list of interior cone orders. The orientation double of $Q$ is the spherical orbifold with cone multiset

$$\{m_1, \dots, m_k, n_1, n_1, \dots, n_l, n_l\}. \tag{B.37.1}$$

The involution on a sphere with a fixed circle is conjugate to a reflection: cut along that circle into two discs and extend a boundary parametrization over either disc, then use the involution to define the parametrization on the other disc. Cone points on its circle become the corners of $Q$, while its other cone points occur in exchanged pairs. Thus the corner and interior data in (B.37.1) exhaust its quotient types. The corner list is considered up to cyclic permutation and reversal; the interior list is unordered.

Use the following presentation of $\Lambda = \pi_1^{\mathrm{orb}}(Q)$:

$$\left\langle x_1, \dots, x_l, e, c_0, \dots, c_k \;\middle|\; \begin{array}{c} x_i^{n_i} = 1, \quad x_1 \cdots x_l e = 1, \\ c_j^2 = 1, \quad (c_{j-1}c_j)^{m_j} = 1, \quad c_k = e c_0 e^{-1} \end{array} \right\rangle. \tag{B.37.2}$$

For $k = 0$ there is one reflection $c_0$, commuting with $e$. The orientation kernel $\Delta = \Lambda^+$ is the orbifold group of (B.37.1). Put $t = c_0$, $Z_j = c_{j-1}c_j$, $P_j = Z_1 \cdots Z_j$, $U_i = x_{l+1-i}^{-1}$, and $V_i = tU_it$. Schreier rewriting, or the doubled fundamental polygon, gives the spherical generating vector

$$(Z_1, \dots, Z_k, U_1, \dots, U_l, V_l^{-1}, \dots, V_1^{-1}), \qquad P_k\, U_1 \cdots U_l\, (V_1 \cdots V_l)^{-1} = 1. \tag{B.37.3}$$

For example $e = U_1 \cdots U_l$ and $P_k = tc_k = tete^{-1} = (V_1 \cdots V_l)e^{-1}$, which proves the product relation with its stated order. Conjugation by $t$ acts on these generators by

$$\psi(Z_j) = P_{j-1} Z_j^{-1} P_{j-1}^{-1}, \qquad \psi(U_i) = V_i, \qquad \psi(V_i) = U_i. \tag{B.37.4}$$

These formulas include every base-path conjugation.

The input [39], whose completeness is established in [40, Theorem 2.2, Corollary 2.3, and the planar calculation], contains every $\mathrm{Aut}(H)$-class of smooth epimorphisms $\theta : \Delta \to H$ in genus three. Its 197 vectors cover all 49 genus-three signature/group pairs, including those with positive quotient genus. Here we use precisely the spherical-quotient pairs. The original input and its attribution are retained in `research_data/actions-genus3.json`; B.33 verifies its orders and surface signatures. A prescribed ordering (B.37.3) is obtained from the published sorted ordering by explicit adjacent Hurwitz moves. This is a bijection of epimorphism sets, since each move has its displayed inverse in the program.

For each vector the program tests whether its kernel is invariant under (B.37.4). More explicitly, it attempts to define

$$\alpha(\theta(a)) = \theta(\psi(a)) \qquad (a \in \Delta) \tag{B.37.5}$$

by propagation on the finite Cayley graph. Agreement on every edge is exactly the condition that (B.37.5) be well defined. Surjectivity then makes it an automorphism and (B.37.4) gives $\alpha^2 = 1$. The resulting full action group is explicitly

$$K = H \rtimes_\alpha \langle t \rangle, \quad t^2 = 1. \tag{B.37.6}$$

Its epimorphism from (B.37.2) sends $x_i$ to their recovered vector entries and $c_j$ to $t\theta(P_j)$. Its kernel is exactly $\ker\theta \subset \Delta$. In particular no choice of a further extension class is omitted: the orbifold kernel determines $\Lambda/\ker\theta$ and hence (B.37.6). Conversely every action in the claim supplies this invariant kernel and therefore occurs.

We now prove the precise fusion calculation. Two epimorphisms from the same orbifold group have the same kernel exactly when a unique isomorphism between their finite images intertwines them: define that isomorphism by $\theta(a) \mapsto \theta'(a)$, which is well defined exactly when the kernels agree. Thus the published $\mathrm{Aut}(H)$-normalization records kernels without loss. A conjugacy of the surface actions descends to a decorated-orbifold homeomorphism of $Q$ carrying one kernel to the other. Conversely such a quotient homeomorphism lifts through the two regular orbifold coverings because it carries their defining kernels to one another; its lift is the required surface conjugacy. This is an exact equivalence, so it remains to calculate the full action of these disc homeomorphisms on the finite kernel set.

A disc homeomorphism permutes its colored boundary corners cyclically, or reverses their cyclic order. After composing with that boundary map it fixes the boundary pointwise up to isotopy. The remaining mapping class is a braid of the interior cone points. To see this, forget the interior points and isotope the boundary-fixed disc map to the identity by the Alexander isotopy; the tracks of the points are its braid, and an isotopy of those tracks gives precisely an isotopy of the punctured disc map by isotopy extension. The colored braid subgroup is generated by pure braid generators and adjacent exchanges among equal colors after the colors are sorted. The program uses the standard words $\sigma_{j-1}\cdots\sigma_{i+1}\sigma_i^2\sigma_{i+1}^{-1}\cdots\sigma_{j-1}^{-1}$ and those equal-color exchanges. On interior generators an adjacent positive move is

$$(x_i, x_{i+1}) \longmapsto (x_i x_{i+1} x_i^{-1}, x_i), \tag{B.37.7}$$

fixing $e$ and every boundary reflection. The inverse is also used when forming the pure braid words.

Here are the boundary moves, including their effect on the chosen reflection. Extend the sequence by $c_{j+k} = e c_j e^{-1}$. One cyclic shift has

$$\begin{aligned} (Z_1,\dots,Z_k) &\longmapsto (Z_2,\dots,Z_k, eZ_1e^{-1}),\\ (x_1,\dots,x_l) &\longmapsto (x_1,\dots,x_l),\\ t &\longmapsto tZ_1, \qquad \alpha \longmapsto \mathrm{Inn}(Z_1^{-1})\alpha. \end{aligned} \tag{B.37.8}$$

A reversal has

$$\begin{aligned} (Z_1,\dots,Z_k) &\longmapsto (Z_k^{-1},\dots,Z_1^{-1}),\\ (x_1,\dots,x_l) &\longmapsto (x_l^{-1},\dots,x_1^{-1}),\\ t &\longmapsto tP_k, \qquad \alpha \longmapsto \mathrm{Inn}(P_k^{-1})\alpha. \end{aligned} \tag{B.37.9}$$

They follow by sending $c_j$ respectively to $c_{j+1}$ and $c_{k-j}$ in (B.37.2); in the reversal $e$ goes to $e^{-1}$. Only the shifts and shifted reversals preserving the corner-order list are self-homeomorphisms of the chosen colored quotient. A fixed braid restores the sorted interior order after a reversal. Equations (B.37.7)–(B.37.9), together with the preceding braid argument, prove both that every programmed fusion is realized and that every actual conjugacy is included.

The program computes the finite orbits under these moves. It identifies two generating vectors under $\mathrm{Aut}(H)$ by their rooted, edge-colored Cayley graphs: starting at the identity, visit vertices in generator order and record each colored successor. Equality defines a unique group isomorphism mapping each generator to the corresponding generator, and such an isomorphism plainly gives equality. Thus this is an exact isomorphism test rather than a coarse group invariant. The computation gives 145 orbits.

For each orbit the second program constructs the regular action of (B.37.6) explicitly from $(h,\epsilon)(h',\epsilon') = (h\alpha^{\epsilon}(h'),\epsilon+\epsilon')$. It checks (B.37.2), every local stabilizer injection, and $|K|\chi^{\mathrm{orb}}(Q) = -4$. The resulting cover is therefore a closed orientable genus-three surface. The finite coset incidence graph then gives every fixed point and oval, multiplier, section, extension, maximal section, centralizer, orbit and stabilizer as in B.26. These checks passed for all 145 classes.

Finally the overlap with B.34 is identified by actual monodromy isomorphisms. In each of B.34's 57 mirror-disc groups the program extracts the positive group and the generators $x_i$, $Z_j$ and conjugation by $c_0$. It applies (B.37.8)–(B.37.9) and the interior braids to put the signature in the same ordering, and locates its rooted colored Cayley graph in the computed kernel orbit. The file `genus3_hyperelliptic_reflection_crosscheck.json` records the resulting bijection with 57 different rows and equality of all eight reported section-count columns. Thus these are exactly the common classes, leaving 88 additional ones. □

The 145 classes have the following order distribution:

| $\lvert K\rvert$ | 4 | 6 | 8 | 12 | 14 | 16 | 18 | 24 |
|---|---|---|---|---|---|---|---|---|
| # | 5 | 3 | 36 | 13 | 3 | 35 | 1 | 15 |
| $\lvert K\rvert$ | 28 | 32 | 42 | 48 | 64 | 96 | 192 | 336 |
| # | 1 | 13 | 1 | 11 | 2 | 4 | 1 | 1. |

(B.37.10)

Together B.34 and B.37 therefore give 149 distinct mixed classes. The additional four are the projective-plane quotients in B.34. The scope of B.37 requires both a spherical positive quotient and a fixed-circle involution on it; the remaining quotient types are listed in the accompanying computation report.

**B.38. [Proved] The free positive involution and the six negative involutions in genus three.** There are exactly twelve conjugacy classes of mixed finite subgroups of $\mathcal{T}_{3,0}$ whose positive subgroup is a free involution. Each is a four-group. Their quotient signatures and numbers are

| underlying quotient | $S_{0,3}$ | $S_{1,1}$ | $N_{1,2}$ | $N_{2,1}$ | $N_{3,0}$ |
|---|---|---|---|---|---|
| number of classes | 4 | 1 | 4 | 2 | 1, |

(B.38.1)

where $S_{h,b}$ is orientable of genus $h$ with $b$ mirror boundary components, and $N_{q,b}$ is nonorientable with $q$ crosscaps and $b$ mirror boundary components. There are no cones or corners. There are also exactly six mixed classes whose positive subgroup is trivial: the negative involutions with quotient signatures

$$S_{0,4},\quad S_{1,2},\quad N_{1,3},\quad N_{2,2},\quad N_{3,1},\quad N_{4,0}. \tag{B.38.2}$$

All eighteen classes, their exact monodromies, complete section posets, $\Phi$, stabilizers, centralizers and multipliers appear in `type_3_0_free_positive_and_anti_catalogue.csv` and its `sections.json` companion.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

*Proof.* Let $X$ be the closed orientable genus-three surface. First let $G^+ = \langle a\rangle$ have order two and act freely, and let $G$ contain negative elements. Then $|G| = 4$. A group of order four is cyclic or a four-group: an element of order four gives the first case; otherwise $(xy)^2 = 1$ for all $x, y$, which gives commutativity and the second case. There is no nontrivial positive point stabilizer. A negative point stabilizer is generated by a reflection, so $Q = X/G$ has no cone or corner points. Its orbifold Euler characteristic is $-4/4 = -1$. The complete list of compact underlying surfaces with this Euler characteristic is exactly (B.38.1).

The cyclic possibility cannot occur. A negative element of order four has no fixed point, since a negative local finite stabilizer element is a reflection of order two. Its square is free by

assumption, so the whole action would be free. The quotient would then be $N_{3,0}$. Its fundamental group has the presentation

$$\langle u_1, u_2, u_3 \mid u_1^2 u_2^2 u_3^2 = 1 \rangle.$$

An orientation-compatible monodromy to $C_4 = \mathbb{Z}/4$ sends each crosscap generator to an odd residue. Each square has image 2, so the relator has image $6 = 2 \pmod 4$, a contradiction. The program checks all eight assignments of odd residues. Thus $G$ is the four-group $V_4 = \mathbf{F}_2^2$, with orientation given by its second coordinate, and $a = (1, 0)$.

We classify all four-group monodromies, including their full topological equivalence relation. The mirror orbifold group of an orientable $S_{h,b}$ has generators $a_i, b_i, e_j, c_j$ and relations

$$c_j^2 = 1, \qquad [e_j, c_j] = 1, \qquad \prod_i [a_i, b_i] \prod_j e_j = 1. \tag{B.38.3}$$

The negative $c_j$ is the reflection on the $j$th boundary, and the positive $e_j$ traverses that boundary. For $N_{q,b}$ replace the handles by crosscap generators $u_i$, and replace the last relation by

$$\prod_i u_i^2 \prod_j e_j = 1. \tag{B.38.4}$$

The orientation character $\varepsilon$ is zero on handles and $e_j$, and one on crosscaps and $c_j$. An orientation-compatible epimorphism to $V_4$ is exactly

$$q_\alpha : \Pi_Q \longrightarrow V_4, \qquad q_\alpha(x) = (\alpha(x), \varepsilon(x)), \quad \alpha \in H^1(\Pi_Q; \mathbf{F}_2) \setminus \{0, \varepsilon\}. \tag{B.38.5}$$

In coordinates write the handle or crosscap values as $\eta_i$, the boundary values as $\beta_j = \alpha(e_j)$, and the reflection values as $\gamma_j = \alpha(c_j)$. The only linear relation is $\sum_j \beta_j = 0$. The two automorphisms of $V_4$ preserving orientation identify $\alpha$ with $\alpha + \varepsilon$.

Every map (B.38.5) is injective on local reflection groups. Its kernel is therefore torsion-free; one proves this either from the universal hyperbolic orbifold cover or by cutting along arcs to its reflection polygon, where every finite stabilizer is a displayed local reflection group. The kernel lies in the orientation kernel, and its Euler characteristic is $4(-1) = -4$. It is consequently the fundamental group of a closed orientable genus-three surface. Conversely every action under consideration yields exactly this construction. Two such actions are conjugate precisely when an orbifold homeomorphism of $Q$ carries one monodromy kernel to the other, or equivalently when it carries the monodromy to the other after an automorphism of $V_4$. This follows by lifting paths in the two covers, with the local reflection charts giving the extension across mirrors. Thus the following finite cohomology-orbit calculation is an exact classification of actions.

We prove the entire mapping class action needed in this calculation. A boundary permutation permutes the pairs $(\beta_j, \gamma_j)$. The interior homology action preserves the mod-two intersection pairing, its radical generated by boundary classes, and the orientation character. It introduces no reflection coordinate: an interior loop is mapped to an interior loop, whereas each reflection generator is mapped to a conjugate of the reflection at the corresponding boundary. Thus the $\gamma_j$ are affected only by boundary permutations and by the global replacement $\alpha \mapsto \alpha + \varepsilon$.

For $S_{0,3}$ the interior homology is generated by its boundary classes, with their single sum relation, so boundary permutations are the whole action. When $\beta = (0, 0, 0)$, surjectivity leaves one nonconstant reflection-color pattern up to permutation and global color reversal. When $\beta$ has two ones, distinguish the boundary with value zero and use global color reversal to make its reflection value zero. The two remaining reflection values have weights zero, one or two. This gives $1 + 3 = 4$ classes.

For $S_{1,1}$ the boundary value is zero. Global color reversal sets its reflection value to zero. The remaining nonzero pair of handle values is a single orbit under $\mathrm{Sp}_2(\mathbf{F}_2)$, generated by the two Dehn twists in a pair of basis curves. This gives one class.

For $N_{1,2}$ write $u$ for its one-sided core and $e$ for either mod-two boundary class. Its intersection form has $u \cdot u = 1$, with $e$ in the radical. An automorphism preserving the boundary classes and the orientation character sends $u$ either to $u$ or to $u + e$. Both possibilities are realized. Push one boundary disc once around the one-sided core and return it as an unparametrized disc; its boundary orientation may reverse. A core arc crossing the track once acquires one boundary circuit. Thus, modulo two, this homeomorphism sends $u$ to $u + e$ and leaves both reflection coordinates unchanged. The boundary exchange is realized away from the core. These maps generate every permitted action on the displayed homology and boundary data. After global color reversal sets $\eta = \alpha(u)$ to zero, the case $\beta = (0, 0)$ has two nonzero reflection patterns up to exchange. The case $\beta = (1, 1)$ has two patterns: the boundary push followed by global reversal interchanges $\gamma = (0, 0)$ and $(1, 1)$, while $(0, 1)$ and $(1, 0)$ are exchanged by the boundary permutation. This gives four classes. The boundary push is necessary for this count.

For $N_{2,1}$ the boundary value is zero. Set its reflection value to zero by global reversal. The intersection form on the two crosscap classes is the identity matrix over $\mathbf{F}_2$; its orthogonal group has exactly the identity and the exchange of those two classes. They are realized by exchanging the two crosscaps. The nonzero crosscap vectors therefore have two orbits, represented by $(1, 0)$ and $(1, 1)$.

For $N_{3,0}$ the mod-two intersection form is the identity matrix in three crosscap classes, and its characteristic vector is $(1, 1, 1)$. An orthonormal basis cannot contain that characteristic vector, since it pairs to one with every vector of odd self-intersection. Its members must therefore be the three standard basis vectors. The full orthogonal group is their permutation group, realized by crosscap permutations. Global addition of $\varepsilon = (1, 1, 1)$ exchanges weights one and two, leaving one orbit after $0, \varepsilon$ are excluded. This proves all five numbers in (B.38.1).

We now compute the sections explicitly. A mirror boundary with data $(\beta_j, \gamma_j)$ has local reflection $c_j = (\gamma_j, 1)$ and boundary monodromy $e_j = (\beta_j, 0)$. Its preimage is described by the two cosets $V_4/\langle c_j \rangle$; one trip around the boundary permutes these cosets by $e_j$. Consequently it gives two fixed ovals when $\beta_j = 0$, and one when $\beta_j = 1$. The two-oval case has an oval stabilizer of order two, $\langle c_j \rangle$, and the one-oval case has stabilizer $V_4 = \langle c_j, e_j \rangle$. Both are one $V_4$-orbit of section classes. There are no point sections because the positive involution is free, and there are no full sections because $V_4$ cannot fix a point without giving a positive local stabilizer. Every reflection section is maximal, has centralizer $\mathbb{Z}$ in the surface group, and has multiplier $1/2$ on its nonidentity element. Thus

$$|\Phi| = \#\{\text{nontrivial partial sections}\} = 2b - \#\{j : \beta_j = 1\}, \qquad |\Phi/V_4| = b. \tag{B.38.6}$$

The program constructs the actual finite coset graph, with three subdivisions per boundary traversal, computes its components and the group action, and verifies every count and stabilizer in this formula.

Finally a mixed group with trivial positive subgroup is $C_2$. Its quotient has orbifold Euler characteristic $-2$ and no cones or corners. An orientable quotient must have mirror boundary; otherwise its orientation character would be trivial and could not give the required negative covering. Solving $2 - 2h - b = -2$ with $b > 0$, or $2 - q - b = -2$ with $q > 0$, gives precisely (B.38.2). For each quotient the monodromy is forced: it is the orientation character, taking every crosscap and local reflection to the nonidentity element and every handle and boundary traversal to zero. Its orientation double is connected, torsion-free, and has Euler characteristic $-4$. The classification of compact quotient surfaces and path lifting prove that each signature gives one conjugacy class and that distinct signatures give distinct classes. There are $b$ fixed ovals, hence $b$ full sections, $b$ nontrivial partial sections, and $b$ maximal classes, each fixed by the whole $C_2$ and with surface-group centralizer $\mathbb{Z}$. The case $N_{4,0}$ has none. These are the six genus-three real types of B.16 in an executed form.

The code `catalogue_genus3_free_positive.py` enumerates every cohomology vector in (B.38.5), applies exactly the proved generators of the equivalence relation, checks surjectivity, all

quotient relations, orientation compatibility, and the covering Euler characteristic, and records every orbit. Its second calculation constructs all six orientation doubles in (B.38.2). This supplies the complete eighteen-row catalogue and its section data. □

**B.39. [Proved] Complete genus-three projective-plane quotient catalogue.** There are exactly 11 conjugacy classes of mixed finite genus-three actions whose positive quotient is a sphere and whose induced involution on that sphere is fixed-point-free. Their full quotients are projective planes with cone points. Four classes are identified with B.34 and seven are additional. Consequently B.37 and the present claim give exactly 156 mixed classes with spherical positive quotient.

*Manuscript:* §18; folios 96–101; scans 193–203; Part B(f) of the task; genus-three projective-plane quotient subcatalogue.

*Proof.* A fixed-point-free sphere involution has projective-plane quotient: its quotient is a closed surface of Euler characteristic one, and its orientation cover is the given sphere. Cone points are exchanged in pairs. Thus each multiplicity in the positive cone multiset must be even. Inspection of the complete input in B.33 leaves exactly O3.4, O3.9, O3.11, O3.15, O3.21, O3.22, and O3.25. Their paired cone orders are respectively $(2,2,2,2)$, $(2,2,2)$, $(4,4)$, $(2,6)$, $(2,4)$, $(2,4)$, and $(2,3)$.

For a projective plane with cone orders $n_1,\dots,n_l$ use

$$\Lambda = \langle x_1,\dots,x_l,d \mid x_i^{n_i}=1,\quad x_1\cdots x_l d^2 = 1\rangle. \tag{B.39.1}$$

Set $X_i = x_i$, $Y_i = dx_i d^{-1}$ and $A = d^2 = (X_1\cdots X_l)^{-1}$. Its orientation double has the spherical vector

$$v = (X_1,\dots,X_l,Y_l^{-1},\dots,Y_1^{-1}),\qquad \prod_i X_i = \prod_i Y_i. \tag{B.39.2}$$

Conjugation by $d$ is

$$X_i \longmapsto Y_i,\qquad Y_i \longmapsto AX_iA^{-1}. \tag{B.39.3}$$

For every published $\mathrm{Aut}(H)$-normalized vector in these seven cases, the program tests that (B.39.3) descends to an automorphism $\alpha$ of its finite positive image $H$. The same Cayley-graph propagation as in B.37 gives an exact kernel-invariance test. It also verifies $\alpha^2 = \mathrm{Inn}(A)$ and $\alpha(A) = A$. The full finite image is then the set $H\times\{0,1\}$ with product

$$(h,\epsilon)(k,\delta) = \big(h\alpha^{\epsilon}(k)A^{\epsilon\delta},\epsilon+\delta \bmod 2\big). \tag{B.39.4}$$

The two identities just verified prove associativity; the element $d=(1,1)$ has square $A$ and induces $\alpha$ on $H$. Thus (B.39.4) is exactly $\Lambda/\ker\theta$, where $\theta$ is the positive epimorphism (B.39.2).

For clarity, the full source fusion is as follows. The colored mapping class group of the punctured projective plane is generated by the colored disc braids and by sliding each puncture once around a one-sided loop. One direct proof cuts the projective plane along a projective line disjoint from the marked disc: a moving puncture crossing the cut is separated off as a one-sided slide, after which all the remaining tracks form a braid in the disc. Equivalently, forgetting punctures successively expresses their point-pushing loops by peripheral loops and the one-sided generator. The former are the pure disc braids. The unmarked projective-plane mapping class group is trivial: isotope the image of a projective line to that line, adjust its parametrization by a projective rotation, and apply the disc Alexander isotopy after cutting. This proves that these point motions exhaust all mapping classes. Color-preserving permutations are supplied by the same equal-color braid generators used in B.37.

The slide of the last puncture has the exact substitutions

$$x_i \longmapsto x_i\quad (i<l),\qquad x_l \longmapsto (x_l d)x_l^{-1}(x_l d)^{-1},\qquad d\longmapsto x_l d. \tag{B.39.5}$$

They are obtained by tracking the meridian through a neighborhood of that puncture and the one-sided loop. The two-cone case, including its geometric puncture-slide interpretation, is verified independently in [27, Lemma 3.19]. Other punctures outside this neighborhood are fixed, so the same formula applies to all $l$ used here. It respects the product relation directly: if $B = x_1 \cdots x_{l-1}$ and $x = x_l$, then

$$B(xdx^{-1}d^{-1}x^{-1})(xd)^2 = Bxd^2 = 1.$$

In the finite positive group its effect is

$$x_l \longmapsto x_l\alpha(x_l)^{-1}x_l^{-1}, \qquad \alpha \longmapsto \mathrm{Inn}(x_l)\alpha, \qquad A \longmapsto x_l\alpha(x_l)A. \tag{B.39.6}$$

To slide any other puncture, a specified sequence of adjacent braids moves it to the last position, applies (B.39.6), and moves it back. The full word preserves its color even when the intermediate braids exchange different colors. Equations (B.39.5)–(B.39.6) and the disc braids therefore give exactly the source mapping-class action. The kernel-to-cover conjugacy argument of B.37 applies without change.

Executing these finite orbits gives respectively $1, 3, 3, 1, 2, 0, 1$ classes in the seven signature/group pairs above. For every class, `catalogue_genus3_projective_quotients.py` constructs the regular representation of (B.39.4), checks every relation in (B.39.1), injectivity of all local cyclic groups, and Euler characteristic $-4$. The finite point-coset calculation gives every partial section and its multiplier, centralizer and extensions. No negative element has a fixed point, since its image on the positive quotient sphere is fixed-point-free. Thus all nontrivial sections in these rows have positive cyclic image and are point sections; their surface-group centralizers are trivial.

Finally the four projective-plane groups in B.34 are compared using their actual cone generators and conjugation by their crosscap element. After a fixed ordering braid, their doubled vectors are identified in the finite kernel orbits by the rooted colored Cayley graph isomorphism test. The resulting exact identifications are

$$\mathrm{G3H}_{002} \leftrightarrow \mathrm{G3P}_{001}, \quad \mathrm{G3H}_{011} \leftrightarrow \mathrm{G3P}_{002}, \quad \mathrm{G3H}_{029} \leftrightarrow \mathrm{G3P}_{008}, \quad \mathrm{G3H}_{039} \leftrightarrow \mathrm{G3P}_{010}. \tag{B.39.7}$$

All seven applicable section-count columns agree under this bijection. Hence precisely seven classes are additional to B.34. The quotient involution on the positive sphere has a fixed circle in B.37 and has empty fixed set here; a conjugacy induces a sphere conjugacy carrying these fixed sets to each other. Thus none of these eleven rows can be conjugate to a B.37 row, proving the total $145 + 11 = 156$. □

Here $p$ counts nontrivial partial section classes, and $m, o$ are $|\Phi|$ and $|K\backslash\Phi|$. The final column $i$ counts negative involutions in the displayed finite group; all of them act freely.

| class | $\lvert K\rvert$ | cone orders | $p$ | $m$ | $o$ | $i$ |
|---|---|---|---|---|---|---|
| 001 | 4 | $(2,2,2,2)$ | 8 | 8 | 4 | 2 |
| 002 | 8 | $(2,2,2)$ | 12 | 12 | 3 | 0 |
| 003 | 8 | $(2,2,2)$ | 12 | 12 | 3 | 2 |
| 004 | 8 | $(2,2,2)$ | 12 | 12 | 3 | 4 |
| 005 | 8 | $(4,4)$ | 8 | 4 | 2 | 0 |
| 006 | 8 | $(4,4)$ | 8 | 4 | 2 | 2 |
| 007 | 8 | $(4,4)$ | 8 | 4 | 2 | 4 |
| 008 | 12 | $(2,6)$ | 12 | 8 | 2 | 2 |
| 009 | 16 | $(2,4)$ | 16 | 12 | 2 | 0 |
| 010 | 16 | $(2,4)$ | 16 | 12 | 2 | 0 |
| 011 | 24 | $(2,3)$ | 20 | 20 | 2 | 4 |

(B.39.8)

Every row has zero full splittings, since its full image contains a negative element and no such element fixes a point. The JSON `type_3_0_projective_quotient_sections.json` includes all finite permutations, local generators, multipliers and section incidences behind this table.

**B.40. [Proved] Complete annulus and Möbius-band quotients in genus three.** There are exactly 57 mixed finite genus-three action classes whose full quotient is an annulus or a Möbius band with its cone and mirror corner structure. There are 39 annulus classes and 18 Möbius-band classes. Their positive quotient is a torus. Every group, generating vector, quotient signature and complete section poset is given in `type_3_0_torus_mirrors_catalogue.csv`, `type_3_0_torus_mirrors_certificates.json`, and `type_3_0_torus_mirrors_sections.json`.

*Manuscript:* §18; folios 96–101; scans 193–203; Part B(f) of the task; genus-three annulus and Mobius quotient cases.

*Proof.* An involution reversing the orientation of a torus has annulus, Möbius-band or Klein-bottle quotient. Indeed its fixed set consists of disjoint circles, and its quotient has Euler characteristic zero; the orientation double is the torus. The classification of compact surfaces leaves the orientable annulus, the nonorientable Möbius band, and the closed nonorientable Klein bottle. Their orientation doubles give respectively two, one, and zero fixed circles. Cone points on a fixed circle become mirror corners, and other cone points are paired to interior cones. The Klein-bottle cases are B.41.

The complete positive input of B.33 leaves the following seven signature/group pairs. In each row $r$ is the number of positive quotient cone points, all of order $m$; the last two columns are the executed numbers of full annulus and Möbius-band classes.

| positive row | $H$ | $m$ | $r$ | annulus | Möbius |
|---|---|---|---|---|---|
| O3.3 | $C_2$ | 2 | 4 | 10 | 3 |
| O3.5 | $C_3$ | 3 | 2 | 5 | 4 |
| O3.7 | $C_4$ | 2 | 2 | 8 | 4 |
| O3.8 | $C_2^2$ | 2 | 2 | 9 | 4 |
| O3.12 | $S_3$ | 3 | 1 | 3 | 1 |
| O3.17 | $D_8$ | 2 | 1 | 3 | 1 |
| O3.18 | $Q_8$ | 2 | 1 | 1 | 1 |

(B.40.1)

Here $D_8$ has order eight. For $l$ interior cones, the total number of corners is $r-2l$. On an annulus distribute them as $k_1 \le k_2$ with $k_1+k_2 = r-2l$; on a Möbius band place all $k = r-2l$ on its single boundary. All their orders are $m$, so these integer choices already classify the colored quotient signatures.

We first give an exhaustive construction of the possible oriented finite groups, rather than assuming a small-group name list. Choose a negative coset representative $t$ in an index-two extension of $H$. Let $\alpha(h) = tht^{-1}$ and $A = t^2$. Its complete extension data satisfy

$$\alpha^2 = \mathrm{Inn}(A), \qquad \alpha(A) = A. \tag{B.40.2}$$

Conversely the product of (B.39.4), with these data, defines that extension:

$$(h,\epsilon)(k,\delta) = (h\alpha^{\epsilon}(k)A^{\epsilon\delta}, \epsilon+\delta \bmod 2). \tag{B.40.3}$$

Every oriented group isomorphism restricts to some $\beta \in \mathrm{Aut}(H)$ and sends $t$ to some $kt'$ in the negative coset. Therefore the entire isomorphism action on (B.40.2) is generated by transport by $\beta$ and by

$$(\alpha, A) \longmapsto (\mathrm{Inn}(k)\alpha,\ k\alpha(k)A). \tag{B.40.4}$$

The converse follows by sending the displayed generators accordingly in (B.40.3), so this is an exact fusion test. The program enumerates $\mathrm{Aut}(H)$ by assigning images to a generating set and

propagating on its complete Cayley graph, enumerates every pair (B.40.2), and takes precisely the finite orbits (B.40.4). For the seven $H$'s in (B.40.1), it obtains respectively $2, 2, 4, 3, 1, 4, 4$ oriented extension classes.

Here are the full source presentations and enumeration conditions. For an annulus, put $X = x_1 \cdots x_l$ and use positive generators $x_i, e_1, e_2$ with

$$x_i^m = 1, \qquad X e_1 e_2 = 1. \tag{B.40.5}$$

On its $j$th boundary there are reflections $c_{j,0}, \ldots, c_{j,k_j}$ satisfying

$$c_{j,s}^2 = 1, \quad (c_{j,s-1} c_{j,s})^m = 1, \qquad c_{j,k_j} = e_j c_{j,0} e_j^{-1}. \tag{B.40.6}$$

For $k_j = 0$, there is one reflection commuting with $e_j$. For a Möbius band there is one negative crosscap generator $d$, one boundary with (B.40.6), and the relation

$$X e d^2 = 1, \qquad e = X^{-1} d^{-2}. \tag{B.40.7}$$

In each finite extension the program assigns every possible positive order-$m$ element to every $x_i$, every positive element to $e_1$ in the annulus, and every negative element to $d$ in the Möbius band. It then assigns every sequence of negative involutions to the mirrors, requiring exact order $m$ for each adjacent product and the stated closing relation. It retains precisely the vectors generating the whole finite group. Two involutions whose product has exact order $m$ generate a dihedral group of order $2m$; thus all local groups inject. The orientation character takes the specified signs on every generator, so the kernel is oriented and contains no reflection. The signature equation gives $|K|\chi^{\mathrm{orb}}(Q) = -4$; its torsion-free surface kernel therefore has genus three. Conversely every action with this quotient yields one of these enumerated epimorphisms.

Target automorphisms are removed by the exact rooted colored Cayley graph test of B.37, now applied to the full NEC generating vector. Its prescribed generator signs force any resulting isomorphism to preserve the orientation character. For source fusion, the following are explicit complete generators. The interior braids are (B.37.7), with all other generators fixed. A cyclic boundary move sends $c_{j,s}$ to $c_{j,s+1}$, extending the reflection sequence by (B.40.6), while fixing $x_i$, $e_j$, and the crosscap generator. After $k_j$ moves it conjugates that boundary's reflection sequence by $e_j$ and fixes the other generators, which supplies the boundary Dehn twist where it is needed.

For the annulus, pushing the last interior point once around the first boundary, with $a = x_l$ and $e = e_1$, has

$$\begin{aligned} a &\longmapsto (ae)a(ae)^{-1}, \\ e_1 &\longmapsto a e_1 a^{-1}, \\ c_{1,s} &\longmapsto a c_{1,s} a^{-1}. \end{aligned} \tag{B.40.8}$$

It fixes the other $x_i$, $e_2$ and the second reflection sequence. This is the square of the Hurwitz exchange between the puncture meridian and the first boundary loop; in particular it preserves $ae_1$ exactly. The interior braids conjugate this to pushes of each other interior point. When $k_1 = k_2$, interchange of the two boundary components is

$$\begin{aligned} e_1 &\longmapsto e_1 e_2 e_1^{-1}, \qquad e_2 \longmapsto e_1, \\ c_{1,s} &\longmapsto e_1 c_{2,s} e_1^{-1}, \quad c_{2,s} \longmapsto c_{1,s}, \end{aligned} \tag{B.40.9}$$

fixing the interior generators. This is their orientation-preserving boundary exchange, with the chosen connecting paths.

The remaining annulus generator reverses the orientation of its fundamental region. For the peripheral list $(w_1, \ldots, w_{l+2}) = (x_1, \ldots, x_l, e_1, e_2)$, write $W_i = w_1 \cdots w_i$. It sends

$$w_i \longmapsto W_{i-1} w_i^{-1} W_{i-1}^{-1}. \tag{B.40.10}$$

For the first boundary its reflection sequence is reversed and conjugated by $X$; for the second it is reversed and conjugated by $Xe_1$. Reversing a cycle means $(c_0, \dots, c_{k-1}) \mapsto (c_k, c_{k-1}, \dots, c_1)$, with $c_k = ec_0e^{-1}$; an empty cycle retains its single reflection. These substitutions verify (B.40.5)–(B.40.6) directly.

For the Möbius band, a one-sided slide of the last interior point is particularly explicit. Set

$$a = x_l, \qquad a' = (ad)a^{-1}(ad)^{-1}.$$

The substitution is

$$\begin{aligned} x_l &\longmapsto a', & d &\longmapsto ad, \\ c_s &\longmapsto a'^{-1} a\, c_s\, a^{-1} a'. \end{aligned} \tag{B.40.11}$$

It fixes the other interior generators and determines $e$ by (B.40.7). To obtain it geometrically, regard the boundary temporarily as the last peripheral loop in the projective-plane presentation: exchange $(a, e)$ by its Hurwitz move, slide the now-last puncture by (B.39.5), and undo that exchange. This gives exactly (B.40.11), including the displayed transport of every boundary reflection. Finally a reversal of the core has

$$\begin{aligned} (x_1, \dots, x_l) &\longmapsto (x_l^{-1}, \dots, x_1^{-1}), \\ d &\longmapsto d^{-1}, \qquad c_s \longmapsto X c_{k-s} X^{-1}. \end{aligned} \tag{B.40.12}$$

Here $e$ goes to $Xe^{-1}X^{-1}$, so the closing relation is preserved.

We justify that this is the whole source action. First prescribe the cyclic permutations, and possible reversal or interchange, of the marked boundary components. The boundary moves above realize exactly these permutations. Upon forgetting the interior points, a remaining annulus map is generated by its core twist, and a remaining Möbius-band map by reversal of its core. Cutting along a spanning arc gives this classification: the annulus permits the integer winding of the arc, while cutting the Möbius band reduces a boundary-fixed map to a disc isotopy. The boundary reversal is (B.40.12). The core twist is already among the full cyclic boundary moves when both boundaries are marked; when a boundary is unmarked its rotation isotopes it away in the unpunctured quotient. The kernel of forgetting interior points consists of their braids. An interior point in an annulus minus the other points has fundamental group generated by the puncture meridians and the core loop; in a Möbius band the corresponding extra generator is one-sided. Forgetting points one at a time, and using isotopy extension, therefore shows that their braid groups are generated by the ordinary interior braids and respectively (B.40.8) or (B.40.11). This also includes any twist that becomes trivial only after the interior points are forgotten. Thus every geometric source automorphism is included, and each included automorphism is realized by the maps just described. The kernel-to-cover lifting argument of B.37 proves that the computed orbits are exactly the surface conjugacy classes.

Execution gives the 57 classes in (B.40.1). There is an independent check on the source generators in every one-cone, no-corner case: the complete primary automorphism lists in [27, Lemmas 3.7 and 3.8] give exactly the same partitions of the entire target-normalized epimorphism sets. The file `check_torus_mirror_generators_output.json` records all six comparisons, comparing full orbit memberships and not just their numbers.

The program then computes all section invariants. Its mirror incidence graph has a separate closed cycle for each quotient boundary, with endpoint gluing by that boundary's actual element $e_j$. At every corner it uses the full local dihedral group, and checks that the fixed subgraph of each reflection has degree two at every vertex. Consequently its connected components are exactly the fixed ovals, including all their intersections with other reflection fixed sets. Cone and corner cosets give the isolated fixed points. The procedure of B.26 then supplies every section, multiplier, centralizer, extension incidence, maximal section, group orbit and stabilizer. All 57 presentations, local injections, Euler characteristics and complete section calculations passed. □

The 57 classes have orders $4, 6, 8, 12, 16$ with multiplicities $13, 9, 25, 4, 6$, respectively. The full quotient surfaces in this claim have boundary, whereas those in B.41 are closed. A conjugacy of actions descends to a homeomorphism of their full quotient orbifolds and therefore preserves boundary. Their positive quotients have genus one, whereas those in B.37 and B.39 have genus zero and those in B.38 have genus two or three. The induced homeomorphism of positive quotients preserves genus. These exact quotient maps prove that the four portions contribute disjoint conjugacy classes.

**B.41. [Proved] The six genus-three actions with Klein-bottle quotient.** There are exactly six mixed finite-subgroup classes in $\mathcal{T}_{3,0}$ whose quotient orbifold has underlying Klein bottle. Their positive subgroups, full groups, cone orders, and section counts are

| $G^+$ | $G$ | cone orders | $f$ | $p$ | $\lvert\Phi\rvert$ | $\lvert\Phi/G\rvert$ |
|---|---|---|---|---|---|---|
| $C_2$ | $C_4$ | $2,2$ | 0 | 4 | 4 | 2 |
| $C_2$ | $C_2^2$ | $2,2$ | 0 | 4 | 4 | 2 |
| $C_3$ | $C_6$ | 3 | 0 | 2 | 2 | 1 |
| $C_4$ | $C_8$ | 2 | 0 | 4 | 4 | 1 |
| $C_4$ | $C_4 \times C_2$ | 2 | 0 | 4 | 4 | 1 |
| $C_2^2$ | $D_8$ | 2 | 0 | 4 | 4 | 1 |

(B.41.1)

Here $f$ and $p$ denote full and nontrivial partial section classes, and $D_8$ has order eight. The files `type_3_0_klein_quotient_catalogue.csv` and `type_3_0_klein_quotient_sections.json` contain every group operation, monodromy, local multiplier, section stabilizer and centralizer. All centralizers in the surface group are trivial. In the primary positive catalogue, these rows lie over O3.3, O3.5, O3.7 and O3.8, respectively in numbers $2, 1, 2, 1$.

*Manuscript:* §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; Part B(f) of the task; no further manuscript assertion.

*Proof.* The orientation double of a Klein bottle is a torus. An interior cone of order $m$ in the quotient has two lifts of order $m$ in that double, and there are no mirrors or corners. The complete positive catalogue in B.33 therefore leaves precisely the positive actions

$$(C_2; 2, 2, 2, 2), \quad (C_3; 3, 3), \quad (C_4; 2, 2), \quad (C_2^2; 2, 2), \tag{B.41.2}$$

with torus underlying quotient. The other torus-positive rows have a single cone and cannot arise by this double-cover comparison. Thus the full groups have orders four, six or eight.

We enumerate all oriented group extensions, rather than presupposing a list of abstract groups. For a fixed positive group $H$ choose a negative representative $t$, put $\alpha(h) = tht^{-1}$ and $A = t^2 \in H$. Associativity is equivalent to

$$\alpha^2 = \mathrm{Inn}(A), \qquad \alpha(A) = A. \tag{B.41.3}$$

Conversely the set $H \times \{0, 1\}$ with normal form $ht^i$ and relations $th = \alpha(h)t$, $t^2 = A$ is a group for every pair in (B.41.3); checking the two possible negative factors verifies associativity. Changing $t$ to $kt$ changes the pair to

$$(\mathrm{Inn}(k)\alpha,\ k\alpha(k)A), \tag{B.41.4}$$

and changing the identification of $H$ by $\beta \in \mathrm{Aut}(H)$ changes it to $(\beta\alpha\beta^{-1}, \beta(A))$. These are all orientation-preserving isomorphisms of the extensions, because any such isomorphism restricts to an automorphism of $H$ and sends a negative representative to one negative coset representative.

For $H = C_2$ this gives $C_4$ and $C_2^2$. For $H = C_3$, the two possible conjugation actions give $C_6$ and $S_3$. Explicitly the identity action allows the square of $t$ to be removed using (B.41.4), while inversion forces $A = 1$. For $H = C_4 = \langle a \rangle$, the identity action has two square classes, $A = 1$ or

$a$, giving $C_4 \times C_2$ and $C_8$. Inversion allows $A = 1$ or $a^2$, giving $D_8$ and $Q_8$. For $H = C_2^2$, the identity action gives $C_2^3$ or $C_4 \times C_2$ according as $A = 1$ or $A \neq 1$. Every nonidentity order-two automorphism of $H$ is conjugate to a transvection. Its fixed subgroup equals the image of $1 + \alpha$, so (B.41.4) removes either allowable square. This gives one further oriented extension, $D_8$, with positive subgroup a four-group. These arguments prove all the extension counts $2, 2, 4, 3$ used in the computation.

First take a Klein bottle with one cone of order $m$. Its orbifold group is

$$\Lambda_m = \langle d_1, d_2 \mid (d_1^2 d_2^2)^m = 1 \rangle. \tag{B.41.5}$$

Put $y = d_1$, $z = d_1 d_2$. Then $y$ is negative, $z$ is positive, and a positive local cone generator is

$$x = (yzy^{-1}z)^{-1}. \tag{B.41.6}$$

A smooth orientation-compatible epimorphism to an oriented finite group $G$ is therefore exactly a pair $(Y, Z)$ such that $Y$ is negative, $Z$ is positive, $\langle Y, Z \rangle = G$, and the image of (B.41.6) has order $m$. Gromadzki–Szepietowski [27, Lemma 3.3] prove that the full outer automorphism group of $\Lambda_m$ is generated by

$$(y, z) \longmapsto (yz, z), \qquad (y, z) \longmapsto (y^{-1}, z), \qquad (y, z) \longmapsto (y, z^{-1}). \tag{B.41.7}$$

Their proof identifies these maps with the Dehn twist and puncture and crosscap slides of the one-pointed Klein bottle. The automorphism assertion concerns $\Lambda_m$ itself and does not impose their later nonorientable-kernel hypothesis. We apply that assertion to the orientation-compatible epimorphisms above; every generator in (B.41.7) preserves the orientation character. Quotienting the finite set of pairs also by $\mathrm{Aut}(G)$ preserving orientation is therefore exactly topological conjugacy of the surface actions.

For $H = C_3$ the cone order is three. In the $S_3$ extension a negative element inverts every positive element, so (B.41.6) is the identity. In $C_6$ it equals $Z^{-2}$; $Z$ must generate the positive $C_3$. The first move in (B.41.7) runs through the entire negative coset, and the last move exchanges the two choices of $Z$. Thus there is one class, with full group $C_6$.

For $H = C_4$ the cone order is two. The negative elements of $D_8$ and $Q_8$ invert $H$, again making (B.41.6) trivial. In the two abelian extensions it is $Z^{-2}$, so $Z$ must generate $C_4$. For $C_4 \times C_2$, multiplication of $Y$ by powers of $Z$ sets $Y$ to the order-two generator of the other factor. For $C_8$, an automorphism sets the odd generator $Y$ to a chosen generator, and inversion of $Z$ identifies its two remaining possibilities. Each gives exactly one class.

For $H = C_2^2$, the elementary abelian extension $C_2^3$ cannot be generated by two elements. In the abelian extension $C_4 \times C_2$, every $Z \in H$ has square one, so (B.41.6) is trivial. Write the remaining extension as

$$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle, \qquad H = \langle r^2, s \rangle.$$

The admissible $Z$ are $s, r^2 s$, and every negative $Y$ gives the cone image $r^2$. An orientation-preserving automorphism exchanges the two choices of $Z$; the first move in (B.41.7) changes a negative reflection to a negative rotation, and inversion then identifies $r$ and $r^{-1}$. All eight pairs are one orbit. This gives the final row in (B.41.1).

It remains to handle the two order-two cones in the first case of (B.41.2). For either extension $G = C_4$ or $C_2^2$, let $h$ be the nonidentity positive element. Both cone generators must map to $h$. The two crosscap generators may independently map to either negative element. Every one of these four assignments is surjective and respects the relation

$$d_1^2 x_1 d_2^2 x_2 = 1. \tag{B.41.8}$$

They are all equivalent. Here is a geometric source move with its exact word comparison, so no unproved transitivity is needed. In a one-crosscap neighborhood of a cone, slide that cone once around the one-sided core. On the free generators $(d, x)$, with exterior boundary word $d^2x$, the move is

$$d \longmapsto dx, \qquad x \longmapsto (dx)^{-1}x^{-1}(dx). \tag{B.41.9}$$

It preserves $d^2x$ exactly and sends the cone generator to a conjugate of its inverse. It is an automorphism: first replace the free basis $(d, x)$ by $(dx, x)$, then invert and conjugate the second member by the first. Cutting the crosscap neighborhood along its core gives the stated cone slide; the unchanged exterior boundary permits extension by the identity to the other crosscap. In either abelian target, where $x$ has order two, this replaces the image of $d$ by its product with $h$ and leaves the cone image unchanged. Apply it independently at the two cones to connect all four assignments. This gives one class for each of $C_4$ and $C_2^2$. Their different exponents distinguish the two full groups.

For each surviving monodromy the local cone groups inject. The orbifold covering subgroup is therefore torsion-free, and it lies in the orientation kernel by construction. The Klein-bottle orbifold Euler characteristic is $-\sum_j(1 - 1/m_j)$; multiplying it by $|G|$ gives $-4$ in all six cases. The covering surface is consequently closed, orientable, and of genus three. This proves existence and faithfulness of every listed action, as well as completeness.

There are no mirrors, so no reflection sections. For each cone generator $x_j$ of prime order $m_j = 2$ or 3, the points of its critical orbit are exactly $G/\langle x_j\rangle$. At the coset represented by $g$, the stabilizer is $g\langle x_j\rangle g^{-1}$. Write $\varepsilon : G \to \{\pm 1\}$ for orientation. The multiplier map $\mu_{j,g}$ on the section represented by that point is

$$\mu_{j,g}(gx_j^k g^{-1}) = \varepsilon(g)\frac{k}{m_j} \pmod{\mathbb{Z}}. \tag{B.41.10}$$

Every such point supplies one nontrivial partial section, which is maximal; its centralizer in the surface group is trivial by B.3. No whole $G$ is a local stabilizer, so there are no full sections. The resulting coset counts are exactly (B.41.1). The program records the permutations of these cosets, their stabilizers and the multiplier of every local element. In the $C_6$ row the two point generators have multipliers $1/3$ and $-1/3$ under the fixed positive generator; the negative coset exchanges the two.

The executed standard-library program `catalogue_genus3_klein.py` independently enumerates all pairs (B.41.3), checks associativity of their finite multiplication tables and the orientation homomorphism, enumerates every smooth generating monodromy, and computes its orbit under the complete source and target automorphism actions. It retains the unsuccessful oriented extensions with their empty monodromy lists, so the exclusions above are part of the reproducible computation. □

# 3 C. Equivariant comparison, drilling, and the topology of special groups

Throughout this part, $U = X \setminus S$ is connected, $X$ is a closed orientable surface, and $S$ is a finite set of cardinality $\nu$. We put $A = \mathrm{Homeo}(X, S)$ with its compact-open topology and $K = \pi_0(A)$. Thus $A$ permits both permutations of $S$ and orientation reversal; $A^0$ fixes $S$ pointwise and preserves orientation. We call the type hyperbolic when $2g - 2 + \nu > 0$. A superscript $u$ means the centralizer of the image of the specified homomorphism $u : \Gamma \to A$. In particular, $(A^0)^u = A^0 \cap A^u$; it is not, by definition, the identity component of $A^u$. We distinguish these two groups until connectedness is proved.

## 3.1 The categories and their exact fidelity conditions

**C.1. [Refuted] The unrestricted full target has too many arrows.** The literal comparison from surfaces and homeomorphisms to the full 2-category of peripheral groupoid diagrams and arbitrary functors is not locally essentially surjective, already for $\Gamma = 1$ and type $(0, 3)$. The correct target for a realization biequivalence is the 2-groupoid of such diagrams with equivalences as 1-arrows.

*Manuscript:* §23; folios 134–136; scans 267–270.

*Proof.* For the three-punctured sphere put $\pi = \langle x, y \rangle$ and take the peripheral diagram

$$\coprod_{i=1}^{3} B\mathbb{Z} \longrightarrow B\pi, \qquad 1 \longmapsto x,\ y,\ (xy)^{-1}.$$

The identity on its three-object index set, the trivial homomorphism on each copy of $\mathbb{Z}$, and the trivial homomorphism $\pi \to \pi$ define a strictly commuting endomorphism of this diagram. Its map on $\pi_{\mathrm{ab}} = \mathbb{Z}^2$ is zero. A homeomorphism induces an automorphism on this abelianization. Natural isomorphisms between functors induce inner conjugations, which act trivially on abelianization. Consequently this endomorphism cannot become isomorphic to the image of a homeomorphism. This is a counterexample to the literal fullness assertion, not to its core version.

Here is the precise core version we use. An admissible surface is a finite disjoint union of finite-type orientable surfaces, each genus-zero component having at least one puncture, as on scan 264, folio 133. The source 2-groupoid has strict $\Gamma$-actions as objects, equivariant homeomorphisms as 1-arrows, and endpoint-preserving homotopy classes of equivariant isotopies as 2-arrows. An object of the target is a diagram of groupoids over $B\Gamma$

$$D_\Gamma \longrightarrow U_\Gamma \longrightarrow B\Gamma.$$

Its homotopy fibre over the base object is a peripheral diagram $\coprod_{i \in I} B\mathbb{Z} \to B\pi$ with the surface presentation $\prod_{j=1}^{g} [a_j, b_j] \prod_{i=1}^{\nu} c_i = 1$; the $i$th map sends 1 to $c_i$. The fibre condition includes the common orientation module and permits the noninjective map in type $(0, 1)$ and the repeated image in type $(0, 2)$. This is the condition $M_{\mathrm{lac}}$. For $I = \varnothing$ we use the closed surface group, with $g \geq 1$.

A target 1-arrow is a pair of groupoid equivalences over $B\Gamma$ together with the invertible natural transformation expressing commutativity of the peripheral square. A 2-arrow is a pair of natural isomorphisms, with identity projection to $B\Gamma$, satisfying that square's compatibility. Composition is composition of functors with pasting of these specified natural transformations. Equivalently these are equivalences of the corresponding opfibrations over $[1] \times B\Gamma$ that preserve the displayed fibres. The topological comparison takes the fundamental groupoids of the homotopy quotients of a $\Gamma$-invariant punctured collar system and of $U$. Two collar systems have a common smaller system; the induced equivalences and natural transformations specify the independence of this choice. No arbitrary functor is silently replaced by an equivalence. □

**C.2. [Proved] The coset category and the quotient correction.** For a hyperbolic peripheral pair $(\pi; (L_i)_{i \in I})$, the category $\mathscr{C}$ with objects $u, p_i$, automorphism groups $\mathrm{Aut}(u) = \pi$, $\mathrm{Aut}(p_i) = 1$, and $\mathrm{Hom}(p_i, u) = \pi / L_i$ determines the pair up to an isomorphism and simultaneous inner conjugation. Its equivalence groupoid has components $\mathrm{Out}_{\mathrm{lac}}(\pi)$ and trivial automorphism groups. For an action with extension $1 \to \pi \to E \to \Gamma \to 1$, the corresponding category over $B\Gamma$ uses the bimodule $E/L_i$, with right action of $E_i/L_i = \Gamma_i$, not $E/E_i$.

*Manuscript:* §23; folios 133–136; scans 264–270.

*Proof.* The stabilizer in $\mathrm{Aut}(u)$ of a selected arrow $p_i \to u$ is $L_i$. Changing that arrow to $aL_i$ conjugates its stabilizer by $a$. An equivalence fixes the unique object with nontrivial noncyclic endomorphism group and permutes the other objects. Its restriction at $u$ is an automorphism $\alpha$ of $\pi$. Equivariance of its maps on Hom-sets says precisely that $\alpha(L_i)$ is conjugate to $L_{\sigma(i)}$. Such a map of transitive sets is determined by a coset whose stabilizer has this property. The equality $N_\pi(L_i) = L_i$ makes that coset unique. A natural transformation between two such equivalences has its component $a \in \pi$ at $u$, its identity components at the $p_i$, and imposes $\alpha' = \mathrm{Int}(a)\alpha$ and the corresponding multiplication of all cosets by $a$. An automorphism of an equivalence therefore lies in $Z(\pi) = 1$. This proves the assertion about the equivalence groupoid and gives both comparison constructions explicitly.

For the geometric exit interpretation, the space of exits from $p_i$ to $u$ is the homotopy fibre of the link map $BL_i \to B\pi$. Its components are $\pi/L_i$ and its fundamental groups are the kernels of $L_i \to \pi$; its higher homotopy groups vanish. The peripheral injection makes those components contractible. Thus the ordinary category above has exactly these exit mapping spaces. This also identifies the exact failure of this ordinary model for the plane: the fibre of $B\mathbb{Z} \to *$ is $B\mathbb{Z}$, not a discrete space. More explicitly the Grothendieck category of $BL_i \to B\pi$ has automorphism groups $L_i, \pi$ and forward Hom-set $\pi$. Its comparison with the exit category is identity at the generic group, trivial at the special endomorphism group, and $\pi \to \pi/L_i$ on forward arrows. Conversely the translation groupoid of the generic action on the exit Hom-set is $(\pi/L_i)//\pi \simeq BL_i$, with its map to $B\pi$. Thus the peripheral diagram is recovered by this explicit link construction; these two displayed categories are not silently identified. On classifying spaces the forward comparison collapses each link to its special point, realizing the filling map $B\pi \to B\pi \cup_{\coprod BL_i} \coprod *$.

In the action category take one special object for each $\Gamma$-orbit of punctures, choosing $i$ in that orbit. The generic object has automorphism group $E$, the special object has automorphism group $\Gamma_i$, and composition on $E/L_i$ is left multiplication by $E$ and right multiplication by $E_i/L_i$. These actions commute. The homotopy fibre over the base object of $B\Gamma$ has special objects indexed by $\Gamma/\Gamma_i$ for each chosen representative, and hence recovers every upstairs puncture. An equivalent unskeletonized model uses all punctures and the nonempty $\Gamma$-transporter sets between those in the same orbit. The projection of a morphism $eL_i$ to $B\Gamma$ is the image $\bar{e} \in \Gamma$; it is well-defined because $L_i \subset \pi$. It is not well-defined on $E/E_i$ when $\Gamma_i \neq 1$: representatives $e$ and $ea$, $a \in E_i$, can have different images in $\Gamma$.

For a geometric example take $U = \mathbf{P}^1 \setminus \{0, \infty, 1, -1\}$, $\Gamma = C_2$ acting by $z \mapsto -z$, and quotient coordinate $w = z^2$. Then $E = \pi_1(\mathbf{P}^1 \setminus \{0, 1, \infty\}) = F_2$. With $a$ a loop about $w = 0$, the upstairs inertia is $L_0 = \langle a^2 \rangle$ and $E_0 = \langle a \rangle$. The right $C_2$ action on $E/\langle a^2 \rangle$ is nontrivial, whereas replacing it by $E/\langle a \rangle$ erases that action. This is the exact morphism obstruction to the quotient-category inference in the prompt. □

**C.3. [Proved] The four levels and the coefficient ladder.** Let $A$ be a locally path-connected topological group whose components are 1-types. For the comparison from strict actions to weak actions in the categorical group $\Pi_1 A$, the first three fidelity conditions are, on each comparison $M \to N$ of transport spaces,

| | |
|---|---|
| 0 | $\pi_1 M \to \pi_1 N$ injective at every point, |
| 1 | $\pi_0 M \to \pi_0 N$ injective and $\pi_1 M \to \pi_1 N$ isomorphic, |
| 2 | $\pi_0 M \to \pi_0 N$ bijective and $\pi_1 M \to \pi_1 N$ isomorphic. |

Level 3 adds essential surjectivity on action objects. When $A^0$ is simply connected, these give exactly scans 279–280: vanishing of $\pi_1((A^0)^u)$; connectedness of $(A^0)^u$; conjugacy of two lifts of the same $\Gamma \to K$ by an element of $A^0$; and existence of a lift.

*Manuscript:* §23; folios 136–137; scans 271–273; §23; folios 140–141; scans 279–281.

*Proof.* The source Hom-groupoid is $\Pi_1 M$. A functor of groupoids is faithful exactly when it is injective on the automorphism group at each object. It is fully faithful exactly when that condition is an isomorphism and two source objects whose images are connected are already connected. It is an equivalence exactly when it is fully faithful and every target component is hit. Applying these three elementary tests proves the table, including the cumulative meaning of each level.

When $\pi_1(A^0) = 0$, the categorical group $\Pi_1 A$ is equivalent to the discrete group $K$. For strict homomorphisms $u, v : \Gamma \to A$ the map of transport spaces is

$$\{a \in A : au(\gamma)a^{-1} = v(\gamma)\ (\gamma \in \Gamma)\} \longrightarrow \{k \in K : k\bar{u}(\gamma)k^{-1} = \bar{v}(\gamma)\}.$$

A nonempty source is a torsor under $A^u$. Its fibres over points of the target are torsors under $(A^0)^u$. The first two lines of the table are therefore the first two conditions stated. For the third, given a target transporter $k$, choose $a \in A$ representing $k$. The two actions $aua^{-1}$ and $v$ have the same projection to $K$; conjugating them by an element of $A^0$ produces the prescribed transporter. Conversely, apply transporter surjectivity to the identity element of $K$. This also verifies the stronger marginal correction on scan 280.

For comparison with the GALOIS ladder, first take the 1-truncations of the transport spaces; fixed spaces are not assumed to inherit the ambient group's higher truncation. On a map of 1-types $M \to N$, set-valued constant coefficients test bijectivity on $\pi_0$; full faithfulness of restriction tests that bijectivity together with surjectivity on $\pi_1$; equivalence tests bijectivity on $\pi_0$ and an isomorphism on $\pi_1$. For groupoid-valued coefficients all three tests coincide with a weak equivalence, because $\pi_i M = \pi_i N = 0$ for $i \geq 2$; the same is true for 2-type coefficients. These assertions follow by applying the long exact sequence to the homotopy fibres: an $r$-connected fibre gives isomorphisms through degree $r$ and surjectivity in degree $r + 1$. Thus level 2 is the $r = 0$ equivalence test on the transport spaces. Levels 0 and 1 do not equal a rung in that ladder: their $\pi_1$-injectivity and $\pi_0$-injectivity tests have been written out above, whereas each rung of the coefficient ladder demands a component bijection. Level 3 concerns objects rather than this map of Hom-spaces. This is a comparison of the specified maps, not an identification based only on the words "faithful" and "equivalence". □

**C.4. [Proved] Coherent actions and rectification on spaces.** For a discrete group $P$, the space of self-homotopy equivalences of $BP$ has components indexed by $\mathrm{Out}(P)$, each with homotopy type $BZ(P)$. In particular, for the centre-free surface groups the map from outer actions to coherent actions is an equivalence of action groupoids. Every coherent action of a discrete group on a space has a strict realization on a space of the same homotopy type. For $P = \mathbb{Z}$ or $\mathbb{Z}^2$, the obstruction class in degree 3 is zero and the coherent actions with a specified linear action are classified by $H^2(\Gamma, P)$.

*Manuscript:* §23; folios 131–132; scans 261–263; §23; folios 135–141; scans 268–281.

*Proof.* The groupoid of functors $BP \to BP$ has homomorphisms $P \to P$ as objects and conjugating elements of $P$ as arrows. Its equivalence subgroupoid has automorphisms as objects. The stabilizer of an automorphism is $Z(P)$, and the orbits are $\mathrm{Out}(P)$. The nerve computes the mapping space because nerves of groupoids represent 1-types. Delooping the resulting categorical group has $\pi_1 = \mathrm{Out}(P)$, $\pi_2 = Z(P)$ and no higher homotopy groups. Its Postnikov class is the usual extension obstruction in $H^3(\mathrm{Out}(P); Z(P))$. For $P$ abelian the map $\mathrm{Aut}(P) \to \mathrm{Out}(P)$ is the identity and gives a strict section; the categorical group is $BP \rtimes \mathrm{Aut}(P)$, so this Postnikov class vanishes. A choice of coherence is then a 2-cocycle with values in the specified $\Gamma$-module $P$; changing it by a coherent equivalence changes the cocycle by a coboundary. This proves the stated classification without asserting a nonzero obstruction for either abelian surface group.

For rectification, represent a coherent action by a fibration $p : Y \to B\Gamma$ with the prescribed homotopy fibre. Pull it back along $E\Gamma \to B\Gamma$. The pullback $Y' = E\Gamma \times_{B\Gamma} Y$ has a strict deck action of $\Gamma$. Its projection to the contractible space $E\Gamma$ has fibre the original homotopy type, so the inclusion of any chosen fibre is a homotopy equivalence. Its homotopy quotient recovers $Y \to B\Gamma$. This construction applies simultaneously to peripheral diagrams by the diagrammatic pullback. It produces a space model; the additional demand that the model be the original surface is exactly the realization problem addressed below. For continuous profinite actions the objects and the homotopy quotient must instead be specified in a continuous/profinite homotopy theory; the preceding construction makes no untyped transfer to that setting. □

### 3.2 Finite actions on hyperbolic surfaces

**C.5. [Proved] Existence, with punctures and orientation reversal.** For a hyperbolic type and every finite group $\Gamma$, each homomorphism $\rho : \Gamma \to K$ lifts to a homomorphism $u : \Gamma \to A$. Its image can be chosen to consist of isometries of a complete finite-area hyperbolic metric on $U$.

*Manuscript:* §23; folios 136, 140–141; scans 270–271, 279–280; §24; folio 147; scan 293.

*Proof.* Let $H = \rho(\Gamma)$, a finite subgroup of the extended mapping class group. Kerckhoff's Theorem 5, together with the explicit extension in §IV, pp. 255–256, of the original paper [41], gives a fixed hyperbolic structure for $H$. That extension explicitly permits cusps; the paper's mapping-class convention explicitly includes orientation-reversing maps. We use the original paper, rather than inferring the punctured case from the announcement.

Choose the isometry in each class of $H$. It is unique. Indeed an isometry isotopic to the identity has a lift to $\mathbb{H}^2$ commuting with the surface deck group. That deck group contains two hyperbolic elements with different endpoint pairs, so an isometry commuting with both fixes at least three points of the ideal boundary and is the identity. The same argument covers a finite-area punctured surface; its non-elementary deck group again contains those elements. Products of the chosen isometries represent products in $H$, and uniqueness therefore makes the choices a homomorphism $H \to A$. Composing with $\rho$ proves the claim. The isometries extend to the end compactification and permute $S$, so the target really is the specified group $A$. □

**C.6. [Proved] Uniqueness up to the identity component.** Let $u, v : \Gamma \to A$ be homomorphisms with $\Gamma$ finite, $U$ hyperbolic, and $\bar{u} = \bar{v} : \Gamma \to K$. There is an $h \in A^0$ such that $hu(\gamma)h^{-1} = v(\gamma)$ for every $\gamma$. This holds for topological actions without a smoothness assumption.

*Manuscript:* §23; folios 136, 140–141; scans 270–271, 279–280.

*Proof.* We first justify the analytic model for either action. For a point $x \in U$ and its finite stabilizer $H$, lift each element of $H$ to the universal cover with the chosen lift of $x$ fixed. Uniqueness of a lift with a specified value makes these lifts a finite group. Compactifying $\widetilde{U} \simeq \mathbb{R}^2$ by one point gives a finite action on $S^2$ with two common fixed points. Kolev's compact-sphere theorem [44, Theorem 1.2] conjugates that action into $O(3)$. A subgroup of $O(3)$ fixing two distinct points is contained in the orthogonal group of the perpendicular plane; choosing a sufficiently small round disc at the first point and pulling it back yields a common invariant disc with an $O(2)$ action. Choosing it small enough makes its translates by distinct cosets of $H$ disjoint. At a puncture we apply the same finite-stabilizer disc argument on the compactification, as in the finite-disc theorem proved from the same source.

The quotient consequently has ordinary charts, cone charts, mirror charts, and corner charts with their finite cyclic or dihedral local groups. A finite triangulation respecting these strata supplies a smooth orbifold structure: smooth each triangle, use collar charts along its edges, and

use the displayed linear quotient charts at its vertices. Pulling the charts to $X$ makes the original finite action smooth for a compatible smooth structure. Average a metric and use its conformal class. Uniformization of the punctured conformal surface gives a finite-type structure for which each operator is conformal or anticonformal. Thus we have conformal structures $c_u, c_v$ on the same marked topological surface.

The extremal quasiconformal map $h : (U, c_u) \to (U, c_v)$ in the identity isotopy class is unique, by the finite-type Teichmüller theorem [5, §3(A)–(D)]. For each $\gamma$, the map $v(\gamma)^{-1}hu(\gamma)$ is orientation preserving: $u(\gamma)$ and $v(\gamma)$ have the same orientation sign. Pre- and postcomposition by these conformal or anticonformal maps leaves the maximal dilatation unchanged. The equality of the two mapping-class homomorphisms says that this map is isotopic to the identity, with the punctures retained. Uniqueness therefore gives $v(\gamma)^{-1}hu(\gamma) = h$. The quasiconformal map extends across the finite punctures; hence $h \in A^0$ and has the asserted conjugacy property. □

**C.7. [Proved] The finite centralizer component, including mirrors.** For a finite action $u$ on a hyperbolic $U$, the group $(A^0)^u$ is contractible. In particular it is the identity component of $A^u$.

*Manuscript:* §§23–24; folios 140–144; scans 279–286.

*Proof.* Replace $\Gamma$ by its image $H$. The preceding local construction makes $H$ a group of hyperbolic isometries. Let $\pi$ be the surface deck group and let $E$ be the full group of lifts of $H$ to $\mathbb{H}^2$. It is a discrete group of isometries, possibly with reflections; its quotient $Q$ is a finite-type hyperbolic orbifold. Write $D$ for descent of a centralizing homeomorphism to the labelled underlying surface of $Q$, with cone points, corner points, mirror boundary, and punctures retained.

We give the injectivity argument at the place where an orientation-preserving-only Birman–Hilden theorem would not suffice. For $f \in (A^0)^u$, the identity isotopy supplies a lift $\widetilde{f}$ that commutes with $\pi$. Since $f$ commutes with $H$, conjugation by $\widetilde{f}$ induces an automorphism $\alpha$ of $E$ that is the identity on $\pi$. For $e \in E$ and $a \in \pi$,

$$\alpha(e)a\alpha(e)^{-1} = \alpha(eae^{-1}) = eae^{-1}.$$

Thus $e^{-1}\alpha(e)$ centralizes $\pi$, and hence is the identity by the distinct-endpoint argument in C.5. Consequently $\alpha = \mathrm{id}_E$. The orbifold Dehn–Nielsen–Baer injectivity theorem, with its mirror and corner structure, makes $D(f)$ isotopic to the identity through homeomorphisms preserving those structures; the relevant precise form is [43, Theorem 2 and Proposition 3]. At the algebraic step in their Lemma 3 we have used the calculation just written: centralizers of individual reflections need not be cyclic, but the centralizer of the non-elementary surface subgroup is trivial. Thus that inapplicable sentence of their proof is not being used. There is an orientation hypothesis in that source when the labelled underlying quotient is orientable, and we verify it here. The canonical $\pi$-equivariant lift $\widetilde{f}$ extends by the identity on $\partial_\infty\mathbb{H}^2$. For a compact surface this follows from bounded displacement on a compact fundamental domain. For a punctured surface use a compact truncation and disjoint invariant horodiscs: displacement is bounded on the truncated part, and preservation of each end sends sufficiently deep cusp horodiscs into the corresponding cusp horodiscs. A sequence escaping through distinct horodiscs has the same ideal limit before and after the map, since their visual diameters tend to zero; a sequence in one horodisc has its cusp endpoint as its only ideal limit. These observations prove the claimed continuous extension. Because $\alpha = \mathrm{id}$, $\widetilde{f}$ preserves each reflection axis. It fixes that axis's two ideal endpoints individually, so preserves its linear orientation, and, being orientation preserving on $\mathbb{H}^2$, preserves its two sides. It therefore preserves every mirror chamber. On such a chamber the quotient orientation is the one induced from $\mathbb{H}^2$; hence $D(f)$ preserves it. Without mirrors an orientable quotient is covered orientably and the same assertion is immediate. Thus the theorem is applied inside its stated homeomorphism group.

Let $\mathcal{H}(Q)^0$ denote the identity component of that labelled homeomorphism group. Every path starting at its identity lifts uniquely, starting at the identity upstairs, to a path centralizing $H$. Lift first on the regular covering locus. The local cyclic and dihedral models extend the lift over cone points, corners and mirrors; continuity of the extended lift follows in the charts from the finite quotient map. The same construction for a parameter family proves the path and homotopy lifting properties. The only ambiguity for an endpoint is a central deck transformation. Such a deck transformation cannot belong to $A^0$ unless it is the identity, by C.5's isometry argument. It follows that descent identifies $(A^0)^u$ with $\mathcal{H}(Q)^0$, as soon as the latter is simply connected; the construction is continuous in local isotopy charts, so this is an isomorphism of topological groups.

We check that this group is contractible, including the small underlying quotients. End-compactify the quotient and retain its finitely many interior and boundary marked points. Yagasaki's compact-pair Lemma 2.1 [74] gives contractibility outside its explicit exceptional pairs. A marked point on a boundary component is covered by Lemma 2.1(ii). With no such point, the exceptional disk has at most one interior marked point, the annulus and Möbius band have none, the sphere has at most two, the projective plane at most one, and the torus and Klein bottle have none. Each of these has nonnegative orbifold Euler characteristic: an interior cone subtracts $1 - 1/m$, an interior puncture subtracts 1, and a corner or boundary puncture would already be a boundary marked point. But $\chi_{\mathrm{orb}}(Q) = \chi(U)/|H| < 0$. None occurs. The homeomorphism group with labels has the same identity component as the group fixing all these finite marks pointwise. Yagasaki's theorem therefore applies. This proves contractibility, and also supplies the simple connectedness used in the lifting construction. □

**C.8. [Proved] The finite core biequivalence and homotopy fixed points.** On hyperbolic surfaces, for every finite $\Gamma$ the corrected comparison of C.1 has all four fidelity levels and is a biequivalence. For a specified action $u$, the comparison

$$A^u \longrightarrow A^{h\Gamma} = \mathrm{Map}_\Gamma(E\Gamma, A), \qquad a \longmapsto (\text{constant map with value } a),$$

where $\Gamma$ acts on $A$ by conjugation through $u$, is a weak homotopy equivalence. In standard CW models it is a homotopy equivalence.

*Manuscript:* §23; folios 136–141; scans 270–281.

*Proof.* The nonequivariant peripheral comparison identifies $K$ with $\mathrm{Out}_{\mathrm{lac}}(\pi)$ and has trivial 2-automorphism groups, by the peripheral Dehn–Nielsen–Baer theorem [22, Theorems 8.1 and 8.8] and $Z(\pi) = 1$. C.7 gives levels 0 and 1, C.6 gives level 2 by the transporter argument of C.3, and C.5 gives level 3. These arguments also treat a nonfaithful homomorphism: its finite image is the action group, and a finite-order homeomorphism in $A^0$ is the identity. Disconnected admissible surfaces reduce to representatives of the finitely many orbits of connected components and their stabilizers, with the stated transporters providing the arrows between choices. This is the desired biequivalence on its hyperbolic part.

The map $A \to K$ is $\Gamma$-equivariant. Its fibres are contractible by the same homeomorphism-group theorem with $H = 1$. Taking derived homotopy fixed points gives a weak equivalence $A^{h\Gamma} \to K^\Gamma = C_K(\bar{u}(\Gamma))$, the latter a discrete space. The constant-map comparison followed by this map is exactly $\pi_0(A^u) \to C_K(\bar{u}(\Gamma))$. Its surjectivity follows by applying C.6 to $aua^{-1}$ and $u$ for any representative $a$ of a centralizing mapping class; its injectivity and its contractible fibres are C.7. Thus the displayed map is a weak equivalence. The source components have CW homotopy type by the ANR statement for quotient homeomorphism groups in [74]; taking the usual CW model of the derived target turns it into a homotopy equivalence. We state the model convention because scan 281 writes "homotopisme" without specifying an infinite-dimensional mapping-space model. □

### 3.3 The exceptional types and infinite operator groups

**C.9. [Proved] Corrected abelian tests and a torus calculation.** For $\pi_1(A^0) \neq 0$, replacement of the four tests by connectedness and uniqueness of lifts of $\Gamma \to K$ loses the coherent action data. The valid tests are C.3 with weak transport spaces. At a specified strict action the target automorphism space is the 1-type $A^{h\Gamma}$ and

$$\pi_1(A^{h\Gamma}, 1) = \pi_1(A, 1)^{\Gamma}.$$

For $A = \mathrm{Homeo}(T^2)$ and $u(C_2) = \{1, -1\}$, the group $(A^0)^u$ has four components, all contractible, whereas the right-hand invariant fundamental group is zero. Thus the naive connectedness test fails; the corrected comparison does not fail in this example.

*Manuscript:* §23; folio 141; scans 280–281.

*Proof.* The homotopy-fixed-point groupoid of a 1-type has loops equal to the fixed automorphisms of its base object, proving the displayed equality. Its components include the $H^1(\Gamma; \pi_1 A)$ coherence data, with possible obstruction for an individual component of $A$. Hence it is not legitimate to test only $\pi_0 A$ when $\pi_1 A \neq 0$.

For the torus, evaluation at 0 gives $A^0 \cong T^2 \times A^0_{\{0\}}$ by $f \mapsto (f(0), t_{-f(0)} f)$. The second factor is contractible by the homeomorphism-group theorem for a torus with a marked point. In the centralizer of $-1$, $f(0)$ is one of the four points of $T^2[2]$. Each is realized by translation. In the fibre $f(0) = 0$, regarding the torus as once punctured makes it hyperbolic; the fibre is contractible by C.7. This proves the four-component assertion. Conjugation by $-1$ acts as $-I$ on $\pi_1(A^0) = \mathbb{Z}^2$, so its invariants vanish. The inclusion of translations $T^2 \hookrightarrow A^0$ is equivariant and a weak equivalence; its homotopy fixed points have components $H^1(C_2; \mathbb{Z}^2_-) = (\mathbb{Z}/2)^2$ and zero higher homotopy groups. The four translations just identified realize exactly these four components.

For the sphere, the peripheral groupoid diagram is empty over $*$ and cannot encode its homotopy type; $\pi_2(S^2) = \mathbb{Z}$ already records the failure of asphericity. The plane, annulus and torus have $A^0 \simeq S^1, S^1, T^2$, respectively; in the plane the peripheral map $B\mathbb{Z} \to *$ retains the circle of automorphism 2-arrows, and in the annulus the two ends remain indexed even though their subgroup images coincide. These exact maps explain why collapsing to a bare outer group would erase data. The complete finite-action comparison for these models, including its parametric centralizer proof, is C.10. □

**C.10. [Proved] Every finite action in the exceptional aspherical types.** For $U = \mathbb{R}^2$, $S^1 \times \mathbb{R}$, or $T^2$, and every finite strict action, the fixed-to-homotopy-fixed comparison in C.8 is a weak equivalence and is a homotopy equivalence on CW models. The corrected peripheral 2-groupoid comparison has all four fidelity levels for these types. The tests are those of C.3; ordinary outer actions alone do not retain the required coherence classes.

*Manuscript:* §23; folios 136–141; scans 270–281.

*Proof.* We prove the parametric assertion, not merely classification of individual actions. Write respectively

$$G = O(2), \qquad G = (\mathbb{R}/\mathbb{Z}) \rtimes (C_2 \times C_2), \qquad G = (\mathbb{R}^2/\mathbb{Z}^2) \rtimes \mathrm{GL}_2(\mathbb{Z}).$$

The annulus element $(a, \epsilon, \eta)$ sends $(\theta, t)$ to $(\epsilon\theta + a, \eta t)$; the first sign acts on the circle factor and the second on the two ends. In every case $G \to A$ is a weak equivalence. Here is a direct check of the identity components. On the torus, evaluation at 0 splits $A^0$ as $T^2$ times its point-fixing subgroup. An identity isotopy can be made point fixing by postcomposing at time $s$ with

translation by the negative of the trace of 0. Thus that subgroup is its own identity component, and is contractible by the punctured-torus case of the homeomorphism-group theorem [74, Theorem 1.1 and Lemma 2.1]. The same adjustment on the annulus uses angular rotation and translation of its real coordinate. Its point-fixing subgroup is the identity group of a hyperbolic three-ended sphere and is contractible, so $A^0 \simeq S^1$. On the plane first fix 0 by translations and then fix 1 using multiplication by a nonzero complex number. The remaining identity group fixes $0, 1, \infty$ and is contractible; hence $A^0 \simeq S^1$ with generator the rotation circle. The component groups are respectively $C_2$, $C_2 \times C_2$, and $\mathrm{GL}_2(\mathbb{Z})$: orientation and end signs give the first two, and the torus action on its lattice gives the third. Cutting the plane or annulus along a ray or spanning arc verifies the stated kernels; the torus case is the genus-one Dehn–Nielsen–Baer theorem.

By the simultaneous finite normal forms proved in B.1 and B.14, conjugate the given finite image $H$ into $G$. We will prove

$$C_G(H) \longrightarrow C_A(H) \tag{C.10.1}$$

is a weak equivalence by examining its fibres. Conjugation by the chosen homeomorphism transports both the fixed and derived-fixed comparisons, so this reduction is an isomorphism of the actual diagrams.

*Torus displacement.* For $f \in C_A(H)$ let $M \in \mathrm{GL}_2(\mathbb{Z})$ be its lattice action and lift it as $F(x) = Mx + d(x)$, with $d$ periodic. Put $b = \int_{T^2} d(x)\,dx$ and $g(x) = Mx + b$ modulo $\mathbb{Z}^2$. Changing the lift changes $b$ by an integral vector and leaves $g$ unchanged. For $\gamma(x) = A_\gamma x + t_\gamma$ the commutation equation is

$$d(A_\gamma x + t_\gamma) - A_\gamma d(x) = (I - M)t_\gamma + k_\gamma, \qquad k_\gamma \in \mathbb{Z}^2.$$

Integration preserves its left-hand change of variables, because the affine operator preserves Haar measure. It gives the same equation with $d$ replaced by $b$. Thus $g$ commutes with $H$ and $g^{-1}f \in A^0$ has a unique lift of zero mean displacement. Consequently multiplication is a homeomorphism of spaces

$$C_G(H) \times \mathcal{K} \;\xrightarrow{\sim}\; C_A(H),$$

where $\mathcal{K}$ consists of these zero-mean maps. The averaging map need not be a group homomorphism; only this explicit space factorization is used. A zero-mean lift commutes exactly with every lift of $H$: the possible integral commutator above has zero average and is therefore zero.

First divide by the finite translation subgroup $N = H \cap T^2$. If $a$ represents a member of $N$ of order $m$, commutation implies $F(x + a) = F(x) + a + k$. Iterating $m$ times and using $ma \in \mathbb{Z}^2$ forces $k = 0$ for a zero-mean identity lift. Hence its displacement is periodic for the enlarged covering lattice. Descent and lift identify $\mathcal{K}$ with the same zero-mean space on $T' = T^2/N$ for $H/N$. This quotient group has faithful linear part, a finite subgroup $P$ of $\mathrm{GL}_2(\mathbb{Z})$.

When $P$ contains a nonidentity positive operator, its positive subgroup $K$ is cyclic, generated by an affine rotation with $I - A$ invertible. It has a nonempty finite fixed set $R$ on $T'$, stable under $H/N$. A zero-mean lift commutes with each lift of that rotation and fixes its unique Euclidean fixed point; thus it fixes every point of $R$. Choose $p \in R$. The point-fixing argument above places our map in $\mathrm{Homeo}(T', p)^0$. C.7, applied to the cyclic action on $T' \setminus \{p\}$, supplies an equivariant identity isotopy. This isotopy fixes all of $R$, since $R$ is finite. Our map therefore belongs to $\mathrm{Homeo}(T', R)^0$. Apply C.7 again, now to the full $H/N$ action on the hyperbolic $T' \setminus R$. Its identity centralizer is exactly $\mathcal{K}$: conversely an equivariant identity isotopy relative to $R$ has commuting identity lifts, whose mean is fixed by $A$ and hence is zero. This proves contractibility of $\mathcal{K}$.

The remaining linear groups are $P = 1$ and a group of order two generated by a reflection. For $P = 1$, the space factorization on the identity component is $\mathrm{Homeo}(T')^0 \cong T' \times \mathcal{K}$. The first factor is a section of the mean map. Since the left side has the homotopy type $T'$, the mean map

is an isomorphism on $\pi_1$ and $\mathcal{K}$ is weakly contractible. It is a retract of the ANR on the left, so it has CW homotopy type and is contractible.

For a reflected or glide-reflected torus the underlying quotient is an annulus, a Möbius band, or a Klein bottle. This exhausts the cases: the action reverses orientation, has no isolated rotation point, and the quotient has Euler characteristic zero; its boundary consists of mirrors. A zero-mean lift $F$ commutes exactly with the Euclidean orbifold group $E$. Each lifted reflection line is preserved, and bounded displacement prevents reversal of its infinite directions. In particular the compact annulus quotient's two boundary circles cannot be interchanged. The explicit straight homotopy

$$F_s(x) = (1-s)F(x) + sx$$

is $E$-equivariant, because every element of $E$ is affine. On each reflection line it stays on that same line. It therefore descends to a homotopy of the quotient as a surface with boundary, preserving its boundary throughout. Its quotient domain is compact, so it is proper. Epstein's Theorem 6.3, with the topological Appendix [20], applies with exactly this proper, boundary-preserving homotopy and gives an identity isotopy. This also handles the boundary-free Klein bottle; Theorem 5.8 of that source separately verifies the Möbius-band case. Thus no unverified inference from an abstract outer automorphism to an exceptional-surface isotopy is needed here.

Descent consequently puts $\mathcal{K}$ in the identity group of the labelled quotient. Every identity isotopy there lifts; among the two lifts exactly one preserves the orientation upstairs. This identifies the identity component of $C_{A^0}(H/N)$ with the quotient identity homeomorphism group. By Yagasaki's actual exception list its homotopy type is $S^1$ in all three cases. The mean factorization on that component is $T_0'^P \times \mathcal{K}$, with $T_0'^P \cong S^1$. Here $P$ acts linearly on the translation torus; this fixed translation circle is present even for an affine glide with no fixed point. The section by translations makes the mean map surjective on $\pi_1$; a surjection $\mathbb{Z} \to \mathbb{Z}$ is an isomorphism. Hence $\mathcal{K}$ is weakly contractible, and is contractible by the same ANR-retract argument. This completes (C.10.1) for the torus, including free glide reflections.

*Annulus displacement.* Lift $f \in C_A(H)$ to $F(\theta,t) = (\epsilon_f\theta + d(\theta,t), F_2(\theta,t))$ and put $b = \int_0^1 d(\theta,0)\,d\theta$. Define $g(\theta,t) = (\epsilon_f\theta + b, \eta_f t)$, where $\eta_f$ is its end sign. The commutation equation, integrated on the invariant core $t = 0$, proves that $g$ commutes with $H$. It gives the same continuous factorization $C_A(H) \cong C_G(H) \times \mathcal{K}$, with zero-mean identity lifts in $\mathcal{K}$. Divide first by the finite rotation subgroup. The resulting faithful sign group is a subgroup of $C_2 \times C_2$.

When it contains $(-1,-1)$, that positive involution has two fixed points. Its subgroup is normal. A zero-mean lift fixes each of the corresponding fixed points in $\mathbb{R} \times \mathbb{R}$. The same two applications of C.7 as in the torus argument, first marking one fixed point and then the whole invariant fixed set, identify $\mathcal{K}$ with a contractible centralizer on a hyperbolic punctured annulus. An equivariant identity lift in that punctured model has mean zero because its angular linear sign is $-1$, proving equality in both directions.

A subgroup without $(-1,-1)$ is trivial, generated by $(-1,1)$, or generated by $(1,-1)$. In the first case the factorization and $A^0 \simeq S^1$ give contractibility as above. An angular reflection has a strip as quotient, with its two mirror boundary lines and two ends. Exact commutation of a zero-mean lift preserves both lines and both ends. The same straight homotopy in $\mathbb{R} \times \mathbb{R}$ is equivariant for all the affine covering operators and preserves the boundary lines. It is proper on the strip quotient: for $t \to +\infty$ both $t$ and $F_2(\theta,t)$ tend uniformly to $+\infty$ on a compact angular fundamental interval, and for $t \to -\infty$ both tend to $-\infty$. Every convex combination has the same uniform property. Epstein's proper boundary-homotopy theorem therefore places the descended map in the identity group. That group is contractible by Yagasaki, so $\mathcal{K}$ is contractible. Finally a radial reflection or radial glide reflection has quotient a half-open annulus or a punctured projective plane, respectively. The straight homotopy is again equivariant; it preserves the fixed mirror line in the reflection case and has the just-verified properness at the quotient's end. Epstein's Theorem 6.3, or Theorem 5.8 for the open Möbius band, gives the

required identity isotopy. Descent and isotopy lifting identify the upstairs centralizer identity component with that group. Yagasaki gives $S^1$ in both cases. The mean factorization has a section by the fixed angular circle, hence again forces $\mathcal{K}$ to be contractible. These cases include every affine finite action and retain the end permutation throughout.

*Plane centralizers.* For a cyclic rotation group of order $n > 1$, a commuting map fixes 0. Descent through $z \mapsto z^n$ is a covering of the orientation-preserving centralizer onto $\mathrm{Homeo}^+(\mathbb{R}^2, 0)$, with deck group $C_n$. The base is connected with homotopy type $S^1$. A loop of one rotation downstairs lifts from the identity to the deck rotation by $2\pi/n$. Its monodromy is therefore onto $C_n$, so the covering centralizer is connected and the upstairs rotation circle induces the isomorphism on $\pi_1$; all higher homotopy groups vanish. A negative map centralizes the rotation only for $n = 2$, as its action on the punctured-plane fundamental group inverts the deck generator. In that case conjugation supplies the second component. These are precisely the components and circle types of $C_{O(2)}(C_n)$.

For a reflection, descent has half-plane quotient. A commuting map has two independent signs: its orientation on the fixed line and its orientation on the plane. The four diagonal orthogonal maps realize them. Each fibre is the identity homeomorphism group of the half-plane, which is contractible by Yagasaki. For a dihedral group with rotation order $n > 2$, a centralizing map preserves plane orientation and each reflection line. Preservation of cyclic order of their rays permits only the identity permutation or simultaneous reversal of every ray; these are realized by $I$ and $-I$. For $n = 2$ the two perpendicular reflection lines permit four independent ray-sign choices, realized by the four diagonal orthogonal maps. After dividing by the matching map, descent is to a sector homeomorphism fixing its vertex and both boundary rays. This group is contractible: the sector is a half-plane with a marked boundary point, and its identity group is covered by Yagasaki's boundary-point case. The boundary rays also make the lift unique. The trivial group case was verified at the start. This proves (C.10.1) for every finite subgroup of $O(2)$.

It remains to identify the derived comparison, including all components. The following elementary torus calculation does so for each of our three $G$. Let $T = V/\Lambda$, with $V$ real and $\Lambda$ a lattice, and let a finite group act on a $T$-torsor by affine maps. Its translation part is a cocycle $b \in Z^1(\Gamma, T)$, where $T$ and $\Lambda$ have the specified linear $\Gamma$-module structures. The homotopy-fixed space is the section space of its Borel fibration. Its obstruction is the class $\delta[b] \in H^2(\Gamma, \Lambda)$ of the extension obtained by lifting the affine maps to $V$. Averaging cochains gives $H^j(\Gamma, V) = 0$ for $j > 0$, so $\delta : H^1(\Gamma, T) \to H^2(\Gamma, \Lambda)$ is an isomorphism. Thus a homotopy fixed point exists exactly when $b$ is a coboundary, which is exactly existence of an actual fixed point. After choosing one,

$$\pi_0(T^{h\Gamma}) = H^1(\Gamma, \Lambda), \qquad \pi_1(T^{h\Gamma}) = \Lambda^\Gamma,$$

and there are no higher groups. The exact sequence

$$0 \to \Lambda^\Gamma \to V^\Gamma \to T^\Gamma \to H^1(\Gamma, \Lambda) \to 0$$

identifies the actual fixed components and their fundamental groups with exactly these groups. The constant-map comparison induces these identifications. Apply this to every invariant component of $G$, each of which is just such an affine torus torsor. Noninvariant components contain neither fixed nor homotopy-fixed points. It follows that $G^\Gamma \to G^{h\Gamma}$ is a weak equivalence. The equivariant weak equivalence $G \to A$, together with (C.10.1), now proves the claimed comparison for $A$ by the commutative square of fixed and derived-fixed spaces.

Finally $BG$ is the split 2-type with fundamental group $\pi_0 G$, second group $\Lambda = \pi_1 G$, and zero Postnikov invariant. Coherent finite actions with a specified homomorphism to $\pi_0 G$ are therefore classified by $H^2(\Gamma, \Lambda)$. The same connecting isomorphism realizes each by a $T$-valued cocycle, hence by a strict homomorphism $\Gamma \to G$. It also identifies coherent equivalences in the identity component with coboundaries, namely conjugacies by translations. The fixed-space calculation supplies their full transport-groupoid comparison; a prescribed nonidentity transporter

component is treated by composing with its representative in $G$. Thus the corrected four fidelity levels all hold. We finish by checking that this is the actual peripheral comparison, not just an abstract replacement of $A$. For the plane the diagram is $B\mathbb{Z} \to *$. Its equivalence 2-group has component group $\{\pm 1\}$, 2-automorphism group $\mathbb{Z}$, and sign action. For the annulus use $B\mathbb{Z} \amalg B\mathbb{Z} \to B\mathbb{Z}$ with the two maps $+1, -1$. An equivalence is specified on components by a generic sign $\epsilon$ and an end permutation of sign $\eta$. Its peripheral sign is forced to be $\epsilon\eta$. The two square conjugators can be removed by peripheral natural transformations; the surviving automorphism of such a diagram equivalence is one diagonal $\mathbb{Z}$, on which the generic sign $\epsilon$ acts. Thus its 2-group has exactly the two-sign component group and lattice action of the annulus $G$ displayed above. For the torus, the empty peripheral diagram over $B\mathbb{Z}^2$ has component group $\mathrm{GL}_2(\mathbb{Z})$ and 2-automorphism group $\mathbb{Z}^2$ with its standard action. In all three cases strict diagram functors with zero square-conjugators give a section, so the Postnikov class is zero. The actual comparison maps a full plane or annulus rotation to the generator traced by a peripheral or generic basepoint, and torus translation loops to the two lattice generators. It is therefore an isomorphism on these specified $\pi_1$ groups, as well as on components. This proves that the models and comparisons used throughout the proof are the ones induced by C.1's peripheral functor. In particular the annulus end permutation has never been forgotten. □

**C.11. [Proved] Infinite cyclic failures at levels one and two.** For every hyperbolic surface containing a disc, there is a strict $\mathbb{Z}$-action for which $(A^0)^u$ is disconnected. There also are two strict $\mathbb{Z}$-actions with the same mapping-class homomorphism that are not conjugate. Thus finite-group hypotheses cannot be removed from levels 1 and 2.

*Manuscript:* §23; folios 140–141; scans 279–281.

*Proof.* In a coordinate disc choose two disjoint small round discs centered at $q_\pm = \pm 1$. Define $f$ to rotate each small disc by angle $\sqrt{2}(1 - r/r_0)$ at radius $r < r_0$, and to be the identity elsewhere. This is a homeomorphism isotopic to the identity through the same formula with the angle multiplied by a parameter. The two centres are isolated fixed points; all other fixed points lie outside the interiors of these discs. At radius $r_0/2$ the rotation angle is $\sqrt{2}/2$, whose ratio to $2\pi$ is irrational, so $f$ has infinite order. Let $h$ rotate the larger disc by angle $\pi$ on a concentric disc containing the two small discs, tapering continuously to angle zero before the outer boundary. It is also isotopic to the identity. It exchanges the two small discs by rigid rotation and therefore commutes with $f$.

Set $u(1) = f$. Every homeomorphism commuting with $f$ preserves its fixed set and sends isolated fixed points to isolated fixed points. Along a path in the centralizer the induced permutation of $\{q_-, q_+\}$ is constant. The identity fixes both and $h$ exchanges them. They therefore belong to different components of $(A^0)^u$. This proves the level-1 failure by the exact test C.3. The action $u$ and the trivial $\mathbb{Z}$-action both project to the trivial homomorphism into $K$, but a conjugate of $f \neq \mathrm{id}$ cannot be the identity. This proves the level-2 failure. The formulas are checked on an explicit grid in the supplied program; the isolation and irrationality arguments prove the assertions beyond that grid. □

**C.12. [Open] The requested infinite-group zero-level example.** For the admissible surfaces and the exact comparison of C.1–C.3, we have not produced an infinite operator group whose level-0 comparison has a nontrivial kernel, nor proved universal injectivity. For a hyperbolic surface the required kernel is $\pi_1(C_{A^0}(u(\Gamma)), 1)$, since the target is zero. The disconnected centralizer in C.11 proves level-1 failure. The enlarged sphere comparison in C.25 has its stated nontrivial kernel, but the unpunctured sphere is excluded from C.1's object class. The requested admissible infinite-group zero-level decision remains unsettled here. C.73 proves the actual prime-end comparison for arbitrary simply connected frontiers, including its dense image and the explicit

proper-image and inverse-topology failures for a non-locally-connected frontier. C.74 gives a transitive actual disk loop with full proper local centralizer $\mathbb{R}$ and proves that the specified change of principal section makes its coordinate interpolation fail to extend at every original boundary point; the original loop contracts. C.75 classifies every actual orbit strip for a one-dimensional proper operator action and computes its finite-endpoint translation characters, with the source stabilizer and all component operators retained. C.76–C.79 construct the controlled lift across all interior frontier lines and the original circle of the actual closed disk. In particular every based loop in the group of homeomorphisms of that disk fixing its boundary pointwise and commuting with the given closed locally compact free proper one-dimensional interior operator group has a based filling in that same relative centralizer. The proof retains a shared increasing base map, both independent endpoint characters, all component operators and the actual inverse topology. C.80 proves the same exact relative filling for every based sphere family and hence the vanishing of all positive based homotopy groups of this specified disk centralizer. The general comparison (C.12.1), arbitrary-frontier descent and nonfixed-frontier pasting remain unresolved. C.81 supplies the exact invariant metric and metric covering for a discrete proper operator group with an actual transitive boundary-fixed commuting subgroup. Compact families with a fixed compact quotient support extend jointly to the original circle, every based sphere family supported over a compact quotient subsurface fills, and fixing any finite configuration of operator orbits preserves every positive based homotopy group. Its explicit escaping-disk sequence proves discontinuity of the actual descent map into the Whitney topology. C.82 constructs a quotient triangulation whose lifted closed stars form a null family in the original disk. Restriction and equivariant pasting identify the pointwise full-skeleton stabilizer with a product of boundary-relative triangle groups, and the coordinatewise Alexander deformation contracts it in the original topology. A controlled deformation of an arbitrary original family into that stabilizer, and the general frontier and descent comparisons, remain to be constructed. C.83 gives an explicit null-homotopic loop in the actual infinite evaluation image, with all operator labels and a null lifted triangulation retained, which has no continuous lift to the original closed-disk group. This records the obstruction to that specified infinite-evaluation route; the general C.12 comparison remains unresolved. C.84 identifies the exact radial discrete-translation centralizer with its weighted sublinear-displacement topology and contracts the entire specified vertically monotone subspace. C.85 constructs a continuous cutoff of arbitrary actual families to common compact quotient support and proves vanishing of all positive homotopy groups for the radial model. C.86 gives inverse continuous group maps to the entire logarithmic-spiral relative centralizer of C.72 and transfers those sphere fillings, with its original boundary topology retained. C.87 uses the exact zero-twist annulus lift, a continuous support height and explicit original-disk normalizers to contract each entire radial and spiral relative centralizer while fixing the identity. C.88 identifies the original general discrete-operator centralizer exactly with interior equivariant maps whose actual invariant-metric displacement vanishes at infinity on the quotient. Restriction and identity extension are inverse isometric group maps; the compact-parameter criterion retains joint interior continuity and a uniform quotient tail. The controlled deformation of arbitrary original families to common compact quotient support remains unresolved. C.89 characterizes the exact restriction image on the compact lifted skeleton and original circle. Three-point-normalized Jordan-domain maps give a global continuous extension section; its displayed product with the C.82 stabilizer gives a strong deformation retraction onto that section image. C.90 proves the vanishing of every positive based homotopy group for every compact quotient in the actual C.81 setting, including nonorientable quotients, by lifting the pointed compact-surface contraction and applying the original-boundary criterion. C.91 strengthens this to a continuous identity-fixed contraction of the entire original centralizer for each such compact quotient. Its literal descent and unique point-fixed lift identify the whole point stabilizer with the pointed identity component downstairs; an explicit global evaluation section supplies the product and its inverse. The noncompact quotient tail, the deformation of the full skeleton embedding image for general quotients, and

the general frontier comparisons remain unresolved.

C.92 proves that every element of the entire original point stabilizer descends into the pointed identity path component for an arbitrary actual quotient. A locally finite based homotopy and the precise orientation comparison meet Epstein's based isotopy theorem. The unique point-fixed lift gives the exact subgroup whose actual lift displacement belongs to $C_0(N)$, with an isometric comparison to the original topology and the exact compact-family uniform-tail criterion. An explicit radial-disk sequence converges downstairs in the compact-open topology while its unique pointed lifts stay more than $2/5$ from the identity. A noncompact deformation with that uniform tail and the general frontier comparisons remain unresolved.

*Manuscript:* §23; folios 140–141; scans 279–281.

The attempted construction using a supported radial rotation produces isolated fixed points and a component obstruction, not a nonzero loop. Adding finitely many marked fixed points still gives aspherical configuration spaces and cannot by that argument produce the required $\pi_2$ of an evaluation base. No implication from the examples for levels 1 and 2 to level 0 is asserted. The independent infinite-group non-realization theorem C.23 gives level-3 failure and does not close this missing zero-level test. In particular, the possibility that level 0 holds for all operator groups has not been ruled out by this return.

**Scope of the remaining infinite-group obligation.** The source fixed in C.1 is the admissible-surface category of scan 264, folio 133: every genus-zero compact component meets the finite puncture set. The unpunctured sphere is therefore absent. The last sentence of Part C(b) asks for an infinite-group example at each fidelity level within this specified comparison. An example on one of its admitted abelian types would meet that existential request; a hyperbolic example would settle the more specific question in C.12. The examples C.11 and C.23 supply the independent obstructions at levels 1, 2, and 3. The level-0 example inside the admitted source remains unprovided.

Here is the exact kernel that is required. For an arbitrary discrete operator group and a strict action $u : \Gamma \to A$, put $H = u(\Gamma)$ and let $N_u$ be its weak automorphism groupoid in $\Pi_1 A$. At its identity object, a 2-automorphism is precisely a loop class in $A$ invariant under conjugation by every $u(\gamma)$. The comparison is thus the inclusion-induced map

$$\pi_1(C_A(H), 1) \longrightarrow \pi_1(N_u, 1) = \pi_1(A, 1)^{\Gamma}. \tag{C.12.1}$$

Pointwise commutation proves that the image lies in these invariants. Every based loop and every based homotopy in the source lies in $A^0$, so its source can equally be written $\pi_1(C_{A^0}(H), 1)$. For hyperbolic types the target vanishes, giving exactly C.12's test. For the plane, annulus, and torus the target is the invariant subgroup of $\mathbb{Z}$, $\mathbb{Z}$, or $\mathbb{Z}^2$, respectively; the required obstruction is a nonzero kernel of (C.12.1).

C.39 proves that every class in this kernel represented by a loop whose values commute pairwise is zero. Its contraction is global on the plane, including arbitrary changes and accumulation of orbit types, and its explicitly lifted filling disk descends to every connected admissible surface. Thus a remaining example must involve a class without a commuting-valued representative. C.40 constructs such a representative for every loop whose generated closure, in the actual compact-open topology, is locally compact. No representative reduction for arbitrary loops is proved here. C.41 proves the exact inclusion injective for every operator group with a dense orbit, without a local compactness assumption on either group. C.36–C.37 separately provide the stated contraction on the closed union of dimension-two proper-image open orbits for general loops. C.42 identifies the actual boundary quotient as an open map and proves that every component of every real lifted fibre is a properly embedded line; it also supplies an exact example with disconnected whole circle fibres. C.43 proves that the real lifted fibre is connected and supplies

actual product coordinates. C.44 gives the five downstairs covering forms, with source a plane or annulus. C.45 gives the exact relation between two quotients and its three planar coordinate domains. C.46 constructs a simultaneous intrinsic contraction when the full set of lifted boundary fibre partitions has exactly two members, retaining every operator and quotient relation. C.47 proves that three independent actual quotients force the full compatible group on the source plane to be locally compact and Lie, with its actual topology. C.48 gives the simultaneous intrinsic contraction for the full one-foliation family, including every component of the proper operator group and the dense lattice case. C.49 identifies the actual closed lifted generated group in the three-quotient case and contracts each whole orbit closure, with exact overlap agreement and finite pasting. C.50 supplies an exact uniform word bound and extends these contractions through arbitrary accumulation, giving a joint contraction on $X_* = \overline{U_2 \cup U_3}$. C.51 extends the intrinsic contractions through the entire frontier when $O/Q_O$ is compact and the full boundary-quotient family has at most two members, with the exact median boundary value and finite pasting. For an empty family every frontier point is then fixed. C.52 supplies exact agreement with the prescribed exponential values on $X_*$ by a natural hybrid target rule and diagonal frontier continuity. C.53 enlarges that region to $X^\dagger$ using all global nonopen quotient maps, and controls every accumulation of all $H$-translates of finitely many remaining cocompact representatives. C.54 proves actual exponential closure contractions whenever there are two metabelian global quotient images, including all diagonal affine weights and the product affine group. C.55 replaces the proper operator image by the full local centralizer, with direct equivariant-map limits and the larger prescribed region $X^\ddagger$. C.56 gives a uniform word bound for all diagonal one-parameter affine groups and extends their fractions through arbitrary accumulation on $X^\sharp$, retaining the full-local relative extension. C.57 determines the exact full local centralizers and quotient spaces for every full two-partition family. C.58 proves a uniform word bound for the full product affine group by an exact antisymmetric commutator correction, and extends every C.54 source through arbitrary accumulation on the larger closed $X^\flat$, with the same relative finite-type extension. C.59 extends localized quotient-preserving maps over all operator translates, with a continuous restriction inverse, and proves that a based two-partition quotient-kernel path is trivial. C.60 contracts every based sphere family in the relative one-partition quotient kernel through arbitrary accumulation, using exact forward and inverse prefix-track bounds; it requires no compact quotient or separate ambient-null assumption. C.61 excludes open-orbit accumulation for equal-level real-fiber intervals. C.62 places every such interval in an exact compact three-word family, excludes ambient escape, and confines all limiting points to the actual orbit of the source limit; at a fixed limit point the intervals shrink uniformly. C.63 constructs a relative continuous lift of the moving principal base by a compact family of small original-path increments and interlaced convex selection. It contracts the localized loop on every invariant union of open orbits with one-dimensional full local centralizer and G-fixed frontier, including arbitrary accumulation and the empty global nonopen-target family. C.64 constructs one compact labelled family controlling every selected open orbit and embeds the full product of locally finite disjoint operator disk groups across G-fixed frontiers, with an explicit Alexander contraction. C.65 proves that finite-word paths fixing the entire principal base act trivially on every nonmetabelian nonopen frontier orbit. The full principal kernel has a central discrete cyclic image there; its actual graph-image quotient is a covering, followed by a continuous bijection onto the principal-base image whose inverse continuity is unproved. C.65 also proves that individually shrinking null evaluation loops have fillings inside the original orbit and small ambient disk. C.66 proves the global filling for every original loop whose nonopen plane locus is zero-dimensional, with all operator dimensions included. Its exact end-extension isomorphism retains the original frontier, infinity, and every operator and deck label. C.67 identifies the full group fixing the complement of a bounded simply connected locally connected-frontier domain with the boundary-fixed disk group through the actual prime-end quotient. It proves contractibility of that full relative group and exactly identifies all original operator centralizers. C.68 extends the exact comparison to

spherical and unbounded plane domains and proves relative centralizer fillings for the specified closed-disk first-kind Fuchsian case. C.69 proves the same relative filling for every specified closed-disk conjugate of a torsion-free Fuchsian action of either kind, without finite-generation assumptions; its partial compactification, boundary correction and compact-parameter extension are explicit. C.70 gives exactly operator-equivariant individual evaluation fillings in compact-family carriers, uniformly shrinking at every fixed frontier point. C.71 proves that every based sphere family in the actual boundary-fixed centralizer of a two-dimensional proper operator group contracts, using a single compact three-word set to retain the original boundary topology. It also computes a cyclic rectangular compactification whose relative centralizer contracts despite failure of the proposed partial-boundary restriction. C.72 gives a spiral compactification with an actual transitive loop-generated group and its full cyclic local centralizer: an explicit interior centralizer map fails to extend at every boundary point, while the same map extends after the displayed interior Fuchsian conjugacy. The original loop in that example has its displayed contraction. Joint continuity in the evaluated point, simultaneous configuration disjointness and arbitrary infinite-domain pasting remain unproved. The remaining requirements include selecting prescribed values on nonfixed limit orbits and lifting nontrivial boundary deformations, untreated noncocompact quotients, arbitrarily many $H$-inequivalent remaining components, and extension through nonopen regions outside these closed sets.

C.25 proves such a kernel for the separately enlarged $\Pi_1 \operatorname{Homeo}(S^2)$ comparison. Its relationship with the admitted punctured comparisons can be computed explicitly. Write $p_\pm$ for the poles of its irrational rotation. In the identity component its centralizer $C$ fixes both poles, and it is the same subgroup in the unmarked, once-marked, and twice-marked sphere homeomorphism groups. Forgetting marks induces

$$\begin{aligned} \pi_1 C = \mathbb{Z} \xrightarrow{1} \pi_1 \operatorname{Homeo}(S^2, \{p_+, p_-\}) = \mathbb{Z} \xrightarrow{1} \pi_1 \operatorname{Homeo}(S^2, \{p_+\}) = \mathbb{Z}, \\ \pi_1 \operatorname{Homeo}(S^2, \{p_+\}) = \mathbb{Z} \xrightarrow{\text{mod } 2} \pi_1 \operatorname{Homeo}(S^2) = \mathbb{Z}/2. \end{aligned} \tag{C.12.2}$$

The same full axis-rotation loop generates the first three groups by the plane and annulus comparisons of C.10; its inclusion into $SO(3)$ gives the last map. Conjugation by the irrational rotation is based homotopic to the identity in each ambient group, so these are also the weak-transporter target groups in (C.12.1). Thus puncturing one or both poles removes C.25's kernel. Every nonpolar point has an infinite orbit, so this action admits no other finite invariant puncture set.

Keeping the peripheral-diagram recipe while adding the sphere gives yet another explicitly related target: its diagram is $\varnothing \to *$, whose equivalence 2-group is terminal. The comparison factors through the functor $\Pi_1 \operatorname{Homeo}(S^2) \to \operatorname{Eq}(\varnothing \to *)$, which sends the identity 2-automorphism group $\mathbb{Z}/2$ to zero. For the irrational action its fundamental-group map is the composite $\mathbb{Z} \to \mathbb{Z}/2 \to 0$; C.25 computes the first arrow. Neither enlargement changes the admitted object class fixed in C.1. Consequently C.25 is retained with its exact stated comparison, while the admissible infinite-group zero-level obligation remains recorded here.

**Exact lift comparison for the remaining zero-level question.** The compact-open topology agrees with the marked compactification topology used in C.1. More precisely, restriction $\operatorname{Homeo}(X, S) \to \operatorname{Homeo}(X \setminus S)$ is a topological-group isomorphism. A homeomorphism extends uniquely to the finite end compactification, giving the algebraic inverse. For its continuity at $f_0$, choose disjoint connected target end neighborhoods $W_j$ of small diameter and complementary compact core $C$. Choose a source core $K$ with $f_0^{-1}(C) \subset \operatorname{int} K$, and connected source tails $V_i$ whose $f_0$-images lie in the prescribed $W_j$. The conditions $f^{-1}(C) \subset \operatorname{int} K$ and $f(z_i) \in W_j$ for one point $z_i \in V_i$ are open: inversion is continuous in the compact-open homeomorphism group of a locally compact, locally connected surface. Connectedness forces $f(V_i) \subset W_j$, since this image avoids $C$ and meets $W_j$. This controls the extended maps uniformly on all tails. Compact-open

control on the remaining compact core controls them on the rest of $X$, proving continuity of the inverse.

Let $U$ be the hyperbolic surface of C.12, let $A^0 = \mathrm{Homeo}(U)^0$, let $H$ be the image of the specified operator group, and let $p : \widetilde{U} \to U$ be the universal covering with deck group $\pi$. Write $\widetilde{H}$ for the group of all lifts of elements of $H$, regarded as actual homeomorphisms of $\widetilde{U}$. There is a continuous injective homomorphism

$$j : A^0 \longrightarrow \mathrm{Homeo}(\widetilde{U}), \qquad p\, j(f) = f\, p,$$

characterized by $j(f)\delta = \delta j(f)$ for every $\delta \in \pi$. It identifies the precise centralizers:

$$j\big(C_{A^0}(H)\big) = j(A^0) \cap C_{\mathrm{Homeo}(\widetilde{U})}(\widetilde{H}).$$

In particular every based loop in the left-hand centralizer lifts to a closed loop of homeomorphisms commuting with the entire lift group.

*Proof.* Lift an isotopy from id to $f$ starting at $\mathrm{id}_{\widetilde{U}}$. Conjugation on the discrete deck group is constant during the lift, so its endpoint commutes with $\pi$. Two such endpoints differ by a deck transformation centralizing $\pi$; they are equal because $Z(\pi) = 1$ by Part A. Uniqueness also makes $j$ a homomorphism and an injection. For continuity, fix $\widetilde{u} \in \widetilde{U}$. The evaluation map $A^0 \to U$ lifts continuously to $\widetilde{U}$, since $A^0$ is contractible and locally path connected by the surface homeomorphism-group theorem. This lift gives $j(f)(\widetilde{u})$, and the local lifting property of a covering gives continuous dependence of the lifted homeomorphism in the compact-open topology.

Suppose $f$ centralizes $H$ and $\widetilde{h}$ is any lift of $h \in H$. Then $[j(f), \widetilde{h}]$ is a deck transformation. Its conjugation on $\pi$ is the commutator of the identity automorphism with the automorphism induced by $\widetilde{h}$, hence is trivial. Centre-freeness makes this deck transformation the identity. This proves one inclusion in the displayed equality; descent proves the other. Finally $j(\mathrm{id}) = \mathrm{id}$, so a based loop has the same endpoint after lifting. □

This comparison does not supply a contraction inside the right-hand centralizer. Choosing a hyperbolic metric and interpolating from $j(f)(x)$ to $x$ along geodesics has two unproved properties: the arbitrary lift group $\widetilde{H}$ need not preserve these geodesics, and an interpolated map need not be a homeomorphism. The first prevents asserting equivariance; the second prevents asserting that the interpolation lies in $j(A^0)$. Neither property follows from the displayed centralizer equality. The universal vanishing of $\pi_1(C_{A^0}(H))$, and the requested counterexample to it, therefore remain unresolved by this argument.

**Current orbitwise reduction and the remaining compatibility problem.** C.26 settles the globally transitive centralizer case on a closed hyperbolic surface and also gives an explicit closed subgroup with an ambient-null but intrinsically nontrivial loop; its latter example is not asserted to be a centralizer. For the canonical plane lift above, C.28 identifies the orbits of the group generated by the loop: points, intrinsic lines, circles, and open surface orbits. C.27 gives an order-natural contraction on each intrinsic line. C.29 proves that every circle restriction has winding zero by the actual plane-loop evaluation map, and contracts that restriction in its circle centralizer. C.31 supplies inverse and track control, and C.34 proves joint continuity of the contractions and their inverses on the entire closed union $Y$ of the nonopen orbits. This is a based contraction through homeomorphisms of $Y$, commuting with every homeomorphism that commutes with the original loop. In particular C.34 settles the hyperbolic centralizer loop whenever its lifted generated group has no open orbit.

The open orbits have also been treated intrinsically. C.30 gives the exact restricted operator group, including its restriction kernel, and proves that its closed image acts freely and properly.

C.32 handles a discrete image, and C.33 treats every locally compact closed image supplied by C.30. In each case the restricted loop has a based null-homotopy inside the centralizer of the full restricted operator group. These results give contractions on all individual open orbits and on the whole nonopen locus. The remaining step is to choose these contractions compatibly across the frontiers of the open components, with joint continuity of the resulting homeomorphisms and their inverses.

C.35 proves that this step cannot require extending the existing median prescription unchanged. Its explicit plane action has the boundary restriction map

$$\mathbb{R}^2 \longrightarrow \mathbb{R}^3, \qquad (A, B) \longmapsto (A, B, A - B).$$

At the displayed parameter the prescribed three edge medians are $(1, 0, 0)$, which lies outside its image because $r_1 - r_2 - r_3 = 1$. Thus that prescribed homeomorphism of $Y$ has no equivariant extension over an adjacent open orbit. The original loop in the same example has the explicit global equivariant contraction given in (C.35.5). C.36 constructs compatible choices on closures of open orbits whose closed proper operator image has dimension two, using the actual open quotient onto their simply transitive Lie centralizers. Scaling the logarithm of each based universal-cover lift respects the exact common boundary action. C.37 supplies a uniform three-word estimate and proves joint continuity, including inverses, on

$$X_2 = \overline{\bigcup_{\dim Q_O = 2} O},$$

including every infinite accumulation of those open orbits. Its bounded residual equations determine the unique logarithmic limit on line orbits. The actual null residual word isotopies and the two-point configuration map determine the real lifted limit on circle orbits, including its winding.

C.49–C.50 subsequently enlarge this closed set to $X_*$, including all open orbits with at least three actual boundary quotient foliations. C.51 proves whole-closure extension for the remaining open orbits with compact operator quotient, using the median frontier prescription and finite pasting. C.52 makes these extensions exactly relative to $X_*$. C.53 enlarges the exponential region to $X^\dagger$ and proves continuity across all accumulations of the operator translates of finitely many remaining cocompact representatives. C.54–C.56 enlarge the treated region through all three-dimensional diagonal affine source groups, with uniform accumulation control, and replace the compact operator quotient condition by compactness of the full local centralizer quotient. C.57 gives the exact plane, open-cylinder, line and open-Möbius quotient alternatives in the remaining noncocompact two-family cases. The outstanding problem includes untreated noncocompact local quotients, infinitely many $H$-inequivalent remaining representatives, and extension through exterior regions consisting of nonopen orbits. The hybrid target prescription has been proved continuous on the treated closed sets; its transverse continuity on all of the nonopen locus has not been proved. C.35 requires allowing changes to the original median prescription. These exact remaining requirements, and the general kernel in (C.12.1), remain unresolved here.

## 3.4 Configuration invariants and drilling

**C.13. [Proved] The invariant braid group and nonabelian cohomology.** Let a finite group act faithfully and orientation preservingly on a hyperbolic $U$, and let $R = U^!$ be its finite set of points with nontrivial stabilizer. Put $V = U \setminus R$, $B = \{f \in A^0 : f|_R = \mathrm{id}\}$, and $B^0 = \mathrm{Homeo}(V)^0$. With the action by conjugation, we have

$$(B/B^0)^\Gamma = 1, \qquad H^1(\Gamma, B^0) = 1, \qquad (A^0)^\Gamma = B^\Gamma \cong \mathrm{Homeo}(V/\Gamma)^0.$$

Here the quotient group identifies canonically with $\pi_1(\mathrm{Conf}_R(U), R)$ by evaluation, and its invariant subgroup is the $H^0$ asked for on scan 285.

*Manuscript:* §24; folios 142–144; scans 282–286.

*Proof.* Evaluation $A^0 \to \mathrm{Conf}_R(U)$ is a locally trivial principal $B$-bundle, by the local extension theorem for embeddings of a finite set. Its total space is contractible. Its homotopy exact sequence gives $\pi_0 B \cong \pi_1 \mathrm{Conf}_R(U)$, with the specified base configuration, and its identity component is the identity component after retaining the new punctures. The construction is equivariant for $\Gamma$, so this is the stated action on the braid group.

An invariant component $b \in B/B^0$ is a mapping class of $V$ in the kernel of filling $R$ that centralizes the action of $\Gamma$ on $V$. By C.6 and its transporter version, choose a representative $f$ on $V$ commuting with $\Gamma$. It extends across the punctures and induces the identity mapping class on $U$. Thus $f \in (A^0)^\Gamma$, and C.7 joins it to the identity by an equivariant isotopy on $U$. Such an isotopy fixes each point of $R$: it preserves each finite critical-point set and starts with the identity permutation. Restriction to $V$ proves $b = 1$.

For a nonabelian 1-cocycle $a_\gamma \in B^0$, the maps $v(\gamma) = a_\gamma u(\gamma)$ form another finite action on $V$. The cocycle equation is exactly the homomorphism equation for $v$. Both actions induce the same homomorphism into the mapping class group of $V$. By C.6 they are conjugate by an element $b \in B^0$; equivalently $a_\gamma = bu(\gamma)b^{-1}u(\gamma)^{-1}$. Hence the cocycle is a coboundary and $H^1(\Gamma, B^0) = 1$.

Finally C.7 gives connectedness. Descent to the quotient with its branch points retained is precisely descent to $V/\Gamma$ with its punctures retained; the identity lift construction in C.7 is the displayed isomorphism. In particular the unprimed $V$ in the transcription of the identity-component line on scan 286 must be compared with the quotient $V'$ of scans 282–285. The two homeomorphism groups need not be isomorphic; the descent map identifies the quotient one. □

**C.14. [Refuted] Finite critical configurations do not include mirrors.** The finite-set configuration formulation of C.13 does not extend by replacing an orientation-preserving action by an arbitrary action and keeping $R = U^!$ equal to all points with nontrivial stabilizer. A reflection has a one-dimensional fixed locus, and deleting finitely many points cannot make that action free.

*Manuscript:* §24; folios 142–148; scans 282–294.

*Proof.* Take complex conjugation on $U = \mathbf{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$. It acts faithfully on this hyperbolic surface. Its fixed set is $\mathbf{P}^1(\mathbb{R}) \setminus \{0, 1, \infty\}$, a union of three open arcs. Every point of these arcs has nontrivial stabilizer. A finite deletion leaves points in each arc. At any remaining fixed point, the lift of conjugation fixing a selected universal-cover point is an element of order two in the lift extension. Therefore the drilled extension still has torsion. This explicit counterexample refutes both the finiteness premise and a torsion-free finite-drilling assertion for such an action. The corrected geometry retains mirror boundary in the quotient or cuts along the mirror locus; it is not the operation of adding finitely many puncture loop classes. The equations C.13 were stated on scans 282–286 after the explicit restriction $\Gamma = \Gamma^+$. □

**C.15. [Proved] The oriented drilling equivalence and the bijection.** Fix a finite group $\Gamma$ and faithful orientation-preserving outer actions on discrete hyperbolic groups with loops. Let $\Phi(\pi, \Gamma)$ be the set of $\pi$-conjugacy classes of nontrivial maximal finite subgroups of the canonical extension. Filling a specified $\Gamma$-stable set $J$ of punctures with nontrivial stabilizers induces an equivalence between the category of such drilled actions and the category of filled actions with a

specified subset $J \subset \Phi(\pi, \Gamma)$. In particular

$$\Phi(\pi', \Gamma) \amalg J \xrightarrow{\sim} \Phi(\pi, \Gamma)$$

is a canonical $\Gamma$-equivariant bijection. Drilling all of $\Phi$ produces a torsion-free extension, uniquely up to the unique exterior isomorphism over the specified filled object.

*Manuscript:* §24; folios 144–147; scans 286–293.

*Proof.* Realize the action by C.5. A finite nontrivial subgroup of the lift group acts on $\mathbb{H}^2$ by orientation-preserving isometries and has a unique fixed point. Its stabilizer at that point is the unique maximal finite subgroup containing it. Conversely every point of $U$ with nontrivial stabilizer gives that subgroup after choosing a lift. Changing the lift conjugates by $\pi$. This identifies $\Phi$ with the finite critical-point set of the realized action, including its $\Gamma$ action. It also proves the assertion about maximality, rather than assuming it for arbitrary finite subgroups containing reflections.

Deleting the points of $J$ gives the drilled surface. Remaining critical points and deleted points partition the original critical-point set, which proves the displayed bijection. The group map is the epimorphism $\pi' \to \pi$ obtained by killing the normal closure of exactly the new peripheral subgroups. Local punctured discs show that its induced map on lift groups sends a cusp stabilizer to the corresponding finite point stabilizer. Thus the geometric bijection is the algebraic map of scan 290, with the same conjugacy ambiguities.

To prove fullness, realize an equivariant outer isomorphism of the filled peripheral groups by an equivariant homeomorphism, using C.6's transporter conclusion. It preserves the selected subset of $\Phi$ by the preceding construction. Its restriction to the complements gives the desired drilled outer isomorphism. To prove faithfulness, represent a drilled equivariant outer automorphism in the kernel by an equivariant homeomorphism of the drilled surface and extend it across the new punctures. It is isotopic to the identity on the filled surface. By C.7 there is an equivariant identity isotopy there. Every point of $J$ stays fixed throughout, since it is a member of the finite critical set, so the isotopy restricts to the drilled surface. The kernel is therefore trivial. These two arguments also prove uniqueness over a fixed filled isomorphism, not merely uniqueness of isomorphism classes.

After all critical points are removed, the finite action is free. Its lift extension acts freely on $\mathbb{H}^2$ and is torsion-free, since a nontrivial finite-order isometry fixes a point. The operations of deleting and filling are inverse by the just-proved full faithfulness. The only use of geometric realization in this proof is C.5 together with the equivariant homeomorphism and isotopy results C.6–C.7; the comparison map on groups and the kernel argument have been specified. □

**C.16. [Refuted] Nonfaithful outer actions obstruct torsion-free drilling.** The unrestricted existence claim on scans 286–288 fails for the trivial outer action of $C_2$ on the group of type $(0, 3)$. The faithful, orientation-preserving version is C.15.

*Manuscript:* §24; folios 144–145; scans 286–288.

*Proof.* Suppose a finite-puncture drilling of that action had torsion-free extension. Realize its finite outer action on the drilled hyperbolic surface by C.5. Filling the specified punctures extends these homeomorphisms to a finite action on the filled three-punctured sphere. The induced mapping class of every element is trivial. The finite-order rigidity argument of C.5 makes every filled homeomorphism the identity. Its restriction to the dense drilled surface is therefore the identity. The upstairs outer action is trivial as well. For a centre-free kernel its canonical extension is $\pi' \times C_2$, which contains $\{1\} \times C_2$. This contradicts torsion-freeness. The argument proves the same obstruction for every nonfaithful outer action after restricting to a nontrivial element of its finite kernel. □

**C.17. [Proved] Torsion-free finite extensions with peripheral data.** Let $\pi$ be a discrete hyperbolic surface group, possibly punctured, and let $1 \to \pi \to E \to \Gamma \to 1$ be a torsion-free extension by a finite group whose outer action preserves the loop structure. Then $E$ is the fundamental group of a finite-type surface, possibly nonorientable. Its peripheral subgroups are the centralizers in $E$ of the peripheral subgroups of $\pi$. With their conjugacy classes it is a surface group with loops in this extended sense. The associated compact surface and boundary groups form a Poincaré duality pair of dimension two with its orientation local system.

*Manuscript:* §24; folios 147–148; scans 292–294; §28; folio 178; scan 353.

*Proof.* The centre-free extension is the pullback of $\mathrm{Aut}(\pi) \to \mathrm{Out}(\pi)$. The centralizer of $\pi$ in $E$ maps isomorphically to the kernel of the outer action: a representative inducing an inner automorphism can be multiplied by the inverse inner element to centralize $\pi$, and this representative is unique because $Z(\pi) = 1$. This centralizer is finite, so torsion-freeness forces it to be trivial. Hence the outer action is faithful.

Apply C.5 and identify $E$ with the group of lifts of the realized action, by uniqueness of the centre-free extension. A nontrivial stabilizer of a point gives a finite subgroup of $E$, so the action on $U$ is free. Its quotient $W = U/\Gamma$ is an ordinary surface and $\mathbb{H}^2 \to W$ is its universal covering. This gives the required isomorphism $E \cong \pi_1(W)$, respecting the specified subgroup $\pi$ and quotient $\Gamma$.

A stabilizer of an end of $U$ cannot contain an orientation-reversing operator. The local finite-group model at its compactifying point would then contain a reflection and fixed points arbitrarily close to that point, contrary to freeness on $U$. Its local action is therefore cyclic rotation. Consequently an upstairs peripheral subgroup is a finite-index subgroup of the downstairs peripheral infinite cyclic group. In the hyperbolic universal cover its centralizer is exactly that full cusp group. These are all the cusps, since every end downstairs has a preimage upstairs. Thus the stated peripheral classes, including their full subgroups rather than merely inclusions, are obtained.

Choose disjoint cusp neighbourhoods in $W$ and truncate them to obtain a compact surface $M$ with boundary. The inclusions of boundary circles induce precisely these cyclic groups. Lift a triangulation to the contractible universal cover. The cellular cap-product map with the relative orientation class gives the chain equivalence between cochains and complementary relative chains; the local sign is the orientation character of $W$. This is the Poincaré–Lefschetz duality pair, and its group-pair chain model is the relative cellular complex because both the universal cover and the lifted boundary components are contractible. This proves the duality-pair assertion rather than inferring it from virtual freeness alone. The exact geometric input is the finite realization theorem; a separate proof from abstract duality axioms is not substituted for this argument. □

**C.18. [Proved] The discrete oriented realization criterion.** For a faithful orientation-preserving finite outer action on a discrete hyperbolic group with loops, the criterion of scan 294 is valid with the word admissible meaning the full-faithfulness and uniqueness conditions of scan 293. The drilling in question is at the canonical set $\Phi(\pi, \Gamma)$, and its extension carries the quotient loop structure described in C.17.

*Manuscript:* §24; folios 147–148; scans 293–294.

*Proof.* A realized action can be drilled at its critical points. C.15 proves that the resulting action is admissible and its extension torsion-free; C.17 identifies that extension with a group with loops. This verifies every requirement of the stated criterion.

Conversely let $\pi' \to \pi$ be the stipulated filling map and let $E'$ be its extension, with the group-with-loops structure required in the criterion. Realize $E'$ by a compact surface with its

boundary circles, then take its covering corresponding to $\pi'$. The comparison of peripheral subgroups in the definition ensures that this cover has exactly the prescribed upstairs loop classes. Its deck group is $\Gamma$, acting freely before filling. The cyclic end stabilizers extend across each new puncture by the local map $z \mapsto z^m$. Filling the selected ends thus gives a $\Gamma$-action on the surface whose peripheral group is $\pi$. The canonical map of lift extensions identifies its outer action with the original one. Admissibility makes this construction independent, up to the uniquely specified exterior isomorphism, of the chosen drilled object. This proves the converse by the typed covering and filling maps. The mirror counterexample C.14 explains the necessary restriction of this formulation. □

### 3.5 Special homeomorphisms: the topology and the extension class

**C.19. [Refuted] Escaping support invalidates the compact-open version.** Let $SA \subset A$ consist of homeomorphisms equal to the identity on some neighbourhood of every puncture, and give it the subspace compact-open topology. The canonical special-group comparison of scan 304 does not identify $\pi_0(SA)$ with the boundary-framed mapping class group. A peripheral boundary twist is connected to the identity inside this $SA$. In particular, the annulus conclusion $\pi_0(SA) = \mathbb{Z}$ on scan 304 is false in this topology.

*Manuscript:* §25; folios 151–153; scans 301–304.

*Proof.* Use a coordinate $z = re^{i\theta}$ on a disc about a puncture at 0. Choose a continuous function $\eta : [0, \infty) \to [0, 1]$ equal to 0 on $[0, 1/3]$ and to 1 on $[2/3, \infty)$. For $0 < s \leq 1$ define

$$h_s(re^{i\theta}) = r \exp(i\theta + 2\pi i \eta(r/s)), \qquad h_0 = \mathrm{id},$$

and extend by the identity outside the coordinate disc. Each $h_s$ is identity near the puncture and outside the annulus $s/3 < r < 2s/3$. It is a boundary twist when viewed on a fixed compact surface obtained by truncating the puncture. Nevertheless $\sup_z |h_s(z) - z| \leq 4s/3$, and the same bound holds for its inverse. Thus $s \mapsto h_s$ is a continuous path in the stated subspace topology from the identity to that twist.

For the annulus every compactly supported orientation-preserving homeomorphism fixes both ends and is represented on a compact annulus by an integer twist; cutting along a spanning arc and applying the Alexander isotopy of a disc proves this integer classification. The isotopy to the corresponding integer twist can be chosen in that fixed compact annulus. The path above then kills that twist in $SA$. Hence $SA$ is path connected in the compact-open topology, proving the explicit contradiction. The programme checks the formula and its uniform bound; the bound itself proves convergence for the full space, not just on a finite sample.

The fixed-boundary formulation below retains a single compact surface throughout the isotopy. That is the required correction. No change of topology is silently made in this refutation. □

**C.20. [Proved] The fixed-boundary repair and its class.** Let $M$ be an oriented compact surface obtained by replacing the $\nu > 0$ punctures of a hyperbolic type by boundary circles. Put $A_\partial = \mathrm{Homeo}^+(M \text{ rel } \partial M)$. Then $A_\partial^0$ is contractible, $\pi_0(A_\partial) \cong S\Gamma_{g,\nu}^{!+}$ for the framed group of E.1–E.2, and there is a central extension

$$1 \longrightarrow \mathbb{Z}^\nu \longrightarrow \pi_0(A_\partial) \longrightarrow \Gamma_{g,\nu}^{!+} \longrightarrow 1.$$

The $i$th kernel generator is the positive boundary twist. Its extension class is the vector of Euler classes of the oriented tangent circle lines at the marked points, with the sign fixed by that generator. For the annulus the kernel is the quotient of $\mathbb{Z}^2$ by the diagonal rotation relation; for the once-punctured sphere it is zero.

*Manuscript:* §25; folios 152–154; scans 302–306.

*Proof.* Allow boundary rotations first, retaining each boundary component setwise. Restriction to the boundary is a locally trivial fibration onto $\prod_i \mathrm{Homeo}^+(S^1)$: extend a small boundary isotopy across collars. Its fibre is $A_\partial$. Each boundary homeomorphism group deformation retracts onto rotations by lifting to increasing maps of $\mathbb{R}$ and linearly interpolating the periodic displacement. Thus the base has homotopy type $(S^1)^\nu$. Capping each boundary by a disc with its centre marked identifies the free-boundary mapping classes with the pure oriented punctured mapping classes. The same collar-extension construction gives the homotopy comparison on identity components.

The identity component of the free-boundary group is contractible in a hyperbolic type by Yagasaki's compact-pair theorem. The homotopy sequence therefore injects $\pi_1((S^1)^\nu) = \mathbb{Z}^\nu$ into the components of the fibre and identifies the quotient as stated. Lifting one rotation in the $i$th collar gives exactly the boundary twist. Its conjugation action is trivial because the boundary circles are retained individually and their orientations are fixed. Choose a point on each boundary circle. A boundary-fixing homeomorphism acts on the framed peripheral diagram of E.2, and an isotopy relative to the boundary gives its prescribed natural isomorphism. This defines $\pi_0(A_\partial) \to S\Gamma^{!+}_{g,\nu}$. On the quotient it is the peripheral Dehn–Nielsen–Baer isomorphism. A positive boundary twist changes the path from the generic basepoint to that boundary point by one peripheral turn; hence the induced map on kernels sends the $i$th boundary generator to the $i$th generator of $T^I$. It is an isomorphism on both kernel and quotient, so the two short exact sequences give the asserted isomorphism of extensions. Contractibility of $A^0_\partial$ follows directly from Lemma 2.1(ii), with the nonempty fixed boundary. The connecting homomorphism is consequently the fundamental-group extension of the bundle of tangent-circle framings at the marked points. Obstruction to a section of that circle bundle is its Euler class, proving the class identification with its sign convention. In complex notation this is the tangent-line first Chern class; the usual cotangent $\psi_i$ has its opposite sign.

For the annulus the identity component of the free-boundary group is $S^1$. Its fundamental-group map to the two boundary-rotation groups is the diagonal when both circles use the same angular coordinate, or the antidiagonal when both use boundary orientations. The cokernel is $\mathbb{Z}$. For the disc this map is an isomorphism $\mathbb{Z} \to \mathbb{Z}$, so the cokernel is zero. These are the exact boundary models underlying the two abelian computations on scans 303–304. □

**C.21. [Proved] The four small extension classes.** With the positive boundary-twist convention of C.20, the special fixed-boundary extension classes for $(0,3)$ and $(0,4)$ are zero. For $(1,1)$ the class is a generator $e$ of $H^2(\mathrm{SL}_2(\mathbb{Z});\mathbb{Z}) = \mathbb{Z}/12$. For $(1,2)$ the vector class is $(q^*e, q^*e)$, where $q : \Gamma^{!+}_{1,2} \to \mathrm{SL}_2(\mathbb{Z})$ forgets the second marked point; its order is 12.

*Manuscript:* §25; folios 153–154; scans 304–306.

*Proof.* The pure oriented mapping class group of a three-punctured sphere is trivial. Filling one puncture of a four-punctured sphere gives the point-pushing group $F_2$, since the target pure group is trivial and the three-punctured surface is hyperbolic. A graph is a classifying space for $F_2$, so $H^2(F_2;\mathbb{Z}^4) = 0$. This proves the first two class computations, including the noncanonical splitting in type $(0,4)$.

The integral matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

have orders 4 and 6 and satisfy $S^2 = R^3 = -I$. Their action on the Farey tessellation gives $\mathrm{PSL}_2(\mathbb{Z}) = C_2 * C_3$: the dual bipartite tree has one edge as fundamental domain, vertex stabilizers $C_2, C_3$, and trivial edge stabilizer. Connectivity and the absence of circuits follow from the Euclidean algorithm for adjacent reduced fractions. Lifting the central element $-I$

gives $\mathrm{SL}_2(\mathbb{Z}) = C_4 *_{C_2} C_6$. The cohomology sequence of this tree of groups identifies $H^2$ with

$$\{(r,s) \in \mathbb{Z}/4 \oplus \mathbb{Z}/6 : r \equiv s \pmod 2\} :$$

the preceding $H^1(C_2;\mathbb{Z})$ and $H^1(C_4;\mathbb{Z}) \oplus H^1(C_6;\mathbb{Z})$ vanish, while the restrictions in degree two reduce modulo 2. This group has twelve elements and is cyclic. The tangent-circle bundle restricts on the square torus fixed by $S$ to the rotation representation by $\pi/2$, and on the hexagonal torus fixed by $R$ to the representation by $\pi/3$. Lifting these circle rotations to $\mathbb{R}$ gives fourth and sixth powers equal to one positive full turn. By the convention of C.20 the Euler class therefore restricts to $(1,1)$, an element of order 12. This proves its generator assertion without an unproved presentation for the boundary mapping class group. The supplied code verifies the matrix orders and the finite abelian-group calculation.

In a complex torus translation canonically identifies the tangent spaces at any two points. These identifications vary with the marked complex torus and commute with all isomorphisms. Hence the two tangent circle bundles on the two-pointed moduli orbifold are isomorphic, and both are pulled back from the one-pointed tangent bundle. This proves the vector formula. It has order dividing 12. A square torus with marked points 0 and $(1+i)/2$ admits the order-four rotation preserving both; a hexagonal torus admits an order-three rotation with at least two marked fixed points. The restrictions of the class to these cyclic subgroups are generators, as can be seen from their nontrivial local rotations, or from C.22 below. Its order is divisible by 4 and 3, and thus is exactly 12. The existence of the displayed fixed points is also verified in the integral torus matrices supplied with the code. □

**C.22. [Proved] Restriction over a finite puncture stabilizer.** For a hyperbolic type, a nontrivial finite subgroup $G \subset \Gamma_i^+$ is cyclic. The extension by the $i$th copy of $T = \mathbb{Z}$ obtained from the special group is nonsplit over $G$; its preimage is infinite cyclic and the kernel is the subgroup of index $|G|$.

*Manuscript:* §25; folios 153–154; scans 305–306.

*Proof.* Realize $G$ by C.5. It fixes the marked point $i$ and preserves orientation. The finite local action is a faithful rotation action: an element trivial on a small disc is the identity as a periodic surface homeomorphism, by the local classification and connectedness. Thus $G$ embeds in the circle rotations and is cyclic. Delete that point and lift a small punctured quotient disc to its universal cover. Its group is $\mathbb{Z}$, while the upstairs punctured disc has subgroup $|G|\mathbb{Z}$. The exact normalizer interpretation proved in E.3 identifies this with the extension pushed out by the $i$th projection. An infinite cyclic group contains no nontrivial finite subgroup, so the extension cannot split. This proves the remark on scan 306 with its stated orientation and puncture-stabilizer hypotheses. □

**C.23. [Proved] An infinite-group failure of existence.** For every closed orientable surface $X_g$ of genus $g \geq 2$, the identity homomorphism from the discrete group $\Gamma = \mathrm{Mod}^+(X_g)$ to its subgroup in $\pi_0 A_g$ has no homomorphic lift to $A_g$. Thus level 3 fails for an infinite operator group already on each such closed hyperbolic surface.

*Manuscript:* §23; folios 140–141; scans 279–280.

*Proof.* Marković–Šarić's Theorem 1.1 [48] states that the projection $\mathrm{Homeo}^+(X_g) \to \mathrm{Mod}^+(X_g)$ has no group section for $g \geq 2$. The theorem is about homeomorphisms, not merely diffeomorphisms; its first-page statement and definitions have been checked in the original paper. A lift of the displayed homomorphism would take every element to an orientation-preserving homeomorphism, since orientation is invariant under isotopy. Its composite with that projection

would be the identity, giving the forbidden section. This is the exact reduction to the source theorem. The group is infinite: a twist about a nonseparating simple curve acts on $H_1(X_g;\mathbb{Z})$ by an integral transvection, whose distinct powers are distinct. □

### 3.6 The integral sphere comparison

**C.24. [Refuted] The sphere comparison already fails for an order-three rotation.** Let $A = \mathrm{Homeo}(S^2)$ with its compact-open topology, and let $\Gamma = C_3$ act on $S^2$ by rotation through $2\pi/3$ about a fixed axis. Give $A$ the resulting conjugation action and base both fixed-point spaces at the identity. The actual comparison has

$$\pi_1(A^{\Gamma}, 1) = \mathbb{Z}, \qquad \pi_1(A^{h\Gamma}, 1) = \mathbb{Z}_3 \times \mathbb{Z}/2, \qquad n \longmapsto (n, n \bmod 2).$$

Consequently the requested integral homotopy equivalence fails in type $(0,0)$. This failure concerns the actual homeomorphism mapping space; it is independent of the earlier loss of the sphere's $\pi_2$ in its fundamental groupoid. For every odd prime $p$, the same construction with $C_p$ gives $\mathbb{Z}_p \times \mathbb{Z}/2$ and the same comparison formula with $p$ in place of 3.

*Manuscript:* §23; folio 141; scans 280–281.

*Proof.* We give the comparison maps and compute their effect, including the integral rather than only completed homotopy group. Write $G = SO(3)$ and $\rho : C_p \hookrightarrow G$ for the rotation subgroup, with $p$ odd. The inclusion $G \hookrightarrow A^+ = \mathrm{Homeo}^+(S^2)$ is $C_p$-equivariant and a weak equivalence. Indeed evaluation on three ordered distinct points is a locally trivial fibration onto $\mathrm{Conf}_3(S^2)$, by isotopy extension. Its fibre is the pure homeomorphism group of the three-punctured sphere. Its component group is trivial, and its identity component is contractible by the surface homeomorphism-group theorem used in C.7. The unique Möbius transformation carrying one ordered triple to another identifies this configuration space with $PSL_2(\mathbb{C})$; polar decomposition retracts this group onto $SO(3)$. The resulting evaluation diagram proves the stated weak equivalence for the actual inclusion. Homotopy fixed points preserve this objectwise weak equivalence, after applying singular complexes. A path through the identity in $A^{hC_p}$ stays in $A^+$, because $EC_p$ is connected and orientation takes values in the discrete group $C_2$. It therefore suffices to calculate $(G^{hC_p}, 1)$.

The identity component of the actual centralizer fixes the two poles individually. Quotient by $z \mapsto z^p$ identifies its descent map to $\mathrm{Homeo}^+(S^2;0,\infty)^0$ as a covering of degree $p$. Every map of the quotient fixing both poles lifts: on the complementary annulus its action on the meridian is $+1$, so it preserves the subgroup $p\mathbb{Z}$ of the covering group. Each lift commutes with the deck group, since the induced automorphism of $\mathbb{Z}/p$ is $+1$; it extends uniquely over the two poles. The usual local lifting neighbourhoods establish that descent is a covering in the compact-open topology. Its total space is connected: the quotient group is connected, and all deck lifts lie on the connected circle of axis rotations. The annulus calculation of C.10 identifies the base with a space of homotopy type $S^1$. The rotation circle maps to its quotient rotation circle by $z \mapsto z^p$; its inclusion in the total centralizer therefore induces an isomorphism on every homotopy group. In particular the actual fundamental group is $\mathbb{Z}$, with generator one complete axis rotation.

Set $Y = BG = BSO(3)$. The adjoint bundle $EC_p \times_{C_p} G \to BC_p$, with action through $\rho$, has section space $G^{hC_p}$. It is also the bundle of gauge automorphisms of the principal $G$-bundle classified by $B\rho$. The universal bundle gives the natural weak equivalence

$$G^{hC_p} \simeq \Omega_{B\rho}\,\mathrm{Map}(BC_p, Y).$$

One way to verify the map is to send a gauge automorphism to the loop of its classifying map obtained by gluing the bundle over the two ends of $BC_p \times [0,1]$. Conversely a loop of classifying

maps gives a bundle over this cylinder with its specified endpoint identifications; trivialization in the interval direction gives the gauge automorphism. The spaces of such trivializations are contractible. This construction is natural for the inclusion $C_G(\rho(C_p)) \hookrightarrow G^{hC_p}$.

We now compute $\pi_2$ of this mapping space using the arithmetic square of the simply connected finite-type space $Y$. Let

$$\widehat{Y} = \prod_q Y_q^\wedge, \qquad V = \Big(\prod_q \mathbb{Z}_q\Big) \otimes \mathbb{Q}.$$

Here $q$ runs over primes and superscript $\wedge$ means homotopical $q$-completion, throughout this proof. We have $Y_\mathbb{Q} \simeq K(\mathbb{Q}, 4)$ and $(\widehat{Y})_\mathbb{Q} \simeq K(V, 4)$. This follows from $SO(3) \simeq \mathbb{RP}^3$: its rational homotopy is concentrated in degree three, and each of its higher homotopy groups is finite. The square

$$\begin{array}{ccc} Y & \longrightarrow & \widehat{Y} \\ \downarrow & & \downarrow \\ K(\mathbb{Q}, 4) & \longrightarrow & K(V, 4) \end{array}$$

is a homotopy pullback. For completeness its homotopy-group verification reduces on each free summand to

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \oplus \prod_q \mathbb{Z}_q \longrightarrow V \longrightarrow 0, \quad (a, (b_q)) \longmapsto (b_q - a)_q.$$

Surjectivity follows by clearing a common denominator and choosing its finitely many required residues by the Chinese remainder theorem; the kernel consists of rational numbers integral at every prime and is $\mathbb{Z}$. On finite summands completion gives the product of their primary parts and rationalization is zero. The long exact sequence of the homotopy pullback and Whitehead's theorem prove the square, also giving the usual arithmetic-square construction in the exact form needed here.

Apply $\mathrm{Map}(BC_p, -)$ to this homotopy pullback and take the component containing $B\rho$. For every rational vector space $W$, positive-degree cohomology $H^k(C_p; W)$ vanishes by averaging. The two rational mapping spaces consequently have vanishing $\pi_2$ and $\pi_3$. The homotopy-pullback exact sequence therefore gives

$$\pi_2 \,\mathrm{Map}(BC_p, Y)_{B\rho} \cong \prod_q \pi_2 \,\mathrm{Map}(BC_p, Y_q^\wedge)_{B\rho}.$$

For $q \neq p$, evaluation is a weak equivalence from this mapping-space component to $Y_q^\wedge$. To check it, use the Postnikov tower: positive-degree $C_p$-cohomology of every $q$-complete homotopy group vanishes since multiplication by $p$ is invertible; at each tower stage only constant maps contribute, and homotopy inverse limits preserve the comparison. Thus only $q = 2$ contributes to $\pi_2$ among these primes, giving $\pi_2(Y_2^\wedge) = \mathbb{Z}/2$.

At $q = p$ we use the exact finite-$p$-group mapping-space theorem: for a finite $p$-group $P$, a compact Lie group $K$, and $\alpha : P \to K$, the map induced by the centralizer identifies

$$\pi_n \,\mathrm{Map}(BP, (BK)_p^\wedge)_{B\alpha} = \big(\pi_n BC_K(\alpha(P))\big)_p^\wedge \quad (n \geq 2).$$

This is the Dwyer–Zabrodsky–Notbohm formula, stated and derived in [37, Theorem 1.9 and its proof, using Theorem 3.2(iii) of their earlier paper]. Its hypotheses apply to the displayed finite cyclic group and $SO(3)$. Since a nontrivial odd-order rotation has centralizer exactly the axis circle $SO(2)$, the contribution is $\mathbb{Z}_p$. This proves $\pi_1(G^{hC_p}, 1) = \mathbb{Z}_p \times \mathbb{Z}/2$.

Naturality of the centralizer map makes the $p$-component of the map from the rotation circle the completion $\mathbb{Z} \to \mathbb{Z}_p$. At the prime 2 it is the map $\pi_1 SO(2) \to \pi_1 SO(3)$, namely reduction modulo 2: lifting a full rotation to unit quaternions gives the endpoint $-1$. This proves the stated formula for the exact comparison. For an explicit missing element at $p = 3$, put $\beta = \sum_{j \geq 1} 3^{j!} \in \mathbb{Z}_3$. Its base-three digits have infinitely many zeros and infinitely many ones,

whereas an ordinary integer has eventually only zeros or eventually only twos. Hence $(\beta, 0)$ is outside the image. This also proves nonsurjectivity without a cardinality argument; for general odd $p$ the same factorial series has digits zero and one, rather than eventually zero or $p-1$. The explicit action, its finite order, the axis centralizer and the finite reductions of this comparison are checked in `code/verify_sphere_comparison.py`. The inverse-limit argument above, rather than a finite computation, proves the nonsurjectivity. □

**C.25. [Refuted] Zero-level fidelity for a faithful infinite cyclic sphere action.** Let $\Gamma = \mathbb{Z}$ act faithfully on $S^2$ by an irrational axis rotation, $u(1)(\theta, t) = (\theta + 2\pi\alpha, t)$ with $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and the usual collapse of the latitude circles at $t = \pm 1$. Let $M = A^\Gamma$ be its actual automorphism space, based at the identity, and let $N$ be its weak automorphism space in the categorical group $\Pi_1 A$, where $A = \mathrm{Homeo}(S^2)$. The induced transporter map has

$$\pi_1(M, 1) = \mathbb{Z} \longrightarrow \pi_1(N, 1) = \mathbb{Z}/2, \qquad n \longmapsto n \bmod 2.$$

Its kernel is $2\mathbb{Z}$, so level 0 fails for this faithful action of an infinite group. This is the exact fundamental-groupoid comparison; C.24 separately retains the higher homotopy of $A$ when computing the untruncated homotopy-fixed space. The hyperbolic zero-level question recorded in C.12 has its stated, stronger surface hypothesis.

*Manuscript:* §23; folios 140–141; scans 279–281.

*Proof.* Commutation with an irrational rotation implies commutation with every axis rotation: its powers are dense in $S^1$, and composition and inversion in the compact-open homeomorphism group are continuous. An equivariant homeomorphism preserves the two poles as a set and sends each other rotation orbit to another orbit. In the orientation- preserving identity component it has exactly the form

$$f(\theta, t) = (\theta + a(t), b(t)), \quad a \in C((-1, 1), \mathbb{R}/2\pi\mathbb{Z}), \quad b \in \mathrm{Homeo}^+([-1, 1]).$$

To prove that this parametrization is onto, equivariance forces the angular function on each orbit to be translation by a unique $a(t)$; the orbit quotient supplies $b$. Orientation forces $b$ to be increasing. Conversely every such pair extends continuously at both poles, because those circle orbits collapse there, and the same argument applies to its inverse. Thus the parametrization is a homeomorphism of spaces with the compact-open topologies. Continuity at a pole imposes no limit on $a(t)$; on any compact subinterval of $(-1, 1)$ it is exactly ordinary compact-open convergence.

The homotopy $a_s(t) = a((1-s)t)$ contracts the angular-function factor to its evaluation circle at 0. The convex interpolation $b_s(t) = (1-s)b(t) + st$ contracts the increasing-homeomorphism factor to the identity: it remains strictly increasing and fixes both endpoints. These homotopies are continuous in the stated topologies and extend continuously to the sphere. The actual identity centralizer therefore has homotopy type $S^1$, with its axis-rotation circle giving the specified generator of $\pi_1 M$.

In the weak transporter groupoid, an automorphism of its identity object is a loop in $A$ fixed by the conjugation action of $\Gamma$. This is exactly the groupoid calculation of C.9, now applied to $\Pi_1 A$ rather than asserting any higher truncation of $A$ itself. By the evaluation and polar-decomposition comparison in C.24, $\pi_1(A, 1) = \pi_1 SO(3) = \mathbb{Z}/2$. Conjugation by an axis rotation acts trivially on this group: joining that rotation to the identity in $SO(3)$ gives a based homotopy of its conjugation map to the identity. Hence $\pi_1 N = \mathbb{Z}/2$.

The transporter comparison sends a loop of equivariant homeomorphisms to that same loop in $A$. Its effect is consequently the usual $\pi_1 SO(2) \to \pi_1 SO(3)$, reduction modulo two. More explicitly, a loop of two complete rotations has degree two under the evaluation $a \mapsto a(0)$ in the

actual centralizer, so it is nonzero there. Its lift to the quaternion double cover $S^3 \to SO(3)$ is the closed loop $t \mapsto \cos(2\pi t) + \mathbf{i}\sin(2\pi t)$, which contracts in the simply connected $S^3$. It is therefore killed by the weak transporter map. This proves the exact nontrivial kernel and the zero-level failure, with no inference from disconnectedness. □

**C.26. [Proved] Transitive centralizers on a closed hyperbolic surface.** Let $X$ be a closed connected orientable surface of genus $g \geq 2$, and give $A = \mathrm{Homeo}(X)$ the compact-open topology. For every subgroup $L \leq A$ acting transitively on $X$, the centralizer $C_A(L)$ is finite. Consequently, for an arbitrary strict action $u : \Gamma \to A$ and a based loop $f_t$ in $C_{A^0}(u(\Gamma))$, the loop is equivariantly null-homotopic whenever the subgroup generated by its values acts transitively on $X$. There also exists a closed torsion-free Polish subgroup $G \leq \mathrm{Homeo}(D^2, \partial D^2)$ with $\pi_1 G = \mathbb{Z}$; extension by the identity embeds it closedly in every hyperbolic $A^0$, where its displayed generator becomes null-homotopic. This last subgroup is not asserted to be a centralizer.

*Manuscript:* 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

*Proof.* Put $K = \overline{L}$ and $N = C_A(K)$. Continuity of composition gives $C_A(L) = C_A(K)$, and $K$ is a closed Polish subgroup of $A$ that still acts transitively. Fix $x \in X$ and write $K_x$ for its stabilizer. The open mapping principle for Polish actions gives a homeomorphism

$$K/K_x \longrightarrow X, \qquad kK_x \longmapsto kx. \tag{C.26.1}$$

Precisely, we use the ordinary-action case of [24, Theorem 3.5]: the acting group is Polish, $X$ is nonmeagre Hausdorff, the action is continuous and transitive, and the domain $K^x$ of the partial-action notation is all of $K$, hence open. The source proves this open mapping statement; no local section of (C.26.1) is being assumed.

Every $n \in N$ is determined by $n(x)$, since $n(kx) = kn(x)$. Its possible values form exactly the closed subset

$$N_K(K_x)/K_x \;\subseteq\; K/K_x = X. \tag{C.26.2}$$

Indeed, $n(x) = kx$ forces $K_x = K_{kx} = kK_xk^{-1}$. Conversely a normalizing $k$ defines the homeomorphism $gK_x \mapsto gkK_x$, commuting with $K$. The normalizer of the closed subgroup $K_x$ is closed: both inclusions in $kK_xk^{-1} = K_x$ are intersections of closed conditions. Its image under (C.26.1) is closed because its inverse image under the quotient map is that same normalizer. Thus (C.26.2) is compact.

Evaluation is a homeomorphism of $N$ onto (C.26.2). Only inverse continuity needs proof. Fix $\varepsilon > 0$. For every $y = g_y x$ choose an identity neighborhood $W_y$ in $K$ such that each $g_y w g_y^{-1}$, $w \in W_y$, moves every point of $X$ by less than $\varepsilon/3$. The open sets $g_y W_y x$ cover $X$; take a finite subcover with elements $g_1, \ldots, g_m$. When $n(x)$ is sufficiently close to $n_0(x)$, each $g_i n(x)$ is within $\varepsilon/3$ of $g_i n_0(x)$. At $z = g_i w x$ in the corresponding set, commutation gives

$$n(z) = g_i w n(x), \qquad n_0(z) = g_i w n_0(x).$$

The two uniformly small movements by $g_i w g_i^{-1}$ and the middle $\varepsilon/3$ bound imply $d(n(z), n_0(z)) < \varepsilon$. This is uniform in $z$ and proves inverse continuity in the compact-open topology. In particular $N$ is compact.

An element of $N$ fixing one point fixes every point, by transitivity and commutation. Every element of $A^0$ has Lefschetz number $\chi(X) = 2 - 2g \neq 0$, hence has a fixed point. Therefore

$$N \cap A^0 = 1.$$

Projection to the discrete group $\pi_0 A$ is thus a continuous injection on $N$. Its image is compact and discrete, hence finite, as asserted.

For the stated loop, put $L = \langle f_t : 0 \le t \le 1 \rangle$. Every $u(\gamma)$ commutes with $L$, so the transitivity hypothesis and the first assertion place $u(\Gamma)$ inside a finite group. Write $H = u(\Gamma)$. The equality $C_{A^0}(u(\Gamma)) = C_{A^0}(H)$ is equality of subspaces, irrespective of the cardinality of $\Gamma$. C.7 applied to the finite group $H$ contracts the loop in this exact centralizer. The assertion therefore does not replace the operator group by a different action without a comparison map. □

**A proved obstruction to a closed-subgroup-only argument.** There is a closed torsion-free Polish subgroup of $\mathrm{Homeo}(D^2, \partial D^2)$ with fundamental group $\mathbb{Z}$. We give the embeddings, because this example explains exactly why closedness and absence of compact circles do not by themselves settle C.12. It is not asserted to be a centralizer.

Let

$$E = \{f \in L^1([0,1], \mathbb{R}) : f \text{ is integer-valued almost everywhere}\}.$$

An $L^1$-convergent sequence has an almost-everywhere convergent subsequence; hence $E$ is closed and Polish. For $0 < s \le 1$ define $D_s f(t) = f(t/s)$ on $[0, s]$ and zero on $(s, 1]$, and put $D_0 f = 0$. These maps remain in $E$ and satisfy $\|D_s f\|_1 = s\|f\|_1$. Strong continuity of dilation follows first for step functions and then by $L^1$ approximation. Thus $(s, f) \mapsto D_s f$ contracts $E$ and each of its norm balls at zero. In particular $E$ is contractible and locally contractible.

The constant integer functions form a closed discrete subgroup $\mathbb{Z}$. The quotient $G = E/\mathbb{Z}$ is Polish, and $E \to G$ is a covering: translates of a norm ball of radius less than $1/3$ by distinct integers are disjoint. It is the universal cover, and

$$\pi_1(G) = \mathbb{Z}, \qquad \gamma(s) = [\mathbf{1}_{[0,s]}] \quad (0 \le s \le 1) \tag{C.26.3}$$

is a generator, because its lift joins zero to the constant function 1. Nevertheless $G$ is torsion-free: $n[f] = 0$ says that $nf$ is a constant integer, so the integer-valued $f$ is constant and $[f] = 0$.

There is a closed isometric group embedding

$$G \longrightarrow V := L^1([0,1], \mathbb{R})/\mathbb{R}, \qquad [f]_{\mathbb{Z}} \longmapsto [f]_{\mathbb{R}}, \tag{C.26.4}$$

where the second denominator consists of all constant real functions. For integer-valued $h$, the convex function $c \mapsto \int |h - c|$ is linear between consecutive integers, tends to infinity at both ends, and attains a minimum at an integer. Hence

$$\inf_{c \in \mathbb{R}} \|h - c\|_1 = \inf_{n \in \mathbb{Z}} \|h - n\|_1,$$

proving the isometry. The quotient metric on $G$ is complete: choose a Cauchy subsequence with successive quotient distances below $2^{-j}$ and adjust each representative by an integer so that the successive $L^1$ distances have that bound. The representatives converge in $E$, and the entire original Cauchy sequence then converges in $G$. This proves closedness in the Banach space $V$.

We spell out a closed linear isometric embedding $T : V \hookrightarrow C_0([0,1], \mathbb{R})$, where the subscript here means vanishing at both endpoints. The weak-star unit ball $B$ of $V^*$ is compact metrizable, and $v \mapsto (\ell \mapsto \ell(v))$ embeds $V$ isometrically into $C(B)$. A continuous surjection from the Cantor set to $B$ exists by nested finite covers of decreasing diameters. Pullback followed by linear interpolation across each complementary interval of the Cantor set embeds $C(B)$ linearly and isometrically in $C([0,1])$. Place this interval in the middle third and interpolate to zero at the two outer endpoints. The resulting map is the required $T$; its image is closed because $V$ is complete.

For $F \in C_0([0,1])$, the formula

$$\Phi_F(x, y) = \big(\tanh(\operatorname{artanh}(x) + F(y)), y\big) \qquad (|x| < 1) \tag{C.26.5}$$

on $[-1, 1] \times [0, 1]$, extended by the identity on the vertical sides, fixes the entire boundary and satisfies $\Phi_F \Phi_Q = \Phi_{F+Q}$. It is a topological-group embedding: tanh is Lipschitz, giving continuity,

and evaluation at $(0, y)$ recovers $\tanh(F(y))$, from which $F$ is recovered uniformly near any given profile. Its image is closed. A homeomorphism limit fixes the boundary and preserves the horizontal leaves; the image of the compact segment $\{0\} \times [0, 1]$ stays a positive distance from the vertical sides. Applying artanh therefore recovers uniformly bounded, uniformly convergent profiles.

Composing (C.26.4), $T$, and (C.26.5) embeds $G$ as a closed subgroup of $\mathrm{Homeo}(D^2, \partial D^2)$. Extension by the identity outside a coordinate disc is a closed topological-group embedding into $A^0$ for any hyperbolic surface: restriction recovers the maps, and being the identity off the disc is a closed condition. Its loop (C.26.3) is ambient-null but nontrivial in $G$. The precise enlargement killing it is already visible: if $F_s = T([\mathbf{1}_{[0,s]}]_{\mathbb{R}})$, then $F_0 = F_1 = 0$ and $(r, s) \mapsto \Phi_{rF_s}$ is a based contraction in the larger real shear group. No operator group cutting out a centralizer that excludes this contraction has been constructed here. Thus this example refutes the auxiliary assertion of fundamental-group injectivity for all closed subgroups, while C.12 retains its stated centralizer question.

**C.27. [Proved] An order-natural contraction of line centralizer loops.** Give $\mathrm{Homeo}(\mathbb{R})$ its compact-open topology, let $H \leq \mathrm{Homeo}(\mathbb{R})$ be an arbitrary subgroup, and put

$$C = C_{\mathrm{Homeo}^+(\mathbb{R})}(H).$$

Every based loop in $C$ has an explicit based null-homotopy in $C$. Orientation-reversing elements of $H$ are allowed. The construction commutes with conjugation by every homeomorphism of the line and requires no finiteness, differentiability, or local compactness of $H$.

*Manuscript:* 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

*Proof.* Let $t \mapsto f_t$ be a compact-open continuous loop in $C$, with $f_0 = f_1 = \mathrm{id}$. For $0 \leq r \leq 1$ define

$$m_r(x) = \min_{0 \leq u \leq r} f_u(x), \qquad M_r(x) = \max_{0 \leq u \leq r} f_u(x). \tag{C.27.1}$$

Both are increasing homeomorphisms. Indeed, for $x < y$, compactness gives

$$\min_{0 \leq u \leq 1} \big(f_u(y) - f_u(x)\big) > 0,$$

which proves strict increase for each minimum and maximum. Their properness is uniform in $r$: compactness of the inverse family $\{f_u^{-1} : 0 \leq u \leq 1\}$ gives, for every bounded target interval, a common bound on all its inverse images. Continuity of (C.27.1), jointly in $r$ and $x$, follows from uniform continuity on compact parameter rectangles. Proper continuous strictly increasing maps of $\mathbb{R}$ are homeomorphisms.

Set

$$F_{s,t}(x) = \mathrm{med}\big(m_{1-s}(x), f_t(x), M_{1-s}(x)\big) = \max\{m_{1-s}(x), \min\{f_t(x), M_{1-s}(x)\}\}. \tag{C.27.2}$$

Finite minima and maxima of increasing homeomorphisms are again increasing homeomorphisms, by the same strict-increase and properness arguments. Formula (C.27.2) is continuous in $(s, t, x)$, uniformly on compact $x$-sets; it is therefore a continuous family in the compact-open homeomorphism group. At $s = 0$, the prefix interval is the entire loop, so $m_1 \leq f_t \leq M_1$ and $F_{0,t} = f_t$. At $s = 1$, $m_0 = M_0 = \mathrm{id}$, giving $F_{1,t} = \mathrm{id}$. Since $u = 0$ occurs in every prefix interval, $m_r \leq \mathrm{id} \leq M_r$, and hence $F_{s,0} = F_{s,1} = \mathrm{id}$ for every $s$.

It remains to check the exact equivariance. An increasing $h \in H$ fixes both envelopes under conjugation, because conjugation commutes with their pointwise minimum and maximum and

fixes every $f_u$. A decreasing $h \in H$ interchanges $m_r$ and $M_r$ instead. Every order-preserving or order-reversing bijection preserves the median of three points. Thus both cases give

$$hF_{s,t}h^{-1} = F_{s,t} \qquad (h \in H, 0 \le s, t \le 1).$$

This proves that the entire based contraction lies in $C$. Replacing the loop by its conjugate by an arbitrary line homeomorphism gives the conjugate of the same formula, proving the asserted naturality. □

The surface problem in C.12 still requires a continuous construction across different orbit strata. Applying (C.27.2) separately after choosing coordinates on individual line orbits does not establish continuity on the surface. No such gluing statement follows from C.27, and none is used here.

**C.28. [Proved] Orbit types of a loop-generated plane group.** Let $t \mapsto f_t$ be a compact-open continuous based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, with $f_0 = f_1 = \mathrm{id}$, and put

$$L = \{f_t : 0 \le t \le 1\}, \qquad P = L \cup L^{-1}, \qquad G = \langle P \rangle.$$

Every $G$-orbit is one of the following: an open subset of the plane; a point; a Jordan curve; or the image of a continuous injective map $\eta : \mathbb{R} \to \mathbb{R}^2$ that restricts to a homeomorphism on every compact interval. In the last case its intrinsic arc order identifies it with the line, but no global homeomorphism from $\mathbb{R}$ to the orbit with its subspace topology is asserted. The loop restricts continuously in the compact-open topology to the intrinsically ordered line. Every homeomorphism commuting with $G$ carries these orbit types to the same types and induces a homeomorphism between the corresponding intrinsic lines, allowing reversal of their orders.

*Manuscript:* 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

*Proof.* Fix $x \in \mathbb{R}^2$ and put

$$K_n = P^n x, \qquad O = Gx = \bigcup_{n \ge 1} K_n. \tag{C.28.1}$$

The sets $K_n$ are nested compact Peano continua. Indeed, $P$ is a continuous image of the wedge of two circles, using the given loop and its inverse; it is compact, connected and locally connected. A finite product of such spaces has the same properties, and the word-evaluation map has image $K_n$. To recall why local connectedness passes through this compact surjection, for an open neighborhood of a point in the image cover its compact inverse fibre by finitely many connected open subsets of its inverse image. Their images all meet that point and lie in its component. The complement of the image of the remaining compact set is a smaller open neighborhood in that component. Thus components of open sets in the image are open. In particular $K_n$ is arcwise connected. The identity lies in $P$, which proves the nesting in (C.28.1). Every element of $G$ can be joined to the identity by a path in $G$, by writing it as a finite word and using the specified paths to its factors.

Suppose first that $O$ contains a Jordan curve $J$. For any $g \in G$ choose a path $g_s$ from id to $g$, so that $g_s J \subseteq O$. Let $D_s$ be the bounded Jordan domain of $g_s J$. If $gJ \ne J$, then $D_0 \triangle D_1$ contains a nonempty open set away from the two boundary curves. Every point $q$ in this open set lies on some $g_s J$: otherwise the winding number of the continuously varying curve $g_s J$ around $q$ would be constant, whereas its endpoint values are 0 and 1. Hence $O$ contains a nonempty open set. Transitivity of $G$ on $O$ carries this open set to a neighborhood of each point of $O$, so $O$ is open. If every $g$ preserves $J$, then for a point $y \in J$ its orbit is contained in $J$; since $Gy = O$ and $J \subseteq O$, we obtain $O = J$. Thus a nonopen orbit containing a Jordan curve is itself that Jordan curve.

We may now consider a nonopen orbit containing no Jordan curve. Each $K_n$ is a dendrite: a compact, connected, locally connected metric space containing no Jordan curve. Arcs between two points are unique, since two distinct arcs would contain a Jordan curve. This also proves uniqueness of arcs in all of $O$: two proposed arcs in $O$ would give the same forbidden curve. Existence follows by putting their endpoints into one $K_n$.

Each $K_n$ has countably many branch points. Choose a countable dense subset $D_n$. Three distinct components of the complement of a branch point are open there, so choose a point of $D_n$ in each. The branch point is the unique common point of the three pairwise connecting arcs for this triple: two branches meet only at their point of separation. Thus each branch point is determined by one of the countably many triples from $D_n$.

A triod point of $O$ is a branch point of some $K_n$. Put the three outer endpoints of its triod into one $K_n$; the unique connecting arcs in $K_n$ must be the arcs of that triod. The triod points of $O$ are consequently countable. Their set is invariant under every element of $G$. If it were nonempty, transitivity would make every point of $O$ a triod point. A nontrivial arcwise connected orbit contains an arc and is uncountable, giving a contradiction. Thus a nontrivial $O$ has no triod points, and each nondegenerate $K_n$ is an arc.

The nested arcs give $O$ a consistent linear arc order, unique up to reversal. To see the absence of additional branches directly, fix two distinct points $a, b$. Every other point is on $[a, b]$, beyond $a$, or beyond $b$, because any fourth alternative would give a triod. There are at most two endpoints of this ordered union. The endpoint set is invariant under $G$, so transitivity and uncountability exclude endpoints. We can therefore pass to a cofinal sequence of arcs $A_j$ such that $A_j$ lies in the relative interior of $A_{j+1}$: extend both endpoints in $O$ and then put the two extensions and the next $K_n$ into a later $K_m$.

Choose compatible homeomorphisms $[-j, j] \to A_j$, extending at each step along the two added arc intervals. Their union is a continuous injective map

$$\eta : \mathbb{R} \longrightarrow \mathbb{R}^2, \qquad \eta(\mathbb{R}) = O, \tag{C.28.2}$$

whose restriction to every compact interval is a homeomorphism onto its image. These assertions follow on each finite interval from one of the compact arc parametrizations. This supplies the stated intrinsic line structure. It does not prove that the inverse of (C.28.2) is continuous for the global subspace topology; accumulations from far along an orbit have not been excluded.

The restricted loop is continuous for that intrinsic line topology. A compact arc in $O$ is contained in some $K_m$, and $f_t(K_m) \subseteq K_{m+1}$ for all $t$. The latter is a compact arc, on which the intrinsic and subspace topologies coincide. Joint continuity of $(t, z) \mapsto f_t(z)$ on each such compact arc therefore gives compact-open continuity of $\eta^{-1} f_t \eta \in \mathrm{Homeo}(\mathbb{R})$. Its order sign is positive throughout the loop: images of two distinct points vary continuously in the line and cannot cross. Finally a commuting homeomorphism $h$ satisfies the exact equality

$$h(K_n(x)) = K_n(hx).$$

It therefore maps compact arcs homeomorphically to compact arcs and induces a homeomorphism of the intrinsic ordered lines; its inverse has the same property. It also preserves the point, Jordan-curve and open alternatives, completing the proof. □

**Exact application and remaining step.** The canonical lift in the paragraph after C.12 sends a hyperbolic centralizer loop to a closed based loop in $\mathrm{Homeo}^+(\widetilde{U})$, where $\widetilde{U} \cong \mathbb{R}^2$. The entire group $\widetilde{H}$ of lifts of the operator image commutes with it and hence with the $G$ of C.28. Thus C.28 applies to this actual lift group, and C.27 provides an order-natural contraction on each intrinsic line orbit after restriction to its stabilizer in $\widetilde{H}$. These are typed restriction and lift maps, but they do not yet give a homotopy through homeomorphisms of the whole plane. Continuity across

distinct orbit strata, compatibility with their frontiers, and a relative contraction on proper open orbits remain unproved. C.26 treats global transitivity on a closed hyperbolic surface; a proper open orbit in C.28 does not satisfy that hypothesis. The admissible infinite-group level-0 question therefore remains C.12.

**C.29. [Proved] Circle centralizers and the winding of a plane loop.** For every subgroup $H \leq \mathrm{Homeo}(S^1)$, put $C = C_{\mathrm{Homeo}^+(S^1)}(H)$, with the compact-open topology. The inclusion induces an injection

$$\pi_1(C, \mathrm{id}) \longrightarrow \pi_1(\mathrm{Homeo}^+(S^1), \mathrm{id}) = \mathbb{Z}, \tag{C.29.1}$$

where the integer is the winding of the image of one point. Its image lies in $\mathbb{Z}^H$, with $H$ acting on $\mathbb{Z}$ by its orientation character. A based loop of winding zero has an explicit contraction in $C$ that is natural under all homeomorphisms between circles. In particular, a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$ preserving a Jordan curve restricts to a winding-zero loop on that curve. Its restriction therefore has this natural contraction against every commuting circle homeomorphism.

*Manuscript:* 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

*Proof.* Write the circle as $\mathbb{R}/\mathbb{Z}$, and set

$$\widetilde{A} = \{F \in \mathrm{Homeo}^+(\mathbb{R}) : F(x+1) = F(x) + 1\}.$$

The quotient map $\widetilde{A} \to \mathrm{Homeo}^+(S^1)$ is a covering with deck transformations $F \mapsto F + n$, $n \in \mathbb{Z}$. Indeed, requiring $F(0)$ to lie in a fixed interval of length less than one gives continuous unique lifts over the corresponding evaluation neighborhood. The homotopy $(r, F) \mapsto (1-r)F + r\,\mathrm{id}$ contracts $\widetilde{A}$: strict increase, the translation equation, and properness hold at every parameter. Thus the target fundamental group in (C.29.1) is precisely $\mathbb{Z}$. For a based loop $f_t$, its unique lift $F_t$ starting at id ends at $x \mapsto x + n$, where $n$ is its point winding.

Suppose that $f_t \in C$ and $n = 0$. Every lift $\widetilde{h}$ of an element $h \in H$, including an orientation-reversing one, commutes with every $F_t$. The commutator projects to the identity on the circle, so it is an integer translation depending continuously on $t$; it is the identity at $t = 0$, and hence at all $t$. Consequently $F_t$ is a based loop in the line centralizer of the entire lift group of $H$. This group also contains the integer translations. Apply C.27, explicitly

$$\begin{aligned} m_r(x) &= \min_{0 \leq u \leq r} F_u(x), \qquad M_r(x) = \max_{0 \leq u \leq r} F_u(x), \\ Q_{s,t}(x) &= \mathrm{med}\big(m_{1-s}(x), F_t(x), M_{1-s}(x)\big). \end{aligned} \tag{C.29.2}$$

Every $Q_{s,t}$ commutes with all lifts and with $x \mapsto x + 1$, so (C.29.2) descends to a contraction through elements of $C$. This proves injectivity in (C.29.1).

The winding of a conjugated loop $hf_th^{-1}$ is the winding of $f_t$ multiplied by the orientation sign of $h$. Since these loops agree for $h \in H$, the image is contained in $\mathbb{Z}^H$. For naturality, let $q$ be a homeomorphism between two circles and choose any lift $\widetilde{q}$ to their universal lines. The based lift of the conjugated loop is $\widetilde{q}F_t\widetilde{q}^{-1}$. An increasing $\widetilde{q}$ carries the two extrema in (C.29.2) to the corresponding extrema; a decreasing one interchanges them. Every line homeomorphism preserves the median of three points. Thus (C.29.2) is carried to exactly the same formula for the conjugated loop. Changing the lift of $q$ by an integer translation does not change the descended contraction.

It remains to prove the stated plane winding comparison. Conjugate by a Schoenflies homeomorphism so that the preserved Jordan curve is the unit circle. Choose two distinct points $a, b$ on it. Evaluation and the difference direction give the continuous maps

$$\mathrm{Homeo}^+(\mathbb{R}^2) \longrightarrow \mathrm{Conf}_2(\mathbb{R}^2) \longrightarrow S^1, \qquad f \longmapsto (f(a), f(b)) \longmapsto \frac{f(b) - f(a)}{|f(b) - f(a)|}. \tag{C.29.3}$$

For the given circle-preserving loop choose continuous angular lifts $\alpha(t), \beta(t)$ with $\alpha(t) < \beta(t) < \alpha(t) + 2\pi$. They both acquire the increment $2\pi n$: their difference remains in $(0, 2\pi)$ and returns to its initial value modulo $2\pi$. The identity

$$e^{i\beta(t)} - e^{i\alpha(t)} = 2i\, e^{i(\alpha(t)+\beta(t))/2} \sin \frac{\beta(t) - \alpha(t)}{2}$$

shows that the winding of (C.29.3) is exactly $n$. A null-homotopy of the ambient plane loop maps under (C.29.3) to a null-homotopy in $S^1$, forcing $n = 0$. This proves the exact comparison and the last assertion. □

**Application to C.12.** The canonical lift of a hyperbolic centralizer loop is null-homotopic as a loop of plane homeomorphisms: it is the image under the continuous canonical lift homomorphism of a loop in the contractible $A^0$. Every Jordan-curve orbit in C.28 is preserved throughout this loop. Its restriction therefore has winding zero by (C.29.3), and (C.29.2) contracts it naturally against its stabilizer in the full operator lift group. Commuting operators carrying one circle orbit to another carry these contractions to each other. This gives a circle-orbit contraction with the exact required equivariance. C.34 subsequently proves joint continuity, including inverses, on the full nonopen locus. C.35 shows that this particular prescription need not admit an equivariant extension over an adjacent open orbit, and C.36 constructs compatible alternatives on finite unions of closures with two-dimensional proper operator image. The remaining global compatibility requirements are stated in C.12, which remains open.

**C.30. [Proved] Properness of a transitive-action centralizer.** Let $V$ be a connected second-countable surface without boundary, and let $K$ be a closed subgroup of $\mathrm{Homeo}(V)$ acting transitively on $V$, with the compact-open topology. Put $N = C_{\mathrm{Homeo}(V)}(K)$. Then $N$ is locally compact, its action on $V$ is free and proper, and for every $x \in V$ evaluation is a homeomorphism onto the closed subset

$$N \xrightarrow{\mathrm{ev}_x} N_K(K_x)/K_x \;\subseteq\; K/K_x \cong V. \tag{C.30.1}$$

Under the normalizer-quotient group structure, this is an anti-isomorphism of topological groups. For an open orbit $O$ of the loop-generated group in C.28, the image in $\mathrm{Homeo}(O)$ of its operator stabilizer lies in such an $N$; the closure of that image acts freely and properly on $O$.

*Manuscript:* 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

*Proof.* The group $K$ is Polish. The open mapping theorem [24, Theorem 3.5], with the ordinary continuous transitive action on the nonmeagre Hausdorff surface $V$, identifies $K/K_x$ homeomorphically with $V$. In particular the orbit map $k \mapsto kx$ is open. We will use the following exact consequence, requiring no local continuous section: for every sequence $x_j \to x$, there are $k_j \to 1$ in $K$ with $k_j x = x_j$. To prove it, choose a decreasing identity-neighborhood basis $W_m$. The open sets $W_m x$ contain $x$, so successive tails of the sequence lie in them. Choose $k_j$ from the corresponding $W_{m(j)}$, with $m(j) \to \infty$.

An element of $N$ is determined by its value at $x$, since $n(kx) = kn(x)$. If $n(x) = ax$, equality of stabilizers gives $aK_xa^{-1} = K_x$. Conversely, for $a$ in this normalizer the map $kK_x \mapsto kaK_x$ is a homeomorphism commuting with $K$. The normalizer is closed: both inclusions defining $aK_xa^{-1} = K_x$ are intersections of closed conditions. It is saturated for the quotient by $K_x$, so its image is a closed subset of $K/K_x = V$. This proves the claimed set-theoretic image and closedness in (C.30.1). If $n(x) = ax$ and $m(x) = bx$, then $(nm)(x) = bax$, proving the stated anti-isomorphism.

We prove inverse continuity of evaluation, including the issue of noncompact $V$. Suppose that $n_j(x) \to n(x)$, with $n_j, n \in N$. For any compact set $B \subset V$, a failure of uniform convergence on

$B$ would provide a subsequence and $y_j \in B$ with $y_j \to y$ witnessing that failure. Choose $a \in K$ with $ax = y$, and then $b_j \to 1$ with $b_j y = y_j$, using open evaluation at $y$. Commutation gives

$$n_j(y_j) = n_j(b_j a x) = b_j a\, n_j(x) \longrightarrow an(x) = n(y). \tag{C.30.2}$$

Also $n(y_j) \to n(y)$, a contradiction. Hence $n_j \to n$ uniformly on every compact set, which is compact-open convergence. Inversion can be checked directly as well. Put $c_j = n^{-1} n_j$, so $c_j x \to x$, and choose $a_j \to 1$ with $a_j x = c_j x$. Because $c_j$ commutes with $a_j$, $c_j^{-1} x = a_j^{-1} x \to x$; applying (C.30.2) to $c_j^{-1}$ gives convergence of the inverses. All the spaces are metrizable, so this sequential argument proves inverse continuity. Thus (C.30.1) is a homeomorphism, and $N$ is locally compact as a closed subspace of the locally compact surface.

If $n \in N$ fixes one point, transitivity and commutation make it fix every point, proving freeness. To prove properness, consider sequences with

$$x_j \longrightarrow x, \qquad n_j x_j \longrightarrow y.$$

Choose $k_j \to 1$ with $k_j x = x_j$. Then

$$n_j x = k_j^{-1} n_j x_j \longrightarrow y. \tag{C.30.3}$$

The image of evaluation at $x$ is closed, so $y = nx$ for a unique $n \in N$. Its inverse continuity gives $n_j \to n$. For any compact subset of $V \times V$, every sequence in its inverse image under $(n, z) \mapsto (z, nz)$ therefore has a convergent subsequence, with limit in the same inverse image. Metrizability turns this sequential compactness into compactness, proving that the action map is proper. Every closed subgroup of $N$ inherits the free proper action.

Finally let $G$ be the group in C.28 and let $O$ be one of its open orbits. It is connected because every element of $G$ is joined to the identity by a path in $G$. The closure

$$K = \overline{\{g|_O : g \in G\}}^{\mathrm{Homeo}(O)}$$

is a closed Polish group still transitive on $O$. Let $\widetilde{H}$ be any group commuting with $G$, and let $\widetilde{H}_O$ be its setwise stabilizer of $O$. The precise restriction homomorphism is

$$\widetilde{H}_O \longrightarrow \mathrm{Homeo}(O), \qquad h \longmapsto h|_O, \quad \ker = \{h \in \widetilde{H}_O : h|_O = \mathrm{id}\}. \tag{C.30.4}$$

Its image commutes with the restriction of $G$ and hence with $K$. The closure of this image is a closed subgroup of $N$, and therefore acts freely and properly on $O$, as claimed. □

**Precise remaining use.** The closure in the last assertion has its subspace topology from $\mathrm{Homeo}(O)$. Formula (C.30.4) does not assert that the original discrete operator group acts freely, since its kernel may be nontrivial; nor does it equip a possibly dense abstract image with the discrete topology and claim properness for that topology. The result supplies the closed locally compact transformation group actually present on an open orbit. No Lie-group classification or quotient-bundle contraction is invoked here. A contraction relative to orbit frontiers, and its continuous assembly with C.27 and C.29 on the one-dimensional strata, remain unproved. Thus C.30 does not change the unresolved status of C.12.

**C.31. [Proved] Track control and continuity at the fixed stratum.** Let $f_t$ be a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let $G$ be its generated group as in C.28, and let $Y$ be the union of the nonopen $G$-orbits. This is a closed subset of the plane. Define $Q_{s,t}$ on $Y$ by C.27 on each intrinsic line orbit, by the descended formula of C.29 on each circle orbit, and by the identity on point orbits. Then every $Q_{s,t}$ is a bijection of $Y$ preserving each orbit, and

$$Q_{s,t}^{-1} = Q_{s,t}[f^{-1}], \qquad Q_{s,t}(x) \in \{f_u(x) : 0 \le u \le 1\}, \qquad Q_{s,t}^{-1}(x) \in \{f_u^{-1}(x) : 0 \le u \le 1\}. \tag{C.31.1}$$

Here $Q[f^{-1}]$ means the same construction applied to the based inverse loop $u \mapsto f_u^{-1}$, which generates the same group. Both evaluation maps $(s,t,x) \mapsto Q_{s,t}^{\pm 1}(x)$ are continuous at every point of $[0,1]^2 \times Z$, where $Z = \{x : f_u(x) = x \text{ for all } u\}$ is the point-orbit stratum. The construction commutes with every homeomorphism commuting with all $f_u$. No continuity at a nonfixed orbit is asserted here.

*Manuscript:* §23; folios140–141; scans279–281; Part C(b) infinite-operator request; fixed-stratum gluing reduction for C.12.

*Proof.* The union of open orbits is open, proving that $Y$ is closed. C.28 supplies exactly the remaining point, circle and intrinsic-line types. The circle restrictions have zero winding by C.29, since the ambient loop is null-homotopic. Thus the formulas defining $Q$ are available on every orbit in $Y$, and are independent of the chosen line or circle coordinates by their proved naturality. Each formula gives a homeomorphism of its single intrinsic line or circle, so together they give a bijection of $Y$.

We prove the inverse identity before making any continuity claim. For increasing homeomorphisms of a line, pointwise inversion reverses order and exchanges finite minimum and maximum:

$$(a \wedge b)^{-1} = a^{-1} \vee b^{-1}, \qquad (a \vee b)^{-1} = a^{-1} \wedge b^{-1}. \tag{C.31.2}$$

For example, at a point $x$ the first inverse is the larger of $a^{-1}(x)$ and $b^{-1}(x)$: at this larger point one of $a, b$ equals $x$, and the other is at least $x$. The same argument works for compact families, whose extrema and inverse extrema are attained. Therefore, with the notation of C.27,

$$m_r^{-1}(x) = \max_{0 \le u \le r} f_u^{-1}(x), \qquad M_r^{-1}(x) = \min_{0 \le u \le r} f_u^{-1}(x). \tag{C.31.3}$$

The median of three line homeomorphisms is the lattice expression $(a \wedge b) \vee (a \wedge c) \vee (b \wedge c)$; its dual expression $(a \vee b) \wedge (a \vee c) \wedge (b \vee c)$ has the same value. Applying (C.31.2) thus proves

$$\operatorname{med}(a,b,c)^{-1} = \operatorname{med}(a^{-1}, b^{-1}, c^{-1}). \tag{C.31.4}$$

Combining (C.31.3)–(C.31.4), and using symmetry of the median in its two envelope entries, gives the first identity in (C.31.1) on every line orbit. On a circle, the inverse of the unique based lift $F_u$ is exactly the unique based lift of $f_u^{-1}$. The same calculation on the universal line descends to the circle. It also holds on point orbits.

At any point the median chooses one of its three entries. Each envelope entry is $f_u(x)$ for a parameter attaining its extremum, and the middle entry is $f_t(x)$ itself. This proves the second assertion of (C.31.1) on lines; projection of the same statement for the based lifts proves it on circles. The inverse identity then proves the third assertion.

Fix $p \in Z$ and an open neighborhood $V$ of $p$ in the plane. Joint continuity of the action, compactness of $[0,1]$, and $f_u(p) = p$ give a neighborhood $W$ of $p$ such that $f_u(W) \subseteq V$ for every $u$. The compact inverse family gives the same assertion for all $f_u^{-1}$ after shrinking $W$. By (C.31.1),

$$Q_{s,t}^{\pm 1}(W \cap Y) \subseteq V \cap Y \quad \text{for all } s,t \in [0,1]. \tag{C.31.5}$$

Since $Q_{s,t}^{\pm 1}(p) = p$, this proves joint continuity at every $(s,t,p)$, uniformly in the two homotopy parameters.

A homeomorphism commuting with all $f_u$ permutes the $G$-orbits, preserves their four types, and maps intrinsic lines by line homeomorphisms by C.28. Naturality in C.27 and C.29 gives $hQ_{s,t}h^{-1} = Q_{s,t}$ on all of $Y$. This proves the asserted equivariance without any continuity assertion at the remaining points. □

**Scope of the gluing step.** Formula (C.31.5) resolves continuity at the fixed stratum for the orbitwise construction on $Y$. It does not yet make the bijections $Q_{s,t} : Y \to Y$ homeomorphisms, because continuity at the nonfixed line and circle strata remains unproved. Nor does it control a contraction on the omitted open orbits. These are the remaining global assembly problems for this route to C.12.

**C.32. [Proved] Discrete proper operator groups on open orbits.** Let $O$ be a connected noncompact second-countable surface without boundary, and let a discrete group $Q \leq \mathrm{Homeo}(O)$ act freely and properly. With compact-open topology on the homeomorphism groups, inclusion induces an injection

$$\pi_1\Big(C_{\mathrm{Homeo}(O)}(Q), \mathrm{id}\Big) \longrightarrow \pi_1(\mathrm{Homeo}(O), \mathrm{id}). \tag{C.32.1}$$

Moreover, a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$ preserving a connected open subset $O \subseteq \mathbb{R}^2$ restricts to a null-homotopic loop in $\mathrm{Homeo}(O)$. Consequently, for a hyperbolic centralizer loop and an open orbit in C.28 whose closed restricted operator image in C.30 is discrete, its restriction has a based equivariant null-homotopy on that orbit. The original restriction kernel is retained; no extension of this null-homotopy to the orbit frontier is asserted.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; discrete-closure open-orbit reduction for C.12.

*Proof.* First consider the free proper discrete action on $O$. It gives an actual covering

$$p : O \longrightarrow V = O/Q. \tag{C.32.2}$$

Here discreteness refers to the topology in the proper-action hypothesis. For completeness, a compact neighborhood of a point meets only finitely many of its translates, by properness. Shrink an open neighborhood inside it to avoid the nonidentity translates among this finite set, using freeness. Distinct translates of the result are disjoint, which gives the evenly covered neighborhoods for (C.32.2). The quotient $V$ is Hausdorff by properness, is a surface without boundary by these local charts, and is second countable because images of a countable open basis of $O$ form an open basis. The connected covering is regular, with deck group exactly $Q$.

We record the homeomorphism-group input precisely. Every second-countable surface admits a PL triangulation. Applied with the compact subpolyhedron consisting of one point $v$, the compact case of [74, Lemma 2.1] and the noncompact case of [74, Theorem 1.1] give

$$\mathrm{Homeo}(V, v)^0 \simeq * \quad \text{for } V \ncong S^2,\ \mathbb{R}P^2,\ \mathbb{R}^2. \tag{C.32.3}$$

These are all exceptions relevant to a surface without boundary with that one-point subpolyhedron. Evaluation $\mathrm{Homeo}(V)^0 \to V$ is a locally trivial bundle: in a coordinate disc a continuously chosen homeomorphism supported in that disc moves its center to a nearby point, giving a local section. Its identity fibre component is exactly the group in (C.32.3). The homotopy exact sequence therefore gives an injection

$$\pi_1(\mathrm{Homeo}(V), \mathrm{id}) \longrightarrow \pi_1(V, v) \quad (V \ncong S^2, \mathbb{R}P^2, \mathbb{R}^2). \tag{C.32.4}$$

The base $V$ of (C.32.2) cannot be $S^2$ or $\mathbb{R}P^2$: their fundamental groups are finite, so every connected covering has finitely many sheets and compact total space, contrary to noncompactness of $O$. If $V = \mathbb{R}^2$, its connected covering is one-sheeted, so $Q = 1$ and (C.32.1) is the identity homomorphism.

Let $f_t$ be a based loop in the centralizer that is null-homotopic in $\mathrm{Homeo}(O)$. It descends to a based loop $b_t$ on $V$, defined by the exact equation

$$b_t p = p f_t. \tag{C.32.5}$$

The family is continuous: joint continuity descends through the open quotient map $[0,1] \times O \to [0,1] \times V$, and the compact-open exponential law applies since $V$ is locally compact. For $x \in O$, the evaluation loop $f_t(x)$ is null-homotopic in $O$. Thus $b_t(p(x)) = p(f_t(x))$ is null-homotopic in $V$. By (C.32.4), $b_t$ is a null-homotopic based loop of homeomorphisms of $V$. Choose a continuous filling $B : D^2 \to \mathrm{Homeo}(V)^0$ with the prescribed boundary $b_t$, and a point $z_* \in \partial D^2$ corresponding to the based endpoint, so that $B_{z_*} = \mathrm{id}$.

The lift is a filling through homeomorphisms and through the exact centralizer; we verify both assertions. Form

$$F : O \times D^2 \longrightarrow V, \qquad F(x,z) = B_z(p(x)). \tag{C.32.6}$$

At $(x_*, z_*)$, the induced fundamental-group image is $p_*\pi_1(O, x_*)$, since $D^2$ is simply connected and $F(x, z_*) = p(x)$. The covering lifting criterion gives a unique continuous lift $\widetilde{B} : O \times D^2 \to O$ with $\widetilde{B}(x, z_*) = x$. Apply the identical construction to $F^-(x,z) = B_z^{-1}(p(x))$, giving a continuous lift $\widetilde{C}$ that is the identity at $z_*$. The maps $(x,z) \mapsto \widetilde{B}_z\widetilde{C}_z(x)$ and $(x,z) \mapsto x$ lift the same map $(x,z) \mapsto p(x)$ and agree at the basepoint. Uniqueness of lifts makes them equal; the reverse composition is equal for the same reason. Hence every $\widetilde{B}_z$ is a homeomorphism, with inverse $\widetilde{C}_z$, and the family is compact-open continuous.

For $q \in Q$, the two maps $\widetilde{B}_z q$ and $q\widetilde{B}_z$ lift the same map $B_z p$ and agree at $z_*$, so they are equal for every $z$. On $O \times \partial D^2$, both $\widetilde{B}_z$ and the specified $f_t$ lift (C.32.5) and agree at the based endpoint. They are therefore equal on that entire connected boundary product. Thus $\widetilde{B}$ fills the original loop in its centralizer. This proves (C.32.1).

Now let $O$ be a connected open subset of the plane preserved by a based ambient-null loop. Its restriction is a continuous loop in $\mathrm{Homeo}(O)^0$. By [74, Theorem 1.1], with the empty subpolyhedron, this identity group is homotopically trivial unless $O$ is the plane or the open annulus; the nonorientable and boundary exceptions in the source cannot occur for $O \subseteq \mathbb{R}^2$. In either remaining case its fundamental group is $\mathbb{Z}$. Choose model coordinates on $O$, a coordinate Jordan circle $J \subset O$, and two distinct points $a, b$ of $J$. A Schoenflies homeomorphism $h$ of the ambient plane sends $J$ to the unit circle. The map

$$\Theta : \mathrm{Homeo}(O) \longrightarrow S^1, \qquad k \longmapsto \frac{h(k(b)) - h(k(a))}{|h(k(b)) - h(k(a))|} \tag{C.32.7}$$

has induced homomorphism $\pi_1 \mathrm{Homeo}(O) \to \mathbb{Z}$ that is an isomorphism. Indeed, the full rotation loop in the chosen plane or annulus model preserves $J$ and has restriction winding one. The explicit computation in C.29 shows that its image under (C.32.7) also has winding one. A homomorphism $\mathbb{Z} \to \mathbb{Z}$ whose image contains 1 is an isomorphism. Formula (C.32.7) extends as a continuous map on all ambient plane homeomorphisms by the same two evaluations. The ambient-null loop therefore has zero image under it, so its restricted fundamental-group class is zero. This proves the second assertion in all cases.

Finally take an open orbit from C.28 and let $Q$ be the closed restricted operator image supplied by C.30. When this group is discrete, C.30 gives a free proper action with precisely that discrete topology, so (C.32.2) applies. The restricted canonical plane loop commutes with $Q$ and is null-homotopic in $\mathrm{Homeo}(O)$ by the second assertion. Its null-homotopy supplied by (C.32.1) commutes with $Q$, hence with the image of the operator stabilizer. The kernel in (C.30.4) acts identically on $O$ throughout. This proves the stated equivariant conclusion with the actual restriction homomorphism. □

**Transport and frontier control.** One may choose the obtained contraction on one representative of an operator orbit of open $G$-orbits and transport it to the others by conjugation. Equivariance under the setwise stabilizer makes this independent of the transporting operator. This establishes well-defined orbitwise families. The covering lift above supplies no uniform

estimate at orbit frontiers. C.33 subsequently treats all positive-dimensional proper closed operator images intrinsically, and C.34 proves joint continuity on the full nonopen locus. C.35 records the obstruction to extending its particular median prescription, while C.36 constructs compatible alternatives on finite unions of closures with two-dimensional proper operator image. The remaining frontier and accumulation requirements are stated in C.12, which remains open.

**C.33. [Proved] All proper operator groups on an open orbit.** Let $O$ be a connected noncompact second-countable surface without boundary. Let $Q$ be a closed locally compact subgroup of $\mathrm{Homeo}(O)$, with its compact-open topology, acting freely and properly. Then inclusion induces an injection

$$\pi_1\Big(C_{\mathrm{Homeo}(O)}(Q), \mathrm{id}\Big) \longrightarrow \pi_1(\mathrm{Homeo}(O), \mathrm{id}). \tag{C.33.1}$$

For every open orbit of the canonically lifted hyperbolic centralizer loop in C.28, its restriction therefore has a based null-homotopy commuting with the entire restricted operator stabilizer. This conclusion retains the restriction kernel in C.30 and imposes no finiteness or discreteness hypothesis on the original operator group. It supplies an intrinsic null-homotopy on each open orbit; continuity across orbit frontiers remains unproved.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; arbitrary proper-closure open-orbit reduction for C.12.

*Proof.* Write $N = C_{\mathrm{Homeo}(O)}(Q)$. The topological group $Q$ acts faithfully on the connected three-manifold $O \times \mathbb{R}$ by

$$q(x, s) = (qx, s). \tag{C.33.2}$$

This action is continuous: on any compact subset of $O \times \mathbb{R}$ the first coordinate is controlled on its compact projection to $O$. Pardon proves that a locally compact group acting faithfully on a connected three-manifold is a Lie group [61, Theorem 1.5 and the reduction preceding it]. Thus $Q$ is a Lie group. Since it is a subspace of the second-countable homeomorphism group, it is second countable. Set $A = Q^0$ and $d = \dim A$. Properness and freeness make $q \mapsto qx$ a proper embedding of $A$ in $O$ for each $x$. Invariance of dimension gives $d \le 2$. The case $d = 0$ is precisely C.32.

If $d = 2$, invariance of domain makes every $A$-orbit open. Connectedness gives a single orbit, and evaluation $A \to O$ is a homeomorphism. For $q \in Q$, choose $a \in A$ with $qx = ax$. Then $a^{-1}q$ fixes $x$, so freeness gives $q = a$ and $Q = A$. Under $O \cong A$ with its left translation action, the elements of $N$ are exactly the right translations. The correspondence is an anti-isomorphism of topological groups, and evaluation $N \to O$ is a homeomorphism. A loop in $N$ null-homotopic in $\mathrm{Homeo}(O)$ has a null evaluation loop in $O$, and hence is null-homotopic in $N$. This proves (C.33.1) in dimension two.

It remains to treat $d = 1$. The connected one-dimensional Lie group $A$ is $\mathbb{R}$ or $\mathbb{R}/\mathbb{Z}$, written additively. Its restricted action is still free and proper. Palais' slice theorem gives a locally trivial principal bundle

$$p_0 : O \longrightarrow B = O/A. \tag{C.33.3}$$

Here all hypotheses of the theorem hold: the surface is completely regular, $A$ is Lie, and a relatively compact neighborhood has relatively compact transporter by properness, which is the Cartan condition. A slice at a point has trivial stabilizer. The resulting local trivialization is the map $(a, s) \mapsto as$ from $A \times S$ onto the open saturated neighborhood of that slice. This is the free-action specialization of [60, Theorem 2.3.3]; its maps and inverse are also given in [45, Theorem 3.7].

We verify the dimension and local topology of the base rather than assuming differentiability of this action. The elementary fact needed is that a metrizable space $S$ for which $S \times (-1, 1)$

is a surface is a one-manifold without boundary. Indeed, projecting path-connected open neighborhoods in the product shows that $S$ is locally path connected. It has no isolated point, since an isolated point would give an open interval in a surface. Every embedded open arc in $S$ is open: its product with $(-1,1)$ maps injectively into a surface, and invariance of domain applies.

Fix $p \in S$. Local path connectedness and the absence of isolated points give an embedded arc $\alpha : [0,1] \to S$ with $\alpha(0) = p$. Choose a path-connected open neighborhood $W$ of $p$ avoiding $\alpha(1)$. If $W$ contains a point outside $\alpha([0,1])$, join $p$ to that point by an embedded arc in $W$. This second arc cannot enter and subsequently leave $\alpha((0,1))$: that open arc has boundary contained in $\{p, \alpha(1)\}$, and an embedded arc in $W$ cannot revisit $p$ or pass through $\alpha(1)$. Thus the two arcs meet only at $p$, and together contain an arc having $p$ as an interior point. Such an open arc is an open neighborhood of $p$ by the preceding paragraph. In the other case $W \subseteq \alpha([0,1))$, and a smaller neighborhood of $p$ is a half interval. Its product with $(-1,1)$ has zero local second homology at $(p,0)$, whereas a surface without boundary has local second homology $\mathbb{Z}$. This is impossible. The elementary fact follows. Applied to the slices in (C.33.3), it proves that $B$ is a one-manifold without boundary. Properness gives a Hausdorff quotient, the open quotient map gives second countability, and connectedness is inherited from $O$. Consequently $B$ is $\mathbb{R}$ or $\mathbb{R}/\mathbb{Z}$.

Every principal $A$-bundle over either of these one-manifolds is trivial. One can see the actual section as follows. Over an interval, local trivializations are joined successively across overlapping subintervals, adjusting one section by its continuous $A$-valued transition function. A locally finite interval exhaustion gives a section on $\mathbb{R}$. Over the circle, first cut at one point and choose a section on the resulting closed interval. Its endpoint discrepancy is an element of the connected group $A$; multiplying the section by a path from 0 to the negative of that element makes the endpoints agree, so that it descends to the circle. The map $(a,b) \mapsto a\,s(b)$, with inverse given in each bundle chart, therefore identifies

$$O \cong A \times B, \qquad c(a,b) = (c+a,b) \quad (c \in A). \tag{C.33.4}$$

The pair $A = B = \mathbb{R}/\mathbb{Z}$ is excluded because $O$ is noncompact.

Let $\Delta = Q/A$. This discrete group acts freely and properly on $B$. Freeness follows directly: an element of $Q$ fixing a point of $B$ sends any point of its fibre to an $A$-translate; after that translate is removed, it fixes a point of $O$ and is the identity. For properness, suppose $b_j \to b$ and $\delta_j b_j \to b'$ in $B$. Local sections give $x_j \to x$ above $b_j$ and $y_j \to y$ above $\delta_j b_j$. Choose a representative $q_j$ of $\delta_j$ and $a_j \in A$ such that $a_j q_j x_j = y_j$. Properness of the $Q$-action gives a convergent subsequence of $a_j q_j$, so their components $\delta_j$ are eventually constant on that subsequence. This proves properness of the discrete action on the locally compact metrizable space $B$.

We spell out this discrete action. On $\mathbb{R}$, an orientation reverser has a fixed point and is excluded by freeness. If $\Delta$ is nontrivial, its orbit of a point is closed and discrete, with a nearest point to the right. The element taking the chosen point to that next point generates $\Delta$: its iterates are unbounded in both directions, and any intervening orbit point would contradict the choice of the nearest one. Choosing coordinates on one fundamental interval and extending by its iterates identifies the action with

$$\Delta = \mathbb{Z}, \qquad \beta(b) = b + 1. \tag{C.33.5}$$

On $\mathbb{R}/\mathbb{Z}$, properness makes $\Delta$ finite and freeness again excludes orientation reversers. The action on one finite orbit preserves its cyclic order, so it identifies $\Delta$ with a cyclic group $C_n$. Coordinates on the successive complementary intervals identify a generator with

$$\beta(b) = b + 1/n \quad \text{in } \mathbb{R}/\mathbb{Z}. \tag{C.33.6}$$

The trivial group is also allowed on either base.

Suppose $\Delta$ is nontrivial and choose a representative $q \in Q$ of the indicated generator. Conjugation by $q$ on $A$ is multiplication by $\lambda$, where $\lambda \in \mathbb{R}^\times$ for $A = \mathbb{R}$, and $\lambda \in \{1,-1\}$ for

$A = \mathbb{R}/\mathbb{Z}$. Equation (C.33.4) therefore expresses its action exactly as

$$q(a, b) = (\lambda a + \tau(b), \beta(b)) \tag{C.33.7}$$

for a continuous function $\tau : B \to A$. We next give a coordinate change removing this translation term, with all component relations retained.

For $B = \mathbb{R}$ and $\Delta = \mathbb{Z}$, choose a continuous function $c : [0, 1] \to A$ whose endpoints satisfy $c(1) = \lambda c(0) + \tau(0)$, and extend it in both directions by

$$c(b + 1) = \lambda c(b) + \tau(b). \tag{C.33.8}$$

The endpoint equation makes the extension continuous at every integer; invertibility of multiplication by $\lambda$ gives the extension towards negative integers. In the new fibre coordinate $a' = a - c(b)$, (C.33.7) becomes $q(a', b) = (\lambda a', b + 1)$.

For $B = \mathbb{R}/\mathbb{Z}$ and $\Delta = C_n$, necessarily $A = \mathbb{R}$. Write $q^n = T_{a_0}$, where $T_{a_0}$ is translation by $a_0$ on the fibre. Conjugation gives $\lambda^n = 1$ and $\lambda a_0 = a_0$. If $\lambda = 1$, replacing $q$ by $T_{-a_0/n} q$ makes $q^n = 1$ and does not change the group generated with $A$. If $\lambda = -1$, then $n$ is even and $a_0 = 0$, so $q^n = 1$ already. Thus the negative value of $\lambda$ is retained. For this finite-order representative define the real-valued function

$$c(b) = \frac{1}{n} \sum_{k=0}^{n-1} \mathrm{pr}_A\left(q^k(0, \beta^{-k} b)\right). \tag{C.33.9}$$

The affine nature of (C.33.7), followed by cyclic reindexing using $q^n = 1$, gives $c(\beta b) = \lambda c(b) + \tau(b)$. The same coordinate change $a' = a - c(b)$ now gives $q(a', b) = (\lambda a', \beta b)$. We have proved, by explicit changes of coordinates, that the nonidentity components can be represented as

$$q(a, b) = (\lambda a, \beta b). \tag{C.33.10}$$

When $\Delta$ is trivial no such additional equation is needed.

Commuting with all translations in (C.33.4) forces a homeomorphism of $A \times B$ to have the form

$$h(a, b) = (a + \phi(b), \psi(b)), \qquad \phi \in C(B, A), \quad \psi \in \mathrm{Homeo}(B). \tag{C.33.11}$$

The inverse is $(a, b) \mapsto (a - \phi(\psi^{-1} b), \psi^{-1} b)$. These formulas and evaluation at the zero section give mutually inverse continuous maps for the compact-open topologies. In the nontrivial-component case, commuting with (C.33.10) is exactly the pair of equations

$$\psi\beta = \beta\psi, \qquad \phi(\beta b) = \lambda \phi(b). \tag{C.33.12}$$

Thus the map (C.33.11) identifies $N$, as a space, with the product of the function space defined by the second equation and the homeomorphism centralizer defined by the first. In particular the two constraints can be contracted separately without changing the group $Q$ being centralized.

Let $h_t$ be a based loop in $N$ null-homotopic in $\mathrm{Homeo}(O)$, and let $(\phi_t, \psi_t)$ be its coordinates. The path $\psi_t$ preserves the orientation of the one-manifold $B$. When $A = \mathbb{R}$, multiplication of $\phi_t$ by $1 - s$ contracts the function-coordinate loop to zero and preserves its equation in (C.33.12), including $\lambda = -1$. For $B = \mathbb{R}$, C.27 contracts $\psi_t$ in the required centralizer, with the trivial operator group when $\Delta = 1$. For $B = \mathbb{R}/\mathbb{Z}$, the evaluation loop $h_t(0, b) = (\phi_t(b), \psi_t(b))$ is null-homotopic in $\mathbb{R} \times (\mathbb{R}/\mathbb{Z})$; hence $\psi_t(b)$ has winding zero. C.29 contracts $\psi_t$ in the centralizer of the finite rotation group in (C.33.6).

When $A = \mathbb{R}/\mathbb{Z}$, necessarily $B = \mathbb{R}$. The based family $\phi_t(b)$ has a unique continuous lift $\varphi : [0, 1] \times \mathbb{R} \to \mathbb{R}$ with $\varphi_0 = 0$. Evaluation of $h_t$ in the annulus $A \times \mathbb{R}$ shows that $\varphi_1(b) = 0$:

this endpoint is the winding number of the first coordinate of the null evaluation loop. If $\Delta = \mathbb{Z}$, equation (C.33.12) says that

$$\varphi_t(b+1) - \lambda\varphi_t(b) \in \mathbb{Z}. \tag{C.33.13}$$

Its left side is a continuous function of $(t, b)$, equal to zero at $t = 0$, and is therefore identically zero. Multiplying $\varphi$ by $1 - s$ and descending to $\mathbb{R}/\mathbb{Z}$ contracts $\phi_t$ while preserving exact equivariance. The same scaling applies when $\Delta = 1$, without (C.33.13). C.27 contracts the remaining $\psi_t$ on the line. All these contractions keep $t = 0, 1$ fixed. Equations (C.33.11)–(C.33.12), including the inverse formula, show that their product is a continuous based contraction in $N$. This completes the proof of (C.33.1).

Finally, for an open loop-group orbit from C.28, C.30 supplies the actual closed locally compact free proper operator image $Q$. The restricted canonical plane loop commutes with $Q$. It is null-homotopic in Homeo$(O)$ by the second assertion of C.32, so (C.33.1) contracts it in the exact centralizer. Every original operator in the setwise stabilizer acts through its restriction to $Q$, and the restriction kernel acts identically on $O$. The stated conclusion follows with that kernel retained. □

**What the open-orbit result provides.** Equations (C.33.11)–(C.33.13) treat every component of the proper operator group, including a circle fibre with a nontrivial $\mathbb{Z}$-action on its line base. There is no replacement of the base-translation centralizer by the homeomorphism group of its quotient circle. As in C.32, the obtained contraction can be transported to other open orbits in the same operator orbit; exact stabilizer equivariance makes the transport independent of choices. This proof supplies no uniform control near the boundary of the open orbit. C.34 subsequently proves joint continuity, including inverses, on the entire closed union of nonopen orbits. C.35 exhibits an exact obstruction to extending that particular median prescription over an adjacent open orbit. C.36 constructs compatible contractions on closures with two-dimensional proper operator image and proves continuity on every finite union of those closures. C.37 supplies the uniform three-word estimate and gives a continuous contraction on the closed union $X_2$ of all those open orbits, including infinite accumulation. Extension across $X_2$ to its exterior and frontier-compatible choices for the dimension-zero and dimension-one operator images remain additional requirements. C.12 retains its unresolved status.

**C.34. [Proved] The median contraction on all nonopen orbits.** Let $f_t$ be a based null-homotopic loop in Homeo$^+(\mathbb{R}^2)$, and use $P, G, Y$ from C.28 and C.31. The orbitwise bijections $Q_{s,t} : Y \to Y$ defined in C.31 are homeomorphisms. Both maps

$$[0,1]^2 \times Y \longrightarrow Y, \qquad (s,t,x) \longmapsto Q_{s,t}^{\pm 1}(x) \tag{C.34.1}$$

are continuous. Thus these formulas give a based compact-open null-homotopy on the entire closed nonopen locus, commuting with every homeomorphism commuting with all $f_t$. In particular, a hyperbolic centralizer loop whose canonically lifted generated group has no open orbit is null-homotopic in the original surface centralizer. No extension over the omitted open orbits is asserted.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; nonopen-stratum assembly in C.12.

*Proof.* We prove continuity at a nonfixed orbit. Continuity at the common fixed set, bijectivity on $Y$, equivariance, and the exact inverse identity have already been proved in C.31.

*Short paths moving in either direction.* Let $q$ be a point of a nonopen nontrivial orbit. It lies on an intrinsic line or a circle by C.28. There are paths $g_s$ from the identity, formed from two factors in $P$, which move $q$ a nonzero amount in either chosen intrinsic direction while keeping its whole track inside any prescribed sufficiently small intrinsic arc about $q$. To see this, $f_t(q)$ is

not identically $q$; otherwise every generator, and hence every element of $G$, would fix $q$. Choose a component $(a, b)$ of $\{t : f_t(q) \neq q\}$ and a sufficiently short initial subinterval $[a, a + \eta]$. The path

$$h_s = f_{a+s\eta} f_a^{-1}, \qquad 0 \leq s \leq 1, \tag{C.34.2}$$

starts at the identity, and $h_s(q) = f_{a+s\eta}(q)$. C.28 gives compact-open continuity on an intrinsic line; the circle restriction is continuous as well. Consequently $h_s$ and $h_s^{-1}$ can be made uniformly close to the identity on an intrinsic neighborhood of $q$. Their endpoint displacements at $q$ have opposite intrinsic signs. One of these two paths has the chosen sign, and its track stays in the prescribed arc. The paths also become uniformly small on each fixed ambient compact set as $\eta \to 0$. Every such $g_s$ is a product of two elements of $P$. For any $z \in K_n(x) = P^n x$,

$$g_s(z) \in K_{n+2}(x). \tag{C.34.3}$$

In particular, the same compact-word argument as in C.28 proves intrinsic continuity of its track on every nonopen orbit, including when $z$ varies along a compact intrinsic subarc.

*A small cap cannot persist at a moving limit point.* We give the local argument used below. Straighten a compact arc of a limit orbit, extending it a little at both ends, so that it is an interval of the horizontal axis in a plane chart. Suppose compact intrinsic arcs $A_j$ in nearby nonopen orbits lie in strips of height tending to zero around this interval. Let $q_j \in A_j$ attain the minimum $m_j$ of the horizontal coordinate on $A_j$, suppose $q_j \to q = (m, 0)$, and suppose both endpoints of $A_j$ have horizontal coordinates at least $m + \delta$ for a fixed $\delta > 0$. Take a rectangle

$$U = (m - \rho, m + \rho) \times (-\varepsilon, \varepsilon), \qquad 0 < \rho < \delta/3, \tag{C.34.4}$$

whose closure stays in the chart. For large $j$, $q_j$ lies in $U$, all of $A_j$ has vertical coordinate of absolute value less than $\varepsilon/2$, and its horizontal minimum is greater than $m - \rho/2$. Starting at $q_j$ in each intrinsic direction towards the two endpoints of $A_j$, take the first point outside $U$. Both first-exit points lie on its right edge. They are distinct points of the embedded arc. Let $J_j$ be the closed subarc between these points. Its interior lies in $U$, and every point of $J_j$ has horizontal coordinate at least $m_j$.

Use (C.34.2), taking the inverse when needed, to choose a path $g_s$ whose $q$-track stays in $U$ and whose endpoint is strictly to the left of $q$. This choice is possible because $q$ is nonfixed. The track is a compact subset of $U$, so joint continuity gives $g_s(q_j) \in U$ for every $s$ and every sufficiently large $j$. Its intrinsic continuity from (C.34.3) now matters: a continuous path in the intrinsic line or circle starting at an interior point of $J_j$ cannot leave $J_j$ without meeting one of its two endpoints. Both endpoints lie outside the open rectangle $U$. Thus the entire $g_s(q_j)$-track stays in $J_j$. At $s = 1$ its horizontal coordinate is at least $m_j$, whereas

$$g_1(q_j) \longrightarrow g_1(q), \qquad \mathrm{pr}_1(g_1(q)) < m = \lim_j m_j. \tag{C.34.5}$$

This contradiction excludes the cap. The argument with horizontal maxima and a path moving to the right is identical. The two first-exit points are the explicit obstruction to escaping the cap along another part of the same orbit.

The same argument excludes a whole Jordan-curve orbit contained in such a shrinking strip and having a nondegenerate interval as its limit: use a global minimum of the horizontal coordinate on the circle, and in its two directions take first exits towards a point whose coordinate is bounded above that minimum. The resulting subarc $J_j$ again has two distinct endpoints outside $U$.

*No excess in subarcs near an intrinsic line.* Fix a point $p$ on an intrinsic line orbit and take $x_j \to p$ in $Y$. For each fixed $n$, compactness of $P^n$ and continuity of word evaluation give

$$K_n(x_j) \longrightarrow K_n(p) \quad \text{in Hausdorff distance.} \tag{C.34.6}$$

The limit is a nondegenerate compact arc. Straighten a slightly larger compact arc of the limit orbit. Then (C.34.6) places the nearby $K_n(x_j)$ in shrinking horizontal strips in that chart. Each $K_n(x_j)$ is an arc, a point, or a whole circle, by C.28 and connectedness. The whole-circle possibility is excluded for all sufficiently large $j$ by the preceding cap argument.

Suppose $u_j, v_j \in K_n(x_j)$ tend to $u, v \in K_n(p)$. Write $[u_j, v_j]_j$ for the subarc contained in $K_n(x_j)$. Then

$$z_j \in [u_j, v_j]_j, \quad z_j \longrightarrow z \quad \Longrightarrow \quad z \in [u, v]. \tag{C.34.7}$$

Indeed, an alleged limit point strictly to the left of both $u, v$ produces a horizontal minimum of $[u_j, v_j]_j$ whose two endpoints stay a fixed distance to its right. A subsequence of these minimum points converges, by (C.34.6), to a point of the nonfixed limit orbit. This is the forbidden cap. A point to the right is excluded by the maximum version. The assertion includes $u = v$: then all of the intervening subarc tends to $u$. This proves the needed continuity of subarcs using the path action; Hausdorff convergence alone has not been used to infer it.

It follows that medians of three points in $K_n(x_j)$ converge to the median of their limits. Each median lies in all three pairwise connecting subarcs. Every subsequential limit therefore lies, by (C.34.7), in all three corresponding limit intervals. Their intersection is the single limit median, including when two or three of the limiting points coincide.

*A circle limit: the precise local covering.* Let the orbit of $p$ be a Jordan curve $C$. Choose a collar annulus $W$ of $C$, with coordinates $(\mathbb{R}/\mathbb{Z}) \times (-1, 1)$ sending $C$ to the middle circle, and write

$$\pi_W : \widetilde{W} = \mathbb{R} \times (-1, 1) \longrightarrow W \tag{C.34.8}$$

for its universal covering. The compact family $f_t$ preserves $C$. Therefore a sufficiently small open disc $V$ about $p$ satisfies $f_t(V) \subset W$ for all $t$. The actual map

$$[0, 1] \times V \longrightarrow W, \qquad (t, x) \longmapsto f_t(x) \tag{C.34.9}$$

has a unique continuous lift $\widehat{f}_t(x)$ after choosing a lift of the initial inclusion $V \to W$. Its endpoint differs from its starting point by an integer deck translation depending continuously on $x \in V$. It is zero at $p$, by C.29's exact plane-to-circle winding comparison, so it is zero on all of the connected disc $V$. Thus these are closed lifted point tracks. For $x_j \to p$ in $V \cap Y$ put

$$T_j(r) = \{\widehat{f}_u(x_j) : 0 \le u \le r\}, \qquad T(r) = \{\widehat{f}_u(p) : 0 \le u \le r\}. \tag{C.34.10}$$

Joint continuity on the compact parameter interval shows $T_j(r_j) \to T(r)$ in Hausdorff distance whenever $r_j \to r$. The set $T(1)$ is a nondegenerate compact interval on the line $\mathbb{R} \times \{0\}$.

We verify the intrinsic meaning of the lifted nearby tracks. An intrinsic line orbit, or a proper component of a circle orbit inside $W$, has an intrinsic interval as its component through $x_j$. Its continuous point track stays in that component and lifts to an injectively parametrized interval in $\widetilde{W}$, with embedded compact subintervals. If the entire circle orbit lies in $W$, it is a Jordan curve. Its winding in $W$ is 0 or $\pm 1$: the planar Jordan winding around the complementary inner disc has exactly these values. Its lifted components are therefore Jordan curves in the first case and intrinsic lines in the second. Consequently $T_j(1)$ is a compact arc, a point, or a whole lifted Jordan curve, with the corresponding intrinsic topology.

The cap argument applies in this covering chart with the original group paths. Here is the exact local map needed to justify that use. At any limiting extremum $q \in T(1)$ choose the small rectangle $U$ of (C.34.4) with horizontal width less than one, so that $\pi_W|_U$ is injective. Choose a path (C.34.2) at $\pi_W(q)$ with the required intrinsic sign and sufficiently small that its images of a compact neighborhood of $\pi_W(q)$ stay in $W$. The map

$$(s, z) \longmapsto g_s(\pi_W(z)) \tag{C.34.11}$$

on that small lifted neighborhood has a unique continuous lift starting with $z \mapsto z$. Its $q$-track lies in $U$ and has the required horizontal displacement. For $q_j \to q$ its whole $q_j$-track also stays in $U$, by compact parameter continuity. Downstairs it is an intrinsically continuous path in the original orbit, by (C.34.3). Its lift cannot leave the cap subarc without meeting one of its first-exit endpoints on $\partial U$. Thus (C.34.5) applies unchanged. No action of $G$ on the entire annulus or its covering has been assumed.

This excludes a whole lifted Jordan curve for large $j$ and proves (C.34.7), and hence median continuity, for compact subarcs of $T_j(1)$. These intrinsic interval operations are exactly the ones used by C.27 and C.29. For a line or a proper circle component the lift of its interval is order preserving or reversing. For an essential circle the annular covering is its universal line covering. For a contractible circle, the now proper arc $T_j(1)$ lies in a single interval of its universal line covering; a path leaving that interval would have to cover the whole circle. In all cases the based circle lift or intrinsic line formula, followed by projection through (C.34.8), gives the original $Q_{s,t}(x_j)$. Median invariance under reversal removes any choice of the interval orientation.

*Continuity of the prefix-clamping formula.* We can now handle a sequence $(s_j, t_j, x_j) \to (s, t, p)$ with $p$ nonfixed. For a line limit use the actual point tracks $T_j(r) = \{f_u(x_j) : 0 \le u \le r\}$ inside $K_1(x_j)$. For a circle limit use (C.34.10). In either case $T_j(1)$ is eventually an intrinsic compact arc, its subarcs satisfy (C.34.7), and its prefix tracks have their stated Hausdorff limits. Set $r_j = 1 - s_j$ and $r = 1 - s$. Let $a_j, b_j$ be the two unordered endpoints of the prefix arc $T_j(r_j)$, allowing equality. Every subsequence has a further subsequence on which both endpoints converge, say to $a, b \in T(r)$. By (C.34.7), the Hausdorff limit of their intervening arc is contained in $[a, b]$. It is also exactly $T(r)$ by the prefix-track continuity. Since $a, b$ belong to $T(r)$, their interval is contained in $T(r)$. Therefore

$$T(r) = [a, b]. \tag{C.34.12}$$

In particular the limiting unordered pair is precisely the endpoint pair of the prefix interval, including a degenerate prefix.

The formula of C.27 or its C.29 lift is the median of $a_j, b_j$ and the current track point at $t_j$. The median continuity just proved and (C.34.12) give its limit as the median of the limiting prefix endpoints and the current point at $t$. This is $Q_{s,t}(p)$, with projection through (C.34.8) in the circle case. Every subsequence has this same limit. The domains are metrizable, so this proves joint continuity on $[0, 1]^2$ times the nonfixed part of $Y$. C.31 supplies the remaining fixed points.

Apply the same argument to the based inverse loop $u \mapsto f_u^{-1}$. It is ambient-null, generates the same group $G$, and has the same closed locus $Y$. The exact identity $Q[f]^{-1} = Q[f^{-1}]$ in C.31 proves joint continuity of the inverse family in (C.34.1). Thus all the orbitwise bijections are homeomorphisms of $Y$. The compact-open continuity follows from the evaluation exponential law, since $Y$ is a closed, locally compact subspace of the plane. The based endpoints and equivariance are those already proved orbitwise in C.31.

Finally, when the canonical lifted group of a hyperbolic surface loop has no open orbit, $Y = \widetilde{U} \cong \mathbb{R}^2$. The obtained plane homeomorphisms commute with the full operator lift group, including all deck transformations. They descend by the exact equations

$$\overline{Q}_{s,t}\, p_U = p_U Q_{s,t} \quad (p_U : \widetilde{U} \to U) \tag{C.34.13}$$

to homeomorphisms of $U$ commuting with every original operator. Their joint continuity and that of their inverses descend through the open covering quotient, giving compact-open continuity downstairs. The endpoints in (C.34.13) are the original loop and the identity, with both based edges fixed. Hence this is its null-homotopy inside the original centralizer. At every parameter it lies in the identity component by these same paths. Equivalently, its lift is the canonical one: two deck-commuting lifts of such a map differ by a central deck transformation, and the hyperbolic

surface group has trivial center. This proves the stated corollary with the actual lift and descent maps. □

**Remaining global extension.** C.34 establishes the full nonopen-locus homeomorphism family and its inverse. C.33 provides intrinsic equivariant contractions on each open orbit. C.35 proves that this particular median family need not extend equivariantly over an adjacent open orbit. C.36 and C.37 construct compatible alternatives on the closed union $X_2$ of open orbits with two-dimensional proper operator image, including every infinite accumulation. Matching that boundary family to the exterior of $X_2$, and treating the other proper operator images at their frontiers, remain the assembly requirements in C.12, whose general status remains open.

**C.35. [Proved] An obstruction to extending the fixed median prescription.** There is a continuous $\mathbb{R}^2$-action on the plane and a based ambient-null loop in its centralizer such that the C.34 median homeomorphisms of the closed nonopen locus do not extend equivariantly over an adjacent open orbit at one parameter. The closed restricted operator image on that open orbit is the free proper translation group $\mathbb{R}^2$. The original loop nevertheless has an explicit equivariant null-homotopy on the entire plane. Thus this example refutes extension of the fixed C.34 median prescription; it is not a failure of the level-0 comparison.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; obstruction to extending a fixed contraction prescription in the C.12 proof route.

*Proof.* Use the closed simplex

$$\Delta = \{(x, y, z) \in \mathbb{R}^3_{\geq 0} : x + y + z = 1\}.$$

For $(a, b) \in \mathbb{R}^2$ define

$$T_{a,b}(x, y, z) = \frac{(e^a x, e^b y, z)}{e^a x + e^b y + z}. \tag{C.35.1}$$

The denominator is positive on $\Delta$. Direct substitution gives $T_{a,b}T_{c,d} = T_{a+c,b+d}$ and $T_{a,b}^{-1} = T_{-a,-b}$, so (C.35.1) is a continuous action by homeomorphisms, fixing the three vertices and preserving each open edge. On the interior the coordinate homeomorphism

$$(x, y, z) \longmapsto (u, v) = \big(\log(x/z), \log(y/z)\big), \qquad (u, v) \longmapsto \frac{(e^u, e^v, 1)}{1 + e^u + e^v} \tag{C.35.2}$$

turns this action into $(u, v) \mapsto (u + a, v + b)$. Its restricted image is therefore the closed free proper translation group on $\mathbb{R}^2$.

Double $\Delta$ along its entire boundary. Since $\Delta$ is a closed disc, this double is a sphere. Apply (C.35.1) on both copies; the formulas agree on their identified boundary and hence give a continuous sphere action. Remove the fixed vertex $(0, 0, 1)$. The remaining surface $M$ is a plane, and the action restricts to it. It is orientation preserving: each map preserves the orientation on either triangle interior, and preserves the two sheets of the oriented double.

Write $Q = \{T_{a,b} : (a, b) \in \mathbb{R}^2\}$ for this operator image. Its orbits in $M$ are the two triangle interiors, the three open edges, and the remaining two vertices. Use the following coordinates on the edges:

| edge | coordinate in $\mathbb{R}$ | translation by $T_{a,b}$ |
|---|---|---|
| $y = 0$ | $\log(x/z)$ | $a$ |
| $x = 0$ | $\log(y/z)$ | $b$ |
| $z = 0$ | $\log(x/y)$ | $a - b$. |

(C.35.3)

Thus the union $Y$ of the three edges and the two vertices is exactly the closed nonopen orbit locus.

Let $w(t) = (a(t), b(t))$ be the piecewise linear based loop with successive values

$$(0,0), \quad (1,1), \quad (1,0), \quad (0,0) \quad \text{at } t = 0, \tfrac{1}{3}, \tfrac{2}{3}, 1, \tag{C.35.4}$$

and put $f_t = T_{a(t),b(t)}$. All its values commute with $Q$, and

$$F_{\sigma,t} = T_{(1-\sigma)a(t),(1-\sigma)b(t)} \tag{C.35.5}$$

is a based equivariant null-homotopy on the entire plane. Continuity, including that of the inverse family, follows from (C.35.1), which is already continuous on the compact double and preserves the removed point. The subgroup generated by the loop is exactly $Q$: its first segment contains all diagonal translations $(r, r)$ for $0 \le r \le 1$, and the differences of points of the second segment contain all $(0, r)$ for $-1 \le r \le 1$. Integer sums and inverses give both full coordinate axes and hence $\mathbb{R}^2$. Consequently C.34 uses precisely the orbit locus $Y$ just described.

At $s = 2/3$ the prefix in the C.27 formula ends at $r = 1 - s = 1/3$. By (C.35.3)–(C.35.4), the three prefix translation intervals are, in their displayed order,

$$[0,1], \quad [0,1], \quad \{0\}. \tag{C.35.6}$$

At $t = 2/3$, the current edge translations are $(1, 0, 1)$. Taking the prescribed medians with the prefix endpoints gives

$$Q^{\mathrm{med}}_{2/3,\,2/3}\Big|_Y : \quad \text{edge translations } (1,0,0). \tag{C.35.7}$$

The vertices stay fixed. These edge maps fit continuously at the vertices, as also guaranteed by C.34.

Every $Q$-equivariant homeomorphism from one triangle interior to itself is a translation in (C.35.2). Indeed, commutation with all translations gives $h(u, v) = (u, v) + h(0, 0)$. If it maps one interior to the other, the same equation holds in the identical coordinates on the target sheet. The sheet exchange fixes the common boundary, so in either case any continuous boundary extension has edge translation parameters

$$R : \mathbb{R}^2 \longrightarrow \mathbb{R}^3, \qquad R(A, B) = (A, B, A - B). \tag{C.35.8}$$

This is the exact restriction morphism. Its image is the kernel of

$$\ell : \mathbb{R}^3 \longrightarrow \mathbb{R}, \qquad \ell(r_1, r_2, r_3) = r_1 - r_2 - r_3. \tag{C.35.9}$$

But $\ell(1, 0, 0) = 1$, so (C.35.7) does not lie in that image. It cannot extend equivariantly even over a single adjacent interior. This proves all assertions, while (C.35.5) gives the explicit global contraction of the original loop. □

**Executed endpoint certificate and scope.** The standard-library rational calculation `code/part_b/c35_median_extension_certificate.py` computes the prefix extrema, the three median values, the restriction matrix in (C.35.8), and the nonzero obstruction (C.35.9). Its JSON output records $(1, 0, 0)$ and residual 1. This certificate verifies the finite endpoint calculation; the preceding simplex, doubling, and restriction arguments prove the topological statement.

C.34 is a homeomorphism construction on the whole nonopen locus, and C.33 contracts the open-orbit restrictions. C.35 proves that a global argument must permit changing the nonopen-locus contraction to satisfy the actual boundary restriction morphisms of the open components. The remaining problem is therefore the existence of compatible equivariant contractions, not extension of the particular median family without modification. C.36 and C.37 meet the actual boundary equations on the closed union $X_2$ of open orbits with two-dimensional proper operator image, including infinite accumulation. Extending this family compatibly to the exterior of $X_2$

and the other proper operator cases remains required. The general C.12 comparison remains unresolved.

**C.36. [Proved] Compatible contractions on two-dimensional proper closures.** Let $f_t$ be a based ambient-null plane loop, let $G$ be its generated group, and let $H$ be any group of plane homeomorphisms commuting with every $f_t$. For an open $G$-orbit $O$, let $Q_O$ be the closure of the restricted setwise $H$-stabilizer, as in C.30. For every such $O$ with $\dim Q_O = 2$, there is a based null-homotopy on the entire closure $\overline{O}$ through homeomorphisms of $\overline{O}$ that commute with its operator stabilizer. These closure contractions agree on every intersection $\overline{O}_1 \cap \overline{O}_2$ and are carried to one another by $H$. They therefore give a jointly continuous family, with jointly continuous inverse, on every finite union of these closures. No continuity across an infinite accumulating family of closures is asserted.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; frontier-compatible dimension-two operator closures in the C.12 proof route.

*Proof. The actual quotient action on the closure.* C.30 makes $Q_O$ a closed locally compact free proper group of homeomorphisms of $O$, and C.33 makes it Lie. Its dimension-two argument shows that $Q_O$ is connected and simply transitive on $O$, and that

$$N_O = C_{\mathrm{Homeo}(O)}(Q_O) \cong Q_O^{\mathrm{op}} \tag{C.36.1}$$

is also a connected, simply transitive two-dimensional Lie group. Evaluation $N_O \to O$ is a homeomorphism. The image of $G$ in $N_O$ is all of $N_O$: for any $n \in N_O$, transitivity of $G$ supplies $g \in G$ with $gx = nx$ at one point $x \in O$, and the injectivity of evaluation on $N_O$ then gives $g|_O = n$.

Put $L = \overline{G}$ in $\mathrm{Homeo}^+(\mathbb{R}^2)$. This is a Polish group. It preserves every open $G$-orbit $O$. Indeed, if $g_j \to l$ with $g_j \in G$, the identities $g_j(\partial O) = \partial O$ and the convergent inverse family give $l(\partial O) = \partial O$. For $x \in O$, $lx \in \overline{O}$ by convergence, and $lx \notin \partial O$ because $l$ preserves that boundary and is injective. Thus $lx \in O$; applying the same argument to $l^{-1}$ proves $l(O) = O$. The elements of $L$ still commute with $H$. Restriction consequently gives a continuous surjective homomorphism

$$r_O : L \longrightarrow N_O. \tag{C.36.2}$$

It is open by the open mapping theorem for Polish groups. Equivalently, apply the open orbit-map principle from [24, Theorem 3.5] to the transitive action of $L$ on the Baire space $N_O$ through (C.36.2).

The kernel of (C.36.2) fixes $O$ pointwise, and hence fixes $\overline{O}$ pointwise. Therefore there is a well-defined action

$$N_O \times \overline{O} \longrightarrow \overline{O}, \qquad (n, z) \longmapsto lz \quad \text{for any } l \text{ with } r_O(l) = n. \tag{C.36.3}$$

This action is jointly continuous: $r_O \times \mathrm{id}$ is an open quotient map, and the continuous $L$-action factors through it. No continuous section of (C.36.2) is required. In fact each $n$ has a preimage in $G$, as proved above. Hence (C.36.3) preserves every original $G$-orbit contained in $\overline{O}$, and its image on such an orbit is exactly the image of $G$. This conclusion does not require a claim that $L$ preserves every other $G$-orbit in the plane.

*The two-dimensional Lie contraction.* We record the Lie groups and the maps being used. A two-dimensional real Lie algebra is either abelian or has a basis $X, Y$ with $[X, Y] = Y$: in the nonabelian case its one-dimensional derived algebra is an ideal, and scaling a complementary basis vector gives that bracket. The simply connected groups are respectively $\mathbb{R}^2$ and the positive affine group of the line. In coordinates $(a, b)$, the latter has product

$$(a, b)(a', b') = (a + a',\ b + e^a b'). \tag{C.36.4}$$

Its exponential map is

$$\exp(u,v) = \left(u, v\,\frac{e^u - 1}{u}\right), \quad \text{with } \frac{e^u - 1}{u} = 1 \text{ at } u = 0. \tag{C.36.5}$$

The scalar factor is positive for every real $u$, so this is a global diffeomorphism. For $\mathbb{R}^2$ the exponential is the identity. The affine group has trivial center, so its connected groups with this Lie algebra have no further covering quotients. In the abelian case the connected group is a quotient of $\mathbb{R}^2$ by a discrete central subgroup. The compact torus is excluded here because $N_O \cong O$ is noncompact. Thus in every case the universal covering group $\widetilde{N}_O$ has a globally defined inverse log to its exponential.

Let $n_O(t) = f_t|_O \in N_O$. By the restriction assertion of C.32 this loop is null-homotopic in $\mathrm{Homeo}(O)$. Evaluation on $O$ and the homeomorphism $N_O \to O$ show that it is null-homotopic in $N_O$. Its unique based lift $\widetilde{n}_O(t)$ to $\widetilde{N}_O$ is therefore a closed loop. With $p_O : \widetilde{N}_O \to N_O$ the covering homomorphism, define

$$R^O_{s,t} = p_O\big(\exp((1-s)\log \widetilde{n}_O(t))\big) \quad \text{acting on } \overline{O} \text{ by (C.36.3).} \tag{C.36.6}$$

This is a continuous family of homeomorphisms with continuous inverse, starts with $f_t$ on $\overline{O}$, ends with the identity, and keeps both based edges fixed. On $O$ it lies in the exact centralizer of $Q_O$. Its commutation with the original setwise operator stabilizer extends to $\overline{O}$ by continuity and density.

*The common boundary image and its topology.* Take two of these closures and a $G$-orbit $B$ in their intersection. It is a point, an intrinsic line, or a circle: an open orbit cannot lie in the boundary of another open orbit. The point case is fixed throughout. For a nontrivial $B$, each $N_i = N_{O_i}$ acts transitively on $B$ by (C.36.3), and both action images are exactly $G|_B$. We verify that this common image has a single Lie topology, rather than identify abstract quotients without their topology.

For $p \in B$, its stabilizer in $N_i$ is closed. The homogeneous space $N_i/(N_i)_p$ maps continuously and bijectively to $B$. Its dimension is one: dimension zero would make the connected orbit a point, and dimension two would make $B$ open in the plane by invariance of domain. Its topology agrees with the intrinsic topology of $B$. Indeed, a compact interval in this homogeneous one-manifold maps to an embedded compact arc. In an intrinsic line orbit that arc is the unique connecting arc supplied by C.28; on a circle the same conclusion holds locally. The interiors of these arcs give intrinsic interval neighborhoods, so the bijective orbit map is a local, and therefore global, intrinsic homeomorphism. Thus the action is continuous also in the intrinsic topology.

Divide $N_i$ by its closed kernel on $B$. The resulting connected Lie group $M_i$ acts faithfully and transitively on that one-manifold, and has dimension at most two. In dimension one it is abelian, so a point stabilizer is normal and is therefore the kernel; faithfulness makes the stabilizer trivial. Evaluation then identifies its action with the translation group of $\mathbb{R}$ or the rotation group of $\mathbb{R}/\mathbb{Z}$. An abelian group of dimension two would give the same freeness conclusion and an orbit of dimension two, so cannot occur faithfully here. In the remaining case $M_i$ is the positive affine group (C.36.4). The identity component of a point stabilizer is a one-dimensional subgroup. It cannot be the translation subgroup, which is normal and would lie in the action kernel. Every other such subgroup is conjugate to the dilation subgroup. Its normalizer is that dilation subgroup itself, as is seen by conjugating its unique fixed point on the line. The full point stabilizer is consequently the dilation subgroup, and the homogeneous action is the usual positive affine action on $\mathbb{R}$.

These descriptions prove topological embedding of each actual image in $\mathrm{Homeo}(B)$ with its intrinsic compact-open topology. For translations or rotations one evaluation recovers the parameter. For the affine group, the images of two distinct ordered points recover the positive slope and the intercept continuously. They also prove that the image groups are closed in that

homeomorphism group. Since both images are the same set $G|_B$, we have obtained a common topological Lie group $M_B$, and continuous quotient homomorphisms

$$\rho_i : N_i \longrightarrow M_B. \tag{C.36.7}$$

The universal cover of $M_B$ is $\mathbb{R}$ or the positive affine group, and its exponential is again a global diffeomorphism.

*Equality on overlaps.* The homomorphisms (C.36.7) lift uniquely to based homomorphisms

$$\widetilde{\rho}_i : \widetilde{N}_i \longrightarrow \widetilde{M}_B. \tag{C.36.8}$$

The two paths $\widetilde{\rho}_i(\widetilde{n}_i(t))$ are based lifts of the identical loop $f_t|_B$, so covering uniqueness makes them equal to one path $\widetilde{b}(t)$. Naturality of the Lie exponential and its global injectivity give

$$\begin{aligned} d\widetilde{\rho}_i(\log \widetilde{n}_i(t)) &= \log \widetilde{b}(t), \\ \widetilde{\rho}_i\big(\exp((1-s)\log \widetilde{n}_i(t))\big) &= \exp\big((1-s)\log \widetilde{b}(t)\big). \end{aligned} \tag{C.36.9}$$

After projection to $M_B$, this says precisely that $R^{O_1}_{s,t}$ and $R^{O_2}_{s,t}$ agree on $B$. The intersection of the two closures is a union of such $G$-orbits, so they agree everywhere on the intersection. This includes circle orbits with their actual based universal-cover lifts; no choice of an unrecorded integer rotation is made.

For $h \in H$, conjugation carries $Q_O$ onto $Q_{hO}$ and $N_O$ onto $N_{hO}$, as follows from the exact restriction of the setwise stabilizers. This is a topological Lie-group isomorphism. It carries $n_O(t)$ to $n_{hO}(t)$, since $h$ commutes with $f_t$. Its based lift and exponential naturality show

$$R^{hO}_{s,t} = hR^{O}_{s,t}h^{-1}. \tag{C.36.10}$$

Thus the compatible families have the asserted operator equivariance.

Finally, on a finite union of these closures, their continuous maps agree on every overlap. The finite closed-set pasting lemma gives joint continuity. Each intersection is preserved, because its $G$-orbits are preserved; hence the inverse maps also agree there. Applying the same pasting argument to the inverse families gives the stated family of homeomorphisms on the finite union. This proves the claim. □

**Exact scope of the closure construction.** The quotients (C.36.2)–(C.36.3) provide continuity on each individual closure, and (C.36.7)–(C.36.9) enforce the actual common boundary restriction relations. In particular they respect the three dependent edge characters in C.35. The present proof uses finite closed-set pasting. C.37 subsequently proves a uniform three-word estimate and extends these contractions continuously, with their inverses, to the closed union $X_2$ of all such open orbits, including infinite accumulation. Extension across $X_2$ to its exterior, including frontier contractions for $\dim Q_O = 0, 1$ and compatible choices elsewhere on $Y$, remains required for a global conclusion in C.12.

**C.37. [Proved] Uniform control on all two-dimensional operator closures.** Let $f_t$, $0 \le t \le 1$, be an ambient-null based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let $G = \langle f_t : 0 \le t \le 1 \rangle$, and let $H$ be any subgroup commuting with every $f_t$. For an open $G$-orbit $O$, let $Q_O$ be the actual closed restricted operator image of C.30. Put

$$U_2 = \bigcup_{\dim Q_O = 2} O, \qquad X_2 = \overline{U_2}. \tag{C.37.1}$$

The contractions of C.36 extend to a jointly continuous based contraction through homeomorphisms of $X_2$, with jointly continuous inverses, commuting with $H$. This includes arbitrary accumulation of the open orbits in (C.37.1).

The uniform estimate used in the proof is the following exact word statement. In either simply connected two-dimensional Lie group $\mathbb{R}^2$ or $\mathrm{Aff}^+(\mathbb{R})$, every continuous path $k : [0,t] \to N$ from 1 to $g$ satisfies

$$\exp(r \log g) = k(b)k(a)^{-1} \quad \text{or} \quad \exp(r \log g) = g\, k(b)k(a)^{-1} \quad (0 \le r \le 1) \tag{C.37.2}$$

for suitable $a, b \in [0,t]$. The same assertion is used in the universal covering group when the actual group is $\mathbb{R} \times S^1$. Consequently every contraction value on an open orbit in $U_2$ is represented by a word of length at most three in the original loop and its inverse, with equality of the specified based universal-cover lifts.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; uniform dimension-two open-orbit closure step for C.12.

*Proof. The three-word estimate.* For a continuous path $\gamma : [0,1] \to \mathbb{R}^2$ with $\gamma(0) = 0$, $\gamma(1) = v$, one has

$$rv = \gamma(b) - \gamma(a) \quad \text{or} \quad rv = v + \gamma(b) - \gamma(a) \qquad (0 \le r \le 1). \tag{C.37.3}$$

Here is an exact measure-theoretic proof that applies to arbitrary paths. For a polygonal path apply the signed-measure splitting theorem of Grandoni–Ravi–Singh–Zenklusen [26, Theorem 4.7 and §4.3.2] to the two signed measures with densities $\gamma_1'$ and $\gamma_2'$. A single circular interval has fraction $r$ of both total measures. Its integral is either a path difference or the total vector plus a path difference, giving (C.37.3). Approximate an arbitrary continuous path uniformly by polygonal paths with the same endpoints. Compactness of the two endpoints and of the two alternatives passes the equation to the limit. The planar case of the cited splitting proof uses at most two cuts of an interval; the two equal-mass halves together have at most three pieces, so one is a single interval. This is the one-circular-interval count required here.

For $N = \mathbb{R}^2$, (C.37.2) is (C.37.3). Write the affine group in the exact coordinates of C.36:

$$(a,b)(a',b') = (a + a',\ b + e^a b'). \tag{C.37.4}$$

Every nonidentity $g = (a,b)$ lies on its unique exponential one-parameter subgroup. If $a \neq 0$, conjugation by $(0, b/(e^a - 1))$ sends $g$ to $(a,0)$. The global coordinate map

$$(a',b') \longmapsto (a'/a,\ e^{-a'} b') \tag{C.37.5}$$

then sends left multiplication by $g^r$ to translation by $(r,0)$, and sends $1, g$ to $(0,0), (1,0)$. If $g = (0,b)$ with $b \neq 0$, use $(a',b') \mapsto (b'/b, a')$ instead. Apply (C.37.3) in these coordinates. A difference $(r,0)$ says $k(b) = g^r k(a)$; the other alternative says $k(a) = g^{1-r} k(b)$. These are exactly (C.37.2). The identity case uses the identity word. This proves the estimate also for the simply connected covering group of every group in C.36.

*The compact family of actual word isotopies.* Use $r = 1 - s$ for the contraction parameter, and set

$$R^O_{r,t} = p_O \exp\bigl(r \log \widetilde{n}_O(t)\bigr), \qquad n_O(t) = f_t|_O, \tag{C.37.6}$$

as in C.36. Consider the following compact family of actual plane isotopies, with the compact-open topology on maps and inverses:

$$W_{\epsilon,a,b,t}(u) = f_{ut}^{\epsilon} f_{ub} f_{ua}^{-1}, \quad 0 \le a, b \le t \le 1, \quad \epsilon \in \{0,1\}, \quad 0 \le u \le 1. \tag{C.37.7}$$

Compactness follows from continuity of the original loop and of multiplication and inversion. Formula (C.37.2), applied to the based lift of its prefix $[0,t]$, represents every $R^O_{r,t}$ by the endpoint of one of (C.37.7). More precisely, the based lift of that word isotopy in $N_O$ has endpoint $\exp(r \log \widetilde{n}_O(t))$. Thus the representation retains the lift, not only the projected homeomorphism.

*The exact groups on nonopen limit orbits.* Each $N_O$ in (C.37.6) is metabelian. Therefore the identity

$$[[g_1, g_2], [g_3, g_4]] = 1 \tag{C.37.8}$$

holds on $U_2$ for all $g_i \in G$, and hence on $X_2$ by continuity. The set $X_2$ is $G$-invariant, so it is a union of whole $G$-orbits. Its open $G$-orbits are precisely those already in $U_2$: any other open orbit has an open neighborhood disjoint from $U_2$.

Let $B \subset X_2$ be a nontrivial nonopen orbit, endowed with the intrinsic line or circle topology of C.28. The image $G_B$ in $\mathrm{Homeo}^+(B)$ is transitive, metabelian, and path connected. For the last assertion, every finite word has a path from the identity obtained from the original paths and their inverses; C.28 gives intrinsic compact-open continuity of all these finite-word paths. We now prove the exact classification needed at $B$:

$$G_B \cong \begin{cases} (\mathbb{R}, +) \text{ or } \mathrm{Aff}^+(\mathbb{R}), & B \text{ a line}, \\ S^1, & B \text{ a circle}, \end{cases} \tag{C.37.9}$$

with the usual transitive actions after conjugacy.

First, a transitive abelian subgroup $A$ of $\mathrm{Homeo}^+(B)$ has abelian closure $\overline{A}$, and this closure acts freely: an element fixing one point commutes with $A$, so fixes its whole transitive orbit. The closure is Polish and transitive. The open orbit-map theorem used in C.26 identifies it homeomorphically with $B$ by evaluation, since its point stabilizer is trivial. In particular it is locally compact. Its faithful product action on the three-manifold $B \times \mathbb{R}^2$ makes it a Lie group by the same Hilbert–Smith input as C.33. It is the one-dimensional connected group $\mathbb{R}$ or $S^1$, and evaluation conjugates its action to translation. Since $A$ was already transitive, injectivity of evaluation also proves $A = \overline{A}$.

If $G_B$ is nonabelian, its derived subgroup $D$ is nontrivial, abelian, normal, and path connected. Path connectedness follows by taking commutator paths and finite products of them. The common fixed set of $D$ is $G_B$-invariant. Transitivity and nontriviality make it empty. Every $D$-orbit is therefore a connected nontrivial interval or circle. Such an orbit has no included endpoint, since that endpoint would be preserved by its transitive orientation-preserving action. All these orbits are open in $B$; connectedness of $B$ forces $D$ to be transitive. The preceding paragraph identifies $D$ with the full translation group.

The orientation-preserving normalizer of translations on the line consists exactly of $x \mapsto cx + d$, $c > 0$. Indeed normalization says $h(x + a) = h(x) + \chi(a)$; continuity and additivity give $\chi(a) = ca$. The multiplier image of the path-connected group $G_B$ is a connected subgroup of $\mathbb{R}_{>0}$. It is nontrivial in the present case, so it is all $\mathbb{R}_{>0}$; because $G_B$ contains every translation, it is the full affine group. On the circle, the orientation-preserving normalizer of the full rotation group is just that rotation group: the induced continuous automorphism of $S^1$ has degree $+1$. This would contradict nonabelianness. These arguments prove (C.37.9), including the topology of its image in the intrinsic homeomorphism group.

For a line $B$, define $R^B_{r,t}$ by the logarithmic formula in this simply connected group. For a circle, use the unique based real lift of $f_t|_B$ to the cover of its rotation group and scale that lift by $r$. Its endpoint is zero by the exact winding comparison C.29. On a point orbit define $R^B_{r,t} = 1$. These definitions are independent of the conjugacies in (C.37.9), since Lie homomorphisms commute with exponential maps and with based universal-cover lifts. They agree with C.36 on every boundary orbit to which that construction already applies.

*Limits from the dense open union.* Take

$$x_i \in O_i \subset U_2, \quad x_i \longrightarrow x \in X_2, \qquad (r_i, t_i) \longrightarrow (r, t). \tag{C.37.10}$$

Choose word isotopies $W_i$ from (C.37.7) representing $R^{O_i}_{r_i,t_i}$ with their specified based lifts. After a subsequence all word parameters converge. Then $W_i$ and their inverses converge uniformly on

compact sets, uniformly in $u$, to a word isotopy $W$ from the same family. Write $w = W(1)$ and $g = f_t$. If $x$ is in an open orbit of $X_2$, eventually $x_i$ is in that same open orbit, where (C.37.6) is continuous. If $x$ is a point orbit, $w(x) = x$ for every possible limiting word, which gives the required limit immediately. It remains to identify $w$ on the intrinsic line or circle $B = Gx$.

We will repeatedly use the following exact passage of identities. For every fixed $h \in G$, $hx_i \in O_i$ and $hx_i \to hx$. An identity of word homeomorphisms on $O_i$, whose word isotopies converge, therefore passes to every point $hx$ of $B$. In particular

$$[w|_B, g|_B] = 1, \tag{C.37.11}$$

because a fractional power commutes with its endpoint on $O_i$.

Fix a positive integer $n$ and set $m_i = \lfloor nr_i \rfloor$. In the universal covering group of $N_{O_i}$ the exact residual is

$$\widetilde{w}_i^{\,n}\, \widetilde{n}_{O_i}(t_i)^{-m_i} = \exp\big((nr_i - m_i)\log \widetilde{n}_{O_i}(t_i)\big). \tag{C.37.12}$$

Its exponent fraction lies in $[0, 1)$. By (C.37.2), the right side is again represented by one of the compact family (C.37.7), with equality of based lifts. Passing to a further subsequence for this fixed $n$, the integer $m_i$ is constant, say $m_n$, and the residual word isotopies converge to $Z_n$. Hence

$$(w|_B)^n (g|_B)^{-m_n} = z_n, \qquad |m_n - nr| \le 1, \qquad z_n = Z_n(1)|_B. \tag{C.37.13}$$

All $z_n$ belong to the same compact set of restricted three-word endpoints, independently of $n$. No simultaneous subsequence for all $n$ is required.

For a line, (C.37.9) and (C.37.11) allow logarithms in $\mathbb{R}$ or $\mathrm{Aff}^+(\mathbb{R})$. In the latter group the centralizer of every nonidentity element is its exponential one-parameter subgroup, as is seen directly from (C.37.4). Thus logarithms add for commuting elements in both groups. The compact set in (C.37.13) has bounded logarithms, giving

$$n \log(w|_B) - m_n \log(g|_B) = \log z_n, \qquad \sup_n \|\log z_n\| < \infty. \tag{C.37.14}$$

Divide by $n$ and let $n$ tend to infinity. It follows that $\log(w|_B) = r \log(g|_B)$. This includes $g|_B = 1$, in which case boundedness forces $w|_B = 1$. The limiting word therefore has exactly the required fractional action on the line.

*The lifted residual bound on a circle.* Identify the action in (C.37.9) on a circle with rotations, and let $\theta(u)$ be the continuous real angle of $f_u|_B$, normalized by $\theta(0) = 0$. The normalization uses rotations $\mathbb{R}/\mathbb{Z}$. C.29 gives $\theta(1) = 0$. Every isotopy in (C.37.7), restricted to $B$, has a canonical real endpoint angle

$$\epsilon\theta(t) + \theta(b) - \theta(a). \tag{C.37.15}$$

Let $A$ be this angle for $W$, let $T = \theta(t)$, and let $C_n$ be the angle for $Z_n$. Writing $M = \max_{0 \le u \le 1} |\theta(u)|$ gives $|C_n| \le 3M$. We must prove the real, rather than merely circle-valued, identity

$$nA - m_n T = C_n. \tag{C.37.16}$$

The word isotopy

$$\Lambda_i(u) = W_i(u)^n f_{ut_i}^{-m_i} Z_i(u)^{-1} \tag{C.37.17}$$

restricts to a based loop in $N_{O_i}$. Equation (C.37.12), with the specified lifts of $W_i$ and $Z_i$, says that its universal-cover lift ends at the identity. Hence that loop is null-homotopic in $N_{O_i}$. Choose $h \in G$ with $hx \ne x$, and evaluate (C.37.17) at the two points $x_i, hx_i$ of $O_i$. For large $i$ they are distinct. The resulting loop in $\mathrm{Conf}_2(\mathbb{R}^2)$ has difference-direction winding zero. The word isotopies converge uniformly on compact sets to $\Lambda(u) = W(u)^n f_{ut}^{-m_n} Z_n(u)^{-1}$. The two-point loops therefore converge uniformly to its evaluation at $x, hx$. This limiting pair stays distinct, so

winding is preserved. On the Jordan circle $B$ its winding, by C.29, is exactly $nA - m_nT - C_n$ (up to a common orientation sign). It is zero, proving (C.37.16).

Using (C.37.13) and $|C_n| \le 3M$ now gives

$$|A - rT| \le \frac{3M + |T|}{n} \quad (n \ge 1). \tag{C.37.18}$$

Consequently $A = rT$. This determines the based lifted fractional action even when $f_t|_B = 1$ but $\theta(t)$ is a nonzero integer. Thus no hidden winding survives the limit from the open orbits.

*Joint continuity, inverses, and equivariance.* We have shown that every convergent subsequence of word parameters in (C.37.10) gives the same endpoint $R^B_{r,t}(x)$. Compactness of (C.37.7) proves

$$R^{O_i}_{r_i,t_i}(x_i) \longrightarrow R^B_{r,t}(x). \tag{C.37.19}$$

This holds for every sequence from the dense set $U_2$ approaching every point of $X_2$, with parameters also varying. It proves joint continuity on all of $X_2$. Explicitly, for a sequence of points of $X_2$, approximate each by a point of $U_2$ so closely that both its distance and its image distance for that fixed parameter are less than $1/i$. Such approximations exist by (C.37.19) with parameters fixed. Applying (C.37.19) to the chosen approximating sequence gives the full continuity assertion.

The inverse loop $f_t^{-1}$ has the same group $G$, the same sets $U_2, X_2$, and the same actual operator images. Its construction gives precisely $(R^B_{r,t})^{-1}$, by inversion of the based lifts. Applying the preceding argument to it proves joint continuity of the inverse family as well. Each member is an orbitwise bijection, hence a homeomorphism of $X_2$. The identities $R_{0,t} = 1$, $R_{1,t} = f_t|_{X_2}$, and $R_{r,0} = R_{r,1} = 1$ give the based contraction.

Finally every $h \in H$ permutes the open $G$-orbits and conjugates their actual closed operator images. It therefore preserves $U_2$ and $X_2$. C.36 gives exact transported equivariance on $U_2$. The established joint continuity extends that equality to its closure $X_2$. This proves all assertions. □

**The remaining global boundary requirement.** C.37 treats the whole closed set $X_2$, including infinite accumulation of its open orbits. Its boundary contraction is the based Lie-group fractional action forced by (C.37.14) and (C.37.16). Extending this family to the complement of $X_2$, compatibly with the dimension-zero and dimension-one proper operator cases, remains a separate requirement. Even where that complement contains only nonopen orbits, its median contraction from C.34 has not been identified with this boundary family. C.35 proves that imposing the fixed median prescription can violate the actual restriction map. Thus the uniform estimate proved here closes the dimension-two accumulation step, while the global centralizer conclusion in C.12 remains open.

**C.38. [Proved] Discrete cyclic operators on nonabelian line and circle orbits.** Let $B$ be a line or circle and let $T \le \mathrm{Homeo}^+(B)$ be transitive and nonabelian. Its full compact-open centralizer

$$N_B = C_{\mathrm{Homeo}(B)}(T) \tag{C.38.1}$$

is discrete and acts freely and properly. On a line it is trivial or infinite cyclic; on a circle it is finite cyclic.

In the plane-loop setting of C.28, let $B$ be a nonopen orbit whose intrinsic image $G|_B$ is nonabelian, and let $H_B = \{h \in H : hB = B\}$ be the operator stabilizer. Its actual restriction has the exact sequence

$$1 \longrightarrow K_B \longrightarrow H_B \xrightarrow{\rho_B} P_B \longrightarrow 1, \quad K_B = \{h \in H_B : h|_B = 1\}, \quad P_B = \rho_B(H_B) \le N_B. \tag{C.38.2}$$

In the intrinsic compact-open topology, $P_B$ is already closed and discrete, acts freely and properly, and

$$p_B : B \longrightarrow B/P_B \tag{C.38.3}$$

is an actual regular covering. Its deck transformations are exactly $P_B$. The quotient is a line for trivial $P_B$ on a line, and otherwise a circle. On a circle $P_B$ is finite and the covering degree is $|P_B|$. The kernel $K_B$ acts identically on $B$ throughout this comparison.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact one-dimensional operator-image reduction at the boundary remaining in C.12.

*Proof.* Let $L$ be the closure of $T$ in the compact-open homeomorphism group of $B$, choose $b \in B$, and put $J = L_b$. The group $L$ is Polish and transitive, and $C(L) = C(T)$ by continuity of composition. The open orbit-map theorem used in C.26 gives $L/J \cong B$. The normalizer $N_L(J)$ is closed, and the same exact evaluation construction as in C.30 gives

$$\mathrm{ev}_b : N_B \xrightarrow{\cong} N_L(J)/J \ \subset B. \tag{C.38.4}$$

The right side is closed. The inverse sends the coset represented by $k$ to the map $lb \mapsto lkb$. The normalizer equation makes this well defined. Its compact-open continuity follows by the same microtransitivity argument as C.30: convergent points of $B$ lift locally through the open orbit map to elements of $L$ tending to the identity. No surface-dimensional step occurs in that argument. The construction also proves freeness, local compactness, and properness. Explicitly, if $n_i b_i = c_i$, $b_i \to b$, and $c_i \to c$, choose $l_i \to 1$ with $l_i b = b_i$. Then $n_i b = l_i^{-1} c_i \to c$, and the closed image and homeomorphism (C.38.4) give a convergent subsequence in $N_B$. This is the required properness test.

Every orientation-reversing homeomorphism of a line or circle has a fixed point. Freeness therefore places $N_B$ in $\mathrm{Homeo}^+(B)$. Its faithful product action on the three-manifold $B \times \mathbb{R}^2$ makes it a Lie group by the verified Hilbert–Smith theorem used in C.33. Equation (C.38.4) bounds its dimension by one. If that dimension were one, every orbit of $N_B^0$ would be open in $B$: its one-dimensional orbit map is injective and hence open by invariance of domain. Connectedness of $B$ would make $N_B^0$ transitive. Freeness would then give $N_B = N_B^0$, since an element of $N_B^0$ with the same evaluation as an element of $N_B$ equals it. This is the translation group $\mathbb{R}$ or the rotation group $S^1$ in evaluation coordinates. Its centralizer is that same abelian group of right translations. But $T \le C(N_B)$, contradicting the assumed nonabelianness of $T$. Consequently $N_B$ is a zero-dimensional Lie group, so is discrete.

A free proper discrete action supplies evenly covered neighborhoods, and its orbit space is a connected one-manifold without boundary. For a line this quotient is a line or a circle. Since the line is simply connected, the first case has trivial deck group and the second has deck group $\mathbb{Z}$. For a circle the quotient is a circle and the covering has finitely many sheets; its deck group is finite cyclic. This proves the first assertion.

In the stated plane-loop application, C.28 proves that each operator stabilizing $B$ induces an actual homeomorphism for its intrinsic line or circle topology. Commutation with $G$ puts its image in (C.38.1). Every subgroup of this discrete group is closed; restriction of a proper free action to a closed subgroup is proper and free. These facts give (C.38.3). Its fibers are precisely the $P_B$-orbits. A deck transformation is determined by one point in the connected covering, and each point of a fiber is obtained by exactly one element of $P_B$, so the whole deck group is exactly $P_B$. The same one-manifold covering classification gives the stated quotient and degree. Equation (C.38.2) retains the original stabilizer and its restriction kernel. □

**Topology and the remaining relative extension.** The topology asserted for $P_B$ is its intrinsic compact-open image topology. The operator group may be regarded as discrete as in the

original problem; no assertion about continuity of the restriction map from an inherited plane topology to a recurrent intrinsic line topology is needed. C.38 gives exact covering quotients on every nonabelian one-dimensional orbit. It supplies no transverse control between different orbits and no relative extension of the C.37 family from $X_2$ to its exterior. Those requirements remain in C.12.

**C.39. [Proved] A global contraction for loops with commuting values.** Let $f_t$, $0 \le t \le 1$, be a based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$ with $f_s f_t = f_t f_s$ for every $s, t$. Suppose the loop is null-homotopic in that ambient group. For every subgroup $H$ commuting with all $f_t$, the loop has a based null-homotopy in $C_{\mathrm{Homeo}^+(\mathbb{R}^2)}(H)$. The homotopy and its inverse family are jointly continuous on the whole plane, including every change and accumulation of orbit types.

Consequently, on every connected admissible surface $U$ of C.1, an element of the kernel of the exact inclusion map

$$\pi_1 C_{\mathrm{Homeo}(U)}(H) \longrightarrow \pi_1 \mathrm{Homeo}(U) \tag{C.39.1}$$

that is represented by a loop with pairwise commuting values is zero. On a hyperbolic surface every such based loop in $C_{\mathrm{Homeo}(U)^0}(H)$ is therefore null-homotopic in that centralizer.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; complete global commuting-value subcase of the exact C.12 level-zero comparison.

*Proof. The actual image on every orbit.* Set $G = \langle f_t \rangle$. This is an abelian group. The orbit classification C.28 applies. On an open orbit $O$, the closure $K_O$ of $G|_O$ in $\mathrm{Homeo}(O)$ is Polish, abelian, connected, and transitive. It acts freely: an element fixing a point commutes with the transitive subgroup $G|_O$ and hence fixes every point. The Polish open orbit-map theorem identifies evaluation $K_O \to O$ homeomorphically, with trivial stabilizer. Thus $K_O$ is locally compact. Its faithful action on $O \times \mathbb{R}$ and the Hilbert–Smith input of C.33 make it a two-dimensional abelian Lie group. Since $O$ is a noncompact open subset of the plane, the possibilities are $\mathbb{R}^2$ and $\mathbb{R} \times S^1$. The original image is already the whole group: for each $k \in K_O$, transitivity of $G|_O$ gives an element with the same evaluation, and freeness makes the two elements equal.

On an intrinsic line or circle orbit $B$, C.28 gives compact-open continuity of the restricted original paths. The same abelian closure and free evaluation argument, now in the intrinsic topology, identifies $G|_B$ with the full translation group $\mathbb{R}$ or rotation group $S^1$. This is also the abelian case of (C.37.9). On a point orbit the image is trivial. Denote the resulting actual Lie image on any orbit $B$ by $K_B$. Its simply connected covering group is a vector group of dimension at most two.

The original restricted loop has a closed based lift $\widetilde{n}_B(t)$ to this vector group. For an open orbit, C.32 gives its null-homotopy in $\mathrm{Homeo}(O)$; evaluation and the homeomorphism $K_O \cong O$ then make its class in $\pi_1 K_O$ zero. For a line there is no covering kernel. For a circle the based lift closes by the exact winding comparison C.29. Define on each orbit

$$R^B_{r,t} = p_B\big(r\,\widetilde{n}_B(t)\big), \qquad 0 \le r, t \le 1. \tag{C.39.2}$$

Here $p_B$ is the Lie-group covering homomorphism, with the trivial interpretation for a point orbit. The action of the right side on $B$ is its actual homeomorphism action. Its endpoints are the identity at $r = 0$ and the original loop at $r = 1$; both based edges $t = 0, 1$ are fixed.

*One compact word family for all orbit types.* In dimension two, C.37 represents every value of (C.39.2) by a word isotopy

$$W_{\epsilon,a,b,t}(u) = f_{ut}^{\epsilon} f_{ub} f_{ua}^{-1}, \quad \epsilon \in \{0, 1\}, \quad 0 \le a, b \le t \le 1, \tag{C.39.3}$$

with equality of its specified based universal-cover endpoint. In dimension one the intermediate value theorem already finds $b \le t$ with $\widetilde{n}_B(b) = r\widetilde{n}_B(t)$. This is a member of (C.39.3) with

$\epsilon = 0, a = 0$. The identity word handles point orbits. Thus the same compact family of original plane isotopies works for every orbit, with the based lifts retained.

Take arbitrary points $x_i \to x$ in the plane and parameters $(r_i, t_i) \to (r, t)$, and write $B_i = Gx_i$, $B = Gx$. Choose representing word isotopies $W_i$ for $R^{B_i}_{r_i,t_i}$. Compactness of the parameter set in (C.39.3) gives a subsequence converging, together with inverses and uniformly in $u$, to a word isotopy $W$. Put $w = W(1)$. If $B$ is open, all sufficiently large $x_i$ are in that same orbit and continuity follows from its Lie action. If $B$ is a point, every possible word limit fixes $x$, giving the required convergence. It remains to identify the limiting word on a line or circle. This argument explicitly allows the $B_i$ to have different types.

For each fixed positive integer $n$, set $m_i = \lfloor nr_i \rfloor$. In the vector covering group of the actual image $K_{B_i}$, one has

$$n\,\widetilde{w}_i - m_i\,\widetilde{n}_{B_i}(t_i) = (nr_i - m_i)\widetilde{n}_{B_i}(t_i). \tag{C.39.4}$$

The right side is represented by another word isotopy $Z_i$ from (C.39.3), with equality of based lifted endpoints. For this fixed $n$, pass to a further subsequence so that $m_i = m_n$ is constant and $Z_i \to Z_n$. Then $|m_n - nr| \le 1$. Identities on $B_i$ pass to the whole orbit $B$ by evaluation at $hx_i \to hx$, for any fixed $h \in G$. In particular, with $g = f_t$ and $z_n = Z_n(1)$,

$$(w|_B)^n (g|_B)^{-m_n} = z_n|_B. \tag{C.39.5}$$

For a line, let $\theta(u)$ be the real translation parameter of $f_u|_B$, and put $M = \max_u |\theta(u)|$. The real parameters $A$ of $w|_B$ and $C_n$ of $z_n|_B$ come from their word paths: each has the form $\epsilon\theta(t) + \theta(b) - \theta(a)$. Consequently $|C_n| \le 3M$, and (C.39.5) is the real identity $nA - m_n\theta(t) = C_n$. Division by $n$ gives $A = r\theta(t)$, as required.

For a circle, use the based real rotation lift $\theta$ of the original loop, so $\theta(0) = \theta(1) = 0$. The word paths again specify real endpoint angles $A, C_n$ with $|C_n| \le 3M$, but (C.39.5) initially gives their relation only modulo integers. The exact lift in (C.39.4) removes this ambiguity. Indeed,

$$\Lambda_i(u) = W_i(u)^n f_{ut_i}^{-m_i} Z_i(u)^{-1} \tag{C.39.6}$$

is a based loop in the actual group $K_{B_i}$ and has a closed based universal-cover lift. It is therefore null-homotopic in that group. Choose $h \in G$ with $hx \ne x$. For large $i$ the two points $x_i, hx_i$ of $B_i$ are distinct. Evaluation of (C.39.6) at them has difference-direction winding zero in $\mathrm{Conf}_2(\mathbb{R}^2)$, regardless of whether $B_i$ is an open, line, or circle orbit. This follows from the actual null-homotopy in $K_{B_i}$ and its continuous evaluation into the plane; C.28 supplies that evaluation for intrinsic line orbits.

The word isotopies and the two evaluation points converge. Their limiting pair stays distinct. The winding therefore remains zero. On the limiting Jordan circle its value, by C.29, is $nA - m_n\theta(t) - C_n$, up to a common orientation sign. Thus the real identity

$$nA - m_n\theta(t) = C_n, \qquad |A - r\theta(t)| \le \frac{3M + |\theta(t)|}{n} \tag{C.39.7}$$

holds for every $n$. It forces $A = r\theta(t)$. This also treats an identity endpoint rotation whose based angle $\theta(t)$ is a nonzero integer. All possible subsequential word limits therefore give the prescribed value (C.39.2).

Compactness of (C.39.3) now proves joint continuity of (C.39.2) on the entire plane. The inverse loop has the same orbits and image groups, and its construction gives precisely the inverse of (C.39.2); the same proof gives joint inverse continuity. Each map is an orbitwise bijection, so these are plane homeomorphisms. Every commuting operator carries the actual abelian Lie image on one orbit isomorphically to that on the transported orbit. These are continuous Lie-group isomorphisms in the specified intrinsic or open-orbit topology. Their based covering lifts are

linear, and hence commute with multiplication by $r$. Formula (C.39.2) is therefore exactly $H$-equivariant. Taking $r = 1 - s$ proves the plane assertion.

*The precise surface lift and descent.* Every connected admissible surface in C.1 has universal covering $p : \widetilde{U} \to U$ with $\widetilde{U} \cong \mathbb{R}^2$. Let $f_t$ represent an element of the kernel in (C.39.1), with its commuting-value property. Choose an ambient filling $A : D^2 \to \mathrm{Homeo}(U)$ whose boundary is this loop and whose specified based boundary point maps to the identity. The actual map

$$D^2 \times \widetilde{U} \longrightarrow U, \qquad (z, x) \longmapsto A(z)(p(x)) \tag{C.39.8}$$

has a unique lift equal to the identity on the specified based fiber. Its domain is simply connected. Lifting the inverse family as well, and using uniqueness, proves that its slices are homeomorphisms with jointly continuous inverses. It is an ambient filling of a closed based plane loop $\widetilde{f}_t$.

For every lift $\widetilde{h}$ of an operator, $[\widetilde{f}_t, \widetilde{h}]$ is a deck transformation. It depends continuously on $t$ and starts at the identity, so it is identically the identity. Similarly $[\widetilde{f}_s, \widetilde{f}_t]$ is a deck transformation depending continuously on $(s, t)$; it is the identity on the axes, and hence everywhere. Thus the lifted loop has commuting values and centralizes the full group of operator lifts, including the deck group. Apply the plane assertion. Its equivariant contraction descends through $p$, with both joint continuities, to the original centralizer on $U$. Its slices lie in the identity component by the displayed paths. This proves (C.39.1). For a hyperbolic $U$ the ambient identity component is contractible, so the ambient kernel condition is automatic. □

**Scope of the global subcase.** This theorem treats loops with commuting values on every orbit at once, including open orbits whose operator image has dimension zero or one. It does not identify every fundamental-group class of an arbitrary centralizer with such a loop. No deformation of a general centralizer loop to this subcase has been constructed here. C.12 remains open for general loops.

**The earlier abelian closed subgroup cannot be a centralizer.** Fix the specific embedding $j : B \hookrightarrow A^0$ of C.26, where $B = L^1([0, 1], \mathbb{Z})/\mathbb{Z}$, $A = \mathrm{Homeo}(U)$ and $U$ is hyperbolic. For every $H \leq A$ commuting with $j(B)$, the induced map $\pi_1(B) \to \pi_1(C_A(H), 1)$ is zero. Indeed the displayed generator of $\pi_1(B) = \mathbb{Z}$ has pairwise commuting values and is ambient-null, so C.39 contracts its image; the induced homomorphism therefore vanishes on the whole group. The based contraction lies in $A^0$, so the same conclusion holds with target $\pi_1(C_{A^0}(H), 1)$. Consequently $j(B)$ cannot equal either of these centralizers, since its topological-group identification with $B$ induces an isomorphism on the nonzero fundamental group. This rules out that exact realization as a zero-level counterexample.

**C.40. [Proved] The locally compact loop-closure case.** Let $U$ be any connected admissible surface of C.1, let $A = \mathrm{Homeo}(U)$ with its compact-open group topology, and let $H \leq A$ be arbitrary. Put $C = C_A(H)$. For a based loop $f : [0, 1] \to C$, form the actual closed subgroup

$$L_f = \overline{\langle f_t : 0 \leq t \leq 1 \rangle}^{A}$$

with the topology induced from $A$. Every class in the kernel of

$$i_* : \pi_1(C, 1) \longrightarrow \pi_1(A, 1)^H \tag{C.40.1}$$

that has a representative with locally compact $L_f$ is zero. Here $i : C \hookrightarrow A$ is the inclusion; its induced map lands in the displayed fixed subgroup because each $h \in H$ fixes the entire representative loop under conjugation.

In particular, the same conclusion holds for loops contained in any closed locally compact subgroup of $C$. If $C$ itself is locally compact, (C.40.1) is injective. On a hyperbolic surface every based loop in $C_{\mathrm{Homeo}(U)^0}(H)$ with locally compact $L_f$ is null-homotopic in that centralizer.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact inclusion map from the centralizer fundamental group to the H-invariant ambient fundamental group.

*Proof. The exact subgroup and its Lie topology.* Centralizers are closed in the Hausdorff topological group $A$. Consequently $L = L_f$ is a closed subgroup of $C$. The subgroup generated by the loop is path connected: each finite product of loop values and their inverses can be joined to the identity by the corresponding product of paths. Its closure $L$ is connected. Thus no identity-component replacement or change of topology is being made. The topology on $L$ is the subspace topology from $A$, and its action on $U$ is continuous and faithful.

Suppose $L$ is locally compact. The product action

$$L \times (U \times \mathbb{R}) \longrightarrow U \times \mathbb{R}, \qquad (l, (x, a)) \longmapsto (lx, a) \tag{C.40.2}$$

is a faithful continuous action on a connected three-manifold. The verified Hilbert–Smith theorem used in C.33, [61, Theorem 1.5 and the introductory reduction], therefore makes $L$ a Lie group with precisely this topology. It is a connected Lie group. All homotopies constructed inside $L$ below are consequently homotopies through actual elements of $C$, continuous in its given topology.

*A based deformation into a compact subgroup.* Borel's principal theorem gives a connected maximal compact subgroup $K$ and closed one-parameter subgroups $H_1, \dots, H_d$, each isomorphic to $\mathbb{R}$, such that multiplication is a homeomorphism

$$K \times \mathbb{R}^d \longrightarrow L, \qquad (k, a_1, \dots, a_d) \longmapsto k\, h_1(a_1) \cdots h_d(a_d). \tag{C.40.3}$$

This is the actual product assertion in [12, §1, p. 271; §§3–4, pp. 277–278]: the product notation is explicitly defined there by uniqueness and continuous dependence of all factors. The maps $h_j : \mathbb{R} \to H_j$ are topological group isomorphisms with $h_j(0) = 1$. The identity has the unique coordinates $(1, 0, \dots, 0)$. Thus

$$D_s\big(k h_1(a_1) \cdots h_d(a_d)\big) = k h_1((1-s)a_1) \cdots h_d((1-s)a_d) \tag{C.40.4}$$

is a continuous strong deformation retraction $D : L \times [0,1] \to L$ onto $K$, fixing the identity. In particular $D_s(f_t)$ is a based homotopy in $C$ from $f$ to a based loop $\beta$ in $K$.

*A commuting representative of the same based class.* Average an inner product on the Lie algebra of $K$ over its compact adjoint action. The resulting positive definite inner product is adjoint invariant, so left translation gives a bi-invariant Riemannian metric on $K$. Its pullback to the universal covering group $p : \widetilde{K} \to K$ is also bi-invariant. For left-invariant vector fields its Levi-Civita connection satisfies

$$\nabla_X Y = \tfrac{1}{2}[X, Y]. \tag{C.40.5}$$

Indeed the Koszul formula and adjoint invariance give this identity directly. Hence all geodesics through the identity are the one-parameter subgroups, and their left translates are defined for all real time. This proves geodesic completeness of the lifted metric. The Hopf–Rinow theorem therefore joins the identity to every point of $\widetilde{K}$ by a geodesic.

Lift $\beta$ from the identity to a path $\widetilde{\beta}$ with endpoint $z \in \ker p$. Choose $X$ in the Lie algebra of $\widetilde{K}$ with $\exp_{\widetilde{K}}(X) = z$, using the geodesic just obtained. The paths $\widetilde{\beta}(t)$ and $\exp_{\widetilde{K}}(tX)$ have the same endpoints in the simply connected space $\widetilde{K}$, so they are homotopic relative to both endpoints. Projection through $p$ is a based homotopy in $K$ from $\beta$ to

$$\gamma(t) = p\big(\exp_{\widetilde{K}}(tX)\big). \tag{C.40.6}$$

This is a based loop, since $p(z) = 1$. It has pairwise commuting values because it is the restriction of a one-parameter subgroup. Every step from $f$ to $\gamma$ stays in $C$.

Now let $[f]$ be in the kernel of (C.40.1). The inclusion of the fixed subgroup $\pi_1(A,1)^H$ into $\pi_1(A,1)$ is injective, so $f$ is ambient-null. The based centralizer homotopy just constructed makes $\gamma$ ambient-null as well. C.39 applies on this same admissible surface to this actual commuting loop and the same operator subgroup $H$. It supplies a based contraction of $\gamma$ inside $C$. Concatenating the two based homotopies contracts $f$ inside $C$.

A loop contained in a closed locally compact subgroup $M \leq C$ has $L_f \subseteq M$ closed; it is therefore locally compact. Taking $M = C$ proves the assertion about a locally compact centralizer. In the hyperbolic case $\mathrm{Homeo}(U)^0$ is contractible by the earlier surface theorem, so the ambient kernel condition holds for every based loop. The constructed contraction lies in the identity path component because it is a based homotopy; it consequently lies in the displayed centralizer inside $\mathrm{Homeo}(U)^0$. □

**Exact remaining scope.** C.40 proves a reduction to commuting values for the specified locally compact closed subgroup. No such based homotopy has been constructed for a general loop whose generated closure is not locally compact. The unrestricted comparison C.12 remains open. In particular, the proof does not replace the induced topology on that closure by a different locally compact topology.

**C.41. [Proved] A dense operator orbit settles the zero-level kernel.** Let $U$ be a connected admissible surface of C.1, let $A = \mathrm{Homeo}(U)$ with its compact-open group topology, and let $H \leq A$ have a dense orbit in $U$. Then the exact inclusion-induced comparison

$$\pi_1(C_A(H),1) \longrightarrow \pi_1(A,1)^H \tag{C.41.1}$$

is injective. In particular, on a hyperbolic surface $\pi_1(C_{A^0}(H),1) = 0$. No local compactness assumption is imposed on $H$ or on the subgroup generated by the values of the loop.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact centralizer inclusion on fundamental groups for a dense operator orbit.

*Proof. The plane argument.* Let $f_t$ be a based ambient-null loop of plane homeomorphisms centralizing a subgroup $H$ with a dense orbit $D$. Put $G = \langle f_t \rangle$. Since the loop is based, its maps preserve orientation, and the orbit results C.28–C.37 apply. Every $h \in H$ transports $G$-orbits according to the exact identity

$$h(Gx) = G(hx). \tag{C.41.2}$$

If there is no open $G$-orbit, C.34 already gives the required global equivariant contraction.

Otherwise let $V$ be the nonempty union of the open $G$-orbits. It is open and $H$-invariant. Density makes $D$ meet $V$, and invariance then gives $D \subseteq V$. Consequently $V$ is dense in the plane.

Fix an open $G$-orbit $O$ and choose $y \in D \cap O$. Let $H_O = \{h \in H : hO = O\}$ be its operator stabilizer. The actual orbit equality is

$$Hy \cap O = H_O y. \tag{C.41.3}$$

Indeed, $hy \in O$ implies that the two $G$-orbits $hO = G(hy)$ and $O$ meet, hence are equal. The reverse inclusion is immediate. Since $Hy = D$ is dense, (C.41.3) is dense in $O$.

Retain the exact restriction homomorphism $H_O \to \mathrm{Homeo}(O)$, with its pointwise-fixing kernel. Let $Q_O$ be the closure of its image in that homeomorphism group. C.30 proves that $Q_O$ is locally compact and acts freely and properly on $O$; C.33 makes it a Lie group of dimension at most two. Properness makes every $Q_O$-orbit closed. In particular $Q_O y$ is a closed subset of $O$ containing the dense set $H_O y$, so

$$Q_O y = O. \tag{C.41.4}$$

Thus this free proper action is transitive. Its orbit map $Q_O \to O$ is a homeomorphism, and $\dim Q_O = 2$.

This holds for every open $G$-orbit. In the notation of C.37, therefore,

$$U_2 = V, \qquad X_2 = \overline{U_2} = \mathbb{R}^2. \tag{C.41.5}$$

The contraction proved there is defined on the whole plane, with jointly continuous inverses, and centralizes every element of the original $H$. This settles both alternatives of the plane argument.

*The actual surface comparison.* Let a based loop $f_t$ represent a kernel class in (C.41.1). The inclusion of the fixed subgroup in $\pi_1(A, 1)$ is injective, so choose an ambient filling $F : D^2 \to A$ of that loop. For the universal covering $p : \widetilde{U} \to U$, every admissible $\widetilde{U}$ is a plane. Lift the actual map

$$D^2 \times \widetilde{U} \longrightarrow U, \qquad (z, x) \longmapsto F(z)(p(x)) \tag{C.41.6}$$

with the identity prescribed on the based boundary fiber. Its domain is simply connected. As in C.39, lifting the inverse family and using uniqueness gives an ambient filling by plane homeomorphisms with jointly continuous inverses. Its boundary is a closed based plane loop $\widetilde{f}_t$.

Let $\widetilde{H}$ be the group of all lifts of elements of $H$, including the deck group. For every $\widetilde{h} \in \widetilde{H}$, the commutator $[\widetilde{f}_t, \widetilde{h}]$ is a deck transformation depending continuously on $t$ and equal to the identity at $t = 0$. It is therefore always the identity. The lifted loop centralizes precisely the same operator lifts required for descent.

Choose $x \in U$ with dense $Hx$ and $\widetilde{x} \in p^{-1}(x)$. One has the exact equality

$$\widetilde{H}\,\widetilde{x} = p^{-1}(Hx). \tag{C.41.7}$$

One inclusion follows by projection. For the other, choose a lift of the specified $h \in H$ and then a deck transformation taking its value at $\widetilde{x}$ to the specified point over $hx$. The right side is dense because $p$ is open: every nonempty open set upstairs projects to a nonempty open set meeting $Hx$.

Apply the plane result to $\widetilde{H}$. Its contraction commutes with all deck transformations and all operator lifts. The equation $\overline{R}_{s,t}p = pR_{s,t}$ therefore defines a based contraction downstairs inside $C_A(H)$. The open covering quotient and its local sections give joint continuity of these maps and their inverses. It contracts the original class in the source of (C.41.1), proving injectivity. For a hyperbolic $U$, the ambient identity component is contractible, so its based centralizer loops all satisfy the kernel condition. The based contraction stays in $A^0$. □

**Remaining scope.** C.41 settles every operator action with a dense orbit. It does not assert that a general operator group has such an orbit or supply an equivariant decomposition into subsets on which this argument applies. The general operator problem without a dense orbit in C.12 remains open.

**C.42. [Proved] The actual boundary quotient has proper line components upstairs.** Let $G$ be generated by a based plane loop as in C.28, let $O$ be an open $G$-orbit, and put $L = \overline{G} \leq \mathrm{Homeo}(\mathbb{R}^2)$. Fix $x \in O$ and $J = L_x$. For $z \in \mathrm{Fix}(J)$ whose $G$-orbit $B = Gz$ is a nontrivial intrinsic line or a circle, the formula

$$\Phi_z(lx) = lz$$

defines the actual continuous open quotient $O = L/J \to B = L/L_z$, with the intrinsic topology on $B$. For the universal coverings $p : \widetilde{O} \to O$ and $q : \mathbb{R} \to B$, every lift $F : \widetilde{O} \to \mathbb{R}$ of $\Phi_z p$ is open and onto, and every connected component of every fiber of $F$ is a properly embedded line in the covering plane $\widetilde{O}$. Every nondegenerate compact connected component of a fiber of $\Phi_z$ is a circle.

Whole fibers of $\Phi_z$ can be disconnected, including for quotients obtained as limits of the original commuting operators.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact boundary quotients in the remaining centralizer extension problem.

*Proof. The quotient and its topology.* Every element of $L$ preserves $O$: forward and inverse limits preserve its boundary, while a forward limit sends an interior point into $\overline{O}$ and cannot send it to that boundary. The Polish orbit-map theorem used in C.26 and C.30 therefore makes $l \mapsto lx$ an open quotient $L \to O$. If $l_1 x = l_2 x$, then $l_2^{-1} l_1 \in J$ fixes $z$, so $\Phi_z$ is well defined and continuous. More precisely, the continuous action map

$$\mathrm{Fix}(J) \times L \longrightarrow \mathbb{R}^2, \qquad (z, l) \longmapsto lz$$

descends through the open quotient $\mathrm{id} \times (L \to O)$. Consequently

$$\mathrm{Map}_L(O, \mathbb{R}^2) \xrightarrow[\cong]{\mathrm{ev}_x} \mathrm{Fix}(J) \tag{C.42.1}$$

is a homeomorphism in the compact-open topology: equivariance proves bijectivity, and the displayed joint action proves continuity of the inverse. For $l \in L$, choose $g \in G$ with $gx = lx$. Then $g^{-1} l \in J$, hence $lz = gz$. Thus the range is exactly $Lz = Gz = B$.

The map into an intrinsic line is continuous in its intrinsic topology. Indeed, the image of a compact coordinate disk in $O$ is a compact Peano continuum in $B$. By C.28 it contains neither a Jordan curve nor a triod, and hence is a point or a compact arc; on such an arc the intrinsic and ambient topologies agree. These disks give neighborhoods at every point of $O$. For a circle the two topologies already agree. Each $l \in L$ restricts to an intrinsic homeomorphism of $B$, because compact arcs are carried to compact arcs. For $b = gz$ one has

$$lb = \Phi_z(lgx). \tag{C.42.2}$$

This makes every point evaluation of the restriction $L \to \mathrm{Homeo}(B)$ continuous. Finite ordered partitions of compact intervals show that pointwise convergence of monotone homeomorphisms to a homeomorphism is uniform on compact intervals; apply the same argument to inverses. Finite cyclic partitions give the circle assertion. Thus restriction is continuous in the compact-open group topology. All these restrictions preserve orientation, since the original loop does and the restriction map is continuous.

Apply the Polish orbit-map theorem to this transitive action on the Polish one-manifold $B$. It identifies $B$ with $L/L_z$ in its stated topology. Since $J \subseteq L_z$, the map $\Phi_z$ is the open quotient $lJ \mapsto lL_z$. In particular, if operators $h_n$ commute with $G$ and $h_n x \to z$, then they commute with $L$ and (C.42.1) gives $h_n|_O \to \Phi_z$ uniformly on compact subsets of $O$.

*Both covering maps and the compatible group.* The universal cover of the connected open planar surface $O$ is a plane. Fix a lift

$$F : \widetilde{O} \longrightarrow \mathbb{R}, \qquad qF = \Phi_z p. \tag{C.42.3}$$

For a line $B$, take $q$ to be an identification. Local covering charts and openness of $\Phi_z p$ show that $F$ is open. Consider the following subgroup, with the topology induced from the indicated product:

$$\begin{aligned} \mathcal{L} = \{(l, a, b) \in L \times \mathrm{Homeo}(\widetilde{O}) \times \mathrm{Homeo}^+(\mathbb{R}) : \quad & pa = (l|_O)p, \\ & qb = (l|_B)q, \qquad Fa = bF\}. \end{aligned} \tag{C.42.4}$$

The three equations are closed conditions, as may be checked by evaluation on countable dense sets. Hence $\mathcal{L}$ is a Polish group, and its actions by $a$ and by $b$ are continuous. Every lift $a$ of

$l|_O$ has a unique compatible lift $b$ of $l|_B$: the maps $Fa$ and $bF$ lift the same map to $B$, and the target deck translation is chosen to make them agree at one point. Uniqueness of lifts then gives equality everywhere. For a line target there is no target deck ambiguity.

The action of $\mathcal{L}$ on $\widetilde{O}$ is transitive. Choose $l$ carrying the projections of two specified points to one another and adjust a source lift by a deck transformation to carry the specified points themselves. The image of $F$ is a nonempty open interval $I$. It is invariant under every compatible $b$. If $B$ is a circle and $I$ has a finite endpoint, every increasing $b$ fixes that endpoint, so its image under $q$ would be fixed by every element of the transitive $L$-action on $B$. This is impossible. Therefore $I = \mathbb{R}$; for a line target surjectivity is already immediate from (C.42.3).

*A component that separates the plane.* Fix $t \in \mathbb{R}$ and write $E = F^{-1}(t)$. Choose $u, v$ with $F(u) < t < F(v)$. The closed set $E$ separates $u$ from $v$. Frankel's planar separation lemma [23, Lemma 2.7, p. 1258] gives a connected component $C$ of $E$ separating them. The lemma applies to unbounded closed sets. For completeness, take the complementary domain $U$ of $E$ containing $u$, and then the component $V$ of $\widetilde{O} \setminus \overline{U}$ containing $v$. The planar Brouwer property makes $\partial V$ connected. It lies in $E$ and separates $u, v$, so the component of $E$ containing $\partial V$ has the required property.

Let $\mathcal{K}$ be the closed subgroup of $\mathcal{L}$ whose $b$-coordinate fixes $t$. It acts transitively on $E$: transitivity on the covering plane finds an element carrying any two points of $E$ to one another, and $Fa = bF$ forces $b(t) = t$. Thus all components of $E$ are ambiently homeomorphic to $C$, and all separate the plane. The stabilizer $\mathcal{K}_C$ is a closed Polish subgroup and acts transitively on $C$. Closedness follows by applying forward and inverse limits to the closed set $C$.

*Accessibility from each fixed complementary domain.* Let $D$ be any component of $\widetilde{O} \setminus C$. Its stabilizer in $\mathcal{K}_C$ is open: for a fixed $d \in D$, every sufficiently small group element sends $d$ into $D$ and therefore sends the whole complementary domain onto $D$. The open orbit-map theorem for $\mathcal{K}_C \curvearrowright C$ makes every orbit of this open stabilizer open in $C$. Since $C$ is connected, the stabilizer acts transitively on $C$.

There is an accessible point of $\partial D$: an interior point and a closest boundary point give a straight end-cut in a coordinate copy of the plane. Its boundary endpoint belongs to $C$. Transporting this end-cut by the domain stabilizer proves that every point of $C$ is accessible from this same $D$. This holds for every complementary domain.

There are exactly two such domains. There are at least two because $C$ separates. For three domains, choose three distinct points $c_1, c_2, c_3 \in C$. Inside each domain, join end-cuts to these points to an embedded tree with these three boundary leaves and one interior branch point. Such a tree is obtained by joining two end-cuts to an arc and stopping a third connecting arc at its first intersection. The three trees have disjoint interiors and share just their specified boundary leaves. Their union embeds $K_{3,3}$ in the plane, contradicting the bipartite planar Euler inequality $9 \leq 2 \cdot 6 - 4$.

*The proper line conclusion.* If $C$ is unbounded, put $C^* = C \cup \{\infty\}$ in the sphere; if $C$ is bounded, put $C^* = C$. This is a compact connected proper subset with exactly two complementary domains. Every point except possibly the added point $\infty$ is accessible from both. Choose a stereographic pole in one complementary domain. Deleting that pole leaves that domain connected, and accessibility at its boundary is unaffected, since end-cuts can be chosen to avoid the pole. In the resulting plane both inaccessible subsets of $C^*$ have at most one point. Polulyakh's converse Jordan theorem [63, Theorem 1.1 and Corollary 1.1] therefore makes $C^*$ a Jordan curve. The bounded alternative is impossible: the real-valued open map $F$ is constant on that Jordan curve, and its restriction to the bounded disk either has an interior extremum or is constant on a nonempty open set. Both contradict openness. Thus $C$ is a Jordan curve with the compactification point removed, a properly embedded line. Transitivity of $\mathcal{K}$ transfers this conclusion to every component of every fiber.

Downstairs, a fiber of $\Phi_z$ is homogeneous under $L_z$, and each of its component stabilizers

is transitive by the same orbit argument. A nondegenerate compact component is therefore homogeneous by ambient neighborhood homeomorphisms. Brechner–Erkama [14, Theorem 2] makes it a Jordan curve.

*An exact disconnected-fiber example.* On $r > 1$ in the plane put $s = \log(r-1)$. Let $\beta_b$ be the time-$b$ flow of $\sin\delta\,\partial_\delta$ on $\mathbb{R}$. It fixes $\pi\mathbb{Z}$, is $2\pi$-periodic as a lifted circle map, and $|\beta_b(\delta) - \delta| \le \pi$. Define

$$\begin{aligned} A_a(r,\theta) &= (r, \theta + a) && (r \le 1), \\ A_a(r,\theta) &= (1 + e^a(r-1), \theta + a) && (r > 1), \\ B_b|_{\{r\le 1\}} &= \mathrm{id}, \\ B_b(s,\theta) &= (\theta + \beta_b(s-\theta), \theta) && (r > 1). \end{aligned} \tag{C.42.5}$$

The last first coordinate is $s$. Its expression is independent of the chosen representative of $\theta$. The bound on $\beta_b - \mathrm{id}$ gives continuity at $r = 1$ and properness at infinity for $B_b$ and $B_{-b}$. The $A$-formulas have these properties directly. They are continuous orientation-preserving one-parameter actions and commute, since $A_a$ preserves $s - \theta$.

The open spiral strip

$$(t,\delta) \longmapsto (1 + e^{t+\delta},\ t \bmod 2\pi), \qquad t \in \mathbb{R}, \quad 0 < \delta < \pi, \tag{C.42.6}$$

is an open orbit $O$. This parametrization is injective because an angular ambiguity is $2\pi k$, while two $\delta$-values differ by less than $\pi$. In these coordinates $A_a$ translates $t$ and $B_b$ sends $\delta$ to $\beta_b(\delta)$. The resulting group is $\mathbb{R}^2$, simply transitive on $O$ and closed in the plane homeomorphism group. Indeed, the open-orbit preservation argument at the start of the proof shows that any homeomorphic limit preserves $O$, using both the forward and inverse limits. Its evaluation at $(0, \pi/2)$ therefore remains inside $O$ and recovers finite parameters $a = t$ and $b = \log\tan(\delta/2)$ continuously. The parameter limits recover the limiting plane homeomorphism, proving closedness. A triangular parameter loop $0 \to (1,0) \to (0,1) \to 0$ generates this group and contracts by scaling both parameters.

For the actual commuting operator subgroup $H = \langle A_{-2\pi}\rangle$, its powers $h_n = A_{-2\pi n}$ converge on compact subsets of $O$ to

$$\Phi(t,\delta) = (1,\ t \bmod 2\pi). \tag{C.42.7}$$

This is exactly the quotient associated to the limit of $h_n(0,\pi/2)$. Its fiber over $\theta_0$ is the disjoint union of the lines $t = \theta_0 + 2\pi k$, $k \in \mathbb{Z}$. Its real lift is $t$. All maps and the ambient contraction have been specified, so this example concerns disconnected whole fibers rather than a nonzero centralizer fundamental-group class. The executed symbolic certificate `code/part_b/c42_spiral_fibre_certificate.py` checks the flow law and differential equation in the exact half-angle coordinate, the $A$–$B$ commutation identities, the integer-labelled whole fibers and their real lift, and the based triangular parameter loop. It uses no numerical approximations; the topological conclusions are proved above. □

**Scope of the reduction.** C.42 proves the component statement for the specified real-valued lift. It does not identify a lifted component with a whole downstairs connected component without checking the actual covering restriction. It does not yet give local product charts for the component decomposition, simultaneous charts for several boundary quotients, or the frontier-compatible contraction needed in C.12. Those global extension questions remain open.

**C.43. [Proved] The lifted boundary quotient is a product projection.** For the actual quotient and covering maps in C.42, every fiber of the real lift $F : \widetilde{O} \to \mathbb{R}$ is connected. There is a homeomorphism

$$T : \mathbb{R} \times \mathbb{R} \longrightarrow \widetilde{O}, \qquad F(T(u,t)) = t. \tag{C.43.1}$$

In particular the components in C.42 form a foliation with genuine local product charts. The original map $\Phi_z : O \to B$ also has local product charts with its specified target coordinate. The construction treats one quotient map at a time; its auxiliary flow and the homeomorphism $T$ carry no asserted equivariance under the original operator family.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; actual boundary-quotient maps in the remaining relative centralizer extension.

*Proof.* Write $X = \widetilde{O}$ and let $\mathcal{L}$ be the closed Polish group (C.42.4), acting transitively on the covering plane $X$. For $v \in X$, let $C_v$ be the connected component of $F^{-1}(F(v))$ containing $v$. By C.42 these sets partition $X$ into properly embedded lines. The equation $Fa = bF$ for an element $(l, a, b) \in \mathcal{L}$ implies

$$a(C_v) = C_{a(v)}. \tag{C.43.2}$$

Thus this is the actual component partition preserved by the given compatible action.

*Regularity and local products.* Let $I \subset C_v$ be a compact arc containing $v$, and let $\epsilon > 0$. In a fixed metric on the covering plane, the condition

$$\sup_{w \in I} d(a(w), w) < \epsilon$$

defines an identity neighborhood in $\mathcal{L}$. The Polish open orbit-map theorem makes its image at $v$ a neighborhood of $v$. Consequently, for every sufficiently near $v'$, some such $a$ sends $v$ to $v'$. Restricting $a$ to $I$ gives an $\epsilon$-close embedding of $I$ into $C_{v'}$ carrying $v$ to $v'$. This verifies precisely the regularity condition of Aarts–Oversteegen [1, Definition 5.3]. Their local-section and flowbox theorems [1, Theorems 5.6–5.7] therefore provide open product neighborhoods $S \times (-1, 1) \to X$, with each vertical plaque contained in one $C_v$.

The transverse factor is an interval locally. Indeed it is metrizable, and its product with an interval is a surface without boundary. The elementary product-cancellation argument proved in C.33 applies: local path connectedness descends to $S$, embedded open arcs in $S$ are open by invariance of domain, and an endpoint would have zero local second homology in the product. Thus $S$ is locally a one-manifold without boundary. Shrink to an interval and write one of these charts as

$$\chi : J \times (-1, 1) \longrightarrow V \subset X.$$

Because every plaque lies in a fiber, $F\chi(s, u) = f(s)$ for a continuous function $f$. For each open $W \subset J$, the set $\chi(W \times (-1, 1))$ is open in $X$, so the openness of $F$ proves that $f(W)$ is open. A continuous open map between intervals is strictly monotone: failure of strict monotonicity produces a local maximum, local minimum, or a constant interval, each contradicting openness. It follows that $f$ is a homeomorphism onto an open interval. Replacing the transverse coordinate by $f(s)$ gives actual charts in which $F$ is the projection.

*The component quotient and its topology.* Let $Y$ be the set of components $C_v$, equipped with the quotient topology, and let $\pi : X \to Y$ be the quotient map. This map is open. To verify this directly, let $U \subset X$ be open and let $v$ belong to the saturation of $U$. Choose $w \in C_v \cap U$ and the compact arc in $C_v$ joining $v$ to $w$. Apply the regularity argument above with an error small enough that the image of $w$ remains in $U$. For every $v'$ in a neighborhood of $v$ the resulting image arc joins $v'$ to a point of $U$ in $C_{v'}$. The saturation of $U$ is therefore open, which proves the assertion about $\pi$.

In a chart where $F$ is projection, distinct plaques have distinct $F$-values. They therefore belong to distinct global components. The transversal maps homeomorphically onto $\pi(V)$: its inverse is the restriction of the continuous factor map $\overline{F}$ in

$$X \xrightarrow{\pi} Y \xrightarrow{\overline{F}} \mathbb{R}, \qquad F = \overline{F}\pi. \tag{C.43.3}$$

Here $\pi(V)$ is open by the preceding paragraph, and the forward inverse in the chosen target coordinate is the continuous transversal section. Thus $Y$ is a one-manifold without boundary in the sense that it is locally an interval; no Hausdorff property has been assumed. The same charts show that $\overline{F}$ is a local homeomorphism.

The quotient $Y$ is connected and second countable: an open quotient carries a countable base to a countable base. Every element of $\mathcal{L}$ induces a homeomorphism of $Y$ by (C.43.2), and transitivity on $X$ makes $Y$ homogeneous. Baillif–Gabard's theorem [3, Theorem 1.1, proved in Section 4] now proves that $Y$ is Hausdorff. The topological hypotheses here are exact: second countability gives Lindelöfness, and the actual quotient charts give the local Euclidean structure required in that theorem.

One can also see the separation mechanism directly from their proof. In a locally Hausdorff space, Zorn's lemma supplies a maximal Hausdorff open set through each point; maximality makes it dense. A countable subcover exists by Lindelöfness. A manifold is locally Baire and hence Baire, so those dense open sets have a common point. That point is separated from every other point inside one member of the subcover. Homogeneity transports this property to every point.

A connected second-countable Hausdorff one-manifold without boundary is a line or a circle. The circle alternative is impossible because $\overline{F}$ is open and has image $\mathbb{R}$ by C.42. Thus $Y$ is a line. Its local homeomorphism $\overline{F} : Y \to \mathbb{R}$ is globally strictly monotone: the local direction cannot change on a connected line. It is onto, hence a homeomorphism. Each fiber of $F$ is consequently one of the proper lines $C_v$.

*An exact global product for this map.* Orient each leaf using the orientation of the covering plane together with the increasing transverse coordinate $F$. On an overlap of two orientation-preserving charts $(u, F)$, the leaf-coordinate transition is increasing. This supplies the coherent orientation required for the Aarts–Oversteegen global-flow theorem [1, Main Theorem 0.5, its proof in Section 4, and Remark 4.4]. It gives a continuous flow

$$R : \mathbb{R} \times X \longrightarrow X$$

whose orbits are exactly the specified leaves. The flow is free, since those leaves are lines rather than circles, and a nontrivial period would make the corresponding orbit a circle.

Take any small transversal section $\sigma : I \to X$ with $F\sigma(t) = t$. The map

$$\Theta_\sigma : \mathbb{R} \times I \longrightarrow F^{-1}(I), \qquad (u, t) \longmapsto R(u, \sigma(t)) \tag{C.43.4}$$

is continuous and bijective: every fiber is exactly one flow orbit, and freeness gives uniqueness of $u$. Both spaces are surfaces, so invariance of domain makes (C.43.4) a homeomorphism. These are actual local trivializations of a principal $\mathbb{R}$-bundle over $\mathbb{R}$.

For completeness, its global section can be constructed with the group coordinate retained. Choose a locally finite interval cover $(I_i)$ with sections $\sigma_i$ and a subordinate partition of unity $(\rho_i)$ whose supports lie in $I_i$. On overlaps, let the continuous real transition functions be defined by

$$\sigma_j(t) = R(a_{ij}(t), \sigma_i(t)).$$

They obey $a_{ij} = a_{ik} + a_{kj}$. Set $b_i(t) = \sum_j \rho_j(t) a_{ij}(t)$ on $I_i$, extending the individual terms by zero outside their supports. The sum is locally finite and $b_i = a_{ik} + b_k$ on overlaps. Consequently $\sigma(t) = R(b_i(t), \sigma_i(t))$ is a well-defined continuous global section with $F\sigma(t) = t$. The same map as (C.43.4), now over all $\mathbb{R}$, gives (C.43.1).

Finally, the identities $qF = \Phi_z p$ from C.42, together with local charts for the two covering maps $p$ and $q$, carry the projection charts to the original map $\Phi_z$. This proves its stated local product property with the actual target coordinate. □

**Scope of the reduction.** The auxiliary flow in the proof is used to trivialize one specified quotient. It has not been shown to commute with the original operators, preserve another boundary quotient, or extend across the frontier of $O$. C.43 therefore strengthens the structure of the necessary boundary maps without completing the simultaneous relative extension in C.12.

**C.44. [Proved] Exact downstairs forms of a nontrivial boundary quotient.** For the actual quotient $\Phi_z : O \to B$ of C.42, the open planar orbit $O$ is a plane or an annulus. Choose a coordinate $B = \mathbb{R}$ when $B$ is a line, or a coordinate $B = \mathbb{R}/\mathbb{Z}$ when it is a circle, and use the corresponding target covering in the definition of $F$. There are source homeomorphisms giving one of the following normal forms for the actual map:

| $O$ | $B$ | $\Phi_z$ | whole fiber |
|---|---|---|---|
| $\mathbb{R} \times \mathbb{R}$ | $\mathbb{R}$ | $(u,t) \mapsto t$ | $\mathbb{R}$ |
| $\mathbb{R} \times \mathbb{R}$ | $\mathbb{R}/\mathbb{Z}$ | $(u,t) \mapsto t + \mathbb{Z}$ | $\coprod_{\mathbb{Z}} \mathbb{R}$ |
| $(\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$ | $\mathbb{R}$ | $([u],t) \mapsto t$ | $\mathbb{R}/\mathbb{Z}$ |
| $(\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$ | $\mathbb{R}/\mathbb{Z}$ | $([u],t) \mapsto t + \mathbb{Z}$ | $\coprod_{\mathbb{Z}} (\mathbb{R}/\mathbb{Z})$ |
| $\mathbb{R} \times (\mathbb{R}/m\mathbb{Z})$ | $\mathbb{R}/\mathbb{Z}$ | $(u,[t]_m) \mapsto [t]_1$ | $\coprod_{j=0}^{m-1} \mathbb{R}, \quad m \geq 1.$ |

(C.44.1)

In the last row the actual induced image $\Phi_{z*}(\pi_1 O)$ is $m\mathbb{Z} \subset \pi_1 B = \mathbb{Z}$; in all other rows that image is zero. Every downstairs fiber component is a properly embedded line or a circle, and the source covering restricts onto each such component as the line covering specified by these formulas.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact quotient and covering constraints in the remaining frontier-extension problem.

*Proof.* Let $\Gamma$ be the deck group of $p : \widetilde{O} \to O$. Use C.43 to identify the covering plane with $\mathbb{R}_u \times \mathbb{R}_t$ so that $F(u,t) = t$. The equation $qF = \Phi_z p$ implies, for each $\gamma \in \Gamma$,

$$F\gamma = F + \eta(\gamma), \tag{C.44.2}$$

where $\eta(\gamma) = 0$ for a line target and $\eta(\gamma) \in \mathbb{Z}$ for a circle target. For the circle, the difference is a continuous integer-valued function on the connected covering plane and is therefore constant. Composition shows that $\eta : \Gamma \to \mathbb{Z}$ is a homomorphism. Its value is exactly the winding number of the image under $\Phi_z$ of the source loop whose lift ends at $\gamma$ times its initial point. Thus its image is the actual induced fundamental-group image.

Write $K = \ker \eta$. Every deck map preserves the orientation pulled back from the planar surface $O$. It has the form

$$\gamma(u,t) = (h_{\gamma,t}(u),\, t + \eta(\gamma)), \tag{C.44.3}$$

and each $h_{\gamma,t}$ is increasing: the base coordinate is increasing, and reversing the fiber direction would reverse the surface orientation. The functions in (C.44.3) and their fiberwise inverses are jointly continuous, by the deck map and its inverse.

*The exact kernel on a real fiber.* The action of $K$ on any fiber $\mathbb{R} \times \{t\}$ is free and proper. Freeness follows from deck freeness. For properness, compact subsets of this fiber are compact in the covering plane, and the deck action has finite transporters of compact pairs. The restriction is faithful, since a deck transformation fixing one point is the identity. An increasing free proper group on a line is trivial or infinite cyclic. Here is the direct argument. Its orbit of a point is closed and discrete. In the nontrivial case choose the element taking that point to its nearest orbit point on the right. Its iterates tend to the two ends, because a finite limiting point would be fixed. Every other orbit point is one of these iterates: an intervening point could be translated back between the original point and its nearest right neighbor. Freeness then identifies every element with the corresponding power. Consequently

$$K = 1 \quad \text{or} \quad K \cong \mathbb{Z}. \tag{C.44.4}$$

Suppose $K = \langle k \rangle \cong \mathbb{Z}$, choosing the generator to move one fiber point to the right. Since the displacement $h_{k,t}(u) - u$ is continuous and nowhere zero on the connected plane, it is positive everywhere. Put $k_t = h_{k,t}$. For $v = n + r$, where $n \in \mathbb{Z}$ and $0 \le r \le 1$, define

$$H(v,t) = \big(k_t^n(r\, k_t(0)),\, t\big). \tag{C.44.5}$$

At integral $v$ the two formulas agree. The intervals with endpoints $k_t^n(0)$ exhaust the line in both directions, so $H$ is a continuous bijection of the plane. Invariance of domain makes it a homeomorphism. It preserves $F$ and satisfies

$$H(v+1,t) = kH(v,t). \tag{C.44.6}$$

Thus the actual intermediate quotient is

$$\widetilde{O}/K \cong (\mathbb{R}/\mathbb{Z}) \times \mathbb{R}, \qquad F_K([v],t) = t. \tag{C.44.7}$$

*The compactness obstruction retaining the quotient map.* The case $K \cong \mathbb{Z}$ cannot coexist with a nonzero image of $\eta$. Indeed that image would be $m\mathbb{Z}$ for some $m \ge 1$. The quotient $\Gamma/K$ is then infinite cyclic, and its chosen generator acts on (C.44.7) by a circle homeomorphism in the first coordinate and by $t \mapsto t + m$ in the second. The compact strip $(\mathbb{R}/\mathbb{Z}) \times [0,m]$ maps onto the full quotient by $\Gamma/K$: each point can be translated into that strip using its second coordinate. It follows that

$$O = \widetilde{O}/\Gamma = (\widetilde{O}/K)/(\Gamma/K)$$

is compact, contrary to its being a nonempty open subset of the plane. Therefore $K \cong \mathbb{Z}$ forces $\eta = 0$ and $\Gamma = K$. Equations (C.44.5)–(C.44.7) give the third and fourth rows of (C.44.1), according to the target.

If $K = 1$ and $\eta = 0$, then $\Gamma = 1$, so C.43 gives the first two rows.

*The remaining nonzero covering degree.* It remains that $K = 1$ and $\eta(\Gamma) = m\mathbb{Z}$, with $m \ge 1$. Then $\Gamma$ itself is infinite cyclic. Choose its generator $\gamma$ with $\eta(\gamma) = m$, and write

$$\gamma(u,t) = (h_t(u), t + m). \tag{C.44.8}$$

We give a coordinate change preserving the real target coordinate and removing $h_t$. For $0 \le t \le m$ set

$$\rho_t(u) = \left(1 - \frac{t}{m}\right) u + \frac{t}{m} h_0(u). \tag{C.44.9}$$

Each $\rho_t$ is an increasing homeomorphism of the line, with $\rho_0 = \mathrm{id}$ and $\rho_m = h_0$. Extend to all real $t$ in both directions by the exact recursion

$$\rho_{t+m} = h_t \circ \rho_t. \tag{C.44.10}$$

The endpoint identity makes the extension jointly continuous at every multiple of $m$. The map $H_1(u,t) = (\rho_t(u), t)$ is a continuous bijection of the plane and hence a homeomorphism. Equations (C.44.8)–(C.44.10) give

$$H_1^{-1} \gamma H_1(u,t) = (u, t+m). \tag{C.44.11}$$

The quotient is therefore $\mathbb{R} \times (\mathbb{R}/m\mathbb{Z})$, and the original map, still satisfying $qF = \Phi_z p$, is precisely $(u,[t]_m) \mapsto [t]_1$. This is the final row of (C.44.1).

All changes of source coordinates preserved $F$; the target was changed only by its initially specified coordinate. The whole-fiber descriptions now follow directly. For a circle target, levels $F = t_0 + j$ are indexed by $j \in \mathbb{Z}$. The deck group identifies their labels by $\eta(\Gamma)$ and acts within each level by $K$. Thus the component labels are $\mathbb{Z}/\eta(\Gamma)$ and each component is $\mathbb{R}/K$ in the explicit coordinates above. For a line target there is only one real level and the component is again $\mathbb{R}/K$. These formulas also prove surjectivity of the restricted source covering onto each downstairs component and give its deck group. In particular, they establish the component assertion without assuming component-surjectivity for an arbitrary restricted covering. □

**Scope of the reduction.** C.44 classifies each of the actual quotient maps separately, including its whole fibers and covering labels. Several boundary points can impose several such maps on the same open orbit. No common product coordinates or homotopy preserving all those maps have been constructed here. That simultaneous compatibility and frontier extension remain part of C.12.

**C.45. [Proved] Two actual boundary quotients give explicit planar coordinates.** Let $\Phi_1 : O \to B_1$ and $\Phi_2 : O \to B_2$ be two actual nontrivial boundary quotients of the same open orbit as in C.42. Use the same source universal covering $p : X \to O$ and the real target coverings $q_i : \mathbb{R} \to B_i$, and choose lifts $q_i F_i = \Phi_i p$. Exactly one of the following alternatives holds. Either there is a homeomorphism $f : \mathbb{R} \to \mathbb{R}$ with

$$F_2 = fF_1, \tag{C.45.1}$$

or the pair $(F_1, F_2)$ is a homeomorphism onto an open domain $\Omega \subset \mathbb{R}^2$ whose horizontal and vertical sections are intervals and whose two projections are onto $\mathbb{R}$. In the second alternative, explicit independent changes of the two real coordinates, and a possible exchange of them, carry $\Omega$ onto one of

$$\mathbb{R}^2, \qquad \{(x,y) : x < y\}, \qquad \{(x,y) : x < y < x+1\}. \tag{C.45.2}$$

The full increasing coordinate-pair groups preserving these three domains are, respectively,

$$\mathrm{Homeo}^+(\mathbb{R})^2, \qquad \{(g,g) : g \in \mathrm{Homeo}^+(\mathbb{R})\}, \qquad \{(g,g) : g \in \mathrm{Homeo}^+(\mathbb{R}),\ g(x+1) = g(x)+1\}. \tag{C.45.3}$$

The actual compatible lifted action maps into the corresponding group. These statements retain both original covering equations and make no identification of their downstairs component labels.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; simultaneous actual boundary-quotient constraints in the remaining extension problem.

*Proof. A common compatible group.* Use the closed subgroup

$$\begin{aligned}\mathcal{P} = \{(l, a, b_1, b_2) \in L \times \mathrm{Homeo}(X) \times \mathrm{Homeo}^+(\mathbb{R})^2 :\ & \\ pa = (l|_O)p, & \qquad (\text{C.45.4})\\ q_i b_i = (l|_{B_i})q_i, \qquad F_i a = b_i F_i \quad (i = 1,2)\}. &\end{aligned}$$

The equations are closed in the product of the indicated Polish groups. For every lift $a$ of $l|_O$, each target lift $b_i$ exists uniquely after its deck translation is chosen at one point, exactly as in C.42. Consequently $\mathcal{P}$ acts transitively on the source covering plane. Both $F_i$ are open onto $\mathbb{R}$ with connected proper-line fibers, by C.43.

Fix $t \in \mathbb{R}$ and set $C = F_1^{-1}(t)$. The closed subgroup of $\mathcal{P}$ with $b_1(t) = t$ acts transitively on this whole line. Its action on the interval $I = F_2(C)$ is also transitive, by the equality $F_2 a = b_2 F_2$. All these target maps are increasing. A nondegenerate interval admitting such a transitive action has no included endpoint, so it is an open interval. The Polish open orbit-map theorem, applied to the actions on $C$ and $I$, makes $F_2|_C : C \to I$ the open map between their homogeneous-space quotients: the point stabilizer in $C$ is contained in the stabilizer of its image in $I$. It is therefore a strictly monotone homeomorphism.

If instead $I$ is a point, transitivity on the source plane transports this constancy to every $F_1$-fiber. There is then a function $f$ satisfying (C.45.1). Since $F_1$ is an open quotient, $f$ is continuous. For every open $W \subset \mathbb{R}$,

$$f(W) = F_2(F_1^{-1}(W))$$

is open. Thus $f$ is open and onto, hence a homeomorphism of the line. This is the first alternative.

In the nonconstant case, $(F_1, F_2)$ is injective, since $F_2$ is injective on each $F_1$-fiber. Invariance of domain makes it a homeomorphism onto an open domain $\Omega$. Each vertical section is the open interval $I$ just proved. Interchanging the two indices gives the horizontal statement; the constant alternative in that direction would contradict the already proved two-dimensional open image. Surjectivity of the two $F_i$ gives both full coordinate projections.

*The endpoint maps are actual homogeneous quotients.* The coordinate action $(b_1, b_2)$ of $\mathcal{P}$ preserves $\Omega$ and is transitive on it. Write its vertical intervals as

$$\Omega_x = (\alpha(x), \beta(x)), \quad -\infty \le \alpha(x) < \beta(x) \le +\infty. \tag{C.45.5}$$

The two endpoints satisfy the exact equations

$$\alpha(b_1 x) = b_2(\alpha(x)), \qquad \beta(b_1 x) = b_2(\beta(x)), \tag{C.45.6}$$

where increasing homeomorphisms fix the appropriate infinite endpoints. Transitivity of the first coordinate action implies that $\alpha$ is either everywhere finite or identically $-\infty$, and that $\beta$ is either everywhere finite or identically $+\infty$.

Each finite endpoint map is a homeomorphism of the real line. For example, choose $x_0$ with $\alpha(x_0) \in \mathbb{R}$. The formula

$$\alpha(b_1 x_0) = b_2(\alpha(x_0))$$

descends the continuous second evaluation through the open first orbit map $\mathcal{P} \to \mathbb{R}$. It proves continuity of $\alpha$ in its actual topology. Its image is nonempty and invariant under the transitive second coordinate action, hence is all $\mathbb{R}$. The stabilizer of $x_0$ in the first action fixes $\alpha(x_0)$ in the second by (C.45.6). The two open orbit maps identify $\alpha$ with the associated open coset-space map. It follows that $\alpha$ is continuous, open, and onto, and therefore strictly monotone and a homeomorphism. The same proof applies to finite $\beta$. This verifies the endpoint topology without assuming regularity of a boundary graph.

*The three domains and the coordinate changes.* If both endpoints are infinite, $\Omega = \mathbb{R}^2$. If precisely one endpoint is finite, use its inverse as the second coordinate. The defining inequality becomes $x < y$ or $y < x$, according to which endpoint and monotonicity occur. In the latter case exchange the two coordinates.

Suppose both endpoints are finite. They have the same monotonicity: an increasing homeomorphism and a decreasing homeomorphism cannot satisfy $\alpha(x) < \beta(x)$ for all real $x$, by their limits at the two ends. When both increase, replace $y$ by $\alpha^{-1}(y)$ and put $\tau = \alpha^{-1}\beta$. When both decrease, replace $y$ by $\beta^{-1}(y)$ and put $\tau = \beta^{-1}\alpha$. In either case the domain is now exactly

$$x < y < \tau(x), \qquad \tau \in \mathrm{Homeo}^+(\mathbb{R}),\ \tau(x) > x. \tag{C.45.7}$$

Choose $x_0$ and any increasing homeomorphism $h : [x_0, \tau(x_0)] \to [0, 1]$. Extend it by $h(\tau^n x) = h(x) + n$. The iterates of $\tau$ exhaust the two ends, since $\tau$ has no fixed point, and the endpoint formulas agree. Thus $h$ is an increasing homeomorphism of $\mathbb{R}$ with $h\tau h^{-1}(x) = x + 1$. Applying $h$ to both coordinates in (C.45.7) gives the final domain in (C.45.2). These are the asserted explicit source coordinate maps; a reversal in one coordinate still conjugates increasing coordinate actions to increasing actions.

Finally, an increasing coordinate pair preserving $x < y$ must preserve its boundary graph $y = x$, which says $g = h$. For $x < y < x + 1$, its lower and upper endpoint graphs are separately preserved, which says $h(x) = g(x)$ and $h(x + 1) = g(x) + 1$. Conversely these identities preserve the indicated domains exactly. The unrestricted plane case has no additional equation. This proves (C.45.3).

All induced actions used above arise from (C.45.4). An element acting trivially on the covering plane may still act elsewhere under its $l$-coordinate; no faithfulness of that global coordinate has been imposed. The original equations $q_i F_i = \Phi_i p$ remain in force, so the separate deck shifts and component labels of C.44 are retained. □

**Scope of the reduction.** C.45 gives the relation between two specified actual boundary maps, including all coordinate changes. A third quotient can impose an additional equation on the group in (C.45.3). Preserving all such equations and obtaining a global frontier-compatible contraction are not consequences of this pairwise classification.

**C.46. [Proved] Two boundary foliations admit a compatible intrinsic contraction.** Use the actual based null-homotopic plane loop $f_t$, its generated group $G$, the open orbit $O$, and $L, x, J$ of C.42. For every $z \in \mathrm{Fix}(J) \cap \partial O$ with nontrivial intrinsic line or circle orbit, lift its quotient $\Phi_z$ to the common source universal covering $p : X \to O$ as in C.42. Let $\mathscr{F}$ be the set of fiber partitions of these real lifts; changing their target coordinates or their lift constants does not change this set. For $|\mathscr{F}| = 2$, the loop $f_t|_O$ has a based compact-open null-homotopy

$$K_{s,t} \in \mathrm{Homeo}(O), \qquad K_{0,t} = f_t|_O, \quad K_{1,t} = K_{s,0} = K_{s,1} = \mathrm{id}\,. \tag{C.46.1}$$

It commutes with every plane homeomorphism commuting with all $f_t$ and preserving $O$. For every indicated point $z$, it also satisfies the exact quotient relation

$$\Phi_z K_{s,t} = Q_{s,t}^{B_z} \Phi_z, \qquad B_z = Gz, \tag{C.46.2}$$

where $Q^{B_z}$ is the natural median contraction of the original restricted loop on that intrinsic line or circle, as in C.27–C.29. Thus all the specified boundary quotient relations hold simultaneously. The asserted homotopy is intrinsic to $O$; continuity at its ambient frontier is a further assertion.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison; simultaneous boundary relations in C12.

*Proof.* Choose representatives $F_1, F_2 : X \to \mathbb{R}$ of the two partitions. C.45 supplies a homeomorphism from $X$ onto one of its three domains $D$, with these two coordinates changed only by independent line homeomorphisms and a possible exchange. Lift the isotopy $f_t|_O$ from the identity to $a_t$ on $X$. For each original boundary target, lift $f_t|_{B_i}$ to its based real path $b_{i,t}$. Covering uniqueness gives

$$F_i a_t = b_{i,t} F_i. \tag{C.46.3}$$

Each $b_{i,t}$ is increasing. For a line target its endpoint is the identity. For a circle target C.29 identifies its endpoint translation with the winding of the original ambient-null plane loop and proves that translation zero. Hence both $b_{i,1}$ are the identity. Since the pair $(F_1, F_2)$ is injective, $a_1 = \mathrm{id}$. We have obtained an actual based loop in the corresponding coordinate-pair group (C.45.3), without discarding a source or target deck shift.

In these coordinates write $a_t = (g_{1,t}, g_{2,t})$. Apply C.27 to each coordinate loop and set

$$\begin{aligned} m_{i,r}(u) &= \min_{0 \le v \le r} g_{i,v}(u), \qquad M_{i,r}(u) = \max_{0 \le v \le r} g_{i,v}(u), \\ R_{i,s,t}(u) &= \mathrm{med}\big(m_{i,1-s}(u), g_{i,t}(u), M_{i,1-s}(u)\big), \\ \widetilde{K}_{s,t}(u,v) &= (R_{1,s,t}(u), R_{2,s,t}(v)). \end{aligned} \tag{C.46.4}$$

The coordinate maps and their inverses are jointly continuous by C.27 and C.31. They preserve the actual domain $D$. For $D = \mathbb{R}^2$ this follows directly from their being line homeomorphisms. For $D = \{u < v\}$, C.45 gives $g_{1,t} = g_{2,t}$, so the two new coordinate maps are identical and preserve the strict inequality. For $D = \{u < v < u + 1\}$ the same equality holds and each $g_{i,t}$ commutes with translation by one. Minima, maxima and the median preserve that equation, so the new equal coordinate maps preserve both inequalities. The inverse family has the same property. Formula (C.46.4) is therefore a based null-homotopy in $\mathrm{Homeo}(X)$ with the asserted endpoint identities.

We establish the operator comparison before descending. Let $h$ commute with all $f_t$ and preserve $O$. It commutes with $L$ by continuity of composition and inversion. The stabilizers of $x$

and $hx$ in $L$ are equal: an element fixes one precisely when it fixes the other, by commutation with $h$ and $h^{-1}$. Choose $a_h \in L$ with $a_h x = hx$. The exact stabilizer identity gives $a_h J a_h^{-1} = J$. For every specified $z$ and every $l \in L$, evaluation gives

$$\Phi_z h(lx) = \Phi_z(la_h x) = la_h z = \Phi_{a_h z}(lx). \tag{C.46.5}$$

The point $a_h z$ lies in $\mathrm{Fix}(J) \cap \partial O$: normalizing $J$ proves the first property, and $a_h$ preserves $O$ and its boundary. Its orbit is the same $Lz = Gz$ as that of $z$, so it has the same intrinsic type. Thus pullback by $h$ permutes the full set $\mathscr{F}$. The inverse $h^{-1}$ gives the inverse permutation. After lifting to $X$, the two sides of (C.46.5) differ at most by the specified target deck translation, which leaves their fiber partitions identical.

Every lift $\widetilde{h}$ of $h|_O$ consequently permutes the two coordinate foliations of $D$. It is a coordinate pair of line homeomorphisms, possibly followed by exchange of the coordinates. For clarity, preservation of a fiber partition defines its map on the corresponding real quotient; the open quotient projection proves continuity of that map and of its inverse. Surjectivity of both coordinate projections then makes it a line homeomorphism. It may be increasing or decreasing. These quotient maps give exactly the stated coordinate form of $\widetilde{h}$.

Moreover $\widetilde{h}$ commutes with $a_t$. Their commutator projects to the identity on $O$ and is a continuous deck-valued function of $t$, equal to the identity at $t = 0$. The discrete deck group makes it the identity for every $t$. C.27's naturality under every line homeomorphism, together with the identical construction in both coordinates, now gives

$$\widetilde{h}\widetilde{K}_{s,t}\widetilde{h}^{-1} = \widetilde{K}_{s,t}. \tag{C.46.6}$$

In particular this holds for every source deck transformation, which is a lift of $h = \mathrm{id}$. Consequently the homotopy descends to $K_{s,t}$ on $O$. The evaluation maps and their inverse maps descend continuously through the open covering quotient. This proves compact-open continuity, the endpoint identities, and the asserted commutation with the original operators.

Finally, take any other specified $z$. Its lifted fiber partition equals one of the two chosen ones. The first alternative of C.45 identifies its real map as a line homeomorphism composed with that coordinate. Naturality of the median formula under this exact reparametrization proves

$$F_z \widetilde{K}_{s,t} = \widetilde{Q}_{s,t}^{B_z^z} F_z.$$

For a circle target the formula commutes with its integer deck translations, so it descends; for a line target it already is the stated target map. Using $q_z F_z = \Phi_z p$ proves (C.46.2). No ambient limit of points approaching $\partial O$ has been inferred from these quotient equations. □

**C.47. [Proved] Three independent boundary quotients force a Lie action.** Let $\Phi_i : O \to B_i$, $1 \le i \le 3$, be three actual boundary quotients of the same open orbit as in C.42. Use their common source universal covering and real target coverings, with

$$p : X \longrightarrow O, \qquad q_i F_i = \Phi_i p, \qquad F_i : X \longrightarrow \mathbb{R}. \tag{C.47.1}$$

Suppose the three lifted fiber partitions are pairwise distinct; equivalently, none of the pairs satisfies the first alternative of C.45. The full group

$$\mathcal{W} = \{a \in \mathrm{Homeo}(X) : F_i a = b_i F_i \text{ for some } b_i \in \mathrm{Homeo}^+(\mathbb{R}),\ 1 \le i \le 3\} \tag{C.47.2}$$

is a closed, locally compact subgroup in the actual compact-open topology. It is a Lie group of dimension two or three and acts transitively on $X$. If any pair $(F_i, F_j)$ has image different from $\mathbb{R}^2$, this action is simply transitive and has dimension two. In particular, the closure in $\mathrm{Homeo}(X)$ of the compatible lifted loop group is locally compact. All three covering equations in

(C.47.1) and the kernel of the global-to-restricted action are retained. The conclusion concerns the action on the source covering plane; it does not assert an extension of an intrinsic homotopy across the frontier of $O$.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; simultaneous actual boundary-quotient constraints in the remaining frontier extension problem.

*Proof. The actual group topology.* Include all three target lifts in the single closed Polish group of tuples $(l, a, b_1, b_2, b_3)$ satisfying

$$pa = (l|_O)p, \qquad q_i b_i = (l|_{B_i})q_i, \qquad F_i a = b_i F_i \quad (1 \le i \le 3). \tag{C.47.3}$$

For each source lift, the three compatible target lifts are obtained exactly as in C.42–C.45. This group acts transitively on $X$, so its image in $\mathcal{W}$ does also. Each $b_i$ in (C.47.2) is unique because $F_i$ is onto. C.43 supplies a continuous section $\sigma_i : \mathbb{R} \to X$ of each $F_i$. Consequently

$$b_i = F_i a \sigma_i, \qquad b_i^{-1} = F_i a^{-1} \sigma_i. \tag{C.47.4}$$

These formulas are continuous for uniform convergence on compact sets, for maps and inverses.

They also prove closedness. If $a_n \in \mathcal{W}$ converges to a homeomorphism $a$ together with its inverse, (C.47.4) gives locally uniform limits $b_i$ and $d_i$ of the target maps and their inverses. The identities $b_{i,n} d_{i,n} = d_{i,n} b_{i,n} = \mathrm{id}$ pass to the limit, since the images of each compact set remain in a common compact set. Thus $d_i = b_i^{-1}$; each $b_i$ is an increasing homeomorphism, and $F_i a = b_i F_i$. Hence $\mathcal{W}$ is a closed Polish subgroup of $\mathrm{Homeo}(X)$.

C.45 identifies $X$ by any two of the $F_i$ with an open domain having interval coordinate sections. In coordinates $(x, y) = (F_1, F_2)$, write $F_3 = w(x, y)$. Every restriction of $w$ to a horizontal or vertical section is strictly monotone. The signs are constant over the connected domain: for two fixed nearby values in one coordinate, the continuous difference of their $w$-values cannot change sign. The coordinate representation of $\mathcal{W}$ is therefore exactly the triples $(g_1, g_2, k)$ satisfying

$$(g_1, g_2)\Omega = \Omega, \qquad w(g_1 x, g_2 y) = k(w(x, y)). \tag{C.47.5}$$

The map from $\mathcal{W}$ to these triples and its inverse are continuous, by (C.47.4) and the pair-coordinate homeomorphism. All topological group assertions below use this topology.

*A strictly interior mean makes an ordered-pair stabilizer trivial.* We first prove an elementary fact used in the non-plane cases. Let

$$D = \{(a, b) : a < b\} \quad \text{or} \quad D = \{(a, b) : a < b < a + 1\}. \tag{C.47.6}$$

Suppose a group of increasing line homeomorphisms preserves $D$ diagonally and preserves a continuous function $M : D \to \mathbb{R}$ with

$$a < M(a, b) < b, \qquad M \text{ injective in each argument separately.} \tag{C.47.7}$$

A group element fixing one pair $(a, b) \in D$ is the identity. Indeed, it fixes all the points obtained by repeatedly inserting $M$ between consecutive fixed endpoints. These subdivisions have mesh tending to zero. To verify this, fix $\varepsilon > 0$. On the compact set of pairs

$$a \le u < v \le b, \qquad v - u \ge \varepsilon,$$

both $M(u, v) - u$ and $v - M(u, v)$ have a positive minimum. Every such pair is still in $D$, including in the band case because $v \le b < a + 1 \le u + 1$. Every subdivision of an interval of length at least $\varepsilon$ therefore shortens each child by a fixed positive amount. No sufficiently deep subdivision can have that length. Continuity now makes the homeomorphism fix $[a, b]$ pointwise.

Let $I$ be the maximal interval of fixed points containing $[a, b]$. A finite right endpoint $c$ belongs to the fixed set. Choose $x < c$ in the interior of $I$ with $(x, c) \in D$. For $y > c$ sufficiently close to $c$, the pair $(x, y)$ remains in $D$ and $M(x, y)$ remains in the interior of $I$. Equivariance gives

$$M(x, gy) = M(gx, gy) = gM(x, y) = M(x, y).$$

Injectivity in the second argument forces $gy = y$, extending $I$ past $c$. The analogous argument, with a fixed $y > c$ and $x < c$ close to a finite left endpoint, uses injectivity in the first argument. Both finite endpoints are impossible. Thus $I = \mathbb{R}$ and $g = \mathrm{id}$.

*A non-plane pair and two full remaining pairs.* Suppose some pair image is not $\mathbb{R}^2$. Relabel and apply the explicit coordinate changes of C.45 so that $\Omega = D$ is one of (C.47.6). The first two target actions then coincide: $g_1 = g_2 = g$; in the band case $g(x + 1) = g(x) + 1$. First suppose the images of $(F_1, F_3)$ and $(F_2, F_3)$ are both the full plane. Each $w$-level is then the graph

$$y = r_z(x), \qquad z \in \mathbb{R}, \tag{C.47.8}$$

where $(x, z) \mapsto r_z(x)$ is continuous and every $r_z$ is a homeomorphism of $\mathbb{R}$ onto $\mathbb{R}$. It is increasing, since $r_z(x) > x$ for all $x$, which a decreasing real-line homeomorphism cannot satisfy. Reverse the $z$-coordinate when necessary. Then $z \mapsto r_z(x)$ is strictly increasing, and its range is $(x, +\infty)$ in the half-plane case and $(x, x + 1)$ in the band case. The sign is the same for every $x$ by connectedness. The action equation is exactly

$$g r_z g^{-1} = r_{kz}. \tag{C.47.9}$$

For $(a, b) \in D$, there is a unique $s = s(a, b)$ with

$$r_s^2(a) = b. \quad \text{Define} \quad M(a, b) = r_s(a). \tag{C.47.10}$$

Here are the endpoint and continuity checks. The left side is continuous and strictly increasing in $s$. As $s \to -\infty$, $r_s(x) \to x$, and hence $r_s^2(a) \to a$. This last limit follows by bounding $r_s(a)$ between $a$ and $a + \delta$ and then using monotonicity of $r_s$ at these two fixed points. As $s \to +\infty$ the square tends to $+\infty$ in the half-plane case. In the band case $r_s(x) \to x + 1$, and the same fixed-point bracketing gives $r_s^2(a) \to a + 2$. The required value $b$ lies strictly between the endpoint limits. Strict monotonicity gives uniqueness, and two nearby $s$-values bracketing a solution continue to bracket it when $(a, b)$ varies slightly; this proves continuity.

The value in (C.47.10) lies strictly between $a$ and $b$. It is strictly increasing in $b$ for fixed $a$. For fixed $b$ and $a_1 < a_2$, the corresponding parameters satisfy $s_1 > s_2$, because $r_s^2$ increases in its argument. If $r_{s_1}(a_1) \geq r_{s_2}(a_2)$, applying these increasing maps once more and using $s_1 > s_2$ gives

$$b = r_{s_1}(r_{s_1}(a_1)) > r_{s_2}(r_{s_2}(a_2)) = b,$$

a contradiction. Thus $M$ is strictly increasing in its first argument as well. Equation (C.47.9) and uniqueness in (C.47.10) prove $M(ga, gb) = gM(a, b)$. The preceding mean argument therefore makes every source point stabilizer trivial: once $g = \mathrm{id}$, surjectivity of $w$ in (C.47.5) also gives $k = \mathrm{id}$.

*A further non-plane pair identifies all three line actions.* It remains, with $\Omega = D$, to consider the case where at least one of the other pair images is not the full plane. One of its finite endpoint homeomorphisms from C.45 gives an exact equation

$$fg = kf. \tag{C.47.11}$$

Replace the third coordinate by $f^{-1}F_3$. Now all three target actions are the same $g$; this remains true when $f$ is decreasing. Denote the new third coordinate again by $w(x, y)$. An equality $w = x$

or $w = y$ at one point would hold everywhere by transitivity and contradict independence. The relative order of the three values is therefore constant.

In the half-plane, the projected group of $g$'s acts transitively on all ordered pairs. Its centralizer in Homeo($\mathbb{R}$) is trivial: the stabilizer of a point $x$ acts transitively on each of $(-\infty, x)$ and $(x, +\infty)$, so its only common fixed point is $x$. A commuting map must send $x$ to that same fixed point. It follows that every finite endpoint of either remaining pair domain, now an equivariant self-homeomorphism of the same line, is the identity.

When $x < w(x, y) < y$, take $M = w$. When $w(x, y) > y$, the $(x, w)$ image is exactly $\{(x, z) : x < z\}$: its finite lower endpoint must be the identity, and a finite upper endpoint would be the same map and leave an empty interval. For each $a < b$, define $M(a, b) = c$ by

$$w(a, c) = b. \tag{C.47.12}$$

The restriction in $c$ is increasing and maps $(a, +\infty)$ onto $(a, +\infty)$; its increase also follows from $w(a, c) > c$ and the limit as $c \to +\infty$. The unique solution satisfies $a < c < b$. When $w(x, y) < x$, the $(y, w)$ image is exactly $\{(y, z) : z < y\}$; define instead

$$w(c, b) = a. \tag{C.47.13}$$

The restriction in $c$ is increasing and maps $(-\infty, b)$ onto $(-\infty, b)$, so again $a < c < b$. Both inverse constructions are continuous by the pair-coordinate homeomorphisms. They are injective in each argument, by the strict injectivity of the corresponding restrictions of $w$. Equation (C.47.5), with all three actions equal, proves their equivariance.

In the band, $g$ commutes with unit translation. The equalities $w = x + n$ and $w = y + n$, for any integer $n$, are excluded by the same transitivity and independence argument. There is consequently one integer $n$, constant on the connected domain, such that

$$x < w(x, y) - n < x + 1.$$

Either $w(x, y) - n < y$ everywhere, or the reverse inequality holds everywhere. In the first case take $M(x, y) = w(x, y) - n$. In the second case use the exact domain homeomorphism

$$(u, v) = (y, x + 1), \qquad M(u, v) = w(v - 1, u) - n. \tag{C.47.14}$$

It sends the band onto itself, and $u < M(u, v) < v$. Both constructions are continuous and injective in each argument. They are equivariant because every $g$ commutes with the integer shifts. We have obtained a mean satisfying (C.47.7) in all cases.

The mean argument proves simple transitivity of $\mathcal{W}$ whenever a pair image is not the full plane. The Polish open orbit-map theorem makes evaluation $\mathcal{W} \to X$ a homeomorphism in this case. Thus $\mathcal{W}$ is locally compact and is a topological two-manifold. Its faithful product action on $X \times \mathbb{R}$ and the Hilbert–Smith theorem in dimension three [61, Theorem 1.5 and the introductory reduction] make it a two-dimensional Lie group.

*Three full pair images give a global ordered loop.* Suppose now that all three pair images are $\mathbb{R}^2$. The function $w : \mathbb{R}^2 \to \mathbb{R}$ is then continuous, and each partial map is a homeomorphism onto $\mathbb{R}$. Reverse input coordinates as necessary to make it increasing in both arguments. At a chosen source point $(a, b)$ put $c = w(a, b)$ and use the explicit coordinates

$$u = w(x, b) - c, \qquad v = w(a, y) - c, \qquad z = w(x, y) - c. \tag{C.47.15}$$

They give a continuous multiplication $m : \mathbb{R}^2 \to \mathbb{R}$ with

$$m(u, 0) = u, \qquad m(0, v) = v, \tag{C.47.16}$$

strictly increasing in both variables, each partial map onto $\mathbb{R}$. Its two division maps are continuous: solving a strictly increasing continuous equation is continuous by fixed bracketing on either side of the solution.

Let $S$ be the stabilizer of $(0,0)$ in $\mathcal{W}$. Setting either argument equal to zero in its action equation shows that the three target maps coincide. Thus, in its actual subspace topology, $S$ is exactly

$$\mathrm{Aut}^+(m) = \{f \in \mathrm{Homeo}^+(\mathbb{R}) : f(0) = 0, \quad f(m(x,y)) = m(fx, fy)\}. \tag{C.47.17}$$

This is a closed subgroup. We next prove its local compactness by uniform estimates for both maps and inverses.

Fix $a > 0$ and $0 < A < B$. Set

$$\mathcal{F}_{A,B} = \{f \in \mathrm{Aut}^+(m) : A \le f(a) \le B\}. \tag{C.47.18}$$

Define $D(x) = m(x,x)$ and $T_c(x) = m(c,x)$. The map $D$ is an increasing homeomorphism, $D(x) > x$ for $x > 0$, and $D(x) < x$ for $x < 0$. The numbers

$$v_n = D^{-n}(a) > 0, \qquad u_n \text{ defined by } m(u_n, v_n) = 0 \tag{C.47.19}$$

therefore converge to zero, with $u_n < 0$. Every automorphism respects these formulas, and hence

$$D^{-n}(A) \le f(v_n) \le D^{-n}(B). \tag{C.47.20}$$

The two bounds tend to zero. If $\ell(t)$ is the unique solution of $m(\ell(t), t) = 0$, then $f(u_n) = \ell(f(v_n))$; continuity of $\ell$ makes this tend to zero uniformly over $\mathcal{F}_{A,B}$ as well.

For $c > 0$, $T_c$ is increasing, onto, and satisfies $T_c(x) > x$. Its integer iterates of zero tend to both ends of $\mathbb{R}$. For every integer $j$,

$$f(T_a^j(0)) = T_{f(a)}^j(0). \tag{C.47.21}$$

To bound $f$ on a given compact interval, enclose it between two such iterates with $j = -N, N$. The right sides of (C.47.21) range over a compact set as $f(a)$ ranges over $[A, B]$; the negative iterates depend continuously on $c$ by the division-map continuity. Monotonicity then bounds all $f$ uniformly on that interval.

For equicontinuity, let $I$ be a compact interval. For fixed $n$, both gaps in

$$m(u_n, x) < x < m(v_n, x) \qquad (x \in I)$$

have positive minima on $I$. Thus a single $\delta_n > 0$ ensures

$$|y - x| < \delta_n,\ x \in I \quad \Longrightarrow \quad m(u_n, x) < y < m(v_n, x).$$

Applying $f$ bounds $f(y)$ between $m(f(u_n), f(x))$ and $m(f(v_n), f(x))$. The already proved compact bound on $f(I)$, the uniform limits $f(u_n), f(v_n) \to 0$, and continuity of $m$ show that both bounds approach $f(x)$ uniformly in $x$ and $f$. This is equicontinuity on every compact interval.

The inverse family has the same estimates with new fixed positive bounds. Choose $n$ so large that $D^{-n}(B) < a$. Then $f(v_n) < a$ and $f^{-1}(a) > v_n > 0$. Choose $N$ with $T_A^N(0) > a$. Since $T_c^N(0)$ increases with $c > 0$, (C.47.21) gives $f(T_a^N(0)) \ge T_A^N(0) > a$, and hence

$$v_n < f^{-1}(a) < T_a^N(0). \tag{C.47.22}$$

Applying the same compact bounds and equicontinuity argument to these positive bounds controls all inverses uniformly.

Arzelà–Ascoli on a countable compact exhaustion now gives, from any sequence in $\mathcal{F}_{A,B}$, locally uniform limits $f$ and $h$ of a common subsequence of maps and inverses. Their compact image bounds permit passage to the composition identities, so $fh = hf = \mathrm{id}$. The multiplication equation and the bounds in (C.47.18) pass to the limit. Thus $\mathcal{F}_{A,B}$ is compact in the homeomorphism-group topology. Taking $A < a < B$ gives a compact neighborhood of the identity in $S$, and proves its local compactness.

Evaluation $f \mapsto f(a)$ is also injective. An automorphism fixing $a$ fixes every $v_n$ and therefore every $T_{v_n}^j(0)$, for $j \in \mathbb{Z}$. For fixed $x$, choose the first such iterate at or to the right of $x$. It lies in $[x, T_{v_n}(x)]$, whose length tends to zero since $v_n \to 0$. These fixed points tend to $x$, so the automorphism fixes every $x$. The compactness of (C.47.18) proves that this continuous evaluation is a proper embedding

$$S \hookrightarrow (0, \infty). \tag{C.47.23}$$

*From the stabilizer to the full group, without changing topology.* The locally compact group $S$ acts faithfully and continuously on $\mathbb{R}^3$ by its action on the first coordinate. Pardon's theorem makes $S$ a Lie group. Its embedding (C.47.23) gives $\dim S \leq 1$. The full group $\mathcal{W}$ is still the closed Polish group already defined; it has not yet been assumed locally compact.

Consider the free right action of $S$ on $\mathcal{W}$. It is a Cartan action in its actual topology. Indeed, choose an identity neighborhood $U$ in $\mathcal{W}$ such that $U \cap S$ has compact closure in $S$, using the proved local compactness and the subspace topology of $S$. Choose an identity neighborhood $V$ with $V^{-1}V \subset U$. Then

$$\{s \in S : Vs \cap V \neq \varnothing\} \subseteq V^{-1}V \cap S$$

has compact closure in $S$, and translates provide the required neighborhoods at every point. The space $\mathcal{W}$ is metrizable and completely regular. The free-action form of Palais' slice theorem therefore gives local sections and local trivializations for

$$\mathcal{W} \longrightarrow \mathcal{W}/S \quad [60, \textit{Theorem } 2.3.3].$$

For clarity, the slice map has the form $\sigma(u)s \mapsto (u, s)$; its inverse is multiplication. If $\theta$ is the equivariant slice function to $S$, the inverse coordinates of $w$ are $(w\theta(w)^{-1}, \theta(w))$, as in the explicit free-action statement [45, Theorem 3.7]. The open orbit-map theorem identifies $\mathcal{W}/S$ homeomorphically with $X = \mathbb{R}^2$. These actual local products of the locally compact spaces $S$ and an open subset of $X$ prove that $\mathcal{W}$ is locally compact. Its faithful action on $X \times \mathbb{R}$ then makes it a Lie group by Pardon again. The same local products give $\dim \mathcal{W} = 2 + \dim S$, which is two or three.

Finally, every compatible lifted loop value belongs to $\mathcal{W}$, and its closure in $\mathrm{Homeo}(X)$ is a closed subgroup of this locally compact group. It is therefore locally compact with precisely the induced homeomorphism-group topology. None of the arguments replaces the $l$-coordinate of (C.47.3) by its action on $X$: the kernel of that restriction can still act elsewhere. The coordinate changes act on the specified real target lifts and leave the original equations (C.47.1) available throughout. This proves the asserted intrinsic reduction and leaves the frontier extension step explicit. □

**C.48. [Proved] One boundary foliation admits a compatible intrinsic contraction.** Use the based ambient-null plane loop and the full family $\mathscr{F}$ of lifted boundary fiber partitions in C.46. For an open orbit $O$ with $|\mathscr{F}| = 1$, there is a based compact-open null-homotopy $K_{s,t}$ of $f_t|_O$ which commutes with its entire original operator stabilizer and satisfies, simultaneously for every specified boundary point,

$$\Phi_z K_{s,t} = Q_{s,t}^{B_z} \Phi_z. \tag{C.48.1}$$

Here $Q^{B_z}$ is precisely the natural median homotopy of C.27–C.29. The original source and target covering maps are retained. This is an intrinsic assertion on $O$; ambient continuity at its frontier is an additional requirement.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and actual boundary quotient compatibility.

*Proof. The actual lifted operator group.* Let $p : X \to O$ be the universal covering and choose one real quotient $F : X \to \mathbb{R}$ representing the unique partition. By C.43, $X$ is a plane, $F$ has connected proper-line fibers, and a product chart identifies it with the second projection. Let $Q$ be the closure, in $\mathrm{Homeo}(O)$, of the restrictions of the original $O$-stabilizing operators. C.30 proves that this exact closed group acts freely and properly. Define its group of all lifts by

$$\widetilde{Q} = \{a \in \mathrm{Homeo}(X) : pa = qp \text{ for some } q \in Q\}. \tag{C.48.2}$$

It includes the source deck group. Its topology is the subspace homeomorphism-group topology, with the maps and their inverses controlled on compact subsets.

We give the needed covering comparisons explicitly. A lift is determined by its value at one point once its downstairs map is specified. For a convergent sequence of downstairs maps and a convergent compatible choice of those values, the lifts converge on compact subsets: cover a compact path from that point by finitely many covering charts and continue the chosen sheet successively; the same argument on finitely many such neighborhoods controls a compact set. Apply it to the inverse maps as well. Conversely a convergent sequence of lifts gives convergence downstairs on each compact set by finitely many local sections of $p$. These observations prove that (C.48.2) is closed and that its projection to $Q$ has the stated topology of lifts.

The action of $\widetilde{Q}$ on $X$ is free. A lift fixing a point has downstairs map fixing a point, hence equal to the identity; the resulting deck map fixing a point is the identity too. It is proper in this topology. Indeed, from $x_n \to x$ and $a_n x_n \to y$, properness downstairs gives a subsequence $q_n \to q$. There is a unique lift $a$ of $q$ with $ax = y$, and the covering-chart argument gives $a_n \to a$. The same argument makes evaluation at a point a homeomorphism onto a closed subset of $X$: convergence of its values supplies convergence of the corresponding downstairs maps and then of the selected lifts. Thus $\widetilde{Q}$ is locally compact. Its faithful product action on $X \times \mathbb{R}$ makes it a Lie group by the already checked Hilbert–Smith theorem of C.33. Its dimension is at most two, since each orbit is properly embedded in $X$.

The original operators preserve the unique partition by (C.46.5). This property passes to their restrictions' closure and to all lifts. More explicitly, choose lifts of an approximating sequence whose values at one point converge to the chosen lift. Equality of $F$-values on a fiber passes to the limit, and the inverse maps give equality of the resulting fiber partitions. Since $F$ is an open quotient, every $a \in \widetilde{Q}$ therefore has a unique induced line homeomorphism $\bar{a}$ with

$$Fa = \bar{a}F. \tag{C.48.3}$$

A continuous section of $F$ gives $\bar{a} = Fa\sigma$, and the inverse formula uses $a^{-1}$; hence this induced homomorphism is continuous in the actual topologies.

Lift the original restricted isotopy from the identity to $a_t$ on $X$. C.32 proves its null evaluation loop in $O$, so $a_1 = \mathrm{id}$. Every element of (C.48.2) commutes with $a_t$: its commutator is a continuous deck-valued function of $t$, initially the identity. Write

$$Fa_t = b_t F. \tag{C.48.4}$$

These are the based target lifts, and $b_0 = b_1 = \mathrm{id}$ by C.29 for a circle target and directly for a line target. The group generated by the paths $b_t$ acts transitively on $\mathbb{R}$. Its orbits are intervals, because every word is joined to the identity by a path of words. A finite endpoint of a proper orbit interval would be fixed by every word. Such a common fixed point would project to a fixed point of the nontrivial transitive boundary orbit, which is impossible. Thus the orbit interval is all of $\mathbb{R}$.

*The base operator image and its kernel.* Take the closed transitive group generated on this real line. The evaluation proof of C.30 applies to the line without change and shows that its full centralizer is locally compact, free and proper. Its product action on $\mathbb{R}^3$ makes it Lie. No decreasing element is possible: it would have a fixed point, contrary to freeness. In dimension one its identity component acts transitively on the line, and freeness excludes additional components; it is the real translation group. In dimension zero the nearest-orbit-point argument of C.44 gives either the identity group or a cyclic translation group. Choose the resulting line coordinate. We have thereby proved, with the comparison (C.48.3), that every base operator is a translation. Write

$$Fa = F + \alpha(a), \qquad \alpha : \widetilde{Q} \longrightarrow (\mathbb{R}, +), \qquad K = \ker \alpha, \tag{C.48.5}$$

after this specified change of target coordinate.

Restriction of $K$ to any fiber is faithful, free and proper. Evaluation gives it the same topology as its closed orbit in that line. The preceding one-dimensional classification therefore gives exactly

$$K = 1, \quad K \cong \mathbb{Z}, \quad \text{or} \quad K \cong \mathbb{R}. \tag{C.48.6}$$

All coordinate changes below are homeomorphisms of the source and reparametrizations of this same target. We construct them and keep the action of every component.

*The real kernel.* For $K \cong \mathbb{R}$, its orbits are the entire fibers. A global section of its principal bundle over the line, constructed as in C.33, gives coordinates $(u, r)$ with $F(u, r) = r$ and

$$k_c(u, r) = (u + c, r). \tag{C.48.7}$$

The quotient $\widetilde{Q}/K$ acts freely and properly on $X/K = \mathbb{R}$, by the exact local-section argument in C.33. Its image under $\alpha$ is consequently closed, and is $0$, $d\mathbb{Z}$ with $d > 0$, or $\mathbb{R}$. For image zero (C.48.7) is the complete operator action. For image $d\mathbb{Z}$, choose a label-$d$ element. Its action is

$$q(u, r) = (\lambda u + \tau(r), r + d), \lambda \in \mathbb{R}^\times. \tag{C.48.8}$$

The scalar is constant because conjugation on the actual group $K$ is a single continuous group automorphism. Choose $c$ on $[0, d]$ with $c(d) = \lambda c(0) + \tau(0)$ and extend it by $c(r+d) = \lambda c(r) + \tau(r)$. In the coordinates $u' = u - c(r)$ the complete additional operator is

$$q(u', r) = (\lambda u', r + d). \tag{C.48.9}$$

This retains negative $\lambda$ as well.

For image $\mathbb{R}$, the group $\widetilde{Q}$ acts transitively and freely on $X$. Both its kernel and quotient are connected lines, so it is connected. A Lie-algebra element mapping to 1 integrates to a section $s_r$ with $\alpha(s_r) = r$. Conjugation on $K$ has the form $s_r k_u s_r^{-1} = k_{e^{cr}u}$ for a real constant $c$. Evaluation gives the exact group coordinates and product

$$X \cong \widetilde{Q} \cong \mathbb{R}^2, \qquad (u, r)(v, z) = (u + e^{cr}v, r + z), \qquad F(u, r) = r. \tag{C.48.10}$$

The formula proves directly that these coordinates are global; $k_u s_r$ is the unique element with the specified base and fiber values. The operator action is left multiplication, and its full centralizer consists of right multiplications.

*The cyclic kernel.* For $K \cong \mathbb{Z}$, the construction (C.44.5) supplies coordinates with $F(u, r) = r$ and generator $k(u, r) = (u + 1, r)$. The quotient is the actual cylinder $X/K = (\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$. The action of $\widetilde{Q}/K$ on it is free and proper: lift convergent point sequences by local sections and correct by elements of $K$, then apply properness upstairs, exactly as in C.33. The compact circle fibers show that its base image is closed. Indeed, a sequence of base translations in a compact interval sends one fixed point into a compact cylinder strip; properness gives a convergent subsequence of the quotient elements. Thus again the image is $0$, $d\mathbb{Z}$, or $\mathbb{R}$.

For image $d\mathbb{Z}$, the additional generator has the form

$$q(u,r) = (g_r(u), r+d), \qquad g_r(u+1) = g_r(u) + \epsilon, \quad \epsilon \in \{1,-1\}. \tag{C.48.11}$$

Here $g_r$ is increasing for $\epsilon = 1$ and decreasing for $\epsilon = -1$. Choose an increasing unit-translation equivariant path $\rho_r$ on $0 \le r \le d$ from $\rho_0 = \mathrm{id}$ to $\rho_d = g_0 \circ (\epsilon\,\mathrm{id})$. The convex interpolation of these two increasing maps is such a path. Extend it in both directions by

$$\rho_{r+d} = g_r \rho_r (\epsilon\,\mathrm{id}). \tag{C.48.12}$$

The endpoint equation makes this extension continuous. Every $\rho_r$ remains increasing and commutes with unit translation. The homeomorphism $(u,r) \mapsto (\rho_r(u), r)$ therefore changes the two generators into exactly

$$k(u,r) = (u+1, r), \qquad q(u,r) = (\epsilon u, r+d). \tag{C.48.13}$$

For base image $\mathbb{R}$, the projection with discrete kernel is a covering homomorphism of Lie groups. Its identity component maps isomorphically onto the simply connected line. It centralizes $K$, since its conjugation map to $\mathrm{Aut}(\mathbb{Z}) = \{1,-1\}$ is continuous and is initially the identity. Thus $\widetilde{Q} = \mathbb{Z} \times \mathbb{R}$. Use this real subgroup to transport the fiber $F^{-1}(0)$ to every other fiber, and use the nearest-orbit-point coordinate for $K$ on that first fiber. These are global coordinates in which

$$k(u,r) = (u+1, r), \qquad s_z(u,r) = (u, r+z). \tag{C.48.14}$$

For base image zero only $k$ remains.

*Trivial kernel and the possible dense base image.* For $K = 1$, the group $\widetilde{Q}$ embeds algebraically in the translation group and is abelian. A positive dimensional identity component maps isomorphically onto $\mathbb{R}$; injectivity excludes other components. The same transport from $F^{-1}(0)$ gives just $s_z$ in (C.48.14). In dimension zero, $X \to X/\widetilde{Q}$ is a regular surface covering with simply connected source. Its deck group is the actual fundamental group of the quotient. A noncompact surface has free fundamental group, and an abelian free group is trivial or infinite cyclic. For a compact quotient the surface classification gives only the torus with abelian torsion-free fundamental group: the sphere and projective plane do not have plane universal cover, the Klein-bottle group is nonabelian, and every remaining closed surface group is nonabelian. Consequently the discrete possibilities are exactly 1, $\mathbb{Z}$, and $\mathbb{Z}^2$.

The cyclic case has a nonzero base translation; choose its positive generator. The construction (C.48.11)–(C.48.13), now without the unit-translation condition, changes it to $(u,r) \mapsto (\epsilon u, r+d)$. The trivial case has no equations. For $\widetilde{Q} \cong \mathbb{Z}^2$ the quotient is a torus, and its base image is dense: an injective homomorphism from $\mathbb{Z}^2$ into $\mathbb{R}$ has two generators with irrational ratio. The foliation descends to an oriented minimal topological foliation of that torus. To verify minimality, the saturation of any lifted fiber under the deck group is $F^{-1}(r + \alpha(\widetilde{Q}))$, which is dense because $F$ is open and the displayed set of levels is dense. The quotient torus is orientable, and its deck maps preserve both the surface and the transverse orientation, so the leaf orientation descends as asserted.

We give the elementary foliation conjugacy needed here, including its maps. It is the minimal-torus conclusion recorded in [50, Proposition 2.5]; no bi-foliation classification or asymptotic-cycle statement is needed. The oriented local flowboxes give a nonsingular continuous flow with these leaves by the construction used in C.43. Choose a small closed transverse interval, with endpoints off the orbit of an interior point, and a first-return arc from that point to a different point of the interval. A flowbox excludes sufficiently short positive returns; minimality supplies a later return, and compactness gives a first one. The endpoints chosen off the orbit put that return in the interval's interior. The arc's interior avoids the interval, and its endpoint differs from its initial point because there is no periodic orbit. Its union with the intervening transverse

subinterval is a simple closed curve. In a thin flow strip along the arc, tilt the arc strictly in the transverse direction pointing from the return point back to the initial point. The tilt may be chosen arbitrarily small, because the two endpoint transverse coordinates are distinct; choose it so that the new terminal point remains strictly between the old endpoints in the return interval. Join the new endpoints inside the transverse interval. Both pieces now have the same strict transverse direction; straightening their two corners in flowboxes gives an embedded closed transversal $\gamma$.

Every leaf meets $\gamma$, since it is dense and a flowbox around $\gamma$ is open. Forward and backward returns exist: on a compact minimal flow each forward and backward limit set is a nonempty closed invariant set, hence the entire torus. Flowboxes and avoidance of $\gamma$ on a compact interior return arc show that the first return time is continuous and positive. The return map $h : \gamma \to \gamma$ is a minimal orientation-preserving circle homeomorphism. The map which follows each point for a fraction of its return time gives the leaf-preserving homeomorphism from its mapping torus

$$[0,1] \times \gamma/((1,z) \sim (0,hz)) \longrightarrow X/\widetilde{Q}. \tag{C.48.15}$$

Surjectivity uses the preceding returns; uniqueness of the last intersection gives injectivity, and the compact source and Hausdorff target give the inverse continuity.

For completeness, choose a probability measure invariant under $h$ by taking a weak limit of finite orbit averages. Minimality gives full support, and the absence of periodic points gives no atoms. Its cumulative distribution is therefore a circle homeomorphism. It conjugates $h$ to a Lebesgue-measure-preserving increasing homeomorphism, hence to a rotation: preservation of the measure of every interval makes the displacement constant. The angle $\theta$ is irrational by minimality. After this conjugacy, the explicit map

$$[s,z] \longmapsto [(s, z + s\theta)] \quad \text{in } \mathbb{R}^2/\mathbb{Z}^2 \tag{C.48.16}$$

is well defined at the identified endpoints and sends the suspended leaves to the lines of slope $\theta$. This proves the required foliation conjugacy.

Lift (C.48.15)–(C.48.16) to the universal covers. It intertwines the original deck action with translations by $\mathbb{Z}^2$, through the induced isomorphism of their fundamental groups. Each original $F$-fiber becomes a whole affine line of slope $\theta$. A linear transverse coordinate $r$ and a complementary linear coordinate $u$ thus give

$$q(u,r) = (u + \beta(q), r + \alpha'(q)) \quad (q \in \widetilde{Q}), \tag{C.48.17}$$

where these translation vectors form a lattice, and $\alpha'(\widetilde{Q})$ is dense. The original function $F$ is a line homeomorphism composed with $r$: both are open quotients with exactly the same connected fibers. This gives the exact target reparametrization, rather than identifying the two quotient functions without a comparison map.

*Contraction in the established normal forms.* Except for the separately displayed transitive case (C.48.10), write the actual lifted loop as

$$a_t(u,r) = (h_t(u,r), b_t(r)). \tag{C.48.18}$$

Each $h_t(\,\cdot\,, r)$ is an increasing line homeomorphism. Its endpoints are the identity family. Let $B_{s,t}$ be the natural median contraction of $b_t$ from C.27, and set

$$H_{s,t}(u,r) = \big((1-s)h_t(u,r) + su,\ B_{s,t}(r)\big). \tag{C.48.19}$$

We check every operator equation rather than changing the operator group. For $k(u,r) = (u+1, r)$ the equation is $h_t(u+1,r) = h_t(u,r) + 1$, which (C.48.19) preserves. For $q(u,r) = (\epsilon u, r + d)$ it is

$$h_t(\epsilon u, r + d) = \epsilon h_t(u,r), \qquad b_t(r+d) = b_t(r) + d. \tag{C.48.20}$$

Convex interpolation preserves the first identity and median naturality preserves the second. For the real kernel (C.48.7), commutation says $h_t(u,r) = u + \phi_t(r)$; with (C.48.9) the further equation is $\phi_t(r+d) = \lambda\phi_t(r)$. Scaling $\phi_t$ by $1-s$ preserves it, including negative $\lambda$. For the real base translations in (C.48.14), the equations are that $h_t$ is independent of $r$ and $b_t(r) = r + c_t$. Both are preserved by (C.48.19).

For the lattice (C.48.17), the dense set of base translations and continuity force $b_t(r) = r + c_t$. The fiber equation is exactly

$$h_t(u + \beta(q), r + \alpha'(q)) = h_t(u,r) + \beta(q). \tag{C.48.21}$$

It is preserved by convex interpolation with $u$. The median of translations is again a translation, so the base equation is preserved as well. The trivial operator case has no extra equation. These cases exhaust (C.48.6).

In (C.48.10), write the loop as right multiplication by $(v_t, c_t)$. Its action is $(u,r) \mapsto (u + e^{cr}v_t, r + c_t)$. The median base contraction is translation by some $C_{s,t}$. Right multiplication by

$$((1-s)v_t, C_{s,t}) \tag{C.48.22}$$

is the required contraction and still commutes with every left multiplication.

Every map in (C.48.19) is a homeomorphism: its base map is a line homeomorphism, and its fiber map is an increasing convex combination of two increasing onto maps. The maps are jointly continuous. Their inverses are jointly continuous as well: on compact parameter sets, monotonicity and two fixed bracketing fiber values bound the inverse, and continuity and uniqueness identify every limit. This is the inverse continuity argument of C.31 applied to the displayed fiber family and the continuous inverse base family. Formula (C.48.22) has the same properties directly from the group law. All endpoint identities are preserved. We have therefore obtained a based null-homotopy commuting with all of $\widetilde{Q}$.

In particular it commutes with the original source deck group, so it descends to a compact-open null-homotopy $K_{s,t}$ on $O$, with continuous inverse family. Its commutation with every original operator follows from (C.48.2), retaining the restriction kernel, which acts identically on $O$. Finally each other boundary real quotient is an exact reparametrization of $F$ by the first case of C.45. Median naturality under that line homeomorphism gives its required quotient identity. For a circle target it commutes with the target integer translations and descends. The equations $q_z F_z = \Phi_z p$ now give (C.48.1) for every $z$. No limit across the ambient frontier has been deduced from this intrinsic construction. □

**C.49. [Proved] Lie contractions on closures with three boundary foliations.** Let $f_t$ be an ambient-null based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let $G$ be its generated group, and let $H$ commute with every $f_t$. For every open $G$-orbit $O$ having at least three distinct actual lifted boundary-quotient fiber partitions of C.42, the group generated by the based lifts of $f_t|_O$ to the universal covering $X \to O$ is closed in $\mathrm{Homeo}(X)$. With this actual topology it is one of

$$\mathbb{R}^2, \qquad \mathrm{Aff}^+(\mathbb{R}), \qquad \mathcal{A}_2 = \mathbb{R}^2 \rtimes \mathbb{R}, \tag{C.49.1}$$

where the last factor acts on $\mathbb{R}^2$ by positive scalar dilations. Each group is simply connected and has a globally invertible exponential map.

There is a jointly continuous based contraction on the whole $\overline{O}$, with jointly continuous inverse, obtained from these actual lifted groups by fractional powers. These contractions preserve every actual boundary-quotient equation, agree on intersections of such orbit closures and with the contraction of C.37 on $X_2$, and are transported to one another by $H$. They consequently give a continuous homeomorphism family on every finite union of these closures and $X_2$. The

conclusion does not assert continuity across an arbitrary infinite accumulating family of the new three-dimensional cases.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; actual simultaneous boundary-quotient and frontier-extension obligations.

*Proof. The actual based source lifts are closed loops and act transitively.* Choose three partitions and lifts $F_i : X \to \mathbb{R}$ as in C.47. Lift the isotopy $f_t|_O$ from the identity to $\widetilde{f}_t \in \mathrm{Homeo}(X)$. The covering homotopy property gives continuity for maps and inverses in their compact-open topology. Every $\widetilde{f}_t$ commutes with every deck transformation: the commutator is a continuous deck-valued function of $t$ and is the identity at $t = 0$. Its target lifts satisfy

$$F_i \widetilde{f}_t = b_{i,t} F_i. \tag{C.49.2}$$

For a real target, $b_{i,1} = \mathrm{id}$ directly. For a circle target, the exact two-point winding comparison of C.29 makes the based real lift return to the identity, since the original plane loop is ambient-null. Thus $b_{i,1} = \mathrm{id}$ for all $i$. The injectivity of $(F_1, F_2)$ from C.45 now gives $\widetilde{f}_1 = \mathrm{id}$.

Let $N_{\mathrm{alg}} = \langle \widetilde{f}_t : 0 \le t \le 1 \rangle$. This actual subgroup is path connected: each finite word is joined to the identity by simultaneously shortening its factor paths. Put

$$P = \{\widetilde{f}_t : 0 \le t \le 1\} \cup \{\widetilde{f}_t^{-1} : 0 \le t \le 1\}.$$

It is compact and contains the identity. The source deck group $\Delta$ is countable, because a fiber of a covering of a second-countable surface is a discrete countable subset of its second-countable covering plane. For $x \in X$, transitivity of $G$ on $O$ gives

$$X = \bigcup_{\delta \in \Delta} \bigcup_{n \ge 1} \delta P^n x. \tag{C.49.3}$$

All sets in this countable union are compact. Baire's theorem gives interior to one of them, hence to $N_{\mathrm{alg}} x$. Transitivity on that orbit then makes the entire orbit open. Every other orbit is a deck translate of it, since the generated lifted group commutes with $\Delta$. These disjoint open orbits partition the connected plane; there is only one. Thus $N_{\mathrm{alg}}$ is transitive on $X$, not just on its projection $O$.

*The connected three-web groups and their exact actions.* Let $\mathcal{W}$ be the full group in C.47 and write $W = \mathcal{W}^0$. The group $W$ is open in $\mathcal{W}$. The open orbit map makes every $W$-orbit open in $X$; normality of $W$ and connectedness of $X$ imply transitivity. For a point $x$, let $J = W_x$. The Lie principal bundle $W \to W/J = X = \mathbb{R}^2$ shows that $J$ is connected: the exact sequence on components has $\pi_1(X) = 0$ and $\pi_0(W) = 0$.

If $\dim W = 2$, then $J$ is discrete and connected, hence trivial. Evaluation identifies $W$ homeomorphically with the simply connected plane. Its Lie algebra and group are the two explicit possibilities $\mathbb{R}^2$ and $\mathrm{Aff}^+(\mathbb{R})$ in C.36. Since $N_{\mathrm{alg}} \subset W$ is already transitive and this action of $W$ is simply transitive, one has $N_{\mathrm{alg}} = W$.

Suppose $\dim W = 3$. Here C.47 puts us in the case where all three pair images are full planes. The connected one-dimensional isotropy $J$ is a subgroup of the ordered-loop automorphism group in that proof. It cannot be a circle. Indeed, every compact group of increasing line homeomorphisms is trivial: the compact orbit of any point has a largest element, which all group elements must fix, so transitivity on that orbit makes it a singleton. Thus $J \cong \mathbb{R}$. The principal bundle over the contractible plane now gives $\pi_1(W) = 0$.

We identify its Lie algebra and homogeneous action explicitly. Let $\mathfrak{w} = \mathrm{Lie}(W)$ and $\mathfrak{j} = \mathrm{Lie}(J)$. For each $i$, the stabilizer $K_i$ of $F_i(x)$ in the induced real target action is a closed subgroup containing $J$. The open orbit map identifies $W/K_i$ with the actual real target. The same principal-bundle component argument makes $K_i$ connected, and its Lie algebra $\mathfrak{k}_i$ has dimension

two. The three $\mathfrak{k}_i$ are distinct: otherwise the corresponding connected subgroups, and hence the quotient fibers through $x$, would coincide, contradicting C.45.

For $0 \neq Z \in \mathfrak{j}$, the induced linear map $\operatorname{ad} Z$ on $\mathfrak{w}/\mathfrak{j}$ preserves the three distinct lines $\mathfrak{k}_i/\mathfrak{j}$. A real linear map on a two-dimensional space preserving three distinct lines is scalar; write that scalar $\lambda$. If $\lambda = 0$, then $[\mathfrak{j}, \mathfrak{w}] \subset \mathfrak{j}$, so $\mathfrak{j}$ is an ideal. Its connected subgroup $J$ is then normal in $W$. As a normal point stabilizer in a transitive action it fixes all of $X$, contrary to the faithful inclusion $W \subset \mathrm{Homeo}(X)$. Therefore $\lambda \neq 0$.

The adjoint map on $\mathfrak{w}$ has the eigenspace decomposition

$$\mathfrak{w} = \mathfrak{j} \oplus V, \qquad \dim V = 2, \qquad [Z, v] = \lambda v \quad (v \in V).$$

For $v, w \in V$, the derivation identity gives $[Z, [v, w]] = 2\lambda[v, w]$. The value $2\lambda$ is absent from the spectrum $\{0, \lambda\}$, so $[V, V] = 0$. Replacing $Z$ by $Z/\lambda$ identifies the Lie algebra with scalar homotheties:

$$[Z, v] = v, \qquad [V, V] = 0. \tag{C.49.4}$$

Since $W$ is simply connected, the corresponding group is exactly $\mathcal{A}_2$, with multiplication

$$(a, v)(b, w) = (a + b,\ v + e^a w), \qquad a, b \in \mathbb{R}, \quad v, w \in \mathbb{R}^2. \tag{C.49.5}$$

Under this isomorphism $J$ is the dilation subgroup $\{(a, 0)\}$. The orbit identification $W/J \to X$ therefore gives an actual equivariant homeomorphism of the source plane with the standard action

$$z \longmapsto v + e^a z. \tag{C.49.6}$$

*The generated subgroup is closed, rather than merely analytic.* Yamabe's theorem makes the path-connected subgroup $N_{\mathrm{alg}} \subset W$ an analytic Lie subgroup [76, pp. 13–14]. This statement initially gives its analytic subgroup structure; it does not identify that structure with the induced topology. We now establish the latter point by classifying the transitive subgroups of (C.49.6).

A transitive analytic subgroup has dimension at least two. In dimension two its Lie algebra is either the translation plane $V$, or has a nonzero projection to the dilation coordinate. In the latter case it is

$$\mathbb{R}(Z + v_0) \oplus L, \tag{C.49.7}$$

where $L \subset V$ is a one-dimensional translation subspace. On the quotient line $V/L$, the generator $Z + v_0$ has flow

$$\bar{z} \longmapsto e^t \bar{z} + (e^t - 1)\bar{v}_0.$$

It fixes $-\bar{v}_0$, while $L$ acts trivially there. Thus the analytic subgroup in (C.49.7) preserves the proper affine line $-v_0 + L$ in the source plane, contradicting transitivity. The remaining two-dimensional subgroup is exactly all translations $V$. A three-dimensional analytic subgroup is open and equals the connected group $\mathcal{A}_2$. Hence, in this case, $N_{\mathrm{alg}} = V$ or $N_{\mathrm{alg}} = \mathcal{A}_2$.

Both are closed with their actual homeomorphism-group topology. One evaluation recovers a translation vector. For a homothety, evaluation at two fixed distinct points recovers its positive scale and translation continuously; a limit that is still a homeomorphism has positive scale. Together with the dimension-two case already treated, this proves that $N = N_{\mathrm{alg}}$ is closed and is one of (C.49.1), with its induced topology. It also proves that choosing a different triple of partitions does not change $N$: it is the same subgroup generated by the same based lifted loop.

For the semidirect products in (C.49.1), with vector dimension one or two, the exponential and its inverse are given by

$$\exp(u, v) = \left(u, \frac{e^u - 1}{u}\, v\right), \qquad \frac{e^u - 1}{u} = 1 \quad (u = 0). \tag{C.49.8}$$

The displayed scalar factor is positive everywhere. For $\mathbb{R}^2$ the exponential is the identity. This proves the asserted global exponential coordinates without an unspecified choice of logarithm.

*The exact Polish quotient extending the action to the closure.* Put $L = \overline{G}$ in the plane homeomorphism group. It preserves $O$ by the forward-and-inverse boundary argument of C.36. Define the closed Polish subgroup

$$E_O = \{(l, n) \in L \times N : pn = (l|_O)p\}. \tag{C.49.9}$$

Projection $E_O \to N$ is onto: each $n \in N$ is an actual finite word in the based lifted loop, so the identical word in the original plane loop supplies its $l \in G$. This continuous surjective homomorphism of Polish groups is open. Its kernel consists of pairs $(l, 1)$ with $l$ fixing $O$ pointwise, and therefore fixing $\overline{O}$ pointwise. Consequently

$$N \times \overline{O} \longrightarrow \overline{O}, \qquad (n, z) \longmapsto lz \quad \big((l, n) \in E_O\big) \tag{C.49.10}$$

is well defined and jointly continuous. Indeed, the product of the open projection with $\mathrm{id}_{\overline{O}}$ is an open quotient, and the continuous $E_O$-action factors through it. This argument keeps the exact source covering and the possibly nontrivial global restriction kernel.

Every $n$ has its chosen preimage in $G$, so (C.49.10) preserves every original $G$-orbit in $\overline{O}$ and has on that orbit exactly the image of $G$. It commutes with the original setwise $H$-stabilizer of $O$, since those preimages commute with that stabilizer. All source deck transformations commute with $N$ because they commute with its generating based lifts, so the restriction of (C.49.10) to $O$ is also the exact descended source action.

The required closure homotopy is now

$$R^O_{s,t} = \exp\big((1 - s) \log \widetilde{f_t}\big) \quad \text{acting by (C.49.10).} \tag{C.49.11}$$

The source loop is closed and based, as proved above. Thus (C.49.11) starts with $f_t|_{\overline{O}}$, ends with the identity, and fixes both based edges. The action and group inversion give joint continuity of the family and its inverse.

*All actual boundary quotients and common boundary images.* Consider any nontrivial nonopen $G$-orbit $B \subset \overline{O}$, with its intrinsic line or circle topology from C.28. The action (C.49.10) is transitive on its underlying set. Its point stabilizer $N_z$ is closed. The continuous bijective orbit map $N/N_z \to B$ has one-dimensional domain. Dimension zero is excluded by connectedness and nontriviality. If its dimension were at least two, restricting to a small two-dimensional coordinate disk and applying invariance of domain in the ambient plane would give interior to $B$, making this whole transitive orbit open.

The bijection agrees with the intrinsic topology. A compact interval in the homogeneous one-manifold maps to an embedded arc. On a non-circular orbit that arc is the unique connecting arc of C.28; on a circle the same statement holds in a sufficiently small interval. Its interior is therefore an intrinsic interval neighborhood. This proves that the orbit identification is a homeomorphism, and that the $N$-action is continuous in the intrinsic topology as well.

Every group in (C.49.1) is metabelian. Its image on $B$ is path connected and equals the actual $G|_B$. The exact metabelian image argument of C.37 therefore identifies this common image, with its induced intrinsic compact-open topology, as

$$M_B = \begin{cases} \mathbb{R} \text{ acting by translations, or } \mathrm{Aff}^+(\mathbb{R}), & B \text{ a line,} \\ \mathbb{R}/\mathbb{Z} \text{ acting by rotations,} & B \text{ a circle.} \end{cases} \tag{C.49.12}$$

These groups are closed in the intrinsic homeomorphism group: one evaluation recovers the translation or rotation, and two recover the affine parameters. The action supplies a continuous surjective Lie homomorphism

$$\rho_B : N \to M_B. \tag{C.49.13}$$

In particular every actual boundary quotient retains its original equivariance equation under the closure action. To verify this directly for $\Phi_z(lx) = lz$, take the preimage $g \in G$ of $n \in N$. The equality $\Phi_z(gy) = g\Phi_z(y)$ follows from the defining quotient formula; replacing $g$ by another preimage changes neither side because its restriction kernel fixes $\overline{O}$. Thus

$$\Phi_z(R^O_{s,t}y) = R^B_{s,t}(\Phi_z(y)) \tag{C.49.14}$$

with the same original line or circle target and its covering equation.

Since $N$ is simply connected, (C.49.13) has a unique based lift $\widetilde{\rho}_B : N \to \widetilde{M}_B$. The latter group is $\mathbb{R}$ or $\mathrm{Aff}^+(\mathbb{R})$, whose exponential is globally invertible. The path $\widetilde{\rho}_B(\widetilde{f}_t)$ is precisely the based lift $\widetilde{b}_t$ of the original $f_t|_B$. Naturality and injectivity of the exponential give the exact identity

$$\widetilde{\rho}_B\left(\exp\big((1-s)\log \widetilde{f}_t\big)\right) = \exp\big((1-s)\log \widetilde{b}_t\big). \tag{C.49.15}$$

This defines $R^B_{s,t}$ in (C.49.14) after projection. On a point orbit the action is the identity.

*Intersections, operators, and the finite-union conclusion.* For two orbit closures, their intersection consists of nonopen $G$-orbits. On each nontrivial one, both images (C.49.12) are the identical group $G|_B$ with the identical intrinsic topology. The two based target-cover paths are the same lift of $f_t|_B$. Formula (C.49.15) therefore makes the closure contractions equal there. It is also exactly the boundary formula in C.36–C.37, so it agrees on every nonopen orbit in the intersection with $X_2$. If $O$ itself belongs to $U_2$, evaluation identifies the simply transitive group of C.36 with $O$ and its universal cover with $X$. The lifted translations are the unique based source lifts used here; their transitive generated group is exactly $N$. The two fractional-power formulas are consequently identical on $O$ and its closure. An open orbit not in $U_2$ is disjoint from $X_2$, because it is an open neighborhood disjoint from $U_2$. These observations prove agreement on the entire intersection with $X_2$.

For $h \in H$ with $hO = O'$, lift $h|_O$ to a source-cover homeomorphism $\widetilde{h} : X \to X'$. Covering uniqueness for the two based isotopies gives

$$\widetilde{h}\,\widetilde{f}_t\,\widetilde{h}^{-1} = \widetilde{f}'_t. \tag{C.49.16}$$

Conjugation is consequently a continuous group isomorphism $N \to N'$. It intertwines their exponential maps and the unique logarithms in (C.49.8). It therefore carries (C.49.11) to the corresponding homotopy on $O'$. Equality extends to the closures. Changing the lift of $h$ by a deck map does not change this identity, since deck maps commute with the generated lifted groups. The actual boundary-quotient families are transported at the same time: with basepoint $hx$, their exact formula is $\Phi^{O'}_{hz}(lhx) = h\Phi^O_z(lx)$. Thus the class of open orbits under consideration is $H$-invariant; no arbitrary selected triple is assumed to be preserved by the operators.

Every closure and $X_2$ is closed in the plane. The matching families, and separately their matching inverse families, paste continuously over a finite union. Each preserves its member closures and is inverted by the pasted inverse, so it is a homeomorphism of that union. These assertions include all original covering and operator labels. Uniform control for infinitely many new homothety orbits, and extension to the remaining one- and two-foliation regions, are not supplied by this finite-pasting argument. □

**C.50. [Proved] Uniform contraction through all the new Lie orbit closures.** In the setting of C.49, let $U_3$ be the union of the open $G$-orbits having at least three distinct actual lifted boundary-quotient fiber partitions, and retain $U_2$ from C.37. Set

$$U_* = U_2 \cup U_3, \qquad X_* = \overline{U_*}. \tag{C.50.1}$$

The contractions of C.36 and C.49 extend to a jointly continuous based contraction through homeomorphisms of $X_*$, with jointly continuous inverse, commuting with $H$. This includes arbitrary accumulation of the three-dimensional homothety orbits.

The uniform algebraic bound is exact. For every continuous path $k : [0,t] \to \mathcal{A}_2$ with $k(0) = 1$, put $P = k([0,t]) \cup k([0,t])^{-1}$ and $g = k(t)$. Then

$$\{\exp(r \log g) : 0 \le r \le 1\} \subseteq P^{1913}. \tag{C.50.2}$$

The identity belongs to $P$, so shorter words can be padded. For $\mathbb{R}^2$ and $\mathrm{Aff}^+(\mathbb{R})$ the three-letter bound of C.37 applies. All word equalities used on an orbit hold in its specified simply connected source group. The bound does not identify an additive vector sum with a noncommutative product: the latter's complete error is corrected explicitly in the proof.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; uniform frontier control for the actual three-quotient Lie reductions.

*Proof. Two additive path identities.* The planar identity of C.37 represents any fraction of the endpoint of a based path in $\mathbb{R}^2$ by at most three signed path values. We also require its three-dimensional version: for a continuous $\gamma : [0,t] \to \mathbb{R}^3$ from 0 to $v$,

$$rv = \sum_{j=1}^{m} \varepsilon_j \gamma(s_j), \qquad m \le 5, \quad \varepsilon_j \in \{1,-1\}, \quad 0 \le s_j \le t. \tag{C.50.3}$$

For a polygonal path, apply the signed-measure theorem [26, Theorem 4.7 and §4.3.2] to its three coordinate derivatives. The chosen set is a union of at most two circular intervals. Its integral is a sum of two path differences, with at most one additional total vector from an interval crossing the identified endpoint. This gives five signed path values. Zero terms $\gamma(0)$ permit padding to a nonempty word when necessary. The interval count in the halving proof is the same corrected count used in C.37: two intervals cut at at most three points give at most five pieces, so one half uses at most two. For the whole interval, three cuts give at most four pieces, so one of the two halves uses at most two intervals; these are also circular intervals. Complementation, halving, and compactness give every fraction. Uniform polygonal approximation with the same endpoints, followed by compactness of the five parameters and the finite sign set, proves (C.50.3) for arbitrary continuous paths.

*A maximizing anchor and its exact translation commutators.* Use the coordinates

$$(a,v)(b,w) = (a+b, v + e^a w), \qquad v, w \in \mathbb{R}^2,$$

and write $k(s) = (a_s, v_s)$. Let $A = \max_{0 \le s \le t} |a_s|$. When $A = 0$, the path lies in the translation plane and the three-letter bound already proves (C.50.2). For $A > 0$, choose $u$ attaining the maximum and let $h = k(u)$ or $k(u)^{-1}$, choosing its sign so that its logarithmic scale is $A$. Thus $h$ is one letter from $P$, not a power of unbounded length. Conjugate all coordinates by the translation which sends its unique fixed point to zero. This is one group automorphism applied to both sides of every forthcoming identity; it adds no letters to any word. In the new coordinates $h = (A, 0)$ and still $|a_s| \le A$. Put

$$D = e^A - 1, \qquad C_s = [h, k(s)] = hk(s)h^{-1}k(s)^{-1} = (0, Dv_s). \tag{C.50.4}$$

The path $C_s$ is continuous in the translation plane and begins at the identity. Every one of its values and inverses uses at most four letters of $P$.

For any fixed $s$ and $|\beta| \le B$, where $B$ is a positive integer, the translation $C_s^{\beta}$ uses at most $12B$ letters of $P$. Indeed, write it as $B$ copies of $C_s^{\beta/B}$. For nonnegative $\beta/B \le 1$, the planar identity applied to the prefix $C|_{[0,s]}$ gives at most three signed commutator values. For a negative fraction invert the resulting word. Substituting (C.50.4) costs at most four letters per commutator. No continuous choice of the parameters in these words is required.

*The large-scale case.* Suppose $A \geq \log 2$, so $D \geq 1$. For $0 \leq q \leq 1$, the translation $T_{qv_s}$ equals $C_s^{q/D}$ and therefore uses at most twelve letters. Writing $D_a = (a, 0)$ gives

$$D_{a_s} = T_{v_s}^{-1} k(s), \tag{C.50.5}$$

which uses at most thirteen. Let $a = a_t$ and $v = v_t$. Define

$$\theta(a, r) = \begin{cases} (e^{ra} - 1)/(e^a - 1), & a \neq 0, \\ r, & a = 0. \end{cases} \qquad g^r = (ra, \theta(a, r)v). \tag{C.50.6}$$

The function $\theta$ is continuous and lies in $[0, 1]$. The intermediate value theorem supplies $s$ with $a_s = ra$, since the path starts at logarithmic scale zero and ends at $a$. Then $g^r = T_{\theta(a,r)v} D_{a_s}$ uses at most $12 + 13 = 25$ letters.

*The small-scale case and its full noncommutative correction.* Now suppose $0 < A \leq \log 2$. Apply (C.50.3) to $\gamma(s) = (a_s, v_s)$, obtaining

$$ra = \sum_{j=1}^{m} \varepsilon_j a_{s_j}, \qquad rv = \sum_{j=1}^{m} \varepsilon_j v_{s_j}, \qquad m \leq 5. \tag{C.50.7}$$

Form the actual product $w = k(s_1)^{\varepsilon_1} \cdots k(s_m)^{\varepsilon_m}$. For an inverse letter, its translation is $-e^{-a_{s_j}} v_{s_j}$. Consequently, with

$$p_j = \begin{cases} \sum_{\ell < j} \varepsilon_\ell a_{s_\ell}, & \varepsilon_j = 1, \\ \sum_{\ell \leq j} \varepsilon_\ell a_{s_\ell}, & \varepsilon_j = -1, \end{cases} \tag{C.50.8}$$

the product is exactly

$$\begin{aligned} w &= \left( ra, \sum_{j=1}^{m} \varepsilon_j e^{p_j} v_{s_j} \right), \\ \mathrm{pr}_v(w) &= rv + \sum_{j=1}^{m} \varepsilon_j (e^{p_j} - 1) v_{s_j}. \end{aligned} \tag{C.50.9}$$

Every $|p_j|$ is at most $mA$. The pure translation which changes (C.50.9) to (C.50.6) by left multiplication is

$$c = (\theta(a, r) - r)v - \sum_{j=1}^{m} \varepsilon_j (e^{p_j} - 1) v_{s_j}. \tag{C.50.10}$$

Using the translation commutators (C.50.4), this is the exact group equation

$$g^r = C_t^{\beta_0} \left( \prod_{j=1}^{m} C_{s_j}^{\beta_j} \right) w, \quad \beta_0 = \frac{\theta(a, r) - r}{D}, \qquad \beta_j = -\varepsilon_j \frac{e^{p_j} - 1}{D}. \tag{C.50.11}$$

The powers before $w$ commute because they are translations.

Here are uniform bounds, including $a = 0$. The integral formula

$$\theta(a, r) = \frac{\int_0^r e^{aq}\, dq}{\int_0^1 e^{aq}\, dq} \tag{C.50.12}$$

holds at zero as well as away from it. For $|a| \leq A$, the inequality $|e^{aq} - 1| \leq D$ holds on $0 \leq q \leq 1$, and the denominator is at least $e^{-A} \geq 1/2$. Subtracting $r$ in (C.50.12), the numerator is the

difference of an integral over $[0,r]$ and $r$ times an integral over $[0,1]$, each with integrand $e^{aq}-1$. Its absolute value is at most $2D$. Therefore $|\beta_0|\le 4$. For every other coefficient,

$$\frac{|e^{p_j}-1|}{D}\le\frac{e^{mA}-1}{e^A-1}=\sum_{\ell=0}^{m-1}e^{\ell A}\le 2^m-1\le 31. \tag{C.50.13}$$

The same bound covers negative $p_j$ because $1-e^{-mA}\le e^{mA}-1$. The word construction following (C.50.4) now applies with $B=4$ and $B=31$. Equation (C.50.11) uses at most

$$5+12(4+5\cdot 31)=1913 \tag{C.50.14}$$

original letters. This proves (C.50.2) in all regimes. The executable certificate accompanying this claim checks the inverse-letter exponents in (C.50.8), the complete product correction, the commutator and conjugacy formulas, and the integer count by exact symbolic arithmetic. The measure selection and the analytic coefficient bounds are proved above, independently of that certificate.

*One compact family of actual word isotopies.* Set $L_0=1913$. For the original plane loop, all words just proved belong to the compact parameter family

$$W_{\varepsilon,s}(u)=\prod_{j=1}^{L_0}f_{us_j}^{\varepsilon_j},\qquad \varepsilon_j\in\{1,-1\},\quad 0\le s_j\le t\le 1,\quad 0\le u\le 1. \tag{C.50.15}$$

Identity letters are obtained by $s_j=0$. Continuity of finite multiplication and inversion in the plane homeomorphism group gives compactness for maps and inverses, uniformly in $u$. On a C.49 orbit, the word equality is in its simply connected actual group $N$. On a C.36 orbit, use the based path in its specified universal covering group and the three-letter C.37 bound, then pad it. Thus the endpoint of the based lift of (C.50.15) is exactly the required fractional power, not just a homeomorphism with an unspecified deck shift.

Use $r=1-s$ for the contraction parameter and write $R^O_{r,t}$ for these fractional actions on an open orbit in $U_*$. C.36 and C.49 already prove their individual closure continuity and finite compatibility. Their groups are metabelian, so $[[g_1,g_2],[g_3,g_4]]=1$ on $U_*$ for every $g_i\in G$, and hence on $X_*$. Every nontrivial nonopen orbit $B\subset X_*$ therefore has the exact intrinsic image $M_B=\mathbb{R}$, $\mathrm{Aff}^+(\mathbb{R})$, or the circle rotation group, by the path-connected metabelian argument of C.37. Define $R^B_{r,t}$ by its unique line-group logarithm or its based real rotation lift, and use the identity on point orbits. The based circle lift ends at zero by C.29. These are the same boundary prescriptions as in C.36 and C.49.

*Limits, with the bounded residual in the specified cover.* Take

$$x_i\in O_i\subset U_*,\quad x_i\to x\in X_*,\qquad (r_i,t_i)\to(r,t). \tag{C.50.16}$$

Choose representations (C.50.15) for $R^{O_i}_{r_i,t_i}$ and pass to convergent word parameters. Let $W$ be the limiting plane isotopy, $w=W(1)$, and $g=f_t$. If $x$ belongs to an open orbit of $X_*$, then that orbit already lies in $U_*$ and eventually all $x_i$ lie in it. Its intrinsic continuity applies. If $x$ is fixed by $G$, every limiting word fixes it. For a remaining intrinsic line or circle orbit $B=Gx$, identities on $O_i$ pass to every point $hx\in B$ by evaluating them at $hx_i$ for fixed $h\in G$. In particular $w|_B$ commutes with $g|_B$.

For a fixed positive integer $n$, set $m_i=\lfloor nr_i\rfloor$. In each specified simply connected group, the residual is exactly

$$\widetilde{w}_i^{\,n}\widetilde{g}_i^{-m_i}=\exp\big((nr_i-m_i)\log\widetilde{g}_i\big). \tag{C.50.17}$$

Its exponent fraction is in $[0,1)$, so it has another representation from the same compact word family. For this fixed $n$, pass to a subsequence on which $m_i=m_n$ and the residual word isotopies converge to $Z_n$. Then

$$(w|_B)^n(g|_B)^{-m_n}=z_n,\qquad |m_n-nr|\le 1,\qquad z_n=Z_n(1)|_B, \tag{C.50.18}$$

and all $z_n$ belong to the same compact family of intrinsic restricted word endpoints. For a line, logarithms of this compact family are bounded. In the affine group commuting nonidentity elements belong to the same exponential one-parameter subgroup; the identity case gives the same additive logarithm equation. Thus

$$n\log(w|_B) - m_n\log(g|_B) = \log z_n.$$

Division by $n$ proves $\log(w|_B) = r\log(g|_B)$.

For a circle, let $\theta(u)$ be the based real angle of $f_u|_B$, and put $M = \max_u|\theta(u)|$. Let $A_W$ be the real endpoint angle of $W|_B$, $T = \theta(t)$, and $C_n$ that of $Z_n|_B$. The fixed word length gives $|C_n| \le L_0 M$. The real, rather than merely circle-valued, equality is

$$nA_W - m_n T = C_n. \tag{C.50.19}$$

To prove it, use the plane word isotopies $W_i(u)^n f_{ut_i}^{-m_i} Z_i(u)^{-1}$. Their restrictions lift to loops ending at the identity by (C.50.17). The specified source groups are simply connected, so these loops are null-homotopic there. Evaluate at the two points $x_i, hx_i$, where $hx \ne x$. Their difference-direction winding is zero. The resulting loops converge uniformly to the analogous two-point loop at $x, hx$, which stays off the diagonal; its winding is also zero. The exact circle comparison of C.29 identifies that winding, up to the common orientation sign, with $nA_W - m_n T - C_n$. This proves (C.50.19). Consequently

$$|A_W - rT| \le \frac{L_0 M + |T|}{n} \quad (n \ge 1),$$

so $A_W = rT$. This also covers an identity endpoint on the circle whose specified real angle $T$ is a nonzero integer.

*Joint continuity and the remaining scope.* Every convergent subsequence of the compact word parameters in (C.50.16) has therefore the prescribed endpoint action at $x$. This proves convergence to $R^B_{r,t}(x)$ for the entire sequence. It gives joint continuity from the dense set $U_*$. For a general sequence in $X_*$, approximate its $i$th point by a point of $U_*$ with both point distance and image distance at that fixed parameter less than $1/i$. The just-proved dense-set limit supplies those approximations and proves full joint continuity.

Apply the same proof to $f_t^{-1}$. It has the same generated group, the same open-orbit families, and the same boundary partitions. Its fractional actions are the orbitwise inverses, so the inverse family is jointly continuous as well. Every member is an orbitwise bijection inverted by this family and is therefore a homeomorphism of $X_*$. The equations $R_{0,t} = 1$, $R_{1,t} = f_t|_{X_*}$, and $R_{r,0} = R_{r,1} = 1$ give the based contraction. The operator transport statements of C.36 and C.49 make $U_*$ and $X_*$ invariant under $H$ and give commutation on $U_*$. Continuity extends that equality to $X_*$.

This settles infinite accumulation for the orbit closures treated by C.49. The remaining global step is a compatible extension from $X_*$ through the regions with fewer actual boundary-quotient foliations and their frontiers. The particular median choice of C.34 cannot be imposed there without proving compatibility, as C.35 demonstrates. The general conclusion in C.12 consequently remains open. □

**C.51. [Proved] Cocompact proper operators control the entire frontier.** Let $f_t$ be the based ambient-null plane loop of C.49, let $G = \langle f_t\rangle$, and let $H$ commute with all its values. For an open $G$-orbit $O$, put

$$H_O = \{h \in H : hO = O\}, \qquad Q_O = \overline{\{h|_O : h \in H_O\}}^{\mathrm{Homeo}(O)}. \tag{C.51.1}$$

Consider all such $O$ for which $O/Q_O$ is compact and the full family $\mathscr{F}_O$ of actual lifted boundary quotient fiber partitions in C.46 has at most two members. For every one of these orbits the restricted loop has a based contraction on its entire closure,

$$\overline{K}^O_{s,t} \in \mathrm{Homeo}(\overline{O}), \qquad \overline{K}^O_{0,t} = f_t|_{\overline{O}}, \qquad \overline{K}^O_{1,t} = \overline{K}^O_{s,0} = \overline{K}^O_{s,1} = \mathrm{id}, \tag{C.51.2}$$

jointly continuous with its inverse. It commutes with $H_O$ and equals exactly the C.34 median contraction on every nonopen orbit in $\partial O$. When $\mathscr{F}_O$ is empty, every point of $\partial O$ is fixed by $G$.

The contractions can be chosen with exact transport under $H$. They agree on all common frontiers and with C.34, and hence paste over the closed nonopen locus together with any finite collection of these closures. This assertion does not prescribe their values on the closed set $X_*$ of C.50: the comparison there with its exponential prescription remains to be proved.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and ambient frontier extension.

*Proof.* Write $Q = Q_O$ and $L = \overline{G}$ in the plane homeomorphism group. C.30 proves that the actual group $Q$ in (C.51.1) is closed, locally compact, free and proper on $O$, with the specified compact-open topology. The restriction kernel of $H_O \to Q$ acts trivially on $O$ and therefore on its closure. We retain that map; we do not replace an element of $Q$ by an unspecified global extension.

*A compact set meeting every operator orbit.* The quotient map $O \to O/Q$ is open. For each point of $O$, choose a relatively compact open neighborhood inside $O$. The images of these neighborhoods cover the compact quotient. Choose finitely many whose images still cover it, and let $C \subset O$ be the union of their closures. Then $C$ is compact and

$$QC = O. \tag{C.51.3}$$

This argument supplies the exact compact fundamental set required below; it uses no fundamental-domain boundary or continuous choice of orbit representatives.

*Every frontier sequence produces an actual quotient.* Take a sequence $x_n \in O$ converging in the plane to $z \in \partial O$. By (C.51.3), write $x_n = q_n y_n$ with $q_n \in Q$ and $y_n \in C$. Pass to a subsequence with $y_n \to y \in C$. For the moment choose $h_n \in H_O$ with

$$|h_n y_n - q_n y_n| < 1/n. \tag{C.51.4}$$

Such a choice follows from the definition of $Q$ as the closure of the actual restrictions in (C.51.1). Here and below $|\cdot|$ is the Euclidean distance in the original ambient plane.

Every $h_n$ commutes with $L$: it commutes with all generators of $G$, and commutation passes to its closure in the plane homeomorphism group. The orbit map

$$L \longrightarrow O, \quad l \mapsto ly \tag{C.51.5}$$

is open by the same Polish orbit-map theorem used in C.30 and C.42. Thus there are $l_n \to 1$ in $L$ with $l_n y = y_n$. For clarity, take a decreasing countable identity neighborhood basis $V_k$ in $L$. Each $V_k y$ is a neighborhood of $y$, so $y_n \in V_k y$ eventually. Choose an increasing sequence of such eventual indices and then $l_n \in V_{k(n)}$ with $k(n) \to \infty$. This proves the stated selection. Commutation now gives the exact equality

$$h_n y = l_n^{-1} h_n y_n \longrightarrow z. \tag{C.51.6}$$

Convergence follows from (C.51.4), $x_n \to z$, and compact-open convergence of $l_n^{-1}$ to the identity on a compact neighborhood of $z$.

Let $J_y = L_y$. For $j \in J_y$, $jh_n y = h_n jy = h_n y$, so (C.51.6) gives $jz = z$. Consequently $z \in \mathrm{Fix}(J_y)$, and C.42's actual evaluation comparison gives

$$h_n|_O \longrightarrow \Phi_z^y \quad \text{uniformly on compact subsets of } O, \qquad \Phi_z^y : O \longrightarrow \mathbb{R}^2, \quad \Phi_z^y(ly) = lz. \tag{C.51.7}$$

This uses the homeomorphism $\mathrm{Map}_L(O, \mathbb{R}^2) \to \mathrm{Fix}(J_y)$ of (C.42.1), including its compact-open topology. In particular its assertion applies when $Gz$ is a single point, in which case $\Phi_z^y$ is constant. There is no convergence claim for arbitrary proper operator maps outside their actual global approximants.

We compare (C.51.7) with the quotient family defined at the fixed basepoint $x$ in C.46. Choose $a \in L$ with $ax = y$, and put $z_0 = a^{-1}z$. Then $z_0 \in \mathrm{Fix}(L_x) \cap \partial O$. For every $l \in L$ the equality

$$\Phi_{z_0}^x(ly) = \Phi_{z_0}^x(lax) = laz_0 = lz = \Phi_z^y(ly) \tag{C.51.8}$$

identifies the actual maps, with the same target orbit. The orbit is contained in $\partial O$, since $L$ preserves $O$ and its boundary. It is nonopen; an open orbit meeting $\partial O$ would be an open neighborhood disjoint from $O$, contradicting the definition of its closure. Thus C.28 makes a nontrivial such orbit an intrinsic line or circle. For that case (C.51.8) is exactly one of the original boundary quotients whose lifts define $\mathscr{F}_O$. It follows in particular that an empty $\mathscr{F}_O$ forces every frontier orbit to be a point: apply the construction to a sequence approaching any prescribed frontier point.

*The intrinsic family and every boundary equation.* For $|\mathscr{F}_O| = 1$ use the contraction $K_{s,t}^O$ of C.48, and for $|\mathscr{F}_O| = 2$ use C.46. Both commute with $H_O$, hence with its restricted closure $Q$, and both have jointly continuous inverses. They retain every quotient equation

$$\Phi K_{s,t}^O = Q_{s,t}^B \Phi, \tag{C.51.9}$$

where $Q^B$ is the C.34 median family on the actual boundary orbit $B$, with its based real covering lift when that orbit is a circle. For an empty family, C.32 makes $f_t|_O$ null in $\mathrm{Homeo}(O)$ and C.33 supplies a based null-homotopy $K^O$ in its exact $Q$-centralizer. All frontier orbits are then points, as just proved, so (C.51.9) for each constant $\Phi$ is automatic.

Define $\overline{K}^O$ to be $K^O$ on $O$ and the C.34 family on $\partial O$. We prove its full joint continuity, rather than continuity only for fixed parameters. Let

$$x_n \in O, \quad x_n \to z \in \partial O, \qquad (s_n, t_n) \to (s, t). \tag{C.51.10}$$

Use $x_n = q_n y_n$ and a subsequence $y_n \to y$ as above. The set

$$S = C \ \cup \bigcup_{(u,v)\in[0,1]^2} (K_{u,v}^O(C) \cup (K_{u,v}^O)^{-1}(C)) \tag{C.51.11}$$

is compact in $O$, by joint continuity on a compact parameter space. Refine the choice of actual operators in (C.51.4) to satisfy

$$\sup_{w\in S} |h_n w - q_n w| < 1/n. \tag{C.51.12}$$

For each $n$ this is a compact-open neighborhood condition on $q_n \in Q$, so density of the specified restrictions supplies $h_n \in H_O$. The previous argument still proves (C.51.6)–(C.51.8).

Commutation with $Q$ and (C.51.12) give

$$\begin{aligned} K_{s_n,t_n}^O(x_n) &= q_n K_{s_n,t_n}^O(y_n), \\ |K_{s_n,t_n}^O(x_n) - h_n K_{s_n,t_n}^O(y_n)| &< 1/n. \end{aligned} \tag{C.51.13}$$

The points $K^O_{s_n,t_n}(y_n)$ converge in $O$ to $K^O_{s,t}(y)$ and lie in the compact set $S$. Uniform convergence (C.51.7), followed by the exact quotient identity (C.51.9), therefore yields

$$\begin{aligned} h_n K^O_{s_n,t_n}(y_n) &\longrightarrow \Phi^y_z K^O_{s,t}(y) \\ &= Q^{Gz}_{s,t}(\Phi^y_z(y)) = Q^{Gz}_{s,t}(z). \end{aligned} \tag{C.51.14}$$

For a point orbit the same computation uses the constant map and gives $z$. Every subsequence in (C.51.10) has a further subsequence of this form because $C$ is compact. The value on the right is independent of all choices. Thus (C.51.14) holds for the entire sequence and proves continuity from $O$ to the prescribed frontier value. Continuity for sequences contained in $\partial O$ is precisely C.34. Splitting a sequence according to its two pieces proves joint continuity everywhere on $[0,1]^2 \times \overline{O}$.

The inverse maps have the same argument, using the inverse images already included in $S$. Equation (C.51.9) implies $\Phi(K^O_{s,t})^{-1} = (Q^B_{s,t})^{-1}\Phi$ by composing with these actual inverse maps. The inverse boundary family is jointly continuous by C.34. This proves the inverse version of (C.51.14), and hence joint continuity of the piecewise inverse on the entire closure. The maps preserve $O$ and each frontier orbit and are bijections there; the piecewise inverses are their actual inverses. They are homeomorphisms. The endpoint and based-edge identities and commutation with $H_O$ hold on both pieces.

*Transport and finite pasting.* The class of orbits in the claim is $H$-invariant: conjugation by $h \in H$ identifies the two exact groups $Q_O, Q_{hO}$, their quotient spaces, and their full boundary-quotient families by C.46. Choose one orbit from each $H$-orbit of these open components, make the construction there, and set

$$\overline{K}^{hO}_{s,t} = h\overline{K}^{O}_{s,t}h^{-1}. \tag{C.51.15}$$

The equality with every stabilizing operator makes this independent of the choice of $h$ carrying the selected representative to a given component. Median naturality in C.27–C.34 makes its frontier restriction the identical prescribed C.34 family.

Distinct open-orbit closures meet only in the closed nonopen locus $Y$. Every one of the new families and its inverse equals the corresponding C.34 map there. The finite closed-set pasting lemma thus gives the asserted maps and inverse maps on $Y$ together with finitely many closures. For every finite collection $\mathcal{I}$, the homeomorphism $h$ carries this union to the union indexed by $h\mathcal{I}$ and intertwines their homotopies by (C.51.15). This is the exact operator-transport comparison on those finite unions.

The proof uses one fixed compact fundamental set inside each treated orbit. It supplies no uniform control over infinitely many different such sets, and it does not identify the median boundary value with the exponential value forced on $X_*$ by C.50. These are explicit remaining comparisons in C.12. □

**C.52. [Proved] Cocompact frontier extensions relative to the exponential region.** Use the actual based ambient-null plane loop $f_t$, the groups $G = \langle f_t \rangle$ and $H$, and the closed set $X_*$ with its contraction $E_{s,t}$ from C.50. For every nonopen $G$-orbit $B$, define a based family $R^B_{s,t}$ as follows:

$$R^B_{s,t} = \begin{cases} \text{id}, & B \text{ a point}, \\ \text{the exponential fraction with parameter } 1-s, & G|_B \text{ metabelian}, \\ \text{the C.27–C.29 median family}, & G|_B \text{ nonmetabelian}. \end{cases} \tag{C.52.1}$$

The exponential fraction uses the actual intrinsic Lie image classified in C.37. On a circle it uses the entire based lift of the original loop to the covering group, including its real angle at each intermediate time.

For every open orbit $O$ disjoint from $X_*$ whose full family $\mathscr{F}_O$ of actual lifted boundary partitions has cardinality at most two, there is an intrinsic based null-homotopy $K^O_{s,t}$ commuting with its original operator stabilizer and satisfying every actual boundary quotient equation

$$\Phi_z K^O_{s,t} = R^{Gz}_{s,t}\Phi_z. \tag{C.52.2}$$

For each such $O$ with compact proper-operator quotient $O/Q_O$, this family extends to a based contraction through homeomorphisms of $\overline{O}$, jointly continuous with its inverse, and equal to (C.52.1) on its entire frontier. It agrees with $E_{s,t}$ on $\overline{O} \cap X_*$.

These extensions can be chosen with exact transport under $H$. They agree on common frontiers. Consequently they paste with the prescribed C.50 contraction on $X_*$ over every finite union

$$X_* \cup \bigcup_{O\in\mathcal{I}} \overline{O}, \tag{C.52.3}$$

where $\mathcal{I}$ is a finite collection of the cocompact orbits just specified. No continuity of (C.52.1) on the entire nonopen locus, and no extension over an infinite collection or over the other noncocompact open orbits, is asserted.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; relative frontier extension preserving the already prescribed exponential region.

*Proof. The precise intrinsic target family.* For a nontrivial intrinsic line or circle orbit, $G|_B$ is path connected and transitive. The proof of (C.37.9) classifies every such metabelian image, with its actual intrinsic homeomorphism-group topology: on a line it is conjugate to all translations or to the full positive affine group, and on a circle to all rotations. That proof uses only these three stated properties of the image, so it applies to every metabelian orbit in (C.52.1), without requiring that orbit to lie in $X_*$.

For a real lift $F : X \to \mathbb{R}$ of an actual boundary quotient, let $b_t$ be its based target path, characterized by $Fa_t = b_t F$ for the based source lift $a_t$. The exact source and target coverings are those of C.42–C.44. The path $b_t$ is a based loop: for a circle target this is the zero-winding conclusion of C.29, and for a line target it follows from the identity endpoint of $f_t$. Its generated group

$$T = \langle b_t : 0 \le t \le 1\rangle \tag{C.52.4}$$

is transitive on $\mathbb{R}$. Indeed its orbits are intervals, as each word is joined to the identity by a path of words. A finite endpoint of such an orbit interval is a common fixed point, which would project to a common fixed point of the nontrivial transitive target action. Thus no proper orbit interval is possible.

For a line target, (C.52.4) is the conjugate of the actual image $G|_B$. For a circle, projection gives a surjection $T \to G|_B$. When the latter image is metabelian, its rotation conjugacy lifts to a real coordinate in which every $b_t$ is a translation. Transitivity then makes $T$ the full translation group. Conversely a metabelian $T$ has metabelian quotient $G|_B$. This proves the exact equivalence used in the case selection below.

Define, in the specified real lift,

$$B_{s,t} = \begin{cases} \exp_T\big((1-s)\log_T b_t\big), & T \text{ metabelian},\\ \text{the C.27 median of the path } b_t, & T \text{ nonmetabelian}. \end{cases} \tag{C.52.5}$$

In the first case $T$ is a simply connected translation or positive affine Lie group with a global exponential and logarithm. Formula (C.52.5) and its inverse are jointly continuous, with all four required based-edge identities. They commute with every target homeomorphism commuting with the original path: in the exponential case each new value lies in $T$, and in the median case this is C.27.

This rule is natural under any line reparametrization, including a decreasing one. Such conjugation identifies the actual groups (C.52.4) topologically, preserves metabelianness, and, in the Lie cases, intertwines exponential maps and their unique logarithms. The median case has the same naturality by C.27. The source of a real lift of a circle may also be changed by a covering conjugacy; the same argument retains the specified based target path. In particular the resulting map on $B$ is exactly (C.52.1), independent of all these coordinates.

For clarity, at a time when $f_t|_B = \mathrm{id}$ on a circle, its based real lift can still be a nonzero integer translation. Formula (C.52.5) scales that real translation. It does not replace it by zero. The endpoint $t = 1$ is zero by C.29. This is the same covering convention as C.50.

*Replacement in the two-foliation construction.* For $|\mathscr{F}_O| = 2$, retain exactly the source cover, the two real quotient coordinates, and the domain $D$ of C.45–C.46. Replace each of the two coordinate median families in (C.46.4) by (C.52.5). On $D = \mathbb{R}^2$ this is a product of increasing line homeomorphisms, with a jointly continuous inverse. On $D = \{u < v\}$ the original two coordinate loops are identical. Their generated groups and their selected families are therefore identical, so the strict inequality is preserved. On $D = \{u < v < u + 1\}$ the same argument applies, and the selected maps commute with unit translation. In the exponential case this holds because that translation centralizes $T$; in the median case it is the original median naturality. Thus both inequalities are preserved, as are they by the inverse family.

Every original stabilizing operator, and every one of its source lifts, permutes the full two-element family of partitions by the exact comparison (C.46.5). In these coordinates it is a pair of line homeomorphisms, possibly followed by exchange. Commutation with the based source loop identifies the corresponding target paths by these same conjugacies. Rule (C.52.5) is natural under them and makes the same choice on conjugate paths. Consequently the modified source family commutes with every such lift, including all source deck maps. It descends to $O$ with continuous inverse.

Every other actual real quotient is a line homeomorphism composed with one of the two selected coordinates, by C.45. The preceding naturality therefore proves its exact lifted equation too. Descending through $q_z F_z = \Phi_z p$ proves (C.52.2), retaining the original source and target covers.

*Replacement in the one-foliation construction.* For $|\mathscr{F}_O| = 1$, keep the actual closed proper operator group, its group of all source lifts, and every coordinate change of C.48. In (C.48.19) replace only the median base map by (C.52.5), leaving the fiber interpolation as displayed:

$$(u, r) \longmapsto \big((1 - s)h_t(u, r) + su,\ B_{s,t}(r)\big). \tag{C.52.6}$$

Here the choice of target coordinate is the exact reparametrization already established in C.48. We check why all of its operator equations survive this replacement.

For the unit fiber translation, the fiber equation $h_t(u + 1, r) = h_t(u, r) + 1$ is unchanged by the base replacement. For an operator $(\epsilon u, r + d)$, the equations are $h_t(\epsilon u, r + d) = \epsilon h_t(u, r)$ and $b_t(r + d) = b_t(r) + d$. The first is preserved by the same convex interpolation; the second says that translation by $d$ centralizes $T$, so it holds for (C.52.5). For the real fiber kernel and a further $(\lambda u, r + d)$, the fiber function is $h_t(u, r) = u + \phi_t(r)$ with $\phi_t(r + d) = \lambda\phi_t(r)$. Scaling $\phi_t$ preserves this equality for either sign of $\lambda$, and the base comparison is again the same one.

In the normal forms with all real base translations or a dense lattice base image, the original base loop is $b_t(r) = r + c_t$. Its selected family is $B_{s,t}(r) = r + (1 - s)c_t$ and commutes with every base operator. The fiber equations, respectively independence of $r$ or the exact affine lattice relation (C.48.21), are preserved by (C.52.6). The trivial operator case imposes no extra equation. These are the C.48 normal forms outside its transitive two-dimensional case. That latter case already lies in $U_2 \subset X_*$ and is absent from the open orbits of the present claim.

Thus the modified family commutes with the exact group of all operator lifts, with its original restriction kernel retained downstairs. The fiber and base inverse-continuity argument

following (C.48.22) applies unchanged to (C.52.6): the base family has continuous inverses by (C.52.5), and fixed bracketing fiber values bound the increasing convex-interpolation inverses. The homotopy descends to $O$, with the asserted endpoint identities. All other actual quotients are reparametrizations of the one selected partition; naturality proves (C.52.2) for them.

For an empty family, C.32–C.33 supply a based null-homotopy in the exact proper-operator centralizer. There is then no nontrivial actual quotient equation to check. This finishes the intrinsic assertion for all three cardinalities.

*Cocompactness and the entire frontier.* Now assume $O/Q_O$ compact and put $Q = Q_O$. The first part of C.51 supplies a compact $C \subset O$ with $QC = O$. It also proves that every nontrivial frontier orbit gives an actual quotient in the specified family, after the explicit basepoint comparison (C.51.8). In particular an empty family makes all frontier points fixed.

Define $\overline{K}^O$ by the just-constructed $K^O$ on $O$ and by (C.52.1) on $\partial O$. We first prove its limit along points of $O$; we do not assume transverse continuity of (C.52.1). Let

$$x_n \in O, \quad x_n \to z \in \partial O, \qquad (s_n, t_n) \to (s, t). \tag{C.52.7}$$

Write $x_n = q_n y_n$ with $y_n \in C$ and pass to a subsequence $y_n \to y \in C$. The compact set consisting of $C$ and all its images and inverse images under $K^O$ is exactly the type of set used in (C.51.11). Approximate $q_n$ uniformly on this set to accuracy $1/n$ by restrictions of actual $h_n \in H_O$, as in (C.51.12). The open evaluation map of $L = \overline{G}$ gives $l_n \to 1$ with $l_n y = y_n$. Exact commutation gives $h_n y = l_n^{-1} h_n y_n \to z$.

The C.42 evaluation homeomorphism, with the basepoint comparison (C.51.8), now gives $h_n|_O \to \Phi_z^y$ uniformly on compact subsets. For a nontrivial target this is an actual quotient covered by (C.52.2); for a fixed target it is constant. Commutation with $Q$ therefore gives precisely

$$\begin{aligned} K^O_{s_n,t_n}(x_n) &= q_n K^O_{s_n,t_n}(y_n) \\ &\longrightarrow \Phi_z^y K^O_{s,t}(y) = R^{Gz}_{s,t}(z). \end{aligned} \tag{C.52.8}$$

The approximation error in the second line is less than $1/n$ before passage to the limit. Compactness of $C$ and uniqueness of the value give (C.52.8) for the entire sequence. This repeats C.51's exact operator comparison with the new quotient equation; no element of the restricted closure is assigned an unproved global extension.

*The diagonal argument at frontier points.* Take now $z_n \in \partial O$ with $z_n \to z$ and $(s_n, t_n) \to (s, t)$. For each fixed triple $(s_n, t_n, z_n)$, (C.52.8) lets us choose $x_n \in O$ such that

$$|x_n - z_n| < 1/n, \qquad \left|K^O_{s_n,t_n}(x_n) - R^{Gz_n}_{s_n,t_n}(z_n)\right| < 1/n. \tag{C.52.9}$$

Then $x_n \to z$, so applying (C.52.8) to this new sequence proves $R^{Gz_n}_{s_n,t_n}(z_n) \to R^{Gz}_{s,t}(z)$. This establishes the needed joint continuity on this entire frontier, even across different target types in (C.52.1). Splitting a sequence into its interior and frontier terms proves joint continuity on $[0,1]^2 \times \overline{O}$.

The same proof applies to the inverse family. The compact approximation set already contains all inverse images, and (C.52.2) implies the exact inverse equation $\Phi_z (K^O_{s,t})^{-1} = (R^{Gz}_{s,t})^{-1} \Phi_z$. Use this equation in (C.52.8), then repeat the diagonal choice (C.52.9) for inverses. Thus the piecewise inverse is jointly continuous. All maps are bijections on $O$ and on each frontier orbit, with these actual inverses; they are homeomorphisms of the closure.

*Agreement with the fixed exponential region.* Every group image on an open orbit of $U_2 \cup U_3$ is metabelian, by C.36 and C.49. Hence, for all $g_1, g_2, g_3, g_4 \in G$, the plane homeomorphism $[[g_1, g_2], [g_3, g_4]]$ fixes $U_2 \cup U_3$ pointwise. Continuity makes it fix $X_*$ pointwise. The set $X_*$ is $G$-invariant. Therefore every nonopen orbit $B \subset X_*$ has metabelian actual image $G|_B$.

An open orbit outside $U_2 \cup U_3$ is disjoint from $X_*$, because it is an open neighborhood disjoint from that union. Thus the intersections $\overline{O} \cap X_*$ occurring here contain only nonopen orbits. On a point orbit both contractions are the identity. On every other such orbit, (C.52.1) and C.50 use the same actual intrinsic Lie image and the same based target-cover path. Exponential naturality therefore gives equality, including the intermediate real circle angles described above.

Distinct open-orbit closures meet only in the nonopen locus. The present construction assigns there the same rule (C.52.1), so their contractions and inverses agree. Conjugation by $h \in H$ identifies the actual target images and their based paths, preserves metabelianness, and intertwines both branches of (C.52.1). Choose the intrinsic construction on one representative of each $H$-orbit of components and transport it by $h$. Commutation with the full stabilizer makes this choice independent of the transporting element, exactly as in (C.51.15).

Finite closed-set pasting now gives the homeomorphism families and inverse families on (C.52.3). A general $h$ intertwines this union with the union indexed by $h\mathcal{I}$; when it preserves the union, it commutes with the displayed contraction. The family on $X_*$ is exactly the prescribed one throughout. The diagonal proof concerns a single fixed closure. It supplies no uniform control over infinitely many different compact fundamental sets, and no compact fundamental set for the remaining noncocompact quotients. These are still retained parts of C.12. □

**C.53. [Proved] Global quotients and finitely many operator representatives.** Use the ambient-null based plane loop $f_t$, its generated group $G$, the commuting operator group $H$, and $L = \overline{G}$ of C.42. For every open $G$-orbit $O$, choose $x \in O$ and put $J = L_x$. Define $\mathscr{F}_O^{\mathrm{all}}$ to be the family of fiber partitions on the common universal covering of $O$ of all real lifts of

$$\Phi_z^x : O \longrightarrow Gz, \qquad \Phi_z^x(lx) = lz, \quad z \in \mathrm{Fix}(J), \quad Gz \text{ nonopen and nontrivial.} \tag{C.53.1}$$

The target points need not belong to $\partial O$. Let $U^\dagger$ be the union of all open orbits with either $\dim Q_O = 2$ or $|\mathscr{F}_O^{\mathrm{all}}| \geq 3$, and set $X^\dagger = \overline{U^\dagger}$. There is a jointly continuous based contraction on $X^\dagger$, with jointly continuous inverse and exact $H$-commutation, extending the prescribed contraction on $X_*$ in C.50. On every nonopen orbit it has the corresponding exponential value of C.52.1.

Now choose finitely many open orbits $O_1, \dots, O_m$ disjoint from $X^\dagger$, each with compact $O_j/Q_{O_j}$, and put

$$U = \bigcup_{j=1}^{m} \bigcup_{h \in H} hO_j, \qquad X = \overline{U}. \tag{C.53.2}$$

The original loop has a based contraction through homeomorphisms of $X^\dagger \cup X$, jointly continuous with its inverse, commuting with $H$ and equal to the specified exponential contraction on $X^\dagger$. On every nonopen orbit in $X$ its value is exactly the hybrid prescription $R_{s,t}^B$ of C.52.1. This assertion allows infinitely many different open components in (C.53.2), with arbitrary accumulation. It retains the stated finite number of $H$-representatives and compactness of their actual operator quotients.

*Manuscript:* Section23, folios140-141, scans279-281; Part C(b), infinite-operator zero-level comparison, global quotient maps and arbitrary accumulation under finitely many operator representatives.

*Proof. The global quotient family and its exact comparisons.* C.42 is already stated for every $z \in \mathrm{Fix}(J)$ with nontrivial line or circle orbit; it imposes no frontier condition on $z$. It gives the actual continuous open map (C.53.1), its intrinsic target topology, the common covering construction, and its proper fiber lines. C.43–C.45 use these maps and their transitive compatible action to give the connected real fibers, product coordinates and the exact three pairwise coordinate domains. Thus each of those conclusions applies to (C.53.1) with the same maps.

Changing the source basepoint also preserves the complete family. If $y = ax$ for $a \in L$, then $L_y = aJa^{-1}$ and, for $z \in \mathrm{Fix}(L_y)$,

$$\Phi^y_z(ly) = lz = \Phi^x_{a^{-1}z}(lax). \tag{C.53.3}$$

The targets are the same actual $G$-orbit, because $Lz = Gz$ for each such quotient, as proved in C.42. This proves the asserted identification of partitions, including the original target coverings.

For $h \in H$ stabilizing $O$, choose $a_h \in L$ with $a_h x = hx$. Exact commutation gives $L_{hx} = L_x$, hence $a_h J a_h^{-1} = J$, and

$$\Phi^x_z h = \Phi^x_{a_h z}. \tag{C.53.4}$$

The point $a_h z$ is fixed by $J$ and lies on the same nonopen orbit as $z$. Therefore (C.53.4) permutes the entire family $\mathscr{F}^{\mathrm{all}}_O$. For $hO = O'$, use basepoint $hx$ in $O'$; then

$$\Phi^{hx}_{hz} h = h\Phi^x_z. \tag{C.53.5}$$

These equalities identify the families under operator transport and prove that their cardinalities are well defined and $H$-invariant.

*Three global quotients give the same actual Lie group.* Choose three distinct partitions in $\mathscr{F}^{\mathrm{all}}_O$. The full three-foliation group of C.47 is constructed exclusively from the real maps in C.43–C.45 and their common transitive action. Its proper-pair or ordered-loop argument therefore applies to these exact maps and gives the same locally compact Lie group of dimension two or three. The based target lifts in C.49.2 are loops: for circle targets C.29 uses the ambient-null plane loop and two points of that circle, independently of the position of $O$; real target endpoints are the identity. The Baire transitivity proof C.49.3, its isotropy calculation C.49.4–C.49.7 and its closedness argument now apply to the same based source lifts. Their actual generated group $N$ is consequently one of $\mathbb{R}^2$, $\mathrm{Aff}^+(\mathbb{R})$, or $\mathcal{A}_2$, with exactly the topology, global exponential and unique logarithm proved in C.49.

We verify the action comparison for every global quotient, rather than only for frontier targets. Use the closed Polish group

$$E_O = \{(l, n) \in L \times N : pn = (l|_O)p\}. \tag{C.53.6}$$

Its projection to $N$ is onto, since every $n$ is an actual word in the lifted loop; the identical global word supplies $l \in G$. It is an open homomorphism of Polish groups. If $(l, 1)$ is in its kernel, $l$ fixes $O$ pointwise. Equivariance of (C.53.1) then makes $l$ fix its entire image $B = Gz$ pointwise:

$$l\Phi^x_z(u) = \Phi^x_z(lu) = \Phi^x_z(u). \tag{C.53.7}$$

C.42 proves continuity of $L \to \mathrm{Homeo}(B)$ in the intrinsic compact-open topology. Its restriction to $E_O$ therefore factors continuously through the open quotient to an actual action of $N$ on $B$. That image is exactly $G|_B$: actual loop words prove both inclusions. It is metabelian and hence is the translation, affine, or rotation Lie image of C.37. The resulting continuous homomorphism and its unique based covering lift give exactly

$$\Phi^x_z K^O_{s,t} = R^B_{s,t}\Phi^x_z, \qquad K^O_{s,t} = \exp_N\big((1-s)\log_N \widetilde{f}_t\big) \text{ descended to } O. \tag{C.53.8}$$

Here $R^B$ is the exponential branch of C.52.1, with the full based real angle for a circle. The original source covering, every original quotient, and every restriction kernel are retained.

*The enlarged closed exponential region.* For a dimension-two proper-operator orbit use the based lifted contraction of C.36–C.37. For an orbit with three global partitions use (C.53.8). Where both constructions apply, their source groups are the same actual group generated by the same based lifted loop, as in C.49, so the maps agree. Operator conjugation transports

these groups and intertwines their exponential maps, by (C.53.5) and the based covering identity C.49.16.

Every source group just specified has the uniform fractional-word bound 1913: C.37 gives the stronger bound three for the first two groups, and C.50.1–C.50.14 give the stated bound for $\mathcal{A}_2$. Each word is a word in the original based path and its inverse. In particular the full argument C.50.15–C.50.19 applies to this enlarged union with precisely the same compact set of actual global word isotopies. We spell out the transfer of its remaining inputs. All the source images are metabelian, so each global commutator-of-commutators fixes $U^\dagger$, hence its closure. At a nonopen limit orbit the actual transitive image is therefore the same translation, affine or rotation group classified in C.37. For every integer $n$, the residual after removing $\lfloor n(1-s)\rfloor$ endpoint powers is a fractional power in the specified simply connected source group, and thus has the same uniform word bound. On a line the bounded residual logarithms force the exponential limit. On a circle the source residual word loop is null in that same simply connected group; its exact two-point winding passes to the limiting orbit, giving the real equality C.50.19, including its integer part. These are all inputs of C.50's limit identification and joint continuity proof. Applying it also to the inverse loop gives a jointly continuous inverse on $X^\dagger$.

The original $U_2 \cup U_3$ lies in $U^\dagger$. On that original union the maps are the same source fractional powers. Continuity extends this equality to $X_*$. The operator identities also extend from the dense union. This proves the first assertion and fixes its prescription for the rest of the proof.

*Compatible intrinsic maps on the remaining representatives.* Each $O_j$ in (C.53.2) has $|\mathscr{F}_{O_j}^{\mathrm{all}}| \le 2$ and $\dim Q_{O_j} \le 1$, by its disjointness from $X^\dagger$. For a family of size one or two, apply the construction of C.52 to this full global family. Here is the exact transfer. Its source coordinates and operator normal forms use C.42–C.45, already verified above for (C.53.1). Its full-partition invariance is (C.53.4). The base-family replacement C.52.5 is natural under every target reparametrization and exchange, and retains every affine operator equation C.52.6. No step of those intrinsic constructions uses the position of the target point on a frontier. They therefore give a based intrinsic contraction $K^j_{s,t}$, jointly continuous with its inverse, commuting with $H_{O_j}$ and with its actual restricted closure $Q_{O_j}$, and satisfying

$$\Phi_z K^j_{s,t} = R^{Gz}_{s,t} \Phi_z \quad \text{for every map in (C.53.1).} \tag{C.53.9}$$

For an empty family, C.32–C.33 give the based intrinsic contraction in that exact proper-operator centralizer. Every constant equivariant quotient also satisfies (C.53.9), since its target is fixed.

Discard repetitions among the finitely many $H$-orbits represented by $O_j$, and transport $K^j$ to each $hO_j$ by $hK^jh^{-1}$. Commutation with the full stabilizer makes this well defined. Its union gives a jointly continuous homeomorphism family $K$ on the open set $U$; components are open, and each carries its intrinsic family and inverse. It commutes with $H$ exactly.

*One compact comparison set for every accumulating sequence.* For each representative, compactness of $O_j/Q_{O_j}$ supplies a compact $C_j \subset O_j$ with $Q_{O_j}C_j = O_j$, by C.51's finite-cover argument. Suppose

$$x_n \in U, \quad x_n \longrightarrow z \in X \setminus U, \qquad (s_n, t_n) \longrightarrow (s,t). \tag{C.53.10}$$

The orbit $Gz$ is nonopen. Indeed an open orbit containing $z$ would contain every sufficiently late $x_n$, and so would itself be one of the components in $U$, contradicting $z \notin U$. There are only finitely many representatives. On a subsequence, write

$$x_n = h_n q_n y_n, \quad h_n \in H, \quad q_n \in Q_{O_j}, \quad y_n \in C_j, \quad y_n \longrightarrow y \in C_j \tag{C.53.11}$$

for one fixed $j$. Fix the compact subset of $O_j$

$$S_j = C_j \ \cup \bigcup_{(u,v)\in[0,1]^2} (K^j_{u,v}(C_j) \cup (K^j_{u,v})^{-1}(C_j)). \tag{C.53.12}$$

Both joint continuity assertions prove compactness.

Choose actual $a_n \in H_{O_j}$ so that

$$\sup_{w \in S_j} |h_n a_n w - h_n q_n w| < 1/n, \qquad g_n = h_n a_n \in H. \tag{C.53.13}$$

For each fixed $n$, the first condition is an open neighborhood of $q_n$ in $\mathrm{Homeo}(O_j)$: composition with the fixed ambient homeomorphism $h_n$ and restriction to the compact set $S_j$ are continuous. The defining density of the actual restrictions $H_{O_j} \to Q_{O_j}$ supplies $a_n$. Thus the estimate is made *after* the possibly distorting operator $h_n$; no uniform modulus of continuity for the sequence $h_n$ is assumed.

The open $L$-orbit map supplies $l_n \to 1$ with $l_n y = y_n$, exactly as in C.51. Since $g_n$ commutes with $L$, (C.53.10)–(C.53.13) give

$$g_n y = l_n^{-1} g_n y_n \longrightarrow z. \tag{C.53.14}$$

Every element of $L_y$ fixes this limit. The evaluation homeomorphism C.42.1 therefore gives

$$g_n|_{O_j} \longrightarrow \Phi_z^y \quad \text{uniformly on compact sets.} \tag{C.53.15}$$

The exact basepoint change (C.53.3) identifies this map with one of the global quotients in (C.53.1) when its target is nontrivial; otherwise it is constant. This is the reason the enlarged family is required in this argument.

Transport and commutation with $Q_{O_j}$ now yield

$$\begin{aligned} K_{s_n,t_n}(x_n) &= h_n q_n K^j_{s_n,t_n}(y_n), \\ \left|K_{s_n,t_n}(x_n) - g_n K^j_{s_n,t_n}(y_n)\right| &< 1/n. \end{aligned} \tag{C.53.16}$$

The inner points converge in $S_j$ to $K^j_{s,t}(y)$. Equations (C.53.15) and (C.53.9) consequently give

$$K_{s_n,t_n}(x_n) \longrightarrow \Phi_z^y K^j_{s,t}(y) = R^{Gz}_{s,t}(z). \tag{C.53.17}$$

Every subsequence admits a further subsequence of this form, by the finite number of $j$ and the compactness of their $C_j$. The value in (C.53.17) is independent of every such choice. Hence the entire sequence has that limit. If a representative has an empty global family, (C.53.15) also proves that every limit reached through its translates is a fixed point.

Define the extension on $X \setminus U$ to be $R^{Gz}_{s,t}(z)$. The diagonal approximation C.52.9 now applies with the dense set $U$ in $X$: for each varying boundary point and fixed parameter pair choose a point of $U$ close to it and with close image, using (C.53.17). Applying (C.53.17) to the resulting sequence proves joint continuity on all of $X$. This argument does not require prior transverse continuity of the hybrid rule.

Use the inverse images already present in $S_j$ and the exact inverse version of (C.53.9) to repeat (C.53.13)–(C.53.17) for inverse maps. The same diagonal approximation then proves joint inverse continuity on $X$. The maps preserve $U$ and each nonopen limit orbit, with the displayed maps as mutual inverses. Thus they are homeomorphisms of $X$. Their based-edge identities and $H$-commutation extend from $U$ by continuity; the restriction to every nonopen orbit is the prescribed $R^B$.

Finally, $U$ is disjoint from $X^\dagger$. Every point of $X \cap X^\dagger$ lies on a nonopen orbit; otherwise its open orbit would meet $U$ and be a component of $U$. The metabelian identity on $X^\dagger$ and the first part of the proof show that both contractions there use precisely the same exponential branch of C.52.1, with the same based covering path. They and their inverses agree on the full intersection. Pasting the two closed sets $X$ and $X^\dagger$ gives the asserted contraction on their union, preserving the prescribed family on $X^\dagger$.

The finite-representative argument does not supply compact comparison sets for arbitrary noncocompact quotients or infinitely many $H$-inequivalent representatives. Nor does it extend the family through an exterior region of nonopen orbits. These remain exact parts of the general C.12 obligation. □

**C.54. [Proved] Two metabelian quotient images give exponential closure contractions.** Use the actual global quotient family of C.53 for the ambient-null based plane loop $f_t$. Let $O$ be an open $G$-orbit admitting two distinct partitions in that family whose actual intrinsic target images are metabelian. On the universal covering $p : X \to O$, the group $N$ generated by the based source lifts is closed in $\mathrm{Homeo}(X)$. With that actual topology, it is one of the following simply connected groups:

$$\begin{gathered}\mathbb{R}^2, \qquad \mathrm{Aff}^+(\mathbb{R}), \qquad N_{a,b} = \mathbb{R}^2 \rtimes \mathbb{R} \quad ((a,b) \neq (0,0)), \\ \mathrm{Aff}^+(\mathbb{R}) \times \mathrm{Aff}^+(\mathbb{R}).\end{gathered} \tag{C.54.1}$$

In $N_{a,b}$ the real parameter $t$ acts on the two translation coordinates by $\mathrm{diag}(e^{at}, e^{bt})$. Thus zero, equal, unequal, and opposite nonzero weights are retained. Each group has the explicit globally invertible exponential map given below.

The based source exponential fractions descend and extend to a jointly continuous based contraction through homeomorphisms of the entire $\overline{O}$, with a jointly continuous inverse. They satisfy every actual global quotient equation, are independent of the selected pair of metabelian quotients, and are transported exactly by $H$. They agree on common orbit frontiers and with the prescribed contraction of C.53 on $X^\dagger$. Consequently they paste over $X^\dagger$ together with any finite collection of these closures. No compactness of $O/Q_O$ is required. Uniform continuity over an arbitrary infinite collection of the new closures is not asserted here.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison; actual noncocompact two-quotient Lie reductions.

*Proof. The actual target groups and the two coordinate domains.* Choose two distinct real lifted quotient maps $F_1, F_2 : X \to \mathbb{R}$, with their original target covers retained. Their based target paths $b_{i,t}$ are loops, by C.29 for circle targets and directly for line targets. As in C.52.4, the generated groups $T_i = \langle b_{i,t} \rangle$ are transitive on the real lines. The exact real-target comparison in C.52 makes each $T_i$ a full translation group or a full positive affine group after a line conjugacy. These are the actual groups with their intrinsic homeomorphism-group topologies.

C.45 identifies the pair of quotient coordinates with one of the three domains $\mathbb{R}^2$, $\{x < y\}$, or $\{x < y < x + 1\}$, by the specified independent line reparametrizations. The based source path has the coordinate form $(b_{1,t}, b_{2,t})$ and is a closed loop, since the target endpoints are the identity and the pair map is injective.

In the half-plane domain the two coordinate paths are identical, and their generated group is the same $T$. A transitive abelian line group is conjugate to translations. Its diagonal action cannot be transitive on an open two-dimensional domain: in translation coordinates the difference of the two coordinates is constant along each orbit. Thus $T$ is the full affine group. Conjugate both coordinates by the same line homeomorphism. The source action becomes

$$(x, y) \longmapsto (e^t x + u, e^t y + u), \qquad x < y. \tag{C.54.2}$$

The exact source change $(x, y) \mapsto (x, \log(y - x))$ identifies this with the simply transitive plane action

$$(x, r) \longmapsto (e^t x + u, r + t). \tag{C.54.3}$$

Its actual generated group is the full $\mathrm{Aff}^+(\mathbb{R})$.

The band domain is impossible under the present metabelian-target hypothesis. Here again the two paths and their group $T$ coincide, and $T$ commutes with unit translation. The full affine line group has trivial centralizer: a map commuting with every translation is a translation, and commuting with a nontrivial dilation makes that translation the identity. Thus $T$ cannot be affine. If $T$ is conjugate to translations, the commuting unit translation is conjugate to translation by a fixed nonzero constant. The band therefore becomes a constant-width band in those same coordinates, and the diagonal translation action again preserves the coordinate difference. It is not transitive on the band, a contradiction.

It remains to classify the full-plane case. Conjugate the two target groups independently to their standard translation or affine actions. The actual source group is then a path-connected subgroup of

$$A = \mathrm{Aff}^+(\mathbb{R})^2 = \{(s_1, s_2, u_1, u_2) : (x_1, x_2) \mapsto (e^{s_1}x_1 + u_1, e^{s_2}x_2 + u_2)\}. \tag{C.54.4}$$

We do not assume that this subgroup is closed.

The based source group is transitive on $X$. Indeed its compact path-and-inverse set $P$ and the countable deck group $\Delta$ satisfy $X = \bigcup_{\delta\in\Delta, n\geq 1} \delta P^n x$, as in C.49.3. Baire gives an open orbit. Every other orbit is a deck translate, because the based source path commutes with every source deck map. Hence all orbits are open, and connectedness of the covering plane gives transitivity. Yamabe's theorem now identifies this path-connected subgroup with the analytic subgroup of its Lie algebra [76, pp. 13–14]. We establish closedness and the induced topology separately.

*All transitive analytic subgroups of the product affine action.* Write the Lie algebra of (C.54.4) as $\mathfrak{a} = D \ltimes V$, where $D = \mathbb{R}^2$ is the pair of log-scale directions, $V = \mathbb{R}^2$ is the translation plane, and

$$[(d, v), (e, w)] = (0, (d_1w_1 - e_1v_1,\ d_2w_2 - e_2v_2)). \tag{C.54.5}$$

Let $\mathfrak{n}$ be the generated group's Lie algebra and put

$$S = \mathrm{pr}_D(\mathfrak{n}), \qquad W = \mathfrak{n} \cap V. \tag{C.54.6}$$

The subspace $W$ is invariant under the diagonal action of $S$. Transitivity makes the tangent evaluation of $\mathfrak{n}$ onto the source plane surjective at every point.

For $\dim S = 0$, this forces $W = V$ and gives all translations. For $\dim S = 1$, choose a lift $Z = ((a, b), v_0)$ of a nonzero generator of $S$. The case $W = 0$ has dimension one and cannot be transitive. If $W = V$, subtracting $v_0$ from $Z$ gives exactly the Lie algebra and connected group $N_{a,b}$ in (C.54.1).

Suppose instead $\dim W = 1$. If $a = b \neq 0$, the induced affine vector field on $V/W$ has nonzero scalar linear part and hence a fixed point. The corresponding affine line in $V$ is invariant under the entire connected subgroup, contradicting transitivity. If $a \neq b$, diagonal invariance makes $W$ one of the two coordinate axes. For $W = \mathbb{R}(1, 0)$, the second-coordinate vector field of $Z$ is $by + (v_0)_2$. If $b \neq 0$, its fixed level is an invariant affine line, again impossible. If $b = 0$ and $(v_0)_2 = 0$, the second coordinate is constant on every orbit. The remaining possibility is

$$b = 0, \qquad a \neq 0, \qquad (v_0)_2 \neq 0. \tag{C.54.7}$$

A translation of the first source coordinate removes $(v_0)_1$, and replacing the second coordinate by $ay/(v_0)_2$ gives exactly (C.54.3). The case $W = \mathbb{R}(0, 1)$ is its coordinate exchange. These are all one-dimensional $S$ cases.

For $\dim S = 2$, invariance under all diagonal matrices makes $W$ either zero, one coordinate axis, or $V$. If $W \neq V$, choose a coordinate $i$ on which every vector of $W$ is zero. For a linear section of $\mathfrak{n} \to S = D$, let $c_i(d)$ be its translation coordinate there. This is independent of the chosen section modulo $W$. Equation (C.54.5), whose bracket must belong to $W$, gives

$$d_ic_i(e) - e_ic_i(d) = 0 \quad (d, e \in D). \tag{C.54.8}$$

Choosing $e_i = 1$ shows $c_i(d) = c\,d_i$ for one constant $c$. Every Lie-algebra vector field then preserves the level $x_i = -c$. Its flows, and hence the connected analytic subgroup that they generate, preserve that affine line. This contradicts transitivity. Thus $W = V$, and $\mathfrak{n} = \mathfrak{a}$, giving the full product affine group.

*Closed groups, actual topology, and logarithms.* The preceding list consists of explicit closed subgroups of $A$. For $N_{a,b}$, the scale vector lies on the closed line $\mathbb{R}(a, b)$ and both translations are arbitrary. The simply transitive affine case (C.54.3) has, in its displayed source coordinates, the closed equations $s_2 = 0$ and $u_2 = s_1$. The translation plane and $A$ itself are closed as well. Each listed group is connected, with exactly the Lie algebra just obtained. Uniqueness of the connected analytic subgroup with a specified Lie algebra therefore identifies the original generated subgroup with that explicit closed group.

The topology of $A$ in $\mathrm{Homeo}(\mathbb{R}^2)$ is its ordinary parameter topology. Evaluation at $(0,0), (1,0), (0,1)$ recovers the translations and the two positive scale factors continuously. A limit which is still a homeomorphism cannot have either scale zero. Conversely convergent parameters give compact-open convergence of the maps and their inverses by the affine formulas. Thus $A$ is closed in this homeomorphism group and has the asserted actual topology. The source conjugacies used above preserve compact-open group topology. This proves the same assertions for the actual $N$ on $X$, rather than only for its analytic Lie-group structure.

Here are the promised exponential maps. Set

$$\sigma(z) = \int_0^1 e^{rz}\,dr = \begin{cases} (e^z - 1)/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \tag{C.54.9}$$

This function is positive for every real $z$. In $N_{a,b}$ the exact group law is

$$(t, u, v)(t', u', v') = (t + t', u + e^{at}u', v + e^{bt}v'), \tag{C.54.10}$$

and

$$\begin{aligned} \exp(t, u, v) &= (t, \sigma(at)u, \sigma(bt)v), \\ \log(t, u, v) &= (t, u/\sigma(at), v/\sigma(bt)). \end{aligned} \tag{C.54.11}$$

The positive affine group has the same formula with one translation coordinate and weight one. For the full product $A$,

$$\begin{aligned} \exp(s_1, s_2, u_1, u_2) &= (s_1, s_2, \sigma(s_1)u_1, \sigma(s_2)u_2), \\ \log(s_1, s_2, u_1, u_2) &= (s_1, s_2, u_1/\sigma(s_1), u_2/\sigma(s_2)). \end{aligned} \tag{C.54.12}$$

The translation exponential is the identity. These formulas follow by integrating the displayed affine vector fields, or directly by differentiating their one-parameter subgroups. Positivity of $\sigma$ proves that each is a global continuous inverse pair, including every zero weight. The displayed group parameter spaces are Euclidean spaces, so every group in (C.54.1) is simply connected.

*The exact action on the closure and on all global quotients.* Use the original global closure $L$ and the specified source covering to form

$$E_O = \{(l, n) \in L \times N : pn = (l|_O)p\}. \tag{C.54.13}$$

This is a closed Polish group. Every $n \in N$ is an actual finite word in the based source lifts; the identical word in the original plane loop gives a preimage in $E_O$. Projection to $N$ is therefore onto and open. Its kernel fixes $O$ pointwise and, by continuity, fixes $\overline{O}$ pointwise. Consequently the continuous action $(l, n, z) \mapsto lz$ descends through the open product quotient $E_O \times \overline{O} \to N \times \overline{O}$. This is the actual jointly continuous $N$-action on the closure. Every value has a global word preimage, so its orbits are precisely the original $G$-orbits there, with the original labels retained.

Define the source family by

$$n_{s,t} = \exp_N\big((1-s)\log_N \widetilde{f}_t\big). \tag{C.54.14}$$

The source deck group commutes with every generator, hence with $N$ itself. Formula (C.54.14) descends through $p$ and acts on $\overline{O}$ through the just-proved quotient action. It and its inverse are jointly continuous and satisfy all based-edge identities.

For any actual global quotient $\Phi_z : O \to B$, equivariance makes the kernel of (C.54.13) fix its entire image: $l\Phi_z(w) = \Phi_z(lw) = \Phi_z(w)$. The intrinsic restriction continuity proved in C.42 therefore descends through the same open quotient to a continuous $N$-action on $B$ with image exactly $G|_B$, as in C.53.7. All groups in (C.54.1) are metabelian. Thus this target image is one of the actual Lie groups in C.37.9. The continuous homomorphism, and its unique based covering lift for a circle, intertwine (C.54.14) with the exponential branch of C.52.1. Hence every actual global quotient equation holds, not only the two equations selected at the start.

*Boundary comparison, choices, and finite pasting.* For a nontrivial nonopen $G$-orbit $B \subset \overline{O}$, the continuous $N$-action just constructed has exactly that orbit. For $z \in B$, its orbit map gives a continuous bijection $N/N_z \to B$ with the ambient topology on the image. Its manifold dimension is one: dimension zero would give a point because $N$ is connected, while dimension at least two would give an injective map of a small two-dimensional disk into the plane with interior image, by invariance of domain. This would make $B$ contain an ambient open subset, contrary to its being nonopen.

On each compact interval of this one-manifold the orbit map is an arc. C.28 identifies its intrinsic interval with that same compact arc. Consequently the orbit bijection is a homeomorphism onto the intrinsic line or circle $B$. The induced action is continuous in that intrinsic topology, and its compact-open group topology, by the continuous-action exponential law. Its image is exactly $G|_B$ and is metabelian. C.37.9 and the same based covering comparison therefore identify the boundary value of (C.54.14) with the precise exponential branch of C.52.1. Fixed boundary points remain fixed.

The group $N$ was defined by the actual based source lifts, independently of the chosen quotient pair. The proof has identified its actual topology and its unique global exponential and logarithm. A different quotient pair therefore gives the identical family. Changing the source covering conjugates the based paths, with their specified identity initial lift, and hence conjugates the identical formula. An operator $h \in H$ gives this same comparison between the source covers of $O$ and $hO$. Its lifted conjugacy identifies the actual generated groups and their paths. Lie homomorphism naturality and uniqueness of logarithms give exact transport of (C.54.14).

Distinct open-orbit closures meet only in nonopen orbits. Their intrinsic images and original based target paths are the same $G|_B$, so the exponential values just proved agree there, including the full real circle lifts. On an open orbit already in $U^\dagger$, the earlier construction uses this same actual group generated by the based source lifts. Thus the two formulas agree on that open orbit and its closure. For a new orbit disjoint from $X^\dagger$, its intersection with $X^\dagger$ consists only of nonopen orbits, where both values are again the same exponential prescription. Finite closed-set pasting applies to the maps and their joint inverses. This proves the stated finite relative contraction without a cocompactness assumption. No uniform word bound for the new three- or four-dimensional groups has been used or asserted. □

**C.55. [Proved] The full local centralizer and its exact global limits.** Retain the ambient-null based plane loop $f_t$, its generated group $G$, its commuting operator group $H$, and $L = \overline{G}$ from C.42. On every open $G$-orbit define the closed topological group

$$\mathcal{C}_O = C_{\mathrm{Homeo}(O)}(G|_O), \qquad H_O \longrightarrow \mathcal{C}_O, \quad h \longmapsto h|_O. \tag{C.55.1}$$

The topology is compact-open, including inverse convergence. The kernel of the displayed homomorphism consists exactly of operators fixing $O$ pointwise. Its image closure is the previously used $Q_O$, a closed subgroup of $\mathcal{C}_O$.

Let $U^{\ddagger}$ be the union of open orbits with $\dim \mathcal{C}_O = 2$ or $|\mathscr{F}_O^{\mathrm{all}}| \geq 3$, using C.53's full global quotient family, and put $X^{\ddagger} = \overline{U^{\ddagger}}$. There is a jointly continuous based contraction on $X^{\ddagger}$, with jointly continuous inverse and commutation with $H$, extending the specified contraction on $X^{\dagger}$ in C.53. Its nonopen-orbit values are the exponential branch of C.52.1.

Choose finitely many open orbits $O_1, \dots, O_m$ disjoint from $X^{\ddagger}$ for which $O_j/\mathcal{C}_{O_j}$ is compact. Set

$$U = \bigcup_{j=1}^{m} \bigcup_{h \in H} hO_j, \qquad X = \overline{U}. \tag{C.55.2}$$

There is a based contraction on $X^{\ddagger} \cup X$, jointly continuous with its inverse, commuting with $H$, and equal to the prescribed contraction on $X^{\ddagger}$. On every nonopen orbit in $X$ it has the exact hybrid value of C.52.1. The intrinsic contraction on each representative commutes with all of $\mathcal{C}_{O_j}$. Thus these conclusions apply with the full local centralizer quotient, without requiring that its elements extend to the plane.

*Manuscript:* Section23; folios140-141; scans279-281; Part C(b), infinite-operator zero-level comparison.

*Proof. The local group, its topology, and its global comparisons.* The centralizer of $G|_O$ is also the centralizer of its closure in Homeo($O$). The latter is a closed Polish group transitive on $O$. C.30 therefore proves that $\mathcal{C}_O$ is locally compact, with a free proper action. C.33's surface Lie argument applies to this very action and makes $\mathcal{C}_O$ a Lie group of dimension at most two. These conclusions use the restriction image and its topology, not a discrete topology on the original operators. Formula (C.55.1) and its kernel follow directly from commutation on the plane. Closure in Homeo($O$) proves the assertion about $Q_O$.

The global group $L$ preserves $O$, by the boundary and inverse-limit argument of C.36. Its restriction is continuous into Homeo($O$): compact subsets of $O$ and their inverse images remain compact subsets of the plane. Hence every $q \in \mathcal{C}_O$ satisfies

$$q(lu) = lq(u) \quad (l \in L,\ u \in O). \tag{C.55.3}$$

For example, choose $g_n \in G$ converging to $l$ globally, restrict on compact subsets of $O$, and pass $qg_n u = g_n qu$ to the limit. This proves (C.55.3) without a plane extension of $q$.

For any $h \in H$, conjugation defines the topological group isomorphism

$$\mathcal{C}_O \longrightarrow \mathcal{C}_{hO}, \qquad q \longmapsto hqh^{-1}. \tag{C.55.4}$$

The domain and codomain here are local homeomorphism groups; the displayed conjugation uses only the actual global operator $h$. It preserves dimension and the compactness of the corresponding orbit quotient.

Fix $x \in O$, put $J = L_x$, and choose $a_q \in L$ with $a_q x = qx$, using transitivity. Equation (C.55.3) and its inverse give $L_{qx} = J$, and therefore $a_q J a_q^{-1} = J$. For every global quotient in C.53.1,

$$\begin{aligned} \Phi_z^x q(lx) &= \Phi_z^x (la_q x) = la_q z, \\ \Phi_z^x q &= \Phi_{a_q z}^x. \end{aligned} \tag{C.55.5}$$

The target $a_q z$ is fixed by $J$ and lies in the same actual nonopen orbit, since C.42 gives $Lz = Gz$. Changing $a_q$ by its stabilizer ambiguity changes no map in (C.55.5). Applying the formula also to $q^{-1}$ proves that the full global family is permuted by $\mathcal{C}_O$. The exact transport between different

orbits is still C.53.5. All source and target coverings in that family are therefore the ones already used in C.43–C.48.

*The enlarged exponential region.* For $\dim \mathcal{C}_O = 2$, the proof of C.36 applies with $\mathcal{C}_O$ as its proper group. More explicitly, its identity-component orbits are open by the free dimension-two action and closed by properness; since $O$ is connected, that component is transitive. Freeness then gives $\mathcal{C}_O = \mathcal{C}_O^0$. The evaluation map identifies it with $O$. Its full centralizer

$$N_O = C_{\mathrm{Homeo}(O)}(\mathcal{C}_O) \cong \mathcal{C}_O^{\mathrm{op}} \tag{C.55.6}$$

is a simply transitive two-dimensional Lie group. The actual restriction image $G|_O$ equals $N_O$: transitivity supplies a word with any prescribed value at one point, and evaluation is injective on $N_O$.

The resulting surjection $L \to N_O$ is continuous and open. Its kernel fixes $\overline{O}$ pointwise, and it fixes the image of every global quotient $\Phi_z$ pointwise by $l\Phi_z(u) = \Phi_z(lu)$. The two actions therefore factor through the open quotient, with their actual topology, exactly as in C.36.3 and C.53.6–C.53.7. The universal covering group is $\mathbb{R}^2$ or $\mathrm{Aff}^+(\mathbb{R})$, with global exponential and logarithm. The original based path lifts to a closed path: C.32 gives its null-homotopy in $\mathrm{Homeo}(O)$, and the evaluation homeomorphism gives the identical statement in $N_O$. Fractional powers in that specified cover define the same closure contraction as C.36.6. On each global nonopen quotient they give the exponential branch of C.52.1, including the based real circle angle.

For $|\mathscr{F}_O^{\mathrm{all}}| \geq 3$, retain exactly the actual group and fractional powers of C.53.6–C.53.8. They also commute with every $q \in \mathcal{C}_O$. Indeed choose any lift $\widetilde{q}$ to the universal cover of $O$. The path $\widetilde{q}\widetilde{f}_t\widetilde{q}^{-1}$ is the based lift of the same path $f_t|_O$, so uniqueness of path lifting identifies it with $\widetilde{f}_t$. It follows that $\widetilde{q}$ commutes with its entire actual generated source group and with the displayed fractional powers. Descent proves the assertion. Where the two constructions apply together, they use the same based path in the same actual simply connected Lie group and agree.

All these source groups have the uniform fractional word bound 1913 of C.50; the two-dimensional groups have the stronger bound three. Their images are metabelian. The compact global word-isotopy family C.50.15 consequently applies to their entire union. For each positive integer $n$, its fractional residual is in the same specified simply connected group and has the same bound. On a limiting line orbit, the bounded logarithm argument C.50.17–C.50.18 determines the fractional limit. On a limiting circle orbit, the null residual word loop has zero two-point winding, so C.50.19 retains its real angle, including the integer part. On fixed points every such word is the identity. These are precisely the inputs of C.50's dense-set limit and diagonal arguments. They prove joint continuity on $X^\ddagger$ and, on applying them to the inverse loop, joint inverse continuity. Equation (C.55.4) and uniqueness of the based lift give the required operator transport and commutation.

Since $Q_O$ is a closed subgroup of $\mathcal{C}_O$, $\dim Q_O = 2$ implies $\dim \mathcal{C}_O = 2$. Thus $U^\dagger \subset U^\ddagger$. The two contractions agree on $U^\dagger$ by their actual source formulas, and hence on $X^\dagger$ by continuity. This proves the first extension assertion with its exact prescription.

*Intrinsic maps commuting with the full local group.* Each remaining representative in (C.55.2) has $\dim \mathcal{C}_{O_j} \leq 1$ and at most two global partitions. Apply the intrinsic construction of C.52 to the proper Lie group $\mathcal{C}_{O_j}$ and to this full family. Every required operator property is now proved: freeness, properness and the topology follow from C.30; the Lie property follows from C.33; and permutation of all quotient partitions is (C.55.5). In the two-partition case, C.46 and C.52 use the exact coordinate changes and target conjugacies. In the one-partition case, C.48's proper-group normal forms and C.52's fiber and base formulas retain every relation of this very local group, including deck relations. With an empty family, C.32–C.33 give the intrinsic centralizer contraction using the same ambient-null restriction. These intrinsic arguments do not use a global extension of an operator.

They give a based family $K^j_{s,t}$ on $O_j$, jointly continuous with its inverse, commuting with $\mathcal{C}_{O_j}$, and satisfying

$$\Phi_z K^j_{s,t} = R^{Gz}_{s,t}\Phi_z \quad \text{for every global quotient.} \tag{C.55.7}$$

Constant equivariant maps satisfy the identical equation, with the fixed-point target prescription. Remove duplicate $H$-representatives and set $K = hK^jh^{-1}$ on $hO_j$. Its independence from the chosen $h$ follows from $H_{O_j}|_{O_j} \subset \mathcal{C}_{O_j}$. The union is a jointly continuous homeomorphism family on the open set $U$, with jointly continuous inverse and exact $H$-commutation.

*Limits of the actual local equivariant maps.* Compactness of $O_j/\mathcal{C}_{O_j}$ supplies a compact $C_j \subset O_j$ with $\mathcal{C}_{O_j}C_j = O_j$. For completeness, the quotient projection is open. Relatively compact open neighborhoods in $O_j$ have open quotient images covering the compact quotient; a finite subcover and the union of their compact closures give $C_j$.

Take $x_n \in U$ converging to $z \in X \setminus U$, and $(s_n, t_n) \to (s, t)$. The orbit $Gz$ is nonopen, because an open orbit containing $z$ would contain all late $x_n$ and would be a component of $U$. After a subsequence, the finite representative set and compactness give one $j$ and

$$x_n = h_n q_n y_n, \quad h_n \in H,\ q_n \in \mathcal{C}_{O_j},\ y_n \in C_j, \qquad y_n \longrightarrow y \in C_j. \tag{C.55.8}$$

The maps

$$m_n : O_j \longrightarrow \mathbb{R}^2, \qquad m_n(u) = h_n q_n u \tag{C.55.9}$$

are actual continuous $L$-equivariant maps by (C.55.3) and global commutation of $h_n$ with $L$. The open $L$-orbit map supplies $l_n \to 1$ in $L$ with $l_n y = y_n$. Consequently

$$m_n(y) = l_n^{-1} x_n \longrightarrow z. \tag{C.55.10}$$

Every element of $L_y$ fixes $z$. C.42's evaluation homeomorphism, applied to the space of all continuous $L$-equivariant maps $O_j \to \mathbb{R}^2$, gives

$$m_n \longrightarrow \Phi^y_z \quad \text{uniformly on compact sets.} \tag{C.55.11}$$

This is a direct use of the actual maps (C.55.9). No element $q_n$ has been extended to the plane or approximated by restrictions of global operators.

The points $K^j_{s_n,t_n}(y_n)$ converge to $K^j_{s,t}(y)$ in a compact subset of $O_j$, by joint continuity and compactness of the parameter square times $C_j$. Transport, commutation, (C.55.11), the basepoint comparison C.53.3, and (C.55.7) give exactly

$$\begin{aligned} K_{s_n,t_n}(x_n) &= h_n q_n K^j_{s_n,t_n}(y_n) \\ &\longrightarrow \Phi^y_z K^j_{s,t}(y) = R^{Gz}_{s,t}(z). \end{aligned} \tag{C.55.12}$$

Every subsequence has a further subsequence of this form, and its limit is the same prescribed value. Thus (C.55.12) holds for the whole sequence. An empty global family makes each such limit orbit a fixed point, since otherwise (C.55.11) itself would supply a member of that family.

Define $K$ on $X \setminus U$ by this prescribed hybrid value. At a varying boundary point and fixed parameter, (C.55.12) supplies a nearby point of $U$ with nearby image. Choosing both distances less than $1/n$ and applying (C.55.12) once more proves joint continuity for boundary-only sequences. Combining interior and boundary subsequences proves joint continuity on the whole square times $X$. The inverse form of (C.55.7) and the compact set of all $(K^j_{s,t})^{-1}(C_j)$ give the identical argument for inverses. Every map preserves the treated orbits and has this displayed inverse; it is therefore a homeomorphism of $X$. The based-edge identities and $H$-commutation extend from the dense open set $U$.

Finally $X \cap X^{\ddagger}$ contains only nonopen orbits. The metabelian identity holds on $X^{\ddagger}$, so the hybrid value there is exactly its prescribed exponential value, with the same based covering lift.

The two families and their inverses agree on the intersection. Finite closed-set pasting gives the contraction on $X^{\ddagger} \cup X$.

This argument still requires finitely many $H$-representatives and compact quotients by their full local centralizers. It supplies no extension through arbitrary noncompact local quotients, infinitely many inequivalent representatives, or the remaining exterior nonopen region. Those parts of C.12 remain under investigation. □

**C.56. [Proved] A uniform word bound for diagonal one-parameter affine groups.** For real weights $\alpha, \beta$, use the group

$$N_{\alpha,\beta} = \mathbb{R} \ltimes \mathbb{R}^2, \qquad (\tau, v)(\tau', w) = (\tau + \tau', v + \operatorname{diag}(e^{\alpha\tau}, e^{\beta\tau})w). \tag{C.56.1}$$

For every continuous path $k : [0,t] \to N_{\alpha,\beta}$ with $k(0) = 1$, put $P = k([0,t]) \cup k([0,t])^{-1}$ and $g = k(t)$. With the explicit uniform integer

$$M = 13\,363\,205, \tag{C.56.2}$$

every exponential fraction $\exp(r \log g)$, $0 \le r \le 1$, belongs to $P^M$. The bound is independent of the weights and the path. Zero, equal and opposite weights are included.

In the plane-loop setting of C.54–C.55, let $U^{\sharp}$ be the union of $U^{\ddagger}$ and every C.54 orbit whose actual generated source group has dimension at most three, and put $X^{\sharp} = \overline{U^{\sharp}}$. The specified exponential contractions extend to a jointly continuous based contraction on $X^{\sharp}$, with jointly continuous inverse and exact $H$-commutation. It extends the prescribed family on $X^{\ddagger}$ and retains the exponential value on every nonopen orbit. The finite-representative extension of C.55 also holds relative to $X^{\sharp}$, for representatives outside this larger set with compact quotients by their full local centralizers.

*Manuscript:* Section23; folios140-141; scans279-281; Part C(b), infinite-operator zero-level comparison.

*Proof. The anchor and the exact product error.* Write $k(s) = (\tau_s, v_s)$ and $A = \max_{0 \le s \le t} |\tau_s|$. For $A = 0$, the path is in the translation plane, and C.37's three-letter additive path identity applies. For $A > 0$, choose a path value or its inverse $h \in P$ whose first coordinate is $A$. Conjugate all coordinates by a single translation so that every nonzero-weight coordinate of $h$ has translation part zero. Its scale in that coordinate differs from one, so this is possible. Leave any zero-weight translation coordinate unchanged. This common conjugacy adds no letters to a word and preserves all group identities.

Put $\alpha_1 = \alpha$, $\alpha_2 = \beta$, $x_i = \alpha_i A$, and $D_i = e^{x_i} - 1$. Direct multiplication gives the translation path

$$C_s = [h, k(s)] = (0, (D_1(v_s)_1, D_2(v_s)_2)). \tag{C.56.3}$$

For a zero weight both contributions to its commutator coordinate vanish, including the unchanged translation part of $h$. Every $C_s$ or inverse costs at most four letters of $P$. Conjugation by $h^j$ multiplies the $i$th translation coordinate by $e^{jx_i}$ for every integer $j$.

Apply C.50.3's three-dimensional additive path identity to $s \mapsto (\tau_s, v_s)$. It supplies $m \le 5$, signs $\varepsilon_j \in \{1, -1\}$ and times $s_j \in [0,t]$ with

$$r\tau_t = \sum_{j=1}^{m} \varepsilon_j \tau_{s_j}, \qquad r v_t = \sum_{j=1}^{m} \varepsilon_j v_{s_j}. \tag{C.56.4}$$

Form the actual product $w = \prod_{j=1}^{m} k(s_j)^{\varepsilon_j}$. For $\varepsilon_j = 1$ put $p_j = \sum_{\ell<j} \varepsilon_\ell \tau_{s_\ell}$; for $\varepsilon_j = -1$ put $p_j = \sum_{\ell \le j} \varepsilon_\ell \tau_{s_\ell}$. The full inverse formula in (C.56.1) proves

$$w = \left( r\tau_t, \left( \sum_{j=1}^{m} \varepsilon_j e^{\alpha_i p_j} (v_{s_j})_i \right)_{i=1,2} \right), \qquad |p_j| \le 5A. \tag{C.56.5}$$

In particular, the additive identity has not been substituted for a noncommutative product.

Define

$$\theta(z,r)=\begin{cases}(e^{rz}-1)/(e^z-1), & z\neq 0,\\ r, & z=0.\end{cases} \tag{C.56.6}$$

C.54.11 gives $g^r=(r\tau_t,(\theta(\alpha_i\tau_t,r)(v_t)_i)_i)$. Put $q=\tau_t/A$, $q_j=p_j/A$, so $|q|\le 1$ and $|q_j|\le 5$. For a real variable $x$, define the smooth functions

$$\begin{aligned}F_0(x)&=\frac{\theta(qx,r)-r}{e^x-1},\\ F_j(x)&=-\varepsilon_j\frac{e^{q_jx}-1}{e^x-1}\quad(1\le j\le m),\end{aligned} \tag{C.56.7}$$

with their continuous values at zero. By (C.56.4)–(C.56.5), the translation taking $w$ to $g^r$ by left multiplication has $i$th coordinate exactly

$$F_0(x_i)D_i(v_t)_i+\sum_{j=1}^{m}F_j(x_i)D_i(v_{s_j})_i. \tag{C.56.8}$$

Terms with $D_i=0$ vanish, just as the original corrections in that coordinate vanish.

*Uniform analytic bounds at every weight.* Use $\sigma(x)=\int_0^1 e^{ux}\,du$ from C.54.9. For every real $x$,

$$\sigma(x)\ge e^{-|x|},\qquad 0\le\sigma'(x)/\sigma(x)\le 1. \tag{C.56.9}$$

For $|b|\le 5$, $(e^{bx}-1)/(e^x-1)=b\sigma(bx)/\sigma(x)$ is smooth at zero, has absolute value at most $5e^{6|x|}$, and has derivative bounded by $30e^{6|x|}$. Indeed its logarithmic derivative when nonzero is the difference of a number of absolute value at most $|b|$ and one in $[0,1]$. The zero function for $b=0$ satisfies the same bounds directly.

For $F_0$, regard $\theta(z,r)$ as the probability of $[0,r]$ for the density $e^{zu}/\sigma(z)$ on $[0,1]$. Differentiating the two integrals gives

$$|\partial_z\theta(z,r)|\le 1,\qquad |\partial_z^2\theta(z,r)|\le 6. \tag{C.56.10}$$

To check the second bound explicitly, its expression in expectations for this density is

$$\begin{aligned}&\mathbb{E}(\mathbf{1}_{[0,r]}u^2)-2\mathbb{E}(\mathbf{1}_{[0,r]}u)\mathbb{E}(u)\\ &\qquad-\mathbb{E}(\mathbf{1}_{[0,r]})\mathbb{E}(u^2)+2\mathbb{E}(\mathbf{1}_{[0,r]})\mathbb{E}(u)^2.\end{aligned}$$

All expectations lie in $[0,1]$, so the absolute sum is at most six. The first derivative is a difference of two numbers in $[0,1]$. Consequently

$$F_0(x)=\frac{q\int_0^1\partial_z\theta(uqx,r)\,du}{\sigma(x)} \tag{C.56.11}$$

has absolute value at most $e^{|x|}$. The derivative of its numerator is at most $6q^2\int_0^1 u\,du\le 3$ in absolute value. Equation (C.56.9) therefore bounds $|F_0'(x)|$ by $4e^{|x|}$. Every function in (C.56.7) thus satisfies

$$|F(x)|\le 5e^{6|x|},\qquad |F'(x)|\le 30e^{6|x|}. \tag{C.56.12}$$

*Bounded interpolation at the two diagonal nodes.* Set

$$\begin{aligned}A_8(x)&=(e^x+e^{-x})^8,\\ B_8(x)&=(e^x-e^{-x})(e^x+e^{-x})^7,\qquad u(x)=\tanh x=B_8(x)/A_8(x).\end{aligned} \tag{C.56.13}$$

For $f(x) = F(x)/A_8(x)$, the estimates just proved give

$$|f(x)| \le 5, \quad |f'(x)| \le 70e^{-2|x|}, \quad u'(x) = \operatorname{sech}^2 x \ge e^{-2|x|}. \tag{C.56.14}$$

We used $A_8(x) \ge e^{8|x|}$ and $A_8'(x)/A_8(x) = 8\tanh x$. Thus $f$ as a function of $u \in (-1,1)$ is Lipschitz with constant seventy.

For distinct $x_1, x_2$ put

$$b = \frac{f(x_1) - f(x_2)}{u(x_1) - u(x_2)}, \qquad a = f(x_1) - bu(x_1).$$

For equal nodes put $b = 0$, $a = f(x_1)$. In either case,

$$|a| \le 75, \quad |b| \le 70, \qquad F(x_i) = aA_8(x_i) + bB_8(x_i) \quad (i = 1, 2). \tag{C.56.15}$$

This includes a zero node and coincident weights. The two node values come from the same function $F$ in (C.56.7), as follows from the single logarithmic scale parameter $\tau$. No uncontrolled divided difference is used.

*Realization by two explicit translation paths.* Expand the Laurent polynomials

$$(z + z^{-1})^8 = \sum_{j=-8}^{8} a_j z^j, \qquad (z - z^{-1})(z + z^{-1})^7 = \sum_{j=-8}^{8} b_j z^j. \tag{C.56.16}$$

Their coefficients are integers with $\sum |a_j| = 256$ and $\sum |b_j| \le 256$. Define continuous translation paths

$$\mathcal{A}_s = \prod_{j=-8}^{8} h^j C_s^{a_j} h^{-j}, \qquad \mathcal{B}_s = \prod_{j=-8}^{8} h^j C_s^{b_j} h^{-j}. \tag{C.56.17}$$

They start at the identity. Their $i$th translation coordinates are $A_8(x_i)D_i(v_s)_i$ and $B_8(x_i)D_i(v_s)_i$, respectively. All factors are translations, so their order of multiplication is immaterial.

A conjugated copy of $C_s^{\pm 1}$ costs at most $2|j| + 4$ letters of $P$. Repetition realizes each integer coefficient. Hence each value or inverse of either path costs at most

$$(2 \cdot 8 + 4)\, 256 = 5120. \tag{C.56.18}$$

The planar additive path identity expresses a fraction in $[0, 1]$ of $\mathcal{A}_s$ by three signed values of $\mathcal{A}|_{[0,s]}$, and likewise for $\mathcal{B}$. For $|a| \le 75$ use seventy-five copies of the fraction $|a|/75$, with inverse for a negative sign. For $|b| \le 70$ use seventy copies. Thus the translation $\mathcal{A}_s^a \mathcal{B}_s^b$ realizes (C.56.15) times the vector $(D_i(v_s)_i)_i$ and costs at most

$$3(75 + 70)\, 5120 = 2\,227\,200. \tag{C.56.19}$$

The word parameters need not be chosen continuously: the translation paths to which the additive identity applies are the actual continuous paths (C.56.17).

Apply this to $F_0$ at $s = t$ and to every $F_j$ at $s = s_j$. At most six such translations give exactly (C.56.8). Their product followed by $w$ is $g^r$, with length at most

$$5 + 6 \cdot 2\,227\,200 = 13\,363\,205. \tag{C.56.20}$$

Identity path values pad shorter words. This proves (C.56.2), including all zero weights.

*Exact algebraic verification.* The executed program `code/root/research_data/c56_diagonal_word_certificate.py` expands the Laurent polynomials, verifies the interpolation identities and coefficient bounds, and checks the complete product correction with rational arithmetic for every signed pattern of length one through five in six weight regimes. It also verifies the

commutator and all Laurent conjugation sums in those cases. The regimes include equal, unequal, opposite and zero weights. The analytic estimates and signed path-selection result have their proofs above and in C.50.3; these finite checks are not used as their proof.

*Accumulation and the relative extension.* On $U^{\ddagger}$, C.55 gives the prescribed source fractions with bound 1913. For a new C.54 orbit of dimension at most three, its actual source is $\mathbb{R}^2$, $\mathrm{Aff}^+(\mathbb{R})$, or (C.56.1). The bounds three and (C.56.2) therefore apply in its specified simply connected source group. C.54 already proves its individual closure action and every global quotient equation. Operator conjugation transports these groups and their based paths. Consequently the fractions agree wherever definitions overlap.

Use $M$ in the compact global word-isotopy family C.50.15. It is a finite union of continuous images of a compact finite cube, for maps and inverses. The metabelian identity holds on $U^{\sharp}$ and hence on $X^{\sharp}$. Every nonopen transitive image there is therefore one of the line translation, line affine, or circle rotation groups classified in C.37.9.

For every positive integer $n$, a sequence of source fractions has the exact residual

$$\exp\big((nr_i - \lfloor nr_i \rfloor)\log \widetilde{f}_{t_i}\big)$$

in the same specified simply connected source. Its exponent lies in $[0,1)$, so the residual has the same uniform bound $M$. Compact word parameters give bounded target residual logarithms on a line. On a circle, the source residual loop is null in that same source group; its two-point winding passes to the limit and is zero. The real equation is exactly C.50.19, with $M$ in its bound. Division by $n$ identifies the limiting fraction, including the integer part of the based circle angle. Every word fixes a fixed target point. These are all inputs of C.50's dense-set limit and diagonal arguments. They prove joint continuity on $X^{\sharp}$.

Apply the same proof to the inverse loop. The resulting family is the orbitwise inverse, so it is jointly continuous and each map is a homeomorphism of $X^{\sharp}$. The based-edge and operator identities extend from the dense $U^{\sharp}$. The fractions agree with the old family on $U^{\ddagger}$; continuity proves agreement on $X^{\ddagger}$.

Finally, representatives outside $X^{\sharp}$ are outside $X^{\ddagger}$, so the intrinsic construction and direct local-equivariant-map limit C.55.7–C.55.12 apply with the same compact fundamental sets and full global quotient equations. Their closure meets $X^{\sharp}$ only in nonopen orbits. The actual image on $X^{\sharp}$ is metabelian, so the hybrid value on the intersection is exactly its prescribed based exponential fraction. Finite closed-set pasting of the maps and inverses proves the last assertion.

The proof above uses one logarithmic scale parameter. C.58 proves the full product affine bound by a separate exact commutator correction in logarithmic coordinates. Arbitrary remaining local quotients, infinitely many inequivalent representatives and the exterior nonopen-region extension remain parts of C.12. □

**C.57. [Proved] The full local centralizer with exactly two global quotients.** Use C.53's full global family and suppose $|\mathscr{F}_O^{\mathrm{all}}| = 2$ for an open $G$-orbit $O$. Let $p : X \to O$ be its original universal covering, $\Delta$ its actual deck group, and $\mathcal{C}_O$ the full local centralizer in C.55. Its group of all source lifts is

$$\widetilde{\mathcal{C}}_O = \{q \in \mathrm{Homeo}(X) : pq = \bar{q}p \text{ for some } \bar{q} \in \mathcal{C}_O\}. \tag{C.57.1}$$

In the exact pair-coordinate domains of C.45 the following list is complete.

For the full-plane domain, put $T_i = \langle b_{i,t} \rangle$ and $D_i = C_{\mathrm{Homeo}(\mathbb{R})}(T_i)$. Each $D_i$ is, after a specified line conjugacy, the identity group, the integer translation group, or the real translation group. In those coordinates,

$$\widetilde{\mathcal{C}}_O = D_1 \times D_2, \qquad \mathcal{C}_O \cong (D_1 \times D_2)/\Delta, \qquad O/\mathcal{C}_O \cong \mathbb{R}^2/(D_1 \times D_2). \tag{C.57.2}$$

All these are topological-group or quotient-space identifications through the original covering. Up to exchange of coordinates, the final quotient is

| $D_1 \times D_2$ | $O/\mathcal{C}_O$ | compact |
|---|---|---|
| $1$ | $\mathbb{R}^2$ | no |
| $\mathbb{Z} \times 1$ | $S^1 \times \mathbb{R}$ | no |
| $\mathbb{R} \times 1$ | $\mathbb{R}$ | no |
| $\mathbb{Z}^2$ | $S^1 \times S^1$ | yes |
| $\mathbb{R} \times \mathbb{Z}$ | $S^1$ | yes |
| $\mathbb{R}^2$ | $\{*\}$ | yes. |

(C.57.3)

For the half-plane domain, $\Delta = 1$ and $\mathcal{C}_O = 1$, so its quotient is a plane. For the band domain $D = \{x < y < x + 1\}$, put $e(x, y) = (y, x + 1)$. Then

$$\widetilde{\mathcal{C}}_O = \langle e \rangle \cong \mathbb{Z}, \qquad \Delta = 1 \text{ or } \langle e^{2m} \rangle \ (m \geq 1), \qquad \mathcal{C}_O \cong \langle e \rangle / \Delta. \tag{C.57.4}$$

Its quotient $O/\mathcal{C}_O$ is always the open Möbius band. Thus every noncocompact case in this two-family list has full lifted local centralizer 1, $\mathbb{Z}$, or $\mathbb{R}$, with quotient a plane, open cylinder, line, or open Möbius band. This identifies the groups and quotient maps; it asserts no further contraction at their frontiers.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison; exact local groups and remaining noncocompact two-quotient domains.

*Proof. The lift group and its one-dimensional factors.* Every element of (C.57.1) commutes with the based source path. Indeed its conjugation of that path is another lift of the same downstairs path, with the same identity initial value. The path and its inverse have compact image. Their generated source group is transitive by the compact-word, countable-deck Baire argument of C.49.3; that argument does not require the generated group to be Lie. The two based target paths are loops by C.29, and their groups $T_i$ are transitive on their real lines, as in C.52.4.

Equation (C.55.5) proves that every local centralizer element permutes the full two-element family of global quotient partitions. Its source lifts consequently have coordinate-pair or coordinate-exchange form, as proved in C.46. The actual source deck maps preserve each selected partition separately: the equation $q_i F_i = \Phi_i p$ says that $F_i \delta$ and $F_i$ differ by the specified target deck map. This retains the original deck group inside (C.57.1).

For any transitive $T \leq \mathrm{Homeo}^+(\mathbb{R})$, its full centralizer $D$ acts freely. An element fixing one point fixes its entire transitive $T$-orbit. A decreasing real-line homeomorphism has a fixed point, so no such element belongs to $D$. The line version of C.30, used with the closed transitive group $\overline{T}$, makes $D$ locally compact, free and proper in its actual topology. The C.48 argument makes it a Lie group of dimension at most one.

In dimension one its identity component has open orbits, hence is transitive on the connected line. Freeness excludes additional components. Evaluation gives the standard real translation action. In dimension zero the orbit of a point is a closed discrete ordered set. A nonidentity free increasing action has points of that orbit on both sides. The element carrying the point to its nearest successor generates the group: its positive iterates cannot have a finite accumulation point, by properness, and there is no orbit point between consecutive iterates. Freeness identifies every group element with the unique iterate having its value at that point. An interpolation on one fundamental interval and its translates conjugates the action to integer translations. The remaining case is the identity group. This proves the asserted exact alternatives for every $D_i$, including their topology.

*The plane: the graph obstruction and every descending pair.* Use the C.45 coordinate map $X \to \mathbb{R}^2$. A coordinate exchange has the form $q(x, y) = (a(y), b(x))$. Commutation with every

source generator would imply

$$ab_{2,t} = b_{1,t}a \quad (0 \le t \le 1). \tag{C.57.5}$$

The proper nonempty graph $\{(a(y), y) : y \in \mathbb{R}\}$ is then preserved by every generator, hence by the transitive generated source group. This is impossible. The argument allows either orientation of $a$; no exchange is omitted.

Every lift in (C.57.1) is therefore a coordinate pair $(a(x), b(y))$. Commutation with the path is exactly $a \in D_1$, $b \in D_2$. Thus the lift group is contained in $P = D_1 \times D_2$. In particular $\Delta \subset P$. The preceding one-dimensional classification makes $P$ abelian. Conversely every pair in $P$ commutes with the original lifted path and with $\Delta$. It therefore descends through the original $p$ to a homeomorphism of $O$ commuting with every $f_t|_O$. Hence it belongs to (C.57.1). This proves equality with $P$.

The source covering gives $O \cong \mathbb{R}^2/\Delta$ and the exact surjection $P \to \mathcal{C}_O$ has kernel $\Delta$. Its quotient topology is the actual compact-open topology: the covering-chart comparison for maps and inverses in C.48 proves this for the group of all lifts, and the group here is exactly that group by (C.57.1). The two maps from $X$ to the orbit quotient have exactly the same fibers:

$$X \longrightarrow O \longrightarrow O/\mathcal{C}_O, \qquad X \longrightarrow X/P. \tag{C.57.6}$$

Both are open quotient maps. Their induced bijection is consequently a homeomorphism, proving (C.57.2). For completeness, the deck group in these formulas is 1 or $\mathbb{Z}$: C.44 already proves that the source of an actual nontrivial global quotient is a plane or an open annulus. Its particular embedding in $P$ remains the original deck embedding, and is not discarded in the formula for $\mathcal{C}_O$.

Choose the line conjugacies just constructed for $D_1, D_2$ independently. They give the six rows of (C.57.3). The quotient maps, in the order displayed there, are $(x, y)$, $([x], y)$, $y$, $([x], [y])$, $[y]$, and the constant map. Here brackets mean the quotient by the specified integer translations. These formulas explicitly identify the quotient spaces and their compactness, independently of which allowed source deck subgroup occurs.

*The half-plane: no pair and no exchange survives.* The normalized source path is $(g_t(x), g_t(y))$ on $x < y$. Write $T = \langle g_t \rangle$. It acts transitively on ordered pairs in this half-plane. For any fixed $x$, its stabilizer $T_x$ is transitive on each of the two open rays about $x$: use ordered pairs $(x, y)$ on the right and $(y, x)$ on the left. Consequently $\mathrm{Fix}(T_x) = \{x\}$.

The coordinate functions of any pair or exchange commuting with this diagonal path centralize $T$. They must be increasing by the free-centralizer argument above. In particular a decreasing exchanged pair, which could otherwise preserve the ordered half-plane, is excluded by commutation and transitivity. An increasing exchanged pair $(a(y), b(x))$ cannot preserve the half-plane: for fixed $x$, taking $y \to +\infty$ makes $a(y) > b(x)$.

For a nonexchanged increasing pair $(a(x), b(y))$, bijection of the vertical ray over $x$ onto the vertical ray over $a(x)$ gives $b(x) = a(x)$. Thus it is $(c, c)$ with $c$ centralizing $T$. The point $c(x)$ is fixed by $T_x$, so $c(x) = x$ for every $x$. The whole lift group is therefore trivial. It contains $\Delta$, which is also trivial. The source is the half-plane itself, homeomorphic to a plane by $(x, y) \mapsto (x, \log(y - x))$, and its full local centralizer is trivial.

*The band: the primitive period and the exact quotient.* Now the source path is diagonal on $x < y < x + 1$, and $T$ commutes with the unit translation $\tau(x) = x + 1$. Transitivity on the band makes $T_x$ transitive on $(x, x+1)$. Commutation with $\tau$ gives transitivity on each $(x+n, x+n+1)$, while every $x + n$ is fixed. Hence

$$\mathrm{Fix}(T_x) = x + \mathbb{Z}. \tag{C.57.7}$$

For $c$ in the full line centralizer of $T$, the point $c(x)$ belongs to (C.57.7), so $c(x) = x + n$. Both $c$ and $\tau^n$ commute with the transitive group $T$. Agreement at one point makes them equal everywhere. Thus the full line centralizer is exactly $\langle \tau \rangle$. In particular the given unit period is primitive; a smaller positive translation would contradict the transitivity of $T_x$ on $(x, x + 1)$.

The two coordinate functions of a lifted local centralizer element are therefore integer translations. A nonexchanged pair $(x+m, y+n)$ preserves the band exactly when $m = n$, giving $e^{2m}$. An exchanged pair $(y+m, x+n)$ preserves it exactly when $n = m+1$, giving $e^{2m+1}$. Decreasing coordinate functions are excluded by the same free-centralizer argument. This proves $\widetilde{\mathcal{C}}_O \subset \langle e \rangle$.

Every source deck map preserves the plane orientation induced from the orientable open surface $O$. The map $e$ reverses orientation, whereas $e^2(x, y) = (x+1, y+1)$. Thus $\Delta \subset \langle e^2 \rangle$, and it has one of the forms stated in (C.57.4). Every power of $e$ commutes with the diagonal path and with this deck group, so it descends. The reverse inclusion in (C.57.4) and the actual topological quotient follow exactly as in the plane case.

Finally the explicit coordinates

$$(u, v) = (x + y,\ y - x - \tfrac{1}{2}) \in \mathbb{R} \times (-\tfrac{1}{2}, \tfrac{1}{2}) \tag{C.57.8}$$

send $e$ to $(u, v) \mapsto (u+1, -v)$. Their inverse is $x = (u - v - \frac{1}{2})/2$, $y = (u + v + \frac{1}{2})/2$. The quotient by $\langle e \rangle$ is therefore the open Möbius band, through this exact glide-reflection quotient map. For $\Delta = \langle e^{2m} \rangle$, the source quotient identifies $u$ modulo $2m$ and is the open annulus; its local centralizer is the cyclic group of order $2m$. With $\Delta = 1$ the source is a plane and that centralizer is infinite cyclic. In both cases the final operator quotient is the same open Möbius band.

The table and the two other domains now give the last classification in the claim. All these coordinate comparisons identify existing groups, covers, and quotients. They do not identify an arbitrary intrinsic isotopy with an ambient extension at their noncompact ends. □

**C.58. [Proved] A uniform word bound for the full product affine group.** Let

$$A = \mathrm{Aff}^+(\mathbb{R}) \times \mathrm{Aff}^+(\mathbb{R}).$$

For a continuous path $k : [0, t] \to A$ with $k(0) = 1$, put $P = k([0, t]) \cup k([0, t])^{-1}$ and $g = k(t)$. Every exponential fraction satisfies the exact inclusion

$$\{\exp(r \log g) : 0 \le r \le 1\} \subseteq P^M, \qquad M = 172\,496\,624\,025\,607. \tag{C.58.1}$$

The integer is independent of the path and all scale coordinates, including zero coordinates.

In the plane-loop setting of C.54–C.56, let $U^\flat$ be the union of $\overline{U^\sharp}$ and all the remaining C.54 open orbits, whose actual generated source is $A$, and set $X^\flat = \overline{U^\flat}$. The prescribed source exponential fractions extend to a jointly continuous based contraction through homeomorphisms of $X^\flat$, with jointly continuous inverse and exact $H$-commutation. They agree with the contraction on $X^\sharp$ and retain the based exponential prescription on every nonopen orbit in $X^\flat$. The finite-representative extension of C.55 also holds relative to $X^\flat$, for representatives outside that set having compact quotients by their full local centralizers.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and uniform frontier control.

*Proof. Seven signed values in logarithmic coordinates.* For every continuous based path $\gamma : [0, t] \to \mathbb{R}^4$ and $0 \le r \le 1$, the signed-measure result [26, Theorem 4.7 and §4.3.2] gives

$$r\gamma(t) = \sum_{j=1}^{m} \varepsilon_j \gamma(s_j), \quad m \le 7, \quad \varepsilon_j \in \{1, -1\}, \quad 0 \le s_j \le t. \tag{C.58.2}$$

For a polygonal path, apply it to the four coordinate derivatives. Three circular intervals contribute six signed endpoint values, and at most one wrap contributes the total endpoint. The

required halving count is explicit: three intervals cut at four points give at most seven pieces, so one half uses at most three intervals. For the whole interval four cuts give at most five pieces, so one half uses at most two. Complementation, halving and compactness supply all fractions. Uniform polygonal approximation and compactness of seven times and their signs give (C.58.2) for continuous paths.

Apply (C.58.2) to the actual path $\log k(s)$, using the global logarithm of C.54. Pad by $\log k(0) = 0$ so that $m = 7$, and put

$$X_j = \varepsilon_j \log k(s_j), \qquad \ell_j = \exp X_j = k(s_j)^{\varepsilon_j}, \qquad w = \ell_1 \cdots \ell_7. \tag{C.58.3}$$

Thus $\sum_j X_j = r \log g$. All inverse letters are included in the signed Lie coordinates $X_j$; the equality $\exp(-X) = (\exp X)^{-1}$ is the one-parameter group identity.

*The full product error and its antisymmetric identity.* In one affine factor use logarithmic scale and group translation coordinates

$$(a, v)(b, z) = (a + b, v + e^a z), \qquad \sigma(a) = \int_0^1 e^{ua}\, du, \qquad \exp(a, u) = (a, \sigma(a)u). \tag{C.58.4}$$

The function $\sigma$ is positive and smooth, with $\sigma(0) = 1$. Write $X_j = (a_j, u_j)$ in this factor, and set $T = \sum_j a_j$ and $P_j = \sum_{\ell<j} a_\ell$. The pure translation taking $w$ to $\exp(\sum_j X_j)$ by *left* multiplication is

$$c = \sum_{j=1}^{7} F_j(a) u_j, \qquad F_j(a) = \sigma(T) - e^{P_j}\sigma(a_j), \qquad a = (a_1, \ldots, a_7). \tag{C.58.5}$$

Indeed the product translation is $\sum_j e^{P_j}\sigma(a_j)u_j$, whereas the target translation is $\sigma(T)\sum_j u_j$. The exponential telescoping identity proves

$$\sum_j a_j F_j(a) = T\sigma(T) - \sum_j e^{P_j}(e^{a_j} - 1) = 0. \tag{C.58.6}$$

For all $i, j$ define the smooth antisymmetric functions

$$f_{ij}(a) = \int_0^1 s\,(\partial_i F_j - \partial_j F_i)(sa)\, ds. \tag{C.58.7}$$

Differentiating (C.58.6) at $sa$ gives $\sum_i a_i \partial_j F_i(sa) = -F_j(sa)/s$ for $s > 0$. Consequently

$$s \sum_i a_i (\partial_i F_j - \partial_j F_i)(sa) = \frac{d}{ds}\{sF_j(sa)\}.$$

The endpoint at zero vanishes by continuity. Integration therefore gives $F_j = \sum_i a_i f_{ij}$ and

$$c = \sum_{i<j} f_{ij}(a)(a_i u_j - a_j u_i). \tag{C.58.8}$$

An explicit form, including its continuous interpretation at $a_j = 0$, is

$$\begin{aligned} f_{ij}(a) &= -\int_0^1 s e^{sP_j}\sigma(sa_j)\, ds \\ &= -\frac{\sigma(P_j + a_j) - \sigma(P_j)}{a_j} \quad (i < j), \qquad f_{ij}\big|_{a_j=0} = -\sigma'(P_j). \end{aligned} \tag{C.58.9}$$

Here the two derivatives of the common term $\sigma(T)$ cancel, and only $\partial_i(e^{P_j})$ remains. Formula (C.58.9) also verifies (C.58.8) by telescoping.

With $[x,y]=xyx^{-1}y^{-1}$, direct multiplication gives

$$[\ell_i,\ell_j]=(0,\sigma(a_i)\sigma(a_j)(a_iu_j-a_ju_i)). \tag{C.58.10}$$

Thus, putting $G_{ij}(a)=f_{ij}(a)/(\sigma(a_i)\sigma(a_j))$, the exact error in (C.58.5) is a sum of actual commutator translations multiplied by $G_{ij}$. In the product group the two factors evaluate *this same function* at their two scale vectors $a^{(1)},a^{(2)}\in\mathbb{R}^7$.

*Bounds smooth across every zero scale.* Let $R=\sum_\ell|a_\ell|$. The integral expression for $\sigma$ bounds it and its first two derivatives at $x$ by $e^{|x|}$. Each $F_j$ and each of its first and second partial derivatives therefore have absolute value at most $2e^R$. Equation (C.58.7) yields

$$|f_{ij}(a)|\le 2e^R,\qquad |\partial_\ell f_{ij}(a)|\le \tfrac{4}{3}e^R.$$

Also $\sigma(x)\ge e^{-|x|}$ and $0\le\sigma'(x)/\sigma(x)\le 1$. Since $i\ne j$, differentiation in one coordinate affects at most one of the two denominator factors. It follows that

$$|G_{ij}(a)|\le 2e^{2R},\qquad |\partial_\ell G_{ij}(a)|<4e^{2R}. \tag{C.58.11}$$

No nonzero scale assumption occurs in these estimates.

*Bounded interpolation at the two seven-dimensional nodes.* For any one of the functions $G=G_{ij}$ put

$$\begin{aligned}\mathcal{A}(a)&=\prod_{\ell=1}^{7}(e^{a_\ell}+e^{-a_\ell})^4,\\ \mathcal{B}_q(a)&=\mathcal{A}(a)\tanh a_q\quad(1\le q\le 7),\\ z_\ell&=\tanh a_\ell,\qquad \phi(z)=G(a)/\mathcal{A}(a).\end{aligned} \tag{C.58.12}$$

The map $a\mapsto z$ is a diffeomorphism onto $(-1,1)^7$. We have $\mathcal{A}\ge e^{4R}$ and $\partial_\ell\log\mathcal{A}=4\tanh a_\ell$. Equation (C.58.11) gives

$$|\phi|\le 2,\quad |\partial_{a_\ell}(G/\mathcal{A})|\le 12e^{-2R},\quad |\partial_{z_\ell}\phi|\le 12. \tag{C.58.13}$$

For the last inequality use $\operatorname{sech}^2 a_\ell\ge e^{-2|a_\ell|}$.

For unequal nodes $z^{(1)},z^{(2)}$, choose $q$ maximizing $|z_q^{(1)}-z_q^{(2)}|$, and define

$$d=\frac{\phi(z^{(1)})-\phi(z^{(2)})}{z_q^{(1)}-z_q^{(2)}},\qquad b=\phi(z^{(1)})-dz_q^{(1)}.$$

Integration along the segment between the nodes in the convex cube $(-1,1)^7$ gives $|d|\le 12\cdot 7=84$ and $|b|\le 86$. For equal nodes take $d=0$ and $b=\phi(z^{(1)})$, with any $q$. In both cases the exact two equations are

$$G(a^{(\nu)})=b\mathcal{A}(a^{(\nu)})+d\mathcal{B}_q(a^{(\nu)})\quad(\nu=1,2),\qquad |b|\le 86,\quad |d|\le 84. \tag{C.58.14}$$

In particular, independent affine scales cause no unbounded divided difference: the complete scale vectors are the nodes of the same smooth function.

*Realization by actual translation paths.* For a fixed pair $i<j$, the path

$$C_{ij}(v)=[k(vs_i)^{\varepsilon_i},k(vs_j)^{\varepsilon_j}],\qquad 0\le v\le 1, \tag{C.58.15}$$

starts at the identity and lies in the translation plane of $A$. Its values and inverses use at most four letters of $P$. Put $Z_\ell=e^{a_\ell}$ in (C.58.12). Both $\mathcal{A}$ and $\mathcal{B}_q$ are integer Laurent polynomials in seven variables. Their exponents have absolute value at most four in each variable, and both

sums of absolute coefficients are at most $2^{28}$. Indeed each factor of $\mathcal{A}$ has coefficient sum $2^4$, while the distinguished factor of $\mathcal{B}_q$ is $(Z_q - Z_q^{-1})(Z_q + Z_q^{-1})^3$ with coefficient norm at most $2^4$.

For a Laurent monomial $Z_1^{n_1} \cdots Z_7^{n_7}$ use the conjugator $h_n = \ell_1^{n_1} \cdots \ell_7^{n_7}$. It uses at most 28 letters. Conjugation by $h_n$ multiplies a translation's $\nu$th coordinate by $\exp(\sum_\ell n_\ell a_\ell^{(\nu)})$, irrespective of the conjugator's own translation. Expand either polynomial and multiply the corresponding integer numbers of conjugated copies of $C_{ij}(v)$ or its inverse. The resulting continuous based translation paths $\mathcal{A}_{ij}(v)$ and $\mathcal{B}_{ij,q}(v)$ multiply the $\nu$th coordinate of $C_{ij}(v)$ by $\mathcal{A}(a^{(\nu)})$ and $\mathcal{B}_q(a^{(\nu)})$, respectively. All their factors are translations and commute. Every value or inverse has the uniform cost

$$L = (2 \cdot 28 + 4)2^{28} = 60 \cdot 2^{28}. \tag{C.58.16}$$

The planar signed-path identity of C.37 represents a fraction of either translation endpoint by three signed values of its actual path. Use 86 copies of the fraction $|b|/86$, with inverse for negative $b$, and 84 copies for $|d|/84$. The resulting translation $\mathcal{A}_{ij}(1)^b \mathcal{B}_{ij,q}(1)^d$ has cost at most $3(86 + 84)L$ and, by (C.58.14), has exactly the two coordinate corrections $G_{ij}(a^{(\nu)})\ \mathrm{pr}_{v_\nu} C_{ij}(1)$. There are at most $\binom{7}{2} = 21$ pairs. Their translations commute, and their product followed by $w$ is exactly $\exp(\sum_j X_j) = g^r$. Thus the total cost is

$$7 + 21 \cdot 3(86 + 84) \cdot 60 \cdot 2^{28} = 172\,496\,624\,025\,607. \tag{C.58.17}$$

Zero path values pad shorter words. The cuts and interpolation choices need not vary continuously; all paths used to obtain their words are the actual continuous paths just specified. The equality is in the original group $A$, with the full product error corrected on the left.

*Exact executable checks.* The accompanying program `code/part_b/c58_affine_product_word_certificate.py` verifies the product, inverse and commutator formulas, the explicit radial coefficient and its telescoping identity, the zero-scale values, the two-node equations, the Laurent coefficient norms and the integer bound. It also checks the entire corrected product for every signed pattern of lengths one through seven in several two-factor scale regimes, retaining zero and canceling prefix scales by exact formal Laurent arithmetic. The signed-measure selection, analytic estimates and topological conclusions have the proofs in this text; the finite checks verify their computable algebra.

*Arbitrary accumulation and the remaining relative statement.* C.54 supplies the actual closed simply connected source group on each new orbit, its entire closure action, its operator transport and every global quotient equation. C.56 supplies the old contraction on $X^\sharp$. The bound in (C.58.17) exceeds all the earlier bounds. Therefore all source fractions on $U^\flat$ belong to one compact family of actual global word isotopies, obtained by replacing the length in C.50.15 by $M$. This is a finite union of continuous images of the compact cube of word parameters; maps and inverses are both included. Each endpoint equality holds in the specified simply connected source, including the based lifts in the older two-dimensional cases.

All these sources are metabelian. Their group identity passes from $U^\flat$ to $X^\flat$, so the nontrivial nonopen intrinsic images there are precisely the line translation, line affine and circle rotation images of C.37. On them use the unique line logarithm or the specified based real circle angle, and fix point orbits.

For a converging sequence of source fractions $w_i = \exp(r_i \log g_i)$, every positive integer $n$ gives the exact source residual

$$w_i^n g_i^{-\lfloor nr_i \rfloor} = \exp\big((nr_i - \lfloor nr_i \rfloor) \log g_i\big). \tag{C.58.18}$$

The residual fraction lies in $[0, 1)$ and has the same fixed bound $M$. The compact word-parameter argument of C.50 now applies without a change of topology or a new boundary prescription. On

a line the limit of the residual logarithms is bounded, and the limiting $w$ commutes with $g$, by passing the source identity to every point of the target orbit. Dividing the logarithmic residual equation by $n$ forces $\log w = r \log g$.

For completeness, the circle comparison retains the real lift. If $A_W, T, C_n$ are the real endpoint angles of the limiting word, original path and residual word, respectively, the equality is $nA_W - m_n T = C_n$, with $|m_n - nr| \le 1$ and $|C_n| \le M \max_u |\theta(u)|$. The source loop $W_i(u)^n f_{ut_i}^{-m_i} Z_i(u)^{-1}$ is null in its specified simply connected source, by (C.58.18). Evaluate it at two points converging to distinct points of the target circle. Its difference-direction winding is zero and stays zero in the limit. C.29 identifies that winding with the integer ambiguity in the displayed real angle equation. Thus $A_W = rT$, including an identity circle endpoint with nonzero specified real angle. At a fixed point every limiting word is the identity.

These are exactly the dense-set limits required by C.50's diagonal approximation on the closure. They prove joint continuity on $X^\flat$. The inverse loop has the same sources and quotient families, and its fractions are the orbitwise inverses, so it supplies a jointly continuous inverse family. The based-edge equations and operator commutation pass from the dense union to its closure. Agreement on $U^\sharp$ gives agreement on $X^\sharp$.

Finally, an outside representative in the last assertion is also outside $X^\ddagger$. C.55's direct equivariant-map evaluation and its compact fundamental set therefore give the same whole-closure limit for its intrinsic contraction. Its intersection with $X^\flat$ consists of nonopen orbits. The actual group image on that intersection is metabelian, so its hybrid target value is exactly the exponential prescription just proved. Finite closed-set pasting of both families and their inverses gives the asserted relative extension.

This includes all C.54 source groups through arbitrary accumulation. It does not establish the remaining relative extension through the noncompact local quotients outside this region, through infinitely many unrelated residual representatives, or through the exterior nonopen locus. C.12 remains open. □

**C.59. [Proved] Localized quotient-kernel maps extend over every operator translate.** Retain the plane-loop setting, $G$, $L = \overline{G}$, the original commuting group $H$, an open $G$-orbit $O$, and the full local centralizer $\mathcal{C}_O$ of C.55. Fix $x \in O$, $J = L_x$, and a compact set $D \subset O$. Let $\mathcal{V}_O[D]$ be the subgroup of $\mathrm{Homeo}(O)$ consisting of maps $a$ satisfying

$$\begin{aligned} aq = qa \quad (q \in \mathcal{C}_O), \qquad & \Phi_z^x a = \Phi_z^x \quad (z \in \mathrm{Fix}(J),\ Gz \text{ nonopen and nontrivial}), \\ & \mathrm{Mov}(a) := \{u : a(u) \ne u\} \subseteq \mathcal{C}_O D. \end{aligned} \tag{C.59.1}$$

Use its actual compact-open group topology, including inverse convergence. There is a continuous injective group homomorphism

$$\begin{aligned} E_D : \mathcal{V}_O[D] &\longrightarrow C_{\mathrm{Homeo}(\mathbb{R}^2)}(H), \\ E_D(a)|_{hO} &= hah^{-1} \quad (h \in H), \\ E_D(a)|_{\mathbb{R}^2 \setminus \bigcup_{h \in H} hO} &= 1. \end{aligned} \tag{C.59.2}$$

Restriction to $O$ is its continuous inverse onto its image. Thus every continuous compact parameter family in $\mathcal{V}_O[D]$ extends jointly with its inverse, even when $O/\mathcal{C}_O$ is noncompact and infinitely many operator translates accumulate.

The identity path component of the kernel of all the quotient equations in (C.59.1) is trivial when the global family has two distinct fiber partitions. The localized perturbations in that component can therefore contribute only in the zero-partition and one-partition cases. The assertion does not identify the entire quotient kernel with its identity path component.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and relative frontier extension.

*Proof. The actual group and the transported maps.* The set $\mathcal{C}_O D$ is closed: the proper action in C.55 has a Hausdorff quotient, the image of $D$ in it is compact and closed, and its full inverse image is $\mathcal{C}_O D$. The support condition is exactly pointwise fixing of the complement. It is closed in the local homeomorphism group, as are the centralizer and quotient conditions. They define a closed subgroup. In particular, $\mathrm{Mov}(a^{-1}) = \mathrm{Mov}(a)$. The inverse satisfies every quotient equation by composing $\Phi^x_z a = \Phi^x_z$ with $a^{-1}$.

Distinct $H$-translates of $O$ are disjoint open $G$-orbits, because each operator commutes with $G$. If $h_1 O = h_2 O$, the restriction of $h_2^{-1} h_1$ lies in the actual image $H_O \to \mathcal{C}_O$ in C.55.1. Equation (C.59.1) makes its conjugation of $a$ trivial. Thus the first rule in (C.59.2) is independent of the chosen operator. The kernel of $H_O \to \mathcal{C}_O$, which fixes $O$ pointwise, causes no additional identification. Algebraically these rules give a bijection of the plane, inverted by the same rules for $a^{-1}$. They respect products and commute with every original operator, with its original label.

*The compact-set limit at every frontier point.* Let $K$ be a compact metric parameter space and $\lambda \mapsto a_\lambda$ a continuous map $K \to \mathcal{V}_O[D]$. Write $U = \bigcup_{h \in H} hO$. Joint continuity inside $U$ follows on each of its open orbits. Consider

$$\lambda_n \to \lambda, \qquad u_n \in U, \qquad u_n \to z \notin U. \tag{C.59.3}$$

Terms fixed by $E_D(a_{\lambda_n})$ already have the required limit. For the remaining terms choose

$$u_n = h_n q_n y_n, \qquad h_n \in H, \quad q_n \in \mathcal{C}_O, \quad y_n \in D. \tag{C.59.4}$$

After a subsequence, $y_n \to y \in D$. The open orbit evaluation $L \to O$ supplies $l_n \to 1$ with $l_n y = y_n$. By C.55.3 each local $q_n$ commutes with $L|_O$, so

$$m_n = h_n q_n : O \longrightarrow \mathbb{R}^2 \quad \text{is } L\text{-equivariant}, \qquad m_n(y) = l_n^{-1} u_n \to z. \tag{C.59.5}$$

The evaluation homeomorphism C.42.1, now based at $y$, therefore gives $m_n \to \Phi^y_z$ uniformly on compact subsets of $O$. Its range is exactly $Gz$. That orbit is nonopen: an open orbit containing $z$ would be a neighborhood disjoint from every distinct open orbit in (C.59.3).

For a point orbit the limit map is constant. Otherwise it is one of the actual global quotient maps, after the exact change of source basepoint. Namely choose $l \in L$ with $lx = y$; then

$$\Phi^y_z = \Phi^x_{l^{-1} z}. \tag{C.59.6}$$

The point $l^{-1} z$ is fixed by $J$ and has the same nonopen orbit type, so (C.59.1) applies. The points $a_{\lambda_n}(y_n)$ converge to $a_\lambda(y)$ and lie in the compact image of $K \times D$. Consequently (C.59.4)–(C.59.6) imply

$$\begin{aligned} E_D(a_{\lambda_n})(u_n) &= h_n q_n a_{\lambda_n}(y_n) \\ &\longrightarrow \Phi^y_z(a_\lambda(y)) = \Phi^y_z(y) = z. \end{aligned} \tag{C.59.7}$$

Every subsequence admits the same limit argument, so this proves convergence for the entire sequence. Sequences outside $U$ are fixed. This proves joint continuity on the whole plane.

The inverse family is continuous locally and has the identical moving-point saturation by (C.59.1). The same proof gives joint continuity of its transported family. Thus the algebraic inverse above is a continuous inverse, and every transported map is a homeomorphism. Compact parameter families give continuous maps into the ambient compact-open group. For continuity of $E_D$ itself, apply this conclusion to a convergent sequence and its limit in $\mathcal{V}_O[D]$, which is a compact metrizable parameter set. Both homeomorphism groups are metrizable, so sequential continuity proves continuity. Restriction to $O$ is continuous and recovers $a$, proving the claimed topological embedding. No compactness of the entire quotient was used; only the one compact set $D$ in (C.59.4) was used.

*The exact two-partition restriction.* Let $a_s$, $0 \le s \le 1$, be a path from the identity fixing all global quotient maps, and choose two maps with distinct partitions. Retain their original source and target covers $p : \widetilde{O} \to O$ and $q_i : \mathbb{R} \to B_i$. The corresponding real lifts $F_i$ satisfy $q_i F_i = \Phi_i p$. C.45 proves that $(F_1, F_2)$ is injective on $\widetilde{O}$. Lift the isotopy $a_s p$ to the based source isotopy $\widetilde{a}_s$, with $\widetilde{a}_0 = 1$. For each $i$ the quotient equation gives

$$q_i F_i \widetilde{a}_s = q_i F_i. \tag{C.59.8}$$

For a line target this is equality of the two real maps. For a circle their difference is an integer deck translation, continuous in $(s, u)$ and zero at $s = 0$. It is therefore zero throughout. Thus both real coordinates of $\widetilde{a}_s(u)$ equal those of $u$. Pair injectivity makes $\widetilde{a}_s = 1$, and descent gives $a_s = 1$. This retains the discrete deck ambiguity for maps outside the identity path component.

For representatives outside the invariant closed set $X^\flat$, (C.59.2) fixes $X^\flat$ pointwise and gives relative perturbations there. It does not fragment the original loop: on a transitive open orbit the union of that loop's moving sets is the whole orbit, so imposing one compact quotient support on it would force a compact quotient. The new assertion supplies the exact extension for localized factors; existence and convergence of a factorization sufficient for C.12 are still unproved. □

**C.60. [Proved] All positive homotopy groups of the relative one-partition kernel vanish.** Use the original plane loop, its generated group $G$, the commuting operator group $H$, and the full global quotient families of C.53. Let $W$ be any $H$-invariant union of open $G$-orbits, each having exactly one partition in that family. Define $\mathcal{K}_W$ to be the subgroup of $C_{\mathrm{Homeo}(\mathbb{R}^2)}(H)$ consisting of maps which fix $\mathbb{R}^2 \setminus W$ pointwise, preserve each of those open orbits, and satisfy

$$\Phi_z a|_O = \Phi_z \quad \text{for every actual global quotient of } O \subset W. \tag{C.60.1}$$

Give it the induced compact-open group topology. Then

$$\pi_n(\mathcal{K}_W, 1) = 0 \qquad (n \ge 1). \tag{C.60.2}$$

More precisely, every based loop $a_t$ in this group has a jointly continuous based contraction $R_{r,t}$ in the same group, with jointly continuous inverse, and with the pointwise control

$$\begin{aligned} R_{r,t}(x) &\in \{a_s(x) : 0 \le s \le t\}, \\ R_{r,t}^{-1}(x) &\in \{a_s^{-1}(x) : 0 \le s \le t\} \qquad (0 \le r, t \le 1). \end{aligned} \tag{C.60.3}$$

Here $R_{0,t} = 1$, $R_{1,t} = a_t$, and $R_{r,0} = R_{r,1} = 1$. Every based sphere family has a based contraction with the analogous control by its entire compact family of point images and inverse images. No compactness of any local quotient, finiteness of operator-orbit representatives, or separate ambient-null hypothesis is required.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and relative one-partition frontier extension.

*Proof. The exact winding input and the source covers.* For $W = \varnothing$ the group is trivial. Otherwise choose a global quotient of one of its orbits. Its target is a nontrivial nonopen $G$-orbit, so it lies outside $W$ and contains two distinct points. Every $a_t$ fixes these two points. The difference-direction map on $\mathrm{Homeo}^+(\mathbb{R}^2)$ induces an isomorphism on fundamental groups, as proved in C.32.7. Its value on this loop is constant. Thus $a_t$ is automatically null-homotopic in the ambient plane homeomorphism group. C.32 consequently makes its restriction to every preserved open orbit null-homotopic in $\mathrm{Homeo}(O)$.

Retain the original source universal covering $p : X \to O$ and one original target covering $q$, with $qF = \Phi p$ as in C.43–C.44. The based lift $\widetilde{a}_t$ of $a_t|_O$ is a closed loop of homeomorphisms

of $X$: the restricted null-homotopy just proved lifts, so its endpoint is the identity deck map. Equation (C.60.1) gives $qF\widetilde{a}_t = qF$. For a circle target the difference between the real lifts is a continuous integer deck translation, initially zero; for a line it is zero directly. In both cases

$$F\widetilde{a}_t = F. \tag{C.60.4}$$

This retains both original covers and their labels. The construction below uses only this closed based source lift and (C.60.4) as its winding input.

*The actual affine fiber coordinates and operator equations.* Put $Q_O = \overline{\mathrm{im}(H_O \to \mathrm{Homeo}(O))}$, with its actual topology and the restriction kernel of C.30. The loop $a_t|_O$ commutes with $Q_O$. Its based lift commutes with every element of the all-lifts group $\widetilde{Q}_O$ of C.48.2: the commutator is deck-valued, continuous in $t$, and initially the identity.

Apply the established C.48 normal forms to this proper group and to the unique actual partition. The transitive base group used there is still the one supplied by the original $G$-action; it has not been replaced by the kernel loop $a_t$. The resulting exact product chart $X \cong \mathbb{R}^2$ has $F(u,v) = v$, and every lifted operator is affine in its fiber coordinate:

$$\widetilde{q}(u,v) = \big(\alpha_{\widetilde{q}}(v)u + \beta_{\widetilde{q}}(v),\, \overline{q}(v)\big), \qquad \alpha_{\widetilde{q}}(v) \neq 0. \tag{C.60.5}$$

This displayed form includes all the cases C.48.7–C.48.17: real and cyclic fiber kernels, discrete and real base images, the affine two-dimensional group, and the normalized lattice. In particular it includes every source deck map. No affine assertion is made for the original ambient coordinates.

By (C.60.4) write $\widetilde{a}_t(u,v) = (h_t(u,v), v)$. For each $t, v$ the function $h_t(\,\cdot\,, v)$ is an increasing onto line homeomorphism, and $h_0(u,v) = h_1(u,v) = u$. Define

$$\widetilde{R}_{r,t}(u,v) = \big((1-r)u + r h_t(u,v),\, v\big). \tag{C.60.6}$$

These are homeomorphisms with jointly continuous inverses by strict monotonicity, properness, and the two-point inverse bracketing argument in C.48. Their operator equations are exact. Namely commutation of $\widetilde{a}_t$ with (C.60.5) reads

$$h_t(\alpha(v)u + \beta(v), \overline{q}(v)) = \alpha(v) h_t(u,v) + \beta(v).$$

Taking the displayed convex combination with the identity proves the same equation for $\widetilde{R}_{r,t}$, including negative $\alpha$. It commutes with the original deck group and descends to $R^O_{r,t}$ on $O$. Every other global real quotient is an exact line reparametrization of $F$, by the dependent case of C.45. It too is fixed, and descent through its specified target cover proves (C.60.1).

Choose the chart on one representative of each $H$-orbit of open orbits and put $R^{hO}_{r,t} = hR^O_{r,t}h^{-1}$. The representative construction commutes with $Q_O$, hence with the actual image of $H_O$. This makes transport independent of the operator chosen, including its restriction kernel. Set $R_{r,t} = 1$ outside $W$.

*Actual prefix tracks control all frontier limits.* Fix $(u,v)$ and $t$. The real path $s \mapsto h_s(u,v)$, $0 \le s \le t$, begins at $u$ and ends at $h_t(u,v)$. Its image contains the whole closed interval between these endpoints. The first coordinate in (C.60.6) lies in that interval. After projection this proves the first inclusion in (C.60.3) on $O$.

For the inverse, write $g(u) = (1-r)u + r h_t(u,v)$ with $v$ fixed. Its strict monotonicity shows that $g^{-1}(u)$ lies between $u$ and $h_t^{-1}(u,v)$: the values of $g$ at these two endpoints bracket $u$. The continuous inverse prefix $s \mapsto h_s^{-1}(u,v)$ contains that interval. Projection proves the second inclusion in (C.60.3). Operator transport preserves both statements because the original loop commutes with $H$. No continuous selection of the prefix parameter $s$ is needed.

For any sequence $x_n \to x \notin W$, joint continuity of the original loop and compactness of its parameter interval give

$$\sup_{0\le s\le 1} |a_s(x_n) - x| \longrightarrow 0, \qquad \sup_{0\le s\le 1} |a_s^{-1}(x_n) - x| \longrightarrow 0. \tag{C.60.7}$$

Here $a_s(x) = a_s^{-1}(x) = x$ throughout. The two track inclusions therefore prove joint continuity of the constructed maps and inverses at $x$, uniformly in $r, t$, regardless of which open orbits contain $x_n$. Inside $W$ a point has its open-orbit neighborhood, where the intrinsic construction is already jointly continuous. Thus both global families are jointly continuous. They are inverse orbitwise bijections and so are homeomorphisms of the plane. They commute with every original operator, fix the exterior, preserve each orbit and every global quotient, and satisfy all based-edge equalities. This proves (C.60.2) for $n = 1$.

*Based sphere families in every higher degree.* Let $n \geq 2$ and let $a : S^n \to \mathcal{K}_W$ be continuous, with $a_{\lambda_*} = 1$. For each original source covering $p : X \to O$ the map

$$X \times S^n \longrightarrow O, \qquad (u, \lambda) \longmapsto a_\lambda(pu) \tag{C.60.8}$$

has simply connected, locally path-connected domain. The covering lifting criterion gives its unique continuous lift with value $u_*$ at $(u_*, \lambda_*)$. On $X \times \{\lambda_*\}$ that lift is the identity. Lift the inverse family from the same basepoint. Composing the two lifts gives a lift of $p$ on the connected product and agrees with the identity at the basepoint; uniqueness proves both inverse identities. Each lifted map is therefore a homeomorphism and both families are jointly continuous.

Commutation with every lifted operator and equality of the real quotient coordinate follow as before: the corresponding deck elements are continuous in the connected parameter space and trivial at $\lambda_*$. Apply (C.60.6) with $h_\lambda$ in place of $h_t$. It is a based sphere contraction, preserves all the exact operator and quotient equations, and transports over $H$ just as above.

For any $\lambda$, choose a path in $S^n$ from $\lambda_*$ to $\lambda$. Its lifted fiber evaluation contains the interval between $u$ and $h_\lambda(u, v)$. The forward interpolation and inverse bracketing consequently give

$$\begin{aligned} R_{r,\lambda}(x) &\in \{a_\mu(x) : \mu \in S^n\}, \\ R_{r,\lambda}^{-1}(x) &\in \{a_\mu^{-1}(x) : \mu \in S^n\}. \end{aligned} \tag{C.60.9}$$

No continuous choice of those parameter paths is used. Compactness of the whole sphere family replaces the time interval in (C.60.7), proving joint frontier continuity of maps and inverses uniformly in $r, \lambda$. The resulting homotopy is based at $\lambda_*$ and constant at $r = 0$, so it descends to a continuous filling on the cone $CS^n \cong D^{n+1}$. This proves (C.60.2) in all positive degrees.

The conclusion supplies based contractions of every sphere family in the relative one-partition quotient kernel, including arbitrary accumulation and noncompact local quotients. It does not supply a lift of a nontrivial deformation of the quotient actions or a decomposition of the original loop into such a lift and this kernel. No contraction of the entire topological group is asserted. Those existence requirements remain part of C.12. □

**C.61. [Proved] Compact prefix families prevent fiber intervals from returning to an open orbit.** Retain the original based plane loop $f_t$, its generated group $G$, and the closed set

$$Y = \mathbb{R}^2 \setminus \bigcup \{O : O \text{ is an open } G\text{-orbit}\}.$$

For each open orbit under consideration, choose one actual global quotient, its original universal source cover $p_O : X_O \to O$, and one real target lift $F_O : X_O \to \mathbb{R}$ as in C.43–C.44. Fix these choices on each orbit. Write $\widetilde{f}_t^O$ for the based source lift of $f_t|_O$.

Suppose $x_n \in O_n$ converges to $x \in Y$, and $s_n, t_n \in [0, 1]$. Choose any source lift $\widetilde{x}_n$ of each $x_n$ and require the exact real equality

$$F_{O_n}(\widetilde{f}_{s_n}^{O_n} \widetilde{x}_n) = F_{O_n}(\widetilde{f}_{t_n}^{O_n} \widetilde{x}_n). \tag{C.61.1}$$

Let $I_n$ be the unique compact interval in that proper real fiber joining those two lifted endpoints, and put $J_n = p_{O_n}(I_n)$. Then every compact set $K \subset \mathbb{R}^2 \setminus Y$ is eventually disjoint from $J_n$:

$$J_n \cap K = \varnothing \quad \text{for all sufficiently large } n. \tag{C.61.2}$$

In particular every finite ambient accumulation point of these interval images lies in $Y$. The statement allows arbitrary variation of the open orbit, either source covering type, and the original target deck labels. It requires equality in (C.61.1), rather than equality modulo a circle target's deck group.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and compact-strip frontier control.

*Proof.* C.43 gives connected proper real-line fibers and an actual product chart $X_O \cong \mathbb{R}^2$ in which $F_O(u,v) = v$. Every based source lift preserves this partition: its real quotient equation has the form $F_O \widetilde{f}_t^O = b_t F_O$ with $b_t$ increasing. The lift commutes with each original source deck map $\delta$. Indeed, $\widetilde{f}_t^O \delta (\widetilde{f}_t^O)^{-1} \delta^{-1}$ is a continuous deck-valued path starting at the identity, and hence is the identity. Also C.44 retains the exact equation

$$F_O \delta = F_O + \eta_O(\delta), \tag{C.61.3}$$

where the label is zero for a line target and is an integer for the specified circle cover. Thus a deck map carries the interval in (C.61.1) to the corresponding interval in its translated real fiber, with no change of endpoint ambiguity.

For any one fixed $O$, the actual family

$$\mathcal{P}_O = \{\widetilde{f}_t^O (\widetilde{f}_s^O)^{-1} : s,t \in [0,1]\} \tag{C.61.4}$$

is compact in $\mathrm{Homeo}(X_O)$, with inverse convergence. This follows from continuity of the original lifted path and multiplication, on the compact parameter square. In the product chart its elements have the form

$$g(u,v) = (h_g(u,v), b_g(v)), \tag{C.61.5}$$

where $h_g(\,\cdot\,, v)$ is increasing and onto. For a convergent sequence $g_n \to g$ in $\mathcal{P}_O$ and $v_n \to v$, these fiber maps are uniformly proper along $v_n$. Explicitly, given $M$, choose $A$ with $h_g(A,v) > M+1$. Then $h_{g_n}(A, v_n) > M$ for all sufficiently large $n$, and monotonicity gives $h_{g_n}(u_n, v_n) > M$ whenever $u_n > A$. The lower bound follows by choosing $A$ with $h_g(A,v) < -M-1$. Consequently

$$\begin{aligned} u_n \longrightarrow +\infty &\implies h_{g_n}(u_n, v_n) \longrightarrow +\infty, \\ u_n \longrightarrow -\infty &\implies h_{g_n}(u_n, v_n) \longrightarrow -\infty. \end{aligned} \tag{C.61.6}$$

These estimates use the whole compact family of lifted homeomorphisms and their actual increasing fiber maps.

Suppose (C.61.2) fails. After a subsequence there are $y_n \in J_n \cap K$ with $y_n \to y \in K$. The point $y$ lies in some open orbit $O$. Eventually $y_n \in O$, while $y_n \in O_n$. Distinct $G$-orbits are disjoint, so $O_n = O$ on this tail. All source covers and the chosen real quotient are now the same fixed maps. Pass to $s_n \to s$ and $t_n \to t$.

Choose $\widetilde{y}_n \in I_n$ over $y_n$. A covering neighborhood at $y$ supplies lifts $\widehat{y}_n \to \widehat{y}$ in one compact source chart. Choose the actual deck map $\delta_n$ with $\delta_n \widetilde{y}_n = \widehat{y}_n$. Put

$$\begin{aligned} a_n &= \delta_n \widetilde{f}_{s_n}^O \widetilde{x}_n, \\ c_n &= \delta_n \widetilde{f}_{t_n}^O \widetilde{x}_n, \qquad g_n = \widetilde{f}_{t_n}^O (\widetilde{f}_{s_n}^O)^{-1}. \end{aligned} \tag{C.61.7}$$

Commutation with the very same deck map $\delta_n$ gives the exact equality $c_n = g_n a_n$. By (C.61.3), the interval $\delta_n I_n$ joins $a_n$ to $c_n$ in one real fiber and contains $\widehat{y}_n$. Write $a_n = (u_n, v_n)$ and $\widehat{y}_n = (w_n, v_n)$. Then $v_n \to v$ and $w_n \to w$ are finite. Equation (C.61.1) also gives $b_{g_n}(v_n) = v_n$.

The first endpoints escape every compact subset of $X_O$. Indeed,

$$p_O(a_n) = f_{s_n}(x_n) \longrightarrow f_s(x).$$

The set $Y$ is $G$-invariant, so $f_s(x) \in Y$. A convergent source subsequence would instead project to a point of $O$. Since $v_n$ is bounded, this proves $|u_n| \to \infty$. Select either the positive or the negative infinite subsequence. Compactness of (C.61.4) gives $g_n \to g$, and (C.61.6) forces the first coordinate of $c_n = g_n a_n$ to escape to the same end. The real interval between these two coordinates cannot contain the bounded $w_n$. This contradicts $\widehat{y}_n \in \delta_n I_n$ and proves (C.61.2).

No uniform choice of charts on different orbits was made: an alleged open accumulation point forces a single actual orbit before any chart is used. The result does not bound accumulation along $Y$, nor exclude escape to ambient infinity. In particular it does not yet give the required control near one specified nonopen orbit or a lift of the nontrivial quotient contraction left open after C.60. □

**C.62. [Proved] Every equal-level prefix interval is contained in a compact three-word image.** Retain the original based plane loop $f_t$ and its generated group $G$. On every open orbit under consideration choose one actual global quotient, the original universal source cover $p_O : X_O \to O$, and the specified real target lift $F_O : X_O \to \mathbb{R}$ of C.43–C.44. Write $\widetilde{f}_t^O$ for the based source lift. For $x \in O$, choose $\widetilde{x} \in p_O^{-1}(x)$ and $s, t \in [0,1]$ satisfying the exact real equality

$$F_O(\widetilde{f}_s^O \widetilde{x}) = F_O(\widetilde{f}_t^O \widetilde{x}). \tag{C.62.1}$$

Let $I$ be the compact interval in that proper real fiber joining the two displayed points. Then

$$\begin{aligned} I &\subset \{\widetilde{f}_u^O (\widetilde{f}_v^O)^{-1} \widetilde{f}_a^O \widetilde{x} : 0 \le a \le v \le u \le 1\}, \\ p_O(I) &\subset \mathcal{W}_3(x), \qquad \mathcal{W}_3(x) = \{f_u f_v^{-1} f_a(x) : 0 \le a \le v \le u \le 1\}. \end{aligned} \tag{C.62.2}$$

The same assertion holds for the original inverse path, with projected word $f_u^{-1} f_v f_a^{-1}$.

Consequently, let $x_n \in O_n$ tend to any $x \in \mathbb{R}^2$, allow the open orbit and the chosen actual quotient to vary, and let $J_n = p_{O_n}(I_n)$ be any intervals obtained from (C.62.1). These intervals remain in one compact subset of the plane and satisfy

$$\sup_{y \in J_n} \operatorname{dist}(y, \mathcal{W}_3(x)) \longrightarrow 0. \tag{C.62.3}$$

Every accumulation point therefore lies in the same $G$-orbit as $x$. In particular, at a point fixed by $G$,

$$\sup_{y \in J_n} |y - x| \longrightarrow 0. \tag{C.62.4}$$

All source and target covering labels in these statements are the original ones. In particular, (C.62.1) cannot be replaced by equality modulo the deck group of a circle target.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and full prefix-fiber interval control.

*Proof. The actual source chart and its two sides.* C.43 provides a homeomorphism

$$\chi : X_O \longrightarrow \mathbb{R}^2, \qquad F_O = \operatorname{pr}_2 \chi. \tag{C.62.5}$$

We use this exact chart for the proof and retain $p_O$. Thus a real fiber is a complete horizontal line, and its two complementary components are precisely its strict sublevel and superlevel sets. Every original source lift preserves the fiber partition. Its induced real target map is increasing, because it is joined to the identity by the based lifted path. The following argument uses these actual homeomorphisms and their actual inverse maps; it does not assume that the auxiliary product flow from C.43 commutes with them.

Swap $s, t$ if necessary so that $s \le t$, and put

$$\gamma(r) = \widetilde{f}_r^{O}\widetilde{x}, \qquad c = F_O\gamma(s) = F_O\gamma(t), \qquad Z = \{r \in [s,t] : F_O\gamma(r) = c\}. \tag{C.62.6}$$

The set $Z$ is compact and contains both endpoints. Every complementary interval $(a, b)$ has endpoints in $Z$, and the continuous function $F_O\gamma(r) - c$ has one strict sign throughout it.

*A single excursion is filled by an actual word triangle.* Fix such an interval $(a, b)$. Suppress the superscript $O$ and define

$$A = \widetilde{f}_a\widetilde{x}, \qquad h_r = \widetilde{f}_r\widetilde{f}_a^{-1} \quad (a \le r \le b), \qquad g = h_b, \qquad B = gA. \tag{C.62.7}$$

In particular $h_a = \mathrm{id}$, $A, B \in F_O^{-1}(c)$, and $h_r A = \gamma(r)$. Write $\beta_r$ for the increasing target map induced by $h_r$. Suppose first that the excursion is positive. Then

$$\beta_r(c) > c \quad (a < r < b), \qquad \beta_r^{-1}(c) < c, \qquad \beta_b(c) = c. \tag{C.62.8}$$

Since $\beta_b$ is increasing and fixes $c$, the two paths

$$r \longmapsto h_r A, \qquad r \longmapsto g h_r^{-1} A \tag{C.62.9}$$

lie strictly on opposite sides of the fiber for $a < r < b$. The first goes from $A$ to $B$ and the second from $B$ to $A$. A negative excursion reverses their two sides and changes nothing else in the argument.

For completeness, these side conditions give a precise winding calculation even when either path has self-intersections. Let $q$ be any point in the open fiber interval between $A$ and $B$, with $A \neq B$. In the chart (C.62.5), each path can be homotoped, relative to its endpoints, to a two-segment path in its own strict half-plane: use straight interpolation with the polygonal path whose middle vertex is directly above or below the midpoint of $A, B$, at unit distance from the fiber on the specified side. Throughout the interpolation all interior parameter values stay strictly on that side and hence avoid $q$; the endpoints also avoid $q$. The resulting two polygonal paths form the boundary of a diamond containing $q$. Their concatenation therefore has winding 1 or $-1$ about $q$. This proves the same nonzero winding for (C.62.9), without imposing simplicity on the original paths.

Now take the closed triangle

$$T_{a,b} = \{(u, v) : a \le v \le u \le b\}$$

and the continuous map

$$\begin{aligned} \Psi(u, v) &= h_u h_v^{-1} A \\ &= \widetilde{f}_u\widetilde{f}_v^{-1}\widetilde{f}_a\widetilde{x}. \end{aligned} \tag{C.62.10}$$

On its edge $v = a$ this is the first path in (C.62.9); on its edge $u = b$ it is the second path, in the order specified there; and on the diagonal $u = v$ it is the constant $A$. Thus its boundary has the nonzero winding just computed about every such $q$. If $q$ were missing from $\Psi(T_{a,b})$, this very triangle would null-homotope that boundary in $X_O \setminus \{q\}$, contradicting the winding. The endpoints are already images of vertices. When $A = B$ the interval is a singleton and the conclusion is immediate. We have proved in all cases that

$$[\gamma(a), \gamma(b)]_{F_O = c} \subset \Psi(T_{a,b}). \tag{C.62.11}$$

The equality in (C.62.10) follows by cancellation of the displayed original lifts, so its three letters have exactly the labels claimed in (C.62.2).

*The full zero set introduces no missing fiber intervals.* Write $\chi\gamma(r) = (\xi(r), \zeta(r))$. On $Z$ retain the real function $\xi|_Z$. On every gap $(a, b)$ of $Z$, extend it by the linear function joining $\xi(a)$ to $\xi(b)$; denote the resulting function on $[s, t]$ by $e$. This extension is continuous, including at accumulation points of gaps. Indeed, on a gap of length less than $\delta$, uniform continuity of $\xi$ bounds the difference between $e(r)$ and $\xi(r)$ by its modulus of continuity at $\delta$. There are only finitely many gaps of length at least $\delta$, and the linear extension is continuous at both endpoints of each of those gaps. Letting $\delta$ decrease proves the assertion at every point of $Z$.

The continuous image $e([s, t])$ contains the entire real interval between $\xi(s)$ and $\xi(t)$. At a parameter in $Z$ its value gives the original point $\gamma(r)$ on the fiber. On a gap its value lies between the two endpoint fiber coordinates, and so lies in the interval already covered by (C.62.11). Consequently the full interval $I$ is covered by the zero-level track and these excursion intervals. A zero-level track point is itself a three-letter value, using $a = v = u = r$. All excursion triangles use $0 \le a \le v \le u \le 1$. This proves the first inclusion in (C.62.2).

*Projection and the original deck equations.* For each source lift we have $p_O \widetilde{f}_r^O = f_r p_O$, and the same equation holds for its inverse. Projecting (C.62.10) therefore gives exactly $f_u f_v^{-1} f_a(x)$, proving the second inclusion in (C.62.2). No sheet selection enters this comparison. More explicitly, each original deck map $\delta$ commutes with every $\widetilde{f}_r^O$, because the deck-valued commutator path starts at the identity. C.44 supplies the retained target equation

$$F_O \delta = F_O + \eta_O(\delta). \tag{C.62.12}$$

Replacing $\widetilde{x}$ by $\delta\widetilde{x}$ therefore carries the entire interval and every word value to their $\delta$-images, with the same translated real level. Projection removes that same deck map. Equality only modulo a circle period would not give the single horizontal fiber or the two sides used in (C.62.8), which explains the exact hypothesis (C.62.1). Applying the already proved argument to the actual continuous path $(\widetilde{f}_r^O)^{-1}$ gives the stated inverse-path inclusion and its projected word $f_u^{-1} f_v f_a^{-1}$.

*Uniform ambient control, across arbitrary orbits.* Let

$$\Delta_3 = \{(a, v, u) \in [0, 1]^3 : a \le v \le u\}, \qquad W((a, v, u), y) = f_u f_v^{-1} f_a(y). \tag{C.62.13}$$

This is one jointly continuous global map, independent of all orbit charts and source covers. The set $E = \{x\} \cup \{x_n : n \ge 1\}$ is compact. Hence $W(\Delta_3 \times E)$ is compact and contains every $J_n$ by (C.62.2). Continuity on this compact product also gives

$$\sup_{\lambda \in \Delta_3} |W(\lambda, x_n) - W(\lambda, x)| \longrightarrow 0. \tag{C.62.14}$$

The distance in (C.62.3) is bounded by this quantity. Every value $W(\lambda, x)$ belongs to $Gx$, so (C.62.3) confines all accumulation to that same orbit and excludes ambient escape. At a $G$-fixed point, every such value is $x$; this proves (C.62.4). The identical compactness argument applies to the inverse-path words.

This strengthens C.61 to compact containment near the specified nonopen orbit and proves the full interval estimate at fixed frontier points. At a nonfixed orbit, the compact set $\mathcal{W}_3(x)$ may contain more than one point. The statement does not yet select the specified median or exponential boundary value, nor construct a jointly continuous lift of the moving quotient deformation. Those two remaining requirements are retained in C.12. □

**C.63. [Proved] A controlled moving-base lift on principal orbits with fixed frontier.** Let $f_t$ be the original based ambient-null plane loop in the centralizer of an arbitrary operator group $H$, and let $G$ be the group generated by its values. Let $W$ be an $H$-invariant union of open $G$-orbits. For every $O \subset W$ suppose that the full local centralizer

$$C_O = C_{\mathrm{Homeo}(O)}(G|_O)$$

has dimension one, with its actual compact-open topology. Suppose also that every point of $\partial W$ is fixed by $G$. Then the actual global loop

$$\widehat{f}_t(x) = \begin{cases} f_t(x), & x \in W, \\ x, & x \notin W \end{cases} \tag{C.63.1}$$

has a based null-homotopy in the $H$-centralizer, fixing the complement of $W$ throughout. Both this homotopy and its inverse are jointly continuous across every accumulation of open orbits.

On each $O$, the induced contraction of the actual principal quotient $O/C_O^0$ is the median contraction of its original based real target lift. The original source cover, every deck label, and the exact $H_O$ restriction kernel are retained. No nontrivial global quotient to a nonopen orbit is required, so the assertion includes the empty global target family. In particular it contracts the original loop when $\mathbb{R}^2 \setminus W$ is fixed by $G$.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and controlled moving-base extension at fixed frontiers.

*Proof. The actual principal coordinates and source labels.* C.30 and C.33 make $C_O$ a free proper Lie transformation group. Put $A = C_O^0$. Their actual principal bundle and section give

$$\Theta : A \times B \xrightarrow{\cong} O, \qquad \Theta(a, b) = a\, s(b), \qquad B = O/A, \tag{C.63.2}$$

where each of $A, B$ is a line or a circle. The pair of circles is excluded because $O$ is noncompact. Let $\eta_A : \mathbb{R} \to A$ and $\eta_B : \mathbb{R} \to B$ be the identity on a line factor and the specified quotient by $\mathbb{Z}$ on a circle factor. The map

$$p_*(u, v) = \Theta(\eta_A(u), \eta_B(v)) \tag{C.63.3}$$

is a universal covering. With basepoints fixed, its unique covering isomorphism with the original $p_O : X_O \to O$ is a homeomorphism $\chi_O : X_O \to \mathbb{R}^2$ satisfying $p_O = p_* \chi_O$. The inverse is the based lift in the opposite direction; uniqueness of based lifts proves both inverse identities. Hence

$$F_O = \mathrm{pr}_2\, \chi_O, \qquad \chi_O \delta \chi_O^{-1}(u, v) = (u + m_A(\delta), v + m_B(\delta)) \tag{C.63.4}$$

for each original deck map $\delta$. A line factor has zero label, and a circle factor has the original integer label under this based covering comparison. At most one factor is a circle.

Every element of $C_O$ normalizes its identity component $A$ and preserves its orbit partition. C.33 proves that $C_O/A$ acts freely and properly on $B$, so no induced base map is decreasing: a decreasing line or circle homeomorphism has a fixed point. Retain the complete affine fiber changes constructed in C.33. On the original universal cover, the full group of lifts of $C_O$ is generated by all fiber translations and, when its real base image is nontrivial, one map of the form

$$T_c(u, v) = (u + c, v), \qquad q(u, v) = (\lambda u + \kappa, v + d), \quad d > 0, \quad \lambda \neq 0. \tag{C.63.5}$$

This also includes the source deck group. For a circle base its lift labels supply the base period; for a circle fiber its integer deck maps are among the $T_c$. The constant $\kappa$ retains the choice of component representative and lift. C.33's fiber gauge removes the nonconstant translation term; a different original lift only changes the constant term. The negative values of $\lambda$ remain allowed. If the base image is trivial, only $T_c$ occurs.

The based source lift of the original loop commutes with every original lift of an operator in $C_O$. Indeed its commutator with such a lift is deck-valued and is initially the identity. Commutation with all $T_c$ gives the exact expression

$$\widetilde{f}_t^O(u, v) = (u + \phi_t^O(v), \beta_t^O(v)), \tag{C.63.6}$$

where $\beta_t^O$ is increasing. Here and below this formula is written through $\chi_O$. By C.32 the ambient-null loop restricts to an ambient-null loop on $O$, so its based source lift closes. Thus

$$\phi_0^O = \phi_1^O = 0, \qquad \beta_0^O = \beta_1^O = \mathrm{id}\,. \tag{C.63.7}$$

When (C.63.5) has the additional generator, its exact commutation equations are

$$\phi_t^O(v+d) = \lambda\phi_t^O(v), \qquad \beta_t^O(v+d) = \beta_t^O(v) + d. \tag{C.63.8}$$

The constant $\kappa$ cancels from this comparison, rather than being discarded from the operator group.

The base maps $\beta_t^O$ have no common fixed point. A common fixed point would project to a point of $B$ fixed by every original generator, contradicting the transitivity of $G$ on $O$ and hence on $B$. For any $a \in \mathbb{R}$ and any $\delta > 0$, the compact increment family

$$\{\beta_s^O(\beta_t^O)^{-1} : |s-t| \le \delta\} \tag{C.63.9}$$

sends $a$ to points on both sides of $a$. To check this for arbitrarily short increments, the path $t \mapsto (\beta_t^O)^{-1}(a)$ is nonconstant. It therefore has unequal values at arbitrarily close times $s, t$; otherwise it is locally constant and then constant on the connected parameter interval. The corresponding increment moves $a$, and its inverse moves $a$ to the opposite side. The parameter set $|s-t| \le \delta$ is convex, contains the diagonal, and gives a path-connected increment family. Its evaluation image consequently contains a neighborhood of $a$. This uses inverse evaluation at the specified point, not a displacement at an unrelated moving point.

*One compact family of global maps and original lifts.* There are at most countably many open orbits, since each contains a member of a fixed countable basis disjoint from all the others. Consider the product of the global plane homeomorphism group and the homeomorphism groups of all the selected original source covers. In this product take the closed subgroup $\mathcal{E}$ of compatible tuples

$$(k, (\widetilde{k}^O)_{O\subset W}), \qquad p_O\widetilde{k}^O = kp_O, \tag{C.63.10}$$

which commute globally with $H$ and in each cover with every retained lift of $C_O$. All these conditions are closed pointwise equalities, including the inverse equalities. Compatibility implies $k(O) = O$, using the surjectivity of $p_O$ and the inverse tuple. Alternatively each original word preserves the closed complement of every $O$; limits of both maps and inverses preserve that complement too. Thus no limiting homeomorphism sends an open orbit into its frontier.

Each factor is Polish in its compact-open homeomorphism topology, with inverses controlled on compact sets. The countable product and its closed subgroup $\mathcal{E}$ are Polish. Choose a complete compatible metric on $\mathcal{E}$ controlling these maps and inverses. For example, after choosing a complete compatible metric, add its value on the inverse pair; the resulting metric is still complete and compatible. The original path

$$L_t = (f_t, (\widetilde{f}_t^O)_O) \tag{C.63.11}$$

is a continuous based loop in this exact group.

We construct positive numbers $\delta_n$ so that all the products

$$L_{s_n}L_{t_n}^{-1}\cdots L_{s_1}L_{t_1}^{-1}L_r, \qquad |s_j - t_j| \le \delta_j, \quad r \in [0,1], \tag{C.63.12}$$

converge uniformly over their entire parameter product as $n \to \infty$. Suppose the first $n-1$ choices have been made; their product image is compact. Continuity of left multiplication, uniform on this compact image, gives an identity neighborhood whose multiplication changes every such product by less than $2^{-n}$ in the chosen metric. Uniform continuity of $(s,t) \mapsto L_sL_t^{-1}$ near its compact diagonal gives $\delta_n$ placing all the next increments in that neighborhood. The successive

product maps are therefore uniformly Cauchy. Completeness gives their uniform limit in $\mathcal{E}$, including every global inverse and every original lifted inverse.

Put

$$T = [0,1] \times \prod_{n\geq 1} \{(s_n, t_n) \in [0,1]^2 : |s_n - t_n| \leq \delta_n\}. \tag{C.63.13}$$

This is compact and path connected; coordinatewise straight segments give paths with their exact labels. The uniform limit just constructed is a continuous map $T \to \mathcal{E}$. Denote its labelled image by $\mathcal{K}$. Let $\mathcal{K}_n$ be the family of products through the first $n$ increments, with $\mathcal{K}_0 = \{L_r\}$. Setting all later increments to the identity shows $\mathcal{K}_n \subset \mathcal{K}$ and $\mathcal{K}_n \subset \mathcal{K}_{n+1}$. Every global value of $\mathcal{K}$ commutes with $H$ and fixes every point fixed by $G$. The source labels in $\mathcal{K}$ are part of the construction even when distinct parameters have the same global homeomorphism.

The C.62 triangle works for any path in such a labelled family. Its proof uses the relative maps $h_r = k_r k_a^{-1}$, so it does not require the selected path itself to start at the identity. Connect any two chosen labels by a path in $T$. Their equal-real-level fiber interval is contained in the corresponding image of

$$\mathcal{K}\mathcal{K}^{-1}\mathcal{K}. \tag{C.63.14}$$

The projected family in (C.63.14) is compact in the global plane homeomorphism group. The same argument for the inverse labelled family uses $\mathcal{K}^{-1}\mathcal{K}\mathcal{K}^{-1}$. No continuous choice of connecting paths or endpoint labels is needed for these uniform containments.

*Enlarging the allowed fiber intervals.* Fix one source orbit and suppress $O$ from the notation. Let $B_{\rho,t}$ be C.27's median of the real path $\beta_t$, using prefix length $\rho$. Thus $B_{0,t} = \mathrm{id}$, $B_{1,t} = \beta_t$, and $B_{\rho,0} = B_{\rho,1} = \mathrm{id}$. For every $v$, its value lies between $v$ and $\beta_t(v)$, since every running minimum and maximum contains the initial value $v$ between them. Hence some original prefix time between 0 and $t$ attains $B_{\rho,t}(v)$.

Each labelled source map $k \in \mathcal{K}$ has the form

$$k(u,v) = (u + c_k(v), b_k(v)), \tag{C.63.15}$$

with its jointly continuous real functions in the label and $v$. For $p = (\rho, t, v)$ define the nonempty compact interval

$$A_n(p) = \mathrm{conv}\{c_k(v) : k \in \mathcal{K}_n,\ b_k(v) = B_{\rho,t}(v)\}. \tag{C.63.16}$$

Nonemptiness follows from the preceding prefix observation, and compactness follows from the closed equality in the compact labelled parameter set. The identical definition with $\mathcal{K}$ gives a compact interval $A_\infty(p)$ containing every $A_n(p)$.

Here is the precise lower-limit property used for selection:

$$A_n(p) \subseteq \liminf_{p' \to p} A_{n+1}(p'). \tag{C.63.17}$$

First take an endpoint value $c_k(v)$ before convexification, with its fixed label $k \in \mathcal{K}_n$ and $b_k(v) = a = B_{\rho,t}(v)$. At nearby parameters $p'$, both $b_k(v')$ and $B_{\rho',t'}(v')$ tend to $a$. By (C.63.9), increments with arbitrarily small time differences move $a$ to both sides of itself. Choose two such increments within the $(n+1)$st allowed family, as close to the identity as desired. Their strict inequalities continue to bracket the desired target at all sufficiently nearby $p'$. A path in that small increment family and the intermediate value theorem therefore give an increment $e_{p'}$ with

$$b_{e_{p'}k}(v') = B_{\rho',t'}(v'). \tag{C.63.18}$$

As the chosen increment neighborhood decreases to the identity, its effect on the fiber coordinate also tends uniformly to zero near the specified source point. Thus the value $c_k(v)$ is approximated by allowed values of $\mathcal{K}_{n+1}$ at every nearby parameter. For a point between two allowed endpoint

values, perform this construction for both endpoints and take the same convex combination. This proves (C.63.17). It does not assert lower semicontinuity of $A_n$ itself.

The operator equations are retained by doing the selection in their actual line bundle. If (C.63.5) has a base generator, all members of $\mathcal{K}$ satisfy

$$c_k(v+d) = \lambda c_k(v), \qquad b_k(v+d) = b_k(v) + d. \tag{C.63.19}$$

The median has the same base equation. Thus the intervals (C.63.16) define convex fiber sets in the real line bundle

$$(v, z) \sim (v + d, \lambda z) \quad \text{over } \mathbb{R}/d\mathbb{Z}, \tag{C.63.20}$$

pulled back over the parameter square. Its continuous fiber norm is $\|[v, z]\| = |\lambda|^{-v/d}|z|$. For trivial base image use the ordinary real line bundle over $[0, 1]^2 \times \mathbb{R}$. Both bases are metrizable and paracompact. Negative $\lambda$ gives the corresponding nonorientable line bundle and preserves its convex fiber sets just as in (C.63.19).

*The relative continuous-selection construction.* On the closed union $D$ of the four boundary faces of the parameter square prescribe

$$c(0, t, v) = 0, \quad c(1, t, v) = \phi_t(v), \quad c(\rho, 0, v) = c(\rho, 1, v) = 0. \tag{C.63.21}$$

This is a continuous section on $D$, including the corners by (C.63.7), and belongs to every allowed interval there. Replace $A_n$ by this prescribed singleton on $D$ and leave it unchanged elsewhere; denote the new fiber sets by $F_n$. Property (C.63.17) survives. At a point of $D$ the required inclusion for approaching points off $D$ is the already proved inclusion for its prescribed value, and for approaching points in $D$ it is continuity of (C.63.21). Away from $D$ a neighborhood avoids $D$.

We give the elementary approximation proof needed for these interlaced intervals. It also applies in the line bundle (C.63.20). Choose positive $\epsilon_n = 2^{-n}$. At each parameter point choose one value of $F_0$ and extend it as a constant local section in a bundle chart. By the lower-limit inclusion, a sufficiently small neighborhood makes this local section have distance less than $\epsilon_1$ from $F_1$. A locally finite partition of unity subordinate to these neighborhoods combines the sections to a continuous section $a_1$ with that same distance bound. Indeed, at a given point the corresponding nearby elements of $F_1$ have their convex combination in $F_1$, so the norm of the error is bounded by the same weighted error bound.

Suppose $a_n$ has distance less than $\epsilon_n$ from $F_n$. At each parameter point choose a value in $F_n$ that close to $a_n$, and extend it in a local bundle chart. Shrink its neighborhood so that the extended section remains within $2\epsilon_n$ of $a_n$ and within $\epsilon_{n+1}$ of $F_{n+1}$; the latter requirement is exactly the lower-limit inclusion. Another locally finite partition of unity gives $a_{n+1}$ with

$$\|a_{n+1} - a_n\| < 2\epsilon_n, \qquad \operatorname{dist}(a_{n+1}, F_{n+1}) < \epsilon_{n+1}. \tag{C.63.22}$$

The summable first bounds give a limit section $c$, uniformly in the fiber norm and hence locally uniformly in every bundle chart. It is continuous. At each fixed parameter, every $F_n$ lies in the closed compact interval $A_\infty$, and the second bound tends to zero; therefore $c \in A_\infty$ at that parameter. On $D$ the sets $F_n$ are the same prescribed singleton, so $c$ equals (C.63.21) exactly. This proves the required relative selection without assuming that the original prefix hulls themselves admit a continuous selection.

*The intrinsic homeomorphisms and all operator equations.* In the original real source chart set

$$K^O_{\rho,t}(u, v) = (u + c(\rho, t, v), B_{\rho,t}(v)). \tag{C.63.23}$$

This is a jointly continuous family of homeomorphisms, with inverse

$$(u, w) \longmapsto (u - c(\rho, t, B^{-1}_{\rho,t}(w)), B^{-1}_{\rho,t}(w)). \tag{C.63.24}$$

The inverse family is jointly continuous because the base median has a jointly continuous inverse. The selected section of (C.63.20) satisfies the first equation of (C.63.19), and the base median satisfies the second. Equations (C.63.5) therefore commute exactly with (C.63.23); all fiber translations commute as well. Consequently every original source deck map commutes, and (C.63.23) descends through $p_O$. It commutes with the entire $C_O$ downstairs. Equations (C.63.21) give the based null-homotopy endpoint conditions.

Choose this construction on one representative of each $H$-orbit of open components in $W$ and transport it by conjugation. The exact restriction $H_O \to C_O$ has the kernel specified in C.30.4. Commutation with its image makes transport independent of the chosen representative operator; its kernel acts identically on $O$. Conjugation carries $C_O^0$ to $C_{hO}^0$ and hence transports the actual principal fibers. Original based source lifts intertwine under a lift of $h$ by uniqueness at time zero. Their finite products and the labelled limits above intertwine too. Thus the original covering labels are retained under this transport.

*The compact-family bounds and fixed-frontier pasting.* At every source point, membership $c(\rho,t,v) \in A_\infty(\rho,t,v)$ places the value (C.63.23) in the real fiber interval between two labelled $\mathcal{K}$-values at that same target level. The endpoints exist because the defining level set in the compact label space is compact. By (C.63.14), after projection the value belongs to

$$\mathcal{K}\mathcal{K}^{-1}\mathcal{K}(x). \tag{C.63.25}$$

For the inverse, put $y = (K^O_{\rho,t})^{-1}x$. The point $x$ is between two labelled values $k_1 y, k_2 y$ in one real fiber. Since both fiber maps are increasing, $y$ lies between $k_1^{-1}x$ and $k_2^{-1}x$ in their common real input fiber. The inverse version of (C.63.14) therefore gives

$$(K^O_{\rho,t})^{-1}(x) \in \mathcal{K}^{-1}\mathcal{K}\mathcal{K}^{-1}(x). \tag{C.63.26}$$

These are inclusions in one compact family of actual global maps, independent of the source orbit, the selected section, and all homotopy parameters. They survive $H$ transport because every global member of $\mathcal{K}$ commutes with $H$.

Define the global $K_{\rho,t}$ by these transported maps on $W$ and the identity outside $W$. It is a bijection: each open orbit is preserved and the complement is fixed. Let $x_n \to z \in \partial W$ and allow the source orbit and both parameters to vary arbitrarily. Every global map in either compact family of (C.63.25)–(C.63.26) fixes $z$, because $z$ is $G$-fixed. Joint continuity of the compact global evaluation family makes all its values at $x_n$ tend uniformly to $z$. The two inclusions thus prove continuity of both the pasted map and its inverse at every such limit. Interior continuity was already proved, and points of the exterior are fixed. Hence the global family consists of homeomorphisms and is jointly continuous with its inverse. It commutes with $H$ by the transported construction and the $H$-invariance of $W$. Its four endpoint faces are precisely those required in (C.63.1).

This establishes an actual moving-base lift and global contraction at fixed frontiers, including arbitrarily many open components and their accumulation. At a nonfixed frontier orbit, a compact global family need not have one evaluation value; the argument does not select the prescribed contraction there. Open components with zero-dimensional full local centralizer are also outside the stated principal-fiber assertion. These remaining cases of C.12 are retained. □

**C.64. [Proved] Compact local control and equivariant disk localization at a fixed frontier.** Let $f_t$ and $G$ be the original based ambient-null plane loop and its generated group, and let $H$ commute with every $f_t$. Let $W$ be an $H$-invariant union of open $G$-orbits. There is a compact path-connected labelled family $\mathcal{K}$ of actual global $H$-centralizing homeomorphisms, containing the identity, such that every member preserves each original open orbit and fixes the $G$-fixed set, and

$$x \in \mathrm{int}_O\big(\mathcal{K}\mathcal{K}^{-1}(x)\big) \qquad (x \in O \subset W). \tag{C.64.1}$$

All original source-cover labels are retained in this family.

Now suppose that $\partial W$ is $G$-fixed and that the full local centralizer on every $O \subset W$ has dimension zero. One can choose arbitrarily small closed embedded disks $D_i \subset O_i$, with points $x_i \in D_i$, for which

$$D_i \subset \mathcal{K}\mathcal{K}^{-1}(x_i), \tag{C.64.2}$$

and every effective stabilizer translate of $D_i$ is disjoint from it. Take any countable such family, with the $D_i$ representing pairwise distinct $H$-orbits of disks, whose distinct $H$-translates are pairwise disjoint and locally finite in $W$. There is an actual continuous group embedding, with continuous inverse on its image,

$$\mathfrak{L} : \prod_i \mathrm{Homeo}(D_i \text{ rel } \partial D_i) \longrightarrow C_{\mathrm{Homeo}(\mathbb{R}^2)}(H), \tag{C.64.3}$$

obtained by conjugating each disk map over its $H$-translates and using the identity elsewhere. Its image is a contractible subgroup. Both the maps and inverses are jointly continuous, including at every accumulation on $\partial W$, uniformly over this whole product of disk maps. In particular every based loop in this subgroup has an explicitly prescribed based contraction in the same global centralizer.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and controlled localization at fixed frontiers.

*Proof. The exact path group.* Use every original universal cover $p_O : X_O \to O$ of the selected open orbits, with the original based lifted loop. The open orbits are countable, by a countable basis of the plane. The closed compatible tuple construction (C.63.10) therefore gives a Polish group of global homeomorphisms and their labelled original lifts. It imposes $p_O \widetilde{k}^O = k p_O$, the inverse equations, global commutation with $H$, and commutation with the retained lifts of the full local centralizers. This construction uses no assumption on their dimension. The original loop belongs to this group: its commutators with fixed operator lifts are deck-valued and initially trivial, and its based lifts close by C.32. Let $L$ be the closure of the subgroup generated by that labelled loop. Its global values preserve every selected open orbit and fix every $G$-fixed point, since these are closed conditions for maps and inverses.

Write $\mathscr{P}(L)$ for the group of continuous paths $\sigma : [0,1] \to L$ with $\sigma(0) = 1$, with pointwise multiplication and the compact-open path topology. A complete compatible metric on $L$ gives a complete uniform path metric. Uniform limits are continuous and based. The space is separable: uniform-continuity scales and finitely many values in a countable dense set give a countable neighborhood coding; choose one actual path from each nonempty code. Multiplication and inversion are continuous uniformly on compact parameter sets. Thus $\mathscr{P}(L)$ is a Polish group.

For $x \in O$, the continuous action by endpoint evaluation is transitive. A finite original word carrying $x$ to a given point of $O$ is the endpoint of a based word path: replace every original time $t_j$ by $st_j$, preserving the order and inverse sign of every letter. These are paths in $L$, with their original source lifts. The precise orbit-open-map theorem used in C.30, [24, Theorem 3.5], consequently makes

$$e_x : \mathscr{P}(L) \longrightarrow O, \qquad e_x(\sigma) = \sigma(1)x \tag{C.64.4}$$

an open surjection. Its source is Polish and its target is the nonmeagre Hausdorff surface $O$, as required there.

*Compact lifting of a compact target.* We give the needed elementary consequence of this open map. Let $p : X \to Y$ be a continuous open surjection from a completely metrizable space to a metric space. For an open $V \subset X$ and a compact $D \subset p(V)$, there is a compact $S \subset V$ with $p(S) = D$.

Here is the full construction, including the equality of images. For each $y \in D$ choose a lift in $V$, and an open metric ball about it whose closure lies in $V$. Images of these balls cover $D$.

A finite subcover and a closed shrinking give compact subsets $D_i$ covering $D$, each contained in the image of its chosen ball $U_i$. A closed shrinking is obtained directly by taking sufficiently small metric neighborhoods in the finite cover of the compact space and then their closures in $D$. Make each ball have diameter at most $1/2$, and each $D_i$ have diameter at most $1/2$, refining the finite cover when needed.

Repeat inside every such pair $(U_i, D_i)$. Lift every point of $D_i$ inside $U_i$; choose child balls with closures in $U_i$ and diameter at most $2^{-2}$; use their open images and a finite closed shrinking to cover $D_i$ by compact child sets contained in those images and of diameter at most $2^{-2}$. Continue with bound $2^{-n}$ at level $n$. Discard empty child sets. This is a finitely branching tree. Along each infinite branch the closed source balls are nested and have diameters tending to zero. Completeness gives one source limit, lying in the initial $V$. The nested nonempty compact target sets similarly give one target limit in $D$. At level $n$ that target limit has a lift in the corresponding source ball, because the whole target set lies in its image. These lifts converge to the source limit. Continuity of $p$ identifies its image with the target limit.

The infinite-branch space is compact, and the source-limit map from it is continuous by the source diameter bounds. Its image $S$ is therefore compact and maps into $D$. Every $y \in D$ belongs to a nested sequence of child sets, so some branch limit maps to $y$. This proves $p(S) = D$, with no continuous endpoint section assumed.

*One compact family, over all selected orbits.* Fix a complete compatible metric on $L$, and use its uniform path metric on $\mathscr{P}(L)$. Choose a compact exhaustion $E_n$ of the open set $W$, whose interiors cover $W$. Let $V_n$ be the open neighborhood of the constant identity path consisting of paths uniformly within $2^{-n}$ of it. For each $x \in E_n$, openness of (C.64.4) supplies a small closed coordinate disk $D_{n,x}$ in $e_x(V_n)$ with $x$ in its interior. These interiors cover $E_n$; retain finitely many, with centers denoted $x_{n,j}$. The preceding compact lifting construction gives compact sets $S_{n,j} \subset V_n$ with

$$\{\sigma(1)x_{n,j} : \sigma \in S_{n,j}\} = D_{n,j}. \tag{C.64.5}$$

Take all labelled values $\sigma(s)$, for $0 \le s \le 1$ and $\sigma$ in these compact sets, and add the identity. Their union $\mathcal{K}$ is compact in $L$: for each $n$ there are finitely many compact value families, and all families with $n \to \infty$ converge uniformly to the identity. The same observation proves sequential compactness directly in the chosen metric. It is path connected, since every one of its values is connected to the identity by the corresponding initial segment of its path, entirely in $\mathcal{K}$. Compactness and these paths hold for the actual labelled tuples, including all source lifts and inverse maps.

For $x$ in the interior of $D_{n,j}$ choose an endpoint $k_x \in \mathcal{K}$ with $x = k_x x_{n,j}$. Every other point of that disk is $k_y x_{n,j}$ for some $k_y \in \mathcal{K}$. Consequently

$$D_{n,j} \subset \mathcal{K}\mathcal{K}^{-1}(x). \tag{C.64.6}$$

Their interiors cover $W$, proving (C.64.1). The projection of $\mathcal{K}$ to global plane homeomorphisms is compact and retains global commutation with $H$ and the fixed set. We use the same notation for that projected family in the pointwise estimates below.

*Disks and the exact operator ambiguity.* For an open orbit with zero-dimensional full local centralizer, C.30 and C.33 make that centralizer a discrete Lie group acting freely and properly. The effective image

$$\overline{H}_O = \{h|_O : h \in H_O\}$$

is a subgroup of this discrete group, and hence is closed and acts freely and properly. Its actual kernel is exactly $\{h \in H_O : h|_O = \mathrm{id}\}$, as in (C.30.4). For any $x \in O$, properness and freeness give a neighborhood whose closure is disjoint from every nonidentity effective translate. To see the shrinking explicitly, start with a compact neighborhood. Properness permits only finitely many translates meeting it. None of the nonidentity elements in this finite set fixes $x$; disjoint

neighborhoods of their distinct images allow a further shrinking with disjoint closures. Intersect also with the interior neighborhood in (C.64.1). A sufficiently small closed coordinate disk $D$ about $x$ then satisfies (C.64.2) and the disjointness assertion.

Use any family specified in the claim. Distinct representatives are taken modulo their whole $H$-orbits. An operator carrying one representative disk to itself stabilizes its source orbit; the preceding disjointness forces its effective restriction to be the identity. Thus it fixes that disk pointwise. For $\psi = (\psi_i)_i$ define

$$\mathfrak{L}(\psi)|_{hD_i} = h\psi_i h^{-1}, \qquad \mathfrak{L}(\psi)(x) = x \quad \text{off the union of the disk interiors.} \tag{C.64.7}$$

Changing $h$ without changing its disk changes it by an operator acting identically on the representative orbit. Formula (C.64.7) is therefore unambiguous. All disk boundaries are fixed; different disk interiors are disjoint. The formula is a bijection with the identical formula for $\psi_i^{-1}$ as inverse, and it commutes with every element of $H$.

*Uniform control at every fixed frontier.* Put

$$Q = \mathcal{K}\mathcal{K}^{-1}, \qquad P = Q^2 = \mathcal{K}\mathcal{K}^{-1}\mathcal{K}\mathcal{K}^{-1}. \tag{C.64.8}$$

These are compact global families, $Q = Q^{-1}$, and every member fixes the $G$-fixed set and commutes with $H$. For $x, y \in D_i$ choose $q_x, q_y \in Q$ with $x = q_x x_i$ and $y = q_y x_i$. Then $y = q_y q_x^{-1} x \in P(x)$. Commutation with $H$ gives the same containment for every translated disk. Hence, including the fixed exterior points by $1 \in P$,

$$\mathfrak{L}(\psi)(x) \in P(x), \qquad \mathfrak{L}(\psi)^{-1}(x) \in P(x) \tag{C.64.9}$$

for all disk maps $\psi$ and all $x \in \mathbb{R}^2$.

Let $x_m \to z \in \partial W$, allowing the disk index, operator translate and every disk map to vary arbitrarily. Compactness of $P$ and joint continuity of evaluation give

$$\sup_{p \in P} |p(x_m) - z| \longrightarrow 0, \tag{C.64.10}$$

because $p(z) = z$ for every $p \in P$. Equations (C.64.9)–(C.64.10) prove joint continuity of the pasted maps and inverses at such points, uniformly over the entire product. At an interior point of $W$, the stipulated local finiteness leaves finitely many disk maps in a neighborhood. Their ordinary disk evaluation and inverse evaluation are jointly continuous, and they agree with the identity on their boundaries. The same assertions hold at exterior interior points, where all maps are the identity. Thus (C.64.7) gives actual global homeomorphisms and joint continuity of both families.

Since the plane is locally compact, joint continuity on a compact set, followed by a finite product-neighborhood subcover, gives continuity into its compact-open map space. Apply this also to inverses. This proves continuity of (C.64.3) for the product topology in its source and the actual homeomorphism topology in its target. Composition is componentwise on every disk, so it is a homomorphism. Restriction to the representative disks is its continuous inverse on the image, and in particular it is injective.

*The actual contraction.* Choose once an actual disk chart $\theta_i : \overline{\mathbb{D}} \to D_i$. On the closed unit disk, for $a \in \mathrm{Homeo}(\overline{\mathbb{D}} \,\mathrm{rel}\, S^1)$ set

$$\mathcal{A}_s(a)(z) = \begin{cases} s\, a(z/s), & |z| \le s, \quad s > 0, \\ z, & |z| \ge s, \end{cases} \qquad \mathcal{A}_0(a) = \mathrm{id}\,. \tag{C.64.11}$$

The two formulas agree on $|z| = s$ because $a$ fixes $S^1$. Its inverse is $\mathcal{A}_s(a^{-1})$. At $s = 0$ the displacement of either map is at most $2s$, uniformly in $a$; at positive $s$ ordinary joint evaluation

proves continuity, also across the joining circle. Therefore (C.64.11) is a continuous contraction of the disk homeomorphism group, with $\mathcal{A}_1(a) = a$ and $\mathcal{A}_s(\mathrm{id}) = \mathrm{id}$. Conjugate this formula by every $\theta_i$, apply it in each factor with the same $s$, and then apply $\mathfrak{L}$. Continuity in the product topology and the proved continuity of $\mathfrak{L}$ give the stated contraction of its image. For a based loop the endpoint identities are preserved by $\mathcal{A}_s(\mathrm{id}) = \mathrm{id}$. Formula (C.64.9) controls the whole contraction and its inverse at every frontier.

This proves actual equivariant disk localization and a contractible global subgroup, with the original stabilizer kernel retained. No factorization of an arbitrary original loop into this subgroup is proved here. Such a controlled factorization or a simultaneous graph deformation, and extension at nonfixed frontier points, remain in C.12. □

**C.65. [Proved] Principal-kernel boundary comparison and controlled planar fillings.** Let $f_t, G$ be the original plane loop and its generated group, and let $O$ be an open $G$-orbit with one-dimensional full local centralizer. Retain the actual principal quotient of C.33,

$$\pi_0 : O \longrightarrow B_0 = O/C_O^0, \qquad \beta : G \longrightarrow \mathrm{Homeo}(B_0). \tag{C.65.1}$$

Let $N$ consist of endpoints $h_1$ of paths

$$\begin{gathered} h_s = \prod_{j=1}^{m} f_{a_j(s)}^{\epsilon_j}, \qquad a_j : [0,1] \to [0,1] \text{ continuous}, \\ \epsilon_j \in \{1,-1\}, \qquad h_0 = \mathrm{id}, \qquad \beta(h_s) = \mathrm{id} \text{ for every } s. \end{gathered} \tag{C.65.2}$$

The finite length and signs may depend on the path. For every nontrivial nonopen $G$-orbit $B \subset \partial O$ whose actual intrinsic restriction image $\rho_B(G)$ is nonmetabelian,

$$\rho_B(N) = 1, \qquad \rho_B : G \longrightarrow \mathrm{Homeo}(B). \tag{C.65.3}$$

In particular two actual finite-word paths $p_s, q_s$ starting at the identity and satisfying $\beta(p_s) = \beta(q_s)$ for every $s$ have exactly the same restriction to each such $B$ at every time.

For the full principal kernel $K_0 = \ker \beta$, the image $D_B = \rho_B(K_0)$ is central in $\rho_B(G)$ and is a closed discrete cyclic subgroup of the intrinsic homeomorphism group of $B$; it is finite when $B$ is a circle. Give

$$E_B = \{(\beta(g), \rho_B(g)) : g \in G\} \subset \mathrm{Homeo}(B_0) \times \mathrm{Homeo}(B)$$

the actual induced product topology, and put $\Delta_B = \{1\} \times D_B$. There are exact maps

$$E_B \longrightarrow E_B/\Delta_B \longrightarrow \beta(G).$$

The first map is a covering with deck group $\Delta_B$; the second is a continuous bijective homomorphism onto the actual induced image $\beta(G)$. No continuity of its inverse is asserted.

There is also the following controlled filling fact for the same planar setting. For any open planar set $V$, any open Euclidean disk $D$, and any connected component $U$ of $V \cap D$, the actual inclusion induces an injection

$$\pi_1(U, x) \longrightarrow \pi_1(V, x) \qquad (x \in U). \tag{C.65.4}$$

Thus a based loop in $U$ null-homotopic in $V$ has a based disk filling entirely in $U$. Applied to shrinking null evaluation loops near a fixed frontier point, this gives fillings inside the original orbit with vanishing ambient diameter.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison, nonfixed-frontier path comparison and controlled planar fillings.

*Proof. The actual kernel paths and their intrinsic topology.* Every $g \in G$ commutes with $C_O^0$, by the definition of the full local centralizer. It therefore induces the actual base homeomorphism in (C.65.1). This defines a homomorphism, with no change of quotient. The set $N$ in (C.65.2) is a subgroup: products and inverses of two displayed paths are again finite-word paths in the exact kernel, and identity letters can be inserted if needed. Conjugation by a fixed original word preserves that kernel and adds only finitely many constant letters. Hence $N$ is normal in $G$. Every intermediate $h_s$ also belongs to $N$, by reparametrizing the path on the interval from 0 to $s$.

We verify the topology used on $B$. On a circular orbit the intrinsic and embedded circle topologies agree. On an intrinsic line orbit, C.28 puts every fixed finite-word parameter track in a compact intrinsic interval, embedded as a compact arc in the plane. More generally, a compact intrinsic interval $J \subset B$ lies in one of C.28's finite-word compact sets: choose its two endpoints there, and use the unique connecting arc in that set. Applying a word of fixed length to every point of $J$ remains in one larger finite-word compact set, hence in one compact embedded intrinsic interval. The ambient joint evaluation map therefore gives joint intrinsic evaluation on the path-parameter interval times $J$. The same argument applies to the inverse word. Consequently each path in (C.65.2) restricts to a continuous path in the actual intrinsic homeomorphism group of $B$, with inverse convergence on compact intervals. All original generators preserve the intrinsic orientation, since their paths start at the identity; the same is true of every word.

It follows in particular that $\rho_B(N)$ is path connected in that intrinsic group: the whole restricting path stays in $\rho_B(N)$ by the preceding reparametrization. This argument uses the exact finite-word paths. It makes no assertion about intrinsic convergence for an arbitrary globally convergent sequence of unbounded words.

*The actual boundary image of the kernel is abelian.* C.33 supplies a principal product chart $O \cong A \times B_0$, where $A = C_O^0$ is the additive line or circle. An element of $\ker \beta$ commutes with all translations by $A$, so its restriction has exactly the form

$$(a, b) \longmapsto (a + \varphi(b), b). \tag{C.65.5}$$

Two such restrictions commute, including when $A$ is a circle. Their actual global commutator is therefore identity on $O$. Continuity makes it identity on $\overline{O}$, and hence on $B$. Thus $\rho_B(N)$ is abelian. It is normal in $\rho_B(G)$ by the already proved normality of $N$. No abstract source kernel has been identified with a boundary kernel; the commutator equality on $O$ and its extension to $\overline{O}$ are the exact comparison.

*A nontrivial path-connected abelian normal image forces a metabelian action.* Put $T = \rho_B(N)$ and $K = \rho_B(G)$, and suppose $T \neq 1$. The common fixed set of $T$ is $K$-invariant because $T$ is normal. Transitivity of $K$ on $B$ makes this fixed set either all of $B$ or empty. It cannot be all of $B$ when $T \neq 1$, so it is empty.

Each $T$-orbit is path connected and nontrivial. A nonconstant path in a line or circle contains a nonempty open arc. Thus each orbit contains such an arc. If an interior point of that arc is the translate of the starting point by an element of $T$, its inverse carries the arc to a neighborhood of the starting point within the same orbit. Hence every $T$-orbit is open. The disjoint open orbits partition the connected one-manifold $B$, so $T$ is transitive.

An abelian transitive group acts freely: an element fixing one point fixes every translate of that point. Let $\overline{T}$ be the closure in the actual intrinsic homeomorphism group. It is still abelian, connected and transitive, and the same argument makes it free. Its full centralizer is exactly $\overline{T}$: a commuting map agrees at one point with some member of the transitive group, and that agreement propagates to every point.

The evaluation argument of C.30, applied in dimension one exactly as in C.48, therefore identifies $\overline{T}$ homeomorphically with $B$ and makes it locally compact. Its faithful product action on $B \times \mathbb{R}^2$ is a continuous action on a connected three-manifold. The Hilbert–Smith input already verified in C.33 makes it a Lie group. It is connected of dimension one, hence is the additive line

or circle. Evaluation is the actual conjugacy of its action with all translations or all rotations, respectively.

Every element of $K$ normalizes this closure: it normalizes $T$, and conjugation is continuous in the intrinsic homeomorphism-group topology. On a line, write the retained translations as $T_a$. For a normalizing homeomorphism $g$,

$$gT_ag^{-1} = T_{\theta_g(a)} \tag{C.65.6}$$

defines a continuous additive automorphism of $\mathbb{R}$, so $\theta_g(a) = \lambda a$. Its orientation gives $\lambda > 0$. Evaluation at zero yields $g(a) = \lambda a + g(0)$. Thus $K$ is a subgroup of the positive affine group. On a circle the same evaluation identity identifies the induced continuous circle-group automorphism with the degree of $g$. That degree is $+1$, so the automorphism is the identity and $g$ is a rotation. In either case $K$ is metabelian. This contradicts the hypothesis in (C.65.3). We conclude that $T = 1$, which proves that equation.

For the claimed path comparison take $h_s = p_s q_s^{-1}$. It is a finite-word path starting at identity and satisfies $\beta(h_s) = \mathrm{id}$ as an equality of whole base homeomorphisms. Each $h_s$ belongs to $N$; applying (C.65.3) gives $\rho_B(p_s) = \rho_B(q_s)$ for every $s$. Equality only at one varying base point would not place this path in $\ker\beta$ and is not used here.

*The full kernel and its exact discrete ambiguity.* Let $k \in K_0$ and $g \in G$. Express $g$ by one finite original word and replace each of its time parameters $a_j$ by $sa_j$. This gives an actual finite-word path $g_s$ from identity to $g$. The path

$$[g_s, k] = g_s k g_s^{-1} k^{-1} \tag{C.65.10}$$

starts at identity, is again a single finite-word path, and has principal-base image identity at every time because $\beta(k) = 1$. Thus every value of (C.65.10) lies in $N$. Equation (C.65.3) gives $[\rho_B(g), \rho_B(k)] = 1$. Consequently $D_B$ is central in $\rho_B(G)$ and lies in its full intrinsic centralizer.

The action $\rho_B(G)$ is transitive and nonmetabelian, hence nonabelian. C.38 identifies that full centralizer, in its actual intrinsic compact-open topology, as a closed discrete group which is trivial or infinite cyclic on a line, and finite cyclic on a circle. Every subgroup of this closed discrete group is closed in the intrinsic homeomorphism group. This proves the asserted closedness, discreteness, and cyclic description of $D_B$. It does not require a topology on the abstract kernel $K_0$.

Projection onto the first coordinate of $E_B$ is a continuous surjective homomorphism

$$p_B : E_B \longrightarrow \beta(G), \qquad \ker p_B = \Delta_B. \tag{C.65.11}$$

Here both image topologies are the induced topologies specified in the statement. Since $D_B$ is closed and discrete in the second factor, $\Delta_B$ is closed and discrete in $E_B$. The preceding centralization makes it central in $E_B$.

Write $q_B : E_B \to E_B/\Delta_B$ for the topological group quotient. This map is open: the inverse image of the image of an open set $V$ is the open set $V\Delta_B$. Choose an identity neighborhood $U$ in $E_B$ with $U^{-1}U \cap \Delta_B = \{1\}$, using discreteness and continuity of multiplication and inversion. The sets $U\delta$, $\delta \in \Delta_B$, are pairwise disjoint, and

$$q_B^{-1}(q_B(U)) = \coprod_{\delta \in \Delta_B} U\delta. \tag{C.65.12}$$

The restriction of $q_B$ to each displayed open set is a homeomorphism onto $q_B(U)$: it is continuous and bijective, and its openness follows from the just proved openness of the quotient map. Translates of these neighborhoods prove the covering assertion.

The space $E_B$ is path connected. Indeed the finite-word path $g_s$ used above is continuous in both displayed image groups. For the frontier factor this was proved with the compact intrinsic

tracks. For the principal factor use the actual local sections of $\pi_0$ in the product chart: evaluation on such a section, followed by $\pi_0$, is jointly continuous on compact base sets, and the inverse word gives the same fact for inverses. This proves the required compact-open path continuity in Homeo$(B_0)$. A deck transformation of a connected covering is determined by its value at one point: the agreement set of two deck transformations is open and closed, by evenly covered neighborhoods. Right translation by $\Delta_B$ is transitive on each fiber. It follows that the entire deck group of $q_B$ is exactly $\Delta_B$.

Finally (C.65.11) induces the continuous bijective homomorphism $\overline{p}_B : E_B/\Delta_B \to \beta(G)$. Continuity is exactly the universal property of the quotient topology. The topological comparison stops at this continuous bijection: no open-map or inverse-continuity assertion for $\overline{p}_B$ has been used or proved.

*A filling inside the original domain and a prescribed ambient disk.* Let $\gamma$ be a based loop in $U$ which is null-homotopic in $V$. Its compact image has an open neighborhood in $U$. A sufficiently fine polygonal approximation $\gamma'$ and the straight homotopies on the small parameter intervals stay there, fixing the basepoint. Choose finitely many vertices generically if necessary, so that its image is a finite planar graph.

Take a small compact connected planar regular neighborhood $M$ of this graph inside $U$. For example, use small vertex disks and thin edge rectangles and then round their corners. Its boundary has one outer Jordan curve and finitely many inner Jordan curves. Write $D_i$ for its bounded complementary Jordan disks. The whole closed outer Jordan disk lies in $D$. Indeed its bounded interior lies in the convex hull of its boundary: a point outside that convex hull is separated by a line and can be joined to infinity without meeting the boundary. The convex hull of a compact subset of the open Euclidean disk $D$ remains inside $D$. In particular every $D_i$ lies in $D$.

Fill into $M$ precisely the inner disks $D_i$ which are entirely contained in $V$, and call the resulting compact planar surface $M'$. Their boundaries already lie in $M$, so the entire closed disk is added. We have the actual inclusions

$$\gamma' \subset M \subset M' \subset V \cap D. \tag{C.65.7}$$

The set $M'$ is connected and meets $U$, hence lies in that same component. For each remaining inner disk choose one point $z_i \in D_i \setminus V$.

The actual inclusion

$$M' \hookrightarrow \mathbb{R}^2 \setminus \{z_i\} \tag{C.65.8}$$

is a homotopy equivalence with an explicit strong deformation retraction. For each remaining closed inner Jordan disk choose a Schoenflies disk coordinate taking $z_i$ to zero and its boundary to the unit circle. On the punctured disk use

$$(r, \theta) \longmapsto ((1 - \tau)r + \tau, \theta), \qquad 0 < r \leq 1. \tag{C.65.9}$$

On the exterior of the outer Jordan disk use its exterior Schoenflies coordinate and the same formula for $r \geq 1$. All these homotopies fix their boundary circles pointwise. With the identity on $M'$, they paste on the finite closed cover of the punctured plane to a strong deformation retraction onto $M'$. Every basepoint in $M'$ stays fixed.

Since each $z_i$ lies outside $V$, the given null-homotopy of $\gamma'$ in $V$ is a null-homotopy in the punctured plane of (C.65.8). Applying the displayed retraction gives a based filling in $M'$. Prepending the based polygonal approximation homotopy gives a filling of $\gamma$ in $U$. This proves (C.65.4).

For original ambient-null evaluation loops converging to a $G$-fixed frontier point, compactness of the original time interval makes their entire tracks shrink to that point. Choose open Euclidean disks containing those tracks with radii tending to zero. Each track lies in one

connected component of its original orbit intersected with that disk, and C.32 supplies its null-homotopy in the original orbit. The just proved injection gives a filling in that same component, with the stated vanishing ambient diameter.

The first part compares specified finite-word kernel paths and records the central discrete ambiguity of the full kernel through an actual covering. It does not construct a new base lift or identify the final continuous bijection with a homeomorphism. It also does not factor arbitrary source germs through a frontier action. The second part chooses each filling separately; it does not give a continuous choice of fillings, simultaneous configuration or edge isotopy extension, or fragmentation of an arbitrary original loop. These remaining requirements of C.12 are retained. □

**C.66. [Proved] Zero-dimensional frontiers give a global centralizer filling.** Let $f_t$ be a based ambient-null loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let $G$ be the group generated by its values, and let $H$ be any group of plane homeomorphisms commuting with every $f_t$. Let $Y$ be the closed union of the nonopen $G$-orbits. When $Y$ is zero-dimensional, the original loop has a based disk filling in $C_{\mathrm{Homeo}(\mathbb{R}^2)}(H)$. Every map of the filling fixes $Y$ pointwise. This includes the empty frontier and imposes no discreteness or local compactness assumption on $H$.

The topological comparison used here is the following actual isomorphism. For every nonempty compact zero-dimensional set $Z \subset S^2$, put $M = S^2 \setminus Z$. With compact-open group topologies, including inverse convergence, restriction is an isomorphism of topological groups

$$R_Z : \mathrm{Homeo}(S^2; Z) \xrightarrow{\cong} \mathrm{Homeo}(M), \qquad \mathrm{Homeo}(S^2; Z) = \{a : a(Z) = Z\}. \tag{C.66.1}$$

Its inverse is the unique end extension. A connected parameter family containing the identity extends to a family fixing every point of $Z$.

Consequently, for a connected admissible surface of C.1, an ambient-null loop in the centralizer of an arbitrary specified operator action is null-homotopic in that same centralizer whenever the nonopen locus of its based plane lift is zero-dimensional. All original operator and deck labels are retained.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison for the zero-dimensional nonopen locus.

*Proof. Finite disjoint Jordan covers of the deleted set.* We first prove the planar compactification statement with all the topology needed for parameter families. End extension for zero-dimensional deleted sets is also used in [4, Proposition 1.5 and Appendix A.1]; the following construction supplies the actual group topology comparison.

Let $Z \subset S^2$ be compact and zero-dimensional. It is a proper subset. Choose a stereographic pole outside $Z$, so $Z$ becomes a compact subset of the coordinate plane. Given an open neighborhood $V$ of $Z$ and $\varepsilon > 0$, there is a finite family of closed Jordan disks $D_1, \dots, D_r$ such that

$$\begin{gathered} D_i \cap D_j = \varnothing \quad (i \neq j), \qquad Z \subset \bigcup_j \operatorname{int} D_j, \qquad D_j \subset V, \qquad D_j \cap Z \neq \varnothing, \\ \operatorname{diam}_{S^2} D_j < \varepsilon. \end{gathered} \tag{C.66.2}$$

Here and below a diameter refers to one fixed compatible metric on the compact sphere.

To construct the disks, partition $Z$ into finitely many disjoint compact relatively clopen sets of arbitrarily small coordinate diameter. Such a partition exists by the clopen basis and compactness: take a finite clopen cover of the required mesh and disjointize it by successive differences. Distinct nonempty pieces have positive distance. Cover them by a finite sufficiently fine collection of coordinate grid cells and take a small polygonal regular neighborhood of that

union. One obtains the neighborhood by separating corner contacts in tiny vertex neighborhoods and rounding the remaining boundary corners. Choose the cell mesh and rounding size below the separation of the clopen pieces. Each connected component of this regular neighborhood then lies in a small neighborhood of one partition piece. Delete any component not meeting $Z$. Its other components are compact connected planar surfaces, with finitely many disjoint Jordan boundary curves, and they still contain $Z$ in their interiors.

Fill each component up to its outer Jordan curve. The resulting disk lies in the convex hull of that component: a point outside the convex hull is separated by a line and belongs to the unbounded component of the complement of its outer curve. Its coordinate diameter is therefore bounded by that of the original component. The outer curves of different components are disjoint. Their closed Jordan disks are either disjoint or nested; keep only the maximal disks. Their closures are pairwise disjoint, and they cover all the original components and hence $Z$. The partition mesh and neighborhood sizes were arbitrary. Choose them small enough that each retained disk, which contains a point of $Z$, lies in $V$ and has spherical diameter below $\varepsilon$. This proves (C.66.2), without a local-connectivity assumption on $Z$.

The complement of $Z$ is path connected. For two points outside $Z$, apply (C.66.2) with a neighborhood avoiding both points. The complement of the interiors of the finitely many disjoint disks is a connected sphere with finitely many disks removed, and it contains both points. A path in this complement avoids $Z$. The same argument inside a Jordan disk $D$ with $\partial D \cap Z = \varnothing$ shows that $\operatorname{int} D \setminus Z$ is path connected: cover the compact set $Z \cap D$ by finitely many smaller disks in its interior, avoiding the two chosen points, and join them in the remaining finitely connected disk.

*The exact end compactification.* Apply (C.66.2) successively to obtain disk families $D_{n,j}$ of mesh less than $2^{-n}$, with every disk at stage $n+1$ contained in the interior of one disk at stage $n$. Every disk meets $Z$. Set

$$K_n = S^2 \setminus \bigcup_j \operatorname{int} D_{n,j}, \qquad U_{n,j} = \operatorname{int} D_{n,j} \setminus Z. \tag{C.66.3}$$

The sets $K_n$ are compact connected planar surfaces contained in $M$, satisfy $K_n \subset \operatorname{int} K_{n+1}$, and exhaust $M$. Indeed each disk meets $Z$ and its diameter tends to zero, so every point outside $Z$ eventually lies outside all the disks. The preceding connectivity argument shows that the $U_{n,j}$ are exactly the components of $M \setminus K_n$. Each is non-relatively-compact in $M$, since its disk meets $Z$.

An end is a nested compatible choice of these components. The corresponding nested closed disks have diameters tending to zero. Their intersection is one point of $Z$: each contains a point of $Z$, and compactness and nesting give an intersection with $Z$. Conversely every point of $Z$ determines one such choice. Thus the map

$$\Xi : \widehat{M} \longrightarrow S^2, \qquad \Xi|_M = \mathrm{id}, \qquad \Xi((U_{n,j(n)})_n) = \bigcap_n D_{n,j(n)} \tag{C.66.4}$$

is a bijection from the end compactification. A basic end neighborhood consisting of $U_{n,j}$ and the ends it contains maps to $\operatorname{int} D_{n,j}$. These disks form a neighborhood basis at their limiting point, by their vanishing diameters. Equation (C.66.4) and its inverse are therefore continuous, and the topology on $M$ is unchanged.

Every $a \in \mathrm{Homeo}(M)$ is proper. Given a target $K_m$, the compact set $a^{-1}(K_m)$ lies inside some $K_n$. The connected set $a(U_{n,j})$ then lies in a unique component of $M \setminus K_m$. Following a nested sequence of source components defines its image end. The same construction for $a^{-1}$ gives the inverse map, and the basic neighborhoods just described prove continuity. This is the unique extension $\widehat{a}$ to $S^2$: $M$ is dense because a zero-dimensional subset has empty interior. The component construction, or uniqueness on this dense set, gives $\widehat{ab} = \widehat{a}\,\widehat{b}$ and $\widehat{a^{-1}} = (\widehat{a})^{-1}$. The extension preserves $Z$ setwise. Conversely a sphere homeomorphism preserving $Z$ restricts to $M$ and is its unique extension. This proves the algebraic bijection in (C.66.1).

*The compact-open and inverse topology.* Restriction in (C.66.1) is continuous on compact subsets of $M$, for maps and inverses. We prove continuity of extension at the identity; the group homomorphism then gives continuity everywhere. Both groups are metrizable, so it suffices to take a sequence $a_i \to \mathrm{id}$ in the actual homeomorphism group of $M$, including $a_i^{-1} \to \mathrm{id}$ on compact sets.

Fix $\varepsilon > 0$ and choose $m$ so all disks $D_{m,j}$ have diameter below $\varepsilon$. The set

$$K_m \cup \bigcup_i a_i^{-1}(K_m) \tag{C.66.5}$$

is compact in $M$. To see this exactly, the convergent sequence $\{\mathrm{id}, a_i^{-1}\}$ is a compact parameter space, and its evaluation on the compact set $K_m$ is a compact image. Choose $N \geq m$ with (C.66.5) contained in $K_N$. Every $a_i(U_{N,j})$ consequently avoids $K_m$. It is connected and hence lies in one component of $M \setminus K_m$.

Choose a point $p_j$ in each of the finitely many $U_{N,j}$. This set lies inside one parent $U_{m,\ell(j)}$. Since $a_i(p_j) \to p_j$, for all sufficiently large $i$ the component containing $a_i(U_{N,j})$ is exactly that parent. The end-extension construction then also sends every end in $D_{N,j}$ into $D_{m,\ell(j)}$. Thus each point of $\operatorname{int} D_{N,j}$, including its deleted points, and its image under $\widehat{a}_i$ lie in the same disk $D_{m,\ell(j)}$ of diameter below $\varepsilon$. On the remaining compact set $K_N$, the original maps $a_i$ converge uniformly to identity. It follows that $\widehat{a}_i \to \mathrm{id}$ uniformly on the whole sphere. Apply the identical argument to the sequence $a_i^{-1}$ to obtain uniform inverse convergence. This proves (C.66.1) as an isomorphism of the stated topological groups.

In particular every continuous parameter family extends jointly with its inverse. When its parameter space is connected and contains a parameter at which the map is identity, evaluation at any $z \in Z$ is a continuous map into the zero-dimensional space $Z$ with connected image. That image is the singleton $\{z\}$. The entire family fixes every deleted point individually. This is the pointwise conclusion used below; the full group in (C.66.1) preserves $Z$ only setwise.

*Application to the original plane loop.* The group $G$ is path connected: any finite word in loop values is joined to identity by scaling each of its time parameters to zero. Its orbits are therefore path connected. A nonopen orbit contained in the zero-dimensional set $Y$ must be a point, so $G$ fixes $Y$ pointwise. Let

$$Z = Y \cup \{\infty\} \subset S^2. \tag{C.66.6}$$

This is compact and zero-dimensional. For completeness, a closed zero-dimensional subset of the plane is locally compact and has a basis of compact relatively open-and-closed sets: inside a compact neighborhood choose a clopen basis element with closure still in that neighborhood. Finite unions of such sets give an increasing compact clopen exhaustion of $Y$. Their complements in $Z$ form a clopen basis at the point at infinity. The old clopen basis at each finite point is unchanged. The empty $Y$ gives the singleton $Z = \{\infty\}$.

The complement $O = \mathbb{R}^2 \setminus Y = S^2 \setminus Z$ is connected by the disk-cover argument. It is a union of disjoint open $G$-orbits. Connectedness makes it one such orbit. Every $h \in H$ transports $G$-orbits to $G$-orbits with their actual labels, since it commutes with $G$. It preserves $Y$ and the unique open orbit $O$.

Let $P$ be the actual restriction image of $H \to \mathrm{Homeo}(O)$, retaining its kernel $\{h : h|_O = 1\}$, and let $Q$ be the closure of $P$ in that homeomorphism group. C.30 makes the full local centralizer of $G|_O$ a closed locally compact group acting freely and properly on $O$. The closed subgroup $Q$ of this full centralizer has these same properties. The restricted original loop commutes with $Q$.

C.32 proves that this restriction is null-homotopic in $\mathrm{Homeo}(O)$, using the specified ambient-null input. C.33 therefore gives an actual based disk filling

$$A : D^2 \longrightarrow C_{\mathrm{Homeo}(O)}(Q), \qquad A|_{\partial D^2} = f_t|_O. \tag{C.66.7}$$

Apply the inverse of (C.66.1) to this whole family. The disk is connected and contains the based identity parameter, so every extended map fixes all of $Z$, including the specified point $\infty$. Restricting these sphere maps to the original plane gives a jointly continuous family of plane homeomorphisms and inverses, fixing $Y$. On $O$ it commutes with each restriction of each original $h \in H$. The two compositions are continuous on the plane and agree on the dense set $O$, so they agree everywhere. The boundary family is the original $f_t$ by the same density argument. This proves the claimed filling with every operator label retained.

*The actual lift and descent for an admissible surface.* Let $p : \mathbb{R}^2 \to U$ be the universal covering of a connected admissible surface, and start with a based centralizer loop which is null-homotopic in its ambient homeomorphism group. Lift that ambient disk family after composition with $p$, choosing the identity at its based parameter. The covering lifting criterion and the corresponding inverse family give actual plane homeomorphisms. Its boundary is a closed ambient-null based plane loop $\widetilde{f}_t$.

Take $H$ to consist of every original lift of every specified surface operator, including all deck transformations. For each such lift $h$, the maps $\widetilde{f}_t h$ and $h\widetilde{f}_t$ lift the same surface family and agree at the based parameter; uniqueness of the covering lift makes them equal for every $t$. Under the zero-dimensional locus assumption the plane result supplies a disk family $K_w$ commuting with every one of these original lifts. Commutation with the deck transformations gives the unique descent

$$\overline{K}_w \, p = p K_w. \tag{C.66.8}$$

The open covering quotient makes the descended map and inverse families jointly continuous. Commutation with each lifted operator descends to commutation with its specified surface operator. The family has the original boundary.

For $U = X \setminus S$ with nonempty finite $S$, the same end-extension topology argument, now using disjoint coordinate disks about the finitely many punctures in $X$, extends this connected based family to $X$ fixing $S$ pointwise. In the compact case there is no extension step. Thus (C.66.8) is a disk filling in the original group $A^0 \cap A^u$, with its original operator action. No replacement of that group by an abstract quotient is made.

The proof handles zero-dimensional $Y$ through its exact end compactification. It does not extend an arbitrary intrinsic filling across a nontrivial line or circle frontier, or across an arbitrary positive-dimensional fixed set. Those remaining extensions in C.12 are unchanged. □

**C.67. [Proved] The exact relative disk group for a locally connected frontier.** Let $O \subset \mathbb{R}^2$ be a nonempty bounded simply connected open set with locally connected frontier. Choose a Riemann map $\phi : \mathbb{D} \to O$ and retain its continuous Carathéodory extension

$$q : \overline{\mathbb{D}} \longrightarrow \overline{O}. \tag{C.67.1}$$

Put

$$\begin{aligned} \mathcal{D}_\partial &= \{a \in \mathrm{Homeo}(\overline{\mathbb{D}}) : a|_{S^1} = \mathrm{id}\}, \\ \mathcal{K}_O &= \{b \in \mathrm{Homeo}(\mathbb{R}^2) : b|_{\mathbb{R}^2 \setminus O} = \mathrm{id}\}. \end{aligned} \tag{C.67.2}$$

The actual map

$$T_q : \mathcal{D}_\partial \longrightarrow \mathcal{K}_O, \qquad T_q(a)|_O = \phi a \phi^{-1}, \qquad T_q(a)|_{\mathbb{R}^2 \setminus O} = \mathrm{id} \tag{C.67.3}$$

is an isomorphism of topological groups with their compact-open and inverse topologies. In particular $\mathcal{K}_O$ is contractible. The frontier need not be a Jordan curve.

For the actual stabilizer $H_O$ of $O$ in any specified group of ambient operators, let

$$\eta_q : H_O \longrightarrow \mathrm{Homeo}(\overline{\mathbb{D}}) \tag{C.67.4}$$

be its original prime-end action in the chosen Riemann coordinate. Its kernel is exactly $\{h \in H_O : h|_O = \mathrm{id}\}$. The isomorphism (C.67.3) restricts to the exact comparison

$$C_{\mathcal{D}_\partial}(\eta_q(H_O)) \xrightarrow{\cong} \mathcal{K}_O \cap C_{\mathrm{Homeo}(\mathbb{R}^2)}(H_O). \tag{C.67.5}$$

Both directions have the actual induced group topologies. The contraction asserted above is for the full relative group $\mathcal{K}_O$; no equivariance of the Alexander contraction under a nontrivial operator is asserted.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and relative extension across a locally connected fixed frontier.

*Proof. The retained quotient and prime-end maps.* The Carathéodory theorem for locally connected boundaries gives (C.67.1); it is a continuous surjection, equals $\phi$ on the open disk, and maps the boundary circle onto $\partial O$. The same conformal coordinate identifies the prime-end compactification of $O$ with the closed disk. We use precisely the crosscut-chain compactification and the extension theorem stated in [7, Sections 1 and 3.1]. The map $q$ may have several preimages for a frontier point. No inverse of $q$ on the frontier is used.

An ambient homeomorphism $h$ preserving $O$ induces a homeomorphism of this prime-end compactification. Here is the relevant functorial construction. It sends a crosscut, with its two distinct frontier endpoints, to the crosscut with their actual images. It sends the corresponding complementary crosscut regions to their actual images. On the compact set $\overline{O}$, $h$ is uniformly continuous. Thus disjoint crosscut closures remain disjoint and diameters tending to zero still tend to zero. Nested fundamental chains are sent to nested fundamental chains. The inclusion relations defining equivalence of chains are preserved, as are those relations under $h^{-1}$. The image-chain rule is therefore a bijection on prime ends. Basic crosscut-region neighborhoods and the inverse rule show that its extension is a homeomorphism. This construction is independent of the auxiliary interior basepoint used to select crosscut sides: a chain tends to the frontier and eventually excludes either chosen interior point.

Conjugating this extension by the fixed prime-end Riemann coordinate gives $\widehat{h} = \eta_q(h)$, with the exact equations

$$\widehat{h}|_{\mathbb{D}} = \phi^{-1} h \phi, \qquad q\widehat{h} = hq. \tag{C.67.6}$$

The second equation holds first on the dense open disk and then everywhere by continuity. Composition of image chains, or uniqueness on the dense disk, proves that $\eta_q$ is a homomorphism. It is identity exactly when $h|_O$ is identity. These are the asserted original restriction labels and kernel.

*A frontier-fixing map fixes the entire prime-end circle.* Let $b \in \mathcal{K}_O$ and let $\widehat{b}$ be its just constructed prime-end extension. It preserves orientation: $b$ is identity on the nonempty open exterior of the bounded set $\overline{O}$, so its global orientation is positive. Write $r = \widehat{b}|_{S^1}$ and $q_\partial = q|_{S^1}$. Since $b$ fixes $\partial O$ pointwise, (C.67.6) gives

$$q_\partial r = q_\partial. \tag{C.67.7}$$

We prove that this exact equation forces $r = \mathrm{id}$.

First, $q_\partial$ is constant on no nonempty open arc. Otherwise, for a point $p$ equal to that constant value, the holomorphic map $\phi - p$ is continuous up to an open circle arc and is zero there. In a local conformal coordinate straightening that arc to a real segment, Schwarz reflection extends it holomorphically across the segment. The identity theorem then makes $\phi - p$ identically zero, contrary to injectivity of $\phi$.

Fix $x_* \in O$. The circle-valued map

$$v(\zeta) = \frac{q_\partial(\zeta) - x_*}{|q_\partial(\zeta) - x_*|} \quad (\zeta \in S^1) \tag{C.67.8}$$

has degree one. Indeed $\phi(sS^1)$, for $s < 1$ sufficiently close to one, is a positively oriented Jordan curve enclosing $x_*$. Uniform convergence of $\phi(s\zeta)$ to $q_\partial(\zeta)$ and the positive distance from $x_*$ to $\partial O$ preserve that winding number in the limit.

Use $\zeta = e^{2\pi it}$ and choose a real lift $R$ of $v$, so $R(t+1) = R(t) + 1$. For an increasing lift $P$ of $r$, equation (C.67.7) implies that $R(P(t)) - R(t)$ is one constant integer. Replacing $P$ by the unique integer translate which subtracts that integer gives

$$R(P(t)) = R(t) \quad (t \in \mathbb{R}). \tag{C.67.9}$$

The function $R(t) - t$ is periodic and bounded. Thus the full orbit $\{P^n(t) : n \in \mathbb{Z}\}$ is bounded for each $t$, since its $R$-value is constant. An increasing real homeomorphism with such an orbit has a fixed point: a nonconstant forward monotone orbit has a finite limit, and continuity fixes that limit. Hence $P$, and therefore $r$, has a fixed point. Moreover, if $r(e^{2\pi it}) = e^{2\pi it}$, then $P(t) = t + k$ for an integer $k$. Periodicity would give $P^n(t) = t + nk$; the established boundedness forces $k = 0$. Thus the fixed set of $r$ is exactly the projection of $\mathrm{Fix}(P)$.

If $r$ were nonidentity, choose a component of the complement of its fixed set. Using the periodic fixed set of $P$, lift that component to an interval between consecutive boundary points of $\mathrm{Fix}(P)$. The interval is bounded, its endpoints are fixed, and $P$ has no fixed point inside. Iterations of every interior point therefore tend to one endpoint. Equation (C.67.7) and continuity make $q_\partial$ constant on that open arc. This contradicts the no-constant-arc property. We conclude that $r = \mathrm{id}$ as claimed.

*The algebraic comparison and its topology.* For $a \in \mathcal{D}_\partial$, the continuous map $qa$ on the closed disk is constant on every fiber of $q$. Interior fibers are single points; on a boundary fiber $a$ fixes every point. The compact quotient $q$ therefore gives a continuous map on $\overline{O}$, equal to the identity on its frontier. The identical construction with $a^{-1}$ gives its inverse. Pasting this map with identity on the closed complement of $O$ defines the actual homeomorphism (C.67.3). The formulas on $O$ give injectivity and preservation of products.

Conversely, for $b \in \mathcal{K}_O$, the prime-end extension $\widehat{b}$ fixes $S^1$ by the preceding argument. Thus $\widehat{b} \in \mathcal{D}_\partial$, and its interior formula gives $T_q(\widehat{b}) = b$ everywhere. This proves bijectivity.

For continuity let $\omega_q$ be the modulus of uniform continuity of $q$ on the compact closed disk. The actual forward maps satisfy

$$\sup_{x \in \mathbb{R}^2} |T_q(a)(x) - T_q(a')(x)| \le \omega_q\left(\sup_{\zeta \in \overline{\mathbb{D}}} |a(\zeta) - a'(\zeta)|\right). \tag{C.67.10}$$

Outside $O$ both sides of the comparison of maps are identity. The same estimate applies to their inverses. Hence $T_q$ is continuous in the actual group topologies.

The source $\mathcal{D}_\partial$ is a closed subgroup of the Polish homeomorphism group of the compact disk. The target $\mathcal{K}_O$ is a closed subgroup of the Polish plane homeomorphism group: it is defined by the pointwise equations on $\mathbb{R}^2 \setminus O$. The source acts continuously and transitively on this Polish target by $(a, b) \mapsto T_q(a)b$. The Polish open orbit-map theorem verified in C.26 and used in C.30 makes its orbit map at identity open. That orbit map is exactly $T_q$. The continuous bijection is therefore open and has a continuous inverse. This proves the full topological group isomorphism, without assuming injectivity of the boundary quotient $q$.

*Contraction of the full relative group.* The Alexander family on $\mathcal{D}_\partial$ is

$$A_s(a)(\zeta) = \begin{cases} s\, a(\zeta/s), & |\zeta| \le s, \quad s > 0, \\ \zeta, & |\zeta| \ge s, \end{cases} \qquad A_0(a) = \mathrm{id}\,. \tag{C.67.11}$$

The boundary-fixing condition gives agreement on $|\zeta| = s$. Every map has inverse $A_s(a^{-1})$, and fixes $S^1$. For $s > 0$ joint uniform continuity follows from uniform continuity on the

compact disk; at zero both the map and its inverse have displacement at most $2s$. Also $\|A_s(a) - A_s(a')\|_\infty \le s\|a - a'\|_\infty$. These observations give joint continuity in $s$ and $a$, including inverses. We have $A_1(a) = a$, $A_0(a) = \mathrm{id}$, and $A_s(\mathrm{id}) = \mathrm{id}$. Transporting (C.67.11) by the proved isomorphism contracts the full group $\mathcal{K}_O$.

*Every original operator relation.* For $h \in H_O$, (C.67.6) implies on the open disk, and therefore by continuity on the relative plane maps,

$$T_q(\widehat{h}\, a\, \widehat{h}^{-1}) = h\, T_q(a)\, h^{-1}. \tag{C.67.12}$$

Outside $O$ both sides are identity, since $h$ preserves $O$ and its complement. Conjugation by $\widehat{h}$ preserves $\mathcal{D}_\partial$ even when $h$ reverses orientation. Equation (C.67.12) proves both directions of the exact centralizer comparison (C.67.5). Restriction of a homeomorphism of topological groups to these subgroups retains their induced topologies. An original operator in the kernel of $H_O \to \mathrm{Homeo}(O)$ has $\widehat{h} = \mathrm{id}$ and conjugates every map of $\mathcal{K}_O$ trivially; that actual kernel has been retained.

The same identity also keeps the labels of operators which transport $O$ to another such domain $O'$. With a chosen Riemann map $\phi'$ and extension $q'$ there, the prime-end transport is the unique disk homeomorphism extending $\phi'^{-1} h \phi$, and satisfies $q'\widehat{h} = hq$. Thus

$$T_{q'}(\widehat{h}\, a\, \widehat{h}^{-1}) = h\, T_q(a)\, h^{-1}. \tag{C.67.13}$$

Composing two original operators composes these exact transports, by their uniqueness on the open disk.

Equations (C.67.5), (C.67.12), and (C.67.13) provide the operator-preserving disk models. The Alexander family was proved to contract the full relative group. Its preservation of the operator centralizers in (C.67.5) has not been established. Nor does this one-domain topology comparison supply uniform control for contractions over infinitely many transported domains. These are retained extension requirements in C.12. □

**C.68. [Proved] Relative first-kind disk centralizers and spherical frontier quotients.** Let $O \subset S^2$ be a simply connected domain whose frontier is a nondegenerate locally connected continuum. Retain a Riemann coordinate and its actual continuous Carathéodory quotient

$$\phi : \mathbb{D} \longrightarrow O, \qquad q : \overline{\mathbb{D}} \longrightarrow \overline{O}. \tag{C.68.1}$$

Writing $\mathcal{D}_\partial$ as in C.67, put

$$\mathcal{K}_O^+ = \{b \in \mathrm{Homeo}^+(S^2) : b|_{S^2 \setminus O} = \mathrm{id}\}. \tag{C.68.2}$$

The map $T_q$, equal to $\phi a \phi^{-1}$ on $O$ and to the identity off $O$, is an isomorphism of topological groups

$$T_q : \mathcal{D}_\partial \xrightarrow{\cong} \mathcal{K}_O^+. \tag{C.68.3}$$

The full group $\mathcal{K}_O^+$ is contractible. No injectivity of $q|_{S^1}$ is assumed.

There is also the following operator comparison. Let $A \le \mathrm{Homeo}(\overline{\mathbb{D}})$, and retain its actual restriction closure

$$Q = \overline{\{a|_{\mathbb{D}} : a \in A\}} \ \le \mathrm{Homeo}(\mathbb{D}). \tag{C.68.4}$$

Suppose that $Q$ is locally compact and acts freely and properly on $\mathbb{D}$. Retain a specified $k \in \mathrm{Homeo}(\overline{\mathbb{D}})$ and a torsion-free Fuchsian group $\Lambda$ of the first kind, with the actual subgroup inclusion

$$(k|_{\mathbb{D}})^{-1} \Lambda (k|_{\mathbb{D}}) \le Q. \tag{C.68.5}$$

Here first kind means that the limit set of the specified disk action is all of $S^1$. Restriction is an isomorphism of topological groups

$$\mathrm{res} : C_{\mathcal{D}_\partial}(A) \xrightarrow{\cong} C_{\mathrm{Homeo}(\mathbb{D})}(Q). \tag{C.68.6}$$

Every based loop in the left-hand group has a based filling in that same group. Neither finite generation nor cocompactness of $\Lambda$ is required.

For original ambient operators stabilizing $O$, take $A$ to be their actual prime-end image under (C.68.1). Equations (C.68.3) and (C.68.6) then give a based filling of every loop in the corresponding relative ambient centralizer whenever the displayed data (C.68.4)–(C.68.5) hold. This retains each original operator and its restriction kernel. For the plane application, retain the specified point $\infty \notin O$; the entire filling fixes that point and restricts to a filling through plane homeomorphisms.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and relative frontier extension.

*Proof. The actual spherical boundary quotient.* The locally connected Carathéodory theorem and the prime-end compactification in [7, Sections 1 and 3.1] give (C.68.1). They apply in a spherical coordinate: the conformal map may be meromorphic in a chosen plane chart, but is a homeomorphism from the open disk onto $O$. The boundary map $q_\partial = q|_{S^1}$ is onto $\partial O$.

This boundary map is constant on no nonempty open arc. Indeed, a constant value $p$ on such an arc admits a target complex coordinate finite near $p$. Continuity of $q$ gives a neighborhood of a smaller closed subarc on which that coordinate of $\phi$ is holomorphic and continuous up to the arc. Subtract its constant boundary value and straighten the source circle locally. Schwarz reflection across the resulting real segment gives a holomorphic extension with a segment of zeros. The identity theorem makes $\phi$ constant on an open subset, and then on the connected disk, a contradiction.

We now prove the needed precise property of its possibly nontrivial fibers. There cannot be four cyclically ordered points $a, b, c, d$ in $S^1$ with

$$q(a) = q(c) = p, \qquad q(b) = q(d) = r, \qquad p \neq r. \tag{C.68.7}$$

Choose an embedded closed crosscut $\alpha$ joining $a$ to $c$ and otherwise contained in $\mathbb{D}$. The map $q$ embeds the interior of this crosscut, and its interior image lies in $O$. Its only coincident endpoint images are the point $p$. Consequently

$$J = q(\alpha) \text{ is a Jordan curve in } S^2, \qquad J \cap \partial O = \{p\}. \tag{C.68.8}$$

The two components of $\mathbb{D} \setminus \alpha$ have connected images disjoint from $J$. Near an interior point of $\alpha$, the map $\phi$ is a local homeomorphism and sends its two local sides to the two local sides of $J$. Thus the two entire component images lie in the different complementary Jordan domains. The cyclic order in (C.68.7) places $b$ and $d$ on the respective boundary arcs of the two source components. Approaching them from those components shows that $r$ lies in the closures of both Jordan domains, and hence in $J$. Since $r \in \partial O$, (C.68.8) gives $r = p$, contrary to (C.68.7). This proves that distinct fibers of $q_\partial$ do not have alternating pairs of points.

Let $u \in \mathrm{Homeo}^+(S^1)$ satisfy

$$q_\partial u = q_\partial. \tag{C.68.9}$$

It follows that $u = \mathrm{id}$. To see this, choose a moved point $\zeta$ and an increasing real lift in which $\zeta < u\zeta < \zeta + 1$. For every $\eta > \zeta$ sufficiently close to $\zeta$, continuity and strict increase give

$$\zeta < \eta < u\zeta < u\eta < \zeta + 1. \tag{C.68.10}$$

The absence of a constant boundary arc permits such an $\eta$ with $q_\partial \eta \neq q_\partial \zeta$. Equation (C.68.9) and (C.68.10) then give the forbidden alternating pairs in two distinct fibers. This proves the assertion.

*The relative spherical group, its topology, and its operators.* Every original sphere homeomorphism $h$ preserving $O$ acts on its prime ends. This is precisely the null-crosscut-chain functor proved in C.67: uniform continuity on the compact sphere preserves null chains, their inclusion relations, and their equivalence, and the inverse gives the inverse prime-end map. In the retained coordinate write this map as $\widehat{h}$. It satisfies

$$\widehat{h}|_{\mathbb{D}} = \phi^{-1} h \phi, \qquad q\widehat{h} = hq. \tag{C.68.11}$$

For $h \in \mathcal{K}_O^+$, the boundary restriction of $\widehat{h}$ preserves orientation and satisfies (C.68.9); it is therefore the identity.

For $a \in \mathcal{D}_\partial$, the continuous map $qa$ is constant on every fiber of $q$: interior fibers are singletons, and all points in each boundary fiber are fixed by $a$. The compact quotient $q$ therefore descends $qa$ to a continuous self-map of $\overline{O}$. The same construction for $a^{-1}$ is its inverse. It is the identity on $\partial O$, so pasting with identity on $S^2 \setminus O$ gives $T_q(a)$. It preserves orientation because its restriction to $O$ does. Equations on $O$ prove injectivity and the homomorphism identity. Equation (C.68.11) and boundary rigidity give its inverse, proving the algebraic bijection (C.68.3).

Uniform continuity of $q$ gives forward continuity of $T_q$: uniform changes of $a$ and $a^{-1}$ on the closed disk give uniform changes of $T_q(a)$ and its inverse on the sphere. Both groups in (C.68.3) are closed Polish subgroups of their homeomorphism groups. The left action of $\mathcal{D}_\partial$ on $\mathcal{K}_O^+$ through $T_q$ is continuous and transitive. The verified Polish open orbit-map theorem used in C.26 and C.30 makes this bijective homomorphism open. Its inverse is continuous, as claimed. Conjugating the explicit Alexander contraction of $\mathcal{D}_\partial$ in C.67 by this isomorphism proves contractibility of the full relative group.

For any original operator $h$ preserving $O$, the exact interior conjugation and the exterior identity give

$$T_q(\widehat{h}\, a\, \widehat{h}^{-1}) = h\, T_q(a)\, h^{-1}. \tag{C.68.12}$$

Orientation reversal by $h$ is allowed. The kernel of $h \mapsto \widehat{h}$ is exactly the kernel of restriction to $O$. Thus (C.68.3) identifies the actual centralizers, with the actual induced topologies and operator labels. No equivariance of the Alexander formula under these operators is used.

*First-kind extension with the specified disk conjugacy.* We use the original first-kind extension theorem in [69, Section 3.B, Proposition 3.5 and Corollary 3.5.1, pp. 31–32]. It states that a homeomorphism of the open disk inducing a geometric isomorphism between first-kind groups extends homeomorphically to the closed disk. Its boundary map is determined by the images of the attracting fixed points of infinite-order elements. The proof constructs the boundary map from nested intervals with hyperbolic-axis endpoints and proves continuity of the actual interior homeomorphism at their limiting boundary points. The hypotheses in that section impose no finite generation or cocompactness requirement.

In particular, an interior homeomorphism $g$ commuting with $\Lambda$ induces its identity group isomorphism. The boundary map supplied by that theorem fixes every one of the stated attracting points; they are dense for a first-kind group. Its boundary extension is therefore the identity:

$$g \in C_{\mathrm{Homeo}(\mathbb{D})}(\Lambda) \quad \Longrightarrow \quad \overline{g} \in \mathcal{D}_\partial. \tag{C.68.13}$$

This is an extension of the given homeomorphism, not a replacement by another representative. Uniqueness follows from density of the open disk.

Now let $g \in C_{\mathrm{Homeo}(\mathbb{D})}(Q)$. By the actual inclusion (C.68.5), the map $kgk^{-1}$ on the open disk commutes with $\Lambda$. Equation (C.68.13) extends it to the closed disk, fixing the boundary. Conjugating this extension by the specified closed-disk homeomorphism $k^{-1}$ gives the unique

extension $\overline{g}$ of $g$, still fixing the boundary. For each $a \in A$, the two continuous maps $\overline{g}\,a$ and $a\,\overline{g}$ agree on the dense open disk, and hence everywhere. Thus $\overline{g} \in C_{\mathcal{D}_\partial}(A)$.

Conversely, a member of $C_{\mathcal{D}_\partial}(A)$ restricts to a homeomorphism commuting with all restrictions of $A$, and then with their actual closure $Q$, by continuity of group operations. This proves surjectivity and injectivity of (C.68.6). Restriction is continuous. The source and target are closed Polish subgroups: commuting with each specified operator is a closed condition, and fixing the boundary is closed. Applying the same open orbit-map theorem to the transitive action through restriction proves continuity of its inverse. Consequently every compact parameter family, including its inverse family, extends continuously through this exact inverse map. No separate assumption about a boundary restriction fibration is needed for this particular isomorphism.

*Based fillings and the original frontier.* Let $t \mapsto f_t$ be a based loop in $C_{\mathcal{D}_\partial}(A)$. As a loop in the full group $\mathcal{D}_\partial$, it is null-homotopic by the Alexander contraction in C.67. Restriction makes it an ambient-null loop in $\mathrm{Homeo}(\mathbb{D})$. The proper-action injection C.33 applies to the exact group $Q$ in (C.68.4). It supplies a based filling

$$B : D^2 \longrightarrow C_{\mathrm{Homeo}(\mathbb{D})}(Q)$$

of that restricted loop. Composing with the continuous inverse in (C.68.6) gives a based filling of the original $f_t$ through boundary-fixed closed-disk homeomorphisms commuting with every original $a \in A$. This proves the asserted loop statement.

For the spherical domain, use $A = \{\widehat{h} : h \in H_O\}$ from (C.68.11). Apply (C.68.6) to the loop obtained through $T_q^{-1}$, and return the resulting filling through $T_q$. Equation (C.68.12) proves commutation with every original operator in $H_O$ throughout the filling; the restriction kernel acts identically on $O$ throughout. The complement of $O$ is fixed pointwise. This passage uses the quotient $q$ and never uses an inverse boundary value of that quotient. It therefore includes slit frontiers and other locally connected frontiers with multiple prime-end accesses.

In the plane application $\infty \notin O$, so $\infty$ belongs to the fixed complement. Restriction from the sphere group fixing $\infty$ gives the original compact-open and inverse topology on plane homeomorphisms; the converse continuity is the one-end compact-core comparison proved in C.66. Hence this is a filling in the required plane group with its specified infinity. □

*Scope of the extension.* The closed-disk homeomorphism $k$ and the first-kind subgroup inclusion (C.68.5) are part of the proved comparison. An arbitrary interior uniformization of $O/Q$ does not supply this $k$ on the original prime-end circle. The argument makes no such substitution. It also supplies no uniform estimate for pasting contractions over infinitely many different transported domains. The remaining general frontier and moving-quotient extension in C.12 stays open.

**C.69. [Proved] Relative disk centralizers for Fuchsian actions of either kind.** Let $\Gamma$ be a torsion-free discrete group of orientation-preserving Möbius transformations of $\mathbb{D}$, with its actual action on $\overline{\mathbb{D}}$. No finite-generation assumption is imposed. Every based loop in

$$C_{\mathcal{D}_\partial}(\Gamma) \tag{C.69.1}$$

has a based filling in that same group. The same assertion holds for the actual action $k^{-1}\Gamma k$, with any specified $k \in \mathrm{Homeo}(\overline{\mathbb{D}})$.

More precisely, let $\mathcal{L}_\Gamma$ be the limit set, and retain the actual spaces and restriction map

$$\Omega_\Gamma = S^1 \setminus \mathcal{L}_\Gamma, \qquad D_\Gamma = \mathbb{D} \cup \Omega_\Gamma, \qquad \mathrm{res}_\Gamma : C_{\mathcal{D}_\partial}(\Gamma) \longrightarrow C_{\mathrm{Homeo}(D_\Gamma, \Omega_\Gamma)}(\Gamma). \tag{C.69.2}$$

The notation on the right means pointwise fixing $\Omega_\Gamma$. The map $\mathrm{res}_\Gamma$ is an isomorphism of topological groups with their compact-open and inverse topologies. This isomorphism extends the full filling, including its inverse family, across the original limit set.

The proof also gives the following exact relative surface comparison. For every second-countable surface $N$ with boundary, inclusion induces an injection

$$\pi_1\big(\mathrm{Homeo}(N,\partial N),\mathrm{id}\big)\longrightarrow\pi_1\big(\mathrm{Homeo}(N),\mathrm{id}\big). \tag{C.69.3}$$

The proof supplies an explicit correction of an ambient parameter-disk filling on a proper collar. It permits infinitely many boundary components and preserves the original parameter-boundary loop exactly.

Through the actual spherical quotient and operator maps of C.68, (C.69.1) gives a relative equivariant filling on a locally connected simply connected domain whenever the original prime-end action is the specified closed-disk conjugate $k^{-1}\Gamma k$. Its original restriction kernel is retained. In the plane application the specified infinity lies outside the domain and is fixed throughout.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and exact relative frontier extension.

*Proof. The original partial compactification and its proper action.* The group $\Gamma=1$ has $D_\Gamma=\overline{\mathbb{D}}$, and (C.69.1) is the Alexander assertion in C.67. We first record facts that also cover every nontrivial $\Gamma$.

For any sequence of distinct elements $\gamma_n\in\Gamma$, a subsequence has points $a,b\in\mathcal{L}_\Gamma$ with

$$\gamma_n(z)\longrightarrow a\quad\text{uniformly on compact subsets of }\overline{\mathbb{D}}\setminus\{b\}. \tag{C.69.4}$$

Here is the explicit disk calculation. Write

$$\gamma_n(z)=e^{i\theta_n}\frac{z-b_n}{1-\bar b_n z},\qquad b_n=\gamma_n^{-1}(0). \tag{C.69.5}$$

Discreteness prevents a sequence of distinct $\gamma_n$ from remaining in a compact subset of the Möbius group. Taking a subsequence gives $b_n\to b\in S^1$ and $e^{i\theta_n}\to e^{i\theta}$. For $z\neq b$, the numerator at the limit equals $-b(1-\bar bz)$. Thus (C.69.4) follows with $a=-e^{i\theta}b$; the denominator is bounded away from zero on each stated compact set. Both $a$ and $b$ are in the limit set, by applying the definition to $\gamma_n(0)$ and $\gamma_n^{-1}(0)$.

The limit set is $\Gamma$-invariant: applying an element to a convergent orbit sequence gives another orbit sequence, and its inverse gives equality. Thus the displayed action preserves $D_\Gamma$. The space $D_\Gamma$ is a second-countable surface with boundary $\Omega_\Gamma$. At a point of $\Omega_\Gamma$ use a small closed-disk boundary chart disjoint from the closed set $\mathcal{L}_\Gamma$. It is contractible: $(z,t)\mapsto(1-t)z$ is an actual contraction in $D_\Gamma$.

The $\Gamma$-action on this space is free and proper. A nonidentity element is hyperbolic or parabolic, because a discrete elliptic element has finite order and $\Gamma$ is torsion free. All its fixed points lie in $\mathcal{L}_\Gamma$, so it fixes no point of $D_\Gamma$. For compact $K\subset D_\Gamma$, a sequence of distinct translates meeting $K$ would contradict (C.69.4): the compact $K$ avoids the repelling point $b$, its images converge uniformly to $a\in\mathcal{L}_\Gamma$, and $K$ has positive distance from that closed limit set. Consequently the exact quotient

$$p:D_\Gamma\longrightarrow N_\Gamma=D_\Gamma/\Gamma \tag{C.69.6}$$

is a regular universal covering of surfaces with boundary, with precisely the deck group $\Gamma$. Evenly covered neighborhoods, Hausdorffness, and second countability follow exactly as in C.32, using boundary half disks when required. Its boundary is $\Omega_\Gamma/\Gamma$, and its interior is $\mathbb{D}/\Gamma$.

*A dense set of controlled boundary accesses.* Let $E_\Gamma$ be the union of $\Omega_\Gamma$ and the attracting fixed points of the hyperbolic elements of $\Gamma$. Then

$$\overline{E_\Gamma}=S^1. \tag{C.69.7}$$

For completeness this also follows from (C.69.4). If $\Gamma$ contains a hyperbolic element $\gamma$, use both of its endpoints, the attracting points of $\gamma$ and $\gamma^{-1}$. For $\zeta \in \mathcal{L}_\Gamma$ choose distinct $\gamma_n$ with $\gamma_n(0) \to \zeta$, and pass to (C.69.4). At least one of the two chosen endpoints differs from $b$. Its images under $\gamma_n$ tend to $\zeta$ and are the attracting points of the actual conjugates of $\gamma$ or $\gamma^{-1}$. Thus the hyperbolic fixed points are dense in the limit set.

If a nontrivial $\Gamma$ has no hyperbolic element, all its elements are parabolic. Two with distinct fixed points would generate a hyperbolic element: conjugate one to $z \mapsto z+1$ in the upper half plane, write the other as a parabolic matrix with nonzero lower-left entry $c$, and observe that the trace of its product with the $n$th power of the first is its original trace plus $nc$. Its absolute value exceeds two for some integer $n$. Thus all fixed points coincide. After conjugating that point to infinity, every element is a real translation. Discreteness makes this group cyclic, and its limit set is a single point. Its $\Omega_\Gamma$ is dense in $S^1$. The trivial group has $\Omega_\Gamma = S^1$. These cases prove (C.69.7).

*The precise cap-side extension.* Let

$$h \in C_{\mathrm{Homeo}(D_\Gamma, \Omega_\Gamma)}(\Gamma). \tag{C.69.8}$$

At a point $p \in \Omega_\Gamma$, it and its inverse already extend continuously with value $p$. At an attracting fixed point $p$ of a hyperbolic $\gamma$, choose a terminal ray of the hyperbolic axis ending at $p$. Partition this ray into the actual translates $\gamma^n A$ of one compact fundamental segment $A$. Its image under $h$ is partitioned into $\gamma^n h(A)$, by commutation. The compact set $h(A)$ has iterates converging uniformly to $p$ under the hyperbolic Möbius map. Thus this image ray ends at the same $p$. The same argument applies to $h^{-1}$. These are accesses with the exact original axis labels.

Fix $\zeta \in S^1$ and $\epsilon > 0$. Choose two points $p, q \in E_\Gamma$ on opposite sides of $\zeta$ in a sufficiently short circular arc $I$ containing $\zeta$. Choose an open crosscut $\alpha \subset \mathbb{D}$ whose closed arc has endpoints $p$ and $q$, as follows. Near an endpoint in $\Omega_\Gamma$, use a short radial tail. Near a hyperbolic fixed endpoint use a terminal axis tail as above. Truncate those two disjoint tails and join their interior endpoints by an embedded arc in a small convex disk cap around $\zeta$. The tails may be chosen in disjoint small endpoint neighborhoods; the joining arc avoids them except at its two endpoints. This gives a crosscut whose closure and the closed boundary arc $\overline{I}$ lie in a convex neighborhood of diameter less than $\epsilon$.

Let $V$ be the component of $\mathbb{D} \setminus \alpha$ adjacent to $I$. The Jordan curve $\overline{\alpha} \cup \overline{I}$ bounds this cap. Its filled region lies in the convex hull of that curve, so

$$\operatorname{diam} V < \epsilon. \tag{C.69.9}$$

The image $\beta = h^{-1}(\alpha)$ has a continuous embedded closure with the same two endpoints $p, q$, by the access calculation. It is therefore another actual crosscut. Let $U$ be its complementary component adjacent to the same boundary arc $I$.

We verify the side, rather than inferring it from the endpoint set. Choose $r \in E_\Gamma \cap I$. There are points $z_n \in \mathbb{D}$ with both $z_n \to r$ and $h^{-1}(z_n) \to r$. For $r \in \Omega_\Gamma$ this follows from continuity at $r$; for a hyperbolic fixed point take $z_n = \gamma^n z_0$ for its attracting map and use $h^{-1}(z_n) = \gamma^n h^{-1}(z_0)$. Eventually $z_n \in V$ and $h^{-1}(z_n) \in U$. Since $h$ maps complementary components of $\beta$ bijectively to those of $\alpha$, it follows that

$$h(U) = V. \tag{C.69.10}$$

The set $U \cup I$ is a neighborhood of $\zeta$ in the closed disk. Equations (C.69.9)–(C.69.10) prove that $h(z) \to \zeta$ as $z \to \zeta$ from the open disk. The same proof applies to the inverse. Thus $h$ extends uniquely to a boundary-fixed homeomorphism of $\overline{\mathbb{D}}$. Commutation with every original $\gamma$ survives by density.

This proves algebraic surjectivity of (C.69.2); injectivity follows from density of the open disk. Restriction is continuous. Both groups are closed Polish groups in their indicated compact-open

homeomorphism groups, since the commutator equations and pointwise boundary constraints are closed. The Polish open orbit-map argument of C.26/C.30 applied to this continuous bijective homomorphism proves continuity of its inverse. In particular this gives joint extension for every compact parameter family and its inverse family, uniformly on the closed disk. It is the full topology comparison, not merely pointwise extension of the individual maps.

*The relative parameter-disk correction on an arbitrary surface.* We prove (C.69.3) independently. Let $f_t$ be a based loop fixing $\partial N$ pointwise, and let $B_z$, $z \in D^2$, be a based ambient filling with that boundary loop. Fix a proper closed collar

$$c : \partial N \times [0,1] \hookrightarrow N. \tag{C.69.11}$$

We recall explicitly that such a collar can be made proper. A locally finite surface triangulation with boundary subcomplex gives a collar by the adjacent boundary-edge rectangles and the finite interval fans at boundary vertices. Their normal parameters agree on the successive common edges and give a product collar. Start with any such collar $c_0$ and choose a compatible proper metric on $N$. A positive continuous function $\delta(b) < 1/2$ can be chosen so that $d(c_0(b,s),b) < 1$ for $0 \le s \le \delta(b)$: local continuity gives the admissible neighborhoods and a locally finite partition of unity gives a positive continuous minorant. The collar $c(b,t) = c_0(b,\delta(b)t)$ is proper. Indeed the $b$ coordinates over a compact target set lie in its closed unit neighborhood, which is compact, and the inverse image is closed in that compact set times $[0,1]$.

Each boundary component is a line or a circle. The connected based family $B_z$ preserves each component and its orientation. Put $\psi_z = B_z|_{\partial N}$. On a line component choose a real coordinate and denote its actual increasing map by $\Psi_z$. On a circle component, lift the joint map $D^2 \times \mathbb{R} \to S^1$ to a real map $\Psi_z$, normalized by $\Psi_{z_*}(u) = u$ at the specified based parameter $z_* \in \partial D^2$. The parameter disk is simply connected, so this is an actual continuous lift. It satisfies

$$\Psi_z(u+1) = \Psi_z(u) + 1, \qquad \Psi_z(u) = u \quad (z \in \partial D^2). \tag{C.69.12}$$

For the second equality, the original boundary loop fixes the boundary circle; the integer discrepancy is continuous on the connected parameter boundary and is zero at $z_*$. Every $\Psi_z$ is increasing. The same normalized lifting of $\psi_z^{-1}$ gives its inverse, by uniqueness.

Choose a continuous function $\chi : [0,1] \to [0,1]$ with $\chi(0) = 1$ and $\chi(t) = 0$ for $t \ge 2/3$. On each real boundary coordinate put

$$E_{z,t}(u) = (1 - \chi(t))u + \chi(t)\Psi_z(u). \tag{C.69.13}$$

This is an increasing proper homeomorphism. On circle components it commutes with integer translation and therefore descends. The families and their inverses are continuous: strict increase is preserved by the convex combination, and compact parameter sets give common properness bounds from the compact inverse family of $\Psi_z$. Extend to $N$ by

$$E_z(c(b,t)) = c(E_{z,t}(b),t), \qquad E_z = \mathrm{id} \text{ outside the collar.} \tag{C.69.14}$$

The notation in (C.69.14) uses the just specified line or circle coordinates. The formula is identity near the outer collar edge, so it pastes to a homeomorphism.

This is also a continuous family for infinitely many boundary components. Properness of the collar reduces each compact target set to a compact subset of $\partial N \times [0,1]$. Only finitely many components meet that subset, since boundary components are open in the boundary one-manifold. All formulas on those components have the proved compact-parameter and inverse continuity. Thus (C.69.14) is continuous in the original compact-open and inverse topology.

We have $E_z|_{\partial N} = \psi_z$ and $E_z = \mathrm{id}$ on the entire parameter boundary. Consequently

$$B'_z = E_z^{-1} B_z \tag{C.69.15}$$

is a based filling fixing $\partial N$ pointwise, with exactly the prescribed loop $f_t$ on $\partial D^2$. This proves (C.69.3) without assuming a boundary-restriction fibration or a global choice of circle lifts over its whole image.

*Filling on the exact quotient and lifting every label.* Return to a nontrivial $\Gamma$ and a based loop $f_t$ in (C.69.1). Restriction through (C.69.2) and descent through (C.69.6) give a based loop $b_t$ of homeomorphisms of $N_\Gamma$ fixing its boundary pointwise, with the exact equation

$$b_t p = p f_t. \tag{C.69.16}$$

Joint continuity and inverse continuity descend through the open covering quotient, as in C.32.

Choose $x \in \mathbb{D}$ and $v = p(x)$ in the interior of $N_\Gamma$. By [74, Theorem 1.1 and Lemma 2.1], the identity component of $\mathrm{Homeo}(N_\Gamma, v)$ is contractible. Here all exceptions are excluded explicitly: the only pointed exceptions for this assertion are the plane, the compact disk, the sphere, and the projective plane. The actual universal cover (C.69.6) has nontrivial infinite fundamental group $\pi_1(N_\Gamma) = \Gamma$ downstairs, so none occurs. An interior coordinate disk gives a continuous local section of evaluation by homeomorphisms supported in that disk. Hence evaluation is a locally trivial bundle onto the interior, and its homotopy exact sequence gives

$$\pi_1(\mathrm{Homeo}(N_\Gamma), \mathrm{id}) \hookrightarrow \pi_1(\mathrm{Int}\, N_\Gamma, v). \tag{C.69.17}$$

The evaluation loop $b_t(v)$ is null-homotopic: it is the projection of the closed loop $f_t(x)$ in the contractible open disk. Thus (C.69.17) supplies an ambient based filling of $b_t$, and (C.69.15) corrects it to a filling $B'_z$ fixing the whole boundary of $N_\Gamma$.

Lift the actual map $(x, z) \mapsto B'_z(p(x))$ on $D_\Gamma \times D^2$ through (C.69.6), with the identity lift at $z_*$. The product is simply connected, so the lift exists and is unique. Lift the inverse family with the same normalization. Their compositions lift the projection $p$ and equal identity at $z_*$, so uniqueness makes them mutual inverses everywhere. For every $\gamma \in \Gamma$, the two compositions of the lifted family with $\gamma$ lift the same map and agree at the based parameter. They therefore agree for all parameters: every deck label is retained.

For $x \in \Omega_\Gamma$, the lifted value remains in the discrete fiber over $p(x)$, because $B'_z$ fixes that boundary point. Connectedness of $D^2$ and the identity value at $z_*$ force it to remain exactly $x$. On the parameter boundary, uniqueness of lifts over the connected product $D_\Gamma \times \partial D^2$ recovers the original $f_t$ exactly. This gives the desired continuous filling in the right side of (C.69.2). Its continuous inverse restriction extends the entire filling to (C.69.1).

The elementary nontrivial groups require no omitted case. For a parabolic generator, the upper-half-plane coordinate $z \mapsto z + 1$ identifies $D_\Gamma$ with the closed upper half plane and $N_\Gamma$ with $S^1 \times [0, \infty)$. For a hyperbolic generator $z \mapsto \lambda z$, $\lambda > 1$, logarithmic polar coordinates $(s, \theta) \in \mathbb{R} \times [0, \pi]$ give the deck translation $s \mapsto s + \log \lambda$ and the compact annular quotient. In both cases the pointed group used in (C.69.17) is contractible by precisely the quoted exception lists. The trivial case was handled at the start.

*The original conjugacy and the frontier transport.* Conjugation by the specified closed-disk homeomorphism $k$ is a topological group isomorphism carrying the centralizer of $k^{-1}\Gamma k$ to (C.69.1). Transport the proved filling by its inverse. Every conjugate operator retains its original label, and the boundary is fixed throughout.

For a spherical domain, compose with the actual $T_q$ and prime-end maps of C.68. The exact conjugation equation (C.68.12) proves the claimed ambient commutation, and the original restriction kernel acts identically on the domain throughout. The complement is fixed. When the specified infinity lies in that complement, the entire filling descends to the original plane homeomorphism group with the topology comparison already proved in C.66. □

*Remaining scope.* The original closed-disk conjugacy is retained in this theorem. The construction does not assert that a uniformizing conjugacy of an arbitrary interior proper action

extends to its original prime-end circle. Nor does it assert uniform pasting over infinitely many different domain conjugacies. Those parts of the general C.12 extension problem remain unresolved.

**C.70. [Proved] Operator-compatible carriers for the original evaluation fillings.** Let $f_t$, $G$, $H$ and the $H$-invariant union $W$ of original open $G$-orbits be as in C.64. There is one compact path-connected family $P$ of actual global $H$-centralizing homeomorphisms, containing the identity and every $f_t$, preserving each original open orbit and fixing every $G$-fixed point, with the following properties.

For a nonempty compact set $K \subset \mathbb{R}^2$, define its filled hull by

$$\mathrm{Fill}(K) = \mathbb{R}^2 \setminus U_\infty(K), \qquad U_\infty(K) \text{ the unbounded component of } \mathbb{R}^2 \setminus K. \tag{C.70.1}$$

For $x \in W$, put $K_x = P^2(x)$. There is a unique component $V_x$ of $\operatorname{int}\mathrm{Fill}(K_x)$ containing the entire set $P(x)$. It is simply connected. Write $O_x$ for the original orbit containing $x$, and let $U_x$ be the component of $V_x \cap O_x$ containing $P(x)$. These actual domains satisfy

$$P(x) \subset U_x \subset O_x \cap \mathrm{Fill}(P^2(x)), \qquad hU_x = U_{hx} \quad (h \in H). \tag{C.70.2}$$

Inclusion induces an injection

$$\pi_1(U_x, x) \longrightarrow \pi_1(O_x, x). \tag{C.70.3}$$

The original evaluation loop $t \mapsto f_t(x)$ therefore has a based disk filling $b_x : D^2 \to U_x$. The fillings can be chosen simultaneously as a set-indexed family satisfying the exact operator equations

$$b_{hx}(z) = h(b_x(z)) \qquad (x \in W,\ h \in H,\ z \in D^2). \tag{C.70.4}$$

Every choice of such fillings obeys the fixed-frontier control

$$\sup_{z \in D^2} |b_{x_n}(z) - y| \longrightarrow 0 \quad \text{for } x_n \to y \text{ with } y \in \mathrm{Fix}(G). \tag{C.70.5}$$

This is a statement about the specified individual evaluation fillings and their simultaneous operator labels. Joint continuity in $x$, avoidance of collisions between distinct evaluation points, and extension to a family of homeomorphisms are not established here.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and controlled fixed-frontier evaluation fillings.

*Proof. The compact family and its actual orbit neighborhoods.* Let $\mathcal{K}$ be C.64's compact path-connected labelled family, and retain its projection to actual global maps. Put

$$P = \mathcal{K}\mathcal{K}^{-1} \ \cup\ \{f_t : 0 \le t \le 1\} \ \cup\ \{f_t^{-1} : 0 \le t \le 1\}. \tag{C.70.6}$$

Each of these three compact sets is path connected and contains the identity; their union has the same properties. Composition and inversion take place in the original global homeomorphism group. Every displayed map commutes with each original $h \in H$, preserves $O_x$, and fixes $\mathrm{Fix}(G)$. The source-cover labels, when required, are the corresponding products and inverses of C.64's retained labels and of the original lifted loop.

By C.64.1, $x$ is in the plane interior of $P(x)$: the orbit is an open subset of the plane. For each $p \in P$, the image under the original homeomorphism $p$ of that neighborhood lies in $pP(x) \subset P^2(x)$. Consequently

$$P(x) \subset \operatorname{int} P^2(x). \tag{C.70.7}$$

The set $P(x)$ is path connected and contains $x$. The set $K_x = P^2(x)$ is also compact and path connected, since it is the continuous evaluation image of $P \times P$.

*Filled hulls and their simply connected interior components.* For any compact $K$, the complement has exactly one unbounded component: outside a sufficiently large closed disk it is connected, and every unbounded component meets that exterior. The filled hull is closed and bounded, hence compact. It is contained in the Euclidean convex hull of $K$. Indeed, a point outside that convex hull is separated from it by a line and can be joined to infinity within the corresponding open half-plane, without meeting $K$.

Every component $V$ of $\operatorname{int}\operatorname{Fill}(K)$ is simply connected. To prove this directly, take a Jordan curve $J \subset V$ and its bounded Jordan disk $D_J$. The connected unbounded set $U_\infty(K)$ is disjoint from $J$, so it lies entirely on its unbounded side. Hence $D_J \subset \operatorname{Fill}(K)$. The interior of $D_J$ is open and is consequently in $\operatorname{int}\operatorname{Fill}(K)$; it belongs to the same component as $J$. Thus the whole disk lies in $V$. For an arbitrary loop in $V$, polygonal approximation inside $V$ and a finite planar regular neighborhood of the resulting graph express it as a product of based conjugates of its inner boundary Jordan curves. Each bounds a disk in $V$ by what we just proved. This gives a based null-homotopy of the original loop in $V$.

Apply this with $K = K_x$. Equation (C.70.7) places the connected set $P(x)$ in a single component $V_x$ of the interior of the filled hull, proving existence and uniqueness of that component. It is simply connected by the preceding argument. Since $P(x) \subset O_x$, it is contained in one component $U_x$ of $V_x \cap O_x$. This proves the inclusions in (C.70.2).

*The actual inclusion on fundamental groups.* The proof of C.65.4 works with a simply connected open plane domain in place of its open Euclidean disk. We give the comparison in this form. Let $O, V$ be open plane domains with $V$ simply connected, and let $U$ be a component of $O \cap V$. Take a based loop $\gamma$ in $U$ which is null-homotopic in $O$. Approximate it by a based polygonal loop within $U$, and take a compact connected planar regular neighborhood $M \subset U$ of its finite graph. The bounded disk inside its outer boundary lies in $V$, by simple connectedness and Jordan separation. In particular all its inner complementary Jordan disks lie in $V$.

Fill precisely those inner disks wholly contained in $O$, obtaining a compact connected planar surface $M' \subset U$. From each remaining inner disk choose a point $z_i$ outside $O$. The inclusion

$$M' \hookrightarrow \mathbb{R}^2 \setminus \{z_i\} \tag{C.70.8}$$

has the explicit strong deformation retraction of C.65.9: choose a Schoenflies coordinate on each inner disk taking its chosen $z_i$ to zero, apply the radial formula $(r, \theta) \mapsto ((1-\tau)r + \tau, \theta)$ there, and apply that formula in the exterior Schoenflies coordinate outside the outer boundary. The identity on $M'$ completes the finite closed pasting. Every boundary and the chosen basepoint stay fixed. The original null-homotopy in $O$ avoids all $z_i$; compose it with this retraction and append the polygonal approximation homotopy. This gives a based filling in $U$. The actual inclusion $U \hookrightarrow O$ is therefore injective on fundamental groups. Taking $O = O_x$ and $V = V_x$ proves (C.70.3).

By C.32, the original homeomorphism loop restricted to $O_x$ is null-homotopic in its homeomorphism group. Evaluation at the original point $x$ is continuous and gives a based null-homotopy of $t \mapsto f_t(x)$ in $O_x$. That entire loop lies in $P(x) \subset U_x$. Equation (C.70.3) therefore gives the asserted filling $b_x$ with image in this actual $U_x$.

*All original operators and uniform control at fixed points.* A plane homeomorphism sends the unbounded component of the complement of a compact set to the unbounded component of the image complement. To see unboundedness, choose a sequence in that component escaping every compact set. Properness of the inverse prevents its image from lying in a compact set. It follows that

$$h \operatorname{Fill}(K) = \operatorname{Fill}(hK). \tag{C.70.9}$$

Every member of $P$ commutes with $h \in H$, so $hP^2(x) = P^2(hx)$ and $hP(x) = P(hx)$. Equation (C.70.9) sends the specified component $V_x$ to $V_{hx}$. Commutation with the original generated group also gives $hO_x = O_{hx}$. Taking the indicated components now proves $hU_x = U_{hx}$ exactly.

Choose one representative $x$ of each $H$-orbit in $W$ and one of its based fillings in $U_x$. Define the filling at $hx$ by $h \circ b_x$. This is independent of the choice of $h$: an operator fixing $x$ fixes every point of $O_x$, since each such point is $gx$ for some $g \in G$ and $h(gx) = g(hx) = gx$. Its image therefore fixes the entire chosen filling. The boundary identity follows from $hf_t(x) = f_t(hx)$. This proves (C.70.4) for every original operator, without any choice of an effective quotient representative.

Finally let $x_n \to y$ with $y \in \mathrm{Fix}(G)$. The compact family $P^2$ fixes $y$ pointwise. Continuity of evaluation on $P^2 \times (\{y\} \cup \{x_n : n \geq 1\})$ gives

$$\sup_{p \in P^2} |p(x_n) - y| \longrightarrow 0. \tag{C.70.10}$$

The Euclidean ball centered at $y$ with this radius contains the convex hull of $P^2(x_n)$, and therefore contains its filled hull and $U_{x_n}$. Every point of every chosen filling lies in that same ball. This proves (C.70.5), uniformly over the disk parameter and independently of every filling choice.

The family just selected is equivariant as a family indexed by points. Its proof imposes no continuity between fillings at distinct source points. It also imposes no configuration-space disjointness. Those requirements, needed for an ambient homeomorphism filling, remain part of C.12. □

**C.71. [Proved] Actual disk comparisons for two-dimensional operators and a cyclic compactification.** Let $Q$ be a closed locally compact subgroup of $\mathrm{Homeo}(\mathbb{D})$, with its actual compact-open and inverse topology, acting freely and properly on $\mathbb{D}$. When $\dim Q = 2$, put

$$E_Q = \{h \in \mathcal{D}_\partial : h|_{\mathbb{D}} \in C_{\mathrm{Homeo}(\mathbb{D})}(Q)\}. \tag{C.71.1}$$

Then every based sphere family $S^n \to E_Q$, $n \geq 1$, has a based contraction in $E_Q$. In particular every based loop fills there. No extension of the $Q$-action to the boundary, and no closed-disk conjugacy for that action, is assumed. If $A \leq \mathrm{Homeo}(\overline{\mathbb{D}})$ has actual restricted closure $Q$, then $E_Q = C_{\mathcal{D}_\partial}(A)$ by continuity on the interior and density in the closed disk.

There is also an explicit cyclic action on the actual closed disk whose interior image is closed, discrete, free and proper, and which admits no closed-disk conjugacy to a torsion-free cyclic Fuchsian action. For its specified invariant partial compactification, the analogue of the restriction isomorphism (C.69.2) is not surjective. The proof gives the actual maps witnessing both assertions. Nevertheless its boundary-fixed centralizer is contractible: it is identified topologically with an explicitly described group of continuous families of increasing real lifts, and linear interpolation supplies the contraction. Its interior orbits are precisely the displayed vertical lines, so this example does not supply an open transitive orbit for that boundary-fixed centralizer.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), admissible infinite-operator level-zero comparison and relative disk centralizers.

*Proof. The actual two-dimensional transformation group.* The Lie conclusion and principal orbit comparison of C.33 apply to $Q$. The identity component has open orbits because the free orbit dimension is two. All these orbits are open; connectedness of $\mathbb{D}$ leaves just one. Given $q \in Q$, transitivity supplies $q_0 \in Q^0$ with $q_0 x = qx$ at one specified $x$. Freeness gives $q = q_0$. Thus $Q = Q^0$, and evaluation

$$\epsilon_x : Q \longrightarrow \mathbb{D}, \qquad q \longmapsto qx \tag{C.71.2}$$

is a homeomorphism. The inverse is the principal orbit coordinate from C.33; alternatively properness makes the continuous orbit bijection a homeomorphism. In particular $Q$ is simply connected.

Its two-dimensional Lie algebra is either abelian or has a basis $X, Y$ with $[X, Y] = Y$: in the nonabelian case the derived algebra is a line and rescaling a complement gives that bracket. The simply connected groups are respectively $\mathbb{R}^2$ and

$$(a, b)(a', b') = (a + a', b + e^a b'), \qquad (a, b) \in \mathbb{R}^2. \tag{C.71.3}$$

This uses the actual Lie-group isomorphisms integrating those bases. Their inverses are the integrations in the opposite direction; the uniqueness for simply connected groups gives both inverse identities. In each case exponential is a global diffeomorphism. In (C.71.3) its explicit formula is $\exp(a, v) = (a, v(e^a - 1)/a)$, with the quotient at $a = 0$ defined to be 1.

Through (C.71.2), a map commuting with all left translations is precisely a right translation. Indeed its value $c$ at the identity forces its value at every $q$ to be $qc$. Right translations and their inverses depend continuously on $c$ in the compact-open topology, and evaluation gives the inverse comparison. Thus the actual interior centralizer is topologically the opposite group

$$N = C_{\mathrm{Homeo}(\mathbb{D})}(Q) \cong Q^{\mathrm{op}}. \tag{C.71.4}$$

Inversion is an isomorphism $Q^{\mathrm{op}} \to Q$, so $N$ has the same two specified simply connected exponential models.

*Compact words retain the given disk boundary.* Let $a : S^n \to E_Q$ be based, with $n \geq 1$, and let $g_z = a(z)|_{\mathbb{D}} \in N$. Define the interior maps

$$R_{r,z} = \exp_N(r \log_N g_z), \qquad 0 \leq r \leq 1. \tag{C.71.5}$$

They are continuous in the actual interior homeomorphism topology. They equal the identity for $r = 0$ and at the sphere basepoint, and equal $g_z$ for $r = 1$.

Put

$$P = a(S^n) \cup a(S^n)^{-1} \subset \mathcal{D}_\partial. \tag{C.71.6}$$

This is compact and contains the identity. For each $z$, choose a path in $S^n$ from its basepoint to $z$. Applying the exact two- or three-letter formula (C.37.2) to its restricted path in the simply connected group $N$ places $R_{r,z}$ in the restriction of $P^3$. No continuous choice of these sphere paths is used. The resulting restriction map

$$P^3 \longrightarrow \mathrm{Homeo}(\mathbb{D}) \tag{C.71.7}$$

is continuous and injective: two closed-disk maps agreeing on the dense interior agree everywhere. Its source is compact and its target is Hausdorff, so it is a homeomorphism onto its image. Consequently (C.71.5) has a unique continuous lift $\overline{R}_{r,z} \in P^3$ through (C.71.7). Continuity here includes the inverse maps, since the source and target carry their homeomorphism-group topologies. All these lifted maps fix the original circle pointwise and centralize $Q$ on the interior. They give the claimed based contraction. This argument uses compact word control to prove boundary continuity; it does not identify the full relative group topology with the topology of an arbitrary subgroup of $N$.

*The cyclic action and its exact disk coordinate.* Write

$$\kappa(x) = \frac{x}{1 + |x|}, \qquad \kappa^{-1}(s) = \frac{s}{1 - |s|} \quad (-1 < s < 1), \qquad S = [-1, 1]^2. \tag{C.71.8}$$

The extension of $\kappa$ takes $-\infty, +\infty$ to $-1, +1$. An explicit homeomorphism $J : S \to \overline{\mathbb{D}}$ and its inverse are

$$\begin{aligned} J(v) &= \frac{\|v\|_\infty}{\|v\|_2} v \quad (v \neq 0), \qquad J(0) = 0, \\ J^{-1}(w) &= \frac{\|w\|_2}{\|w\|_\infty} w \quad (w \neq 0), \quad J^{-1}(0) = 0. \end{aligned} \tag{C.71.9}$$

The formulas are inverse on each ray and continuous at zero. They take the square boundary onto the unit circle. Retain the actual interior coordinate $p = J \circ (\kappa \times \kappa) : \mathbb{R}^2 \to \mathbb{D}$.

On the real plane take $T(x, y) = (x, y + 1)$. Its extension on $S$ is

$$\overline{T}(s,t) = (s, \tau(t)), \qquad \tau(t) = \kappa(\kappa^{-1}(t) + 1) \quad (|t| < 1), \qquad \tau(-1) = -1, \quad \tau(1) = 1. \tag{C.71.10}$$

The specified closed-disk generator is $J\overline{T}J^{-1}$. The group $Q = p\langle T\rangle p^{-1}$ is free and proper on the interior: a compact real-plane set has bounded $y$-coordinate and meets only finitely many of its integer translates. It is closed and discrete in the actual interior homeomorphism group, since evaluation at $(0, 0)$ sends $T^m$ to $(0, m)$; a convergent sequence of such maps has eventually constant integer parameter. Conjugation by $p$ transports these assertions, with its inverse giving the opposite comparison.

The fixed set of $\overline{T}$ on $\partial S$ is exactly the union of the two closed horizontal edges. Indeed $\tau(t) > t$ for $-1 < t < 1$. Under a purported closed-disk conjugacy $k$ to a generator $\gamma$ of a torsion-free cyclic Fuchsian action, the equation $k(J\overline{T}J^{-1}) = \gamma k$ would imply

$$kJ\big(\mathrm{Fix}(\overline{T}) \cap \partial S\big) = \mathrm{Fix}(\gamma) \cap S^1. \tag{C.71.11}$$

The left side consists of two nondegenerate arcs, whereas the right side has one point for a parabolic element or two points for a hyperbolic element. Since $kJ$ is injective, (C.71.11) is impossible.

There is a direct failure of the analogous partial restriction comparison as well. Set

$$S_* = [-1, 1] \times (-1, 1), \qquad \Omega_* = \{-1, 1\} \times (-1, 1). \tag{C.71.12}$$

The $\langle\overline{T}\rangle$ action on $S_*$ is free and proper. Through $(s, y) \mapsto (s, \kappa(y))$, its actual quotient is the closed annulus $[-1, 1] \times (\mathbb{R}/\mathbb{Z})$, with deck map $y \mapsto y + 1$ and boundary $\Omega_*/\langle\overline{T}\rangle$. The horizontal translation $L(x, y) = (x + 1, y)$ extends on $S$ as $\overline{L}(s, t) = (\tau(s), t)$. It commutes with $\overline{T}$, and its restriction to $S_*$ fixes $\Omega_*$ pointwise. It does not fix the horizontal edges pointwise. Thus

$$C_{\mathrm{Homeo}(S,\partial S)}(\overline{T}) \longrightarrow C_{\mathrm{Homeo}(S_*,\Omega_*)}(\overline{T}) \tag{C.71.13}$$

is not surjective: any preimage of $\overline{L}|_{S_*}$ would equal $\overline{L}$ on the dense set $S_*$ and hence on $S$. The maps $J$ and $J^{-1}$ transport this same counterexample to the specified disk.

*The actual boundary-fixed centralizer of this action.* Let $\mathcal{V}$ consist of continuous functions $F : \mathbb{R}^2 \to \mathbb{R}$, written $F_x(y)$, such that each $F_x$ is an increasing homeomorphism of $\mathbb{R}$, and

$$F_x(y + 1) = F_x(y) + 1, \qquad \lim_{|x|\to\infty} \sup_{0\le y\le 1} |F_x(y) - y| = 0. \tag{C.71.14}$$

Its product and inverse are the pointwise fiber composition and inverse. Give it the metric topology

$$d_{\mathcal{V}}(F, G) = \sup_{x,y} |F_x(y) - G_x(y)| + \sup_{x,y} |F_x^{-1}(y) - G_x^{-1}(y)|. \tag{C.71.15}$$

Both suprema are finite. The displacements are continuous and periodic in $y$, and vanish as $|x| \to \infty$, so they are bounded globally. The inverse displacements have the same suprema at each $x$ and the same vanishing property. Joint continuity of $(x, y) \mapsto F_x^{-1}(y)$ follows directly: on a compact $x$-interval the periodic displacements are uniformly bounded, which confines all inverse arguments to a compact interval; strict monotonicity identifies every subsequential limit.

There is an exact group isomorphism

$$\begin{aligned} &\mathcal{V} \longrightarrow C_{\mathrm{Homeo}(S,\partial S)}(\overline{T}), \\ &F \longmapsto H_F, \qquad H_F(\kappa(x), \kappa(y)) = (\kappa(x), \kappa(F_x(y))), \end{aligned} \tag{C.71.16}$$

with the identity on $\partial S$. To prove surjectivity, write the real-plane restriction of a map on the right as $(X(x,y),Y(x,y))$. Commutation gives $X(x,y+m)=X(x,y)$ and $Y(x,y+m)=Y(x,y)+m$ for each integer $m$. Letting $m\to+\infty$ and using the actual fixed top edge forces $X(x,y)=x$. Bijectivity now gives a homeomorphism $F_x(y)=Y(x,y)$ of each fiber, and the integer equation forces it to be increasing. Continuity at the two fixed vertical edges, uniformly for $y\in[0,1]$, is exactly the vanishing condition in (C.71.14).

Conversely (C.71.14) defines a homeomorphism of the real plane with the stated inverse. For bounded $x$ its displacement is uniformly bounded, so it extends as the identity at the horizontal edges. For $|x|\to\infty$ its displacement tends uniformly to zero, giving the identity at the vertical edges and at the corners. These assertions also apply to the inverse. They prove the claimed extension and bijection.

The isomorphism (C.71.16) is topological. In the forward direction $\kappa$ is globally Lipschitz, so each supremum in (C.71.15) controls the uniform square displacement of the corresponding maps or inverses. Conversely suppose $H_{F_j}$ and their inverses converge uniformly to $H_F$ and its inverse. For real $y\in[0,1]$, the values $F_x(y)$ lie in one fixed compact real interval, uniformly in $x$. Uniform convergence of their $\kappa$-coordinates keeps all $F_{j,x}(y)$ in a slightly larger compact real interval. Uniform continuity of $\kappa^{-1}$ there gives uniform convergence of these actual values. Periodicity extends it to every real $y$. Applying the same argument to the inverse maps gives exactly (C.71.15).

Finally define

$$\mathcal{H}_r(F)_x(y)=(1-r)F_x(y)+ry,\qquad 0\le r\le 1. \tag{C.71.17}$$

It is strictly increasing, proper, and has the same integer equivariance and vanishing condition. Forward continuity in (C.71.15) is immediate. For inverse continuity, each fixed element of $\mathcal{V}$ and its inverse is uniformly continuous in the fiber variable, uniformly in $x$: its displacement is continuous on $\mathbb{R}\times(\mathbb{R}/\mathbb{Z})$ and vanishes at the two $x$-ends. The elementary increasing-map inverse estimate therefore turns uniform forward convergence to a member of $\mathcal{V}$ into uniform inverse convergence. This applies to (C.71.17) when $(r,F)$ varies. Hence it is a continuous contraction of the full topological group to the identity, fixing the identity throughout. Conjugation by $J$ gives the stated disk contraction.

Every such map preserves the real $x$-coordinate. For each specified $x$ and $c\in\mathbb{R}$, a continuous compactly supported function $b$ with $b(x)=c$ gives $F_u(y)=y+b(u)\in\mathcal{V}$. Thus the exact group orbits in the interior are the vertical lines. This records the transitivity scope of the example explicitly.

The exact certificate `code/part_b/c71_disk_compactification_certificate.py` checks the rational compactification and inverse, the translation conjugacies and commutation, the displayed boundary and quotient maps, and nonlinear periodic fiber models together with their interpolation and inverse formulas. The full topological comparisons and contractions are proved above. This claim leaves the general zero- and one-dimensional proper-operator disk comparison unresolved. □

**C.72. [Proved] A spiral boundary obstruction with the full discrete local centralizer.** There is an explicit based loop of closed-disk homeomorphisms fixing the boundary pointwise whose generated group $G$ is transitive on the interior and has full interior centralizer equal to a specified closed discrete free proper cyclic group $Q=\langle T_*\rangle$. Every forward $Q$-orbit has the whole boundary circle as its limit set. Nevertheless the actual restriction map

$$C_{\mathcal{D}_\partial}(Q)\longrightarrow C_{\mathrm{Homeo}(\mathbb{D})}(Q) \tag{C.72.1}$$

is not surjective: the proof displays an interior centralizer map with no continuous extension at any boundary point. Here $Q$ on the left means its specified closed-disk action.

The proof also supplies an explicit interior conjugacy $j$ from $Q$ to a parabolic Fuchsian group. Under that conjugacy the same obstructed map does extend as a boundary-fixed homeomorphism. Thus the map $h \mapsto jhj^{-1}$ between the two interior centralizers does not identify their actual boundary-relative subgroups. The exact failed inclusion is witnessed below; no comparison of unnamed compactifications is used.

The original loop in this example has an explicit boundary-fixed contraction. The result therefore refutes the displayed extension comparison, including the versions supplemented by transitivity and the full-circle orbit limit set; it supplies no failure of the level-zero loop comparison in C.12.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), admissible infinite-operator level-zero comparison and the actual relative disk extension problem.

*Proof. The actual spiral coordinate and its inverse.* Regard the real plane as $\mathbb{C}$, with $z = x + iy$, and define

$$\begin{aligned} p(z) &= \frac{z}{1+|z|}\exp\bigl(i\log(1+|z|)\bigr), \\ p^{-1}(w) &= \frac{w}{1-|w|}\exp\bigl(i\log(1-|w|)\bigr) \qquad (w \in \mathbb{D}). \end{aligned} \tag{C.72.2}$$

Both maps take zero to zero. The radial coordinates are $r \mapsto r/(1+r)$ and $\rho \mapsto \rho/(1-\rho)$, which are inverse. The angular changes in the two displayed formulas are opposite because $1 + r = (1-\rho)^{-1}$. They prove both inverse identities and continuity, including at zero. Thus (C.72.2) is an actual homeomorphism of the plane with the disk interior. It is used to specify the action on the given closed disk; no boundary extension of $p$ is asserted.

The following elementary estimate retains its boundary topology. Suppose $|a| \le M$ and $r = |z| > 2M + 1$. Write $r' = |z + a|$. The reverse triangle inequality gives $|r' - r| \le M$, and

$$\begin{aligned} \left|\frac{z+a}{r'} - \frac{z}{r}\right| &\le \frac{2M}{r-M}, \\ |\log(1+r') - \log(1+r)| &\le \frac{M}{1+r-M}, \\ \left|\frac{r'}{1+r'} - \frac{r}{1+r}\right| &\le \frac{M}{(1+r-M)(1+r)}. \end{aligned} \tag{C.72.3}$$

The first inequality follows by adding and subtracting $z/r'$, and the second by the mean value theorem. Combining them with $|e^{iu} - e^{iv}| \le |u - v|$ gives

$$|p(z+a) - p(z)| \le \frac{4M}{r-M}. \tag{C.72.4}$$

In particular, any plane homeomorphism $h$ for which both $h(z) - z$ and $h^{-1}(z) - z$ are bounded gives a closed-disk homeomorphism

$$\overline{h}(w) = \begin{cases} php^{-1}(w), & w \in \mathbb{D}, \\ w, & w \in S^1. \end{cases} \tag{C.72.5}$$

For a compact continuous parameter family with a common bound on both displacements, (C.72.4) gives joint boundary continuity of the maps and inverses. Interior continuity is just the actual conjugation in (C.72.2). These observations prove continuity in the full closed-disk homeomorphism topology.

*The actual cyclic and translation actions.* For $a \in \mathbb{C}$ put $L_a(z) = z + a$ and denote its extension (C.72.5) by $\overline{L}_a$. The equations

$$\overline{L}_a\overline{L}_b = \overline{L}_{a+b}, \qquad \overline{L}_a^{-1} = \overline{L}_{-a} \tag{C.72.6}$$

hold first on the interior and then on the boundary. The parameter map $a \mapsto \overline{L}_a$ is continuous by (C.72.4), and its inverse on its image is $h \mapsto p^{-1}(h(0))$. Its image is closed: if $\overline{L}_{a_n}$ converges to a closed-disk homeomorphism $h$, then $h(0)$ is interior and $a_n \to p^{-1}(h(0))$. Thus this is a specified closed translation subgroup, topologically $\mathbb{R}^2$, acting transitively on $\mathbb{D}$ and fixing $S^1$ pointwise.

Set

$$T(x,y) = (x, y+1), \qquad T_* = \overline{L}_i, \qquad Q = \langle T_* \rangle. \tag{C.72.7}$$

Its interior image is closed and discrete: evaluating the conjugate $T^m$ at the real origin gives $(0,m)$. It is free and proper there, since compact real-plane sets have bounded vertical coordinates and meet only finitely many integer translates. These assertions are transported by the actual $p, p^{-1}$. Every element of $Q$ fixes the entire boundary.

Every forward orbit of $T_*$ has every point of $S^1$ as a limit. Fix $z = x + iy$ and $\zeta = e^{i\alpha}$. For large positive integers $k$ choose an integer $n_k$ nearest to $\exp(2\pi k + \alpha - \pi/2)$. Then

$$\begin{aligned} \arg(x + i(y+n_k)) &\longrightarrow \pi/2, \\ \log(1 + |x + i(y+n_k)|) - (2\pi k + \alpha - \pi/2) &\longrightarrow 0, \\ |p(x + i(y+n_k))| &\longrightarrow 1. \end{aligned} \tag{C.72.8}$$

The second limit follows since the integer rounding error is at most $1/2$ and $|x+i(y+n_k)|/n_k \to 1$. Substitution in (C.72.2) gives $p(T^{n_k} z) \to \zeta$. Conversely $|T^n z| \to \infty$, so every accumulation point is on the circle. This proves the stated full-circle forward limit set for each original point.

*A based loop with precisely this full local centralizer.* Use the continuous functions

$$\varphi(y) = \operatorname{dist}(y, \mathbb{Z}), \qquad \psi(x) = \max(0, 1 - |x|). \tag{C.72.9}$$

The first has zero set exactly $\mathbb{Z}$ and period one; the second is nonzero and compactly supported. For $0 \le u \le 1$ set

$$\begin{aligned} P_1(u)(x,y) &= (x+u, y), & P_2(u)(x,y) &= (x, y+u), \\ P_3(u)(x,y) &= (x + u\varphi(y), y), & P_4(u)(x,y) &= (x, y + u\psi(x)). \end{aligned} \tag{C.72.10}$$

Each is a homeomorphism, with inverse obtained by replacing $u$ by $-u$ in the same formula. They all commute with $T$. Their displacements and those of their inverses have a common bound of one, so their extensions $\overline{P}_j(u)$ in (C.72.5) are continuous paths in $C_{\mathcal{D}_\partial}(Q)$.

For $(j-1)/4 \le t \le j/4$ define the triangular amplitude

$$a_j(t) = \min\{\, 8(t - (j-1)/4),\, 2 - 8(t - (j-1)/4) \,\}, \qquad f_t = \overline{P}_j(a_j(t)). \tag{C.72.11}$$

At every joining parameter the amplitude is zero, so these are an actual continuous based loop. Replacing $a_j(t)$ by $(1-r)a_j(t)$ gives its explicit based contraction in the same relative centralizer. In particular its ambient class is zero, with no appeal to a different group.

Let $G = \langle f_t : 0 \le t \le 1 \rangle$. It contains all real horizontal and vertical translations in the $p$-coordinate: the intervals $[0,1]$ in the first two paths generate the two additive real groups, using products and inverses. Thus $G$ is transitive on $\mathbb{D}$. An interior homeomorphism centralizing $G$ must commute with all these translations. Conjugation by $p^{-1}$ and evaluation at zero then force it to be a translation $L_{(a,b)}$. Its commutation with $P_3(1)$ says $\varphi(y+b) = \varphi(y)$ for every $y$. At $y = 0$ this gives $b \in \mathbb{Z}$. Commutation with $P_4(1)$ says $\psi(x+a) = \psi(x)$ for all $x$. If $a \neq 0$, iteration would give $\psi(na) = \psi(0) = 1$ for arbitrarily large $|na|$, contradicting its compact support. Hence $a = 0$. Conversely these integer vertical translations commute with every path (C.72.10). We have proved the exact equality

$$C_{\mathrm{Homeo}(\mathbb{D})}(G|_{\mathbb{D}}) = Q|_{\mathbb{D}}. \tag{C.72.12}$$

In particular the effective full local operator group has dimension zero and is the named closed discrete proper group, in precisely the topology of the original comparison. The larger $C_{\mathcal{D}_\partial}(Q)$ is also transitive, since it contains the same translation subgroup.

*The interior centralizer map that cannot extend.* The real-plane map

$$A(x,y) = (2x, y) \tag{C.72.13}$$

is a homeomorphism and commutes with $T$. Thus $A_* = pAp^{-1}$ belongs to the target of (C.72.1). Fix again $\zeta = e^{i\alpha}$ and, for sufficiently large $k$, put

$$\begin{aligned} z_k &= (e^{2\pi k+\alpha} - 1, 0), \\ z'_k &= (0, e^{2\pi k+\alpha-\pi/2} - 1). \end{aligned} \tag{C.72.14}$$

Both $p(z_k)$ and $p(z'_k)$ approach $\zeta$, with the angular value exactly $\alpha$ modulo $2\pi$. On the vertical sequence $A$ is the identity, so $p(Az'_k) \to \zeta$. On the horizontal sequence its image angle is

$$2\pi k + \alpha + \log(2 - e^{-(2\pi k+\alpha)}). \tag{C.72.15}$$

Its radius tends to one, so $p(Az_k) \to e^{i\log 2}\zeta$. These limits differ, since $0 < \log 2 < 1 < 2\pi$. Thus $A_*$ has no continuous extension at $\zeta$. As $\zeta$ was arbitrary, it has no continuous extension at any boundary point. This proves the failure of surjectivity in (C.72.1).

*The specified interior Fuchsian comparison.* Let $\mathbb{H} = \{u + iv : v > 0\}$ and retain the actual homeomorphisms

$$\begin{aligned} B(x,y) &= y + ie^x, & B^{-1}(u+iv) &= (\log v, u), \\ C(\zeta) &= \frac{\zeta - i}{\zeta + i}, & C^{-1}(w) &= i\frac{1+w}{1-w}, \\ j &= CBp^{-1} : \mathbb{D} \longrightarrow \mathbb{D}. \end{aligned} \tag{C.72.16}$$

The displayed inverse formulas give $j^{-1} = pB^{-1}C^{-1}$. Direct substitution yields $BTB^{-1}(\zeta) = \zeta + 1$. Hence $j$ conjugates $Q|_{\mathbb{D}}$ to the parabolic Fuchsian group generated by $C(\zeta \mapsto \zeta + 1)C^{-1}$. This interior conjugacy cannot extend to a closed-disk homeomorphism: such an extension would take the whole fixed boundary circle of $T_*$ bijectively onto the single boundary fixed point of that parabolic generator. This is the same direct fixed-set equation as (C.71.11), with both maps specified here.

For the obstructed map itself, the same comparison gives

$$BAB^{-1}(u+iv) = u + iv^2. \tag{C.72.17}$$

This homeomorphism of $\mathbb{H}$ extends as the identity on $\mathbb{R} \cup \{\infty\}$. At finite real points this follows from $v^2 \to 0$, and at infinity from $|u + iv^2| \to \infty$ whenever $|u + iv| \to \infty$. The inverse $u + iv \mapsto u + i\sqrt{v}$ has the same property. Consequently $jA_*j^{-1}$ is in the actual boundary-fixed centralizer of the parabolic closed-disk action, although $A_*$ is absent from the original boundary-relative group. Conjugation by $j$ is an isomorphism of the interior centralizers, with inverse conjugation by $j^{-1}$, but it does not carry the original relative subgroup onto the parabolic one. This proves the precise comparison failure claimed in the statement.

All paths and their explicit contraction fix the circle, so a specified closed-disk embedding in a surface transports them and extends them by the identity outside that disk. The inverse families have the same extension. This places the example in the original local-orbit setting without changing its operator or loop labels. The contraction in (C.72.11) continues to be the actual contraction there.

The exact certificate `code/part_b/c72_spiral_comparison_certificate.py` checks the radial inverse and phase formulas, the two displayed asymptotic phases symbolically, the

specified interior conjugacy and centralizer map, and the periodic shear, inverse, and loop identities with rational arithmetic. The uniform boundary estimates, complete orbit limit sets, full local-centralizer equality and topological comparison failures are proved above. □

**C.73. [Proved] The exact prime-end comparison for arbitrary simply connected frontiers.** Let $O \subset S^2$ be a simply connected domain with nondegenerate frontier, and fix a conformal map $\phi : \mathbb{D} \to O$. Retain the actual groups

$$K_O = \{h \in \mathrm{Homeo}^+(S^2) : h|_{S^2 \setminus O} = \mathrm{id}\}, \qquad \mathcal{D}_\partial = \mathrm{Homeo}(\overline{\mathbb{D}}, S^1). \tag{C.73.1}$$

Every $h \in K_O$ induces the identity on the prime-end circle of $O$. Its actual prime-end extension defines a continuous injective homomorphism

$$\begin{aligned} &\iota_O : K_O \longrightarrow \mathcal{D}_\partial, \\ &\iota_O(h)|_{\mathbb{D}} = \phi^{-1} h \phi. \end{aligned} \tag{C.73.2}$$

All groups have the compact-open and inverse topologies. The image of (C.73.2) is dense. For locally connected frontier it is the topological isomorphism of C.68. For every non-locally-connected frontier its image is proper, and its inverse on that image is not continuous. The proof supplies the actual sequences and a disk map witnessing these two failures.

Every actual ambient operator preserving $O$ induces its own prime-end map, and (C.73.2) intertwines conjugation by the original operator with conjugation by that prime-end map. In particular it identifies an ambient relative centralizer with the intersection of its image and the corresponding disk centralizer, with the original operator labels and restriction kernel retained. This is a continuous injective comparison; no injection on fundamental groups or descent of a disk null-homotopy is asserted.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), actual prime-end and ambient relative centralizer comparisons.

*Proof. The prime-end space and the precise access data.* Let $O^{\mathrm{P}}$ denote the prime-end compactification. The exact extension of $\phi$ is a homeomorphism $\phi^{\mathrm{P}} : \overline{\mathbb{D}} \to O^{\mathrm{P}}$. The crosscut-chain definition and the neighborhood basis are those in [7, §3.1]. We use the spherical version, with diameters taken in the compact sphere. The explicit spherical definition and functorial action are also recorded in [58, §2, pp. 1571–1572]. No continuous landing map on the whole prime-end circle is assumed.

Here are the exact crosscut facts needed below. A genuine crosscut is an arc whose interior lies in $O$ and whose two distinct endpoints lie on $\partial O$. In the prime-end disk its closure is an arc with two distinct prime-end endpoints; their principal sets are the respective singleton endpoint sets. Each side of the crosscut determines the corresponding open circle interval. An access arc landing at an actual frontier point has a unique endpoint in the prime-end disk, and that endpoint has the singleton principal set consisting of the landing point. Equivalently the radial image under $\phi$ lands at that point. These precise statements appear in [58, p. 1575 and p. 1581, before the proof proper of Lemma 7]; they apply to spherical frontier points, including infinity.

Let $A \subset S^1$ be the set of accessible prime ends in the specified coordinate. For $a \in A$, write $\ell(a)$ for its unique principal point, so

$$\lim_{r \uparrow 1} \phi(ra) = \ell(a). \tag{C.73.3}$$

Every nonempty open circle interval $I$ contains two members of $A$ with distinct values of $\ell$. To see this without postulating continuity of $\ell$ on $I$, first choose a prime-end open neighborhood whose closure meets the circle only inside $I$. The crosscut-neighborhood basis supplies a crosscut

side contained in this neighborhood. The two prime-end endpoints of its closure lie in $I$, and the preceding genuine crosscut statement gives them distinct actual landing points. In particular $A$ is dense, and $\ell|_{A\cap I}$ is never constant.

An ambient homeomorphism $h$ preserving $O$ takes null chains to null chains: it preserves crosscuts, their sides and cofinal inclusions, and uniform continuity on $S^2$ preserves vanishing diameters. Its inverse gives the inverse action. This is the actual induced homeomorphism $h^{\mathrm{P}}$ of $O^{\mathrm{P}}$. For $h \in K_O$, let $u$ be its action on the circle through $\phi^{\mathrm{P}}$. It preserves orientation. Indeed its interior action preserves the oriented relative class of the closed disk; naturality of the boundary isomorphism to the oriented circle class gives degree $+1$ on the boundary. The access statements give

$$u(A) = A, \qquad \ell(ua) = \ell(a). \tag{C.73.4}$$

The second equality uses the actual fact that $h$ fixes every point of $\partial O$.

*The unlinked access fibres and circle rigidity.* There cannot be four distinct accessible prime ends $a, b, c, d$ in cyclic order such that

$$\ell(a) = \ell(c) = p, \qquad \ell(b) = \ell(d) = q, \qquad p \neq q. \tag{C.73.5}$$

For a complete separation argument, take radial tails at $a$ and $c$ in the closed unit disk and join their interior endpoints by a simple arc inside the disk, disjoint from those tails except at the two joining endpoints. For example, an arc on a smaller concentric circle gives such a joining arc. Their union is a genuine disk crosscut $\alpha$ from $a$ to $c$. By (C.73.3), $\phi(\alpha \cap \mathbb{D})$ closes at both ends to the same point $p$. It is injective on its interior and does not contain $p$ there. The resulting map from the circle obtained by identifying the two endpoints is a continuous bijection onto its image. Compactness makes this image a Jordan curve $J \subset S^2$, and

$$J \cap \partial O = \{p\}. \tag{C.73.6}$$

The two sides of $\mathbb{D} \setminus \alpha$ map to opposite components of $S^2 \setminus J$. This can be checked at any interior point of $\alpha$, where the open-set homeomorphism $\phi$ takes its two local sides to the two local Jordan sides; connectedness then gives the same assignment for each entire component. The radial tails at $b$ and $d$ lie on opposite disk sides. Their landing point $q$ therefore lies in the closures of both Jordan domains, whose intersection is $J$. Equation (C.73.6) would force $q = p$. This proves the impossibility of (C.73.5), without any assertion about other impressions.

Suppose $u$ is nonidentity. Its moved set is open and meets $A$, so choose a moved $a \in A$. Choose real representatives so that $a < u(a) < a + 1$. There is a small open interval $I$ just after $a$ such that every $b \in I$ satisfies $a < b < u(a) < u(b) < a + 1$. The nonconstancy proved above supplies $b \in A \cap I$ with $\ell(b) \neq \ell(a)$. Equation (C.73.4) now gives exactly the forbidden four-tuple $a, b, u(a), u(b)$. Consequently $u = \mathrm{id}$. This proves the prime-end rigidity assertion and defines (C.73.2) on every element of $K_O$. The group equations follow from the interior conjugacy and density of the interior. Injectivity also follows from that conjugacy and the specified identity outside $O$.

*Continuity in the actual ambient topology.* The group $K_O$ is a closed subgroup of $\mathrm{Homeo}^+(S^2)$, hence Polish; $\mathcal{D}_\partial$ is Polish as well. The graph

$$\mathcal{G} = \{(h, g) \in K_O \times \mathcal{D}_\partial : g|_{\mathbb{D}} = \phi^{-1} h \phi\} \tag{C.73.7}$$

is closed. Namely, convergence of both coordinates preserves the displayed equality at every interior point; density and the fixed circle then determine $g$ uniquely. Thus $\mathcal{G}$ is a closed Polish subgroup. Its projection to $K_O$ is a continuous bijective homomorphism, by the existence and uniqueness already proved. The Polish-group open-mapping consequence of the orbit theorem used in C.30 makes that projection open: apply the theorem to the transitive left action of $\mathcal{G}$ on the Polish group $K_O$ through the projection. Its inverse is therefore continuous. Composing

that inverse with the second projection proves continuity of $\iota_O$, including for compact families of maps and inverses. This proves no continuity of its inverse on its image.

Every compactly supported disk homeomorphism $g$ belongs to the image. Its interior conjugate $\phi g \phi^{-1}$ has compact support in $O$, so extension by the identity gives an element of $K_O$. Such disk maps are dense in $\mathcal{D}_\partial$: for $0 < r < 1$, define

$$g^{[r]}(z) = \begin{cases} r\, g(z/r), & |z| \le r, \\ z, & |z| \ge r. \end{cases} \tag{C.73.8}$$

The boundary-fixed equation makes these formulas agree. Their inverse has the same formula with $g^{-1}$. Uniform continuity of $g, g^{-1}$ on the closed disk gives $g^{[r]} \to g$ with inverses as $r \uparrow 1$. Thus the image in (C.73.2) is dense.

*A nontrivial impression gives both failures of comparison.* In the fixed coordinate the impression of $\zeta \in S^1$ can be written

$$I(\zeta) = \bigcap_{\epsilon > 0} \overline{\phi\big(\mathbb{D} \cap B(\zeta, \epsilon)\big)} \quad \text{in } S^2. \tag{C.73.9}$$

This is the usual impression: the disk neighborhoods and the crosscut neighborhoods are cofinal bases in $O^{\mathrm{P}}$. The sets in (C.73.9) are nested compact connected sets, and their intersection is nonempty and contained in $\partial O$.

A non-locally-connected frontier has a nontrivial impression. Here is also a direct verification of this consequence. If every $I(\zeta)$ were a singleton, assign that value to $\phi(\zeta)$. Any interior sequence tending to $\zeta$ would have all its spherical cluster points in that singleton, so would converge to it. For a sequence of boundary points $\zeta_n \to \zeta$, approximate each assigned value by an interior value at a point within $1/n$ of $\zeta_n$. The same interior-sequence argument then proves continuity on the circle and on the whole closed disk. Every frontier point is a value on the circle, by compactness applied to an interior approximating sequence. The frontier would therefore be a continuous image of the compact locally connected circle, hence locally connected, by [7, Lemma 3.2]. This contradiction proves the assertion, and also keeps the spherical topology explicit.

Fix distinct $a, b \in I(\zeta)$. Choose successively positive numbers $\epsilon_n \to 0$ and points $u_n, v_n \in \mathbb{D} \cap B(\zeta, \epsilon_n)$ such that

$$\phi(u_n) \longrightarrow a, \qquad \phi(v_n) \longrightarrow b. \tag{C.73.10}$$

Their existence follows from (C.73.9). The lens $\mathbb{D} \cap B(\zeta, \epsilon_n)$ is convex and open, so a small closed neighborhood of the segment $[u_n, v_n]$ is a convex topological disk $D_n$ contained in that lens, with both points in its interior. After choosing $D_n$, choose the next $\epsilon_{n+1}$ smaller than its distance from $\zeta$ and the distances of all earlier disks from $\zeta$. This retains (C.73.10) and makes the $D_n$ pairwise disjoint. They form a locally finite family in $\mathbb{D}$, with sole possible accumulation point $\zeta$ in the closed disk, and $\operatorname{diam} D_n \le 2\epsilon_n$.

Choose a homeomorphism $g_n$ supported in $D_n$ taking $u_n$ to $v_n$. There is a concrete such map: on each segment from $u_n$ to a boundary point $z \in \partial D_n$, send $u_n + t(z - u_n)$ to $v_n + t(z - v_n)$, $0 \le t \le 1$. Convexity and interiority give the unique ray coordinates and the inverse formula with $u_n, v_n$ interchanged. It fixes $\partial D_n$ and extends by the identity outside $D_n$. Therefore

$$\sup |g_n - \mathrm{id}\,| \le 2\epsilon_n, \qquad \sup |g_n^{-1} - \mathrm{id}\,| \le 2\epsilon_n. \tag{C.73.11}$$

Let $h_n \in K_O$ be its compactly supported ambient conjugate. Then $\iota_O(h_n) = g_n$ tends to the identity in $\mathcal{D}_\partial$, but $h_n$ does not tend to the identity: it moves $\phi(u_n)$ to $\phi(v_n)$, whose limiting points in (C.73.10) are distinct. This proves the claimed failure of inverse continuity on the actual image.

There is also one map witnessing nonsurjectivity. Set $g = g_n$ on each $D_n$ and set $g = \mathrm{id}$ elsewhere on $\overline{\mathbb{D}}$. Local finiteness gives a homeomorphism in the interior, and the shrinking supports give continuity at $\zeta$ and every other boundary point. The same reasoning applies to the piecewise inverses, so $g \in \mathcal{D}_\partial$. If $g = \iota_O(h)$ for an $h \in K_O$, the interior conjugacy would give $h(\phi(u_n)) = \phi(v_n)$. Continuity at the fixed frontier point $a$ would then imply $a = b$, a contradiction. Thus this specified $g$ is outside the image. Together with density this proves the stated proper-image assertion. For locally connected frontier, C.68 gives the full opposite comparison and its topology; no such comparison is inferred here at a nontrivial impression.

*Original operators and the remaining filling problem.* Let $H_O$ be any actual ambient operator group preserving $O$. Denote its same-label prime-end action in the chosen coordinate by

$$\widehat{h} = (\phi^{\mathrm{P}})^{-1} h^{\mathrm{P}} \phi^{\mathrm{P}} \quad (h \in H_O). \tag{C.73.12}$$

An operator may reverse orientation. Its conjugation still preserves $K_O$, and the interior equations and density give exactly

$$\iota_O(hkh^{-1}) = \widehat{h}\, \iota_O(k)\, \widehat{h}^{-1} \quad (k \in K_O,\ h \in H_O). \tag{C.73.13}$$

Consequently

$$\iota_O\big(C_{K_O}(H_O)\big) = \iota_O(K_O) \cap C_{\mathcal{D}_\partial}(\widehat{H}_O). \tag{C.73.14}$$

For the reverse implication, the disk equation first gives the ambient commutation on $O$; on its complement both relative maps are the identity. This proves the original same-label equation everywhere. An operator acting trivially on $O$ induces the identity prime-end map and already commutes with every element of $K_O$, so the restriction kernel is retained by (C.73.13).

The density proved above concerns $\iota_O(K_O)$, not an unproved density statement for the centralizer intersection. Likewise a disk contraction need not stay in this image, and (C.73.11) shows why continuity of a putative inverse transfer requires its own argument. These exact comparisons leave the general equivariant frontier-filling obligation of C.12 open. □

**C.74. [Proved] An exact obstruction to principal-coordinate interpolation.** There is an explicit based loop $f_\tau \in \mathcal{D}_\partial$ whose generated group $G$ is transitive on $\mathbb{D}$ and whose full interior centralizer is a closed free proper group $Q \cong \mathbb{R}$. The loop has an explicit based contraction in $\mathcal{D}_\partial$ commuting with this actual $Q$.

There are two specified principal coordinates for the same $Q$-action, with their exact transition map and inverse. In the second coordinate, write the loop as

$$\widetilde{f}_\tau(s,t) = (s + u_\tau(t), v_\tau(t)).$$

The coordinate centralizer homotopy

$$(s,t) \longmapsto (s + r u_\tau(t), (1-r)t + r v_\tau(t)) \qquad (0 \le r \le 1) \tag{C.74.1}$$

is a continuous family of interior homeomorphisms, but its map at $r = 1/2, \tau = 1/2$ has no continuous extension at any point of the original boundary circle. Thus this literal coordinate interpolation does not give the required relative contraction.

The same map admits no uniform bound by filled orbits of any compact family of actual boundary-fixed maps: for every compact $P \subset \mathcal{D}_\partial$ and every positive integer $N$, it fails the inclusion

$$A_*(x) \in \mathrm{Fill}\big(P^N(x)\big) \quad \text{for all } x \in \mathbb{D}, \tag{C.74.2}$$

where Fill is the actual planar filled hull of C.70 after the standard disk is embedded in the plane. The comparison concerns the stated interpolation and the original maps; the displayed loop itself remains contractible.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact actual-boundary comparison for one-dimensional proper operator groups.

*Proof. The actual disk coordinate and three loop paths.* Use the exact spiral coordinate of C.72, now writing its real source variables as $(w,t)$:

$$\begin{aligned} p(z) &= \frac{z}{1+|z|}\exp\bigl(i\log(1+|z|)\bigr),\\ p^{-1}(\xi) &= \frac{\xi}{1-|\xi|}\exp\bigl(i\log(1-|\xi|)\bigr), \qquad z = w+it. \end{aligned} \tag{C.74.3}$$

These are inverse homeomorphisms between the real plane and $\mathbb{D}$ by their polar formulas, including at zero.

Recall the precise compact-family boundary control proved in C.72. If $|z'-z|\le M$ and $|z|>2M+1$, its direction, radial and logarithm estimates give

$$|p(z')-p(z)| \le \frac{4M}{|z|-M}. \tag{C.74.4}$$

Consequently every plane homeomorphism with bounded displacement, and with bounded inverse displacement, has a unique conjugate extension fixing $S^1$ pointwise. A compact parameter family with a common bound and jointly continuous interior maps and inverses has a jointly continuous extension with inverses. The proof is the uniform estimate (C.74.4) near the circle and the actual interior conjugacy elsewhere.

Put $\psi(t)=\max\{0,1-|t|\}$ and define

$$\begin{aligned} P_{1,a}(w,t) &= (w+a,t),\\ P_{2,a}(w,t) &= (w,t+a),\\ P_{3,a}(w,t) &= (w+a\psi(t),t) \qquad (0\le a\le 1). \end{aligned} \tag{C.74.5}$$

Their inverses have the same respective formulas with $-a$. Their forward and inverse displacements are at most one. Denote their actual extensions through (C.74.3) by $\overline{P}_{j,a}\in\mathcal{D}_\partial$. For $\tau\in[(j-1)/3,j/3]$, set

$$\alpha_j(\tau)=\min\{6(\tau-(j-1)/3),\,2-6(\tau-(j-1)/3)\}, \qquad f_\tau=\overline{P}_{j,\alpha_j(\tau)}. \tag{C.74.6}$$

Each amplitude is zero at the two endpoints and one at the midpoint. The three pieces therefore form one continuous based loop, with its inverse continuous as well.

There is already an exact relative contraction: replace each amplitude in (C.74.6) by $r\alpha_j(\tau)$. The common displacement bound one still applies. At $r=0$ this is the identity loop and at $r=1$ it is the original loop; all joins and the loop basepoint stay the identity. All three families commute with horizontal translations, so this contraction retains those actual operator equations.

*Transitivity and the full actual interior centralizer.* The values of the first path generate every horizontal translation, using products and inverses of parameters in $[0,1]$. The values of the second generate every vertical translation in the same way. Thus $G$ contains all translations of the $(w,t)$-plane and is transitive.

Any homeomorphism commuting with all plane translations is itself a translation: if its value at zero is $(a,b)$, commutation forces its value at $(w,t)$ to be $(w+a,t+b)$. Commutation with $P_{3,1}$ then says

$$\psi(t+b)=\psi(t) \quad (t\in\mathbb{R}). \tag{C.74.7}$$

At $t=0$, this gives $\psi(b)=1$, whose unique solution is $b=0$. Conversely every horizontal translation commutes with all three families. Consequently the full interior centralizer, including possible orientation-reversing candidates, is exactly

$$C_{\mathrm{Homeo}(\mathbb{D})}(G|_{\mathbb{D}})=Q=\{p\,q_a\,p^{-1}:a\in\mathbb{R}\}, \qquad q_a(w,t)=(w+a,t). \tag{C.74.8}$$

There are no orientation-reversing elements in this explicitly computed group.

The map $a \mapsto q_a$ is a topological isomorphism onto its actual interior image: evaluation at zero gives the inverse, and the formulas give continuity of the maps and inverses on every compact set. This image is closed, since a homeomorphic limit evaluated at zero has a finite real parameter and hence is the corresponding translation. The action is free. It is proper because two compact source sets have bounded first coordinates, which bound every translation parameter taking one set into the other. Conjugation by $p$ and its displayed inverse preserve these assertions. Thus $Q$ has dimension exactly one and the topology required in C.33. Its actual extensions to the closed disk are themselves boundary-fixed, by (C.74.4).

*Two actual principal sections and their transition.* The quotient by $Q$ is the real coordinate $t$. A first section is $\sigma_0(t) = p(0,t)$, with the principal coordinate

$$\Theta_0(w,t) = p(w,t).$$

A second section of the same quotient is $\sigma_1(t) = p(-t^3,t)$, and its principal coordinate is

$$\begin{aligned} \Theta_1(s,t) &= p(s-t^3,t), \\ H = \Theta_0^{-1}\Theta_1 &: (s,t) \longmapsto (s-t^3,t), \\ H^{-1} &: (w,t) \longmapsto (w+t^3,t). \end{aligned} \tag{C.74.9}$$

These are the actual coordinate maps and their inverse identities. In both, the same labelled element $pq_ap^{-1}$ acts by addition of $a$ to the first coordinate. In particular the second coordinate satisfies every C.33 principal bundle requirement; no operator has been changed by the section choice.

At the midpoint $\tau = 1/2$ of the second loop path, the original map is $pP_{2,1}p^{-1}$. More generally its vertical-translation path in the second coordinate is

$$F_a(s,t) = \big(s + (t+a)^3 - t^3, t+a\big) = H^{-1}P_{2,a}H(s,t). \tag{C.74.10}$$

Its literal interpolation with the identity in that coordinate is

$$M_{r,a}(s,t) = \big(s + r((t+a)^3 - t^3), t + ra\big). \tag{C.74.11}$$

It is an interior homeomorphism commuting with every horizontal translation. The corresponding calculation in the first coordinate is exact:

$$HM_{r,a}H^{-1}(w,t) = \big(w + 3r(1-r)a^2t + r(1-r^2)a^3,\ t + ra\big). \tag{C.74.12}$$

In particular $r = 1/2, a = 1$ gives

$$A(w,t) = \left(w + \frac{3}{4}t + \frac{3}{8}, t + \frac{1}{2}\right), \qquad A^{-1}(w,t) = \left(w - \frac{3}{4}t, t - \frac{1}{2}\right). \tag{C.74.13}$$

The actual candidate on the disk is $A_* = pAp^{-1}$. Formula (C.74.13) is also an explicit inverse calculation; it establishes the interior assertion before any boundary comparison.

*Two sequences at each original boundary point.* Fix $\zeta = e^{i\theta} \in S^1$ and take sufficiently large positive integers $n$. Define

$$\begin{aligned} R_n &= e^{2\pi n + \theta} - 1, \qquad & X_n &= (R_n, 0), \\ S_n &= e^{2\pi n + \theta - \pi/2} - 1, \qquad & Y_n &= (0, S_n). \end{aligned} \tag{C.74.14}$$

The polar formula (C.74.3) gives $p(X_n) \to \zeta$ and $p(Y_n) \to \zeta$. For the horizontal sequence, $A(X_n) - X_n = (3/8, 1/2)$ is bounded, so (C.74.4) gives $p(A(X_n)) \to \zeta$.

For the vertical sequence there is the exact factorization

$$A(Y_n) = \left(S_n + \frac{1}{2}\right)\left(\frac{3}{4}, 1\right). \tag{C.74.15}$$

Its argument is $\pi/2 - \arctan(3/4)$ and its norm is $\frac{5}{4}(S_n + \frac{1}{2})$. Thus the logarithmic radial change tends to $\log(5/4)$, and

$$p(A(Y_n)) \longrightarrow e^{i\delta}\zeta, \qquad \delta = \log(5/4) - \arctan(3/4). \tag{C.74.16}$$

The two limits are distinct without any numerical approximation. The integral formulas give $0 < \log(5/4) < 1/4$ and $\arctan(3/4) > (3/4)/(1 + (3/4)^2) = 12/25$. Consequently

$$-\frac{\pi}{2} < \delta < -\frac{23}{100} < 0. \tag{C.74.17}$$

In particular $e^{i\delta} \neq 1$. The two sequences (C.74.14) prove that $A_*$ has no continuous extension at this specified $\zeta$. Since $\zeta$ was arbitrary, this holds at every point of the original circle.

*The exact obstruction to the proposed filled-word bound.* Let $P \subset \mathcal{D}_\partial$ be any compact family, and fix $N \geq 1$. The product family $P^N$ is compact. Joint continuity of evaluation on this compact family, and its pointwise-fixed boundary, imply

$$\sup_{g \in P^N} |g(p(Y_n)) - \zeta| \longrightarrow 0. \tag{C.74.18}$$

Every planar filled hull is contained in the Euclidean convex hull of its compact set, as proved in C.70. Therefore every point of $\mathrm{Fill}(P^N(p(Y_n)))$ also tends uniformly to $\zeta$. Its proposed member $A_*(p(Y_n))$ instead tends to $e^{i\delta}\zeta$. This disproves (C.74.2), for these actual maps and for every such compact family and finite word length. Extending each original boundary-fixed map by the identity outside the closed disk makes the same assertion in the original plane homeomorphism group.

The original contraction (C.74.6) with scaled amplitudes remains valid and keeps the same operators. The failed construction is exactly the interpolation (C.74.1) in the specified changed principal section. A section-adapted filling or a controlled selection still requires its own boundary argument; this example does not furnish a nontrivial centralizer fundamental group.

The executable `code/part_b/c74_principal_interpolation_certificate.py` checks the coordinate and inverse identities, the group equations, the exact interpolation formula, the two ray calculations and the rational constants in (C.74.17). The limit and compact-family arguments above are topological proofs, with no floating-point verification used. □

**C.75. [Proved] The actual orbit strips for a one-dimensional proper operator action.** Let $O$ be a plane, including an actual open disk, and let $Q \leq \mathrm{Homeo}(O)$ be closed, locally compact, free and proper, with $\dim Q = 1$. Let $f_\tau$ be a continuous based loop in its centralizer, and put $G = \langle f_\tau \rangle$. Retain the actual principal coordinate of C.33, with its inverse, in which

$$\begin{aligned} O &\cong \mathbb{R}_s \times \mathbb{R}_t, \\ f_\tau(s,t) &= (s + u_\tau(t), \beta_\tau(t)), \\ A &= \{t : \beta_\tau(t) = t \text{ for every } \tau\}. \end{aligned} \tag{C.75.1}$$

The set $A$ is closed. On every component $I$ of $\mathbb{R} \setminus A$, precisely one of the following two descriptions holds. Either $\mathbb{R} \times I$ is one open $G$-orbit, or there is a continuous function $b_I : I \to \mathbb{R}$ such that all the orbits in that strip are exactly

$$\{(c + b_I(t), t) : t \in I\}, \qquad c \in \mathbb{R}. \tag{C.75.2}$$

In the latter case the actual coordinate $(s,t) \mapsto (s - b_I(t), t)$, with inverse $(w,t) \mapsto (w + b_I(t), t)$, changes every restricted generator to $(w,t) \mapsto (w, \beta_\tau(t))$. The effective restricted action and its base image are topologically isomorphic in their actual compact-open and inverse topologies.

At each $a \in A$ the orbits on $\mathbb{R} \times \{a\}$ are individual points when every $u_\tau(a)$ vanishes, and otherwise that entire horizontal line is one orbit. This lists all orbits. In particular every finite interior frontier line of an open orbit is the actual proper line at an endpoint of its base interval. There are no circular orbits. The graph in (C.75.2) is proper in its indicated strip; extension of $b_I$ to an endpoint is not asserted.

There is a further exact comparison for a finite endpoint $a \in A$ of a base interval $I$. Write $g(s,t) = (s + u_g(t), \beta_g(t))$ for the original element $g \in G$, and put

$$c_a(g) = u_g(a), \qquad G_t = \{g \in G : \beta_g(t) = t\} \quad (t \in I). \tag{C.75.3}$$

The character $c_a$ is continuous. There are exactly three alternatives: $c_a = 0$; or $c_a \neq 0$ and $c_a(G_t) = 0$; or $c_a(G_t) = \mathbb{R}$. In the second alternative there is an actual homeomorphism $b : I \to \mathbb{R}$ satisfying

$$b(\beta_g(t)) = b(t) + c_a(g) \qquad (g \in G,\ t \in I). \tag{C.75.4}$$

In the third alternative every real endpoint displacement is realized by an actual source element fixing the chosen base point $t$. All component-operator equations and the kernel of the comparison action used below are retained. No continuous choice of these correcting source elements, or frontier filling, is inferred.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact one-dimensional proper-operator orbit and frontier reduction.

*Proof. The principal coordinate and the original group words.* C.33 gives a Lie group with identity component $Q^0$, and its actual product principal bundle. Because $O$ is a plane, neither factor in that product can be a circle: a circle factor gives a nontrivial fundamental-group factor, whereas the plane has trivial fundamental group. Thus $Q^0 = \mathbb{R}$, its base is $\mathbb{R}$, and an actual section gives the coordinate and inverse in (C.75.1). Every labelled element of $Q^0$ acts by $(s,t) \mapsto (s + c, t)$.

Commutation with these translations forces every $g \in G$ to have the stated form. The base homeomorphism is increasing because each generator is joined to the identity. The inverse map is exactly

$$(s,t) \longmapsto (s - u_g(\beta_g^{-1}(t)), \beta_g^{-1}(t)). \tag{C.75.5}$$

For the composition convention $gh = g \circ h$,

$$\beta_{gh} = \beta_g \beta_h, \qquad u_{gh}(t) = u_h(t) + u_g(\beta_h(t)). \tag{C.75.6}$$

The functions and their inverse maps are jointly continuous on every compact parameter family.

Put $P = \{f_\tau\} \cup \{f_\tau^{-1}\}$, retaining the actual maps. It is compact, path connected, and contains the identity. Every $P^n$ is compact and path connected, and $\bigcup_n P^n = G$. More explicitly an element $g = f_{\tau_1}^{\epsilon_1} \cdots f_{\tau_m}^{\epsilon_m}$ has the actual path

$$r \longmapsto f_{r\tau_1}^{\epsilon_1} \cdots f_{r\tau_m}^{\epsilon_m}, \qquad 0 \leq r \leq 1, \tag{C.75.7}$$

from the identity to $g$, entirely in $P^m$. No path in a larger closure is substituted.

The fixed set of all base generators is exactly $A$, and it is also the fixed set of their generated base group. The base orbit of any $t \notin A$ is an open interval. Indeed choose a based path of base maps moving that point. That path and its inverse give points on both sides of it; their evaluation intervals give a neighborhood in the orbit. The same argument at each point of the orbit makes the orbit open. It is an interval because it is the increasing union of the connected

compact evaluation images of $P^n$. A finite endpoint of this orbit is fixed by every increasing base map preserving the orbit. Hence the orbit is exactly the component $I$ of $\mathbb{R} \setminus A$ containing $t$.

*The precise interval consequence of the word triangle.* For a continuous path $h_r$ of actual maps of the form in (C.75.1), starting at the identity, suppose that $h_1(s,t)$ has second coordinate $t$. The entire horizontal interval joining $(s,t)$ to $h_1(s,t)$ is contained in the orbit of $(s,t)$ under the group generated by that path. This is the exact C.62 word-triangle argument with second-coordinate quotient. Its hypotheses here can be checked directly: the source is the actual product plane, the real fibers are proper horizontal lines, the induced base maps are increasing, and the endpoint equality is real equality. No original-open-orbit hypothesis enters that triangle proof.

More explicitly, subdivide the path by the closed set of parameters at which its second-coordinate evaluation equals $t$. On an excursion $(a,b)$, put $A_0 = h_a(s,t)$ and $k_r = h_r h_a^{-1}$. The paths $k_r A_0$ and $k_b k_r^{-1} A_0$ have opposite strict second-coordinate signs relative to the original fiber. Their concatenation has winding 1 or $-1$ about every interior point of the interval between their two endpoints, as in C.62.9. The triangle $(v,u) \mapsto k_u k_v^{-1} A_0$ therefore contains that interval. Interpolating on the closed zero set and these excursion gaps, with uniform continuity on their compact parameter set, gives the complete endpoint interval, exactly as in C.62. Every displayed value is an actual three-word value from the path. For (C.75.7) it consequently belongs to $P^{3m}(s,t)$.

*The two possible strip actions.* For each $t \in I$, define the actual additive subgroup

$$D_t = \{u_g(t) : g \in G_t\} \leq \mathbb{R}. \tag{C.75.8}$$

Additivity and inverses follow from (C.75.5–6). If $h$ takes the base point $t$ to $t'$, conjugation by that actual map takes a shift $c \in D_t$ to the same shift $c$ in $D_{t'}$: its first-coordinate translations cancel. Thus $D_t = D_{t'}$ throughout this base orbit.

If $D_t$ contains a nonzero $c$, take its actual source element $g$ and the path (C.75.7). The preceding triangle puts every point of the interval from $(0,t)$ to $(c,t)$ in the original orbit. Each such orbit element fixes the base point, so $D_t$ contains the full interval from zero to $c$. An additive subgroup containing a nondegenerate interval is all of $\mathbb{R}$. Transitivity on the base interval, followed by these arbitrary fiber translations, proves that $\mathbb{R} \times I$ is one orbit.

Suppose instead that $D_t = 0$. Choose $t_0 \in I$ and $x_0 = (0, t_0)$. Every fiber over $I$ meets $Gx_0$ exactly once: existence follows from base transitivity, and two such points would give a nonzero element of $D_t$ after composing their source words. Write that point as $(b_I(t), t)$.

This function is continuous with its actual, not a postulated, graph topology. The compact set $P^n(x_0)$ projects continuously and bijectively to a compact interval $I_n$. A compact-to-Hausdorff bijection is a homeomorphism. The intervals $I_n$ increase and exhaust $I$. For each $t \in I$, choose $t_- < t < t_+$ in $I$. Both belong to some $I_n$, so $[t_-, t_+] \subset I_n$. Thus the inverse projection, and hence $b_I$, is continuous on a neighborhood of every $t$.

Every horizontal translation commutes with $G$. Its translates of $Gx_0$ are all the other orbits, proving (C.75.2). Invariance of this actual graph under $g$ gives

$$u_g(t) = b_I(\beta_g(t)) - b_I(t). \tag{C.75.9}$$

This verifies both conjugacy directions of the asserted coordinate change. The graph is closed in $\mathbb{R} \times I$, and its projection is a homeomorphism onto $I$, so it is proper in that strip.

Under this coordinate, the effective restricted group is exactly $\{\, \mathrm{id}_{\mathbb{R}} \times \beta \,:\, \beta \in \beta(G)|_I \,\}$. Projection to the base has the explicit continuous inverse $\beta \mapsto \mathrm{id} \times \beta$, also on inverse maps, in the compact-open topologies. Evaluation on the section $\{0\} \times I$ proves continuity in the other direction. This proves the stated topology of the effective images. It makes no quotient-topology claim for restriction from the original global group.

For $a \in A$, the first-coordinate shifts form the image of the path-connected group $G$ under the continuous evaluation $g \mapsto u_g(a)$. This is a connected additive subgroup of $\mathbb{R}$, hence zero

or $\mathbb{R}$. The zero alternative is exactly the vanishing of every generator $u_\tau(a)$. This proves the remaining orbit description. The frontier inside the product plane of an open strip is exactly the horizontal lines at its finite endpoints.

*The endpoint character with all source labels retained.* Fix the indicated finite endpoint $a$ and interval $I$. By (C.75.6), $c_a$ is an additive character, continuous in the original group topology. Form the labelled comparison action on a second product:

$$\begin{aligned} T_a : G &\longrightarrow \mathrm{Homeo}(\mathbb{R}_z \times I), \\ T_a(g)(z,t) &= (z + c_a(g), \beta_g(t)), \\ \ker T_a &= \{g : c_a(g) = 0,\ \beta_g|_I = \mathrm{id}\}. \end{aligned} \tag{C.75.10}$$

The displayed kernel is kept; no faithfulness or closedness of this image is assumed. Every finite word and every path (C.75.7) has precisely its displayed image under $T_a$.

The strip proof just given therefore applies to this actual image action. Its fiber stabilizer at $t$ is exactly $c_a(G_t)$, so it is zero or all of $\mathbb{R}$. In the latter case the same triangle supplies actual source words with every intermediate real displacement, and their finite products give every real displacement. They fix this specified base point; a continuous choice of these words has not been constructed.

In the zero case the actual graph function from that argument satisfies

$$b(\beta_g(t)) = b(t) + c_a(g). \tag{C.75.11}$$

Suppose that $c_a$ is nonzero. Its path-connected additive image is then $\mathbb{R}$, so $b(I) = \mathbb{R}$ by base transitivity and (C.75.11). We check that $b$ is injective. If two distinct arguments had the same value, either $b$ would be constant on the interval between them, or it would have an interior local maximum or minimum there. An open interval of constancy is transported by the transitive base action to neighborhoods of every point, which would make $b$ constant. A local maximum is similarly transported to every point. A continuous real function having a local maximum at every point has countable image: each of its values is the maximum on one rational closed interval inside a local maximum neighborhood. Its connected image must therefore be a singleton. The identical argument with minima handles the other case. All alternatives contradict $b(I) = \mathbb{R}$. Thus $b$ is a continuous injective surjection from an interval to $\mathbb{R}$, hence a strictly monotone homeomorphism. This proves (C.75.4) with its exact inverse coordinate.

*Component operators and the remaining boundary step.* C.33 retains either no nontrivial component operator or one labelled generator

$$q(s,t) = (\lambda s + \kappa, t + d), \qquad d > 0, \quad \lambda \neq 0. \tag{C.75.12}$$

The original commutation equations, with the constant $\kappa$ retained, are

$$\beta_g(t + d) = \beta_g(t) + d, \qquad u_g(t + d) = \lambda u_g(t). \tag{C.75.13}$$

They give $A + d = A$, carry each strip and its orbit description to the translated strip, and send its endpoint character to $\lambda c_a$. The actual maps $q$ and $q^{-1}$ prove both transport directions. Negative $\lambda$ is included.

All finite interior frontier strata in this reduction are consequently the named horizontal lines, with the original translation characters just computed. An actual disk coordinate can still have complicated limits at its original outer circle, as C.74 demonstrates. Neither the graph coordinate nor the endpoint correction surjectivity supplies a jointly continuous filling at that circle or across the nonfixed interior frontier lines. Those extension steps remain part of C.12. □

**C.76. [Proved] Simultaneous endpoint prescriptions with compact original-boundary control.** Let $Q$ be the actual closed, locally compact, free proper one-dimensional group on $O = \operatorname{int} \overline{D}$ of C.75. Let $f_\tau$ be a based loop of homeomorphisms of the actual closed disk, fixing its boundary pointwise and commuting with $Q$ on $O$. Use the actual principal coordinate and its inverse from C.75, with coordinates $(s,t)$, common base fixed set $A$, and generated group $G$.

Retain the following components $I$ of $\mathbb{R} \setminus A$: those for which every finite-endpoint character $c_a$ is zero, and those having a finite endpoint $a$ with

$$c_a \neq 0, \qquad c_a(G_t) = 0 \quad (t \in I). \tag{C.76.1}$$

The first prescription includes an interval without finite endpoints. Let $Z \subset \overline{D}$ consist of the original boundary circle, the actual lines over $A$, and all retained strips. This is a closed subset, preserved by the loop and by every actual interior operator wherever that operator is defined. The restricted loop has a based contraction through homeomorphisms of $Z$, with jointly continuous inverses, fixing the original circle and commuting with every $q \in Q$ on $Z \cap O$.

On every line over $a \in A$, the contraction is exactly

$$(s,a) \longmapsto (s + r u_\tau(a), a), \qquad 0 \leq r \leq 1. \tag{C.76.2}$$

On a strip retained by (C.76.1), choose C.75's actual homeomorphism $b : I \to \mathbb{R}$ for one such endpoint. The base contraction is

$$B_{r,\tau}(t) = b^{-1}\big(b(t) + r c_a(f_\tau)\big). \tag{C.76.3}$$

Both finite endpoint values, when there are two, are prescribed simultaneously by (C.76.2). All original component operators, including negative fiber multipliers, are retained. One compact family of original closed-disk homeomorphisms controls the forward and inverse evaluations over all retained strips and all parameters. Hence the construction includes infinitely many strips accumulating at an interior line or at the original circle.

More precisely, let $\mathscr{R}_Z$ be the subgroup of the original relative $Q$-centralizer preserving $Z$, and let

$$\begin{aligned} \mathscr{A}_Z &= \{k \in \operatorname{Homeo}(Z) : k|_{\partial \overline{D}} = \mathrm{id},\ kq = qk \text{ on } Z \cap O\}, \\ \rho_Z &: \mathscr{R}_Z \longrightarrow \mathscr{A}_Z, \qquad \rho_Z(h) = h|_Z. \end{aligned} \tag{C.76.4}$$

These are the actual compact-open and inverse topologies, and $\rho_Z$ is a continuous homomorphism. The contraction just asserted fills the actual loop $\rho_Z(f_\tau)$ in $\mathscr{A}_Z$. When every complementary interval is retained, $Z = \overline{D}$ and this is a contraction in the original relative centralizer itself. No lifting of the constructed filling through $\rho_Z$ is asserted for a proper $Z$.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), actual endpoint-prescribed and original-boundary-controlled continuation of the dimension-one proper-operator reduction.

*Proof. The endpoint image and its actual short increments.* For any one strip with $k \in \{1,2\}$ finite endpoints, write $c = (c_{a_1}, \dots, c_{a_k})$ and $V = c(G) \subset \mathbb{R}^k$. This is the image of the actual group $G$, before taking any closure. It is a real vector subspace. Indeed, every $g \in G$ has the finite-word path (C.75.7) from the identity. Its character image is a path from 0 to $c(g)$ in $V$. The planar three-word statement (C.37.3), using a zero second coordinate for $k = 1$, places every real fraction $rc(g)$, $0 \leq r \leq 1$, in $V$. Addition and negation then give all real multiples. This proves the assertion and, in particular, that $V$ is closed in $\mathbb{R}^k$. It is the real span of $\{c(f_\tau) : 0 \leq \tau \leq 1\}$: one inclusion follows from generation, and the other from the vector-space property just proved.

For $\delta > 0$, retain the actual compact family

$$\begin{aligned} J_\delta &= \{f_s f_t^{-1} : s,t \in [0,1],\ |s-t| \leq \delta\}, \\ E_\delta &= J_\delta^6, \qquad D_\delta = c(J_\delta). \end{aligned} \tag{C.76.5}$$

Both parameter families are compact and path connected, contain the identity, and include all smaller choices of $\delta$. Their global maps and inverses tend uniformly to the identity as $\delta \downarrow 0$, in the actual closed-disk homeomorphism topology. The set $D_\delta$ is compact, symmetric and path connected. It spans $V$: divide $[0,\tau]$ into intervals of length at most $\delta$ and telescope $c(f_\tau)$ into their character increments.

Choose a basis $v_1,\dots,v_d$ of $V$ in $D_\delta$, where $d \le 2$. For each $i$ a path in $D_\delta$ joins 0 to $v_i$. Again (C.37.3) gives $[-v_i, v_i] \subset D_\delta + D_\delta + D_\delta$; symmetry supplies the negative fractions. Consequently

$$\Big\{\sum_{i=1}^{d} a_i v_i : |a_i| \le 1\Big\} \subset c(E_\delta). \tag{C.76.6}$$

For $d < 2$ use identity factors to reach six increments. For $d = 0$ the assertion is the identity image. Thus $c(E_\delta)$ contains a neighborhood of 0 in the actual vector image $V$, for every $\delta > 0$. This is a statement about specified compact sets of actual source words. It does not choose a continuous inverse to $c$.

Put $P = \{f_\tau\} \cup \{f_\tau^{-1}\}$. The same three-word statement, now on the original prefix character path, gives the uniform inclusion

$$rc(f_\tau) \in c(P^3) \quad (0 \le r, \tau \le 1). \tag{C.76.7}$$

The single family $P^3$ works for every strip and both endpoint characters, including varying ranks of $V$.

*One compact family in the original relative group.* Take the closure $L$ of $G$ in the group of actual closed-disk homeomorphisms fixing the circle and commuting with $Q$ on the interior. This is a Polish group: the ambient compact-open homeomorphism topology controls maps and inverses, and the stated pointwise equations are closed. Every member of $L$ still has the principal form

$$k(s,t) = (s + u_k(t), \beta_k(t)), \tag{C.76.8}$$

with $\beta_k$ increasing, fixing $A$ pointwise and preserving every complementary interval. These are closed conditions for maps and inverses. Each finite-endpoint evaluation $c_a(k) = u_k(a)$ extends continuously to $L$. For any particular endpoint vector its image on $L$ is still $V$: density gives inclusion in the closed space $V$, and the original image already equals $V$. The translation equation (C.75.4) also passes to $L$ on each retained strip by pointwise continuity. No assertion about a quotient topology on $c(G)$ is used.

Choose a complete compatible metric on $L$. Exactly as in the compact-product argument of C.63, choose $\delta_n > 0$ so that left multiplication by any member of $E_{\delta_n}$ changes every previously formed product by less than $2^{-n}$. This is possible by compactness of the previous product family and the uniform identity limit in (C.76.5). The families

$$K_n = E_{\delta_n} \cdots E_{\delta_1} P^3, \qquad K_0 = P^3, \tag{C.76.9}$$

are nested, and their infinite products converge uniformly over the entire product of their compact parameter spaces. Completeness gives a compact limit family $K \subset L$ containing every $K_n$. It is path connected: parameterize $P$ by concatenating the original loop and its inverse, and use the convex time bands in (C.76.5). The product of these parameter spaces is path connected, and its uniform product map into $L$ is continuous. This includes convergence of the actual inverse maps.

Every member of $K$ fixes the original circle and commutes with every labelled $q \in Q$. It preserves all principal strips, whether retained or not. On a given strip the functions $u_k(t), \beta_k(t)$ in (C.76.8) are jointly continuous in its compact label and its real coordinate.

*Base prescriptions and nonempty fiber intervals.* On a strip with all finite-endpoint characters zero, use C.27's based median contraction $B_{r,\tau}$ of the original base loop. Its value is between $t$

and $\beta_\tau(t)$, and hence is attained by an original prefix value. Define

$$A_n(r,\tau,t) = \operatorname{conv}\{u_k(t) : k \in K_n,\ \beta_k(t) = B_{r,\tau}(t)\} \tag{C.76.10}$$

on this strip. These are nonempty compact intervals.

On a strip satisfying (C.76.1), choose one of its eligible endpoints $a$ and its actual coordinate $b$. Use (C.76.3), and include *all* finite endpoints in the vector $c$. Define instead

$$A_n(r,\tau,t) = \operatorname{conv}\{u_k(t) : k \in K_n,\ c(k) = rc(f_\tau)\}. \tag{C.76.11}$$

Equation (C.76.7) proves nonemptiness. For every label occurring in (C.76.11), the $a$ coordinate and the exact translation relation give

$$\beta_k(t) = B_{r,\tau}(t). \tag{C.76.12}$$

Thus (C.76.11) really is an interval of displacements in one specified principal fiber. Every one of its labels has the prescribed displacement at *both* finite endpoints. The base map (C.76.3) lies between $t$ and $\beta_\tau(t)$ even when $b$ is decreasing. It is an increasing homeomorphism of $I$ with jointly continuous inverse and all four based parameter faces.

In both definitions use $K$ in place of $K_n$ to obtain a nonempty compact interval $A_\infty$. The four desired fiber prescriptions are

$$\begin{aligned} &U(0,\tau,t) = 0, \qquad U(1,\tau,t) = u_\tau(t),\\ &U(r,0,t) = U(r,1,t) = 0. \end{aligned} \tag{C.76.13}$$

They belong to every $A_n$ on the respective faces.

*The exact lower-limit comparison and relative selection.* For each of (C.76.10) and (C.76.11) one has

$$A_n(p) \subset \liminf_{p' \to p} A_{n+1}(p'), \qquad p = (r,\tau,t). \tag{C.76.14}$$

For (C.76.10), the short-increment argument of C.63 applies directly. The base group is transitive on $I$ by C.75. At every base point, arbitrarily short original increments and their inverses move that point to both sides. The inverse-evaluation argument (C.63.9) proves this assertion for the actual specified point. These increments occur in $E_{\delta_{n+1}}$, with five identity factors added. The intermediate value theorem gives a correction of the nearby target level, as close to the identity as desired.

For (C.76.11), take a label $k \in K_n$ with $c(k) = rc(f_\tau)$. At a nearby parameter the required character correction is the actual vector

$$v(p') = r'c(f_{\tau'}) - c(k) \in V, \qquad v(p') \longrightarrow 0. \tag{C.76.15}$$

Choose any arbitrarily small $\delta' \le \delta_{n+1}$. By (C.76.6), every sufficiently nearby $p'$ has an actual $e_{p'} \in E_{\delta'}$ with $c(e_{p'}) = v(p')$. Then $e_{p'}k \in K_{n+1}$ has precisely the new required character vector. Equation (C.76.12) supplies its base level at $t'$ automatically. As $\delta' \downarrow 0$, its fiber displacement at $t'$ tends uniformly to that of $k$ near the chosen source point. This proves the lower-limit inclusion for every unconvexified value. Applying it to two endpoints and retaining their convex coefficient proves (C.76.14) for the whole interval. No continuous choice of $e_{p'}$ was used.

Replace the allowed sets by the prescribed singletons (C.76.13) on the closed parameter boundary. The lower-limit property survives by the same two comparisons as in C.63: off that boundary it is (C.76.14), and along the boundary it is continuity of the prescribed values. Here are the precise selection ingredients. Choose at each parameter a value of the $n$th interval, extend it locally as a constant section, and use (C.76.14) to make it arbitrarily close to the next interval. A locally finite partition of unity combines these approximations, since every fiber set is convex.

Starting with error $2^{-1}$, inductively choose continuous approximations $U_n$ with distance less than $2^{-n}$ from the $n$th modified interval and with $|U_{n+1} - U_n| < 2^{1-n}$. Their locally uniform limit is continuous, belongs to the closed compact interval $A_\infty$ at every parameter, and equals each prescribed singleton exactly. This is C.63's relative selection construction with the lower-limit hypothesis now proved for the simultaneous character constraints.

*Every original component operator.* If $Q$ is connected, all its operators are the original fiber translations and the selected maps commute with them. Otherwise retain C.33's actual component generator

$$q(s,t) = (\lambda s + \kappa, t + d), \quad d > 0, \quad \lambda \neq 0. \tag{C.76.16}$$

All members of $L$, and hence all labels in $K$, obey

$$u_k(t + d) = \lambda u_k(t), \qquad \beta_k(t + d) = \beta_k(t) + d. \tag{C.76.17}$$

Consequently $A + d = A$, and the retained intervals are permuted with all their exact endpoint criteria: the character at $a + d$ is $\lambda c_a$. For a proper interval $I \neq \mathbb{R}$, no nonzero power of the base translation preserves $I$. Choose the construction on one representative of each interval orbit and transport it by the actual powers of $q$. For the endpoint coordinate the transported choice is $b'(t + d) = \lambda b(t)$, with any additive constant permitted; (C.76.3) then transports exactly. The fiber displacement transports by $U'(r, \tau, t + d) = \lambda U(r, \tau, t)$. An operator stabilizing that proper interval belongs to $Q^0$ and is an actual fiber translation, so this transport is independent of its representative.

The only interval which can stabilize under a nonzero component power is $I = \mathbb{R}$, when $A$ is empty. It is retained by the zero-endpoint rule. Perform its relative selection in the line bundle $(t, z) \sim (t + d, \lambda z)$ exactly as in (C.63.20). Its continuous fiber norm is $|\lambda|^{-t/d}|z|$; the preceding approximation bounds are taken in this norm. The convex fiber sets and their lower-limit property are well-defined there by (C.76.17), and the median commutes with the actual base translation. This covers the negative values of $\lambda$ as well. The constant $\kappa$ cancels in the commutation equation and has not been removed from the operator label.

*The homeomorphisms and their compact evaluation bounds.* On each retained strip put

$$\begin{aligned} K_{r,\tau}(s,t) &= (s + U(r,\tau,t), B_{r,\tau}(t)), \\ K_{r,\tau}^{-1}(s,v) &= \big(s - U(r,\tau,B_{r,\tau}^{-1}(v)), B_{r,\tau}^{-1}(v)\big). \end{aligned} \tag{C.76.18}$$

These are exact inverse homeomorphisms, jointly continuous on each strip. Use (C.76.2) on lines over $A$ and the identity on the original circle. The formulas preserve each piece of $Z$ and have the four required based faces. Their operator commutation follows from (C.76.16)–(C.76.17) and the transported construction.

At every source point of a retained strip the selected value lies between two $K$-values in one real fiber, by (C.76.10)–(C.76.12). Connect their labels by a path in the compact parameter space for $K$. The C.62 word triangle, in precisely the form (C.63.14), gives

$$K_{r,\tau}(x) \in KK^{-1}K(x), \qquad K_{r,\tau}^{-1}(x) \in K^{-1}KK^{-1}(x). \tag{C.76.19}$$

For the inverse inclusion, the two increasing fiber maps bracket the inverse point by their own inverses in their common input fiber, as in (C.63.26). On a line over $a \in A$, the real path $\sigma \mapsto u_\sigma(a)$ joins 0 to $u_\tau(a)$ on the prefix $[0, \tau]$. It attains $ru_\tau(a)$. Thus the forward value in (C.76.2) is an original prefix value, and its inverse is an original inverse prefix value. Since $P \subset K$, the two bounds (C.76.19) hold there as well. The bounds persist under all component transports because each actual member of $K$ commutes with $q$ on $O$.

*All finite endpoint limits, including varying strips.* Suppose $t_n \to a \in A$, with $t_n$ in retained strips $I_n$, and allow both homotopy parameters to vary. Choose a finite endpoint $a_n$ of $I_n$ lying

between $t_n$ and $a$. Then $a_n \to a$. For a zero-character strip every $k \in K$ satisfies $u_k(a_n) = 0$. For a character-prescribed strip every label used in its defining interval satisfies

$$u_k(a_n) = r_n u_{\tau_n}(a_n), \tag{C.76.20}$$

because the vector constraint includes both finite endpoints. This same equality holds for the zero-character strip, whose right-hand side is zero. Compactness of $K$ in the actual group makes $u_k(t_n) - u_k(a_n) \to 0$ uniformly in $k$: the principal coordinate and its inverse are continuous on a compact neighborhood of the finite line point under consideration. Convexification preserves this estimate. It follows that

$$U(r_n, \tau_n, t_n) - r_n u_{\tau_n}(a_n) \longrightarrow 0. \tag{C.76.21}$$

The original compact loop makes its latter endpoint values jointly continuous. Both base prescriptions are between $t$ and $\beta_\tau(t)$, so their base values tend to $a$ uniformly in the parameters. Equations (C.76.20)–(C.76.21) give exactly (C.76.2) at the limiting line, also when $a$ is not itself an endpoint of one fixed interval.

For inverses, a base map lying pointwise between the identity and $\beta_\tau$ has inverse lying pointwise between the identity and $\beta_\tau^{-1}$. This follows by applying the two increasing maps to the inequalities at its inverse argument. Hence an inverse base argument approaching $a$ also tends to $a$ uniformly. Applying (C.76.21) at that argument proves the inverse limit with displacement $-r u_\tau(a)$. Along $A$ itself, both limits follow directly from (C.76.2).

*The original circle and the exact restriction map.* The two word families in (C.76.19) are compact families of the original closed-disk maps, and every member fixes the original circle pointwise. For $x_n \to \zeta$ on that circle, their evaluation values at $x_n$ tend uniformly to $\zeta$. Thus (C.76.19) proves joint continuity of both the constructed maps and their inverses at every original boundary point, with no restriction on the escaping principal coordinates or on the varying strip index. The circle is fixed throughout. The set $Z$ is closed because its complement in the disk is exactly the union of the excluded open strips. The preceding bijections and their continuous inverses are therefore actual homeomorphisms of this specified compact space, and the compact-parameter evaluation criterion gives continuity into its actual homeomorphism group.

Restriction to a closed invariant subset is continuous for maps and inverses, so (C.76.4) is the asserted continuous group homomorphism. Its source contains the original loop. The formulas construct the claimed filling of that loop's restriction; no assertion that every intermediate map extends through an excluded strip is made. The excluded intervals have a nonzero finite-endpoint character but none satisfying (C.76.1). C.75 identifies their nonzero endpoint-stabilizer images as the full real line. Choosing a continuous, boundary-controlled lift there remains a separate step in C.12. In particular no source-word surjectivity alone has been substituted for that lift. □

**C.77. [Proved] All abelian base strips and the exact remaining character action.** Keep the actual closed disk, proper one-dimensional operator group $Q$, loop, principal coordinate, and common base fixed set $A$ of C.76. For a component $I$ of $\mathbb{R} \setminus A$, let

$$B_I = \{\beta_g|_I : g \in G\} \leq \mathrm{Homeo}^+(I) \tag{C.77.1}$$

be its actual effective base image, with its induced compact-open and inverse topology. Retain every strip whose finite-endpoint characters are all zero, and every strip for which $B_I$ is abelian. Let $Z_{\mathrm{ab}}$ be their union with all lines over $A$ and the original circle. The restriction of the original loop to this closed set has a based contraction through its actual homeomorphisms, fixing the original circle and commuting with $Q$ on its interior part. The inverse family is jointly continuous. One compact family of original closed-disk maps controls all evaluations, including arbitrary accumulation of strips.

For an abelian base image there are an actual increasing homeomorphism $b_I : I \to \mathbb{R}$ and a continuous source character $d_I : G \to \mathbb{R}$ such that

$$b_I(\beta_g(t)) = b_I(t) + d_I(g), \qquad d_I(G) = \mathbb{R}. \tag{C.77.2}$$

The prescribed base contraction is $b_I^{-1}(b_I(t) + rd_I(f_\tau))$, and each finite endpoint has the simultaneous displacement $ru_\tau(a)$. This includes abelian base strips with full-real endpoint-stabilizer character images. The exact restriction homomorphism is the map (C.76.4) with $Z_{\mathrm{ab}}$ in place of $Z$; no lifting through excluded strips is asserted. The set $Z$ of C.76 is contained in $Z_{\mathrm{ab}}$ by the actual translation comparison (C.75.4).

There is also an exact description of the remaining character comparison. On a strip with nonabelian $B_I$, write $c : G \to V \subset \mathbb{R}^k$ for the vector of its finite-endpoint characters, $0 \le k \le 2$. Then

$$c(G_t) = V \quad (t \in I), \qquad T(g)(z,t) = (z + c(g), \beta_g(t)) \tag{C.77.3}$$

defines a transitive action on the actual space $V \times I$. Let $L$ be the actual closed relative group used in C.76, and let $\mathscr{P}(L)$ be its Polish based path group. For each $t \in I$, endpoint evaluation gives an open surjection and the indicated homeomorphism

$$\begin{aligned} e_t : \mathscr{P}(L) &\longrightarrow V \times I, \\ e_t(\sigma) &= (c(\sigma(1)), \beta_{\sigma(1)}(t)), \\ \mathscr{P}(L)/J_t &\xrightarrow{\cong} V \times I, \\ \sigma J_t &\longmapsto e_t(\sigma). \end{aligned} \tag{C.77.4}$$

where $J_t = e_t^{-1}(0,t)$ is the actual stabilizer. Every sufficiently small compact target set has a compact preimage in any specified open path neighborhood whose image contains it. No continuous section, nor uniform compact preimage for targets escaping towards a strip endpoint, is inferred.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), enlargement of the actual one-dimensional proper-operator filling and exact remaining character/base quotient.

*Proof. The actual topology of an abelian base image.* C.75 proves that $B_I$ is transitive on $I$. Take its closure $\overline{B}_I$ in the actual homeomorphism group of $I$. It is abelian and still transitive. Its action is free: an element fixing one point commutes with the transitive original image and consequently fixes every point. The verified Polish orbit-open-map theorem used in C.30 identifies evaluation $\overline{B}_I \to I$ homeomorphically, since its stabilizer is trivial. The closure is locally compact. The one-dimensional group comparison in (C.37.9) now identifies its actual action, through this evaluation homeomorphism, with the translation action of $\mathbb{R}$. This is exactly C.37's transitive abelian line argument, including its faithful action and topology hypotheses. Because $B_I$ was already transitive, freeness of the closure shows $B_I = \overline{B}_I$: the original element with a specified evaluation value is the only element of the closure with that value.

Choose the orientation of its real translation coordinate so that $b_I$ is increasing. Then

$$d_I(g) = b_I(\beta_g(t_0)) - b_I(t_0) \tag{C.77.5}$$

for any fixed $t_0 \in I$ gives (C.77.2). It is a character, is independent of the chosen point $t_0$, and is continuous on the actual source group by the displayed evaluation formula. Its kernel is exactly $\{g : \beta_g|_I = \mathrm{id}\}$. Its image is all of $\mathbb{R}$, by transitivity. Thus the continuous source map and the topological isomorphism of its effective base image have both been specified; a quotient-topology claim for the original source restriction is unnecessary.

*Three actual characters and their compact representatives.* For an abelian base strip put

$$a_I(g) = (d_I(g), c_{a_1}(g), \dots, c_{a_k}(g)) \in \mathbb{R}^{k+1}, \qquad k \le 2, \tag{C.77.6}$$

including every finite endpoint, and write $V_I = a_I(G)$. The three-dimensional additive path identity (C.50.3) applies after adding zero coordinates when needed. For every actual finite-word path from 1 to $g$, it places each real fraction of $a_I(g)$ in $V_I$ using at most five signed path values. Hence $V_I$ is a real vector subspace of dimension at most three, before passing to any closure. It is the real span of the original character path and is closed in this Euclidean space. With $P = \{f_\tau\} \cup \{f_\tau^{-1}\}$ one has

$$ra_I(f_\tau) \in a_I(P^5) \quad (0 \le r, \tau \le 1). \tag{C.77.7}$$

Retain the actual short-increment family $J_\delta$ of (C.76.5), and now put $E_\delta = J_\delta^{15}$. Telescoping the original character path proves that $a_I(J_\delta)$ spans $V_I$ for every $\delta > 0$. It is compact, symmetric and path connected. Choose a basis of at most three vectors in this set. The identity (C.50.3) puts the full segment from minus to plus each basis vector in five signed values from that set; symmetry absorbs the signs. Adding the at most three segments therefore proves

$$a_I(E_\delta) \text{ contains a neighborhood of 0 in the actual } V_I. \tag{C.77.8}$$

Identity factors pad shorter products. The numbers five and fifteen do not depend on the strip, the dimension of its image, or the parameter values. All original maps and inverses in $E_\delta$ tend uniformly to the identity as $\delta \downarrow 0$.

The character $d_I$ extends continuously to $L$ by (C.77.5). The full translation equation is a closed pointwise relation and still holds there. Each endpoint character extends by its actual line evaluation. Density and closedness of $V_I$ show $a_I(L) = V_I$, with no replacement of the original image by a larger vector space. Construct the one compact path-connected global family $K$ from

$$K_0 = P^5, \qquad K_n = E_{\delta_n} \cdots E_{\delta_1} P^5 \tag{C.77.9}$$

as in (C.76.9). Complete compatible metrics, summable left-product errors, compact parameter products, and inverse control apply verbatim to these actual finite families. Every element of $K$ is a homeomorphism of the original closed disk fixing its circle and commuting with the original operators.

*The selection, every endpoint, and the original circle.* On an abelian base strip use the base prescription of the claim. Define the allowed fiber intervals by

$$A_n(r, \tau, t) = \operatorname{conv}\{u_k(t) : k \in K_n,\ a_I(k) = ra_I(f_\tau)\}. \tag{C.77.10}$$

Equation (C.77.7) proves nonemptiness. Every label in this set has exactly the prescribed base map by its $d_I$ coordinate, and exactly the prescribed displacement at every finite endpoint by the other coordinates. For nearby parameters, the needed correction is $r' a_I(f_{\tau'}) - a_I(k) \in V_I$. Equation (C.77.8), with an arbitrarily smaller $\delta' \le \delta_{n+1}$, realizes that correction by an actual arbitrarily globally small increment. Thus the complete lower-limit proof (C.76.14)–(C.76.15) holds with $a_I$ in place of $c$. It uses no selected continuous inverse of $a_I$.

On a retained nonabelian strip, all finite-endpoint characters are zero by its definition. Use the median base prescription and (C.76.10). The original single short increments still lie in the new fifteenfold family, so its lower-limit proof is unchanged. The four parameter-face values (C.76.13) belong to every defining interval. The relative partition-of-unity approximation and summable limiting selection of C.76 therefore give a continuous fiber displacement $U$. It lies in the compact limiting interval and has those four faces exactly. Use the same explicit homeomorphism and inverse formulas (C.76.18).

Every actual component operator still has the form $q(s, t) = (\lambda s + \kappa, t + d)$ of C.75. Transport the chosen abelian base coordinate by $b_{I+d}(t + d) = b_I(t)$. This makes the transported base character equal to $d_I$, while each transported endpoint character is $\lambda c_a$. Equations (C.77.2) and (C.77.10) therefore commute with the actual $q$, including negative $\lambda$. For proper intervals choose orbit representatives and transport the fiber selections by multiplication by $\lambda$, as in C.76. No

nonzero component power stabilizes a proper interval. If $I = \mathbb{R}$ and the base image is abelian, the actual base translation $t \mapsto t + d$ commutes with every $\beta_g$. In the $b_I$ coordinate it therefore commutes with all real translations and is itself a translation: $b_I(t + d) = b_I(t) + e$ for one $e > 0$. The fractional base prescription consequently satisfies $B_{r,\tau}(t + d) = B_{r,\tau}(t) + d$. The vector $a_I$ then has only its $d_I$ coordinate, independent of the evaluation point. Its intervals (C.77.10) obey the exact multiplier equation $A_n(r, \tau, t + d) = \lambda A_n(r, \tau, t)$. Perform the selection in the actual line bundle (C.63.20). This retains the claimed fractional base prescription and every component equation, including negative $\lambda$. Thus every interval and every original operator label are covered.

At a finite endpoint $a$, every label defining an abelian-strip fiber interval has $u_k(a) = ru_\tau(a)$; on a zero-character strip the same equality has both sides zero. These are exactly the hypotheses used in (C.76.20)–(C.76.21). Compactness of the single global $K$ gives the same uniform endpoint estimate over varying intervals. Every base prescription lies between the identity and the original base map; their inverses have the corresponding inverse bounds. Consequently the homeomorphisms and inverses glue to (C.76.2) on all lines over $A$, including accumulation points which are not endpoints of any one fixed interval.

Each selected fiber value is between two $K$-values at the same real base level. The labelled triangle comparison therefore gives the same bounds

$$K_{r,\tau}(x) \in KK^{-1}K(x), \qquad K_{r,\tau}^{-1}(x) \in K^{-1}KK^{-1}(x). \tag{C.77.11}$$

On the lines over $A$, these follow from the original prefix path as in C.76. Compactness of these actual original families, all fixing the original circle, proves joint boundary continuity and inverse continuity uniformly over all strips. This verifies the claimed contraction and the continuous restriction map in the actual topology, by the final argument of C.76.

Every interval retained by (C.76.1) has an actual base image conjugate to all real translations by (C.75.4) and the nonzero character's full real image. Its base image is therefore abelian, proving the specified inclusion $Z \subset Z_{\mathrm{ab}}$. On the subgroup of $\mathrm{Homeo}(Z_{\mathrm{ab}})$ preserving $Z$, restriction to $Z$ is a continuous homomorphism for maps and inverses. Every constructed map belongs to that subgroup, since it preserves each strip and each line over $A$. This actual restriction sends the new filling to a filling of the same original restricted loop. No identity between the two independently selected interior homotopies is required or asserted.

*The nonabelian base and the actual transitive kernel.* Now let $B_I$ be nonabelian, and put $D = [G, G]$. This is a path-connected subgroup of the actual source group: every $g, h \in G$ has the finite-word paths (C.75.7), so their commutator has a commutator path from the identity; finite products give paths to every element of $D$. Its base image is nontrivial, normal in $B_I$, and path connected. Its common fixed set is $B_I$-invariant, and hence is either empty or all of $I$, since $B_I$ is transitive. Nontriviality excludes the latter.

For every point of $I$, an actual commutator-product path moves it to a different point; its inverse path supplies movement to the other side. The evaluation interval therefore contains a neighborhood of that point. Each $\beta(D)$-orbit is path connected and open. These disjoint orbits partition the connected interval $I$, so there is exactly one. Thus $\beta(D)$ is transitive. All endpoint characters vanish on $D$, because they are additive characters.

For $v \in V = c(G)$ choose an actual $g \in G$ with $c(g) = v$. For any desired $t' \in I$, choose $h \in D$ with $\beta_h(\beta_g(t)) = t'$. Then $hg$ has character $v$ and the desired base value. Taking $t' = t$ proves $c(G_t) = V$, and arbitrary $t'$ proves (C.77.3). The inverse of its displayed map is exactly $(z, t) \mapsto (z - c(g), \beta_g^{-1}(t))$. Both map and inverse depend continuously on the actual source element on every compact target set, by continuity of the character and of base restriction. Its action kernel is exactly

$$\{g \in G : c(g) = 0, \beta_g|_I = \mathrm{id}\}. \tag{C.77.12}$$

This proves the statement for the actual source words, not only for their closure.

The vector space $V$ and the endpoint characters extend to $L$ as in C.76. Endpoint action gives a continuous action of the Polish based path group $\mathscr{P}(L)$ on $V \times I$. Every original word $hg$ just used is reached by its finite-word path, so this path-group action is transitive. The target is the nonmeagre Hausdorff manifold $V \times I$. The exact orbit-open-map theorem already verified in C.30 makes $e_t$ in (C.77.4) open. Its fibers are the actual cosets of its closed stabilizer $J_t$. The displayed quotient map is consequently a continuous open bijection and hence a homeomorphism. The kernel of the whole path-group action consists of the paths whose endpoints satisfy (C.77.12), now read in $L$; it has not been suppressed.

Apply C.64's proved compact-covering lemma to this open map, using the complete compatible path metric. For every compact target set lying in the image of an assigned open path neighborhood, it provides a compact preimage entirely inside that neighborhood. The union of the values of these compact paths is also compact in the original relative group, including the inverse maps. This proves the final compactness assertion. The argument supplies no single compact family for a collection of target points escaping to a finite endpoint or to infinity in the base coordinate. The original-boundary-controlled selection on these remaining nonabelian strips is therefore still a precise unresolved step in C.12. □

**C.78. [Proved] Rank-one endpoint images with a controlled joint character and base lift.** Keep the actual disk, proper one-dimensional interior operator group $Q$, original relative loop, principal coordinate, and set $A$ of C.76–C.77. For each component $I$ of $\mathbb{R} \setminus A$, let $B_I$ be its actual base image, and let $V_I \subset \mathbb{R}^k$, $0 \le k \le 2$, be the actual vector image of all finite-endpoint characters. Retain the strips satisfying

$$B_I \text{ is abelian} \quad \text{or} \quad \dim V_I \le 1. \tag{C.78.1}$$

Let $Z_1$ consist of these strips, all lines over $A$, and the original boundary circle. The original loop restricted to this closed set has a based contraction through its actual homeomorphisms, fixing the circle and commuting with each original interior operator. Both the contraction and its inverse are jointly continuous. On every line over $a \in A$ its value is exactly $(s, a) \mapsto (s + ru_\tau(a), a)$.

The construction includes arbitrary accumulation of strips at interior lines and at the original circle. It gives an explicit continuous restriction homomorphism as in (C.76.4), with $Z_1$ as target subset. When every strip is retained it contracts the original loop in its original relative centralizer. For a proper $Z_1$ no lift through the excluded strips is asserted. Every excluded strip is a bounded interval $I = (a, b)$ with nonabelian base image and

$$(c_a, c_b)(G) = \mathbb{R}^2. \tag{C.78.2}$$

Two exact constructions supply the new assertion. Compact convex averages of actual source maps give a continuous increasing base deformation. The C.62 triangle in the scalar-character/base comparison plane realizes every such value by one compact three-word family of actual original maps. A second selection uses compact local pair transports for the exact Polish path-group actions. No continuous section of an orbit or character map is assumed.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact rank-one endpoint-image filling and remaining two-character bounded-strip obstruction.

*Proof. The first compact source family.* Use the original closed relative group $L$ of C.76, which is the closure of the group generated by the original loop in its actual closed-disk topology. Every member has the principal form

$$k(s, t) = (s + u_k(t), \beta_k(t)), \tag{C.78.3}$$

preserves each interval and each line over $A$, fixes the original circle, and commutes with $Q$. Its maps and inverse maps are controlled by a complete compatible metric. Put $P = \{f_\tau\} \cup \{f_\tau^{-1}\}$.

Construct the compact path-connected family $K$ and its nested compact families $K_n$ exactly as in (C.77.9), starting with $K_0 = P^5$ and using fifteen short increments at each stage. Their compact parameter products and uniform limits take place in the actual group $L$.

The newly retained strips have nonabelian $B_I$ and $\dim V_I = 1$. Choose one nonzero endpoint character $c = c_a$ on such a strip. Every other finite-endpoint character is a real multiple of $c$: this is exactly the assertion that the actual vector image $V_I$ is a line. The relation holds on $L$ by continuity. The scalar image is $c(G) = c(L) = \mathbb{R}$. For $p = (r, \tau) \in [0,1]^2$, put

$$F_n(p) = \{k \in K_n : c(k) = rc(f_\tau)\}, \qquad F_\infty(p) = \{k \in K : c(k) = rc(f_\tau)\}. \tag{C.78.4}$$

These are nonempty compact sets. Nonemptiness already follows from the original prefix, since the real path $c(f_\sigma)$ joins 0 to $c(f_\tau)$. All labels in a given set have the prescribed fraction of *every* finite endpoint character.

*A compact space of positive averages, with its exact topology.* Choose a countable dense sequence $\varphi_j$ in the unit ball of $C(K, \mathbb{R})$ and define

$$\chi(k) = (2^{-j}\varphi_j(k))_{j\geq 1} \in \ell^2, \qquad \mathcal{P}(F) = \overline{\mathrm{conv}}\chi(F) \quad (F \subset K \text{ compact}). \tag{C.78.5}$$

The map $\chi$ is continuous and injective: the dense continuous functions separate distinct points. Its inverse on its image is continuous by compactness. The closed convex hull $\mathcal{P}(K)$ is compact. Indeed, a finite $\epsilon$-net in $\chi(K)$ places its convex hull within $\epsilon$ of the convex hull of a finite set. That latter hull is a compact finite-dimensional polytope. Thus the full hull is totally bounded; its closure in the complete Hilbert space is compact. The same proof applies to every $\mathcal{P}(F)$.

Every $q \in \mathcal{P}(K)$ defines a positive average $\Lambda_q(h)$ of a continuous function $h \in C(K)$. To give the exact definition, approximate $q$ by finite convex combinations of the vectors $\chi(k)$ and take the corresponding averages of $h(k)$. This has a unique limit: approximate $h/\max(1, \|h\|_\infty)$ uniformly by one of the $\varphi_j$, whose average is the specified $j$th coordinate multiplied by $2^j$. The uniform approximation bounds the discrepancy for any two approximating convex combinations. It also proves independence of the approximating sequence. The resulting functional is linear, positive, sends the constant 1 to 1, and has $|\Lambda_q(h)| \leq \|h\|_\infty$. The same approximation proves joint continuity in $q$ and in $h$ with its uniform norm. For $q \in \mathcal{P}(F)$ one moreover has

$$\min_F h \leq \Lambda_q(h) \leq \max_F h. \tag{C.78.6}$$

Thus no representation theorem for measures or additional selection theorem is needed.

There is a continuous choice

$$q_p \in \mathcal{P}(F_\infty(p)), \qquad q_{0,\tau} = \chi(1), \quad q_{1,\tau} = \chi(f_\tau), \quad q_{r,0} = q_{r,1} = \chi(1). \tag{C.78.7}$$

Here is its full lower-limit input. If $k \in F_n(p)$, the correction $r'c(f_{\tau'}) - c(k)$ tends to zero. C.77's short-increment neighborhood statement realizes it by an actual arbitrarily small increment $e$ in the next allowed family. The same increment corrects every label of $F_n(p)$, since those labels have the same character. For a finite convex combination of their $\chi$-images, replace every $k$ by $ek$. Uniform continuity of $\chi$ on the compact source family makes the new combination arbitrarily close. Approximation by finite convex combinations therefore proves

$$\mathcal{P}(F_n(p)) \subset \liminf_{p' \to p} \mathcal{P}(F_{n+1}(p')). \tag{C.78.8}$$

Replace these convex sets on the four parameter faces by the prescribed singletons in (C.78.7). The lower-limit inclusion survives by continuity of those prescribed values. Apply the elementary relative approximation argument of (C.63.22), now in the Hilbert space: extend one selected vector locally as a constant, use a partition of unity, and make successive errors less than $2^{-n}$ with successive differences less than $2^{1-n}$. Every approximation stays in the convex compact set

$\mathcal{P}(K)$. Completeness gives a continuous limit. At each parameter all approximating allowed sets lie in the compact closed $\mathcal{P}(F_\infty(p))$, so the limit lies there; on the parameter faces the error to the prescribed singleton tends to zero. This proves (C.78.7).

*The base average and its actual three-word realization.* On the new strip define

$$B_p(t) = \Lambda_{q_p}\big(k \mapsto \beta_k(t)\big). \tag{C.78.9}$$

It is jointly continuous: on each compact real coordinate interval, $t \mapsto (k \mapsto \beta_k(t))$ is continuous into $C(K)$ in its uniform norm. For $t_1 < t_2$, compactness gives

$$\min_{k\in K}\big(\beta_k(t_2) - \beta_k(t_1)\big) > 0. \tag{C.78.10}$$

Positivity of the average makes $B_p$ strictly increasing. It is proper onto $I$: for each $v \in I$, the compact set $\{\beta_k^{-1}(v) : k \in K\}$ is bounded away from both endpoints of $I$. Beyond its upper bound every $\beta_k(t)$ exceeds $v$, and below its lower bound every $\beta_k(t)$ is less than $v$. The same statements hold for their average. Thus $B_p$ is an increasing homeomorphism of $I$. Strict increase and joint continuity give a jointly continuous inverse, by bracketing its value between two fixed nearby input points. The four faces in (C.78.7) give the required identity and original base maps exactly.

Use the actual comparison-plane action

$$\begin{aligned} T_I(k)(z,t) &= (z + c(k), \beta_k(t)),\\ T_I(k)^{-1}(z,t) &= (z - c(k), \beta_k^{-1}(t)). \end{aligned} \tag{C.78.11}$$

The source $\mathbb{R} \times I$ is a plane; its proper real fibers for the target $z$ are the vertical copies of $I$. Every displayed map preserves that partition and induces an increasing translation on its real target. By (C.78.6), the value $B_p(t)$ lies between the smallest and largest values $\beta_k(t)$ for $k \in F_\infty(p)$. Both extrema have actual compact labels in $K$, and those endpoints have the same $z$ coordinate $rc(f_\tau)$ in (C.78.11).

Connect their labels by a path in $K$ and apply the C.62 triangle and zero-set argument in this comparison plane, using $z$ as the exact real target. Its proof only requires proper real fibers and increasing target maps; both have just been checked. It does not require affine maps along the fibers. The resulting three-letter word retains its actual labels in $L$ and realizes the desired comparison point. Thus, with

$$M_0 = KK^{-1}K, \tag{C.78.12}$$

for every $p, t$ there is an actual $w \in M_0$ such that $c(w) = rc(f_\tau)$ and $\beta_w(t) = B_p(t)$. This is a pointwise source realization in one compact family. No continuous choice of $w$ has been asserted.

*All comparison spaces and compact local pair transports.* For the strips retained already in C.77 use its base prescription. An abelian base strip uses the joint character $a_I$ of (C.77.6) and target space $a_I(G)$ with its translation action. Its source point is 0 and desired target is $ra_I(f_\tau)$; $P^5 \subset M_0$ realizes it. A nonabelian strip with zero endpoint image uses target space $I$, its actual base action, source point $t$, and C.27's median target $B_p(t)$; an original prefix in $P$ realizes that target. The new rank-one strips use (C.78.11), source $(0, t)$, and target $(rc(f_\tau), B_p(t))$. Their compact initial realization is (C.78.12).

There are countably many such spaces, one for each retained interval. Denote them by $X_i$. Each is a Euclidean vector space or a product of a line with an interval, or an interval. The actual based path group $\mathscr{P}(L)$ acts transitively on each: the vector action uses the original character image, the interval action uses C.75, and the comparison-plane action uses (C.77.3)–(C.77.4). All their orbit maps are open by the same verified Polish orbit theorem. Every action uses the original endpoint character or base map; any action kernel remains its literal kernel in $\mathscr{P}(L)$.

We require the following exact strengthening of C.64's compact lifting construction. Given an identity neighborhood $U$ in $L$, there is a compact path-connected family $R \subset U$ containing 1 with

this property: for every $y \in X_i$ and every identity neighborhood $V$ in $L$, there is a neighborhood $N$ of $y$ such that

$$\text{for } y_1, y_2 \in N \text{ there is } e \in R \cap V \text{ with } e \cdot y_1 = y_2. \tag{C.78.13}$$

No continuity of the choice $e$ is part of the assertion.

To prove it, choose an identity neighborhood $U_0$ with $U_0 U_0^{-1} \subset U$. Exhaust the disjoint countable union of the $X_i$ by nested compact sets $E_m$ whose interiors cover it. Use based path neighborhoods whose values lie in $U_0$ and within $\epsilon_m$ of 1, where $\epsilon_m \downarrow 0$ in a compatible metric on $L$. At each point of $E_m$, openness of its actual orbit map supplies a small closed coordinate ball $D_{m,j}$ within the image of that path neighborhood. A finite collection of their interiors covers $E_m$. C.64's proved compact-covering lemma gives a compact family of these small paths whose endpoints map the chosen center exactly onto $D_{m,j}$. Let $S$ be the union of all their path values, together with 1.

This $S$ is compact and path connected: at each level there are finitely many compact path-value families, and the families with $m \to \infty$ converge uniformly to 1. It lies in $U_0$. Set $R = SS^{-1}$, so $R$ is compact, path connected, and contained in $U$. Given $y, V$, take $m$ sufficiently large that $y \in E_m$ and that products of two level-$m$ values, one inverted, lie in $V$. Choose a ball $D_{m,j}$ whose interior contains $y$. For two points in that interior choose endpoint labels $s_1, s_2$ from the same compact path family mapping its one center to those two points. Then $e = s_2 s_1^{-1} \in R \cap V$ maps the first to the second. This proves (C.78.13) simultaneously for every comparison space. Both maps and inverses are controlled in the original group.

*The second compact product and the actual fiber lift.* Starting with $M_0$ in (C.78.12), choose an identity neighborhood $U_n$ so small that its left multiplication changes every previously formed product by less than $2^{-n}$ in a complete compatible metric on $L$. Choose $R_n \subset U_n$ by (C.78.13), and put

$$M_n = R_n \cdots R_1 M_0. \tag{C.78.14}$$

Uniform Cauchy convergence on the product of the compact parameter families gives a compact path-connected limit family $M \subset L$ containing every $M_n$. Inverse convergence is included by the actual group topology.

On every retained strip define the fiber interval using precisely the appropriate target equality:

$$A_n(p,t) = \operatorname{conv}\{u_k(t) : k \in M_n,\ k \cdot x_i(t) = y_i(p,t)\}. \tag{C.78.15}$$

Here $(x_i(t), y_i(p,t))$ are the actual source and target points specified above: respectively the character-vector pair, the base pair, or the scalar-character/base pair. These sets are nonempty and compact by the initial realizations in $M_0$. Their defining equality ensures the same actual target base level $B_p(t)$ in each case.

For a fixed allowed $k \in M_n$ at $(p,t)$, the two comparison points $k \cdot x_i(t')$ and $y_i(p',t')$ tend to the same point as $(p',t') \to (p,t)$. Property (C.78.13), applied to $R_{n+1}$ and an arbitrarily small $V$, supplies an actual $e \in R_{n+1} \cap V$ carrying the first point to the second. Then $ek \in M_{n+1}$ has exactly the new target equality. Its fiber displacement approaches that of $k$ as $V$ decreases to the identity, uniformly near the specified source point. Using the same convex coefficient for two corrected endpoint values proves

$$A_n(p,t) \subset \liminf_{(p',t') \to (p,t)} A_{n+1}(p',t'). \tag{C.78.16}$$

The four prescribed fiber values are again (C.76.13), represented by 1 or $f_\tau$ in $M_0$. The full relative interval selection argument of C.76 therefore supplies a continuous $U(p,t)$ in the limiting compact fiber interval, with those faces exactly. Use (C.76.18) for the actual homeomorphism and its displayed inverse.

*Operator transport and every finite endpoint.* Retain the original component map $q(s,t) = (\lambda s + \kappa, t + d)$ of C.75, when it exists. Choose all constructions on representatives of its orbits of

proper intervals. For a new rank-one strip the chosen endpoint character transports to $\lambda c$. Its level sets (C.78.4) for the transported strip are therefore identical subsets of the same global $K$. Transport the selected positive average by using the same vector $q_p$ in $\mathcal{P}(K)$. Equation (C.76.17), linearity and preservation of the constant function by $\Lambda_{q_p}$ give $B'_p(t+d) = B_p(t) + d$ exactly. The allowed actual fiber intervals transform by multiplication by $\lambda$. Transport their selection by that multiplier; the additive $\kappa$ cancels in the actual commutation equation. For the older retained strips use precisely the character or median transport verified in C.77. A nonzero component power never stabilizes a proper interval. The interval $I = \mathbb{R}$ has no endpoint characters and is one of those older cases; its possible stabilizer is handled by the actual line-bundle selection of C.77. Negative $\lambda$ and every original component label are consequently retained.

For the new base averages one has the uniform pointwise bounds

$$\begin{aligned} \min_{k\in K} \beta_k(t) \le B_p(t) \le \max_{k\in K} \beta_k(t), \\ \min_{k\in K} \beta_k^{-1}(v) \le B_p^{-1}(v) \le \max_{k\in K} \beta_k^{-1}(v). \end{aligned} \tag{C.78.17}$$

The first is positivity. For the second, an input below all $\beta_k^{-1}(v)$ has every $\beta_k$-value below $v$, and an input above all of them has every value above $v$. The older base prescriptions and inverses have the same bounds, since they lie between the original map and the identity, both in $K$.

Now let retained interval points $t_n$ tend to $a \in A$, allowing the interval and all parameters to vary. Compactness of $K$ and its inverse family, all fixing the base point $a$, makes both bounds (C.78.17) tend to $a$. Choose a finite endpoint $a_n$ of the source interval between $t_n$ and $a$; then $a_n \to a$. Every label in (C.78.15) has $u_k(a_n) = r_n u_{\tau_n}(a_n)$: for an abelian strip this is its vector constraint; for the new strip it follows from its scalar constraint and the proportionality of both endpoint characters; for a zero-character strip both sides vanish. Uniform continuity over the one compact family $M$ gives $u_k(t_n) - u_k(a_n) \to 0$, uniformly in its allowed labels. Convexification preserves the estimate. This proves the specified fractional endpoint displacement, including varying intervals, just as in (C.76.21). The inverse base bounds in (C.78.17) permit the identical argument at the inverse input, with the negative fiber displacement. Thus both families glue continuously to every line over $A$.

*The original circle and the exact remaining region.* Each selected actual fiber value is between two $M$-values at the same real target level. The original source-plane C.62 triangle and inverse bracketing give

$$K_{r,\tau}(x) \in MM^{-1}M(x), \qquad K_{r,\tau}^{-1}(x) \in M^{-1}MM^{-1}(x). \tag{C.78.18}$$

On lines over $A$, the prescribed scalar fraction is attained by an original prefix value and its inverse, so these bounds hold there too. Both displayed word families are compact families of the original closed-disk homeomorphisms and fix the circle pointwise. They therefore send a sequence tending to an original boundary point uniformly to that same point. This proves joint continuity and inverse continuity at the original circle, with arbitrarily varying intervals and unbounded principal coordinates.

The resulting piecewise bijections preserve all the retained pieces, have the displayed continuous inverses, and are actual homeomorphisms of the closed set $Z_1$. Their compact-parameter evaluations give continuity in its actual homeomorphism topology. The restriction subgroup and homomorphism are precisely those of (C.76.4) with this named closed subset. On the subgroup preserving the older subset $Z_{\mathrm{ab}}$, further restriction is a continuous homomorphism; every constructed map preserves that subset. This supplies the exact comparison with the earlier restricted loop, without asserting identity of independently chosen fillings.

If a strip remains excluded, its base image is nonabelian and its endpoint vector image has dimension two. It must therefore have two finite endpoints, and their joint image is all of $\mathbb{R}^2$, proving (C.78.2). The scalar-character comparison plane used in (C.78.11) does not retain both independent characters of such a strip. No vertical interval containment in the corresponding

three-dimensional comparison space has been proved here, and no filling through that excluded region is inferred. The other remaining cases of C.12 are unchanged. □

**C.79. [Proved] Every relative disk loop for a one-dimensional proper operator action fills.** Let $O = \operatorname{int} \overline{D}$ be the interior of the actual closed disk. Let $Q \leq \mathrm{Homeo}(O)$ be closed, locally compact, free and proper, with $\dim Q = 1$. Use the actual relative centralizer

$$\mathscr{C}_Q = \{h \in \mathrm{Homeo}(\overline{D}) : h|_{\partial \overline{D}} = \mathrm{id},\ hq = qh \text{ on } O \text{ for every } q \in Q\} \tag{C.79.1}$$

with its compact-open topology on maps and inverses. Every continuous based loop $f_\tau$ in this group has a continuous based contraction in the same group. In particular

$$\pi_1(\mathscr{C}_Q, 1) = 0. \tag{C.79.2}$$

The construction uses the original disk boundary and every original operator. Its map and inverse families are jointly continuous there, including arbitrary accumulation of principal strips. It does not require a specified Fuchsian disk comparison, and does not replace the original compactification by another one.

The two-endpoint step retains one common increasing base map and one compact global source family in both endpoint selections. Their fiber displacements are blended in that same original real fiber. This gives an actual filling on the whole disk; no extension through an omitted strip is left in the stated dimension-one disk assertion.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), complete actual closed-disk relative loop filling for one-dimensional free proper interior operator groups.

*Proof. The exact strips and the common preliminary family.* Use the actual principal coordinate and inverse of C.75. Write

$$f_\tau(s,t) = (s + u_\tau(t), \beta_\tau(t)), \qquad A = \mathrm{Fix}\langle \beta_\tau \rangle. \tag{C.79.3}$$

Retain the original group $G = \langle f_\tau \rangle$ and its closure $L$ in the actual relative group (C.79.1). The complete compatible metric, all component operators, and the jointly continuous principal-coordinate functions on compact source families are exactly those in C.76–C.78. There are countably many complementary intervals.

C.78 fills all their types except bounded intervals $I = (a,b)$ with nonabelian effective base image and actual joint character image

$$c_I = (c_a, c_b) : G \longrightarrow \mathbb{R}^2, \qquad c_I(G) = \mathbb{R}^2. \tag{C.79.4}$$

We construct all strip fillings together, using one compact family, so that their accumulation estimates remain uniform. Begin with precisely the preliminary compact $K$ and $K_n$ of C.78: $K_0 = P^5$, where $P$ is the original loop and its inverse, followed by fifteenfold short-increment families whose infinite products converge in $L$. This construction is independent of the strip.

*A full-vector average gives the one common base.* On a strip (C.79.4), replace (C.78.4) by the full two-character fibers

$$F_n(r,\tau) = \{k \in K_n : c_I(k) = r c_I(f_\tau)\}, \qquad F_\infty(r,\tau) = \{k \in K : c_I(k) = r c_I(f_\tau)\}. \tag{C.79.5}$$

The planar additive identity (C.37.3) gives nonempty fibers already in $P^3 \subset P^5$. For nearby parameters the required correction is the actual vector $r' c_I(f_{\tau'}) - r c_I(f_\tau)$. C.76's six-short-increment neighborhood covers this vector when it is sufficiently small; it is a subfamily of the fifteenfold increments used here. The same increment corrects every label in the given fiber. Thus the Hilbert convex-average proof (C.78.5)–(C.78.8) applies to (C.79.5) without change: its

lower-limit argument needs only a common arbitrarily small character correction, not a scalar target.

Choose its continuous positive average $q_p$, $p = (r, \tau)$, supported in the precise sense $q_p \in \mathcal{P}(F_\infty(p))$, with the same four exact parameter faces. Put

$$B_p(t) = \Lambda_{q_p}(k \mapsto \beta_k(t)). \tag{C.79.6}$$

The compact positive-gap and inverse-envelope arguments (C.78.9)–(C.78.10) make this a jointly continuous family of increasing homeomorphisms of $I$, with jointly continuous inverses. It has the original base map at $r = 1$ and the identity on the other based faces. Crucially, this is the *same* $B_p$ for both endpoints.

*Two actual scalar comparison planes.* For $e = a, b$ retain the actual source action

$$\begin{aligned} T_e(k)(z,t) &= (z + c_e(k), \beta_k(t)), \\ T_e(k)^{-1}(z,t) &= (z - c_e(k), \beta_k^{-1}(t)). \end{aligned} \tag{C.79.7}$$

In each of these planes use $z$ as the real target, whose fibers are the proper copies of $I$. Every map has an increasing target translation and an increasing fiber map. The minimum and maximum labels for the positive average (C.79.6) belong to $F_\infty(p)$ and hence have *both* endpoint characters prescribed. Apply the C.62 triangle separately in the two planes, along a path in the same compact $K$ joining those labels. It follows that the single compact family

$$M_0 = KK^{-1}K \tag{C.79.8}$$

realizes $(rc_a(f_\tau), B_p(t))$ under $T_a$ at $(0,t)$, and separately realizes $(rc_b(f_\tau), B_p(t))$ under $T_b$ there. The realizing labels need not be equal. No realization retaining both characters at once is claimed at this step.

Each scalar action is transitive. Indeed, C.77 proves that the actual source derived subgroup has transitive base image and vanishes under every endpoint character; each character in (C.79.4) has full real image. Thus the precise based path-group orbit maps of both scalar actions are open, with the actual kernels and stabilizers described in (C.77.4) and (C.77.12).

Add these two comparison spaces per interval to the countable collection in C.78. Apply its proved compact local pair-transport construction (C.78.13) to the whole collection. The second infinite-product construction (C.78.14) then yields *one* compact path-connected global family $M \subset L$ and nested compact families $M_n$ starting with (C.79.8). This includes the comparison spaces for all the older strip types as well. All maps and inverses are original closed-disk maps fixing the original circle.

*The two fiber selections and their exact blend.* For $e = a, b$, define

$$A_n^e(p,t) = \operatorname{conv}\{u_k(t) : k \in M_n,\ c_e(k) = rc_e(f_\tau),\ \beta_k(t) = B_p(t)\}. \tag{C.79.9}$$

They are nonempty compact intervals by (C.79.8). The local pair transports in their respective scalar comparison planes prove the interlaced lower-limit property exactly as in (C.78.16). The four relative face values are represented by the identity and the original $f_\tau$. The same relative interval selection therefore supplies two continuous displacements $U_a(p,t)$ and $U_b(p,t)$, each in its limiting compact interval, with those four faces exactly. Their actual base map is the common $B_p$.

Choose the explicit cutoff

$$\begin{aligned} \theta_I(t) &= \max\left(0, \min\left(1, 3\frac{t-a}{b-a} - 1\right)\right), \\ U(p,t) &= (1 - \theta_I(t))U_a(p,t) + \theta_I(t)U_b(p,t). \end{aligned} \tag{C.79.10}$$

It is zero on a neighborhood of $a$ and one on a neighborhood of $b$. Define

$$\begin{aligned} H_p(s,t) &= (s + U(p,t), B_p(t)), \\ H_p^{-1}(s,v) &= (s - U(p, B_p^{-1}(v)), B_p^{-1}(v)). \end{aligned} \tag{C.79.11}$$

These are exact inverse homeomorphisms, jointly continuous on the open strip. Both selections have the same four faces, so their blend does too.

Every value of $U_a$ or $U_b$ lies between actual $M$-label displacements at that same base level. Their convex blend lies in the convex hull of the union of those labels. The minimum and maximum of the compact set of *all* $M$-labels at that level therefore bracket $U$. Applying the original source-plane C.62 triangle and its inverse bracketing gives directly

$$H_p(x) \in MM^{-1}M(x), \qquad H_p^{-1}(x) \in M^{-1}MM^{-1}(x). \tag{C.79.12}$$

There is no new word-family bound inferred from an unrelated coordinate interpolation: both selected displacements and their blend have the identical actual real base level. The older strip constructions use the same $M$ and have the same two bounds by C.78.

*All component operators.* The original component generator is $q(s,t) = (\lambda s + \kappa, t + d)$, with $d > 0$ and $\lambda \neq 0$, when it exists. For the translated interval $I + d$, the two endpoint characters are multiplied by $\lambda$, so the fibers (C.79.5) are the same actual subsets of $K$. Transport the positive average by that identical label choice. Then $B'_p(t + d) = B_p(t) + d$, exactly as in C.78. Transport both fiber selections by $U'_{e+d}(p, t + d) = \lambda U_e(p,t)$. The explicit cutoff satisfies $\theta_{I+d}(t + d) = \theta_I(t)$, and hence the blended displacement has the same multiplier equation. This verifies actual commutation with $q$, including negative $\lambda$; the original constant $\kappa$ cancels from the equation and remains in the operator. No nonzero component power stabilizes a bounded interval, so transport from interval representatives is independent of the chosen operator. The older interval types, including $I = \mathbb{R}$ with its possible component stabilizer, retain their exact C.78 line-bundle and transport equations.

*The two finite-endpoint sequence alternatives.* On every line over $a \in A$, prescribe $H_p(s,a) = (s + ru_\tau(a), a)$. Consider a sequence of source points with $t_n \to a \in A$, allowing all parameters and all interval indices to vary. The older strip types have the joint limit already proved in C.78, with the same compact $M$. For the new bounded strips, every subsequence has a further subsequence of one of these two forms.

First, one fixed interval $I = (a_0, b_0)$ occurs throughout. The limiting point of $A$ must be $a_0$ or $b_0$. Near $a_0$ the cutoff is identically zero, so $U = U_a$ and every defining label has $u_k(a_0) = ru_\tau(a_0)$. Near $b_0$ it is identically one, and the same statement uses $U_b$ and $b_0$. Uniform continuity over the compact actual $M$ proves the required limit of $U$.

Second, the bounded intervals $I_n = (a_n, b_n)$ are pairwise distinct. Then both $a_n$ and $b_n$ tend to $a$. For example, if along a subsequence $b_n \geq a + 2\epsilon$ while $t_n \to a$, then for all sufficiently large $n$ the fixed point $a + \epsilon$ lies in $I_n$, contradicting disjointness of distinct components. The analogous argument with $a - \epsilon$ handles the other endpoints. Thus uniform continuity of the one compact family $M$ gives, respectively,

$$\begin{aligned} U_a(p_n, t_n) - r_n u_{\tau_n}(a_n) &\longrightarrow 0, \\ U_b(p_n, t_n) - r_n u_{\tau_n}(b_n) &\longrightarrow 0. \end{aligned} \tag{C.79.13}$$

For the first estimate use the labels in the limiting interval of $A_n^a$, and for the second use those for $A_n^b$; they need not be the same labels. Each estimate persists under its convexification. The original compact loop makes both endpoint terms approach the prescribed value at $a$, uniformly in the parameters. Any convex combination in (C.79.10) has the same limit. This proves the desired limit even when the cutoff values vary without converging. The two subsequence alternatives establish the assertion for the original arbitrary sequence.

The base average and its inverse satisfy the compact envelopes (C.78.17), as do the older prescriptions. Therefore their values tend to $a$ uniformly whenever the source or inverse input tends to $a$. For the inverse, put $v_n = B_{p_n}^{-1}(t_n)$. Then $v_n \to a$ and it stays in the same interval $I_n$. Apply the same two alternatives at $v_n$ in (C.79.11). This gives the negative prescribed fiber displacement and proves inverse continuity at the limiting line. Along $A$ itself, both limits follow from the original continuous function $ru_\tau(a)$.

*The actual whole-disk filling.* Use (C.79.11) on every new strip, the simultaneously constructed C.78 maps on every other strip, the prescribed map on each line over $A$, and the identity on the original boundary circle. These form a bijection of the actual closed disk with the displayed piecewise inverse, since every base interval and every line over $A$ is preserved. Interior joint continuity of maps and inverses is exactly the strip and endpoint comparison just proved.

On a line over $A$, the scalar fraction $ru_\tau(a)$ is attained by an original prefix value; its negative is attained by the corresponding inverse. Thus the compact bounds (C.79.12) hold there too. Both global word families in (C.79.12) are compact families of actual closed-disk homeomorphisms fixing the original circle. As a point approaches any original circle point, all their evaluations approach that same point uniformly. This proves joint boundary continuity of the pasted maps and their inverses, including arbitrarily varying strips and unbounded principal coordinates.

Hence the family consists of actual homeomorphisms of $\overline{D}$ and is continuous in the topology of (C.79.1). Its operator equations were checked for every actual component and identity-component operator. Its four parameter faces are $H_{0,\tau} = 1$, $H_{1,\tau} = f_\tau$, and $H_{r,0} = H_{r,1} = 1$. This is the asserted based contraction, proving (C.79.2).

The proof is a loop statement for the specified actual disk group. It asserts neither contractibility of that entire group nor a homotopy lift through C.73's prime-end injection for a different original domain. Those exact comparison and broader C.12 requirements remain as previously stated. □

**C.80. [Proved] Every based sphere family fills in the one-dimensional proper disk centralizer.** Let $O = \operatorname{int}\overline{D}$ be the actual open disk, and let $Q \le \mathrm{Homeo}(O)$ be closed, locally compact, free and proper, with $\dim Q = 1$. Retain the actual group and topology

$$\mathscr{C}_Q = \{h \in \mathrm{Homeo}(\overline{D}) : h|_{\partial\overline{D}} = \mathrm{id},\ hq = qh \text{ on } O \text{ for every } q \in Q\}, \tag{C.80.1}$$

using compact-open convergence of maps and inverses. For every $n \ge 1$ and every continuous based map $f : (S^n, *) \to (\mathscr{C}_Q, 1)$, there is a continuous map $H : [0,1] \times S^n \to \mathscr{C}_Q$ satisfying

$$\begin{gathered} H_{0,\xi} = 1, \qquad H_{1,\xi} = f_\xi, \qquad H_{r,*} = 1. \\ \text{Consequently} \quad \pi_n(\mathscr{C}_Q, 1) = 0 \quad (n \ge 1). \end{gathered} \tag{C.80.2}$$

Both evaluation families $H_{r,\xi}(x)$ and $H_{r,\xi}^{-1}(x)$ are jointly continuous on the original closed disk. Every original operator is retained. This proves the stated based homotopy groups, without asserting contractibility of the whole group or describing its components.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), all positive-degree based homotopy groups of the actual closed-disk relative centralizer for one-dimensional free proper interior operator groups.

*Proof. The actual source group and the sphere parameter.* Put $\Sigma = S^n$ and $P = f(\Sigma) \cup f(\Sigma)^{-1}$. This is a compact path-connected family containing 1. Use $G = \langle P \rangle$ and its closure $L$ in the actual group (C.80.1). The latter is a closed subgroup of the Polish closed-disk homeomorphism group: the boundary equations and all commutator equations on $O$ are closed under convergence of maps and inverses. Thus $L$ has the complete compatible group topology used in C.76–C.79.

C.33's actual principal coordinate, with its inverse, has $Q^0$ acting by real translations in the first coordinate. Hence every $k \in L$ has the form

$$\begin{aligned} k(s,t) &= (s + u_k(t), \beta_k(t)), \\ k^{-1}(s,t) &= (s - u_k(\beta_k^{-1}(t)), \beta_k^{-1}(t)). \end{aligned} \tag{C.80.3}$$

The base maps are increasing. This holds first for $f_\xi$, since a path in $\Sigma$ from $*$ to $\xi$ gives an actual based path of maps, and then for $G$ and its closure. Every finite word in $P$ has a based path within the same finite power of $P$: choose a parameter path for each letter and multiply the resulting paths, with the indicated inverses. A jointly continuous choice of those paths is unnecessary.

Let $A = \{t : \beta_{f_\xi}(t) = t \text{ for all } \xi\}$. It is closed. Every complementary component $I$ is a transitive base interval for $G$. Indeed a based path moving a point and its inverse give a neighborhood in its base orbit; the orbit is connected because the increasing union $\bigcup_m P^m(t)$ is connected. A finite endpoint of that interval orbit is fixed by all its increasing base maps. This proves the stated component equality, exactly as in C.75, for this actual sphere-generated $G$. Every $k \in L$ fixes $A$ pointwise on the base and preserves each $I$.

The endpoint characters are the actual evaluations $c_a(k) = u_k(a)$, $a \in A$. They are continuous characters by the composition formula (C.75.6). All other source facts needed from C.75–C.77 have the same precise hypotheses here. For an abelian effective base image, its closure is abelian, transitive and free; the verified evaluation homeomorphism identifies the actual image with real translations, as in C.77. This gives an increasing $b_I : I \to \mathbb{R}$ and the continuous character

$$d_I(k) = b_I(\beta_k(t_0)) - b_I(t_0), \qquad b_I\beta_k(t) = b_I(t) + d_I(k). \tag{C.80.4}$$

For a nonabelian base image, $[G,G]$ is path connected: each commutator and each finite product of commutators has a based path from the finite-word paths above. Its nontrivial base image is normal. Its common fixed set is $G$-invariant and therefore empty by base transitivity. Its interval orbits are open by the same path argument. They partition the connected interval $I$, so it is transitive there. All endpoint characters vanish on $[G,G]$. Thus every scalar character/base comparison used below is exactly the transitive action in (C.77.3), with its stated kernel.

*Short increments with the complete original topology.* Give $\Sigma$ its round geodesic metric $d$. For $0 < \delta < \pi$ put

$$J_\delta = \{f_\xi f_\eta^{-1} : d(\xi,\eta) \le \delta\}, \qquad E_\delta = J_\delta^{15}. \tag{C.80.5}$$

The pair space in this formula is compact and path connected. The unique shortest geodesic from $\xi$ to $\eta$ contracts its first entry to the second, continuously on this pair space and without increasing their distance. The diagonal is a copy of the path-connected sphere. Consequently $J_\delta$ is compact, symmetric, path connected, and contains 1. Uniform continuity of $f$ and of multiplication and inversion on its compact image proves $J_\delta \to \{1\}$ in the actual topology as $\delta \downarrow 0$.

For any endpoint-character vector, or the vector consisting of $d_I$ and all finite endpoint characters, there are at most three coordinates. Its actual image $V$ is a real vector subspace. To check this before closure, apply the five-letter additive path identity (C.50.3) to a finite-word based path ending at any specified $g \in G$. This puts every fraction of its vector in the actual image. Addition, negatives and integer multiples then give the whole real span. Applying the same identity to a path from $*$ to $\xi$ within $\Sigma$ also gives

$$ra(f_\xi) \in a(P^5), \qquad a(E_\delta) \text{ contains a neighborhood of 0 in } V. \tag{C.80.6}$$

For the second assertion, subdivide a parameter path into pieces of length less than $\delta$. Telescoping shows that $a(J_\delta)$ spans $V$. Choose at most three basis vectors in this compact symmetric path-connected image. The same five-letter identity supplies the full segment from minus to plus each

basis vector; their sum contains a neighborhood of zero. This uses at most fifteen factors. Every such character extends continuously to $L$, and closedness of $V$ gives $a(L) = V$.

Choose the successive $\delta_m$ in (C.80.5) as in C.77–C.78, so multiplication by the next compact increment changes every preceding compact product by less than $2^{-m}$ in a complete compatible metric. Starting with $K_0 = P^5$, the resulting nested families $K_m$ lie in one compact path-connected limit family $K \subset L$. The product parameter space is a product of compact path-connected spaces, and uniform Cauchy convergence gives its continuous map into $L$. These statements include convergence of the inverse maps. One such $K$ serves every strip.

*The exact relative parameter conditions.* Use the compact metric parameter space

$$T = [0,1] \times \Sigma, \qquad D = (\{0,1\} \times \Sigma) \cup ([0,1] \times \{*\}). \tag{C.80.7}$$

On $D$ the prescribed source labels are 1 at $r = 0$, $f_\xi$ at $r = 1$, and 1 at $\xi = *$. They agree on all intersections because $f_* = 1$. They are continuous on $D$ and belong to $K_0$.

For every nonabelian strip with nonzero endpoint vector $c_I$, form

$$F_m(p) = \{k \in K_m : c_I(k) = r c_I(f_\xi)\}, \qquad p = (r, \xi) \in T. \tag{C.80.8}$$

These compact fibers are nonempty by (C.80.6). Use exactly C.78's embedding $\chi : K \to \ell^2$ and compact convex sets $\mathcal{P}(F)$, with their continuous positive averages $\Lambda_q$. The actual character correction from $p$ to $p'$ is $r' c_I(f_{\xi'}) - r c_I(f_\xi)$. By (C.80.6), an arbitrarily small next increment realizes it and corrects every label in the old fiber at once. The finite-convex-combination argument (C.78.8) gives the interlaced lower-limit inclusion for these Hilbert convex sets.

The relative selection proof of C.63 and C.78 applies on $T$, with $D$ as its closed relative subset. Explicitly, replace the allowed sets on $D$ by the continuous singleton labels $\chi(1)$ or $\chi(f_\xi)$ just specified. At a point outside $D$ a neighborhood avoids $D$; at a point of $D$ the old lower-limit inclusion and continuity of these singleton values give the relative inclusion. The locally finite partitions of unity and summable successive approximations used in C.63 require a metrizable paracompact domain, not a rectangular parameter boundary. Both $T$ and $T \times I$ have that property. Their limits lie in the closed limiting compact-family convex sets, and their errors to the prescribed singletons tend to zero. We therefore obtain continuous positive averages $q_p \in \mathcal{P}(F_\infty(p))$ with

$$q_{0,\xi} = \chi(1), \qquad q_{1,\xi} = \chi(f_\xi), \qquad q_{r,*} = \chi(1). \tag{C.80.9}$$

*All base prescriptions and their compact realizations.* For an abelian base strip use $B_p(t) = b_I^{-1}(b_I(t) + r d_I(f_\xi))$. For a nonabelian base strip with nonzero endpoint vector use $B_p(t) = \Lambda_{q_p}(k \mapsto \beta_k(t))$. In the rank-two case the average uses the full vector in (C.80.8), so both scalar endpoint selections below have the same base. For a nonabelian strip with zero endpoint vector, use instead the formula

$$B_p(t) = (1 - r)t + r\beta_{f_\xi}(t). \tag{C.80.10}$$

This last choice removes any dependence on an interval ordering of the sphere parameters. It is an increasing homeomorphism of $I$: strict inequalities are preserved by its convex combination, and both summands tend to the same respective endpoints of $I$. Its inverse is jointly continuous by strict monotone bracketing. For each fixed $p, t$, a parameter path from $*$ to $\xi$ shows by the intermediate value theorem that an original label in $f(\Sigma) \subset P$ realizes $B_p(t)$ at $t$. No continuous choice of this path or label is used.

The other two prescriptions and their inverses are jointly continuous by C.78's compact strict gap and inverse-envelope arguments. All three have exactly the relative conditions (C.80.7). They and their inverses lie between the corresponding compact $K$-evaluation envelopes.

Use the same countable comparison spaces as in C.78–C.79: the joint character vector space for an abelian strip; $I$ for a zero endpoint vector; and each required scalar plane with the actual

action

$$\begin{aligned} T_e(k)(z,t) &= (z + c_e(k), \beta_k(t)), \\ T_e(k)^{-1}(z,t) &= (z - c_e(k), \beta_k^{-1}(t)). \end{aligned} \tag{C.80.11}$$

For a rank-one vector choose a nonzero endpoint character; the other endpoint character, when present, is its actual real multiple. For a rank-two vector use both planes, separately. Their desired targets are exactly $(rc_e(f_\xi), B_p(t))$ at source $(0,t)$. Two extremal labels for the average have the same full endpoint vector. A path in $K$ joins them. The C.62 triangle in either scalar plane, with real target $z$, realizes the entire intervening fiber interval in the single actual family $M_0 = KK^{-1}K$. The vector targets and (C.80.10) already have labels in $P^5$ and $P$, respectively. Thus $M_0$ is a common initial realizing family for every listed comparison target.

All these actions of the actual Polish based path group $\mathscr{P}(L)$ are transitive. For the scalar planes this follows from the transitive derived base image and the full nonzero character image proved above; the other cases use the actual vector image or base orbit. The orbit map is endpoint action on the named source point. Its stabilizer is exactly the paths whose endpoint fixes that point; the action kernel retains the zero character and identity-base equations. The verified Polish orbit theorem makes these precise orbit maps open.

C.78's compact local pair-transport construction therefore supplies, for every small identity neighborhood, a compact family of actual maps that realizes arbitrarily small pair transports near each fixed comparison point. It is built from compact preimages under these path-group orbit maps and includes all countably many spaces at once. Its proof uses no parameter ordering. The second compact-product construction gives one compact path-connected family $M \subset L$, with nested $M_m$ starting at $M_0$, exactly as in (C.78.13)–(C.78.14).

*Relative fiber selection and the common-base splice.* For each target equality just listed, take the convex interval of $u_k(t)$ over labels $k \in M_m$ satisfying that equality. These intervals are nonempty. Small pair transports correct an allowed label at $(p,t)$ to the precise nearby target at $(p',t')$ and converge to the identity. This is exactly the lower-limit proof (C.78.16). Replace the intervals on $D \times I$ by the values 0, $u_{f_\xi}(t)$, and 0, in the order prescribed in (C.80.7). The same relative approximation proof gives continuous selections with these values exactly.

On a rank-two interval $I = (a,b)$ perform the two scalar selections using the same $B_p$ and the same $M$. Denote them by $U_a, U_b$, and use precisely the cutoff and blend of (C.79.10):

$$\begin{aligned} \theta_I(t) &= \max\left(0, \min\left(1, 3\frac{t-a}{b-a} - 1\right)\right), \\ U &= (1-\theta_I)U_a + \theta_I U_b. \end{aligned} \tag{C.80.12}$$

All other intervals have their single selected $U$. Define the maps and their actual inverses by

$$\begin{aligned} H_p(s,t) &= (s + U(p,t), B_p(t)), \\ H_p^{-1}(s,v) &= (s - U(p, B_p^{-1}(v)), B_p^{-1}(v)). \end{aligned} \tag{C.80.13}$$

Every defining label in a fiber selection has that identical actual base level. The rank-two blend stays between the extrema of their union at that level. The original source-plane C.62 triangle and its inverse bracketing therefore give, uniformly over all parameters and strips,

$$H_p(x) \in MM^{-1}M(x), \qquad H_p^{-1}(x) \in M^{-1}MM^{-1}(x). \tag{C.80.14}$$

The required connecting paths lie in the compact path-connected family $M$; no continuous choice of them enters these set containments.

*Every original component operator and every limit.* Retain C.75's actual component generator $q(s,t) = (\lambda s + \kappa, t + d)$ when it exists, with $d > 0$ and $\lambda \neq 0$. Choose constructions on interval

representatives and transport them. Endpoint characters multiply by $\lambda$; the full-vector fibers are the same actual label sets after that multiplication. Transport their averages by those same labels, and their fiber selections by $U'(p,t+d)=\lambda U(p,t)$. All base prescriptions satisfy $B'_p(t+d)=B_p(t)+d$. In particular (C.80.10) has this identity by its displayed affine formula. The cutoff in (C.80.12) has it as well. For $I=\mathbb{R}$ stabilized by a component translation, the actual line-bundle selection of C.76–C.78 retains the equation with $\lambda$, using its positive norm $|\lambda|^{-t/d}|z|$. The relative approximation works in that same bundle over $T$ times the quotient base. Thus every operator equation, including negative $\lambda$ and the unchanged constant $\kappa$, holds exactly.

On each $a\in A$ put $H_p(s,a)=(s+ru_{f_\xi}(a),a)$. The finite-line limit proofs of C.78–C.79 are uniform for the new compact parameter space $T$. In detail, compactness of $M$ and $f(\Sigma)$ gives uniform continuity of their principal displacements on bounded base intervals. Labels constrained at a finite endpoint have precisely the specified fraction there. For rank two, a fixed interval approaching an endpoint uses the corresponding selection because the cutoff is identically zero or one nearby. For pairwise distinct bounded intervals approaching a finite line, both endpoints approach that line, by the disjoint-interval argument of C.79. Each scalar selection therefore has the same limiting prescribed displacement; so does their blend. Every arbitrary sequence has a subsequence of one of those two forms. The compact inverse base envelopes first give the same limiting input line for (C.80.13), and the identical argument then proves its inverse limit. These assertions allow $r$ and $\xi$ to vary.

On a line over $A$, the continuous image $\{u_{f_\eta}(a):\eta\in\Sigma\}$ is an interval containing 0 and $u_{f_\xi}(a)$. It therefore contains $ru_{f_\xi}(a)$; the corresponding inverse label realizes the negative displacement. Thus the compact bounds (C.80.14) hold on these lines too. Set $H_p=1$ on the original boundary circle. Both word families there are compact families of original closed-disk maps fixing that circle. All their evaluations at a point tending to a given circle point converge uniformly to that same point. This proves both required boundary limits, even through varying strips and unbounded principal coordinates.

The piecewise formulas preserve each base interval and line and are inverse bijections. The preceding limits make both maps jointly continuous on $T\times\overline{D}$. Because $\overline{D}$ is compact, this is continuity $T\to\mathscr{C}_Q$ in its specified map-and-inverse topology. The relative selections give exactly the three equations in (C.80.2). Collapsing $\{0\}\times S^n$ gives the cone on $S^n$, homeomorphic to the closed $(n+1)$-disk, and the basepoint ray remains the identity. This is the asserted based filling.

The proof uses path connectedness and compactness of $S^n$ for $n\geq 1$. It makes no assertion about $\pi_0$ or a single contraction of all of $\mathscr{C}_Q$. It also supplies no homotopy lift through the prime-end injection of C.73 for a different original compactification; the remaining general C.12 extension question retains that precise scope. □

**C.81. [Proved] Exact metric, compact-support, and finite-configuration comparisons for discrete operators.** Let $O=\operatorname{int}\overline{D}$ be the actual open disk, let $d$ be a compatible metric on its closed disk, and let $Q\leq\operatorname{Homeo}(O)$ be closed, discrete, free and proper. Retain the actual relative centralizer

$$\mathscr{C}_Q=\{h\in\operatorname{Homeo}(\overline{D}):h|_{\partial\overline{D}}=\mathrm{id},\ hq=qh \text{ on } O \text{ for every } q\in Q\} \tag{C.81.1}$$

with compact-open convergence of maps and inverses. Assume that an actual subgroup $G\leq\mathscr{C}_Q$ is transitive on $O$. Write $N=O/Q$ and retain the original quotient covering $\pi:O\to N$. No extension of the operators $q$ to the circle is assumed. The following comparisons hold.

The metric

$$\begin{aligned} d_Q(x,y)&=\sup_{q\in Q}d(qx,qy),\\ \rho(\pi x,\pi y)&=\inf_{q\in Q}d_Q(x,qy). \end{aligned} \tag{C.81.2}$$

gives the original topology on $O$ and its quotient topology on $N$. The specified covering is locally isometric for these metrics and each fiber is uniformly discrete. Restriction embeds $\mathscr{C}_Q$ as a closed subgroup of the uniform $d_Q$-homeomorphisms commuting with $Q$, and is an isometry for the sum of the uniform map and inverse distances and the corresponding actual closed-disk distances.

Every compact parameter family of equivariant interior homeomorphisms and their inverses, supported over one fixed compact subset of $N$, extends jointly by the identity to the original circle. In particular, every based sphere family in $\mathscr{C}_Q$ whose descents are supported in the interior of one compact connected subsurface $N_0 \subset N$ with nonempty boundary has a based filling in $\mathscr{C}_Q$, retaining that quotient support and every original label. Here sphere dimension is any $n \geq 1$.

For finite $A \subset O$ with distinct quotient labels put $\mathscr{C}_Q(A) = \{h \in \mathscr{C}_Q : h|_A = \mathrm{id}\}$. For $x \notin QA$, the actual evaluation map is a locally trivial principal bundle

$$\mathscr{C}_Q(A \cup \{x\}) \longrightarrow \mathscr{C}_Q(A) \xrightarrow{h \mapsto h(x)} O \setminus QA. \tag{C.81.3}$$

Its source and fiber retain their original closed-disk topologies. Every inclusion $\mathscr{C}_Q(A) \hookrightarrow \mathscr{C}_Q$ induces an isomorphism on $\pi_n$ at 1 for every $n \geq 1$.

When $N$ is noncompact, descent from the subgroup of actual lifts of compactly supported quotient isotopies is not continuous from its induced $\mathscr{C}_Q$ topology to the Whitney topology on the quotient homeomorphism group. The proof gives an explicit convergent sequence and one positive continuous majorant excluding all its descents.

These are finite-configuration and fixed compact quotient-support comparisons. They do not provide a coherent infinite-configuration homotopy or a controlled fragmentation of every original family into compact quotient supports.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact discrete-operator topology, fixed quotient-support sphere fillings, and finite-configuration continuation for the remaining level-zero question.

*Proof. The invariant metric and the actual open evaluation map.* Let $L$ be the closure of $G$ in (C.81.1). It is a Polish group, remains transitive on $O$, and commutes with every $q$. Closedness of the commutator equations uses only the interior: if $h_j \to h$ with inverses, then $h_j x \to hx \in O$, so continuity of $q$ gives $qh_j x \to qhx$. The verified Polish orbit theorem in C.30 makes $L \to O$, $g \mapsto gx$, open.

The first formula (C.81.2) is finite because $d$ is bounded on the compact disk. The identity term separates points, and the triangle inequality holds term by term. Reindexing by right multiplication proves $Q$-invariance. Also $d(x,y) \leq d_Q(x,y)$. For the reverse topology comparison, take $x_j \to x$ in $O$. Openness of evaluation supplies $g_j \to 1$ in $L$ with $g_j x = x_j$: choose a decreasing identity-neighborhood basis $U_m$, let $m(j) \to \infty$ so that $x_j \in U_{m(j)}x$, and choose one such $g_j$. No section has been assumed. Commutation gives

$$d_Q(x_j, x) = \sup_{q \in Q} d(g_j qx, qx) \leq \sup_{z \in \overline{D}} d(g_j z, z) \longrightarrow 0. \tag{C.81.4}$$

This proves equality of the two interior topologies, and in particular separability of the displayed invariant metric space.

*The specified quotient is a metric covering.* Properness, discreteness and freeness give an open neighborhood $U$ of any $x$ whose distinct $Q$-translates are disjoint. Choose $r > 0$ with $B_{d_Q}(x, r) \subset U$. Then $d_Q(x, qx) \geq r$ for $q \neq 1$, and invariance makes the entire fiber $Qx$ uniformly $r$-discrete.

The formula for $\rho$ is independent of both representatives by invariance and reindexing. Symmetry and the triangle inequality follow from those of $d_Q$. A zero value gives $q_j y \to x$ for

some $q_j$; the closed-orbit property of the proper action then gives $x \in Qy$. Thus it is a metric. The equality

$$\pi B_{d_Q}(x, \epsilon) = B_\rho(\pi x, \epsilon) \tag{C.81.5}$$

shows that it gives the quotient topology, and $\rho(\pi x, \pi y) \le d_Q(x, y)$. For $y, z \in B_{d_Q}(x, r/5)$ and $q \ne 1$,

$$d_Q(y, qz) \ge r - 2r/5 = 3r/5, \qquad d_Q(y, z) < 2r/5. \tag{C.81.6}$$

Consequently $\rho(\pi y, \pi z) = d_Q(y, z)$. The inverse image of $\pi B_{d_Q}(x, r/5)$ is precisely the disjoint union of the translates of that ball, each mapped isometrically onto this quotient neighborhood. These are all three metric-cover conditions in [75, Definition 2.1], proved here for the actual map $\pi$. The separation constant may depend on the fiber; no compact quotient is inferred.

*The exact isometry of the relative source group.* Let $\mathscr{U}_Q$ denote the group of uniform homeomorphisms of $(O, d_Q)$ commuting with $Q$, with the uniform topology on maps and inverses. An $h \in \mathscr{C}_Q$ belongs to this group after restriction. Indeed its modulus of uniform continuity on the original compact disk applies to every pair $qx, qy$; commutation then bounds $d(hqx, hqy)$ uniformly in $q$. This is uniform $d_Q$-continuity of $h$. Apply the identical argument to $h^{-1}$.

For $h, k \in \mathscr{C}_Q$ the exact formula is

$$\begin{aligned} \sup_{x \in O} d_Q(hx, kx) &= \sup_{x \in O,\ q \in Q} d(hqx, kqx) \\ &= \sup_{z \in \overline{D}} d(hz, kz). \end{aligned} \tag{C.81.7}$$

The last equality uses $q = 1$ and density of $O$. Repeat it with both inverses. Thus restriction is an isometry for the stated sum metrics, and its inverse on its image is continuous. The source metric is complete: simultaneous uniform limits of maps and inverses on the compact disk are continuous inverse maps, and all boundary and commutator equations are closed. Its isometric image is therefore closed in $\mathscr{U}_Q$.

This does not give an extension for arbitrary members of $\mathscr{U}_Q$. For the explicit case $Q = \{1\}$ and the standard round disk, $d_Q = d$ and a nonidentity rotation is a uniform interior homeomorphism. Its boundary value is the same rotation, so it has no preimage under this particular restriction map. The named relative image condition must consequently be retained in any use of a uniform-homeomorphism deformation.

*Compact cores and one common original source bound.* Fix $x_0 \in O$. C.64's proved compact-covering lemma for the actual open evaluation map gives, for every compact $K \subset O$, a compact family $S \subset L$ with $K \subset S(x_0)$. One covers $K$ by finitely many evaluation neighborhoods and takes the union of their compact preimages. For compact $K, K'$ choose such $S, T$, respectively. For $z = sx_0 \in K$ and $z' = tx_0 \in K'$, commutation gives

$$\begin{aligned} qz' &= tqx_0 = ts^{-1}(qz), \\ qK' &\subset TS^{-1}(qz) \quad (q \in Q,\ z \in K). \end{aligned} \tag{C.81.8}$$

Every member of the one compact family $TS^{-1}$ is an original circle-fixed map. Thus $z_j \in K$ and $q_j z_j \to y$ on the original circle imply uniform convergence of the whole sets $q_j K'$ to $y$. No boundary extension or convergence of $q_j$ is used.

Now take a compact parameter family $h_p$ of $Q$-equivariant interior maps and inverses, identity outside $\pi^{-1}(K_0)$ for a fixed compact $K_0 \subset N$. Choose compact $C \subset O$ with $\pi C = K_0$, using finitely many local sections of the covering. Include in $C'$ all of $C$, $h_p(C)$ and $h_p^{-1}(C)$. Joint interior continuity and compactness make $C'$ compact. Since $\pi^{-1}(K_0) = QC$, (C.81.8) supplies one compact original family $P = TS^{-1} \cup \{1\}$ with

$$\begin{gathered} h_p(x) \in P(x), \qquad h_p^{-1}(x) \in P(x) \\ (p \text{ any parameter},\ x \in O). \end{gathered} \tag{C.81.9}$$

Inside $QC$ use equivariance and (C.81.8); outside it use the identity. Compact evaluation of this boundary-fixed $P$ proves both joint circle limits. The pasted maps and inverses are inverse homeomorphisms of the actual disk, continuous in the actual topology of (C.81.1).

*A based sphere with fixed compact quotient support.* Let $f : (S^n, *) \to (\mathscr{C}_Q, 1)$, $n \geq 1$, have descents $b_\xi$ identity outside $\operatorname{Int} N_0$, where $N_0$ is as in the statement. Descent and its inverse family are continuous by the open covering quotient and its local sections. The restrictions to $N_0$ lie in the identity component of $\mathrm{Homeo}(N_0 \text{ rel } \partial N_0)$: the sphere is path connected and its basepoint is the identity.

That component is contractible by [74, Lemma 2.1(ii)]. The nonempty entire boundary is fixed, so all its compact-surface exceptions are excluded. A contraction can be normalized to fix the identity throughout by replacing its value $C_r(h)$ by $C_r(h)C_r(1)^{-1}$, with the parameter directed from 1 to $h$. Applied to $b_\xi$ and extended by the identity off $N_0$, it gives a continuous family

$$B_{0,\xi} = 1, \qquad B_{1,\xi} = b_\xi, \qquad B_{r,*} = 1 \tag{C.81.10}$$

of homeomorphisms of $N$ supported in $N_0$.

Lift the homotopy $B_{r,\xi}\pi$ from its specified initial lift $H_{0,\xi} = \mathrm{id}_O$. The covering homotopy property gives a joint lift on $[0,1] \times S^n \times O$; it does not require the sphere parameter to be simply connected. Lift the inverse homotopy as well. The two composites lift the identity and start at the identity, hence are the identity by uniqueness of lifts. For each original $q$, the maps $H_{r,\xi}q$ and $qH_{r,\xi}$ lift the same family with identical initial value $q$. They agree. The basepoint family in (C.81.10) lifts to the constant identity.

At $r = 1$, both $H_{1,\xi}$ and $f_\xi$ lift the same $b_\xi$. Their relative deck label is locally constant in $\xi$ and equals 1 at $*$. Connectedness of the sphere therefore gives $H_{1,\xi} = f_\xi$ for every parameter. This proves agreement with the original family upstairs, rather than only after descent. Points above $N \setminus \operatorname{Int} N_0$ have constant base tracks and initial identity lifts, so they stay fixed. Formula (C.81.9) applies to this compact parameter family and its inverse. The resulting joint original-circle extension is the asserted based filling in $\mathscr{C}_Q$ with every deck label retained.

*Continuous local sections for actual orbit evaluation.* For finite $A$ as in (C.81.3), its $Q$-saturation is closed and locally finite in $O$. Commutation identifies $\mathscr{C}_Q(A)$ with the subgroup fixing that saturation pointwise. Choose $x \notin QA$ and a small quotient disk $U$ about $\pi x$, with compact closure disjoint from $\pi A$, that is evenly covered. A local coordinate bump translation moves $\pi x$ to each sufficiently nearby $y$, using a fixed compact subdisk as support. One explicit local formula is $z \mapsto z + \chi(z)v$, where $\chi = 1$ near the center, $\chi = 0$ outside that subdisk, and $|v| \operatorname{Lip}(\chi) < 1/2$. The two-sided Lipschitz estimate makes it injective with continuously varying inverse; properness and invariance of domain make it onto. The parameter $av$, $0 \leq a \leq 1$, gives its based supported isotopy.

Lift that isotopy from the identity, retaining the sheet containing $x$. It fixes $QA$, commutes with every $q$, and its endpoint maps $x$ to the corresponding local lift of $y$. The compact quotient-support proof above gives its continuity with inverses in the original closed-disk topology. Denote this actual local section by $\sigma_y$, now with $y$ denoting the upstairs point, and choose $\sigma_x = 1$.

The evaluation map is onto even from its identity path component. Given any other $x' \in O \setminus QA$, choose a path from $x$ to $x'$ in that connected punctured plane. Cover its compact projected path by finitely many quotient coordinate disks avoiding $\pi A$. Their short supported point-moving isotopies compose to move the point along that path. The upstairs path is the chosen one by uniqueness of the original covering lift. All supports lie in one compact quotient set, so their lifted family extends by (C.81.9) and fixes $QA$. Its endpoint sends $x$ to $x'$ in the asserted original group.

Set $F = \mathscr{C}_Q(A \cup \{x\})$. The local bundle chart and its inverse are exactly

$$\begin{aligned} h &\longmapsto (h(x), \sigma_{h(x)}^{-1} h), \\ (y, k) &\longmapsto \sigma_y k \qquad (k \in F). \end{aligned} \tag{C.81.11}$$

They are continuous inverse maps in the specified topology. Translation supplies the other charts. Thus (C.81.3) is a locally trivial principal bundle. Its metrizable manifold base makes it numerable, so the usual based homotopy lifting and exact sequence apply to these actual maps.

*The exact positive-degree effect of fixing finitely many orbits.* For the first orbit the base in (C.81.3) is the contractible plane $O$. Its homotopy exact sequence gives an isomorphism on every positive homotopy group after fixing that first orbit. Suppose $|Q| \geq 2$ and $A$ is now nonempty. The surface $M = O \setminus QA$ has at least two punctures. It is aspherical: a compact sphere image lies in a compact planar subsurface with nonempty boundary; that subsurface retracts onto a finite graph and has no homotopy in degrees at least two.

Moreover $\mathrm{Homeo}(M)_0$ is contractible by [74, Theorem 1.1]. Its exceptions are explicitly excluded here. The surface is oriented, and forgetting all but two punctures gives a surjection onto a free group of rank two: the two small meridians and their connecting paths can be chosen to avoid the remaining locally finite punctures. Thus it is neither the plane nor an annulus, and cannot be the nonorientable exception.

Every loop in $\mathscr{C}_Q(A)$ restricts continuously, with inverses, to a loop in $\mathrm{Homeo}(M)_0$. Its evaluation at $x$ factors through that restriction, and hence has zero class in $\pi_1(M, x)$. Asphericity and the exact sequence of (C.81.3) now give

$$\pi_n(\mathscr{C}_Q(A \cup \{x\}), 1) \xrightarrow{\cong} \pi_n(\mathscr{C}_Q(A), 1) \qquad (n \geq 1). \tag{C.81.12}$$

For $n = 1$, injectivity uses $\pi_2(M) = 0$ and surjectivity uses the just proved zero evaluation homomorphism. The higher degrees use both adjacent vanishing groups of $M$. Induction proves the asserted finite-set comparison for $|Q| \geq 2$. For $Q = \{1\}$, [74, Lemma 2.1(ii)] applies directly on the compact disk with its boundary and the finite fixed set. The identity components have zero positive homotopy groups on both sides, giving the same inclusion isomorphism.

This proves that any sphere class has a representative fixing any specified finite number of orbit labels, through an actual based homotopy in the original relative group. No uniform choice or limit over an infinite configuration has been used.

*The explicit comparison with Whitney topology.* Descent $h \mapsto \bar{h}$ from $\mathscr{C}_Q$ is continuous into the compact-open and inverse topology of $\mathrm{Homeo}(N)$: local covering sections prove joint continuity on each compact quotient set, also for the descended inverse maps. Assume now that $N$ is noncompact. Choose points $y_j$ escaping every compact set and pairwise disjoint, locally finite coordinate disks $V_j$ around them. Choose lifts $x_j$. Shrink each disk so that its lift $U_j$ at $x_j$ has compact closure, disjoint $Q$-translates, and is contained in a $d_Q$-ball of radius less than $1/(4j)$. Then every $qU_j$ has original $d$-diameter less than $1/j$.

Choose a disk homeomorphism $b_j$ supported in $V_j$, isotopic to the identity there, with $b_j(y_j) \neq y_j$. Lift its supported isotopy from the identity. The endpoint $h_j$ preserves each lifted disk $qU_j$ and is the identity elsewhere. It has the actual circle-fixed extension already proved. Consequently

$$\begin{aligned} \sup_{z \in \overline{D}} d(h_j z, z) &< 1/j, \\ \sup_{z \in \overline{D}} d(h_j^{-1} z, z) &< 1/j. \end{aligned} \tag{C.81.13}$$

Thus $h_j \to 1$ in $\mathscr{C}_Q$ and, by (C.81.7), in its exact uniform $d_Q$ topology as well.

Put $a_j = \rho(b_j(y_j), y_j) > 0$. There is a continuous positive $\epsilon$ on $N$ with $\epsilon(y_j) < a_j/2$: on each locally finite disk use a bump function interpolating between that positive chosen value at $y_j$ and the value 1 outside the disk. These locally finite formulas give a positive continuous global function. The one Whitney neighborhood

$$\{b : \rho(b(y), y) < \epsilon(y) \text{ for every } y \in N\} \tag{C.81.14}$$

contains none of the $b_j$. Hence the specified descent homomorphism, even on the subgroup of actual lifts of compactly supported quotient isotopies with its induced $\mathscr{C}_Q$ topology, is not continuous to this Whitney topology.

The definition and fragmentation results in [46, p. 395 and §2, Proposition 2.9, Corollary 2.10] use that majorant topology and compactly supported isotopies. Formula (C.81.13) and the single excluded neighborhood (C.81.14) prove the exact comparison obstruction to their direct use for an arbitrary original continuous family. Likewise the metric-cover deformation in [75, Theorem 1.1] has a compact quotient hypothesis and does not itself establish preservation of our named relative image. The uniform controlled fragmentation and the coherent infinite-configuration limit remain the stated general discrete-case requirements. □

**C.82. [Proved] A null lifted triangulation and its actual skeleton stabilizer.** Retain the actual disk, discrete operator group, transitive relative subgroup, quotient covering and metrics of C.81. There is a locally finite triangulation $\mathcal{T}$ of $N = O/Q$ whose full lift $\widetilde{\mathcal{T}}$ through the original covering $\pi : O \to N$ has the following property: for every $\epsilon > 0$, only finitely many lifted closed vertex stars have $d$-diameter at least $\epsilon$. In particular, this holds for the lifted closed triangles. Every such star is mapped injectively by $\pi$.

Write $\Lambda = |\widetilde{\mathcal{T}}^{(1)}|$ for the actual lifted one-skeleton and set

$$\mathscr{S}_{Q,\Lambda} = \{h \in \mathscr{C}_Q : h|_\Lambda = \mathrm{id}\}. \tag{C.82.1}$$

Choose one closed lifted triangle $D_\sigma$ for each two-simplex $\sigma$ of $\mathcal{T}$. Restriction and $Q$-equivariant pasting are mutually inverse topological group isomorphisms

$$\mathscr{S}_{Q,\Lambda} \;\cong\; \prod_{\sigma \in \mathcal{T}^{(2)}} \mathrm{Homeo}(D_\sigma \text{ rel } \partial D_\sigma). \tag{C.82.2}$$

Here $\mathcal{T}^{(2)}$ indexes the two-simplices, the product has the product topology, and the left side retains the original closed-disk topology of maps and inverses. Coordinatewise Alexander deformation gives a continuous contraction of this entire actual subgroup to 1, fixing 1 throughout.

The conclusion concerns the pointwise skeleton stabilizer. A deformation of every original based sphere family in $\mathscr{C}_Q$ into that stabilizer, with joint original-boundary control, is not supplied by this statement or by the finite-label comparisons of C.81.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact null lifted triangulation and contractible skeleton stabilizer in the remaining discrete-operator level-zero question.

*Proof. The quotient charts and the chosen subdivision.* The quotient $N$ is a second-countable surface, possibly nonorientable. Indeed the original free proper discrete action gives the actual covering charts, and the open quotient of the countable base of $O$ is a countable base of $N$. Radó's original triangulation theorem, [65, §3, Hilfssatz 2, pp. 111–114], gives a locally finite triangulation of this surface. Its proof explicitly establishes local finiteness before subdividing the finite polygons into triangles. We retain the homeomorphism from its geometric realization onto the actual $N$.

Choose a continuous proper function $a : N \to [0, \infty)$. To recall its construction, take compact sets $K_j \subset \operatorname{int} K_{j+1}$ exhausting $N$, and continuous functions $\varphi_j$ equal to zero on $K_j$ and one

outside $\operatorname{int} K_{j+1}$, with values in $[0,1]$. The locally finite sum $a=\sum_j \varphi_j$ is proper: outside $K_{m+1}$ its first $m$ summands equal one. In the compact case the constant function zero suffices.

For each $y\in N$ choose a relatively compact open coordinate disk $V_y$ containing $y$ so small that every sheet above $V_y$ is mapped isometrically onto $V_y$ for the metrics $d_Q,\rho$ of C.81, and

$$\begin{aligned} |a(z)-a(y)|&<1 \quad (z\in V_y),\\ \operatorname{diam}_\rho(V_y)&<\frac{1}{2+a(y)}. \end{aligned} \tag{C.82.3}$$

Such disks are obtained by shrinking the actual isometric covering neighborhoods of C.81. No coordinate on the original circle is changed in this step.

Apply Haver's subdivision lemma with $k=0$ to the initial triangulation, its actual map to $N$, and the cover $\{V_y\}$ [33, Lemma 1, p. 458]. It gives a locally finite subdivision $\mathcal{T}$ for which every closed vertex star is contained in one $V_y$. The notation $N(v,\mathcal{T})$ in that lemma is the union of the closed simplices meeting the vertex, so it is precisely the closed star used here.

Lift this actual triangulation through $\pi$. On each covering chart the lifted simplices and all their incidences are the pullbacks under its inverse section. On overlapping charts these pullbacks agree by covering uniqueness. They therefore give a triangulation $\widetilde{\mathcal{T}}$ of $O$, with the original $Q$-action by simplicial deck maps. Every lifted closed vertex star lies in one sheet over a $V_y$, so $\pi$ restricts to a homeomorphism on that star. If its quotient vertex is $v$, then (C.82.3) gives

$$\operatorname{diam}_{d_Q}\left(\overline{\operatorname{st}}(\widetilde{v})\right)<\frac{1}{1+a(v)}. \tag{C.82.4}$$

The lift is locally finite, directly from the covering charts and local finiteness downstairs. Alternatively, a compact subset of $O$ projects to a compact set meeting finitely many quotient stars; properness allows only finitely many translates of their compact chosen lifts to meet it.

*Nullity in the original closed disk.* We use the precise compact-core conclusion of C.81. For a compact set $C\subset O$, choose a compact family $T$ of actual boundary-fixed commuting homeomorphisms with $C\subset T(x_0)$. Such a family exists by the compact-covering lemma applied to the open evaluation map of the closure of the transitive subgroup. For $z,z'\in C$ choose $s,t\in T$ with $z=sx_0$, $z'=tx_0$. For every original operator $q\in Q$,

$$qz'=ts^{-1}(qz), \qquad qC\subset TT^{-1}(qz). \tag{C.82.5}$$

The compact family $TT^{-1}$ consists of actual maps fixing every point of the original circle. Hence

$$z_j\in C,\quad q_jz_j\longrightarrow b\in\partial\overline{D} \quad\Longrightarrow\quad q_jC\longrightarrow\{b\} \text{ uniformly in } d. \tag{C.82.6}$$

For completeness, failure of uniformity would give $h_j\in TT^{-1}$ with $d(h_jq_jz_j,b)$ bounded away from zero. A convergent subsequence of $h_j$ in the actual map topology contradicts $h_jb=b$. No extension of $q_j$ across the circle appears in this argument.

Suppose infinitely many distinct lifted closed stars have diameter at least $\epsilon>0$. Choose a point $x_j$ in each. Compactness of $\overline{D}$ gives a convergent subsequence. Its limit cannot lie in $O$, because the lifted stars are locally finite there. Thus $x_j\to b$ on the original circle.

Let $v_j$ be their quotient vertices. An unbounded subsequence of $a(v_j)$ has a further subsequence tending to infinity. Equation (C.82.4), together with $d\le d_Q$, contradicts the lower diameter bound on that subsequence. Otherwise the $v_j$ lie in one compact sublevel of $a$. Local finiteness gives only finitely many such vertices. After taking a subsequence, the stars are therefore $q_jC$ for one chosen compact lifted star $C$. Writing $x_j=q_jz_j$ with $z_j\in C$, (C.82.6) gives the same contradiction. This proves the asserted finite-large-star property.

Every closed triangle is contained in each of its three closed vertex stars. Only finitely many triangles belong to any one star. The finite-large-star property consequently implies the

finite-large-triangle property. It also gives the exact boundary form that will be used below: for any lifted triangles $E_j$ and points $x_j \in E_j$,

$$x_j \longrightarrow b \in \partial\overline{D} \quad \Longrightarrow \quad \sup_{z\in E_j} d(z,b) \longrightarrow 0. \tag{C.82.7}$$

Indeed each fixed finite list of compact triangles is eventually avoided, and all others have arbitrarily small diameter.

*The actual product and both inverse maps.* Choose one lift $D_\sigma$ for each quotient triangle. A deck transformation preserving $D_\sigma$ is the identity: $\pi$ is injective there, so that transformation fixes its points, and the original action is free.

For a tuple $(h_\sigma)$ of the factors in (C.82.2), define $\mathcal{P}(h_\sigma)$ on $\overline{D}$ by

$$\begin{cases} qh_\sigma q^{-1} & \text{on } qD_\sigma, \quad q \in Q, \\ \mathrm{id} & \text{on } \Lambda \cup \partial\overline{D}. \end{cases} \tag{C.82.8}$$

The first prescription is unambiguous on every open face. On common edges and vertices all prescriptions are the identity. Local finiteness gives continuity on $O$ by finite pasting. Equation (C.82.7) gives continuity at the original circle: the image of a point in a triangle stays in that same triangle. Replacing every $h_\sigma$ by its inverse gives a continuous map by exactly the same proof, and the two composites are the identity face by face and on the skeleton and circle. Thus (C.82.8) is an actual closed-disk homeomorphism. Reindexing its formula by left multiplication in $Q$ proves commutation with every original operator.

Conversely, an element $h$ of (C.82.1) preserves each closed lifted triangle. Its Jordan boundary is fixed pointwise. Of the two complementary components in $O$, only its interior has compact closure in $O$. An interior homeomorphism preserves this property, so it cannot exchange the two components. Restriction therefore defines

$$\mathcal{R}(h) = (h|_{D_\sigma})_\sigma. \tag{C.82.9}$$

The commutator equations imply the formula (C.82.8) on every other lift. It follows that $\mathcal{PR}$ and $\mathcal{RP}$ are the identity. Both maps respect the group operations. This proves the asserted group isomorphism with its exact maps.

Restriction $\mathcal{R}$ is continuous for maps and inverses, since each $D_\sigma$ is compact and invariant. For continuity of $\mathcal{P}$, fix $\epsilon > 0$. Only finitely many actual lifted triangles have diameter at least $\epsilon/4$. On every other triangle, two pasted maps, and separately their inverses, take each point into the same set of diameter less than $\epsilon/4$. On the finitely many large triangles, uniform convergence of the corresponding finitely many factors gives uniform convergence after their specified conjugations by $q$. This uses continuity of that fixed $q$ on a compact triangle, and requires no boundary extension of $q$. The finitely many factor neighborhoods therefore control both uniform distances on all of $\overline{D}$. This is exactly continuity from the product topology to the original map-and-inverse topology. It proves (C.82.2) as a topological group isomorphism.

*A contraction in the actual topology.* Choose a homeomorphism from the standard closed unit disk to each $D_\sigma$. In that coordinate, for a boundary-fixed disk homeomorphism $h$ put

$$A_r(h)(z) = \begin{cases} rh(z/r), & |z| \le r, \quad r > 0, \\ z, & |z| \ge r, \end{cases} \qquad A_0(h) = \mathrm{id}\,. \tag{C.82.10}$$

The two formulas agree at $|z| = r$, and $A_r(h)^{-1} = A_r(h^{-1})$. For positive $r$, evaluation is jointly continuous, including across that seam; compactness of the standard disk gives the required compact-open continuity. At $r = 0$ the map and inverse displacements are each at most $2r$, uniformly in $h$. This proves joint continuity for the full map-and-inverse group topology. Also $A_1(h) = h$ and $A_r(\mathrm{id}) = \mathrm{id}$.

Take the product of these deformations and transport it through the mutually inverse maps $\mathcal{P}, \mathcal{R}$. A map into a product is continuous exactly when its coordinate maps are continuous. Thus this gives a continuous map $[0,1] \times \mathscr{S}_{Q,\Lambda} \to \mathscr{S}_{Q,\Lambda}$ from the constant identity map to the identity map, fixing the group identity throughout. All original operator labels, the pointwise skeleton condition, and the original circle condition remain exact.

The construction proves an actual infinite product comparison because the nullity estimate supplies uniform control outside finitely many triangles. It does not assert an infinite-configuration lifting theorem for restriction from $\mathscr{C}_Q$. In particular, C.81's finite-orbit evaluation bundles have not been passed to an inverse limit. The deformation of a general compact parameter family to this skeleton stabilizer, with controlled vertex and edge motions, remains the next step in the discrete-operator part of C.12. □

**C.83. [Proved] An actual equivariant infinite-evaluation path without a closed-disk lift.** There is an actual closed disk and a closed, discrete, free proper interior operator group $Q \cong \mathbb{Z}$ satisfying every hypothesis of C.81, including an actual transitive boundary-fixed subgroup of its centralizer, with the following property.

For a specified ordered, locally finite, $Q$-invariant vertex set $V$, let $\mathscr{E}_V$ be the actual image of simultaneous labelled evaluation

$$\mathrm{ev}_V : \mathscr{C}_Q \longrightarrow \mathscr{E}_V. \tag{C.83.1}$$

Give $\mathscr{E}_V$ the subspace topology of ordered coordinate convergence together with vague convergence of counting measures in the specified interior plane coordinate. The map (C.83.1) is continuous and onto. There is an explicit based null-homotopic loop in $\mathscr{E}_V$ which has no continuous lift to $\mathscr{C}_Q$ starting at 1. Every point of that loop is the evaluation of a displayed actual boundary-fixed, $Q$-commuting homeomorphism. All original $Q$-labels are retained.

The vertex set can be taken to be the full lift of a quotient triangulation having the null-star property of C.82. Thus, for these exact source and target topologies, infinite evaluation has neither the path-lifting property nor the Serre or Hurewicz fibration property. The proof does not assert a nontrivial homotopy class in the actual centralizer.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact infinite-configuration topology and lifting obstruction in the remaining discrete-operator family-to-skeleton step.

*Proof. The actual disk, operators and transitive source.* Let $\overline{D}$ be the closed Euclidean unit disk, let $O = \operatorname{int} \overline{D}$, and use its original Euclidean distance $d$. The specified interior coordinate and inverse are

$$\begin{aligned} j : \mathbb{R}^2 &\longrightarrow O, \qquad j(z) = \frac{z}{1+\|z\|}, \\ j^{-1} : O &\longrightarrow \mathbb{R}^2, \qquad j^{-1}(w) = \frac{w}{1-\|w\|}. \end{aligned} \tag{C.83.2}$$

For $a \in \mathbb{R}^2$ write $T_a(z) = z + a$. Set $q = jT_{(1,0)}j^{-1}$ and $Q = \langle q \rangle$, with exactly these powers as its labels. The conjugacy (C.83.2) proves freeness and properness: two compact subsets of the plane meet under only finitely many integer horizontal translations. It also proves that $Q$ is a closed discrete subgroup of $\mathrm{Homeo}(O)$. Indeed, evaluation at $j(0)$ of any convergent sequence of its elements bounds the integer translations in the plane, and a convergent subsequence of integers is eventually constant.

Each $jT_aj^{-1}$ extends by the identity on the original circle. We will use the following explicit estimate repeatedly. For $z, v \in \mathbb{R}^2$,

$$\|j(z+v) - j(z)\| \le \frac{2\|v\|}{1+\|z\|}. \tag{C.83.3}$$

To prove it, put $r = \|z\|$ and $r' = \|z + v\|$ and write the difference as $v/(1 + r) + (z + v)((1 + r')^{-1} - (1 + r)^{-1})$. The reverse triangle inequality bounds the second summand by $\|v\|/(1 + r)$. The same estimate applies to inverse translations. Uniform bounds for $a$ on compact parameter sets therefore give joint continuity of the extended maps and inverses.

Let $G$ consist of these extended translations. Their displayed composition law makes $G$ an actual subgroup of $\mathscr{C}_Q$, and $G$ is transitive on $O$. Here $\mathscr{C}_Q$ is exactly the group in C.81.1, with metric

$$D(h, k) = \sup_{w \in \overline{D}} \|h(w) - k(w)\| + \sup_{w \in \overline{D}} \|h^{-1}(w) - k^{-1}(w)\|. \tag{C.83.4}$$

This is its original compact-open topology on maps and inverses. In particular, no interior-only topology is substituted.

*The complete label set and target topology.* In the plane put $v_{k,l} = (k/4, l)$ for $(k, l) \in \mathbb{Z}^2$, and let $V = \{j(v_{k,l})\}$. Enumerate $\mathbb{Z}^2$ by increasing $\max\{|k|, |l|\}$, using lexicographic order within each finite shell. This fixes one literal enumeration by the positive integers. The operator $q$ acts on its labels by $(k, l) \mapsto (k + 4, l)$.

An ordered configuration is a sequence of distinct points $z_i \in \mathbb{R}^2$ for which every compact planar set contains only finitely many $z_i$. Write $\mu_z = \sum_i \delta_{z_i}$ and use

$$\begin{aligned} d_{\mathrm{prod}}(z, w) &= \sum_{i \geq 1} 2^{-i} \min\{1, \|z_i - w_i\|\}, \\ d_\Sigma(z, w) &= d_{\mathrm{prod}}(z, w) + d_V(\mu_z, \mu_w), \end{aligned} \tag{C.83.5}$$

where $d_V$ metrizes vague convergence: $\mu_m \to \mu$ means $\int \varphi \, d\mu_m \to \int \varphi \, d\mu$ for every continuous compactly supported planar function $\varphi$. This is the sum topology specified in [68, Definition 2.6 and equation (2), pp. 5–6]. Only that definition is used here.

The actual evaluation is

$$\begin{aligned} \big(\mathrm{ev}_V(h)\big)_{k,l} &= j^{-1} h j(v_{k,l}), \qquad \mathscr{E}_V = \mathrm{ev}_V(\mathscr{C}_Q), \\ z_{k+4,l} &= z_{k,l} + (1, 0) \quad (z \in \mathscr{E}_V). \end{aligned} \tag{C.83.6}$$

The last equation follows from the original commutation relation with $q$, not from discarding its labels. A homeomorphism of the plane sends this grid to a locally finite set, because its inverse sends compact sets to compact sets. Thus every evaluation in (C.83.6) belongs to the stated configuration space.

We verify continuity of this exact map. Suppose $h_m \to h$ in (C.83.4). Each labelled coordinate converges. For a compact planar set $K$, uniform convergence of the inverse maps implies that all $h_m^{-1}(j(K))$, for sufficiently large $m$, lie in one compact subset of $O$: take a small closed neighborhood of the compact set $h^{-1}(j(K))$ still inside $O$. Only finitely many vertices of $V$ lie in that neighborhood. Hence the evaluation of any fixed compactly supported test function is eventually a fixed finite sum of continuous coordinate evaluations. This proves vague convergence and then continuity for (C.83.5).

The grid also has the required triangulation interpretation. Divide its rectangles of width $1/4$ and height 1 by the same diagonal, and transport this plane triangulation through $j$. Horizontal translation by 1 preserves it. Each closed vertex star has horizontal width $1/2 < 1$, so its projection to $O/Q$ is injective. The planar stars have one common diameter bound. Formula (C.83.3) shows that their images have diameters tending to zero as their vertices escape in the plane; only finitely many vertices lie in any fixed compact set. This is precisely the finite-large-star property used in C.82.

*The actual up-and-back paths.* Define the continuous tent function $b(u) = \max\{0, 1 - 4|u|\}$. For $n \geq 1$ and $s \in [0, 1]$ put

$$\begin{aligned} F_{n,s}(x, y) &= (x + sn\, b(y - n), y), \\ F_{n,s}^{-1}(x, y) &= (x - sn\, b(y - n), y). \end{aligned} \tag{C.83.7}$$

These formulas prove both inverse identities and commutation with every $T_{(m,0)}$. Their displacement is at most $n$. Consequently (C.83.3) extends $H_{n,s} = jF_{n,s}j^{-1}$ by the identity to the original circle, with a jointly continuous inverse. For each fixed $n$, $s \mapsto H_{n,s}$ is continuous in (C.83.4). To check the uniform assertion, use (C.83.3) outside one large planar disk and ordinary uniform continuity on its compact closure. The identical argument works for the displayed inverse. Thus $H_{n,s} \in \mathscr{C}_Q$ and $H_{n,0} = 1$.

Only row $l = n$ of the vertex set moves:

$$\big(\mathrm{ev}_V(H_{n,s})\big)_{k,l} = \begin{cases} (k/4 + sn, n), & l = n, \\ (k/4, l), & l \neq n. \end{cases} \tag{C.83.8}$$

These equations retain every label in (C.83.6), including labels in the same operator orbit.

Let $I_n = [2^{-n-1}, 2^{-n}]$ and write

$$\lambda_n(t) = 2^{n+1}t - 1, \qquad s_n(t) = 2\min\{\lambda_n(t), 1 - \lambda_n(t)\}, \quad t \in I_n, \qquad t_n = 3 \cdot 2^{-n-2}. \tag{C.83.9}$$

The amplitude is zero at both endpoints of $I_n$ and one at $t_n$. Define

$$c(t) = \begin{cases} \mathrm{ev}_V(H_{n,s_n(t)}), & t \in I_n,\ n \geq 1, \\ \mathrm{ev}_V(1), & t = 0 \text{ or } 1/2 \leq t \leq 1. \end{cases} \tag{C.83.10}$$

The formulas agree at all overlapping interval endpoints.

We check both components of the topology at $t = 0$. Any finite collection of fixed labels avoids the moving row once $n$ is sufficiently large. Its coordinates then equal their original values for every $s \in [0, 1]$; the remaining product metric terms are bounded by the tail of $\sum_i 2^{-i}$. For a compact planar set $K$, its $y$-coordinates are bounded. For all sufficiently large $n$, the moving row and all its images are disjoint from $K$, since (C.83.7) preserves $y$. Every test function supported in $K$ therefore has exactly its original counting-measure evaluation, uniformly in $s$. This proves vague convergence. Equivalently, for a metric defined by a weighted countable test family, first fix finitely many supports and then bound the remaining summands by their weight tail. It proves continuity in $d_\Sigma$ at zero. Continuity on each $I_n$ and at its other endpoints has already been proved. Thus (C.83.10) is a continuous based loop in the actual subspace $\mathscr{E}_V$. Its based null-homotopy in this same subspace is the exact map

$$C(r,t) = \begin{cases} \mathrm{ev}_V(H_{n,(1-r)s_n(t)}), & t \in I_n,\ n \geq 1, \\ \mathrm{ev}_V(1), & t = 0 \text{ or } 1/2 \leq t \leq 1, \end{cases} \qquad 0 \leq r \leq 1. \tag{C.83.11}$$

The preceding finite-coordinate and test-support arguments are uniform in $r$, since every displayed amplitude still belongs to $[0, 1]$. Together with continuity on each parameter rectangle and equality at its interval endpoints, they prove joint continuity of $C$. Its faces are $C(0,t) = c(t)$, $C(1,t) = \mathrm{ev}_V(1)$ and $C(r,0) = C(r,1) = \mathrm{ev}_V(1)$.

*The lower bound for every proposed lift.* Suppose a continuous map $\widetilde{c} : [0,1] \to \mathscr{C}_Q$ satisfied $\widetilde{c}(0) = 1$ and $\mathrm{ev}_V(\widetilde{c}(t)) = c(t)$. At the peak time $t_n \to 0$, the literal label $(0, n)$ forces

$$\widetilde{c}(t_n)j(0,n) = j(n,n),$$
$$\sup_{w \in \overline{D}} \|\widetilde{c}(t_n)w - w\| \ \geq\ \|j(n,n) - j(0,n)\| \ \geq\ \frac{n}{1 + \sqrt{2}\,n} \ >\ \frac{1}{3}. \tag{C.83.12}$$

The penultimate inequality uses the first coordinate alone. The last follows from $\sqrt{2} < 2$ and $n/(1 + 2n) \geq 1/3$. It holds for every homeomorphism having the required labelled evaluation, regardless of its behavior between vertices. It contradicts continuity at zero in (C.83.4).

This proves the claimed failure of path lifting. The lifting axiom of either named fibration, applied to a one-point initial space, would supply precisely that lift, so it also fails. No contraction of an original compact sphere family has been disproved: the loop just constructed lives in the explicitly stated evaluation-image topology and has the displayed based null-homotopy there. In particular this calculation does not decide the remaining general centralizer homotopy question in C.12. □

**C.84. [Proved] The exact radial translation topology and a contractible monotone subspace.** Retain the actual radial coordinate $j(z) = z/(1 + \|z\|)$, its inverse, the original closed unit disk, and the operator $q = jT_{(1,0)}j^{-1}$ of C.83. Conjugation $h \mapsto j^{-1}hj$ is a topological group isomorphism from the actual relative centralizer $\mathscr{C}_Q$ onto the following explicitly topologized group $\mathscr{P}$. Its elements are plane homeomorphisms $F = (X_F, Y_F)$ commuting with $T_{(1,0)}$ and satisfying

$$\lim_{R\to\infty} \sup_{\substack{0\le x\le 1\\ |y|\ge R}} \frac{\|F(x,y) - (x,y)\|}{|y|} = 0. \tag{C.84.1}$$

Every inverse satisfies the same condition. The exact topology on $\mathscr{P}$ is given by the finite metric

$$\begin{aligned} \|F - H\|_* &= \sup_{\substack{0\le x\le 1\\ y\in\mathbb{R}}} \frac{\|F(x,y) - H(x,y)\|}{1 + |y|}, \\ \delta_*(F,H) &= \|F - H\|_* + \|F^{-1} - H^{-1}\|_*. \end{aligned} \tag{C.84.2}$$

In (C.84.1) the limit is uniform over every compact family in the actual group.

Let $\mathscr{M} \subset \mathscr{P}$ consist of those $F$ for which, at every $x$, $y \mapsto Y_F(x,y)$ is an increasing homeomorphism of $\mathbb{R}$. The proof gives an explicit continuous contraction of this entire subspace, fixing the identity, through the same subspace. Conjugating back through $j$ gives the contraction in the original closed-disk topology, preserving its circle and every original operator label. The inclusion $\mathscr{M} \hookrightarrow \mathscr{P}$ is proper, as witnessed by an explicit periodic core rotation below.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact actual topology and controlled contracting subspace in the radial discrete-cyclic operator model.

*Proof. The exact set correspondence and its compact-family bound.* For a commuting plane homeomorphism write

$$F(x,y) = (x + u_F(x,y),\ y + w_F(x,y)). \tag{C.84.3}$$

Both displacement functions are periodic of period one in $x$. We first prove necessity of (C.84.1) uniformly over a compact family of actual disk maps.

Take any sequence $F_i$ from such a family and points $(x_i, y_i)$ with $0 \le x_i \le 1$ and $|y_i| \to \infty$. Choose integers $m_i^+, m_i^-$ such that

$$x_i + m_i^+ = y_i + e_i^+, \qquad x_i + m_i^- = -y_i + e_i^-, \qquad |e_i^\pm| \le \tfrac{1}{2}. \tag{C.84.4}$$

Pass to a subsequence on which the sign of $y_i$ is constant. The two radial input sequences approach the corresponding two diagonal points of the original circle. Compactness of the actual family, whose members fix every circle point, forces the output sequences to approach those same points. To justify the uniform assertion directly, any alleged failure has a subsequence of disk maps converging uniformly; its limiting map fixes the specified boundary point.

Periodicity makes the two output second coordinates equal to $Y_i = Y_{F_i}(x_i, y_i)$. The limiting directions have nonzero second coordinate. Thus the following ratios are defined eventually and satisfy

$$\frac{X_{F_i}(x_i,y_i) + m_i^+}{Y_i} \longrightarrow 1, \qquad \frac{X_{F_i}(x_i,y_i) + m_i^-}{Y_i} \longrightarrow -1. \tag{C.84.5}$$

Subtracting and using $(m_i^+ - m_i^-)/y_i \to 2$ gives $Y_i/y_i \to 1$. Either ratio then gives $u_{F_i}(x_i, y_i)/y_i \to 0$. Consequently the normalized norm in (C.84.1) tends to zero along every such sequence. This proves the required uniform compact-family limit. Apply the same proof to the compact family of actual inverse maps.

Conversely, suppose a commuting plane homeomorphism satisfies (C.84.1). For any $\epsilon > 0$ its displacement is at most $\epsilon|y|$ outside one horizontal band, uniformly in all $x$ by periodicity. Within that band the displacement has a finite bound, by continuity on its compact quotient. It follows that

$$\frac{\|F(z) - z\|}{1 + \|z\|} \longrightarrow 0 \quad (\|z\| \to \infty). \tag{C.84.6}$$

The exact radial estimate (C.83.3) therefore proves that $jFj^{-1}$ extends continuously by the identity to the original circle. This extension is a continuous bijection of the compact disk onto itself, so its inverse is continuous. It is an actual member of $\mathscr{C}_Q$. Applying necessity to that inverse also proves its version of (C.84.1). Restriction and extension by the specified identity are mutually inverse, and conjugation preserves multiplication and every original power of $q$. This proves the set and group correspondence without assuming a different compactification.

*Both inverse topology maps.* The expressions in (C.84.2) are finite: on compact $y$-bands they are bounded by periodicity, and on the tails this follows from (C.84.1). The displayed formula is a metric.

Suppose actual disk maps converge, with their inverses, to an actual map. The sequence together with its limit is a compact family, so the uniform version of (C.84.1) bounds both weighted tails in (C.84.2). On each compact rectangle $[0, 1] \times [-R, R]$, conjugation through $j$ and $j^{-1}$ gives uniform convergence in plane coordinates. The images remain in one compact interior set, where $j^{-1}$ is uniformly continuous. Combining this compact-rectangle convergence with the tail bound proves convergence of $\|F_i - F\|_*$. The same proof for inverses gives convergence in $\delta_*$.

For the other direction fix $H \in \mathscr{P}$ and set

$$c_H = \sup_{(x,y)\in\mathbb{R}^2} \frac{1 + |y|}{1 + \|H(x, y)\|} < \infty. \tag{C.84.7}$$

The finiteness follows from $|Y_H(x, y)| \geq |y|/2$ on sufficiently far tails, and boundedness of the numerator on the remaining band. Apply (C.83.3) at the base point $H(x, y)$ to obtain

$$\sup_{z\in\mathbb{R}^2} \|jF(z) - jH(z)\| \leq 2c_H \|F - H\|_*. \tag{C.84.8}$$

The periodicity of $F - H$ permits exactly the supremum in (C.84.2). The inverse version of (C.84.8) proves convergence in the original disk map-and-inverse metric. These two metrizable-space arguments prove the asserted topological group isomorphism.

*Exact factorization on the monotone subspace.* For $F \in \mathscr{M}$ define

$$\begin{aligned} V_F(x, y) &= (x, Y_F(x, y)), & H_F &= FV_F^{-1}, \\ H_F(x, y) &= (h_{F,y}(x), y), & F &= H_F V_F. \end{aligned} \tag{C.84.9}$$

The inverse of $V_F$ is continuous. Indeed, for a converging sequence of $x$-coordinates and target heights, two fixed height values bracketing the limiting inverse bracket all nearby inverses. Every convergent subsequence between those brackets must solve the same strictly monotone equation. Shrinking the brackets proves convergence of the inverse. This gives an actual plane homeomorphism. Its commutation with $T_{(1,0)}$ is literal periodicity, and its displacement satisfies (C.84.1). Thus $V_F \in \mathscr{P}$, and so $H_F \in \mathscr{P}$ as well.

The formula for $H_F$ fixes the height. For each height its first coordinate is a real-line homeomorphism and satisfies $h_{F,y}(x + 1) = h_{F,y}(x) + 1$. It is therefore increasing. For $0 \leq r \leq 1$

put

$$\begin{aligned} V_{F,r}(x,y) &= (x,(1-r)y + rY_F(x,y)),\\ H_{F,r}(x,y) &= ((1-r)x + rh_{F,y}(x),y),\\ A_r(F) &= H_{F,r}V_{F,r}. \end{aligned} \tag{C.84.10}$$

Every displayed one-dimensional convex combination is strictly increasing and proper, and its inverse depends continuously on the other coordinates by the same bracketing argument. The maps retain the translation equations exactly. Their displacements are scalar multiples of the respective original fiber displacements, so (C.84.1) holds uniformly for $r \in [0,1]$. They are actual elements of $\mathscr{P}$.

The second coordinate of $A_r(F)$ is the increasing function $(1-r)y + rY_F(x,y)$. Thus $A_r(F) \in \mathscr{M}$. The faces satisfy $A_0(F) = 1$, $A_1(F) = F$, and $A_r(1) = 1$.

*Full family continuity, including inverses.* The map $F \mapsto V_F$ is continuous in weighted forward distance, by its formula and (C.84.2). Equation (C.84.8) converts this to uniform convergence of the actual disk maps. For homeomorphisms of this compact disk, uniform forward convergence to a homeomorphism implies uniform inverse convergence. Explicitly, if $\omega_{B^{-1}}$ is a modulus of continuity of the fixed inverse,

$$\sup \|B_i^{-1} - B^{-1}\| \le \omega_{B^{-1}}(\sup \|B_i - B\|). \tag{C.84.11}$$

This follows by applying $B^{-1}$ to $B(B_i^{-1}z)$ and $z$. Consequently $F \mapsto V_F$ is continuous in the full actual group topology. The actual group operations make $F \mapsto H_F$ continuous as well.

On either fiber subgroup the weighted forward difference of the interpolations at $(r,F)$ and $(s,H)$ is bounded by $r\|F-H\|_* + |r-s|\|H-1\|_*$. Equations (C.84.8) and (C.84.11) therefore give joint continuity of each factor in the original map-and-inverse topology. Their product is jointly continuous. This proves the contraction of the entire subspace, rather than only an individual pathwise construction.

*The explicit scope witness.* For each integer $k$, take the disk of radius $1/10$ centered at $(k+1/2,0)$. These disks are pairwise disjoint. In polar coordinates about each center use the identical homeomorphism

$$(\rho,\theta) \longmapsto (\rho,\theta+\alpha(\rho)), \qquad \alpha(\rho) = \begin{cases} \pi, & 0 \le \rho \le 1/20,\\ \pi(2-20\rho), & 1/20 \le \rho \le 1/10. \end{cases} \tag{C.84.12}$$

Use the identity elsewhere. The inverse subtracts the same angle, and the boundary values agree. Local finiteness proves that this is a plane homeomorphism. It commutes with the original unit translation and its displacement is at most $1/5$. Hence it belongs to $\mathscr{P}$. On the vertical line through any core center it sends the heights $-1/40, 1/40$ to $1/40, -1/40$, respectively. It consequently lies outside $\mathscr{M}$.

The contraction (C.84.10) retains every original operator and boundary label on its stated subspace. C.85–C.87 below construct the continuous cutoff for arbitrary original families and a contraction on the whole group $\mathscr{P}$. □

**C.85. [Proved] A parametric cutoff and all positive homotopy groups in the radial cyclic model.** Retain the exact coordinate $j$, original disk, operator $q = jT_{(1,0)}j^{-1}$, and topological group $\mathscr{P}$ of C.84. There is a continuous map

$$\begin{aligned} \mathcal{C} : \mathscr{P} \times [1,\infty] \longrightarrow \mathscr{P}, \qquad & \mathcal{C}_R(F) = \mathcal{C}(F,R),\\ \mathcal{C}_\infty(F) = F, \qquad & \mathcal{C}_R(\mathrm{id}) = \mathrm{id}\,. \end{aligned} \tag{C.85.1}$$

For every finite $R$, $\mathcal{C}_R(F)$ agrees with $F$ on $|y| \le R$ and is the identity outside a finite horizontal band. The band can be chosen uniformly for $F$ in any compact family and $R$ in any compact

finite interval. Moreover, $\mathcal{C}_R(F) \to F$ as $R \to \infty$ uniformly on compact families in the original closed-disk map-and-inverse topology. Consequently every based sphere family in the actual relative centralizer $\mathscr{C}_Q$ has a based filling there:

$$\pi_n(\mathscr{C}_Q, 1) = 0 \qquad (n \geq 1). \tag{C.85.2}$$

Every original boundary point and every original integer-translation label is retained. This statement concerns the specified radial cyclic action.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), continuous actual cutoff and all positive homotopy groups for the specified radial discrete-cyclic model.

*Proof. The actual upper component and its marked conformal lift.* Write $z = x + iy$ and put

$$\begin{aligned} &E(z) = e^{2\pi i z}, \qquad m = m_R(F) = \max_{0 \leq x \leq 1} |F(x,R) - (x,R)|, \\ &M = 1 + m, \qquad B = 2(M+1) = 2m + 4, \qquad L = 100B. \end{aligned} \tag{C.85.3}$$

All these quantities refer to the actual plane coordinates of C.84. The proper line $\Gamma = F(\mathbb{R} \times \{R\})$ is periodic under $T_{(1,0)}$ and lies in $R - m \leq y \leq R + m$. Its upper complementary component is $\Omega = F(\{y > R\})$. Indeed, the sublinear displacement condition makes the image of the positive end approach the positive end, which identifies this component exactly.

The curve $\gamma(x) = E(F(x,R))$, $x \in \mathbb{R}/\mathbb{Z}$, is a Jordan curve. Equality of two values means that their plane images differ by an integer translation; injectivity of $F$ and its exact translation relation give equality of the source labels modulo one. Its winding about zero is one, since the real part of $F(x+1,R)$ exceeds that of $F(x,R)$ by one. Its bounded Jordan domain $D$ therefore contains zero. The preimage of $\gamma$ under $E$ is exactly $\Gamma$. Points sufficiently high above $\Gamma$ map into $D$, so $E^{-1}(D \setminus \{0\}) = \Omega$.

Let $f : \overline{\mathbb{D}} \to \overline{D}$ be the conformal Jordan-domain map normalized by

$$f(0) = 0, \qquad f(1) = E(F(0,R)). \tag{C.85.4}$$

Existence and uniqueness follow from the Riemann map and its Jordan boundary extension, followed by the unique rotation of the source circle fixing the displayed boundary value. The function $g(w) = f(w)/w$, with $g(0) = f'(0)$, is holomorphic and nonzero in the disk and is continuous and nonzero on its closure. Choose its continuous logarithm by the exact equation $\log g(1) = 2\pi i F(0,R)$. Define, for $\operatorname{Im} z \geq 0$,

$$\begin{aligned} \phi(z) &= z + \frac{1}{2\pi i} \log g(E(z)), \\ d(z) &= \phi(z) - z - iR. \end{aligned} \tag{C.85.5}$$

The displayed logarithm exists because $g$ is nonzero on the simply connected closed disk; its holomorphic logarithm in the interior extends continuously to the boundary. Formula (C.85.5) fixes the entire lift, including its integer label: $\phi(0) = F(0,R)$ and $\phi(z+1) = \phi(z) + 1$.

It is a conformal bijection from the upper half-plane onto $\Omega$. For injectivity, equality of its values first gives equality of $E(z)$ by the injectivity of $f$, and then the period relation removes the resulting integer. For surjectivity, given $w \in \Omega$, lift $f^{-1}(E(w))$ to the upper half-plane and change that lift by the unique integer making its $\phi$-image equal to $w$. The same formulas give its continuous boundary extension and inverse.

There is an increasing real homeomorphism $\beta$ satisfying

$$\begin{aligned} \phi(x) = F(\beta(x), R), \qquad \beta(0) = 0, \qquad \beta(x+1) = \beta(x) + 1, \\ \alpha = \beta^{-1}, \qquad |\beta(x) - x| \leq 1, \qquad |\alpha(x) - x| \leq 1. \end{aligned} \tag{C.85.6}$$

These statements follow by applying the actual inverse of $F$ to the boundary map. Its increasing orientation follows from the period-one relation for a real homeomorphism. The two bounds follow on $[0,1]$ from the fixed endpoints and then on all of $\mathbb{R}$ by periodicity.

The function $d$ in (C.85.5) is the pullback of a holomorphic function on the disk, continuous on its closure; in particular the end $y=\infty$ is its removable point $w=0$. On the real boundary, $|d(x)| \le m+1 = M$ by (C.85.6). The maximum principle and the Cauchy estimate on a radius-$y/2$ disk give

$$|d(z)| \le B, \qquad |d'(z)| \le \frac{2B}{y} \quad (y>0). \tag{C.85.7}$$

Thus the bound includes the entire original half-plane, without a new boundary compactification assumption.

*A cutoff which remains a homeomorphism.* Fix the following $C^1$ function:

$$\chi(s) = \begin{cases} 0, & s \le 1, \\ 3(s-1)^2 - 2(s-1)^3, & 1 \le s \le 2, \\ 1, & s \ge 2. \end{cases} \qquad 0 \le \chi \le 1, \quad |\chi'| \le 2. \tag{C.85.8}$$

Put

$$\widetilde{\phi}(z) = z + iR + (1 - \chi(y/L))d(z). \tag{C.85.9}$$

For $0 < y < L$ this is the original conformal map $\phi$. For $y \ge L$, the operator norm of the real derivative of its perturbation of $z+iR$ is at most

$$|d'(z)| + \frac{2|d(z)|}{L} \le \frac{4B}{L} = \frac{1}{25} < 1. \tag{C.85.10}$$

Hence its derivative is invertible there and preserves orientation: the straight segment from the identity derivative to this derivative consists of invertible matrices. The cutoff has matching first derivatives at $L$ and $2L$. It follows that $\widetilde{\phi}$ is a local homeomorphism everywhere in the open half-plane.

Its image stays in $\Omega$. Below $L$ this follows from $\widetilde{\phi} = \phi$; above $L$ its height is at least $R + L - B > R + m$, above the entire boundary curve. It is proper as a map to $\Omega$. Indeed, its distance from $z+iR$ is bounded by $B$. An unbounded source sequence therefore has unbounded image. A bounded source sequence escaping compact subsets of the open half-plane has a subsequence approaching its real boundary, whose image approaches $\Gamma$ and cannot remain in a compact subset of $\Omega$.

A proper local homeomorphism has an open and closed image here and is a covering onto that image. To recall the latter assertion directly, a fiber is compact and discrete, hence finite; choose disjoint local inverse charts at its points. Properness excludes additional preimages over a sufficiently small target neighborhood, giving an evenly covered neighborhood. Connectedness of $\Omega$ gives surjectivity. A target of height greater than $R + 2L + B$ has exactly one preimage: every preimage has $y > 2L$, where (C.85.9) is exactly $z + iR$. The covering therefore has one sheet. Its boundary extension is a homeomorphism of the corresponding closed half-planes as well. It is a continuous bijection, and the same bounded-distance estimate makes it proper, including on the real boundary. Its inverse is consequently continuous.

Set $\eta(y) = \min\{y, 1\}$ for $y \ge 0$ and

$$\begin{aligned} A(x,y) &= \big((1-\eta(y))\alpha(x) + \eta(y)x,\ y\big), \\ E^+_{F,R}(x, R+y) &= \widetilde{\phi}(A(x,y)). \end{aligned} \tag{C.85.11}$$

Each first-coordinate function of $A$ is increasing, proper, and period one. Its inverse is continuous by the two-sided monotone bracketing argument of C.84. Thus $A$ is a homeomorphism of the

closed half-plane, commutes with the original unit translation, and has displacement at most one. The inverse of $E^+_{F,R}$ is explicitly the inverse of $\widetilde{\phi}$, followed by $A^{-1}$ and by $(x,y)\mapsto(x,R+y)$. It is a homeomorphism onto $\overline{\Omega}$, and

$$\begin{aligned} E^+_{F,R}(x,R) &= F(x,R),\\ E^+_{F,R}(x,y) &= (x,y) \quad (y\ge R+2L),\\ |E^+_{F,R}(x,y)-(x,y)| &\le B+1 = 2m+5. \end{aligned} \tag{C.85.12}$$

The identity above $R+2L$ uses $2L>1$, so both cutoffs have already become the identity there.

Let $J(x,y)=(x,-y)$. This exact reflection commutes with $T_{(1,0)}$, and $F\mapsto JFJ$ preserves $\mathscr{P}$ and its weighted metric. Define $E^-_{F,R}=JE^+_{JFJ,R}J$ on $y\le -R$. The complete cutoff is

$$\mathcal{C}_R(F)(x,y)=\begin{cases} E^-_{F,R}(x,y), & y\le -R,\\ F(x,y), & -R\le y\le R,\\ E^+_{F,R}(x,y), & y\ge R. \end{cases} \tag{C.85.13}$$

The shared values on both boundary lines agree. The images are respectively the lower component, the middle strip image under $F$, and the upper component of the two actual image lines. They cover the plane with disjoint interiors. The inverses just described, together with $F^{-1}$ on the middle image, therefore paste to the actual inverse of (C.85.13). Every piece commutes with the original unit translation. Outside a finite band the map is the identity; on that band its displacement is bounded by periodicity. It thus belongs to $\mathscr{P}$ by C.84. For $F=\mathrm{id}$, the normalized conformal map is $f(w)=e^{-2\pi R}w$, so $d=0$, $\alpha=\mathrm{id}$, and every piece in (C.85.13) is exactly the identity.

*Finite-parameter continuity, with the chosen labels.* Here is the precise conformal parameter input. Radó's original theorem [64, Section 1, p. 182; proof pp. 183–186] states that normalized Riemann maps with value zero and positive derivative there converge uniformly on the entire closed disk exactly when their enclosing Jordan curves converge in Fréchet distance. We use that theorem for the specific Jordan curves $\gamma$ above. The remaining normalization and topology comparisons are as follows.

Take $F_i\to F$ in $\mathscr{P}$ and finite $R_i\to R$. Their maps and inverses converge uniformly on compact plane sets by C.84. Hence $E(F_i(x,R_i))\to E(F(x,R))$ uniformly for $0\le x\le 1$; this is an explicit Fréchet correspondence. Radó's theorem gives uniform convergence of the positive-derivative maps on the closed disk. Their boundary preimages of $E(F_i(0,R_i))$ converge as well: any convergent subsequence of those circle points must have image $E(F(0,R))$, whose limiting preimage is unique. Rotating by these precise circle points proves uniform convergence $f_i\to f$ with normalization (C.85.4).

The functions $g_i=f_i/w$ also converge uniformly on the closed disk. Indeed, the difference quotient $(f_i-f)/w$ has a removable value at zero and its maximum modulus is bounded by $\|f_i-f\|_\infty$ on the unit circle. Since $g$ is nonzero on the compact disk, $g_i/g\to 1$ uniformly. The small logarithm of this ratio tends uniformly to zero. It equals $\log g_i-\log g$ with our actual branches: this equality holds at $w=1$ for large $i$ by $F_i(0,R_i)\to F(0,R)$, and hence everywhere by connectedness. Consequently $d_i\to d$ uniformly on the entire closed half-plane.

The quantities $m_i,B_i,L_i$ converge. The functions $\chi(y/L_i)$ converge uniformly for $y\ge 0$ to $\chi(y/L)$: they agree above one common finite height, and convergence is uniform on the remaining compact interval. Thus $\widetilde{\phi}_i\to\widetilde{\phi}$ uniformly on the closed half-plane. On $[0,1]$ the exact boundary inverses

$$\beta_i(x)=\mathrm{pr}_1\,F_i^{-1}(\phi_i(x)) \tag{C.85.14}$$

converge uniformly to $\beta$. Monotone bracketing and periodicity then give uniform convergence of $\alpha_i=\beta_i^{-1}$ to $\alpha$ on $\mathbb{R}$ in their displacement coordinates. The maps $A_i$ converge uniformly to $A$ on

the closed half-plane. Each fixed $\widetilde{\phi}$ is uniformly continuous there: its perturbation is periodic in $x$, continuous on one compact vertical strip modulo that period, and zero above $2L$. It follows that $\widetilde{\phi}_i A_i \to \widetilde{\phi} A$ uniformly.

The two reflected versions have the same properties. When the interfaces $\pm R_i$ move, the formulas agree exactly on each interface, so their pasting is jointly continuous in the parameters and the source point. All cutoffs in this convergent sequence are the identity outside one common horizontal band. Uniform convergence on its compact period rectangle gives weighted forward convergence in C.84. The radial estimate gives uniform convergence of the original disk maps. Their inverses converge uniformly too: for homeomorphisms $a_i, a$ of the compact disk,

$$\|a_i^{-1} - a^{-1}\|_\infty \le \omega_{a^{-1}}(\|a_i - a\|_\infty), \tag{C.85.15}$$

where $\omega_{a^{-1}}$ is a modulus of continuity of the fixed inverse. C.84 now gives convergence in the full actual group topology. This proves finite-$R$ continuity, including both inverse topology maps.

*The infinite interface and compact families.* For a compact $K \subset \mathscr{P}$ put

$$a_K(R) = \sup_{\substack{F \in K,\ 0 \le x \le 1 \\ |y| \ge R}} \frac{|F(x,y) - (x,y)|}{|y|}. \tag{C.85.16}$$

C.84 proves $a_K(R) \to 0$. The two boundary displacement maxima at $\pm R$ are at most $R a_K(R)$. On the middle band (C.85.13) equals $F$. On either tail (C.85.12) and its reflection therefore give

$$\sup_{F \in K} \|\mathcal{C}_R(F) - F\|_* \le 3 a_K(R) + \frac{5}{1+R} \longrightarrow 0. \tag{C.85.17}$$

The constants $c_F$ from (C.84.7) are bounded uniformly for $F \in K$: the common tail bound gives $|Y_F(x,y)| \ge |y|/2$ off one band, and the numerator there is bounded on the remaining band. Formula (C.84.8) thus turns (C.85.17) into uniform forward convergence of the original disk maps, uniformly over $K$. The compact family of original inverses has a common modulus of continuity. Applying (C.85.15) with that common modulus proves the same uniform assertion for inverses.

This also proves joint continuity at $R = \infty$. For a sequence $F_i \to F$, $R_i \to \infty$, the set $\{F, F_1, F_2, \ldots\}$ is compact, so the estimate just proved gives $\mathcal{C}_{R_i}(F_i) \to F$ in the actual topology. Together with the finite-parameter argument this proves (C.85.1). In particular compact quotient-support elements are dense, with the displayed continuous cutoff maps providing the parameter comparison.

*The based sphere and its exact fixed-support filling.* Let $a : (S^n, *) \to (\mathscr{C}_Q, 1)$, $n \ge 1$, and write $F_\xi = j^{-1} a_\xi j$. For $0 < s \le 1$ use $j \mathcal{C}_{1/s}(F_\xi) j^{-1}$, and at $s = 0$ use $a_\xi$. Formula (C.85.1) proves that this is a continuous based homotopy in the original source group. Its last family $a'_\xi = j \mathcal{C}_1(F_\xi) j^{-1}$ is supported, in the quotient $N = \mathbb{R}^2/\langle T_{(1,0)} \rangle$, in the interior of one fixed compact annulus $N_0$. Such an annulus exists because the two values of $L$ in (C.85.3) are bounded on the compact sphere family. Choose its two boundary circles strictly beyond that common support band.

Apply the exact compact-support sphere filling of C.81 to the descended family $b_\xi$ on $N_0$. Its hypotheses hold here: the actual extended plane translations form a transitive boundary-fixed group commuting with the original $Q$, as in C.83; the quotient support is the specified compact annulus with its entire boundary fixed. In the notation of (C.81.10), this gives $B_{0,\xi} = 1$, $B_{1,\xi} = b_\xi$, $B_{r,*} = 1$. Lift $B_{r,\xi}\pi$ from the identity, where $\pi : \mathbb{R}^2 \to N$ is the original covering. Lifting the inverse family proves the inverse equations; uniqueness of lifts proves commutation with every original integer translation. At $r = 1$ the difference from $\mathcal{C}_1(F_\xi)$ is an integer deck label, locally constant on $S^n$ and zero at $*$. It is zero everywhere. The whole lifted family and its inverse fix the complement of the same band. C.81's original-boundary extension, or the bounded periodic displacement estimate here, gives their joint closed-disk continuity.

Writing this exact lifted disk family as $H_{r,\xi}$, a complete based null-homotopy is

$$K_{s,\xi} = \begin{cases} a_\xi, & s = 0, \\ j\mathcal{C}_{1/(2s)}(F_\xi)j^{-1}, & 0 < s \le \frac{1}{2}, \\ H_{2-2s,\xi}, & \frac{1}{2} \le s \le 1. \end{cases} \tag{C.85.18}$$

The two middle values agree with the same original family $a'_\xi$. The basepoint stays the identity, $K_{0,\xi} = a_\xi$, and $K_{1,\xi} = 1$. This proves (C.85.2) in the original topological group, with both maps and inverses controlled throughout.

The conclusion includes all compact based sphere families for this exact radial cyclic action. The construction uses its literal period-one coordinate, its radial boundary topology, and the sublinear comparison of C.84. The general discrete-operator deformation and general C.12 remain open. □

**C.86. [Proved] The exact spiral centralizer comparison and its based sphere fillings.** Retain the actual radial coordinate $j$ of C.83–C.85 and the actual spiral coordinate $p$ of C.72:

$$j(z) = \frac{z}{1+|z|}, \qquad p(z) = \frac{z}{1+|z|}e^{i\log(1+|z|)}, \qquad \mathcal{R}(z) = iz. \tag{C.86.1}$$

Let $\mathscr{C}_{\mathrm{rad}}$ be the relative closed-disk centralizer of $jT_1j^{-1}$, and let $\mathscr{C}_{\mathrm{sp}}$ be the relative closed-disk centralizer of the original C.72 operator $pT_ip^{-1}$. All these operators have their specified identity extensions on the circle. There is an actual topological group isomorphism

$$\begin{gathered} \Theta : \mathscr{C}_{\mathrm{rad}} \longrightarrow \mathscr{C}_{\mathrm{sp}}, \\ \Theta(h)|_{\mathbb{D}} = p\mathcal{R}j^{-1}hj\mathcal{R}^{-1}p^{-1}, \qquad \Theta(h)|_{S^1} = \mathrm{id}\,. \end{gathered} \tag{C.86.2}$$

Its inverse is given by the inverse interior conjugation and the same identity boundary extension. The proof establishes both topology maps in the original map-and-inverse topology. In particular,

$$\pi_n(\mathscr{C}_{\mathrm{sp}}, 1) = 0 \qquad (n \ge 1). \tag{C.86.3}$$

The interior coordinate $p\mathcal{R}j^{-1}$ in (C.86.2) has no continuous extension at the boundary point 1. The group comparison and the stated fillings are established by the exact sublinearity and topology calculations below.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact actual-relative-centralizer comparison and based sphere fillings for the specified spiral discrete-cyclic model.

*Proof. A global spiral displacement estimate.* For all $z, v \in \mathbb{C}$ one has

$$|p(z+v) - p(z)| \le \frac{4|v|}{1+|z|}. \tag{C.86.4}$$

Put $r = |z|$, $r' = |z+v|$, and $M = |v|$. For $M \le (1+r)/2$, the reverse triangle inequality gives $\min\{1+r, 1+r'\} \ge (1+r)/2$. The mean value theorem for the logarithm and (C.83.3) therefore give

$$\begin{aligned} |\log(1+r') - \log(1+r)| &\le \frac{2M}{1+r}, \\ |p(z+v) - p(z)| &\le |j(z+v) - j(z)| + |j(z)|\,|e^{i\log(1+r')} - e^{i\log(1+r)}| \\ &\le \frac{4M}{1+r}. \end{aligned} \tag{C.86.5}$$

For $M > (1+r)/2$, both spiral values lie in the unit disk, so their distance is at most $2 < 4M/(1+r)$. This proves the global estimate, including at zero.

We first compare the horizontal operator $pT_1p^{-1}$ with $jT_1j^{-1}$. The final rotation will restore the literal vertical label of C.72.

*Necessity, uniformly on compact actual families.* Let $K$ be a compact family of original closed-disk homeomorphisms fixing the circle and commuting with $pT_1p^{-1}$. Write their actual plane conjugates as $F = p^{-1}hp$. Each $F$ commutes with the unit horizontal translation. Compactness and boundary fixation give the exact uniform limit

$$\sup_{h\in K,\ |z|\geq R} |pF(z) - p(z)| \longrightarrow 0. \tag{C.86.6}$$

Indeed, $|p(z)| \to 1$. Any failing sequence has a subsequence on which $p(z)$ approaches one circle point and $h$ converges uniformly to a map fixing that point. Its displayed difference then tends to zero.

Take arbitrary $F_i$ from these conjugates and $(x_i, y_i)$ with $0 \leq x_i \leq 1$ and $|y_i| \to \infty$. Put

$$U_i = \frac{X_{F_i}(x_i, y_i) - x_i}{y_i}, \qquad V_i = \frac{Y_{F_i}(x_i, y_i)}{y_i}. \tag{C.86.7}$$

For any fixed real $c$ choose the actual integer $m_i(c)$ so that

$$x_i + m_i(c) = cy_i + e_i(c), \qquad |e_i(c)| \leq \tfrac{1}{2}. \tag{C.86.8}$$

The input and output at this integer translate are exactly

$$\begin{aligned} z_i(c) &= y_i(c+i) + e_i(c), \\ F_i(z_i(c)) &= y_i(c + U_i + iV_i) + e_i(c). \end{aligned} \tag{C.86.9}$$

Here $i$ in the complex expressions is the imaginary unit; subscripts retain the sequence index.

First $(U_i, V_i)$ is bounded. Suppose a subsequence has norm tending to infinity. The two outputs for $c = 0$ and $c = 1$ differ by the real integer $m_i(1) - m_i(0)$, of absolute value at most $|y_i| + 1$. The norm of the output at zero, divided by $|y_i|$, tends to infinity. Applying (C.86.4) to these two outputs shows that their spiral images have distance tending to zero. In contrast, the distance of the two spiral input images tends to a positive number. To see its exact value, define

$$\Psi(w) = \frac{w}{|w|} e^{i \log |w|} \quad (w \neq 0). \tag{C.86.10}$$

After removal of the common unit factor $\operatorname{sign}(y_i)e^{i \log |y_i|}$, the two input phases tend to $\Psi(i)$ and $\Psi(1+i)$. Their angle difference is $\delta = \pi/4 - (\log 2)/2$, with $0 < \delta < \pi/4$; the first inequality follows from $\log 2 < 1 < \pi/2$. Their distance is $2\sin(\delta/2) > 0$. Formula (C.86.6) would force the input and output distances to have the same limit, a contradiction. This proves boundedness without presupposing a normalized nonzero output limit.

Pass to a subsequence with $(U_i, V_i) \to (U, V)$. For every real $c$ with $c + U + iV \neq 0$, formula (C.86.9) and removal of the same common unit factor give, by (C.86.6),

$$\Psi(c + U + iV) = \Psi(c + i). \tag{C.86.11}$$

This is equality of circle-valued functions. No equality of real arguments, or choice of an integer multiple of $2\pi$, has been imposed. On every open interval where the left argument is nonzero, differentiate the circle-valued identity and divide by its nonzero value. The exact logarithmic derivative is

$$\frac{d}{dc}\Psi(c + U + iV)/\Psi(c + U + iV) = i\frac{c + U - V}{(c + U)^2 + V^2}. \tag{C.86.12}$$

Thus

$$(c+U-V)(c^2+1)=(c-1)((c+U)^2+V^2). \tag{C.86.13}$$

The identity holds as a polynomial because it holds on an open interval. This includes the possible limit $V=0$: only the single value $c=-U$ was omitted, and no conclusion at that zero was needed. Comparing the coefficient of $c^2$ gives $U+V=1$; comparing constants then gives $U-V+U^2+V^2=2U^2=0$. Consequently $U=0$ and $V=1$.

Every convergent subsequence of the bounded pair has this same limit. Hence

$$\frac{|F_i(x_i,y_i)-(x_i,y_i)|}{|y_i|}=\sqrt{U_i^2+(V_i-1)^2}\longrightarrow 0. \tag{C.86.14}$$

Since the original sequences were arbitrary, this proves precisely the uniform compact-family sublinearity condition (C.84.1) for the spiral relative centralizer. Apply the same argument to the compact inverse family to retain its inverse condition.

*Sufficiency and the two inverse topology maps.* Conversely, let $F\in\mathscr{P}$ as defined in C.84. Periodicity and its sublinearity give $|F(z)-z|/(1+|z|)\to 0$ at infinity, as proved in (C.84.6). Formula (C.86.4) shows that $pFp^{-1}$ extends by the identity to the original circle. The extension is a continuous bijection of the compact disk and thus a homeomorphism. It commutes with the specified horizontal spiral operator; its inverse is the extension of $pF^{-1}p^{-1}$. We have proved an exact group bijection

$$\begin{aligned} I_p:\mathscr{P}&\longrightarrow C_{\mathcal{D}_\partial}(pT_1p^{-1}),\\ I_p(F)|_{\mathbb{D}}=pFp^{-1},&\qquad I_p(F)|_{S^1}=\mathrm{id}, \end{aligned} \tag{C.86.15}$$

whose inverse is restriction followed by $p^{-1}(\cdot)p$.

For its forward continuity fix $H\in\mathscr{P}$. The finite constant $c_H$ of (C.84.7) and (C.86.4), applied at $H(z)$, give

$$\|I_p(F)-I_p(H)\|_\infty\le 4c_H\|F-H\|_*. \tag{C.86.16}$$

The estimate uses the exact periodic displacement and the original source height in the definition of $\|\cdot\|_*$. Applying it to $F^{-1},H^{-1}$ proves continuity for the original inverses as well.

For the inverse topology map, a convergent sequence of actual disk maps together with its limit is compact. The necessity proof above gives uniform sublinear tails for the corresponding plane maps and their inverses. On each compact period rectangle, conjugation by $p,p^{-1}$ gives uniform convergence: the disk images stay in one compact interior set, where $p^{-1}$ is uniformly continuous. Combining that convergence with the common tail bounds gives convergence of the weighted forward differences and of the weighted inverse differences. Thus the inverse of (C.86.15) is continuous in $\delta_*$. These metrizable-space arguments prove the asserted topological group isomorphism with the actual topology.

*The original vertical label, the nonextendible coordinate, and the filling.* Both $j$ and $p$ commute with the unit complex rotation $\mathcal{R}$. Its closed-disk extension $\overline{\mathcal{R}}(w)=iw$ conjugates the horizontal spiral operator to the original vertical operator of C.72. The exact C.84 isomorphism $I_j$ and (C.86.15) therefore give

$$\Theta(h)=\overline{\mathcal{R}}\,I_p(I_j^{-1}(h))\,\overline{\mathcal{R}}^{-1}. \tag{C.86.17}$$

This is (C.86.2) on the interior and the identity on the boundary. Its inverse is $I_jI_p^{-1}(\overline{\mathcal{R}}^{-1}(\cdot)\overline{\mathcal{R}})$. The two inverse identities and both continuous topology maps follow from the specified component maps; every original integer operator label is preserved by the displayed rotation and conjugations.

For completeness, the interior coordinate itself does not extend at the indicated boundary point. Put $r_k=e^{2\pi k}-1$ and $s_k=e^{(2k+1)\pi}-1$. Both $j(r_k)$ and $j(s_k)$ tend to 1, whereas

$$\begin{aligned} p\mathcal{R}j^{-1}(j(r_k))&=i\frac{r_k}{1+r_k}\longrightarrow i,\\ p\mathcal{R}j^{-1}(j(s_k))&=-i\frac{s_k}{1+s_k}\longrightarrow -i. \end{aligned} \tag{C.86.18}$$

Thus the group isomorphism above is proved by (C.86.6)–(C.86.17), with its own actual boundary continuity, rather than by extending that coordinate.

Given a based sphere family in $\mathscr{C}_{\mathrm{sp}}$, apply the continuous inverse $\Theta^{-1}$, then the exact C.85 filling, and finally $\Theta$ at every homotopy parameter. This is a continuous based filling in the original spiral relative centralizer, for both maps and inverses, and proves (C.86.3). It covers the entire relative centralizer of C.72, including families beyond its explicitly contracted example. The general discrete-operator case and general C.12 remain open. □

**C.87. [Proved] Global contractions of the radial and spiral relative centralizers.** The entire topological groups $\mathscr{P}$ of C.84, $\mathscr{C}_{\mathrm{rad}}$ of C.85, and $\mathscr{C}_{\mathrm{sp}}$ of C.86 are contractible by continuous contractions fixing the identity throughout. The proof constructs one contraction on the whole group from the parametric C.85 cutoff, a continuous support height, an exact compact-annulus lift comparison, and the compact boundary-relative identity-component contraction. Every original disk boundary point and every specified integer operator label is retained, including in the inverse topology. The compact-annulus comparison identifies its identity component exactly with the kernel of the displayed integer boundary-shift homomorphism.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), global actual contractions of the two specified discrete-cyclic relative centralizers.

*Proof. The exact compact-annulus comparison.* Put $A = (\mathbb{R}/\mathbb{Z}) \times [-1,1]$ and let $\mathscr{H}$ be its orientation-preserving homeomorphism group fixing both boundary circles pointwise. Each $b \in \mathscr{H}$ induces the identity on the fundamental group, since the lower boundary is fixed. It has a unique lift $\widetilde{b} : \mathbb{R} \times [-1,1] \to \mathbb{R} \times [-1,1]$ fixing $(0,-1)$. Its restriction to the whole lower boundary is the identity, and uniqueness of lifts gives $\widetilde{b}T_1 = T_1\widetilde{b}$. On the upper boundary it has the form

$$\widetilde{b}(x,-1) = (x,-1), \qquad \widetilde{b}(x,1) = (x+\tau(b),1), \qquad \tau(b) \in \mathbb{Z}. \tag{C.87.1}$$

The integer is independent of $x$ by connectedness. The canonical lift of a composite is the composite of the canonical lifts, so $\tau$ is a homomorphism. The maps

$$b_k([x],y) = ([x + k(y+1)/2], y) \quad (k \in \mathbb{Z}) \tag{C.87.2}$$

fix both boundary circles and have $\tau(b_k) = k$; they give a literal homomorphic section.

The canonical lift varies continuously in the compact-open map-and-inverse topology, uniformly on one period rectangle. To verify this directly, let $b_i \to b$. For large $i$, the two circle coordinates of $b_i(z)$ and $b(z)$ differ by a unique small real value $\delta_i(z) \in (-1/4, 1/4)$, continuous in $z \in A$ and tending uniformly to zero. The difference of their lifted first coordinates minus $\delta_i$ is a continuous integer on the connected covering strip. It is zero on the lower boundary and hence zero everywhere. Their second coordinates converge uniformly as well. Apply the same argument to inverses. In particular $\tau(b_i) = \tau(b)$ eventually, since $\delta_i = 0$ on the upper boundary. Thus $\tau$ is continuous with discrete target.

We now prove its exact kernel. Suppose $\tau(b) = 0$ and let $\gamma(y) = ([0], y)$ be the spanning arc. Its image has a lift $\widetilde{b}(0,y)$ with endpoints $(0,-1)$ and $(0,1)$. The straight homotopy in the convex strip from this lifted arc to $(0,y)$ fixes these two endpoints; its interior stays in the interior of the strip. Its projection proves that $b\gamma$ and $\gamma$ are homotopic with their endpoints fixed. Epstein's actual relative arc theorem [20, Theorem 3.1, pp. 89–90; Appendix A1–A2 and their proofs, pp. 105–106] gives an ambient isotopy fixed pointwise on both boundary circles carrying $b\gamma$ to $\gamma$. The Appendix supplies the stated topological arc case: A2's proof first makes the arc piecewise linear near its endpoints and then applies the construction in the proof of A1.

Any remaining parametrization of $\gamma$ is an increasing interval homeomorphism $\rho$ fixing its endpoints. In a rectangular neighborhood with coordinates $(u,y)$ and $\gamma = \{u = 0\}$, choose

$0 \le \zeta \le 1$ with $\zeta(0) = 1$ and $\zeta = 0$ on the two side edges. The formula

$$(u, y) \longmapsto \big(u, (1 - s\zeta(u))y + s\zeta(u)\rho^{-1}(y)\big) \quad (0 \le s \le 1) \tag{C.87.3}$$

is an isotopy of that rectangle, fixed on its entire boundary, and has a continuous inverse by monotone bracketing. Extend it by the identity. After this correction the composed annulus map fixes $\gamma$ pointwise. Cutting along $\gamma$ produces a disk map fixing its boundary pointwise. The disk Alexander isotopy descends to an isotopy of the annulus relative to its boundary and $\gamma$. Thus $b$ belongs to the identity path component of $\mathscr{H}$. Conversely the continuous integer $\tau$ vanishes on the identity component. We have proved the exact equality

$$\ker \tau = \mathscr{H}_0. \tag{C.87.4}$$

The arc isotopies in this argument establish membership in that component; no parameter choice of those isotopies is used below.

Let

$$\mathscr{E} = \{F \in \mathscr{P} : F(x, y) = (x, y) \text{ for } |y| \ge 1\}. \tag{C.87.5}$$

Restriction and descent identify $\mathscr{E}$ with $\mathscr{H}_0$. The actual inverse map $\Lambda : \mathscr{H}_0 \to \mathscr{E}$ takes the lower-fixed lift and extends it by the identity outside the strip. Equation (C.87.4) fixes the upper line as well, so this pasting is an actual plane homeomorphism. Its inverse is $\Lambda(b^{-1})$ and its translation labels are unchanged. Its displacement is bounded and periodic, so it lies in $\mathscr{P}$. These are inverse topological group maps. Restriction and descent are continuous on the compact period rectangle by C.84. The canonical lift continuity just proved, together with its identity extension, gives continuity of $\Lambda$ in both weighted terms of (C.84.2).

The compact boundary-relative identity component is contractible by [74, Lemma 2.1(ii), pp. 2–3]. Here the whole nonempty boundary of the compact annulus is fixed pointwise, as required there. Choose one continuous contraction $C_s$ of $\mathscr{H}_0$ with $C_0(b) = b$ and $C_1$ constant. Replacing it by $C_s(b)C_s(1)^{-1}$ fixes the identity throughout and makes the last map the constant identity. Transport it through $\Lambda$ and denote the resulting single continuous contraction on the whole of $\mathscr{E}$ by

$$\widehat{\Gamma} : [0, 1] \times \mathscr{E} \to \mathscr{E}, \quad \widehat{\Gamma}_0(G) = G, \quad \widehat{\Gamma}_1(G) = 1, \quad \widehat{\Gamma}_s(1) = 1. \tag{C.87.6}$$

*The actual vertical normalizers.* For $w > 0$ put $D_w(x, y) = (x, wy)$ and

$$\begin{aligned} \mathcal{N}_w(F) &= D_w F D_w^{-1}, \\ \mathcal{N}_w(F)(x, y) &= \big(X_F(x, y/w),\ wY_F(x, y/w)\big). \end{aligned} \tag{C.87.7}$$

The exact inverse parameter is $w^{-1}$ and $D_w$ commutes with the original $T_1$. Sublinearity in (C.84.1) is preserved by the displayed coordinate formula. For fixed $w$ the weighted metric satisfies

$$\delta_*(\mathcal{N}_w(F), \mathcal{N}_w(H)) \le \max\{w, w^{-1}\}\, \delta_*(F, H). \tag{C.87.8}$$

Indeed the Euclidean operator norm of $D_w$ is $\max\{1, w\}$, and the change of source height has ratio $(1 + |y|/w)/(1 + |y|) \le \max\{1, w^{-1}\}$. Apply the same estimate to the exact inverse conjugations of $F^{-1}$ and $H^{-1}$.

For joint continuity in $w$ retain the original closed disk. For $0 \le \rho \le 1$ and $|u| = 1$ the actual normalizer is

$$\overline{D}_w(\rho u) = \frac{\rho D_w u}{1 - \rho + \rho|D_w u|}, \qquad \overline{D}_w(0) = 0, \qquad \overline{D}_w^{-1} = \overline{D}_{1/w}. \tag{C.87.9}$$

For $\rho < 1$ it is exactly $jD_w j^{-1}$, and on the circle it is $D_w u/|D_w u|$. On any compact positive interval of $w$, the denominator is bounded away from zero. The formula is jointly continuous,

also at $\rho = 0$, and so is the displayed inverse. It gives joint continuity into the closed-disk homeomorphism group. It commutes with the specified operator $jT_1j^{-1}$. Conjugation by it preserves pointwise boundary fixation. Transporting that conjugation through the exact C.84 isomorphism proves joint continuity of (C.87.7) as a map $(0,\infty)\times\mathscr{P}\to\mathscr{P}$, with both original inverse topology maps included.

*A single contraction on the entire group.* Let $L_+(F) = L(F,1)$ be the positive cutoff height of (C.85.3), and let $L_-(F) = L(JFJ,1)$ use the reflection $J$ of C.85. Both are continuous functions of $F$. Put

$$\begin{aligned} W(F) &= 3 + 2\max\{L_+(F), L_-(F)\},\\ G(F) &= \mathcal{N}_{1/W(F)}(\mathcal{C}_1(F)). \end{aligned} \tag{C.87.10}$$

The cutoff is the identity beyond heights $1 + 2L_+(F)$ and $-1 - 2L_-(F)$. Since each of their absolute values is at least two less than $W(F)$, the map $G(F)$ is the identity near both boundary lines of the fixed strip and outside it. Thus $G(F)\in\mathscr{E}$. It depends continuously on $F$ by C.85 and (C.87.7). The exact original lift fixes both boundary lines, so the annulus component used in (C.87.6) is indeed the zero-$\tau$ component just proved, for every $F$.

Define on the whole group

$$\mathcal{H}_s(F) = \begin{cases} F, & s = 0,\\ \mathcal{C}_{1/(2s)}(F), & 0 < s \le \frac{1}{2},\\ \mathcal{N}_{W(F)}\big(\widehat{\Gamma}_{2s-1}(G(F))\big), & \frac{1}{2} \le s \le 1. \end{cases} \tag{C.87.11}$$

C.85 gives joint continuity of the first part, including at zero. At $s = 1/2$ the second part equals $\mathcal{N}_{W(F)}G(F) = \mathcal{C}_1(F)$, so the two parts agree exactly. Every component map in the last expression is continuous on its full stated domain. The function $W$ is locally bounded and positive; in particular the last part near any fixed $F$ has support in one common horizontal band, and its inverse has that same support. The normalizer comparison already proves the required original map-and-inverse continuity. Pasting gives a single continuous map $[0,1]\times\mathscr{P}\to\mathscr{P}$. Its faces are

$$\mathcal{H}_0(F) = F, \qquad \mathcal{H}_1(F) = 1, \qquad \mathcal{H}_s(1) = 1. \tag{C.87.12}$$

For the last equality use $\mathcal{C}_R(1) = 1$ and $\widehat{\Gamma}_s(1) = 1$; no normalization of $W(1)$ is required.

Conjugate (C.87.11) through the C.84 topological group isomorphism to obtain the contraction on $\mathscr{C}_{\mathrm{rad}}$. Conjugate that contraction through the two actual topology maps $\Theta$ and $\Theta^{-1}$ of C.86 to obtain the contraction on $\mathscr{C}_{\mathrm{sp}}$. These formulas act on the whole respective groups, fix the identity, and preserve the original boundary and integer operator labels throughout. The general discrete-operator case and general C.12 remain open. □

**C.88. [Proved] The exact vanishing-displacement image and its compact-family topology.** Retain all the actual hypotheses and maps of C.81: the original closed disk $\overline{D}$, its interior $O$, a compatible metric $d$ on $\overline{D}$, a closed discrete free proper operator group $Q$ on $O$, and a transitive subgroup of the original boundary-fixed centralizer $\mathscr{C}_Q$. Write $\pi : O \to N = O/Q$ and retain $d_Q(x,y) = \sup_{q\in Q} d(qx,qy)$. For an interior homeomorphism $h$ commuting with every actual $q$, put

$$\delta_h(\pi x) = d_Q(hx,x), \qquad \mathscr{V}_Q = \{h \in C_{\mathrm{Homeo}(O)}(Q) : \delta_h \in C_0(N)\}. \tag{C.88.1}$$

Here $C_0(N)$ means continuous real-valued functions whose absolute values are uniformly small off compact subsets of $N$. Then restriction and identity extension are mutually inverse topological group maps

$$\begin{aligned} &R : \mathscr{C}_Q \longrightarrow \mathscr{V}_Q, \qquad R(a) = a|_O,\\ &E : \mathscr{V}_Q \longrightarrow \mathscr{C}_Q, \quad E(h)|_O = h, \quad E(h)|_{\partial\overline{D}} = \mathrm{id}\,. \end{aligned} \tag{C.88.2}$$

The topology on $\mathscr{V}_Q$ is the sum of the uniform $d_Q$ distances of maps and inverses. These maps are isometries for the corresponding original closed-disk distances. Every member of $\mathscr{V}_Q$ and its inverse is automatically uniformly $d_Q$-continuous; no uniform-continuity assumption is added to (C.88.1).

There is also an exact compact-parameter criterion. Let $P$ be a compact metric space and let $h_p$ be actual equivariant interior homeomorphisms such that $(p,x)\mapsto h_p(x)$ is continuous on $P\times O$. Their identity extensions form a continuous $P$-family of original closed-disk homeomorphisms and inverses precisely when

$$\forall \epsilon>0\ \exists K\subset N \text{ compact} \qquad \sup_{\substack{p\in P\\ \pi x\notin K}} d_Q(h_p x,x)<\epsilon. \tag{C.88.3}$$

In that case the inverse family has the same uniform vanishing property. Thus (C.88.3), with the actual interior maps retained, expresses the original boundary and compact-family requirements exactly. It supplies no cutoff deformation by itself.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact actual-boundary image and compact-parameter criterion for the remaining discrete-operator cutoff.

*Proof. The actual displacement functions and algebra.* C.81 proves that $d_Q$ induces the original topology of $O$, that $\pi$ is the specified covering onto the locally compact surface $N$, and that $d_Q$ is $Q$-invariant. Consequently $x\mapsto d_Q(hx,x)$ is continuous. Commutation and invariance give

$$d_Q(hqx,qx)=d_Q(qhx,qx)=d_Q(hx,x). \tag{C.88.4}$$

The quotient map therefore gives exactly the continuous function in (C.88.1). The map $h$ descends to the homeomorphism $\bar h(\pi x)=\pi(hx)$, whose displayed inverse is $\overline{h^{-1}}$. Symmetry and the triangle inequality give

$$\begin{aligned} \delta_{h^{-1}} &= \delta_h\circ \bar h^{-1},\\ \delta_{hk} &\le \delta_h\circ \bar k+\delta_k. \end{aligned} \tag{C.88.5}$$

A homeomorphism of $N$ and its inverse carry compact sets to compact sets. Thus its pullback preserves $C_0(N)$. Equations (C.88.5) prove that $\mathscr{V}_Q$ is a group, before using the boundary extension.

*Necessity, including the common tail for compact families.* Let $a_p$, $p\in P$, be a continuous compact family in $\mathscr{C}_Q$. Joint evaluation on $P\times\overline{D}$ is continuous. Since every $a_p$ fixes every boundary point, for any $\epsilon>0$ there is a compact set $C\subset O$ such that

$$d(a_p z,z)<\epsilon \quad (p\in P,\ z\in O\setminus C). \tag{C.88.6}$$

For completeness, failure would give $p_j\to p$ after taking a subsequence and $z_j$ escaping all interior compact sets. A further subsequence converges to $z\in\partial\overline{D}$, and joint continuity gives $a_{p_j}z_j\to z$, contradicting the displayed failure.

The set $K=\pi C$ is compact. If $\pi x\notin K$, then $qx\notin C$ for every $q\in Q$. The original commutator equations imply

$$\begin{aligned} d_Q(a_p x,x) &= \sup_{q\in Q} d(a_p qx,qx)\\ &\le \epsilon \qquad (p\in P,\ \pi x\notin K). \end{aligned} \tag{C.88.7}$$

Starting with $\epsilon/2$ gives the strict bound in (C.88.3). The same proof applies to the compact inverse family. Taking $P$ to be a point proves that $R$ maps into $\mathscr{V}_Q$.

*One common control for compact quotient cores.* We recall the exact C.81 comparison needed for the reverse direction. Let $C, C' \subset O$ be compact. For a fixed $x_0 \in O$, openness of evaluation for the actual transitive Polish source, and its compact-covering lemma, give compact families $S, T$ of original boundary-fixed maps with $C \subset S(x_0)$ and $C' \subset T(x_0)$. For $c = sx_0 \in C$ and $c' = tx_0 \in C'$, commutation gives

$$qc' = tqx_0 = ts^{-1}(qc). \tag{C.88.8}$$

All choices on the right lie in the single compact actual family $TS^{-1}$. It follows that, for $c_j \in C$ and $q_j c_j \to z \in \partial\overline{D}$,

$$q_j C' \longrightarrow \{z\} \quad \text{uniformly.} \tag{C.88.9}$$

Indeed joint evaluation of the compact family $TS^{-1}$ fixes $z$ throughout. No limiting map or original-circle extension of the operators $q_j$ has been used.

*Sufficiency and all parameter and inverse maps.* Suppose the interior family in the statement satisfies (C.88.3), and define $E(h_p)$ by the literal formula (C.88.2). Interior continuity is given. To prove joint continuity at the original circle, take $p_j \to p$ and $x_j \to z$ with $z \in \partial\overline{D}$. Terms already on the circle are fixed. If the desired convergence fails for the remaining interior terms, retain a sequence on which it fails by a fixed positive distance.

The surface $N$ has a compact exhaustion. After passage to a subsequence, either $\pi x_j$ leaves every compact subset of $N$, or all $\pi x_j$ belong to one compact $K \subset N$. This dichotomy follows by taking infinitely many terms in one exhaustion set when there is such a set; otherwise the sequence itself leaves every set. In the escaping case, (C.88.3) gives

$$d(h_{p_j} x_j, x_j) \le d_Q(h_{p_j} x_j, x_j) \longrightarrow 0, \tag{C.88.10}$$

which contradicts failure.

In the compact-core case choose a compact $C \subset O$ with $\pi C = K$. Finitely many local covering sections over relatively compact neighborhoods of points of $K$ give such a $C$. Write $x_j = q_j c_j$, $c_j \in C$. The joint interior continuity and compactness of $P \times C$ make

$$C' = \{h_p(c) : p \in P,\ c \in C\} \tag{C.88.11}$$

compact in $O$. By (C.88.9), $q_j C' \to \{z\}$ uniformly. But $h_{p_j} x_j = q_j h_{p_j} c_j$ belongs to $q_j C'$, again a contradiction. This proves joint boundary continuity in every case.

Each $E(h_p)$ is bijective: its interior restriction is a homeomorphism onto $O$ and its boundary restriction is the identity. The actual map

$$\mathcal{E} : P \times \overline{D} \longrightarrow P \times \overline{D}, \qquad (p, x) \longmapsto (p, E(h_p)x) \tag{C.88.12}$$

is consequently a continuous bijection between compact Hausdorff spaces. Its inverse is continuous and has the exact formula $(p, y) \mapsto (p, E(h_p)^{-1} y)$. Thus both the family and its inverses are continuous in the original closed-disk topology. The original operator equations hold on $O$ unchanged. Applying the proved necessity to this inverse family gives its uniform $C_0$ tails. This proof even obtains joint inverse continuity without assuming it for the initial interior family.

With $P$ a point, this establishes that $E$ maps every member of $\mathscr{V}_Q$ into $\mathscr{C}_Q$. The two formulas (C.88.2) compose to the identity on their exact domains. Extension commutes with composition and inversion, since these identities hold on $O$ and both sides fix the original circle.

*The complete topology comparison.* A modulus of uniform continuity of $E(h)$ on $\overline{D}$ applies to each pair $qx, qy$. Commutation then proves uniform continuity of $h$ for $d_Q$. Apply the same argument to $E(h)^{-1}$. For $h, k \in \mathscr{V}_Q$, the actual equality is

$$\begin{aligned} \sup_{x \in O} d_Q(hx, kx) &= \sup_{\substack{x \in O \\ q \in Q}} d(E(h)qx, E(k)qx) \\ &= \sup_{z \in \overline{D}} d(E(h)z, E(k)z). \end{aligned} \tag{C.88.13}$$

The last equality uses $q = 1$ and density of $O$. Repeating it for both inverses proves the asserted isometry and both continuous maps. In particular the displacement map to $C_0(N)$ satisfies

$$\|\delta_h - \delta_k\|_\infty \le \sup_{x \in O} d_Q(hx, kx). \tag{C.88.14}$$

All image and topology requirements are therefore expressed using the specified covering and the original operator labels.

The remaining general cutoff problem is to deform an arbitrary original based sphere family to a family with one compact quotient support while retaining joint interior continuity, all equivariance equations and the uniform tail condition (C.88.3). C.81 fills the resulting supported family. C.88 provides the exact boundary criterion for such a deformation; it does not construct that deformation or replace its uniform tail control by convergence on compact subsets of $N$. □

**C.89. [Proved] A global section for the actual compact skeleton image.** Retain the actual hypotheses of C.81 and the null lifted triangulation of C.82. Write $\Lambda$ for its full one-skeleton and put

$$X = \Lambda \cup \partial\overline{D}. \tag{C.89.1}$$

This is a compact connected subset of the original closed disk. Define $\mathscr{B}$ to be the following space of embeddings $b : X \to \overline{D}$, with uniform convergence on this actual compact domain: $b$ fixes the original circle, maps $\Lambda$ into $O$, and satisfies $b(qx) = qb(x)$ for every $x \in \Lambda$ and every original $q \in Q$. For each lifted closed triangle $\Delta$, its boundary image is positively oriented relative to the orientation of $\partial\Delta$, and the Jordan disk $D_b(\Delta)$ bounded by this image satisfies

$$\operatorname{int} D_b(\Delta) \cap b(X) = \varnothing. \tag{C.89.2}$$

Then $\mathscr{B}$ is exactly the image of the original restriction map $\mathscr{C}_Q \to \operatorname{Emb}(X, \overline{D})$. That map has a global continuous section $\mathfrak{s} : \mathscr{B} \to \mathscr{C}_Q$ with $\mathfrak{s}(\mathrm{id}_X) = \mathrm{id}$. Its construction retains all three labelled vertices of each representative triangle and every original deck operator, including orientation-reversing operators.

If $\mathscr{K} = \mathscr{S}_{Q,\Lambda}$ is the pointwise stabilizer in C.82, the maps

$$\begin{aligned} \Phi : \mathscr{B} \times \mathscr{K} &\longrightarrow \mathscr{C}_Q, \qquad & \Phi(b,k) &= \mathfrak{s}(b)k, \\ \Psi : \mathscr{C}_Q &\longrightarrow \mathscr{B} \times \mathscr{K}, \qquad & \Psi(h) &= \big(h|_X, \mathfrak{s}(h|_X)^{-1}h\big) \end{aligned} \tag{C.89.3}$$

are mutually inverse homeomorphisms of spaces, for the stated skeleton topology and the original map-and-inverse group topologies. The C.82 contraction of $\mathscr{K}$ supplies a strong deformation retraction of $\mathscr{C}_Q$ onto $\mathfrak{s}(\mathscr{B})$. Thus the remaining based-family deformation can be expressed exactly on this compact equivariant skeleton image. No contraction of $\mathscr{B}$ is asserted here.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), global section and compact-skeleton image topology for the remaining discrete-operator family deformation.

*Proof. The compact skeleton and its face labels.* The lifted triangulation is locally finite on $O$, so $\Lambda$ is closed there. Adding the original circle makes $X$ closed in $\overline{D}$, hence compact. Its one-skeleton is connected: any two vertices are joined by a finite edge path in the connected triangulated surface, and each edge belongs to that same connected set. Every original boundary point is in the closure of $\Lambda$. Indeed choose interior points tending to it and the closed triangles containing them. If these triangles did not leave all finite sets of triangles, a subsequence would lie in one compact interior triangle. Otherwise C.82 nullity makes their diameters tend to zero, so choosing one point on each boundary gives the same original boundary limit. It follows that $X = \overline{\Lambda}$ is connected.

Fix once and for all an oriented disk coordinate $\iota : \overline{D} \to \overline{\mathbb{D}}$. Use $d_E(x,y) = |\iota(x) - \iota(y)|$ in the conformal constructions below. This coordinate does not change the original operators: each final formula is transported back by $\iota^{-1}$. Uniform continuity of $\iota$ and $\iota^{-1}$ on their compact domains gives moduli

$$d_E \le \omega_1(d), \qquad d \le \omega_2(d_E), \tag{C.89.4}$$

so uniform convergence in these two metrics, including the inverse metrics on the original group, is the same. C.82 nullity also holds for $d_E$.

Choose one lifted closed triangle $\Delta_\sigma$ for each quotient two-simplex $\sigma$. Choose its three vertices $(v_{\sigma,0}, v_{\sigma,1}, v_{\sigma,2})$ in positive cyclic order in the coordinate $\iota$. The stabilizer of any lifted triangle under $Q$ is trivial. Its projection is one-to-one, so an operator preserving the triangle fixes the unique lift there of each labelled quotient vertex; freeness then makes the operator the identity. Every lifted triangle consequently has a unique expression $q\Delta_\sigma$.

*The intrinsic face conditions and the extensions.* For $b \in \mathscr{B}$, the curve $b(\partial\Delta)$ is a Jordan curve contained in $O$. Its bounded Jordan disk in the $\iota$ coordinate lies in $O$: the disk is contained in the convex hull of its compact boundary, which is contained in $\mathbb{D}$.

The interiors of the disks $D_b(\Delta)$ are pairwise disjoint. To see this, the connected interior of one such disk avoids the boundary of every other by (C.89.2). If two interiors meet, this connected interior therefore lies inside the other disk. The first boundary is then contained in that closed disk. Condition (C.89.2) forces it to lie entirely in the second boundary. A Jordan curve contained in another Jordan curve is the whole curve: after deleting a missing point, a proper such subset would embed a circle in a real interval, which is impossible by the two local directions at every point. Thus the two boundaries coincide. Injectivity of $b$ and the simplicial triangulation force the original triangles to coincide. This proves the claimed disjointness.

Set $\zeta_j = e^{2\pi i j/3}$. For each representative triangle fix the conformal Jordan-disk map

$$f_\sigma : \overline{\mathbb{D}} \longrightarrow \iota(\Delta_\sigma), \qquad f_\sigma(\zeta_j) = \iota(v_{\sigma,j}). \tag{C.89.5}$$

For $b \in \mathscr{B}$, let $f_{\sigma,b}$ be the conformal map onto $\iota(D_b(\Delta_\sigma))$ with $f_{\sigma,b}(\zeta_j) = \iota(b(v_{\sigma,j}))$. The positive cyclic order supplies the unique three-point normalization. All these maps include their Jordan boundary homeomorphisms. Define the exact circle homeomorphism

$$\eta_{\sigma,b} = f_{\sigma,b}^{-1} \circ \iota \circ b \circ \iota^{-1} \circ f_\sigma \mid_{\mathbb{S}^1} . \tag{C.89.6}$$

It fixes each $\zeta_j$. Its radial extension is $\widehat{\eta}_{\sigma,b}(r\zeta) = r\eta_{\sigma,b}(\zeta)$, with value zero at zero. Its inverse is the radial extension of $\eta_{\sigma,b}^{-1}$. The required map on the representative triangle and its inverse are

$$\begin{aligned} F_{\sigma,b} &= \iota^{-1} f_{\sigma,b} \widehat{\eta}_{\sigma,b} f_\sigma^{-1} \iota, \\ F_{\sigma,b}^{-1} &= \iota^{-1} f_\sigma \widehat{\eta}_{\sigma,b}^{-1} f_{\sigma,b}^{-1} \iota. \end{aligned} \tag{C.89.7}$$

It agrees with $b$ on the entire boundary, including all labelled vertices, not only at those vertices.

For the unique expression $\Delta = q\Delta_\sigma$, define

$$\mathfrak{s}(b)|_\Delta = qF_{\sigma,b}q^{-1}, \qquad \mathfrak{s}(b)|_{\partial\overline{D}} = \mathrm{id}\,. \tag{C.89.8}$$

Boundary compatibility follows from the actual equation $bq = qb$ on $\Lambda$. A plane homeomorphism $q$ maps the compact Jordan disk bounded by a curve to the compact Jordan disk bounded by its image. Consequently the image in (C.89.8) is exactly $D_b(q\Delta_\sigma)$. No conformality of $q$ is used. An orientation-reversing $q$ occurs on both sides of the conjugation, so the extension is still orientation preserving. The unique triangle labels give, for every $q_0 \in Q$, $\mathfrak{s}(b)q_0 = q_0\mathfrak{s}(b)$ on every triangle and on $\Lambda$.

*Pasting on the original closed disk.* The formula is continuous in the interior by local finiteness. For a sequence approaching the original circle through infinitely many triangles, the source

triangle diameters tend to zero. Uniform continuity of $b$ on $X$ then makes their boundary-image diameters tend to zero, with the correct original boundary limit. The convex-hull observation gives the exact diameter equality

$$\operatorname{diam}_{d_E} D_b(\Delta) = \operatorname{diam}_{d_E} b(\partial\Delta). \tag{C.89.9}$$

Thus the entire face images have that same limit. This proves continuity of (C.89.8) on the original circle.

The disjoint-face result and injectivity on $X$ make this a continuous injection of $\overline{D}$ into itself. It is surjective. Its interior image is open by invariance of domain and closed relative to $O$: the whole compact image is closed and its boundary image is precisely the original circle. The interior image is nonempty and $O$ is connected. The compact-to-Hausdorff bijection is a homeomorphism, with inverse given by the inverses in (C.89.7), their original $Q$-transports, and the identity on the circle. It therefore belongs to $\mathscr{C}_Q$ and restricts to $b$.

Conversely, any $h \in \mathscr{C}_Q$ restricts to an embedding satisfying every condition defining $\mathscr{B}$. The map fixes the circle and hence preserves the disk orientation. It carries every original triangle onto the Jordan disk bounded by the image of its boundary, with interior disjoint from $h(X)$. This proves the asserted exact image equality. When $b = \mathrm{id}_X$, uniqueness of the three-point normalization gives $f_{\sigma,b} = f_\sigma$ and $\eta_{\sigma,b} = \mathrm{id}$. Every formula (C.89.8) is then the identity, proving the required normalization of the section.

*The full parameter topology of the section.* Suppose $b_n \to b$ uniformly on $X$. For each representative triangle the parametrized Jordan boundaries converge uniformly, hence in Fréchet distance. Here is the exact use of Radó's verified parameter theorem [64, §1, p. 182; proof, pp. 183–186]. Choose one point inside the limiting Jordan domain. It remains inside the nearby domains by the winding number of the uniformly convergent curves about that point. Normalize the Riemann maps there with positive derivative. Radó gives uniform convergence on the closed disk. The preimages of each of the three marked boundary vertices also converge: any subsequential limit maps to the specified limiting vertex, whose preimage is unique. The disk automorphism carrying the three fixed $\zeta_j$ to these three preimages varies continuously. This follows, for example, from the cross-ratio formulas for the unique Möbius map of distinct cyclic triples. Composing with this automorphism gives exactly $f_{\sigma,b_n}$. Therefore $f_{\sigma,b_n} \to f_{\sigma,b}$ uniformly on the closed disk. The normalization is unique, so the temporary interior point has not introduced any extra choice into (C.89.7).

Equation (C.89.6) now gives uniform convergence of the circle maps $\eta_{\sigma,b_n}$. One may verify the inverse composition without presuming a common target domain: for any convergent circle input sequence, compactness of the possible preimages and uniform convergence of the Riemann maps force every preimage limit to be the unique one in (C.89.6). The same argument with varying inputs proves uniform convergence. The radial extensions converge uniformly as well. Formula (C.89.7) consequently converges uniformly on each fixed representative triangle. For any other fixed triangle, (C.89.8) gives the same conclusion: the fixed map $q$ is uniformly continuous on a compact interior neighborhood of the convergent image faces.

It remains to control all triangles at once. The compact family $\{b, b_1, b_2, \dots\}$ is equicontinuous on $X$. By C.82 nullity, for any $\epsilon > 0$ all but finitely many source triangles have both $b$-boundary and $b_n$-boundary diameter less than $\epsilon$, uniformly in $n$. For a boundary point $a_\Delta$ of any such triangle and $x \in \Delta$, (C.89.9) gives

$$\begin{aligned} d_E(\mathfrak{s}(b_n)x, \mathfrak{s}(b)x) &\le \operatorname{diam}_{d_E} b_n(\partial\Delta) + d_E(b_n a_\Delta, b a_\Delta) \\ &\quad + \operatorname{diam}_{d_E} b(\partial\Delta) \\ &< 2\epsilon + \sup_{a \in X} d_E(b_n a, ba). \end{aligned} \tag{C.89.10}$$

On the finitely many remaining faces the proved individual convergence applies. This proves uniform convergence of the entire original disk maps. Their inverses converge uniformly too: for

homeomorphisms $u_n, u$ of a compact metric space,

$$\sup_y d_E(u_n^{-1}y, u^{-1}y) \le \omega_{u^{-1}}\Big(\sup_x d_E(u_n x, ux)\Big). \tag{C.89.11}$$

This follows by writing $y = u_n x$ and applying the modulus of $u^{-1}$ to $u_n x, ux$. Together with (C.89.4), these estimates prove continuity into the actual map-and-inverse group. Both domain and target are metrizable, so the sequential argument proves continuity of the global section.

*The product and the actual deformation.* Restriction is continuous for the stated uniform topology on $X$. The second coordinate of $\Psi(h)$ fixes $X$ pointwise and hence lies in $\mathscr{K}$. Composition and inversion in the actual closed-disk group are continuous. Thus both maps in (C.89.3) are continuous; their displayed formulas compose to the identity.

Let $A_t : \mathscr{K} \to \mathscr{K}$ be the single C.82 contraction, with $A_0(k) = k$, $A_1(k) = 1$ and $A_t(1) = 1$. The exact strong deformation retraction is

$$\mathcal{R}_t(h) = \mathfrak{s}(h|_X)\, A_t\Big(\mathfrak{s}(h|_X)^{-1} h\Big). \tag{C.89.12}$$

Its initial value is $h$, its final value is $\mathfrak{s}(h|_X)$, and it fixes that entire section image throughout. Every factor retains the original boundary values and each individual $Q$ label. This supplies the claimed global family extension and product comparison. Constructing a controlled deformation of arbitrary based families in $\mathscr{B}$ to $\mathrm{id}_X$, or to one compact quotient support for C.81, remains the outstanding general-to-skeleton step. □

**C.90. [Proved] Cocompact discrete operators in the original boundary topology.** Retain all the actual disk, operator, transitivity, covering and topology hypotheses of C.81. For a compact quotient $N = O/Q$, every based sphere family in the original relative centralizer has a based filling in that same group:

$$\pi_n(\mathscr{C}_Q, 1) = 0 \qquad (n \ge 1). \tag{C.90.1}$$

The filling fixes the original circle pointwise, commutes with each individual original $q \in Q$, and is continuous for both maps and inverses. The quotient may be nonorientable; in particular the torus and Klein-bottle cases are included. No boundary extension of an operator or of an interior conjugacy is required.

More precisely, let $p \in N$, let $x \in O$ be a specified lift of $p$, and write

$$\mathscr{C}_Q(x) = \{h \in \mathscr{C}_Q : h(x) = x\}, \qquad \mathscr{H}_p = \mathrm{Homeo}(N \text{ rel } \{p\})^0. \tag{C.90.2}$$

The superscript on $\mathscr{H}_p$ denotes its identity component with its compact-open group topology. Every based sphere family in $\mathscr{C}_Q(x)$ descends into $\mathscr{H}_p$. A single identity-fixed contraction of $\mathscr{H}_p$, lifted through the original covering from the identity, fills that sphere with its original endpoint maps retained. The inclusion $\mathscr{C}_Q(x) \hookrightarrow \mathscr{C}_Q$ is the positive-homotopy isomorphism of C.81, and transfers this filling to (C.90.1).

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), zero-level kernel and original-boundary discrete-operator fillings for compact quotient.

*Proof. The precise pointed quotient group.* The original quotient is a connected surface without boundary, since the free proper action gives covering neighborhoods modeled on open disks. It is compact by the present data. The covering $\pi : O \to N$ has contractible simply connected source $O$, so $\pi_2(N) \cong \pi_2(O) = 0$. In particular $N$ is neither the sphere nor the projective plane: their universal covers are the sphere and their second homotopy groups are $\mathbb{Z}$.

Use the quotient triangulation of C.82. It is finite by compactness and local finiteness. A subdivision in a coordinate disk can make the specified point $p$ a vertex, so that $\{p\}$ is a compact subpolyhedron of this PL surface. Apply the verified compact-surface result [74, Lemma 2.1(i), p. 2; proof, p. 3] to the exact pair $(N, \{p\})$. The exceptions with boundary cannot occur. Among its closed-surface exceptions, the sphere and projective-plane pairs have just been excluded, while the torus and Klein-bottle exceptions have empty fixed set. Our fixed set is the one specified point. Thus the cited statement proves contractibility of $\mathscr{H}_p$ for every quotient in the present assertion, including the nonorientable cases.

Choose a contraction $C_t : \mathscr{H}_p \to \mathscr{H}_p$ with $C_0(b) = b$ and $C_1(b) = c$ independent of $b$. Its endpoint need not initially be the identity. The normalized formula

$$J_r(b) = C_{1-r}(b)\, C_{1-r}(1)^{-1} \qquad (0 \le r \le 1) \tag{C.90.3}$$

is continuous in this actual topological group and has $J_0(b) = 1$, $J_1(b) = b$, and $J_r(1) = 1$. Every factor fixes $p$.

*Descent of the actual pointed sphere.* Let $a : (S^n, *) \to (\mathscr{C}_Q(x), 1)$, where $n \ge 1$. Commutation with each deck operator defines its exact descent

$$b_\xi(\pi z) = \pi(a_\xi z), \qquad b_\xi^{-1}(\pi z) = \pi(a_\xi^{-1} z). \tag{C.90.4}$$

These formulas are independent of the chosen lift $z$. They are mutually inverse, and $b_\xi(p) = p$. Local sections of the original covering prove joint continuity in $\xi$ and the downstairs point for both formulas. Since $N$ is compact, this is continuity of $\xi \mapsto b_\xi$ in the stated map-and-inverse topology. The path connected sphere has $b_* = 1$, so its entire image lies in $\mathscr{H}_p$. We make no assertion about descents of other components of $\mathscr{C}_Q(x)$.

Put $B_{r,\xi} = J_r(b_\xi)$ and retain the actual homotopy

$$(r, \xi, z) \longmapsto B_{r,\xi}(\pi z) \quad \text{from } [0,1] \times S^n \times O \text{ to } N. \tag{C.90.5}$$

The covering homotopy property lifts it uniquely from $H_{0,\xi} = \mathrm{id}_O$. This supplies a jointly continuous family $H_{r,\xi} : O \to O$. Apply the same covering property to $B_{r,\xi}^{-1}\pi$, from the identity, to obtain $K_{r,\xi}$. The two composites $H_{r,\xi}K_{r,\xi}$ and $K_{r,\xi}H_{r,\xi}$ lift the constant homotopy $\pi$ and start at the identity. Uniqueness of the lifted homotopy makes both composites the identity. Thus $H_{r,\xi}$ is a homeomorphism with jointly continuous inverse $K_{r,\xi}$.

For every original $q \in Q$, the maps $H_{r,\xi}q$ and $qH_{r,\xi}$ lift the same homotopy $B_{r,\xi}\pi$ and start at the same map $q$. They agree by uniqueness. This retains each individual deck label. The track $H_{r,\xi}(x)$ lies over $B_{r,\xi}(p) = p$ and starts at $x$; discreteness of that fiber gives $H_{r,\xi}(x) = x$ throughout. For the sphere basepoint, the downstairs homotopy is constantly the identity, so $H_{r,*} = \mathrm{id}_O$. At $r = 1$, both $H_{1,\xi}$ and $a_\xi|_O$ lift $b_\xi\pi$ and take $x$ to $x$. Uniqueness of a lift with specified value on the connected space $O$ proves exact endpoint agreement:

$$H_{0,\xi} = 1, \qquad H_{1,\xi} = a_\xi|_O, \qquad H_{r,*} = 1, \qquad H_{r,\xi}(x) = x. \tag{C.90.6}$$

*Extension in the original group topology.* Use C.88 with the compact metric parameter space $P = [0,1] \times S^n$ and the actual interior family $H_{r,\xi}$ just constructed. Its required uniform quotient tail holds with $K = N$: the set of $z$ with $\pi z \notin K$ is empty. Thus the explicit identity extensions

$$\overline{H}_{r,\xi}(z) = \begin{cases} H_{r,\xi}(z), & z \in O, \\ z, & z \in \partial\overline{D} \end{cases} \tag{C.90.7}$$

form a continuous family in the original $\mathscr{C}_Q$, including its inverse topology. The inverse family is the identity extension of $K_{r,\xi}$. Equations (C.90.6) now hold on the entire original closed disk, so

this is a based filling of $a$ in the pointed original group itself. No change of compactification has occurred at any step.

*Return to an arbitrary sphere.* C.81 gives the actual locally trivial evaluation bundle

$$\mathscr{C}_Q(x) \longrightarrow \mathscr{C}_Q \xrightarrow{h\mapsto h(x)} O. \tag{C.90.8}$$

Its base is contractible. Its exact based homotopy sequence therefore makes the displayed inclusion an isomorphism on $\pi_n$ for every $n \geq 1$, precisely the finite-orbit comparison already proved in C.81. For an arbitrary based sphere $f$ in $\mathscr{C}_Q$, choose its representative $a$ in that stabilizer and a based homotopy $U_t$ from $f$ to $a$ supplied by this isomorphism. Both families use the original group topology. The complete contraction of the given sphere is

$$F_{t,\xi} = \begin{cases} U_{2t,\xi}, & 0 \leq t \leq \frac{1}{2}, \\ \overline{H}_{2-2t,\xi}, & \frac{1}{2} \leq t \leq 1. \end{cases} \tag{C.90.9}$$

The two middle values are the same original map $a_\xi$. The entire homotopy is based, begins at $f$, ends at the identity, and retains the claimed boundary and operator conditions for maps and inverses. This proves (C.90.1).

Compactness is used here exactly in the C.88 tail condition with $K = N$ and in the pointed compact-surface contraction. For a noncompact quotient, that choice of $K$ is unavailable; the controlled quotient tail for the lifted family remains part of the unresolved general discrete-operator deformation. □

**C.91. [Proved] A contraction of the whole cocompact original centralizer.** Retain the actual hypotheses of C.81 and the compact quotient $N = O/Q$ of C.90. Thus $O = \operatorname{int} \overline{D}$ is the original open disk, $Q$ is closed, discrete, free and proper, and an actual subgroup of the boundary-fixed centralizer $\mathscr{C}_Q$ is transitive on $O$. Fix $x \in O$, put $p = \pi x$, and write $\mathscr{F} = \mathscr{C}_Q(x)$ and $\mathscr{H}_p = \operatorname{Homeo}(N \text{ rel } \{p\})^0$. Give these groups their stated compact-open map-and-inverse topologies. There are exact topological group isomorphisms

$$\begin{aligned} D &: \mathscr{F} \longrightarrow \mathscr{H}_p, \quad D(h)(\pi z) = \pi(hz), \\ L &: \mathscr{H}_p \longrightarrow \mathscr{F}, \qquad L(b)|_{\partial\overline{D}} = \mathrm{id}\,. \end{aligned} \tag{C.91.1}$$

The interior restriction of $L(b)$ is exactly the unique lift of $b\pi$ that fixes $x$. In particular this identifies the entire actual point stabilizer, not only its identity component. The proof establishes that every one of its descents lies in the indicated pointed component. The evaluation map admits a continuous section $s : O \to \mathscr{C}_Q$ with $s(y)x = y$ and $s(x) = 1$. Consequently the mutually inverse homeomorphisms

$$\begin{aligned} \Phi &: O \times \mathscr{F} \longrightarrow \mathscr{C}_Q, \qquad \Phi(y,k) = s(y)k, \\ \Psi &: \mathscr{C}_Q \longrightarrow O \times \mathscr{F}, \qquad \Psi(h) = (hx, s(hx)^{-1}h) \end{aligned} \tag{C.91.2}$$

give an identity-fixed contraction of the whole original $\mathscr{C}_Q$. All maps in the contraction fix the original circle and commute with every individual original $q \in Q$. The assertion includes the torus, Klein bottle, and all the other compact quotient surfaces permitted by these data.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), original-boundary discrete-operator zero-level kernel and its full cocompact contraction.

*Proof. A specified plane coordinate used only for homotopies and cells.* Choose a disk coordinate $\theta : \overline{D} \to \{u \in \mathbb{C} : |u| \leq 1\}$ and put $a = \theta(x)$. Define

$$\begin{aligned} m_a(u) &= \frac{u-a}{1-\overline{a}u}, & m_a^{-1}(v) &= \frac{v+a}{1+\overline{a}v}, \\ \psi(z) &= \frac{m_a(\theta(z))}{1-|m_a(\theta(z))|}, & \psi^{-1}(w) &= \theta^{-1} m_a^{-1}\Big(\frac{w}{1+|w|}\Big). \end{aligned} \tag{C.91.3}$$

These displayed inverse formulas give a homeomorphism $\psi : O \to \mathbb{R}^2$ with $\psi(x) = 0$. They are a coordinate on the specified interior; all operator equations below continue to use the original $Q$ and the original quotient covering. In particular the coordinate does not replace the original boundary topology of the group.

*Every actual pointed descent is pointed homotopic to the identity.* For $h \in \mathscr{F}$, the formulas

$$b(\pi z) = \pi(hz), \qquad b^{-1}(\pi z) = \pi(h^{-1}z) \tag{C.91.4}$$

are well-defined, mutually inverse continuous maps, because $h$ and $h^{-1}$ commute with every $q$. Both fix $p$. For clarity, the induced fundamental-group comparison retains the original labels. A loop $\gamma$ at $p$ has a unique lift $\widetilde{\gamma}$ starting at $x$, ending at some $qx$. Its image $h\widetilde{\gamma}$ lifts $b\gamma$, starts at $x$, and ends at $h(qx) = qh(x) = qx$. Lifts with the same endpoints are homotopic rel endpoints in the simply connected $O$. Projecting such a homotopy shows $[b\gamma] = [\gamma]$ in the actual $\pi_1(N, p)$.

Here is a direct based homotopy, so no classification of pointed mapping classes is being presumed from that computation. Take a finite triangulation of $N$ in which $p$ is a vertex, as in C.90, and lift it through $\pi$. Every lifted simplex maps homeomorphically onto its quotient simplex. A deck map preserving such a lift fixes a point of it by injectivity of that restriction and hence is the identity. Choose one lift of every quotient simplex, with the chosen lift of $p$ equal to $x$. We construct a $Q$-equivariant homotopy

$$W : O \times [0,1] \longrightarrow O, \qquad W_0 = h|_O, \quad W_1 = \mathrm{id}_O, \quad W_t(qx) = qx. \tag{C.91.5}$$

On a chosen vertex $v$, use the path $\psi^{-1}((1-t)\psi(hv) + t\psi(v))$. For $v = x$ this is constantly $x$. Define the paths on every translate by the literal formula $W_t(qv) = qW_t(v)$.

For a chosen lifted edge $e$, the paths on its two endpoints, together with $h|_e$ at time zero and $\mathrm{id}\,|_e$ at time one, give a continuous map on the boundary of the square $e \times [0,1]$. This extends across the square. An explicit extension for any boundary map $f : S^j \to O$ is obtained in cone coordinates by

$$(u, r) \longmapsto \psi^{-1}(r\psi(f(u))), \qquad 0 \le r \le 1, \tag{C.91.6}$$

with the cone vertex sent to $x$; a fixed ball-to-prism homeomorphism supplies the square coordinates here. Continuity at the cone vertex follows because $f(S^j)$ is compact and $\psi(f(S^j))$ is bounded. Transport this extension to every translate of $e$ by $q$. It agrees with the already defined vertex paths.

For a chosen lifted triangle, the two end faces and the three edge homotopies define a continuous map on the boundary of its triangular prism. Apply (C.91.6) with $j = 2$ and a fixed ball-to-prism homeomorphism, and transport the extension to all its $Q$-translates. The boundary prescriptions agree on every common face. The lifted triangulation is locally finite, so the closed-cell pasting lemma on $O \times [0,1]$ proves continuity of $W$. Each equivariance equation holds on every cell and hence on all of $O$. The vertex prescription proves the final fixed-orbit equation in (C.91.5).

It follows that

$$B_t(\pi z) = \pi W_t(z) \tag{C.91.7}$$

is a well-defined continuous homotopy from $b$ to $\mathrm{id}_N$ fixing $p$ at every time. Continuity follows, for example, by using local sections of the original covering. This homotopy is proper: its domain $N \times [0,1]$ is compact and the inverse image of a compact subset of the Hausdorff surface $N$ is closed in that domain. The surface $N$ has empty boundary, so the boundary-pair condition in Epstein's theorem is satisfied as well.

Apply exactly [20, Theorem 6.3, p. 99; proof, pp. 100–101]. This is the theorem for a proper homotopy of the boundary pair; its last sentence retains a basepoint fixed by the homotopy. Its proof explicitly treats a non-PL homotopy on p. 101 by a basepoint-preserving isotopy and

PL approximation, using the Appendix; the stated topological conclusion therefore applies to our actual $b$. We obtain an isotopy from $b$ to $\mathrm{id}_N$ fixing $p$. On the compact $N$, the isotopy is continuous in the compact-open topology of maps, and its inverse family is continuous as well: the level map on $N \times [0,1]$ is a continuous bijection of a compact space onto a Hausdorff space. Thus $b \in \mathscr{H}_p$ in the stated group topology. This proves the asserted component membership for every $h \in \mathscr{F}$, without first restricting $h$ to a component.

*The exact pointed compact-surface contraction and all exceptions.* As in C.90, $N$ is a closed connected PL surface with universal cover $O$. The covering isomorphism on second homotopy groups gives $\pi_2(N) = 0$, whereas $S^2$ and $\mathbb{RP}^2$ have second homotopy group $\mathbb{Z}$. These two closed-surface cases are therefore excluded by the actual covering. For the pair $(N, \{p\})$, Yagasaki's [74, Lemma 2.1(i), p. 2; proof, p. 3] proves contractibility of $\mathscr{H}_p$. The exceptional disk, annulus, and Möbius-band pairs have boundary and cannot be our $N$. The exceptional torus and Klein-bottle pairs have empty fixed set, whereas our fixed set is $\{p\}$. The remaining listed closed exceptions belong to the already excluded sphere and projective plane. No orientability assumption occurs in this application. Likewise Epstein's Theorem 6.3, used above with the proper based homotopy (C.91.7), includes the closed nonorientable surfaces. We have not used the distinct nonproper-homotopy statement of his Theorem 6.4(b) or its open or closed Möbius-band qualification.

Write $C_t$ for a contraction of $\mathscr{H}_p$, with $C_0(b) = b$ and constant endpoint $C_1(b) = c$. The exact normalization already used in C.90 is

$$J_r(b) = C_{1-r}(b)C_{1-r}(1)^{-1}, \qquad J_0(b) = 1, \quad J_1(b) = b, \quad J_r(1) = 1. \tag{C.91.8}$$

Every value and every factor fixes $p$.

*The inverse lift and the two actual topologies.* For $b \in \mathscr{H}_p$, lift the homotopy $J_r(b)\pi : O \to N$ from the identity lift at $r = 0$. Let $H_r(b)$ be the resulting interior family. Lift $J_r(b)^{-1}\pi$ from the identity to obtain $K_r(b)$. The two compositions $H_r(b)K_r(b)$ and $K_r(b)H_r(b)$ lift the constant homotopy $\pi$ and start at the identity; uniqueness makes both compositions the identity. For every literal $q \in Q$, the two maps $H_r(b)q$ and $qH_r(b)$ lift the same homotopy and start at the same map $q$, so they agree. The track of $x$ lies in the discrete fiber over $p$ and starts at $x$, hence remains $x$.

C.88 applies to this compact parameter interval with $K = N$: the required tail is empty. It supplies the continuous original-circle identity extensions of $H_r(b)$ and $K_r(b)$, which remain inverse. Define $L(b)$ as the extension of $H_1(b)$. It is therefore an actual member of $\mathscr{F}$. It descends to $b$. Any two lifts of $b\pi$ that take $x$ to $x$ agree on the connected $O$; their boundary extensions agree too. This proves

$$DL(b) = b, \qquad LD(h) = h. \tag{C.91.9}$$

The same uniqueness gives $L(bc) = L(b)L(c)$, so both maps in (C.91.1) are group maps with the displayed inverse.

The descent topology has a direct metric estimate. Use the actual metrics $d, d_Q, \rho$ of C.81 and set

$$d_{\mathscr{C}}(h,k) = \sup_{z \in \overline{D}} d(hz, kz) + \sup_{z \in \overline{D}} d(h^{-1}z, k^{-1}z),$$
$$d_{\mathscr{H}}(b,c) = \sup_{u \in N} \rho(bu, cu) + \sup_{u \in N} \rho(b^{-1}u, c^{-1}u).$$

They induce the asserted map-and-inverse topologies because their underlying spaces are compact. The quotient inequality and the exact C.81 isometry give

$$d_{\mathscr{H}}(Dh, Dk) \le d_{\mathscr{C}}(h,k). \tag{C.91.10}$$

Thus $D$ is continuous in the original group topology.

For the inverse, take any sequence $b_n \to b$ in $\mathscr{H}_p$ and use the compact metric parameter space $P = \{0\} \cup \{1/n : n \geq 1\}$ with $b_0 = b$. The family $J_r(b_u)$ and its inverse are continuous on $[0,1] \times P \times N$. Lift them jointly from the identity through the original covering, exactly as above. The resulting interior families have every original commutation equation and fix $x$. C.88, now with compact parameter $[0,1] \times P$ and again $K = N$, supplies joint continuity of their identity extensions and inverses on the original closed disk. At $r = 1$ this gives $L(b_n) \to L(b)$ in $d_{\mathscr{C}}$. Both groups in (C.91.1) are metrizable subspaces of the indicated compact-domain homeomorphism groups, so this sequential argument proves continuity of $L$. It proves the entire asserted topological group isomorphism, rather than only a set-theoretic lifting rule.

In particular the single formula

$$A_t(k) = L(J_{1-t}(Dk)), \quad A_0(k) = k, \quad A_1(k) = 1, \quad A_t(1) = 1 \tag{C.91.11}$$

is a continuous identity-fixed contraction of the whole point stabilizer $\mathscr{F}$ in its original topology.

*A global section with explicit local formulas.* C.81 supplies actual continuous local sections of $h \mapsto hx$ over an open cover $\{U_i\}$ of $O$. Write a chosen one on $U_i$ as $\sigma_i : U_i \to \mathscr{C}_Q$, so that $\sigma_i(y)x = y$; these maps already have the original map-and-inverse topology by C.81.11. Start with the standard locally finite unit-grid triangulation of $\mathbb{R}^2$, with 0 a vertex. Apply exactly the $k = 0$ subdivision lemma used in C.82, [33, Lemma 1, p. 458], to this complex, the actual map $\psi^{-1}$ into $O$, and the cover $\{U_i\}$. It gives a locally finite subdivision whose closed vertex stars map into members of that cover. Each closed simplex is in the closed star of any one of its vertices, so its image lies in some $U_i$. Pull the subdivided triangles back to $O$ by $\psi^{-1}$. The original vertex 0 remains a vertex under subdivision, and its image is $x$.

At every vertex $v$, choose once an actual $s(v) \in \mathscr{C}_Q$ taking $x$ to $v$, using the surjectivity in C.81. Choose $s(x) = 1$. We extend this section over the edges and then over the two-simplices. Suppose its values on the boundary of a closed simplex $\tau$ have been defined. Choose $i(\tau)$ with $\tau \subset U_{i(\tau)}$ and put

$$a_\tau(u) = \sigma_{i(\tau)}(u)^{-1} s(u) \in \mathscr{F} \qquad (u \in \partial\tau). \tag{C.91.12}$$

In the affine coordinates of the plane simplex, let $c_\tau$ be its barycenter. Every noncentral point has a unique expression $y = (1-r)c_\tau + ru$ with $u \in \partial\tau$ and $0 < r \leq 1$; this notation is transported to $O$ by $\psi^{-1}$. Define the extension by the exact formulas

$$\begin{aligned} s(y) &= \sigma_{i(\tau)}(y) A_{1-r}(a_\tau(u)) \quad (0 < r \leq 1), \\ s(c_\tau) &= \sigma_{i(\tau)}(c_\tau). \end{aligned} \tag{C.91.13}$$

The second factor always fixes $x$, so $s(y)x = y$. At $r = 1$ its value is exactly the already prescribed boundary value. At the center, continuity follows from $A_1 = 1$ uniformly on the compact set $a_\tau(\partial\tau)$: the restriction of the continuous map $A$ to $[0,1] \times a_\tau(\partial\tau)$ is uniformly continuous in any compatible metric on its image. This checks the possible polar-coordinate singularity without imposing any uniform condition at infinity. The edge extensions fit at their endpoints, and the face extensions retain every edge value. Local finiteness and the closed-cell pasting lemma therefore give one global continuous $s : O \to \mathscr{C}_Q$ in the actual group topology. The original value $s(x) = 1$ is retained.

Now the maps in (C.91.2) are defined on the displayed spaces: $s(hx)^{-1}h$ fixes $x$. Their two composites are identities by multiplication and $s(y)x = y$. Evaluation, multiplication, inversion, and $s$ are continuous in the original topology, so these are mutually inverse homeomorphisms. This also proves that the constructed section is a section of the actual C.81 bundle, with its original fiber and topology.

*The contraction of the whole original group.* In the specified coordinate (C.91.3), put

$$R_t(y) = \psi^{-1}((1-t)\psi(y)). \tag{C.91.14}$$

This is a continuous contraction of $O$ to $x$, fixing $x$ throughout. For $h \in \mathscr{C}_Q$ let $y = hx$ and $k = s(y)^{-1}h$. The required global contraction and its pointwise inverse map are

$$\begin{aligned} \mathcal{C}_t(h) &= s(R_t(y))A_t(k), \\ \mathcal{C}_t(h)^{-1} &= A_t(k)^{-1}s(R_t(y))^{-1}. \end{aligned} \tag{C.91.15}$$

All factors belong to the original $\mathscr{C}_Q$, so every value fixes its original circle pointwise and commutes with each original $q$. The formulas and (C.91.2), (C.91.11), (C.91.14) prove joint continuity in the map-and-inverse topology. At time zero the first formula is $s(y)k = h$; at time one it is $s(x) \cdot 1 = 1$. For $h = 1$, we have $y = x$, $k = 1$, so $\mathcal{C}_t(1) = 1$ for every $t$. This proves the claimed contraction of the entire group. □

**C.92. [Proved] Arbitrary pointed descents and the exact original lift topology.** Retain all the actual data of C.81: the original closed disk $\overline{D}$, its interior $O$, the compatible closed-disk metric $d$, the closed discrete free proper group $Q$ of operators on $O$, the boundary-fixed centralizer $\mathscr{C}_Q$ with a transitive actual subgroup, and the specified quotient covering $\pi : O \to N = O/Q$. The quotient $N$ is allowed to be noncompact throughout this note. Fix $x \in O$, put $p = \pi x$, and set $\mathscr{F} = \mathscr{C}_Q(x)$. Write $\mathscr{I}_p$ for the identity path component of Homeo($N$ rel $\{p\}$) in the compact-open topology of maps and inverses. Thus a path in this definition has jointly continuous maps and inverses on $[0,1] \times N$.

Every member of the entire original $\mathscr{F}$ descends into $\mathscr{I}_p$. Below we prove this assertion without assuming component membership upstairs. We also identify its exact image inside $\mathscr{I}_p$, with inverse maps and both relevant topologies. The original compact-quotient contraction C.91 remains unchanged.

More precisely, let $\lambda(b)$ be the unique lift of $b\pi$ fixing $x$ for $b \in \mathscr{I}_p$. The image of $D$ is exactly the subgroup $\mathscr{B}_p$ defined by $\delta_b(\pi z) = d_Q(\lambda(b)z, z) \in C_0(N)$. Descent and extension of this lift are inverse group maps and are isometric for the original sum metric and the lifted sum metric in (C.92.15). On a compact parameter space their exact continuity criterion is the uniform tail (C.92.17). The explicit noncompact example (C.92.18)–(C.92.21) gives a sequence in this exact image converging in the quotient compact-open topology whose pointed lifts stay more than 2/5 from the identity in the original metric.

*Manuscript:* Section 23, folios 140-141, scans 279-281; Part C(b), exact original-boundary pointed descent, lift topology and the remaining noncompact level-zero kernel.

*Proof. Literal descent and the original orbit labels.* For $h \in \mathscr{F}$, define

$$D(h)(\pi z) = \pi(hz), \qquad D(h)^{-1}(\pi z) = \pi(h^{-1}z). \tag{C.92.1}$$

Commutation with every actual $q$ makes these formulas well-defined. The quotient property makes them continuous, and substitution proves they are inverse maps. They fix $p$. Composition gives $D(hk) = D(h)D(k)$. Two such maps with equal descents are lifts of the same map on the connected $O$, agree at $x$, and hence agree everywhere; their boundary values are both the identity. Thus $D$ is injective.

For a loop $\gamma$ at $p$, let its lift starting at $x$ end at $qx$. The image under $h$ of that lift starts at $x$ and ends at $h(qx) = qh(x) = qx$. It lifts $D(h)\gamma$. Since $O$ is simply connected, the two lifted paths with these endpoints are homotopic rel endpoints. Their projections show $D(h)_*[\gamma] = [\gamma]$ in the specified $\pi_1(N, p)$. In particular every original orbit label is retained.

*A based homotopy on the locally finite lifted triangulation.* Here is a direct homotopy, including its maps, rather than an inference of pointed isotopy from the preceding loop calculation. Choose

a disk coordinate $\theta : \overline{D} \to \{|u| \le 1\} \subset \mathbb{C}$, put $a = \theta(x)$, and use the mutually inverse maps

$$\begin{aligned} m_a(u) &= (u-a)/(1-\overline{a}u), \quad m_a^{-1}(v) = (v+a)/(1+\overline{a}v), \\ \psi(z) &= \frac{m_a(\theta z)}{1-|m_a(\theta z)|}, \qquad \psi^{-1}(w) = \theta^{-1} m_a^{-1}\left(\frac{w}{1+|w|}\right). \end{aligned} \tag{C.92.2}$$

They give $\psi : O \to \mathbb{R}^2$ with $\psi(x) = 0$. The original covering and every operator $q$ remain the ones in (C.92.1).

Use C.82's locally finite triangulation of the quotient surface, refined locally so that $p$ is a vertex, and lift it by $\pi$. Every lifted closed simplex maps homeomorphically to its quotient simplex. The lifted family is locally finite, by the local covering charts. Any deck operator preserving a lifted simplex fixes it pointwise, by injectivity of the projection on that simplex, and is therefore the identity by freeness. The lifts of a given quotient simplex form one $Q$-orbit: match one point in two lifts by the unique operator in its fiber and then use uniqueness of the lift on the connected simplex. Choose one lift of each simplex and choose $x$ as the lift of the vertex $p$.

For a chosen vertex $v$ define $W_t(v) = \psi^{-1}((1-t)\psi(hv) + t\psi(v))$, and define $W_t(qv) = qW_t(v)$ on all translates. The path at $x$ is constant. For a chosen edge, its endpoint paths and its prescribed maps $h$ and the identity at times zero and one give a map $f$ on the boundary of its square prism. For a chosen triangle the already defined edge homotopies and the two end faces likewise give a map on the boundary of its triangular prism.

For completeness, use the following explicit extension on either convex coordinate prism $P$. Choose an interior point $c$. Every point other than $c$ is uniquely $c + r(u-c)$ with $u \in \partial P$ and $0 < r \le 1$. Define

$$\mathrm{Cone}_f(c + r(u-c)) = \psi^{-1}\big(r\psi(f(u))\big), \qquad \mathrm{Cone}_f(c) = x. \tag{C.92.3}$$

These radial coordinates are continuous away from $c$: on each ray the boundary parameter is the first intersection with one of the finitely many facet hyperplanes of the convex prism. Equivalently, for a unit vector $v$ and inequalities $a_i \cdot y \le b_i$ defining $P$, that parameter is $\min_{a_i \cdot v > 0}(b_i - a_i \cdot c)/(a_i \cdot v)$. This is positive and continuous on the unit sphere. Continuity of (C.92.3) at $c$ follows from boundedness of $\psi(f(\partial P))$. The value on $\partial P$ is exactly $f$.

Apply (C.92.3) first to each chosen edge prism and then to each chosen triangle prism, using the coordinates of the triangulation, and transport each extension by the literal operator $q$. The boundary prescriptions agree on common faces. Local finiteness gives continuity by closed-cell pasting on $O \times [0,1]$. Thus we have actual maps

$$\begin{aligned} W : O \times [0,1] \to O, \qquad & W_0 = h|_O, \quad W_1 = \mathrm{id}_O, \\ W_t(qz) = qW_t(z), \qquad & W_t(qx) = qx, \\ B_t(\pi z) = \pi W_t(z), \qquad & B_0 = D(h), \quad B_1 = \mathrm{id}_N, \quad B_t(p) = p. \end{aligned} \tag{C.92.4}$$

The last family is continuous by the local sections of $\pi$. This is a based homotopy for every original $h \in \mathscr{F}$. The argument uses local finiteness and does not require the triangulation to be finite.

*The orientation maps, including a nonorientable quotient.* We verify every orientation qualification in the source theorem. The connecting map

$$\partial : H_2(\overline{D}, \partial\overline{D}; \mathbb{Z}) \longrightarrow H_1(\partial\overline{D}; \mathbb{Z}) \tag{C.92.5}$$

is an isomorphism by the exact sequence of the contractible disk. Naturality and $h|_{\partial\overline{D}} = \mathrm{id}$ give $\partial h_* = \partial$, so $h_*$ is the identity on the left group.

Let $j_x$ be the map from $H_2(\overline{D}, \partial\overline{D})$ to $H_2(\overline{D}, \overline{D} \setminus \{x\})$ induced by inclusion of pairs. It is an isomorphism. Indeed, in the coordinate $\kappa = m_a\theta$, the formula

$$R_s(z) = \kappa^{-1}\left(\left(1 - s + \frac{s}{|\kappa(z)|}\right)\kappa(z)\right), \quad z \neq x, \quad 0 \leq s \leq 1, \tag{C.92.6}$$

deformation retracts $\overline{D} \setminus \{x\}$ onto the boundary while fixing the boundary. The natural diagram of the two exact sequences then proves the assertion about $j_x$. Let $e_x : H_2(O, O \setminus \{x\}) \to H_2(\overline{D}, \overline{D} \setminus \{x\})$ be the excision isomorphism. Naturality for $h$, which fixes $x$, gives, with $\eta_x = e_x^{-1} j_x$,

$$h_{x*} = \eta_x \, h_* \, \eta_x^{-1} = \mathrm{id} \quad \text{on } H_2(O, O \setminus \{x\}; \mathbb{Z}). \tag{C.92.7}$$

Choose one specified evenly covered neighborhood $V_0$ of $p$ and its sheet $U_0$ containing $x$. Write $i_U, i_V$ for the excision maps from the local homology groups of $U_0, V_0$ to those of $O, N$, respectively. The precise orientation comparison at $x$ and $p$ is the isomorphism

$$\omega_x = i_V \circ (\pi|_{U_0})_* \circ i_U^{-1} : H_2(O, O \setminus \{x\}) \longrightarrow H_2(N, N \setminus \{p\}). \tag{C.92.8}$$

Choose a smaller open neighborhood $U \subset U_0 \cap h^{-1}(U_0)$ of $x$. On $\pi U$ the literal local formula is

$$D(h)|_{\pi U} = (\pi|_{U_0}) \circ h|_U \circ (\pi|_U)^{-1}. \tag{C.92.9}$$

Excision and naturality applied to this formula prove $D(h)_{p*} = \omega_x h_{x*} \omega_x^{-1} = \mathrm{id}$. Thus the local orientation at the fixed point is preserved. The use of a sheet in (C.92.8) is essential to the definition: the global map $\pi$ sends every other point $qx$ in this fiber to $p$, so the required local comparison is the displayed excision composite.

On a connected orientable $N$, the sign of a homeomorphism on the continuously chosen local fundamental classes is locally constant and hence constant. Its sign at $p$ is $+1$ by (C.92.7)–(C.92.9), so $D(h)$ preserves orientation globally. This verifies the plane and open-cylinder cases. On an open Möbius band the same local-homology identity verifies preservation of orientation near the basepoint. Each point of $N$ has an open-disk covering chart, so $\partial N = \varnothing$; the closed-cylinder and closed Möbius-band cases in the source statement therefore do not occur as this $N$. No orientation of a nonorientable $N$, and no orientation assumption on the original $q$, has entered the local calculation.

*The exact Epstein result and the topology of the resulting path.* Use [20, Theorem 6.4(b), p. 101; proof, pp. 101–103]. Its based statement is the following: a self-homeomorphism of a surface whose boundary components are compact, based homotopic to the identity, is isotopic to the identity fixing that basepoint, with orientation preservation required for the plane and the open or closed cylinder, and local orientation preservation at the basepoint required for the open or closed Möbius band. Equations (C.92.4) and (C.92.7)–(C.92.9) verify these hypotheses; the boundary-component requirement holds because $\partial N$ is empty.

The exceptional Möbius cases are proved in Lemma 6.5 on p. 102. For the general case the proof on p. 103 concatenates isotopies fixing successively larger compact submanifolds and explicitly constructs a level-preserving homeomorphism of $N \times [0, 1]$. The topological version is included in the statement. Appendix Theorem A4 on p. 107 supplies its basepoint-fixed passage to a PL homeomorphism; the corresponding topological proof paragraph on p. 101 records the relative approximation step. The hypotheses of Theorem 6.4(b) are met by the displayed based homotopy itself; a properness assertion for (C.92.4) on a noncompact $N$ is not used.

There is a small topology point in using this source: Definition 1.4 on p. 84 defines an isotopy of embeddings. The proof used here supplies homeomorphisms throughout. For the general case this is explicit on p. 103. The plane and open-cylinder steps in Theorems 5.4 and 5.7, pp. 96–97, use Theorem 5.1, p. 95: its level maps for $t < 1$ are conjugates of the displayed

homeomorphism, and at $t = 1$ are the identity. The compact disk step is the explicit level-preserving homeomorphism of Theorem 5.2, pp. 95–96. The open-Möbius proof in Lemma 6.5 uses a compact Möbius core and the same half-cylinder construction on its complement. In the cylinder step our based homotopy also retains the generator of its fundamental group, as proved after (C.92.1). After the based core-circle adjustment in the source proof, the map fixes the core pointwise; local orientation preservation makes it preserve its two sides, so the two half-cylinder constructions apply to the actual complementary components. The topological approximation from A4 is an ambient isotopy and likewise consists of homeomorphisms.

Write the resulting level-preserving homeomorphism as $(y,t) \mapsto (b_t(y),t)$, where $b_0 = D(h)$, $b_1 = \mathrm{id}_N$, and $b_t(p) = p$. Its inverse is exactly $(y,t) \mapsto (b_t^{-1}(y),t)$ and is continuous. This inverse continuity can also be checked without a compactness assumption on $N$: extend the homeomorphism family constantly for $t < 0$ and $t > 1$. Its level map is a continuous bijection of the boundaryless 3-manifold $N \times \mathbb{R}$. Invariance of domain makes it open, so its inverse is continuous; restrict back to $[0,1]$. The compact-open exponential law on the locally compact surface $N$ therefore gives a continuous path of maps and inverses. Consequently $D(h) \in \mathscr{I}_p$ for every $h \in \mathscr{F}$. This also places each such descent in the pointed identity component; no equality of all components or all possible descents has been presumed.

*The exact pointed lift and its compact-open continuity.* For a pointed homeomorphism $b$ of $N$, define the actual path and its specified lift by

$$\beta_{b,z}(t) = b\Big(\pi\psi^{-1}(t\psi(z))\Big), \qquad \lambda(b)(z) = \widetilde{\beta}_{b,z}(1), \quad \widetilde{\beta}_{b,z}(0) = x. \tag{C.92.10}$$

The covering homotopy property applied with parameter $(b,z)$ gives joint continuity of the lifted paths. Indeed evaluation of $b$ is continuous for the compact-open topology because $N$ is locally compact, and the displayed formula is jointly continuous before lifting. It gives $\pi\lambda(b) = b\pi$ and $\lambda(b)(x) = x$. Uniqueness of lifts on the connected $O$ makes it the unique lift with that value. The same uniqueness proves the exact identities

$$\lambda(bc) = \lambda(b)\lambda(c), \qquad \lambda(b^{-1}) = \lambda(b)^{-1}, \qquad \lambda(\mathrm{id}_N) = \mathrm{id}_O\,. \tag{C.92.11}$$

For the inverse identity, the two composites lift $\pi$ and fix $x$, and so are the identity. Thus these are actual homeomorphisms. Applying the joint argument in (C.92.10) to $b^{-1}$ also proves joint inverse continuity. The exponential law on $O$ proves that $\lambda$ is continuous from the pointed compact-open map-and-inverse topology to the compact-open map-and-inverse topology on $O$.

For $b \in \mathscr{I}_p$, take a pointed path $b_t$ from the identity to $b$. The continuous path $\lambda(b_t)(qx)$ lies in the discrete fiber $Qx$ and starts at $qx$. It is therefore constantly $qx$. Now $\lambda(b)q$ and $q\lambda(b)$ lift the same map $b\pi$ and agree at $x$, so uniqueness gives

$$\lambda(b)q = q\lambda(b) \quad \text{for every original } q. \tag{C.92.12}$$

This proves the individual operator equations without choosing a deck representative modulo any subgroup.

Descent is continuous in the other direction as well. The map $(h,z) \mapsto \pi(hz)$ on $\mathscr{F} \times O$ is continuous and is constant on the fibers of $\mathrm{id}_{\mathscr{F}} \times \pi$. This product map is an open quotient map, since $\pi$ is open. It therefore induces the continuous joint map $(h,y) \mapsto D(h)y$. Repeat the argument with $h^{-1}$ and use the exponential law on $N$. This proves continuity of $D : \mathscr{F} \to \mathscr{I}_p$ in the quotient compact-open map-and-inverse topology.

*The exact original image and its isometric topology.* Retain $d_Q(z,w) = \sup_{q\in Q} d(qz,qw)$ from C.81 and put

$$\begin{aligned} \delta_b(\pi z) &= d_Q(\lambda(b)z, z), \\ \mathscr{B}_p &= \{b \in \mathscr{I}_p : \delta_b \in C_0(N)\}, \\ L(b)|_O &= \lambda(b), \qquad L(b)|_{\partial\overline{D}} = \mathrm{id} \quad (b \in \mathscr{B}_p). \end{aligned} \tag{C.92.13}$$

The displacement is well-defined and continuous by (C.92.12) and $Q$-invariance of $d_Q$. C.88 gives its original boundary extension exactly when this displacement belongs to $C_0(N)$. For $h \in \mathscr{F}$, uniqueness gives $\lambda(D(h)) = h|_O$, so C.88 puts $D(h)$ in $\mathscr{B}_p$. Conversely C.88 extends every $\lambda(b)$ in (C.92.13) to an element of $\mathscr{F}$, and its descent is $b$. Thus

$$D : \mathscr{F} \longrightarrow \mathscr{B}_p, \qquad L : \mathscr{B}_p \longrightarrow \mathscr{F}, \qquad LD = \mathrm{id}_{\mathscr{F}}, \quad DL = \mathrm{id}_{\mathscr{B}_p} \tag{C.92.14}$$

are the exact inverse group maps. For example the subgroup assertion also follows directly from $\delta_{bc} \le \delta_b \circ c + \delta_c$ and $\delta_{b^{-1}} = \delta_b \circ b^{-1}$: a homeomorphism and its inverse carry compact sets to compact sets, so these pullbacks preserve $C_0(N)$.

Define a metric on this exact image by the actual lifts:

$$d^{\uparrow}(b,c) = \sup_{z\in O} d_Q(\lambda(b)z, \lambda(c)z) + \sup_{z\in O} d_Q(\lambda(b)^{-1}z, \lambda(c)^{-1}z). \tag{C.92.15}$$

It is finite since $d$ is bounded on the compact original disk. It is a metric by injectivity of the lift. Equivariance, reindexing, and density of the interior give

$$\begin{aligned} \sup_{z\in O} d_Q(\lambda(b)z, \lambda(c)z) &= \sup_{z\in O,\ q\in Q} d(\lambda(b)qz, \lambda(c)qz) \\ &= \sup_{w\in \overline{D}} d(L(b)w, L(c)w). \end{aligned} \tag{C.92.16}$$

The same identity for the inverse maps proves that (C.92.14) are isometric topological group isomorphisms for the original closed-disk sum metric and $d^{\uparrow}$. In particular the identity from this lifted metric on $\mathscr{B}_p$ to its quotient compact-open topology is continuous, by the descent argument above.

For a compact metric parameter space $P$ and a continuous map $u \mapsto b_u$ into $\mathscr{B}_p$ with its quotient compact-open topology, (C.92.10) gives joint interior continuity of the actual pointed lifts and their inverses. C.88 now gives the exact criterion for $u \mapsto L(b_u)$ to be continuous in the original topology:

$$\forall \epsilon > 0\ \exists K \subset N \text{ compact} \quad \sup_{\substack{u\in P \\ y\notin K}} \delta_{b_u}(y) < \epsilon. \tag{C.92.17}$$

Its inverse family then satisfies the same uniform tail condition. Thus the compact-family requirement is retained on the unique lifts in (C.92.10), with every deck label fixed.

When $N$ is compact, taking $K = N$ makes (C.92.17) automatic, and $C_0(N) = C(N)$ in the definition used here. Then $\mathscr{B}_p = \mathscr{I}_p$. For a convergent sequence $b_n \to b$ in the quotient topology, apply (C.92.17) to $\{0\} \cup \{1/n : n \ge 1\}$ and the lifts from (C.92.10). This gives $L(b_n) \to L(b)$ in the original metric. Both topologies are metrizable, so this proves continuity of the inverse comparison in the compact case. C.91's pointed-component contraction makes that entire component path connected, giving $\mathscr{I}_p = \mathrm{Homeo}(N \text{ rel } \{p\})^0$ there. These are precisely C.91's original maps and compact hypothesis.

*An exact noncompact sequence, including its deck labels.* The following actual C.81 example records why the last compact argument cannot be replaced just by quotient compact-open convergence. In this example $\overline{D}$ is the closed unit disk with its Euclidean metric $d$. Use the original radial disk coordinate

$$j : \mathbb{R}^2 \to \{|w| < 1\}, \quad j(z) = \frac{z}{1+|z|}, \qquad j^{-1}(w) = \frac{w}{1-|w|}, \tag{C.92.18}$$

and take $Q = \langle jTj^{-1} \rangle$ with $T(s,t) = (s+1, t)$. The quotient coordinate is $\pi(j(s,t)) = ([s], t)$ on $N = (\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$. The actual extended translations $j(z \mapsto z + a)j^{-1}$ are transitive, fix the original circle, and commute with every displayed operator. The action is closed, discrete, free and proper by the explicit plane coordinate, so this is an instance of the retained hypotheses.

For an integer $R \geq 2$ put $\beta(v) = \max(0, 1 - 4|v|)$ and

$$F_{R,t}(s,y) = (s + tR\beta(y-R), y), \qquad F_{R,t}^{-1}(s,y) = (s - tR\beta(y-R), y), \quad 0 \leq t \leq 1. \quad \text{(C.92.19)}$$

They commute with every $T^n$, fix $(0,0)$, and have quotient support in $(\mathbb{R}/\mathbb{Z}) \times [R - \frac{1}{4}, R + \frac{1}{4}]$. Both displacements in the plane are bounded by $R$. For $|w| \leq R$ and $|z| > R$, direct subtraction gives

$$|j(z+w) - j(z)| \leq \frac{2R}{1 + |z| - R}. \quad \text{(C.92.20)}$$

Thus $jF_{R,t}j^{-1}$ and its inverse extend jointly over $[0,1]$ by the identity on the original circle. Let $b_R$ be the descent of this endpoint. Then $b_R \in \mathscr{B}_p$, with $x = j(0,0)$. Its unique pointed lift is exactly $jF_{R,1}j^{-1}$. Both $b_R$ and $b_R^{-1}$ are eventually the identity on any specified compact subset of $N$, so $b_R \to 1$ in the quotient compact-open map-and-inverse topology.

At $y_R = ([0], R)$ the quotient value is $b_R(y_R) = y_R$, because the horizontal translation there is the integer $R$. The actual pointed lift, however, sends $j(0,R)$ to $j(R,R)$, and therefore

$$\rho(b_R(y_R), y_R) = 0, \qquad \delta_{b_R}(y_R) \geq |j(R,R) - j(0,R)| \geq \frac{R}{1 + \sqrt{2}R} > \frac{2}{5},$$
$$d^{\uparrow}(b_R, 1) > \frac{2}{5}. \quad \text{(C.92.21)}$$

For the strict scalar bound, put $R = 1 + s$, $s \geq 0$. Then $5R - 2 > 0$ and $(5R-2)^2 - 8R^2 = 1 + 14s + 17s^2 > 0$. The point-fixing lift in (C.92.10) cannot be replaced by a different deck translate: a nonidentity translate sends $x$ to a different point of its fiber. This is a sequence on the exact image $\mathscr{B}_p$ for which the inverse compact-open comparison fails to be continuous. It also fails the compact-family tail (C.92.17) on the convergent-sequence parameter space, since the $y_R$ escape all compact sets.

The proved comparison is consequently the original-image description (C.92.13)–(C.92.17), together with pointed isotopy of every actual descent. A deformation of a noncompact quotient family still requires (C.92.17) for the actual lift and its inverse throughout the deformation. No such deformation is constructed in this note. C.91 continues to assert the whole-group contraction under its existing compact-quotient hypothesis.

□

# 4 D. Profinite extensions and the limits of the realization problem

We write $P = \widehat{\pi}_0$, where $\pi_0 = \pi_1(U,p)$ for a connected orientable surface of finite type with $\chi(U) < 0$. All maps between profinite groups in this section are continuous. A lift of an outer action means a homomorphism inducing the prescribed outer homomorphism, with its labels in the operator group retained. For a finite group $G$ acting faithfully on $U$, $E$ denotes the group of lifts to the universal cover, not a mapping class group. The completion map $E \to \widehat{E}$ is injective: $E$ is a finitely generated linear hyperbolic orbifold group and is residually finite. We use $E^+$ for its orientation-preserving subgroup. The passages answered here are §31, folios 198–199, scans 391–394; the inserted $G = G^+$ on scan 391 is material.

**D.1. [Proved] Exact completion of the lift extension.** The completion of $1 \to \pi_0 \to E \xrightarrow{q} G \to 1$ is the exact sequence

$$1 \longrightarrow P \longrightarrow \widehat{E} \xrightarrow{\widehat{q}} G \longrightarrow 1.$$

It is the extension attached to the completed outer action. An invariant discretization therefore supplies a canonical comparison of extensions, with identity maps on $P$ and $G$.

*Manuscript:* 31; 198–199 verso; 391–394.

*Proof.* Every subgroup of finite index in $\pi_0$ contains a subgroup normal and of finite index in $E$: take its core in $E$, using $[E : \pi_0] < \infty$. Thus $\pi_0$ has its full profinite topology induced from $E$, and its closure is $P$. The kernel of $\widehat{q}$ is that closure, since $\pi_0 = q^{-1}(1)$ is open. Conjugation in the completed extension is the continuous extension of conjugation in the dense subgroup $E$. The uniqueness of a center-free kernel extension, proved in Part A, identifies this extension with the pullback of $\mathrm{Aut}(P) \to \mathrm{Out}(P)$. The identification fixes the named kernel and quotient, so the assertion of canonicity has no unmentioned conjugator. □

**D.2. [Proved] Finite subgroups of completed surface orbifold groups, including reflections.** Every finite subgroup of $\widehat{E}$ is conjugate to a finite subgroup of $E$. Two finite subgroups of $E$ conjugate in $\widehat{E}$ are conjugate in $E$. These finite subgroups are cyclic rotation groups or dihedral groups, with a single reflection included as the dihedral case of order two.

*Manuscript:* 31; 198–199 verso; 391–394.

*Proof.* We specify the two inputs and the reduction. Bridson–Conder–Reid, Theorem 5.1 [15], applies to the finitely generated Fuchsian group $E^+$: every finite subgroup of $\widehat{E^+}$ is conjugate into $E^+$. Boggi–Zalesskii, Theorem 6.3 [11], gives conjugacy and normalizer comparison for finite $p$-subgroups of $E$:

$$N_{\widehat{E}}(A) = \overline{N_E(A)}.$$

Here its hypotheses hold because $E$ has a finite-index surface group (a free group in the noncompact case), hence is virtually compact special; its good-group and finiteness hypotheses are precisely those in Theorem 6.3. We use only this prime-power theorem, not an assertion obtained by commuting arbitrary intersections with closure.

Put $K^+ = K \cap \widehat{E^+}$ for a finite $K \le \widehat{E}$. The first input makes $K^+$ cyclic. When $K^+ = 1$, the orientation character embeds $K$ in $C_2$, and the prime-power input handles $K$. Otherwise choose the unique subgroup $A$ of order $p$ in $K^+$, for one prime $p \mid |K^+|$. It is characteristic in $K^+$ and normal in $K$. Conjugate $A$ into $E^+$. A nontrivial elliptic rotation of the hyperbolic plane has one fixed point $x$. Its normalizer in the discrete group $E$ stabilizes $x$, and conversely every element of $E_x$ normalizes the unique subgroup of order $p$ in its cyclic rotation subgroup. Thus $N_E(A) = E_x$ is finite cyclic or dihedral. The displayed comparison gives $N_{\widehat{E}}(A) = E_x$, so $K \le E_x \le E$.

For injectivity, suppose $K, L \le E$ and $tKt^{-1} = L$, with $t \in \widehat{E}$. The orientation character gives $tK^+t^{-1} = L^+$. In the case $K^+ = 1$ apply prime-power conjugacy separation. Otherwise choose $A \le K^+$ as above and $B = tAt^{-1} \le L^+$. Prime-power conjugacy separation supplies $e \in E$ with $eAe^{-1} = B$. Now $e^{-1}t$ normalizes $A$, hence belongs to $N_E(A) \subset E$. Therefore $t \in E$, after this adjustment. This proves the required conjugacy assertion. Finally finite discrete hyperbolic-plane isometry groups stabilize a point and embed in $O(2)$; the listed cyclic and dihedral possibilities follow. □

**D.3. [Proved] The comparison of full and partial section classes.** For every subgroup $H \le G$, completion induces a bijection

$$\mathrm{Sec}(E_H \to H)/\pi_0 \;\longrightarrow\; \mathrm{Sec}(\widehat{E}_H \to H)/P, \qquad E_H = q^{-1}(H).$$

Both sets are finite. In particular the first conjecture on scan 393 holds for geometrically realized actions, with orientation-reversing operators allowed.

*Manuscript:* 31; 198–199 verso; 391–394; 18; 99–101; 197–203.

*Proof.* We prove the change from total-group conjugacy to kernel conjugacy, since omission of this step changes the question. A profinite section has a finite image $K$ mapping isomorphically to $H$. Apply D.2 to $E_H$, and choose $t \in \widehat{E}_H$ with $tKt^{-1} \leq E_H$. Choose $e \in E_H$ with $q(e) = \widehat{q}(t)$. Then $e^{-1}t \in P$, and conjugation by this element puts $K$ in $E_H$. The restriction of $q$ identifies the resulting subgroup with $H$, so it defines a discrete section with the correct labels.

Conversely, let discrete section images $K, L$ be conjugate by an element of $P$. D.2, applied in $E_H$, gives an $e \in E_H$ with $eKe^{-1} = L$. Let $l \in L$ be the unique element satisfying $q(l) = q(e)$. Then $l^{-1}e \in \pi_0$ still conjugates $K$ to $L$. Because the projection is injective on each section image, this equality of subgroups is equality of the labelled section homomorphisms after conjugation. This proves injectivity.

There are finitely many $E_H$-conjugacy classes of finite subgroups: they occur in the finite list of cone stabilizers, corner stabilizers, and mirror components of a finite-type quotient orbifold, and each finite stabilizer has finitely many subgroups. Each such class splits into at most $[E_H : \pi_0] = |H|$ kernel-conjugacy classes. The asserted finiteness follows. □

**D.4. [Proved] The exact profinite fixed subgroup of a section.** Let $s : H \to E_H$ be a nontrivial discrete section. Its fixed subgroup $P^{s(H)} = C_P(s(H))$ is trivial when $s(H)$ contains a nontrivial rotation. For a reflection $s(H) = \langle t \rangle$, it is 1 for a noncompact real component and is the closure of the primitive oval subgroup, isomorphic to $\widehat{\mathbb{Z}}$, for a compact real component. The same assertions hold for every profinite section by D.3.

*Manuscript:* 31; 199; 393; 33; 222 verso–225; 440–445.

*Proof.* For a rotation choose $A$ of prime order inside $s(H)^+$. D.2's proof gives $N_{\widehat{E}}(A) = E_x$, a finite group. Thus $C_P(s(H)) \leq E_x \cap P = 1$, because $P$ is torsion-free. For a reflection the prime-power centralizer comparison gives

$$C_{\widehat{E}}(t) = \overline{C_E(t)}, \qquad C_P(t) = \overline{C_{\pi_0}(t)}.$$

The second equality uses that $P$ is open and $P \cap E = \pi_0$: the intersection of the closure of a subgroup with an open subgroup is the closure of their intersection. On the universal cover, $C_{\pi_0}(t)$ is the group of deck transformations preserving and commuting with reflection in its fixed line. It acts freely on that line. Its quotient is the corresponding real component of $U$. The deck group is trivial for a line and infinite cyclic for a circle; the circle is an essential primitive simple curve. Its induced profinite topology is full, so its closure is $\widehat{\mathbb{Z}}$.

This proves, rather than assumes, the correction to the expected answer in Part D(a): a dihedral section with a nontrivial rotation has fixed subgroup 1. The value $\widehat{\mathbb{Z}}$ belongs to a reflection in an oval. It is not the fixed subgroup of a larger dihedral stabilizer containing that reflection. □

**D.5. [Refuted] Unique maximal sections require a rotation.** A finite subgroup of $\widehat{E}$ containing a nontrivial rotation is contained in a unique maximal finite subgroup. The assertion for every nontrivial finite subgroup is false, already for type $(0, 3)$ with reflections.

*Manuscript:* 18; 99–101; 197–203; 31; 198–199; 391–393.

*Proof.* For the positive assertion conjugate the subgroup into $E$ using D.2. Its rotation part has a subgroup $A$ of prime order. Every finite overgroup has cyclic orientation-preserving part, so normalizes this same $A$. All finite overgroups therefore lie in the finite group $N_{\widehat{E}}(A) = E_x$, which is itself the unique maximal one.

For the counterexample use the $(2, 3, \infty)$ reflection group

$$W = \langle s_0, s_1, s_2 \mid s_i^2 = 1, (s_0 s_1)^2 = (s_1 s_2)^3 = 1 \rangle = V_4 *_{C_2} S_3.$$

Map it onto $S_3 \times C_2$, with $r^3 = a^2 = \tau^2 = 1$, $ara = r^{-1}$ and $\tau$ central, by $s_0 \mapsto \tau$, $s_1 \mapsto a\tau$, $s_2 \mapsto ra\tau$. The vertex groups inject. The kernel is an orientable torsion-free group of index 12, Euler characteristic $-1$, with three cusps; hence it is $\pi_{0,3}$. The two maximal finite subgroups $\langle s_0, s_1\rangle$ and $\langle s_1, s_2\rangle$ contain the same reflection $s_1$ and have distinct images in the quotient. They remain distinct and are not kernel-conjugate after completion. Their generated subgroup is $W$, so no finite overgroup contains both. Part B supplies the finite certificate and the geometric kernel computation. □

**D.6. [Proved] A well-defined transfer of the geometric rotation invariant.** For a realized $E$, its rotation invariant on finite-order elements transfers uniquely to the torsion of $\widehat{E}$ by conjugating each torsion element into $E$. This assertion uses the orientation module, so that conjugation by an orientation-reversing element acts by $-1$ on its values. Its values embed canonically in $\widehat{T} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$, where $\widehat{T} = T \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}}$.

*Manuscript:* 17; 77–79 verso; 155–160; 31; 198–199; 391–393.

*Proof.* Use D.2 for existence. Explicitly, for torsion $h \in \widehat{E}$ choose $t \in \widehat{E}$ with $u = tht^{-1} \in E$ and set

$$\mu_{\widehat{E}}(h) = \varepsilon(t)\mu_E(u),$$

where $\varepsilon : \widehat{E} \to \{\pm 1\}$ is the orientation character, acting on $\widehat{T}$. Suppose $u, v \in E$ and $v = tut^{-1}$ in $\widehat{E}$. For a rotation apply the proof of D.2 to an order-$p$ subgroup of $\langle u\rangle$: first align it by discrete conjugation, then put the remaining conjugator in the finite discrete stabilizer. Thus the elements themselves, and not just their cyclic subgroups, are discretely conjugate, with the same orientation character of the conjugator. For reflections apply prime-power element conjugacy separation, or D.2 because an order-two subgroup has a unique nonidentity element. This establishes independence with the indicated action on $T$. A rotation of order $n$ has a value in $T/nT \subset T \otimes \mathbb{Q}/\mathbb{Z}$; the identification $T/nT = \widehat{T}/n\widehat{T}$ defines the claimed inclusion. □

**D.7. [Proved] Equivariance of the profinite rotation invariant.** Let $E$ be the lift group of a finite action on an orientable hyperbolic surface of finite type, and put $P = \widehat{\pi} \triangleleft \widehat{E}$. Every automorphism $\alpha$ of $\widehat{E}$ preserving $P$ and its indexed peripheral structure satisfies

$$\mu_{\widehat{E}}(\alpha(u)) = \alpha_T(\mu_{\widehat{E}}(u)) \qquad (u \in \widehat{E} \text{ of finite order}).$$

Here $\alpha_T$ is the induced automorphism of the orientation line $\widehat{T}$ defined in A.2, and $\mu_{\widehat{E}}$ is the invariant of D.6. The assertion holds for compact and noncompact surfaces, and for actions containing orientation-reversing elements.

*Manuscript:* 17; 77–79 verso; 155–160; 31; 198–199; 391–393.

*Proof.* Choose an orientation, so that $\alpha_T$ is multiplication by some $\lambda \in \widehat{\mathbb{Z}}^\times$. Conjugation by an element of $\widehat{E}$ acts on $\widehat{T}$ by its geometric orientation sign. Naturality of this action and commutativity of the automorphism group of a rank-one module show that $\alpha$ preserves that sign. Thus $\alpha$ restricts to an automorphism of $\widehat{B}$, where $B = E^+$ is the orientable Fuchsian subgroup. A nonidentity negative torsion element is a reflection and has invariant $1/2$; multiplication by a profinite unit fixes this value. It remains to treat torsion in $\widehat{B}$.

We first make explicit the behavior of orientation under finite covers. For a degree-$d$ connected finite cover of oriented surfaces, the map on orientation homology takes the fundamental class upstairs to $d$ times the fundamental class downstairs. For punctured surfaces this follows directly in A.2's kernel model: the peripheral factors above any one puncture map by their positive local degrees, whose sum is $d$. For closed surfaces it is the degree formula for fundamental classes. The

same formulas hold over $\widehat{\mathbb{Z}}$, either by this peripheral matrix or by the natural finite-coefficient cohomology comparison. Consequently, if $\alpha$ carries an open surface subgroup $V$ to $V'$, the multipliers on their orientation lines and on $P$ satisfy $d\lambda_V = d\lambda$. Multiplication by a positive integer is injective on $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, so $\lambda_V = \lambda$. This proves the required compatibility with further finite covers; it does not divide by $d$ in a finite coefficient ring.

*The noncompact case.* Let $L$ be a peripheral subgroup of $\pi$, and let $K$ be the primitive cusp subgroup of $B$ containing it. Write $K = \langle w \rangle$ and $L = \langle w^n \rangle$, with compatible positive generators. We have

$$N_{\widehat{B}}(\overline{L}) = \overline{K} \simeq \widehat{\mathbb{Z}}.$$

Here is the reduction from Part A, rather than an assumption that arbitrary normalizers commute with completion. For an element of the normalizer, choose a representative in $B$ of its coset modulo $P$. That coset stabilizes the profinite peripheral class. Distinct discrete peripheral classes remain distinct in $P$, so the same coset stabilizes the discrete peripheral class. Adjust its representative by an element of $\pi$ to put it in $N_B(L) = K$. The remaining factor belongs to $N_P(\overline{L}) = \overline{L}$, as proved in Part A. This proves the displayed equality.

The loop condition on $\alpha|_P$ sends a positive generator of $\overline{L}$ to a $P$-conjugate of the $\lambda$th power of the corresponding generator of another peripheral subgroup $\overline{L}'$. After making this conjugation, $\alpha$ sends $\overline{K}$ to $\overline{K}'$ and has the form $w \mapsto (w')^a$. The indices $[\overline{K} : \overline{L}] = [\overline{K}' : \overline{L}'] = n$ are equal because $\alpha$ preserves $P$. Thus $na = n\lambda$ in $\widehat{\mathbb{Z}}$, and hence $a = \lambda$. Every primitive orbifold cusp therefore has precisely the same multiplier $\lambda$.

Write the signature of $B$ as $(h; m_1, \dots, m_r; s)$ with $s \geq 1$. Choose cone generators $c_j$, of orders $m_j$, and cusp generators $w_i$ in a common orientation. The abelianized presentation is

$$\widehat{B}^{\mathrm{ab}} = \left( \widehat{\mathbb{Z}}^{2h} \oplus \bigoplus_{j=1}^{r} \mathbb{Z}/m_j \, c_j \oplus \bigoplus_{i=1}^{s} \widehat{\mathbb{Z}} \, w_i \right) / \left\langle \sum_j c_j + \sum_i w_i \right\rangle.$$

D.2 applied to $B$ makes $\alpha$ permute the maximal cone subgroups. Hence, up to conjugacy in $\widehat{B}$, $\alpha(c_j) = c_{\tau(j)}^{k_j}$, where $m_{\tau(j)} = m_j$ and $k_j$ is a unit modulo $m_j$. The preceding paragraph gives $\alpha(w_i) \sim w_{\sigma(i)}^{\lambda}$. Apply $\alpha$ to the displayed relation and subtract $\lambda$ times that relation. This yields

$$\sum_j (k_j - \lambda) c_{\tau(j)} = 0.$$

Eliminating $w_s$ exhibits the cone factors as an independent direct sum, so $k_j \equiv \lambda \pmod{m_j}$ for every $j$. The local formula of B.12 now proves the assertion on every cone subgroup and, by conjugation in $\widehat{B}$, on all its torsion.

*The compact case.* First suppose that there are at least two cone points. We use Wilkes's cohomological calculation with its exact scope: for a closed orientable hyperbolic orbifold group $B$ with cone generators $c_j$ of orders $m_j$, an automorphism of $\widehat{B}$ sending each $c_j$ to a conjugate of $c_j^{k_j}$ acts on $H^2(\widehat{B}; \mathbb{Z}/n)$, for every $n$, by one scalar $\kappa \in \widehat{\mathbb{Z}}$ satisfying $\kappa \equiv k_j \pmod{m_j}$. This is [70, Proposition 5.7], numbered Proposition 6.6 in arXiv:1512.05587v1. Its proof concerns arbitrary automorphisms of the indicated form; no lifting to a Seifert group is used in that calculation. In particular the scalar is independent of $n$.

For transparency, the part of the calculation that supplies both facts is as follows. After taking coinvariants, the relation cells $y_0, y_j$ have boundaries $\partial y_0 = \sum_j x_j$ and $\partial y_j = m_j x_j$. The automorphism sends $x_j$ to $k_j x_j$, $y_j$ to $k_j y_j$, and $y_0$ to $\kappa y_0 + \sum_j \rho_j y_j$. Commutation with the boundary gives $\kappa + m_j \rho_j = k_j$ in $\widehat{\mathbb{Z}}$. The dual coboundaries are $y_0^* + m_j y_j^*$, which shows that the induced action is multiplication by $\kappa$ for every finite coefficient group. The completed resolution and the naturality of this calculation are established in the cited proposition and its use of Proposition 2.9 there.

The cone permutation of our $\alpha$ is realized by an orientation-preserving homeomorphism of the underlying surface carrying the marked cone points to the correspondingly ordered points. It induces an orbifold automorphism $\beta$ with the same permutation and power 1 on each cone. Put $\psi = \beta^{-1}\alpha$ and apply the preceding result to $\psi$. Let

$$\eta_n \in H^2(\widehat{B}; \mathbb{Z}/n)$$

be the image of the oriented fundamental cohomology class of the underlying closed surface under the canonical map from the orbifold classifying space to its underlying space. Equivalently, in the displayed resolution it is $[y_0^*]$, up to the common orientation sign. The finite-coefficient class is transported to continuous cohomology by goodness of Fuchsian groups, as in Wilkes's calculation. Since $\beta$ preserves orientation, $\beta^*\eta_n = \eta_n$. Thus $\alpha^*\eta_n = \psi^*\beta^*\eta_n = \kappa\eta_n$. This argument does not require $\beta$ to preserve the chosen subgroup $P$.

Set $N = [B : \pi]$. Restriction to the surface subgroup gives

$$\mathrm{res}_P(\eta_n) = N\omega_n, \qquad \omega_n \in H^2(P; \mathbb{Z}/n)$$

with $\omega_n$ the positive surface class: the underlying map from the covering surface to the orbifold's underlying surface is an orientation-preserving branched map of degree $N$. Naturality of restriction and A.2's duality yield

$$N\kappa\,\omega_n = N\lambda\,\omega_n \quad \text{for every } n.$$

Indeed $\alpha_*[U] = \lambda[U]$ implies $\langle \alpha^*\omega_n, [U]\rangle = \langle \omega_n, \alpha_*[U]\rangle = \lambda$. Taking all finite coefficients gives $N(\kappa - \lambda) = 0$ in $\widehat{\mathbb{Z}}$, hence $\kappa = \lambda$. The local congruences are therefore $k_j \equiv \lambda \pmod{m_j}$, exactly as required.

We remove the single-cone case from the resolution input explicitly. When $r = 1$, the underlying genus $h$ is positive and $B^{\mathrm{ab}} = \mathbb{Z}^{2h}$, because the one cone generator is zero in abelianization. The characteristic subgroup

$$B_1 = \ker\bigl(B \longrightarrow B^{\mathrm{ab}}/2B^{\mathrm{ab}}\bigr)$$

has index $2^{2h}$ and contains every conjugate of the cone group. Its orbifold has $2^{2h}$ cone points of the same order. The subgroup $\widehat{B_1}$ and $P \cap \widehat{B_1}$ are $\alpha$-invariant. The argument for at least two cones applies there, and the finite-cover calculation at the beginning identifies its surface multiplier with $\lambda$. For $r = 0$ there is no nonidentity torsion in $\widehat{B}$, by D.2. Thus no exceptional compact signature remains.

Finally, choose positive local cone generators, for which $\mu(c_j) = 1/m_j$ in the chosen orientation line. The proved congruence says $\mu(\alpha(c_j)) = \lambda/m_j$; its powers have the same property. Every positive torsion element is conjugate in $\widehat{B}$ to one of these powers by D.2. Together with the reflection case this proves the stated equation on all torsion of $\widehat{E}$. □

*Why the earlier rank-one argument was insufficient.* Integral relative orbifold cohomology can have a torsion summand: for $C_3 * C_3$ with cusp $\langle ab\rangle$ it has the presentation $\langle u, v, w \mid 3u = w = 3v\rangle \cong \mathbb{Z} \oplus \mathbb{Z}/3$. The proof above uses the cusp relation in the noncompact case and the action on the full finite-coefficient cohomology in the compact case. It therefore supplies the missing local-power congruences without choosing an unproved canonical generator through that torsion summand.

## 4.1 Completion of geometric mapping class groups and full loop automorphisms

**D.8. [Proved] All finite subgroups in the geometric completion in the three requested types.** Let $M = \Gamma_{g,\nu}$ include puncture permutations and orientation reversal. For $(g, \nu) =$

$(0,3),(0,4),(1,1)$, completion induces a bijection on conjugacy classes of finite subgroups of $M$ and $\widehat{M}$. Every finite subgroup of $\widehat{M}$ is solvable. For every finite $K \leq M$,

$$N_{\widehat{M}}(K) = \overline{N_M(K)}, \qquad C_{\widehat{M}}(K) = \overline{C_M(K)}.$$

These assertions concern $\widehat{M}$ with its given embedding in $\mathrm{Out}_{\mathrm{lac}}(P)$.

*Manuscript:* 30; 194 verso; 384; 30; 196 verso; 388.

*Proof.* For the three-punctured sphere $M = S_3 \times C_2$ is finite. For the punctured torus $M = \mathrm{GL}_2(\mathbb{Z})$; its principal congruence subgroup of level 3 is a torsion-free free group of finite rank. For the four-punctured sphere the subgroup fixing each puncture and preserving orientation is a free group of rank two and has finite index. Thus all three $M$ are finitely generated virtually free groups.

Bridson–Piwek, Proposition 5.10 [16], states for every finitely generated virtually free group that finite subgroup conjugacy classes are preserved bijectively by profinite completion. Its proof uses the finite graph of finite groups for $M$: completion gives the profinite fundamental group of the same graph, a finite subgroup fixes a vertex of its profinite Bass–Serre tree, and subgroup conjugacy separability distinguishes the resulting discrete subgroups. This applies without an orientation restriction and proves the first assertion. The normalizer equality is Ribes–Zalesskii, Theorem 2.6 [66], for finitely generated subgroups of finitely generated virtually free groups. To deduce the centralizer equality, the homomorphism $N_{\widehat{M}}(K) \to \mathrm{Aut}(K)$ has finite discrete target. Its kernel is open; the intersection of a dense subgroup with this open kernel is dense in the kernel. This is exactly $C_M(K)$.

For solvability, a finite subgroup of $M_{0,4}$ embeds in $S_4 \times C_2$ by the torsion-free pure oriented kernel; one in $\mathrm{GL}_2(\mathbb{Z})$ embeds in $\mathrm{GL}_2(\mathbb{F}_3)$ by the level-3 kernel. The former group is solvable. The latter has normal series with factors inside $C_2$, $A_4$, and the central $C_2$, since $\mathrm{PGL}_2(\mathbb{F}_3) = S_4$; it too is solvable. The finite case $S_3 \times C_2$ is immediate. Transfer by the proved conjugacy bijection gives solvability in $\widehat{M}$. □

**D.9. [Proved] The nonsolvable discrete possibilities in genus at most two.** A nonsolvable finite subgroup of the extended mapping class group of a punctured orientable surface of genus at most two is realized on a sphere and has group $A_5$ or $A_5 \times C_2$. Its puncture set is invariant under the corresponding icosahedral action. In particular it has at least twelve punctures. This classification does not assert the absence of additional nonsolvable finite subgroups in completions for all larger puncture numbers.

*Manuscript:* 30; 194 verso; 384.

*Proof.* Realize the finite subgroup by Part C. On the sphere its orientation-preserving subgroup is cyclic, dihedral, $A_4$, $S_4$, or $A_5$, by the finite rotation-group classification. The first four are solvable and an extension of one of them by an orientation group of order at most two is solvable. The orthogonal normalizer of the icosahedral rotation group is $A_5 \times \{\pm I\}$: a normalizing orthogonal transformation preserves its rotation axes and its icosahedral cellulation, and composition with an icosahedral rotation leaves either $I$ or $-I$. A nonsolvable full group is consequently $A_5$ or $A_5 \times C_2$. The smallest nontrivial orbit has twelve vertices; an invariant puncture set has at least twelve points.

On a torus the kernel of the action on $H_1$ consists of a finite translation group, which is abelian. The orientation-preserving linear quotient is cyclic of order $1, 2, 3, 4$, or $6$, and the extended linear quotient is cyclic or dihedral. These groups and their extensions by the translation group are solvable.

In genus two the hyperelliptic involution is central in the full automorphism group of the invariant Riemann surface. Adjoin it when it is absent. The quotient is a sphere with six branch points. A nonsolvable orientation-preserving quotient would contain $A_5$, whose smallest orbit has twelve points, contradicting preservation of the six branch points. The quotient is therefore solvable; so is its central extension by $C_2$ and its extension by orientation reversal. Punctures can only reduce the finite group. This proves the asserted list and its restriction to the discrete setting. □

**D.10. [Proved] Finite torsion in the odd-prime normalized pro-$p$ centralizer.** Let $p$ be odd, let $F^{(p)}$ be the free pro-$p$ group on $x, y$, and define $GT_{0,p}$ by the normalized automorphisms

$$F_{\lambda,f}(x) = x^{\lambda}, \qquad F_{\lambda,f}(y) = f^{-1}y^{\lambda}f, \quad \lambda \in \mathbb{Z}_p^{\times}, \quad f \in (F^{(p)})',$$

satisfying the two-cycle and three-cycle equations. Every finite subgroup of $GT_{0,p}$ is cyclic of order dividing $p-1$. There is a subgroup of order $p-1$, and every finite subgroup is conjugate in $GT_{0,p}$ to a subgroup of it. For $p \geq 5$ this supplies finite loop-preserving outer automorphism groups which are not conjugate to geometric groups of the three-punctured sphere.

*Manuscript:* 25 bis; 157 verso–160; 312–317; 30; 194 verso; 384.

*Proof.* The normalized description is faithful: a conjugator between two normalized automorphisms first belongs to $C(x) = \langle x \rangle$; the normalization $f \in F'$ and the $y$ equation remove its exponent. The same observation identifies these automorphisms with the pure outer centralizer of the three puncture permutations. The equations are closed conditions, composition is the usual substitution law, and their inverses satisfy the same equations. Hence this is a profinite group. The Galois action on $\mathbf{P}^1_{\overline{\mathbb{Q}}} - \{0, 1, \infty\}$ gives elements of this group with multiplier the $p$-adic cyclotomic character. This character surjects onto $\mathbb{Z}_p^{\times}$: the finite cyclotomic extensions have Galois groups $(\mathbb{Z}/p^n)^{\times}$, and their inverse limit is $\mathbb{Z}_p^{\times}$.

The kernel of $\mathrm{Aut}(F^{(p)}) \to \mathrm{Aut}(F^{(p)}/\Phi(F^{(p)}))$ is pro-$p$; the exact statement and its application are also given in Herfort–Zalesskii, proof of Theorem 6.3 [35]. For the normalized group the action on this Frattini quotient is the scalar $\lambda \bmod p$. Thus the kernel of

$$GT_{0,p} \longrightarrow \mathbb{F}_p^{\times}$$

is pro-$p$, and the map is surjective. The profinite Schur–Zassenhaus theorem supplies a complement $C_{p-1}$ and conjugacy of complements in the inverse image of any subgroup of $\mathbb{F}_p^{\times}$. One can see the passage from the finite theorem directly: apply the finite theorem to every quotient by an open normal subgroup of the pro-$p$ kernel, then take the inverse limit of the nonempty finite sets of compatible complements and conjugators. The finite intersection property follows by passing to a common finite quotient.

We check the necessary absence of torsion in the kernel of the multiplier. A free pro-$p$ group has its injective Magnus map into $\mathbb{Z}_p\langle\langle X, Y \rangle\rangle^{\times}$, with $x \mapsto 1 + X$, $y \mapsto 1 + Y$. The map is injective because reduction modulo $p$ is the usual faithful free pro-$p$ Magnus map; it extends continuously from the free group for the $(p, X, Y)$-adic topology on the power-series algebra. The ordinary homogeneous degree is used only in the following leading-term calculation, not as the topology for that extension. An automorphism acting identically on $F^{(p)}_{\mathrm{ab}} = \mathbb{Z}_p^2$ acts identically on the degree-one quotient of this power-series algebra. For a nonidentity such automorphism choose the least degree $d \geq 2$ of a nonzero correction to $X$ or $Y$. The degree-$d$ correction to its $n$th iterate is $n$ times that correction: substitution of the higher terms can change only degrees above $d$. Each homogeneous coefficient module is a torsion-free $\mathbb{Z}_p$-module, so no positive iterate is the identity. Consequently the multiplier is injective on a finite subgroup.

The torsion subgroup of $\mathbb{Z}_p^\times$ is cyclic of order $p-1$. Therefore every finite subgroup maps injectively to $\mathbb{F}_p^\times$. It is a complement in the inverse image of its image, so the preceding conjugacy theorem proves the full assertion. Finally the geometric pure group of type $(0,3)$ consists of 1 and complex conjugation, with multipliers 1 and $-1$. A generator with multiplier a primitive $(p-1)$st root cannot be conjugated into it; the multiplier is unchanged by conjugation even in the full loop-preserving outer group. The first example has order four at $p=5$. □

**D.11. [Proved] Executed finite checks and their exact scope.** The program `code/root/finite_checks.py` verifies the two-cycle and three-cycle equations in every Heisenberg group $H(\mathbb{F}_p)$ for $p=5,7,11$ and every $\lambda\in\mathbb{F}_p^\times$, using

$$f(a,b)=[a,b]^{(\lambda^2-1)/24},\qquad m=(\lambda-1)/2.$$

In particular it verifies an order-four normalized automorphism at $(p,\lambda \bmod p)=(5,2)$ and an order-three one at $(7,2)$. The multipliers of the corresponding infinite finite-order subgroups are Teichmüller lifts of these residues; the integer $2\in\mathbb{Z}_5^\times$ itself has infinite order. D.10, not the finite checks alone, proves the existence of compatible infinite pro-$p$ lifts.

*Manuscript:* 25 bis; 159 verso–160; 316–317; 30; 194 verso; 384.

*Proof.* The program defines all $p^3$ elements as triples and uses $(a,b,c)(a',b',c')=(a+a',b+b',c+c'+ab')$. It enumerates inverses and powers and evaluates each side of both equations in that group. Its output records every multiplier and both truth values. Since all commutators are central, the associated automorphism is $(a,b,c)\mapsto(\lambda a,\lambda b,\lambda^2 c)$; the program checks multiplication preservation and computes its order. Division by 24 is only used at the three displayed primes. No assertion of this formula at 2 or 3 is made. At $p=3$, D.10 gives only order-two prime-to-$p$ torsion. At $p=2$, the Magnus argument still proves that a finite subgroup injects into $\{\pm1\}$, but the complement argument does not settle conjugacy of involutions in $GT_{0,2}$. □

**D.12. [Open] Torsion in the full profinite $GT_0$ and the full low-type groups.** D.29 refutes the finite geometric-recognition assertion for $(g,\nu)=(0,4)$ by an actual involution with a complete preserved peripheral system. We do not settle whether every nonidentity torsion element of the full profinite $GT_0$ is conjugate to $(-1,1)$, or the full finite-subgroup recognition questions for $(g,\nu)=(0,3),(1,1)$. The geometric-completion assertion in all three low types is proved in D.8, and the odd-prime analogue is decided in D.10. D.32 proves that every finite subgroup of the full pure three-cusp group, including $GT_0$, has order at most two; it gives the faithful map $G\to S_3\times C_2$ for the full three-cusp group. D.34, D.36, D.41 and D.42 identify the global orientation character of every such finite group: every involution has its global 2-adic sign, and every order-three element has multiplier 1. D.35 identifies the exact remaining third-cusp conjugator equation, proves full generation from it, and identifies its conjugator sets in the pure group and in $GT_0$. D.38 gives a complete characteristic $A_5^{19}$ quotient test, including proved exclusions of finite candidates from full lifts. D.44 solves the coboundary and the first symmetry equation simultaneously in the full group; the third-cusp equation and hence existence of the conjugator for each actual full action remain unresolved. D.33 proves that full homology detects every finite subgroup in type $(1,1)$ and that its order divides 24. D.43 proves that every involution in this full punctured-torus group has its global orientation sign. D.45 identifies the exact characteristic binary-icosahedral refinement and its full finite solution counts; the same cusp-triangle survivors pass. D.46 proves that every cyclic order-four outer action on the punctured-torus group has an actual order-four lift, using the three full section classes of its positive square. D.47 checks the exact conjugator coordinates in the binary refinement: precisely 10,125,000 of the 324,000,000 coordinate pairs pass, with the count unchanged by finite invertibility and both symmetry relations. D.48 proves that a full punctured-torus order-three action has exactly two section classes and a torsion-free cusp

normalizer whose cube exponent is a 3-adic unit. D.49 computes the standard pure involution centralizer and the exact two-element fibres of its conjugacy orbit map. D.50 classifies the finite two-subgroups of the full punctured-torus group as exactly $1, C_2, C_4, C_2 \times C_2, D_8$, exhibits all five, and proves the global multiplier character on every four-group or dihedral group of order eight. D.51 identifies the second torsion-free normal index-three kernel of every order-three torus extension, with its exact type-$(1,3)$ intersection cover and three labelled peripheral normalizers. D.52 gives the twelve possible abstract finite-subgroup types, realizes nine of them, and proves the reflection multiplier for every subgroup of type $S_3, D_{12}, D_{24}$. D.53 proves the unique full labelled order-four section class and its exact normalizer. D.54 excludes every nonzero prime-to-three invariant character for a cubic torus action by an explicit peripheral norm comparison in a cyclic cover. D.55 supplies the corresponding square comparison, proves that every positive involution acts as $-I$ at every prime, and proves faithfulness of every primary homology map on every finite subgroup. The finite torus abstract types are consequently exactly the nine realized types $1, C_2, C_3, C_4, C_2 \times C_2, C_6, S_3, D_8, D_{12}$; the three extra types in the earlier allowed list are excluded. D.56 computes the prime-section permutations of every cyclic-six action and proves that its full extension has no order-six subgroup or finite-order representative of the outer generator. D.57 proves actual labelled lifts for every four-group and order-eight dihedral action, with the exact three-or-one and one full section counts. D.58 determines the exact negative-involution section counts and their full fixed-subgroup mod-two cohomology, and proves that the two internal cusp-reflection classes fuse under full conjugacy. D.59 identifies the negative cusp section as the class with zero full fixed-subgroup mod-two first cohomology, proves that its entire fixed group has zero ambient abelian image, and proves that every actual cusp-reflection transporter generates the full fixed homology line. Its odd cyclic covers have exactly the displayed cusp cycles and full section counts, and the other fixed group, when present, surjects onto that homology line. D.60 identifies the quotient by the closed normal closure of the cusp reflection with the exact full homology coinvariants, and the quotient by all torsion with $\widehat{\mathbb{Z}}$. It constructs a section in $F$, and in the two-class case a section inside the other fixed group, giving an actual embedded $\widehat{\mathbb{Z}} \times C_2$. D.61 proves that the peripheral commutator belongs to the closed derived subgroup of no proper closed subgroup of the full rank-two free group. It consequently gives exact negative-conjugator parameter spaces with no additional generation condition, retaining both cusp-reflection indices and all full unit multipliers. Existence of these parameters, pure-involution orbit exhaustion, full fixed-group recognition, the remaining scalar cubic torus multipliers, and geometric conjugacy remain unresolved; D.70 below settles the quartic multiplier.

D.62 gives the exact full negative-conjugator transporter cosets. D.63–D.65 compute the full standard negative centralizers and their homological characters. D.66 computes the exact scalar-cover norm preimage and primary abelianized-extension splittings. D.67 gives the standard Klein-group and dihedral normalizers and all finite overgroups of the specified two-section negative involution. D.68 identifies the unique invariant double cover of every actual quartic action, its integral homology maps, and the full invariant dihedral quotient. D.69 determines the exact rotation image of the positive-square fixed subgroup in that finite quotient, its two labelled section classes, its fixed-kernel first cohomology and the full integral positive-square action on the nontrivial deck summand. D.70 uses the actual closed filling and its pro-two quotient to exclude every negative odd quartic coordinate by a degree-three restriction contradiction. Hence every order-four full torus element has global multiplier one, and every finite two-subgroup has its multiplier in the two global signs. D.71 constructs the actual cubic Frobenius cover, proves the full integral regular module on its nontrivial deck summand, and determines the image and first mod-three cohomology of the relevant section fixed subgroup. D.72 gives an explicit surjective quasi-isomorphism from the actual normalized pants cochain algebra to the three-section Boolean graded algebra; its acyclic kernel proves the stated strong Massey vanishing and exact finite unitriangular representation image. The arbitrary negative conjugator equations, scalar cubic multiplier and full low-type recognition remain unresolved. D.73 proves the exact

compatible $q$-adic cubic Frobenius quotient and its full fixed-group splitting, the integral module and section constraints at every $q$-power level, the limiting fixed-kernel character vanishing, and an actual two-torsion-generator obstruction. D.74 proves that all three full pants section fixed subgroups are trivial. Its actual finite-coefficient relative duality, full Brown maps and coset-module witness give self-centralizing sections and real projectivity. The full coboundary $h$ in D.35 is consequently unique, including D.44's first symmetry. The exact third-cusp equation with this $h$, arbitrary negative torus conjugacy, and the scalar cubic realization or exclusion remain unresolved. D.75 proves full triviality of every negative torus cusp-reflection fixed subgroup, uniqueness of the actual cusp-fusion transporter, and both singleton transporter formulas for every full unit. In the one-section-class case its extension is real projective. Its exact remaining equation is the conjugacy (D.75.5); the other fixed group in the two-section-class case retains its full homology surjection without a procyclic recognition conclusion. D.76 identifies the actual pants section augmentation module with $\mathbb{F}_2[[E]]^2$, using all finite coefficient modules and compatible finite-quotient isomorphisms. Its specified incidence map has cokernel equal to the augmentation kernel of $\mathbb{F}_2[[E/H]]$, where $H$ is the closed subgroup generated by the three specified section representatives. Thus its invertibility records exactly $H = E$. Four explicit involutions generate $E$, with the duplicated class and the resulting three-involution subgroup both written out. Generation by the specified triple and the original third-cusp equation remain unresolved. D.77 constructs a complete three-class involution triple in the standard full pants extension whose two kernel generators give automorphisms of every maximal pro-$p$ quotient, while an actual $A_5$ quotient sends the triple to $D_{10}$ and its kernel to $C_5$. Its specified incidence cokernel maps onto a five-dimensional augmentation module. The same quotient proves failure of conjugacy to every unit power of the original second cusp. That retained cusp condition prevents this example from deciding the distinguished triple of D.76 or the equation (D.74.40). D.78 proves that every open subgroup of the other negative torus fixed group has mod-two first cohomology of dimension one. Its specified dyadic procyclic subgroup is a Sylow subgroup, and the full fixed group has the actual split coordinates $B = K \rtimes \widehat{\mathbb{Z}}$ with a pro-odd kernel $K$. The finite-coefficient inflation and section-restriction maps retain $Q^K \hookrightarrow Q$. The original finite-coset invariant map detects every nonidentity element of $K$; its surjectivity and full procyclic recognition remain unresolved.

*Manuscript:* 25 bis; 157 verso–160; 312–317; 30; 194 verso; 384.

Here is the exact obstruction reached, rather than an extrapolation from computation. The homomorphism from a free profinite group to the product of its maximal pro-$p$ quotients does not detect all its finite quotients. Our program constructs $A_5$ from a 5-cycle and a 3-cycle, enumerates its sixty elements, and verifies that its commutator subgroup has sixty elements. Thus $A_5$ is a nontrivial two-generated finite quotient invisible to every nilpotent quotient. The pro-$p$ complements constructed in D.10 do not give a compatible solution of the defining equations in such quotients.

We also checked the $f = 1$ attack: the three-cycle relation becomes the power-product relation of Lochak–Schneps, Lemma 9 [47]; their finite modular quotients force the global alternatives $\lambda = 1$ or $-1$. A choice of independent signs in the factors of $\widehat{\mathbb{Z}}$ is not a solution of this relation. The finite nilpotent search therefore supplies no full-profinite counterexample. D.32 excludes every element of odd prime order in the full pure three-cusp group and every element of finite order greater than two. The smallest unresolved full-$GT_0$ case is therefore an additional involution up to conjugacy; D.34 proves that its multiplier is globally $-1$, supplies its three section classes and a commutator coboundary, and leaves the conjugator inside the specified loop group to be determined. D.23 supplies the exact order-bounded compatibility criterion over all isolated finite shadows. D.24 executes nonnilpotent refinement tests and proves that specific order-three shadows cannot lift with order three; D.25 computes the entire dihedral inverse limit and the exact residual multiplier-one norm equation. D.28 proves the additional prime-order section

and fixed-subgroup constraint, including its exact maximal pro-$p$ comparison. The finite-cover argument of D.32 now excludes all odd-order full lifts, beyond those bounded computations; the involution conjugacy problem retains its exact isolated-refinement formulation. D.29 uses an arbitrary Galois pants isomorphism and its inverse on two different pieces; this produces a finite involution in type $(0,4)$ without requiring that pants isomorphism itself have finite order. For genus at most two and arbitrary puncture number, the solvable comparison is supplied by the checked argument in F.7. D.31 now proves that every nonsolvable finite subgroup in these completions is abstractly $A_5$ or $A_5 \times C_2$ on a punctured sphere, and gives exactly the geometric positive puncture-orbit sizes. Its proof also excludes every nonsolvable finite subgroup in the genus-one and genus-two completions. What remains is geometric conjugacy of the actual $A_5$ embeddings and the corresponding centralizer comparison of F.7.

The later tests also close two specific attempted constructions of a lower-type counterexample. D.39 constructs an actual rank-two Haran involution but proves, using its unique section class and the trivial centralizer of that section, that it preserves neither hyperbolic loop marking under any full profinite change of basis. D.37 gives the exact four-cusp descent condition for a negative transposition. D.40 excludes the required additional deck symmetry for the actual D.29 example by the unequal homomorphism counts 10200 and 4920 in $\mathrm{PSL}_2(\mathbb{F}_{11})$; the argument also excludes deck maps moving the original separator. These are proved obstructions for the specified constructions. They do not provide the conjugator still required in D.35.

## 4.2 Descent along a finite covering

Put $A(P) = \mathrm{Aut}_{\mathrm{lac}}(P)$ and let a discretization mean the image of a loop-preserving isomorphism $\widehat{\pi_0} \to P$ restricted to $\pi_0$. Thus it is an actual dense subgroup with its peripheral classes. Define $\mathrm{Discr}(P)$ using actual subgroups, and define $\mathrm{Discr}^{\mathrm{out}}(P)$ by quotienting this set by conjugation by $P$. This convention is necessary for an outer action: an outer automorphism acts canonically on the second set. Write $J = \mathrm{Aut}_{\mathrm{lac}}(\pi_0)$, $B = \mathrm{Out}_{\mathrm{lac}}(P)$ and $M = \mathrm{Out}_{\mathrm{lac}}(\pi_0)$ for their embedded images.

**D.13. [Proved] Restriction and normal descent with the peripheral step included.** For an open subgroup $V \le P$, restriction of the corresponding finite covering defines an injective map

$$\mathrm{Discr}(P) \longrightarrow \mathrm{Discr}(V), \qquad \Delta \longmapsto \Delta \cap V.$$

For normal $V$, its image consists exactly of the lattices $\Lambda \subset V$ whose outer deck action $P/V \to \mathrm{Out}_{\mathrm{lac}}(V)$ is induced by orientation-preserving loop automorphisms of $\Lambda$. The cyclic peripheral subgroups of the descended lattice have closures equal to the specified peripheral subgroups of $P$.

*Manuscript:* 28; 178–179; 352–354.

*Proof.* We prove the self-normalizer input, including the class-preserving step. Paris [62, Theorems 1.7 and 3.2] proves conjugacy $p$-separability for closed orientable surface groups and free groups. Thus two elements of $\Lambda$ conjugate in $\widehat{\Lambda}$ are already conjugate in $\Lambda$.

For a free group $F_r = \langle x_1, \dots, x_r \rangle$, let $\varphi$ preserve every conjugacy class. After an inner adjustment we have $\varphi(x_1) = x_1$. Write $\varphi(x_i) = w_i x_i w_i^{-1}$. Choose a reduced representative $w$ for the double coset $\langle x_1 \rangle w_i \langle x_i \rangle$, so that $w_i = x_1^a w x_i^b$ and $w$ begins with neither $x_1$ nor $x_1^{-1}$ and ends with neither $x_i$ nor $x_i^{-1}$. The word $x_1 w x_i w^{-1}$ is cyclically reduced with length $2 + 2|w|$. Since $\varphi(x_1 x_i)$ is conjugate to $x_1 x_i$, we obtain $w = 1$. Hence $\varphi(x_i) = x_1^{n_i} x_i x_1^{-n_i}$. For distinct $i, j \ge 2$, conjugating $\varphi(x_i x_j)$ by $x_1^{-n_i}$ gives $x_i x_1^{n_j - n_i} x_j x_1^{n_i - n_j}$; cyclic length is two exactly when $n_i = n_j$. Thus all $n_i$ agree, and $\varphi$ is inner. The rank-two case has only one exponent and gives the same conclusion.

For a closed orientable surface group of genus $g \geq 2$, Dehn–Nielsen–Baer realizes the outer class of a class-preserving automorphism by a mapping class $f$. It fixes every oriented closed-curve class and acts identically on $H_1$, so preserves orientation. Choose a finite filling Alexander system of simple curves. The Alexander method ([22, §2.3]) makes the subgroup fixing all those curve classes finite: its possible permutations of intersection points and edges form a finite group; the kernel fixes the resulting graph and is isotopic to the identity on its complementary discs. Therefore $f$ has finite order. Realize the cyclic group generated by $f$ using Part C. For an action of order $d$, rational cochains of the quotient are the invariant cochains upstairs, by an equivariant triangulation and averaging over the finite stabilizers. The quotient's first Betti number is consequently $2g$, because the action on $H_1$ is trivial. Its underlying genus is $g$. Riemann–Hurwitz now gives

$$2 - 2g = d\left(2 - 2g - \sum_j (1 - 1/m_j)\right).$$

For $d > 1$ the right side is strictly smaller than $2 - 2g < 0$, a contradiction. Thus $f = 1$ and the automorphism is inner.

Now let $h \in N_{\widehat{\Lambda}}(\Lambda)$. Its conjugation automorphism is class-preserving by the first paragraph, hence is conjugation by an element of $\Lambda$ by the two arguments just given. Dividing $h$ by that element centralizes the dense lattice and therefore all of $\widehat{\Lambda}$. Part A gives trivial centre, so $h \in \Lambda$. This proves $N_{\widehat{\Lambda}}(\Lambda) = \Lambda$ with the exact comparisons needed in LM3 B.8.

For normal $V$, a descended lattice $\Delta$ normalizes $\Lambda = \Delta \cap V$. Conversely, an element of $N_P(\Lambda)$ has a coset in the finite group $P/V$. The intersection $N_P(\Lambda) \cap V$ is $\Lambda$ by the just-proved input. Given that the deck action is induced on $\Lambda$, form the pullback discrete extension $E_\Lambda$ of $P/V$ by $\Lambda$ inside $\mathrm{Aut}_{\mathrm{lac}}(\Lambda)$. We construct its map into $P$ explicitly. An element of the pullback is a pair $(g, \varphi)$, with $g \in P/V$, $\varphi \in \mathrm{Aut}_{\mathrm{lac}}(\Lambda)$, and prescribed outer class. Extend $\varphi$ continuously to $\widehat{\varphi} \in \mathrm{Aut}(V)$ and choose a lift $p_g \in P$ of $g$. Equality of outer classes gives a $v \in V$ with $\widehat{\varphi} = \mathrm{Inn}(vp_g)|_V$. This $v$ is unique, since $C_P(V) = 1$ by Part A. Send $(g, \varphi)$ to $vp_g$. Changing $p_g$ changes $v$ inversely and leaves the product unchanged; multiplication is respected because both products induce the same conjugation on $V$ and have the same quotient label. This defines an injective homomorphism: an element in its kernel has $g = 1$ and induces the identity on the dense subgroup $\Lambda$. Its image is exactly $N_P(\Lambda)$, by restricting a normalizing conjugation to $\Lambda$ and applying the same uniqueness. The discrete group is torsion-free, as a subgroup of $P$. The finite-extension realization theorem of Part C gives a surface lattice $\Delta = E_\Lambda$; its orientation is positive because the deck action on $V$ has multiplier $+1$. Its completion is $P$ by D.1.

We supply the peripheral argument left incomplete in LM3 B.10. Let $C \leq \Lambda$ be a primitive peripheral subgroup, and let $Q = C_P(\overline{C})$ be the maximal procyclic peripheral subgroup in $P$ containing $\overline{C}$. Put $D = C_\Delta(C)$. The quotient surface supplied by Part C makes $D$ its primitive cusp subgroup. We have $\overline{D} \leq Q$ and

$$\overline{D} \cap V = \overline{C} = Q \cap V.$$

The image of $D$ in $P/V$ is the stabilizer of the $\Lambda$-conjugacy class of $C$. To see this, adjust a representative of a stabilizing coset by an element of $\Lambda$ to normalize $C$; orientation preservation changes normalization into centralization. Conversely a centralizing element fixes the class. The identical argument in the profinite group identifies the image of $Q$ with the stabilizer of the $V$-conjugacy class of $\overline{C}$. The map from discrete cusp classes to their profinite classes is bijective and equivariant by peripheral noncommensurability and the finite covering construction of Part A. These two stabilizers are therefore equal. The two closed subgroups $\overline{D} \leq Q$ have the same intersection with the open subgroup $V$ and the same image in $P/V$, so they are equal. This proves the promised loop identification.

In particular a descended lattice is forced to equal $N_P(\Lambda)$, proving injectivity for normal $V$. For arbitrary $V$, take $W = \bigcap_{p \in P} pVp^{-1}$, an open normal subgroup, and use the actual

intersection map $\Lambda \mapsto \Lambda \cap W$. Equality after restriction to $V$ implies equality after restriction to $W$, hence equality of the original lattices by the normal case. This proves injectivity without assuming $V$ normal. □

**D.14. [Proved] The image for a nonnormal finite covering.** Let $V \leq P$ be open, $W = \mathrm{core}_P(V)$ and $\Lambda \in \mathrm{Discr}(V)$. Set $\Lambda_W = \Lambda \cap W$. The image condition for $\Lambda$ is that the full deck action of $P/W$ on $W$ is induced by orientation-preserving loop automorphisms of $\Lambda_W$. This condition is independent of a choice of coset representatives. The unique extension is $N_P(\Lambda_W)$, and its intersection with $V$ is $\Lambda$.

*Manuscript:* 28; 178–179; 352–354.

*Proof.* Necessity follows by restriction of an actual lattice in $P$. For sufficiency apply the normal descent statement D.13 to $W \triangleleft P$, obtaining $\Delta = N_P(\Lambda_W)$. Both $\Delta \cap V$ and $\Lambda$ are discretizations of $V$ with intersection $\Lambda_W$ with the normal open subgroup $W$ of $V$. The injectivity already proved inside $V$ gives $\Delta \cap V = \Lambda$. The deck action is an outer action, so changing a representative multiplies its automorphism by an inner automorphism of $W$; its membership in the embedded geometric outer subgroup is unaffected. This proves the asserted independence and the exact nonnormal image description. □

**D.15. [Proved] The explicit nonextending discretization in scan 354.** Let $U = \mathbf{P}^1 - \{0, 1, \infty\}$ and $U' = \mathbf{P}^1 - \{0, \infty, 1, -1\}$, with the double cover $z \mapsto z^2$. Put $P = \widehat{\pi_1(U)}$, let $V \triangleleft P$ be its index-two subgroup, and let $\Lambda = \pi_1(U') \subset V$ be the geometric discretization. There is an explicit continuous loop automorphism $\gamma$ of $V$ for which $\gamma(\Lambda)$ is a discretization of $V$ that does not extend to $P$, although it has the same prediscretization as $\Lambda$.

*Manuscript:* 28; 178–179; 352–354.

*Proof.* The deck involution $s : z \mapsto -z$ fixes $0, \infty$ and exchanges $1, -1$. Let $a$ be a simple curve separating $\{0, 1\}$ from $\{\infty, -1\}$ and let $b = s(a)$; write $T_a, T_b$ for positive Dehn twists. They give a free basis of the pure oriented mapping class group $M_{0,4}^{!+} \simeq F_2$: in the ordered moduli space $\mathcal{M}_{0,4} \simeq \mathbf{P}^1 - \{0, 1, \infty\}$ they are monodromies around two distinct boundary points. One can verify the completed embedding directly from A.12: filling the fourth puncture identifies this pure group with the point-pushing group $\pi_{0,3}$, and the peripheral-normalization map sends its completion into $\mathrm{Aut}(\widehat{\pi}_{0,3})$ as $\mathrm{Inn}(\widehat{\pi}_{0,3})$. The latter is isomorphic to $\widehat{\pi}_{0,3}$ by triviality of its centre. The composite is injective, so the completion embeds already in $\mathrm{Out}_{\mathrm{lac}}(V)$. Thus the pure group's closure is its free profinite completion, with abelianization $\widehat{\mathbb{Z}}^2$ on the indicated twists.

Choose the fully specified profinite integer

$$\alpha_2 = 0 \in \mathbb{Z}_2, \qquad \alpha_\ell = 1 \in \mathbb{Z}_\ell \quad (\ell \text{ odd}), \qquad \alpha \in \widehat{\mathbb{Z}}.$$

It is not an integer: its 2-adic coordinate would force that integer to be zero, contrary to its 3-adic coordinate. Let $u = T_a^\alpha$ in the closure of $M_{0,4}^{!+}$. Since $s$ preserves orientation, $sT_as^{-1} = T_b$, and the continuous power maps give

$$u^{-1}su = s\,T_b^{-\alpha}T_a^\alpha.$$

The pure factor on the right has abelianization $(\alpha, -\alpha)$. Every element of the discrete pure group has integral abelianization. Moreover $M_{0,4} \cap \overline{M_{0,4}^{!+}} = M_{0,4}^{!+}$: the puncture permutation and orientation sign are detected by finite quotients. Consequently $u^{-1}su$ does not belong to the discrete extended mapping class group.

Choose any continuous automorphism $\gamma \in \mathrm{Aut}_{\mathrm{lac}}(V)$ representing $u$. Its ambiguity is conjugation by an element of $V$, which does not change membership in the outer geometric subgroup. The geometric outer subgroup associated to $\gamma(\Lambda)$ is $uM_{0,4}u^{-1}$. The calculation says that the deck involution is absent from this subgroup. The necessary direction of D.13 therefore proves that $\gamma(\Lambda)$ does not extend to $P$. Since $u$ lies in the closure of the pure geometric group, its coset modulo the closed geometric stabilizer is the identity, proving equality of the prediscretizations.

The code in `code/root/finite_checks.py` verifies compatible Chinese-remainder residues of $\alpha$ and the puncture partitions interchanged by $s$. Nonintegrality and the infinite completed-group conclusion follow from the coordinate argument above. This spells out and verifies the proposed element in LM3's review, §4.2. □

*Manuscript:* §28, folios 178–179, scans 352–354, especially *Un tel $\gamma$ existe toujours.* We have proved the indicated index-two example; the word *toujours* is not used to assert the construction for every finite covering.

**D.16. [Proved] A precise fixed-set formula for every actual action.** Let $A$ be a group, $J \leq A$, and $G \leq A$ a finite subgroup. Under the left action of $G$ on $A/J$,

$$(A/J)^G = \{aJ : a^{-1}Ga \leq J\}.$$

It is the disjoint union of the $C_A(G)$-orbits

$$\coprod_{[a]} C_A(G)/(aC_J(a^{-1}Ga)a^{-1}),$$

where $[a]$ ranges over $J$-conjugacy classes of the labelled embeddings $g \mapsto a^{-1}ga$ occurring in $J$. This formula applies to actual discretizations with $(A, J) = (A(P), \mathrm{Aut}_{\mathrm{lac}}(\pi_0))$, to outer discretizations with $(A, J) = (B, M)$, and to prediscretizations with $(A, J) = (B, \overline{M})$.

*Manuscript:* 28; 178–181; 352–358.

*Proof.* The fixed-coset condition is $gaJ = aJ$ for every $g$, exactly the displayed inclusion. Two fixed cosets $aJ, bJ$ are in the same centralizer orbit exactly when there are $c \in C_A(G)$ and $j \in J$ with $b = caj$. Substitution gives $b^{-1}gb = j^{-1}a^{-1}gaj$ for every labelled $g$. Conversely this equality makes $bj^{-1}a^{-1}$ centralize $G$. The stabilizer of $aJ$ is $C_A(G) \cap aJa^{-1} = aC_J(a^{-1}Ga)a^{-1}$, proving the orbit formula.

The homogeneous-space identifications use the extension of each discrete loop automorphism to completion and its injectivity. The stabilizer of an actual lattice in $A(P)$ is exactly $J$. Its inner-conjugacy quotient is $B/M$. Since the closure of $\mathrm{Inn}(\pi_0)$ is $\mathrm{Inn}(P)$, the closed based stabilizer contains all inner automorphisms and its quotient is $\overline{M}$; hence the prediscretization set is $B/\overline{M}$. These identifications also show explicitly why a bare outer element is not assigned an action on the set of actual subgroups without choosing a lift. □

## 4.3 Torsion-free extensions and recognition

**D.17. [Proved] Torsion-free extensions with an invariant discretization.** Let $1 \to P \to \mathcal{E} \to G \to 1$ be a profinite extension with $G$ finite, $P = \widehat{\pi}_0$ hyperbolic, and the outer action preserving the loop structure and a discretization. A torsion-free $\mathcal{E}$ is the profinite completion of a torsion-free discrete surface group, with the descended peripheral system. The quotient surface may be nonorientable.

*Manuscript:* 28; 178 verso; 353; 24; 147 verso; 293.

*Proof.* First the outer action is faithful. Its kernel is the image of $C_{\mathcal{E}}(P)$: an element acting inner can be divided by its unique inner element of $P$. The centralizer meets $P$ trivially, so embeds in the finite quotient $G$ and is finite. Torsion-freeness makes it trivial. Construct the discrete extension $E$ of the stabilized lattice with this faithful outer action, using centre-freeness. By D.1, its completion is the given $\mathcal{E}$ with the identity on kernel and quotient. The discrete extension embeds in its completion because it is the finite extension realized by Part C and hence residually finite. Torsion-freeness of $\mathcal{E}$ therefore implies torsion-freeness of $E$. In a realization, a nontrivial point stabilizer gives a nontrivial finite-order element of the lift group. Thus $G$ acts freely on the surface associated to $\pi_0$, and its quotient is a surface. Its fundamental group is $E$ by lifting paths. The primitive cusp group in $E$ is the centralizer of the covering cusp group; D.13 proves equality of their closures with the specified procyclic peripherals. Orientation-reversing free deck transformations yield the nonorientable quotient and its orientation character. This is the exact scope of the descent argument. □

**D.18. [Proved] The unmarked pro-$p$ extension statement for punctured surfaces.** A finitely generated torsion-free pro-$p$ group $\mathcal{E}$ containing an open free pro-$p$ group $P$ of rank $r$ is free pro-$p$. Its rank is

$$1 + \frac{r-1}{[\mathcal{E} : P]}.$$

In particular the displayed quantity is an integer. This proves the underlying-group assertion for punctured pro-$p$ surfaces, without identifying the peripheral system. More generally, every torsion-free profinite group with an open normal pro-$p$ subgroup is itself pro-$p$. Thus D.19–D.20 apply to arbitrary finite-quotient extensions of their pro-$p$ kernels; the quotient is forced to be a $p$-group.

*Manuscript:* 28; 178 verso; 353; 24; 147 verso; 293.

*Proof.* Herfort–Zalesskii, Theorem 1.1 [35], writes such a finitely generated virtually free pro-$p$ group as the fundamental pro-$p$ group of a finite graph of finite $p$-groups, with vertex groups embedded in $\mathcal{E}$. Torsion-freeness makes all vertex and edge groups trivial. The fundamental pro-$p$ group is therefore the free pro-$p$ group on the cycles of the graph. The pro-$p$ Schreier rank formula gives $r - 1 = [\mathcal{E} : P](\mathrm{rank}(\mathcal{E}) - 1)$, proving the formula. The theorem retains no designated peripheral subgroups in its graph-of-groups conclusion; their identification is a further assertion, not supplied by this calculation.

We prove the assertion about a general finite quotient. Write $G = \mathcal{E}/P$ and fix a prime $q \neq p$. For every open normal $U \triangleleft \mathcal{E}$ contained in $P$, the kernel of $\mathcal{E}/U \to G$ is a finite $p$-group. A Sylow $q$-subgroup of $\mathcal{E}/U$ therefore has order $|G|_q$ and maps isomorphically to a Sylow $q$-subgroup of $G$. Fix one such subgroup $Q \leq G$. The finite nonempty sets of Sylow $q$-subgroups above $Q$ admit compatible choices under quotient maps: an image is again a Sylow subgroup, and any specified one lifts by taking a Sylow subgroup of its inverse image. Compactness gives an inverse-limit subgroup of $\mathcal{E}$ isomorphic to $Q$, since all these transition maps on the chosen subgroups are isomorphisms. Torsion-freeness forces $Q = 1$. Hence every prime divisor of $|G|$ is $p$, and every finite quotient of $\mathcal{E}$ is a $p$-group. This is exactly the assertion that $\mathcal{E}$ is pro-$p$. □

**D.19. [Proved] Recognition of the peripheral pro-$p$ extension.** Let $p$ be any prime, let $P$ be the pro-$p$ completion of an orientable hyperbolic surface group of type $(g, \nu)$ with $\nu > 0$, and let

$$1 \longrightarrow P \longrightarrow \mathcal{E} \longrightarrow G \longrightarrow 1$$

be an exact sequence of pro-$p$ groups with $G$ finite. Assume that $\mathcal{E}$ is torsion-free and its outer action preserves the $P$-conjugacy classes of the peripheral subgroups. Choose one $L_j$ from each

$\mathcal{E}$-orbit of these classes. Then

$$\mathcal{N}_j = N_{\mathcal{E}}(L_j) = C_{\mathcal{E}}(L_j) \simeq \mathbb{Z}_p,$$

and the pair $(\mathcal{E}; \{\mathcal{N}_j\})$ is continuously isomorphic, preserving the indexed peripheral conjugacy classes, to the pro-$p$ completion of a compact surface with its boundary circles. Its interior is orientable for odd $p$. At $p = 2$ the interior may be nonorientable; its orientation character is the induced action on the orientation line of $P$.

*Manuscript:* 28; 178 verso; 353; 24; 147 verso; 293.

*Proof.* Each $\mathcal{N}_j$ is closed, and $\mathcal{N}_j \cap P = N_P(L_j) = L_j$ by Part A. Thus $\mathcal{N}_j$ is a torsion-free pro-$p$ group with an open subgroup $\mathbb{Z}_p$. It is finitely generated, by lifting generators of its finite quotient and adjoining a generator of $L_j$. Apply D.18 to $\mathcal{N}_j$: the free pro-$p$ rank formula gives rank one. Consequently $\mathcal{N}_j \simeq \mathbb{Z}_p$, and it centralizes $L_j$.

We identify the restricted peripheral family, including its multiplicities. The restriction of $(\mathcal{E}; \{\mathcal{N}_j\})$ to $P$ contains the subgroups

$$P \cap e\mathcal{N}_j e^{-1} = eL_j e^{-1}, \qquad e \in P\backslash\mathcal{E}/\mathcal{N}_j.$$

The map from this double-coset set to the orbit of the $P$-conjugacy class of $L_j$ is surjective by definition. It is injective: an equality $e'L_j e'^{-1} = aeL_j e^{-1}a^{-1}$ with $a \in P$ is equivalent to $e'^{-1}ae \in N_{\mathcal{E}}(L_j) = \mathcal{N}_j$, exactly equality of the double cosets. Thus the restricted family consists of all original peripheral classes, each once.

Wilkes, [71, Proposition 5.11], proves finite-index ascent and descent of Poincaré duality pairs at $p$ for a $p$-torsion-free profinite ambient group, with compatible dualizing modules. The pair $(P; \{L_i\})$ is orientable of dimension two: this follows from its standard free pro-$p$ surface presentation and the converse direction of [73, Theorem 3.3]. We have checked the exact restricted family and torsion hypothesis. Hence $(\mathcal{E}; \{\mathcal{N}_j\})$ is a Poincaré duality pair of dimension two.

Its dualizing module has underlying group $\mathbb{Q}_p/\mathbb{Z}_p$, with action through a continuous character $\chi : \mathcal{E} \to \mathbb{Z}_p^\times$ whose restriction to $P$ is trivial. Naturality of duality under conjugation identifies its character with the action on the surface orientation line, using the inverse convention on the dual cohomology line. The image of $\chi$ is a finite $p$-subgroup because it factors through $G$: it is trivial at odd $p$ and is trivial or $\{\pm 1\}$ at $p = 2$. Thus inversion of the character does not affect the answer. Its restriction to each $\mathcal{N}_j$ is trivial, by the boundary-duality restriction in [71, Proposition 5.9] and the trivial duality character of $\mathbb{Z}_p$.

D.18 makes $\mathcal{E}$ free pro-$p$. Write the peripheral count as $b + 1$ and the rank as $n + b$. The condition $n \geq 0$ in the classification follows from the relative cohomology sequence: the cokernel of

$$H^1(\mathcal{E}; \mathbb{F}_p) \longrightarrow \bigoplus_{j=0}^{b} H^1(\mathcal{N}_j; \mathbb{F}_p)$$

has dimension one, since $H^2(\mathcal{E}; \mathbb{F}_p) = 0$ and the relative top group has dimension one. Hence $b \leq \operatorname{rank}(\mathcal{E})$. The classification in [73, Theorem 3.3] gives a free basis $x_1, \ldots, x_n, s_1, \ldots, s_b$, with each $s_i$ generating a conjugate of $\mathcal{N}_i$, and a generator $s_0$ of $\mathcal{N}_0$ in one of these forms:

$$\begin{aligned}
&\chi = 1: && s_0 = [x_1, x_2]_- \cdots [x_{n-1}, x_n]_- \, s_1 \cdots s_b, \\
&\chi \neq 1,\ n \text{ odd}: && s_0 = x_1^2 [x_2, x_3]_- \cdots [x_{n-1}, x_n]_- \, s_1 \cdots s_b, \\
&\chi \neq 1,\ n \text{ even}: && s_0 = x_1^2 [x_1, x_2]_- [x_3, x_4]_- \cdots [x_{n-1}, x_n]_- \, s_1 \cdots s_b.
\end{aligned}$$

Here $[x, y]_- = x^{-1}y^{-1}xy$ matches that source's convention. The first line has $n$ even. The other two have $p = 2$; their parameter $2^f$ is zero, since $\chi$ has no infinite principal-unit part.

These are surface words, with explicit changes of basis. Inverting each handle generator changes the commutator convention in the first line. The second line gives one crosscap and orientable handles. For the third, put

$$a = x_1 x_2^{-1}, \qquad b' = x_2, \qquad x_1 = ab', \quad x_2 = b'.$$

Then $x_1^2[x_1, x_2]_- = a^2(b')^2$, which gives two crosscaps and orientable handles. Replacing $s_0$ by its inverse gives the usual boundary relation and preserves its subgroup. Sending the discrete surface generators to these free pro-$p$ generators induces the asserted isomorphism of completed pairs, with exactly the indexed peripheral conjugacies already specified. The free rank formula gives $\chi(\mathcal{E}) = \chi(P)/[\mathcal{E} : P] < 0$, so its interior is hyperbolic. This proves the peripheral statement at every prime, including 2 and 3. □

**D.20. [Proved] Recognition of the closed pro-$p$ extension.** Let $P$ be the pro-$p$ completion of a closed orientable surface group of genus $g \geq 2$. A torsion-free pro-$p$ group $\mathcal{E}$ containing $P$ as an open normal subgroup is the pro-$p$ completion of a closed surface group. The surface is orientable at odd $p$. At $p = 2$ a nonorientable surface is allowed, and its orientation character is the action on the orientation line of $P$.

*Manuscript:* 28; 178 verso; 353; 24; 147 verso; 293.

*Proof.* We give the absolute finite-index duality calculation. The orientable surface presentation makes $P$ a Poincaré duality pro-$p$ group of dimension two, with

$$H^i(P, \mathbb{Z}_p[[P]]) = \begin{cases} \mathbb{Z}_p, & i = 2, \\ 0, & i \neq 2, \end{cases}$$

and trivial right $P$-action on the top group. This is the trivial-character case of the Demushkin classification in [73, Theorem 1.9].

Serre's theorem [67, §1, Theorem] equates the $p$-cohomological dimensions of an open subgroup and a profinite group without elements of order $p$. Therefore $\mathrm{cd}_p(\mathcal{E}) = 2$. Wilkes [71, Proposition 1.26] gives finite-index ascent and descent of type $p$-$FP_\infty$ for profinite groups. The surface resolution gives this property for $P$, and hence for $\mathcal{E}$.

At finite index, induction and coinduction of compact modules agree, and continuous Shapiro gives

$$\begin{aligned} H^i(\mathcal{E}, \mathbb{Z}_p[[\mathcal{E}]]) &= H^i\Big(\mathcal{E}, \mathrm{coind}_P^{\mathcal{E}} \mathbb{Z}_p[[P]]\Big) \\ &\simeq H^i(P, \mathbb{Z}_p[[P]]). \end{aligned}$$

The first coefficient isomorphism and compact Shapiro are [71, Propositions 1.16 and 1.21, and the remark after 1.21]. We make their module compatibility explicit. Put $\Lambda_H = \mathbb{Z}_p[[H]]$ and let $\mathrm{pr}_P : \Lambda_{\mathcal{E}} \to \Lambda_P$ retain the $P$-coset component. The coefficient isomorphism is

$$\Lambda_{\mathcal{E}} \longrightarrow \mathrm{Coind}_P^{\mathcal{E}} \Lambda_P, \qquad \lambda \longmapsto (x \mapsto \mathrm{pr}_P(x\lambda)).$$

Here coinduction consists of left-$P$-linear functions and its left $\mathcal{E}$-action is $(ef)(x) = f(xe)$. For representatives $t$ of the finite set $P\backslash\mathcal{E}$, the inverse is $f \mapsto \sum_t t^{-1} f(t)$. These continuous maps are left-$\mathcal{E}$-linear and right-$P$-linear. Evaluation at 1 in Shapiro is natural for the right-$P$ coefficient endomorphisms. Consequently the top right duality character restricts to that of $P$. The cohomology criterion for Poincaré duality now makes $\mathcal{E}$ a duality pro-$p$ group of dimension two with dualizing module $\mathbb{Q}_p/\mathbb{Z}_p$. Its orientation character $\chi : \mathcal{E} \to \mathbb{Z}_p^\times$ has finite $p$-group image, trivial at odd $p$ and contained in $\{\pm 1\}$ at $p = 2$.

Its one-dimensional degree-two cohomology and nondegenerate degree-one cup product make it an infinite Demushkin group. The classification [73, Theorem 1.9] gives, for these characters, exactly the three words of D.19 with the boundary letters omitted. The same invertible free-basis change converts the even-rank nonorientable word to the surface form. Hence an actual closed surface presentation induces a continuous isomorphism of its pro-$p$ completion with $\mathcal{E}$. Finite-index multiplicativity for the Euler characteristic of the finite two-dimensional resolution gives

$$\chi(\mathcal{E}) = \frac{2-2g}{[\mathcal{E} : P]} < 0.$$

The resulting surface is hyperbolic, and conjugation-naturality of the dualizing module gives the stated orientation. □

**D.21. [Refuted] A full profinite normalizer shortcut.** The assertion that every torsion-free profinite group with an open subgroup $\widehat{\mathbb{Z}}$ is itself $\widehat{\mathbb{Z}}$ is false. The pro-$p$ version used in D.19 is proved in D.18.

*Manuscript:* 28; 178 verso; 353.

*Proof.* Define

$$N = (\mathbb{Z}_3 \rtimes \mathbb{Z}_2) \times \prod_{\ell \neq 2,3} \mathbb{Z}_\ell, \qquad (a,b)(a',b') = (a + (-1)^{b \bmod 2} a',\ b + b')$$

on the first two factors, and use addition on the remaining factors. This is a profinite group with the open normal subgroup

$$L = (\mathbb{Z}_3 \times 2\mathbb{Z}_2) \times \prod_{\ell \neq 2,3} \mathbb{Z}_\ell \simeq \widehat{\mathbb{Z}}$$

of index two. A finite-order element projects trivially to the torsion-free quotient $\mathbb{Z}_2 \times \prod_{\ell \neq 2,3} \mathbb{Z}_\ell$, and then lies in the torsion-free kernel $\mathbb{Z}_3$; it is therefore the identity. Yet conjugation by $(0,1)$ sends $(1,0)$ to $(-1,0)$, so $N$ is nonabelian. The code in `code/root/finite_checks.py` checks its quotient $C_3 \rtimes C_4$, the index-two cyclic subgroup of order six, and this noncommutation. The proof in the inverse limit establishes torsion-freeness; no finite quotient is claimed to be torsion-free.

The multiplier on $L$ is $-1$ in its 3-adic coordinate and $+1$ in every other coordinate. Thus passing separately to pro-$p$ groups loses precisely this interaction between two different primes. D.29 uses this $N$ as an explicit retract in its full peripheral counterexample. Its cusp normalizers are all procyclic, so that example does not depend on making $N$ itself a cusp normalizer. □

**D.22. [Refuted] Full profinite recognition without an invariant discretization.** The assertion that every torsion-free finite extension of a full profinite hyperbolic surface group with preserved loop structure is a completed surface pair is false. D.29 constructs a counterexample with kernel of type $(0,4)$ and quotient $C_2$, including the exact peripheral comparisons and a finite quotient excluding every free profinite presentation. D.30 constructs a counterexample with closed genus-two kernel and quotient $C_2$; its degree-two cohomology at two odd primes excludes every surface completion. Recognition with an invariant discretization holds by D.17. The separate pro-$p$ recognition theorems, with their precise group and peripheral hypotheses, are proved in D.18–D.20.

*Manuscript:* 28; 178 verso; 353; 24; 147 verso; 293.

*Proof.* The basis-derived four-cusp presentation, faithful outer involution, continuous twisted extension and torsion-free retraction are given in D.29. Its order-eighteen quotient needs three

generators, whereas an index-two extension of a free rank-three group would have rank two if it were free. The exact finite-cover comparison excludes closed surface completions. Thus the counterexample satisfies the full peripheral hypothesis and fails the claimed conclusion.

For the closed case, D.30 gives the extension by its continuous twisted multiplication and by an equivalent profinite presentation. Its torsion-free maximal pro-2 quotient detects the parity of every element, proving that every torsion element lies in its torsion-free surface kernel. The two restriction isomorphisms give $H^2(\mathcal{E};\mathbf{F}_3)=\mathbf{F}_3$ and $H^2(\mathcal{E};\mathbf{F}_5)=0$. The cohomology of every orientable, nonorientable, or punctured surface completion is computed in D.30 and excludes this pair. This settles the closed hyperbolic assertion in its smallest genus as well.

This also identifies the failed step in the earlier recognition attempt. Separate pro-$p$ comparisons do not produce a single full profinite surface presentation. The mixed-prime multiplier in D.29 is an actual finite-order loop automorphism, and all its cusp normalizers are procyclic. The preserved global surface relation therefore does not restore the proposed conclusion. The first possible punctured nonabelian rank is three: D.27 excludes proper torsion-free finite extensions of $\widehat{F}_2$. The explicit free-basis comparison in D.29 settles the formerly undecided peripheral marking on D.26's rank-three extension.

We checked Collas–Philip–Yamaguchi [17], Definition 2.1, Theorems 1.3 and 2.4, against the supplied source. Their profinite F-groups already have the stated orbifold presentation. Theorem 2.4 constructs a torsion-free open normal subgroup, and Theorem 1.3 bounds the derived length of a finite quotient in its nonperfect case. The maps go from a prescribed presentation to an open subgroup; they do not assert the reverse reconstruction from an arbitrary finite extension. D.29 refutes that reverse assertion under the present full profinite hypotheses. The positive conclusions of D.17–D.20 retain their proved maps and hypotheses throughout. □

## 4.4 A full profinite lifting criterion and nonnilpotent finite tests

**D.23. [Proved] Order-preserving lifting and conjugacy in the full shadow system.** Let $\mathcal{I}$ be the cofiltered poset of isolated finite-index normal subgroups $N \trianglelefteq B_3$ contained in the pure braid group $PB_3$, and put $Q_N = GT(N)$. The finite groups and their reduction homomorphisms $R_{N,H} : Q_N \to Q_H$ for $N \subseteq H$ are those of [19, Proposition 3.14, Theorem 5.2 and Proposition 5.5]. We use the continuous isomorphism

$$\Psi : GT_0 \xrightarrow{\sim} \varprojlim_{N\in\mathcal{I}} Q_N, \qquad (\lambda, f) \longmapsto \big([(\lambda-1)/2, f]_N\big)_N.$$

For a positive integer $d$, write $T_d(Q) = \{q \in Q : q^d = 1\}$. For each $H \in \mathcal{I}$, the image of $T_d(GT_0)$ in $Q_H$ is exactly

$$\bigcap_{\substack{N\in\mathcal{I}\\ N\subseteq H}} R_{N,H}\big(T_d(Q_N)\big). \tag{D.23a}$$

An element of order exactly $d$ in this intersection has a lift of order exactly $d$. Every element of order exactly $d$ in $GT_0$ has an image of order exactly $d$ at some isolated level. For $a, b \in GT_0$, conjugacy in $GT_0$ is equivalent to conjugacy of $\Psi(a)_N$ and $\Psi(b)_N$ in every $Q_N$.

*Manuscript:* §31; folios 198–199 verso; scans 391–394; §30; folio 194 verso; scan 384; Part D(b) of the task: full GT_0 torsion.

*Proof.* The cited theorem identifies the two profinite groups, including their topologies. Its Proposition 5.1 makes the coordinates cofinal in the finite quotients of $\widehat{\mathbb{Z}}$ and $\widehat{F}_2$. The map between the normalization in that paper and ours is exactly $(\widehat{m}, \widehat{f}) \mapsto (2\widehat{m}+1, \widehat{f})$; Proposition 5.5 checks the equations and the multiplication under this map. Thus its inverse limit concerns the full $GT_0$ of Part E.

Fix $q \in Q_H$ in the right side of (D.23a). For every isolated $N \subseteq H$ define the finite nonempty set

$$S_N = \{v \in Q_N : v^d = 1,\ R_{N,H}(v) = q\}.$$

Every reduction $R_{N,M}$ restricts to a function $S_N \to S_M$. The inverse limit of these sets is nonempty. To verify the exact compactness step, any finite list of coordinates and compatibility conditions is dominated by one isolated subgroup $L \subseteq H \cap N_1 \cap \cdots \cap N_t$; Proposition 3.14 supplies an isolated refinement of the intersection. Choose an element of $S_L$ and project it to the specified coordinates. This satisfies all the finitely many conditions. The closed conditions in the compact product $\prod_N S_N$ consequently have the finite intersection property. A compatible element maps under $\Psi^{-1}$ to $g \in GT_0$ with $g^d = 1$ and $g_H = q$. The reverse inclusion follows by taking the coordinates of such a $g$. The levels below $H$ are cofinal, so the constructed family is a full inverse-limit point.

When $q$ has order $d$, the order of $g$ both divides $d$ and is a multiple of the order of its image $q$, hence equals $d$. Conversely, for $g$ of order $d$, choose, for each prime $p \mid d$, a finite coordinate detecting $g^{d/p} \neq 1$. A common isolated refinement detects all these powers. The order of its image divides $d$ and divides none of the proper divisors $d/p$, so it is $d$.

For the conjugacy assertion, the sets

$$C_N(a, b) = \{v \in Q_N : v\Psi(a)_N v^{-1} = \Psi(b)_N\}$$

are finite nonempty sets with restriction maps. The same finite intersection argument gives a compatible conjugator, which $\Psi^{-1}$ sends to a conjugator in $GT_0$. Necessity follows by projection. The construction does not assume surjectivity of a reduction map or that every element of $Q_N$ comes from $GT_0$. □

**D.24. [Proved] Dihedral tests with an order bound and the full dyadic quotient.** The program `code/part_a/gt_shadow_refinement.py` executes the finite order-two and order-three tests below. Each entry counts nonidentity elements with order dividing $d$ at the lower level, and then those admitting a lift with order dividing $d$ at the upper level.

| $n$ | $q$ | $d$ | Lower elements | Elements admitting such a lift |
|---|---|---|---|---|
| 6 | 18 | 2 | 7 | 7 |
| 6 | 18 | 3 | 2 | 0 |
| 18 | 54 | 3 | 8 | 0 |
| 54 | 162 | 3 | 8 | 0 |
| 14 | 98 | 3 | 14 | 14 |
| 4 | 8 | 2 | 3 | 1 |
| 8 | 16 | 2 | 7 | 2 |
| 16 | 32 | 2 | 11 | 4 |
| 32 | 64 | 2 | 19 | 8 |

The levels are isolated dihedral shadows $Q_{K^{(n)}}$. Their subsystem with $n = 2^\alpha$, $\alpha \geq 2$, has inverse limit

$$H_2 = \{(k, (-1)^a 5^b) \in \mathbb{Z}_2 \rtimes \mathbb{Z}_2^\times : a \in \{0, 1\},\ b \in \mathbb{Z}_2,\ k \equiv b \pmod 2\}. \tag{D.24a}$$

Every finite subgroup of $H_2$ is trivial or cyclic of order two, and all its order-two subgroups are conjugate to $\langle(0, -1)\rangle$. There is a continuous surjection $GT_0 \to H_2$. The family of all dihedral levels is not cofinal in $\mathcal{I}$; an explicit $A_5$ quotient below proves this.

*Manuscript:* §31; folios 198–199 verso; scans 391–394; §30; folio 194 verso; scan 384; Part D(b) of the task: finite-quotient torsion program.

*Proof.* Write $D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1} \rangle$. The homomorphism

$$\psi_n : PB_3 = \langle x, y \rangle \times \langle c \rangle \longrightarrow D_n^3, \quad x \longmapsto (r, s, s), \quad y \longmapsto (rs, r, rs), \quad c \longmapsto 1$$

defines $K^{(n)} = \ker \psi_n$. For even $n \geq 4$, the explicit finite model of [13, Theorem 4.3 and Proposition 4.5] is

$$Q_{K^{(n)}} \simeq \left\{ (k, u) \in (\mathbb{Z}/(n/2)) \rtimes (\mathbb{Z}/(2n))^\times : \ k \equiv \nu(u) \pmod 2 \text{ when } 4 \mid n \right\}, \tag{D.24b}$$

where $\nu(u) = (u-1)/4$ for $u \equiv 1 \pmod 4$ and $\nu(u) = (u+1)/4$ for $u \equiv 3 \pmod 4$. Multiplication is $(k, u)(l, v) = (k+ul, uv)$, with the indicated moduli. For $n \mid q$, reduction is $(k \bmod (q/2), u \bmod 2q) \mapsto (k \bmod (n/2), u \bmod 2n)$.

The code constructs the actual subgroup $G_n = \langle (r, s, s), (rs, r, rs) \rangle \leq D_n^3$ for $n = 4, 6, 8, 12, 14$. It computes its commutator subgroup and exhausts all pairs $(m, f)$ with $0 \leq m < n$, $\gcd(2m + 1, n) = 1$ and $f \in G_n'$. It tests the two cycle equations

$$\theta(f) f = 1, \qquad \tau^2(y^m f)\tau(y^m f) y^m f = 1,$$

where $\theta(x) = y$, $\theta(y) = x$, $\tau(x) = y$ and $\tau(y) = (xy)^{-1}$. It checks descent of these maps, and of $x \mapsto x^{2m+1}$, $y \mapsto f^{-1} y^{2m+1} f$, on every labelled edge of the finite Cayley graph, and checks that the resulting maps are bijective. The accepted pairs are exactly those in (D.24b), of respective orders $4, 12, 16, 24, 84$. This checks the reduction to the quoted finite model on the tested groups, with no reliance on floating point arithmetic. Projection of $G_6, G_{12}, G_{14}$ onto $D_3, D_3, D_7$ exhibits finite quotients in which the odd-order rotation and order-two reflection do not commute; their nilpotence would force these coprime Sylow subgroups to commute. These tests therefore include nonnilpotent finite quotients of $\widehat{F}_2$.

For the refinement counts, the code enumerates every element of (D.24b), computes its $d$th power using the affine multiplication, reduces every upper root, and counts the exact fibres. For example, at $n = 6$ the group is $\mathrm{Aff}(\mathbb{F}_3) \times C_2$, and its two order-three elements are the nonzero translations. At $q = 18$, an element of order dividing three has multiplier $u \equiv 1, 4, 7 \pmod 9$ and translation divisible by 3. Both its affine coordinates reduce to the identity modulo 3. Thus neither order-three element at $n = 6$ has an order-dividing-three lift, although the unbounded reduction map is surjective. At $n = 14$ the multipliers $2, 4$ modulo 7 have order three. Each lifts to its order-three root modulo 49, and each of the seven lifts of a translation coordinate is allowed because $1 + u + u^2 = 0$ there. This gives seven order-three lifts of each of the fourteen lower elements, as the exhaustive output records. The same program gives the remaining rows and the full table in its accompanying CSV.

The compatible affine coordinate reductions identify the dyadic inverse limit with (D.24a): the residue of $b$ modulo 2 is the condition in (D.24b), and the inverse limits of the two coordinate rings are $\mathbb{Z}_2$ and $\mathbb{Z}_2^\times$. The surjection from $GT_0$ is [13, Theorem 5.7 and Proposition 5.8]; more precisely, that theorem proves surjectivity already after restriction along $G_{\mathbb{Q}} \to GT_0$.

The translation kernel $\mathbb{Z}_2$ is torsion-free, so a finite subgroup of $H_2$ injects into the finite-order subgroup of $\mathbb{Z}_2^\times$. This subgroup is $\{1, -1\}$: the parametrization $\mathbb{Z}_2^\times = \{\pm 1\} \times 5^{\mathbb{Z}_2}$ has torsion-free second factor, as follows from $v_2(5^{2^j} - 1) = j + 2$ and passage to the inverse limit. Every nonidentity finite-order element of $H_2$ is consequently $(k, -1)$ with $k \in 2\mathbb{Z}_2$. Put $t = k/2$ and choose $v = 5^{t \bmod 2}$. Then $(t, v) \in H_2$, and direct multiplication gives

$$(t, v)(0, -1)(t, v)^{-1} = (2t, -1) = (k, -1).$$

This proves the full finite-subgroup and conjugacy assertions for this infinite quotient, rather than extrapolating the finite dyadic counts.

Finally every $G_n$ is metabelian: the commutator subgroup of $D_n^3$ lies in its abelian rotation subgroup, and $G_n'$ is a subgroup of that rotation subgroup. Define the epimorphism $\eta : F_2 \to A_5$

by $x \mapsto (1\ 2\ 3\ 4\ 5)$ and $y \mapsto (1\ 2\ 3)$. The code enumerates its image and derived subgroup; both have order 60, proving directly that this image is a nontrivial perfect group. Proposition 5.1 of [19] gives an isolated $N$ with $N \cap F_2 \subseteq \ker \eta$. An inclusion $K^{(n)} \subseteq N$ would give the surjection

$$G_n = F_2/(K^{(n)} \cap F_2) \longrightarrow F_2/\ker \eta = A_5.$$

Its image would be metabelian, contradicting the computed nontrivial perfect derived subgroup. Hence no such inclusion exists. This proves the asserted failure of cofinality by an explicit quotient and the exact factoring map. □

These tests give explicit finite obstructions for D.12. D.32 proves, by its full fixed-subgroup and finite-cover argument, that $GT_0$ has no nonidentity element of odd order and that every finite subgroup has order at most two, detected by the dyadic multiplier. D.34 further proves that a nonidentity involution has the global multiplier $-1$. The fourteen order-three shadows at $K^{(14)}$ survive the executed test through $K^{(98)}$. Combining D.32 with D.23 shows that each fails the order-three lifting condition at some isolated refinement; a common refinement excludes all fourteen. This existence conclusion does not identify that refinement or alter the recorded finite counts. D.25 explains why the entire dihedral subsystem alone cannot supply the exclusion. The remaining full $GT_0$ question is conjugacy of its involutions. D.35 gives the exact third-cusp equation and the finite conjugator criterion, and D.44 supplies a coboundary satisfying its first symmetry. D.42 also proves the global multiplier restriction for every finite subgroup of the full three-cusp loop-outer group.

**D.25. [Proved] All-prime dihedral compatibility and the multiplier-one obstruction.** The inverse limit $D$ over all the dihedral levels of D.24 has the following explicit isomorphism of profinite groups:

$$D \xrightarrow{\sim} \left( \prod_{p \text{ odd}} \mathrm{Aff}(\mathbb{Z}_p) \right) \times H_2. \tag{D.25a}$$

There is a canonical continuous homomorphism $GT_0 \to D$. The group $D$ contains an element of every prime order. In particular, imposing order three throughout this entire dihedral system still permits a multiplier whose 7-adic coordinate is a primitive cube root and whose other prime coordinates are 1.

Put $F = \widehat{F}_2 = \langle x, y \rangle$ and define the continuous epimorphism and its closed kernel by

$$\kappa : F \longrightarrow \prod_p F_2^{(p)}, \qquad R = \ker \kappa.$$

Write $\lambda : GT_0 \to \widehat{\mathbb{Z}}^\times$ for the multiplier. For every integer $d \geq 2$, the elements $a \in \ker \lambda$ with $a^d = 1$ correspond exactly to the following elements $f \in R$. The continuous endomorphism

$$\alpha_f : F \to F, \qquad x \mapsto x, \quad y \mapsto f^{-1} y f$$

must be an automorphism and satisfy

$$\theta(f) f = 1, \qquad \tau^2(f)\tau(f) f = 1, \qquad \prod_{j=0}^{d-1} \alpha_f^j(f) = 1, \tag{D.25b}$$

where the product is in increasing order of $j$, $\theta(x) = y$, $\theta(y) = x$, $\tau(x) = y$, and $\tau(y) = (xy)^{-1}$. Every such finite-order element has identity image in $D$. The kernel $R$ itself maps onto the explicit perfect group $A_5$ used in D.24. The finite-order solutions of (D.25b) in the stated automorphism domain are exactly $f = 1$, by D.32's subsequent full fixed-subgroup argument. Thus the nontrivial prime-order points of the dihedral inverse limit exhibited here do not lift with that order to $GT_0$ when the prime is odd.

*Manuscript:* §25 bis; folios 157 verso–160; scans 312–317; §30; folio 194 verso; scan 384; §31; folios 198–199 verso; scans 391–394.

*Proof.* It suffices to use the even levels $n = 2^\alpha n_0$ with $\alpha \geq 1$ and $n_0$ odd; every odd level agrees with its double. Apply the Chinese remainder isomorphism separately to the translation coordinate modulo $n/2$ and to the multiplier coordinate modulo $2n$ in (D.24b). The odd factors give $\mathrm{Aff}(\mathbb{Z}/n_0)$, and the powers of two give the finite groups defining $H_2$. These coordinate maps commute with every reduction $n \mid q$. Conversely, specified odd coordinates and a specified point of $H_2$ reconstruct every finite pair $(k, u)$ by the Chinese remainder theorem. They satisfy the parity condition because that condition involves only the power-of-two coordinates. The resulting inverse maps are continuous, being continuous at every finite coordinate. This proves (D.25a), including its topology. Restriction of the full inverse-limit point in D.23 defines the indicated map from $GT_0$; surjectivity onto $D$ is not used or asserted.

Let $q$ be a prime, and choose a prime divisor $p$ of $2^q - 1$. The residue of 2 in $\mathbb{F}_p^\times$ has order $q$: its order divides $q$, and order one would give $p \mid 2 - 1$. Hence $q \mid p - 1$, in particular $p$ is odd and $p \neq q$. The element 2 mod $p$ lifts to a unique $q$th root $\zeta \in \mathbb{Z}_p^\times$. Explicitly, after a root $a$ modulo $p^j$ is chosen, substitute $a + p^j t$, $0 \leq t < p$, in $X^q - 1$ modulo $p^{j+1}$. Its coefficient of $t$ is $p^j q a^{q-1}$, and $qa^{q-1}$ is a unit modulo $p$; there is exactly one permitted value of $t$. The compatible limit has order exactly $q$ because its residue already does. Put $(0, \zeta)$ in the $p$-factor of (D.25a), and put the identity in all the other factors. This gives the asserted element of order $q$ in $D$.

For $q = 3$, take $p = 7$. The executed program `code/part_a/gt_shadow_all_prime.py` constructs ten successive roots of $X^2 + X + 1$ starting with 2 mod 7, and uses the exact Chinese remainder formulas to construct compatible shadow coordinates at sixteen levels through $n = 2744$. At $n = 14$ its affine pair is $(0, 9 \bmod 28)$; at $n = 98$ it is $(0, 177 \bmod 196)$. Their orders are three and the latter reduces to the former. The preceding root-lifting proof constructs the infinite point, while the program verifies the displayed finite coordinates. Thus the equations tested over all dihedral levels admit prime-dependent torsion multipliers. They do not force the global alternatives 1 and $-1$.

We next prove the statement about $R$. Each projection $F \to F_2^{(p)}$ is the canonical maximal pro-$p$ quotient map. Their product is onto: in any finite collection of finite $p$-group quotients at distinct primes, its image surjects to each factor. Its order is consequently divisible by every factor order, hence by their product, since these orders are pairwise coprime. The image is the full product. These finite products form the finite quotients of $\prod_p F_2^{(p)}$, so the compact image of $F$ is the whole product. Also $R \subset F'$, since the abelianization $F^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ is detected by the abelianizations of all these pro-$p$ quotients.

Let $a = (1, f) \in GT_0$ have finite order dividing $d$. It induces a finite-order automorphism of each free pro-$p$ quotient with trivial action on its abelianization. The Magnus leading-term argument in D.10, which also applies at $p = 2$, makes this automorphism the identity. Its normalized formula then gives $f_p \in C_{F_2^{(p)}}(y) \cap (F_2^{(p)})' = 1$ by A.7 and the primitive abelianization coordinate of $y$. Thus $f \in R$.

The two first equations of (D.25b) are precisely the defining cycle equations at multiplier one; the required invertibility is stated explicitly. To obtain the last equation, put $w_d = f\alpha_f(f) \cdots \alpha_f^{d-1}(f)$. Induction on $d$ gives

$$\alpha_f^d(x) = x, \qquad \alpha_f^d(y) = w_d^{-1} y w_d.$$

Therefore $\alpha_f^d = 1$ is equivalent to $w_d \in C_F(y)$. But $w_d \in F'$ and $C_F(y) \cap F' = \overline{\langle y \rangle} \cap F' = 1$, again by A.7 and abelianization. This proves the necessity of the last equation and its sufficiency. The normalized identification of $GT_0$ with these automorphisms gives the asserted bijection. The kernel of the multiplier on $D$ is a subgroup of the additive translation group $\prod_p \mathbb{Z}_p$, which is torsion-free. Consequently every finite-order multiplier-one element maps to the identity of $D$.

Finally let $\eta : F \to A_5$ be the epimorphism from D.24. The quotient $A_5/\eta(R)$ is a finite quotient of $F/R = \prod_p F_2^{(p)}$ and is therefore nilpotent. It is also perfect, since the executed permutation calculation in D.24 proves $A_5' = A_5$. A finite nilpotent perfect group is trivial: its lower central series would otherwise be constantly equal to the nontrivial group and could not terminate. Hence $\eta(R) = A_5$. This identifies a nontrivial part of the residual group in the exact equation (D.25b). The full three-cusp argument in D.32 proves that this equation, together with its stated cycle and invertibility requirements, has only the identity solution. The surjection $R \to A_5$ and the compatible points of $D$ remain the exact quotient calculations proved above. □

**D.26. [Refuted] Unmarked recognition at index two and the peripheral obstruction.** The assertion that a torsion-free profinite group with an open free profinite subgroup of rank three is free is false, already at index two. We give the rank-three instance of [29, Proposition, pp. 350–352] as an explicit extension

$$1 \longrightarrow H = \widehat{F}(l, a, b) \longrightarrow E \longrightarrow C_2 \longrightarrow 1. \tag{D.26a}$$

It has no section. Its outer involution preserves no loop structure of type $(1, 2)$ on $H$. Every preserved type-$(0, 4)$ structure would have a double-transposition cusp permutation, multiplier $-\epsilon$, and canonical character $\chi(l) = 1$, $\chi(a) = \chi(b) = -1$ sending all four cusp generators to $-1$. Two explicit type-$(0, 4)$ candidates fail, with finite $S_3$ and $S_4$ certificates. D.29 supplies a different complete type-$(0, 4)$ marking, satisfying exactly these necessary conditions, and thereby turns this extension into a counterexample to the loop-preserving assertion D(d).

*Manuscript:* §28; folios 178–179; scans 352–354; Part D(d) of the task: full profinite recognition.

*Proof.* Choose $\epsilon, \kappa \in \widehat{\mathbb{Z}}$ by their prime coordinates:

$$\epsilon_3 = -1, \quad \epsilon_p = 1 \ (p \neq 3), \qquad \kappa_2 = 1, \quad \kappa_p = 0 \ (p \neq 2).$$

Thus $\epsilon$ is a unit, $\epsilon^2 = 1$ and $\epsilon\kappa = \kappa$. Define the continuous endomorphism of the free profinite group $H$ by

$$\alpha(l) = l^{\epsilon}, \qquad \alpha(a) = b, \qquad \alpha(b) = l^{\kappa} a l^{-\kappa}, \qquad c = l^{\kappa}. \tag{D.26b}$$

Substitution on the three free generators gives $\alpha^2 = \mathrm{Inn}(c)$ and $\alpha(c) = c$. In particular $\mathrm{Inn}(c^{-1})\alpha$ is an inverse, so $\alpha$ is an automorphism.

Define $E$ on the compact space $H \times \{0, 1\}$ by

$$(h, i)(h', j) = \big(h\alpha^i(h')c^{\lfloor (i+j)/2 \rfloor}, i + j \bmod 2\big). \tag{D.26c}$$

The two identities just checked prove associativity by the eight triples of quotient labels. When the first two labels are 1, the required interchange is $\alpha^2(h'')c = ch''$; the triple $(1, 1, 1)$ additionally uses $\alpha(c) = c$. The other triples follow directly. The identity and inverses follow by the same formulas. All operations are continuous, so this is a profinite group. Writing $t = (1, 1)$ gives $tht^{-1} = \alpha(h)$ and $t^2 = c$. The projection to the second coordinate gives (D.26a).

The closed subgroup $N = \overline{\langle l, t \rangle}$ has the explicit isomorphism

$$N \simeq (\mathbb{Z}_3 \rtimes \mathbb{Z}_2) \times \prod_{p \neq 2,3} \mathbb{Z}_p,$$

where the parity of the $\mathbb{Z}_2$ coordinate acts by inversion on $\mathbb{Z}_3$. Namely, the normal form $l^u t^i$ maps to $(u_3, 2u_2 + i, (u_p)_{p \neq 2,3})$. This is a continuous bijection preserving the displayed relations and multiplication; its inverse reads the parity $i$ and then all coordinates of $u$. The projection of $N$ to $\mathbb{Z}_2 \times \prod_{p \neq 2,3} \mathbb{Z}_p$ has torsion-free kernel $\mathbb{Z}_3$ and torsion-free target, hence $N$ is torsion-free.

There is a continuous retraction $E \to N$ given on the generators by $l \mapsto l$, $t \mapsto t$, $a, b \mapsto 1$; (D.26b) verifies its compatibility with the group law. An element of finite order maps trivially under this retraction and therefore has trivial image in $C_2$. It lies in $H$, which is torsion-free by A.7, so it is the identity. Thus $E$ is torsion-free and the extension has no section. The induced outer action is faithful: its nonidentity element acts nontrivially on $H^{\mathrm{ab}}$.

For comparison with Haran's construction, the homomorphism $N \amalg \widehat{\mathbb{Z}} \to E$ from the free profinite product, identity on $N$ and sending the other generator to $a$, is onto since $b = tat^{-1}$. In the opposite direction send $l, a, b, t$ to $l, a, tat^{-1}, t$ in that free product. Equations (D.26b) hold there, using the two relations in $N$, so (D.26c) gives a continuous homomorphism in the opposite direction. The composites fix every generator. These are mutually inverse isomorphisms.

To prove that $E$ is not free, form the finite group

$$Q = (\mathbb{F}_3^2) \rtimes C_2, \qquad (v, i)(w, j) = (v + (-1)^i w, i + j).$$

The formulas $l \mapsto (e_1, 0)$, $a \mapsto (e_2, 0)$, $t \mapsto (0, 1)$, and $b \mapsto (-e_2, 0)$ give an epimorphism $E \to Q$: $\epsilon \equiv -1 \pmod 3$ and $\kappa \equiv 0 \pmod 3$ verify all relations. This group requires three generators. Two elements of $\mathbb{F}_3^2$ do not generate its $C_2$ quotient. When one of two proposed generators lies outside $\mathbb{F}_3^2$, it is an involution. Multiplying the other by it when necessary puts that other generator in $\mathbb{F}_3^2$. Conjugation then only inverts this one vector, so the generated subgroup has order at most six. Three generators, as displayed, do generate $Q$. Were $E$ a free profinite group of rank $r$, its open index-two subgroup would have rank $2r - 1$ by the finite coset Schreier rewriting. Since $H$ has rank three, this would give $r = 2$, contradicting its quotient $Q$. The program `code/part_a/haran_rank3.py` checks all $18^3$ associativity instances and all $18^2$ ordered pairs in $Q$. Their generated subgroup orders are $1, 2, 3, 6, 9$, with multiplicities $1, 27, 32, 216, 48$; none is 18.

We now check the peripheral condition rather than discarding it. On the basis $[l], [a], [b]$ of $H^{\mathrm{ab}} = \widehat{\mathbb{Z}}^3$, the outer involution has matrix and determinant

$$M = \begin{pmatrix} \epsilon & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \det M = -\epsilon. \tag{D.26d}$$

This determinant has value $+1$ at the prime 3 and $-1$ at every other prime, and is invariant under a change of basis. For a preserved type-$(1, 2)$ peripheral structure, let $W$ be the primitive rank-one cusp submodule in $H^{\mathrm{ab}}$. Write $\lambda$ for the action on the orientation line and $\delta \in \{1, -1\}$ for the sign of the permutation of the two punctures. The action on $W$ is multiplication by $\lambda\delta$. The quotient $H^{\mathrm{ab}}/W$ is the first homology of the filled torus. The orientation comparison of A.2 identifies its determinant with $\lambda$. The square of the outer action is the identity, so $\lambda^2 = 1$. Multiplicativity of determinant in the exact sequence of these free modules gives

$$\det M = (\lambda\delta)\lambda = \delta.$$

This contradicts (D.26d). Thus no such type-$(1, 2)$ peripheral structure is preserved.

For type $(0, 4)$ this determinant argument permits a prime-dependent orientation multiplier. We test the specific free basis $l, a, b^\epsilon$, which is a basis because the power substitution on $b$ has inverse $b \mapsto b^\epsilon$. Set

$$P = lab^\epsilon, \quad L_1 = \overline{\langle l \rangle}, \quad L_2 = \overline{\langle a \rangle}, \quad L_3 = \overline{\langle b^\epsilon \rangle}, \quad L_4 = \overline{\langle P^{-1} \rangle}.$$

This is a full type-$(0, 4)$ peripheral system. The first three classes transform as required with multiplier $\epsilon$ and permutation (2 3). The fourth would require $\alpha(P)$ to be conjugate to $P^\epsilon$. The actual words are

$$\alpha(P) = l^\epsilon b l^\kappa a^\epsilon l^{-\kappa}, \qquad P^\epsilon = (lab^\epsilon)^\epsilon.$$

Map the free group $H$ continuously onto $S_3$ by

$$l \longmapsto (1\ 2), \qquad a \longmapsto (2\ 3), \qquad b \longmapsto (1\ 2\ 3).$$

At its exponent six, $\epsilon \equiv 5$ and $\kappa \equiv 3$. Direct multiplication gives $P \mapsto 1$, $\alpha(P) \mapsto (1\ 2\ 3)$, and $P^{\epsilon} \mapsto 1$. These images are not conjugate, so the fourth loop class fails. The program checks these values and all $6^3$ possible assignments, finding twenty-four nonconjugacy certificates.

There is a basis-independent restriction on any remaining type-$(0,4)$ structure. Write $\sigma \in S_4$ for its cusp permutation and $\lambda \in \widehat{\mathbb{Z}}^{\times}$ for its multiplier. The square of the outer action is the identity, so $\sigma^2 = 1$ and $\lambda^2 = 1$. Its action on $H^{\mathrm{ab}}$ is $\lambda$ times the action of $\sigma$ on $\widehat{\mathbb{Z}}^4/\widehat{\mathbb{Z}}(1,1,1,1)$. Comparing traces with (D.26d) gives

$$\epsilon = \lambda(|\operatorname{Fix}(\sigma)| - 1). \tag{D.26e}$$

The identity permutation would give $3\lambda = \epsilon$, which is impossible modulo 3. A single transposition forces $\lambda = \epsilon$; a double transposition forces $\lambda = -\epsilon$.

For each such peripheral structure there is a canonical character $\chi : H \to \{1, -1\}$ which is $-1$ on all four cusp generators. The product relation makes it well-defined, and a free peripheral basis makes it nonzero. It is preserved by $\alpha$, since $\lambda$ is odd and the cusp classes are permuted. In the basis $l, a, b$ the three nonzero invariant characters are exactly

$$\chi_1 = (-1, 1, 1), \qquad \chi_2 = (1, -1, -1), \qquad \chi_3 = (-1, -1, -1). \tag{D.26f}$$

We compute the action on continuous cohomology with coefficients $\mathbb{Q}_p(\chi)$ for an odd prime $p$; equivalently one may use the inverse limit of the finite coefficient modules and then tensor with $\mathbb{Q}_p$. A crossed homomorphism $u$ is determined freely by $(L, A, B) = (u(l), u(a), u(b))$. Put $\eta = \chi(l)$ and $\zeta = \chi(a) = \chi(b)$, and define

$$e = \begin{cases} \epsilon_p & \eta = 1, \\ 1 & \eta = -1, \end{cases} \qquad k = \begin{cases} 0 & \eta = 1, \\ 1 & \eta = -1. \end{cases}$$

Here $u(l^{\epsilon}) = eL$ and $u(l^{\kappa}) = kL$. For $\eta = 1$ this is multiplication by the $p$-coordinate of the exponent; for $\eta = -1$ the geometric sum is $(1 - (-1)^v)/2 = 1$ for either odd exponent $v = \epsilon, \kappa$. Substitution in (D.26b) gives

$$\alpha^* u \quad \longleftrightarrow \quad A_{\chi,p} \begin{pmatrix} L \\ A \\ B \end{pmatrix}, \qquad A_{\chi,p} = \begin{pmatrix} e & 0 & 0 \\ 0 & 0 & 1 \\ (1-\zeta)k & \eta & 0 \end{pmatrix}. \tag{D.26g}$$

The coboundary line is spanned by $(\eta - 1, \zeta - 1, \zeta - 1)$, and this nonzero vector is fixed by the displayed matrix. Thus

$$\det\bigl(\alpha^* | H^1(H, \mathbb{Q}_p(\chi_i))\bigr) = \begin{cases} 1 & i = 1, 3, \\ -\epsilon_p & i = 2. \end{cases} \tag{D.26h}$$

We identify the determinant required by the peripheral structure through actual restriction and filling maps. Let $K = \ker \chi$. Under the specified marking it is the completion of the double cover of the four-punctured sphere in which every cusp has nontrivial monodromy. The cover has four punctures and genus one, since its Euler characteristic is $-4$. The deck involution fixes each of the four peripheral homology classes. Their span is the three-dimensional invariant subspace of $H_1(K, \mathbb{Q}_p)$: restriction and averaging identify the invariant cohomology with $H^1(H, \mathbb{Q}_p)$, of dimension three. Filling the four punctures therefore identifies the anti-invariant part with the

first homology of the filled torus. The restriction of crossed homomorphisms induces the precise comparison

$$H^1(H,\mathbb{Q}_p(\chi)) \xrightarrow{\sim} H^1(K,\mathbb{Q}_p)^- \xleftarrow{\sim} H^1(\widehat{\pi_1(T^2)},\mathbb{Q}_p). \tag{D.26i}$$

The first isomorphism follows by averaging over the order-two deck group, which is invertible in $\mathbb{Q}_p$; the second is the dual of the filling quotient on the anti-invariant part. Both commute with the induced action of $\alpha$. Upstairs cusp generators are squares of the downstairs generators. A conjugator in $H$ can be moved into $K$ by multiplying it by the corresponding downstairs cusp generator when necessary; that generator centralizes its own square. Thus $\alpha$ has the same multiplier $\lambda$ on this covering pair. The transfer of the relative orientation class, followed by filling, is the comparison of A.2; it identifies the determinant on the filled torus with $\lambda_p$. Consequently the determinant in (D.26h) must be $\lambda_p$.

At $p=3$ all three determinants in (D.26h) are 1. This excludes the single-transposition case, which by (D.26e) would have $\lambda_3=\epsilon_3=-1$. For the remaining double transposition, $\lambda=-\epsilon$; at $p=5$ its value is $-1$. Formula (D.26h) then excludes $\chi_1$ and $\chi_3$, leaving exactly $\chi_2$. This proves the necessary conditions stated at the beginning for every marking.

One can satisfy the first two cusp conditions in this remaining case by the explicit free basis

$$x=a,\qquad y=b^{-\epsilon},\qquad z=b^{\epsilon}l,\qquad w=(xyz)^{-1}=(al)^{-1}. \tag{D.26j}$$

The inverse substitution is $a=x$, $b=y^{-\epsilon}$, $l=yz$, proving that this is a free basis. It has canonical character $\chi_2$, and $\alpha(x)$ is $y^{-\epsilon}$ while $\alpha(y)$ is conjugate to $x^{-\epsilon}$. For permutation $(x\ y)(z\ w)$ and multiplier $-\epsilon$, the next condition would require

$$\alpha(z)\sim w^{-\epsilon},\quad \text{equivalently}\quad a^{\epsilon}l^{\epsilon}\sim(al)^{\epsilon}. \tag{D.26k}$$

Indeed $\alpha(z)=ca^{\epsilon}c^{-1}l^{\epsilon}$ is conjugate to $a^{\epsilon}l^{\epsilon}$ because $c=l^{\kappa}$ commutes with $l$. In $S_4$ take $a=(2\ 3\ 4)$, $l=(1\ 2\ 3\ 4)$ and $b=1$. Its exponent is twelve, with $\epsilon\equiv 5$. The left member of (D.26k) is the transposition $(1\ 4)$, whereas the right member is the four-cycle $(1\ 3\ 2\ 4)$. They cannot be conjugate. The supplied program checks all $24^2$ pairs in $S_4$, finding 96 such nonconjugacy witnesses. It also checks all six matrices (D.26g) over $\mathbb{F}_3$ and $\mathbb{F}_5$, their fixed coboundary lines and the determinants on their two-dimensional cohomology quotients.

These calculations reduce the permissible type-$(0,4)$ markings to double transpositions with the specified multiplier and character. The second candidate (D.26j) is excluded, but D.29 constructs another marking by gluing a genuine three-cusp automorphism to its inverse. The exact basis change $b'=\tau(a)$ in that construction identifies its extension with (D.26a). Thus an invariant marking does exist; the finite failures above concern the two displayed candidates and the proved exclusions of the other permutation and character types. □

**D.27. [Proved] The full-profinite rank divisibility and prime-order lifting test.** For a torsion-free profinite extension

$$1\longrightarrow \widehat{F}_r\longrightarrow E\longrightarrow G\longrightarrow 1,\qquad 2\le r<\infty,\quad |G|<\infty,$$

the integer $|G|$ divides $r-1$. In particular a torsion-free finite extension of $\widehat{F}_2$ equals its prescribed kernel. Every homomorphism $C_p\to \mathrm{Out}(\widehat{F}_2)$, for any prime $p$, has a homomorphic lift to $\mathrm{Aut}(\widehat{F}_2)$; a faithful one has a lift of order exactly $p$. This last assertion holds in the full profinite category, without an invariant discretization and without first passing to a pro-$p$ quotient.

*Manuscript:* §28; folios 178–179; scans 352–354; §31; folios 198–199 verso; scans 391–394.

*Proof.* We give the normal-subgroup version of Jarden's index argument [38, Theorems 1.2 and 4.1, pp. 496–498], with the quotient comparisons written out. Put $F=\widehat{F}_r$. For every prime

$q$, $\mathrm{cd}_q(F) = 1$. The torsion-free finite-index dimension theorem of Serre [67, §1], whose torsion hypothesis applies to $E$ itself, gives $\mathrm{cd}_q(E) = 1$. The same is true of every open subgroup of $E$. Thus these groups are projective profinite groups. Here we use the precise projectivity criterion recalled in [38, §2]: a profinite group is projective exactly when its cohomological dimension at every prime is at most one; equivalently, all its finite embedding problems have weak solutions. This is the criterion for the full profinite category, including finite embedding problems whose kernels are nonabelian.

Fix a prime divisor $p$ of $|G|$, and let $p^a$ be its exact $p$-part. Choose a Sylow $p$-subgroup $G_p \leq G$ and write $E_p$ for its inverse image in $E$. Define $O^p(J)$, for a profinite group $J$, as the kernel of its maximal pro-$p$ quotient. The characteristic subgroup $O^p(F)$ of $F$ is normal in $E_p$. There are exact continuous quotient maps

$$1 \longrightarrow F/O^p(F) \longrightarrow E_p/O^p(F) \longrightarrow G_p \longrightarrow 1.$$

The middle group is pro-$p$, since both its kernel and quotient are pro-$p$. Therefore $O^p(E_p) \leq O^p(F)$. Conversely the restriction to $F$ of the maximal pro-$p$ quotient of $E_p$ is a pro-$p$ quotient of $F$, so $O^p(F) \leq O^p(E_p)$. This proves

$$O^p(E_p) = O^p(F), \qquad [E_p^{(p)} : F^{(p)}] = p^a. \tag{D.27a}$$

The quotient $E_p^{(p)}$ is projective as a pro-$p$ group. Indeed a finite embedding problem with $p$-group target and covering group pulls back to one for $E_p$; its projective solution factors through $E_p^{(p)}$ because its target is a $p$-group. A projective pro-$p$ group is free pro-$p$. It has finite rank $d_p$ because $E_p$ is finitely generated. The subgroup $F^{(p)}$ in (D.27a) is free pro-$p$ of rank $r$, with its full induced topology, as the exact quotient construction proves. Finite-coset Schreier rewriting therefore gives

$$r - 1 = p^a(d_p - 1). \tag{D.27b}$$

Consequently each exact prime-power divisor $p^a$ of $|G|$ divides $r - 1$. These divisors are pairwise coprime, proving the first assertion. At $r = 2$, this gives $|G| = 1$.

Now let $\rho : C_p \to \mathrm{Out}(\widehat{F}_2)$ be a prescribed continuous outer action, with the finite group given its discrete topology. Form the pullback

$$E_\rho = \{(a, g) \in \mathrm{Aut}(\widehat{F}_2) \times C_p : [a] = \rho(g)\}.$$

This is a profinite group. Centre-freeness identifies its kernel over $C_p$ with $\mathrm{Inn}(\widehat{F}_2) = \widehat{F}_2$ by the injective map $x \mapsto (\mathrm{Inn}(x), 1)$. If $E_\rho$ were torsion-free, the already proved rank-two assertion would force its quotient to be trivial, contradicting $|C_p| = p$. Hence $E_\rho$ contains a nonidentity torsion element $v$. Its finite cyclic subgroup intersects the torsion-free kernel trivially, so projection embeds it in $C_p$. Its order is therefore exactly $p$, and it projects isomorphically onto $C_p$. The inverse of that projection is a section with the prescribed labels. Composing with the first projection $E_\rho \to \mathrm{Aut}(\widehat{F}_2)$ gives the asserted lift. For faithful $\rho$, the lift is faithful because its outer projection is. No choice of conjugacy class of this section is claimed; the argument proves existence.

Applied to a prime-order element of $GT_0$, this gives a genuine finite-order loop-preserving automorphism representing its outer class. It does not impose an unproved restriction on its multiplier, or identify its fixed subgroup. In particular it neither solves the residual norm equation D.25 nor proves conjugacy to the geometric involution. It does prove that the nontrivial torsion-free recognition test of D.22 first reaches free rank three, where (D.27b) permits index two and D.26 supplies an explicit unmarked example. □

**D.28. [Proved] Prime-order sections and the full fixed subgroup.** Let $p$ be a prime and let $\rho : C_p \to GT_0$ be a continuous homomorphism whose multiplier has $p$-adic component 1. This condition is automatic for odd $p$. Let $\alpha$ be the normalized automorphism of $F = \widehat{F}(x, y)$ associated to the chosen generator of $C_p$, and put

$$E = F \rtimes_\alpha C_p, \qquad S = \{1\} \times C_p, \qquad C = C_F(S).$$

There is exactly one $F$-conjugacy class of sections of $E \to C_p$. Each of the three peripheral normalizers contains a section. The inclusion $C \hookrightarrow F$ induces an isomorphism of maximal pro-$p$ quotients

$$C^{(p)} \xrightarrow{\sim} F^{(p)}. \tag{D.28a}$$

This assertion concerns the maximal pro-$p$ quotient of the actual closed fixed subgroup. It does not identify that subgroup with $F$. An explicit order-three automorphism below has a proper fixed subgroup and exactly the isomorphism (D.28a), while its third peripheral class fails in a finite dihedral quotient.

*Manuscript:* §31; folios 198–199verso; scans 391–394; §30; folio 194verso; scan 384; Part D(a) of the task: actual full GT_0 torsion.

*Proof.* The normalized map $GT_0 \to \mathrm{Aut}(F)$ is a homomorphism: for the pair $(\lambda, f)$ it sends $x \mapsto x^\lambda$, $y \mapsto f^{-1}y^\lambda f$, and composition gives precisely the product $f\alpha(g)$ in the second coordinate. Consequently $\alpha^p = 1$. On the maximal pro-$p$ quotient its multiplier is 1; the Magnus argument of D.10 and D.25 makes this finite-order normalized automorphism the identity. Thus the quotient of the semidirect product is the explicit map

$$\pi : E \longrightarrow F^{(p)} \times C_p, \qquad (h, s^j) \longmapsto (\bar{h}, s^j). \tag{D.28b}$$

For odd $p$, a finite-order unit in $\mathbb{Z}_p^\times$ of order dividing $p$ is 1: reduction modulo $p$ eliminates the prime-to-$p$ part, and the logarithm on $1 + p\mathbb{Z}_p$ eliminates the remaining torsion. This explains the automatic condition.

Write $c_1 = x$, $c_2 = y$, $c_3 = (xy)^{-1}$, and $L_i = \overline{\langle c_i \rangle}$. Preservation of the three peripheral classes gives an exact sequence

$$1 \longrightarrow L_i \longrightarrow N_E(L_i) \longrightarrow C_p \longrightarrow 1; \tag{D.28c}$$

its kernel is $L_i$ by A.7. Choose $a_i$ over the specified generator of $C_p$. Then

$$a_i c_i a_i^{-1} = c_i^\lambda, \qquad a_i^p = c_i^{m_i}, \qquad (\lambda - 1)m_i = 0. \tag{D.28d}$$

The first component of $\pi(a_i)$ centralizes $\bar{c}_i$, since $\lambda_p = 1$. The centralizer of this primitive element in the free pro-$p$ group is $\overline{\langle \bar{c}_i \rangle}$. Hence $\pi(a_i) = (\bar{c}_i^{v_i}, s)$ for $v_i \in \mathbb{Z}_p$, and comparison of $p$th powers yields

$$(m_i)_p = pv_i. \tag{D.28e}$$

Let $N_\lambda = 1 + \lambda + \cdots + \lambda^{p-1}$. In the $q$-adic coordinate where $\lambda_q \neq 1$, the equation $\lambda_q^p = 1$ and the integral-domain property give $(N_\lambda)_q = 0$ and $(m_i)_q = 0$. Where $\lambda_q = 1$, the norm is $p$; division by $p$ is possible for $q \neq p$, and (D.28e) supplies it at $q = p$. Define $u_i \in \widehat{\mathbb{Z}}$ coordinatewise by $(u_i)_q = -(m_i)_q/p$ in the latter coordinates and zero in the former. This gives $N_\lambda u_i = -m_i$. The element $b_i = a_i c_i^{u_i}$ has order $p$, by (D.28d), and projects to the prescribed generator. Thus every normalizer in (D.28c) contains a section $S_i$.

Let $e_p \in \widehat{\mathbb{Z}}$ be the idempotent with $p$-coordinate 1 and all other coordinates 0. Each $S_i$ centralizes $c_i^{e_p}$. Its image in $F^{\mathrm{ab}} \otimes \mathbb{F}_p$ is the nonzero primitive vector $[c_i]$. It follows that

$$\dim_{\mathbb{F}_p} H^1(C_F(S_i), \mathbb{F}_p) \geq 1. \tag{D.28f}$$

This inference uses restriction of a linear functional on $F^{\mathrm{ab}}/p$ that is nonzero on the displayed element; it does not infer first homology from the mere presence of a pro-$p$ subgroup.

We now use the finite-cohomology form of the profinite Brown theorem, quoted as Theorem 2 and applied in [42, §1.1, equations (1)–(6)]. There is no subgroup $C_p \times C_p$ in $E$, because every finite subgroup injects into $E/F = C_p$. The Lyndon–Hochschild–Serre spectral sequence and $\mathrm{cd}_p(F) = 1$ give, for every $n > 1$,

$$\dim_{\mathbb{F}_p} H^n(E, \mathbb{F}_p) \le 1 + \dim_{\mathbb{F}_p} H^1(F, \mathbb{F}_p) = 3.$$

For each subgroup $T$ of order $p$ in $E$, projection to $C_p$ shows that its normalizer equals its centralizer: a normalizing element must induce the identity on the generator of the quotient. Dividing by the unique element of $T$ with the same quotient gives

$$N_E(T) = C_F(T) \times T.$$

The Brown restriction map is a dense embedding into the product of the cohomology of these normalizers. Its finite-dimensional source makes it an isomorphism and makes the number of factors finite. Each closed subgroup $C_F(T)$ has $p$-cohomological dimension at most one. The direct-product cohomology calculation therefore gives the exact finite sum and bound

$$\sum_{[T]} \Big(1 + \dim_{\mathbb{F}_p} H^1(C_F(T), \mathbb{F}_p)\Big) = \dim_{\mathbb{F}_p} H^n(E, \mathbb{F}_p) \le 3, \tag{D.28g}$$

where $[T]$ runs through $E$-conjugacy classes of order-$p$ subgroups. These are also their $F$-conjugacy classes: for any such $T$, the decomposition $E = FT$ gives the same conjugacy orbit under $E$ and $F$.

By (D.28f), two distinct cusp-section classes would contribute at least four to (D.28g). Thus all three $S_i$ are conjugate. Conjugating $S_1$ and $S_2$ to a single representative places conjugates of $x^{e_p}$ and $y^{e_p}$ in its fixed subgroup. Their images in $F^{\mathrm{ab}}/p$ are still the independent vectors $[x], [y]$. This gives two independent restricted linear functionals, so that common factor contributes at least three. Bound (D.28g) is exhausted. There are no other classes, and its fixed subgroup has $\dim H^1(-, \mathbb{F}_p) = 2$. Every order-$p$ subgroup projects isomorphically onto $C_p$ and determines one labelled section; hence there is exactly one conjugacy class of sections. In particular these conclusions apply to $S$.

The image of $C$ in $F^{(p)}$ is closed and surjects on its Frattini quotient by the two independent cusp vectors, so it is all of $F^{(p)}$. Since $C$ is a closed subgroup of a free profinite group, it is projective. Pulling a finite pro-$p$ embedding problem back to $C$ shows that $C^{(p)}$ is projective pro-$p$, and hence free pro-$p$. Its rank equals $\dim H^1(C, \mathbb{F}_p) = 2$. The resulting surjection from a rank-two free pro-$p$ group onto $F^{(p)}$ is an isomorphism: after identifying free bases it is a surjective endomorphism of a finitely generated profinite group, which is injective. This proves (D.28a) in the claimed direction.

Here is an explicit limitation of that last comparison. Choose $u \in \widehat{\mathbb{Z}}^\times$ with $u_q = 1$ for $q \neq 7$, and $u_7$ the order-three root with $u_7 \equiv 2 \pmod 7$. The Hensel construction is given in D.25. The substitutions

$$\beta(x) = x^u, \qquad \beta(y) = y^u \tag{D.28h}$$

have inverse given by exponent $u^{-1}$ and have order exactly three. The fixed subgroup $C_\beta$ is proper, since $\beta(x) \neq x$ in the cyclic quotient of order seven. But it contains $x^{e_3}$ and $y^{e_3}$, so its map to $F^{(3)}$ is onto. The same Brown bound, now using this one section and these two independent vectors, gives $\dim H^1(C_\beta, \mathbb{F}_3) = 2$. The preceding projectivity argument proves $C_\beta^{(3)} \cong F^{(3)}$, despite $C_\beta \neq F$.

The precise missing cusp condition is detected by

$$D_7 = \langle r, s \mid r^7 = s^2 = 1, srs = r^{-1} \rangle, \qquad x \mapsto s, \quad y \mapsto sr.$$

Here $xy \mapsto r$, $u \equiv 9 \pmod{14}$, and the two involutions are unchanged under exponent $u$. Thus $x^u y^u \mapsto r$ whereas $(xy)^u \mapsto r^2$. The conjugacy class of $r$ consists of $r, r^{-1}$, so these are not conjugate. Consequently (D.28h) is not an element of the three-cusp loop-outer group, and in particular not a $GT_0$ torsion construction. The program `code/part_a/gt_torsion_sections.py` executes this finite certificate and the coordinate norm-solvability tests used above. D.32 uses the third cusp in a finite-cover rank argument to strengthen (D.28a) to $C = F$ under the three-cusp hypotheses, and thereby excludes the nontrivial prime-order actions considered here. □

**D.29. [Refuted] A preserved four-puncture structure on the torsion-free Haran extension.** The full profinite recognition assertion in D(d) is false, including its peripheral hypothesis. There is an exact sequence of profinite groups

$$1 \longrightarrow H = \widehat{\pi}_{0,4} \longrightarrow E \longrightarrow C_2 \longrightarrow 1 \tag{D.29.1}$$

in which $E$ is torsion-free and its faithful outer action preserves a complete type-$(0,4)$ loop structure, while $E$ is the completion of no finite-type surface group, orientable or nonorientable. The extension is explicitly isomorphic to D.26's group. Its outer involution has a mixed-prime orientation multiplier and is conjugate to no subgroup of the geometric completion inside $\mathrm{Out}_{\mathrm{lac}}(\widehat{\pi}_{0,4})$. For this involution both the discretization and prediscretization fixed sets of F.1 are empty.

*Manuscript:* 28; 178–179; 352–354; 30; 194–196; 383–387; 31; 198–199 verso; 391–394.

*Proof.* Define units and an exponent by their prime coordinates:

$$\epsilon_3 = -1, \qquad \epsilon_q = 1 \ (q \neq 3), \qquad \kappa_2 = 1, \quad \kappa_q = 0 \ (q \neq 2), \qquad \lambda = -\epsilon. \tag{D.29.2}$$

Thus $\epsilon^2 = \lambda^2 = 1$ and $\epsilon\kappa = \kappa$. In particular $\lambda_3 = 1$ and $\lambda_q = -1$ for every other prime.

We first obtain the required pants isomorphism, without any finite-order requirement on it. The compatible assignments $\zeta_n \mapsto \zeta_n^{\lambda \bmod n}$ define an automorphism of the cyclotomic extension $\mathbb{Q}^{\mathrm{cyc}}/\mathbb{Q}$. An automorphism of an algebraic extension extends to its algebraic closure: extend its embedding along algebraic generators, or take a maximal extension of the embedding and adjoin a root of a missing minimal polynomial to prove maximality gives all of $\overline{\mathbb{Q}}$. We therefore choose an element $\sigma \in G_{\mathbb{Q}}$ with cyclotomic character $\lambda$. The Galois action on the three-punctured projective line commutes outerly with its rational $S_3$ action and preserves each cusp; on its local inertia it has exponent $\lambda$. Harbater–Schneps's normalized description [31, §0.2, Main Theorem (a), Theorem A] therefore supplies an element $(\lambda, f) \in GT_0$, with $f \in \widehat{F}(x,y)'$, whose automorphism is

$$x \longmapsto x^\lambda, \qquad y \longmapsto f(x,y)^{-1} y^\lambda f(x,y). \tag{D.29.3}$$

Its third cusp $(xy)^{-1}$ goes to a conjugate of its $\lambda$th power. This is the only property of $f$ needed below, besides invertibility of the displayed automorphism. Every choice of the extension $\sigma$ has these properties, and every one gives the construction and conclusions that follow.

Let $H = \widehat{F}(l,a,b)$ and define its closed free factors

$$P_1 = \widehat{F}(l,a), \qquad P_2 = \widehat{F}(l,b), \qquad L = \overline{\langle l \rangle}.$$

Their shared-factor universal property is $H = P_1 *_L P_2$: homomorphisms out of the two free factors agreeing on $l$ are exactly arbitrary continuous assignments of $l, a, b$. Give $P_1$ the ordered pants basis $(l^{-1}, a)$ and $P_2$ the ordered pants basis $(l, b)$. Formula (D.29.3), between these bases, defines a continuous isomorphism $\tau : P_1 \to P_2$ with

$$\begin{aligned} \tau(l) &= l^\epsilon, \\ \tau(a) &= f(l,b)^{-1} b^\lambda f(l,b), \\ \tau(a^{-1}l) &\sim_{P_2} (b^{-1}l^{-1})^\lambda. \end{aligned} \tag{D.29.4}$$

The symbol $\sim_{P_2}$ means conjugacy by an element of $P_2$; in particular it specifies a conjugacy in $H$. Because $\lambda^{-1} = \lambda$, the inverse isomorphism has the same peripheral multiplier.

Put $c = l^{\kappa}$ and prescribe

$$\alpha|_{P_1} = \tau, \qquad \alpha|_{P_2} = \mathrm{Inn}(c)\tau^{-1}. \tag{D.29.5}$$

Both maps send the shared generator $l$ to $l^{\epsilon}$: $\tau^{-1}(l) = l^{\epsilon}$ and $c$ commutes with $l$. They consequently extend to a continuous endomorphism of $H$. Direct substitution on the two free factors gives

$$\alpha(c) = c, \qquad \alpha^2 = \mathrm{Inn}(c). \tag{D.29.6}$$

For example, on $P_1$ the second iterate is $\mathrm{Inn}(c)\tau^{-1}\tau$; on $P_2$ it is $\mathrm{Inn}(\alpha(c))\tau\tau^{-1}$. These identities also prove invertibility, with inverse $\mathrm{Inn}(c^{-1})\alpha$.

Here is the full peripheral structure, with its relation:

$$\begin{aligned}(u_1, u_2, u_3, u_4) &= (a,\ a^{-1}l,\ b,\ b^{-1}l^{-1}),\\ u_1u_2u_3u_4 &= 1.\end{aligned} \tag{D.29.7}$$

The first three entries form a free profinite basis, since $a = u_1$, $l = u_1u_2$, and $b = u_3$ are its continuous inverse substitutions. Thus (D.29.7) is precisely a type-$(0, 4)$ presentation, not just four procyclic subgroups whose generators happen to satisfy a relation. Equations (D.29.4)–(D.29.5) give

$$\alpha(u_i) \sim_H u_{s(i)}^{\lambda}, \qquad s = (1\ 3)(2\ 4). \tag{D.29.8}$$

For the first two entries this is (D.29.4), and for the last two it is the inverse comparison, followed by conjugation by $c$. Hence the action preserves every required loop source, its conjugacy ambiguity, and the common orientation multiplier. It permutes distinct cusp classes nontrivially, so its outer class is nonidentity. Equation (D.29.6) gives that class exact order two.

On the profinite space $H \times \{0, 1\}$ define

$$(h, i)(k, j) = \big(h\alpha^i(k)c^{ij},\ i + j \bmod 2\big). \tag{D.29.9}$$

Associativity follows by the two carry identities in (D.29.6); checking the binary indices gives precisely $\alpha^2(k)c = ck$ and $\alpha(c) = c$. The element $t = (1, 1)$ satisfies $t^2 = c$ and $tht^{-1} = \alpha(h)$. The inclusion $h \mapsto (h, 0)$ and parity projection give (D.29.1), with the profinite product topology and continuous operations.

We make the comparison with D.26 explicit. Set $b' = \tau(a)$. The pair $(l^{\epsilon}, b')$ is a free basis of $P_2$, and replacing its first member by its $\epsilon$th power gives the free basis $(l, b')$. Thus $(l, a, b')$ is a free basis of $H$, and

$$\alpha(l) = l^{\epsilon}, \qquad \alpha(a) = b', \qquad \alpha(b') = cac^{-1}. \tag{D.29.10}$$

The map from D.26's named free basis to $(l, a, b')$, with $t \mapsto t$, is a continuous isomorphism of its entire twisted extension onto $E$. It is an isomorphism of extensions with identity map on $C_2$. Formula (D.29.7) therefore supplies the previously missing preserved peripheral system on that exact group. In particular D.26's necessary double-transposition and double-cover character conditions are attained.

For clarity, we give the torsion and finite-quotient arguments also in the present basis. The retraction $r : H \to \widehat{\mathbb{Z}}$ sends $l$ to 1 and $a, b$ to 0. Since both $\tau(a)$ and $\tau^{-1}(b)$ lie in the normal closures of the respective nonshared generators, $r\alpha = \epsilon r$. It extends to a retraction of $E$ onto the group $N = \widehat{\mathbb{Z}} \times \{0, 1\}$ with law (D.29.9), action $\epsilon$, and square parameter $\kappa$. An element $(z, 1)$ in $N$ has square $((1 + \epsilon)z + \kappa, 0)$, whose 2-coordinate is $2z_2 + 1$ and is never zero. Since its open subgroup $\widehat{\mathbb{Z}}$ is torsion-free, every torsion element outside it would have order two; hence $N$ is torsion-free. A torsion element of $E$ has trivial image in $N$, so has even parity and lies in the torsion-free $H$. It is therefore the identity.

Let $V = \mathbb{F}_3^2$, with basis $v_1, v_2$, and send

$$l \longmapsto v_1, \qquad a \longmapsto v_2, \qquad b \longmapsto -v_2. \tag{D.29.11}$$

Here $\lambda_3 = 1$ and $\epsilon_3 = -1$; conjugators vanish in the abelian target. The induced action of $\alpha$ is $-1$ on $V$, and $c$ has image zero because $\kappa_3 = 0$. Sending $t$ to the involution consequently extends (D.29.11) to a continuous epimorphism

$$E \longrightarrow V \rtimes_{-1} C_2. \tag{D.29.12}$$

The target has order eighteen and requires three generators. Two positive elements do not generate the quotient $C_2$; one positive and one negative element generate at most one line of $V$; two negative elements reduce to the preceding case by taking their product. The three displayed images do generate it. The independent program `haran_quotient_independent_audit.py` checks all $18^3$ associativity cases, all $18^2$ ordered pairs, all ordered triples, and the actual matrix identity in (D.29.11). The ordered-pair subgroup-size histogram is

| generated subgroup order | 1 | 2 | 3 | 6 | 9 |
|---|---|---|---|---|---|
| ordered pairs | 1 | 27 | 32 | 216 | 48. |

(D.29.13)

These calculations agree with the independently executed D.26 certificate under (D.29.10).

A free profinite $E$ of finite rank $d$ would have an index-two free subgroup of rank $1 + 2(d-1)$. The actual rank of $H$ is three, forcing $d = 2$, contrary to (D.29.12). Thus $E$ is not free. This excludes all punctured finite-type surface completions. For a putative isomorphism from a closed surface completion to $E$, the inverse image of $H$ is an open index-two subgroup. Its intersection with the dense surface group has index two, and completion of this intersection is exactly that open subgroup by the finite-index topology argument of D.1. It is the completion of the fundamental group of the corresponding closed double cover. Every aspherical closed surface completion has cohomological $p$-dimension two, whereas $\widehat{F}_3$ has dimension one. The spherical and projective-plane groups and their completions are finite and cannot give $H$. This excludes every closed case as well.

Finally A.2 supplies the continuous orientation homomorphism

$$\mathrm{or} : \mathrm{Out}_{\mathrm{lac}}(H) \longrightarrow \widehat{\mathbb{Z}}^{\times}.$$

It is unchanged by conjugation. Every discrete geometric class has value 1 or the global unit $-1$, and the closure of the geometric group still has values in that closed two-element subgroup. The value of $[\alpha]$ is the mixed unit $\lambda$ in (D.29.2), which belongs to neither value. No conjugate of $\langle [\alpha] \rangle$ is therefore contained in the geometric completion. The transporter formulas of D.16 and F.1 now give both empty fixed sets claimed above.

The cusp normalizers themselves cause no exception to this argument. Since $s$ fixes none of the four cusp classes, their normalizers in $E$ map trivially to $C_2$; each equals its normalizer in $H$, namely its own procyclic loop group by A.7. The failure is present even with these procyclic normalizers retained. □

**D.30. [Refuted] A closed genus-two counterexample to full profinite recognition.** The assertion that every torsion-free finite extension of a full profinite closed hyperbolic surface group is a completed surface group is false. There is an exact extension

$$1 \longrightarrow H = \widehat{\pi}_1(\Sigma_2) \longrightarrow E \longrightarrow C_2 \longrightarrow 1 \tag{D.30.1}$$

with $E$ torsion-free and with a faithful outer action, such that

$$H^2(E; \mathbb{F}_3) = \mathbb{F}_3, \qquad H^2(E; \mathbb{F}_5) = 0. \tag{D.30.2}$$

No profinite completion of a discrete surface group has these two cohomology groups. The exact maximal pro-$p$ quotients are

$$E^{(2)} \cong \widehat{\pi_1(N_3)}^{(2)}, \qquad E^{(3)} \cong \mathbb{Z}_3^2, \qquad E^{(p)} \cong \widehat{F}_2^{(p)} \quad (p \neq 2,3), \tag{D.30.3}$$

where $N_3$ is the closed nonorientable surface of genus three. Thus the counterexample also gives the precise maps through which its separate pro-$p$ surface presentations arise. Its outer involution is conjugate to no element of the extended geometric completion. Its fixed discretization and prediscretization sets are both empty.

*Manuscript:* 28; 178 verso; 353; 24; 147 verso; 293; Part D(d) of the task.

*Proof.* We use the commutator convention $[x,y] = xyx^{-1}y^{-1}$. Set

$$\epsilon_3 = -1, \quad \epsilon_p = 1 \ (p \neq 3), \qquad \kappa_2 = 1, \quad \kappa_p = 0 \ (p \neq 2), \qquad \lambda = -\epsilon \tag{D.30.4}$$

in $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. These elements satisfy $\epsilon^2 = \lambda^2 = 1$ and $\epsilon\kappa = \kappa$.

*An actual punctured-torus isomorphism.* Take the elliptic curve $Y^2 = 4X^3 - 4X$ over $\mathbb{Q}$, with its origin removed and its rational Weierstrass tangential base point. Nakamura's full profinite construction [57, §1, equation (1.2), and §2.5, p. 424] gives an actual homomorphism

$$G_{\mathbb{Q}} \longrightarrow \mathrm{Aut}\widehat{\langle x,y,z \mid [x,y]z = 1\rangle}, \qquad \sigma(z) = z^{\chi(\sigma)}. \tag{D.30.5}$$

Here $\chi$ is the full cyclotomic character. The local coordinate at the rational puncture and its Kummer covers identify its inertia with $\widehat{\mathbb{Z}}(1)$, which explains the exact power equation. Every finite cyclotomic extension has Galois group $(\mathbb{Z}/n)^\times$; the resulting surjections and compactness show that $\chi : G_{\mathbb{Q}} \to \widehat{\mathbb{Z}}^\times$ is onto. Choose $\sigma$ with $\chi(\sigma) = \lambda$. Renaming the target generators gives a continuous isomorphism

$$\tau : F_1 = \widehat{F}(x,y) \xrightarrow{\sim} F_2 = \widehat{F}(u,v), \qquad \tau([x,y]) = [u,v]^\lambda. \tag{D.30.6}$$

This uses a single Galois element of prescribed multiplier. The construction below proves its outer involution directly from $\tau$ and $\tau^{-1}$.

*The automorphism of the closed surface group.* Use its defining profinite presentation

$$H = \widehat{\langle x,y,u,v \mid [x,y][u,v] = 1\rangle}, \qquad l = [x,y]^{-1} = [u,v], \qquad c = l^\kappa. \tag{D.30.7}$$

Define $\alpha$ on the first pair by the two words of $\tau$, and on the second pair by the two words of $\mathrm{Inn}(c)\tau^{-1}$. Equation (D.30.6) and $\lambda^{-1} = \lambda$ give

$$\alpha([x,y]) = l^\lambda, \qquad \alpha([u,v]) = c\,[x,y]^\lambda c^{-1} = l^{-\lambda}.$$

Their product is one, so this substitution defines a continuous endomorphism of the presented group $H$. It gives $\alpha(l) = l^\epsilon$ and $\alpha(c) = c$. Substitution on each of the four generators, using the actual inverse $\tau^{-1}$, then gives

$$\alpha^2 = \mathrm{Inn}(c). \tag{D.30.8}$$

Consequently $\mathrm{Inn}(c^{-1})\alpha$ is its inverse. These identities use the universal property of the displayed presentation, so no assertion about efficiency of a profinite amalgam is required.

Define $E$ on the compact space $H \times \{0,1\}$ by

$$(h,i)(k,j) = \big(h\alpha^i(k)c^{\lfloor (i+j)/2 \rfloor}, i + j \bmod 2\big). \tag{D.30.9}$$

For quotient labels $(1,1,0)$ associativity is precisely $\alpha^2(k)c = ck$; for $(1,1,1)$ it also uses $\alpha(c) = c$. The other six label triples follow by direct substitution. The identity is $(1,0)$; the inverse of

$(h,0)$ is $(h^{-1},0)$, and that of $(h,1)$ is $(\alpha^{-1}(h^{-1}c^{-1}),1)$. All operations are continuous. Thus $E$ is a profinite group and its projection gives (D.30.1). Write $t=(1,1)$, so that $tht^{-1}=\alpha(h)$ and $t^2=c$.

*An exact presentation and torsion-freeness.* Set $u'=\tau(x)$ and $v'=\tau(y)$. This is a free basis of the formal second punctured-torus group, with

$$[u',v']=[u,v]^{\lambda}=l^{\lambda},\qquad u'=txt^{-1},\quad v'=tyt^{-1}.$$

The relations for $t$ on $u',v'$ follow from $t^2=c$. Eliminating these two generators therefore gives the exact profinite presentation

$$E\cong\langle x,y,t\mid tlt^{-1}=l^{\epsilon},\quad t^2=l^{\kappa}\rangle\widehat{\ },\qquad l=[x,y]^{-1}. \tag{D.30.10}$$

For completeness, the homomorphism to (D.30.9) sends $x,y,t$ to the named elements and is onto because their conjugates $u',v'$ generate the second factor. In the reverse direction, recover $u,v$ by applying the words of $\tau^{-1}$ to $txt^{-1},tyt^{-1}$. Their commutator is $[txt^{-1},tyt^{-1}]^{\lambda}=l$, by (D.30.10). Thus the surface relator holds, and the conjugation relations on both pairs follow from $t^2=c$ and the defining words of $\alpha$. The two maps fix all named generators and are inverse.

The universal property of the maximal pro-2 quotient substitutes $\epsilon_2=\kappa_2=1$ into (D.30.10). Its first relation follows from the second, giving

$$E^{(2)}\cong\langle x,y,t\mid [x,y]t^2=1\rangle_{\text{pro-2}}. \tag{D.30.11}$$

This is the rank-three Demushkin presentation with $q=2$ and $f=\infty$: cyclically move $t^2$ to the beginning and replace $x,y$ by their inverses to match the commutator convention of [73, §1.4 and Theorem 1.9]. The converse in that theorem proves the Demushkin assertion. Its abelianization is $\mathbb{Z}_2^2\times C_2$, so it is infinite. The same source's preceding cohomological-dimension statement gives $\mathrm{cd}_2(E^{(2)})=2$. A subgroup of order two has infinite 2-cohomological dimension, so $E^{(2)}$ is torsion-free.

There is also an explicit free-basis identification with the three-crosscap word. The substitution

$$x\mapsto ad^{-1}b^{-1}a^{-1},\qquad y\mapsto abdaba^{-1},\qquad t\mapsto abd \tag{D.30.12}$$

sends $[x,y]t^2$ to $a^2b^2d^2$. Its inverse is

$$a\mapsto xt,\qquad b\mapsto t^{-1}x^{-1}t^{-1}yxt,\qquad d\mapsto t^{-1}x^{-1}y^{-1}t^2.$$

Free reduction checks the relator and both composites. The maps extend to mutually inverse pro-2 isomorphisms, proving the first comparison in (D.30.3).

The parity quotient $E\to C_2$ factors through $E^{(2)}$ and sends $t$ to its nonidentity element and $x,y$ to the identity. Every finite-order element of $E$ maps trivially to the torsion-free $E^{(2)}$, hence has trivial parity and lies in $H$. The latter is torsion-free by A.7. This proves torsion-freeness of $E$ itself and, in particular, the absence of a section of (D.30.1).

For an odd prime $p$, (D.30.10) gives $t^2=1$ in $E^{(p)}$, so $t=1$. At $p=3$ the remaining relation is $l=l^{-1}$, which forces $l=1$, and the quotient is $\langle x,y\mid [x,y]=1\rangle_{\text{pro-3}}=\mathbb{Z}_3^2$. At every other odd prime the remaining relation is the identity $l=l$. The universal properties in both directions give the free pro-$p$ group on $x,y$. This proves all of (D.30.3).

*The orientation line and the obstruction to every surface.* Let $A\in\mathrm{GL}_2(\widehat{\mathbb{Z}})$ be the abelianization matrix of $\tau$. In the finite Heisenberg group modulo $n$, the commutator of two elements is the central generator raised to the determinant of their two abelianization columns. Applying (D.30.6) there gives $\det A=\lambda \pmod n$ for every $n$, hence $\det A=\lambda$ in $\widehat{\mathbb{Z}}$. On the basis $([x],[y],[u],[v])$ of $H^{\mathrm{ab}}$, inner conjugation disappears and the matrix of $\alpha$ is

$$M=\begin{pmatrix}0 & A^{-1}\\ A & 0\end{pmatrix}.$$

For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, the identity $A^t J A = (\det A) J$ gives

$$M^t \operatorname{diag}(J, J) M = \lambda \operatorname{diag}(J, J), \tag{D.30.13}$$

because $\lambda^{-1} = \lambda$. The companion identity $M \operatorname{diag}(J, J) M^t = \lambda \operatorname{diag}(J, J)$ also holds by $AJA^t = (\det A)J$. Since the matrix of $\alpha^*$ on degree-one cohomology is $M^t$, this last identity gives the multiplier of its cup pairing with all dual maps specified. The completed surface cup-product comparison of A.2 is natural for automorphisms. Its nondegenerate degree-one pairing therefore shows that $\alpha^*$ acts by $\lambda_p$ on the one-dimensional $H^2(H; \mathbb{F}_p)$. The matrix $M$ exchanges two nonzero direct summands, so $\alpha$ is not inner. Equation (D.30.8) thus gives a faithful outer $C_2$ action.

The orientation-line character is a continuous homomorphism

$$\operatorname{Out}(H) \longrightarrow \widehat{\mathbb{Z}}^{\times}.$$

Its restriction to the extended geometric mapping class group has image the two diagonal elements $+1, -1$, and continuity gives the same bound for its closure. Conjugation in $\operatorname{Out}(H)$ preserves this character because its target is abelian. Our element has value $\lambda$, with $\lambda_3 = 1$ and $\lambda_5 = -1$, so no conjugate lies in that closure. D.16's exact transporter formulas therefore give empty fixed discretization and prediscretization sets for this actual involution.

For odd $p$, averaging over $C_2$ makes all its higher cohomology with $\mathbb{F}_p$-coefficients vanish. The continuous Hochschild–Serre spectral sequence of (D.30.1) consequently gives

$$H^2(E; \mathbb{F}_p) \xrightarrow[\text{restriction}]{\sim} H^2(H; \mathbb{F}_p)^{C_2}. \tag{D.30.14}$$

At $p = 3$ the scalar is $\lambda_3 = 1$, while at $p = 5$ it is $\lambda_5 = -1$. This proves (D.30.2).

Finally, the completion of a closed orientable surface group of positive genus has one-dimensional $H^2(-; \mathbb{F}_p)$ at every prime, by the surface comparison of A.2 (including its torus calculation). A closed nonorientable surface of genus at least two has an orientable double cover. Exact completion at index two and the same restriction map (D.30.14) compute its degree-two cohomology at odd primes as the invariants of the orientation reversal, hence zero. The sphere group is trivial and the projective-plane group is $C_2$, so both have zero degree-two cohomology at odd primes. A surface with boundary or a noncompact surface has a free fundamental group, whose profinite completion has cohomological dimension at most one. These cases exhaust the surface groups and exclude the pair of values in (D.30.2).

The independently executed program `code/part_b/closed_extension_d30_audit.py` checks the free-word isomorphism in (D.30.12), the intersection matrices and the finite cochain computation of the two invariant lines. The profinite construction, presentation comparisons and torsion-freeness are proved above; the finite computations verify the stated identities. □

**D.31. [Proved] Abstract finite groups in the geometric completions of genus at most two.** Let $M^+_{g,n} = \Gamma^+_{g,n}$ allow permutations of the $n$ punctures, and let $M^{\pm}_{g,n}$ also allow orientation reversal. Every finite nonsolvable subgroup of $\widehat{M^+_{0,n}}$ is isomorphic to $A_5$. Its permutation action on the punctures has

$$n = 12a + 20b + 30c + 60m, \qquad a, b, c \in \{0, 1\}, \quad m \geq 0, \tag{D.31.1}$$

where the summands are the actual orbit sizes. In particular, it has at least twelve punctures. Every finite nonsolvable subgroup of $\widehat{M^{\pm}_{0,n}}$ is isomorphic to $A_5$ or $A_5 \times C_2$. Every finite subgroup of $\widehat{M^{\pm}_{g,n}}$ for $g = 1, 2$ is solvable.

These statements determine the abstract groups and, in the positive sphere case, their puncture-permutation representations. The surjectivity of the map from geometric $A_5$ conjugacy

classes to completed $A_5$ conjugacy classes, and the completed $A_5$ centralizer comparison of F.7, remain to be proved.

*Manuscript:* §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D(b): finite subgroups of the geometric completion and full completion comparison.

*Proof.* For $n = 0, 1, 2$ the positive sphere mapping class groups are respectively $1, 1, C_2$; including orientation reversal gives groups of order at most four. Their completions are the same finite solvable groups. The sphere arguments below therefore take $n \geq 3$.

We give the completed exact sequences first, so that every use of a subgroup or quotient has its specified completion map.

*A completion and configuration-group preparation.* For the forgetting sequences used below, the kernel $N$ is finitely generated and the quotient $Q$ is good. These properties imply that $\widehat{N} \to \widehat{E}$ is injective. Here is the required argument. Take a characteristic finite-index subgroup $N_0$ of $N$; these are cofinal because $N$ is finitely generated. In $E/N_0$ the kernel $B = N/N_0$ is finite. Its centralizer has finite index and fits into the central extension

$$1 \longrightarrow Z(B) \longrightarrow C_{E/N_0}(B) \longrightarrow Q_0 \longrightarrow 1,$$

where $Q_0$ has finite index in $Q$. Goodness passes to $Q_0$. The class in $H^2(Q_0, Z(B))$ is therefore represented by a continuous cohomology class of $\widehat{Q}_0$, hence is inflated from a finite quotient. Restriction to the kernel of that finite quotient kills it. The central extension splits over this finite-index subgroup; a chosen complement meets $B$ trivially and has finite index in $E/N_0$. Its core is a finite-index normal subgroup still meeting $B$ trivially. Pulling this core back to $E$ gives a finite quotient whose restriction to $N$ has kernel exactly $N_0$. This proves the full induced topology and thus

$$1 \longrightarrow \widehat{N} \longrightarrow \widehat{E} \longrightarrow \widehat{Q} \longrightarrow 1. \tag{D.31.2}$$

The last equality follows directly from the finite quotients of $E/N$.

The braid groups on the plane, on finitely punctured surfaces, and on the closed surfaces of genus one or two used here are good and have finite cohomological dimension. We recall how the ingredients apply here. The ordered configuration-space fibrations of Fadell–Neuwirth [21, Theorems 1 and 3] forget one point and have fibre the surface with finitely many points removed. The base surface and the fibres are aspherical; hence their configuration spaces are aspherical by the homotopy exact sequence. The fibres' groups are finitely generated free groups once a point has been removed. Starting with the good surface group, the preceding finite-kernel argument and the Lyndon–Hochschild–Serre comparison spectral sequences prove goodness inductively. The required cohomology of each fibre with finite coefficients is finite. For the plane the initial group is trivial; for the closed surfaces its goodness is the usual surface-group goodness also used in [11]. The unordered group has the ordered one as a finite-index subgroup, so goodness follows by the finite-index comparison and Shapiro's lemma. The configuration spaces are finite-dimensional aspherical manifolds, which give finite cohomological dimension. Goodness therefore makes each completed braid group torsion-free: its cohomology with every finite $p$-primary coefficient module vanishes above that dimension, whereas a subgroup $C_p$ has infinite $p$-cohomological dimension. This also follows directly from the prime-subgroup comparison in [11, Theorem A].

*Finite subgroups of a completed puncture stabilizer are cyclic.* For $n \geq 4$, capping the boundary of a disc by a once-punctured disc gives the exact sequence

$$1 \longrightarrow \langle \Delta^2 \rangle \cong \mathbb{Z} \longrightarrow B_{n-1} \longrightarrow (M^+_{0,n})_p \longrightarrow 1. \tag{D.31.3}$$

Here the distinguished puncture is $p$, and the kernel is the central full twist; this is [6, Lemma 2.2], whose proof specifies the capping map. Braid abelianization sends each Artin generator to 1 and $\Delta^2$ to $(n-1)(n-2)$. Thus the composite

$$\widehat{\mathbb{Z}} \longrightarrow \widehat{B}_{n-1} \longrightarrow \widehat{\mathbb{Z}}$$

is multiplication by this nonzero integer and is injective. Consequently completion of (D.31.3) is the exact central extension

$$1 \longrightarrow \widehat{\mathbb{Z}} \longrightarrow \widehat{B}_{n-1} \longrightarrow \widehat{(M^+_{0,n})_p} \longrightarrow 1. \tag{D.31.4}$$

The last group identifies with the stabilizer of $p$ in $\widehat{M^+_{0,n}}$: its discrete source has finite index, and stabilizing $p$ is the inverse image of a subgroup in the finite puncture-permutation quotient.

Let $K$ be a finite subgroup of that completed stabilizer and let $E_K$ be its inverse image in $\widehat{B}_{n-1}$. The subgroup $A = \widehat{\mathbb{Z}}$ is central and open in $E_K$, of index $d = |K|$. Continuous transfer over its finitely many cosets gives

$$\mathrm{Ver} : E_K \longrightarrow A, \qquad \mathrm{Ver}(a) = da \quad (a \in A). \tag{D.31.5}$$

Its kernel intersects $A$ trivially, so injects into the finite group $E_K/A$. It is finite, and is trivial because $E_K$ is a subgroup of the torsion-free completed braid group. Transfer is therefore a continuous injection into $\widehat{\mathbb{Z}}$. Its compact image is a closed procyclic subgroup. Hence $E_K$, and then its finite quotient $K$, are procyclic and cyclic respectively. For $n = 3$ the same capping map is $1 \to 2\mathbb{Z} \to B_2 = \mathbb{Z} \to C_2 \to 1$, and the conclusion is immediate. For $n = 1, 2$ the positive point stabilizer is trivial.

It follows in particular that a finite subgroup of a completed positive sphere mapping class group preserving a specified set of $1 \le b \le 3$ punctures is solvable. The kernel of its permutation map to $S_b$ fixes a puncture and is cyclic, while $S_b$ is solvable. This assertion also applies to every finite-index decorated subgroup through its actual inclusion into the completed sphere group.

*The prime-power inputs and elimination of a solvable radical.* The good-group prime-power theorem [11, Theorem A] conjugates each finite $p$-subgroup of $\widehat{M^+_{0,n}}$ to a geometric one and gives its completed normalizer. The checked prime-power topology argument in F.7 identifies that normalizer with the full completion of its discrete normalizer. Thus, for a nontrivial normal $p$-subgroup $A$ of a finite $G \le \widehat{M^+_{0,n}}$, conjugate $A$ into $M^+_{0,n}$ and use the exact covering-normalizer sequence

$$1 \longrightarrow A \longrightarrow N_{M^+_{0,n}}(A) \longrightarrow Q_A \longrightarrow 1. \tag{D.31.6}$$

The group $Q_A$ is a finite-index decorated subgroup of the mapping class group of the quotient sphere. Its specified branch set has two points for a cyclic rotation group, and three for a noncyclic rotation group. This branch set includes the images of ramification points that were already punctures upstairs; the ramification indices are retained in the covering decoration. Completion of (D.31.6) and the normalizer comparison give an actual injection

$$G/A \hookrightarrow \widehat{Q}_A. \tag{D.31.7}$$

It preserves that set of two or three branch points. The puncture stabilizer argument makes $G/A$ solvable, and then $G$ is solvable.

For a nontrivial finite solvable normal subgroup $R$ of $G$, the last nonzero derived term of $R$ is abelian. Its nontrivial $p$-primary component $A$ is characteristic in $R$, hence normal in $G$. The preceding argument again makes $G$ solvable. Consequently a nonsolvable finite $G$ has trivial solvable radical. Only the prime-power comparison was used in this step.

The same prime-power theorem and the finite rotation-group list give cyclic odd Sylow subgroups of $G$, and cyclic or dihedral Sylow 2-subgroups, including $C_2^2$ in the latter case. There is also the precise normalizer constraint

$$\mathrm{im}\big(N_G(C_p) \longrightarrow \mathrm{Aut}(C_p)\big) \subset \{1, -1\}. \tag{D.31.8}$$

Indeed normalize a geometric rotation. A representative of a discrete normalizer either fixes its two poles or exchanges them. The oriented local rotation numbers are opposite, so the induced

automorphism is respectively 1 or $-1$. The normalizer comparison preserves this image because $\mathrm{Aut}(C_p)$ is finite. The same argument with the local orientation sign proves (D.31.8) in the extended geometric completion as well. Equivalently one can apply the prime-power normalizer theorem to the good index-two extended mapping class group; the discrete representatives still induce only these two exponents.

We record the elementary finite-group reduction needed below. A finite group all of whose Sylow subgroups are cyclic is solvable. For the least prime $p$ dividing its order, its cyclic Sylow group $P$ has $N(P)/C(P)$ of order both prime to $p$ and dividing $p^{a-1}(p-1)$, where $|P| = p^a$. Minimality of $p$ forces this quotient to be trivial. Transfer to $P$ has restriction $x \mapsto x^{[G:P]}$ and is onto. To check this formula without a fusion assumption, any two elements of the abelian Sylow group $P$ conjugate in $G$ are conjugate by $N(P)$: the two Sylow subgroups containing the conjugate element are conjugate inside its centralizer. Thus they are equal when $P \le Z(N(P))$. The coset-cycle formula for transfer now gives the stated exponent. Its kernel is a normal $p$-complement. Induction on the group order proves solvability. It follows that every odd-order subgroup here is solvable, and that a nonsolvable $G$ has a noncyclic dihedral Sylow 2-subgroup.

*The verified Gorenstein–Walter application.* Let $S$ be such a Sylow 2-subgroup and $z \in Z(S)$ an involution. The quotient of $C_G(z)$ by $\langle z \rangle$ in (D.31.7) preserves the two branch points of the rotation quotient. Its pointwise branch stabilizer is cyclic. Its inverse image $C_0 \le C_G(z)$ has index at most two and is a central extension of a finite cyclic group by $\langle z \rangle$, hence is abelian: a lift of a quotient generator and the central kernel generate it. The odd-order part of $C_0$ is characteristic in $C_0$, normal in $C_G(z)$, and cyclic because all odd Sylow subgroups are cyclic. It is a normal abelian 2-complement of $C_G(z)$.

We can therefore apply the original, precisely matching theorem [25, Theorem I, p. 553]. It states that $G$ has a normal odd-order subgroup $K$ such that $G/K$ is one of

$$\mathrm{PSL}_2(q), \quad \mathrm{PGL}_2(q) \quad (q \text{ odd}), \qquad A_7, \qquad \text{a Sylow 2-subgroup.} \tag{D.31.9}$$

The source explicitly includes $C_2^2$ among dihedral groups. All its hypotheses have just been verified. The odd group $K$ is solvable by the preceding transfer argument, so the triviality of the solvable radical gives $K = 1$.

The group $A_7$ contains the commuting independent cycles (1 2 3) and (4 5 6), contradicting cyclic odd Sylow subgroups. A 2-group is solvable. Write $q = p^f$ in either matrix case. The unipotent subgroup $u(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ for $t \in \mathbb{F}_q$ is elementary abelian of order $p^f$, so $f = 1$. In $\mathrm{PSL}_2(p)$ the diagonal matrices $\mathrm{diag}(a, a^{-1})$ act on this $C_p$ by $u(t) \mapsto u(a^2 t)$, supplying $(p-1)/2$ distinct automorphisms. In $\mathrm{PGL}_2(p)$ the matrices $\mathrm{diag}(a, 1)$ supply all $p-1$ automorphisms. Constraint (D.31.8) leaves $p \le 5$ in the first case and $p \le 3$ in the second. At $p = 3$, the faithful projective-line action embeds these groups into $S_4$, so they are solvable. The remaining group is $\mathrm{PSL}_2(5) \cong A_5$.

For this last isomorphism the exact map is conjugation on the five Sylow 2-subgroups. The supplied matrix enumeration finds fifteen involutions, partitions them into the five four-groups, and verifies that the conjugation image has order sixty and consists of even permutations. With the recorded ordering of the five subgroups, the matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

map respectively to (1 2 4 3 5) and (2 4)(3 5). Thus the map is injective and its image is exactly $A_5$, with its sources and targets fixed.

*The puncture representation.* The pure positive sphere group is the ordered configuration group of $n-3$ points on the three-punctured sphere: fix the first three marked points and use their contractible identity homeomorphism component in the evaluation fibration. It is good of

finite cohomological dimension by the preceding configuration argument, so its completion is torsion-free. This is exactly the kernel of the completed map to $S_n$, by the finite-index completion sequence. Thus every finite subgroup acts faithfully on the puncture set.

For a finite $A_5$ subgroup, each puncture stabilizer is cyclic by (D.31.4)–(D.31.5). The cyclic subgroups of $A_5$ have orders $1, 2, 3, 5$, so its transitive puncture orbits are precisely $A_5/1$, $A_5/C_2$, $A_5/C_3$, and $A_5/C_5$. The last three have sizes $30, 20, 12$. For each $d = 2, 3, 5$, an element generating $C_d$ fixes $|N_{A_5}(C_d)/C_d| = 2$ points in the corresponding orbit and none in the other three types. Prime-power conjugacy makes its puncture permutation that of a rotation of the sphere, which has at most two fixed punctures. Each exceptional orbit occurs at most once. This proves (D.31.1), including its actual permutation representation, and gives the lower bound twelve.

*The extended sphere group.* Let $G \leq \widehat{M^{\pm}_{0,n}}$ be finite and nonsolvable. Its positive subgroup $G^+$ has index at most two and is nonsolvable, so is $A_5$. The centralizer $C_G(A_5)$ intersects $A_5$ trivially and injects into $G/G^+$, so has order at most two. With a nontrivial centralizer, its generator is negative and $G = A_5 \times C_2$. With a trivial centralizer, conjugation embeds $G$ into $\mathrm{Aut}(A_5) \cong S_5$ with image containing the inner $A_5$, leaving $A_5$ or $S_5$. The automorphism isomorphism is induced by the action on the five subgroups $A_4 \leq A_5$. An automorphism fixing all five makes $g$ and its image induce the same permutation of them for every $g \in A_5$; the faithful natural action of $A_5$ then makes that automorphism the identity. Conjugation by $S_5$ is faithful as well and supplies one hundred twenty automorphisms, proving the isomorphism. In $S_5$, the normalizer of a 5-cycle acts on it by all four nonzero exponents modulo five. This violates the extended version of (D.31.8). Therefore only $A_5$ and $A_5 \times C_2$ occur as abstract nonsolvable groups.

*Forgetting punctures in genus two.* Evaluation on the unordered puncture set, with isotopy extension, gives the discrete forgetting sequence

$$1 \longrightarrow B_n(S_2) \longrightarrow M^+_{2,n} \longrightarrow M^+_{2,0} \longrightarrow 1. \tag{D.31.10}$$

The identity homeomorphism component of $S_2$ is contractible, as used in Part C, so there is no preceding evaluation-loop kernel. The quotient is good, the braid kernel is finitely generated, and (D.31.2) completes this sequence exactly. The completed kernel is torsion-free by the configuration argument. Every finite subgroup of $\widehat{M^+_{2,n}}$ therefore injects under forgetting into $\widehat{M^+_{2,0}}$. The hyperelliptic quotient is the exact sequence

$$1 \longrightarrow C_2 \longrightarrow M^+_{2,0} \longrightarrow M^+_{0,6} \longrightarrow 1; \tag{D.31.11}$$

see [6, Propositions 3.2–3.3]. The central involution is detected on homology modulo three, so it injects into the completion. Finite-kernel completion then makes (D.31.11) exact with hats. Every finite subgroup has solvable image in $\widehat{M^+_{0,6}}$ by (D.31.1), and kernel contained in $C_2$. It is solvable. The same holds in the extended group because the positive subgroup has index at most two. The case $n = 0$ uses (D.31.11) directly.

*Forgetting punctures in genus one.* For $n \geq 1$, evaluation gives

$$1 \longrightarrow N_n \longrightarrow M^+_{1,n} \longrightarrow \mathrm{SL}_2(\mathbb{Z}) \longrightarrow 1, \qquad N_n = B_n(T^2)/\mathbb{Z}^2. \tag{D.31.12}$$

Here $\mathbb{Z}^2$ is the image of the translation loops in the identity homeomorphism component, identified by evaluation as in C.9. It is central in the braid group: applying a loop of translations to every configuration of a braid gives the square homotopy interchanging the two loop products. Its injection can be checked on the ordered configuration space by evaluating the first strand. To verify its full topology in the unordered braid group, sum the homology classes of all strand paths:

$$B_n(T^2) \longrightarrow H_1(T^2, \mathbb{Z}) = \mathbb{Z}^2.$$

Endpoints are permuted, so the sum is a cycle and gives a homomorphism. On diagonal translation loops it is multiplication by $n$. The resulting completed injection proves the exact central extension

$$1 \longrightarrow \widehat{\mathbb{Z}}^2 \longrightarrow \widehat{B_n(T^2)} \longrightarrow \widehat{N}_n \longrightarrow 1. \tag{D.31.13}$$

The completed braid group is torsion-free by the configuration argument. For completeness the ordered torus configuration has the explicit product coordinates

$$(x_1, \dots, x_n) \longmapsto (x_1, x_2 - x_1, \dots, x_n - x_1) \in T^2 \times \mathrm{Conf}_{n-1}(T^2 \setminus \{0\}),$$

so its goodness also follows directly from that of $\mathbb{Z}^2$ and the iterated punctured-surface configuration groups.

For a finite $K \leq \widehat{N}_n$, pull it back in (D.31.13). Transfer to the central open $\widehat{\mathbb{Z}}^2$ has restriction $a \mapsto |K|a$. Its kernel again injects into $K$ and is finite, so torsion-freeness makes transfer injective. The pullback, and hence $K$, is abelian. Since $\mathrm{SL}_2(\mathbb{Z})$ is good and $N_n$ is finitely generated, (D.31.2) also completes (D.31.12) exactly. The image of every finite subgroup lies in a finite subgroup of $\widehat{\mathrm{SL}_2(\mathbb{Z})}$, which is cyclic of order $1, 2, 3, 4$, or $6$ by D.8's virtually free comparison. The finite kernel is abelian by (D.31.13), proving solvability. For orientation reversal the positive subgroup has index at most two, which preserves this conclusion. For $n = 0$ use D.8 directly with $\mathrm{GL}_2(\mathbb{Z})$.

The executed program `code/part_a/sphere_completion_restriction.py` checks the projective-matrix automizer calculations at $p = 3, 5, 7, 11, 13$, constructs the displayed $\mathrm{PSL}_2(5) \to A_5$ isomorphism, compares the $C_5$ normalizer actions in $A_5$ and $S_5$, and checks the exceptional coset fixed-point counts and the $C_3^2$ subgroup of $A_7$. The general reductions are the completed exact sequences and transfer arguments above. □

**D.32. [Proved] The three-cusp rank test and finite pure loop groups.** Let $F = \widehat{F}(x, y)$, with the three named peripheral generators $c_1 = x$, $c_2 = y$, and $c_3 = (xy)^{-1}$. Let $P$ be the subgroup of $\mathrm{Out}(F)$ preserving each of their three procyclic conjugacy classes. Its common peripheral multiplier is denoted by $\lambda$. Every finite subgroup of $P$ maps injectively, by $\lambda_2$, to $\{1, -1\} \leq \mathbb{Z}_2^\times$. In particular it has order at most two, and a nonidentity element of finite order has order two and $\lambda_2 = -1$. These statements hold for $GT_0 \leq P$ as well.

For the full three-cusp loop-outer group, allowing puncture permutations, every finite subgroup $G$ has the explicit faithful representation

$$G \longrightarrow S_3 \times C_2, \qquad g \longmapsto (\sigma_g, \lambda_2(g)). \tag{D.32a}$$

This is an assertion about the displayed homomorphism and abstract finite groups. Conjugacy of its original action to the geometric action with this image is not asserted here. D.34 determines the global multiplier of a surviving involution; its geometric conjugacy remains to be proved.

*Manuscript:* 30; folios 194–196; scans 383–387; 31; folios 198–199 verso; scans 391–394; Part D(b) of the task.

*Proof.* We first prove the closed-subgroup fact which uses all three cusps. For a prime $p$, let $e_p \in \widehat{\mathbb{Z}}$ have $p$-coordinate one and all other coordinates zero. Suppose a closed subgroup $C \leq F$ has $\dim_{\mathbb{F}_p} H^1(C, \mathbb{F}_p) = 2$ and contains, for each $i = 1, 2, 3$, a conjugate

$$g_i c_i^{e_p \mathbb{Z}_p} g_i^{-1}. \tag{D.32b}$$

We prove $C = F$.

The comparison with open subgroups is the actual restriction map

$$\operatorname*{colim}_{U \text{ open},\ C \leq U \leq F} H^1(U, \mathbb{F}_p) \xrightarrow{\sim} H^1(C, \mathbb{F}_p), \tag{D.32c}$$

where the transition maps are restrictions as $U$ decreases. Indeed a continuous character of $C$ kills $C \cap V$ for some open normal $V \triangleleft F$. It extends to $CV$ by the quotient $CV/V = C/(C \cap V)$. Conversely a character on $U$ which vanishes on $C$ vanishes after restriction to its open kernel, which contains $C$. This proves both surjectivity and injectivity of (D.32c).

If $C$ is open, finite-coset Schreier rewriting identifies it with a free profinite group of rank $1 + [F : C]$. Its given first-cohomology dimension therefore forces $[F : C] = 1$. Suppose instead that $C$ has infinite index. The $p$-valuations of the finite indices $[F : U]$, for open $U$ containing $C$, must be unbounded. To prove this, assume they were bounded and choose $U_0$ attaining their maximum. For $C \leq V \leq U_0$, every further restriction between such open subgroups has prime-to-$p$ index and is injective on $H^1(-, \mathbb{F}_p)$: a nonzero character has an index-$p$ kernel and cannot vanish on a subgroup of prime-to-$p$ index. Thus (D.32c) contains the first cohomology of every such $V$. The indices $[F : V]$ are unbounded, since their intersection is $C$ and $[F : C] = \infty$. Their Schreier dimensions $1 + [F : V]$ are therefore unbounded, a contradiction.

Choose an open $U \geq C$ for which $p \mid [F : U]$. Put $\Delta = F(x, y)$, the named dense discrete subgroup of $F$, and $U_0 = U \cap \Delta$. The finite covering of the pair of pants defined by $U_0 \leq \Delta$ has degree $n = [F : U]$, and its completed fundamental group is exactly $U$: finite-index subgroups of a discrete group carry the full induced profinite topology. For each $g_i$ choose $h_i \in \Delta$ in its left $U$-coset and write $g_i = u_i h_i$, with $u_i \in U$. The subgroup

$$U \cap g_i \overline{\langle c_i \rangle} g_i^{-1} = u_i \overline{\langle h_i c_i^{d_i} h_i^{-1} \rangle} u_i^{-1} \tag{D.32d}$$

is consequently the completion, up to the displayed inner conjugation, of a particular boundary subgroup in this finite cover. Here $d_i$ is its boundary covering degree. Inclusion (D.32b) proves $p \nmid d_i$.

The three selected boundaries lie over the three different original boundaries, so they are distinct. There is at least one more boundary. Otherwise each original boundary would have just one lift, and its degree would be $n$, contrary to $p \nmid d_i$ and $p \mid n$. If the cover has genus $h$ and $b \geq 4$ boundary components, its presentation is

$$\left\langle a_1, b_1, \ldots, a_h, b_h, d'_1, \ldots, d'_b \;\middle|\; \prod_{j=1}^{h} [a_j, b_j]\, d'_1 \cdots d'_b = 1 \right\rangle.$$

Thus the only relation between its boundary homology classes over $\mathbb{F}_p$ is their sum. The three selected classes are independent, and evaluation on them gives a surjection $H^1(U, \mathbb{F}_p) \to \mathbb{F}_p^3$.

This remains a surjection after restriction of these characters to $C$ and evaluation on the three elements in (D.32b). To check the comparison explicitly, the element $g_i c_i^{e_p} g_i^{-1}$ is the $e_p/d_i$ power of the boundary generator in (D.32d). Inner conjugation does not change a character, and its evaluation is therefore multiplied by the nonzero scalar $d_i^{-1} \in \mathbb{F}_p$. We obtain three independent characters of $C$, contradicting its dimension two. This proves the closed-subgroup assertion.

We now apply it to finite loop actions, retaining the precise normalization map. If an outer class fixes the three peripheral classes, its exponents on $c_1, c_2, c_3$ are equal: in $F^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$, the relation $[c_1] + [c_2] + [c_3] = 0$ gives equality of the two coordinates. Conjugate a representative $\beta$ to have $\beta(x) = x^\lambda$, and write $\beta(y) = k^{-1} y^\lambda k$. Write $[k] = (a, b)$ in the named abelian basis. Conjugating this representative by $x^a$ gives the normalized pair

$$\alpha(x) = x^\lambda, \qquad \alpha(y) = f^{-1} y^\lambda f, \qquad f = y^{-b} k x^{-a} \in F'. \tag{D.32e}$$

Multiplication on the left of the conjugator by $y^{-b}$ changes neither image of $y$ nor the automorphism. Two normalized representatives differ by conjugation by an element centralizing $x^\lambda$, hence by $x^v$ by A.7. Their two conjugators then differ by left multiplication by a power of $y$ and right multiplication by $x^{-v}$. Since both are in $F'$, their two abelian coordinates force both

powers to be zero. Normalization is therefore unique. A product of normalized pairs has second coordinate $f\alpha(g) \in F'$, so (D.32e) is a homomorphic section into $\mathrm{Aut}(F)$. Its image is precisely the Belyi group defined in [31, p. 3120, equation (2)]; the argument has not identified that entire group with its subgroup $GT_0$.

Let an element of $P$ have prime order $p$ with $\lambda_p = 1$. Its normalized automorphism has order $p$. The proof of D.28 uses exactly the properties just established: finite order of the normalized automorphism, the common multiplier, and preservation of the three cusps. Neither of the extra $GT_0$ symmetry equations is used in its norm computation or Brown restriction argument. In particular that proof gives a single conjugacy class of sections in $F \rtimes C_p$ and the actual fixed subgroup $C = C_F(C_p)$ with $\dim H^1(C, \mathbb{F}_p) = 2$. Each of the three peripheral sections centralizes the whole $c_i^{e_p\mathbb{Z}_p}$; conjugating those sections to the specified $C_p$ places the three groups (D.32b) in this same $C$. The closed-subgroup assertion gives $C = F$, so the normalized automorphism is the identity. This contradicts its prime order.

For odd $p$, an element of order dividing $p$ in $\mathbb{Z}_p^\times$ equals one, by reduction modulo $p$ and the torsion-free group $1 + p\mathbb{Z}_p$. Thus no odd prime occurs in the order of a finite subgroup of $P$. The finite image of $\lambda_2$ lies in $\{1, -1\}$, since $1 + 4\mathbb{Z}_2$ is torsion-free. Its kernel in any finite subgroup has no element of order two by the preceding argument, and no element of odd prime order either. Cauchy's theorem makes that kernel trivial. This proves the first assertion.

For a class permuting cusps, compose it with the inverse of the corresponding positively oriented geometric permutation. The preceding abelian calculation gives its common multiplier too, and composition on cusp generators proves that both $\sigma$ and $\lambda$ are homomorphisms. On a finite subgroup, the kernel of (D.32a) is a finite subgroup of $P$ on which $\lambda_2 = 1$, hence is trivial. All subgroups of the target already have their geometric actions on the three-punctured sphere: use an equilateral triple on the equator, the positive rotational $S_3$, and reflection in that equator for the commuting $C_2$. This proves the claimed abstract group bound and its attainment, while keeping the action-conjugacy question separate through the actual map (D.32a).

The program `code/part_c/three_cusp_rank_d32.py` enumerates the finite covers used for a validation through degree six. It computes the Schreier graph, every boundary cycle, the unique boundary-homology relation, and explicit dual characters for the selected boundaries, including the $d_i^{-1}$ restriction scalars. The proof above applies to arbitrary degrees and closed subgroups; no bounded enumeration is used in place of it. □

**Source check for the fixed-second-coordinate assertion.** Lochak–Schneps, Lemma 9 on p. 583 of [47], asserts that a fixed $f$ occurs only with the two multipliers $\lambda$ and $-\lambda$. The displayed class-two identity $\lambda^2 = 24r + 1$ determines the squares of the prime coordinates. The second proof in that source adds the required full-group equation, whose elementary verification is as follows.

Comparing $(\lambda, f)$ and $(\mu, f)$ by the exact group law gives $(\lambda^{-1}\mu, 1)$. Set $n = (\lambda^{-1}\mu - 1)/2 \in \widehat{\mathbb{Z}}$. The three-cycle relation is

$$x^n(xy)^{-n}y^n = 1. \tag{LS.1}$$

In the class-two Heisenberg quotients its central coordinate gives $n(n + 1) = 0$. Thus in each integral domain $\mathbb{Z}_p$ one has $n_p = 0$ or $n_p = -1$. These choices are linked by actual finite quotients: for every odd prime $q$, evaluate (LS.1) in

$$D_q = \langle r, s \mid r^q = s^2 = 1,\ srs = r^{-1} \rangle, \qquad x = s, \quad y = r.$$

Both $x$ and $xy$ have order two. If $n_2 = 0$, (LS.1) is $r^{n_q} = 1$, forcing $n_q = 0$. If $n_2 = -1$, it is $r^{1+n_q} = 1$, forcing $n_q = -1$. Hence $n$ is globally zero or globally minus one, proving $\mu = \lambda$ or $\mu = -\lambda$. Both occur, because multiplication on the right by $(-1, 1)$ leaves the second coordinate unchanged.

This verifies the source's fixed-$f$ conclusion by exact quotient morphisms. Powers of a general torsion element have second coordinates $f\alpha(f)\cdots\alpha^{j-1}(f)$; the argument does not identify those coordinates with each other. D.32 supplies the separate argument restricting finite subgroups, and leaves the stated involution questions unchanged.

**D.33. [Proved] Finite punctured-torus actions are detected by their primary homology.** Put $F = \widehat{F}(x,y)$ and $c = [x,y]$, and let $T = \mathrm{Out}_{\mathrm{lac}}(F; \overline{\langle c \rangle})$ be the full once-punctured-torus loop-outer group. For every prime $p$, the homomorphism induced by the named abelianization,

$$A_p : T \longrightarrow \mathrm{GL}_2(\mathbb{Z}_p), \tag{D.33a}$$

is injective on every finite $p$-subgroup of $T$. The full abelianization homomorphism $T \to \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is therefore injective on every finite subgroup. Every such subgroup has order dividing 24.

The fixed-subgroup calculation underlying these assertions is also explicit. For an automorphism $\alpha$ of $\widehat{F}_r$ with $\alpha^p = 1$ whose induced automorphism on the maximal pro-$p$ quotient is the identity, put $E = \widehat{F}_r \rtimes_\alpha C_p$. There is one kernel-conjugacy class of sections of $E \to C_p$, and for its fixed subgroup $C$ the actual restriction map is an isomorphism

$$H^1(\widehat{F}_r, \mathbb{F}_p) \xrightarrow{\sim} H^1(C, \mathbb{F}_p). \tag{D.33b}$$

For the punctured-torus loop structure and $r = 2$, its cusp forces $C = F$ and hence $\alpha = 1$.

*Manuscript:* 30; folios 194–196; scans 383–387; 31; folios 198–199 verso; scans 391–394; Part D(b) of the task.

*Proof.* We first prove (D.33b), keeping the multiplicative information in the cohomology comparison. Write $P = \widehat{F}_r^{(p)}$ and $S = \{1\} \times C_p \leq E$. The quotient homomorphism is

$$\pi : E \longrightarrow P \times C_p, \qquad (h, s^i) \longmapsto (\bar{h}, s^i). \tag{D.33c}$$

Restriction along $\widehat{F}_r \to P$ is an isomorphism on mod-$p$ cohomology: in degree one both sides are the same $r$-dimensional character group, and higher cohomology vanishes because both groups have $p$-cohomological dimension one. The Hochschild–Serre comparison for (D.33c) is consequently an isomorphism on its second pages and gives a ring isomorphism

$$\pi^* : H^*(P \times C_p, \mathbb{F}_p) \xrightarrow{\sim} H^*(E, \mathbb{F}_p). \tag{D.33d}$$

All maps here are induced by the specified quotient, not by an identification of the two groups. In particular $\dim H^n(E, \mathbb{F}_p) = r + 1$ for every $n > 1$.

Let $u \in H^2(C_p, \mathbb{F}_p)$ be the nonzero periodic class, and inflate it to either side of (D.33d). The Künneth calculation for the direct product gives, for every $m \geq 1$, an injective map

$$H^1(P, \mathbb{F}_p) \longrightarrow H^{2m+1}(E, \mathbb{F}_p), \qquad \chi \longmapsto \pi^*(\chi u^m). \tag{D.33e}$$

For a subgroup $Q \leq E$ of order $p$, its projection to $C_p$ is an isomorphism, and $N_E(Q) = C_{\widehat{F}_r}(Q) \times Q$, exactly as in D.28. The first coordinate of $\pi(Q)$ is trivial because $P$ is torsion-free. Thus every character $\pi^*\chi$ in (D.33e) vanishes on $Q$. Its restriction to this normalizer lies entirely in $H^1(C_{\widehat{F}_r}(Q), \mathbb{F}_p)$, and multiplication by $u^m$ places it in

$$H^1(C_{\widehat{F}_r}(Q), \mathbb{F}_p) \otimes H^{2m}(Q, \mathbb{F}_p). \tag{D.33f}$$

Apply the profinite Brown restriction theorem in the precise form verified in [42, Theorem 2 and §1.1]. There is no $C_p \times C_p$ in $E$, since a finite subgroup injects into $E/\widehat{F}_r = C_p$. Its open subgroup $\widehat{F}_r$ has $p$-cohomological dimension one. The finite-dimensional source makes the

theorem's dense restriction embedding an isomorphism onto a finite product. If $k$ is the number of conjugacy classes and $d_Q = \dim H^1(C_{\widehat{F}_r}(Q), \mathbb{F}_p)$, its dimension formula is

$$k + \sum_{[Q]} d_Q = r + 1. \tag{D.33g}$$

The injection (D.33e), followed by this same restriction map, lands in the summands (D.33f). Therefore $\sum d_Q \geq r$. Since $k \geq 1$, (D.33g) forces $k = 1$ and $d_Q = r$. For the specified $S$, (D.33e)–(D.33f) then say exactly that restriction $H^1(\widehat{F}_r, \mathbb{F}_p) \to H^1(C_{\widehat{F}_r}(S), \mathbb{F}_p)$ is injective; both spaces have dimension $r$, proving (D.33b). As in D.28, conjugacy under $E$ equals conjugacy under the kernel, because $E = \widehat{F}_r Q$ for any section $Q$.

Now take $r = 2$ and assume that $\alpha$ preserves the conjugacy class of $L = \overline{\langle c \rangle}$, where $c = [x, y]$. Write its peripheral multiplier as $\lambda$. In a class-two Heisenberg quotient, the image of the commutator under an abelianization matrix $A$ is the $\det(A)$ power of the commutator. The finite Heisenberg quotients for all moduli therefore give the exact identity

$$\lambda = \det(\alpha^{\mathrm{ab}}). \tag{D.33h}$$

In the present calculation $\alpha$ is the identity on $P$, so $\lambda_p = 1$.

The cusp normalizer has the exact sequence

$$1 \longrightarrow L \longrightarrow N_E(L) \longrightarrow C_p \longrightarrow 1. \tag{D.33i}$$

The kernel is $L$ by A.7. For a lift $a$ of the chosen generator, write $aca^{-1} = c^\lambda$ and $a^p = c^v$, so $(\lambda - 1)v = 0$. The first coordinate of $\pi(a)$ centralizes the image $\bar{c}$ in $P$ and belongs to $\overline{\langle \bar{c} \rangle}$, by the pro-$p$ peripheral centralizer assertion of A.7. Thus it is $\bar{c}^w$ and $v_p = pw$. This uses the whole cusp subgroup. Its primitive pro-$p$ parameter can also be checked directly: in the mod-$p$ Magnus expansion, $[x, y] - 1$ starts with $XY - YX$. A $p$th power starts in degree divisible by $p$. At $p = 2$, the possible degree-two square is $(aX + bY)^2$; its $X^2$ and $Y^2$ coefficients prevent it from equaling $XY + YX$. Hence the commutator is not a $p$th power.

Since $\alpha^p = 1$, its multiplier satisfies $\lambda^p = 1$. Set $N_\lambda = 1 + \lambda + \cdots + \lambda^{p-1}$. At a prime $q$ with $\lambda_q \neq 1$, both $(N_\lambda)_q$ and $v_q$ are zero. Where $\lambda_q = 1$ the norm is $p$; $v_q$ is divisible by $p$, using $v_p = pw$ at $q = p$ and invertibility of $p$ at all other primes. Consequently there is $z \in \widehat{\mathbb{Z}}$ with $N_\lambda z = -v$. The element $ac^z$ has order $p$ and centralizes $c^{e_p \mathbb{Z}_p}$. The unique section-conjugacy class just proved places a conjugate $gc^{e_p \mathbb{Z}_p} g^{-1}$ inside $C = C_F(S)$.

Every character of $C$ vanishes on $gc^{e_p} g^{-1}$: by (D.33b) it is the restriction of a character of $F$, and $c$ is a commutator. We prove that this forces $C = F$. Its mod-$p$ first-cohomology dimension is two. If $C$ is open, Schreier gives $1 + [F : C] = 2$. If $C$ is proper and not open, the restriction-colimit proof of D.32 shows that its index has infinite $p$-part. Choose an open $U \geq C$ with $p \mid n = [F : U]$, and take the degree-$n$ covering of the punctured torus corresponding to $U \cap F(x, y)$. The chosen cusp has a boundary lift of degree $d$ prime to $p$, since its subgroup contains the full chosen $c^{e_p \mathbb{Z}_p}$. The comparison with a discrete boundary generator is exactly (D.32d), with a dense discrete representative for the profinite conjugator $g$. There must be at least two boundary lifts: a unique lift would have degree $n$, divisible by $p$. The selected boundary has nonzero homology in this cover, since the sole relation among the at least two boundary classes is their sum. A character of $U$ detecting this boundary restricts to a character of $C$ detecting $gc^{e_p} g^{-1}$; the value is multiplied by $d^{-1} \bmod p$, as computed in D.32. This is the required contradiction. Hence $C = F$ and $\alpha = 1$.

To apply this to (D.33a), let an outer element have order $p$ and $A_p = I$. D.27 supplies an actual order-$p$ automorphism representing it. Its induced pro-$p$ automorphism has identity abelianization. The Magnus argument in D.10 and D.25 makes this finite-order automorphism the identity: a first nonzero homogeneous deviation in the $\mathbb{Z}_p$ Magnus algebra would be multiplied

by $p$ under the $p$th iterate, and the homogeneous module has no $p$-torsion. Thus (D.33c) applies to this lift directly; there is no subsequent conjugation which could change its order. The cusp argument proves that the lift, and hence its outer class, is the identity. The kernel of (D.33a) on a finite $p$-subgroup cannot contain an order-$p$ element, so it is trivial. Likewise, the kernel of full abelianization on any finite subgroup would contain an element of some prime order and is trivial.

We finish with the numerical bound on these finite groups. For $p > 3$, a matrix of order $p$ over $\mathbb{Q}_p$ has a primitive $p$th root among its eigenvalues. The polynomial $\Phi_p(t)$ is irreducible of degree $p-1$ over $\mathbb{Q}_p$, as seen by Eisenstein after $t \mapsto t+1$. It cannot be the factor of a degree-two characteristic polynomial. Thus no such prime divides the order of a finite subgroup of $T$. A finite 3-subgroup injects by $A_3$ into $\mathrm{GL}_2(\mathbb{Z}_3)$ and then injects modulo 3: the kernel $1 + 3M_2(\mathbb{Z}_3)$ is torsion-free by the matrix logarithm. Since $|\mathrm{GL}_2(\mathbb{F}_3)| = 48$, its order divides 3.

For completeness, a finite 2-subgroup $K$ of $\mathrm{GL}_2(\mathbb{Q}_2)$ has order at most eight. The same Eisenstein argument for $\Phi_{2^a}$ excludes order $2^a$ for $a \geq 3$. Every matrix of order four has minimal polynomial $t^2 + 1$, so its square is $-I$. Consequently the projective image of $K$ has exponent two. Extend the matrices to the algebraic closure $\overline{\mathbb{Q}}_2$; the scalar kernel is unchanged. An exponent-two subgroup of $\mathrm{PGL}_2(\overline{\mathbb{Q}}_2)$ has order at most four. To prove this last bound, conjugate a nonidentity element to $z \mapsto -z$. Every commuting transformation preserves its pair of fixed points, so is $z \mapsto az$ or $z \mapsto a/z$. The scaling subgroup has at most two elements under the exponent-two condition and has index at most two. The scalar kernel of $K$ is contained in $\{I, -I\}$, the finite 2-torsion of $\mathbb{Q}_2^\times$. Thus $|K| \leq 8$. Apply this to a Sylow 2-subgroup through $A_2$. The Sylow bounds and the exclusion of other primes give $|G| \mid 8 \cdot 3 = 24$.

The program `code/part_c/torus_homology_d33.py` checks the Magnus degree-two coefficients, the periodic cohomology multiplication used in (D.33e), and the one-cusp restriction argument on every connected finite cover through degree six. The matrices and boundary characters are obtained from actual Schreier walks. The bound and comparison maps above are proved for the full profinite groups, independently of the bounded checks. □

**D.34. [Proved] Global signs and the cusp cycle of a pure involution.** Every nonidentity finite-order element of the full pure three-cusp loop-outer group $P$ of D.32 is an involution whose common multiplier is the global element $-1 \in \widehat{\mathbb{Z}}^\times$. In particular this holds in $GT_0$, and the multiplier-one kernel in either group is torsion-free.

For its normalized lift

$$\alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1}y^{-1}f, \qquad f \in F', \tag{D.34a}$$

there are precisely three $F$-conjugacy classes of sections of $F \rtimes_\alpha C_2 \to C_2$. Each of their fixed subgroups has zero first cohomology with $\mathbb{F}_2$ coefficients. The two section classes in each cusp normalizer are the endpoints of one edge of a triangle. Moreover there is $h \in F'$ with

$$f = h^{-1}\alpha(h). \tag{D.34b}$$

Thus $\alpha$ inverts both $x$ and $hyh^{-1}$. The endomorphism $x \mapsto x$, $y \mapsto hyh^{-1}$ induces an automorphism of every maximal pro-$p$ quotient. Its surjectivity on the full $F$, and conjugacy of the original involution to complex conjugation in the pure loop group, are not conclusions of this statement.

*Manuscript:* 30; folios 194–196; scans 383–387; 31; folios 198–199 verso; scans 391–394; Part D(b) of the task.

*Proof.* We first prove a genus-zero incidence calculation for an arbitrary number $n \geq 3$ of cusps. Let

$$F_n = \langle c_1, \ldots, c_n \mid \widehat{c_1 \cdots c_n} = 1 \rangle$$

and let $\beta$ be an actual involution preserving each peripheral conjugacy class, with common multiplier $\mu$ and $\mu_2 = -1$. Put $E_n = F_n \rtimes_\beta C_2$. The equality $\beta^2 = 1$ gives $\mu^2 = 1$; hence each $\mu_q$ is 1 or $-1$.

For each cusp $L_i = \overline{\langle c_i \rangle}$, A.7 gives an exact sequence

$$1 \longrightarrow L_i \longrightarrow N_{E_n}(L_i) \longrightarrow C_2 \longrightarrow 1. \tag{D.34c}$$

Choose a lift $a_i$ of the nonidentity element and write $a_i c_i a_i^{-1} = c_i^{\mu}$ and $a_i^2 = c_i^{v_i}$. Then $(\mu - 1)v_i = 0$. In every coordinate with $\mu_q = -1$ this forces $(v_i)_q = 0$, including $q = 2$. In a coordinate with $\mu_q = 1$ the prime $q$ is odd, so division by two is possible. Thus $(1 + \mu)u_i = -v_i$ has a solution in $\widehat{\mathbb{Z}}$, and $b_i = a_i c_i^{u_i}$ is an involution.

The set of its further order-two translates by $L_i$, modulo conjugation by $L_i$, is exactly

$$\frac{\ker(1 + \mu : \widehat{\mathbb{Z}} \to \widehat{\mathbb{Z}})}{(1 - \mu)\widehat{\mathbb{Z}}} \;\simeq\; \mathbb{F}_2. \tag{D.34d}$$

At an odd prime with $\mu_q = -1$, multiplication by two is onto, and a coordinate with $\mu_q = 1$ contributes zero to the kernel. The remaining coordinate is $\mathbb{Z}_2/2\mathbb{Z}_2$. Representatives for the two classes are $b_i$ and $c_i^{e_2} b_i$.

Since $\beta$ is pure and the cusp classes span homology, its action on $V = H_1(F_n, \mathbb{F}_2)$ is the identity. Therefore the actual mod-two abelianization of the semidirect product is

$$H_1(E_n, \mathbb{F}_2) = V \oplus \mathbb{F}_2. \tag{D.34e}$$

The two involutions just exhibited have images differing by the nonzero vector $[c_i]$ in $V$. They cannot be conjugate in $E_n$ or in $F_n$. Thus the two local classes in (D.34d) give two distinct global section classes.

Build a finite graph with one edge for each of the $n$ cusps and with these two global classes as its endpoints. This graph uses some of the global section classes; we now bound all of them. The Brown calculation of D.28, with free rank $n - 1$ and prime two, gives

$$\sum_{[Q]} (1 + \dim_{\mathbb{F}_2} H^1(C_{F_n}(Q), \mathbb{F}_2)) = \dim_{\mathbb{F}_2} H^k(E_n, \mathbb{F}_2) \le n \qquad (k > 1). \tag{D.34f}$$

Every order-two subgroup is a section, and its total-group and kernel conjugacy classes agree, because $E_n = F_n Q$. Hence there are at most $n$ vertices available for the graph. An edge has no loop by (D.34e).

Assign to every vertex its image $(v, 1)$ in (D.34e). The sum of the edge vectors around a graph cycle is zero, since its successive vertex differences cancel. But the $n$ vectors $[c_1], \ldots, [c_n]$ have just one nonzero linear relation over $\mathbb{F}_2$, namely $[c_1] + \cdots + [c_n] = 0$. Consequently any simple cycle in the graph uses all $n$ edges. The graph cannot be a forest: a forest with $n$ edges needs at least $n + 1$ vertices, more than (D.34f) permits. Thus it is exactly a simple $n$-cycle. It has $n$ vertices, so there are no additional section classes outside this graph, and (D.34f) gives

$$H^1(C_{F_n}(Q), \mathbb{F}_2) = 0 \quad \text{for every section } Q. \tag{D.34g}$$

This construction is invariant under any automorphism of the extension preserving its kernel and peripheral classes, with any permutation of those classes: such an automorphism sends the normalizer and its two classes in (D.34c)–(D.34d) to those of the image cusp.

Return to a nontrivial finite-order element of $P$. D.32 gives its order two, its normalized actual lift $\alpha$, and $\lambda_2 = -1$. We show that $\lambda_q = -1$ for every odd prime $q$. The only other possibility is $\lambda_q = 1$. Suppose this occurs and use the exact cyclic quotient

$$\chi : F = \widehat{F}(x, y) \longrightarrow C_q, \qquad \chi(x) = 1, \quad \chi(y) = 0, \qquad N = \ker \chi. \tag{D.34h}$$

The common multiplier gives $\chi\alpha = \lambda\chi = \chi$, since its reduction modulo $q$ is one. In particular $N$ is invariant. The corresponding degree-$q$ covering of the pair of pants has one boundary over $x$, one over $z = (xy)^{-1}$, and $q$ over $y$, with representative generators

$$x^q, \qquad z^q, \qquad x^i y x^{-i} \quad (0 \le i < q). \tag{D.34i}$$

The two first degrees are $q$ and all others are one. Its Euler characteristic is $-q$ and it has $q+2$ boundaries, so its genus is zero. Finite-index completion identifies its full profinite group with the specified subgroup $N$.

The restriction of $\alpha$ is pure on this entire peripheral structure, with the same multiplier $\lambda$. The first two boundaries are the unique lifts of their original boundaries. For the other $q$ boundaries use the actual normalized formula

$$\alpha(x^i y x^{-i}) = x^{\lambda i} f^{-1} y^{\lambda} f x^{-\lambda i}. \tag{D.34j}$$

Here $\chi(f) = 0$, because $f \in F'$, and $\lambda i \equiv i \pmod q$. The conjugator in (D.34j) therefore belongs to the same $N$-coset as $x^i$: explicitly $k_i = x^{\lambda i} f^{-1} x^{-i} \in N$ satisfies $\alpha(x^i y x^{-i}) = k_i (x^i y x^{-i})^{\lambda} k_i^{-1}$. This is the required conjugacy inside $N$. The action on $H_1(N, \mathbb{Z}_2)$ is consequently the scalar $-1$, so the restriction remains a nontrivial involution.

Put $E = F \rtimes_\alpha C_2$ and $E_N = N \rtimes_\alpha C_2$. The quotient maps give

$$E/N = C_q \times C_2. \tag{D.34k}$$

In particular $E_N$ is normal in $E$. Conjugation by $x$ defines an automorphism of $E_N$ preserving its map to $C_2$, and thus acts on its $N$-conjugacy classes of sections. Its $q$th power is conjugation by $x^q \in N$, so this action has order dividing $q$ on those classes and on the cusp incidence graph. On the cusp edges (D.34i), it fixes the $x^q$ and $z^q$ classes and cycles the $q$ classes over $y$. The second fixed class is the unique boundary over $z$, or follows directly from its cyclic coset decomposition. Thus its graph action has order exactly $q$.

By the preceding calculation the graph is a cycle with $q+2$ edges. Its automorphism group has order $2(q+2)$: an image of a vertex and a choice of which neighbor is its successor determine the automorphism. The odd prime $q$ does not divide $2(q+2)$. This contradicts the displayed order-$q$ graph action. Therefore $\lambda_q = -1$ for every odd $q$, proving the global multiplier assertion. This argument used the whole pure group $P$, without an additional $GT_0$ symmetry assumption.

At $n = 3$ the incidence graph is the stated triangle. Write $s$ for the chosen section in $E = F \rtimes C_2$. The normalized group law gives $f\alpha(f) = 1$, so $fs$ is also an involution and it normalizes $y$ by inversion. The three involutions

$$s, \quad x^{e_2} s, \quad y^{e_2} f s \tag{D.34l}$$

have the distinct images $(0,0,1)$, $(1,0,1)$, and $(0,1,1)$ in (D.34e). They represent all three classes. Since $f \in F'$, the image of $fs$ is $(0,0,1)$, so it is conjugate to $s$ by an element of $F$. Choose $h$ with $fs = h^{-1} s h$. This is exactly (D.34b). In abelianization it says $0 = (-2)[h]$ in the torsion-free group $\widehat{\mathbb{Z}}^2$, hence $h \in F'$. Substitution gives

$$\alpha(hyh^{-1}) = (hf)(f^{-1}y^{-1}f)(f^{-1}h^{-1}) = hy^{-1}h^{-1}.$$

The two displayed inverted generators have the same abelian images as $x, y$. On each maximal pro-$p$ quotient their images therefore generate the Frattini quotient and hence the whole group. The induced endomorphism of that finitely generated free pro-$p$ group is surjective and thus an automorphism. No full-profinite surjectivity is inferred from those separate comparisons.

The program `code/part_c/pure_involution_d34.py` checks the normalizer orbit calculation, the unique binary cusp relation, and the exact cyclic-cover boundary and deck permutations for several odd primes. The exclusion of all odd primes is the unrestricted divisibility argument above. □

**D.35. [Proved] One exact conjugator equation and its finite obstructions.** Let $P$ be the full pure three-cusp group of D.32, let $F = \widehat{F}(x, y)$, and put $\delta(x) = x^{-1}$, $\delta(y) = y^{-1}$. For a nonidentity involution $\alpha \in P$, use its normalized lift from D.34,

$$\alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1}y^{-1}f, \qquad f \in F'.$$

Define the compact solution space

$$\begin{aligned} \mathcal{S}_\alpha = \{(\mu, h, u) \in \widehat{\mathbb{Z}}^\times \times F' \times F : \quad & h^{-1}\alpha(h) = f, \\ & x^\mu h y^\mu h^{-1} = u(xy)^\mu u^{-1}\}. \end{aligned} \tag{D.35a}$$

The assignment

$$(\mu, h, u) \longmapsto [\beta_{\mu,h}], \qquad \beta_{\mu,h}(x) = x^\mu, \quad \beta_{\mu,h}(y) = hy^\mu h^{-1}, \tag{D.35b}$$

has image exactly

$$\mathrm{Conj}_P(\delta, \alpha) = \{\beta \in P : \ \beta\delta\beta^{-1} = \alpha\}. \tag{D.35c}$$

Every map (D.35b) is an automorphism of the full profinite group. The fibers of the assignment are exactly $(\mu, h, u) \mapsto (\mu, h, u(xy)^a)$ for $a \in \widehat{\mathbb{Z}}$. It therefore induces a homeomorphism from the corresponding quotient of $\mathcal{S}_\alpha$ to (D.35c).

For $\alpha \in GT_0$, the inclusion $GT_0 \hookrightarrow P$ gives the equality of actual conjugator sets

$$\mathrm{Conj}_{GT_0}(\delta, \alpha) = \mathrm{Conj}_P(\delta, \alpha). \tag{D.35d}$$

The defining Harbater–Schneps relations for the conjugating pair follow from the proof below; they are not discarded. Finally, the existence of a solution of (D.35a) is equivalent to the nonemptiness of its explicit finite solution sets (D.35h) at every characteristic finite quotient of $F$.

*Manuscript:* §31; folios 198-199verso; scans 391-394; §30; folio 194verso; scan 384; Part D(b): actual full GT_0 torsion and conjugacy.

*Proof.* For $(\mu, h, u) \in \mathcal{S}_\alpha$, the free universal property first gives a continuous endomorphism $\beta_{\mu,h} : F \to F$ with the displayed generator images. Write $H = \beta_{\mu,h}(F)$, a closed subgroup because $F$ is compact. It is generated by two elements, so $\dim H^1(H, \mathbb{F}_2) \le 2$. The restrictions of the two standard characters of $F$ to these generators are the two coordinate characters: $\mu \bmod 2 = 1$ and conjugation does not change abelianization. These restrictions are independent, proving

$$\dim H^1(H, \mathbb{F}_2) = 2. \tag{D.35e}$$

The unit exponent $\mu$ shows that $H$ contains $\overline{\langle x \rangle}$ and $h\overline{\langle y \rangle}h^{-1}$. The second equation in (D.35a) puts $u\overline{\langle xy \rangle}u^{-1}$ in $H$ as well. In particular $H$ contains the three full cusp $\mathbb{Z}_2$ subgroups required by D.32. Its closed-subgroup rank test, applied to (D.35e), gives $H = F$. A surjective endomorphism of a finitely generated profinite group is injective, so $\beta_{\mu,h}$ is an automorphism. Its images of the three cusps, including the exact displayed equation for the third, place its outer class in $P$.

The first equation in (D.35a) is $\alpha(h) = hf$. Substitution on the two free generators gives

$$\alpha\beta_{\mu,h}(x) = x^{-\mu} = \beta_{\mu,h}\delta(x), \qquad \alpha\beta_{\mu,h}(y) = hy^{-\mu}h^{-1} = \beta_{\mu,h}\delta(y). \tag{D.35f}$$

These are equalities of actual automorphisms, proving that the image of (D.35b) lies in (D.35c).

Conversely, let $\beta \in P$ satisfy (D.35c), and take its unique normalized representative from D.32. Write it as $\beta_{\mu,h}$ with $h \in F'$. The normalization is a homomorphic section $P \to \mathrm{Aut}(F)$, so the outer conjugacy equality lifts to the actual equality $\alpha\beta_{\mu,h} = \beta_{\mu,h}\delta$. Its value on $y$ implies that $h^{-1}\alpha(h)f^{-1}$ centralizes $y^{-\mu}$. By A.7 it is a power of $y$. It also lies in $F'$, so its abelianization

forces this power to be zero. Thus $h^{-1}\alpha(h) = f$. The third-cusp condition on $\beta$ gives a conjugator $u$ in the second equation of (D.35a). This proves that the image is exactly (D.35c).

The normalized pair $(\mu, h^{-1})$ is unique. For this fixed pair the two third-cusp conjugators $u, u'$ differ by right multiplication by the centralizer of $(xy)^\mu$, which is $\overline{\langle xy \rangle}$ by A.7. These are precisely the asserted fibers. The assignment is continuous in the profinite automorphism topology, and its domain is compact. Its quotient is therefore homeomorphic to its Hausdorff image.

For an alternative form using D.34, fix $h_0 \in F'$ with $h_0^{-1}\alpha(h_0) = f$, and put $C = C_F(\alpha)$. All solutions of the first equation of (D.35a) are exactly $h = ch_0$ with $c \in C$: $\alpha(h) = hf$ and $\alpha(h_0) = h_0 f$ give $\alpha(hh_0^{-1}) = hh_0^{-1}$, and the reverse substitution proves the converse. Since $\alpha$ acts by $-1$ on $F^{\mathrm{ab}}$, every $c \in C$ has $2[c] = 0$, hence $[c] = 0$. Thus $C \subset F'$ and this substitution respects the stated domain of (D.35a). The remaining exact equation is

$$x^\mu c h_0 y^\mu h_0^{-1} c^{-1} = u(xy)^\mu u^{-1}. \tag{D.35g}$$

We prove (D.35d) by identifying the relevant centralizer, rather than assuming that a conjugator in $P$ lies in $GT_0$. Let $P^1 = \ker(\lambda : P \to \widehat{\mathbb{Z}}^\times)$. First, $C_{P^1}(\delta) = 1$. Indeed a normalized element $(1, k) \in P^1$ commuting with $\delta$ has $k = \delta(k)$, by uniqueness of the normalized representative and its group law. The following explicit mutually inverse homomorphisms identify the semidirect product:

$$\begin{aligned} C_2 \amalg C_2 \amalg C_2 &\xrightarrow{\sim} F \rtimes_\delta \langle s \rangle, & (s_0, s_1, s_2) &\longmapsto (s, xs, ys), \\ (x, y, s) &\longmapsto (s_1 s_0, s_2 s_0, s_0). \end{aligned}$$

The orders and conjugation relations follow by direct substitution. A free profinite factor has trivial intersection with a conjugate by an element outside that factor; the original full-profinite statement is recalled in [36, Theorem 5 of the inspected author version]. Consequently the centralizer of $s_0$ in this free product is $\langle s_0 \rangle$. Its intersection with $F$ is trivial. Hence $C_F(\delta) = 1$, so $k = 1$ and $C_{P^1}(\delta) = 1$ as claimed.

Let $\Sigma \cong S_3$ be the geometric cusp-permutation subgroup of the full loop-outer group, generated by the outer classes $\theta(x) = y$, $\theta(y) = x$ and $\omega(x) = y$, $\omega(y) = (xy)^{-1}$. The exact identification $GT_0 = C_P(\Sigma)$ follows from [31, Propositions 3–4 and Theorem 2]: commutation with $\theta$ is relation (I), and, after (I), commutation with $\omega$ is relation (II). Both $\alpha$ and $\delta$ centralize $\Sigma$. For any $\beta \in \mathrm{Conj}_P(\delta, \alpha)$ and $\sigma \in \Sigma$, form the actual element of the outer group $r_\sigma = \beta^{-1}\sigma\beta\sigma^{-1}$. The pure subgroup $P$ is normal in the full loop-outer group, and the common multiplier is a homomorphism there. Therefore $r_\sigma \in P^1$. Moreover

$$\begin{aligned} r_\sigma \delta r_\sigma^{-1} &= \beta^{-1}\sigma\beta(\sigma^{-1}\delta\sigma)\beta^{-1}\sigma^{-1}\beta \\ &= \beta^{-1}\sigma\alpha\sigma^{-1}\beta = \beta^{-1}\alpha\beta = \delta. \end{aligned}$$

The proved equality $C_{P^1}(\delta) = 1$ gives $r_\sigma = 1$. Thus $\beta$ centralizes $\Sigma$, so its normalized pair satisfies both (I) and (II) and lies in $GT_0$. This proves (D.35d).

Here is the exact finite form of the remaining existence problem. For each open characteristic subgroup $N \triangleleft F$, put $Q_N = F/N$ and $m_N = \exp(Q_N)$. The automorphism $\alpha$ induces $\alpha_N$. Use bars for the images of $x, y, f$ in $Q_N$ and define

$$\begin{aligned} \mathcal{S}_{\alpha,N} = \{(a, b, v) \in (\mathbb{Z}/m_N)^\times \times Q_N' \times Q_N : \quad & b^{-1}\alpha_N(b) = \bar{f}, \\ & \bar{x}^a b \bar{y}^a b^{-1} = v(\bar{x}\bar{y})^a v^{-1}\}. \end{aligned} \tag{D.35h}$$

These are finite sets, obtained entirely from the specified finite quotient and its induced automorphism. Reduction sends $\mathcal{S}_\alpha$ to every $\mathcal{S}_{\alpha,N}$. Conversely, a point of a nonempty $\mathcal{S}_{\alpha,N}$ lifts to $\widehat{\mathbb{Z}}^\times \times F' \times F$: units lift by the Chinese remainder theorem, and the image of $F'$ in $Q_N$ is exactly $Q_N'$. The lifts satisfying the two equations modulo $N$ form a nonempty closed subset of this

compact space. Every finite collection of such conditions is implied by the one at a common characteristic refinement. Their finite intersection property therefore gives a point satisfying both equations in $F$ itself. Open characteristic subgroups are cofinal for a finitely generated profinite group, so equality at all these quotients is equality in $F$. This proves the asserted finite criterion.

The need for the full third-cusp equation can be checked by a short executed finite certificate. Put $d = [x, y] = xyx^{-1}y^{-1}$ and take the actual commutator word

$$w = yd^{-1}y^{-1}x^{-1}d^{-1}x \in F'.$$

Consider the endomorphism $\rho(x) = x$, $\rho(y) = wyw^{-1}$. Its abelianization is the identity, so the Frattini argument makes its induced endomorphism on every maximal pro-$p$ quotient an automorphism. In the finite quotient

$$F \twoheadrightarrow A_5, \qquad x \longmapsto (1\,2\,3), \quad y \longmapsto (3\,4\,5),$$

the program `code/part_a/pure_involution_conjugator.py` enumerates the generated group of order 60 and computes

$$\bar{w} = (1\,4\,3\,2\,5), \qquad \overline{wyw^{-1}} = (1\,2\,3).$$

The image of $\rho(F)$ in this quotient therefore has order 3, proving $\rho(F) \neq F$. Moreover $\overline{xy}$ has order 5, whereas $\overline{xwyw^{-1}}$ has order 3; the third-cusp equation fails in this same quotient. This word is a certificate concerning the generation test. It is not asserted to solve the coboundary equation for a nonstandard involution. No nonemptiness or emptiness assertion for the spaces (D.35a) of arbitrary involutions is inferred from this finite example. □

**D.36. [Proved] The global multiplier of a positive transposition.** Let $B$ be the full three-cusp loop-outer group, and let $g \in B$ be an involution whose 2-adic multiplier is $\lambda_2(g) = 1$. Its puncture permutation is a transposition, and its common multiplier is the global element $1 \in \widehat{\mathbb{Z}}^\times$.

Consequently a finite subgroup of $B$ whose map (D.32a) has image $S_3 \times C_2$ has the usual global orientation character. The same is true when that image is a $V_4$ containing the pure involution, or is the positively oriented $S_3 \times \{1\}$. These statements identify the actual multiplier homomorphism; they do not assert conjugacy of those finite actions.

*Manuscript:* §30; folios194–196; scans383–387; §31; folios198–199verso; scans391–394; Part D(b) of the task.

*Proof.* We first record the genus-zero extension of the actual automorphism calculation already proved in D.32–D.33. Let $F_n = \widehat{\pi}_{0,n}$ with $n \geq 3$, and let $\beta$ be a pure automorphism with $\beta^p = 1$ and $p$-adic peripheral multiplier one. Then $\beta = 1$. Indeed $F_n$ is free of rank $r = n - 1$, and its cusp classes span its abelianization. Its induced pro-$p$ automorphism has identity abelianization and finite order, so the $\mathbb{Z}_p$ Magnus argument of D.33 makes it the identity. The ring calculation (D.33c)–(D.33g) gives one conjugacy class of sections of $F_n \rtimes_\beta C_p \to C_p$, and its actual fixed subgroup $C$ has

$$\dim H^1(C, \mathbb{F}_p) = n - 1. \tag{D.36a}$$

The norm calculation of D.28 applies to every cusp: its image is a primitive element of the free pro-$p$ group, the first coordinate of its normalizer lift centralizes that element, and the lift's $p$th power has $p$-coordinate divisible by $p$. Coordinatewise division of the norm therefore supplies a section in each peripheral normalizer. Uniqueness of the section class places conjugates of all $n$ full peripheral $\mathbb{Z}_p$ subgroups inside the same $C$.

The finite-cover rank proof of D.32 now applies with all $n$ cusps. If $C$ is open, its first-cohomology dimension is $1 + [F_n : C](n - 2)$, which equals $n - 1$ only at index one. If it has

infinite index and finite $p$-part of that index, the same exact restriction colimit and eventual prime-to-$p$ injections give unbounded first-cohomology dimension. Otherwise choose an open $U \geq C$ of $p$-divisible index. The $n$ selected boundary lifts all have degrees prime to $p$ and lie over different original boundaries. They cannot be all the boundaries: a single lift over any original boundary would have degree $[F_n : U]$. With at least $n+1$ boundaries, the $n$ selected boundary homology classes are independent. Their dual characters restrict to $C$ with the nonzero multipliers $d_i^{-1} \bmod p$ computed in (D.32d). Thus $\dim H^1(C, \mathbb{F}_p) \geq n$, contradicting (D.36a). Hence $C = F_n$ and $\beta = 1$. This proof concerns an actual finite-order automorphism; it does not presume a lift for arbitrary finite outer actions in larger rank.

Return to $g$. D.32 excludes a nonidentity pure involution with $\lambda_2 = 1$, so its permutation must be a transposition. Relabel the original cusp generators as $x, y, z = (xy)^{-1}$ so that $x$ is fixed and $y, z$ are exchanged. D.27 provides an actual order-two lift $\alpha$ of this outer action. Its peripheral multiplier satisfies $\lambda^2 = 1$, so all its prime coordinates are 1 or $-1$.

Suppose $\lambda_q = -1$ for an odd prime $q$, and use the exact cyclic quotient

$$\chi : F \longrightarrow C_q, \qquad \chi(x) = 0, \quad \chi(y) = 1, \quad \chi(z) = -1, \qquad N = \ker \chi. \tag{D.36b}$$

The puncture permutation and the multiplier give $\chi\alpha = \chi$: on $y$ the exponent $-1$ at $q$ is multiplied by $\chi(z) = -1$, and similarly for $z$. Thus $N$ is invariant and the induced action on its cyclic deck quotient is the identity. The degree-$q$ cover is a sphere with $q+2$ cusps. There are $q$ degree-one cusps over $x$, and one degree-$q$ cusp over each of $y, z$.

Here are the exact covering and filling generators. Finite-coset Schreier rewriting with coset representatives $1, y, \ldots, y^{q-1}$ gives the free basis of $N$

$$a_i = y^i x y^{-i} \quad (0 \leq i < q), \qquad b = y^q.$$

The remaining boundary generator is

$$d = z^q = b^{-1} a_{q-1}^{-1} \cdots a_0^{-1}, \qquad a_0 \cdots a_{q-1} b d = 1. \tag{D.36c}$$

For example the latter relation is the actual free-word identity $(xy)^q = a_0 \cdots a_{q-1} y^q$. Completing the discrete Schreier comparison identifies this presented free profinite group with the specified open subgroup $N$.

The lift $\alpha$ fixes the conjugacy class of every $a_i$ inside $N$. To see this without choosing an unproved normalization, write $\alpha(x) = h^{-1} x^\lambda h$ for some $h \in F$. Since $\chi\alpha(y) = 1$, its action on the indices of these cusp classes is the translation $i \mapsto i - \chi(h)$ in $C_q$. Its square is the identity, so $2\chi(h) = 0$ and $\chi(h) = 0$. Explicitly

$$k_i = \alpha(y)^i h^{-1} y^{-i} \in N, \qquad \alpha(a_i) = k_i a_i^\lambda k_i^{-1}. \tag{D.36d}$$

The two unique boundary classes over $y, z$ are exchanged, with multiplier $\lambda$.

Fill exactly these two cusps. The map is the quotient by the closed normal subgroup of $N$ generated by $b, d$:

$$N \longrightarrow K = N / \overline{\langle\langle b, d \rangle\rangle}_N = \langle a_0, \ldots, a_{q-1} \mid \widehat{a_0 \cdots a_{q-1} = 1} \rangle \cong \widehat{\pi}_{0,q}. \tag{D.36e}$$

Both presentation maps are explicit: kill $b$ and send the remaining $a_i$ to the named peripheral generators; the inverse identifies $a_{q-1}$ with $(a_0 \cdots a_{q-2})^{-1}$. These substitutions verify both universal properties and compositions. Since $\alpha$ exchanges the normal generators' peripheral classes, the closed normal subgroup in (D.36e) is invariant. Consequently $\alpha$ induces an actual automorphism $\bar{\alpha}$ of $K$ with $\bar{\alpha}^2 = 1$. By (D.36d) it is pure, with the same multiplier $\lambda$.

The proved genus-zero calculation at $p = 2$ applies: $q \geq 3$ and $\lambda_2 = 1$, so $\bar{\alpha} = 1$. But its $q$-adic multiplier on each surviving nontrivial cusp is $-1$, which is not the identity on the cusp's

$\mathbb{Z}_q$ subgroup. Equivalently, it acts as $-I$ on the nonzero $q$-adic abelianization of the genus-zero group $K$. This contradiction excludes every such $q$. Together with $\lambda_2 = 1$ it proves $\lambda = 1$ globally.

For the finite-group consequences, use the faithful map (D.32a). In a $V_4$ with pure kernel, the pure involution has multiplier $-1$ by D.34, and the positive transposition has multiplier 1 by the present proof. These generate it. The positively oriented $S_3$ is generated by its positive transpositions. Finally $S_3 \times C_2$ is generated by those transpositions and its pure involution. Multiplicativity of the common multiplier proves each asserted character identity.

The program `code/part_c/positive_transposition_d36.py` checks (D.36c) as an exact free-word identity, the finite cyclic cusp permutations and their involutive translations, and the two presentation substitutions in (D.36e) for several odd primes. The proof above supplies the all-prime assertion. □

**D.37. [Proved] Negative transpositions and the specified four-cusp deck action.** Every involution of the full three-cusp loop-outer group whose puncture permutation is a transposition and whose multiplier has $\lambda_2 = -1$ gives a faithful $V_4$ action on a specified four-cusp double cover. One generator is the standard free positive deck involution; another has double-transposition permutation and multiplier $\lambda$, and its center-free extension is torsion-free. Conversely every such $V_4$ action containing the specified standard deck action descends to a negative transposition on the three-cusp group. The constructions are inverse as marked extensions.

For the double-transposition permutation of D.29, neither of the commuting single transpositions preserves any separating two-pants partition whose pants the double transposition exchanges. Thus an additional deck action in this construction necessarily moves the separator. This is an obstruction to preserving that decomposition, and asserts no obstruction to an action moving it.

*Manuscript:* Part D(b) and D(d) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394.

*Proof.* We specify both the markings and all the extension maps. Put

$$F = \widehat{F}(x, y), \qquad z = (xy)^{-1}, \qquad \chi : F \longrightarrow C_2, \quad (\chi(x), \chi(y)) = (0, 1), \qquad N = \ker \chi. \tag{D.37a}$$

Schreier rewriting identifies $N$ with the full free profinite group on

$$A_0 = x, \qquad A_1 = yxy^{-1}, \qquad B = y^2, \qquad D = z^2 = B^{-1}A_1^{-1}A_0^{-1}. \tag{D.37b}$$

Its four peripheral generators satisfy $A_0A_1BD = 1$. These are the actual degree-one lifts of $x$ and the degree-two lifts of $y, z$; completion of the discrete index-two Schreier isomorphism gives precisely the specified inclusion $N \subset F$. Let $\delta \in \mathrm{Out}(N)$ be conjugation by $y$. Its named representative is

$$A_0 \longmapsto A_1, \qquad A_1 \longmapsto BA_0B^{-1}, \qquad B \longmapsto B, \qquad \delta^2 = \mathrm{Inn}(B). \tag{D.37c}$$

Here the final equality concerns the named automorphism; its outer class has order two, multiplier 1, and permutation $(A_0\ A_1)$. Its center-free extension is the explicitly identified group $F$, hence is torsion-free. In the converse below, "the specified deck action" means this action transported by a given peripheral isomorphism. It does not mean an arbitrary positive outer involution with a torsion-free extension.

Let $g$ be a negative transposition, labeled to fix $x$ and exchange $y, z$. Its center-free extension $E$ contains $F$ as an open subgroup of index two. The peripheral normalizer surjects onto $\langle g \rangle$, and A.7 gives its kernel $\overline{\langle x \rangle} = \widehat{\mathbb{Z}}$. For a lift $v$ in this normalizer write

$$vxv^{-1} = x^{\lambda}, \qquad v^2 = x^a, \qquad (\lambda - 1)a = 0, \qquad \lambda^2 = 1. \tag{D.37d}$$

At every prime where $\lambda_p = -1$ the last commutation identity gives $a_p = 0$. At every prime where $\lambda_p = 1$ the prime is odd, since $\lambda_2 = -1$, so 2 is invertible. Choose $b_p = -a_p/2$ in these latter coordinates and $b_p = 0$ in the former. Then $s = x^b v$ has order two. Thus

$$E = F \rtimes_\alpha \langle s \rangle, \qquad \alpha^2 = 1, \qquad \alpha(x) = x^\lambda. \tag{D.37e}$$

This constructs the required finite-order lift inside the peripheral normalizer, without a further choice of inner normalization. The multiplier and the cusp permutation give $\chi\alpha = \chi$. Therefore $N$ is normal in $E$, and $E/N = C_2 \times C_2$, with basis the images of $y, s$.

There is exactly one $F$-conjugacy class of order-two subgroups of $E$. Indeed the action on $H^1(F, \mathbb{F}_2)$ is the two-dimensional nontrivial unipotent module $J_2$, isomorphic to the regular $\mathbb{F}_2[C_2]$ module. Consequently $H^i(C_2, H^1(F, \mathbb{F}_2)) = 0$ for $i > 0$. The Hochschild–Serre spectral sequence, with the split inflation on its bottom row, gives

$$\dim H^n(E, \mathbb{F}_2) = 1 \qquad (n \geq 3). \tag{D.37f}$$

The full profinite Brown restriction theorem used in D.28 identifies this high-degree cohomology with the product over the order-two conjugacy classes. Each class contributes $1 + \dim H^1(C_F(T), \mathbb{F}_2)$, since its normalizer is $C_F(T) \times C_2$; see [42, Theorem 2 and §1.1]. The section in (D.37e) exists, so (D.37f) forces exactly one class and zero first mod-two cohomology for its centralizer. Conjugacy by $E$ agrees here with conjugacy by $F$: for any section $T$, one has $E = FT$, and the $T$ factor centralizes that subgroup itself. Every finite-order element of $E$ injects into $E/F$, because $F$ is torsion-free, and consequently has order two. All nonidentity torsion therefore maps to the one line $\langle \bar{s} \rangle$ in the abelian group $E/N$.

Conjugation gives a faithful action of $E/N$ on $N$ in $\mathrm{Out}(N)$. To check the kernel, first $C_F(N) = 1$: an element centralizing both $x$ and $y^2$ lies in $\langle x \rangle \cap \langle y \rangle = 1$ by A.7. For $e \in C_E(N)$ and $f \in F$, the commutator $[e, f]$ lies in $F \cap C_E(N)$ and hence is trivial. Thus $e$ centralizes $F$; faithfulness of the original nonidentity outer action and $Z(F) = 1$ give $e = 1$.

Write $a$ for the outer restriction of $\alpha$ to $N$. It fixes the classes of $A_0, A_1$ and exchanges $B, D$. For the second fixed class use the explicit conjugator $\alpha(y)y^{-1} \in N$:

$$\alpha(A_1) = (\alpha(y)y^{-1}) A_1^\lambda (\alpha(y)y^{-1})^{-1}. \tag{D.37g}$$

The unique lifts of the other two cusps are exchanged inside $N$. Thus $\delta$ and $a$ commute in $\mathrm{Out}(N)$, and

$$\tau = \delta a, \qquad \sigma(\tau) = (A_0\ A_1)(B\ D), \qquad \mathrm{or}(\tau) = \lambda. \tag{D.37h}$$

The preimage $E_\tau$ of the diagonal line $\langle \bar{y} + \bar{s} \rangle$ in $E/N$ is the center-free extension of this outer involution. It is torsion-free: all nonidentity torsion has image $\bar{s}$, and its intersection with $N$ is already torsion-free. For an explicit representative set

$$\beta = \mathrm{Inn}(y)\alpha|_N, \qquad c = y\alpha(y) \in N; \qquad \beta^2 = \mathrm{Inn}(c), \quad \beta(c) = c. \tag{D.37i}$$

Then $E_\tau$ is exactly the twisted extension generated by $N$ and $ys$, whose square is $c$; the multiplication rule is the rule of (D.29.9) with these named $\beta, c$.

Conversely let a marked four-cusp group carry the specified $\delta$ and an involution $\tau$ with the permutation and multiplier in (D.37h), satisfying $[\delta, \tau] = 1$. The distinct permutations make $\Gamma = \langle \delta, \tau \rangle \cong V_4$. Take its center-free extension

$$K = \{(\varphi, \gamma) \in \mathrm{Aut}(N) \times \Gamma : [\varphi] = \gamma\}. \tag{D.37j}$$

Here $N$ embeds by its inner automorphisms. The preimage $K_\delta$ of $\langle \delta \rangle$ is identified with the exact group $F$ in (D.37a) by

$$f \longmapsto (\mathrm{Inn}(f)|_N, \delta^{\chi(f)}). \tag{D.37k}$$

Injectivity follows from $C_F(N) = 1$; surjectivity follows because an automorphism in either of these two outer classes differs from 1 or $\mathrm{Inn}(y)|_N$ by an inner automorphism from $N$. The quotient $K/F = C_2$ therefore gives an outer action on this very $F$. It is faithful, since any element of $K$ centralizing $F$ centralizes its inner copy of $N$, and an automorphism of the center-free $N$ with that property is the identity.

We check descent of the peripheral powers, not only of the four unordered procyclic subgroups. The two classes $A_0, A_1$ become the one $F$-class of $x$. The other classes are $B = y^2$ and $D = z^2$. For a representative of $\tau$, the induced automorphism $\psi$ of $F$ satisfies $\psi(y)^2 = hz^{2\lambda}h^{-1}$ for some $h \in N$. A.7 implies $C_F(hz^{2\lambda}h^{-1}) = h\langle z\rangle h^{-1}$, so $\psi(y) = hz^{\mu}h^{-1}$ and $2\mu = 2\lambda$. Multiplication by two is injective on $\widehat{\mathbb{Z}}$; hence $\mu = \lambda$. The same proof applies to $z$, and the $x$ exponent is already given by $A_0$. Thus the descended action fixes $x$, exchanges $y, z$, and has exactly the original $\lambda$, in particular $\lambda_2 = -1$.

For completeness its normalizer construction (D.37d)–(D.37e) and (D.37f) also recover the torsion-free assertion upstairs, without making that assertion an extra converse hypothesis. An order-two lift in the normalizer of $\langle x\rangle$ fixes $A_0$, so its image in $\Gamma$ is $\delta\tau$, not $\tau$. All torsion has this same image by (D.37f). The preimage of $\langle\tau\rangle$ is therefore torsion-free. The conjugation maps (D.37j)–(D.37k) identify the recovered extension with the original one on every element; conversely restriction to $N$ recovers the given $\Gamma$ and both named generators. This proves the asserted inverse correspondence of marked extensions.

We finish by making the obstacle for the D.29 gluing explicit. Write its ordered cusps as $1, 2, 3, 4$, so that $\sigma(\tau) = (1\ 3)(2\ 4)$. The unordered two-pants partitions and their images are

| partition | $\tau$ on its two blocks | (1 3) or (2 4) |
|---|---|---|
| 12 \| 34 | exchanges | 14 \| 23 |
| 13 \| 24 | preserves each | 13 \| 24 |
| 14 \| 23 | exchanges | 12 \| 34. |

(D.37l)

The only single transpositions commuting with $(1\ 3)(2\ 4)$ are $(1\ 3)$ and $(2\ 4)$: a commuting permutation must send its two two-cycles to two-cycles, and a transposition can do this only by exchanging the members of one of them. Hence a deck involution of the required permutation cannot act on the same two-vertex pants graph as the inverse-pair gluing, under any relabeling or choice of its exchanged partition.

One can nevertheless write the required geometric deck map in the D.29 free basis. For $(u_1, u_2, u_3, u_4) = (a, a^{-1}l, b, b^{-1}l^{-1})$, the change of four-cusp marking

$$A_0 = a, \quad A_1 = (a^{-1}l)b(a^{-1}l)^{-1}, \quad B = a^{-1}l, \quad D = b^{-1}l^{-1} \tag{D.37m}$$

satisfies $A_0A_1BD = 1$. Its inverse substitutions are $a = A_0$, $l = A_0B$, $b = B^{-1}A_1B$. Transporting (D.37c) through these maps gives

$$\Delta(l) = a^{-1}lb, \qquad \Delta(a) = a^{-1}lbl^{-1}a, \qquad \Delta(b) = a, \qquad \Delta^2 = \mathrm{Inn}(a^{-1}l). \tag{D.37n}$$

It has multiplier 1 and permutation $(1\ 3)$. The existence question for D.29 is the existence of a deck action of this marked geometric kind commuting outerly with its actual $\alpha$; (D.37l) alone does not decide it.

In particular failure of commutation for one selected deck representative cannot settle the question. In the geometric specialization $\lambda = -1$, $\epsilon = 1$, $\kappa = 1$, $f = 1$, D.29 gives

$$\alpha(l) = l, \quad \alpha(a) = b^{-1}, \quad \alpha(b) = la^{-1}l^{-1}.$$

For (D.37n), the words $\alpha\Delta(l) = bl^2a^{-1}l^{-1}$ and $\Delta\alpha(l) = a^{-1}lb$ are not conjugate even in the full profinite group: the finite quotient $l \mapsto (1\ 2\ 3)$, $a, b \mapsto (2\ 3)$ in $S_3$ sends the first to 1 and the

second to a nonidentity three-cycle. These two automorphisms therefore do not commute outerly. Yet another geometric deck does commute:

$$\begin{aligned} \Delta_0(l) = bla^{-1}, \qquad \Delta_0(a) = b, \qquad \Delta_0(b) = blal^{-1}b^{-1}, \\ \Delta_0^2 = \mathrm{Inn}(bl), \qquad \alpha\Delta_0 = \mathrm{Inn}(b^{-1})\Delta_0\alpha. \end{aligned} \tag{D.37o}$$

Its origin is also explicit. In the geometric extension $E = \widehat{F}(a,t)$, put $l = t^2$ and $b = ta^{-1}t^{-1}$. The involution $\eta(a) = a^{-1}$, $\eta(t) = a^{-1}t$ preserves the index-two kernel. Restricting $\mathrm{Inn}(t)\eta$ to that kernel gives exactly (D.37o). The formulas prove its peripheral permutation and multiplier as well as the two displayed composition identities. Thus this geometric specialization supplies an explicit example of the separator being moved. To verify directly that $\Delta_0$ is a deck of the specified geometric kind, adjoin its lift $r$ with $r^2 = bl$. The extension presentation eliminates to $\widehat{F}(a,r)$ by $b = rar^{-1}$ and $l = b^{-1}r^2 = ra^{-1}r$. The peripheral generators downstairs are $a, a^{-1}r, r^{-1}$, whose product is one. Their lifts are the two classes of $a, b$, the square $(a^{-1}r)^2 = a^{-1}l$, and the square $r^{-2}$, conjugate to $b^{-1}l^{-1}$. These substitutions identify its four-cusp inclusion with (D.37a)–(D.37b).

The executable `code/part_c/negative_transposition_d37.py` verifies the free-word covering and inverse substitutions, the deck squares, all the equations in (D.37o), the two-dimensional mod-two cohomology calculation, and the complete permutation calculation (D.37l) under all relabelings. The extension and torsion arguments above prove the statements for the full profinite groups. □

**D.38. [Proved] A nonnilpotent finite obstruction on the universal $A_5$ quotient.** Let $R$ be the intersection of the kernels of all continuous epimorphisms $\widehat{F}(x,y) \twoheadrightarrow A_5$. This is an open characteristic subgroup, and the quotient specified by the nineteen generating-pair representatives in the accompanying program is

$$Q = F/R \xrightarrow{\sim} A_5^{19}. \tag{D.38a}$$

Write $X, Y, Z = (XY)^{-1}$ for the images of the three cusps, and $\theta, \omega$ for the automorphisms of $Q$ induced by $x \leftrightarrow y$ and $x \mapsto y,\ y \mapsto (xy)^{-1}$. The image of every full $GT_0$ involution belongs to the following explicitly computed finite set:

$$\begin{aligned} \mathcal{T} = \{t \in Q : \quad & t^2 = 1, \quad tXt^{-1} = X^{-1}, \\ & \omega(t) = Y^{-1}\theta(t)\}. \end{aligned} \tag{D.38b}$$

Its normalized coordinates are $\alpha_Q = \mathrm{Inn}(t)$ and $f_Q = \theta(t)t^{-1}$. Every tuple in $\mathcal{T}$ satisfies the finite normalized relations (I), (II), and $f_Q\alpha_Q(f_Q) = 1$, and defines an actual automorphism of $Q$ of order two.

The executed exact enumeration gives

| finite set or test | cardinality |
|---|---|
| $\mathcal{T}$ | 10 125 000 |
| tuples surviving the full cusp-triangle obstruction | 5 062 500 |
| tuples conjugate to standard inversion in the finite pure model (D.38f) | 2 531 250 |
| triangle survivors outside that finite conjugacy set | 2 531 250. |

(D.38c)

Thus 5 062 500 explicit finite tuples satisfying the finite involution equations have no full pure-involution lift. The last row is disjoint from the image of the standard full conjugacy class. No tuple in that last row is asserted to have a full lift.

*Manuscript:* §31; folios 198-199verso; scans 391-394; Part D(b): actual full GT_0 torsion and conjugacy.

*Proof.* The program enumerates all $60^2$ ordered pairs in $A_5$, tests generation by closure under multiplication, and obtains 2 280 generating pairs. Simultaneous conjugation by all 120 elements of $S_5$ has nineteen free orbits on this set. These are all $\mathrm{Aut}(A_5)$-orbits: an automorphism permutes the five Sylow-2 subgroups, and equivariance with the faithful inner $A_5$ action makes that permutation representation of $\mathrm{Aut}(A_5)$ injective; conjugation by $S_5$ already supplies its 120 elements, as checked in D.31. Two epimorphisms to $A_5$ have the same kernel exactly when they differ by an automorphism of $A_5$. The nineteen kernels are therefore distinct. The joint quotient map is a subdirect product of nineteen copies of the nonabelian simple group $A_5$. The elementary subdirect-product argument shows that a proper subdirect product of copies of one simple group identifies two coordinates by an automorphism: induct on the number of factors and apply the normal-subgroup description of a direct product of nonabelian simple groups to the last projection. Such an identification would equate two of the nineteen kernels. This proves surjectivity of the specified map to $A_5^{19}$ and hence (D.38a). The kernel is characteristic because every automorphism of $F$ permutes all epimorphisms to $A_5$.

For each representative pair $(X_i, Y_i)$ the program also records the ordered triple of conjugacy classes of $X_i, Y_i, Z_i$ in $A_5$, allowing the simultaneous exchange of the two classes of elements of order five by odd conjugation in $S_5$. All nineteen resulting triples are different. Every triple contains an element of order five, and has at least two elements of odd order. These assertions are checked by complete finite enumeration, with representatives and component permutations retained in the output.

By D.34 a full pure involution has multiplier $-1$. Inversion preserves each conjugacy class of $A_5$: this is immediate for involutions, and reversal of a 3-cycle or 5-cycle is achieved by an even permutation. An automorphism of $A_5^{19}$ permutes its minimal normal simple factors and acts on each factor by an element of $\mathrm{Aut}(A_5) = S_5$. Preservation of the three cusp classes therefore forces its factor permutation to preserve the recorded class triple. The nineteen distinct triples force the permutation to be the identity. Since each triple contains an order-five entry, the component automorphisms must have even parity; odd conjugation exchanges its two order-five classes. Thus the induced automorphism of $Q$ is $\mathrm{Inn}(t)$ for a unique $t \in Q$. The group $Q$ has trivial center, so its involutivity gives $t^2 = 1$.

The exact Harbater–Schneps automorphism identities from [31, Propositions 3–4], with $m = (-1-1)/2 = -1$, descend to $Q$:

$$\theta\alpha_Q\theta^{-1}\alpha_Q^{-1} = \mathrm{Inn}(f_Q), \qquad \omega\alpha_Q\omega^{-1}\alpha_Q^{-1} = \mathrm{Inn}(Y^{-1}f_Q).$$

Center-freeness yields $f_Q = \theta(t)t^{-1}$ and $\omega(t)t^{-1} = Y^{-1}f_Q$. Together with the normalized equation on $X$ these give precisely (D.38b).

Conversely take $t$ in (D.38b) and define $f = \theta(t)t^{-1}$. Then $\theta(t)$ inverts $Y$, so

$$tYt^{-1} = f^{-1}Y^{-1}f, \qquad \theta(f)f = 1, \qquad f\,\mathrm{Inn}(t)(f) = 1.$$

The second relation in (D.38b) involving $\omega$ can be rewritten $f = Y\omega(t)t^{-1}$. Apply $\omega$ and $\omega^2$ and use $\omega^3 = 1$:

$$\omega(f) = Z\omega^2(t)\omega(t)^{-1}, \qquad \omega^2(f) = Xt\omega^2(t)^{-1}.$$

Substitution now proves the complete finite three-cycle relation by cancellation,

$$\omega^2(f)Z^{-1}\omega(f)Y^{-1}fX^{-1} = 1. \tag{D.38d}$$

The tuple $X$ has components of odd order, so $tXt^{-1} = X^{-1}$ excludes $t = 1$ in $Q$. Thus $\mathrm{Inn}(t)$ has exact order two. This verifies the assertions for every member of the finite set, independently of any lifting.

We next impose an obstruction from the full group. For each component the extension quotient is explicitly

$$A_5 \rtimes_{\mathrm{Inn}(t_i)} C_2 \xrightarrow{\sim} A_5 \times C_2, \qquad (g, s^j) \longmapsto (gt_i^j, s^j). \tag{D.38e}$$

The three vertices of the full cusp triangle in D.34 are represented by $s$, $xs$, and $yfs$. Their first coordinates under (D.38e) are $t_i$, $X_i t_i$, and $Y_i f_i t_i$. None can be the identity in $A_5$. Indeed that would make the image of the corresponding section central in $A_5 \times C_2$. This section class is incident to two different cusp edges of the full triangle. At either edge choose its representative in the cusp normalizer. Its image is still central, because centrality is unchanged by conjugation. It must therefore both act trivially on the cusp image and act by inversion there. Both incident cusp images would have order at most two. No cusp image is trivial for a generating $A_5$ pair. Two involutions generate a dihedral group, while any two entries of the generating triple generate $A_5$. This is a contradiction. Possible fusion of distinct section classes in the finite quotient does not change this argument.

For all components together, this obstruction is exactly $t_i \notin \{1, X_i\}$ for every $i$. The third vertex condition follows from these: $f_i t_i = \theta(t)_i$ and $\theta(X) = Y$, so $Y_i \theta(t)_i = 1$ is another occurrence of $t_j = X_j$ under the component action of $\theta$.

Here are the complete finite counts, with the factorization used to execute them. The relation $\omega(t) = Y^{-1}\theta(t)$ joins components in twelve independent sets, using indices $0, \ldots, 18$:

$$\{0,2\},\{1\},\{3,5\},\{4\},\{6,7\},\{8\},\{9,17\},\{10\},\{11,14\},\{12,18\},\{13,15\},\{16\}.$$

Exhaustive propagation of each initial component value through its specified $S_5$ conjugation maps gives respectively

$$3,3,3,3,4,2,5,5,5,5,5,5$$

assignments. The equations on a component set involve no other set, so their Cartesian product has exactly 10 125 000 elements. The triangle obstruction changes only the count for $\{6,7\}$, from four to two. Its surviving cardinality is therefore 5 062 500. All forty-eight component assignments are executed and checked, including the original equations (D.38d) and the finite norm; the Cartesian-product count is exact.

For precision, the finite conjugacy model in (D.38c) consists of the following actual automorphisms of $Q$ with a common unit exponent:

$$\begin{aligned}\mathcal{P}_Q = \{(\mu,\gamma) \in (\mathbb{Z}/30)^\times \times \mathrm{Aut}(Q) : \quad & \gamma(X) = X^\mu, \\ & \gamma(Y) \sim_Q Y^\mu, \quad \gamma(Z) \sim_Q Z^\mu\}.\end{aligned} \tag{D.38f}$$

Every normalized full pure automorphism maps to this set, since $R$ is characteristic. The recorded class triples again imply that an element of (D.38f) acts on the $i$th factor by conjugation by some $g_i \in S_5$, without permuting factors. Its parity is even for $\mu \bmod 5 \in \{1,4\}$ and odd for $\mu \bmod 5 \in \{2,3\}$. The remaining condition is exactly $g_i X_i g_i^{-1} = X_i^\mu$; with that parity, the $Y_i$ and $Z_i$ class conditions follow.

The standard inversion induces $\mathrm{Inn}(d)$ on $Q$, where the program supplies each simultaneous inverter $d_i$ of $(X_i, Y_i)$. The conjugacy condition is the finite equation $t_i = g_i d_i g_i^{-1}$ for every $i$. The program tests all 120 choices of $g_i$ for each of the eight units modulo thirty. There are exactly two distinct compatible patterns, each with component counts

$$3,3,3,3,1,1,5,5,5,5,5,5.$$

They are disjoint, so their total cardinality is $2 \cdot 1\,265\,625 = 2\,531\,250$. This proves the last two rows of (D.38c). It proves an upper constraint on the image of the known full conjugacy class, not surjectivity from that class onto the finite model.

The complete implementation is `code/part_a/a5_involution_quotient.py`; its output retains the nineteen pairs, the $\theta, \omega$ component maps, the standard inversion tuple, every component count, and an explicit rejected tuple. No family of lifts through all further characteristic quotients has been supplied for a surviving tuple. □

**D.39. [Proved] A rank-two Haran involution admitting neither hyperbolic loop type.** Let $\epsilon, \kappa \in \widehat{\mathbb{Z}}$ have the coordinates

$$\epsilon_3 = -1, \quad \epsilon_q = 1 \ (q \neq 3), \qquad \kappa_2 = 1, \quad \kappa_q = 0 \ (q \neq 2).$$

There is an actual involution of the rank-two free profinite group $H = \widehat{F}(a, b)$ given by

$$\alpha(a) = b^{-1}a^{\epsilon}b, \qquad \alpha(b) = b^{-1}a^{\kappa}. \tag{D.39a}$$

Its split extension has the explicit identification

$$H \rtimes_{\alpha} C_2 \xrightarrow{\sim} N \amalg C_2, \qquad N = (\mathbb{Z}_3 \rtimes \mathbb{Z}_2) \times \prod_{q \neq 2,3} \mathbb{Z}_q, \tag{D.39b}$$

where parity in $\mathbb{Z}_2$ acts by inversion on $\mathbb{Z}_3$. Its actual fixed subgroup $C_H(\alpha)$ is trivial. The outer involution preserves no three-cusp marking and no once-punctured-torus marking of $H$, under any full profinite change of free basis.

For the natural once-punctured-torus marking $\overline{\langle [a, b] \rangle}$, the peripheral condition also fails: a quotient of order 21 separates $\alpha([a, b])$ from every conjugate of $[a, b]^{-\epsilon}$.

*Manuscript:* §31; folios 198-199verso; scans 391-394; Part D: full rank-two loop actions and precise peripheral conditions.

*Proof.* The relations $\epsilon^2 = 1$ and $\epsilon\kappa = \kappa$ give, by substitution on the specified free generators,

$$\alpha^2(a) = (a^{-\kappa}b)(b^{-1}ab)(b^{-1}a^{\kappa}) = a,$$
$$\alpha^2(b) = (a^{-\kappa}b)(b^{-1}a^{\epsilon\kappa}b) = b.$$

Thus the universal continuous endomorphism defined by (D.39a) is an automorphism with square one. On full abelianization its matrix is

$$A = \begin{pmatrix} \epsilon & \kappa \\ 0 & -1 \end{pmatrix}, \qquad \det A = -\epsilon, \qquad \operatorname{tr} A = \epsilon - 1. \tag{D.39c}$$

It is nonidentity on abelianization, so both the automorphism and its outer class have exact order two.

For the exact identification (D.39b), use the normal form for $N$ verified in D.26:

$$N = \overline{\langle l, t \rangle}, \qquad tlt^{-1} = l^{\epsilon}, \qquad t^2 = l^{\kappa}.$$

The map $l^u t^i \mapsto (u_3, 2u_2 + i, (u_q)_{q \neq 2,3})$, $i \in \{0, 1\}$, is its displayed isomorphism with the group in (D.39b). The parity character sends $l$ to zero and $t$ to one. In $N \amalg \langle r \mid r^2 = 1 \rangle$ extend this character by $r \mapsto 1$. The substitutions

$$a \longmapsto l, \qquad b \longmapsto tr, \qquad s \longmapsto r$$

define a homomorphism from $H \rtimes_{\alpha} \langle s \rangle$ to that free profinite product. To check its relations, note that $r = (tr)^{-1}t$, so

$$rlr = (tr)^{-1}l^{\epsilon}(tr), \qquad r(tr)r = (tr)^{-1}l^{\kappa}.$$

In the reverse direction send $l \mapsto a$, $t \mapsto bs$, and $r \mapsto s$. Here $(bs)^2 = b\alpha(b) = a^{\kappa}$ and $(bs)a(bs)^{-1} = b\alpha(a)b^{-1} = a^{\epsilon}$. The normal-form multiplication of $N$ therefore gives its continuous homomorphism to the semidirect product, and the free product property extends it over $r$. The two composites fix every displayed generator, proving the isomorphism. In particular the parity kernel is exactly the free profinite group on the stated generators $a, b$, and the extension has the actual section $\langle r \rangle$.

The free factor $\langle r\rangle$ is malnormal by the full-profinite free-factor intersection property [36, Theorem 5 of the inspected author version]. Its centralizer in $N \amalg C_2$ is therefore precisely $\langle r\rangle$. Intersecting with the parity kernel proves $C_H(\alpha) = 1$.

In fact every involution $u \in H \rtimes_\alpha C_2$ has $C_H(u) = 1$. To prove this with the specified index-two kernel, reduce (D.39c) modulo two. The resulting matrix is $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Thus its dual action on $H^1(H, \mathbb{F}_2)$ is the regular two-dimensional $\mathbb{F}_2[C_2]$ module $J_2$. Its positive-degree $C_2$ cohomology vanishes: both differentials of the periodic resolution are $1 + \alpha$, and their image equals their kernel, both of dimension one. Since $H$ has cohomological dimension one and the extension is split, the Hochschild–Serre spectral sequence gives

$$\dim H^n(H \rtimes_\alpha C_2, \mathbb{F}_2) = 1 \qquad (n \geq 3). \tag{D.39g}$$

The full profinite Brown restriction theorem [42, Theorem 2 and §1.1], applied exactly as in (D.37f), expresses this cohomology as the product over involution classes. Every class contributes $1 + \dim H^1(C_H(u), \mathbb{F}_2)$, because its normalizer is $C_H(u) \times C_2$. The existing section $\langle s\rangle$ therefore accounts for the unique conjugacy class. Conjugacy by the extension agrees with conjugacy by $H$: for any involution $u$ the decomposition $E = H\langle u\rangle$ absorbs its second factor, which centralizes $u$. Every involution is consequently $H$-conjugate to $s$, and the already proved equality $C_H(s) = 1$ gives

$$C_H(u) = 1 \quad \text{for every involution } u \in E. \tag{D.39h}$$

There is a basis-independent obstruction to a three-cusp structure. An involution on three cusps has either the identity permutation or a transposition. In the former case its homology is a scalar matrix $\lambda I$ and has trace $2\lambda$, whose 2-coordinate is nonzero. This contradicts the zero 2-coordinate of the trace in (D.39c). For a transposition the reduced permutation module has trace zero, and multiplication by a common unit keeps that trace zero. This contradicts the 3-coordinate $-2$ of (D.39c). Trace is unchanged by conjugating the homology matrix under a change of free basis. Hence neither peripheral permutation can occur.

There is also a basis-independent obstruction to a once-punctured-torus marking. Take any full profinite free basis $a', b'$ of $H$ and put $c' = [a', b']$, $L = \overline{\langle c'\rangle}$. Preservation of the conjugacy class of $L$ would give the multiplier $\nu = \det A = -\epsilon$. The exact comparison is the one in (D.33h): in every finite class-two Heisenberg quotient the commutator exponent is the determinant of the abelianization matrix, so equality for all moduli identifies these elements of $\widehat{\mathbb{Z}}^\times$. Changing the free basis conjugates the matrix and leaves its determinant unchanged.

The normalizer $N_E(L)$ surjects onto $E/H = C_2$, and its kernel is $L$ by A.7 transported through this free-basis isomorphism. For a lift $v$ of the nonidentity element, write

$$vc'v^{-1} = (c')^\nu, \qquad v^2 = (c')^d, \qquad (\nu - 1)d = 0. \tag{D.39i}$$

Here $\nu_3 = 1$ and $\nu_q = -1$ for every $q \neq 3$. Therefore $d_q = 0$ for $q \neq 3$, including $q = 2$, since multiplication by two is injective on $\mathbb{Z}_q$. Choose $j \in \widehat{\mathbb{Z}}$ with $j_3 = -d_3/2$ and $j_q = 0$ otherwise. The lift $u = (c')^j v$ then satisfies

$$u^2 = (c')^{(1+\nu)j+d} = 1, \qquad u(c')^{e_3 w}u^{-1} = (c')^{e_3 w} \quad (w \in \mathbb{Z}_3), \tag{D.39j}$$

where $e_3$ is the idempotent supported at the prime three. The lift $u$ has exact order two because its image in $E/H$ is nonidentity. The peripheral injection $\widehat{\mathbb{Z}} \xrightarrow{\sim} L$ makes $(c')^{e_3\mathbb{Z}_3}$ a nontrivial subgroup of $C_H(u)$, contradicting (D.39h). This excludes every once-punctured-torus marking, including those obtained from arbitrary full profinite free bases.

For an additional finite certificate in the original basis, put $c = [a, b] = aba^{-1}b^{-1}$.

Use the finite group

$$Q = \mathbb{F}_7 \rtimes C_3, \qquad (v, i)(w, j) = (v + 2^i w,\ i + j), \tag{D.39d}$$

where the second coordinate is modulo three and 2 has order three in $\mathbb{F}_7^\times$. The assignment

$$a \mapsto (0,1), \qquad b \mapsto (1,1) \tag{D.39e}$$

is an epimorphism: $a, b$ have order three, their commutator is $(1,0)$, and these elements generate the normal $\mathbb{F}_7$ and the $C_3$ quotient. Since $\epsilon \equiv -1 \pmod 3$ and $\kappa \equiv 0 \pmod 3$, (D.39a) has the following exact images:

| word | $c$ | $\alpha(a)$ | $\alpha(b)$ | $\alpha(c)$ | $c^{-\epsilon}$ |
|---|---|---|---|---|---|
| image | $(1,0)$ | $(5,2)$ | $(3,2)$ | $(1,0)$ | $(6,0)$. |

(D.39f)

The last entry uses $(-\epsilon)_7 = -1$. Conjugacy on the normal $\mathbb{F}_7$ in (D.39d) is multiplication by a power of two. The two relevant orbits are exactly

$$\{(1,0),(2,0),(4,0)\}, \qquad \{(3,0),(5,0),(6,0)\}.$$

They are disjoint. Thus the images of $\alpha(c)$ and $c^{-\epsilon}$ are not conjugate, proving failure of the natural torus cusp.

The program `code/part_a/haran_torus_test.py` executes this order-21 quotient and an independent matrix check inside $\mathrm{PSL}_2(\mathbb{F}_7)$. For the latter it enumerates all 168 elements and uses

$$a = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix} \mod \{\pm I\}.$$

Both have order three, $c = \left(\begin{smallmatrix} 1 & 1 \\ 3 & 4 \end{smallmatrix}\right)$ has order seven, and $\alpha(c) = \left(\begin{smallmatrix} 0 & 3 \\ 2 & 2 \end{smallmatrix}\right)$. There are seven conjugators from $c$ to this image and none from $c^{-1}$. The matrix calculation agrees with (D.39f). The finite calculation detects the original cusp explicitly. The argument (D.39g)–(D.39j) supplies the exclusion for every alternative marking. The program also verifies the regular mod-two module used in (D.39g). □

**D.40. [Proved] A finite homomorphism-count obstruction to the extra deck symmetry.** The mixed-multiplier four-cusp involution constructed in D.29 commutes with no single-transposition involution of multiplier 1 in its full loop-outer group. In particular it admits none of the specified geometric deck actions required for the descent of D.37, including actions moving the two-pants separator.

More precisely, for the exact profinite group

$$E = \overline{\langle a, l, t \mid tlt^{-1} = l^{\epsilon},\ t^2 = l^{\kappa} \rangle}, \qquad p = a^{-1}l, \tag{D.40a}$$

with $\epsilon, \kappa$ as in D.29 and $\lambda = -\epsilon$, there is no continuous automorphism sending the conjugacy classes of $a, p$ to those of $a^\lambda, p$, respectively. There is also none sending those classes to $a, p^\lambda$. The obstruction is the exact inequality

$$\begin{aligned} \#\{\phi : E \to Q : \phi(a), \phi(p) \in C\} &= 10200, \\ \#\{\phi : E \to Q : \phi(a) \in C^{-1}, \phi(p) \in C\} &= 4920, \end{aligned} \tag{D.40b}$$

where all homomorphisms are continuous, $Q = \mathrm{PSL}_2(\mathbb{F}_{11})$, and $C$ is the conjugacy class of $\left(\begin{smallmatrix} 0 & 1 \\ 10 & 2 \end{smallmatrix}\right)$ modulo scalar matrices $\{\pm I\}$.

*Manuscript:* Section 24; folio 147 verso; scan 293. Section 30; folios 194-196; scans 383-387. Section 31; folios 198-199 verso; scans 391-394. Part D(b) and D(d).

*Proof.* We first identify the group and its peripheral marking. In the basis $(l, a, b')$ of (D.29.10), adjoining the outer lift $t$ gives the relations

$$tlt^{-1} = l^{\epsilon}, \quad tat^{-1} = b', \quad tb't^{-1} = l^{\kappa}al^{-\kappa}, \quad t^2 = l^{\kappa}.$$

Eliminating $b'$ gives (D.40a): the third relation follows from the second and fourth. Conversely $b' = tat^{-1}$ defines the inverse homomorphism from (D.40a) to this presentation. Both composites fix the named generators. Thus this is an isomorphism of profinite groups, and

$$E = N_{\mathrm{bad}} \amalg \widehat{\mathbb{Z}}(a) \tag{D.40c}$$

with the first factor generated by $l, t$. The index-two subgroup $H$ is the kernel of the character sending $t$ to 1 and $a, l$ to 0 in $C_2$.

The four $H$-peripheral classes of D.29 are represented by

$$a, \quad p = a^{-1}l, \quad b, \quad b^{-1}l^{-1}.$$

Since $\alpha$ interchanges the first and third classes, and the second and fourth, with multiplier $\lambda$, and conjugation by $t$ is $\alpha$, these are equally the classes of

$$a, \quad p, \quad (tat^{-1})^{\lambda}, \quad (tpt^{-1})^{\lambda}. \tag{D.40d}$$

Here $\lambda^2 = 1$, and the original cusp conjugators belong to $H$; thus (D.40d) records actual conjugacy classes under the specified inclusion into $E$. This description is independent of the Galois conjugator $f$ used to construct the two pants isomorphisms.

For a finite group $Q$ define the central function

$$F_Q(L) = \#\{T \in Q : TLT^{-1} = L^{\epsilon},\ T^2 = L^{\kappa}\}. \tag{D.40e}$$

Every assignment $a \mapsto A$, $p \mapsto P$ forces $l \mapsto AP$. The presentation (D.40a) then gives exactly $F_Q(AP)$ continuous homomorphisms to $Q$. For two conjugacy classes $C, D$ put

$$K_Q(C, D) = \sum_{A \in C,\ P \in D} F_Q(AP). \tag{D.40f}$$

Conjugating $L$ conjugates all its possible $T$, so $F_Q$ is central. Since $PA = A^{-1}(AP)A$, this also proves $K_Q(C, D) = K_Q(D, C)$.

Precomposition by a continuous automorphism of $E$ is a bijection on its continuous homomorphisms to $Q$. An automorphism with the first pair of class images in the claim would therefore give $K_Q(C, D) = K_Q(C^{\lambda}, D)$ for every pair $C, D$. The inverse map has the same power effect because $\lambda^2 = 1$, so this is equality of the two full sets, without a choice of conjugators in the target. The second proposed class effect would similarly give $K_Q(C, D) = K_Q(C, D^{\lambda})$.

We compute these integers in one completely specified finite group. Enumerate the determinant-one matrices over $\mathbb{F}_{11}$, and identify each matrix with its negative, using their lexicographically smaller representative. Matrix multiplication followed by this identification constructs all 660 elements of $Q$. The class $C$ has size 60 and order 11. Its inverse class is distinct and has the same size. Since $\lambda_{11} = -1$, we have $C^{\lambda} = C^{-1}$. The following table gives the full product counts needed in (D.40f). The representative of each nonidentity column is $\left(\begin{smallmatrix} 0 & 1 \\ 10 & r \end{smallmatrix}\right)$; the last column is the identity. We write $N_{C,D}(L) = \#\{(A, P) \in C \times D : AP \in [L]\}$.

| $r$ | 0 | 1 | 2 | 3 | 4 | 5 | 9 | $I$ |
|---|---|---|---|---|---|---|---|---|
| $\lvert L\rvert$ | 2 | 3 | 11 | 5 | 5 | 6 | 11 | 1 |
| class size | 55 | 110 | 60 | 132 | 132 | 110 | 60 | 1 |
| $F_Q(L)$ | 0 | 6 | 1 | 1 | 1 | 0 | 1 | 56 |
| $N_{C,C}(L)$ | 0 | 1320 | 120 | 660 | 660 | 0 | 840 | 0 |
| $N_{C^{-1},C}(L)$ | 660 | 0 | 120 | 660 | 660 | 1320 | 120 | 60 |

(D.40g)

Multiplying either of the last two rows by the $F_Q$ row and summing gives respectively

$$7920 + 120 + 660 + 660 + 840 = 10200, \qquad 120 + 660 + 660 + 120 + 3360 = 4920.$$

These unequal integers prove the first nonexistence assertion. Symmetry of $K_Q$ proves the second.

All entries of (D.40g) are verified by the finite enumeration in `code/root/haran_pair_counts.py`. It constructs the multiplication, inverse, power and conjugacy tables directly from the displayed matrices, computes the residues of $\epsilon, \kappa$ modulo each element order, and tests both equations in (D.40e) for every pair $(L, T)$. It then counts each product in the two displayed class pairs. Two independent computations, `code/part_a/haran_pair_count_audit.py` and `code/part_c/deck_obstruction_psl11_independent.py`, instead generate the faithful twelve-point permutation action by $z \mapsto z + 1$ and $z \mapsto -1/z$ on $\mathbf{P}^1(\mathbb{F}_{11})$. The first enumerates allowed $(L, T)$ before setting $P = A^{-1}L$; the second also enumerates the triples directly. Both reproduce (D.40b). The attached outputs include the full integer data.

Finally suppose a positive single-transposition involution $\delta$ commuted with the D.29 involution $\tau$. Their distinct peripheral permutations make $\langle \delta, \tau \rangle \simeq V_4$. Its center-free pullback extension inside $\mathrm{Aut}(H)$ contains $E$ as the normal preimage of $\langle \tau \rangle$, using the canonical conjugation identification of center-free extensions. Conjugation by any lift of $\delta$ therefore defines a continuous automorphism $\eta$ of this exact $E$. By D.37's permutation computation, $\delta$ is either (1 3) or (2 4) in the labeling (D.40d). Its positive multiplier is 1. In the first case $\eta$ sends $a$ to an $E$-conjugate of $a^\lambda$ and $p$ to an $E$-conjugate of $p$; in the second case the roles are reversed. Both have just been excluded. The argument imposes no preservation of the original pants separator and no order condition on the chosen lift of $\delta$. It proves the stated obstruction for this actual four-cusp construction; the general negative three-cusp conjugacy question remains in D.12. □

**D.41. [Proved] The global multiplier of a negative transposition.** An involution of the full three-cusp loop-outer group whose puncture permutation is a transposition and whose 2-adic multiplier is $-1$ has the global multiplier $-1 \in \widehat{\mathbb{Z}}^\times$. Together with D.34 and D.36, this identifies the multiplier of every involution in that full group with its global 2-adic sign. Consequently every finite subgroup generated by involutions has the usual global orientation character.

*Manuscript:* Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394.

*Proof.* Label the original peripheral generators as $x, y, z = (xy)^{-1}$ so that the involution fixes $x$ and exchanges $y, z$. Write its common multiplier as $\lambda$. It satisfies $\lambda^2 = 1$, so every coordinate $\lambda_p$ is 1 or $-1$. The peripheral-normalizer construction (D.37d)–(D.37e) supplies an actual order-two automorphism $\alpha$ of $F = \widehat{F}(x, y)$ representing it and satisfying $\alpha(x) = x^\lambda$.

Suppose a positive odd-prime coordinate exists, and let $q$ be the least odd prime with $\lambda_q = 1$. Use the exact cyclic quotient

$$\chi : F \longrightarrow C_q, \qquad \chi(x) = -2, \quad \chi(y) = \chi(z) = 1, \qquad N = \ker \chi. \tag{D.41a}$$

The relation $xyz = 1$ is respected. Since $\lambda = 1$ modulo $q$ and $\alpha$ exchanges the last two peripheral classes, $\chi\alpha = \chi$. The cover corresponding to $N$ is the completion of the specified connected cyclic cover of the pair of pants. Each of its three original cusps has one lift of degree $q$: all three displayed monodromies are nonzero in $C_q$. Its Euler characteristic is $-q$, so its genus is

$$g = (q - 1)/2. \tag{D.41b}$$

Fill the two cusp lifts represented by $y^q, z^q$. The quotient and its deck extension are explicit:

$$\begin{aligned} R &= F/\overline{\langle\langle y^q, z^q \rangle\rangle}_F = \langle y, z \mid y^q = z^q = 1 \rangle_{\mathrm{prof}} = C_q \amalg C_q, \\ K &= \ker(R \longrightarrow C_q, \ y, z \longmapsto 1) \\ &= N/\overline{\langle\langle y^q, z^q \rangle\rangle}_N \cong \widehat{\pi}_{g,1}. \end{aligned} \tag{D.41c}$$

Here $x = z^{-1}y^{-1}$, and the one remaining cusp is $c = x^q$. To verify the equality of normal subgroups used in (D.41c), note $F = N\langle y\rangle = N\langle z\rangle$. Conjugating $y^q$ or $z^q$ by any element of $F$ therefore gives an $N$-conjugate of the same element. Thus its closed $F$-normal closure is its closed $N$-normal closure. The completion comparison is also exact: start with the discrete cyclic surface cover, fill those two boundaries, and complete the displayed quotient presentation. Finite-index subgroup topology identifies its completed kernel with the specified kernel in $C_q \amalg C_q$. The inverse presentation substitutions are the original $y, z$ and $x = z^{-1}y^{-1}$.

The automorphism $\alpha$ preserves this normal subgroup, because it exchanges the two filled peripheral classes. It therefore induces an actual automorphism of $R$, preserves $K$, and induces an involution $\beta$ of $K$ with

$$\beta(c) = c^{\lambda}. \tag{D.41d}$$

Let $T$ be conjugation by $y$ on $K$. In $R$, $y^q = 1$, so $T^q = 1$ as an actual automorphism. The equality $\chi\alpha = \chi$ gives $\alpha(y)y^{-1} \in K$; hence $\beta T \beta^{-1}$ and $T$ differ by an inner automorphism of $K$. In particular their maps on abelianization commute.

We compute that abelianization as a deck module, without requiring a comparison of traces. Reidemeister–Schreier rewriting with representatives $1, y, \dots, y^{q-1}$ gives

$$u_i = y^i z y^{-(i+1)} \quad (i \in \mathbb{Z}/q), \qquad u_0 u_1 \cdots u_{q-1} = 1, \qquad T(u_i) = u_{i+1}. \tag{D.41e}$$

Exponents of $y$ here are interpreted in $R$, where $y^q = 1$. The first $q-1$ elements form a free profinite basis; eliminate the last by $u_{q-1} = (u_0 \cdots u_{q-2})^{-1}$. These substitutions give both maps in the Schreier presentation. The remaining boundary also has the exact word

$$c = x^q = \prod_{j=0}^{q-1} u_{q-1-2j}^{-1}, \qquad \text{indices modulo } q. \tag{D.41f}$$

Indeed $z = u_0 y$ gives $x = z^{-1}y^{-1} = y^{-1}u_0^{-1}y^{-1} = u_{q-1}^{-1}y^{-2}$. Multiplying its $q$ successive translates by $y^{-2}$ gives exactly (D.41f), because $y^{2q} = 1$.

For any prime $p$ put $V = H_1(K, \mathbb{Q}_p)$ and let $A$ be the map induced by $\beta$. From (D.41e) we obtain the exact module isomorphism

$$V \cong \mathbb{Q}_p[C_q]/\mathbb{Q}_p(1 + T + \cdots + T^{q-1}) \cong \mathbb{Q}_p[X]/\Phi_q(X), \qquad A^2 = 1, \quad AT = TA. \tag{D.41g}$$

Every irreducible cyclotomic factor occurs with multiplicity one.

The one-boundary surface structure in (D.41c) supplies a nondegenerate alternating pairing, and its relation to (D.41d) is algebraic. In a surface free basis the boundary word is the product of the $g$ commutators, up to inversion and conjugation. Its degree-two image in the class-two free pro-$p$ quotient is a nondegenerate bivector $\omega \in \bigwedge^2 V$. Equation (D.41d) gives $(\bigwedge^2 A)\omega = \lambda_p \omega$. The deck map preserves its peripheral exponent, so $(\bigwedge^2 T)\omega = \omega$. If $\Omega$ is the nonsingular skew matrix of $\omega$, these are precisely

$$A\Omega A^{\mathsf{t}} = \lambda_p \Omega, \qquad T\Omega T^{\mathsf{t}} = \Omega.$$

Consequently the alternating form with matrix $\Omega^{-1}$ satisfies

$$\langle Av, Aw\rangle = \lambda_p \langle v, w\rangle, \qquad \langle Tv, Tw\rangle = \langle v, w\rangle. \tag{D.41h}$$

This proves the multiplier comparison also at $p = 2$: one may first take the integral commutator calculation in each finite class-two quotient and then tensor with $\mathbb{Q}_2$. It uses no geometric realization of $\beta$.

Now let $p \neq q$ be a prime with $d = \operatorname{ord}_q(p)$ even. The irreducible factors of $\Phi_q$ over $\mathbb{Q}_p$ have root sets the Frobenius orbits

$$\{\zeta^{jp^k} : 0 \le k < d\}, \qquad j \in (\mathbb{Z}/q)^{\times}. \tag{D.41i}$$

For completeness these roots lie in the unramified extension with residue field $\mathbb{F}_{p^d}$: the primitive $q$th roots in that field lift uniquely since $p \neq q$, and Frobenius lifts their map $z \mapsto z^p$. The least such degree is $d$, proving the factor description. Since $d$ is even, $p^{d/2} = -1$ modulo $q$, and each of these root sets is closed under inversion.

Write $V = \bigoplus V_i$ for those irreducible summands. By (D.41g), the commutant of $T$ is the product of their fields $\mathbb{Q}_p(\zeta^j)$. Therefore $A|_{V_i}$ is multiplication by an element of a field whose square is one, and so is the scalar 1 or $-1$ on that entire summand. Each $V_i$ is nondegenerate for the form in (D.41h). To see this explicitly after a splitting-field extension, a $T$-eigenvector with eigenvalue $\zeta^j$ can pair only with the eigenvalue $\zeta^{-j}$, since the pairing is $T$-invariant. Both belong to $V_i$ by (D.41i); different summands are orthogonal. Global nondegeneracy thus restricts to nondegeneracy on $V_i$. Multiplication by either scalar 1 or $-1$ preserves that restricted form. Comparing with (D.41h) gives

$$\lambda_p = 1 \quad \text{whenever } p \neq q \text{ and } \operatorname{ord}_q(p) \text{ is even.} \tag{D.41j}$$

There is a prime divisor $p$ of $q-1$ having this even order. Otherwise every prime factor of $q-1$ would lie in the odd-order subgroup of the cyclic group $(\mathbb{Z}/q)^\times$. Their product, with multiplicities, would also have odd order. But that product is $q-1 = -1$ modulo $q$, of order two. This contradiction proves the assertion. Such a prime satisfies $p < q$. For $p = 2$ the hypothesis gives $\lambda_p = -1$; for an odd $p < q$ the minimal choice of $q$ gives the same value. Both contradict (D.41j).

Thus no positive odd-prime coordinate exists, and $\lambda = -1$ globally. D.34 handles a pure involution, while D.36 handles an involution with positive 2-adic multiplier. All involutions are therefore accounted for. Multiplicativity proves the last assertion for every subgroup generated by them. This conclusion determines these actual multiplier homomorphisms; it does not assert geometric conjugacy of the finite actions or settle the multiplier of an isolated order-three subgroup.

The program `code/part_c/negative_multiplier_d41.py` checks the exact Schreier boundary word, the induced cyclotomic deck matrices and the nondegenerate degree-two boundary form in examples, and checks the prime-factor and Frobenius-orbit calculation. The displayed argument proves the assertion for every odd prime, rather than only for the executed range. □

**D.42. [Proved] Order-three pants actions have global multiplier one.** Every element of order three in the full three-cusp loop-outer group has orientation multiplier $1 \in \widehat{\mathbb{Z}}^\times$. Consequently every finite subgroup of that full group has multiplier image contained in the two global units $\{1, -1\}$, and its multiplier is determined by its 2-adic coordinate.

The closed-surface calculation used in the proof is also unconditional. For $g \geq 2$ and every prime $p$, the homomorphism

$$\operatorname{Aut}(\widehat{\pi}_{g,0}) \longrightarrow \operatorname{GL}(H_1(\widehat{\pi}_{g,0}, \mathbb{Z}_p)) \tag{D.42a}$$

is injective on every finite $p$-subgroup. Here the source consists of actual continuous automorphisms. The order-three outer action in the first assertion is supplied with an actual lift by the exact rank-two lifting result D.27.

*Manuscript:* §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D: full loop-outer actions of type (0,3).

*Proof.* We first prove (D.42a), keeping the two completion maps and the fixed subgroup explicit.

*The pro-p automorphism calculation.* Let $P = \pi_{g,0}^{(p)}$ be the pro-$p$ completion. The closed lower central terms $\gamma_n P$ satisfy

$$\bigcap_n \gamma_n P = 1, \qquad P/\gamma_n P \text{ is torsion-free for every } n. \tag{D.42b}$$

These are precisely [55, Propositions 2.14–2.15, pp. 1515–1516], applied to the orientable surface presentation. The second proposition checks the primitive degree-two initial form of its single defining relation; the first passes the torsion-free nilpotent quotients to the pro-$p$ completion by the stated canonical quotient maps. In particular every $\gamma_n P/\gamma_{n+1}P$, as a subgroup of $P/\gamma_{n+1}P$, is torsion-free.

Put $K_n = \ker(\mathrm{Aut}(P) \to \mathrm{Aut}(P/\gamma_{n+1}P))$ for $n \geq 1$. For $b \in K_n$, the formula

$$d_n(b)(\bar{z}) = b(z)z^{-1} \pmod{\gamma_{n+2}P} \tag{D.42c}$$

defines a continuous homomorphism from $P^{\mathrm{ab}}$ to $\gamma_{n+1}P/\gamma_{n+2}P$. Indeed this last quotient is central modulo $\gamma_{n+2}P$, which makes the formula additive in $z$ and trivial on commutators. An automorphism acting trivially on $P^{\mathrm{ab}}$ acts trivially on every lower central graded quotient: these quotients are generated by the iterated commutators of its abelianization generators. Consequently $d_n$ is a group homomorphism on $K_n$, and its kernel is exactly $K_{n+1}$. Its target is torsion-free by (D.42b). A finite-order element of $K_1$ therefore lies successively in every $K_n$ and is the identity. Thus the kernel of $\mathrm{Aut}(P) \to \mathrm{GL}(H_1(P, \mathbb{Z}_p))$ is torsion-free.

*The full fixed subgroup.* Write $\Pi = \widehat{\pi}_{g,0}$ and let $a \in \mathrm{Aut}(\Pi)$ satisfy $a^p = 1$ and act trivially on $H_1(\Pi, \mathbb{Z}_p)$. Its induced automorphism on $P = \Pi^{(p)}$ is the identity by the preceding calculation. There is therefore a specified quotient homomorphism

$$E = \Pi \rtimes_a C_p \longrightarrow P \times C_p, \qquad (h, s^i) \longmapsto (\bar{h}, s^i). \tag{D.42d}$$

The natural map $\Pi \to P$ induces an isomorphism on mod-$p$ cohomology, including products. Both groups have cohomological $p$-dimension two, first cohomology $\mathbb{F}_p^{2g}$, and second cohomology $\mathbb{F}_p$ with the standard nondegenerate surface cup pairing. The map is the identity on the generator characters, so the pairing shows that it is also an isomorphism in degree two; higher degrees vanish. These are the surface duality comparisons of A.2 and A.7, with the full and pro-$p$ formations respectively. Hochschild–Serre comparison for the named map (D.42d) now gives a ring isomorphism

$$H^*(P \times C_p, \mathbb{F}_p) \xrightarrow{\sim} H^*(E, \mathbb{F}_p). \tag{D.42e}$$

In particular $\dim H^n(E, \mathbb{F}_p) = 2g + 2$ for $n > 2$.

Choose a nonzero surface class $\eta \in H^2(P, \mathbb{F}_p)$ and a nonzero periodic class $u \in H^2(C_p, \mathbb{F}_p)$. For every $m \geq 1$, the class $\eta u^m \in H^{2m+2}(E, \mathbb{F}_p)$ is nonzero under (D.42e). Every subgroup $S \leq E$ of order $p$ projects isomorphically to $C_p$, because $\Pi$ is torsion-free. Its first coordinate in $P \times C_p$ is trivial because $P$ is torsion-free. Its normalizer is exactly

$$N_E(S) = C_\Pi(S) \times S. \tag{D.42f}$$

An element normalizing $S$ fixes each of its elements by projection to the abelian quotient $C_p$; splitting off the uniquely determined $S$ component proves (D.42f).

Apply the full profinite Brown restriction theorem [42, Theorem 2 and §1.1] with virtual cohomological dimension two. Its hypotheses hold: $\Pi$ is an open subgroup of cohomological $p$-dimension two, and there is no $C_p \times C_p$ in $E$, since every finite subgroup injects into $E/\Pi = C_p$. The finite-dimensional source makes its dense restriction embedding an isomorphism onto a finite product of the normalizer cohomologies. Restriction of the nonzero class $\eta u^m$ is nonzero in at least one factor. For that factor put $C = C_\Pi(S)$. Because the first coordinate of $S$ in (D.42d) is trivial, its restriction is exactly

$$(\eta|_C) \otimes (u^m|_S) \in H^2(C, \mathbb{F}_p) \otimes H^{2m}(S, \mathbb{F}_p). \tag{D.42g}$$

Thus $\eta|_C \neq 0$. The restriction $H^1(\Pi, \mathbb{F}_p) \to H^1(C, \mathbb{F}_p)$ is injective: for any nonzero $v$, nondegeneracy supplies $w$ with $v \smile w$ a nonzero multiple of $\eta$, and its restriction cannot vanish. Hence

$\dim H^1(C,\mathbb{F}_p) \geq 2g$ and $\dim H^2(C,\mathbb{F}_p) \geq 1$. For every normalizer (D.42f), the Künneth formula in degree $n > 2$ contributes

$$1 + \dim H^1(C_\Pi(S),\mathbb{F}_p) + \dim H^2(C_\Pi(S),\mathbb{F}_p). \tag{D.42h}$$

This holds also for odd $p$: the exterior generator of $H^*(C_p,\mathbb{F}_p)$ and its degree-two polynomial generator give one dimension in each nonnegative degree. At $p = 2$ the degree-one polynomial generator has the same dimension consequence. The one factor in (D.42g) already accounts for all $2g + 2$ dimensions. There is exactly one conjugacy class of sections, and for its centralizer

$$\dim H^1(C,\mathbb{F}_p) = 2g, \qquad H^2(C,\mathbb{F}_p) = \mathbb{F}_p. \tag{D.42i}$$

Conjugacy under $E$ equals conjugacy under $\Pi$, since $E = \Pi S$ for each section $S$ and the $S$ factor centralizes that section. Consequently these conclusions apply to $C = C_\Pi(a)$ for the specified section in (D.42d).

We now prove directly that this closed subgroup equals $\Pi$. Continuous cohomology has the restriction comparison

$$H^j(C,\mathbb{F}_p) = \operatorname*{colim}_{C \leq U \leq_o \Pi} H^j(U,\mathbb{F}_p). \tag{D.42j}$$

For an explicit cochain comparison, choose an open normal subgroup $V$ of $\Pi$ so small that the given continuous finite-valued cochain on $C$ factors through $C/(C \cap V)$ in every variable. The isomorphism $CV/V = C/(C \cap V)$ inflates it to the open subgroup $CV$. Cocycle equalities descend to that finite quotient; for a coboundary choose a common $V$ also for its primitive cochain. This proves (D.42j) with the displayed restriction maps. Each $U$ is the completion of its finite covering surface, of genus $g_U = 1 + [\Pi : U](g-1)$. For $V \leq U$ the restriction on $H^2$ multiplies the fundamental class by $[U : V]$. A nonzero class surviving in (D.42j) in degree two therefore has a representative at some $U$ for which every smaller open $V \geq C$ has $p \nmid [U : V]$. For these subgroups the restriction on $H^1$ is injective: corestriction composed with restriction is multiplication by this prime-to-$p$ index. If $C$ had infinite index, the indices $[\Pi : V]$ would be unbounded, making

$$\dim H^1(V,\mathbb{F}_p) = 2 + 2[\Pi : V](g-1)$$

unbounded inside the fixed finite-dimensional $H^1(C,\mathbb{F}_p)$, contrary to (D.42i). Thus $C$ is open. The same genus formula and (D.42i) give $[\Pi : C] = 1$. It follows that $a = 1$. The kernel of (D.42a) on a finite $p$-subgroup cannot contain an element of order $p$, proving its asserted injectivity.

*The order-three cyclic cover.* Let $g$ have order three in the full pants group, with multiplier $\lambda$. By D.32 its cusp permutation is a three-cycle. Label $x, y, z = (xy)^{-1}$ so that it sends their classes successively to $y, z, x$, with the common power $\lambda$. D.27 gives an actual order-three representative $\alpha \in \mathrm{Aut}(\widehat{F}(x,y))$. The equality $\lambda^3 = 1$ implies $\lambda_3 = 1$: an order-three unit of $\mathbb{Z}_3$ reduces to one, and for $v \in \mathbb{Z}_3^\times$, $r \geq 1$, the valuation of $(1 + 3^r v)^3 - 1$ is $r + 1$.

Suppose a coordinate $\lambda_q$ were nonidentity. Then $q \neq 3$, and its reduction $\zeta \in \mathbb{F}_q^\times$ has exact order three; in particular $q \equiv 1 \pmod 3$ and $q \geq 7$. Define the surjective character

$$\chi : F \longrightarrow C_q = \mathbb{F}_q, \qquad \chi(x) = \zeta, \quad \chi(y) = 1, \quad \chi(z) = \zeta^2. \tag{D.42k}$$

The relation is respected because $1 + \zeta + \zeta^2 = 0$. The three scalar identities $\zeta \cdot 1 = \zeta$, $\zeta \cdot \zeta^2 = 1$, and $\zeta \cdot \zeta = \zeta^2$ show exactly that $\chi\alpha = \chi$; all inner conjugators disappear under $\chi$. Let $N = \ker \chi$. The specified degree-$q$ cover has one cusp above each of the original three, since all three values in (D.42k) are nonzero. Its type is $(h, 3)$ with $h = (q-1)/2 \geq 3$, by $2 - 2h - 3 = -q$. Fill these three cusps and put

$$\Pi_h = N/\overline{\langle\langle x^q, y^q, z^q \rangle\rangle}_N = \widehat{\pi}_{h,0}. \tag{D.42l}$$

Completion of the finite-cover presentation gives this exact quotient map. The peripheral powers under $\alpha$ preserve the indicated closed normal subgroup, so it induces an actual automorphism $b$

of $\Pi_h$ with $b^3 = 1$. Conjugation by $y$ gives the deck automorphism $T$ of this quotient; $T^q = 1$, because $y^q$ has been killed. The relation $\chi\alpha = \chi$ says that $bT$ and $Tb$ differ by an inner automorphism. Their actions on homology therefore commute.

Here is an explicit calculation of that deck module. First fill only $y^q, z^q$ upstairs. The group downstairs is $D = C_q(y) \amalg C_q(z)$, with its map to $C_q$ given by $y \mapsto 1$, $z \mapsto r = \zeta^2$. Its kernel $K$ has Schreier generators

$$v_i = y^i z y^{-i-r} \quad (i \in \mathbb{Z}/q), \qquad v_0 v_r v_{2r} \cdots v_{(q-1)r} = 1. \tag{D.42m}$$

The displayed relation contains every generator once and eliminates one; these are the completed discrete Schreier presentation and maps. Conjugation by $y$ sends $v_i$ to $v_{i+1}$. The remaining cusp $x^q = (z^{-1}y^{-1})^q$ rewrites as

$$\prod_{j=0}^{q-1} v_{-r-j(r+1)}^{-1}. \tag{D.42n}$$

Since $r + 1 \neq 0$, it also uses every index once. Its abelianization is the negative of the already imposed sum relation. Filling it changes no first homology. It follows, for the actual map $K \to \Pi_h$, that

$$H_1(\Pi_h, \mathbb{Q}_3) = \mathbb{Q}_3[C_q]/\mathbb{Q}_3(1 + T + \cdots + T^{q-1}) \simeq \mathbb{Q}_3[X]/\Phi_q(X). \tag{D.42o}$$

This is the regular augmentation module, with each irreducible factor occurring once.

Because $q \neq 3$, reduction of $\Phi_q$ modulo three is square-free. Hensel lifting its irreducible factors expresses the last algebra in (D.42o) as a product of finite unramified extensions of $\mathbb{Q}_3$. None contains a nonidentity cube root of one: reduction to the residue field, whose multiplicative group has order $3^d - 1$, first gives residue one; the same valuation calculation used for $\mathbb{Z}_3$ then applies, since three remains a uniformizer. The centralizer of $T$ on its cyclic module (D.42o) is exactly that product algebra. Thus the commuting operator $b_*$, with $b_*^3 = 1$, equals one in every factor. The torsion-free lattice $H_1(\Pi_h, \mathbb{Z}_3)$ embeds in (D.42o), so $b_*$ is also the identity on that lattice. The first part of this proof, applied with $p = 3$, now gives $b = 1$ on the full $\Pi_h$.

Finally the quotient filling comparison in A.2 identifies the relative orientation line of $(N; \langle x^q \rangle, \langle y^q \rangle, \langle z^q \rangle)$ with the closed orientation line of $\Pi_h$. The three cusp maps have the common power $\lambda$, so $b$ acts on the latter line by $\lambda$ as well. Its being the identity forces $\lambda_q = 1$, contradicting the chosen coordinate. There is no such $q$, proving $\lambda = 1$.

For the finite-subgroup assertion, D.32 embeds every finite subgroup in $S_3 \times C_2$. Its elements therefore have orders $1, 2, 3$, or 6. D.34, D.36, and D.41 give the global sign for every involution; the preceding proof handles order three. An element $v$ of order six is the product of its commuting powers $v^3$ and $v^4$, of orders two and three respectively. Multiplicativity gives the assertion for every element, with its prescribed 2-adic sign.

The program `code/part_a/pants_order_three.py` executes the character invariance, both Schreier index cycles, the cyclic deck matrix and its cyclotomic polynomial, and the characteristic-three factorization checks for both primitive cube roots at every prime $q \equiv 1 \pmod 3$ through 97. The arguments above prove the result for every prime and for the full profinite groups. □

**D.43. [Proved] Global orientation of punctured-torus involutions.** Every involution in the full once-punctured-torus loop-outer group has the global orientation multiplier 1 or $-1$, with the same sign as its 2-adic determinant. Consequently the orientation character of every finite subgroup generated by involutions has values in the global two-element subgroup $\{1, -1\} \subset \widehat{\mathbb{Z}}^\times$.

*Manuscript:* Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394.

*Proof.* Write $F = \widehat{F}(x, y)$ and $c = [x, y]$ for the torus cusp. D.27 supplies an actual order-two representative $\alpha$ of the given outer involution. Its peripheral multiplier is

$$\lambda = \det(\alpha|_{F^{\mathrm{ab}}}) \in \widehat{\mathbb{Z}}^{\times}, \qquad \lambda^2 = 1, \tag{D.43a}$$

by the exact Heisenberg comparison of D.33. Thus every $\lambda_p$ is 1 or $-1$.

*The characteristic Heisenberg cover.* For an odd prime $q$, take the characteristic quotient

$$F \longrightarrow G_q, \qquad G_q = \langle X, Y, Z \mid X^q = Y^q = Z^q = 1,\ [X, Y] = Z,\ [X, Z] = [Y, Z] = 1 \rangle, \tag{D.43b}$$

with $x \mapsto X$, $y \mapsto Y$, $c \mapsto Z$. Collection of these words identifies $G_q$ with the order-$q^3$ Heisenberg group. Explicitly it is $\mathbb{F}_q^2 \times \mathbb{F}_q$, with multiplication

$$(v, z)(w, t) = (v + w, z + t + \tfrac{1}{2}\omega(v, w)), \qquad \omega((a, b), (a', b')) = ab' - ba'. \tag{D.43c}$$

Take $X = ((1, 0), 0)$, $Y = ((0, 1), 0)$ and $Z = ((0, 0), 1)$. The kernel $N_q$ is the closed subgroup generated by the laws $\gamma_3 F$ and all $q$th powers, so it is characteristic. More precisely it is their closed normal closure: the exponent-$q$, class-two quotient has the normal forms $X^a Y^b Z^d$ in (D.43b), and (D.43c) verifies that all $q^3$ forms are distinct. This proves the characteristic-quotient identification.

The corresponding discrete cover of a once-punctured torus has $q^2$ cusps, one for each coset of $\langle Z \rangle$ in $G_q$, each of degree $q$. Fill all these cusps. Completing the finite-cover presentation gives the exact quotient

$$N_q \longrightarrow \Pi_q = N_q / \overline{\langle\langle hc^q h^{-1} : h \in F \rangle\rangle}_{N_q} \cong \widehat{\pi}_{g_q, 0}, \qquad 2g_q = 2 + q^3 - q^2. \tag{D.43d}$$

Representatives for the $q^2$ cosets suffice in the normal closure. The genus formula follows by multiplying the punctured torus's Euler characteristic by $q^3$ and then adding its $q^2$ filled discs. The original deck group induces its actual homology representation on this closed surface. Since $\alpha$ preserves $N_q$ and permutes all the filled peripheral classes, it induces an actual involution $\beta_q$ of $\Pi_q$.

For every prime $p$, put $V_p = H_1(\Pi_q, \mathbb{Q}_p)$, let $\rho$ be its $G_q$ representation, and let $A$ be the map induced by $\beta_q$. If $\theta$ is the automorphism induced by $\alpha$ on $G_q$, the precise deck comparison is

$$A^2 = 1, \qquad A\rho(h)A^{-1} = \rho(\theta(h)), \qquad \theta(Z) = Z^{\lambda \bmod q}. \tag{D.43e}$$

Indeed representatives in $F$ for the two sides differ only by an element of $N_q$, whose conjugation is inner on $\Pi_q$ and hence trivial on its first homology.

The nondegenerate alternating surface pairing satisfies

$$\langle Av, Aw \rangle = \lambda_p \langle v, w \rangle, \qquad \langle \rho(h)v, \rho(h)w \rangle = \langle v, w \rangle. \tag{D.43f}$$

Here is an exact verification of the first multiplier. The involution permutes the odd number $q^2$ of lifted cusps, so one cusp class is fixed. Fill every other cusp first. This is an invariant closed normal quotient, and gives a surface group of genus $g_q$ with one cusp, which $\alpha$ preserves with power $\lambda$. Filling its last cusp does not change first homology. The degree-two boundary bivector calculation in (D.41h) therefore proves the first identity in (D.43f) for precisely $A$ on $V_p$. The second identity comes from the orientation-preserving geometric deck maps of the specified finite surface cover. Thus no realization of $\alpha$ as a homeomorphism has been used.

We compute the whole deck module by finite cover chains. The cover before filling retracts onto the regular cover of a two-loop rose, so its first homology has representation class $1 + \mathbb{Q}_p[G_q]$. The relative second homology of the attached discs is $\mathbb{Q}_p[G_q/\langle Z \rangle]$. The exact homology sequence of this filling, including the invariant orientation class in degree two, gives

$$[V_p] = 2[1] + [\mathbb{Q}_p[G_q]] - [\mathbb{Q}_p[G_q/\langle Z \rangle]]. \tag{D.43g}$$

All these maps are deck-equivariant. They compute the homology of the completed surface groups through the finite-index and surface-homology comparisons already used in A.2 and D.42.

Over a characteristic-zero field containing a primitive $q$th root $\zeta$, let $V_j$ be the subspace where $Z$ acts as $\zeta^j$, for $j \in \mathbb{F}_q^\times$. The last permutation module in (D.43g) is trivial on $Z$, and the regular representation has each of its $q$ central eigenvalues with multiplicity $q^2$. Semisimplicity therefore gives

$$\dim V_j = q^2. \tag{D.43h}$$

The $G_q$-invariant alternating form pairs $V_j$ only with $V_{-j}$, and gives a nondegenerate pairing between them.

*One negative odd-prime sign determines every sign.* Suppose $\lambda_q = -1$ at an odd prime $q$. For any prime $p$, extend scalars to $\mathbb{Q}_p(\zeta)$. By (D.43e), $A$ exchanges $V_j$ and $V_{-j}$. Choose bases of these two spaces and write $P$ for their nonsingular pairing matrix, $B$ for $A : V_j \to V_{-j}$ and $C$ for $A : V_{-j} \to V_j$. Then $CB = BC = 1$, and alternation in (D.43f) gives

$$-B^{\mathsf{t}} P^{\mathsf{t}} C = \lambda_p P. \tag{D.43i}$$

The dimension $q^2$ is odd. Taking determinants in (D.43i) yields $-1 = \lambda_p^{q^2}$, hence $\lambda_p = -1$ by (D.43a). This works also at $p = q$ and $p = 2$, since the calculation is over characteristic-zero fields. Thus

$$\lambda_q = -1 \text{ for an odd } q \quad \Longrightarrow \quad \lambda = -1. \tag{D.43j}$$

*The positive* 3*-adic case.* It remains to prove that $\lambda_3 = 1$ forces $\lambda_2 = 1$. The matrix of $\alpha$ on $H_1(F, \mathbb{Z}_3)$ has square one and determinant one. Its two eigenvalues over $\mathbb{Q}_3$ are consequently both 1 or both $-1$, and it is diagonalizable, so the matrix is $I$ or $-I$.

In the group $G_3$ of (D.43c), the resulting $\theta$ has the form

$$\theta(v, z) = (Mv, z + \ell(v)), \qquad M = I \text{ or } -I, \tag{D.43k}$$

with $\ell$ a linear functional. For $M = I$, $\theta^2 = 1$ implies $2\ell = 0$ and $\ell = 0$. For $M = -I$, conjugation of $\theta$ by the inner automorphism of $(u, 0)$ replaces $\ell(v)$ by $\ell(v) - 2\omega(u, v)$. Choose $u$ with $\omega(u, v) = \ell(v)/2$. Lifting $(u, 0)$ to $F$ and conjugating the actual automorphism $\alpha$ by that inner automorphism preserves its order two and its peripheral multiplier. It gives exactly

$$\theta = 1 \quad \text{or} \quad \theta = \iota, \qquad \iota(v, z) = (-v, z). \tag{D.43l}$$

This is conjugation of an actual automorphism, not multiplication by an arbitrary inner lift.

Work now over the unramified quadratic field $L = \mathbb{Q}_2(\zeta_3)$, with $\sigma(\zeta_3) = \zeta_3^{-1}$. For $j = 1, 2$ the irreducible $G_3$ module $W_j$ with central character $\zeta_3^j$ has basis $e_r$, $r \in \mathbb{F}_3$, and matrices

$$Xe_r = e_{r+1}, \qquad Ye_r = \zeta_3^{-jr} e_r, \qquad Ze_r = \zeta_3^j e_r. \tag{D.43m}$$

These matrices satisfy (D.43b). The distinct eigenvalues of $Y$ and the cyclic action of $X$ show irreducibility. The diagonal projectors for $Y$, followed by powers of $X$, span the full matrix algebra. Hence the central block of $L[G_3]$ is $M_3(L)$. Together with (D.43g)–(D.43h), this gives

$$V_j = W_j \otimes_L U_j, \qquad \dim U_j = 3. \tag{D.43n}$$

The matrix $J = I$ implements $\theta = 1$ on $W_j$. For $\theta = \iota$, take $Je_r = e_{-r}$. This rational permutation matrix implements $X \mapsto X^{-1}$, $Y \mapsto Y^{-1}$, fixes $Z$, and has $J^2 = I$ and $J^{\mathsf{t}} J = I$. Thus (D.43e) and the full matrix-algebra commutant give

$$A|_{V_j} = J \otimes B_j, \qquad B_j^2 = I. \tag{D.43o}$$

Choose a multiplicity basis for $U_1$ and its $\sigma$-image for $U_2$. Since the original representation and $A$ are defined over $\mathbb{Q}_2$, we then have $B_2 = \sigma(B_1)$. The canonical invariant pairing between $W_1$ and $W_2$ is the ordinary dot product in (D.43m). Consequently the nondegenerate cross-pairing in (D.43f) has the form $I_3 \otimes S$ for an invertible 3-by-3 matrix $S$ on the multiplicity spaces. Equation (D.43f) becomes

$$B_1^{\mathsf{t}} S\, \sigma(B_1) = \lambda_2 S. \tag{D.43p}$$

Because $B_1^2 = I$, its determinant is 1 or $-1$, both fixed by $\sigma$. Taking determinants in (D.43p) therefore gives $1 = \lambda_2^3$. Together with $\lambda_2^2 = 1$, this proves $\lambda_2 = 1$.

If $\lambda_2 = -1$, the last conclusion forces $\lambda_3 = -1$, so (D.43j) gives $\lambda = -1$. If $\lambda_2 = 1$, (D.43j) excludes every negative odd-prime coordinate, and $\lambda = 1$. This proves the claim. Multiplicativity proves the stated consequence for subgroups generated by involutions; no conclusion about their conjugacy is inferred.

The program `code/part_c/torus_involution_d43.py` checks the characteristic Heisenberg presentations, the finite-cover chain ranks and central-character dimensions, and every $G_3$ involution with determinant one in (D.43k), including the exact inner-conjugacy formula in (D.43l). It also verifies the Schrödinger matrices and their rational implementer. The field and determinant arguments above prove the assertion for the full profinite groups. □

**D.44. [Proved] A first-symmetry choice for the involution coboundary.** Let $\alpha = (-1, f) \in GT_0$ be an involution in its normalized actual action on $F = \widehat{F}(x, y)$:

$$\alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1} y^{-1} f, \qquad f \in F'.$$

Write $\theta(x) = y$, $\theta(y) = x$. There exists $h \in F'$ satisfying both exact equations

$$f = h^{-1}\alpha(h), \qquad h\theta(h) = 1. \tag{D.44a}$$

Thus the coboundary in D.35 can be chosen with the first Harbater–Schneps symmetry. For every unit $\mu \in \widehat{\mathbb{Z}}^\times$, the associated continuous endomorphism $\rho_{\mu,h}(x) = x^\mu$, $\rho_{\mu,h}(y) = h y^\mu h^{-1}$ satisfies $\alpha\rho_{\mu,h} = \rho_{\mu,h}\delta$, where $\delta$ inverts $x, y$. Its full peripheral-conjugator condition is precisely D.35's remaining equation $x^\mu h y^\mu h^{-1} \sim_F (xy)^\mu$.

At the characteristic finite quotient $A_5^{19}$ of D.38, this first-symmetry condition is soluble for every one of the 5 062 500 tuples that pass the triangle test. This finite equation therefore gives no further exclusion among them.

*Manuscript:* §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394.

*Proof.* The normalized involution and the first Harbater–Schneps equation give

$$f\alpha(f) = 1, \qquad \theta(f) f = 1. \tag{D.44b}$$

Checking the two free generators gives the identity of actual automorphisms

$$\theta\alpha\theta = \mathrm{Inn}(f)\alpha. \tag{D.44c}$$

Indeed both sides send $x$ to $f x^{-1} f^{-1}$ and $y$ to $y^{-1}$.

Form $E = F \rtimes_\alpha \langle s \mid s^2 = 1\rangle$. Define an automorphism $\Theta$ of this exact extension by

$$\Theta|_F = \theta, \qquad \Theta(s) = fs. \tag{D.44d}$$

The first equation of (D.44b) preserves $s^2 = 1$; (D.44c) preserves the conjugation relation; and the second equation of (D.44b) gives $\Theta^2(s) = \theta(f) f s = s$. Thus $\Theta^2 = 1$ on all of $E$. Adjoin its named involution $\tau$ and put

$$G = E \rtimes_\Theta \langle \tau \rangle, \qquad G/F = C_2 \times C_2. \tag{D.44e}$$

The two quotient coordinates are represented by $s, \tau$.

The preimage of the $\tau$ line is the normal subgroup $G_\theta = F \rtimes_\theta \langle\tau\rangle$. It has the explicit free-product identification

$$G_\theta \xrightarrow{\sim} C_2(\tau) \amalg \widehat{\mathbb{Z}}(v), \qquad x \mapsto v, \quad y \mapsto \tau v \tau. \tag{D.44f}$$

The inverse sends $v$ to $x$ and $\tau$ to the named involution; both composites fix the indicated generators. The free-factor intersection theorem [36, Theorem 5 of the inspected author version] therefore gives $C_F(\tau) = 1$.

There is exactly one $F$-conjugacy class of involutions in $G_\theta$. For a direct verification, its mod-two $H^1(F, \mathbb{F}_2)$ module is the regular two-dimensional $C_2$ module, since $\theta$ exchanges its two basis characters. Its positive-degree $C_2$ cohomology vanishes. The split Hochschild–Serre calculation therefore gives $\dim H^n(G_\theta, \mathbb{F}_2) = 1$ for $n \geq 3$. Brown restriction [42, Theorem 2 and §1.1] gives one involution class, by exactly the dimension argument of (D.37f). Conjugacy by $G_\theta$ equals conjugacy by $F$, since $G_\theta = F\langle u\rangle$ for every involution $u$ and its second factor centralizes that element.

Consider the closed normalizer $V = N_G(\langle\tau\rangle)$. For every $g \in G$, the conjugate $g\tau g^{-1}$ belongs to $G_\theta$, and its image in $G/F$ is still the $\tau$ coordinate. The preceding uniqueness gives an $F$-conjugator taking it back to $\tau$. It follows that $V \to G/F$ is onto. Its kernel is $C_F(\tau) = 1$, so

$$V \xrightarrow{\sim} C_2 \times C_2. \tag{D.44g}$$

In particular $V$ is a finite subgroup, contains $\tau$, and has a unique involution $u$ mapping to the $s$ coordinate. That involution belongs to $E$ and commutes with $\tau$.

The three $F$-conjugacy classes of involutions in $E$ were determined in D.34. They are represented by

$$s, \quad xs, \quad yfs. \tag{D.44h}$$

Their distinct mod-two abelianization vectors also identify the classes. Formula (D.44d) sends $s$ to $fs$, which belongs to the class of $s$, and gives the exact transposition

$$\Theta(xs) = yfs, \qquad \Theta(yfs) = xs. \tag{D.44i}$$

Thus the class of $s$ is the unique fixed class under $\Theta$. Since $u$ commutes with $\tau$, its class is fixed, so $u = ksk^{-1}$ for some $k \in F$. Conjugate the finite subgroup (D.44g) by $k^{-1}$. It now contains $s$ and

$$k^{-1}\tau k = h\tau, \qquad h = k^{-1}\theta(k). \tag{D.44j}$$

The square of this second involution gives $h\theta(h) = 1$. Its commutation with $s$ gives $hf = \alpha(h)$, which is precisely the first equation of (D.44a). On abelianization, this equation reads $-2h^{\mathrm{ab}} = f^{\mathrm{ab}} = 0$. Multiplication by two is injective on $\widehat{\mathbb{Z}}^2$, hence $h \in F'$. Finally substitution in $\alpha(hy^\mu h^{-1})$ using $\alpha(h) = hf$ proves $\alpha\rho_{\mu,h}(y) = \rho_{\mu,h}(y^{-1})$; the $x$ identity is immediate.

We check the finite scope explicitly. Write $Q = A_5^{19}$, and use D.38's component description of $\theta$ and its involution tuple $t$. The triangle test makes every component $t_i$ a nonidentity involution of $A_5$. The equations to solve in this quotient are

$$h^{-1}th = \theta(t), \qquad h\theta(h) = 1, \qquad f = \theta(t)t^{-1}. \tag{D.44k}$$

They separate over the one-element and two-element orbits of $\theta$ on the factors.

For a two-element orbit, write $\theta(v)_i = gv_jg^{-1}$ and $\theta(v)_j = g^{-1}v_ig$, with $g \in S_5$. All nonidentity involutions of $A_5$ are conjugate within $A_5$. Choose $h_i \in A_5$ satisfying $h_i^{-1}t_ih_i = gt_jg^{-1}$, and set $h_j = g^{-1}h_i^{-1}g$. These substitutions give both equations of (D.44k) on that orbit.

For a fixed factor, $\theta$ is conjugation by an involution $g \in S_5$, allowing $g = 1$. Indeed its square is the identity and $C_{S_5}(A_5) = 1$. Choose an element $v$ with $v^2 = 1$, commuting with $t_i$ and having

the same parity as $g$. When $g$ is even choose $v = 1$. When $g$ is odd, take either transposition in the disjoint-transposition expression for $t_i$. Then $h_i = vg^{-1} \in A_5$ satisfies both equations in (D.44k). Thus every triangle survivor has a finite first-symmetry solution, including the 2 531 250 survivors failing D.38's finite peripheral-conjugacy test. No lifting of these finite solutions is asserted.

The program `code/part_a/pure_involution_first_symmetry.py` executes the local construction for all 390 fixed-factor cases and all 27 000 two-factor cases. It checks both equations in (D.44k), which cover every factor orbit in D.38. □

**D.45. [Proved] A characteristic binary-icosahedral refinement.** Let $F = \widehat{F}(x, y)$ and let $R$ be D.38's characteristic kernel with $F/R = A_5^{19}$. Intersecting the kernels of all epimorphisms $F \twoheadrightarrow S = \mathrm{SL}_2(\mathbb{F}_5)$ gives an open characteristic subgroup $\widetilde{R} \subset R$ and the following exact quotient and reduction map:

$$\widetilde{Q} = F/\widetilde{R} \xrightarrow{\sim} S^{19} \times C_2^2, \qquad \pi : \widetilde{Q} \longrightarrow A_5^{19}, \quad ((g_i), a) \longmapsto (g_i \bmod \{\pm I\})_i. \tag{D.45a}$$

For the 10 125 000 finite tuples $t$ of D.38, impose at this specified refinement the normalized multiplier $-1$ action, both Harbater–Schneps cycle equations, and the exact norm $f\alpha(f) = 1$. Exactly 5 062 500 base tuples admit such a refined $f$-coordinate. They are precisely D.38's cusp-triangle survivors, and each has 64 admissible $f$-coordinates. Thus the total number of refined coordinate pairs is 324 000 000. For a fixed $t$, these 64 coordinates define the same actual automorphism of $\widetilde{Q}$, of order two. In particular this named characteristic refinement detects exactly D.38's first excluded half. All 2 531 250 triangle survivors outside D.38's finite pure conjugacy set still pass this refinement.

*Manuscript:* §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D of the task: full GT_0 torsion and involution conjugacy.

*Proof.* All matrix calculations below are in $\mathbb{F}_5$. The program enumerates the 120 determinant-one matrices and verifies that their commutator subgroup has order 120. Their center is $\{I, -I\}$, and their projective quotient is identified explicitly with D.38's $A_5$ by a generating pair of orders two and three with product of order five. The resulting bijection is checked against all $60^2$ products. Conjugation by the 120 projective invertible matrices gives all automorphisms of $S$. Indeed $\mathrm{Aut}(S) \to \mathrm{Aut}(A_5)$ is injective: an element of its kernel has the form $g \mapsto c(g)g$, where $c : S \to \{\pm I\}$ is a homomorphism, and perfection makes $c = 1$. D.38 gives $|\mathrm{Aut}(A_5)| = 120$, and the enumerated projective conjugations already give 120 distinct automorphisms of $S$.

Choose lifts $X_i, Y_i \in S$ of the nineteen generating pairs in D.38, and put $Z_i = (X_iY_i)^{-1}$. Every pair of lifts generates $S$: its image generates $A_5$, so a proper image would have index two in $S$, contrary to perfection. The program also checks all $4 \cdot 19 = 76$ choices directly. Two maps to $A_5$ with a given kernel differ by an automorphism, and every automorphism lifts uniquely to $S$. Over each of the nineteen $A_5$ kernels there are exactly four $S$ kernels, from the independent sign choices for the lifts of $x$ and $y$. If two such lifts had the same kernel, the induced automorphism of $S$ would act trivially on $A_5$; the proved injectivity would identify the lifts. Consequently these are exactly the 76 epimorphism kernels defining $\widetilde{R}$.

The joint map

$$x \longmapsto ((X_i), e_x), \qquad y \longmapsto ((Y_i), e_y), \qquad e_x, e_y \text{ the basis of } C_2^2, \tag{D.45b}$$

is onto. To see this without a rank inference, its projection $D$ to $S^{19}$ maps onto $A_5^{19}$ by D.38. Thus $DZ(S^{19}) = S^{19}$; centrality gives $[S^{19}, S^{19}] = [D, D] \leq D$, so perfection gives $D = S^{19}$. The commutator subgroup of the image of (D.45b) therefore contains $S^{19} \times \{0\}$, and its projection to

$C_2^2$ is onto. This proves surjectivity of (D.45b). Every other sign lift of the $i$th pair is obtained by composing (D.45b) with

$$((g_j), a) \longmapsto g_i(-I)^{\ell(a)}, \qquad \ell \in \mathrm{Hom}(C_2^2, C_2).$$

The intersection of these four kernels, for all $i$, is trivial: the zero character first makes each $g_i = I$, and all characters then make $a = 0$. This proves (D.45a), including the specified kernel. It is characteristic because every automorphism of $F$ permutes its epimorphisms to $S$.

Here are the complete finite equations used in the refinement. The named word substitutions $\theta, \omega$ of D.38 descend to $\widetilde{Q}$. On its characteristic perfect subgroup $S^{19}$ they permute the factors by D.38's recorded maps and apply the unique lifted $S$ automorphisms just described. Write $\widetilde{t}_i \in S$ for any lift of $t_i$ and put

$$f_{0,i} = \theta(\widetilde{t})_i \widetilde{t}_i^{-1}, \qquad f_i = (-I)^{b_i} f_{0,i}, \qquad b_i \in \mathbb{F}_2. \tag{D.45c}$$

These are all the $2^{19}$ lifts of $f_Q = \theta(t)t^{-1}$ into $\widetilde{Q}' = S^{19}$. For every such lift, set

$$\begin{aligned} c_{x,i} &= X_i^{-1}(\widetilde{t}_i X_i \widetilde{t}_i^{-1})^{-1}, \\ c_{y,i} &= (f_i^{-1} Y_i^{-1} f_i)(\widetilde{t}_i Y_i \widetilde{t}_i^{-1})^{-1}. \end{aligned} \tag{D.45d}$$

D.38's normalization implies $c_{x,i}, c_{y,i} \in \{I, -I\}$. The actual automorphism induced by the normalized pair is exactly

$$\alpha((g_i), (a_x, a_y)) = \big((\widetilde{t}_i g_i \widetilde{t}_i^{-1} c_{x,i}^{a_x} c_{y,i}^{a_y})_i, (a_x, a_y)\big). \tag{D.45e}$$

This formula is a continuous automorphism; its restriction to the first factor is an automorphism, and its two correction characters are central. It sends the two generators (D.45b) to the required $X^{-1}$ and $f^{-1}Y^{-1}f$. Since $\widetilde{t}_i^2$ is central and both correction characters have exponent two, its square is the identity. Its reduction is the nonidentity order-two automorphism of D.38, so its order is exactly two. Multiplying $f_i$ by a central sign changes neither (D.45d) nor (D.45e). Conversely these images of the two generators determine the automorphism uniquely.

The program tests the following three exact equations in $S^{19}$ for all choices (D.45c):

$$\theta(f)f = 1, \qquad \omega^2(f)Z^{-1}\omega(f)Y^{-1}fX^{-1} = 1, \qquad f\alpha(f) = 1. \tag{D.45f}$$

On $f$, which has zero $C_2^2$ coordinate, formula (D.45e) is conjugation by $\widetilde{t}$. Thus every operation in (D.45f) is a specified matrix multiplication or a specified factor map; the norm is tested as an equation, not inferred from the order of the finite automorphism.

The factor maps for $\theta, \omega$ have six joint orbits. Exhaustion within these orbits gives the following complete counts, with D.38's zero-based factor indices:

| factor orbit | base tuples | numbers of admissible central lifts |
|---|---|---|
| $\{0, 2, 10\}$ | 15 | 2 for each of 15 |
| $\{1, 9, 17\}$ | 15 | 2 for each of 15 |
| $\{3, 5, 6, 7, 11, 14\}$ | 60 | 0 for 30; 4 for 30 |
| $\{4, 12, 18\}$ | 15 | 2 for each of 15 |
| $\{8, 13, 15\}$ | 10 | 2 for each of 10 |
| $\{16\}$ | 5 | 1 for each of 5. |

(D.45g)

The rejected thirty assignments in the six-factor orbit are exactly its triangle failures; the other five orbits have no triangle failure. The equations on one joint orbit involve no other orbit, so their solution sets have the displayed Cartesian-product decomposition. The surviving count is $15 \cdot 15 \cdot 30 \cdot 15 \cdot 10 \cdot 5 = 5\,062\,500$, and the number of central lifts per survivor is $2 \cdot 2 \cdot 4 \cdot 2 \cdot 2 \cdot 1 = 64$. This proves every asserted count. The executed implementation is `code/part_a/a5_schur_refinement.py`; its output records each orbit, the lift histogram, and sample coordinates. These computations impose a necessary finite refinement on full involutions. They assert no full lift of a surviving finite coordinate pair. □

**D.46. [Proved] Three positive involution sections and order-four lifting.** Let $T$ be the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$. For every involution $b \in T$ whose 2-adic orientation is positive, its centre-free extension

$$1 \longrightarrow F \longrightarrow E_b \longrightarrow \langle b \rangle \longrightarrow 1$$

has exactly three $F$-conjugacy classes of order-two sections. For every such section $S$,

$$H^1(C_F(S), \mathbb{F}_2) = 0. \tag{D.46a}$$

Every cyclic order-four subgroup of $T$ has a homomorphic lift to $\mathrm{Aut}(F)$, with its prescribed outer action. Thus every element of order four in $T$ has an actual order-four representative.

*Manuscript:* Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394.

*Proof. The homology of a positive involution.* D.27 gives an actual order-two representative $\beta$ of $b$, and identifies $E_b$ with $F \rtimes_\beta \langle s \rangle$. The map of D.33 on the finite 2-group $\langle b \rangle$ is faithful. The matrix of $\beta$ on $H_1(F, \mathbb{Z}_2)$ is therefore nonidentity. Its square is the identity and its determinant is 1. Over $\mathbb{Q}_2$ it is diagonalizable; the determinant condition makes its two eigenvalues equal. It follows that

$$\beta_2^{\mathrm{ab}} = -I. \tag{D.46b}$$

This uses the punctured-torus statement D.33 to exclude the identity matrix; no analogous exclusion for an arbitrary unmarked rank-two action is assumed.

*The full cohomology algebra and its Bockstein.* Let $u : E_b \to \mathbb{F}_2$ be the quotient character, with $u(s) = 1$. Choose a basis $\chi_1, \chi_2$ of $H^1(F, \mathbb{F}_2)$ and extend these characters to $E_b$ by declaring $\chi_i(s) = 0$. These extensions exist because (D.46b) is the identity modulo 2. The Hochschild–Serre spectral sequence has precisely the two rows

$$E_2^{a,0} = \mathbb{F}_2 u^a, \qquad E_2^{a,1} = H^1(F, \mathbb{F}_2)\, u^a. \tag{D.46c}$$

The transgression on $E_2^{0,1}$ is zero since the two characters have just been extended to $E_b$. Its other possible differentials are the products of this transgression with powers of $u$, and hence are zero as well. There are no further nonzero rows. Consequently, for every $n \geq 1$,

$$H^n(E_b, \mathbb{F}_2) = \langle u^n, u^{n-1}\chi_1, u^{n-1}\chi_2 \rangle_{\mathbb{F}_2}, \qquad \dim H^n(E_b, \mathbb{F}_2) = 3. \tag{D.46d}$$

Independence follows from the two associated graded pieces in (D.46c). Multiplication by $u$ is an isomorphism in every degree at least one: it is an isomorphism on those associated graded pieces.

For each $i$, lift $\chi_i|_F$ to a continuous homomorphism $\varphi_i : F \to \mathbb{Z}/4$. Equation (D.46b) gives $\varphi_i(\beta(h)) = -\varphi_i(h)$. The actual continuous one-cochain on $E_b$

$$\widetilde{\chi}_i(hs^\epsilon) = \varphi_i(h) \quad (\epsilon = 0, 1)$$

therefore satisfies

$$\begin{aligned} \delta\widetilde{\chi}_i(hs^\epsilon, ks^\eta) &= \varphi_i(h) + \varphi_i(k) - \varphi_i(h\beta^\epsilon(k)) \\ &= 2\epsilon\, \varphi_i(k) \quad \text{in } \mathbb{Z}/4. \end{aligned} \tag{D.46e}$$

For the coefficient sequence $0 \to \mathbb{F}_2 \xrightarrow{2} \mathbb{Z}/4 \to \mathbb{F}_2 \to 0$, this says that the Bockstein of $\chi_i$ is $u\chi_i$. It is also $\chi_i^2$. For completeness, the lift of any $\mathbb{F}_2$ character which takes its values in $\{0, 1\} \subset \mathbb{Z}/4$ has coboundary, divided by 2, equal to the cup square of that character; comparison of two lifts changes this cocycle only by a coboundary. Thus

$$\chi_i^2 = u\chi_i. \tag{D.46f}$$

Invert the central class $u$ and take degree zero:

$$R = \bigl(H^*(E_b, \mathbb{F}_2)[u^{-1}]\bigr)^0.$$

It has dimension three, with basis $1, z_1 = \chi_1/u, z_2 = \chi_2/u$ by (D.46d). Equation (D.46f) makes both $z_i$ idempotent. In particular $R$ is a quotient of the algebra

$$\mathbb{F}_2[Z_1, Z_2]/(Z_1^2 - Z_1, Z_2^2 - Z_2) \cong \prod_{(a_1,a_2)\in\mathbb{F}_2^2} \mathbb{F}_2, \quad P \longmapsto (P(a_1, a_2))_{(a_1,a_2)}. \tag{D.46g}$$

The displayed evaluation map is an isomorphism: the four products of $Z_i$ and $1 - Z_i$ are its four coordinate idempotents. Every ideal of a finite product of fields is a product of ideals in those fields. Every quotient is therefore reduced, so $R$ has no nonzero nilpotent element.

*Brown restriction and the three full sections.* Every finite subgroup of $E_b$ injects into $E_b/F = C_2$, since $F$ is torsion-free. Thus the nontrivial elementary abelian 2-subgroups are exactly its order-two sections, with no inclusions between distinct ones. For such a section $S$ put $C_S = C_F(S)$. Its normalizer is the exact direct product

$$N_{E_b}(S) = C_S \times S. \tag{D.46h}$$

Indeed every normalizer element centralizes $S$, since $\mathrm{Aut}(C_2) = 1$; multiplying it by its unique image in $S$ leaves an element of $C_S$.

The full-profinite Brown restriction theorem used in D.28 and D.33 applies because $F$ is open in $E_b$ and has 2-cohomological dimension one [42, Theorem 2 and §1.1]. In sufficiently high degrees it identifies cohomology with the product of the cohomologies of (D.46h); the identification is the actual restriction map and preserves cup products. Each factor contributes a copy of $\mathbb{F}_2$ in every positive degree, so (D.46d) in particular makes the number of factors finite. Since $C_S$ is a closed subgroup of $F$, it has $\mathrm{cd}_2(C_S) \le 1$. After the same localization, restriction identifies $R$ as a ring with

$$\prod_{[S]} (\mathbb{F}_2 \oplus H^1(C_S, \mathbb{F}_2)\, u^{-1}). \tag{D.46i}$$

The second summand in each factor is an ideal with square zero, since $H^2(C_S, \mathbb{F}_2) = 0$. Here $u|_{C_S} = 0$ and $u|_S$ is the nonzero character, so the localization in (D.46i) is exactly the localization already used in (D.46g). Reducedness of $R$ forces every second summand to vanish. Its dimension three then proves that there are exactly three section classes, and proves (D.46a).

These are also precisely the $F$-conjugacy classes. To see this directly, let $e \in E_b$ and choose $s' \in S$ with the same image as $e$ in $E_b/F$. Then $eSe^{-1} = (es'^{-1})S(es'^{-1})^{-1}$, and $es'^{-1} \in F$.

*An invariant section in the order-four extension.* Now let $a \in T$ have order four, and form its centre-free extension

$$E = \{(\phi, j) \in \mathrm{Aut}(F) \times C_4 : [\phi] = a^j\}. \tag{D.46j}$$

Its kernel is identified with $F$ by $h \mapsto (\mathrm{Inn}(h), 0)$. Let $E_2$ be the inverse image of $\langle a^2\rangle$. The orientation of $a^2$ at 2 is positive: the only roots of unity of order dividing four in $\mathbb{Z}_2^\times$ are $1, -1$, whose squares are 1. Equivalently, D.33 embeds $\langle a\rangle$ in $\mathrm{GL}_2(\mathbb{Z}_2)$, where its square is $-I$. The first part of the proof thus applies to $E_2$.

Conjugation by $E$ permutes the three $F$-conjugacy classes of sections of $E_2$. Every element of $E_2$ acts trivially on this class set, by the preceding paragraph, so the permutation action factors through $E/E_2 = C_2$. An action of $C_2$ on a set of three elements has a fixed element. Choose a section $S$ in that class. If $e \in E$ projects to the specified generator of $C_4$, there is $f \in F$ with $eSe^{-1} = fSf^{-1}$. Hence $f^{-1}e \in N_E(S)$ still projects to that generator. Put $N = N_E(S)$ and $C = C_F(S)$. We have proved the exact sequences

$$\begin{gathered} 1 \longrightarrow C \longrightarrow N \longrightarrow C_4 \longrightarrow 1, \qquad S \subseteq Z(N), \\ 1 \longrightarrow C \longrightarrow J := N/S \longrightarrow C_2 \longrightarrow 1. \end{gathered} \tag{D.46k}$$

The embedding of $C$ in the second line is injective because $C \cap S = 1$.

*Projectivity supplies the order-four section.* Equation (D.46a) implies

$$C^{(2)} = 1 : \tag{D.46l}$$

every nontrivial pro-2 group has a nontrivial finite 2-group quotient, which in turn has a quotient $C_2$. In particular every homomorphism from $J$ to a finite 2-group kills $C$ and factors through the last $C_2$ in (D.46k). That quotient is nontrivial, so

$$J^{(2)} = C_2. \tag{D.46m}$$

We show that $J$ has a nonidentity torsion element. If it were torsion-free, its open subgroup $C$ has $\mathrm{cd}_p(C) \le 1$ at every prime $p$, as a closed subgroup of the free profinite group $F$. Serre's torsion-free finite-index dimension theorem, in exactly the form used in D.27, would give $\mathrm{cd}_p(J) \le 1$ at every prime. Thus $J$ would be projective as a full profinite group. Its maximal pro-2 quotient would then be a projective, hence free, pro-2 group: pull a finite 2-group embedding problem back to $J$, solve it by projectivity, and factor the solution through $J^{(2)}$. This contradicts (D.46m), since $C_2$ is not a free pro-2 group.

Let $j \in J$ be a nonidentity torsion element. Its cyclic subgroup injects into $J/C = C_2$, because $C$ is torsion-free. Consequently $j$ has order two and lies outside $C$. Choose $n \in N$ over $j$. Then $n^2 \in S$, whereas its image in $C_4$ is an odd element and has order four. It follows that $n^2$ is the nonidentity element of $S$, and $n$ has order exactly four. After replacing $n$ by $n^{-1}$ if necessary, its projection is the specified generator of $C_4$. Thus $C_4 \to E$, sending that generator to $n$, is a section of (D.46j). Projection to $\mathrm{Aut}(F)$ gives the claimed homomorphic lift with its original labels.

The program `code/part_c/torus_order4_lift_d46.py` checks the finite cochain formula (D.46e), its comparison with the cup square, and the explicit Boolean-algebra evaluation in (D.46g). The dimension and normalizer arguments above prove the statement for the full profinite extensions. No claim of conjugacy to a geometric order-four action is made. D.43 applied to $a^2$ does also give $\lambda(a)^2 = 1$ globally, so the remaining orientation issue for $a$ concerns mixed signs, rather than fourth roots of unity. □

**D.47. [Proved] Conjugator coordinates at the binary refinement.** Use the specified quotient $\widetilde{Q} = S^{19} \times C_2^2$, $S = \mathrm{SL}_2(\mathbb{F}_5)$, of D.45, and its 324 000 000 refined pairs $(t, f)$. Every one of these pairs satisfies D.44's finite first-symmetry coboundary condition

$$h^{-1}\alpha(h) = f, \qquad h\theta(h) = 1, \qquad h \in S^{19}. \tag{D.47a}$$

Exactly 10 125 000 of the pairs satisfy the two finite conjugator equations of D.35 at this quotient. The count stays the same after requiring both actual invertibility of the finite substitution and both Harbater–Schneps symmetries for the conjugating pair. There are 5 062 500 pairs with admissible unit parameters congruent to a square modulo five, and 5 062 500 with parameters congruent to a nonsquare. These two sets are disjoint.

Thus 313 875 000 refined coordinate pairs lie outside the image of the standard full $GT_0$ involution conjugacy class. For each of D.38's 2 531 250 conjugator-compatible base tuples, exactly four of its 64 refined $f$-coordinates pass this test; for each of the other 2 531 250 base tuples, none does. These are statements about the exact normalized coordinates: D.45's 64 coordinates for a fixed $t$ define the same finite automorphism. No surviving pair is asserted to have a full involution or conjugator lift.

*Manuscript:* §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D of the task: full GT_0 involution conjugacy.

*Proof.* Write $X_i, Y_i, Z_i, \widetilde{t}_i$ and $\alpha$ for the matrices and actual automorphism fixed in D.45. Since $S$ has exponent dividing 60, the power parameter in D.35 factors through $(\mathbb{Z}/60)^\times$. For the second Harbater–Schneps symmetry we retain the finer, explicit parameter map

$$a \in (\mathbb{Z}/120)^\times \longmapsto (a \bmod 60,\ m = (a-1)/2 \bmod 60)\,. \tag{D.47b}$$

It is well-defined, and every such $a$ is the reduction of a unit in $\widehat{\mathbb{Z}}$. Changing the representative of $a$ by 120 changes $m$ by 60, so all subsequent equations are independent of the choice of integer representative.

The image of $F'$ is exactly $S^{19}$ by D.45. For fixed $(t, f)$ the first conjugator equation therefore separates into the nineteen exact equations

$$h_i^{-1}\widetilde{t}_i h_i \widetilde{t}_i^{-1} = f_i, \qquad h_i \in S. \tag{D.47c}$$

Each has exactly four solutions. The program verifies this by enumeration of all 120 matrices. It also follows from the order-four conjugacy class in $S$: $\widetilde{t}_i$ has square $-I$, and the norm equation in D.45 gives $(f_i\widetilde{t}_i)^2 = \widetilde{t}_i^2 = -I$. The thirty matrices with square $-I$ form one conjugacy class with centralizer of order four, as the same matrix enumeration verifies.

For each parameter $a$ mod 60 and each of these four matrices, the second equation of D.35 is tested by the exact matrix conjugacy condition

$$X_i^a h_i Y_i^a h_i^{-1} \sim_S (X_i Y_i)^a. \tag{D.47d}$$

Indeed the $C_2^2$ coordinates of both sides are $e_x + e_y$, since $a$ is odd, and that direct factor is central. Conjugacy in $\widetilde{Q}$ is therefore exactly these nineteen independent matrix conjugacies. The program computes every conjugacy class in $S$ by conjugation by all 120 matrices; (D.47d) is tested against those complete classes.

The finite invertibility test is explicit as well. Reduce the ordered pair $(X_i^a, h_i Y_i^a h_i^{-1})$ to $A_5$ and test whether it belongs to the $i$th recorded $\mathrm{Aut}(A_5)$ orbit of generating pairs in D.38. This is sufficient: its unique component automorphism lifts to an automorphism $\gamma_i$ of $S$, and the two central differences from $\gamma_i(X_i), \gamma_i(Y_i)$ are corrected by characters of $C_2^2$, exactly as in (D.45d)–(D.45e). The resulting automorphism of $\widetilde{Q}$ has the prescribed two generator images. It is necessary for a finite pure automorphism with these images: its reduction preserves the three specified power-conjugacy classes, and D.38's distinct passports force it to preserve each simple factor. Its pair must therefore be in that recorded orbit. This tests the actual finite substitution, with its characteristic quotient map retained.

For completeness the program also imposes both symmetries on the conjugating pair, not just its three peripheral images. Put $k = h^{-1}$ and use $m$ from (D.47b). The equations tested are

$$h\theta(h) = 1, \qquad \omega^2(k) Z^m \omega(k) Y^m k X^m = 1. \tag{D.47e}$$

All $h, k, f$ coordinates lie in the perfect factor $S^{19}$; the actions of $\theta, \omega$ there are the specified lifted factor maps of D.45. The $C_2^2$ coordinate of the second equation is $m(e_x + e_y + e_z) = 0$. Thus these are exact equations at the indicated finite quotient, including the half-exponent required by relation (II).

The first equation of (D.47e) is exhausted without assuming independent choices where factors are exchanged. If $\theta(v)_i = G_i v_j G_i^{-1}$, a choice of $h_i$ forces

$$h_j = G_i^{-1} h_i^{-1} G_i. \tag{D.47f}$$

For a fixed factor it must satisfy the same equation with $j = i$. The program enumerates one of the four candidates from (D.47c) for each orbit of $\theta$, uses (D.47f), and checks membership in the corresponding candidate sets. It verifies that (D.47a) is soluble for all the refined pairs. For the

full conjugator test it then checks (D.47d), finite invertibility, and the second equation of (D.47e) for every remaining choice and all 32 units modulo 120.

Here is the entire factorization of the result. Let $U_+$ and $U_-$ denote the subsets of $(\mathbb{Z}/120)^\times$ whose residues modulo five are respectively squares and nonsquares; each has 16 elements. For each of D.45's six joint factor orbits, the table gives the number of refined pairs having the specified set of admissible parameters.

| factor orbit | $U_+$ | $U_-$ | $U_+ \cup U_-$ | $\varnothing$ |
|---|---|---|---|---|
| $\{0, 2, 10\}$ | 15 | 15 | 0 | 0 |
| $\{1, 9, 17\}$ | 15 | 15 | 0 | 0 |
| $\{3, 5, 6, 7, 11, 14\}$ | 30 | 30 | 0 | 60 |
| $\{4, 12, 18\}$ | 15 | 15 | 0 | 0 |
| $\{8, 13, 15\}$ | 10 | 10 | 0 | 0 |
| $\{16\}$ | 0 | 0 | 5 | 0. |

(D.47g)

The admissible global parameter set is the intersection of the six local sets, because the parameter is common while the factor equations have the exact orbit decomposition of D.45. It is nonempty precisely when the first five nonempty choices all select the same one of $U_+, U_-$. Each choice gives $15 \cdot 15 \cdot 30 \cdot 15 \cdot 10 \cdot 5 = 5\,062\,500$ pairs. Their sum is 10 125 000, and subtraction from D.45's 324 000 000 gives the stated 313 875 000 failures. The program records the same table when only (D.47c)–(D.47d) are imposed, and when finite invertibility and the first symmetry are added. Thus the asserted equality of these counts is itself checked at every factor orbit. It also records the counts for each base tuple before multiplying over the six orbits. Their Cartesian product gives the exact histogram

| admissible $f$-coordinates per base tuple | number of base tuples |
|---|---|
| 0 | 2 531 250 |
| 4 | 2 531 250. |

(D.47h)

Every full conjugator in D.35 reduces to solutions of (D.47c)–(D.47d), and D.35's equality of the $P$ and $GT_0$ conjugator sets also supplies (D.47e). Consequently failure of the displayed finite test excludes that exact coordinate pair from the image of the standard full conjugacy class. The executed implementation is `code/part_a/pure_involution_binary_conjugacy.py`. Its output retains all unit sets, the six local tables, the per-base histogram, and the separate first-symmetry counts. The proof uses only this finite refinement and makes no assertion that one of its surviving pairs lifts through all characteristic refinements of $F$. □

**D.48. [Proved] The two sections of a punctured-torus order-three action.** Let $a$ have order three in the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$, and let $E_a$ be its centre-free extension. There are exactly two $F$-conjugacy classes of order-three sections in $E_a$. Each such section $S$ satisfies

$$H^1(C_F(S), \mathbb{F}_3) = 0. \tag{D.48a}$$

The normalizer in $E_a$ of the torus cusp $L = \overline{\langle [x, y] \rangle}$ is torsion-free. If $v$ is any element of this normalizer projecting to the prescribed generator $a$, and

$$v[x, y]v^{-1} = [x, y]^\lambda, \qquad v^3 = [x, y]^\kappa,$$

then its 3-adic coordinate satisfies $\kappa_3 \in \mathbb{Z}_3^\times$.

*Manuscript:* Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394.

*Proof. The actual action and its abelian quotient.* D.27 gives an actual order-three representative $\alpha$ of $a$, so $E_a = F \rtimes_\alpha \langle s \rangle$. Let $A$ be its matrix on $H_1(F, \mathbb{Z}_3)$. It has order exactly three by D.33. Over $\mathbb{Q}_3$ the polynomial $X^2 + X + 1$ is irreducible: a primitive cube root would give a square root of $-3$, whose 3-adic valuation is odd. Since $A^3 = I$ and $A \neq I$, it follows that

$$A^2 + A + I = 0, \qquad \det(A - I) = 3, \qquad \operatorname{coker}(A - I) \cong C_3. \tag{D.48b}$$

The last equality follows by Smith reduction over $\mathbb{Z}_3$, since the determinant has valuation one. Abelianizing the actual semidirect product gives the exact maximal pro-3 abelian quotient

$$E_a^{\mathrm{ab},(3)} \cong C_3^2. \tag{D.48c}$$

One factor is (D.48b), the other is the specified section quotient.

The reduction of $A$ modulo 3 is a nontrivial unipotent Jordan block. Indeed, if $A = I + 3B$, then $A^3 = I$ implies $B(I + 3B + 3B^2) = 0$ after division by 9. The second factor is invertible, giving $B = 0$, contrary to $A \neq I$. The induced action on $M = H^1(F, \mathbb{F}_3)$ is therefore also one nontrivial Jordan block of dimension two. Writing $D$ for the action matrix on this dual space, the exact cyclic resolution has successive operators

$$D - I, \quad 1 + D + D^2 = 0, \quad D - I, \quad 0, \ldots, \qquad \operatorname{rank}(D - I) = 1. \tag{D.48d}$$

Consequently $H^j(C_3, M)$ has dimension one for every $j \geq 0$.

*Bockstein injectivity and periodicity.* Let $u \in H^1(E_a, \mathbb{F}_3)$ be the quotient character, and let $\chi$ be a nonzero invariant character of $F$, extended to $E_a$ by $\chi(s) = 0$. They form a basis of $H^1(E_a, \mathbb{F}_3)$ by (D.48c). For the coefficient sequence

$$0 \longrightarrow \mathbb{F}_3 \xrightarrow{3} \mathbb{Z}/9 \longrightarrow \mathbb{F}_3 \longrightarrow 0$$

write $\mathfrak{b}$ for the Bockstein. Every homomorphism $E_a \to \mathbb{Z}/9$ factors through (D.48c), so its values are in $3(\mathbb{Z}/9)$ and its reduction modulo 3 is zero. The coefficient long exact sequence therefore makes

$$\mathfrak{b} : H^1(E_a, \mathbb{F}_3) \longrightarrow H^2(E_a, \mathbb{F}_3) \quad \text{injective.} \tag{D.48e}$$

Put $v = \mathfrak{b}(u)$ and $w = \mathfrak{b}(\chi)$. The notation $v$ in this cohomology calculation denotes a class, not the normalizer element in the statement. The class $v$ is the inflation of the usual degree-two period class of $C_3$; it is nonzero on the section $\langle s \rangle$. The classes $v, w$ are linearly independent by (D.48e).

The Hochschild–Serre spectral sequence has only rows zero and one. By (D.48d), at each positive total degree its $E_2$ page has two one-dimensional terms. Thus its degree-two abutment has dimension at most two. The two independent Bockstein classes make its dimension exactly two. The only possible differential is $d_2 : E_2^{j,1} \to E_2^{j+2,0}$. At $j = 0$ it vanishes because $\chi$ extends to $E_a$, and at $j = 1$ it vanishes because the one-dimensional term $E_2^{1,1}$ must survive to obtain dimension two in degree two.

Multiplication by the period class $v$ identifies each cyclic-resolution term with the term two places later. On the row with coefficient $M$ this also holds starting at $j = 0$, since the norm in (D.48d) is zero, so $H^0(C_3, M) \to H^2(C_3, M)$ is the identity on $\ker(D - I)$. On the row with trivial coefficients it is the usual period isomorphism. The multiplicative spectral sequence therefore makes every remaining $d_2$ a multiple of one of the two already vanishing differentials. It collapses, and multiplication by $v$ is an isomorphism on its two associated graded pieces at every positive total degree. Hence

$$\begin{gathered} \dim H^n(E_a, \mathbb{F}_3) = 2 \quad (n \geq 1), \\ v : H^n(E_a, \mathbb{F}_3) \xrightarrow{\sim} H^{n+2}(E_a, \mathbb{F}_3) \quad (n \geq 1), \\ wv^m \neq 0 \quad (m \geq 0). \end{gathered} \tag{D.48f}$$

*Exactly two full section classes.* Every finite subgroup of $E_a$ injects into its quotient $C_3$, because $F$ is torsion-free. For an order-three section $S$ the normalizer is the exact direct product

$$N_{E_a}(S) = C_F(S) \times S. \tag{D.48g}$$

Conjugation by a normalizer element induces the identity on $S$: projection to the abelian quotient $C_3$ identifies every element of $S$ with its label, so that label cannot change under conjugation. Multiplying by the section element with the same quotient label then leaves precisely $C_F(S)$.

The full-profinite Brown restriction theorem [42, Theorem 2 and §1.1], applied with the open subgroup $F$ of 3-cohomological dimension one, identifies sufficiently high cohomology with the product over (D.48g). Since $C_F(S)$ is a closed subgroup of $F$, its 3-cohomological dimension is at most one. Writing $k$ for the number of section classes, the dimensions in (D.48f) therefore give

$$k + \sum_{[S]} \dim H^1(C_F(S), \mathbb{F}_3) = 2. \tag{D.48h}$$

In particular the class set is finite.

It cannot consist of a single class. Otherwise take that class to be the chosen $S = \langle s \rangle$. On its normalizer (D.48g), the character $\chi$ is pulled back from $C_F(S)$, since $\chi(s) = 0$. Its Bockstein therefore restricts to zero:

$$w|_{N_{E_a}(S)} = \mathrm{pr}^*_{C_F(S)}\, \mathfrak{b}(\chi|_{C_F(S)}) = 0, \tag{D.48i}$$

because $H^2(C_F(S), \mathbb{F}_3) = 0$. Every nonzero class $wv^m$ from (D.48f) would then have zero Brown restriction, a contradiction for large $m$. Thus $k \geq 2$; (D.48h) makes $k = 2$ and makes both summands zero, proving (D.48a). As in D.46, $E_a$-conjugacy equals $F$-conjugacy: for any section $S$ choose its unique element with the same quotient label as a given conjugator and multiply the conjugator by its inverse.

*The peripheral normalizer.* We spell out the boundary consequence of the vanishing in (D.48a). A closed subgroup $C \leq F$ with $H^1(C, \mathbb{F}_3) = 0$ cannot contain a conjugate of the full 3-part $\langle [x, y]^{e_3} \rangle \cong \mathbb{Z}_3$ of the torus cusp. First its supernatural index in $F$ must have infinite 3-part. Otherwise choose an open $U_0 \geq C$ whose index has maximal 3-valuation. For every open $V$ with $C \leq V \leq U_0$, the index $[U_0 : V]$ is prime to 3. Transfer makes $H^1(U_0, \mathbb{F}_3) \to H^1(V, \mathbb{F}_3)$ injective. The restriction-colimit comparison of D.32 would then inject the nonzero group $H^1(U_0, \mathbb{F}_3)$ into $H^1(C, \mathbb{F}_3)$, a contradiction.

Choose an open $U \geq C$ with $3 \mid [F : U]$. Its corresponding finite cover of the punctured torus has a lift of the selected cusp of degree prime to 3, because its completed boundary subgroup contains the selected full $\mathbb{Z}_3$. The exact discrete boundary comparison uses the dense discrete representative for the profinite conjugator, as in D.32 and D.33. There are at least two boundary lifts: if there were only one, its degree would be $[F : U]$. The selected boundary class is nonzero in $H_1(U, \mathbb{F}_3)$, since the only relation among the boundary classes is their sum. A boundary-dual character restricts nontrivially to the selected $\mathbb{Z}_3$ in $C$; the scalar comparison is the inverse of its covering degree modulo 3. This again contradicts $H^1(C, \mathbb{F}_3) = 0$ and proves the assertion.

For $L = \overline{\langle [x, y] \rangle}$, the peripheral normalizer has the exact sequence

$$1 \longrightarrow L \longrightarrow N_{E_a}(L) \longrightarrow C_3 \longrightarrow 1. \tag{D.48j}$$

Surjectivity follows from preservation of the cusp class, and its kernel is $L$ by A.7. The multiplier satisfies $\lambda^3 = 1$, and $\lambda_3 = 1$, since $\mathbb{Z}_3^\times$ has no nontrivial root of unity of order three. An order-three section in (D.48j) would therefore centralize the full cusp $\mathbb{Z}_3$. Its fixed subgroup in $F$ would contradict the boundary consequence of (D.48a). Every torsion subgroup of (D.48j) injects into $C_3$, so this proves that the normalizer is torsion-free.

Finally take the element called $v$ in the statement, with $v^3 = [x, y]^\kappa$. Commuting $v$ with its own cube gives $(\lambda - 1)\kappa = 0$. Where $\lambda_p \neq 1$, the coordinate $\kappa_p$ and the norm $1 + \lambda_p + \lambda_p^2$ are zero. Where $\lambda_p = 1$, that norm is 3. If $\kappa_3$ were divisible by 3, the exact norm equation

$$(1 + \lambda + \lambda^2)z = -\kappa \quad \text{in } \widehat{\mathbb{Z}} \tag{D.48k}$$

would consequently have a solution: divide by 3 at the positive coordinates, and use zero at the other coordinates. The element $v[x, y]^z$ would then have order three and project to $a$, contradicting the torsion-free conclusion. Thus $\kappa_3$ is a unit, as asserted.

The program `code/part_c/torus_order3_sections_d48.py` checks the integral and mod-3 matrix calculations, the abelian character and Bockstein input in its finite semidirect quotient, and the cyclic resolution operators. The full restriction and finite-cover arguments above establish the section and normalizer assertions. They do not assert global triviality of the order-three multiplier or geometric conjugacy. □

**D.49. [Proved] The standard centralizer and the two-element conjugator fibres.** Let $P$ be the full pure three-cusp group of D.32 and $\delta(x) = x^{-1}$, $\delta(y) = y^{-1}$. Its full centralizer and normalizer are

$$C_P(\delta) = N_P(\langle\delta\rangle) = \langle\delta\rangle. \tag{D.49a}$$

Define the actual conjugacy orbits

$$\mathcal{O}_P = \{\beta\delta\beta^{-1} : \beta \in P\}, \qquad \mathcal{O}_{GT_0} = \{\beta\delta\beta^{-1} : \beta \in GT_0\}.$$

Every fibre of $P \to \mathcal{O}_P$ is exactly the right coset $\{\beta, \beta\delta\}$, and

$$\mathcal{O}_{GT_0} = \mathcal{O}_P \cap GT_0. \tag{D.49b}$$

For $\alpha \in \mathcal{O}_P$, its two conjugators have the exact normalized formulas

$$\begin{array}{ll} x \mapsto x^\mu, & y \mapsto hy^\mu h^{-1}, \\ x \mapsto x^{-\mu}, & y \mapsto hy^{-\mu}h^{-1}, \end{array} \qquad \mu \in \widehat{\mathbb{Z}}^\times, \quad h \in F'. \tag{D.49c}$$

Here $h$ is unique, and the unordered pair $\{\mu, -\mu\}$ is unique. In D.35's solution space, a choice $(\mu, h, u)$ therefore gives all the solutions over $\alpha$ as

$$(\mu, h, u(xy)^a), \qquad (-\mu, h, x^{-\mu}u(xy)^a), \qquad a \in \widehat{\mathbb{Z}}. \tag{D.49d}$$

These assertions concern the specified orbit map; they do not assert that it contains every full pure or $GT_0$ involution.

*Manuscript:* §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D of the task: full GT_0 involution conjugacy.

*Proof.* Lochak–Schneps prove that complex conjugation is self-centralizing in the usual $GT$ [47, Proposition 4(ii), with Lemma 9]. Their argument uses the first two defining relations and applies to $GT_0$ as well. We give the passage from the full group $P$ explicitly, together with the normalized fibre formulas.

Let $\beta \in C_P(\delta)$ have normalized pair $(\mu, k)$, so $\beta(x) = x^\mu$ and $\beta(y) = k^{-1}y^\mu k$, with $k \in F'$. D.32's normalization is a homomorphic section into $\mathrm{Aut}(F)$, so the commutation is an actual automorphism equality. Its value on $y$ makes $k\delta(k)^{-1}$ centralize $y^\mu$. The peripheral centralizer is $\langle y\rangle$ by A.7, while $k\delta(k)^{-1} \in F'$. Abelianization therefore gives $k = \delta(k)$. The exact free-product identification of D.35, $F \rtimes_\delta C_2 \cong C_2 \amalg C_2 \amalg C_2$, gives $C_F(\delta) = 1$. Hence $k = 1$.

We also need the precise inclusion in $GT_0$. For each geometric cusp permutation $\sigma \in \Sigma$, form $r_\sigma = \beta^{-1}\sigma\beta\sigma^{-1}$. It lies in $P^1 = \ker\lambda$, and commutes with $\delta$, because both $\beta$ and $\sigma$ do. D.35

proved $C_{P^1}(\delta) = 1$, so $r_\sigma = 1$. The exact identity $GT_0 = C_P(\Sigma)$ of D.35 therefore puts this $\beta$ in $GT_0$.

Its second Harbater–Schneps relation, with $k = 1$ and $m = (\mu - 1)/2 \in \widehat{\mathbb{Z}}$, is

$$(xy)^{-m} y^m x^m = 1, \qquad \text{equivalently} \quad (xy)^m = y^m x^m. \tag{D.49e}$$

For every positive integer $n$ use the explicit finite Heisenberg quotient with coordinates $(a, b, c) \in (\mathbb{Z}/n)^3$ and multiplication

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab'), \quad x = (1, 0, 0), \quad y = (0, 1, 0).$$

The two sides of (D.49e) have coordinates

$$(m, m, m(m+1)/2), \qquad (m, m, 0), \tag{D.49f}$$

where the binomial expression is the continuous integer-valued function on $\widehat{\mathbb{Z}}$. These identities first follow for nonnegative integers by multiplication, and then for all $m \in \widehat{\mathbb{Z}}$ by continuity; modulo $n$ they depend only on $m$ mod $2n$. Equality in every such quotient gives $m(m+1)/2 = 0$ in $\widehat{\mathbb{Z}}$, and hence

$$\mu^2 = 1. \tag{D.49g}$$

Since $k = 1$, the actual square of $\beta$ fixes both $x$ and $y$ by (D.49g). Thus $\beta^2 = 1$. D.34 gives the global multiplier $-1$ for a nonidentity pure involution; the identity has multiplier 1. The normalized pair is therefore $(1, 1)$ or $(-1, 1)$, proving $C_P(\delta) = \langle \delta \rangle$. A normalizer of its order-two subgroup must centralize its sole nonidentity element, proving the normalizer equality in (D.49a).

Two elements $\beta, \beta'$ have the same image under the orbit map exactly when $\beta^{-1}\beta' \in C_P(\delta)$. This proves its two-element fibres. For an orbit element lying in $GT_0$, D.35's proved equality of conjugator sets places both conjugators in $GT_0$; the reverse inclusion is immediate. This proves (D.49b).

Write one conjugator in D.35's convention as $\beta_{\mu,h}$. Right composition with $\delta$ gives exactly $\beta_{-\mu,h}$, so (D.49c) follows with the same $h$. They are distinct because $\mu \neq -\mu$ in $\widehat{\mathbb{Z}}^\times$. Moreover actual normalized conjugation identifies $C_F(\alpha)$ with $\beta(C_F(\delta)) = 1$. The description of all coboundaries in (D.35g) then makes $h$ the unique solution of $h^{-1}\alpha(h) = f$ in $F'$. The two-element fibre proves uniqueness of the unordered multiplier pair.

Finally put $A = x^\mu$, $B = hy^\mu h^{-1}$ and suppose $AB = u(xy)^\mu u^{-1}$. The exact word identity

$$A^{-1}B^{-1} = A^{-1}(AB)^{-1}A$$

shows that $x^{-\mu}u$ is a third-cusp conjugator for the opposite multiplier. D.35's centralizer calculation gives all other third-cusp conjugators by right multiplication by $(xy)^a$. This proves (D.49d). The program `code/part_a/standard_involution_centralizer.py` checks (D.49f) over sixteen finite moduli and checks the displayed word identity. The full conclusion (D.49a) uses the proved global-sign theorem D.34, beyond those finite checks. □

**D.50. [Proved] The finite two-subgroups of the full punctured-torus group.** Let $T = \mathrm{Out}_{\mathrm{lac}}(\widehat{F}(x, y); \overline{\langle [x, y] \rangle})$ be the full once-punctured-torus loop-outer group. The possible abstract isomorphism types of its finite 2-subgroups are exactly

$$1, \quad C_2, \quad C_4, \quad C_2 \times C_2, \quad D_8, \tag{D.50a}$$

where $D_8$ has order eight. All five types occur. For a subgroup of type $C_2 \times C_2$ or $D_8$, the full multiplier takes its values in the globally constant signs $\{1, -1\} \subset \widehat{\mathbb{Z}}^\times$. In $C_2 \times C_2$ there is one positive involution and two negative involutions. In $D_8$ the rotation subgroup has multiplier 1 and the four reflections have multiplier $-1$.

For an arbitrary cyclic subgroup of order four, a generator $a$ satisfies the already proved restrictions

$$\lambda(a)^2 = 1, \qquad \lambda(a)_2 = 1, \qquad \lambda(a^2) = 1. \tag{D.50b}$$

The abstract classification (D.50a) does not establish conjugacy of arbitrary such subgroups, or replace (D.50b) by a global multiplier determination.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof.* D.33 makes the actual homology homomorphism

$$A_2 : T \longrightarrow \mathrm{GL}_2(\mathbb{Z}_2) \tag{D.50c}$$

injective on every finite 2-subgroup $K$ and proves $|K| \leq 8$. We classify its image using that same specified injection.

No matrix in $\mathrm{GL}_2(\mathbb{Q}_2)$ has order eight: an eigenvalue of order eight has minimal polynomial $X^4+1$ over $\mathbb{Q}_2$, which is irreducible by Eisenstein after $X \mapsto X+1$. The same argument excludes higher powers of two. Every matrix $R$ of order four has minimal polynomial $X^2+1$, since this polynomial is irreducible over $\mathbb{Q}_2$ and $R^2 \neq I$. Consequently

$$R^2 = -I. \tag{D.50d}$$

An exponent-two subgroup is abelian, because $(ab)^2 = a^2 = b^2 = 1$ implies $ab = ba$. Its matrices are commuting diagonalizable involutions over $\mathbb{Q}_2$, and simultaneous diagonalization embeds it in $\{\mathrm{diag}(\epsilon_1, \epsilon_2) : \epsilon_i = \pm 1\}$. Its order is at most four. Thus an image of order eight contains an $R$ of order four.

The matrix centralizer of $R$ is the multiplicative group of the field $\mathbb{Q}_2[R] \cong \mathbb{Q}_2(i)$. For completeness, choose a nonzero vector $v$: the vectors $v, Rv$ are independent, since an eigenvalue in $\mathbb{Q}_2$ would contradict the irreducibility of $X^2+1$. A commuting endomorphism is determined by its value on $v$ and is therefore $aI + bR$. Every element of the finite 2-subgroup in this field has order at most four by the preceding matrix argument. It is consequently one of the at most four roots of $X^4 - 1$. Since $\langle R \rangle$ already has four elements,

$$C_K(R) = \langle R \rangle \tag{D.50e}$$

for a subgroup of order eight.

Choose $S \in K \setminus \langle R \rangle$. The index-two subgroup $\langle R \rangle$ is normal; by (D.50e) conjugation by $S$ induces its nonidentity automorphism. Hence

$$SRS^{-1} = R^{-1}, \qquad S^2 \in \{I, R^2\}. \tag{D.50f}$$

The second inclusion follows because $S^2 \in \langle R \rangle$ and $S$ fixes its own square, whereas it inverts $R$. The alternative $S^2 = I$ gives the dihedral presentation of order eight. We exclude the alternative $S^2 = R^2 = -I$ explicitly.

In the basis $(v, Rv)$ used above one has

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The equation $SR = -RS$ forces

$$S = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \qquad S^2 = (a^2 + b^2)I. \tag{D.50g}$$

Thus the second alternative would require $a^2 + b^2 = -1$ in $\mathbb{Q}_2$. Here is the exact valuation obstruction. Put $m = \min(v_2(a), v_2(b))$ and write $a = 2^m u$, $b = 2^m v$, with $u, v \in \mathbb{Z}_2$ and at least one a unit. If both are units, $u^2 + v^2 \equiv 2 \pmod 8$, so this sum has valuation one. If exactly one is a unit, the sum has valuation zero. On the other hand the equation says

$$u^2 + v^2 = -2^{-2m}; \tag{D.50h}$$

the valuation of its right side is even. Both-unit values are therefore impossible. The other case forces $m = 0$; reduction modulo four then gives $1 = -1$, also impossible. Zero entries cause no exception to this argument. This excludes the quaternion alternative and leaves only $D_8$ at order eight. Orders four, two, and one give the remaining groups in (D.50a).

We specify actual full-group realizations. On $\widehat{F}(x, y)$ define

| | $x$ | $y$ |
|---|---|---|
| $r$ | $y$ | $x^{-1}$ |
| $s$ | $y$ | $x$. |

(D.50i)

The free universal property gives continuous endomorphisms, and the displayed relations $r^4 = s^2 = 1$, $srs = r^{-1}$, verified on $x, y$, give their inverses and make them automorphisms. For $c = [x, y]$ direct multiplication gives

$$r(c) = x^{-1}cx, \qquad s(c) = c^{-1}. \tag{D.50j}$$

Thus their outer classes lie in $T$. Their homology matrices are

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{D.50k}$$

and the eight matrices $R^i S^j$ $(0 \le i < 4,\ 0 \le j < 2)$ are distinct. These maps therefore exhibit an embedded $D_8$ in $T$. The subgroups $\langle r \rangle$, $\langle r^2, s \rangle$, $\langle s \rangle$, and the identity realize $C_4$, $C_2 \times C_2$, $C_2$, and 1.

We finish with the asserted multiplier data. D.43 makes every involution's multiplier a global sign. A positive nonidentity involution has determinant 1 in (D.50c). Its order-two matrix is diagonalizable in characteristic zero, and faithfulness excludes $I$; its matrix is consequently $-I$. There is at most one such involution in any finite 2-subgroup. For $C_2 \times C_2$, simultaneous diagonalization identifies its full image with the four sign matrices above, giving exactly one positive and two negative involutions. For $D_8$, the square of a rotation is positive by (D.50d), so every other involution is negative. The reflections generate $D_8$, and a rotation is a product of two reflections. The homomorphic multiplier is therefore globally 1 on the rotations and globally $-1$ on the reflections.

Finally let $a$ have order four in an arbitrary cyclic subgroup. Its matrix in (D.50c) has minimal polynomial $X^2 + 1$ and determinant one. Thus $\lambda(a)_2 = 1$ and $\lambda(a^2)_2 = 1$. D.43 applied to the involution $a^2$ gives $\lambda(a^2) = 1$ globally. The multiplier homomorphism then gives (D.50b). No assertion of conjugacy follows from this matrix classification.

The finite certificate `code/part_c/torus_two_subgroups_d50.py` verifies the full-word maps (D.50i)–(D.50j), their eight distinct matrices, and the unit square computations modulo eight used in (D.50h). The passage from that finite calculation to all of $\mathbb{Q}_2$ is the valuation argument just given. □

**D.51. [Proved] The second index-three kernel and its three peripheral normalizers.** Let $a$ have order three in the full punctured-torus loop-outer group, put $F = \widehat{F}(x, y)$, $c = [x, y]$,

and let $E$ be its centre-free extension. Besides $F$, there is exactly one other torsion-free normal subgroup $U$ of index three in $E$. Write $V = F \cap U$. Then

$$\begin{gathered}[F:V] = [U:V] = 3, \qquad E/V \cong C_3^2, \\ V = \widehat{\pi_1(S_{1,3})}, \qquad U \text{ is projective}, \qquad U^{(3)} \cong \widehat{F}_2^{(3)}.\end{gathered} \tag{D.51a}$$

The cusp normalizer $N = N_E(\langle c \rangle)$ is contained in $U$. The three cusp subgroups $L_0, L_1, L_2$ of the displayed degree-three cover have normalizers $N_i = N_U(L_i)$ with the exact sequences

$$1 \longrightarrow L_i \longrightarrow N_i \longrightarrow U/V \longrightarrow 1. \tag{D.51b}$$

They are three distinct $U$-conjugacy classes, cyclically permuted by $E/U$. All have the given multiplier $\lambda$ under the same generator label in $U/V$.

For a prime $q \neq 3$ at which $\lambda_q$ is a nonidentity cube root of unity, the $U/V$-character multiplicities of $H_1(V, \mathbb{Q}_q)$, ordered as $(1, \lambda_q, \lambda_q^2)$, are exactly one of

$$(1,3,0), \qquad (0,2,2). \tag{D.51c}$$

Correspondingly $U^{(q)}$ is free pro-$q$ of rank one or zero. These are constraints on the actual subgroup $U$; they do not identify it with a full free profinite group.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The section and cusp lines in the actual abelian quotient.* Use D.27 to choose an actual order-three representative $\alpha$ and write $E = F \rtimes_\alpha \langle s \rangle$. Let $u : E \to \mathbb{F}_3$ be its quotient character, with $u(s) = 1$. D.48 gives exactly two full section classes, $S_0 = \langle s \rangle$ and $S_1$, and gives $E^{\mathrm{ab},(3)} \cong C_3^2$.

Let $M$ be the reduction of the homology matrix of $\alpha$ modulo three. By D.48 it is a nonidentity unipotent matrix with $(M-I)^2 = 0$. On $\mathbb{F}_3^2$ write $\omega((a,b),(a',b')) = ab' - ba'$. Choose $w$ with $e = (M-I)w \neq 0$. Then $(e,w)$ is a basis, $Me = e$, $Mw = w + e$, and $d = \omega(e,w) \neq 0$. The linear functional

$$\chi(r) = \omega(e,r)/d \tag{D.51d}$$

is $M$-invariant. It gives a character of $F$, which we extend to $E$ by $\chi(s) = 0$. Thus $(u, \chi) : E \to \mathbb{F}_3^2$ is the actual abelian quotient above.

The generator of $S_1$ labelled by $u = 1$ has $\chi$-value $b_1 \neq 0$. Here is the required separation, using the maps already proved in D.48 rather than an assumption about section labels. The Bockstein of $\chi$ has nonzero high periodic multiples. On a section normalizer its restriction is the Bockstein of $\chi$ restricted to the section: the fixed-group first cohomology vanishes by D.48. If $\chi$ vanished on both section classes, all those multiples would have zero full Brown restriction. This contradicts D.48f. It follows that $b_1 \neq 0$.

To locate the cusp line, use the characteristic class-two exponent-three quotient

$$Q = \mathbb{F}_3^2 \times \mathbb{F}_3, \qquad (r,z)(r',z') = (r + r', z + z' + \tfrac{1}{2}\omega(r,r')). \tag{D.51e}$$

The images of $x, y$ are the two standard vectors with central coordinate zero, and $c$ maps to $Z = (0,1)$. Since $\lambda_3 = 1$, the induced automorphism is

$$\theta(r,z) = (Mr, z + \ell(r)) \tag{D.51f}$$

for a linear functional $\ell$. For an element $(r,z)s$ of $Q \rtimes_\theta C_3$ with quotient label one, direct multiplication gives

$$((r,z)s)^3 = Z^{d\chi(r)^2 + \ell(e)\chi(r)}. \tag{D.51g}$$

To check the formula, write $r = be+tw$, so $\chi(r) = t$. The successive vector parts are $r, r+te, r+2te$, whose sum is zero. The sum of the three central correction terms is $\ell((2I + M)r) = t\ell(e)$. The three alternating cross terms, multiplied by $1/2$, sum to $dt^2$ in $\mathbb{F}_3$. This proves (D.51g) for every element, including the arbitrary central coordinate $z$.

The image of the actual generator of $S_1$ has cube one. Equation (D.51g) and $b_1 \neq 0$ therefore give $\ell(e) = -db_1$. Thus its cube polynomial is

$$db(b - b_1). \tag{D.51h}$$

Choose $v \in N$ with $u(v) = 1$, as in D.48. It satisfies $v^3 = c^\kappa$ with $\kappa_3$ a unit. Its image in (D.51e) has nonzero cube exponent. The two roots of (D.51h) are 0 and $b_1$; the only other element of $\mathbb{F}_3$ is $-b_1$. Consequently

$$(u, \chi)(S_0) = \langle (1, 0) \rangle, \quad (u, \chi)(S_1) = \langle (1, b_1) \rangle, \quad (u, \chi)(N) = \langle (1, -b_1) \rangle. \tag{D.51i}$$

The last equality uses that $N$ is generated topologically by $v$ and $\langle c \rangle$, and both characters kill $c$.

*The full kernels and peripheral labels.* Define the continuous characters and subgroups

$$\psi = \chi + b_1 u, \qquad U = \ker \psi, \qquad V = \ker u \cap \ker \chi. \tag{D.51j}$$

These are actual subgroups of $E$. The character $\psi$ is nonzero on both section classes in (D.51i), and it kills $N$. Every nonidentity finite subgroup of $E$ is one of the order-three section groups, up to conjugacy. Hence $U$ is torsion-free. The four lines in $E^{\mathrm{ab},(3)} = \mathbb{F}_3^2$ give all normal index-three subgroups of $E$. Two contain respectively $S_0$ and $S_1$. The other two are exactly $\ker u = F$ and $\ker \psi = U$, proving the uniqueness assertion. Also $E = U \rtimes S_0$, since $\psi|_{S_0}$ is an isomorphism onto $\mathbb{F}_3$.

The restriction of $\chi$ to $F$ is nonzero and kills its commutator cusp. Choose a word $f$ in the dense discrete group $F(x, y)$ with $\chi(f) = 1$. The degree-three covering of the punctured torus corresponding to $\ker(\chi|_{F(x,y)})$ has three boundary lifts of degree one. Its Euler characteristic is $-3$; thus its type is $(1, 3)$, and its fundamental group has rank four. The induced profinite subgroup is exactly $V$, with cusp representatives

$$L_i = f^i \langle c \rangle f^{-i}, \qquad N_i = f^i N f^{-i} \quad (0 \leq i < 3). \tag{D.51k}$$

The full peripheral normalizer assertion A.7 gives $N_E(L_i) = N_i$. Since $N \subset U$ and $U$ is normal in $E$, all $N_i$ lie in $U$. Their kernels under $U \to U/V$ are exactly $L_i$, because $N_i \cap F = L_i$; their images are all of $U/V$, because $u(v) = 1$. This proves (D.51b), with the common generator label supplied by $u$.

For the class assertions, the actual quotient $(u, \chi)$ identifies $E/V$ with $\mathbb{F}_3^2$, and the image of every $N_i$ is the same line $U/V$. The three $V$-conjugacy classes of the cusp lifts are therefore the three cosets of this line in $E/V$. The subgroup $U/V$ fixes each class, so they remain distinct under $U$. The quotient $E/U$ acts regularly on these three cosets, proving its asserted cyclic permutation. This also proves that conjugation by a generator of $U/V$ fixes each cusp class with multiplier $\lambda$.

The open subgroup $V$ of $U$ is free profinite of rank four and $U$ is torsion-free. Serre's finite-index cohomological-dimension theorem, used in D.27, makes $U$ projective. Its maximal pro-three quotient is projective as a pro-three group, by the finite embedding problem characterization, and is therefore free pro-three. The pro-three topology on $V$ is exactly the one induced from $U$: taking the intersection of the three conjugates of a normal three-power-index subgroup of $V$ produces one normal in $U$, still of three-power index. Thus $V^{(3)}$ embeds as an open subgroup of index three in $U^{(3)}$. Schreier's formula now reads

$$4 = 1 + 3(\operatorname{rank} U^{(3)} - 1),$$

and gives rank two. This proves (D.51a).

*The exact character restriction at another prime.* The three oriented boundary classes of $V$ have their sole relation given by their sum. They form a two-dimensional submodule of $H_1(V, \mathbb{Q}_q)$ on which $U/V$ acts by $\lambda_q$. Filling the three boundaries gives the compact degree-three cover of the original torus. Its map on homology is the inclusion of a rank-two sublattice of index three, hence an isomorphism over $\mathbb{Q}_q$, and over $\mathbb{Z}_q$ when $q \neq 3$. Conjugation by $v$ projects to the original outer action $a$ on the base; its inner factor acts trivially on that base homology. Therefore, if $A_q$ denotes the original matrix, the exact boundary sequence and averaging over $C_3$ give

$$H_1(V, \mathbb{Q}_q) \cong \lambda_q^{\oplus 2} \oplus (\mathbb{Q}_q^2, A_q) \quad \text{as } U/V\text{-modules.} \tag{D.51l}$$

This comparison is induced by the stated boundary inclusion and covering homology map.

At the specified prime, $A_q^3 = I$ and $\det A_q = \lambda_q$. The three cube roots $1, \lambda_q, \lambda_q^2$ are distinct in $\mathbb{Q}_q$. Diagonalization gives exactly the two eigenvalue multisets

$$\{1, \lambda_q\}, \qquad \{\lambda_q^2, \lambda_q^2\}. \tag{D.51m}$$

Adding the two boundary characters in (D.51l) proves (D.51c). For $q \neq 3$, the same decomposition holds over $\mathbb{Z}_q$: the averaging idempotents have unit denominator three, and the cube-root differences are units. Hochschild–Serre for $V \triangleleft U$ then identifies $H^1(U, \mathbb{F}_q)$ with the invariant subspace of $H^1(V, \mathbb{F}_q)$, of dimension one or zero respectively. The projectivity of $U$ makes its maximal pro-$q$ quotient free pro-$q$, with this number of generators. This proves the last assertion.

The program `code/part_c/torus_order3_sibling_d51.py` checks (D.51g) for every central-linear automorphism above every nonidentity unipotent matrix over $\mathbb{F}_3$, the resulting section and cusp lines, and the finite eigenvalue patterns in (D.51m). These finite calculations validate the displayed formulas. The full subgroup, topology, and projectivity conclusions are proved by the maps above. □

**Exact remaining recognition problem.** The construction supplies a specified torsion-free projective group $U$ with an open full free subgroup of rank four and three specified normalizers. It supplies no identification of $U$ with $\widehat{F}_2$. At the primes in (D.51c), its maximal pro-$q$ quotient has rank zero or one, whereas the maximal pro-three quotient has rank two. The existence or exclusion of such a full $U$ with these three normalizers remains the step needed for this approach to determine every torus order-three multiplier.

**D.52. [Proved] An abstract allowed list for every finite full torus subgroup.** Let $T$ be the full once-punctured-torus loop-outer group of D.50. Every finite subgroup of $T$ has one of the following twelve abstract types:

$$\begin{gathered} 1,\ C_2,\ C_3,\ C_4,\ C_2 \times C_2,\ C_6,\ S_3,\ D_8, \\ C_{12},\ D_{12},\ \mathrm{Dic}_{12},\ D_{24}. \end{gathered} \tag{D.52a}$$

Here $D_{2n}$ has order $2n$, and $\mathrm{Dic}_{12} = C_3 \rtimes C_4$, with a generator of $C_4$ inverting $C_3$. The nine types obtained by omitting $C_{12}, \mathrm{Dic}_{12}, D_{24}$ all occur. Occurrence of those three remaining types is not determined by this allowed list. The subsequent full covering arguments in D.54–D.55 exclude all three.

Every subgroup of type $S_3, D_{12}$ or $D_{24}$ has the global usual reflection multiplier: rotations have multiplier 1 and reflections have multiplier $-1$. In a subgroup of type $\mathrm{Dic}_{12}$, the subgroup of order three has multiplier 1, and every generator $b$ of a cyclic subgroup of order four satisfies

$$\lambda(b)^2 = 1, \qquad \lambda(b)_2 = 1, \qquad \lambda(b)_3 = -1. \tag{D.52b}$$

For a subgroup of type $C_{12}$, its order-four generator instead has $\lambda(b)_2 = \lambda(b)_3 = 1$, and still has $\lambda(b)^2 = 1$. These restrictions retain the unresolved cyclic order-three and order-four multiplier questions.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The two primary representations and their actual kernels.* For a finite subgroup $G \leq T$, D.33 gives

$$|G| \mid 24, \qquad A_p : G \longrightarrow \mathrm{GL}_2(\mathbb{Z}_p), \qquad \det A_p(g) = \lambda(g)_p. \tag{D.52c}$$

It makes $A_p$ injective on every finite $p$-subgroup. Put $K = \ker(A_2|_G)$. An element of order two in $K$ would contradict that injection. Cauchy's theorem and (D.52c) therefore give

$$K = 1 \quad \text{or} \quad K \cong C_3. \tag{D.52d}$$

We classify the actual image $H = A_2(G)$ and then the extension by this specified kernel. We do not identify $G$ with a representation at one prime when the kernel is nontrivial.

*Finite images over $\mathbb{Q}_2$.* The two-subgroups of $H$ have the types in D.50. Indeed a Sylow two-subgroup of the preimage of any such subgroup maps isomorphically onto it, since the kernel has odd order. The matrix arguments in D.50 also explicitly exclude the quaternion group of order eight.

Suppose 3 divides $|H|$. Its number of Sylow three-subgroups is either one or four. We exclude four. Conjugation gives a transitive homomorphism $H \to S_4$. Its kernel has order at most six. If that kernel had order divisible by three, its unique Sylow three-subgroup would be characteristic in the kernel and normal in $H$, contrary to the assumed four Sylow subgroups. Thus the transitive image has order divisible by both four and three, and is $A_4$ or $S_4$. The kernel consequently has order one or two.

There is no faithful two-dimensional representation of $A_4$ over $\mathbb{Q}_2$. Its normal Klein four subgroup would be simultaneously diagonalizable and, by faithfulness, would be the full group of diagonal sign matrices. One of its three involutions would therefore be $-I$. Conjugation fixes this scalar, whereas the order-three elements of $A_4$ cyclically permute all three involutions. This is a contradiction. It excludes the cases with trivial kernel, whose images are $A_4$ or $S_4$. In the remaining case $|H| = 24$ and the image is $A_4$ with a kernel of order two. The preimage $P$ of its Klein four subgroup is a normal subgroup of order eight, hence is $D_8$ by D.50. But $|\operatorname{Aut}(D_8)| = 8$: a generator of its rotation subgroup has two possible images, and a chosen reflection has four. Every element of order three therefore centralizes $P$. Multiplication by a Sylow three-subgroup gives $H = P \times C_3$, which has only one Sylow three-subgroup. This contradiction proves that $H$ has a normal subgroup $P_3$ of order three.

Let $R$ generate $P_3$. Its minimal polynomial is $X^2 + X + 1$, irreducible over $\mathbb{Q}_2$ by reduction modulo two. The basis $(v, Rv)$ for any nonzero vector $v$ identifies the matrix centralizer with

$$L_2^{\times}, \qquad L_2 = \mathbb{Q}_2(\zeta_3). \tag{D.52e}$$

Under this identification, a commuting endomorphism is multiplication by $a + b\zeta_3$. The full normalizer consists exactly of these multiplication maps and their compositions with $\sigma(\zeta_3) = \zeta_3^{-1}$: conjugation must send $R$ to $R$ or $R^{-1}$, and division by $\sigma$ in the second case gives a commuting map.

The field $L_2$ contains no element of order four. To check this directly, the equation $(a+b\zeta_3)^2 = -1$ has coefficient equations

$$a^2 - b^2 = -1, \qquad b(2a - b) = 0. \tag{D.52f}$$

They require either $a^2 = -1$ or $a^2 = 1/3$ in $\mathbb{Q}_2$. Both are impossible: these are units congruent respectively to seven and three modulo eight, whereas an odd square is congruent to one. Since $|H| \mid 24$, it follows that $C = H \cap L_2^\times$ is $C_3$ or $C_6$. Here its two-part is contained in $\{1, -1\}$, and its three-part is $P_3$, so no general classification of local roots of unity is required.

If $H = C$, these are the two possible groups. Otherwise choose $B = a\sigma \in H \setminus C$. Then

$$B^2 = N_{L_2/\mathbb{Q}_2}(a)\, I \in C \cap \mathbb{Q}_2^\times = \{I, -I\} \cap C. \tag{D.52g}$$

The last scalar restriction follows because a finite-order scalar of order dividing twenty-four in $\mathbb{Q}_2$ is a sign: there are no primitive cube or fourth roots in $\mathbb{Q}_2$. For $C = C_3$, equation (D.52g) gives $S_3$. For $C = C_6$, its two values give respectively $D_{12}$ and $\mathrm{Dic}_{12}$. Thus the images with order divisible by three have the complete allowed list

$$C_3,\ C_6,\ S_3,\ D_{12}, \mathrm{Dic}_{12}\,. \tag{D.52h}$$

*The three-adic normalizer and its determinant.* We will need a statement about any normal subgroup $P_3 \cong C_3$ of the actual $G$. The map $A_3$ is injective on $P_3$. Its generator has minimal polynomial $X^2 + X + 1$ over $\mathbb{Q}_3$, irreducible since its discriminant $-3$ has odd valuation. The same centralizer and normalizer calculation identifies

$$C_{\mathrm{GL}_2(\mathbb{Q}_3)}(A_3(P_3)) = L_3^\times, \qquad N_{\mathrm{GL}_2(\mathbb{Q}_3)}(A_3(P_3)) = L_3^\times \rtimes \langle\sigma\rangle, \quad L_3 = \mathbb{Q}_3(\zeta_3). \tag{D.52i}$$

This field also has no element of order four: (D.52f) would require $a^2 = -1$, impossible modulo three, or $a^2 = 1/3$, impossible by valuation. Thus every finite-order element in its centralizer with order dividing twenty-four lies in $\mu_6$ and has norm one.

There is also no element of norm $-1$ in $L_3$. Here is an explicit calculation, including possible nonintegral coordinates. The norm of $a + b\zeta_3$ is $a^2 - ab + b^2$. Put $m = \min(v_3(a), v_3(b))$ and write $a = 3^m u, b = 3^m v$, where $u, v \in \mathbb{Z}_3$ and at least one is a unit. When $u + v \not\equiv 0 \pmod 3$, the norm of $u + v\zeta_3$ has valuation zero and residue $(u+v)^2 = 1$ modulo three. When $u + v \equiv 0 \pmod 3$, both are units; writing $u = -v + 3k$ gives

$$u^2 - uv + v^2 = 3(v^2 - 3vk + 3k^2), \tag{D.52j}$$

of valuation one. The norm $-1$ would have valuation zero. The second case cannot give that valuation, since $2m + 1 \neq 0$. The first forces $m = 0$ and residue one, whereas $-1$ has residue two. This proves the asserted norm obstruction.

An element of the finite image $A_3(G)$ outside the centralizer has the form $B = a\sigma$, and $B^2 = N(a)I$ is a finite-order scalar, hence a sign. The norm obstruction forces $B^2 = I$. In the basis $(1, \zeta_3)$, the determinant of $\sigma$ is $-1$, so $\det B = -N(a) = -1$. Together with the centralizer calculation, this proves the exact character identity

$$\lambda(g)_3 = \begin{cases} 1, & g \text{ centralizes } P_3, \\ -1, & g \text{ inverts } P_3. \end{cases} \tag{D.52k}$$

The argument uses faithfulness only on $P_3$, not on all of $G$.

*Reconstructing the actual finite subgroup.* If $K = 1$ in (D.52d), the injection into $H$ gives precisely D.50 and (D.52h). Now let $K = P_3 \cong C_3$. A Sylow two-subgroup $Q$ of $G$ has order $|G|/3$ and maps isomorphically onto $H$. The actual multiplication map

$$P_3 \rtimes Q \longrightarrow G, \qquad (a, q) \longmapsto aq, \tag{D.52l}$$

is an isomorphism, with action given by conjugation. D.50 lists the five possibilities for $Q$. For $Q = 1, C_2, C_4$, their homomorphisms to $\mathrm{Aut}(C_3) = C_2$ give $C_3$; $C_6$ or $S_3$; and $C_{12}$ or $\mathrm{Dic}_{12}$, respectively.

For $Q = C_2 \times C_2$ or $D_8$, D.50 already determines its global multiplier. Equation (D.52k) therefore makes its action on $P_3$ exactly the reflection character of D.50. Write $a$ for a generator of $P_3$. For $Q = C_2 \times C_2$, write $z$ for its positive involution and $s$ for a negative one. The element $az$ has order six, is inverted by $s$, and generates $a, z$ through $(az)^4 = a$ and $(az)^3 = z$. This gives an explicit $D_{12}$ presentation. For $Q = D_8 = \langle r, s\rangle$, where $r$ is its order-four rotation, the element $ar$ has order twelve, is inverted by $s$, and satisfies

$$(ar)^4 = a, \qquad (ar)^9 = r. \tag{D.52m}$$

It gives the explicit $D_{24}$ presentation. These actual maps finish the allowed list.

*Multipliers and realized cases.* In every group just obtained that inverts its normal $C_3$, the multiplier on that $C_3$ is one: a conjugation-invariant homomorphism into the abelian group $\widehat{\mathbb{Z}}^\times$ takes an inverted element to its own inverse, and a value of order dividing three with square one is one. In $S_3, D_{12}, D_{24}$ each reflection inverts $C_3$ and hence has three-adic multiplier $-1$ by (D.52k). D.43 makes its full multiplier $-1$. Every rotation is a product of two reflections, proving the stated global reflection character. For $\mathrm{Dic}_{12}$, an order-four generator inverts $C_3$, so (D.52k) gives its negative three-coordinate. D.50 gives its positive two-coordinate and global square one. For $C_{12}$ the same generator centralizes $C_3$, giving instead the positive three-coordinate. This proves all the claimed multiplier restrictions.

D.50 realizes its five types by explicit full profinite free-group automorphisms. The other four realized types have the following simultaneous exact surface model. On $\mathbb{R}^2/\mathbb{Z}^2$ the integral matrices

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{D.52n}$$

fix the origin and satisfy $R^6 = S^2 = I$, $SRS = R^{-1}$. Their twelve matrices $R^i S^j$ are distinct. Removing the origin therefore gives an actual $D_{12}$ action on a once-punctured torus. Passing to its fundamental group, then to its full profinite completion and outer automorphisms, gives a homomorphism to $T$. A small peripheral circle is carried to an ellipse of local degree $\det R = 1$ or $\det S = -1$, so this homomorphism preserves the stated full cusp with the indicated multiplier. The induced homology matrices are exactly $R, S$ in $\mathrm{GL}_2(\widehat{\mathbb{Z}})$; their distinctness proves that the homomorphism is injective. The subgroups $\langle R^2\rangle, \langle R\rangle, \langle R^2, S\rangle$, and $\langle R, S\rangle$ realize $C_3, C_6, S_3, D_{12}$.

The program `code/part_c/torus_finite_types_d52.py` checks the matrix relations, the local norm and square residues, the eight automorphisms of $D_8$, and the explicit finite semidirect-product identifications in (D.52l)–(D.52m). The prime-wise kernel and valuation arguments above prove the full allowed list. No assertion of occurrence or nonoccurrence of its three remaining types is inferred from those finite checks. □

**D.53. [Proved] The unique labelled order-four section class.** Let $a$ have order four in the full punctured-torus loop-outer group, and let

$$1 \longrightarrow F \longrightarrow E \xrightarrow{\pi} C_4 = \langle a\rangle \longrightarrow 1, \qquad F = \widehat{F}_2,$$

be its centre-free extension. Its homomorphic sections form exactly one $F$-conjugacy class, with the generator label $a$ retained. For an actual lift $b \in E$ with $b^4 = 1$ and $\pi(b) = a$, put $B = \langle b\rangle$. Then

$$N_E(B) = C_F(b) \times B, \qquad H^1(C_F(b), \mathbb{F}_2) = 0. \tag{D.53a}$$

Conjugation by a lift of $a$ fixes exactly one of the three full section classes for $a^2$ and interchanges the other two. Consequently $E$ has two conjugacy classes of subgroups of order two; exactly one of them is contained in its class of cyclic subgroups of order four.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof.* D.46 supplies the actual lift $b$ with its prescribed label. Put $T = \langle b^2 \rangle$ and

$$E_2 = \pi^{-1}(\langle a^2 \rangle) = F \rtimes T. \tag{D.53b}$$

The action of $b^2$ on $H_1(F, \mathbb{Z}_2)$ is $-I$ by D.46. Thus

$$E_2^{\mathrm{ab},(2)} \cong C_2^3. \tag{D.53c}$$

Use coordinates $(v, u) \in \mathbb{F}_2^2 \times \mathbb{F}_2$, where $u$ is the quotient character and the first two coordinates are the extended characters $\chi_1, \chi_2$ of D.46. The three full section classes of $E_2$ give three distinct points of the affine plane $u = 1$. Indeed, the degree-zero localized algebra in D.46g–i has dimension three, is generated by $\chi_1/u, \chi_2/u$, and restriction to the three section normalizers identifies it with $\mathbb{F}_2^3$. Each restriction is evaluation at $(\chi_1(t), \chi_2(t))$ for the nonidentity element $t$ of that section. If two such points agreed, the two restriction coordinates would agree on both generators and hence on the entire algebra, contradicting the stated product isomorphism.

Let $R = A_2(a) \in \mathrm{GL}_2(\mathbb{Z}_2)$. D.50 gives $R^2 = -I$. Moreover $R \not\equiv I \pmod 2$. For otherwise $R = I + 2D$ with an integral matrix $D$, so $R^2 = -I$ would give $I + 2D + 2D^2 = 0$, an impossibility modulo two. Conjugation by $b$ on (D.53c) fixes $u$ and induces the actual affine map

$$v \longmapsto \overline{R}v + d \quad \text{on } u = 1 \tag{D.53d}$$

for a vector $d \in \mathbb{F}_2^2$. The linear part follows from the restriction of conjugation to the specified subgroup $F$; the translation is the image of a chosen point of $u = 1$. Conjugation by an element of $E_2$ acts trivially on the three $F$-conjugacy classes, as shown in D.46. Thus (D.53d) permutes their three points with order dividing two. Three distinct points of $\mathbb{F}_2^2$ affinely span it. Fixing all three would therefore make this affine map the identity, contrary to $\overline{R} \neq I$. Its permutation is consequently a transposition, with exactly one fixed class. The class of $T = \langle b^2 \rangle$ is that fixed class.

Put $C = C_F(T)$ and $N = N_E(T)$. Since $b$ normalizes $T$, the image of $N$ under $\pi$ is all of $C_4$. Its kernel is exactly $C$, because an element of $F$ normalizing $T$ centralizes its nonidentity element. The chosen section gives the actual semidirect-product identification

$$N = C \rtimes \langle b \rangle, \qquad T \subseteq Z(N), \qquad J := N/T = C \rtimes \langle \bar{b} \rangle, \quad |\bar{b}| = 2. \tag{D.53e}$$

In the last expression the embedding of $C$ is injective since $C \cap T = 1$. Conjugation by $b$ on $C$ has square the identity because $b^2$ centralizes $C$.

As a closed subgroup of the full free profinite group $F$, the group $C$ is projective and has $\mathrm{cd}_2(C) \leq 1$. D.46 gives $H^1(C, \mathbb{F}_2) = 0$. All its positive-degree mod-two cohomology therefore vanishes. Hochschild–Serre for the last extension in (D.53e) identifies inflation as

$$H^*(C_2, \mathbb{F}_2) \xrightarrow{\cong} H^*(J, \mathbb{F}_2). \tag{D.53f}$$

This uses the displayed full group $C$, not an identification of $C$ with its maximal pro-two quotient.

We spell out the full Brown hypotheses for this new normalizer quotient. The subgroup $C$ is open in $J$ and has finite 2-cohomological dimension. It is torsion-free. Every finite subgroup of $J$ consequently injects into $J/C = C_2$; in particular $J$ has no subgroup $C_2^2$. Theorem 2 of [42, §1.1] therefore supplies, in degrees $n > 1$, the dense restriction embedding into the product over the conjugacy classes of order-two subgroups of $J$. By (D.53f) its source has dimension one. Each product factor is nonzero: for a section $S$ of $J \to C_2$, its normalizer is

$$C_C(S) \times S, \tag{D.53g}$$

and the restriction from the $S$ factor already supplies one dimension. Thus the product has finitely many factors and the finite dense image is the entire product. Since $\mathrm{cd}_2(C_C(S)) \le 1$, the dimensions give the exact equality

$$1 = \sum_{[S]} (1 + \dim H^1(C_C(S), \mathbb{F}_2)). \tag{D.53h}$$

There is a section $\langle \bar{b} \rangle$. It follows that it is the unique conjugacy class and that $H^1(C_C(\bar{b}), \mathbb{F}_2) = 0$. These are also $C$-conjugacy classes: an element of $J$ can be multiplied by the appropriate element of a given section to leave a conjugator in $C$.

Now let $B'$ be any subgroup of order four in $E$. Its intersection with $F$ is trivial, so it is cyclic and $\pi|_{B'}$ is an isomorphism onto $C_4$. Choose its unique generator $b'$ with $\pi(b') = a$. Its square generates an order-two section $T'$ of $E_2$. Conjugation by $b'$ has the same action on the section classes of $E_2$ as conjugation by $b$, because $b'b^{-1} \in F$. The class of $T'$ is fixed, and is therefore the unique class of $T$. An $F$-conjugation puts $T'$ equal to $T$, and then $B' \subseteq N_E(T) = N$. Its image in $J = N/T$ has order two. By (D.53h), a further conjugation by an element of $C$ puts that image equal to $\langle \bar{b} \rangle$.

The exact inverse image of this subgroup under $N \to N/T$ has order four, because its kernel is the specified subgroup $T$ of order two. It contains $\langle b \rangle$ and hence equals $\langle b \rangle$. Thus the conjugated $B'$ equals $B$. All conjugations used lie in $F$ and preserve $\pi$, so its labelled generator $b'$ is sent to $b$, proving uniqueness of the homomorphic section with labels.

The fixed group $C_C(\bar{b})$ in (D.53h) is exactly $C_F(b)$. For an element of $C$ fixed by $\bar{b}$, its commutator with $b$ lies in $T$ and also in $C$, and $T \cap C = 1$. The converse is immediate, and an element commuting with $b$ already centralizes $T$. This proves the cohomology statement in (D.53a). Finally, an element normalizing $B$ must fix its generator $b$: conjugation preserves its label in the abelian quotient $C_4$, and $\pi|_B$ is injective. Multiplication by the matching power of $b$ then leaves an element of $C_F(b)$, proving the direct-product normalizer in (D.53a).

The two orbits of the transposition in (D.53d) are exactly the two $E$-conjugacy classes of order-two subgroups. Only its fixed class can be the square of a subgroup of order four, and $B$ realizes that containment. This proves the final assertion.

The finite program `code/part_c/torus_order4_sections_d53.py` checks the mod-four matrix obstruction and every invariant three-point affine configuration above a nonidentity order-two matrix in $\mathrm{GL}_2(\mathbb{F}_2)$. The normalizer and restriction arguments prove the full section result. No conjugacy of the outer action $a$ to a prescribed geometric action is asserted here. □

**D.54. [Proved] No fixed character for a cubic torus action.** Let $a$ have order three in the full once-punctured-torus loop-outer group. For every prime $q \neq 3$, it fixes no nonzero continuous character

$$\chi : F \longrightarrow C_q, \qquad F = \widehat{F}(x, y). \tag{D.54a}$$

Consequently the reduction of $A_q(a)$ modulo $q$ has no eigenvalue one. For every prime $p$, the endomorphism $A_p(a) - I$ of $\mathbb{Z}_p^2$ is injective. The full fixed group of every actual order-three representative is therefore contained in the full commutator subgroup $F'$.

In particular, $A_2$ is injective on every finite subgroup of the full torus group. The allowed list of D.52 loses $C_{12}$ and $D_{24}$. At a nontrivial cubic multiplier coordinate $q \neq 3$, the remaining homology matrix is exactly

$$A_q(a) = \lambda(a)_q^2 I. \tag{D.54b}$$

This statement does not exclude the remaining scalar cubic multiplier case.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof.* Use D.27 to choose an actual representative $\alpha$ of order three and put $E = F \rtimes_\alpha \langle s \rangle$, where $s$ has quotient label $a$. Write $c = [x, y]$ and $L = \langle c \rangle$. D.48 supplies a labelled cusp-normalizer element $v$ with

$$v \in N_E(L), \qquad v^3 = c^\kappa, \qquad vcv^{-1} = c^\lambda, \qquad \kappa_3 \in \mathbb{Z}_3^\times. \tag{D.54c}$$

It also gives $\lambda_3 = 1$ and

$$I + A_3(a) + A_3(a)^2 = 0 \quad \text{on } \mathbb{Z}_3^2. \tag{D.54d}$$

Suppose $\chi$ in (D.54a) were nonzero and invariant. Its extension

$$\widetilde{\chi} : E \longrightarrow C_q, \qquad \widetilde{\chi}(hs^j) = \chi(h) \tag{D.54e}$$

is an actual continuous homomorphism. Put

$$N = \ker \chi, \qquad E_N = \ker \widetilde{\chi} = N \rtimes \langle s \rangle. \tag{D.54f}$$

Both have their indicated index $q$ in $F$ and $E$. Since $\chi(c) = 0$, (D.54c) gives $3\widetilde{\chi}(v) = 0$. The prime $q$ differs from three, so $v \in E_N$. Thus $v = ns$ for a specified $n \in N$.

The restriction of $\chi$ to the dense discrete free group is onto $C_q$. Its covering surface is the degree-$q$ cover of the punctured torus with boundary monodromy zero. It has $q$ boundary lifts of degree one, Euler characteristic $-q$, and hence type $(1, q)$. Its full fundamental-group completion is exactly $N$, by the finite-index completion map. Choose a discrete word $f$ with $\chi(f) = 1$. Its boundary representatives are

$$c_i = f^i c f^{-i} \quad (0 \le i < q). \tag{D.54g}$$

Their homology classes have exactly the single relation $\sum_i [c_i] = 0$. In particular each $[c_i]$ is nonzero in $H_1(N, \mathbb{F}_3)$, because $q \ge 2$. Equivalently, omitting any other boundary from the usual genus-one presentation puts $c_i$ into a free basis.

The action on every boundary class has multiplier $\lambda$ and fixes its label. This follows from the actual $v \in E_N$: the element $n_i = (vf^iv^{-1})f^{-i}$ belongs to $N$, and $vc_iv^{-1} = n_i c_i^\lambda n_i^{-1}$. Conjugation by $s = n^{-1}v$ differs from this by an inner automorphism of $N$. Thus $\alpha$ fixes all $q$ boundary homology classes over $\mathbb{Z}_3$.

We give the required splitting of the entire homology module, including its comparison with the base torus. Let $D = F/N = C_q$ be the deck group and set

$$M = H_1(N, \mathbb{Z}_3), \qquad e = \frac{1}{q} \sum_{d \in D} d. \tag{D.54h}$$

The action of $\alpha$ on $M$ commutes with the deck action: it induces the identity on $D$, and two lifts of a deck transformation differ by an inner automorphism of $N$ on this homology. Consequently $e$ commutes with $\alpha$.

Filling the $q$ boundaries gives the compact degree-$q$ torus cover. Explicitly its lattice is the kernel $\Lambda \le \mathbb{Z}^2$ of the character $\mathbb{Z}^2 \to C_q$ induced by $\chi$, and its map to the base torus is $\mathbb{R}^2/\Lambda \to \mathbb{R}^2/\mathbb{Z}^2$. The deck maps are translations and act trivially on the filled homology. The boundary submodule

$$B_\partial = \mathbb{Z}_3[D]/\mathbb{Z}_3\Big(\sum_{d \in D} d\Big) \tag{D.54i}$$

has no deck-invariant vector, because $q$ is a unit in $\mathbb{Z}_3$. The exact boundary sequence and (D.54h) therefore give

$$(1 - e)M = B_\partial, \qquad eM \cong H_1(\mathbb{R}^2/\Lambda, \mathbb{Z}_3) \cong H_1(\mathbb{R}^2/\mathbb{Z}^2, \mathbb{Z}_3). \tag{D.54j}$$

The last isomorphism is induced by the displayed covering: its lattice inclusion has index $q$, a 3-adic unit. These maps intertwine the induced automorphisms, so $\alpha|_{eM}$ is exactly $A_3(a)$. On $(1-e)M$ it is the identity by (D.54g). Hence

$$I + \alpha + \alpha^2 = 0 \quad \text{on } H_1(N, \mathbb{F}_3) : \tag{D.54k}$$

on the first summand this is (D.54d), and on the boundary summand the norm is multiplication by three.

The cube of the actual element $ns$ now gives the contradiction:

$$\begin{aligned} [v^3] &= [n\,\alpha(n)\,\alpha^2(n)] \\ &= (I + \alpha + \alpha^2)[n] = 0 \quad \text{in } H_1(N, \mathbb{F}_3), \\ [c^\kappa] &= (\kappa_3 \bmod 3)[c] \neq 0. \end{aligned} \tag{D.54l}$$

Equations (D.54c) and (D.54l) concern the same two elements of $N$. Thus no invariant character (D.54a) exists.

For $q \neq 3$, a matrix and its transpose have the same rank, so a one-eigenvector would supply a nonzero invariant character. It has just been excluded. In particular $A_q(a) - I$ is invertible over $\mathbb{Z}_q$. At $q = 3$ its determinant is three, by D.48, so it is still injective. If $h$ is fixed by an actual representative, its image in every $\mathbb{Z}_p^2$ is therefore zero. The common kernel of the actual abelianization map $F \to \widehat{\mathbb{Z}}^2 = \prod_p \mathbb{Z}_p^2$ is $F'$, proving the fixed-group assertion.

At a nontrivial cubic multiplier coordinate the two eigenvalue multisets were exactly $\{1, \lambda_q\}$ and $\{\lambda_q^2, \lambda_q^2\}$ in D.51. The first is now excluded. A finite-order matrix in characteristic zero is diagonalizable, so the second gives the actual scalar matrix (D.54b).

Finally let $G$ be any finite torus subgroup. D.33 gives $|G| \mid 24$. A nontrivial kernel of $A_2|_G$ would contain a prime-order element. Its prime can only be two or three. D.33 excludes an element of order two, and (D.54a), with $q = 2$, excludes an element of order three acting trivially on two-adic homology. Thus $A_2|_G$ is injective. The faithful-image part (D.52h), together with D.50, now excludes $C_{12}$ and $D_{24}$ from that allowed list.

The program `code/part_c/torus_cover_norm_d54_d55.py` builds the cyclic covering homology, its individual boundary vectors and the deck-average splitting for several primes $q \neq 3$, and verifies the forced norm contradiction over $\mathbb{F}_3$. The displayed maps prove it for every such prime. □

**D.55. [Proved] Full primary faithfulness and the nine finite torus types.** Every positive involution $b$ in the full once-punctured-torus loop-outer group has

$$A_p(b) = -I \quad \text{for every prime } p. \tag{D.55a}$$

Consequently, for each prime $p$, the actual homology map $A_p$ is injective on every finite subgroup of that group. The possible finite abstract subgroup types are exactly the following nine, all realized:

$$1,\ C_2,\ C_3,\ C_4,\ C_2 \times C_2,\ C_6,\ S_3,\ D_8,\ D_{12}. \tag{D.55b}$$

The groups $D_8, D_{12}$ have orders eight and twelve respectively.

For an arbitrary order-four element $a$,

$$A_p(a)^2 = -I \quad \text{for every } p. \tag{D.55c}$$

At a negative multiplier coordinate $q$ it is necessarily a scalar matrix $i_q I$ or $-i_q I$, where $i_q^2 = -1$ in $\mathbb{Q}_q$; thus $q \equiv 1 \pmod 4$. Together with (D.54b), this retains the precise scalar cubic and

quartic multiplier cases. The abstract classification (D.55b) does not settle those scalar cases or the conjugacy of arbitrary finite subgroups.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The positive involution's full cusp normalizer.* Let $b$ be any positive involution. D.43 gives its global multiplier 1. D.27 gives an actual representative $\beta$ of order two, and put $E_b = F \rtimes_\beta \langle s \rangle$. The first half of D.46 applies to every such positive involution; it does not require $b$ to have first been exhibited as an order-four square. It gives

$$A_2(b) = -I, \qquad H^1(C_F(S), \mathbb{F}_2) = 0 \quad \text{for every order-two section } S. \tag{D.55d}$$

We recall the prime-independent part of the closed-subgroup argument in D.48. If $C \le F$ is closed and $H^1(C, \mathbb{F}_p) = 0$, it cannot contain the full subgroup $c^{e_p\mathbb{Z}_p}$ of the torus cusp, where $c = [x, y]$. If the $p$-part of $[F : C]$ were finite, choose an open neighborhood $W \ge C$ with maximal such part. All subsequent indices are prime to $p$, so transfer makes the restriction maps on first cohomology injective. Their direct limit is $H^1(C, \mathbb{F}_p)$, contradicting the nonzero first cohomology of the open full free group $W$. If the $p$-part is infinite, choose an open $W \ge C$ of index divisible by $p$. Its selected cusp lift has degree prime to $p$, because it contains the full cusp $\mathbb{Z}_p$. There must be at least two boundary lifts: a unique lift would have degree $[F : W]$. A boundary character of this finite cover then restricts nontrivially to that cusp $\mathbb{Z}_p$ and hence to $C$, again a contradiction. These are the actual open-subgroup restriction and finite-cover maps used in D.48, with an arbitrary prime $p$ in place of three.

Apply this with $p = 2$ to (D.55d). An involution in $N_{E_b}(\langle c \rangle)$ would be a section and would centralize the entire cusp, since the global multiplier is 1. Its full fixed group would therefore contain the cusp $\mathbb{Z}_2$, which has just been excluded. Thus this cusp normalizer is torsion-free. Choose its labelled element $v$:

$$vcv^{-1} = c, \qquad v^2 = c^\kappa. \tag{D.55e}$$

If $\kappa_2$ were even, the equation $2z = -\kappa$ would be soluble in $\widehat{\mathbb{Z}}$: at two by that divisibility, and at every odd prime because two is a unit. Then $(vc^z)^2 = 1$, and its nontrivial quotient label would make it an involution in the cusp normalizer. Consequently

$$\kappa_2 \in \mathbb{Z}_2^\times. \tag{D.55f}$$

The use of the global multiplier 1 in (D.55e) is essential to this exact norm equation.

*The primitive boundary rules out every odd fixed character.* Let $q$ be odd and suppose a nonzero $\beta$-invariant character $\chi : F \to C_q$ existed. Extend it by $\widetilde{\chi}(s) = 0$ to $E_b$, and put $N = \ker \chi$ and $E_N = \ker \widetilde{\chi} = N \rtimes \langle s \rangle$. Equation (D.55e) gives $2\widetilde{\chi}(v) = 0$, so $v \in E_N$. Write $v = ns$ with $n \in N$.

The exact cover and its maps are those of (D.54g)–(D.54j), now over $\mathbb{Z}_2$. It has type $(1, q)$. The deck-average idempotent $e = q^{-1} \sum_{C_q} d$ commutes with $\beta$, identifies $eH_1(N, \mathbb{Z}_2)$ with the base torus homology by the index-$q$ lattice inclusion, and has boundary augmentation as its complement. On the first summand $\beta = -I$ by (D.55d). On the second it is the identity: the actual $v \in E_N$ fixes each cusp label and centralizes its cusp generator up to $N$-conjugacy, exactly as in (D.54g). Hence

$$I + \beta = 0 \quad \text{on } H_1(N, \mathbb{F}_2). \tag{D.55g}$$

The same two elements in (D.55e) then have incompatible homology:

$$[v^2] = [n\,\beta(n)] = 0, \qquad [c^\kappa] = (\kappa_2 \bmod 2)[c] \neq 0. \tag{D.55h}$$

The second class is nonzero because $c$ is a primitive boundary in the cover with $q \ge 3$ cusps. This excludes the invariant character.

At every odd prime $q$ the matrix $A_q(b)$ squares to $I$ and has determinant 1. Its characteristic-zero eigenvalues are both 1 or both $-1$. The first possibility makes it $I$ and supplies a nonzero invariant character modulo $q$, contrary to (D.55h). The second makes it $-I$. Together with (D.55d), this proves (D.55a).

*Faithfulness at every prime and the exact abstract list.* Fix a prime $p$ and a finite subgroup $G$ of the full torus group. By D.33 its order divides twenty-four. A nontrivial kernel of $A_p|_G$ would contain an element of order two or three. A positive involution has image $-I$ by (D.55a). A negative involution has determinant $-1$ at every prime by D.43, so also cannot lie in the kernel. An order-three element has nonidentity image at $p = 3$ by D.33, and at every other prime by D.54's fixed-character obstruction. The kernel is therefore trivial, as claimed.

D.54 already removed $C_{12}$ and $D_{24}$ from D.52. The only additional type in that allowed list was $\mathrm{Dic}_{12}$. Its normal subgroup of order three would be faithfully represented over $\mathbb{Q}_3$, and an order-four element inverting it would also have order four by the just-proved faithfulness. But the exact field-normalizer calculation (D.52i)–(D.52k) proves that every finite-order matrix inverting this $C_3$ has square $I$: the alternative would require norm $-1$ in $\mathbb{Q}_3(\zeta_3)$. This contradiction excludes $\mathrm{Dic}_{12}$. The nine remaining types all have the explicit realizations of D.50 and D.52, proving (D.55b).

For an arbitrary order-four $a$, its square is a positive involution, so (D.55a) gives (D.55c). The full multiplier still has square one by D.50. At a negative coordinate the two eigenvalues of $A_q(a)$ are roots of $X^2 + 1$ whose product is $-1$, so they are equal. The matrix is diagonalizable in characteristic zero, hence scalar $i_q I$ or $-i_q I$. The two-coordinate is positive by D.50. For an odd $q$ a square root of $-1$ in $\mathbb{Q}_q$ reduces to one in $\mathbb{F}_q$, which requires $q \equiv 1 \pmod 4$. This proves the stated quartic restriction without excluding its remaining scalar case.

The program `code/part_c/torus_cover_norm_d54_d55.py` also checks the square norm and primitive-boundary comparison over $\mathbb{F}_2$ in explicit cyclic odd-degree covering homology modules. The full subgroup, covering and normalizer arguments above give the all-prime conclusion. □

**D.56. [Proved] The prime sections of a cyclic-six torus action.** Let $a$ have order six in the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$, and let

$$E = \{(\phi, j) \in \mathrm{Aut}(F) \times C_6 : [\phi] = a^j\} \tag{D.56a}$$

be its centre-free extension, with kernel identified with $F$ by the inner-automorphism map. There are exactly three $F$-conjugacy classes of order-two sections over $\langle a^3 \rangle$ and exactly two $F$-conjugacy classes of order-three sections over $\langle a^2 \rangle$. Conjugation by an element with quotient label $a^2$ cycles the three former classes; conjugation by an element with quotient label $a^3$ exchanges the two latter classes.

Consequently $E$ has exactly one conjugacy class of subgroups of order two and one of order three, and these are its only nontrivial finite subgroup types. In particular the full extension (D.56a) does not split, and $a$ has no finite-order representative in $\mathrm{Aut}(F)$. For each prime section $S$ of order $p = 2$ or 3,

$$N_E(S) = C_F(S) \times S, \qquad H^1(C_F(S), \mathbb{F}_p) = 0. \tag{D.56b}$$

The assertions concern these specified extension and section maps; they do not identify $E$ with a free profinite product.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), finite actions and actual lifts in type (1,1).

*Proof.* Put

$$E_2 = \pi^{-1}(\langle a^3 \rangle), \qquad E_3 = \pi^{-1}(\langle a^2 \rangle), \quad \pi : E \longrightarrow \langle a \rangle. \tag{D.56c}$$

The involution $a^3$ is positive. Indeed D.55 embeds $\langle a\rangle$ faithfully in $\mathrm{GL}_2(\mathbb{Q}_2)$. An order-six matrix there has its order-three part in the irreducible quadratic field $\mathbb{Q}_2(\zeta_3)$; its commuting involution is the nonidentity scalar $-I$. For completeness, the centralizer is that field because the cubic part has irreducible polynomial $X^2+X+1$, and its only elements squaring to one are $1,-1$. Faithfulness excludes the former for $a^3$. Thus $a^3$ has determinant one at two, and D.43 gives its global positive sign. D.27 supplies the actual prime-order sections of both groups in (D.56c).

*The three involution classes and their affine coordinates.* Choose an order-two section generator $t$ in $E_2$ and identify $E_2 = F \rtimes \langle t\rangle$. By D.46 and D.55, its action on two-adic homology is $-I$, and its maximal elementary abelian two-quotient has coordinates

$$(u,\chi_1,\chi_2) : E_2 \longrightarrow \mathbb{F}_2^3, \quad u(t)=1, \quad \chi_1(t)=\chi_2(t)=0, \tag{D.56d}$$

where $\chi_1|_F, \chi_2|_F$ form a basis of $H^1(F,\mathbb{F}_2)$. Every section generator has $u=1$, so its coordinate is a point of the affine plane $\mathbb{F}_2^2$. The three $F$-conjugacy classes of D.46 give three distinct such points. One can read the distinctness directly from that proof: Brown restriction identifies the localized degree-zero ring with $\mathbb{F}_2^3$, and this ring is generated by $\chi_1/u$ and $\chi_2/u$. Equal values on two section classes would prevent these two elements from generating the corresponding two coordinate factors.

Conjugation by an element $e \in E$ with $\pi(e)=a^2$ preserves $u$ and acts on the plane (D.56d) by

$$r \longmapsto Rr + d, \tag{D.56e}$$

where the linear part is the reduction of $A_2(a^2)$ in the chosen homology coordinates. D.54, at $q=2$, says that $R-I$ is invertible. Also $R^3=I$, so $R$ is a nonidentity cubic matrix over $\mathbb{F}_2$ and $I+R+R^2=0$. The affine map (D.56e) has one fixed point, and its other three points form a single orbit of length three: translate its unique fixed point to zero and use the corresponding linear statement. Its invariant three-point set of section classes must be that orbit. This proves the asserted cyclic permutation. The action on the class set factors through $E/E_2 = C_3$, because conjugation by $E_2$ is already $F$-conjugation on its sections, as proved in D.46.

*The two cubic classes and their affine coordinates.* Choose an order-three section generator $s$ in $E_3$ with label $a^2$. Use the actual abelian quotient of D.48 and the separation proved in D.51:

$$(u_3,\chi) : E_3 \longrightarrow \mathbb{F}_3^2, \qquad u_3(s)=1, \quad \chi(s)=0. \tag{D.56f}$$

The two $F$-conjugacy classes of section generators with prescribed label $u_3=1$ have $\chi$-values $0$ and $b$, with $b\neq 0$. For an element $e\in E$ with label $a^3$, conjugation preserves $u_3$ because $E/F$ is abelian. On the restriction of $\chi$ to $F$ it acts by $-1$, since D.55 gives $A_3(a^3)=-I$. The difference between the conjugated character and $-\chi$ vanishes on $F$, so is a scalar multiple of $u_3$. On the line $u_3=1$ the action consequently has the exact form

$$r \longmapsto -r + d'. \tag{D.56g}$$

Preserving the two-point set $\{0,b\}$ forces $d'=b$: the alternative of fixing both would give $2b=0$ in $\mathbb{F}_3$. Thus this action exchanges the two classes. Conjugation by $E_3$ is $F$-conjugation on its sections, by D.48, so the action factors through $E/E_3 = C_2$.

*The full finite subgroups and the failure of lifting.* Every finite subgroup of $E$ injects into $E/F = C_6$, because $F$ is torsion-free. It is therefore cyclic of order one, two, three or six. A subgroup of order six would normalize its order-two subgroup. Its element with quotient label $a^2$ would then fix that subgroup's $F$-conjugacy class, contradicting (D.56e). Thus no subgroup of order six exists. The two affine actions show respectively that all order-two subgroups are conjugate in $E$ and all order-three subgroups are conjugate in $E$.

For a chosen section $S$ of order two, the stabilizer of its $F$-conjugacy class in $E/F$ is exactly $\langle a^3\rangle$; for order three it is exactly $\langle a^2\rangle$. The image of $N_E(S)$ equals this stabilizer: a conjugator

whose class is fixed can be multiplied by an element of $F$ to normalize $S$ itself. Thus the full normalizer is already its normalizer in $E_p$. Equations (D.46h) and (D.48g) identify it as $C_F(S)\times S$ and give the cohomology assertion in (D.56b).

The first projection of (D.56a) is injective and identifies $E$ with the inverse image of $\langle a\rangle$ in $\mathrm{Aut}(F)$. Indeed the outer class of the first coordinate determines its label $j$. A homomorphic lift of $\langle a\rangle$ would give a subgroup of order six in this exact inverse image, which has been excluded. More generally, a finite-order representative $\phi$ of $a$ would satisfy $\phi^6 \in \mathrm{Inn}(F)$. The latter group is isomorphic to the torsion-free group $F$, so $\phi^6 = 1$, making $\phi$ a forbidden element of order six. This proves the labelled nonlifting assertion while preserving the actual prime-order lifts in (D.56c).

The program `code/part_a/torus_order6_sections_d56.py` checks every affine cubic action on $\mathbb{F}_2^2$ with nonidentity cubic linear part, both nonzero cubic section separations in (D.56f), and the corresponding quotient stabilizers. The extension and cohomological arguments above establish that these are the actions on the full section class sets. □

**D.57. [Proved] Full lifts of the noncyclic torus two-groups.** Let $Q$ be a subgroup of the full once-punctured-torus loop-outer group, isomorphic to $C_2 \times C_2$ or to $D_8$ of order eight. Put

$$E_Q = \{(\phi, q) \in \mathrm{Aut}(F) \times Q : [\phi] = q\}, \qquad F = \widehat{F}(x, y). \tag{D.57a}$$

The projection $E_Q \to Q$ has a homomorphic section with exactly its specified outer labels. For $Q = D_8$ there is exactly one $F$-conjugacy class of such sections.

For $Q = C_2 \times C_2$, let $z$ be its positive involution and let $r$ be either of its two negative involutions. There are exactly three section classes when

$$A_2(r) \bmod 2 = I, \tag{D.57b}$$

and exactly one otherwise. This choice does not depend on which negative involution is called $r$. For the image $H$ of every full section,

$$N_{E_Q}(H) = C_F(H) \times H, \qquad H^1(C_F(H), \mathbb{F}_2) = 0. \tag{D.57c}$$

Together with D.27 and D.46, this supplies actual lifts for every finite two-subgroup in the list of D.50. It is a statement about labelled automorphism lifts, not a geometric conjugacy classification.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), finite two-groups and actual labelled lifts in type (1,1).

*Proof. The invariant prime-section classes for a four-group.* Suppose first that $Q = C_2 \times C_2$. By D.50 and D.55 its faithful dyadic representation has exactly one nonidentity positive element $z$, with $A_2(z) = -I$; the two other involutions are negative. In the inverse image $E_z$ of $\langle z\rangle$, D.46 supplies exactly three $F$-conjugacy classes of sections $T$ of order two, each with

$$H^1(C_F(T), \mathbb{F}_2) = 0. \tag{D.57d}$$

As in (D.56d), their elementary abelian coordinates are three distinct points of the affine plane $\mathbb{F}_2^2$.

Conjugation by an element of $E_Q$ with label $r$ induces an affine map

$$v \longmapsto Mv + d, \qquad M = A_2(r) \bmod 2. \tag{D.57e}$$

Its square is the identity on the plane: the square of the conjugating element lies in $F$, and inner automorphisms act trivially on the elementary abelian quotient of $E_z$. Its three-point set of section classes is invariant.

If $M = I$, a nonzero translation would have only two-element orbits, so could not preserve a three-point set. Thus $d = 0$, and all three classes are fixed. If $M \neq I$, it is the unique two-dimensional unipotent type over $\mathbb{F}_2$, with

$$\ker(I + M) = \mathrm{im}(I + M), \quad \dim \ker(I + M) = 1.$$

The square identity gives $(I + M)d = 0$, so (D.57e) has two fixed points and one orbit of size two. An invariant three-point subset must contain that orbit and one fixed point. Thus exactly one section class is fixed in this case. Since $A_2(zr) = -A_2(r)$, the reductions of the two negative involutions are equal, proving the independence in (D.57b).

Choose any fixed class and one of its section groups $T$. Its normalizer $N = N_{E_Q}(T)$ maps onto $Q$: adjust a conjugator with label $r$ by an element of $F$ to normalize $T$, and use $T$ itself for the label $z$. Its kernel is $C = C_F(T)$, since an element of $F$ normalizing the order-two group $T$ centralizes it. Moreover $T$ is central in $N$. The quotient $J = N/T$ therefore has the exact sequence

$$1 \longrightarrow C \longrightarrow J \longrightarrow Q/\langle z\rangle \longrightarrow 1, \qquad Q/\langle z\rangle = C_2. \tag{D.57f}$$

The inclusion of $C$ in this quotient is injective, because $C \cap T = 1$.

*The invariant order-four class for a dihedral group.* Now suppose that $Q = D_8$ and let $R \cong C_4$ be its rotation subgroup. By D.53 the inverse image $E_R$ has exactly one $F$-conjugacy class of labelled order-four sections $B$, and

$$H^1(C_F(B), \mathbb{F}_2) = 0. \tag{D.57g}$$

Every element of $E_Q$ preserves this class, since $R$ is normal in $Q$. The normalizer $N = N_{E_Q}(B)$ thus maps onto $Q$, by the same adjustment of a conjugator by an element of $F$. Its kernel is $C = C_F(B)$: an element of $F$ normalizing $B$ fixes the label of every element of $B$, and therefore centralizes $B$. The subgroup $B$ is normal in $N$, and the quotient $J = N/B$ has

$$1 \longrightarrow C \longrightarrow J \longrightarrow Q/R \longrightarrow 1, \qquad Q/R = C_2. \tag{D.57h}$$

Again $C$ embeds, because $C \cap B = 1$. Conjugation on $B$ in this construction is the actual conjugation on its specified rotation labels in $D_8$.

*Torsion and exact finite preimages in the two constructed quotients.* In each of (D.57f) and (D.57h), the group $C$ is a closed subgroup of $F$, and hence is torsion-free with $\mathrm{cd}_p(C) \leq 1$ at every prime. Equations (D.57d) and (D.57g) give $C^{(2)} = 1$. Every map from $J$ to a finite two-group therefore kills $C$, and the displayed nontrivial quotient gives

$$J^{(2)} = C_2. \tag{D.57i}$$

The same full cohomological-dimension argument used in D.27 and D.46 proves that $J$ has torsion. Indeed, a torsion-free $J$ would be projective, since $C$ is open and has the stated cohomological dimensions. Its maximal pro-two quotient would then be a free pro-two group, contrary to (D.57i). Every finite subgroup of $J$ injects into $J/C = C_2$. Consequently $J$ has an involution $j$ outside $C$.

Take the full inverse image $H$ of $\langle j\rangle$ in $N$. In the four-group construction its kernel is $T$, so $|H| = 4$; in the dihedral construction its kernel is $B$, so $|H| = 8$. In either case $H \cap F = 1$. To check this in the actual maps, an element of $H \cap F$ has image in $\langle j\rangle \cap C = 1$ in $J$, and then lies in $T \cap F = 1$ or in $B \cap F = 1$ respectively. Thus the projection $H \to Q$ is injective. Its order equals $|Q|$, so it is an isomorphism with the prescribed labels. Its inverse is the required section. This exact injection also fixes the isomorphism type of the finite lift; no alternative central extension is substituted for $Q$.

*Uniqueness above each invariant class.* The section just constructed identifies $N = C \rtimes H$ and

$$J = C \rtimes \langle j \rangle. \tag{D.57j}$$

The Hochschild–Serre sequence, with $H^1(C, \mathbb{F}_2) = 0$ and $\mathrm{cd}_2(C) \leq 1$, gives the actual inflation isomorphism

$$H^*(J, \mathbb{F}_2) \cong H^*(C_2, \mathbb{F}_2). \tag{D.57k}$$

The full Brown restriction theorem [42, Theorem 2 and §1.1] applies: $C$ is open of finite two-cohomological dimension and no finite subgroup of $J$ can be a four-group, since it injects into $C_2$. In high degree the source of its dense restriction map is the finite one-dimensional space in (D.57k). Restriction to the displayed section in (D.57j) is already nonzero in every degree, so this map is injective. Its image is therefore closed as well as dense, so the restriction is an isomorphism onto the product of the section-normalizer cohomologies. For a section $S \leq J$ its normalizer is $C_C(S) \times S$, by the quotient labels. Each such factor contributes

$$1 + \dim H^1(C_C(S), \mathbb{F}_2)$$

in every sufficiently high degree. There is consequently exactly one conjugacy class of sections in $J$, and its fixed group has zero first cohomology over $\mathbb{F}_2$. These are also exactly the $C$-conjugacy classes: multiply a given conjugator by the section element with its quotient label.

Full sections of $E_Q$ containing the chosen $T$ or $B$ lie in $N$, and are exactly the full inverse images of the order-two sections of $J$. This follows from their orders and the same kernel calculation used above. They therefore form one $C$-conjugacy class. Every full four-group section has a positive subgroup $T$ in one of the invariant classes of (D.57e); distinct such classes cannot be identified by $F$-conjugacy. The numbers of full section classes are thus exactly three and one as asserted. Every full dihedral section has a rotation subgroup in the unique class of D.53, so the dihedral count is exactly one.

The fixed group $C_C(j)$ in the Brown calculation is precisely $C_F(H)$ in the original extension. For example, commuting with $j$ modulo $T$ or $B$ means that a commutator lies in that finite subgroup. The commutator also lies in $F$, so is trivial. The elements of $C$ already commute with $T$ or $B$. This proves the cohomology part of (D.57c). Finally an element normalizing $H$ can be multiplied by the inverse of the unique element of $H$ with its outer label. The result lies in $F$ and normalizes $H$, and therefore centralizes each element of $H$ by preservation of labels. This gives the exact direct product in (D.57c).

The program `code/part_a/torus_two_group_lifts_d57.py` checks all affine involutions of $\mathbb{F}_2^2$ together with their invariant three-point sets and verifies the three-versus-one fixed-class count. The full projectivity, Brown and finite-preimage arguments above provide the labelled lifts and their uniqueness statements. □

**D.58. [Proved] Negative torus involution sections and cusp fusion.** Let $b$ be a negative involution of the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$, put $c = [x, y]$, and let $E_b$ be its centre-free extension. Write $A_2$ for its action on $H_1(F, \mathbb{Z}_2)$. The $F$-conjugacy classes of order-two sections and their full fixed-subgroup cohomology are exactly

| $A_2$ mod 2 | number of section classes | $\{\dim_{\mathbb{F}_2} H^1(C_F(S), \mathbb{F}_2)\}_{[S]}$ |
|---|---|---|
| $I$ | 2 | $\{1, 0\}$ |
| $\neq I$ | 1 | $\{0\}$. |

(D.58a)

For every such section, $N_{E_b}(S) = C_F(S) \times S$. The cusp normalizer is the full profinite dihedral group

$$N_{E_b}(\overline{\langle c \rangle}) = \overline{\langle c \rangle} \rtimes \langle v \rangle, \qquad v^2 = 1, \quad vcv^{-1} = c^{-1}. \tag{D.58b}$$

Its two internal conjugacy classes of order-two sections become the same $F$-conjugacy class in $E_b$. In particular, for every choice of the cusp reflection $v$, there exists $g \in F$ with

$$gvg^{-1} = cv, \qquad c = [g, v] = g\,(vgv^{-1})^{-1}. \tag{D.58c}$$

The dimensions in (D.58a) describe continuous cohomology of the full closed fixed subgroups.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The actual extension and the two integral module cases.* D.27 supplies an actual order-two representative $\alpha$ and the exact identification $E_b = F \rtimes_\alpha \langle s \rangle$. D.43 gives its global multiplier $\lambda = -1$, so $A_2^2 = I$ and $\det A_2 = -1$. Let $u : E_b \to \mathbb{F}_2$ be the quotient character, $u(s) = 1$.

First suppose $A_2 \equiv I \pmod 2$. On the free $\mathbb{Z}_2$-module $H^1(F, \mathbb{Z}_2)$, the induced involution $D$ also satisfies $D \equiv I \pmod 2$. The endomorphism

$$P = (I - D)/2 \tag{D.58d}$$

is integral and satisfies $P^2 = P$. Its image and kernel are direct summands of rank one, because $D$ has the two eigenvalues $1, -1$ over $\mathbb{Q}_2$. Thus choose continuous characters $\varphi_+, \varphi_- : F \to \mathbb{Z}_2$ which form a basis and satisfy

$$\varphi_+\alpha = \varphi_+, \qquad \varphi_-\alpha = -\varphi_-. \tag{D.58e}$$

This is a decomposition of the specified cohomology module. All subsequent maps use these actual characters; no free generating pair is changed.

Their mod-2 reductions extend to characters $a, b : E_b \to \mathbb{F}_2$ by $a(s) = b(s) = 0$. The Hochschild–Serre spectral sequence has the two rows $\mathbb{F}_2 u^i$ and $\mathbb{F}_2^2 u^i$. The degree-one characters extend, so their transgression vanishes. All other possible differentials are its products with powers of $u$ and also vanish. The associated graded pieces give, for every $n \geq 1$,

$$H^n(E_b, \mathbb{F}_2) = \langle u^n, u^{n-1}a, u^{n-1}b \rangle_{\mathbb{F}_2}. \tag{D.58f}$$

Multiplication by $u$ is an isomorphism in degrees at least one, since it is an isomorphism on both associated graded pieces.

*The two Bocksteins and the actual localized ring.* Reduce (D.58e) modulo four and use the continuous one-cochains

$$\widetilde{a}(hs^\epsilon) = \varphi_+(h) \bmod 4, \qquad \widetilde{b}(hs^\epsilon) = \varphi_-(h) \bmod 4.$$

The first is a homomorphism, while the second has coboundary

$$\delta\widetilde{b}(hs^\epsilon, ks^\eta) = 2\epsilon\varphi_-(k) \pmod 4. \tag{D.58g}$$

For the coefficient sequence $0 \to \mathbb{F}_2 \xrightarrow{2} \mathbb{Z}/4 \to \mathbb{F}_2 \to 0$, the Bockstein of a degree-one character is its cup square. This follows directly by taking its $\{0, 1\}$-valued lift; changing the lift changes the divided coboundary by a coboundary. Thus (D.58g) gives

$$a^2 = 0, \qquad b^2 = ub. \tag{D.58h}$$

Write $ab = \epsilon ua + \delta ub + \gamma u^2$ using (D.58f), with coefficients in $\mathbb{F}_2$. Restriction to the chosen section $\langle s \rangle$ gives $\gamma = 0$. Multiplication by $a$ and (D.58h) then give

$$0 = a^2 b = \delta\epsilon\, u^2 a + \delta^2 u^2 b.$$

The two displayed degree-three classes are independent by (D.58f), so $\delta = 0$. Consequently the degree-zero localization $R = (H^*(E_b, \mathbb{F}_2)[u^{-1}])^0$ has the exact presentation

$$R = \mathbb{F}_2[A, B]/(A^2,\ B^2 - B,\ AB - \epsilon A), \qquad A = a/u, \quad B = b/u, \quad \dim_{\mathbb{F}_2} R = 3. \tag{D.58i}$$

For either value of $\epsilon$ the map

$$\begin{aligned} R &\longrightarrow \mathbb{F}_2[\nu]/(\nu^2) \times \mathbb{F}_2, \\ 1 &\longmapsto (1, 1), \\ A &\longmapsto (\nu, 0), \\ B &\longmapsto (\epsilon, 1 - \epsilon) \end{aligned} \tag{D.58j}$$

is a ring isomorphism. It preserves the stated relations and sends the three-dimensional basis $1, A, B$ to a basis. In particular $R$ has exactly two primitive idempotents, and its nilpotent ideal has dimension one.

*The nonidentity action modulo two.* Now suppose $A_2 \bmod 2 \neq I$. The dual two-dimensional module $M = H^1(F, \mathbb{F}_2)$ is the regular $\mathbb{F}_2[C_2]$-module: $D - I = D + I$ has rank one, with image equal to kernel. The periodic cyclic resolution therefore gives

$$\dim M^{C_2} = 1, \qquad H^i(C_2, M) = 0 \quad (i > 0).$$

Choose the nonzero invariant character on $F$ and extend it to $\chi : E_b \to \mathbb{F}_2$ with $\chi(s) = 0$. This again makes the only possible transgression zero. The two-row spectral sequence now gives

$$H^1(E_b, \mathbb{F}_2) = \langle u, \chi \rangle, \qquad H^n(E_b, \mathbb{F}_2) = \mathbb{F}_2 u^n \quad (n \geq 2). \tag{D.58k}$$

Both $u\chi$ and $\chi^2$ restrict to zero on $\langle s \rangle$, whereas $u^2$ restricts nontrivially. Thus $u\chi = \chi^2 = 0$. In this case the degree-zero localization is

$$(H^*(E_b, \mathbb{F}_2)[u^{-1}])^0 = \mathbb{F}_2. \tag{D.58l}$$

*Full Brown restriction and the section labels.* Every finite subgroup of $E_b$ injects into $E_b/F = C_2$, because $F$ is torsion-free. In particular $E_b$ has no subgroup $C_2 \times C_2$. For a section $S$, conjugation in its normalizer induces the identity on $S$. Multiplying a normalizer element by the section element with its quotient label leaves an element of $C_S = C_F(S)$. This proves the exact direct product

$$N_{E_b}(S) = C_S \times S. \tag{D.58m}$$

The closed subgroup $C_S \leq F$ has $\mathrm{cd}_2(C_S) \leq 1$.

The full-profinite Brown restriction theorem applies with the open subgroup $F$ of cohomological 2-dimension one [42, Theorem 2 and §1.1]. For $n > 1$ its restriction map to the product of the cohomologies of (D.58m) is a dense embedding. The source is finite by (D.58f) or (D.58k); therefore that map is an isomorphism onto a finite product, as in the cited proof. These are the actual restriction maps and preserve cup products. Since $u$ restricts trivially on $C_S$ and nontrivially on $S$, localization gives the ring isomorphism

$$R \xrightarrow{\text{res}} \prod_{[S]} \left( \mathbb{F}_2 \oplus H^1(C_S, \mathbb{F}_2) u^{-1} \right). \tag{D.58n}$$

The second summand in each factor has square zero, because $H^2(C_S, \mathbb{F}_2) = 0$. Each factor is local, with that summand as its nilpotent ideal and with residue field $\mathbb{F}_2$.

If $A_2 \bmod 2 \neq I$, (D.58l) and (D.58n) give one factor and a zero second summand. If $A_2 \bmod 2 = I$, (D.58j) gives two factors, one with a one-dimensional second summand and one with a zero second summand. This proves all the numbers in (D.58a). The classes in this

comparison are also precisely the $F$-conjugacy classes: for a conjugator $e \in E_b$, multiply it by the inverse of the unique element of $S$ having the same quotient label. This replaces $e$ by an element of $F$ without changing $eSe^{-1}$.

In the identity-modulo-two case the two classes are distinguished by the value of $b$ on their nonidentity section element. Indeed the two residue-field evaluations of (D.58j) give $B = 0$ and $B = 1$. Restriction to a section is exactly that residue-field evaluation in (D.58n). Also $a$ vanishes on every section element, since $a^2 = 0$ whereas the square of the nonzero character of $C_2$ is nonzero.

*The exact cusp reflections and their fusion.* Put $L = \overline{\langle c \rangle} \cong \widehat{\mathbb{Z}}$. Preservation of the cusp class and A.7 give

$$1 \longrightarrow L \longrightarrow N_{E_b}(L) \longrightarrow C_2 \longrightarrow 1. \tag{D.58o}$$

Choose $v$ over its nonidentity label. The global multiplier $-1$ gives $vcv^{-1} = c^{-1}$. Write $v^2 = c^{\kappa}$. Commutation of $v$ with its own square implies $c^{\kappa} = c^{-\kappa}$, hence $2\kappa = 0$ in $\widehat{\mathbb{Z}}$ and $\kappa = 0$. This proves (D.58b), with the multiplication and specified label of its finite factor retained.

Every section of this dihedral group is $\langle c^t v \rangle$, $t \in \widehat{\mathbb{Z}}$. Conjugation by $c^z$ changes $t$ to $t + 2z$, and conjugation by $v$ changes it to $-t$. Its two internal classes are therefore represented by $v$ and $cv$. These elements have the same image under every abelian character of $E_b$, since $c = [x, y] \in [F, F]$. When there is only one full section class they are already $F$-conjugate. When there are two, their equal $b$-values and the preceding residue-field distinction again put them in the same $F$-class. An $F$-conjugator sends the unique nonidentity element to the unique nonidentity element, giving the first equation of (D.58c). Multiplication by $v^{-1}$ gives its other two exact equations.

The executable `code/part_c/torus_negative_involution_d58.py` checks the mod-four one-cochains, the cyclic resolution modules, the two explicit localized algebra isomorphisms, and geometric word examples of the cusp equation. The full-group conclusions above follow from restriction and the actual normalizer maps. They assert no full triviality or procyclicity of the fixed subgroups from their first cohomology dimensions. □

**D.59. [Proved] The negative cusp section and its fixed homology line.** Let $b$ be a negative involution in the full once-punctured-torus loop-outer group, let $F = \widehat{F}(x, y)$ and $c = [x, y]$, and choose a cusp reflection $v$ in its centre-free extension $E$. Thus

$$E = F \rtimes \langle v \rangle, \qquad v^2 = 1, \qquad vcv^{-1} = c^{-1}.$$

Put $\alpha = \mathrm{Inn}(v)|_F$, $C = C_F(v)$, and

$$M = H_1(F, \widehat{\mathbb{Z}}), \qquad M^+ = \ker(\alpha_* - I : M \to M).$$

Then the cusp section is specifically the class with

$$H^1(C, \mathbb{F}_2) = 0, \qquad C \subseteq \overline{[F, F]}. \tag{D.59a}$$

For every actual transporter $g$ supplied by D.58,

$$gvg^{-1} = cv, \tag{D.59b}$$

its homology class is a generator of the full fixed line:

$$\widehat{\mathbb{Z}} \xrightarrow{\cong} M^+, \qquad t \longmapsto t[g]. \tag{D.59c}$$

This line is a direct summand of $M$.

More precisely, let $m \geq 3$ be odd and let $\chi : F \to C_m$ be any surjective continuous $\alpha$-invariant character. Extend it to $\widetilde{\chi} : E \to C_m$ by $\widetilde{\chi}(v) = 0$, and put $F_m = \ker \chi$, $E_m = \ker \widetilde{\chi}$. Then

$$C \subset F_m, \qquad \chi(g) \text{ generates } C_m, \qquad E_m = F_m \rtimes \langle v \rangle. \tag{D.59d}$$

The $m$ cusp-dihedral edges form one cycle on the $m$ $F_m$-section classes above the original cusp class. Their fixed groups are conjugates of the same full group $C$. The complete section counts in this cover are

| $\alpha_* \bmod 2$ | cusp classes in $E_m$ | other classes in $E_m$ |
|---|---|---|
| $I$ | $m$, each with $H^1(-,\mathbb{F}_2)=0$ | 1, with $\dim H^1(-,\mathbb{F}_2)=1$ |
| $\neq I$ | $m$, each with $H^1(-,\mathbb{F}_2)=0$ | 0. |

(D.59e)

In the first row, if $D = C_F(w)$ is the fixed group of the other full section class, its actual homology-image map is surjective:

$$D \longrightarrow M^+. \tag{D.59f}$$

These statements specify full subgroup cohomology and the indicated ambient homology maps. They make no identification of $C$ with the trivial group, of $D$ with a procyclic group, or of $g$ with a member of a full free profinite basis.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof.* D.43 gives the global multiplier $-1$. D.58 gives the stated actual reflection, the transporter (D.59b), and exactly one or two full section classes according to the two rows above. Their normalizers are the actual direct products $C_F(S) \times S$. Write $A_2$ for the action on $H_1(F,\mathbb{Z}_2)$.

*The odd-degree cover and its boundary injection.* The displayed extension of $\chi$ is a homomorphism because $\chi\alpha = \chi$. Its kernel is precisely the split extension in (D.59d). Every order-two element of $E$ lies in that kernel, since a homomorphism to the odd cyclic group kills such an element.

The restriction of $\chi$ to the dense discrete free group is onto. Its degree-$m$ covering of the punctured torus has $m$ degree-one boundary lifts, because $\chi(c) = 0$. Filling those boundaries gives the corresponding unramified degree-$m$ cover of the compact torus. Consequently its type is $(1,m)$, its completed group is exactly $F_m$, and $F_m \cong \widehat{F}_{m+1}$. Choose a discrete word $f$ with $\chi(f) = 1$, and put

$$c_i = f^i c f^{-i}, \qquad v_i = f^i v f^{-i}, \qquad 0 \le i < m. \tag{D.59g}$$

Each $v_i$ belongs to $E_m$, has order two, and inverts $c_i$.

Here is the exact homology comparison required for the section argument. Let $\Delta = F/F_m = C_m$ be the deck group and let $P = H_1(F_m,\mathbb{Z}_2)$. The boundary sequence is

$$0 \longrightarrow B \longrightarrow P \longrightarrow H_1(\overline{S}_m,\mathbb{Z}_2) \longrightarrow 0, \qquad B = \mathbb{Z}_2^m/\mathbb{Z}_2(1,\dots,1). \tag{D.59h}$$

The image of the $i$th standard generator of $B$ is $[c_i]$. Thus these classes have their sole relation $\sum_i [c_i] = 0$. Deck transformations act trivially on the compact torus homology. Since $m$ is odd, the actual idempotent

$$e = \frac{1}{m}\sum_{\delta\in\Delta}\delta \tag{D.59i}$$

is defined on $P$. It annihilates $B$ and induces the identity on the last term of (D.59h). It follows that

$$P = eP \oplus (1-e)P, \qquad (1-e)P = B. \tag{D.59j}$$

The covering map identifies $eP$ with the base torus homology: on integral homology its image is the index-$m$ covering lattice, and that inclusion is an isomorphism over $\mathbb{Z}_2$.

The automorphism $\alpha$ induces the identity on $\Delta$. Its action on $P$ therefore commutes with the deck action; a discrepancy between two chosen lifts is an inner automorphism of $F_m$ and has trivial action on $P$. It preserves every individual boundary class with multiplier $-1$. Explicitly,

$\alpha(f^i)f^{-i} \in F_m$ conjugates $c_i^{-1}$ to $\alpha(c_i)$. Thus $\alpha = -I$ on $B$. Its action on $eP$ is the actual base matrix $A_2$, through the covering homology map just specified.

Reduction modulo two makes $\alpha$ the identity on $B/2B$. The abelianization of the split extension gives the exact comparison

$$H_1(E_m, \mathbb{F}_2) \cong \big(H_1(F_m, \mathbb{F}_2)\big)_\alpha \oplus \mathbb{F}_2[v]. \tag{D.59k}$$

The direct summand $B/2B$ survives injectively in these coinvariants by (D.59j). In particular the images $d_i = [c_i] \in H_1(E_m, \mathbb{F}_2)$ still have precisely their one total-sum relation. Their span has dimension $m - 1$.

*The actual section graph and its transporter labels.* Let $h = |\chi(C)|$. The $F_m$-conjugacy classes lying above the original full cusp-section class are exactly the vertices

$$C_m/\chi(C). \tag{D.59l}$$

Indeed $f^i v f^{-i}$ and $f^j v f^{-j}$ are $F_m$-conjugate precisely when $i - j \in \chi(C)$: a conjugator $n \in F_m$ gives $f^{-j} n f^i \in C$, and the converse follows by reversing that formula. All original order-two sections lie in $E_m$, so this is the full class splitting, not a selection of section images in a finite quotient.

For each $i$, take the edge with endpoints $\langle v_i \rangle$ and $\langle c_i v_i \rangle$. Both endpoints are actual order-two sections. Equation (D.59b) identifies the second endpoint with the conjugate by $f^i g$ of $\langle v \rangle$. Thus, with $\delta = \chi(g)$, this edge has the exact vertex labels

$$i + \chi(C) \quad \text{and} \quad i + \delta + \chi(C). \tag{D.59m}$$

There are $m$ labelled edges and $m/h$ vertices. Assign to a vertex the homology class in $H_1(E_m, \mathbb{F}_2)$ of its nonidentity section element. This is independent of the conjugate representative. The sum of the two endpoint values of the $i$th edge is exactly $d_i$, by the group equality $c_i v_i = c_i \cdot v_i$.

For a finite graph with $r$ vertices and $k$ connected components, its edge differences span a subspace of dimension at most $r - k$ in any vector space of vertex values. One proves this by choosing one vertex in each component and expressing every edge difference in the differences from those chosen vertices. Apply this to the actual values above. By (D.59k),

$$m - 1 \le m/h - k. \tag{D.59n}$$

Since $h \ge 1$ and $k \ge 1$, equality is forced: $h = 1$ and $k = 1$. The edges now join $i$ to $i + \delta$ in $C_m$. Connectedness says exactly that $\delta$ generates $C_m$. Since $m$ is odd and at least three, these edges form a single $m$-cycle. This proves all assertions in (D.59d) and the cusp part of the graph assertion. Because $C \subset F_m$, the fixed group of each such section is the corresponding conjugate of the same full group $C$.

*Which full section is the cusp section.* We compute the high-degree cohomology of $E_m$ using the actual module (D.59j). Over $\mathbb{F}_2$ its dual consists of $m - 1$ trivial boundary summands and the dual base module. When $A_2 \bmod 2 = I$ this last module is two trivial summands; otherwise it is the regular two-dimensional $\mathbb{F}_2[C_2]$-module. The two-row Hochschild–Serre sequence has no nonzero differential into the quotient row: the actual section makes inflation from $C_2$ injective in every degree. There are no other possible differentials. The periodic cyclic resolution therefore gives, for every $n \ge 2$,

$$\dim_{\mathbb{F}_2} H^n(E_m, \mathbb{F}_2) = \begin{cases} m + 2, & A_2 \bmod 2 = I, \\ m, & A_2 \bmod 2 \ne I. \end{cases} \tag{D.59o}$$

The full-profinite Brown restriction theorem applies with the open subgroup $F_m$ of 2-cohomological dimension one [42, Theorem 2 and §1.1]. Every nontrivial finite subgroup has

order two. For a section $T$, its normalizer in $E_m$ is $C_{F_m}(T) \times T$, with $\operatorname{cd}_2 C_{F_m}(T) \leq 1$. The finite source in (D.59o) makes the dense Brown restriction an isomorphism onto its finite product, so each section class contributes

$$1 + \dim H^1(C_{F_m}(T), \mathbb{F}_2) \tag{D.59p}$$

to (D.59o). Conjugacy by $E_m$ and by $F_m$ agrees: multiply an $E_m$-conjugator by the element of its source section with the same quotient label.

D.58 already gives $H^1(C, \mathbb{F}_2) = 0$ when $A_2 \bmod 2 \neq I$. In the other case its dimension is zero or one. Choose an invariant surjective character to $C_3$: the integral involution on $\mathbb{Z}_3^2$ has distinct eigenvalues $1, -1$, so projection onto its fixed direct summand supplies this character. The graph just proved gives three cusp classes with the same group $C$. A dimension-one $H^1(C, \mathbb{F}_2)$ would make them contribute six to (D.59o), whose value is five. Thus the dimension is zero also in this case, proving the first assertion of (D.59a).

If $A_2 \bmod 2 \neq I$, D.58 has no other full section class, so the $m$ cusp classes exhaust $E_m$. If $A_2 \bmod 2 = I$, let $D = C_F(w)$ be the fixed group of the other full class. As observed above, $w \in E_m$. The same exact class-splitting formula gives $m/|\chi(D)|$ classes, each with fixed group conjugate to $D \cap F_m$. Subtracting the $m$ cusp contributions from (D.59o) yields

$$\frac{m}{|\chi(D)|}\bigl(1 + \dim H^1(D \cap F_m, \mathbb{F}_2)\bigr) = 2. \tag{D.59q}$$

The first factor is odd. It is therefore one, and the cohomology dimension is one. This proves (D.59e), as well as $\chi(D) = C_m$ for every such odd character.

*The full ambient homology images.* Every element of $C$ has homology in $M^+$. At an odd prime $p$, the two idempotents $(I \pm A_p)/2$ split $\mathbb{Z}_p^2$ into its rank-one fixed and negative summands. For every $r \geq 1$, projection onto a chosen generator of the fixed summand followed by reduction modulo $p^r$ is an actual invariant surjective character $F \to C_{p^r}$. Equation (D.59d) makes the image of $C$ vanish under every one of these maps, and hence in the entire fixed $\mathbb{Z}_p$-line. At the prime two, a nonzero compact image of $C \to M_2^+ \cong \mathbb{Z}_2$ would be $2^r\mathbb{Z}_2$ for some $r$, and its quotient by twice itself would give a nonzero continuous character $C \to \mathbb{F}_2$. The already proved $H^1(C, \mathbb{F}_2) = 0$ excludes that. Thus $C$ has zero image in every primary factor of $M$, which is exactly $C \subseteq \ker(F \to M) = \overline{[F, F]}$.

From $c = g\alpha(g)^{-1}$, the homology of $g$ belongs to $M^+$. At every odd prime the same characters and the generator assertion of (D.59d) say that $[g]$ generates the fixed $\mathbb{Z}_p$-line. We check the dyadic factor directly.

In the maximal pro-two group, let $D_2, D_3$ be the second and third Zassenhaus terms. The quotient $D_2/D_3$ is the three-dimensional $\mathbb{F}_2$-space with basis

$$X^2, \quad Y^2, \quad Z = [X, Y].$$

For completeness this is the central subgroup of the finite characteristic quotient with coordinates $(a, b, k) \in (\mathbb{Z}/4)^2 \times \mathbb{F}_2$ and product

$$(a, b, k)(a', b', k') = (a + a',\, b + b',\, k + k' + ba'). \tag{D.59r}$$

The generators are $(1, 0, 0), (0, 1, 0)$ and their commutator is $(0, 0, 1)$. The subgroup with $a, b$ even has the stated basis. Equivalently its kernel is the closed subgroup generated by fourth powers, squares of commutators, and triple commutators, so the quotient is characteristic.

The induced action on this degree-two space depends only on $\overline{A} = A_2 \bmod 2$. A vector $aX + bY$ has square $aX^2 + bY^2 + abZ$, and $\overline{A}$ fixes $Z$. If $\overline{A} = I$, the degree-two action is the identity. Otherwise an actual linear basis change over $\mathbb{F}_2$ gives $X \mapsto X$, $Y \mapsto Y + X$. It preserves $Z$ since every invertible two-dimensional matrix over $\mathbb{F}_2$ has determinant one. In these coordinates

$$\operatorname{im}(I + \overline{A}|_{D_2/D_3}) = \langle X^2 + Z \rangle. \tag{D.59s}$$

In either case $Z$ is not in that image. If $[g]$ vanished modulo two, its pro-two image would lie in $D_2$, and the exact equation $c = g\alpha(g)^{-1}$ would give $Z = (I + \overline{A})[g]_2$ in $D_2/D_3$. This contradicts (D.59s). Thus $[g] \bmod 2 \neq 0$.

The kernel of $A_2 - I$ on $\mathbb{Z}_2^2$ is a saturated rank-one submodule: if $2z$ lies in it, then $2(A_2 - I)z = 0$ in a torsion-free module. It is therefore generated by a primitive vector, and the nonzero reduction of $[g]$ makes its coefficient in that generator a unit. Together with the odd-primary conclusions, this proves (D.59c). For each prime choose a complementary basis vector to $[g]_p$. Taking their product gives an actual decomposition $M = \widehat{\mathbb{Z}}[g] \oplus \widehat{\mathbb{Z}}e$ and a continuous homology retraction $M \to \widehat{\mathbb{Z}}$ sending $[g]$ to one.

Finally suppose $A_2 \bmod 2 = I$. Equation (D.59q) makes the ambient image of $D$ surject onto every odd-primary fixed line. At two, use the actual positive character $\varphi_+$ and its reduction $a$ in D.58e–D.58j. The nonzero nilpotent class $a/u$ restricts under the full Brown map to the one-dimensional $H^1(D, \mathbb{F}_2)$ factor; the cusp factor has no such summand by (D.59a). Hence $a|_D \neq 0$. The integral idempotent decomposition of D.58 makes $\varphi_+$ primitive on the fixed dyadic line. Since the ambient image of $D$ lies in that line, its image contains a dyadic unit and is all of the line. The total image is a compact subgroup of $M^+ \cong \widehat{\mathbb{Z}}$ surjecting onto every primary factor. Its image in each finite cyclic quotient is therefore the full quotient; compactness proves surjectivity onto $\widehat{\mathbb{Z}}$. This establishes (D.59f).

The executable `code/part_c/torus_cusp_cycle_d59.py` checks the exact finite incidence ranks, boundary relations, cyclic-module dimensions and the degree-two quotient calculation. The full group and topology comparisons, including the Brown restriction and all ambient homology maps, are proved above. □

**D.60. [Proved] The exact torsion quotient of a negative torus extension.** Use D.59's notation

$$E = F \rtimes \langle v \rangle, \quad F = \widehat{F}(x, y), \quad c = [x, y], \quad v^2 = 1, \quad \alpha = \mathrm{Inn}(v)|_F, \quad M = F^{\mathrm{ab}},$$

and choose an actual transporter $g$ with $gvg^{-1} = cv$. For a subset of a profinite group, double angle brackets below mean its closed normal closure in the specified ambient group. There are exact continuous quotient identifications

$$\begin{aligned} E/\langle\langle v \rangle\rangle_E &\xrightarrow{\cong} M/(\alpha_* - I)M, \\ E/\langle\langle \mathrm{Tor}(E) \rangle\rangle_E &\xrightarrow{\cong} \widehat{\mathbb{Z}}, \end{aligned} \tag{D.60a}$$

where $\mathrm{Tor}(E)$ is the set of finite-order elements. More explicitly,

| $A_2 \bmod 2$ | $E/\langle\langle v \rangle\rangle_E$ | $\chi(g)$ |
|---|---|---|
| $I$ | $\widehat{\mathbb{Z}} \times C_2$ | 1 |
| $\neq I$ | $\widehat{\mathbb{Z}}$ | 2. |

(D.60b)

Here $\chi : E \to \widehat{\mathbb{Z}}$ is the second quotient in (D.60a), with the normalization specified in the proof. Its restriction to $F$ is onto. It has a continuous section whose image lies in $F$, so

$$E = \langle\langle \mathrm{Tor}(E) \rangle\rangle_E \rtimes s(\widehat{\mathbb{Z}}). \tag{D.60c}$$

In the first row, let $w$ represent the other full section class. One can choose that section $s$ inside $C_F(w)$, and the resulting map

$$\widehat{\mathbb{Z}} \times C_2 \longrightarrow E, \qquad (t, \epsilon) \longmapsto s(t)w^{\epsilon} \tag{D.60d}$$

is a continuous embedding. These statements specify the quotient, splitting, and embedded subgroup; they give no further free-product presentation of $E$.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The two quotient maps before any module decomposition.* Write $R = \langle\langle v \rangle\rangle_E$ and $Q = E/R$. The map $F \to Q$ is onto, since $E$ is generated by $F$ and $v$. D.58's actual transporter gives

$$c = [g, v] = g\alpha(g)^{-1}. \tag{D.60e}$$

Consequently $c$ dies in $Q$. The profinite quotient $F/\langle\langle [x, y] \rangle\rangle_F$ is exactly $\widehat{\mathbb{Z}}^2 = M$: its two generating images commute, while the abelianization map to $\widehat{\mathbb{Z}}^2$ gives the inverse universal map. Thus the specified map $F \to Q$ factors through $M$. Since $v$ dies, $\alpha(f)$ and $f$ have the same image for every $f \in F$. This supplies a continuous surjection

$$M/(\alpha_* - I)M \longrightarrow Q. \tag{D.60f}$$

The image $(\alpha_* - I)M$ is compact and hence closed.

In the other direction, the formula

$$E \longrightarrow M/(\alpha_* - I)M, \qquad fv^\epsilon \longmapsto [f] \bmod (\alpha_* - I)M \tag{D.60g}$$

is a continuous homomorphism. Indeed the product in the specified semidirect extension is $f\alpha^\epsilon(h)v^{\epsilon+\eta}$, and $\alpha_*$ acts trivially on this quotient. Formula (D.60g) kills $v$, so it factors through $Q$. Its composite with (D.60f) is the identity on the images of $F$; the reverse composite has the same property. Those images generate both quotients, proving the first isomorphism in (D.60a). Equivalently, all of its full-group relations can be retained as

$$Q = F/\langle\langle c,\ \alpha(f)f^{-1}\ (f \in F) \rangle\rangle_F. \tag{D.60h}$$

The relation $c$ is already implied by the displayed relation with $f = g$, through (D.60e).

*Explicit coordinates for both integral module cases.* Put $R_0 = \widehat{\mathbb{Z}}$. D.59 proves that $e_+ = [g]$ generates the entire fixed line in $M$ and that this line is a direct summand. Choose a complementary basis vector $e$ in the *homology module*, so $M = R_0 e_+ \oplus R_0 e$. No free generating pair of $F$ is changed. Since the global determinant of $\alpha_*$ is $-1$ by D.43, its matrix in these coordinates is

$$A = \begin{pmatrix} 1 & a \\ 0 & -1 \end{pmatrix}, \qquad (A - I)M = R_0(a, -2) \tag{D.60i}$$

for a specified $a \in R_0$. The two cases of (D.60b) are exactly whether $a \in 2R_0$ or $a_2 \in \mathbb{Z}_2^\times$.

In the first case write $a = 2b$. The following map is onto:

$$R_0^2 \longrightarrow R_0 \times C_2, \qquad (r, t) \longmapsto (r + bt,\ t \bmod 2). \tag{D.60j}$$

Its kernel is exactly $R_0(a, -2)$. Indeed $t = 2z$ and $r + bt = 0$ are equivalent to $(r, t) = (-az, 2z)$. Conversely each multiple of $(a, -2)$ has zero image. The map is onto by choosing $t = 0$ or 1 for the prescribed second coordinate and then choosing $r$ for the first. It therefore gives the first-row quotient homeomorphism, since a continuous bijection between compact Hausdorff spaces is a homeomorphism.

In the second case the map

$$R_0^2 \longrightarrow R_0, \qquad (r, t) \longmapsto 2r + at \tag{D.60k}$$

is onto. At every odd prime its coefficient 2 is a unit; at the prime two its coefficient $a_2$ is a unit. These coordinatewise solutions give an element of $R_0^2$ for every prescribed value. Its kernel is again exactly $R_0(a, -2)$: if $2r + at = 0$, the dyadic unit $a_2$ forces $t_2 \in 2\mathbb{Z}_2$, hence $t \in 2R_0$. Write $t = -2z$. The equation becomes $2(r - az) = 0$, and $R_0$ has no additive two-torsion, so $r = az$. The converse follows by substitution. This gives the second-row quotient homeomorphism.

For $[f] = re_+ + te$, define $\chi(fv^\epsilon)$ to be $r + bt$ in the first case and $2r + at$ in the second. Formulas (D.60g), (D.60j), and (D.60k) prove that these are continuous homomorphisms. They give exactly the values $\chi(g) = 1$ and 2, respectively, and $\chi|_F$ is onto in both cases.

*The second section and the precise finite factor.* The free profinite group $F$ is torsion-free. Every nontrivial finite-order element of $E$ therefore has order two and generates a section: its finite cyclic subgroup injects into $E/F = C_2$. D.58 gives one full $F$-conjugacy class in the second row and exactly two in the first.

In the first row, the dyadic coordinate character $[f] = re_+ + te \mapsto t_2$ is a primitive anti-invariant character $\varphi_- : F \to \mathbb{Z}_2$, since (D.60i) sends $t$ to $-t$. Its mod-two extension

$$\beta : E \longrightarrow \mathbb{F}_2, \qquad \beta(fv^\epsilon) = t_2 \bmod 2 \tag{D.60l}$$

is a homomorphism and vanishes on $v$. It is exactly the D.58 section-distinguishing character normalized to vanish on this cusp reflection. To make that normalization explicit, if D.58 was written using another initial section $s$, its character $b$ has the same restriction to $F$ as (D.60l), and

$$\beta = b + b(v)u, \tag{D.60m}$$

where $u : E \to C_2$ is the original quotient character. Primitive choices of the anti-invariant integral character differ by a dyadic unit and have the same mod-two restriction. D.58 proves that the two section classes have opposite $b$-values. Equation (D.60m) thus gives $\beta(v) = 0$ and $\beta(w) = 1$ for every representative $w$ of the other class.

The image of $w$ in the quotient (D.60j) has order dividing two. The additive group $R_0$ is torsion-free, so this image has first coordinate zero. Equation (D.60l) is its second coordinate and has value one. Thus that image is exactly $(0, 1)$. The closed normal subgroup generated by all torsion in the first row is consequently the inverse image of $\{0\} \times C_2$ under $E \to Q$. In the second row all torsion is conjugate to $v$, so its closed normal closure is already $R$. This proves in both cases

$$\ker \chi = \langle\langle \mathrm{Tor}(E) \rangle\rangle_E \tag{D.60n}$$

and proves the second isomorphism in (D.60a). It also identifies the exact relation between the two quotient maps, including the entire finite factor in the first row.

*Actual splittings and the embedded commuting factor.* Choose $h \in F$ with $\chi(h) = 1$, possible because $\chi|_F$ is onto. The continuous profinite power map

$$s : R_0 \longrightarrow E, \qquad t \longmapsto h^t \tag{D.60o}$$

satisfies $\chi(s(t)) = t$. It is therefore injective, and compactness identifies its image with $R_0$ topologically. Every $z \in E$ has the unique factorization

$$z = \big(zs(\chi(z))^{-1}\big)s(\chi(z)), \qquad zs(\chi(z))^{-1} \in \ker \chi.$$

This proves (D.60c), with the given maps.

In the first row, D.59 proves that $D = C_F(w) \to M^+$ is onto. Choose $h \in D$ with $[h] = [g]$. Then $\chi(h) = 1$, so the same construction gives a section $s$ inside $D$. The formula (D.60d) is a homomorphism because $h$ commutes with $w$. If its value is the identity, application of $\chi$ gives $t = 0$, and then $w^\epsilon = 1$ gives $\epsilon = 0$. It is a continuous embedding by compactness. This establishes every stated splitting and subgroup without identifying their generated subgroup with the whole extension.

The executable `code/part_c/torus_torsion_quotient_d60.py` checks the displayed quotient maps and their exact kernels in finite coefficient rings. The full quotient identifications follow from the two universal maps and the actual transporter relation above. □

**D.61. [Proved] Full generation and exact negative conjugator equations.** Put $F = \widehat{F}(x,y)$ and $c = [x,y]$. Every closed subgroup $J \le F$ satisfying $c \in \overline{[J,J]}$ is $F$ itself. Consequently every continuous endomorphism of $F$ carrying $c$ to a conjugate of $c^{\mu}$, with $\mu \in \widehat{\mathbb{Z}}^{\times}$, is an automorphism.

Let $b$ be any negative full once-punctured-torus outer involution, with the specified extension

$$E = F \rtimes \langle v \rangle, \quad v^2 = 1, \quad vcv^{-1} = c^{-1}.$$

Let $A_2$ be its action on $H_1(F, \mathbb{Z}_2)$. Set $v_i = c^i v$ and $\alpha_i = \mathrm{Inn}(v_i)|_F$, for $i = 0, 1$. On a separate free profinite group $F_0 = \widehat{F}(X,Y)$, with $c_0 = [X,Y]$, use the two standard actual negative involutions

| | $X$ | $Y$ |
|---|---|---|
| $\beta_0$ | $YXY^{-1}$ | $Y^{-1}$ |
| $\beta_1$ | $Y$ | $X$. |

(D.61a)

For $j = 0, 1$, let $\mathcal{T}_j(b)$ be the space of peripheral outer isomorphisms $[\phi] : F_0 \to F$ conjugating $[\beta_j]$ to $b$. The peripheral requirement is $\phi(c_0) \sim_F c^{\mu}$ for a full unit $\mu$. Identifying $F_0$ with $F$ by $X \mapsto x$, $Y \mapsto y$ identifies this with the usual outer conjugator space.

For $A_2 \bmod 2 = I$, define

$$\begin{aligned} \mathcal{S}_{0,i} = \{(\mu, h, w) : \mu \in \widehat{\mathbb{Z}}^{\times},\ h \in F, \quad w^2 = 1,\ w \in E \setminus F, \\ w \text{ belongs to the other full section class}, \\ [h, w] = 1, \quad [h, v_i] = c^{\mu}\}. \end{aligned} \tag{D.61b}$$

Here *other* means the class distinct from the common cusp-reflection class of D.58. There is a continuous bijection, with continuous inverse,

$$\mathcal{S}_{0,0} \sqcup \mathcal{S}_{0,1} \xrightarrow{\cong} \mathcal{T}_0(b), \qquad (\mu, h, w) \longmapsto [\, X \mapsto h,\ Y \mapsto v_i w\,]. \tag{D.61c}$$

For $A_2 \bmod 2 \neq I$, define

$$\mathcal{S}_{1,i} = \{(\mu, f) \in \widehat{\mathbb{Z}}^{\times} \times F : [f, \alpha_i(f)] = c^{\mu}\}. \tag{D.61d}$$

The corresponding continuous bijection is

$$\mathcal{S}_{1,0} \sqcup \mathcal{S}_{1,1} \xrightarrow{\cong} \mathcal{T}_1(b), \qquad (\mu, f) \longmapsto [\, X \mapsto f,\ Y \mapsto \alpha_i(f)\,]. \tag{D.61e}$$

Equivalently, writing $z = f v_i$, the last equation is

$$z \in E \setminus F, \qquad [z^2, v_i] = c^{\mu}. \tag{D.61f}$$

The two basis images are then $zv_i, v_i z$. Thus no separate generation condition occurs in either parameter space. The equations and their outer conjugator spaces are identified exactly; nonemptiness for an arbitrary $b$ remains unproved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The boundary test for a full closed subgroup.* Suppose $J$ were proper. A proper open subgroup $U \le F$ contains $J$: separate a point outside the compact closed subgroup $J$ in a finite quotient and take the inverse image of the image of $J$. In particular $c \in U$. Write $\Pi = F(x,y)$ for the dense discrete punctured-torus group. Its subgroup $\Pi_U = \Pi \cap U$ has the same finite index $d = [F : U] > 1$, and its full profinite completion is exactly $U$.

In the associated connected covering surface, the boundary lift through the base point has degree one because $c \in \Pi_U$. The degrees of all boundary lifts sum to $d$. There must therefore

be a second boundary component. In the homology of an orientable surface with at least two boundary components, the boundary classes have only their total-sum relation. Each individual boundary class is nonzero over every $\mathbb{F}_p$. The chosen class is the class of $c$, with its specified base path. Its image in $H_1(U,\mathbb{F}_p) = H_1(\Pi_U,\mathbb{F}_p)$ is nonzero. On the other hand $c \in \overline{[J,J]} \subseteq \overline{[U,U]}$, so that image is zero. This contradiction proves $J = F$.

Now let $\rho : F \to F$ be an endomorphism with $\rho(c) = qc^{\mu}q^{-1}$ for a full unit $\mu$. Its image is compact and closed. In the conjugate image $J = q^{-1}\rho(F)q$, the element $c^{\mu}$ belongs to $\overline{[J,J]}$. Taking its $\mu^{-1}$ power puts $c$ in the same closed commutator subgroup. The result just proved makes $\rho$ onto. A finitely generated profinite group is Hopfian: the inverse-image operation of a surjective endomorphism permutes the finite set of open normal subgroups of each given index, so its kernel lies in every such subgroup. Their intersection is trivial. Thus $\rho$ is an automorphism. The same argument applies to a homomorphism $F_0 \to F$ after the explicit basis identification in the claim.

*The exact two standard extensions.* The substitutions in (D.61a) square to the identity and send $c_0$ to $c_0^{-1}$. Let $E_j = F_0 \rtimes_{\beta_j} \langle v_0\rangle$. For $j = 0$ put $w_0 = Y^{-1}v_0$. There are mutually inverse continuous maps

$$\begin{aligned}(\widehat{\mathbb{Z}}\langle X\rangle \times \langle w_0\rangle) \amalg \langle v_0\rangle &\longrightarrow E_0,\\ (X, w_0, v_0) &\longmapsto (X, Y^{-1}v_0, v_0),\\ E_0 &\longrightarrow (\widehat{\mathbb{Z}}\langle X\rangle \times C_2) \amalg C_2,\\ (X, Y, v_0) &\longmapsto (X, v_0w_0, v_0).\end{aligned} \tag{D.61g}$$

Both displayed finite generators have order two. The equations $w_0^2 = 1$ and $[X, w_0] = 1$ follow directly from the action $\beta_0$. Conversely $Y = v_0w_0$ gives $v_0Yv_0 = Y^{-1}$ and $v_0Xv_0 = YXY^{-1}$. Substitution on the displayed generators proves that the two maps are inverse. The quotient label sends $X$ to zero and $w_0, v_0$ to one in $C_2$. The two finite factors represent distinct section classes: the character taking $w_0$ to one and $X, v_0$ to zero separates them.

For $j = 1$ put $z_0 = Xv_0$. The exact maps are

$$\begin{aligned}\langle v_0\rangle \amalg \widehat{\mathbb{Z}}\langle z_0\rangle &\longrightarrow E_1,\\ (v_0, z_0) &\longmapsto (v_0, Xv_0),\\ E_1 &\longrightarrow C_2 \amalg \widehat{\mathbb{Z}},\\ (X, Y, v_0) &\longmapsto (z_0v_0, v_0z_0, v_0).\end{aligned} \tag{D.61h}$$

Here $v_0$ swaps the two displayed images of $X, Y$, so all semidirect relations hold. The two composites fix the generators. The quotient label is one on both $v_0$ and $z_0$. These maps specify the full profinite products through their universal properties.

*Normalization of every peripheral outer conjugator.* Take $[\phi] \in \mathcal{T}_j(b)$. After an inner change in $F$, choose a representative satisfying

$$\phi(c_0) = c^{\mu}. \tag{D.61i}$$

The unit $\mu$ is unique: a conjugator between two full unit powers of $c$ normalizes $L = \overline{\langle c\rangle}$; the full peripheral normalizer inside $F$ is $L$ itself, which centralizes $c$.

The centre-free extension construction gives the unique isomorphism $\Phi : E_j \to E$ restricting to $\phi$ and inducing the identity on $C_2$. Explicitly, conjugation by $\phi$ carries the full inverse image of $\langle[\beta_j]\rangle$ in $\mathrm{Aut}(F_0)$ to the full inverse image of $\langle b\rangle$ in $\mathrm{Aut}(F)$. The embeddings of these extensions are faithful: an element centralizing their free kernel has trivial quotient label and then belongs to the trivial centre of that kernel. This gives both the extension map and its inverse.

Put $t = \Phi(v_0)$. It has order two, has nonzero quotient label, and inverts $c^{\mu}$ by (D.61i). As $\mu$ is a unit, it also inverts $c$. D.58's actual cusp normalizer therefore gives the unique expression

$$t = c^z v, \qquad z \in \widehat{\mathbb{Z}}. \tag{D.61j}$$

Let $i \in \{0,1\}$ be the residue of $z$ modulo two and put $r = (i-z)/2 \in \widehat{\mathbb{Z}}$. Replacing $\Phi$ by $\mathrm{Inn}(c^r)\Phi$ retains (D.61i) and sends (D.61j) to $c^{z+2r}v = v_i$. Thus every outer class has a representative with the exact two normalization equations

$$\phi(c_0) = c^\mu, \qquad \Phi(v_0) = v_i. \tag{D.61k}$$

This representative and the index $i$ are unique within the outer class. Indeed two representatives satisfying the first equation differ by an inner conjugation by an element of $C_F(c^\mu) = L$. Conjugation by $c^r$ changes the reflection exponent by $2r$. Equality of two indices in $\{0,1\}$ modulo two first makes the indices equal, and then $2r = 0$ gives $r = 0$. All peripheral centralizers and normalizers used here are the full specified subgroups.

*The two-class parameter map and its inverse.* Suppose first that $A_2 \bmod 2 = I$. For a tuple in (D.61b), put $k = v_i w \in F$. Commutation of $h$ with $w$ gives the exact identity

$$[h,k] = [h,v_i] = c^\mu. \tag{D.61l}$$

The homomorphism $X \mapsto h$, $Y \mapsto k$ is consequently an automorphism $F_0 \to F$ by the boundary test. Also

$$\alpha_i(k) = k^{-1}, \qquad \alpha_i(h) = khk^{-1}, \tag{D.61m}$$

the second equation following by inserting $whw = h$. Equations (D.61m) prove $\phi\beta_0 = \alpha_i\phi$. Since $\alpha_i$ represents $b$, this is the asserted peripheral outer conjugator. Its actual extension sends $v_0$ to $v_i$ and $w_0$ to $w$.

Conversely, normalize any member of $\mathcal{T}_0(b)$ by (D.61k). Put $h = \phi(X)$ and $w = \Phi(w_0)$. The source relations in (D.61g) give $w^2 = 1$, $w \notin F$, and $[h,w] = 1$. That isomorphism carries the two distinct source section classes to the two distinct target classes, so $w$ is in the other class. Since $\phi(Y) = v_i w$, equation (D.61i) gives (D.61l), hence (D.61b). These two constructions recover the exact tuple and the normalized representative. This proves bijectivity in (D.61c).

*The one-class parameter map and its inverse.* Now let $A_2 \bmod 2 \neq I$. A tuple in (D.61d) defines $\phi(X) = f$ and $\phi(Y) = \alpha_i(f)$. Its commutator is $c^\mu$, so the boundary test again proves full invertibility. The equality $\alpha_i^2 = 1$ makes $\phi\beta_1 = \alpha_i\phi$. This gives the indicated outer conjugator and an extension sending $v_0$ to $v_i$. Conversely, for the normalized representative of any member of $\mathcal{T}_1(b)$, the same intertwining equation forces $\phi(Y) = \alpha_i(\phi(X))$. Taking $f = \phi(X)$ and using (D.61i) yields exactly (D.61d).

Finally $z = fv_i$ gives $z^2 = f\alpha_i(f)$ and

$$[z^2, v_i] = f\alpha_i(f)\big(\alpha_i(f)f\big)^{-1} = [f, \alpha_i(f)]. \tag{D.61n}$$

Conversely $f = zv_i$ and $\alpha_i(f) = v_i z$. This proves the alternative parameterization and both basis-image formulas in (D.61f).

*Topology and the remaining existence obligation.* Each parameter space is closed in the indicated compact product: the full other section class in (D.61b) is the compact image of its conjugation map, and all equations are continuous. The maps to automorphisms are continuous on generator images, and hence so are their maps to the peripheral outer isomorphism space. The latter space is Hausdorff and closed in the profinite outer group, after the explicit basis identification. The continuous bijections just proved are therefore homeomorphisms.

The two integral homology actions in (D.61a) have respectively identity and nonidentity reduction modulo two. Conjugacy preserves this distinction, so only the indicated standard action can occur in each row. The formulas supply full actual generators for every tuple in the displayed spaces. They establish no tuple for an arbitrary negative full outer involution. The remaining problem is the existence of solutions of those precise equations, with their retained peripheral element and reflection index. The executable `code/part_c/torus_negative_parameters_d61.py` checks the exact standard free-word and free-product identities and the boundary test on all labelled connected covers of degrees two through five. □

**D.62. [Proved] Primitive homology centralizers and the exact cusp-transporter root test.** Let $F = \widehat{F}(x, y)$ and let $g \in F$ have primitive abelianization: there is a continuous homomorphism $\rho : F \to \widehat{\mathbb{Z}}$ with $\rho(g) = 1$. Then, for every nonzero integer $n$,

$$C_F(g^n) = \overline{\langle g \rangle} \cong \widehat{\mathbb{Z}}, \qquad N_F(\overline{\langle g \rangle}) = \overline{\langle g \rangle}. \tag{D.62a}$$

This conclusion uses homology primitivity; it does not require a free basis containing $g$.

Now use the exact negative torus extension of D.58–D.61:

$$\begin{gathered} E = F \rtimes \langle v \rangle, \qquad v^2 = 1, \qquad \alpha = \mathrm{Inn}(v)|_F, \qquad c = [x, y], \qquad \alpha(c) = c^{-1}, \\ C = C_F(v), \qquad gvg^{-1} = cv. \end{gathered} \tag{D.62b}$$

D.59 supplies the required primitive abelianization of this $g$ and the exact inclusion $C \subseteq \overline{F'}$. For every $g' \in gC$, the following two conditions are equivalent:

$$\begin{gathered} \alpha(g') \text{ is conjugate to } g' \text{ in } F; \\ C_E(g') \text{ contains an element outside } F. \end{gathered} \tag{D.62c}$$

Their consequences and converse maps are exact:

1. In the two-section-class case of D.58, they are equivalent to the existence of an involution $w \in E \setminus F$ centralizing $g'$. That involution is unique, belongs to the other section class, and

   $$C_E(g') = \overline{\langle g' \rangle} \times \langle w \rangle. \tag{D.62d}$$

   The maps

   $$X \longmapsto g', \qquad Y \longmapsto vw, \qquad v_0 \longmapsto v \tag{D.62e}$$

   give a full isomorphism from the two-class standard extension in D.61 to $E$, preserving the quotient labels and carrying $[X, Y]$ to $c$.

2. In the one-section-class case of D.58, they are equivalent to the existence of $z \in E \setminus F$ with $z^2 = g'$. This root is unique, and

   $$C_E(g') = \overline{\langle z \rangle} \cong \widehat{\mathbb{Z}}, \qquad F \cap C_E(g') = \overline{\langle z^2 \rangle} = \overline{\langle g' \rangle}. \tag{D.62f}$$

   The maps

   $$X \longmapsto zv, \qquad Y \longmapsto vz, \qquad v_0 \longmapsto v \tag{D.62g}$$

   give the corresponding full isomorphism from the one-class standard extension, with the same exact peripheral and quotient labels.

Thus the multiplier-one, $v_0 \mapsto v$ portion of the D.61 conjugator problem is exactly the existence of a member $g' \in gC$ satisfying (D.62c). The explicit continuation (D.62q)–(D.62w) treats every full unit multiplier and both actual reflection labels, and gives the same criterion for the entire D.61 conjugator space. Neither equivalence asserts that its required conjugacy occurs.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The primitive-homology centralizer.* The equation $\rho(g) = 1$ first proves that $a \mapsto g^a$ embeds $\widehat{\mathbb{Z}}$ as the indicated closed procyclic subgroup. Put $D = C_F(g^n)$. We show $\ker(\rho|_D) = 1$.

Suppose $k \neq 1$ belongs to this kernel. The procyclic group generated by $k$ has a nontrivial $p$-primary component for some prime $p$. Since $F$ is torsion-free, that component is infinite procyclic, isomorphic to $\mathbb{Z}_p$. Let $k_p$ be a topological generator of it. The $p$-primary component

of $g^n$ also generates a copy of $\mathbb{Z}_p$: under $\rho$ it is $n\mathbb{Z}_p$ in the $p$ coordinate, and $n \neq 0$ in $\mathbb{Z}_p$. These two procyclic groups commute, because $k$ centralizes $g^n$. Their intersection is trivial: $\rho$ is zero on the first and injective on the second. Multiplication consequently gives a continuous injection

$$\mathbb{Z}_p \times \mathbb{Z}_p \longrightarrow F. \tag{D.62h}$$

Its image is closed by compactness. This contradicts $\mathrm{cd}_p(F) = 1$, since a closed subgroup has no larger $p$-cohomological dimension and $\mathrm{cd}_p(\mathbb{Z}_p^2) = 2$. Thus $\rho|_D$ is injective.

Every $h \in D$ satisfies $hg^{-\rho(h)} \in \ker(\rho|_D)$, because $g \in D$. Hence $h = g^{\rho(h)}$, proving the centralizer equality. For a normalizing element $h$, write $hgh^{-1} = g^\lambda$ with $\lambda \in \widehat{\mathbb{Z}}^\times$. Applying $\rho$ gives $1 = \lambda$. The normalizer equality now follows from the centralizer equality with $n = 1$. No change from homology primitivity to free-basis primitivity has been made.

*The two exact labels of a central extension.* Fix $g' \in gC$. Then

$$[g', v] = c, \qquad \alpha(g') = c^{-1}g', \qquad [g']_{\mathrm{ab}} = [g]_{\mathrm{ab}}. \tag{D.62i}$$

The first equality uses the defining commutation of $C$ with $v$, and the last uses $C \subseteq \overline{F'}$. In particular (D.62a) applies to $g'$. An outside element has the unique form $hv$, $h \in F$, and

$$(hv)g'(hv)^{-1} = h\alpha(g')h^{-1}. \tag{D.62j}$$

This is the actual comparison proving (D.62c), in both directions.

Suppose its conditions hold and choose $t \in C_E(g') \setminus F$. The kernel of the parity map on $C_E(g')$ is exactly $\overline{\langle g' \rangle}$ by (D.62a). Thus $t^2 = (g')^\kappa$ for a unique $\kappa \in \widehat{\mathbb{Z}}$, and $t$ commutes with $g'$. The entire centralizer has the presentation and coordinate maps

$$\begin{aligned} &P_\kappa = \langle G, T \mid [G,T] = 1,\ T^2 = G^\kappa \rangle_{\mathrm{prof}} \xrightarrow{\sim} C_E(g'), \\ &G \longmapsto g', \qquad T \longmapsto t, \\ &(a, \epsilon) \longmapsto (g')^a t^\epsilon \quad (a \in \widehat{\mathbb{Z}},\ \epsilon \in \{0,1\}). \end{aligned} \tag{D.62k}$$

Surjectivity is the two-coset description. For injectivity, a relation with $\epsilon = 1$ has nonzero parity, whereas a relation with $\epsilon = 0$ is detected by the primitive character of $g'$. The coordinate multiplication is

$$(a, \epsilon)(b, \delta) = \big(a + b + \epsilon\delta\kappa,\ \epsilon + \delta \bmod 2\big). \tag{D.62l}$$

These maps also give the specified compact topology.

An outside element $(g')^a t$ is an involution precisely when $2a + \kappa = 0$ in $\widehat{\mathbb{Z}}$. This equation has a solution exactly when $\kappa_2 \in 2\mathbb{Z}_2$; at every odd prime division by two is unique. In that case

$$w = (g')^{-\kappa/2}t, \qquad (a, \epsilon) \longmapsto (g')^a w^\epsilon$$

is the exact isomorphism $\widehat{\mathbb{Z}} \times C_2 \to C_E(g')$. Its outside involution is unique.

If $\kappa_2$ is odd, choose $b \in \widehat{\mathbb{Z}}$ with

$$b_2 = 0, \qquad b_p = (1 - \kappa_p)/2 \quad (p \text{ odd}), \qquad \kappa' = \kappa + 2b. \tag{D.62m}$$

The element $\kappa'$ is a full unit: its dyadic coordinate is $\kappa_2$ and every odd coordinate is one. Set

$$t' = (g')^b t, \qquad \nu = (\kappa')^{-1}, \qquad z = (t')^\nu. \tag{D.62n}$$

Then $(t')^2 = (g')^{\kappa'}$, $z^2 = g'$, and $z$ has outside parity because $\nu$ is odd at two. Moreover $t' = z^{\kappa'}$, so $z$ generates the entire centralizer. If $z^a = 1$, then $(g')^a = (z^2)^a = 1$, whence $a = 0$. This proves the isomorphism with $\widehat{\mathbb{Z}}$. Multiplication by two is injective there, proving uniqueness of

the specified square root. The displayed odd-prime corrections retain the full profinite group, not only its pro-two quotient.

*Which central extension occurs.* In the two-class case, the explicit torsion quotient $\chi : E \to \widehat{\mathbb{Z}}$ of D.60 has $\chi(g) = 1$ and vanishes on $C$. Thus $\chi(g') = 1$. Applying it to $t^2 = (g')^\kappa$ gives $\kappa = 2\chi(t)$, so $\kappa_2$ is even. The preceding split case gives the involution $w$. It cannot be $F$-conjugate to $v$: otherwise $g' \in C_F(w)$ would lie in a conjugate of $C \subseteq \overline{F'}$, contradicting its primitive abelianization. By D.58 it therefore belongs to the other actual section class.

In the one-class case, the same contradiction excludes every outside involution centralizing $g'$, because all such involutions are $F$-conjugate to $v$. Thus $\kappa_2$ must be odd. Equations (D.62m)–(D.62n) give precisely the root and centralizer in (D.62f). Conversely either an outside centralizing involution or an outside square root is an outside centralizer element, so these implications are equivalences.

*Full basis and extension maps.* For the two-class case, put $X' = g'$ and $Y' = vw$. Both lie in $F$, and $[g', w] = 1$ gives

$$[X', Y'] = [g', v] = c, \qquad \alpha(X') = Y'X'(Y')^{-1}, \qquad \alpha(Y') = (Y')^{-1}. \tag{D.62o}$$

For the one-class case put $X' = zv$ and $Y' = vz$. Again both lie in $F$, and

$$[X', Y'] = [z^2, v] = c, \qquad \alpha(X') = Y', \qquad \alpha(Y') = X'. \tag{D.62p}$$

In both cases the full boundary test of D.61 proves that $X \mapsto X'$, $Y \mapsto Y'$ is an automorphism of the full rank-two free profinite group. The displayed intertwining identities extend it to the asserted isomorphism with $v_0 \mapsto v$; its inverse on the kernel is the inverse free-group automorphism, and on an element $fv^\epsilon$ it is that inverse applied to $f$, followed by $v_0^\epsilon$. This proves the exact extension inverse and retains both parity labels.

Conversely, an isomorphism in this normalized portion of D.61 sends $X$ in the two-class standard model, or $XY$ in the one-class standard model, to an element $g'$ with $[g', v] = c$. The set of all such elements is precisely $gC$: equality of the two conjugations of $v$ says that $g^{-1}g'$ centralizes $v$. The image of the standard other reflection in the first model, or of its element $Xv_0$ in the second, supplies the centralizer element in (D.62c). This proves both directions of the final assertion.

The centralizer calculation and root formula give no proof that $\alpha(g')$ is $F$-conjugate to $g'$ for any member of this coset. The continuation below retains every unit multiplier and the other reflection label through explicit additional transporter maps. □

**Every unit multiplier and both actual reflection labels.** The preceding criterion extends to the full parameter spaces of D.61 by explicit transporter maps. No peripheral marking is rescaled. Retain $c, v, g, C$ from (D.62b), and put

$$\begin{gathered} v_i = c^i v, \qquad \alpha_i = \mathrm{Inn}(v_i)|_F, \qquad C_i = C_F(v_i), \qquad i \in \{0, 1\}, \\ C_0 = C, \qquad C_1 = gCg^{-1}. \end{gathered} \tag{D.62q}$$

For each actual unit $\mu \in \widehat{\mathbb{Z}}^\times$, define

$$g_{\mu,0} = c^{(\mu-1)/2} g, \qquad g_{\mu,1} = c^{(\mu+1)/2} g^{-1}. \tag{D.62r}$$

Both exponents are elements of $\widehat{\mathbb{Z}}$: the dyadic coordinate of $\mu$ is odd, and division by two is available at every odd prime. The conjugation identities

$$c^a v c^{-a} = c^{2a} v, \qquad gvg^{-1} = cv, \qquad g^{-1}(cv)g = v$$

give, with no change of the named cusp,

$$\begin{aligned} g_{\mu,i} v_i g_{\mu,i}^{-1} &= c^{\mu} v_i, \\ \{a \in F : [a, v_i] = c^{\mu}\} &= g_{\mu,i} C_i, \\ [a]_{\mathrm{ab}} &= (-1)^i [g]_{\mathrm{ab}} \quad (a \in g_{\mu,i} C_i). \end{aligned} \tag{D.62s}$$

The coset equality follows by dividing two transporters. The homology equality follows from $c \in \overline{F'}$ and $C_i \subseteq \overline{F'}$, the latter being (D.59a) and its displayed conjugate. In particular every one of these transporters is homologically primitive; its exact homology is independent of the unit $\mu$.

For a specified $a \in g_{\mu,i} C_i$ we now have

$$\alpha_i(a) = c^{-\mu} a, \qquad \alpha_i(a) \sim_F a \iff C_E(a) \not\subset F. \tag{D.62t}$$

The equivalence is the exact map $h \mapsto h v_i$ of (D.62j). In the two-class case, D.60 gives $\chi(a) = (-1)^i$. Thus $t^2 = a^{\kappa}$ for an outside centralizer element implies $2\chi(t) = (-1)^i \kappa$. The even-dyadic argument (D.62k)–(D.62l) supplies the unique outside commuting involution $w$. It is in the other full section class because $a$ has nonzero primitive homology whereas every cusp-class fixed group lies in $\overline{F'}$. The exact basis and extension maps are

$$X \longmapsto a, \qquad Y \longmapsto v_i w, \qquad v_0 \longmapsto v_i, \qquad [X, Y] \longmapsto c^{\mu}. \tag{D.62u}$$

In the one-class case, that same fixed-group argument excludes every outside involution in $C_E(a)$. The odd-dyadic construction (D.62m)–(D.62n) gives the unique $z \in E \setminus F$ with $z^2 = a$, and the exact maps are

$$X \longmapsto z v_i, \qquad Y \longmapsto v_i z, \qquad v_0 \longmapsto v_i, \qquad [X, Y] \longmapsto c^{\mu}. \tag{D.62v}$$

In both cases D.61's boundary test proves full invertibility because $\mu$ is a full unit. The inverse extension maps are the inverse kernel map together with the specified label-1 reflection, as in (D.62o)–(D.62p).

Conversely, every normalized conjugator in D.61 has some specified $\mu$ and $i$. In the two-class model its image of $X$, and in the one-class model its image of $XY$, belongs to the precise coset in (D.62s). The image of the standard other reflection or of $X v_0$ supplies the outside centralizer element, respectively. Therefore

$$\mathcal{T}_j(b) \neq \varnothing \iff \begin{cases} \text{there are } \mu \in \widehat{\mathbb{Z}}^{\times},\ i \in \{0,1\},\ a \in g_{\mu,i} C_i \\ \text{with } c^{-\mu} a \sim_F a. \end{cases} \tag{D.62w}$$

Here $\mathcal{T}_j(b)$ is the exact peripheral conjugator space defined in D.61, with its standard model chosen by the section count. Equations (D.62u)–(D.62v) and the converse give the actual maps proving this equivalence. They retain the unit exponent $c^{\mu}$ and both reflection labels throughout. The existence in the right-hand set remains unresolved; none of these identities asserts it for an arbitrary full negative action.

The executable `code/part_c/torus_unit_transporters_d62.py` checks both reflection-label identities in the full free-product word normal form for thirty-two odd integer exponents. These identities hold before imposing the boundary condition; the full unit hypothesis is retained where D.61 proves invertibility. The full centralizer and root statements are proved above by their exact subgroup and coefficient maps.

**D.63. [Proved] The exact homology characters of a negative torus centralizer.** Let $\Omega$ be the full once-punctured-torus peripheral outer group, and let $b \in \Omega$ be a negative involution. Keep the actual extension and peripheral labels of D.59–D.62:

$$\begin{gathered} F = \widehat{F}(x, y), \quad c = [x, y], \quad E = F \rtimes \langle v \rangle, \quad \alpha = \mathrm{Inn}(v)|_F, \quad v^2 = 1, \quad \alpha(c) = c^{-1}, \\ M = F_{\mathrm{ab}}, \quad A = \alpha_*, \quad C = C_F(v), \quad g v g^{-1} = c v, \quad M^+ = \ker(A - I) = \widehat{\mathbb{Z}}[g]. \end{gathered} \tag{D.63a}$$

In particular, D.59 says that $[g]$ is primitive and $C \subseteq \overline{F'}$. Put $Z_b = C_\Omega(b)$. There are actual continuous characters

$$\sigma : Z_b \longrightarrow \{1, -1\}, \qquad \lambda : Z_b \longrightarrow \widehat{\mathbb{Z}}^\times \tag{D.63b}$$

such that $\lambda$ is the peripheral multiplier, $\beta_*[g] = \sigma(\beta)[g]$, and the full homology matrix of every $\beta \in Z_b$ is exactly

$$\rho(\beta) = \frac{\sigma(\beta)}{2}\big((1 + \lambda(\beta))I + (1 - \lambda(\beta))A\big) \quad \text{in } \mathrm{GL}_2(\widehat{\mathbb{Z}}). \tag{D.63c}$$

Both half-coefficients in this formula belong to $\widehat{\mathbb{Z}}$: a full unit is odd at two, and two is invertible at every odd prime. In particular,

$$\lambda(\beta) = 1 \quad \Longrightarrow \quad \rho(\beta) = \sigma(\beta)I. \tag{D.63d}$$

Write

$$\Xi_b = \{(\sigma(\beta), \lambda(\beta)) : \beta \in Z_b\},$$
$$I_b = Z_b \cap \ker\big(\Omega \longrightarrow \mathrm{GL}_2(\widehat{\mathbb{Z}})\big).$$

The exact character sequence is

$$1 \longrightarrow I_b \longrightarrow Z_b \xrightarrow{(\sigma,\lambda)} \Xi_b \longrightarrow 1. \tag{D.63e}$$

Its kernel has the following description by actual automorphisms, with inverse maps given in the proof:

$$I_b \;\cong\; J_b := \{\psi \in \mathrm{Aut}(F) : \psi(c) = c,\ \psi\alpha = \alpha\psi\}. \tag{D.63f}$$

Every member of $J_b$ automatically acts trivially on $M$. The group $I_b$ is torsion-free. This description does not assert that $I_b$ is trivial, or that $\Xi_b$ is the whole product in (D.63b).

Every finite subgroup $K \leq Z_b$ has the exact faithful homology embedding

$$K \;\hookrightarrow\; \{I, -I, A_2, -A_2\} \cong C_2 \times C_2, \qquad \beta \longmapsto \rho_2(\beta). \tag{D.63g}$$

Here $A_2$ is the original dyadic involution matrix. Thus its possible abstract types are $1, C_2, C_2^2$; a finite subgroup containing $b$ has order two or four. This bound makes no assertion that a four-element subgroup occurs for every $b$.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The extension map for a centralizing outer class.* The map

$$E \longrightarrow \mathrm{Aut}(F), \qquad fv^\epsilon \longmapsto \mathrm{Inn}(f)\alpha^\epsilon \tag{D.63h}$$

identifies $E$ with the full inverse image of $\langle b \rangle$ under $\mathrm{Aut}(F) \to \mathrm{Out}(F)$. Indeed an element of that inverse image is $\mathrm{Inn}(f)\alpha^\epsilon$ by its outer label. The map is injective because $F$ is centre-free and $b$ has order two. It is a homeomorphism onto its image by compactness. This also gives $C_E(F) = 1$.

Let $\beta \in Z_b$, and choose an actual representative $\psi \in \mathrm{Aut}(F)$. Conjugation by $\psi$ preserves the inverse image in (D.63h), since its outer class centralizes $b$. Transport through (D.63h) gives an actual automorphism $\Psi : E \to E$, with $\Psi|_F = \psi$ and the identity map on $E/F = C_2$. This constructs the required extension directly from the centralizer; no conjugator from a standard extension has been presumed.

Write the peripheral equation as $\psi(c) = uc^\mu u^{-1}$, with $\mu = \lambda(\beta) \in \widehat{\mathbb{Z}}^\times$. Conjugate $\Psi$ by $u^{-1}$ to obtain $\Psi(c) = c^\mu$. The element $t = \Psi(v)$ has order two, has outside label, and inverts $c^\mu$.

Taking its $\mu^{-1}$ power shows that it inverts $c$. The full cusp normalizer from D.58 is $\langle c\rangle \rtimes \langle v\rangle$. Consequently

$$t = c^z v \quad (z \in \widehat{\mathbb{Z}})$$

with a unique exponent. Let $i \in \{0,1\}$ be the reduction of $z$ modulo two. Conjugating the entire extension map by $c^{(i-z)/2}$ now gives

$$\Psi(c) = c^{\mu}, \qquad \Psi(v) = v_i := c^i v. \tag{D.63i}$$

The half-exponent is defined at every prime, with even numerator at two. All these conjugations lie in $F$, so the outer class and its homology action have been preserved.

For completeness, (D.63i) determines the actual representative uniquely once its outer class is fixed. An inner comparison preserving $c^{\mu}$ belongs to $C_F(c^{\mu}) = \langle c\rangle$, by the full peripheral centralizer calculation. Write it as $c^a$. It changes the exponent of $v_i$ by $2a$. Two exponents in $\{0,1\}$ can therefore agree only with the same $i$, and then $2a = 0$ gives $a = 0$. The multiplier itself is unique by the same full peripheral normalizer calculation.

*The actual transporter determines the fixed-line sign.* Apply $\Psi$ to $gvg^{-1} = cv$. The resulting equation is

$$\Psi(g) v_i \Psi(g)^{-1} = c^{\mu} v_i. \tag{D.63j}$$

D.62 gives the full solution cosets, retaining the specified generator $c$:

| $i$ | $\Psi(g)$ belongs to | $[\Psi(g)]$ in $M$ |
|---|---|---|
| 0 | $c^{(\mu-1)/2} g\, C$ | $[g]$ |
| 1 | $c^{(\mu+1)/2} g^{-1}\, gCg^{-1}$ | $-[g]$. |

(D.63k)

The last column uses $C \subseteq \overline{F'}$. Thus the action on the fixed line is specifically $\sigma(\beta) = (-1)^i$. Because $[g]$ generates the full direct summand $M^+$, its actual action defines a continuous character; composition multiplies these two signs.

*The full matrix and the exact kernel.* The peripheral multiplier is also $\det \rho(\beta)$. One can verify the equality at each prime $p$ by mapping $F$ to the group of upper unitriangular three-by-three matrices over $\mathbb{Z}_p$, with $x, y$ the two elementary generators. The commutator coordinate of their images under $\psi$ is $\det \rho_p(\beta)$. The equation $\psi(c) = c^{\mu}$ makes it $\mu_p$; a peripheral conjugator would have the same effect because that commutator coordinate is central. This proves the equality in every $\mathbb{Z}_p$ and hence in $\widehat{\mathbb{Z}}$.

The matrices $\rho_p(\beta)$ and $A_p$ commute, since their outer classes commute. On $M \otimes_{\widehat{\mathbb{Z}}} \mathbb{Q}_p$, the matrix $A_p$ has the two distinct eigenvalues $1, -1$. Each eigenspace has dimension one, since its determinant is the negative multiplier $-1$. The fixed-space eigenvalue of $\rho_p(\beta)$ is $\sigma$, and its determinant is $\mu_p$. Its other eigenvalue is consequently $\sigma\mu_p$. This proves (D.63c) over $\mathbb{Q}_p$. The two half-coefficients are integral as explained in the statement, and $\mathbb{Z}_p^2 \to \mathbb{Q}_p^2$ is injective. The same matrix equality therefore holds over $\mathbb{Z}_p$ for every $p$, proving the displayed full formula and (D.63d).

More explicitly, extend $[g]$ to a basis of $M$. In that basis there is an $a \in \widehat{\mathbb{Z}}$ with

$$A = \begin{pmatrix} 1 & a \\ 0 & -1 \end{pmatrix}, \qquad \rho(\beta) = \sigma \begin{pmatrix} 1 & a(1-\mu)/2 \\ 0 & \mu \end{pmatrix}. \tag{D.63l}$$

This is the same formula on the actual integral lattice, including its possibly nonsplit dyadic eigenspaces. It does not replace that lattice by a direct sum of the two integral eigenlattices.

The formula proves that the kernel of $(\sigma, \lambda)$ is exactly $I_b$. Its image is the stated compact group $\Xi_b$, so (D.63e) is an exact sequence of profinite groups. In particular all commutators in $Z_b$ lie in $I_b$. On $Z_b^1 = \ker \lambda$ it restricts to the exact sequence

$$1 \longrightarrow I_b \longrightarrow Z_b^1 \xrightarrow{\sigma} \sigma(Z_b^1) \longrightarrow 1, \qquad \sigma(Z_b^1) \le \{1, -1\}. \tag{D.63m}$$

No larger image is inferred.

For $\beta \in I_b$, the unique normalized representative (D.63i) has $\mu = 1$ and $i = 0$. Its extension fixes $v$ and $c$, so its restriction belongs to $J_b$. Conversely, a member $\psi \in J_b$ extends by $v \mapsto v$ to $E$, and its outer class centralizes $b$. It is already normalized with $\mu = 1, i = 0$. Equations (D.63c) and (D.63k) give $\psi_* = I$, so that outer class belongs to $I_b$. These two maps are inverse. Composition of two representatives fixing both $c$ and $v$ again fixes both, so they are group homomorphisms. They are homeomorphisms because $J_b$ is a closed subgroup of the profinite group $\mathrm{Aut}(F)$ and the outer projection is a continuous bijection onto $I_b$. Equivalently, an inner automorphism in $J_b$ would be conjugation by $c^a$ and would require $c^a = \alpha(c^a) = c^{-a}$, giving $a = 0$. This also verifies directly that no inner ambiguity remains in (D.63f).

D.55 proves faithfulness of every primary homology representation on every finite subgroup of $\Omega$. A nonidentity torsion element of $I_b$ would generate a finite subgroup killed by that representation. This proves that $I_b$ is torsion-free. It does not remove infinite-order elements of the actual group (D.63f).

*Finite centralizers.* For $\beta$ in a finite $K \leq Z_b$, its dyadic multiplier is a finite-order element of $\mathbb{Z}_2^\times$, hence is 1 or $-1$. Indeed $\mathbb{Z}_2^\times = \{1, -1\} \times (1 + 4\mathbb{Z}_2)$, and the latter factor is torsion-free by the injective 2-adic logarithm. Formula (D.63c) at two therefore puts its matrix in $\{I, -I, A_2, -A_2\}$. These four matrices are distinct and form a Klein four group: $A_2^2 = I$ and $\det A_2 = -1$, which excludes $A_2 = \pm I$. D.55 makes the displayed map injective on $K$. This proves (D.63g) and all the stated finite bounds. The proof leaves exactly the kernel in (D.63f) and the actual image $\Xi_b$ to be determined. □

**D.64. [Proved] Standard negative torus centralizers and their actual kernels.** Let $b_0, b_1$ be the two standard negative outer involutions on $F_0 = \widehat{F}(X, Y)$:

| | $X$ | $Y$ |
|---|---|---|
| $\alpha_0$ | $YXY^{-1}$ | $Y^{-1}$ |
| $\alpha_1$ | $Y$ | $X$. |

(D.64a)

Both have the named cusp $c_0 = [X, Y]$ and $\alpha_j(c_0) = c_0^{-1}$. For $b_0$, put $\zeta_0(X) = X^{-1}$, $\zeta_0(Y) = Y$. Its full peripheral outer centralizer and normalizer are exactly

$$C_\Omega(b_0) = N_\Omega(\langle b_0 \rangle) = \langle b_0, [\zeta_0] \rangle \cong C_2^2. \tag{D.64b}$$

Thus every fibre of the actual orbit map $\theta \mapsto \theta b_0 \theta^{-1}$ is precisely a right coset of this specified four-element group. Conjugation by $\theta$ gives the exact corresponding centralizer on that orbit.

For both $j = 0, 1$, the IA kernel $I_{b_j}$ of D.63 is trivial. Consequently the full centralizer of $b_1$ has the faithful character embedding

$$C_\Omega(b_1) \hookrightarrow \{1, -1\} \times \widehat{\mathbb{Z}}^\times, \qquad \beta \longmapsto (\sigma, \lambda). \tag{D.64c}$$

In particular it is abelian. For the actual automorphism $\zeta_1(X) = X^{-1}$, $\zeta_1(Y) = Y^{-1}$,

$$\begin{aligned} C_\Omega(b_1) \cap \ker \lambda &= \langle [\zeta_1] \rangle \cong C_2, \\ \{\beta \in C_\Omega(b_1) : \lambda(\beta) \in \{1, -1\}\} &= \langle b_1, [\zeta_1] \rangle \cong C_2^2. \end{aligned} \tag{D.64d}$$

This is also the entire torsion subgroup of this abelian centralizer. The remaining multiplier calculation for (D.64c) is supplied by D.65 below.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The exact standard extensions.* D.61 gives the mutually inverse full presentation maps

$$\begin{aligned} E_0 &= (\widehat{\mathbb{Z}}\langle X\rangle \times C_2\langle w\rangle) \amalg C_2\langle v\rangle, \\ Y &= vw, \qquad w = Y^{-1}v, \\ E_1 &= C_2\langle v\rangle \amalg \widehat{\mathbb{Z}}\langle z\rangle, \\ X &= zv, \qquad Y = vz, \qquad z = Xv. \end{aligned} \tag{D.64e}$$

The parity homomorphisms have $\epsilon(X) = 0, \epsilon(v) = \epsilon(w) = 1$ in the first row and $\epsilon(v) = \epsilon(z) = 1$ in the second. Their kernels are the actual $F_0$ in (D.64a), and conjugation by $v$ gives $\alpha_j$. The full free-factor intersection theorem used in D.35 gives

$$C_{E_0}(v) = \langle v\rangle, \qquad C_{E_0}(w) = \langle X\rangle \times \langle w\rangle, \qquad C_{E_1}(v) = \langle v\rangle. \tag{D.64f}$$

For example a centralizer element outside the specified factor would put the nontrivial involution in the intersection of that factor with its conjugate. The intersection theorem excludes that possibility; the indicated factor is already abelian where needed [36, Theorem 5 of the inspected author version]. In particular $C_{F_0}(v) = 1$ in both rows.

The actual fusion transporters are

$$g_0 = X, \qquad g_1 = z^2 = XY, \qquad g_j v g_j^{-1} = c_0 v. \tag{D.64g}$$

The identities follow by substitution in (D.64e). Both have primitive homology, and $c_0 = [g_j, v]$ in their respective extensions.

*Vanishing of the two actual IA kernels.* Use D.63f to represent an IA centralizer element by its unique actual automorphism fixing $c_0$ and commuting with $\alpha_j$. Its extension fixes $v$. The exact multiplier-one transporter formula in D.62, together with $C_{F_0}(v) = 1$, makes it fix $g_j$ as well.

For $E_0$, D.62 says that $w$ is the unique outside involution centralizing $g_0 = X$. The extension must fix it. It fixes all three generators $X, w, v$ of $E_0$, and is therefore the identity. For $E_1$, D.62 says that $z$ is the unique outside square root of $g_1 = z^2$. The extension must fix it, and hence fixes the two generators $z, v$. This proves $I_{b_0} = I_{b_1} = 1$ using actual extension generators and their uniqueness, rather than generation inferred from homology. The faithful map (D.64c) and abelianness follow from D.63e.

*All multipliers in the two-class standard model.* Let $\beta \in C_\Omega(b_0)$. Normalize its actual extension by D.63:

$$\Psi(c_0) = c_0^{\mu}, \qquad \Psi(v) = c_0^{i} v, \qquad i \in \{0, 1\}.$$

There is no fixed-subgroup ambiguity in its transporter image, since the two groups $C_i$ in D.62 are trivial by (D.64f). Thus, with $a = \Psi(X)$,

$$a = \begin{cases} c_0^{(\mu-1)/2} X, & i = 0, \\ c_0^{(\mu+1)/2} X^{-1}, & i = 1. \end{cases} \tag{D.64h}$$

The involution $t = \Psi(w)$ belongs to the other full $F_0$-section class: $\Psi$ permutes those two classes, and already preserves the cusp class by its normalized image of $v$. Therefore $t = hwh^{-1}$ for some actual $h \in F_0$. Since $a$ centralizes $t$, (D.64f) gives

$$a \in h(C_{E_0}(w) \cap F_0)h^{-1} = h\langle X\rangle h^{-1}.$$

Write $a = hX^u h^{-1}$. Its primitive homology in D.62 is $(-1)^i[X]$, so $u = (-1)^i$ in the full $\widehat{\mathbb{Z}}$. This is the exact conjugacy

$$a \sim_{F_0} X^{(-1)^i}. \tag{D.64i}$$

For every integer $n \geq 2$, use the explicit finite dihedral quotient

$$\begin{aligned} q_n : E_0 &\longrightarrow D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ srs = r^{-1} \rangle, \\ X &\longmapsto r, \qquad v \longmapsto s, \qquad w \longmapsto 1. \end{aligned} \tag{D.64j}$$

It respects the defining relations in (D.64e) and is surjective. The exact peripheral image is $q_n(c_0) = r^2$. Both lines of (D.64h) consequently give $q_n(a) = r^{\mu}$. By (D.64i), this rotation is conjugate to $r^{(-1)^i}$. The conjugacy class of either $r$ or $r^{-1}$ in this finite group is $\{r, r^{-1}\}$. Hence

$$\mu \bmod n \in \{1, -1\} \quad \text{for every } n \geq 2. \tag{D.64k}$$

These simultaneous congruences force one global sign $\mu = 1$ or $\mu = -1$ in $\widehat{\mathbb{Z}}$. For a direct compactness proof, the sets $\{\varepsilon \in \{1, -1\} : \mu \equiv \varepsilon \pmod{n}\}$ are nonempty and have the finite intersection property, since the condition for a least common multiple implies every specified finite set of conditions. Their intersection supplies the one sign at all moduli.

Together with $I_{b_0} = 1$, D.63 now bounds its full centralizer by four elements. All four occur through actual extension maps. The map fixing $v, w$ and sending $X$ to $X^{-1}$ is an order-two automorphism of $E_0$. On the kernel it is precisely $\zeta_0$ in the statement. It commutes with $\mathrm{Inn}(v)$ and

$$\zeta_0(c_0) = X^{-1} c_0^{-1} X,$$

so it is peripheral of multiplier $-1$. The two commuting involutions have respective homology matrices $\mathrm{diag}(1, -1)$ and $\mathrm{diag}(-1, 1)$. They give four distinct outer classes. This proves (D.64b). A normalizer of an order-two subgroup fixes its sole nonidentity element and is its centralizer. Finally equality of the two orbit images of $\theta, \theta'$ is exactly the statement that $\theta^{-1}\theta'$ belongs to this computed centralizer, proving the stated actual orbit fibres.

*The exact multiplier-one conclusion in the one-class model.* The automorphisms $\alpha_1$ and $\zeta_1$ commute on both free generators and have order two. Moreover

$$\zeta_1(c_0) = (XY)^{-1} c_0 (XY), \qquad (\zeta_1)_* = -I. \tag{D.64l}$$

Thus $\zeta_1$ has multiplier one and fixed-line sign $-1$, while $b_1$ has multiplier $-1$ and fixed-line sign 1. They provide the four indicated elements. The injective characters in (D.64c) allow at most two elements with multiplier one. Multiplication by $b_1$ gives exactly the two with multiplier $-1$. This proves (D.64d). Every torsion element of this abelian centralizer generates a finite subgroup; together with the displayed four-element subgroup it still generates a finite group. D.63g bounds that group by order four. Consequently the displayed four elements are exactly all its torsion. This argument establishes the stated character embedding and its torsion subgroup. D.65 below computes its remaining multiplier image through the specified cone-filling quotients.

The program `code/part_c/standard_negative_centralizers_d64.py` checks the exact free-group identities in (D.64a), (D.64g), and (D.64l), and checks both dihedral transporter formulas over a finite range. The global conclusion (D.64k) is the proved all-modulus statement above. □

**D.65. [Proved] A uniform square obstruction and the remaining standard centralizer.** For each integer $n \geq 2$, put

$$G_n = \langle z, v \mid v^2 = 1,\ [z^2, v]^n = 1 \rangle, \qquad g = z^2, \quad c = [g, v]. \tag{D.65a}$$

These are discrete groups; $\widehat{G}_n$ denotes their full profinite completions. For an integer $r$ with $\gcd(2r + 1, n) = 1$, the exact square criterion is

$$c^r g \text{ is a square in } \widehat{G}_n \quad \Longleftrightarrow \quad r \equiv 0 \text{ or } -1 \pmod{n}. \tag{D.65b}$$

For either allowed residue the square root already lies in the displayed discrete group.

Consequently the full peripheral outer centralizer of the standard one-section negative torus involution $b_1 = [\alpha_1]$, $\alpha_1(X) = Y, \alpha_1(Y) = X$, is

$$C_\Omega(b_1) = N_\Omega(\langle b_1\rangle) = \langle b_1, [\zeta_1]\rangle \cong C_2^2, \qquad \zeta_1(X) = X^{-1}, \quad \zeta_1(Y) = Y^{-1}. \tag{D.65c}$$

In particular all its multiplier values are the two global units $1, -1$. Together with D.64 this computes both standard negative centralizers, and every fibre of each actual standard conjugacy-orbit map has exactly four elements, with the explicit right-coset description. It does not assert that the two standard orbits exhaust all negative involutions.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. Two exact discrete graph-of-groups presentations.* Let

$$\begin{aligned} D_n &= \langle c, v \mid c^n = v^2 = 1,\ vcv = c^{-1}\rangle, \qquad w = cv, \\ H_n &= \langle D_n, g \mid gvg^{-1} = w\rangle. \end{aligned} \tag{D.65d}$$

The first group is the finite dihedral group of order $2n$. Its two specified order-two subgroups are $\langle v\rangle, \langle w\rangle$. The second line is their HNN extension, with the actual edge map $v \mapsto w$. The HNN normal form embeds $D_n$ in $H_n$. There is an actual exponent character

$$\ell : H_n \longrightarrow \mathbb{Z}, \qquad \ell(g) = 1, \quad \ell(c) = \ell(v) = 0. \tag{D.65e}$$

In particular $\langle g\rangle$ is infinite cyclic.

Eliminating $c$ gives the mutually inverse presentation maps

$$\begin{aligned} H_n &\longrightarrow \langle G, V \mid V^2 = 1,\ [G,V]^n = 1\rangle, \quad (g, v, c) \longmapsto (G, V, [G, V]), \\ (G, V) &\longmapsto (g, v). \end{aligned} \tag{D.65f}$$

Indeed $gvg^{-1} = cv$ gives $[g, v] = c$. Conversely $V[G, V]V = [G, V]^{-1}$ follows from $V^2 = 1$, so the defining dihedral relation is retained. Now form the discrete amalgam with its stated injective edge maps:

$$P_n = H_n *_{\langle g\rangle = \langle z^2\rangle} \langle z\rangle. \tag{D.65g}$$

Its inverse presentation maps with (D.65a) send

$$(g, v, c, z) \longmapsto (z^2, v, [z^2, v], z), \qquad (z, v) \longmapsto (z, v). \tag{D.65h}$$

All defining relators have just been verified in (D.65f), and each composite fixes its generators. Thus $P_n = G_n$ through these explicit maps. In particular both amalgam factors embed, and $c$ has exactly order $n$. The following global character will be used after completion:

$$\vartheta : G_n \longrightarrow \mathbb{Z}, \qquad \vartheta(z) = 1, \quad \vartheta(v) = 0. \tag{D.65i}$$

Its restriction to $H_n$ is $2\ell$.

*Length-one conjugacy and its full discrete centralizer.* In the Bass–Serre tree of (D.65d), the vertex group is $D_n$ and the edge groups are the specified order-two subgroups. Both $g$ and $a_r = c^r g$ have cyclically reduced stable letter length one. Their successive positive and negative powers are reduced, so they translate their axes by one edge, and both axes contain the vertex stabilized by $D_n$.

Suppose $hgh^{-1} = a_r$ in $H_n$. The element $h$ carries the axis of $g$ to the axis of $a_r$. Since $a_r$ translates by one edge, some power of $a_r$ carries the vertex $hD_n$ to $D_n$. Multiplying $h$ by that power therefore gives a conjugator $d \in D_n$ with the same equation. Rearrange it as

$$gd^{-1}g^{-1} = d^{-1}c^r \in D_n. \tag{D.65j}$$

Britton's normal form forces $d^{-1} \in \langle v \rangle$. There are exactly two possibilities. For $d = 1$ the equation gives $r = 0 \bmod n$. For $d = v$ the exact relation is

$$vgv = c^{-1}g, \tag{D.65k}$$

and gives $r = -1 \bmod n$. Conversely those two displayed conjugators work. Thus

$$c^r g \sim_{H_n} g \quad \Longleftrightarrow \quad r \equiv 0, -1 \pmod{n}. \tag{D.65l}$$

Assume now that $r$ is neither residue. The character (D.65e) shows that $a_r$ cannot be $H_n$-conjugate to any element of the amalgam edge group $\langle g \rangle$: such a conjugacy would have to be to $g$ itself, which (D.65l) excludes. In the tree of (D.65g), $a_r$ fixes the $H_n$ vertex and fixes no adjacent edge. If it fixed another vertex, the connecting path would give an adjacent fixed edge. Hence this is its unique fixed vertex. Every element centralizing $a_r$ must preserve that vertex, so

$$C_{G_n}(a_r) = C_{H_n}(a_r). \tag{D.65m}$$

This is an argument in the discrete tree; it assumes no fixed-path assertion for a profinite tree.

For $h \in C_{H_n}(a_r)$, commuting with a nontrivial translation preserves and orients its axis. Again its vertices form a single orbit under the length-one translation $a_r$. Thus $h = a_r^m k$ for an integer $m$ and a $k \in D_n$ centralizing $a_r$. The same Britton reduction as in (D.65j) forces $k \in \langle v \rangle$. Its nonidentity possibility is excluded by the exact dihedral calculation

$$a_r v a_r^{-1} = c^{2r+1} v \neq v, \tag{D.65n}$$

where the inequality uses $\gcd(2r + 1, n) = 1$ and $n \geq 2$. We have proved the complete discrete equality

$$C_{G_n}(a_r) = \langle a_r \rangle. \tag{D.65o}$$

*The actual closed surface cover.* The homomorphism $\epsilon : G_n \to C_2$ sending both $z$ and $v$ to the nontrivial element has kernel

$$\Pi_n = \langle X, Y \mid [X, Y]^n = 1 \rangle. \tag{D.65p}$$

Here the exact isomorphism $G_n \cong \Pi_n \rtimes \langle v \rangle$ has

$$X = zv, \quad Y = vz, \qquad z = Xv, \qquad vXv = Y, \quad vYv = X. \tag{D.65q}$$

Substitution gives $[z^2, v] = [X, Y]$. Conversely swapping $X, Y$ preserves the relation in (D.65p), because it inverts their commutator. The indicated maps between the presentation and this semidirect product therefore invert each other, proving both the kernel and its inclusion.

Let $Q_n$ be the finite Heisenberg group

$$Q_n = (\mathbb{Z}/n)^3, \qquad (a, b, t)(a', b', t') = (a + a', b + b', t + t' + ab').$$

The map

$$\Pi_n \longrightarrow Q_n, \qquad X \longmapsto (1, 0, 0), \quad Y \longmapsto (0, 1, 0), \quad [X, Y] \longmapsto (0, 0, 1) \tag{D.65r}$$

is onto and has a commutator of exactly order $n$. To identify its kernel, start with the actual degree-$n^3$ regular cover of a once-punctured torus given by the same homomorphism from its free group. The boundary permutation is multiplication by that central element of order $n$. It has exactly $n^2$ cycles, each of length $n$, so the covering surface has $n^2$ boundary components, each mapping with degree $n$. Fill each by a disk. Van Kampen kills exactly the normal closure of the corresponding conjugates of $[X, Y]^n$. These are exactly the Reidemeister–Schreier relators

for the kernel of (D.65r): the original normal closure is generated by its conjugates over the $n^3$ cosets, and the $n$ cosets in a boundary cycle give conjugates of the same boundary relator. Thus the resulting closed surface group is precisely that kernel, with the inclusion supplied by the covering construction. Its Euler characteristic and genus are

$$\chi = -n^3 + n^2, \qquad h_n = 1 + \frac{n^2(n-1)}{2}. \tag{D.65s}$$

It has index $2n^3$ in $G_n$. This proves directly that $G_n$ is virtually a closed surface group.

*The full completion bridge.* Martino's theorem states that every group having a surface subgroup of finite index is conjugacy separable [49, Theorem 3.7 of the inspected author version]. Every finite-index subgroup of $G_n$ has the same property, by intersection with the surface group just constructed. Consequently $G_n$ is hereditarily conjugacy separable, and in particular residually finite.

We give the precise centralizer consequence and its maps. For any such group $L$ and $a \in L$,

$$C_{\widehat{L}}(a) = \overline{C_L(a)} \quad \text{inside } \widehat{L}. \tag{D.65t}$$

Let $x$ belong to the left-hand side. For a finite-index normal subgroup $N \lhd L$, choose $h \in L$ with $x = h\widehat{n}$, $\widehat{n} \in \overline{N}$. The commuting equation gives $h^{-1}ah = \widehat{n}a\widehat{n}^{-1}$. Its image modulo $N$ is therefore the image of $a$, so the actual element $h^{-1}ah$ belongs to $L_N = N\langle a\rangle$. Both actual elements lie in the finite-index subgroup $L_N$, and they are conjugate in its full completion, since $\overline{N} \subseteq \widehat{L_N}$. Conjugacy separability of $L_N$ gives a conjugator there. Write it as $na^k$, with $n \in N$. It follows that $h^{-1}ah = nan^{-1}$, so $hn \in C_L(a)$ and $x \in C_L(a)\overline{N}$. Intersecting these neighborhoods over $N$ puts $x$ in the right-hand side of (D.65t). The reverse inclusion follows from the closed commuting equation. Finite-index subgroups carry their full induced profinite topologies, which is the only completion inclusion used in this proof.

Apply (D.65t) to (D.65o). The continuous extension of (D.65i) satisfies $\widehat{\vartheta}(a_r) = 2$. Hence the map

$$\widehat{\mathbb{Z}} \longrightarrow \overline{\langle a_r\rangle} \subseteq \widehat{G}_n, \qquad t \longmapsto a_r^t \tag{D.65u}$$

is injective: its composite with $\widehat{\vartheta}$ is multiplication by two on $\widehat{\mathbb{Z}}$. It is onto the indicated procyclic closure by definition. Thus (D.65t) is the exact full procyclic centralizer $\langle a_r\rangle \cong \widehat{\mathbb{Z}}$, with no missing prime coordinates. If $q^2 = a_r$ in $\widehat{G}_n$, then $q$ centralizes $a_r$ and equals $a_r^t$. The equation would give $2t = 1$ in $\widehat{\mathbb{Z}}$, which is impossible modulo two. This excludes a square for the two disallowed residue conditions above. For $r = 0$ the element is $g = z^2$. For $r = -1$, (D.65k) gives $c^{-1}g = vgv = (vzv)^2$. Since $c$ has order $n$, these formulas prove both directions of (D.65b).

*Full units and both original reflection labels.* Return to the full standard extension $E_1 = C_2\langle v\rangle \amalg \widehat{\mathbb{Z}}\langle z\rangle$ in D.64, with $g = z^2$ and $c_0 = [g, v]$. For each $n$, the generator maps give an actual continuous epimorphism

$$q_n : E_1 \longrightarrow \widehat{G}_n, \qquad z \longmapsto z, \quad v \longmapsto v. \tag{D.65v}$$

Their dense discrete image generates $G_n$, so compactness makes the map onto its full completion. Let $\beta \in C_\Omega(b_1)$ have the unique normalized extension from D.63. D.64 gives $C_{F_0}(v_i) = 1$, so its image $a = \Psi(g) = \Psi(z)^2$ is exactly

$$a = \begin{cases} c_0^{(\mu-1)/2} g, & i = 0, \\ c_0^{(\mu+1)/2} g^{-1}, & i = 1, \end{cases} \qquad \mu = \lambda(\beta) \in \widehat{\mathbb{Z}}^\times. \tag{D.65w}$$

First take $i = 0$ and put $r = (\mu - 1)/2 \in \widehat{\mathbb{Z}}$. Choose an integer representative $r_n$ of $r$ mod $n$. Its image in (D.65v) is $c^{r_n}g$ and is a square. The integer $2r_n + 1$ is prime to $n$, because it is the reduction of the full unit $\mu$. Criterion (D.65b) gives $r_n \equiv 0, -1 \pmod n$. Equivalently,

$$\mu \equiv 1 \text{ or } -1 \pmod{2n} \quad \text{for every } n \geq 2. \tag{D.65x}$$

The finite-intersection argument of D.64k now forces a single global sign $\mu = 1$ or $\mu = -1$.

For $i = 1$, inversion and the specified conjugation give the exact comparison

$$g^{-1}a^{-1}g = c_0^{-(\mu+1)/2}g = c_0^{((-\mu)-1)/2}g. \tag{D.65y}$$

This element is the square of $g^{-1}\Psi(z)^{-1}g$. It is therefore precisely the already treated first-label expression for the full unit $-\mu$, and again $\mu \in \{1, -1\}$. No reflection label or named cusp has been rescaled.

D.64 proves that the character pair is injective for $b_1$, and explicitly realizes all four pairs with these two multipliers. This gives (D.65c). The normalizer and actual orbit-fibre conclusions follow exactly as in D.64. The proof restricts the standard centralizers; the separate existence of a conjugator for an arbitrary negative involution remains the full equation in D.62.

The program `code/part_c/standard_oneclass_square_d65.py` checks the defining word maps, the finite dihedral identities in the axis calculation, and the exact Heisenberg covering data. The square obstruction in the full completion uses the proved discrete centralizer and the specified hereditary-conjugacy-separability bridge, beyond those finite checks. □

**D.66. [Proved] The exact norm in a scalar cyclic torus cover.** Let $a$ be an actual element of order $d = 3$ or $4$ in the full once-punctured-torus loop-outer group, and put $p = 3$ or $2$, respectively. At every scalar coordinate retained by D.54–D.55, write

$$A_q(a) = uI, \qquad \begin{cases} u = \lambda_q^2, & \lambda_q \neq 1, \quad d = 3, \\ u^2 = -1, & \lambda_q = -1, \quad d = 4. \end{cases} \tag{D.66a}$$

Thus $q \equiv 1 \pmod{d}$, and the reduction $\bar{u} \in \mathbb{F}_q^\times$ has exact order $d$. For every nonzero character $\chi : F = \widehat{F}(x, y) \to C_q$, its kernel $N$ is the full completion of the actual degree-$q$ cover of type $(1, q)$. The complete primary homology comparison is

$$H_1(N, \mathbb{Z}_p) \cong (\mathbb{Z}_p^2, A_p(a)) \ \oplus \ \mathbb{Z}_p[C_d]^{(q-1)/d}. \tag{D.66b}$$

Here the action is the one induced by a labelled cusp-normalizer element above $a$; the two displayed summands are the actual deck-average handle summand and boundary summand.

Label the boundary classes by $b_i$, $i \in \mathbb{F}_q$, with the distinguished original cusp labelled 0. The boundary module has the exact relation $\sum_i b_i = 0$, and the action is $b_i \mapsto b_{\bar{u}i}$. If $R$ contains one representative of each nonzero $\langle \bar{u} \rangle$-orbit, then

$$w = -\sum_{r \in R} b_r, \qquad (1 + T + \cdots + T^{d-1})w = b_0, \tag{D.66c}$$

where $T$ denotes this action. In particular the distinguished unit cusp-power class is an integral norm, at every primary precision. More fully, for $M = H_1(N, \mathbb{Z}_p)$,

$$H^2(C_d, M) = 0, \qquad H^1(C_d, M) \cong C_p. \tag{D.66d}$$

The actual cyclic extension over this cover, after quotienting its kernel to $M$, has the explicit splitting given below. These statements identify exactly why the invariant-character norm contradiction of D.54–D.55 does not extend to the scalar coordinates (D.66a). They neither construct an action having such a coordinate nor decide the remaining scalar multiplier or conjugacy questions.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The actual extension and its finite quotient.* Choose an actual representative $\alpha$ of order $d$, using D.27 for $d = 3$ and D.46 for $d = 4$, and set

$$E = F \rtimes_\alpha \langle s \rangle, \quad c = [x, y], \quad L = \langle c \rangle.$$

The chosen $s$ has quotient label $a$. Choose a cusp-normalizer element $v$ with the same quotient label. The exact relations are

$$vcv^{-1} = c^\lambda, \qquad v^d = c^\kappa, \qquad \lambda_p = 1, \quad \kappa_p \in \mathbb{Z}_p^\times. \tag{D.66e}$$

For $d = 3$ these are D.48. For $d = 4$, the element $v^2$ lies over the positive involution $a^2$. Its square exponent is a two-adic unit by D.55's full cusp-normalizer calculation. Also $\lambda_2 = 1$: the order-four matrix on $\mathbb{Z}_2^2$ has square $-I$, characteristic polynomial $X^2 + 1$, and determinant one. The peripheral multiplier equals that determinant.

Let $\bar{u}$ be as in (D.66a), and write $C_q$ additively. Both conjugation by $s$ and by $v$ satisfy

$$\chi(vhv^{-1}) = \bar{u}\,\chi(h) \qquad (h \in F). \tag{D.66f}$$

Indeed they induce the same homology map, and every character factors through the actual abelianization of $F$. Thus $N = \ker \chi$ is normal in $E$. Since $c \in N$, the class of $v$ in $E/N$ has exact order $d$. Choose a discrete word $f$ with $\chi(f) = 1$. The two mutually inverse finite quotient maps are

$$\begin{aligned} C_q \rtimes_{\bar{u}} C_d &\longrightarrow E/N, \qquad (i, j) \longmapsto f^i v^j N, \\ E/N &\longrightarrow C_q \rtimes_{\bar{u}} C_d, \quad hv^j N \longmapsto (\chi(h), j), \quad 0 \le j < d. \end{aligned} \tag{D.66g}$$

The multiplication on the left is $(i, j)(k, l) = (i + \bar{u}^j k, j + l)$; the relation $v^d \in N$ and (D.66f) verify both maps and their inverse identities.

Put $E_N = N\langle v \rangle$, the inverse image of $\{0\} \rtimes C_d$ in this exact quotient. It has index $q$ in $E$, with kernel $N$ and quotient $C_d$. For completeness it has an actual order-$d$ section. Write the image of the original $s$ as $(j, 1)$ and choose the integer residue

$$k = \frac{j}{\bar{u} - 1} \quad \text{in } \mathbb{F}_q.$$

The denominator is nonzero. Conjugation by $(k, 0)$ sends $(j, 1)$ to $(j + (1 - \bar{u})k, 1) = (0, 1)$. Consequently $s_0 = f^k s f^{-k}$ belongs to $E_N$, has exact order $d$, and gives $E_N = N \rtimes \langle s_0 \rangle$ with the specified quotient label. No character extension $E \to C_q$ with trivial action is being substituted for (D.66g).

*The actual cover and the boundary labels.* The restriction of $\chi$ to the dense discrete free group is onto. Its punctured-torus cover has degree $q$ and trivial cusp monodromy, hence $q$ degree-one boundary lifts. Euler characteristic $-q$ gives genus one. The finite-index completion map identifies its full fundamental-group completion with precisely $N$. The chosen boundary representatives are

$$c_i = f^i c f^{-i} \in N \qquad (0 \le i < q). \tag{D.66h}$$

They have distinct cover labels. In homology their classes $b_i$ have exactly the single relation $\sum_i b_i = 0$. For $j$ the representative of $\bar{u}i$ modulo $q$, put $n_i = (vf^i v^{-1}) f^{-j} \in N$, by (D.66f). Then the actual conjugacy equation is

$$vc_i v^{-1} = n_i c_j^\lambda n_i^{-1}. \tag{D.66i}$$

In $p$-primary homology $\lambda_p = 1$, so its action is exactly $Tb_i = b_{\bar{u}i}$. The label zero is fixed, and every nonzero label has orbit length $d$.

*Deck averaging and both inverse boundary-module maps.* Let $D = F/N = C_q$ act on $M = H_1(N, \mathbb{Z}_p)$ by conjugation of deck lifts, and set

$$e = \frac{1}{q} \sum_{h \in D} h. \tag{D.66j}$$

The prime $q$ differs from $p$. Although $T$ need not commute with individual deck transformations, it permutes them by $h \mapsto h^{\bar{u}}$; hence it commutes with $e$. Filling all $q$ boundaries gives the actual compact torus cover associated to the lattice $\Lambda = \ker(\mathbb{Z}^2 \to C_q)$, with covering map $\mathbb{R}^2/\Lambda \to \mathbb{R}^2/\mathbb{Z}^2$. Its deck maps are translations, so they act trivially on the filled homology. The boundary module is

$$B_\partial = \mathbb{Z}_p\{b_i : i \in \mathbb{F}_q\}/\mathbb{Z}_p\Big(\sum_i b_i\Big). \tag{D.66k}$$

It has zero deck average, since $q$ is a unit. The boundary sequence therefore gives the exact identifications

$$(1-e)M = B_\partial, \qquad eM \xrightarrow{\cong} H_1(\mathbb{R}^2/\Lambda, \mathbb{Z}_p) \xrightarrow{\cong} H_1(\mathbb{R}^2/\mathbb{Z}^2, \mathbb{Z}_p). \tag{D.66l}$$

The second map is induced by the index-$q$ lattice inclusion; its determinant is a $p$-adic unit. Both maps intertwine the induced homology actions, so $T|_{eM} = A_p(a)$ under these specified identifications.

Write $m = (q-1)/d$. With $R$ as in the statement, define

$$\begin{aligned} \Phi : \bigoplus_{r\in R} \mathbb{Z}_p[C_d]e_r \longrightarrow B_\partial, &\qquad T^j e_r \longmapsto b_{\bar{u}^j r}, \\ \Psi : B_\partial \longrightarrow \bigoplus_{r\in R} \mathbb{Z}_p[C_d]e_r, &\quad b_{\bar{u}^j r} \longmapsto T^j e_r, \\ &\quad b_0 \longmapsto -\sum_{r\in R}\sum_{j=0}^{d-1} T^j e_r. \end{aligned} \tag{D.66m}$$

The displayed image of the total boundary relation is zero. It also makes $\Psi$ equivariant at the fixed label zero. Every nonzero label has a unique expression $\bar{u}^j r$, so these maps are mutually inverse on all their generators. This proves the free cyclic-module assertion in (D.66b). Applying $\Phi$ to the negative sum of the $e_r$ proves (D.66c) integrally.

*The exact cyclic cohomology and abelianized extension.* On the handle summand $H = eM$, D.48 and D.55 give respectively

| $d$ | $p$ | identity on $H$ | $\det(T-I)\vert_H$ |
|---|---|---|---|
| 3 | 3 | $T^2+T+I=0$ | 3 |
| 4 | 2 | $T^2=-I$ | 2 |

(D.66n)

Thus the cyclic norm $\mathcal{N} = 1 + T + \cdots + T^{d-1}$ is zero on $H$, while $H^T = 0$. The cokernel of $T - I$ on $H \cong \mathbb{Z}_p^2$ is $C_p$: its determinant has valuation one, so its two invariant factors have valuations zero and one. On each free summand $\mathbb{Z}_p[C_d]$, the invariants are exactly the multiples of the sum of its $d$ basis elements, which are exactly its norm image. The kernel of the norm is the augmentation ideal, the image of $T - I$. The periodic cyclic cochain complex consequently gives

$$M^T/\mathcal{N}M = 0, \qquad \ker(\mathcal{N})/(T-I)M \cong C_p,$$

which is precisely (D.66d).

Here is an explicit splitting retaining the cusp exponent. Let $K = \ker(N \to M)$ and $P = E_N/K$. This is an actual quotient, since $K$ is characteristic in $N$. Write the kernel $M$ additively, and still denote the image of $v$ by $v$. Its relations are

$$vmv^{-1} = Tm, \qquad v^d = \kappa_p b_0. \tag{D.66o}$$

They are complete: every element of $P$ has a unique normal form $mv^j$, $0 \le j < d$, and these relations give its multiplication. By (D.66c), the element

$$t = (-\kappa_p w)v$$

has $t^d = 0$ in the additive kernel, and has exact order $d$ because of its quotient label. The inverse isomorphisms are

$$\begin{aligned} M \rtimes_T C_d \longrightarrow P, &\qquad (m, j) \longmapsto m t^j, \\ P \longrightarrow M \rtimes_T C_d, &\qquad m \longmapsto (m, 0), \\ &\qquad v \longmapsto (\kappa_p w, 1). \end{aligned} \tag{D.66p}$$

The norm identity verifies the image of $v^d$, and the two composites fix the indicated kernel and generator. Reducing these maps modulo every $p^n$ gives the same splitting at that exact precision. All module solutions of $\mathcal{N}z = -\kappa_p b_0$ form $-\kappa_p w + \ker \mathcal{N}$.

The actual section $s_0$ constructed above has one such vector relative to $v$, but this calculation does not assert that every vector in that coset lifts to a finite-order element before the quotient by $K$. In the invariant-character proof all boundary labels were fixed and the primary norm vanished modulo $p$. Here the explicit free boundary summands (D.66m) give the precise integral preimage for the same distinguished cusp class. Thus there is no remaining norm obstruction from this cover module, without any assertion that a scalar-coordinate full action exists.

The executable certificate `code/part_c/torus_scalar_cover_norm_d66.py` checks the finite quotient conjugation, the two inverse boundary-module maps, the deck conjugacy and averaging, the integral norm preimage, and the handle matrices for the first admissible cubic and quartic primes. The arbitrary-prime assertions follow from the explicit maps above, not from this finite enumeration. □

**D.67. [Proved] The standard negative normalizers and their finite overgroups.** Retain the two standard negative involutions $b_0, b_1$ and the actual automorphisms $\alpha_j, \zeta_j$ of D.64. Put

$$R(X) = Y, \quad R(Y) = X^{-1}, \quad r = [R], \quad \delta = r^2 = [\zeta_1], \quad V_j = C_\Omega(b_j), \quad D = \langle r, b_0 \rangle. \tag{D.67a}$$

Then $D$ is the specified geometric dihedral group of order eight, and

$$\begin{aligned} b_1 = r b_0, &\qquad V_0 = \langle \delta, b_0 \rangle, \qquad V_1 = \langle \delta, b_1 \rangle, \qquad V_0 \cap V_1 = \langle \delta \rangle, \\ N_\Omega(V_0) = N_\Omega(V_1) = D, &\qquad N_\Omega(D) = D, \qquad C_\Omega(D) = \langle \delta \rangle. \end{aligned} \tag{D.67b}$$

The complete list of finite subgroups of $\Omega$ containing this specified $b_0$ is

$$\langle b_0 \rangle, \qquad V_0, \qquad D. \tag{D.67c}$$

Thus $D$ is its unique maximal finite overgroup. Conjugation by any $\theta \in \Omega$ transports this exact list and all the normalizer and centralizer equalities to the corresponding specified conjugate groups. The statement does not assert that every negative involution is conjugate to $b_0$ or $b_1$.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. Actual word maps for the dihedral group.* Let $B(X) = X$, $B(Y) = Y^{-1}$. Its relation with the cusp-normalized representative is

$$\alpha_0 = \mathrm{Inn}(Y) B. \tag{D.67d}$$

All of $R, B, \alpha_j, \zeta_j$ are actual automorphisms of the free group and extend to its full completion. The following identities are identities of automorphisms, checked on the two free generators:

$$\begin{aligned} R^4 = 1, &\qquad R^2 = \zeta_1, \qquad BRB = R^{-1}, \\ RB = \alpha_1, &\qquad R\alpha_0 = \mathrm{Inn}(X^{-1})\alpha_1, \\ R\alpha_0 R^{-1} = \mathrm{Inn}(X^{-1})\zeta_0, & \\ R\zeta_0 R^{-1} = \mathrm{Inn}(Y^{-1})\alpha_0, & \\ R\alpha_1 R^{-1} = \zeta_1 \alpha_1, &\qquad \alpha_0 \zeta_0 = \mathrm{Inn}(Y)\zeta_1. \end{aligned} \tag{D.67e}$$

Their peripheral maps, with $c_0 = [X, Y]$, include

$$R(c_0) = X^{-1}c_0X, \qquad B(c_0) = Y^{-1}c_0^{-1}Y. \tag{D.67f}$$

Consequently the indicated outer classes belong to the full peripheral group. The homology matrices of $R$ and $B$ are respectively

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{D.67g}$$

Their eight matrices $R_*^k B_*^\epsilon$, $0 \le k < 4$, $0 \le \epsilon < 2$, are distinct. The actual relations (D.67e) give at most eight elements, while these matrices give at least eight distinct outer classes. Hence $D \cong D_8$ with exactly these labels. The linear maps $(u, v) \mapsto (-v, u)$ and $(u, v) \mapsto (u, -v)$ on the torus, punctured at the origin, realize these homology matrices. The discrete once-punctured-torus identification $\mathrm{Out}(F(X, Y)) \cong \mathrm{GL}_2(\mathbb{Z})$ proved in A.13 identifies their induced outer classes with the displayed word maps. This supplies the asserted actual geometric realization.

By D.64–D.65 the full centralizers are exactly $V_0 = \langle b_0, [\zeta_0]\rangle$ and $V_1 = \langle b_1, \delta\rangle$, both of order four. Equations (D.67d)–(D.67e) give $[\zeta_0] = \delta b_0$ and $b_1 = rb_0$, proving the first line of (D.67b) and identifying their intersection. They are the two specified index-two Klein groups in $D$.

*Their full normalizers.* The unique nonidentity multiplier-one element of each $V_j$ is $\delta$; its two other nonidentity elements have multiplier $-1$. The multiplier homomorphism on $\Omega$ is conjugacy invariant. Thus conjugation by a member of $N_\Omega(V_j)$ fixes $\delta$ and either fixes or exchanges the other two elements. Its image in $\mathrm{Aut}(V_j)$ therefore has order at most two. Its kernel is $C_\Omega(V_j)$, and the exact inclusions

$$V_j \le C_\Omega(V_j) \le C_\Omega(b_j) = V_j$$

make this kernel precisely $V_j$. It follows that the full normalizer has at most eight elements. The element $r$ realizes the nontrivial exchange in each case by (D.67e), and the group $D$ already normalizes both $V_j$. This proves both full normalizer equalities.

The mod-two homology map on $D$ has exact kernel $V_0$: $B_* \equiv I$ and $R_* \equiv \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ over $\mathbb{F}_2$. Every element of $N_\Omega(D)$ preserves this kernel, since its actual homology map conjugates the mod-two matrices and preserves the identity matrix. It therefore belongs to $N_\Omega(V_0) = D$. Thus $N_\Omega(D) = D$. The full centralizer is now its ordinary group center, which is $\langle\delta\rangle$.

*Every finite overgroup of the specified two-class reflection.* Let $K \le \Omega$ be finite and contain $b_0$. D.55 gives the complete nine-type list $1, C_2, C_3, C_4, C_2^2, C_6, S_3, D_8, D_{12}$. The first and third types cannot contain an involution. The unique involution of a cyclic order-four group is positive by D.55; that of a cyclic order-six group is positive by D.56. Hence neither can be $b_0$.

We exclude $S_3$ and $D_{12}$ by the exact mod-two representation. Every element $h$ of order three has mod-two homology matrix of exact order three by D.54. In $S_3$ every involution inverts its normal $C_3$. In $D_{12}$ the central involution is the positive involution of a cyclic-six subgroup, so a negative involution is a reflection and also inverts the normal $C_3$. Thus either type would give

$$b_0hb_0^{-1} = h^{-1}, \qquad |h| = 3. \tag{D.67h}$$

But $A_2(b_0) \bmod 2 = I$. Applying the actual homology map and reducing modulo two would make the order-three matrix equal to its inverse, a contradiction.

For $K \cong C_2^2$, every element commutes with $b_0$, so $K \le V_0$, and equality follows from their orders. For $K \cong D_8$, its central involution is the square of an order-four element, hence is positive. It commutes with $b_0$, so it is the unique positive nonidentity element $\delta$ of $V_0$. The group $\langle\delta, b_0\rangle = V_0$ has index two in $K$ and is normal. Consequently $K \le N_\Omega(V_0) = D$, and equality again follows from the orders. The remaining cyclic-two group is exactly $\langle b_0\rangle$. This proves (D.67c).

Finally $K \mapsto \theta K \theta^{-1}$ and its inverse are exact bijections between finite overgroups of $b_0$ and of $\theta b_0 \theta^{-1}$. The same conjugation identifies each full normalizer and centralizer. This gives the asserted transport without any claim that these specified orbits exhaust all negative actions.

The executable certificate `code/part_c/standard_negative_overgroups_d67.py` checks the actual word identities, all eight signed permutation automorphisms and their multiplication, both labelled Klein subgroups, and the mod-two order-three inversion obstruction. The full upper bounds use the proved centralizers and nine-type list, not a finite enumeration of the ambient profinite group. □

**D.68. [Proved] The exact dihedral quotient of a quartic torus action.** Let $a$ have order four in the full once-punctured-torus loop-outer group. Choose an actual order-four representative $\alpha$ as in D.46, and write

$$F = \widehat{F}(X, Y), \qquad c = [X, Y], \qquad E = F \rtimes_\alpha \langle s \rangle.$$

There is exactly one nonzero character $\chi : F \to C_2$ fixed by $a$. Extend it to $\widetilde{\chi} : E \to C_2$ by $\widetilde{\chi}(s) = 0$. Every cusp-normalizer element $v$ with quotient label $a$ has

$$\widetilde{\chi}(v) = 1. \tag{D.68a}$$

There is an explicitly specified discrete free basis $x, y$ with $[x, y] = c$, $\chi(x) = 1$, and $\chi(y) = 0$. For this basis the closed normal subgroup

$$K = \langle\langle x^2, (xy)^2 \rangle\rangle_F \tag{D.68b}$$

is $\alpha$-invariant. The actual quotient and its inverse presentation maps identify

$$\begin{gathered} F/K \cong \mathcal{D} = \widehat{\mathbb{Z}} \rtimes C_2 = \langle r, t \mid t^2 = 1, trt = r^{-1} \rangle, \\ x \longmapsto t, \qquad y \longmapsto r, \qquad c \longmapsto r^{-2}. \end{gathered} \tag{D.68c}$$

Here $r$ generates the full $\widehat{\mathbb{Z}}$ rotation subgroup. The induced automorphism is exactly

$$\overline{\alpha}(r) = r^{-\lambda}, \qquad \overline{\alpha}(t) = r^j t, \qquad j_2 \in \mathbb{Z}_2^\times, \qquad (1 - \lambda) j = 0. \tag{D.68d}$$

It has order two. For every odd prime $q$ the further specified quotient $\mathcal{D} \to D_{2q}$ sends $r$ to a rotation of order $q$ and $t$ to a reflection. On that quotient the induced action is the identity exactly at the coordinates $\lambda_q = -1$; at $\lambda_q = 1$ it is an involution inverting the rotation. In particular each remaining scalar quartic coordinate has an actual nonabelian quotient $F \twoheadrightarrow D_{2q}$ fixed pointwise by this representative.

The proof also gives the full rank-three homology matrix of the actual index-two cover, with all cusp labels retained. These conclusions constrain every actual quartic action; they do not construct a negative multiplier coordinate or settle its remaining exclusion or conjugacy problem.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The specified mod-two character and a cusp-preserving basis.* By D.55 the positive square of $a$ acts as $-I$ on $H_1(F, \mathbb{Z}_p)$ for every prime. Hence

$$A_p(a)^2 = -I, \qquad \lambda^2 = 1, \qquad \lambda_2 = 1. \tag{D.68e}$$

The last assertion also follows directly from the characteristic polynomial $Z^2 + 1$ over $\mathbb{Q}_2$ and the determinant interpretation of the cusp multiplier. The matrix $A_2(a)$ cannot be identity modulo two: otherwise $A_2(a) = I + 2B$ would have square identity modulo four, contradicting its square

$-I$. Its reduction is therefore a nonidentity unipotent matrix over $\mathbb{F}_2$. Its fixed dual line has exactly one nonzero vector. This is the stated character $\chi$. Conjugation on $F$ does not change a character, so this character is defined by the outer action as well.

Here are the basis maps and their actual inverses, according to the values of $\chi$ on $X, Y$:

| $(\chi(X), \chi(Y))$ | $x$ | $y$ | $X$ | $Y$ |
|---|---|---|---|---|
| $(1,0)$ | $X$ | $Y$ | $x$ | $y$ |
| $(0,1)$ | $XYX^{-1}$ | $X^{-1}$ | $y^{-1}$ | $yxy^{-1}$ |
| $(1,1)$ | $X$ | $YX^{-1}$ | $x$ | $yx$. |

(D.68f)

Free reduction verifies both composites and $[x, y] = [X, Y]$ in every row. The same substitutions give mutually inverse maps on full free profinite groups. No peripheral conjugation or change of multiplier has been introduced.

Put $N = \ker \chi$. Its actual discrete double cover of the punctured torus has type $(1, 2)$. Schreier rewriting with transversal $1, x$ gives its full free profinite basis

$$u = x^2, \qquad B = y, \qquad C = xyx^{-1}, \qquad c = CB^{-1}. \tag{D.68g}$$

The second based peripheral generator is $xcx^{-1}$; its homology class is the negative of that of $c$. Write $M = H_1(N, \widehat{\mathbb{Z}})$ in the ordered basis $[u], [B], [C]$ and put $b = [C] - [B]$. The actual deck action $J$, induced by conjugation by $x$, is

$$J[u] = [u], \qquad J[B] = [C], \qquad J[C] = [B]. \tag{D.68h}$$

Before abelianization the last identity is $xCx^{-1} = uBu^{-1}$. Homology of the actual inclusion $N \hookrightarrow F$ is the map

$$I : M \longrightarrow H_1(F, \widehat{\mathbb{Z}}), \qquad I = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \ker I = \widehat{\mathbb{Z}} b. \tag{D.68i}$$

It factors through filling both cusps of the covered torus and then the index-two closed-torus lattice inclusion. Its image is precisely that lattice. Thus after tensoring each primary coordinate with $\mathbb{Q}_p$, the $+1$ eigenspace of $J$ is the handle space and the $-1$ eigenspace is precisely $\mathbb{Q}_p b$.

*The cusp root has the nonzero character label.* Choose $v$ above $a$ in the normalizer of the actual cusp subgroup. The cusp relations, including their primary exponent, are

$$vcv^{-1} = c^{\lambda}, \qquad v^4 = c^{\kappa}, \qquad \kappa_2 \in \mathbb{Z}_2^{\times}. \tag{D.68j}$$

The unit assertion is D.55 applied to $v^2$, which lies over the positive involution $a^2$. The invariant character extends to $E$ by sending $s$ to zero. Since $\chi$ is invariant, $N$ is normal in $E$. Let $T$ be conjugation by $v$ on $M$. The element $vxv^{-1}$ has the same nonzero character as $x$, so its action on $M$ is also $J$. Consequently $T$ commutes with $J$.

On the boundary line $T$ is multiplication by $\lambda$. On the filled handle quotient it induces the actual matrix $A(a)$. Equations (D.68e) and (D.68i), first over $\mathbb{Q}_p$ and then in the torsion-free integral module for each $p$, prove

$$T^2 = -J. \tag{D.68k}$$

At the two-primary coordinate $Tb = b$, and therefore

$$\mathcal{N}_4(T) = (I + T)(I + T^2) = (I + T)(I - J) = 2(I - J), \qquad \mathcal{N}_4(T) M_2 = 2\mathbb{Z}_2 b. \tag{D.68l}$$

If $\widetilde{\chi}(v)$ were zero, write $v = ns$ with $n \in N$. Conjugation by $v$ and by $s$ then agree on $M$. The actual identity $(ns)^4 = n\alpha(n)\alpha^2(n)\alpha^3(n)$ would give $\kappa_2 b = \mathcal{N}_4(T)[n]$, contrary to (D.68l) and the unit exponent in (D.68j). This proves (D.68a).

Let $S$ denote the action of $\alpha$ on $M$. Writing $v = f_0 s$, the proved label means $\chi(f_0) = 1$, so conjugation by $f_0$ on $M$ is exactly $J$. Thus the actual actions satisfy

$$T = JS, \qquad S = JT, \qquad S^2 = -J, \qquad Sb = -\lambda b. \tag{D.68m}$$

In particular the primary norm of $S$ at two is zero.

The entire matrix is explicit. Write the base homology matrix in the basis $[x], [y]$ as $A(a) = \left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right)$. Then

$$T = \begin{pmatrix} a_{11} & a_{12}/2 & a_{12}/2 \\ a_{21} & (a_{22}+\lambda)/2 & (a_{22}-\lambda)/2 \\ a_{21} & (a_{22}-\lambda)/2 & (a_{22}+\lambda)/2 \end{pmatrix}, \qquad S = JT. \tag{D.68n}$$

Every half coefficient belongs to $\widehat{\mathbb{Z}}$: the invariant mod-two character makes $a_{12}$ even, and $a_{22}$ and $\lambda$ are odd. To verify the matrix without choosing an integral splitting, use $TJ = JT$, $Tb = \lambda b$, and the actual equality $IT = A(a)I$. These equations determine the displayed entries uniquely.

*The full dihedral quotient and both presentation maps.* Define the actual character

$$\ell : N \longrightarrow \widehat{\mathbb{Z}}, \qquad \ell(u) = 0, \quad \ell(B) = 1, \quad \ell(C) = -1. \tag{D.68o}$$

It is surjective, and $\ell \circ \mathrm{Inn}(x) = -\ell$. Thus its kernel is normal in $F$. Equation (D.68n) gives, as an equality of characters of $N$,

$$\ell\alpha = -\lambda\ell. \tag{D.68p}$$

Since $-\lambda$ is a unit, $\ker \ell$ is $\alpha$-invariant as a subgroup, in both directions.

The homomorphism $F \to \mathcal{D}$ sending $x$ to $t$ and $y$ to $r$ restricts on $N$ to $n \mapsto r^{\ell(n)}$. It is onto, and its kernel is $\ker \ell$. Indeed an element outside $N$ maps to the nontrivial reflection coset, while the rotation parameter detects every element of $N/\ker \ell$. This is also precisely the quotient (D.68b): in that quotient $x^2 = 1$ and $(xy)^2 = 1$, so $xyx = y^{-1}$. Conversely the images $x = t, y = r$ in $\mathcal{D}$ satisfy these two relators. The inverse homomorphism sends $r$ to $yK$ and $t$ to $xK$. Both composites fix the named generators. The rotation subgroup is full $\widehat{\mathbb{Z}}$, also directly witnessed by the surjective character $\ell$. This proves all assertions about the actual kernel and maps in (D.68b)–(D.68c).

Equation (D.68p) specifies the image of $r$ under the induced automorphism. Preservation of $\chi$ specifies the reflection coset of the image of $t$. Hence there is a unique $j \in \widehat{\mathbb{Z}}$ with the two formulas in (D.68d). Their square sends $r$ to $r$ and $t$ to $r^{(1-\lambda)j}t$. Since $\alpha^4 = 1$, its fourth power gives $2(1-\lambda)j = 0$ in $\widehat{\mathbb{Z}}$. Multiplication by two is injective, proving the last equation of (D.68d) and $\overline{\alpha}^2 = 1$.

For the retained cusp label one can compute $j$ more precisely. The quotient extends to the actual map

$$E \longrightarrow \mathcal{D} \rtimes_{\overline{\alpha}} C_4, \qquad f \longmapsto \pi(f), \quad s \longmapsto s. \tag{D.68q}$$

Write $\pi(f_0) = r^m t$, using $\chi(f_0) = 1$. In this quotient the exact square is

$$\pi(v)^2 = r^{(1+\lambda)m-j} s^2, \qquad \pi(v)^4 = r^{2((1+\lambda)m-j)} = r^{-2\kappa}.$$

Injectivity of multiplication by two in the rotation parameter gives

$$j = (1+\lambda)m + \kappa. \tag{D.68r}$$

At two, this reads $j_2 = 2m_2 + \kappa_2$, so $j_2$ is odd. In particular the induced action is nonidentity and has exact order two. The formula retains the original exponent and does not rescale the cusp.

Finally reduce the full rotation parameter modulo an odd prime $q$. If $\lambda_q = -1$, equation $(1-\lambda)j = 0$ gives $j_q = 0$ already in $\mathbb{Z}_q$. Both displayed generators of $D_{2q}$ are fixed. If $\lambda_q = 1$, the rotation is inverted and is nontrivial because $q$ is odd, so the induced action has exact order two. The cusp has image $r^{-2}$, which still generates the order-$q$ rotation subgroup. Thus these are actual peripheral finite quotients, not quotients inferred only from a dimension count.

The executed certificate `code/part_c/torus_quartic_dihedral_d68.py` checks the discrete basis maps, cusp and deck words, the double-cover matrices and integral norm identity, the standard quartic representative, and the exact dihedral formulas at finite primary coordinates. It does not assert that a permitted coordinate model lifts to a full torus automorphism. □

**D.69. [Proved] A negative quartic coordinate forces a full rotation image.** Use the actual quartic representative, adapted basis, and dihedral quotient of D.68. For every odd prime $q$ at which $\lambda_q = -1$, write

$$\rho : F \twoheadrightarrow Q = D_{2q}, \qquad N = \ker \rho, \qquad \beta = \alpha^2, \qquad \sigma = s^2, \qquad C = C_F(\sigma), \qquad C_0 = C \cap N. \tag{D.69a}$$

The specified rotations form $R = \langle \rho(y) \rangle \cong C_q$. Then restriction of the actual quotient map gives the exact sequence and dimension

$$1 \longrightarrow C_0 \longrightarrow C \xrightarrow{\rho|_C} R \longrightarrow 1, \qquad \dim_{\mathbb{F}_2} H^1(C_0, \mathbb{F}_2) = q - 1. \tag{D.69b}$$

The cover defining $N$ is the full completion of an actual surface of type $(q, 2)$. There are exactly two $N$-conjugacy classes of order-two sections in $N \rtimes \langle \sigma \rangle$; representatives are $\langle \sigma \rangle$ and $\langle x\sigma x^{-1} \rangle$. Their exact normalizers are

$$C_0 \times \langle \sigma \rangle, \qquad x(C_0 \times \langle \sigma \rangle)x^{-1}. \tag{D.69c}$$

They are also the two order-two subgroup classes in $N \rtimes \langle s \rangle$. Let $V_2 = H^1(N, \mathbb{Z}_2)$, let $e = q^{-1} \sum_{g \in R} g$ be the actual deck-averaging operator, and put $W_2 = (1-e)V_2$. On this rank-$2(q-1)$ integral summand the positive square acts identically:

$$\beta^*|_{W_2} = I. \tag{D.69y}$$

The statement determines the indicated quotient of the centralizer of the positive square. It does not determine the image of $C_F(s)$ or the full structure of either centralizer, and it does not exclude or realize a negative quartic coordinate.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The actual covers and the two cyclic extensions.* By D.68 the homomorphism $\rho$ has

$$\rho(x) = t, \qquad \rho(y) = r, \qquad \rho(c) = r^{-2}, \qquad \rho\alpha = \rho, \qquad Q = \langle r, t \mid r^q = t^2 = 1,\ trt = r^{-1} \rangle. \tag{D.69d}$$

The last equality $\rho\alpha = \rho$ is the proved identity action at this multiplier coordinate. Thus $N$ is normal in $E = F \rtimes \langle s \rangle$, and the actual quotient, with inverse coordinate maps, is

$$\begin{aligned} E/N &\longrightarrow Q \times C_4, \quad fs^jN \longmapsto (\rho(f), s^j), \\ Q \times C_4 &\longrightarrow E/N, \qquad (z, s^j) \longmapsto fs^jN \quad (\rho(f) = z). \end{aligned} \tag{D.69e}$$

The second map is independent of the chosen lift $f$. The multiplication rule follows from $\rho\alpha = \rho$, so these maps are inverse homomorphisms. Let

$$E_N = N \rtimes \langle s \rangle, \qquad E_{2,N} = N \rtimes \langle \sigma \rangle, \qquad F_0 = \ker \chi = \rho^{-1}(R). \tag{D.69f}$$

These are the inverse images of the named factors under (D.69e), with their actual generator labels.

The original peripheral monodromy $r^{-2}$ has order $q$. Its right action on the $2q$ cosets has two cycles, both of length $q$. Thus the finite covering surface has two punctures. Its Euler characteristic is $-2q$, so its genus is $q$. The inclusion of its discrete fundamental group has finite index in the original free group and therefore has its full induced profinite topology. This identifies its full completion with $N$ and gives $\operatorname{rank}(N) = 2q + 1$. The intermediate group $F_0$ is the specified type-$(1, 2)$ double cover with Schreier basis $u = x^2, B = y, C = xyx^{-1}$ from D.68. Its quotient $F_0/N = R$ sends these generators to $1, r, r^{-1}$, respectively.

*The exact integral deck module.* Put $\Lambda = \mathbb{Z}_2[R]$ and $\nu = 1 + r + \cdots + r^{q-1}$. The actual degree-$q$ graph cover of the three-petal rose for $F_0$ has chain boundary

$$\partial : \Lambda^3 \longrightarrow \Lambda, \qquad (a, b, d) \longmapsto (r - 1)b + (r^{-1} - 1)d. \tag{D.69g}$$

The two explicit inverse maps on its cycle module are

$$\begin{aligned} \Lambda^2 \oplus \mathbb{Z}_2 &\longrightarrow \ker \partial, \qquad (a, d, k) \longmapsto (a, r^{-1}d + k\nu, d), \\ \ker \partial &\longrightarrow \Lambda^2 \oplus \mathbb{Z}_2, \quad (a, b, d) \longmapsto (a, d, k), \quad b - r^{-1}d = k\nu. \end{aligned} \tag{D.69h}$$

The scalar $k$ exists uniquely because the annihilator of $r - 1$ in $\Lambda$ consists exactly of constant-coefficient multiples of $\nu$. This follows directly by comparing the $q$ cyclic coefficients. Hence

$$H_1(N, \mathbb{Z}_2) \cong \mathbb{Z}_2 \oplus \Lambda^2 \quad \text{as actual deck modules.} \tag{D.69i}$$

Dualizing gives the same multiplicities in $V_2 = H^1(N, \mathbb{Z}_2) = \operatorname{Hom}_{\text{cont}}(N, \mathbb{Z}_2)$. Let $e = q^{-1} \sum_{g \in R} g$ and

$$V = H^1(N, \mathbb{F}_2), \qquad W_2 = (1 - e)V_2, \qquad W = W_2/2W_2. \tag{D.69j}$$

Thus $eV_2$ has rank three and $W_2$ has rank $2(q - 1)$. Restriction and transfer for the index-$q$ inclusion $N \subset F_0$, with $q$ invertible in $\mathbb{Z}_2$, identify $eV_2$ with the actual $H^1(F_0, \mathbb{Z}_2)$. These maps are equivariant for the original $\alpha$.

Let $S$ act on $V_2$ by pullback under $\alpha^{-1}$, and write $B = S^2$ for the pullback under $\beta$. Because $\alpha$ induces identity on $Q$, $S$ commutes with every deck operator from $R$: the two actual conjugations differ only by conjugation by an element of $N$, which acts trivially on this homology. Choose the finite unramified extension $\mathcal{O}$ of $\mathbb{Z}_2$ containing the $q$th roots of unity. Its explicit character projectors are $q^{-1} \sum_{g \in R} \psi(g)^{-1} g$. Every nontrivial character summand of $W_2 \otimes \mathcal{O}$ is free of rank two by (D.69i). On its residue field $S^4 = 1$ makes $S$ unipotent. A two-dimensional unipotent matrix in characteristic two has square identity. Consequently

$$B|_W = I. \tag{D.69k}$$

On the deck-trivial rank-three summand, let $S_{\text{hom}}$ be the matrix induced by $\alpha$ on the retained homology basis of $F_0$. D.68 gives $S_{\text{hom}}^2 = -J$. In the dual basis the present pullback convention is $S = (S_{\text{hom}}^{-1})^{\mathsf{T}}$; therefore $B = S^2 = -J^{\mathsf{T}}$, since $J^2 = I$. The reduction of $J^{\mathsf{T}} - I$ has rank one. Hence $S$ has one Jordan block of size three there, and $B$ has blocks of sizes two and one. On all of $V$, therefore,

$$\operatorname{rank}(B - I) = 1. \tag{D.69l}$$

*Which full section classes occur in these covers.* D.46 gives three $F$-conjugacy classes of order-two sections over $\beta$, with zero mod-two first cohomology for each of their full fixed groups. The proof of D.53 gives their three distinct affine labels in $H_1(F, \mathbb{F}_2)$. Use $\sigma = s^2$ as label zero. Conjugation by $s$ now has zero translation part, since it fixes $\sigma$ itself. Its linear part is $A_2(a)$ modulo two, a nonidentity transvection. Its unique fixed line is $\ker \chi$. D.53 proves that its action fixes exactly the class of $\sigma$ and interchanges the other two classes. Their two labels consequently

have $\chi$-value one. An element $f\sigma$ in either class has $\chi(f) = 1$, so its image under (D.69e) has nontrivial $Q$ coordinate. Neither class meets $E_{2,N}$. Every finite subgroup of $E_N$ injects into its quotient $C_4$, since $N$ is torsion-free. Its order-two subgroups lie in $E_{2,N}$. Thus the only global class meeting either group is the class of $\langle\sigma\rangle$.

Put $C = C_F(\sigma)$. Its mod-two first cohomology vanishes by D.46, so $\chi|_C = 0$. The image $\rho(C)$ is contained in $R$ and is therefore either 1 or $R$. The exact class parametrization is

$$Q/\rho(C) \longrightarrow \{N\text{-conjugacy classes of sections in } E_{2,N}\}, \qquad \rho(f)\rho(C) \longmapsto [\, f\langle\sigma\rangle f^{-1}\,]. \tag{D.69m}$$

Indeed two such subgroups are $N$-conjugate precisely when their conjugators differ by an element of $N$ and an element of $N_F(\langle\sigma\rangle) = C$. Every conjugate belongs to $E_{2,N}$ because its image in $Q \times C_4$ is the same central subgroup. These are also its $E_{2,N}$-conjugacy classes: multiply any conjugator by the matching element of the section to leave a conjugator in $N$. Conjugation by $s$ on (D.69m) sends $\rho(f)$ to $\rho(\alpha(f)) = \rho(f)$ and fixes $\sigma$. Hence (D.69m) also parametrizes the $E_N$ classes.

*The trivial-image alternative and its Brown algebra.* Suppose $\rho(C) = 1$. Then $C \subset N$ and (D.69m) has exactly $2q$ classes. The normalizer in $E_{2,N}$ of the representative $f\langle\sigma\rangle f^{-1}$ is exactly

$$f(C \times \langle\sigma\rangle)f^{-1}. \tag{D.69n}$$

The product map sends $(h, \sigma^\epsilon)$ to $fh\sigma^\epsilon f^{-1}$; its inverse reads the unique $F$ coordinate and the specified quotient coordinate after conjugation by $f^{-1}$. As a closed subgroup of the full free profinite group $F$, $C$ has $\mathrm{cd}_2(C) \le 1$. Since $H^1(C, \mathbb{F}_2) = 0$, all its positive-degree mod-two cohomology vanishes.

The full Brown restriction theorem [42, Theorem 2 and §1.1] applies to $E_{2,N}$: its open subgroup $N$ has 2-cohomological dimension one, every finite subgroup injects into $C_2$, and the only nontrivial elementary abelian two-subgroups are the order-two sections, with no inclusions between distinct ones. The two-row Hochschild–Serre spectral sequence has finite modules in every degree, since $N$ is finitely generated and $C_2$ is finite. Thus $H^n(E_{2,N}, \mathbb{F}_2)$ is finite in every degree; the theorem's dense restriction embedding is therefore onto its Hausdorff target. The actual restriction map is a ring isomorphism in sufficiently high degrees onto the product of the cohomologies of (D.69n). There are exactly $2q$ factors, each one-dimensional in each positive degree. For the quotient character $u : E_{2,N} \to \mathbb{F}_2$, localizing that restriction and taking degree zero gives

$$\mathcal{R} = \left(H^*(E_{2,N}, \mathbb{F}_2)[u^{-1}]\right)^0 \cong \mathbb{F}_2^{2q}. \tag{D.69o}$$

Here $u$ is zero on each fixed group and nonzero on its order-two section. This is the actual localization of the displayed restriction maps.

We need the following precise injection into this algebra. Every $\theta \in W$ is invariant under $\beta$ by (D.69k), and extends to a character of $E_{2,N}$ by declaring $\theta(\sigma) = 0$. The Hochschild–Serre spectral sequence has only rows zero and one. The actual section $\langle\sigma\rangle \to E_{2,N}$ makes inflation from $H^*(C_2, \mathbb{F}_2)$ injective in every degree. Thus every bottom-row class survives, and all possible differentials, whose targets are in that row, are zero. On $W$ the row-one module is $W[u]$, since $\beta$ acts trivially modulo two. The associated graded class of $u^m\theta$ is the nonzero element $\theta u^m$ whenever $\theta \neq 0$. It follows that

$$W \longrightarrow \mathcal{R}, \qquad \theta \longmapsto \theta/u \quad \text{is injective.} \tag{D.69p}$$

*The exact mod-four character cochain.* Since $B = I$ on $W_2/2W_2$, the integral operator $(I - B)/2$ is defined on $W_2$. Let $D$ be its reduction modulo two. Lift $\theta \in W$ to a continuous character $\varphi : N \to \mathbb{Z}/4$ from the integral summand $W_2$. The cochain $\widetilde{\theta}(n\sigma^\epsilon) = \varphi(n)$ has exact coboundary

$$\begin{aligned} \delta\widetilde{\theta}(n\sigma^\epsilon, m\sigma^\eta) &= \varphi(n) + \varphi(m) - \varphi(n\beta^\epsilon(m)) \\ &= \epsilon(\varphi(m) - \varphi(\beta(m))) = 2\epsilon(D\theta)(m) \quad \text{in } \mathbb{Z}/4. \end{aligned} \tag{D.69q}$$

Changing the lift by twice a character does not change $D\theta$, because $\beta$ is identity modulo two on $W$. Thus its Bockstein, which equals its cup square, is the actual identity

$$\theta^2 = uD\theta. \tag{D.69r}$$

Every element of the product algebra (D.69o) is idempotent. Applying this to $\theta/u$ and using (D.69r) and the injection (D.69p) gives $D\theta = \theta$. Consequently

$$B|_{W_2} \equiv -I \pmod 4. \tag{D.69s}$$

*The incompatible quartic section count.* Extend (D.69s) to the unramified ring $\mathcal{O}$ used above. On a nontrivial character summand, if the two-dimensional reduction of $S$ were identity, write $S = I + 2L$ over $\mathcal{O}$. Then $S^2 \equiv I \pmod 4$, contrary to $S^2 = B \equiv -I \pmod 4$. Hence every nontrivial character summand of $V \otimes \overline{\mathbb{F}}_2$ has one Jordan block of size two for $S$. The deck-trivial summand has one block of size three, as already proved. There are $q-1$ nontrivial characters. In the periodic $C_4$ resolution the two maps are $S - I$ and $1 + S + S^2 + S^3 = (S-I)^3$. Each of these blocks contributes exactly one dimension in every positive cohomological degree. Descent of dimension from the splitting field gives

$$\dim H^j(C_4, V) = q \quad (j > 0). \tag{D.69t}$$

The two-row spectral sequence for $E_N$ therefore gives the upper bound

$$\dim H^n(E_N, \mathbb{F}_2) \le q + 1 \quad (n \ge 2). \tag{D.69u}$$

On the other hand, the exact full normalizer in $E$ of $\langle\sigma\rangle$ is $C \rtimes \langle s\rangle$, as in D.53. For each representative in (D.69m), its conjugate lies entirely in $E_N$ under the present assumption $\rho(C) = 1$: (D.69e) sends its fixed group to identity and its conjugated section generator to the named $C_4$ factor. Thus its $E_N$ normalizer is exactly

$$f(C \rtimes \langle s\rangle)f^{-1}. \tag{D.69v}$$

The map $(h, s^j) \mapsto fhs^jf^{-1}$ and its coordinate inverse give this semidirect-product identification with labels. All positive-degree cohomology of $C$ is zero, so inflation identifies the cohomology of each normalizer with $H^*(C_4, \mathbb{F}_2)$. Brown applies here as well: $N$ is the open dimension-one subgroup, finite subgroups inject into $C_4$, and there is no elementary abelian subgroup of rank two. Its elementary two-subgroup classes are exactly the $2q$ classes in (D.69m); conjugation by $s$ does not fuse them. The same two-row finite-module spectral sequence makes $H^n(E_N, \mathbb{F}_2)$ finite, so this dense restriction embedding is again onto. Restriction therefore gives $\dim H^n(E_N, \mathbb{F}_2) = 2q$ in high degrees. This contradicts (D.69u), since $q > 1$. The alternative $\rho(C) = 1$ is impossible. We have proved the surjectivity in (D.69b).

*The exact remaining fixed-group cohomology.* Now $\rho(C) = R$. Formula (D.69m) has exactly two cosets, represented by 1 and $t = \rho(x)$. Intersecting the full normalizers with $E_{2,N}$ gives precisely (D.69c), since its fixed kernel is $C_0 = C \cap N$. Both are conjugate by the original element $x$ and have the same first-cohomology dimension.

The split spectral sequence for $E_{2,N}$ still collapses by the actual section argument above. By (D.69l) the operator $B - I$ on its $(2q+1)$-dimensional module $V$ has rank one. For $C_2$ both periodic differentials are $B - I$, so

$$\dim H^j(C_2, V) = (2q+1) - 2 = 2q - 1 \quad (j > 0), \qquad \dim H^n(E_{2,N}, \mathbb{F}_2) = 2q \quad (n \ge 2). \tag{D.69w}$$

Brown restriction to the two normalizers (D.69c) then gives the exact equality

$$2q = 2(1 + \dim H^1(C_0, \mathbb{F}_2)). \tag{D.69x}$$

In particular this first cohomology is finite-dimensional, and its dimension is $q-1$, as claimed. No assertion that $C_0$ is itself a full free profinite group, or that $C_F(s)$ has the same finite image, is used.

*The full integral action on the nontrivial deck summand.* We now use the proved image $\rho(C) = R$, not the trivial-image alternative discarded above. Choose an actual $h \in C$ with $\rho(h) = r$. Conjugation by $h$ preserves $E_{2,N}$ and fixes $\sigma$ itself. Its $q$th power is conjugation by $h^q \in C_0 \subset E_{2,N}$, so the resulting cohomology action factors through $R$. On $V_2$ it is exactly the deck action: replacing $h$ by any other lift of $r$ changes conjugation by an inner automorphism of $N$. The extension of $\theta \in W$ which is zero on $\sigma$ is consequently $R$-equivariant, and the quotient character $u$ is fixed.

By the same finite-dimensional Brown restriction argument, the actual localized degree-zero ring is now

$$\mathcal{R} \cong (\mathbb{F}_2 \oplus H^1(C_0, \mathbb{F}_2)u^{-1}) \times (\mathbb{F}_2 \oplus H^1(xC_0x^{-1}, \mathbb{F}_2)u^{-1}). \tag{D.69z}$$

In each factor the product of two degree-one fixed-group classes vanishes, because its full fixed group has 2-cohomological dimension at most one. Thus the two displayed augmentation ideals are square zero; their product $\mathcal{I}$ is the nilradical of $\mathcal{R}$, and $\mathcal{R}/\mathcal{I} = \mathbb{F}_2^2$. The action of $R$ fixes both section classes, which are the two cosets in $Q/R$. Its action on this residue algebra is therefore identity.

The injection (D.69p) still holds, since its split spectral-sequence proof used no assumption on $\rho(C)$. It is now the $R$-equivariant map $W \to \mathcal{R}$, $\theta \mapsto \theta/u$. The composite with $\mathcal{R}/\mathcal{I}$ vanishes: deck averaging is zero on $W$ and identity on that residue algebra, and $q$ is invertible in $\mathbb{F}_2$. Hence its image is contained in $\mathcal{I}$. Both vector spaces have dimension $2(q-1)$ by (D.69j) and (D.69x), so it identifies $W$ with that square-zero ideal. In particular $(\theta/u)^2 = 0$ for every $\theta \in W$. The exact cochain identity (D.69r) and the same injection give $D\theta = 0$ for every $\theta$. Thus $B|_{W_2} \equiv I \pmod 4$.

Finally $B^2 = I$ on the free $\mathbb{Z}_2$-module $W_2$. Writing $B = I + 4L$ gives $L(I + 2L) = 0$ after cancelling the nonzero scalar 8 in its torsion-free endomorphism module. The matrix $I + 2L$ is invertible over $\mathbb{Z}_2$, because its reduction is identity. Therefore $L = 0$ and $B = I$ on $W_2$. This is precisely (D.69y), since $\beta^{-1} = \beta$. Together with the preceding dual-basis calculation, it determines the integral square as $-J^{\mathsf{T}}$ on $eV_2$ and identity on $W_2$.

The executable certificate `code/part_c/torus_quartic_bad_prime_d69.py` checks the exact finite covering cycles and module maps, the two section-label possibilities, the cyclic resolution dimensions, and the mod-four cochain identity. The full normalizer and Brown arguments above supply the stated centralizer conclusion. □

**D.70. [Proved] The quartic torus multiplier is globally one.** Every element of order four in the full loop-outer group of the once-punctured torus has multiplier $1 \in \widehat{\mathbb{Z}}^\times$. Consequently every finite two-subgroup of that full group has multiplier image in the two global units $\{1, -1\}$, and the multiplier is determined by its 2-adic coordinate. The conclusion concerns the full group, without a geometric-membership assumption. It determines the quartic multiplier; the conjugacy of an arbitrary quartic action to a specified geometric action is not asserted.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. An exact degree-three restriction obstruction.* We first give the cohomological calculation used below. Let $P$ be the pro-two completion of an orientable closed surface group of genus $g \geq 2$, and let $b$ be an actual involution of $P$ whose action on $H^1(P, \mathbb{F}_2)$ is identity and whose action on $H^1(P, \mathbb{Z}_2)$ is nonidentity. Write

$$G = P \rtimes_b \langle \tau \rangle, \qquad \tau^2 = 1, \qquad u : G \longrightarrow \mathbb{F}_2, \quad u(h\tau^\epsilon) = \epsilon. \tag{D.70a}$$

Then any two $b$-fixed continuous characters $\theta, \psi : P \to \mathbb{Z}_2$ have zero reduced cup product in $H^2(P, \mathbb{F}_2)$.

Here are all maps and hypotheses for this assertion. The group $P$ is a torsion-free Demushkin pro-two group: its cohomological dimension is two, its first cohomology has dimension $2g$, and its second cohomology is $\mathbb{F}_2$ with the nondegenerate surface cup pairing. The integral character module is $\mathbb{Z}_2^{2g}$. These are the surface duality comparisons of A.2 and A.7, used with both completions in D.42.

Every $\xi \in H^1(P, \mathbb{F}_2)$ has the actual character extension

$$\widetilde{\xi} : G \longrightarrow \mathbb{F}_2, \qquad \widetilde{\xi}(h\tau^\epsilon) = \xi(h), \tag{D.70b}$$

because $b$ fixes $\xi$. The action on $H^2(P, \mathbb{F}_2)$ is also identity, since this space is one-dimensional over $\mathbb{F}_2$. The multiplicative Hochschild–Serre spectral sequence therefore has its actual second page

$$E_2^{i,j} = H^i(C_2, \mathbb{F}_2) \otimes H^j(P, \mathbb{F}_2) = \mathbb{F}_2[u] \otimes H^j(P, \mathbb{F}_2), \qquad 0 \le j \le 2. \tag{D.70c}$$

The quotient class $u$ is permanent: the section $C_2 \to G$, $\tau^\epsilon \mapsto \tau^\epsilon$, makes inflation injective. All classes in $E_2^{0,1}$ are permanent by (D.70b). The surface class in $E_2^{0,2}$ is the cup product of two of them, so it is permanent as well. These classes generate the entire second-page algebra. Thus every differential vanishes, and multiplication by $u$ on its row-two term gives the nonzero class $u\eta \in E_\infty^{1,2}$ for each nonzero surface class $\eta$.

The fixed integral characters have the stronger actual extensions

$$\widetilde{\theta} : G \longrightarrow \mathbb{Z}_2, \qquad \widetilde{\psi} : G \longrightarrow \mathbb{Z}_2, \qquad \widetilde{\theta}(h\tau^\epsilon) = \theta(h), \quad \widetilde{\psi}(h\tau^\epsilon) = \psi(h). \tag{D.70d}$$

Every finite subgroup of $G$ injects into $G/P = C_2$, because $P$ is torsion-free. For each order-two subgroup $T = \langle z \rangle$, the two characters in (D.70d) vanish on $T$: twice their values on $z$ is zero in the torsion-free group $\mathbb{Z}_2$. There is the exact direct-product normalizer

$$\begin{aligned} C_P(T) \times T &\longrightarrow N_G(T), \qquad (h, z^\epsilon) \longmapsto hz^\epsilon, \\ N_G(T) &\longrightarrow C_P(T) \times T, \qquad w \longmapsto (wz^{-\epsilon}, z^\epsilon), \qquad \epsilon = u(w). \end{aligned} \tag{D.70e}$$

Indeed every normalizer element fixes the unique nonidentity element of $T$, and its quotient label determines $\epsilon$ uniquely. These are inverse continuous homomorphisms.

Write $z = n\tau$. Its actual action on $P$ is $\mathrm{Inn}(n)b$, with square identity. It has the same integral homology action as $b$, so it is nonidentity by the stated hypothesis. The hypotheses of [42, Corollary 2 and its proof in §2] therefore hold for this particular action: $P$ is a pro-two Poincare duality group of dimension two and type $\mathrm{FP}_\infty$, and the acting subgroup has order two in $\mathrm{Aut}(P)$. The corollary proves that the full fixed subgroup $C_P(T)$ is a free pro-two group. In particular

$$H^2(C_P(T), \mathbb{F}_2) = 0. \tag{D.70f}$$

For clarity, the infinite-index step in that proof uses its Lemma 1 for an actual nontrivial finite two-group of automorphisms of a torsion-free pro-two group; it then applies the dimension bound for an infinite-index subgroup of a Demushkin group. It is applicable separately to every section $T$.

Let bars denote the reductions of (D.70d) modulo two. Under (D.70e), both restricted characters are pulled back from $C_P(T)$, since they vanish on $T$. Their cup product is zero by (D.70f). It follows that

$$\mathrm{res}_{N_G(T)}(u \smile \overline{\widetilde{\theta}} \smile \overline{\widetilde{\psi}}) = 0 \quad \text{for every } T. \tag{D.70g}$$

The full Brown restriction theorem [42, Theorem 2] is injective in degree three here: $P$ is open of cohomological dimension two, and $G$ has no $C_2 \times C_2$ because its finite subgroups inject into $C_2$.

Thus the class in (D.70g) is zero in $H^3(G,\mathbb{F}_2)$. If $\bar\theta \smile \bar\psi = \eta \neq 0$ on $P$, its associated graded class under (D.70c) is the nonzero element $u\eta \in E_\infty^{1,2}$, a contradiction. This proves the asserted zero-cup property of the integral fixed characters. Only the injectivity part of Brown restriction is needed in this calculation.

*The actual full cusp filling and its pro-two quotient.* Now take an order-four full torus action and the actual lift $\alpha$ of D.46. D.43 and D.50 give $\lambda^2 = 1$ and $\lambda_2 = 1$. Suppose an odd coordinate satisfies $\lambda_q = -1$. D.68 gives its actual dihedral quotient, and D.69 gives the actual type-$(q,2)$ cover

$$N = \ker(\rho : F \twoheadrightarrow D_{2q}), \qquad \beta = \alpha^2, \qquad R = \langle \rho(y) \rangle \cong C_q. \tag{D.70h}$$

In particular $q \equiv 1 \pmod 4$, so $q \geq 5$. Let $\Gamma_N$ be the corresponding discrete finite-index subgroup of the original punctured-torus group, with its original covering surface. Use the two actual based peripheral generators $d_0 = c^q$ and $d_1 = xc^qx^{-1}$ of this cover. The second base path is the lift of the original word $x$ to the other peripheral orbit. Their full procyclic subgroups in $N$ are the two covering cusps. Let

$$K = \overline{\langle\langle d_0, d_1 \rangle\rangle}_N, \qquad \Pi = N/K, \qquad P = \Pi^{(2)}, \qquad p : N \twoheadrightarrow \Pi \twoheadrightarrow P. \tag{D.70i}$$

These are the specified quotient maps. Filling the two punctures of the discrete cover gives the closed genus-$q$ surface group $\Gamma_N/\langle\langle d_0, d_1 \rangle\rangle$. Its full completion is exactly $\Pi$: a map from either group to a finite group is precisely a map from $\Gamma_N$ killing $d_0, d_1$. The finite-index subgroup $\Gamma_N$ has its full induced profinite topology in $F$, as in D.69, so this identifies that universal finite quotient with the displayed $N/K$. Restricting the same finite targets to two-groups gives the displayed maximal pro-two quotient $P$. Consequently $P$ is the genus-$q$ pro-two closed surface group used in the preceding calculation.

The preservation of $K$ has explicit conjugators in $N$. Retain the cusp root $v = f_0 s$ of D.68, so $\alpha(c) = f_0^{-1} c^\lambda f_0$. For $g \in \{1, x\}$ choose $g' \in \{1, x\}$ with $\rho(g')R = \rho(\alpha(g) f_0^{-1})R$. Since $\rho(c) = r^{-2}$ generates $R$, there is an integer $k$ for which $\rho(\alpha(g) f_0^{-1}) = \rho(g' c^k)$. The actual element $n = \alpha(g) f_0^{-1} c^{-k} (g')^{-1}$ then belongs to $N$, and direct multiplication gives

$$\alpha(gc^q g^{-1}) = n\big(g' c^q (g')^{-1}\big)^\lambda n^{-1}.$$

The inverse automorphism gives the reverse inclusion. Thus $\alpha(K) = K$ with the original two labels and their full unit powers retained. The maximal pro-two kernel of $\Pi$ is characteristic, so $\alpha$, and hence $\beta$, induce actual automorphisms through both arrows in (D.70i). Denote the latter induced automorphism of $P$ by $b$; it satisfies $b^2 = 1$. No section is changed in this passage.

*The dual inclusion and the closed integral deck splitting.* The actual pullback of characters gives an injection

$$p^* : L = H^1(P, \mathbb{Z}_2) \hookrightarrow V_2 = H^1(N, \mathbb{Z}_2). \tag{D.70j}$$

Its image is exactly the characters which vanish on $d_0, d_1$. One direction follows from (D.70i). Conversely such a character kills $K$, factors through $\Pi$, and then through $P$ because its target $\mathbb{Z}_2$ is pro-two. This proves both directions of the assertion for the exact displayed maps. The same argument holds with target $\mathbb{F}_2$, and reduction identifies $L/2L$ with $H^1(P, \mathbb{F}_2)$.

The rotation deck action on the punctured cover extends over each of its two filled punctures. It preserves orientation and gives the actual action of $R$ on $L$ and its surface cup form. As in D.69 put

$$e = \frac{1}{q} \sum_{r' \in R} r', \qquad W_2 = (1 - e)V_2. \tag{D.70k}$$

Both cusp restriction functionals on $V_2$ are $R$-invariant. Indeed $R$ fixes each of the two cusp labels, and its oriented deck action changes a based peripheral generator only by conjugation in $N$. Consequently each such functional vanishes on $W_2$, so (D.70j) identifies $W_2$ with $(1 - e)L$.

On the deck-trivial part, restriction and transfer identify $eV_2$ with $H^1(F_0, \mathbb{Z}_2)$, where $F_0 = \rho^{-1}(R)$ is the type-$(1,2)$ double cover of D.68. Its retained basis is $u = x^2, B = y, C = xyx^{-1}$. The original boundary is $c = CB^{-1}$, and its two covering boundary homology classes are opposite. Restriction to $N$ sends each peripheral evaluation to $q$ times the corresponding evaluation on $F_0$. Since $q$ is a unit in $\mathbb{Z}_2$, the condition in (D.70j) on this part is exactly

$$\xi(B) = \xi(C). \tag{D.70l}$$

Thus $eL$ has rank two, and the actual direct sum of integral character modules is

$$L = eL \oplus W_2, \qquad \operatorname{rank}(W_2) = 2q - 2. \tag{D.70m}$$

D.68 and the exact dual convention in D.69 give $\beta^* = -J^{\mathsf{T}}$ on $eV_2$, with $J$ fixing $u$ and interchanging $B, C$. On (D.70l), $J^{\mathsf{T}}$ is identity. D.69 gives $\beta^* = I$ on $W_2$. Equivariance of (D.70j) therefore gives the full integral action

$$b^*|_{eL} = -I, \qquad b^*|_{W_2} = I. \tag{D.70n}$$

It follows that $b$ has order exactly two, its integral homology action is nonidentity, and its action on $L/2L$ is identity. All three assertions follow from the actual rank-two summand in (D.70n); none is inferred from the order of the original outer class alone.

*The unimodular fixed cup pair and the contradiction.* Choose the oriented surface generator of $H^2(P, \mathbb{Z}_2) \cong \mathbb{Z}_2$. The cup pairing

$$\langle\, ,\, \rangle : L \times L \to \mathbb{Z}_2 \tag{D.70o}$$

is alternating and unimodular. Every deck operator $r'$ preserves this generator, so $\langle r'\xi, \eta\rangle = \langle \xi, (r')^{-1}\eta\rangle$. Summing over $R$ proves that the actual integral idempotent $e$ is self-adjoint:

$$\langle e\xi, \eta\rangle = \langle \xi, e\eta\rangle. \tag{D.70p}$$

Hence (D.70m) is an orthogonal direct sum. Each restricted form is unimodular: in bases of the two integral direct summands, the whole Gram matrix is block diagonal with unit determinant, so each block determinant is a unit in $\mathbb{Z}_2$. In particular the rank-$2q-2$ form on $W_2$ is nonzero and unimodular. Choose $\theta, \psi \in W_2$ with $\langle\theta, \psi\rangle$ a unit. Their reductions have nonzero cup product in $H^2(P, \mathbb{F}_2)$, while (D.70n) makes both characters fixed by the actual $b$. The degree-three restriction obstruction (D.70a)–(D.70g) applies to exactly this $P, b$, and says that their reduced cup product is zero. This contradiction excludes the chosen coordinate $\lambda_q = -1$.

There is no negative odd coordinate. Together with $\lambda^2 = 1$ and $\lambda_2 = 1$, this proves $\lambda = 1$ in the full product $\widehat{\mathbb{Z}}^\times = \prod_p \mathbb{Z}_p^\times$. The possible finite two-subgroups are exactly $1, C_2, C_4, C_2 \times C_2, D_8$ by D.50. D.43 gives the global sign for their involutions; the present result gives multiplier one for their order-four elements. These statements cover every element of each of the listed groups and prove the final assertion. □

*Scope.* This result closes the quartic multiplier branch left in D.50 and D.68. It uses the exact rotation-cover conclusion of D.69 and the actual closed filling (D.70i). It supplies no conjugacy to a geometric quartic representative and no recognition of either full fixed subgroup in D.69. The remaining scalar order-three branch is unaffected by this degree-three calculation.

**D.71. [Proved] The Frobenius cover of a scalar cubic torus coordinate.** Let $a$ be an order-three element of the full once-punctured-torus loop-outer group, with actual order-three representative $\alpha$ and labelled extension $E = F \rtimes_\alpha \langle s\rangle$. At every prime $q \neq 3$ with $\lambda_q \neq 1$, there is the following actual finite quotient, after the explicit based peripheral change of generators in the proof:

$$\begin{gathered} Q = \langle r, t \mid r^q = t^3 = 1,\ trt^{-1} = r^\ell\rangle, \qquad \ell = \lambda_q \bmod q \in \mathbb{F}_q^\times, \\ \rho : F \twoheadrightarrow Q, \qquad \rho(x) = t, \quad \rho(y) = r, \quad \rho\alpha = \rho. \end{gathered} \tag{D.71a}$$

Here $\ell$ has order three and $|Q| = 3q$. Put $R = \langle r \rangle$, $N = \ker \rho$, $C = C_F(s)$, and $C_0 = C \cap N$. The actual cover defining $N$ has type $((3q-1)/2, 3)$, and

$$1 \longrightarrow C_0 \longrightarrow C \xrightarrow{\rho|_C} R \longrightarrow 1, \qquad H^1(C_0, \mathbb{F}_3) = 0. \tag{D.71b}$$

There are exactly three $N$-conjugacy classes of order-three sections in $N \rtimes \langle s \rangle$. With their original quotient-generator labels, their representatives and exact normalizers are

$$\begin{aligned} T_i &= x^i \langle s \rangle x^{-i}, \\ N_{N \rtimes \langle s \rangle}(T_i) &= x^i (C_0 \times \langle s \rangle) x^{-i} \qquad (0 \le i < 3). \end{aligned} \tag{D.71c}$$

On the integral character module $V_3 = H^1(N, \mathbb{Z}_3)$, retain the actual rotation projector and its complement

$$e = \frac{1}{q} \sum_{g \in R} g, \qquad W_3 = (1-e)V_3. \tag{D.71d}$$

For the action $S$ given by pullback under $\alpha^{-1}$, the module $W_3$ is free of rank $q-1$ over $\mathbb{Z}_3[\langle S \rangle] \cong \mathbb{Z}_3[C_3]$. The isomorphism is supplied by lifting cyclic vectors for the actual action. The averaged rank-four module $eV_3$ reduces to two Jordan blocks of size two modulo three. These are exact quotient, lattice, and fixed-group constraints. No exclusion or realization of the scalar cubic coordinate is asserted.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The actual index-three cover and its based generators.* Use D.48 and D.51 for the two section classes of $E$, with $\langle s \rangle$ the chosen one. There is an invariant character $\chi : F \to \mathbb{F}_3$, extended by $\chi(s) = 0$, which can be scaled uniquely so that

$$\chi(v) = 1, \qquad \chi(s_1) = -1, \qquad vcv^{-1} = c^{\lambda}, \qquad v^3 = c^{\kappa}, \quad c = [X, Y], \tag{D.71e}$$

where $v$ is a cusp-normalizer element labelled by $s$, and $s_1$ is the labelled generator of the other full section class. Indeed D.51 gives opposite nonzero values for the cusp line and the other section line. Also $\kappa_3$ is a unit by D.48.

Here is an explicit peripheral change of basis with $\chi(x) = 1, \chi(y) = 0$, without rescaling the now specified character. For any current basis $(X, Y)$ the transformation

$$Q_0(X, Y) = (XYX^{-1}, X^{-1}), \qquad Q_0^{-1}(x, y) = (y^{-1}, yxy^{-1}) \tag{D.71f}$$

has the displayed inverse and preserves $[X, Y]$ exactly. Its action on the two character values is $(a, b) \mapsto (b, -a)$. Among its four iterates, at least one has first character value 1, since $(a, b) \neq (0, 0)$. Choose the first such iterate $(x, Y')$, and let $d \in \{0, 1, -1\}$ represent $\chi(Y')$. Set $y = Y'x^{-d}$. This shear has inverse $Y' = yx^d$ and $[x, Y'x^{-d}] = [x, Y']$. These actual discrete free-group automorphisms, with the stated inverses, therefore supply the required basis in the full completion and keep $c = [x, y]$ exactly.

Let $F_0 = \ker \chi$. Its actual type-$(1, 3)$ covering has the Schreier basis and peripheral representatives

$$\begin{aligned} u &= x^3, \qquad B_i = x^i y x^{-i} \quad (0 \le i < 3), \\ c_i &= x^i c x^{-i} = B_{i+1} B_i^{-1}, \qquad B_3 = u B_0 u^{-1}. \end{aligned} \tag{D.71g}$$

These identities are actual words, including the final value $i = 2$. On $M = H_1(F_0, \widehat{\mathbb{Z}})$, deck conjugation by $x$ gives $J$, which fixes $u$ and cyclically permutes the $B_i$. The inclusion homology map is the actual map

$$I : M \longrightarrow H_1(F, \widehat{\mathbb{Z}}), \qquad I(u) = 3[x], \quad I(B_i) = [y]. \tag{D.71h}$$

Its kernel is the rank-two boundary augmentation, generated by the three differences $B_{i+1} - B_i$ with their sole sum relation. Filling the three punctures gives the actual degree-three closed torus cover; (D.71h) factors through its integral lattice inclusion.

Let $S_{\mathrm{hom}}$ be induced by $\alpha$ on $M$. Since $\chi\alpha = \chi$, it commutes with $J$: the conjugating elements $\alpha(x)$ and $x$ differ by an element of $F_0$, whose inner action on $M$ is trivial. Write $v = f_0 s$. Equation (D.71e) gives $\chi(f_0) = 1$, so the homology action $T$ of conjugation by $v$ is

$$T = JS_{\mathrm{hom}}. \tag{D.71i}$$

It fixes each of the three peripheral classes with multiplier $\lambda$. To check its labels, conjugation by $v$ preserves $\chi$ on $F$, so $vx^i v^{-1}$ differs from $x^i$ by an element of $F_0$, while $vcv^{-1} = c^{\lambda}$. Thus

$$S_{\mathrm{hom}}|_{\ker I} = \lambda J^{-1}. \tag{D.71j}$$

There is also a fully specified matrix for this comparison. If $A = (a_{ij})$ is the original homology matrix in the basis $(x, y)$ and $\mathbf{1} = (1, 1, 1)^{\mathsf{T}}$, then

$$S_{\mathrm{hom}} = \begin{pmatrix} a_{11} & (a_{12}/3)\mathbf{1}^{\mathsf{T}} \\ a_{21}\mathbf{1} & \lambda J_3^{-1} + ((a_{22} - \lambda)/3)\mathbf{1}\mathbf{1}^{\mathsf{T}} \end{pmatrix}, \tag{D.71k}$$

where $J_3$ is the cycle on $(B_0, B_1, B_2)$. The numerator divisibilities hold at 3: $\chi\alpha = \chi$ gives $a_{11} \equiv 1$ and $a_{12} \equiv 0$ modulo three; the determinant is $\lambda$, with $\lambda_3 = 1$, so $a_{22} \equiv 1 \equiv \lambda$ there. At every other prime division by three is invertible. To prove the formula, the $u$-column is determined by $J$-invariance and (D.71h); the remaining three columns are determined by (D.71j), commutation with $J$, and $IS_{\mathrm{hom}} = AI$. These determinations can first be read over each $\mathbb{Q}_p$; the stated integral divisibilities and torsion-free lattices give (D.71k) in the full module.

*The invariant character and the actual Frobenius quotient.* Fix the specified bad prime $q$. The unit $\lambda_q$ is a primitive cube root. It reduces to one modulo $q$ only for the identity cube root, because $q \neq 3$ and the three roots have distinct reductions. Thus $\ell = \lambda_q \bmod q$ has order three and $q \equiv 1 \pmod 3$. Define the continuous character

$$\ell_q : F_0 \longrightarrow \mathbb{Z}_q, \qquad \ell_q(u) = 0, \quad \ell_q(B_i) = \lambda_q^i. \tag{D.71l}$$

Its values sum to zero, and $\ell_q J = \lambda_q \ell_q$. Over $\mathbb{Z}_q$, the row therefore annihilates the deck-trivial summand and is supported on the boundary augmentation. Equation (D.71j), or directly (D.71k), gives the exact equation

$$\ell_q \alpha = \ell_q. \tag{D.71m}$$

Put $N = \ker(\ell_q \bmod q)$ inside $F_0$. It is normal in $F$: conjugation by $x$ scales the character by the unit $\ell$, and conjugation by $F_0$ does not change a character. It is also $\alpha$-invariant in both directions by (D.71m). Its quotient in $F_0$ has order $q$, so $[F : N] = 3q$.

In $F/N$, the images of $x, y$ satisfy $x^3 = u = 1$, $y^q = 1$, and $xyx^{-1} = y^{\ell}$. The latter equality is the equality of their images under $\ell_q \bmod q$ in $F_0/N$. The presented group in (D.71a) has the explicit multiplication

$$(a, i)(b, j) = (a + \ell^i b,\ i + j) \quad \text{on } \mathbb{F}_q \times \mathbb{F}_3, \qquad r = (1, 0), \quad t = (0, 1). \tag{D.71n}$$

It has order $3q$. The inverse coordinate homomorphisms are $r \mapsto yN, t \mapsto xN$ and $xN \mapsto t, yN \mapsto r$. The relations and the already computed index make them inverse, proving exactly (D.71a)'s group and quotient, rather than just a surjection to a smaller group.

The induced automorphism on $Q$ fixes $r$ by (D.71m) and preserves the quotient label in $Q/R$ by $\chi\alpha = \chi$. It consequently sends $t$ to $r^j t$ for some $j \in \mathbb{F}_q$. Its third iterate sends $t$ to $r^{3j}t$. Since

the original $\alpha$ has order three and $q \neq 3$, $j = 0$. Thus $\rho\alpha = \rho$. The actual extension quotient and inverse are

$$\begin{aligned} E/N &\longrightarrow Q \times C_3, \quad fs^iN \longmapsto (\rho(f), s^i), \\ Q \times C_3 &\longrightarrow E/N, \qquad (z, s^i) \longmapsto fs^iN \quad (\rho(f) = z). \end{aligned} \tag{D.71o}$$

Independence of the lift and multiplication follow from $\rho\alpha = \rho$.

The cusp has image $\rho(c) = [t, r] = r^{\ell-1}$ of order $q$. Its action on the $3q$ cosets has three cycles of length $q$. The corresponding discrete cover therefore has three punctures and Euler characteristic $-3q$, hence genus $(3q-1)/2$. Its finite-index subgroup carries its full induced profinite topology, giving exactly the stated full surface completion $N$.

*The integral rotation module, including its inverse maps.* Write $\Lambda_3 = \mathbb{Z}_3[R]$ and $\nu = \sum_{i=0}^{q-1} r^i$. Let $a, b \in \{1, \dots, q-1\}$ be the integer representatives of $\ell, \ell^2$, and put $P_k = 1 + r + \cdots + r^{k-1}$. The actual degree-$q$ graph cover of the four-generator rose for $F_0$ has boundary

$$\partial : \Lambda_3^4 \longrightarrow \Lambda_3, \qquad (A, B, D, E) \longmapsto (r-1)B + (r^a - 1)D + (r^b - 1)E. \tag{D.71p}$$

Its exact cycle-coordinate maps are

$$\begin{aligned} \Lambda_3^3 \oplus \mathbb{Z}_3 &\longrightarrow \ker \partial, \qquad (A, D, E, k) \longmapsto (A, -P_aD - P_bE + k\nu, D, E), \\ \ker \partial &\longrightarrow \Lambda_3^3 \oplus \mathbb{Z}_3, \quad (A, B, D, E) \longmapsto (A, D, E, k), \\ &\qquad\qquad B + P_aD + P_bE = k\nu. \end{aligned} \tag{D.71q}$$

The scalar $k$ exists uniquely: the annihilator of $r-1$ consists exactly of constant-coefficient multiples of $\nu$, as comparison of the cyclic coefficients proves. Thus these are inverse integral module maps and

$$H_1(N, \mathbb{Z}_3) \cong \mathbb{Z}_3 \oplus \Lambda_3^3. \tag{D.71r}$$

Dualizing gives the same deck multiplicities in $V_3$. Restriction and transfer for the actual index-$q$ subgroup $N \subset F_0$ identify $eV_3$ with $H^1(F_0, \mathbb{Z}_3)$. The maps are equivariant under the original $\alpha$; the remaining module $W_3$ has rank $3(q-1)$.

On $H_1(F_0, \mathbb{Q}_3)$, the boundary part of $S_{\mathrm{hom}}$ is $J^{-1}$ by (D.71j). Its two eigenvalues are the primitive cube roots. The filled rank-two quotient has the original nonidentity cubic matrix $A_3$ by the lattice map (D.71h). D.48 proves its polynomial is $X^2 + X + 1$. Consequently the same polynomial annihilates the whole rank-four module: first over $\mathbb{Q}_3$, by the two commuting $J$-summands, and then over $\mathbb{Z}_3$ by torsion-freeness. For the dual convention $S = (S_{\mathrm{hom}}^{-1})^{\mathsf{T}}$ this gives

$$I + S + S^2 = 0 \quad \text{on } eV_3. \tag{D.71s}$$

The ring $\mathbb{Z}_3[z]/(z^2 + z + 1)$ is a discrete valuation ring: with $\pi = z - 1$ its defining polynomial is $\pi^2 + 3\pi + 3$, and $(3) = (\pi^2)$. The rank-four $\mathbb{Z}_3$-lattice in (D.71s) is a torsion-free module over this ring, so is free of rank two. Its reduction modulo three is therefore exactly two Jordan blocks of size two.

*The integral fixed-cup obstruction at three.* We use the following calculation with the actual pro-three closed-surface group. Let $P$ be such a group of genus at least two and let $b$ be an actual automorphism of order three whose integral homology action is nonidentity. Any two $b$-fixed characters $\theta, \psi : P \to \mathbb{Z}_3$ have cup pairing divisible by three.

Indeed put $G = P \rtimes_b \langle \tau \rangle$. Both characters extend by $\widetilde{\theta}(h\tau^i) = \theta(h)$ and $\widetilde{\psi}(h\tau^i) = \psi(h)$. They kill every subgroup of order three, since $\mathbb{Z}_3$ is torsion-free. Such a subgroup $T = \langle n\tau \rangle$ has the exact normalizer $C_P(T) \times T$, by splitting off its unique element with a prescribed quotient label, as in (D.70e). Its actual action on $P$ is a nonidentity inner twist of $b$, because inner twists do not change the given nonidentity integral homology matrix. The Demushkin conclusion of [42, Corollary 2 and §2] therefore makes $C_P(T)$ free pro-three. The restricted product of the two character classes is zero on every normalizer.

If their reduced cup were a nonzero surface class $\eta$, the actual product of the extensions would lift $\eta$ to $H^2(G,\mathbb{F}_3)$. It is consequently a permanent nonzero class in the highest row $E_2^{0,2}$ of the Hochschild–Serre spectral sequence. The action on $H^2(P,\mathbb{F}_3)$ is identity, since an order-three scalar in $\mathbb{F}_3^\times$ is identity. Let $z \in H^2(C_3,\mathbb{F}_3)$ be its period generator. Its inflation is permanent by the actual section, so $z\eta$ is a nonzero permanent element of $E_2^{2,2}$. There is no incoming differential to this highest row, and its outgoing differentials vanish by multiplicativity. The class $z\widetilde{\theta}\widetilde{\psi}$ is therefore nonzero in $H^4(G,\mathbb{F}_3)$. Its restrictions to all normalizers are zero. This contradicts Brown injectivity in degree four [42, Theorem 2]: $P$ is open of cohomological dimension two, and finite subgroups of $G$ inject into $C_3$, so there is no $C_3 \times C_3$. This proves the divisibility assertion. It requires no assumption that $b$ is identity on mod-three homology.

*The actual filled character module and its paired lattices.* Fill the three actual cusps $d_i = x^i c^q x^{-i}$ of $N$ and put

$$P = \left(N/\overline{\langle\langle d_0, d_1, d_2\rangle\rangle}\right)^{(3)}, \qquad p : N \twoheadrightarrow P. \tag{D.71t}$$

The same finite-target universal property as in (D.70i) identifies the quotient before maximal pro-three completion with the full completion of the closed covering surface. Thus $P$ is the genus-$(3q-1)/2$ closed-surface pro-three group. The normal closure is $\alpha$-invariant with the original labels: for $g = x^i$, write $\alpha(g)f_0^{-1} = nx^jc^k$ with $n \in N$, choosing $j \in \{0,1,2\}$ by its coset in $Q/R$ and $k$ by the rotation coordinate. This is possible because $\rho(c)$ generates $R$. Then $\alpha(gc^qg^{-1}) = nd_j^\lambda n^{-1}$. The inverse proves equality of the closed normal subgroups. Hence $\alpha$ induces an actual automorphism of $P$.

Pullback gives the exact dual injection $H^1(P,\mathbb{Z}_3) \hookrightarrow V_3$ whose image is the characters vanishing on all $d_i$: such a character factors through the full quotient and then through its maximal pro-three quotient, and the converse follows from (D.71t). The three cusp evaluations are $R$-invariant, since the rotations fix their labels and their positive generators up to conjugation. Thus $W_3$ lies in this closed character module. Its complement there is the averaged rank-two closed-torus module. The original covering homology map identifies the action on the filled rank-two homology over $\mathbb{Q}_3$ with the nonidentity matrix $A_3$. On its dual the pullback convention is $(A_3^{-1})^{\mathsf{T}}$ after that rational identification. It is nonidentity as well. Thus the induced automorphism of $P$ has order exactly three and is nonidentity on integral homology.

The rotation action extends over the three filled punctures and preserves the integral surface orientation. The odd projector $e$ is therefore self-adjoint for the unimodular alternating cup form, exactly as in (D.70p), with $q$ a unit in $\mathbb{Z}_3$. Consequently its complementary closed summand $W_3$ has a unimodular alternating form. The automorphism $S$ preserves this form: its action on the integral surface class has order dividing three in $\mathbb{Z}_3^\times$, which has no nonidentity element of order three.

Choose a finite unramified extension $\mathcal{O}/\mathbb{Z}_3$ containing the $q$th roots of unity. For a character $\psi : R \to \mathcal{O}^\times$, the actual projector is $e_\psi = q^{-1}\sum_{g\in R}\psi(g)^{-1}g$. Each nontrivial character lattice

$$V_\psi = e_\psi(W_3 \otimes \mathcal{O}) \tag{D.71u}$$

has rank three by (D.71r). The automorphism $S$ commutes with every rotation operator because $\rho\alpha = \rho$, so preserves each such lattice. The cup form pairs $V_\psi$ perfectly with $V_{\psi^{-1}}$ and pairs it trivially with every other character lattice. The inverse character is distinct, since $q$ is odd.

Let $K_3$ be the fraction field of $\mathcal{O}$. The polynomial $X^2+X+1$ is irreducible over $K_3$: a square root of $-3$ would have half-integral valuation in this unramified extension. Since $S^3 = 1$, its fixed subspace on the rank-three $K_3$-space has dimension one or three. The fixed-cup divisibility just proved holds after extension to $\mathcal{O}$. Indeed $\mathcal{O}$ is finite free over $\mathbb{Z}_3$, so kernels of $S - I$ commute with this flat scalar extension, and the bilinear divisibility passes to their scalar extensions.

Fixed dimension three is impossible. It would make $S = I$ on $V_\psi$, and preservation of the perfect paired form would give $S = I$ on $V_{\psi^{-1}}$. The perfect integral pairing would then give two

fixed characters with unit cup pairing, contradicting that divisibility. Thus both fixed lattices have rank one. Choose primitive generators $u_\psi$ and $u_{\psi^{-1}}$. They are primitive in the full lattices: each fixed kernel is saturated, since the image of $S-I$ is torsion-free. Semisimplicity over $K_3$ and preservation of the pairing make the pairing of the two fixed lines nonzero.

Unimodularity supplies $w \in V_\psi$ with $\langle w, u_{\psi^{-1}}\rangle = 1$. Writing the actual norm as $\mathcal{N} = I+S+S^2$, we have $\mathcal{N}w = b\,u_\psi$ for some $b \in \mathcal{O}$, and

$$b\,\langle u_\psi, u_{\psi^{-1}}\rangle = \langle \mathcal{N}w, u_{\psi^{-1}}\rangle = 3. \tag{D.71v}$$

Fixed-cup divisibility says the displayed pairing lies in $3\mathcal{O}$. Since $\mathcal{O}$ is unramified, (D.71v) forces that pairing to be three times a unit and forces $b$ to be a unit. The norm therefore maps $V_\psi$ onto its primitive fixed line.

Modulo three, the norm is $(S-I)^2$, and its rank on this three-dimensional block is one. As $(S-I)^3 = 0$, the block is exactly one Jordan block of size three. This also supplies an actual integral cyclic basis: lift a vector whose three cyclic images form a basis modulo three. The columns $w, Sw, S^2w$ then have unit determinant, so the map $\mathcal{O}[C_3] \to V_\psi$ sending the group generator to $S$ and 1 to $w$ is an isomorphism.

For descent, no unspecified module form is used. After the splitting residue-field extension, the norm on $W_3/3W_3$ has rank $q-1$ in dimension $3(q-1)$. Ranks are unchanged by this field extension. Since every Jordan block has size at most three and each size-three block contributes one to the norm rank, all blocks over $\mathbb{F}_3$ are of size three. Choose a basis of the actual $\mathbb{F}_3[C_3]$-module by cyclic vectors and lift them to $W_3$. The resulting explicit action map

$$\mathbb{Z}_3[C_3]^{q-1} \longrightarrow W_3, \qquad (f_i(S))_i \longmapsto \sum_i f_i(S)w_i \tag{D.71w}$$

reduces to an isomorphism modulo three. It is a map between free $\mathbb{Z}_3$-modules of the same finite rank, so its matrix has unit determinant and is invertible. Its inverse is $C_3$-equivariant. This proves the integral freeness in the statement.

*The three full section classes and the fixed-kernel map.* Put $E_N = N \rtimes \langle s\rangle$. D.48 gives $H^1(C, \mathbb{F}_3) = 0$, so $\chi(C) = 0$ and $\rho(C) \subset R$. Its image is either 1 or $R$. The other global section class has nonzero $\chi$-value by (D.71e), so none of its conjugates meets $E_N$. Every conjugate of $\langle s\rangle$ does meet $E_N$, because its image in (D.71o) is the same central order-three subgroup. There is the exact class bijection

$$Q/\rho(C) \longrightarrow \{N\text{-conjugacy classes of sections in } E_N\}, \qquad \rho(f)\rho(C) \longmapsto [f\langle s\rangle f^{-1}]. \tag{D.71x}$$

Two conjugators give the same class precisely when their ratio differs by an element of $N$ and an element of $N_F(\langle s\rangle) = C$. Conjugacy under $E_N$ gives the same classes, by removing the matching section label from a conjugator.

Let $V = H^1(N, \mathbb{F}_3)$. On $eV$ the action is two Jordan blocks of size two, while (D.71w) makes the complementary module free over $\mathbb{F}_3[C_3]$. The periodic cyclic resolution has maps $S-I$ and $I+S+S^2 = (S-I)^2$. Each size-two block contributes one dimension in every positive degree; the free blocks contribute zero. Thus

$$\dim H^j(C_3, V) = 2 \quad (j>0), \qquad \dim H^n(E_N, \mathbb{F}_3) \le 3 \quad (n \ge 2). \tag{D.71y}$$

The last bound is the actual two-row Hochschild–Serre sequence, since $N$ has cohomological dimension one.

Brown restriction applies to $E_N$: $N$ is open of dimension one, and every finite subgroup injects into $C_3$, so there is no elementary abelian subgroup of rank two. The finite-dimensional source in (D.71y) makes the dense restriction embedding onto its Hausdorff target. If $\rho(C) = 1$, (D.71x) supplies $3q$ normalizer factors, already contradicting (D.71y). Therefore $\rho(C) = R$,

proving the surjectivity in (D.71b). The classes are exactly its three cosets, represented by $1, t, t^2$. Intersecting the full direct-product normalizers with $E_N$ gives exactly (D.71c). The maps $(h, s^j) \mapsto x^i h s^j x^{-i}$ and their inverses, which read the fixed-group coordinate and the specified quotient label after conjugation by $x^{-i}$, prove these identifications as full topological groups.

Each fixed kernel $x^i C_0 x^{-i}$ is a closed subgroup of the free profinite group $N$ and has 3-cohomological dimension at most one. Kunneth and Brown therefore give, in high degrees,

$$3\big(1 + \dim H^1(C_0, \mathbb{F}_3)\big) = \dim H^n(E_N, \mathbb{F}_3) \le 3. \tag{D.71z}$$

All dimensions are nonnegative, so $H^1(C_0, \mathbb{F}_3) = 0$ and the high-degree dimension is exactly three. This completes the proof of the fixed-kernel and section assertions.

The executable certificate `code/part_c/torus_cubic_frobenius_d71.py` checks all eight based character adaptations and their word inverses, the integral covering matrices, the Frobenius quotients and cusp cycles for the first admissible primes, the graph-cycle maps with their inverses, and every rank-three order-three matrix over $\mathbb{F}_3$. The integral fixed-cup argument and the full Brown normalizers above supply the conclusions beyond those finite identities. □

**D.72. [Proved] The full mod-two cochain comparison for a pants involution.** Let $\alpha$ be the actual normalized representative of any nonidentity finite-order element of the full pure three-cusp loop-outer group. Retain all the data of D.34:

$$F = \widehat{F}(x, y), \qquad \alpha(x) = x^{-1}, \quad \alpha(y) = f^{-1} y^{-1} f, \quad f \in F', \qquad E = F \rtimes_\alpha \langle s \rangle. \tag{D.72a}$$

Write $e_2 \in \widehat{\mathbb{Z}}$ for the element whose 2-adic coordinate is one and whose other primary coordinates are zero. The three actual section generators and their label maps are

$$s_0 = s, \qquad s_1 = x^{e_2} s, \qquad s_2 = y^{e_2} f s, \qquad \iota_i : C_2 \longrightarrow E, \quad \tau \longmapsto s_i. \tag{D.72b}$$

Let $\mathcal{C}^\bullet(E)$ be the differential graded $\mathbb{F}_2$-algebra of normalized continuous group cochains with trivial coefficients, with its ordinary cup multiplication. Put

$$\mathcal{A} = \mathbb{F}_2[t_0, t_1, t_2]/(t_i t_j : i \ne j), \qquad |t_i| = 1, \qquad d_{\mathcal{A}} = 0. \tag{D.72c}$$

The following is an actual surjective quasi-isomorphism of unital differential graded algebras:

$$\begin{aligned} q : \mathcal{C}^\bullet(E) &\longrightarrow \mathcal{A}, \\ q_0(a) &= a, \\ q_n(c) &= \sum_{i=0}^{2} c(s_i, \dots, s_i) t_i^n \quad (n > 0). \end{aligned} \tag{D.72d}$$

Thus these are the complete mod-two cohomology ring and a specified formality comparison, without an assumption that the full centralizers of the sections are trivial.

Every sequence of at least three positive-degree cohomology classes whose neighboring cup products are zero has a defined higher Massey product containing zero. There is also the following exact finite-matrix image calculation. For $n \ge 1$, let $U_{n+1}(\mathbb{F}_2)$ be the group of upper unitriangular matrices, and define

$$\mathcal{X}_n = \{(\chi_1, \dots, \chi_n) \in H^1(E, \mathbb{F}_2)^n : \ \chi_j \chi_{j+1} = 0 \ (1 \le j < n)\}.$$

The map reading the first superdiagonal gives a surjection

$$\mathrm{Hom}_{\mathrm{cont}}(E, U_{n+1}(\mathbb{F}_2)) \longrightarrow \mathcal{X}_n. \tag{D.72e}$$

It has exactly the displayed image. A representing homomorphism can be chosen with the section values

$$s_i \longmapsto I + \sum_{j=1}^{n} \chi_j(s_i) E_{j,j+1} \quad (0 \le i < 3). \tag{D.72f}$$

Writing $F_0 = 0, F_1 = 1, F_{k+2} = F_{k+1} + F_k$ for the ordinary Fibonacci numbers, the number of possible superdiagonal tuples is

$$|\mathcal{X}_n| = F_{n+2}^3. \tag{D.72g}$$

No equality of full profinite groups, or solution of the third-cusp conjugator equation, is inferred from these cochain and matrix comparisons.

*Manuscript:* Section 31, folios 198-199 verso, scans 391-394; Part D(b), full pure three-cusp involution recognition and the exact remaining conjugator equation.

*Proof. The three literal restrictions and their normalizers.* D.34 proves that the elements in (D.72b) have order two and represent precisely the three $F$-conjugacy classes of sections. The chosen generator of each quotient is retained. Put $C_i = C_F(s_i)$ and $N_i = N_E(\langle s_i \rangle)$. There are inverse continuous homomorphisms

$$C_i \times C_2 \longrightarrow N_i, \quad (h, \tau^\epsilon) \longmapsto h s_i^\epsilon, \qquad w \longmapsto (w s_i^{-\epsilon}, \tau^\epsilon), \tag{D.72h}$$

where $\epsilon$ is the actual image of $w$ in $E/F = C_2$. A normalizer fixes the unique nonidentity element of its order-two subgroup, so these formulas have the stated domains and are inverse. Each $C_i$ is a closed subgroup of the free profinite group $F$; consequently it has 2-cohomological dimension at most one. D.34 gives $H^1(C_i, \mathbb{F}_2) = 0$. Hence all its positive-degree cohomology with these coefficients vanishes.

The Hochschild–Serre sequence for $C_i \times C_2 \to C_2$ has just its row zero. Inflation and restriction to its displayed section therefore give inverse isomorphisms

$$H^k(N_i, \mathbb{F}_2) \underset{\longleftarrow}{\overset{\sim}{\rightleftarrows}} H^k(C_2, \mathbb{F}_2) \qquad (k \ge 0). \tag{D.72i}$$

The Brown restriction theorem [42, Theorem 2 and the finite-dimensional argument in §1.1] applies to $E$: $F$ is open of cohomological dimension one, and every finite subgroup injects into $E/F$, so $E$ has no $C_2 \times C_2$. The two-row spectral sequence for $F \to E \to C_2$ bounds $\dim H^k(E, \mathbb{F}_2)$ by three for $k > 1$. Its dense restriction embedding is therefore onto its Hausdorff target. With the three representatives and (D.72i), it gives the actual isomorphisms

$$(\iota_0^*, \iota_1^*, \iota_2^*) : H^k(E, \mathbb{F}_2) \xrightarrow{\sim} \mathbb{F}_2^3 \qquad (k > 1). \tag{D.72j}$$

*Degree one and cochains in every degree.* Let $u : E \to \mathbb{F}_2$ be the quotient character. Let $\xi_x, \xi_y : E \to \mathbb{F}_2$ read the two mod-two abelianization coordinates on $F$, and be zero on $s$. These are actual characters because $\alpha$ is minus identity on the full abelianization and therefore identity modulo two. Every character of $E$ is uniquely determined by its values on $x, y, s$, and each triple of values is permitted by the semidirect relations. Thus $u, \xi_x, \xi_y$ form a basis, with the exact restriction table

| | $s_0$ | $s_1$ | $s_2$ |
|---|---|---|---|
| $u$ | 1 | 1 | 1 |
| $\xi_x$ | 0 | 1 | 0 |
| $\xi_y$ | 0 | 0 | 1. |

(D.72k)

Its inverse basis is $\eta_0 = u + \xi_x + \xi_y, \eta_1 = \xi_x, \eta_2 = \xi_y$; each $\eta_i$ restricts to one on $s_i$ and zero on the other two generators. Consequently (D.72j) also holds in degree one. Degree zero is the common scalar field, as in (D.72d).

For an order-two group with generator $\tau$, a normalized cochain in degree $k > 0$ is determined by its value on $(\tau, \dots, \tau)$. The differential is zero: all internal terms have an identity argument, and the two endpoint terms cancel in characteristic two. The cup product multiplies the two specified values. Its normalized cochain algebra is therefore literally $\mathbb{F}_2[t]$, with $t$ in degree one and zero differential. The three restrictions agree in degree zero and give exactly (D.72d), including its cup multiplication and its unit. This proves that $q$ is a differential graded algebra homomorphism. The cohomology calculations above prove that it is a quasi-isomorphism.

Surjectivity is also literal on cochains. Define the finite quotient character

$$\rho = (u, \xi_x, \xi_y) : E \longrightarrow \mathbb{F}_2^3, \qquad v_0 = (1,0,0), \quad v_1 = (1,1,0), \quad v_2 = (1,0,1). \tag{D.72l}$$

For $k > 0$, the continuous normalized cochain

$$e_{i,k}(g_1, \dots, g_k) = \begin{cases} 1, & \rho(g_1) = \cdots = \rho(g_k) = v_i, \\ 0, & \text{otherwise} \end{cases} \tag{D.72m}$$

satisfies $q(e_{i,k}) = t_i^k$. The values $v_i$ are distinct and nonzero, so this formula is normalized and gives all three basis vectors. It factors through the stated finite quotient; no extension of a function from an unspecified subset is needed. Therefore $\mathcal{K}^\bullet = \ker q$ is an acyclic differential graded ideal, by the cohomology exact sequence of this degreewise surjection.

The target is the graded algebra associated with the eight-element Boolean ring $\mathbb{F}_2^3$ in [59, Definition 6.1]. The displayed restriction is a direct formality proof for these particular extensions; it does not apply the real-projective hypothesis to $E$.

*All defining systems and their continuous lifts.* Take $n \geq 3$ positive-degree classes $a_1, \dots, a_n$, with $a_j \in H^{d_j}(E, \mathbb{F}_2)$, whose neighboring products are zero. Their images in $\mathcal{A}$ have the same property, by the proved cohomology isomorphism. Choose cocycles $c_{j,j+1}$ representing them. Their $q$-values are exactly the specified images, because the target differential is zero.

We now choose cochains $c_{i,j}$ for every gap $j - i \geq 2$, including $(i,j) = (1, n+1)$, with

$$|c_{i,j}| = \sum_{k=i}^{j-1} d_k - (j - i - 1), \qquad q(c_{i,j}) = 0, \qquad dc_{i,j} = \sum_{i<k<j} c_{i,k} \smile c_{k,j}. \tag{D.72n}$$

Suppose all smaller gaps have been chosen. The right side is a cocycle: applying $d$ and using the equations for smaller gaps makes every triple product appear twice, so the terms cancel. Its $q$-image is zero. At gap two this is exactly the assumed neighboring product. At a larger gap, each summand has a factor from a previously chosen higher entry with $q$-value zero. The cocycle is therefore in $\mathcal{K}^\bullet$. Acyclicity gives the required $c_{i,j}$ inside that kernel. All its degrees are positive because every $d_j \geq 1$. All cochains in this construction are continuous, since the complex and its kernel consist of continuous cochains throughout.

Omitting the final entry gives a defining system for the higher Massey product. Including it in (D.72n) exhibits its value as a coboundary. Thus the product is defined and contains zero, with every degree and every actual lifting complex specified.

*The exact upper unitriangular image.* For a tuple in $\mathcal{X}_n$, apply the same construction with $d_j = 1$ and $c_{j,j+1} = \chi_j$. The gap recursion also applies for $n = 2$: its single higher entry is obtained from acyclicity of $\mathcal{K}^\bullet$ in exactly the displayed degree. All entries are now normalized one-cochains. Set

$$M(g)_{i,i} = 1, \qquad M(g)_{i,j} = c_{i,j}(g)(i < j), \qquad M(g)_{i,j} = 0(i > j). \tag{D.72o}$$

For a one-cochain the equation in (D.72n) reads

$$c_{i,j}(gh) = c_{i,j}(g) + c_{i,j}(h) + \sum_{i<k<j} c_{i,k}(g) c_{k,j}(h).$$

These are exactly all entries of $M(gh) = M(g)M(h)$. The finite matrix-valued map is continuous and has the required superdiagonal. Its higher entries vanish on every $s_i$ because they lie in $\ker q$, proving (D.72f). Conversely any continuous unitriangular representation has $dc_{j,j+2} = \chi_j \smile \chi_{j+1}$, so its superdiagonal belongs to $\mathcal{X}_n$. This proves both directions of the image statement. For $n = 1$ the map is just the character itself.

Finally use the isomorphism (D.72k) to record each $\chi_j$ by its three section values. The neighboring cup product vanishes exactly when those two three-bit vectors have disjoint supports, by (D.72j). Each of the three coordinates therefore gives a binary string of length $n$ with no two consecutive ones. Their number $b_n$ satisfies $b_0 = 1, b_1 = 2$ and $b_n = b_{n-1} + b_{n-2}$ by its final digit. Hence $b_n = F_{n+2}$, and the three independent coordinates give (D.72g).

The same three restriction maps for the standard involution $\alpha_0(x) = x^{-1}, \alpha_0(y) = y^{-1}$ give the explicit zigzag

$$\mathcal{C}^\bullet(E) \xrightarrow{q} \mathcal{A} \xleftarrow{q_{\mathrm{std}}} \mathcal{C}^\bullet(F \rtimes_{\alpha_0} C_2). \tag{D.72p}$$

Both arrows are the specified quasi-isomorphisms of differential graded algebras. This is the exact comparison established here. It leaves the third-cusp equation of D.35 and the full section-centralizer equations unchanged. □

**D.73. [Proved] The full $q$-adic cubic quotient and its section kernels.** Retain the actual action, based generators, character and cusp labels of D.71 at a prime $q \neq 3$ with $\lambda_q \neq 1$. Write $\omega = \lambda_q \in \mathbb{Z}_q^\times$, so that $\omega^2 + \omega + 1 = 0$ and $\omega - 1$ is a unit. There is an actual quotient

$$\begin{gathered} Q_\infty = \mathbb{Z}_q \rtimes_\omega C_3, \qquad (a,i)(b,j) = (a + \omega^i b, i + j), \\ \rho_\infty : F \twoheadrightarrow Q_\infty, \qquad \rho_\infty(x) = t = (0,1), \quad \rho_\infty(y) = r = (1,0), \quad \rho_\infty \alpha = \rho_\infty. \end{gathered} \tag{D.73a}$$

The labelled extension quotient is $E/\ker \rho_\infty = Q_\infty \times \langle s \rangle$. For $n \geq 1$, let $\rho_n$ be reduction modulo $q^n$, and put

$$\begin{gathered} N_n = \ker \rho_n, \qquad N_\infty = \ker \rho_\infty, \qquad C = C_F(s), \\ C_n = C \cap N_n, \qquad D = C \cap N_\infty. \end{gathered} \tag{D.73b}$$

The finite-level cover has type $((3q^n - 1)/2, 3)$, and the following are exact:

$$\begin{gathered} \rho_\infty(C) = \mathbb{Z}_q, \qquad H^1(C_n, \mathbb{F}_3) = 0 \quad (n \geq 1), \qquad H^1(D, \mathbb{F}_3) = 0, \\ 1 \longrightarrow D \longrightarrow C \xrightarrow{\rho_\infty|_C} \mathbb{Z}_q \longrightarrow 1. \end{gathered} \tag{D.73c}$$

The last sequence splits by a specified procyclic-power construction in the proof. In particular $E$ contains the actual subgroup $\mathbb{Z}_q \times \langle s \rangle$ supplied by that section.

For both $n \geq 1$ and $n = \infty$, there are exactly three $N_n$-conjugacy classes of sections in $E_n = N_n \rtimes \langle s \rangle$. They have representatives

$$T_i = x^i \langle s \rangle x^{-i} \quad (0 \leq i < 3), \tag{D.73d}$$

and their full normalizers are, respectively, $x^i(C_n \times \langle s \rangle)x^{-i}$ and $x^i(D \times \langle s \rangle)x^{-i}$. At each finite level, for the actual rotation projector $e_n$ on $V_n = H^1(N_n, \mathbb{Z}_3)$, the pullback action of $\alpha^{-1}$ gives

$$(1 - e_n)V_n \cong \mathbb{Z}_3[C_3]^{\,q^n - 1}; \qquad e_n V_n \bmod 3 \cong J_2 \oplus J_2. \tag{D.73e}$$

The first isomorphism is obtained by lifting cyclic vectors, with the actual action retained.

There is also an exact finite obstruction to two-section generation: the closed image of every continuous homomorphism

$$C_3 \amalg C_3 \longrightarrow E \tag{D.73f}$$

is proper. This is proved in the actual quotient $Q_1 \times C_3$, with both full section classes labelled below. These statements retain the remaining cubic-coordinate question: they do not assert

existence or exclusion of that coordinate, or a presentation of the full extension by its section subgroups.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The quotient before reduction at $q$.* The based character change in D.71 keeps $c = [x, y]$ exactly and gives $\chi(x) = 1, \chi(y) = 0$. Its index-three cover has

$$F_0 = \ker \chi, \qquad u = x^3, \quad B_i = x^i y x^{-i} \ (0 \le i < 3), \qquad B_3 = uB_0u^{-1}. \tag{D.73g}$$

Use the continuous additive character already constructed integrally in (D.71l)–(D.71m):

$$\ell_q : F_0 \longrightarrow \mathbb{Z}_q, \qquad \ell_q(u) = 0, \quad \ell_q(B_i) = \omega^i, \qquad \ell_q\alpha = \ell_q. \tag{D.73h}$$

In particular its image is all of $\mathbb{Z}_q$, since it contains 1. Conjugation by $x$ multiplies it by $\omega$; conjugation by $F_0$ fixes it. Thus its kernel is a closed normal subgroup of $F$, preserved in both directions by the actual automorphism $\alpha$.

The universal homomorphism from the full free profinite group which sends $x$ to $t$ and $y$ to $r$ in (D.73a) is onto: the closure of the powers of $r$ is the entire normal $\mathbb{Z}_q$, and $t$ gives the complement. Its restriction to $F_0$, followed by the rotation coordinate, has exactly the four values (D.73h), so its kernel is $\ker \ell_q$. The inverse quotient map is

$$Q_\infty \longrightarrow F/\ker \ell_q, \qquad (a, i) \longmapsto y^{e_q a} x^i \ker \ell_q, \tag{D.73i}$$

where $e_q a \in \widehat{\mathbb{Z}}$ has $q$-coordinate $a$ and every other primary coordinate zero. Here $y^{e_q a} \ker \ell_q$ is the rotation $a$, $x^3 \in \ker \ell_q$, and conjugation by $x$ is multiplication by $\omega$. These verify multiplication and both inverse compositions, including the labels.

The induced automorphism fixes $r$ and sends $t$ to $r^j t$ for some $j \in \mathbb{Z}_q$. Since it fixes every rotation, its actual cube sends $t$ to $r^{3j}t$. The equality $\alpha^3 = 1$ and invertibility of 3 in $\mathbb{Z}_q$ give $j = 0$. This proves $\rho_\infty\alpha = \rho_\infty$. The extension comparison and its inverse are consequently

$$\begin{aligned} E/N_\infty &\longrightarrow Q_\infty \times C_3, \quad fs^jN_\infty \longmapsto (\rho_\infty(f), s^j), \\ Q_\infty \times C_3 &\longrightarrow E/N_\infty, \qquad (z, s^j) \longmapsto fs^jN_\infty \quad (\rho_\infty(f) = z). \end{aligned} \tag{D.73j}$$

The choice of $f$ changes neither coset, and the maps are inverse continuous homomorphisms.

*The fixed subgroup and its explicit procyclic section.* D.48 gives $H^1(C, \mathbb{F}_3) = 0$. Therefore $\chi(C) = 0$, so $\rho_\infty(C)$ lies in the rotation $\mathbb{Z}_q$. It is closed, since it is a continuous image of the compact group $C$. D.71 proves that its reduction modulo $q$ is all of $C_q$. Every nonzero closed subgroup of the additive group $\mathbb{Z}_q$ is $q^a\mathbb{Z}_q$: choose an element with least valuation, whose integer powers are dense in that subgroup. Reduction modulo $q$ thus forces $a = 0$. This proves the surjection in (D.73c).

Choose $h \in C$ with $\rho_\infty(h) = r$. For $a \in \mathbb{Z}_q$ define

$$\iota(a) = h^{e_q a}. \tag{D.73k}$$

Continuous powers in the actual procyclic closure of $\langle h \rangle$ show that this is a continuous homomorphism. Its composite with $\rho_\infty$ is $a \mapsto r^a$, so it is injective and splits the displayed surjection. In particular the domain is the full $\mathbb{Z}_q$, rather than a finite cyclic quotient. With the conjugation action given by this actual section there are inverse maps

$$\begin{aligned} D \rtimes_\iota \mathbb{Z}_q &\longrightarrow C, \qquad (d, a) \longmapsto d\iota(a), \\ C &\longrightarrow D \rtimes_\iota \mathbb{Z}_q, \qquad h' \longmapsto \big(h'\iota(\rho_\infty(h'))^{-1}, \rho_\infty(h')\big). \end{aligned} \tag{D.73l}$$

In these expressions the value of $\rho_\infty$ on $C$ means its additive rotation coordinate. Every $\iota(a)$ commutes with $s$. Thus $(a, s^j) \mapsto \iota(a)s^j$ gives the asserted embedded direct product, with inverse given by the rotation and original quotient-generator labels.

*Every finite power cover, with its integral lattice.* Put $m = q^n$, and let $\omega_n$ denote the reduction of $\omega$ modulo $m$. The quotient $Q_n$ has order $3m$, with exact multiplication obtained by reducing (D.73a); the actual quotient maps between levels are reduction of the first coordinate. The cusp $c = [x, y]$ has image $r^{\omega_n - 1}$, which generates the rotation $C_m$, since $\omega - 1$ is a $q$-adic unit. It has three cycles of length $m$ on the $3m$ covering sheets. Thus $N_n$ has three cusps and Euler characteristic $-3m$, giving the stated type $((3m-1)/2, 3)$. A set of actual cusp representatives is

$$d_i = x^i c^m x^{-i} \quad (0 \le i < 3). \tag{D.73m}$$

Here the graph-cycle computation in D.71 has the following literal version for $m$. Set $\Lambda_m = \mathbb{Z}_3[C_m]$, $\nu_m = 1 + r + \cdots + r^{m-1}$, and let $a, b \in \{1, \dots, m-1\}$ represent $\omega_n, \omega_n^2$. The four Schreier generators $u, B_0, B_1, B_2$ have rotation images $0, 1, a, b$. The graph boundary and its inverse cycle coordinates are

$$\begin{aligned} \partial(A, B, D', E') &= (r-1)B + (r^a - 1)D' + (r^b - 1)E', \\ (A, D', E', k) &\longmapsto (A, -P_a D' - P_b E' + k\nu_m, D', E'), \\ P_j &= 1 + r + \cdots + r^{j-1}. \end{aligned} \tag{D.73n}$$

The inverse reads $A, D', E'$ and the unique $k$ for which $B + P_a D' + P_b E' = k\nu_m$. Indeed the annihilator of $r - 1$ in $\Lambda_m$ is exactly $\mathbb{Z}_3\nu_m$, by comparing adjacent coefficients. These are actual inverse $\Lambda_m$-module maps, so

$$H_1(N_n, \mathbb{Z}_3) \cong \mathbb{Z}_3 \oplus \Lambda_m^3. \tag{D.73o}$$

Duality gives the same rotation multiplicities for $V_n$. The projector $e_n = m^{-1} \sum_{g \in C_m} g$ is integral, and restriction and transfer identify $e_n V_n$ with the actual rank-four module $H^1(F_0, \mathbb{Z}_3)$. They are equivariant for $\alpha$, since the actual quotient action is the identity. Its two Jordan-two blocks are exactly those computed in (D.71s).

For completeness, the paired-lattice argument applies at every level with the following precise quotient maps. Fill the three actual cusps (D.73m), then take the maximal pro-three quotient:

$$P_n = \left(N_n / \overline{\langle\langle d_0, d_1, d_2 \rangle\rangle}\right)^{(3)}. \tag{D.73p}$$

It is the closed orientable genus-$(3m-1)/2$ surface pro-three group by the same finite-target filling universal property used in D.71. Write the cusp normalizer element as $v = f_0 s$; then $\alpha(c) = f_0^{-1} c^\lambda f_0$. For $g = x^i$, the quotient $Q_n / C_m$ selects a unique $j \in \{0, 1, 2\}$; since $\rho_n(c)$ generates $C_m$, there is $k \in \mathbb{Z}$ and $h \in N_n$ such that

$$\alpha(g) f_0^{-1} = h x^j c^k, \qquad \alpha(g c^m g^{-1}) = h d_j^\lambda h^{-1}. \tag{D.73q}$$

The inverse automorphism gives the opposite inclusion of the normal closures. Thus the actual action descends to (D.73p). The injection $H^1(P_n, \mathbb{Z}_3) \to V_n$ has as its image exactly the characters killing the three $d_i$. All cusp evaluations are rotation-invariant, so $W_n = (1 - e_n)V_n$ lies in this image. The averaged filled summand has rank two, with the nonidentity integral action dual to the base torus matrix $A_3$; this is the same restriction/transfer comparison and cusp vanishing test as in D.71, with $m$ a unit in $\mathbb{Z}_3$. The rotation deck transformations extend positively over the filled cusps. Hence $e_n$ is self-adjoint for the unimodular alternating cup form, and its orthogonal summand $W_n$ is integrally unimodular.

Choose a finite unramified extension $\mathcal{O}/\mathbb{Z}_3$ containing all $m$-th roots of unity. The idempotents

$$e_\psi = \frac{1}{m} \sum_{g \in C_m} \psi(g)^{-1} g \tag{D.73r}$$

split $W_n \otimes \mathcal{O}$ into rank-three lattices for all nontrivial $\psi$. Since $m$ is odd, $\psi \neq \psi^{-1}$; the cup form pairs precisely those two lattices, perfectly. The action $S$, pullback under $\alpha^{-1}$, commutes with all these projectors. Its action on $H^2(P_n, \mathbb{Z}_3)$ is trivial: it has order dividing three, whereas $\mathbb{Z}_3^\times$ has no nontrivial element of order three. Thus $S$ preserves the cup form used here.

The integral fixed-cup lemma proved in D.71 applies to the actual action on $P_n$. Its nonidentity hypothesis, including every inner-twisted section action, follows from the retained rank-two averaged action just specified. Thus pairings of two $S$-fixed integral characters are divisible by three, and the same is true after flat extension to $\mathcal{O}$. The polynomial $X^2 + X + 1$ remains irreducible over the fraction field of this unramified extension. On each rank-three character lattice the fixed rank is therefore one or three. Fixed rank three on one lattice would give the identity there and on its perfectly paired inverse lattice, contradicting that divisibility and the perfect form. The fixed rank is consequently one on both. Choose primitive fixed generators $u, u'$. For $w$ with $\langle w, u' \rangle = 1$, write

$$(1 + S + S^2)w = bu, \qquad b\langle u, u' \rangle = 3. \tag{D.73s}$$

The divisibility and unramified valuation show that $\langle u, u' \rangle$ is three times a unit and $b$ is a unit. Thus the norm has rank one modulo three on this rank-three lattice, giving one Jordan block $J_3$. After scalar extension all nontrivial character summands have this property. Hence over $\mathbb{F}_3$ the norm has rank $m - 1$ on a space of dimension $3(m - 1)$. Every Jordan block has size at most three; the rank forces every one to have size three. Lift cyclic vectors for these actual blocks. The resulting homomorphism $\mathbb{Z}_3[C_3]^{m-1} \to W_n$ reduces to an isomorphism modulo three, hence has unit determinant and an equivariant inverse. This proves (D.73e) at every finite level, without a descent or generation assumption about the full groups.

*The exact section classes and their fixed kernels.* The other full section generator $s_1$ has $\chi(s_1) = -1$, whereas $\chi(s) = 0$, as in (D.71e). Consequently none of its $F$-conjugates belongs to $E_n$, for finite $n$ or for $n = \infty$. Every $F$-conjugate of $s$ does belong, because the image of $s$ in (D.73j) is central. The criterion for two of these sections to be $N_n$-conjugate gives the exact bijections

$$\begin{aligned} Q_n/\rho_n(C) &\longrightarrow \{N_n\text{-classes of sections in } E_n\}, \\ \rho_n(f)\rho_n(C) &\longmapsto [f\langle s \rangle f^{-1}]. \end{aligned} \tag{D.73t}$$

The identical maps apply to $Q_\infty$. Indeed a conjugacy between the labelled generators holds precisely when the corresponding elements of $F$ differ by a left $N_n$-factor and a right $C$-factor. An $E_n$-conjugator may be multiplied by an element of the source section to remove its quotient label, so it gives the same classes as $N_n$-conjugacy. The proved $\rho_\infty(C) = \mathbb{Z}_q$ makes every coset space in (D.73t) the three cosets represented by $1, t, t^2$. Any normalizer fixes the labelled generator, because its conjugation action on $E/F = C_3$ is trivial. Removing its quotient power gives exactly the normalizers stated after (D.73d).

At finite level, the LHS sequence for $N_n \to E_n \to C_3$, together with (D.73e), bounds

$$\dim H^j(E_n, \mathbb{F}_3) \leq 3 \qquad (j \geq 2). \tag{D.73u}$$

In fact the two $J_2$ blocks each contribute one to positive cyclic-group cohomology, while the regular blocks contribute zero. The subgroup $N_n$ is open with cohomological dimension one, and every finite subgroup of $E_n$ injects into $C_3$. The full profinite Brown restriction theorem [42, Theorem 2 and §1.1] therefore applies, with no $C_3 \times C_3$ subgroup. The displayed bound makes its dense restriction embedding onto. Each $C_n$ is closed in $F$, of 3-cohomological dimension at most one. The three actual direct-product normalizers give

$$3(1 + \dim H^1(C_n, \mathbb{F}_3)) = \dim H^j(E_n, \mathbb{F}_3) \leq 3 \qquad (j \geq 2). \tag{D.73v}$$

This proves the finite-level vanishing.

It also proves the vanishing for $D$, as follows directly on continuous characters. Let $\xi : D \to \mathbb{F}_3$ be continuous. Choose an open normal subgroup $U$ of $C$ with $D \cap U \subseteq \ker \xi$. Then $\xi$ descends to $D/(D \cap U) \cong DU/U$ and inflates to a character of $DU$. The quotient $C/D = \mathbb{Z}_q$ identifies the open subgroup $DU$ with the preimage of some $q^n\mathbb{Z}_q$, including possibly $n = 0$. Thus $DU = C_n$ when $n \geq 1$, or $DU = C$ when $n = 0$. All these character groups vanish, by (D.73v) and D.48. Hence $\xi = 0$. The normalizer and class statements at $n = \infty$ were already proved by the actual coset maps; no finite-type surface assertion for $N_\infty$ is made.

*The finite image of any two torsion generators.* Reduce (D.73j) modulo $q$, writing $Q_1 = \langle r, t \rangle$. Every nonidentity finite-order element of $E$ has order three, since $F$ is torsion-free and its image in $E/F$ is injective. Replace it by its inverse if needed to give the specified quotient-generator label $s$. The two full section classes then have images

$$s, \qquad r^a t^{-1} s \quad (a \in \mathbb{F}_q). \tag{D.73w}$$

For the first class this follows from centrality of $s$ in the quotient; for the other it follows from $\chi(s_1) = -1$, and conjugation by rotations gives exactly the displayed possibilities because $\omega_1 - 1$ is a unit.

Two elements from the first class generate at most $\langle s \rangle$. One element from each class generates the order-nine abelian subgroup $\langle s, r^a t^{-1} s \rangle$, which omits the order-$q$ rotation. Two from the second class lie in the kernel of the explicit homomorphism

$$Q_1 \times C_3 \longrightarrow C_3, \qquad (r^a t^i, s^j) \longmapsto i + j. \tag{D.73x}$$

This is again a proper subgroup. The conclusions also hold if one image is the identity. Thus every subgroup generated by two finite-order elements has proper image in this particular finite quotient. A homomorphism from the full free profinite product in (D.73f) has precisely such generator images, proving its proper-image assertion. No converse generation statement is used.

The exact producer `code/part_c/torus_cubic_qadic_d73.py` checks the compatible prime-power roots, quotient coordinate maps, three cusp cycles, graph-module inverse formulas, procyclic primary splitting coordinates, and all two-section image types in the specified finite cases. It passes 155,189 exact checks over fourteen prime-power cases and 3,252 section-image pairs. The infinite quotient, fixed-kernel vanishing and integral lattice conclusions are the actual maps and cohomology arguments above. □

**D.74. [Proved] Full section centralizers from twisted relative duality.** Retain the actual normalized involution, full profinite group, and labelled extension of D.34:

$$\begin{aligned} F = \widehat{F}(x, y), \qquad z = (xy)^{-1}, \qquad E = F \rtimes_\alpha \langle s \rangle, \\ \alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1} y^{-1} f, \qquad f \in F'. \end{aligned} \tag{D.74.1}$$

Thus $\alpha^2 = 1$, and its multiplier at every prime is the global element $-1$. Put

$$s_0 = s, \qquad s_1 = x^{e_2} s, \qquad s_2 = y^{e_2} f s, \qquad S_i = \langle s_i \rangle, \qquad C_i = C_F(s_i). \tag{D.74.2}$$

Then

$$C_i = 1, \qquad N_E(S_i) = S_i \quad (0 \leq i < 3). \tag{D.74.3}$$

These equalities concern the full profinite groups. The extension $E$ is real projective in Haran's sense, with the actual finite embedding-problem formulation given below. There is exactly one $h \in F$ satisfying $f = h^{-1}\alpha(h)$; it lies in $F'$. For $\alpha \in GT_0$, this same unique element satisfies $h\theta(h) = 1$, where $\theta$ exchanges $x$ and $y$.

The proof gives an exact finite-coefficient boundary restriction calculation. D.35's remaining peripheral conjugator problem is the displayed equation (D.74.40) with this unique $h$. Its solution and a free-product presentation of $E$ are not assertions of this result.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Section 24, folio 147 verso, scan 293; Part D(b), full loop-outer finite actions in type (0,3).

*Proof. The three actual peripheral extensions.* Let $L_1 = \overline{\langle x \rangle}$, $L_2 = \overline{\langle y \rangle}$, and $L_3 = \overline{\langle z \rangle}$, and set $D_j = N_E(L_j)$. A.7 and D.34 give the exact maps

$$\begin{aligned} 1 \longrightarrow L_j \longrightarrow D_j \xrightarrow{u} C_2 \longrightarrow 1, \\ \widehat{\mathbb{Z}} \rtimes_{-1} C_2 \longrightarrow D_j, \qquad (a, \tau^\epsilon) \longmapsto c_j^a b_j^\epsilon, \end{aligned} \tag{D.74.4}$$

where $(c_1, c_2, c_3) = (x, y, z)$ and $b_j$ is an actual involution in $D_j$ with $u(b_j) = 1$. The inverse of the second map reads $\epsilon = u(d)$ and then the unique exponent of $db_j^{-\epsilon} \in L_j$. Here and throughout, $u : E \to C_2 = \mathbb{F}_2$ is the specified quotient map.

The two $D_j$-conjugacy classes of subgroups of order two are represented by $\langle b_j \rangle$ and $\langle c_j^{e_2} b_j \rangle$. Each has normalizer equal to itself in $D_j$: an element of $L_j$ commuting with either involution has exponent $a$ with $2a = 0$, and hence is the identity. Every finite subgroup of $E$ injects into $E/F$, because $F$ is torsion-free. D.34 identifies all three $E$-conjugacy classes of nontrivial finite subgroups as $S_0, S_1, S_2$ and proves that the six occurrences just listed form the endpoints of a triangle. Consequently each $S_i$ occurs exactly twice among those six classes.

The full normalizers are already specified without any assumption on $C_i$. There are inverse continuous maps

$$\begin{aligned} C_i \times S_i \longrightarrow N_E(S_i), \quad (h, s_i^\epsilon) \longmapsto h s_i^\epsilon, \\ N_E(S_i) \longrightarrow C_i \times S_i, \qquad w \longmapsto (w s_i^{-\epsilon}, s_i^\epsilon), \quad \epsilon = u(w). \end{aligned} \tag{D.74.5}$$

A normalizer fixes the unique nonidentity element of $S_i$, which proves that the second formula has its stated target. These are the normalizer maps that will occur in every restriction below.

*One actual coefficient module for the relative theory.* Write $k = \mathbb{F}_2$. Form the compact permutation $k[[E]]$-module and its augmentation kernel

$$P = \bigoplus_{j=1}^{3} k[[E/D_j]], \qquad 0 \longrightarrow \Delta \longrightarrow P \xrightarrow{\varepsilon} k \longrightarrow 0. \tag{D.74.6}$$

There is an actual $F$-equivariant homeomorphism

$$F/L_j \longrightarrow E/D_j, \qquad aL_j \longmapsto aD_j. \tag{D.74.7}$$

It is injective since $F \cap D_j = L_j$ and surjective since $E = FD_j$; compactness gives its inverse continuity. Thus $\Delta|_F$ is precisely the augmentation kernel for the three specified full peripheral subgroups of $F$. The additional $E$-action is retained, rather than replacing it by an action fixing the $L_j$ pointwise. In the coordinates of (D.74.7), it is

$$e \cdot (aL_j) = e a b_j^{-u(e)} L_j. \tag{D.74.8}$$

The displayed representative lies in $F$. Changing $a$ by $a\ell$, $\ell \in L_j$, gives the same coset because $b_j$ normalizes $L_j$. This is the action transported from $E/D_j$, so it obeys the group law.

For a finite continuous $E$-module $M$ over $k$, put

$$\mathscr{H}_M = \mathrm{Hom}_{k,\mathrm{cont}}(\Delta, M), \qquad (e\varphi)(d) = e\varphi(e^{-1}d). \tag{D.74.9}$$

This is a discrete continuous $E$-module. A continuous map from the compact module $\Delta$ to a finite discrete module factors through a finite quotient; the compact-open topology on this Hom space is discrete, and each such map has an open stabilizer. In particular ordinary continuous group cohomology with coefficients $\mathscr{H}_M$ is defined.

The relative groups used here are, in every positive degree, the following specified ordinary groups:

$$R^n(E,M) := H^n(E,\{D_j\};M) = H^{n-1}(E,\mathscr{H}_M) \qquad (n \geq 1). \tag{D.74.10}$$

The corresponding formula for $F$ uses precisely $\Delta|_F$ and the $L_j$ by (D.74.7). For clarity, applying continuous Hom to (D.74.6) gives the exact sequence of discrete $E$-modules

$$0 \longrightarrow M \longrightarrow \mathrm{Hom}_{k,\mathrm{cont}}(P,M) \longrightarrow \mathscr{H}_M \longrightarrow 0. \tag{D.74.11}$$

Exactness follows also directly because (D.74.6) splits as a sequence of compact $k$-vector spaces. Shapiro on the three summands gives the actual long exact sequence

$$\cdots \longrightarrow H^n(E,M) \xrightarrow{r} \bigoplus_{j=1}^{3} H^n(D_j,M) \xrightarrow{\delta} R^{n+1}(E,M) \xrightarrow{j} H^{n+1}(E,M) \longrightarrow \cdots. \tag{D.74.12}$$

Here $r$ consists of literal subgroup restrictions. These are the augmentation and Hom constructions of [72, Definition 2.1, Propositions 2.3–2.4 and 2.10]. In particular the spectral sequence below will be the ordinary extension spectral sequence for the actual coefficient module (D.74.9).

*The pairing and its equivariance.* Let $M^\vee = \mathrm{Hom}_k(M,k)$, with $(e\ell)(m) = \ell(e^{-1}m)$. There is the actual $E$-equivariant coefficient pairing

$$M \otimes_k \mathscr{H}_{M^\vee} \longrightarrow \mathscr{H}_k, \qquad m \otimes \varphi \longmapsto \bigl[d \longmapsto \varphi(d)(m)\bigr]. \tag{D.74.13}$$

Its cup products identify, under (D.74.10), with the ordinary absolute-relative cup products. This identification can be made on the completed bar resolution: tensor that resolution over $k$ with $\Delta$, with diagonal group action, and use the continuous adjunction

$$\mathrm{Hom}_F(P_\bullet(F)\widehat{\otimes}_k\Delta, M) \simeq \mathrm{Hom}_F(P_\bullet(F), \mathscr{H}_M). \tag{D.74.14}$$

The augmentation makes the tensor complex a resolution of $\Delta$; its terms are projective, since the diagonal action on a free group-module factor can be untwisted and compact $k$-vector spaces are projective over $k$. The Alexander–Whitney diagonal followed by (D.74.13) is the stated cup product. This also proves compatibility with the coefficient sequence (D.74.11) and its connecting maps. Equivalently these compatibilities are the natural cup constructions in [72, Section 3, especially Propositions 3.1 and 3.5].

The full profinite pair $(F,\{L_j\})$ is the completion of the ordinary pair of pants with its three boundary subgroups. Its mod-two duality is the actual cup duality

$$\begin{aligned} H^0(F,M) \otimes H^2(F,\{L_j\};M^\vee) &\longrightarrow k, \\ H^1(F,M) \otimes H^1(F,\{L_j\};M^\vee) &\longrightarrow k, \end{aligned} \tag{D.74.15}$$

and both pairings are perfect. One direct comparison is restriction along the dense inclusions $F(x,y) \to F$ and $\langle c_j\rangle \to L_j$. For finite coefficient modules these induce the usual cohomology isomorphisms for a free group and for an infinite cyclic group. The relative long exact sequences give the relative isomorphisms, with the same cup products by bar restriction. Thus the perfect pairings are those of the compact surface with its boundary, including its original peripheral maps. The completed surface-pair statement and its cup formulation are also [72, Proposition 6.16 and Proposition 5.7].

All these pairings are $E/F$-equivariant. Indeed (D.74.13) is $E$-equivariant, and the target

$$H^1(F,\mathscr{H}_k) = H^2(F,\{L_j\};k) = k \tag{D.74.16}$$

has trivial $E/F$-action. This last assertion is literal: the target is one-dimensional over $k$, so its automorphism group is trivial. Its fundamental coclass is the boundary-sum coclass; the

conjugations in (D.74.8) act by inversion on each peripheral exponent and hence act trivially on every mod-two boundary character. No geometric representative for $\alpha$ is needed for this equivariance.

Put $A = H^0(F, M)$ and $B = H^1(F, M)$. They are finite $k[C_2]$-modules. Equations (D.74.10) and (D.74.15) give the following *equivariant* identifications:

$$H^0(F, \mathscr{H}_{M^\vee}) = B^\vee, \qquad H^1(F, \mathscr{H}_{M^\vee}) = A^\vee. \tag{D.74.17}$$

Every higher row is zero, since $\mathrm{cd}_2(F) = 1$.

*The two transgressions have equal rank.* For a finite $k[C_2]$-module $V$, write $T = 1 + \sigma$ and

$$\overline{V} = \ker T / \operatorname{im} T. \tag{D.74.18}$$

The normalized positive-degree cyclic cochain complex has value group $V$ in every degree and differential $T$. Thus $H^a(C_2, V) = \overline{V}$ for $a \geq 1$. Multiplication by the standard class $u \in H^1(C_2, k)$ is the identity under these identifications. The pairing

$$\overline{V} \otimes \overline{V^\vee} \longrightarrow k \tag{D.74.19}$$

induced by evaluation is perfect. To check this, the annihilator of $\ker T$ is $\operatorname{im} T^\vee$, and the annihilator of $\operatorname{im} T$ is $\ker T^\vee$. The radicals on the kernels are therefore precisely the images being divided out. On invariant representatives, the cyclic cup formula is evaluation itself, since every factor $\sigma^a$ acts trivially on such a representative.

Apply the ordinary Hochschild–Serre spectral sequence to $F \to E \to C_2$, first with coefficient $M$ and then with coefficient $\mathscr{H}_{M^\vee}$. Its $E_2$ terms in the two cases are

$$H^a(C_2, H^b(F, M)), \qquad H^a(C_2, H^b(F, \mathscr{H}_{M^\vee})). \tag{D.74.20}$$

Both have exactly the two possibly nonzero rows $b = 0, 1$. In positive columns their rows are, respectively,

| | $b = 0$ | $b = 1$ |
|---|---|---|
| absolute | $\overline{A}$ | $\overline{B}$ |
| shifted relative | $\overline{B^\vee}$ | $\overline{A^\vee}$. |

(D.74.21)

The coefficient pairing (D.74.13), on the same filtered bar resolutions, supplies a product of these spectral sequences into that for $\mathscr{H}_k$. The Alexander–Whitney product preserves the quotient filtration and obeys the cochain Leibniz identity; therefore its $d_2$ obeys that identity as well. Its products on the fibre rows are exactly (D.74.15)–(D.74.17).

Only $d_2$ can be nonzero. In any sufficiently positive columns denote the two maps by

$$d : \overline{B} \longrightarrow \overline{A}, \qquad d' : \overline{A^\vee} \longrightarrow \overline{B^\vee}. \tag{D.74.22}$$

They do not depend on a further column shift, under the cyclic identifications, because they commute with multiplication by $u$. Take an absolute row-one element $b$ and a shifted-relative row-one element $a^\vee$, both in positive columns. Their product on the fibre row two is zero: $H^2(F, \mathscr{H}_k) = 0$. The $d_2$ Leibniz identity in the target fibre row one is consequently

$$\langle db, a^\vee \rangle + \langle b, d' a^\vee \rangle = 0. \tag{D.74.23}$$

By (D.74.19) these are perfect pairings in the positive quotient columns. Hence $d'$ is the transpose of $d$ for the specified pairings, and they have a common rank $r$.

There are no later differentials. It follows that the two stable total dimensions are

$$\begin{aligned} \dim H^n(E, M) &= \dim \overline{A} + \dim \overline{B} - 2r, \\ \dim H^{n-1}(E, \mathscr{H}_{M^\vee}) &= \dim \overline{B^\vee} + \dim \overline{A^\vee} - 2r = \dim H^n(E, M) \end{aligned} \tag{D.74.24}$$

for all sufficiently large $n$. The degree shift in the second line is accounted for by the explicitly proved one-periodicity (D.74.18) of the positive quotient columns. More precisely, multiplication by the actual inflated $u \in H^1(E,k)$ is an isomorphism in sufficiently high degree on each group in (D.74.12): on the two-row spectral sequences it is an isomorphism on both surviving positive-column rows, and hence on their finite filtrations. The same proof applies to each $D_j$, using its open subgroup $L_j$.

*The rank of the actual boundary restriction.* Suppose now that $M$ has a specified invariant perfect bilinear pairing with itself, so that $M \simeq M^\vee$ as $E$-modules. This will be the permutation pairing used below. For precision, take degree-zero localizations by the actual class $u$:

$$\begin{aligned} \mathcal{A}_M &= H^\bullet(E,M)[u^{-1}]_0, \\ \mathcal{B}_M &= \left(\bigoplus_j H^\bullet(D_j,M)\right)[u^{-1}]_0, \\ \mathcal{R}_M &= R^\bullet(E,M)[u^{-1}]_0. \end{aligned} \tag{D.74.25}$$

These are finite vector spaces; each is an actual sufficiently high cohomology group under the proved multiplication-by-$u$ maps. All maps in (D.74.12) commute with these multiplications. Localization, or equivalently the direct limit of that exact sequence under those maps, gives the exact cyclic sequence

$$\mathcal{A}_M \xrightarrow{\bar r} \mathcal{B}_M \xrightarrow{u^{-1}\bar\delta} \mathcal{R}_M \xrightarrow{\bar j} \mathcal{A}_M. \tag{D.74.26}$$

For example the middle arrow sends a class $b/u^n$ to $\delta(b)/u^{n+1}$. Equation (D.74.24) gives $\dim \mathcal{R}_M = \dim \mathcal{A}_M$. If the ranks of the three arrows in (D.74.26) are $a, b, c$, exactness says

$$\dim \mathcal{A}_M = a + c, \qquad \dim \mathcal{B}_M = a + b, \qquad \dim \mathcal{R}_M = b + c. \tag{D.74.27}$$

Consequently $a = b$, and therefore

$$\operatorname{rank} \bar r = \tfrac{1}{2} \dim \mathcal{B}_M. \tag{D.74.28}$$

This is a rank equality for the actual restriction map with the original boundary labels.

*Brown restriction and the two occurrences of each section.* The full finite-coefficient Brown restriction theorem [42, Theorem 2 and Section 1.1] applies to $E$: $F$ is an open subgroup of 2-cohomological dimension one, and $E$ has no $C_2 \times C_2$. The two-row sequence (D.74.20) shows that $H^n(E,M)$ is finite for finite $M$. Thus the theorem's dense restriction embedding is onto its Hausdorff target. The three full normalizers are exactly those in (D.74.5). It follows, for $n > 1$, that

$$H^n(E,M) \xrightarrow{\sim} \bigoplus_{i=0}^{2} H^n(N_E(S_i),M). \tag{D.74.29}$$

The same theorem applies to each $D_j$, whose open subgroup $L_j$ has dimension one and whose two section normalizers are the finite sections themselves. Its cohomology is finite by its two-row cyclic spectral sequence. Consequently

$$H^n(D_j,M) \xrightarrow{\sim} H^n(T_{j,0},M) \oplus H^n(T_{j,1},M), \qquad n > 1, \tag{D.74.30}$$

where $T_{j,0} = \langle b_j \rangle$ and $T_{j,1} = \langle c_j^{e_2} b_j \rangle$.

Choose, for each of these six occurrences, an actual $a_{j,\epsilon} \in F$ with

$$T_{j,\epsilon} = a_{j,\epsilon} S_{i(j,\epsilon)} a_{j,\epsilon}^{-1}. \tag{D.74.31}$$

Such an $F$-conjugator exists by D.34, which identifies $E$- and $F$-conjugacy for sections. Its coefficient transport on cochains is the literal map

$$b(t_1,\dots,t_n) \longmapsto a_{j,\epsilon}^{-1} b(a_{j,\epsilon} t_1 a_{j,\epsilon}^{-1}, \dots, a_{j,\epsilon} t_n a_{j,\epsilon}^{-1}). \tag{D.74.32}$$

Canonical inner conjugation with this coefficient transport acts identically on group cohomology. For example, on the homogeneous bar resolution it is induced by simultaneous right multiplication of all vertices by $a_{j,\epsilon}^{-1}$. The bar prism between that map and the identity has terms

$$\sum_{\ell=0}^{n}(g_0,\dots,g_\ell,g_\ell a_{j,\epsilon}^{-1},\dots,g_n a_{j,\epsilon}^{-1}); \tag{D.74.33}$$

its boundary identity is the required cochain homotopy after applying equivariant Hom.

Thus (D.74.29)–(D.74.32) identify $\bar r$ in (D.74.26) with three duplicated restriction maps. More explicitly, let

$$U_i = H^\bullet(N_E(S_i),M)[u^{-1}]_0, \qquad V_i = H^\bullet(S_i,M)[u^{-1}]_0, \qquad r_i : U_i \longrightarrow V_i \tag{D.74.34}$$

be induced by the actual inclusion $S_i \subset N_E(S_i)$. The triangle incidence and the coefficient transports give

$$\mathcal{A}_M \simeq \bigoplus_i U_i, \qquad \mathcal{B}_M \simeq \bigoplus_i (V_i \oplus V_i), \qquad \bar r((w_i)_i) = ((r_i w_i, r_i w_i))_i. \tag{D.74.35}$$

There is no unidentified automorphism of a coefficient module in this display; its two copies use exactly (D.74.32). In particular

$$\operatorname{rank}\bar r = \sum_i \operatorname{rank} r_i, \qquad \dim \mathcal{B}_M = 2\sum_i \dim V_i. \tag{D.74.36}$$

Together with (D.74.28) these equalities force every $r_i$ to be surjective.

*A finite permutation coefficient detects every nontrivial commuter.* Fix $i$ and suppose $1 \neq c \in C_i$. Choose an open normal subgroup $N$ of $E$ with $N \subset F$ and $c \notin N$. This is possible by separating $c$ in a finite continuous quotient and then intersecting its kernel with $F$. Write

$$J = E/N, \qquad Q = F/N, \qquad \bar S_i = S_i N/N, \qquad M = k[J/\bar S_i]. \tag{D.74.37}$$

The quotient still has the specified map to $C_2$, so $\bar S_i$ has order two, $Q \cap \bar S_i = 1$, and $J = Q\bar S_i$. The basis pairing $\langle \delta_a, \delta_b\rangle = 1$ for $a = b$ and zero otherwise is a perfect $E$-invariant self-duality of $M$. Therefore all of (D.74.25)–(D.74.36) apply to this exact coefficient module.

For $n \geq 1$, the cyclic calculation (D.74.18) identifies $H^n(S_i, M)$ with the vector space on those cosets in $J/\bar S_i$ fixed by $S_i$. Indeed a two-point orbit gives the regular $k[C_2]$-module, whose invariant vector is also its norm image, so it contributes zero. A singleton orbit contributes its basis vector in every positive degree. This identification commutes with multiplication by $u$ and hence also describes $V_i$.

The coset $\bar S_i$ is fixed by $S_i$. The element $\bar c \in Q$ is nonidentity and centralizes $\bar S_i$, so $\bar c\bar S_i$ is another fixed coset, different from $\bar S_i$. Consequently the nonzero class of $\delta_{\bar S_i}$ in $V_i$ is moved by $c$, to the different class of $\delta_{\bar c\bar S_i}$.

On the other hand the image of the restriction $r_i : U_i \to V_i$ is contained in $C_i$-invariants. For $c \in C_i$, the cochain transport $b(g_1,\dots,g_n) \mapsto c\, b(c^{-1}g_1c,\dots,c^{-1}g_nc)$ is inner on $N_E(S_i)$ and acts identically on that group's cohomology, by (D.74.33) with $a = c^{-1}$. On $S_i$ its group arguments are unchanged, so it acts there precisely by $c$ on $M$. Naturality of restriction therefore gives the asserted invariance also after localization. The class of $\delta_{\bar S_i}$ is not invariant, contradicting the surjectivity of $r_i$ proved in (D.74.36). This proves $C_i = 1$. Equation (D.74.5) then gives $N_E(S_i) = S_i$, as claimed.

*The exact real-projectivity consequence.* Every involution in $E$ is conjugate to one of the $s_i$, so its full centralizer is its own order-two subgroup. Also $F$ is an actual open projective subgroup of index two. The set of involutions is closed: it is $\{t \in E : t^2 = 1\} \cap (E \setminus F)$, with $E \setminus F$ clopen.

These verify all the hypotheses of Haran's characterization [30, Proposition 2.2]. Thus $E$ is real projective. In the formulation of [59, Definition 1.1 and Theorem 2.1], this has the following precise meaning. For finite $A, B$, a surjection $\pi : B \to A$, and a continuous $\varphi : E \to A$, assume that each involution $t$ with $\varphi(t) \neq 1$ has an involution $b_t \in B$ with $\pi(b_t) = \varphi(t)$. There is then a continuous homomorphism

$$\gamma : E \longrightarrow B, \qquad \pi\gamma = \varphi. \tag{D.74.38}$$

This statement keeps the given finite quotient maps; it does not require or conclude surjectivity of $\gamma$.

*Uniqueness of the actual coboundary.* D.34 supplies $h \in F'$ with $f = h^{-1}\alpha(h)$. If $h'$ is any other solution in $F$, then

$$\alpha(h'h^{-1}) = h'f(hf)^{-1} = h'h^{-1}. \tag{D.74.39}$$

The element on the right belongs to $C_F(s) = 1$, so $h' = h$. For $\alpha \in GT_0$, D.44 supplies a solution with $h\theta(h) = 1$. Uniqueness shows that this is the same element, with no remaining fixed-subgroup adjustment.

Keeping the exact original cusps and every unit exponent, D.35's conjugator set is now parametrized by the solutions $(\mu, u)$ of

$$\begin{gathered} \mu \in \widehat{\mathbb{Z}}^{\times}, \qquad u \in F, \\ x^{\mu} h y^{\mu} h^{-1} = u(xy)^{\mu} u^{-1}. \end{gathered} \tag{D.74.40}$$

Its associated maps are the actual maps $x \mapsto x^{\mu}$, $y \mapsto hy^{\mu}h^{-1}$ of D.35. The closed-subgroup test there proves that every solution gives a full automorphism; its converse gives every peripheral conjugator. The proof above has removed the $C_F(s)$ variable from that equation. It has not established the existence of a pair $(\mu, u)$ satisfying it. □

**Scope and proof record.** The finite coefficient used at the last step is the actual coset permutation module (D.74.37); no pro-2 quotient is substituted for $F$ or $E$. The two transgressions in (D.74.22) are not assumed to vanish. Their equality of ranks is proved by the specified cup pairing, and the relative-to-absolute dimension comparison is then passed through the actual relative long exact sequence. The argument uses D.34's three global section classes and its labelled triangle, including the fact that each class occurs twice. The resulting full self-centralizing property and real-projectivity are proved above. Recognition and the existence in (D.74.40) require their own subsequent maps.

**D.75. [Proved] The full negative torus cusp centralizer and unique transporters.** Let $b$ be any negative involution of the full once-punctured-torus loop-outer group, with the actual cusp-normalized extension of D.58–D.59:

$$\begin{gathered} F = \widehat{F}(x, y), \qquad c = [x, y], \qquad E = F \rtimes \langle v \rangle, \\ v^2 = 1, \qquad vcv^{-1} = c^{-1}, \qquad \alpha = \mathrm{Inn}(v)|_F. \end{gathered} \tag{D.75.1}$$

Then

$$C_F(v) = 1, \qquad N_E(\langle v \rangle) = \langle v \rangle. \tag{D.75.2}$$

Every cusp reflection $c^t v$, $t \in \widehat{\mathbb{Z}}$, has the corresponding full trivial fixed subgroup in $F$. In particular there is a unique $g \in F$ with

$$gvg^{-1} = cv. \tag{D.75.3}$$

Its homology is the primitive generator of the full fixed line specified in D.59.

For every full unit $\mu \in \widehat{\mathbb{Z}}^{\times}$ and both literal reflection labels $v_i = c^i v$, $i = 0, 1$, the entire transporter set is a singleton:

$$\begin{aligned} \{a \in F : [a, v_i] = c^{\mu}\} &= \{a_{\mu,i}\}, \\ a_{\mu,0} = c^{(\mu-1)/2} g, \qquad a_{\mu,1} &= c^{(\mu+1)/2} g^{-1}. \end{aligned} \tag{D.75.4}$$

If $\alpha_* \bmod 2 \neq I$, the extension has one section class and is real projective. If $\alpha_* \bmod 2 = I$, the other section class retains the fixed group and surjective homology map of D.59; no full procyclic identification of that group is asserted here.

With the standard model selected by these section counts, the complete D.61 peripheral-conjugator existence problem is exactly

$$\text{there exist } \mu \in \widehat{\mathbb{Z}}^{\times},\ i \in \{0,1\} \text{ with} \quad c^{-\mu} a_{\mu,i} \sim_F a_{\mu,i}. \tag{D.75.5}$$

The maps proving this equivalence are retained below. Existence in (D.75.5), and recognition of an arbitrary negative action by one of the standard models, remain unproved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Section 24, folio 147 verso, scan 293; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The one actual cusp module.* Let $u : E \to C_2 = \mathbb{F}_2$ be the quotient character, $u(v) = 1$, and set $L = \overline{\langle c \rangle}$ and $D = N_E(L)$. D.58 gives the actual inverse homomorphisms

$$\begin{aligned} \widehat{\mathbb{Z}} \rtimes_{-1} C_2 &\longrightarrow D, \qquad (t, \tau^{\epsilon}) \longmapsto c^t v^{\epsilon}, \\ D &\longrightarrow \widehat{\mathbb{Z}} \rtimes_{-1} C_2, \qquad d \longmapsto (t, u(d)), \qquad d v^{-u(d)} = c^t. \end{aligned} \tag{D.75.6}$$

Its two internal section classes have representatives $S = \langle v \rangle$ and $S' = \langle cv \rangle$. Their normalizers in $D$ are $S$ and $S'$: an element $c^a$ commuting with a reflection has $2a = 0$, so $a = 0$. D.58 supplies an actual $g \in F$ with $S' = gSg^{-1}$ and (D.75.3). We will prove its uniqueness after proving (D.75.2).

Put $k = \mathbb{F}_2$, and form the actual compact module

$$0 \longrightarrow \Delta \longrightarrow k[[E/D]] \xrightarrow{\varepsilon} k \longrightarrow 0. \tag{D.75.7}$$

The explicit restriction map

$$F/L \longrightarrow E/D, \qquad fL \longmapsto fD \tag{D.75.8}$$

is a homeomorphism: $E = FD$ and $F \cap D = L$ give bijectivity, and both spaces are compact Hausdorff. In these coordinates the full $E$-action is

$$e \cdot (fL) = efv^{-u(e)} L. \tag{D.75.9}$$

The representative is in $F$, and its coset is independent of the representative of $fL$, because $v$ normalizes $L$. Thus $\Delta|_F$ is exactly the augmentation module of the full completed once-punctured-torus pair $(F, L)$, with the original commutator cusp and the transported $E$-action.

For a finite continuous $E$-module $M$ over $k$, define

$$\mathscr{H}_M = \mathrm{Hom}_{k,\mathrm{cont}}(\Delta, M), \qquad (e\varphi)(d) = e\varphi(e^{-1}d). \tag{D.75.10}$$

This is the same actual discrete coefficient construction as (D.74.9), now for (D.75.7). The augmentation sequence, its continuous Hom sequence, and Shapiro give

$$\begin{aligned} R^n(E, M) = H^n(E, D; M) &= H^{n-1}(E, \mathscr{H}_M) \quad (n \geq 1), \\ \cdots \longrightarrow H^n(E, M) \xrightarrow{r} H^n(D, M) &\longrightarrow R^{n+1}(E, M) \longrightarrow H^{n+1}(E, M) \longrightarrow \cdots . \end{aligned} \tag{D.75.11}$$

All maps here are the actual coefficient maps and subgroup restrictions. No other section normalizer has been introduced into this relative pair.

*The exact duality and half-rank comparison.* The pair $(F, L)$ in (D.75.8) is the full completion of a compact oriented genus-one surface with its one boundary. Apply the surface-pair duality of [72, Propositions 5.7 and 6.16 of the accepted version] to this specified pair. Equivalently the dense discrete subgroup and its cyclic boundary give the absolute and relative cohomology comparisons, with their actual cup products, as in (D.74.15). For $A = H^0(F, M)$ and $B = H^1(F, M)$, they give

$$H^0(F, \mathscr{H}_{M^\vee}) = B^\vee, \qquad H^1(F, \mathscr{H}_{M^\vee}) = A^\vee. \tag{D.75.12}$$

The identifications are equivariant for the actual $E/F$ action. The coefficient evaluation map is

$$M \otimes \mathscr{H}_{M^\vee} \longrightarrow \mathscr{H}_k, \qquad m \otimes \varphi \longmapsto [d \longmapsto \varphi(d)(m)]. \tag{D.75.13}$$

Its target relative orientation group $H^1(F, \mathscr{H}_k) = H^2(F, L; k) = k$ has trivial action. In boundary coordinates this also follows from $vcv^{-1} = c^{-1}$, which fixes the mod-two boundary character. All higher fibre rows vanish because $\mathrm{cd}_2(F) = 1$.

The two ordinary Hochschild–Serre sequences for $M$ and $\mathscr{H}_{M^\vee}$ therefore have exactly the rows in (D.74.21), with these $A$ and $B$. Their products are induced by (D.75.13). The positive-column cyclic complex has differential $1 + \sigma$, with the perfect kernel/image pairing and period-one $u$ map of (D.74.18)–(D.74.19). The product of the two fibre-row-one classes lies in $H^2(F, \mathscr{H}_k) = 0$. Thus the same explicit $d_2$ derivation identity gives

$$\langle db, a^\vee \rangle = \langle b, d' a^\vee \rangle. \tag{D.75.14}$$

These transgressions have equal rank, without either one being assumed zero.

For a finite self-dual $M$, the high-degree absolute and relative dimensions are consequently equal. Multiplication by the actual $u$ is an isomorphism in sufficiently high degree on all terms of (D.75.11), by the same two-row calculation. Taking degree-zero localizations, write

$$\mathcal{A} = H^\bullet(E, M)[u^{-1}]_0, \quad \mathcal{B} = H^\bullet(D, M)[u^{-1}]_0, \quad \mathcal{R} = R^\bullet(E, M)[u^{-1}]_0. \tag{D.75.15}$$

They are finite-dimensional, and the original long exact sequence becomes $\mathcal{A} \to \mathcal{B} \to \mathcal{R} \to \mathcal{A}$, with the connecting map divided by $u$. Its dimensions satisfy $\dim \mathcal{A} = \dim \mathcal{R}$. The rank identities (D.74.27), applied to these actual three maps, give

$$\operatorname{rank} \bar{r} = \tfrac{1}{2} \dim \mathcal{B}. \tag{D.75.16}$$

This use of D.74's coefficient argument has replaced its three-cusp augmentation module by (D.75.7); it has not replaced the torus peripheral structure or its involution matrix.

*Both incidence transports and the other section factor.* Let $C = C_F(v)$. Before proving it trivial, the exact normalizer is $N_E(S) = C \times S$, via multiplication and the inverse $e \mapsto (ev^{-u(e)}, v^{u(e)})$. In the identity-modulo-two case let $T$ be the other full section class of D.58 and write $N_E(T) = C_F(T) \times T$ with the same labelled coordinate maps. In the other case there is no such factor.

The finite-coefficient Brown theorem applies to $E$ and to $D$: their open subgroups $F$ and $L$ have cohomological dimension one, and every finite subgroup injects into $C_2$. Their high cohomology is finite by the two-row sequences with finite coefficients. Thus the actual dense restriction embeddings of [42, Theorem 2 and Section 1.1] are isomorphisms. Put

$$U = H^\bullet(N_E(S), M)[u^{-1}]_0, \qquad V = H^\bullet(S, M)[u^{-1}]_0. \tag{D.75.17}$$

Brown identifies $\mathcal{A}$ with $U$ together with the other normalizer factor when present. It identifies $\mathcal{B}$ with the two section factors for $S, S'$.

Transport the first local factor by the identity, and the second by the actual $g$ of (D.75.3). The second cochain transport is

$$b(t_1,\dots,t_n) \longmapsto g^{-1}b(gt_1g^{-1},\dots,gt_ng^{-1}). \tag{D.75.18}$$

As proved by the explicit bar prism (D.74.33), inner conjugation with exactly this coefficient transport acts identically on global cohomology. Consequently, if $r_S : U \to V$ is literal restriction, the boundary map is exactly

$$\bar{r} : U \oplus U_{\text{other}} \longrightarrow V \oplus V, \qquad (a, a_{\text{other}}) \longmapsto (r_S a, r_S a). \tag{D.75.19}$$

Here $U_{\text{other}} = 0$ in the one-class case. The additional global section factor has no boundary occurrence and its map in this display is zero. Both local sections occur in the cusp class, by D.58's actual $F$-fusion. Thus (D.75.16) says $\operatorname{rank} r_S = \dim V$, proving surjectivity of $r_S$ for every finite self-dual coefficient module.

Suppose $1 \neq k_0 \in C$. Choose an open normal $N \triangleleft E$, with $N \subset F$ and $k_0 \notin N$, and use the exact finite module

$$J = E/N, \qquad \bar{S} = SN/N, \qquad M = k[J/\bar{S}]. \tag{D.75.20}$$

Its basis pairing is invariant and perfect. In positive $S$-cohomology the singleton $S$-fixed cosets are a basis; two-point orbits have equal invariant and norm subspaces and contribute zero. The cosets $\bar{S}$ and $\bar{k}_0\bar{S}$ are distinct and both fixed. They are distinct because $\bar{k}_0 \in F/N$ is nonidentity and $(F/N) \cap \bar{S} = 1$. The class of $\delta_{\bar{S}}$ is therefore moved by $k_0$.

Every class in the image of $r_S$ is $C$-invariant: the cochain map $b \mapsto k_0 b(k_0^{-1}(\,\cdot\,)k_0)$ is inner on $N_E(S)$, whereas on $S$ it is exactly the coefficient action of $k_0$. This contradicts the proved surjectivity. Therefore $C = 1$, proving (D.75.2). All cusp reflections are $F$-conjugate to $v$: inside $D$ each is conjugate to $v$ or $cv$, and these two fuse by $g$. Their fixed subgroups in $F$ are therefore all trivial. Dividing any two solutions of (D.75.3) gives an element of $C$, so that solution is unique.

*Every unit and both literal cusp reflections.* The two half-exponents in (D.75.4) belong to $\widehat{\mathbb{Z}}$, because a full unit is odd at two and division by two is defined at every other prime. The identities

$$c^a v c^{-a} = c^{2a}v, \qquad gvg^{-1} = cv, \qquad g^{-1}(cv)g = v \tag{D.75.21}$$

show directly that $a_{\mu,i} v_i a_{\mu,i}^{-1} = c^{\mu} v_i$. Dividing two such transporters gives an element of $C_F(v_i) = 1$. This proves the full singleton assertion (D.75.4). Since $c$ is a commutator, D.59's primitive homology statement gives

$$[a_{\mu,i}]_{\text{ab}} = (-1)^i [g]_{\text{ab}}, \qquad \alpha_i(a_{\mu,i}) = c^{-\mu} a_{\mu,i}, \quad \alpha_i = \operatorname{Inn}(v_i)|_F. \tag{D.75.22}$$

In particular D.62's full primitive-homology centralizer theorem applies to every one of these elements, without a free-basis primitivity assumption.

For completeness retain the maps relating (D.75.5) to the peripheral conjugator. Fix $(\mu, i)$ and write $a = a_{\mu,i}$. Every outside element is uniquely $hv_i$, $h \in F$, and

$$(hv_i)a(hv_i)^{-1} = h(c^{-\mu}a)h^{-1}. \tag{D.75.23}$$

Thus the required $F$-conjugacy is exactly the existence of an outside centralizer element. Its kernel under $u$ is the full group $\overline{\langle a \rangle} \cong \widehat{\mathbb{Z}}$ by D.62. Choose an outside centralizer element $t$; then $t^2 = a^{\kappa}$, and the complete centralizer has the actual coordinates $a^r t^{\epsilon}$, as in (D.62k)–(D.62l).

In the two-class case the D.60 character $\chi : E \to \widehat{\mathbb{Z}}$ has $\chi(a) = (-1)^i$. Hence $2\chi(t) = (-1)^i\kappa$ and $\kappa_2$ is even. There is the unique outside commuting involution

$$w = a^{-\kappa/2}t. \tag{D.75.24}$$

It belongs to the other class, because a cusp-class involution has trivial fixed subgroup in $F$ and $a \neq 1$.

In the one-class case every outside involution has trivial fixed subgroup in $F$, so no such involution centralizes $a \neq 1$. Thus $\kappa_2$ is odd. The full root is obtained by the exact primewise correction

$$\begin{aligned} b_2 = 0, \qquad b_p = (1-\kappa_p)/2 \quad (p \neq 2), \\ \kappa' = \kappa + 2b \in \widehat{\mathbb{Z}}^\times, \qquad z = (a^b t)^{(\kappa')^{-1}}, \qquad z^2 = a. \end{aligned} \tag{D.75.25}$$

The centralizer is the full procyclic group $\overline{\langle z \rangle}$; multiplication by two in this group is injective, so this root is unique. These are the full-coordinate maps of D.62, rather than a root only in a maximal pro-two quotient.

The resulting extension maps from the specified D.61 standard models are

| | $X$ | $Y$ | $v_0$ |
|---|---|---|---|
| two classes | $a$ | $v_i w$ | $v_i$ |
| one class | $z v_i$ | $v_i z$ | $v_i$. |

$$[X, Y] \longmapsto c^\mu. \tag{D.75.26}$$

The commutator identity in the first row uses $[a, w] = 1$; in the second it uses $z^2 = a$ and (D.75.4). Conjugation by $v_i$ has exactly the standard D.61 action on the two displayed basis images. D.61's actual boundary test makes the kernel map a full isomorphism. The inverse extension map is its inverse on $F$ and sends the specified $v_i$ back to $v_0$. Conversely every D.61 normalized peripheral conjugator has a unit $\mu$ and one of these labels $i$. Its image of $X$ in the two-class model, or of $XY$ in the one-class model, is the unique element in (D.75.4). The image of its other reflection or of $X v_0$, respectively, gives the outside centralizer element in (D.75.23). This proves (D.75.5) in both directions with all cusp and quotient labels.

*The one-class consequence and the other fixed group.* If $\alpha_* \bmod 2 \neq I$, D.58 gives only the cusp section class. Every involution is therefore self-centralizing in $E$. The actual open subgroup $F$ is projective of index two, and the involutions form the closed subset $\{t : t^2 = 1\} \cap (E \setminus F)$. Haran's Proposition 2.2 [30] applies. The resulting real-projectivity has exactly the finite embedding-problem maps spelled out in (D.74.38).

If $\alpha_* \bmod 2 = I$, D.58 gives the other full section class $T = \langle w_1 \rangle$. D.59 retains the exact assertions

$$\begin{aligned} N_E(T) = C_F(w_1) \times T, \qquad \dim_k H^1(C_F(w_1), k) = 1, \\ C_F(w_1) \longrightarrow \ker(\alpha_* - I : H_1(F, \widehat{\mathbb{Z}}) \to H_1(F, \widehat{\mathbb{Z}})) \quad \text{is onto.} \end{aligned} \tag{D.75.27}$$

This factor is precisely the unused factor in (D.75.19). The proof does not identify it with a procyclic group. In either case, the remaining recognition question is the exact conjugacy in (D.75.5); its existence has not been inferred from the uniqueness of the transporter. □

**D.76. [Proved] The full section augmentation module and its incidence map.** Retain the actual pants extension $E = F \rtimes_\alpha \langle s \rangle$ of D.74, with $F = \widehat{F}(x, y)$, and put $k = \mathbb{F}_2$, $R = k[[E]]$, and $R_F = k[[F]]$. Choose one actual subgroup $T_i = \langle t_i \rangle$ in each of its three section classes, $0 \leq i < 3$. The compact left $R$-module

$$\Delta_T = \ker \left( \bigoplus_{i=0}^{2} k[[E/T_i]] \xrightarrow{\varepsilon} k \right) \tag{D.76.1}$$

is projective and has a full continuous module isomorphism

$$R^2 \xrightarrow{\cong} \Delta_T. \tag{D.76.2}$$

The proof constructs this isomorphism from compatible isomorphisms at every finite group-ring quotient. For every finite continuous $E$-module $M$ over $k$,

$$\dim_k \operatorname{Hom}_E(\Delta_T, M) = 2 \dim_k M. \tag{D.76.3}$$

There is a separate, specified incidence map. Let $e_i$ denote the identity coset in the $i$th summand of (D.76.1), and put $H = \overline{\langle T_0, T_1, T_2 \rangle}$. Then

$$\begin{aligned} &d_T : R^2 \longrightarrow \Delta_T, && (a,b) \longmapsto a(e_1 - e_0) + b(e_2 - e_0), \\ &\operatorname{coker} d_T \xrightarrow{\cong} \ker\bigl(k[[E/H]] \xrightarrow{\varepsilon} k\bigr). \end{aligned} \tag{D.76.4}$$

Under any isomorphism (D.76.2), $d_T$ is a square module endomorphism. It is invertible exactly when these three actual involutions generate $E$. This criterion retains the chosen subgroup maps; the existence of (D.76.2) does not assert invertibility of the specified $d_T$.

In the normalized D.74 coordinates, four actual involutions already generate $E$:

$$s, \quad xs, \quad fs, \quad yfs. \tag{D.76.5}$$

The first and third belong to the same section class, with the exact unique transporter $fs = h^{-1}sh$ from D.74. For the three actual involutions

$$s, \qquad xs, \qquad hyh^{-1}s, \tag{D.76.6}$$

the subgroup $H \cap F$ in (D.76.4) is exactly $\overline{\langle x, hyh^{-1} \rangle}$. Generation by (D.76.6), and D.74's remaining third-cusp equation, are not asserted by this module calculation.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Section 24, folio 147 verso, scan 293; Part D(b), the full loop-outer finite-action and actual peripheral-recognition problem for type (0,3).

*Proof. All actual coset and restriction maps.* Every $T_i$ maps isomorphically to $E/F = C_2$, so $E = FT_i$ and $F \cap T_i = 1$. Hence

$$F \longrightarrow E/T_i, \qquad a \longmapsto aT_i \tag{D.76.7}$$

is an $F$-equivariant homeomorphism: it is a continuous bijection between compact Hausdorff spaces. It identifies the restriction of the $i$th permutation module with the actual regular module $R_F$. If $T_i = a_i S_i a_i^{-1}$ for a representative in D.74, the full $E$-equivariant coset comparison is $gT_i \mapsto ga_iS_i$, with inverse $gS_i \mapsto ga_i^{-1}T_i$. Thus all changes of the three representatives retain explicit maps.

Write $I_F = \ker(R_F \to k)$. In the coordinates (D.76.7), the mutually inverse $R_F$-linear maps

$$\begin{aligned} I_F \oplus R_F \oplus R_F &\longrightarrow \Delta_T, && (a,b,c) \longmapsto (a+b+c, b, c), \\ \Delta_T &\longrightarrow I_F \oplus R_F \oplus R_F, && (a_0,a_1,a_2) \longmapsto (a_0+a_1+a_2, a_1, a_2) \end{aligned} \tag{D.76.8}$$

give the actual restricted module. The augmentation ideal of the full free group is free on $x-1, y-1$:

$$R_F^2 \longrightarrow I_F, \qquad (a,b) \longmapsto a(x-1) + b(y-1). \tag{D.76.9}$$

Here is a direct verification in the completed category. For every finite $F$-module $M$, a continuous linear map $I_F \to M$ is exactly a continuous crossed homomorphism $F \to M$, by $g \mapsto g - 1$ and $d(gh) = d(g) + g\,d(h)$. The images of $x, y$ of such a crossed homomorphism are arbitrary elements of $M$: use the free profinite universal property on the two lifts in the finite semidirect product $M \rtimes \operatorname{im}(F \to \operatorname{Aut} M)$. This also proves uniqueness. Consequently (D.76.9) induces a bijection

on continuous Hom into every finite module. Applying those bijections to the finite quotients of its source supplies compatible inverse maps; finite quotients separate points. Thus (D.76.9) is the asserted full continuous isomorphism. In particular

$$\Delta_T|_F \cong R_F^4. \tag{D.76.10}$$

It follows that $\Delta_T$ is finitely generated over $R$ as well. No finiteness statement for a hypothetical discrete group has been used.

*A direct degree-one bridge for every coefficient module.* Let $u : E \to C_2 = k$ be the original quotient character, and let $M$ be any finite continuous left $E$-module over $k$. Give $W_M = M \otimes_k k[E/F]$ the diagonal action. The exact coefficient sequence is

$$\begin{aligned} 0 \longrightarrow M \xrightarrow{j} W_M \xrightarrow{\epsilon} M \longrightarrow 0, \\ j(m) = m \otimes (F + sF), \qquad \epsilon(m \otimes gF) = m. \end{aligned} \tag{D.76.11}$$

The actual induction comparison and its inverse are

$$\begin{aligned} R\widehat{\otimes}_{R_F} M \longrightarrow W_M, \qquad & g \otimes m \longmapsto gm \otimes gF, \\ W_M \longrightarrow R\widehat{\otimes}_{R_F} M, \quad & m \otimes gF \longmapsto g \otimes g^{-1}m. \end{aligned} \tag{D.76.12}$$

Changing $g$ to $gf$ with $f \in F$ leaves the second formula unchanged by the tensor relation. Both composites are the identity. Since the subgroup has finite index, induction is also coinduction. Explicitly, $g \otimes m$ corresponds to the function on $E$ whose value at $a$ is $(ag)m$ when $ag \in F$, and zero otherwise; the coinduced action is right translation of the argument. Shapiro therefore gives $H^2(E, W_M) = H^2(F, M) = 0$.

The connecting map from (D.76.11) is precisely cup product by the named $u$. To check it without a choice of quotient generator, lift a 1-cocycle $a : E \to M$ by $\widetilde{a}(g) = a(g) \otimes F$. Its coboundary at $(g, h)$ is

$$j\big(u(g)\, g\, a(h)\big). \qquad \text{Thus} \qquad \partial[a] = [u \smile a]. \tag{D.76.13}$$

The coefficient long exact sequence proves that $u \smile - : H^1(E, M) \to H^2(E, M)$ is onto.

D.74 gives the actual equalities $N_E(T_i) = T_i$. The finite-coefficient Brown theorem applies in degree two: the open group $F$ has $\mathrm{cd}_2(F) = 1$, all finite subgroups inject into $E/F$, and the three $T_i$ are exactly its full section classes. The two-row Hochschild–Serre sequence with this finite module makes $H^2(E, M)$ finite, so the dense embedding of [42, Theorem 2 and Section 1.1] is onto. Hence

$$H^2(E, M) \xrightarrow{\cong} \bigoplus_{i=0}^{2} H^2(T_i, M). \tag{D.76.14}$$

On every actual $T_i$, $u$ is its nonzero quotient character. The cyclic complex has differential $1 + t_i$ in each positive degree. Cup product by $u$ identifies $H^1(T_i, M)$ with $H^2(T_i, M)$. Given three degree-one local classes, use (D.76.14) on their cups with $u$, then use surjectivity in (D.76.13). Restriction and cup commute, and local periodicity gives exactly the original three classes. Thus

$$H^1(E, M) \longrightarrow \bigoplus_i H^1(T_i, M) \text{ is onto}, \qquad H^2(E, \{T_i\}; M) = 0. \tag{D.76.15}$$

*Projectivity with compact module maps.* For this specified augmentation kernel, the continuous Hom construction and the relative long exact sequence are the same ones used in D.74, now with $T_i$ in place of the peripheral normalizers. The augmentation sequence splits as compact $k$-vector spaces. Consequently

$$\mathrm{Ext}^1_R(\Delta_T, M) = H^1\big(E, \mathrm{Hom}_{k,\mathrm{cont}}(\Delta_T, M)\big) = H^2(E, \{T_i\}; M) = 0 \tag{D.76.16}$$

for every finite $M$. These are the compact/discrete Hom and relative augmentation identifications of [72, Section 1 and Definition 2.1]. They may also be read directly as follows. A compact module extension of $\Delta_T$ by finite $M$ has a continuous $k$-linear splitting; the difference of its translates is a continuous cocycle with values in the displayed Hom module. Vanishing of its class is exactly an equivariant splitting.

For completeness, finite-coefficient vanishing here does give projectivity in the compact module category. Choose a surjection $p : R^m \to \Delta_T$, possible by (D.76.10), and let $K = \ker p$. For every open $R$-submodule $U$ of $R^m$ there is an exact finite module sequence

$$0 \longrightarrow (K+U)/U \longrightarrow R^m/U \longrightarrow \Delta_T/p(U) \longrightarrow 0. \tag{D.76.17}$$

Its pullback along $\Delta_T \to \Delta_T/p(U)$ splits by (D.76.16). It gives a continuous $R$-linear map $\ell_U : \Delta_T \to R^m/U$ lifting that quotient map. The possible maps are a nonempty finite set, since $\Delta_T$ is finitely generated and the target is finite. Reduction for a finer $U$ carries such a lift to a lift at every coarser $U$. Any finitely many compatibility requirements are realized at a common finer submodule. Compactness of the product of these finite sets supplies compatible $\ell_U$ for all $U$. Their inverse limit is an actual continuous $R$-linear section of $p$. Therefore $\Delta_T$ is a direct summand of $R^m$, and is projective.

*The full coefficient count and finite free quotients.* Apply $\mathrm{Hom}_R(\Delta_T, -)$ to (D.76.11). Projectivity gives an exact short sequence whose first and last spaces are $\mathrm{Hom}_R(\Delta_T, M)$. Frobenius reciprocity and (D.76.10) identify its middle space with $\mathrm{Hom}_{R_F}(R_F^4, M)$. The reciprocity maps, using the coinduced functions in (D.76.12), are evaluation at 1 and its inverse

$$q \longmapsto \big[d \longmapsto (a \longmapsto q(ad))\big]. \tag{D.76.18}$$

These are inverse continuous equivariant maps. All three Hom spaces are finite. Their dimensions give $2 \dim \mathrm{Hom}_R(\Delta_T, M) = 4 \dim M$, which is (D.76.3).

For an open normal subgroup $N \triangleleft E$, write

$$R_N = k[E/N], \qquad P_N = R_N \widehat{\otimes}_R \Delta_T = \Delta_T / \overline{\langle (n-1)d : n \in N,\ d \in \Delta_T \rangle}. \tag{D.76.19}$$

This is a finite projective left $R_N$-module: base change of the direct-summand description above is a direct summand of the finite free module $R_N^m$. For every simple $R_N$-module $M$, (D.76.3) says

$$\dim_k \mathrm{Hom}_{R_N}(P_N, M) = 2 \dim_k M = \dim_k \mathrm{Hom}_{R_N}(R_N^2, M). \tag{D.76.20}$$

Let $J_N$ be the Jacobson radical of the finite algebra $R_N$. Both Hom spaces factor through the respective semisimple tops. For every simple $M$, its multiplicity in such a top is the dimension of its Hom space divided by $\dim_k \mathrm{End}_{R_N}(M)$. Thus (D.76.20) identifies the two semisimple tops. Lift that identification to a map $R_N^2 \to P_N$. Its cokernel has zero top, so is zero by Nakayama's lemma, since $J_N$ is nilpotent. Projectivity splits this surjection. The complementary summand again has zero top and hence is zero. This proves $P_N \cong R_N^2$ as actual finite modules.

Let $\mathcal{I}_N = \mathrm{Iso}_{R_N}(R_N^2, P_N)$. These are nonempty finite sets. For $N' \subseteq N$, the canonical base-change identity $R_N \otimes_{R_{N'}} P_{N'} = P_N$ sends an isomorphism at $N'$ to one at $N$. Every finite list of compatibility equations has a solution obtained at a common finer open normal subgroup. The finite intersection property therefore gives a compatible family $(\phi_N)_N$. The kernels in (D.76.19) form a neighbourhood basis for $\Delta_T$: every finite module quotient has an open normal subgroup acting trivially, and its kernel contains the corresponding coinvariant kernel. Consequently

$$\Delta_T = \varprojlim_N P_N, \qquad R^2 = \varprojlim_N R_N^2, \qquad \phi = \varprojlim_N \phi_N \tag{D.76.21}$$

give the isomorphism (D.76.2), with inverse $\varprojlim_N \phi_N^{-1}$. This argument uses neither surjectivity of the maps between the finite isomorphism sets nor an assumed preferred basis.

*The specified incidence map and its exact cokernel.* Put $P_T = \bigoplus_i k[[E/T_i]]$. The map

$$P_T \longrightarrow k[[E/H]], \qquad gT_i \longmapsto gH \tag{D.76.22}$$

is well defined, continuous and equivariant. It kills the image of $d_T$. In $P_T/\operatorname{im} d_T$, the three identity cosets are equal. Each $t_i$ fixes the common class, because it fixes $e_i$ and $e_i = e_0$ there. Continuity makes their closed generated group $H$ fix that class. Thus $gH \mapsto [ge_0]$ is a well-defined continuous inverse to the map induced by (D.76.22). Checking the generators $gT_i$ and $gH$ proves both compositions are the identity. All images are closed, since the source modules are compact. The augmentations commute with these maps. Taking their kernels gives exactly the cokernel isomorphism in (D.76.4).

The augmentation kernel of $k[[E/H]]$ vanishes exactly when $E/H$ is a singleton. If two distinct cosets exist, some finite clopen quotient separates them; the difference of their two basis elements survives in its finite permutation module. Therefore $d_T$ is onto exactly when $H = E$. Choose $\phi$ in (D.76.21) and set $A_T = \phi^{-1} d_T : R^2 \to R^2$. If this map is onto, its reduction at every $N$ is an onto endomorphism of the finite module $R_N^2$, hence is invertible. Taking the compatible inverse limits makes $A_T$ invertible. The converse is immediate. This proves the stated exact incidence criterion. It supplies no reason to replace $d_T$ by the unrelated isomorphism $\phi$.

*The literal generators and the remaining removal target.* The normalized action gives $f\alpha(f) = 1$ and $\alpha(y) = f^{-1}y^{-1}f$. Direct substitution shows that every element in (D.76.5) squares to one. They generate the original generators through

$$x = (xs)s, \qquad y = (yfs)(fs), \qquad s = s. \tag{D.76.23}$$

D.74's unique equation $h^{-1}\alpha(h) = f$ gives $fs = h^{-1}sh$. The other two involutions have respectively the second and third section classes of D.34: replacing $x^{e_2}s$ by $xs$, and $y^{e_2}fs$ by $yfs$, is conjugation by $x^{(1-e_2)/2}$ and $y^{(1-e_2)/2}$. These full half-exponents exist because $1 - e_2$ has zero dyadic component. Thus the four generators have exactly the duplicated class stated in the claim.

Set $Y = hyh^{-1}$. The coboundary equation gives $\alpha(Y) = Y^{-1}$, and $h(yfs)h^{-1} = Ys$. Hence the three elements in (D.76.6) represent the three actual classes. The subgroup $K = \overline{\langle x, Y\rangle}$ is $\alpha$-invariant, and

$$\overline{\langle s, xs, Ys\rangle} = K\langle s\rangle, \qquad K \cap \langle s\rangle = 1, \qquad H \cap F = K. \tag{D.76.24}$$

These follow from $x = (xs)s$, $Y = (Ys)s$ and the literal quotient label $u(s) = 1$. Equations (D.76.4) and (D.76.24) specify the exact three-generator removal question. Even its solution would require the original third-cusp equation to give the peripheral conjugator of D.35. Neither that equation nor a new group presentation is inferred from the free module (D.76.2). □

**D.77. [Proved] A proper three-section subgroup with every primary image full.** There is an explicit choice of one representative of each of the three section classes in the standard pants extension for which the specified incidence map of D.76 is not onto, although the chosen orientation-kernel generators have full image in every maximal pro-$p$ quotient.

In the full profinite coproduct

$$E_0 = C_2(a) \amalg C_2(b) \amalg C_2(c), \qquad F_0 = \ker\bigl(E_0 \longrightarrow C_2,\ a, b, c \longmapsto 1\bigr), \tag{D.77.1}$$

put

$$x = ba, \quad y = ca, \quad \delta = \operatorname{Inn}(a)|_{F_0}, \quad g = x[x, y]x^{-1}, \quad Y = g\, y\, \delta(g)^{-1}. \tag{D.77.2}$$

Then $F_0 = \widehat{F}(x, y)$, $\delta$ inverts $x, y$, and

$$\begin{aligned} T_0 = \langle a\rangle, \qquad T_1 = \langle b\rangle, \qquad T_2 = \langle gcg^{-1}\rangle, \\ H = \overline{\langle T_0, T_1, T_2\rangle} = K \rtimes \langle a\rangle, \qquad K = \overline{\langle x, Y\rangle} = H \cap F_0. \end{aligned} \tag{D.77.3}$$

The pair $x, Y$ induces the identity on $F_0^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ and generates $F_0^{(p)}$ for every prime $p$, but $K$ is a proper subgroup of $F_0$ and $H$ is a proper subgroup of $E_0$. The corresponding section augmentation module is the free module of rank two in D.76. Its specified incidence cokernel has a quotient of dimension five over $\mathbb{F}_2$.

The same finite quotient proving properness proves that $Y$ is not conjugate to any full unit power of the original $y$. Thus this example retains the three section labels and all primary generation tests, and explicitly fails the additional original-cusp alignment in D.76's distinguished triple.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Section 24, folio 147 verso, scan 293; Part D(b), the full loop-outer finite-action and actual peripheral-recognition problem for type (0,3).

*Proof. The full extension and inverse maps.* Let $F = \widehat{F}(x, y)$ and let $\delta$ be the actual automorphism inverting its named generators. The mutually inverse full profinite homomorphisms are

$$\begin{aligned} E_0 &\longrightarrow F \rtimes_\delta C_2(a), \quad (a, b, c) \longmapsto (a, xa, ya), \\ F \rtimes_\delta C_2(a) &\longrightarrow E_0, \qquad\quad (x, y, a) \longmapsto (ba, ca, a). \end{aligned} \tag{D.77.4}$$

The three images in the first row square to one. The second row preserves $a(ba)a = (ba)^{-1}$ and $a(ca)a = (ca)^{-1}$. Both composites fix every indicated generator. Their quotient characters to $C_2$ agree on those generators, so their restriction identifies the actual kernel in (D.77.1) with the full $F$, including its topology.

The subgroups $\langle a\rangle, \langle b\rangle, \langle c\rangle$ are the three section classes for this standard inversion extension, as in D.34. Their images in the actual abelian quotient

$$E_0 \longrightarrow C_2^3, \qquad a \longmapsto e_0, \quad b \longmapsto e_1, \quad c \longmapsto e_2 \tag{D.77.5}$$

are distinct and unchanged by conjugation. Replacing $\langle c\rangle$ by its conjugate $T_2$ therefore retains exactly one representative of each class.

Since $c = ya$, direct multiplication gives

$$gcg^{-1} = gyag^{-1} = gy\delta(g)^{-1}a = Ya, \qquad \delta(Y) = Y^{-1}. \tag{D.77.6}$$

Thus $K$ is $\delta$-invariant. The identities $x = ba$ and $Y = (gcg^{-1})a$ show that the group generated by the three $T_i$ is exactly $K\langle a\rangle$. The quotient label of $a$ is nonzero, whereas $K \le F_0$; hence the intersection and semidirect-product statements in (D.77.3) follow.

*The actual all-prime images.* The word $g = x[x, y]x^{-1}$ lies in $F_0'$. Taking full abelianizations in (D.77.2) gives

$$x^{\mathrm{ab}} = (1, 0), \qquad Y^{\mathrm{ab}} = (0, 1). \tag{D.77.7}$$

For each prime $p$, the continuous map $x \mapsto x,\ y \mapsto Y$ induces an endomorphism of the free pro-$p$ group $F_0^{(p)}$. Its image maps onto the elementary abelian quotient $F_0^{(p)}/\Phi(F_0^{(p)})$ by (D.77.7). To verify full generation here, a proper closed image would have proper image in some finite $p$-group quotient. A maximal subgroup of that finite $p$-group has index $p$, so a nonzero continuous character to $\mathbb{F}_p$ would vanish on both specified generators. That contradicts (D.77.7). The induced endomorphism is therefore onto.

It is also injective. For completeness, a finitely generated pro-$p$ group has only finitely many homomorphisms to the finitely many $p$-groups of any bounded order. The intersections of their kernels are open fully invariant subgroups and separate points. An onto endomorphism induces a bijection on each of these finite quotients. Its kernel is therefore contained in their trivial intersection. This proves the asserted automorphism on every maximal pro-$p$ quotient

by the actual specified maps. Equation (D.77.6) also gives $\delta\rho = \rho\delta$ for the full endomorphism $\rho(x) = x,\ \rho(y) = Y$.

*The finite quotient detecting both proper inclusions.* Use permutations of $\{1,2,3,4,5\}$ with multiplication $(uv)(i) = u(v(i))$. Define the full continuous quotient $q : E_0 \to A_5$ on the three order-two factors by

$$q(a) = (1\,2)(3\,4), \quad q(b) = (1\,3)(2\,5), \quad q(c) = (1\,4)(2\,3). \tag{D.77.8}$$

The coproduct universal property gives this homomorphism. The exact finite multiplication and generation calculation gives

$$\begin{gathered} q(x) = (1\,5\,2\,3\,4), \qquad q(y) = (1\,3)(2\,4), \qquad q(g) = (1\,3\,2), \\ q(gcg^{-1}) = q(a), \qquad q(Y) = 1, \\ \langle q(a), q(b), q(c)\rangle = \langle q(x), q(y)\rangle = A_5. \end{gathered} \tag{D.77.9}$$

These equalities are also exhaustively verified by the finite producer below: closure under multiplication lists exactly sixty elements in each of the last two generated groups. In particular the restriction $q|_{F_0}$ is onto.

The images of the chosen subgroups are instead

$$q(H) = \langle q(a), q(b)\rangle = D_{10}, \qquad q(K) = \langle q(x)\rangle = C_5. \tag{D.77.10}$$

Indeed $q(x) = q(b)q(a)$ has order five and $q(a)$ conjugates it to its inverse. The five rotations and five reflections are distinct, proving the first order in (D.77.10); the second is immediate. Both are proper subgroups of the specified image $A_5$. This proves $H \neq E_0$ and $K \neq F_0$ in the full profinite groups.

*The specified incidence cokernel and cusp comparison.* Let $k = \mathbb{F}_2$, $R = k[[E_0]]$, and use the actual $T_i$ from (D.77.3). D.76 applies to give $\Delta_T \cong R^2$ and

$$\begin{aligned} &d_T : R^2 \longrightarrow \Delta_T, \qquad (r_1, r_2) \longmapsto r_1(e_1 - e_0) + r_2(e_2 - e_0), \\ &\operatorname{coker} d_T \cong \ker\bigl(k[[E_0/H]] \to k\bigr). \end{aligned} \tag{D.77.11}$$

There is a specified continuous surjection of coset spaces

$$E_0/H \longrightarrow A_5/D_{10}, \qquad zH \longmapsto q(z)D_{10}. \tag{D.77.12}$$

It is well defined by (D.77.10) and is onto by (D.77.9). The induced permutation-module surjection respects augmentation. Its restriction maps the kernel in (D.77.11) onto $\ker(k[A_5/D_{10}] \to k)$, which has dimension $6 - 1 = 5$. This is an explicit nonzero finite quotient of the actual incidence cokernel.

Finally $q(y)$ has order two, while $q(Y) = 1$. For every $\mu \in \widehat{\mathbb{Z}}^\times$, its residue modulo two is one, so $q(y^\mu) = q(y)$ still has order two. A conjugacy $Y = wy^\mu w^{-1}$ in $F_0$ would induce a conjugacy between elements of orders one and two under $q|_{F_0}$, which is impossible. This proves the claimed failure of original-y alignment through the same full homomorphism. D.76's distinguished element $hyh^{-1}$ retains that alignment by its defining equation; this example does not solve its remaining generation or third-cusp problem.

The producer `code/part_c/pants_incidence_limit.py` executes twenty exact checks: the free-word expression for $g$, its two zero abelian coordinates, $\delta(Y) = Y^{-1}$, the identity abelianization of $x, Y$, and all finite permutations, element orders and generated image orders in (D.77.8)–(D.77.10). The preceding Frattini argument supplies the all-prime assertion; no finite list of primes is substituted. □

**D.78. [Proved] The other negative fixed subgroup and its actual pro-odd kernel.** Retain the two-section-class situation of D.75:

$$F = \widehat{F}(x,y), \quad c = [x,y], \quad E = F \rtimes \langle v \rangle, \quad vcv^{-1} = c^{-1}, \quad gvg^{-1} = cv.$$

The transporter $g$ is the unique original D.75 transporter. Put $S = \langle v \rangle$, let $T = \langle w \rangle$ be an actual representative of the other section class, and put

$$B = C_F(w), \qquad N_E(T) = B \times T, \qquad k = \mathbb{F}_2.$$

D.59 gives $\dim_k H^1(B,k) = 1$. Retain the actual D.60 quotient $\chi : E \to \widehat{\mathbb{Z}}$, normalized by $\chi(g) = 1$. Its restriction to $B$ is onto. Choose $a \in B$ with $[a]_{\mathrm{ab}} = [g]_{\mathrm{ab}}$, as in D.60, so $\chi(a) = 1$. Define the actual closed groups

$$A = \overline{\langle a \rangle}, \qquad K = \ker(\chi|_B), \qquad O = \ker(B \xrightarrow{\chi} \widehat{\mathbb{Z}} \to \mathbb{Z}_2).$$

Every open subgroup $B_0$ of the full $B$ satisfies

$$\dim_k H^1(B_0,k) = 1. \tag{D.78.1}$$

The actual subgroup $A_2 = \overline{\langle a^{e_2} \rangle}$ is a Sylow pro-2 subgroup of $B$, and the restriction of $\chi_2$ to it is an isomorphism onto $\mathbb{Z}_2$. Both $O$ and $K$ are pro-odd. In particular the full original fixed group has the exact inverse coordinates

$$\begin{aligned} K \rtimes \widehat{\mathbb{Z}} &\longrightarrow B, \qquad (k,t) \longmapsto k a^t, \\ B &\longrightarrow K \rtimes \widehat{\mathbb{Z}}, \qquad b \longmapsto (b a^{-\chi(b)}, \chi(b)). \end{aligned} \tag{D.78.2}$$

The action is conjugation by the specified $a^t$. For every finite continuous 2-primary $B$-module $Q$, the actual inflation map is an isomorphism

$$H^n(\widehat{\mathbb{Z}}, Q^K) \xrightarrow{\sim} H^n(B,Q). \tag{D.78.3}$$

Under $A \xrightarrow{\chi} \widehat{\mathbb{Z}}$, restriction to $A$ is precisely the coefficient-inclusion map

$$H^n(\widehat{\mathbb{Z}}, Q^K) \longrightarrow H^n(\widehat{\mathbb{Z}}, Q). \tag{D.78.4}$$

The action on $Q$ in the right-hand group is explicitly $t \cdot q = a^t q$; on $Q^K$ it is the quotient action of $B/K$. These actions agree on the indicated submodule. For every finite continuous $E$-set $X$, put $Q_X = k[X^T]$ with its actual $B$-action. The proof supplies perfect finite-coefficient pairings and the exact comparison $\mathcal{U}_T(k[X]) \cong H^0(B, Q_X) \oplus H^1(B, Q_X)$. The full recognition $B = A$ is equivalent to surjectivity for every such $X$ of the specified inclusion $Q_X^B \to Q_X^A$. Every nonidentity element of $K$ has the explicit finite-coset coefficient witness given below. Triviality of $K$ and that invariant-surjectivity assertion remain unproved. The original cusp $c$, transporter $g$, and D.75.5 remain unchanged.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Section 24, folio 147 verso, scan 293; Part D(b), full loop-outer finite actions in type (1,1).

*Proof. The actual perfect localized duality.* Use the original cusp normalizer $D = N_E(\overline{\langle c \rangle})$ and the exact augmentation kernel and coefficient modules of D.75.7–13. For a finite continuous $E$-module $M$, write

$$\begin{aligned} \mathcal{A}_M &= H^\bullet(E,M)[u^{-1}]_0, \\ \mathcal{B}_M &= H^\bullet(D,M)[u^{-1}]_0, \\ \mathcal{R}_M &= H^\bullet(E,D;M)[u^{-1}]_0, \end{aligned}$$

where $u : E \to C_2 = k$ is the original quotient character. The maps

$$\mathcal{A}_M \xrightarrow{r} \mathcal{B}_M \xrightarrow{u^{-1}\delta} \mathcal{R}_M \xrightarrow{j} \mathcal{A}_M \tag{D.78.5}$$

are the actual localized restriction, connecting and relative-to-absolute maps. They are linear over the localized coefficient ring $\mathcal{A}_k$.

The D.75.14 transgression calculation gives more than equality of ranks. Put $U_M = H^0(F, M)$ and $V_M = H^1(F, M)$. In positive quotient columns the absolute rows are $\overline{U}_M, \overline{V}_M$, whereas the shifted relative rows for $M^\vee$ are $\overline{V_M^\vee}, \overline{U_M^\vee}$, with the kernel/image cyclic pairings of D.74.19. The only possible differentials are transpose maps

$$d : \overline{V}_M \to \overline{U}_M, \qquad d' : \overline{U_M^\vee} \to \overline{V_M^\vee}.$$

Consequently the pairings on the two surviving pairs of rows are perfect:

$$\operatorname{coker} d \;\times\; \ker d' \longrightarrow k, \qquad \ker d \;\times\; \operatorname{coker} d' \longrightarrow k. \tag{D.78.6}$$

These are the actual pairings induced by evaluation and the fibre surface-pair fundamental class, not chosen vector-space identifications. For sufficiently large absolute degree $n$ and relative degree $m$, the surviving filtration sequences are exactly

$$0 \longrightarrow \operatorname{coker} d \longrightarrow H^n(E, M) \longrightarrow \ker d \longrightarrow 0,$$
$$0 \longrightarrow \operatorname{coker} d' \longrightarrow H^m(E, D; M^\vee) \longrightarrow \ker d' \longrightarrow 0.$$

The first subspace has quotient-column filtration $n$, and the second has filtration $m - 1$ in the shifted relative complex.

Here is the required trace on the target. For trivial coefficient $k$, the absolute transgression is zero: the actual section $v$ makes inflation from $C_2$ injective in every degree. Its transpose in the shifted relative sequence is therefore zero as well. The top fibre row $H^1(F, \mathscr{H}_k) = H^2(F, \langle c \rangle; k) = k$ survives. For sufficiently large $n$ its quotient edge map supplies

$$H^n(E, D; k) \longrightarrow E_\infty^{n-2,1} = H^{n-2}(C_2, k) = ku^{n-2} \longrightarrow k, \tag{D.78.7}$$

where the last map reads the coefficient of the specified power of $u$. These maps commute with a further multiplication by $u$. Thus they give a specified trace $\mathrm{tr} : \mathcal{R}_k \to k$.

Cup product followed by this trace gives

$$\mathcal{A}_M \times \mathcal{R}_{M^\vee} \longrightarrow k. \tag{D.78.8}$$

Its associated two-row pairings are exactly (D.78.6), in opposite filtration positions. The product of the two filtration subspaces just displayed has quotient filtration at least $n + m - 1$, whereas the trace reads column $n + m - 2$; that product therefore has trace zero. An absolute vector with nonzero image in $\ker d$ pairs nontrivially with the relative subspace $\operatorname{coker} d'$. A nonzero absolute vector in $\operatorname{coker} d$ pairs nontrivially with a relative vector mapping to $\ker d'$. These statements follow from the two perfect pairings in (D.78.6), and the same argument applies with the sides reversed. This proves nondegeneracy on the full finite filtered spaces, not only on their associated graded spaces. Passing through the specified $u$ maps proves that (D.78.8) is perfect. The construction is natural in the coefficient maps and balanced over $\mathcal{A}_k$, by the original cup-product and connecting-map identities.

*The other Brown factor and its exact relative lift.* D.75 proves $C_F(v) = 1$. Brown restriction therefore gives the actual coefficient decomposition

$$\mathcal{A}_M \xrightarrow{\sim} \mathcal{V}_S(M) \oplus \mathcal{U}_T(M), \qquad \begin{cases} \mathcal{V}_S(M) = H^\bullet(S, M)[u^{-1}]_0, \\ \mathcal{U}_T(M) = H^\bullet(B \times T, M)[u^{-1}]_0. \end{cases} \tag{D.78.9}$$

The two cusp factors are transported by 1 and the original $g$, with exactly the coefficient transport of D.75.18. The restriction in (D.78.5) is consequently

$$\mathcal{V}_S(M) \oplus \mathcal{U}_T(M) \longrightarrow \mathcal{V}_S(M) \oplus \mathcal{V}_S(M), \qquad (z,t) \longmapsto (z,z). \tag{D.78.10}$$

In particular the other factor has zero boundary restriction for every finite module, including every finite permutation module.

For coefficient $k$, let $e \in \mathcal{A}_k$ be the Brown inverse of $(0,1)$, the identity in the other normalizer factor and zero in the cusp factor. It is a specified idempotent. Its restriction to both cusp factors is zero, so $e\mathcal{B}_M = 0$. Applying this idempotent to (D.78.5) gives an actual isomorphism

$$j : e\mathcal{R}_M \xrightarrow{\sim} e\mathcal{A}_M = \mathcal{U}_T(M). \tag{D.78.11}$$

Its inverse is unambiguous: lift an element through $j$, then multiply its lift by $e$. Two lifts differ by a connecting image, which $e$ kills.

The pairing (D.78.8) splits orthogonally under $e$ and $1-e$; its restriction to $e\mathcal{A}_M \times e\mathcal{R}_{M^\vee}$ is perfect. Transporting it through (D.78.11) gives a perfect pairing

$$\mathcal{U}_T(M) \times \mathcal{U}_T(M^\vee) \longrightarrow k, \qquad (z,z') \longmapsto \tau\big(\mathrm{ev}_*(z \smile z')\big), \tag{D.78.12}$$

where $\tau = \operatorname{tr} j^{-1} : \mathcal{U}_T(k) \to k$ uses (D.78.11). The formula follows because $j$ commutes with cup multiplication by an absolute class; it therefore retains the actual normalizer cup product and the actual coefficient evaluation.

*Arbitrary finite permutation coefficients.* Let $X$ be any finite continuous $E$-set and $M = k[X]$, with its invariant basis pairing. The $T$-fixed points give a specified $B \times T$-submodule

$$Q_X = k[X^T] \subseteq M. \tag{D.78.13}$$

Projection onto these basis elements is an inverse on this summand. The complementary summand consists of the two-point $T$-orbits and is free as a $k[T]$-module. It is $B$-stable because $B$ centralizes $T$. Its positive $T$-cohomology vanishes. The extension spectral sequence for $T \to B \times T \to B$, with $\mathrm{cd}_2 B \leq 1$, shows that this complementary summand contributes no cohomology in degree at least two. Thus inclusion of $Q_X$ induces the exact localized coefficient comparison.

The action of $T$ on $Q_X$ is trivial. The direct-product resolution, or the two-row complex with zero $T$ differential, gives the actual isomorphism

$$\begin{aligned} H^0(B, Q_X) \oplus H^1(B, Q_X) &\xrightarrow{\sim} \mathcal{U}_T(M), \\ (q, z) &\longmapsto q + u^{-1} \inf_B(z). \end{aligned} \tag{D.78.14}$$

At a positive degree $n \geq 2$, this formula means $qu^n + \inf_B(z)u^{n-1}$, followed by the coefficient inclusion (D.78.13). Projection to the fixed-point summand and the direct product cochain decomposition supply its inverse. Restriction to the literal subgroup $T$ is exactly

$$H^0(B, Q_X) \oplus H^1(B, Q_X) \longrightarrow Q_X, \qquad (q,z) \longmapsto q. \tag{D.78.15}$$

Thus its image is $Q_X^B$, with precisely that subgroup action retained.

The $H^1(B, Q_X)$ summand in (D.78.14) is totally isotropic for (D.78.12): the product of two such summands is a coefficient image of $H^2(B, Q_X \otimes Q_X) = 0$. Since (D.78.12) is perfect on the finite self-dual module $M$, it follows that

$$\dim H^1(B, Q_X) \leq \dim H^0(B, Q_X). \tag{D.78.16}$$

Decompose the actual finite $B$-set $X^T$ into its orbits, with stabilizers $B_1, \ldots, B_r$. The usual finite-coset Shapiro maps give

$$H^0(B, Q_X) = k^r, \qquad H^1(B, Q_X) = \bigoplus_{i=1}^{r} H^1(B_i, k). \tag{D.78.17}$$

Every $B_i$ is open in $B$, and $\chi(B_i)$ has finite index in $\widehat{\mathbb{Z}}$ because $\chi(B) = \widehat{\mathbb{Z}}$. Write this image as $d_i\widehat{\mathbb{Z}}$, with $d_i$ a positive integer. The actual map

$$B_i \xrightarrow{\chi} d_i\widehat{\mathbb{Z}} \xrightarrow{\div d_i} \widehat{\mathbb{Z}} \longrightarrow \mathbb{F}_2 \tag{D.78.18}$$

is a nonzero continuous character. Division here is the continuous inverse of multiplication by $d_i$ onto its indicated image. Thus each summand in (D.78.17) has dimension at least one. Together with (D.78.16) this proves equality and dimension exactly one for each stabilizer.

Every open subgroup $B_0 \leq B$ occurs as such a stabilizer. Choose an open normal $N \triangleleft E$ contained in $F$ with $N \cap B \subseteq B_0$. The group $U = B_0TN$ is open in $E$, since $B_0T$ is a subgroup and $N$ is normal. The quotient label $u$ and $N \subset F$ show $U \cap B = B_0$. In the actual finite $E$-set $X = E/U$, the identity coset is $T$-fixed and its $B$-stabilizer is $B_0$. This proves (D.78.1), with no passage to a maximal pro-2 quotient.

The two summands in (D.78.14) now have equal dimension. The mixed part of the perfect pairing (D.78.12) is therefore perfect as well. Indeed a vector of $H^1$ pairing trivially with $H^0$ pairs trivially with both summands and must vanish, and the dimensions agree. Hence the actual cup pairings, with the degree-correct trace $\tau_1(z) = \tau(u^{-1}\inf_B z)$, are

$$H^0(B, Q_X) \times H^1(B, Q_X^\vee) \longrightarrow H^1(B, k) \xrightarrow{\tau_1} k \tag{D.78.19}$$

and their reversed versions are perfect. The fixed-point basis pairing identifies $Q_X^\vee$ with $Q_X$. On the one-dimensional $H^1(B, k)$, $\tau_1$ is nonzero: otherwise that square-zero summand would lie in the radical of (D.78.12) for $M = k$. Thus $\tau_1$ takes the original nonzero class $\chi|_B$ mod 2 to one. This records the original finite-coefficient duality, including the actual stabilizers and coefficient actions.

*The full pro-odd kernel consequence.* The map $t \mapsto a^t$ is an actual continuous section of $\chi|_B$. In particular $A_2 = \overline{\langle a^{e_2} \rangle}$ is a copy of $\mathbb{Z}_2$ mapping isomorphically onto the dyadic quotient. Choose a Sylow pro-2 subgroup $P$ of $B$ containing $A_2$. There cannot be two linearly independent continuous characters $P \to k$. Their joint map factors through $P/(P \cap N_0)$ for some open normal $N_0 \triangleleft B$. It therefore extends to the open subgroup $PN_0$ by the quotient isomorphism $PN_0/N_0 = P/(P \cap N_0)$, contradicting (D.78.1).

It follows that $P$ is procyclic. One elementary verification is that every nontrivial finite 2-group quotient has at most one independent character to $C_2$, hence a unique maximal subgroup; any element outside that maximal subgroup generates the finite group. Thus every finite quotient is cyclic, and compatible generators give a procyclic inverse limit. Since $P$ maps onto $\mathbb{Z}_2$, it is $\mathbb{Z}_2$, and that surjection is an isomorphism. Its subgroup $A_2$ already maps onto the same quotient, so $P = A_2$.

Every Sylow pro-2 subgroup of $B$ is conjugate to $A_2$. The homomorphism $\chi_2$ is invariant under conjugation and is injective on all these subgroups. Hence $O = \ker \chi_2$ has trivial Sylow pro-2 subgroup and all its finite quotients have odd order. It is pro-odd. The actual subgroup $K = \ker(\chi|_B)$ lies in $O$, so is pro-odd too. The multiplication and inverse formulas (D.78.2) follow directly from the chosen section $a^t$ and the original character $\chi$.

*The surviving coefficient map and the original scope.* For any finite continuous 2-primary $B$-module $Q$, average over the finite odd image of $K$ on $Q$. Its odd order is invertible on $Q$. The averaging map

$$e_K : Q \longrightarrow Q^K, \qquad q \longmapsto \frac{1}{|\overline{K}|} \sum_{h \in \overline{K}} hq \tag{D.78.20}$$

is independent of the chosen finite image through which the action factors. It commutes with $B$, since $K$ is normal. It makes the invariants functor exact; higher 2-primary cohomology of $K$ vanishes. The extension spectral sequence for the actual sequence $K \to B \xrightarrow{\chi} \widehat{\mathbb{Z}}$ therefore gives (D.78.3). Restriction to the specified section $A$ commutes with inflation and is induced by the

inclusion $Q^K \hookrightarrow Q$, proving (D.78.4). Evaluation, coefficient maps and cup products commute with this comparison.

For $Q = Q_X = k[X^T]$, the invariant space is concretely $k[K\backslash X^T]$, mapped to $Q_X$ by its orbit sums. Every such orbit has odd size, so the original basis pairing restricts to the ordinary basis pairing on these sums. The action of the quotient is the action of the specified powers of $a$ on the $K$-orbits. This is the actual finite-coefficient comparison; it retains the action of $K$ through its orbit partition.

There is an exact finite-permutation detection of the surviving kernel. The equality $B = A$ is equivalent to surjectivity, for every finite continuous $E$-set $X$, of the actual restriction map

$$Q_X^B \longrightarrow Q_X^A. \tag{D.78.21}$$

The forward implication follows from equality of the subgroups. For the reverse implication, take a nonidentity $k_0 \in K$. The compact subgroup $AT$ is closed, and $k_0 \notin AT$: its $u$-label is zero, so membership would put it in $A$, where $\chi$ is injective. Choose an open normal $N \triangleleft E$ contained in $F$ such that $k_0 \notin ATN$. For the finite set $X = E/(ATN)$, the identity coset is fixed by both $A$ and $T$, whereas $k_0$ moves it. The corresponding basis vector in $Q_X$ lies in $Q_X^A$ and not in $Q_X^B$. Thus (D.78.21) fails for that actual finite coefficient.

The cusp restriction (D.78.10) vanishes on the entire other factor. The perfect duality proves (D.78.1)–(D.78.3), but has supplied no surjectivity for (D.78.21). This is the retained coefficient map that still detects the full pro-odd kernel. The coefficient comparison retains the actual $K$-orbit partition and supplies no replacement of $Q_X^K$ by $Q_X$. The original cusp $c$, unique transporter $g$, and D.75 conjugacy equation are unchanged throughout.

□

# 5 E. Special groups and two operator groups

## 5.1 E(a). Special groups from conjugators and from framed diagrams

We give the constructions of §25, folios 149–154, scans 296–306, with their precise hypotheses. Let $\pi$ be a discrete hyperbolic surface group or its full profinite completion, and let $I$ be its finite peripheral index set. Choose representatives $L_i$ and coherent parametrizations $c_i : T \to L_i$, where $T = \mathbb{Z}$ or $\widehat{\mathbb{Z}}$. The self-centralizing and self-normalizing properties needed below are the results of Part A. Put $\mathcal{E} = \mathrm{Aut}_{\mathrm{lac}}(\pi)$ and $\Gamma = \mathrm{Out}_{\mathrm{lac}}(\pi)$. Every $\alpha \in \mathcal{E}$ has a permutation $\sigma_\alpha$ of $I$ and an automorphism $\chi_\alpha$ of $T$.

**E.1. [Proved] The conjugator construction and the product kernel.** Define $S\mathcal{E}$ to consist of pairs $(\alpha, (g_i)_{i\in I})$ with $\alpha \in \mathcal{E}$, $g_i \in \pi$, and

$$\alpha(c_i(t)) = g_i^{-1} c_{\sigma_\alpha(i)}(\chi_\alpha t) g_i \quad (t \in T).$$

The multiplication is

$$(\beta, (h_i))(\alpha, (g_i)) = \big(\beta\alpha, (h_{\sigma_\alpha(i)}\, \beta(g_i))_{i\in I}\big).$$

There are exact sequences and a canonically embedded normal subgroup

$$1 \to T^I \to S\mathcal{E} \to \mathcal{E} \to 1, \qquad \iota : \pi \hookrightarrow S\mathcal{E}, \quad a \mapsto (\mathrm{Int}(a), (a^{-1})_i).$$

The groups $T^I$ and $\iota(\pi)$ commute and intersect trivially. Consequently

$$S\Gamma = S\mathcal{E}/\iota(\pi), \qquad 1 \to T^I \times \pi \to S\mathcal{E} \to \Gamma \to 1, \qquad 1 \to T^I \to S\Gamma \to \Gamma \to 1.$$

The superscript ! takes the inverse image of the kernel of the permutation action on $I$. All maps in the profinite case are continuous maps of profinite groups, and all kernels and quotients are closed.

*Manuscript:* §25; folios 149–151; scans 296–300.

*Proof.* Apply $\beta$ to the defining equation for $\alpha$. The two conjugating factors multiply as $\beta(g_i)^{-1}h^{-1}_{\sigma_\alpha(i)}$, which is the inverse of the factor displayed in the product. Associativity follows either by substitution in that formula or by associativity of composition of the specified conjugations. The identity is $(\mathrm{id}, (1)_i)$, and the inverse has automorphism $\alpha^{-1}$ and $j$th conjugator $\alpha^{-1}(g^{-1}_{\sigma_\alpha^{-1}(j)})$.

Surjectivity to $\mathcal{E}$ is the definition of preserving the loop classes. In its kernel every $g_i$ centralizes $L_i$, hence belongs to $L_i$; this identifies the kernel with $T^I$. Substituting $(\mathrm{Int}(a), (a^{-1})_i)$ proves that $\iota$ is a homomorphism and that conjugation by $(\alpha, (g_i))$ takes $\iota(a)$ to $\iota(\alpha(a))$. Its injectivity follows from $Z(\pi) = 1$, including when $I$ is empty. To check commutation with the kernel, the two products with $(\mathrm{id}, (\ell_i))$, $\ell_i \in L_i$, both have conjugators $\ell_i a^{-1}$. Their intersection is trivial because projection to $\mathcal{E}$ sends $\iota(a)$ to $\mathrm{Int}(a)$. Finally an element lying over the identity in $\Gamma$ has an inner automorphism component and can be multiplied by a unique $\iota(a)^{-1}$ to put that component equal to the identity. This proves that the whole kernel is the displayed direct product, not merely an abstract extension of its two factors.

For a finitely generated profinite $\pi$, its automorphism group has the usual profinite topology obtained from characteristic open quotients. The defining equations of $S\mathcal{E}$ are closed equations in $\mathcal{E} \times \pi^I$. Thus it is profinite, the multiplication and all maps above are continuous, and the compact subgroups giving the quotients are closed. The same algebraic proof gives the exact continuous sequences. □

**E.2. [Proved] Agreement with autoequivalences of the framed diagram.** The group $S\Gamma$ of E.1 is the group of isomorphism classes of autoequivalences of

$$B_I \longrightarrow \coprod_{i \in I} BL_i \longrightarrow B\pi$$

with the common orientation action retained. Adding a fixed base object in $B\pi$ gives $S\mathcal{E}$. Thus the two constructions on scans 296–300 agree, with the comparison maps and inner ambiguities exactly as in E.1.

*Manuscript:* §25; folios 150–151; scans 298–300.

*Proof.* An autoequivalence gives a permutation of the selected objects, the specified automorphism $\chi$ on their cyclic automorphism groups, and an automorphism $\alpha$ at the generic base object. The natural isomorphism making the second square commute is, at its $i$th object, an element $g_i \in \pi$. Its naturality equation is exactly the defining equation of E.1. Pasting the squares gives the product there, including the index $\sigma_\alpha(i)$.

Initially the first square also contains an isomorphism at each selected peripheral object. A natural transformation at that object changes this isomorphism by the corresponding element of $L_i$. Use it to make the first-square isomorphism the identity. After this gauge choice there is no remaining peripheral-object natural-transformation ambiguity. A natural transformation at the generic object is an element $a \in \pi$ and changes the representative by $\iota(a)$, with the convention of E.1. These are precisely the equivalences being divided out in $S\Gamma$. Fixing the generic base object and its comparison isomorphism removes that last ambiguity and gives $S\mathcal{E}$. Conversely every pair in E.1 defines the three functors and their two commutativity squares. The two constructions undo each other on the chosen representatives and on the specified natural transformations, so the group isomorphisms are canonical relative to the initial peripheral representatives. Changing those representatives transports the diagrams by their explicit conjugating isomorphisms. □

**E.3. [Proved] The puncture-stabilizer pushout is the normalizer group.** Let $\Gamma_i \subset \Gamma$ stabilize $i$. Restrict the extension $S\Gamma \to \Gamma$ to $\Gamma_i$ and push its kernel out by $\mathrm{pr}_i : T^I \to T$. The resulting extension is canonically

$$1 \to L_i \xrightarrow{\mathrm{Int}} \{\alpha \in \mathrm{Aut}_{\mathrm{lac}}(\pi) : \alpha(L_i) = L_i\} \longrightarrow \Gamma_i \to 1.$$

*Manuscript:* §25; folios 153–154; scans 305–306.

*Proof.* For a representative $(\alpha, (g_j))$ with $\sigma_\alpha(i) = i$, put

$$F_i[(\alpha, (g_j))] = \mathrm{Int}(g_i)\alpha.$$

The defining equation shows that this automorphism normalizes the actual subgroup $L_i$. Replacing the pair by its generic-basepoint change replaces $\alpha$ by $\mathrm{Int}(a)\alpha$ and $g_i$ by $g_i a^{-1}$; the displayed value is unchanged. Hence the map descends to $S\Gamma_i$. The multiplication formula gives

$$\mathrm{Int}(h_i\beta(g_i))\beta\alpha = \mathrm{Int}(h_i)\beta\,\mathrm{Int}(g_i)\alpha,$$

so it is a homomorphism. An automorphism normalizing $L_i$ has a representative with $g_i = 1$, proving surjectivity. On the kernel $T^I$ the map is $\ell_i \mapsto \mathrm{Int}(\ell_i)$; its kernel is exactly the other factors because $Z(\pi) = 1$. This is the universal pushout claimed. The normalizer equality in Part A gives the last kernel $L_i$ in the displayed sequence. This is the compatibility of scans 305–306, with its actual group of automorphisms rather than an unspecified outer class. □

**E.4. [Refuted] The product-kernel claim requires the rigidity hypotheses.** The formula $\ker(S\mathcal{E} \to \Gamma) = T^I \times \pi$ does not hold unchanged for all abelian types. For the unpunctured torus, $I = \varnothing$, $S\mathcal{E} = \mathrm{GL}_2(\mathbb{Z}) = \Gamma$ and the kernel is trivial, whereas $\pi = \mathbb{Z}^2$. For the annulus the inner basepoint subgroup intersects the peripheral kernel, and the special outer kernel is a quotient of $T^2$ by one diagonal copy of $T$.

*Manuscript:* §25; folios 149–153; scans 296–304.

*Proof.* For the torus, all inner automorphisms of $\mathbb{Z}^2$ are the identity and there are no conjugators indexed by punctures. Thus the pair construction contains only the automorphism itself, giving the stated identity map and its trivial kernel.

For the annulus, both peripheral images are the whole group $\pi = \mathbb{Z}$. The conjugator kernel is $\pi^2$, but the generic-basepoint change $a$ is $(\mathrm{id}, (a^{-1}, a^{-1}))$ and lies in that same kernel. Dividing by these changes gives $\mathbb{Z}^2/\Delta\mathbb{Z} \cong \mathbb{Z}$, with the antidiagonal convention after orienting both boundary loops. Keeping the two end labels and the independent orientation data supplies the appropriate extended quotient; their omission would create an additional error. For the plane the peripheral homomorphism $\mathbb{Z} \to 1$ is not injective, so the normalizer calculation used in E.1 is unavailable. Its correct fixed-boundary positive special group is trivial, as C.20 computes. These explicit maps identify the failed hypotheses in each case. □

## 5.2 E(b). Two commuting operator groups

We retain the names $\theta$ and $\omega$ for the standard order-two and order-three lifts on $\widehat{F}_2$; Grothendieck's $\sigma_0$ and $\rho$ denote the corresponding choices on scans 316–317, §25 bis, folios 159–160. A change of base path conjugates a lift by $\widehat{F}_2$. We keep this ambiguity in the statements below.

**E.20. [Proved] The two lifting conditions cut out exactly the multiplier-1 subgroup modulo 3.** In $B = \mathrm{Out}_{\mathrm{lac}}(\widehat{\pi}_{0,3})$ let $S_3$ be the geometric positive subgroup. Its centralizer is $GT_0$, with the normalized description $(\lambda, f)$ of Harbater–Schneps. Every element of $GT_0$ has an

automorphism representative commuting with the chosen order-two lift $\theta$. The elements with an automorphism representative commuting with the chosen order-three lift $\omega$ are exactly

$$GT_0^{(3)} = \{(\lambda, f) \in GT_0 : \lambda \equiv 1 \pmod 3\}.$$

This subgroup has index two. The two representatives need not be the same representative; this is the two-condition formulation on scans 316–317.

*Manuscript:* 25 bis; 157 verso–160; 312–317.

*Proof.* Harbater–Schneps, Main Theorem (a) and Theorem A(i) [31], identify the centralizer and state precisely

$$[\theta, F] = \mathrm{Inn}(f), \qquad [\omega, F] = \mathrm{Inn}(y^m f), \qquad m = (\lambda - 1)/2,$$

for $F(x) = x^\lambda$, $F(y) = f^{-1} y^\lambda f$. Here $[a, b] = aba^{-1}b^{-1}$. A representative $\mathrm{Inn}(h)F$ commutes with $\theta$ precisely when

$$\theta(h) f = h,$$

since $F_2$ has trivial centre. It commutes with $\omega$ precisely when $\omega(h) y^m f = h$. These are the two exact coboundary equations. We determine their solvability without imposing the pentagon equation.

The positive lift extension of the $S_3$ action on the three-punctured sphere is

$$E = C_2 * C_3 = \langle s, r \mid s^2 = r^3 = 1 \rangle, \qquad E \longrightarrow S_3,$$

whose kernel is the peripheral free group $F_2$. Its cusp generator can be chosen as $e = sr$; $e^2$ is a primitive cusp in that kernel. Complete this exact sequence by D.1. An outer centralizer element, represented by $F$, acts on $\widehat{E}$ by conjugating its image in $\mathrm{Aut}(\widehat{F}_2)$ and induces the identity on $S_3$. It sends $e^2$ to a $\widehat{F}_2$-conjugate of $e^{2\lambda}$. After that conjugation it sends $e$ into $C_{\widehat{E}}(e^{2\lambda}) = \overline{\langle e \rangle}$. To prove this equality, put $L = \overline{\langle e^2 \rangle}$. A centralizing element preserves its $\widehat{F}_2$-conjugacy class, so its image in $S_3$ stabilizes the associated puncture. That stabilizer is $\langle q(e) \rangle$ of order two. Multiplication by $e$ when necessary puts the element in $\widehat{F}_2$, where Part A gives $C_{\widehat{F}_2}(e^{2\lambda}) = L$. This proves the equality and identifies the whole centralizer with $\widehat{\mathbb{Z}}$. Doubling is injective in that group, so the image of $e$ is $e^\lambda$.

Consequently this automorphism acts on $\widehat{E}_{\mathrm{ab}} = C_6$ by multiplication by $\lambda$. Every order-two element is conjugate to $s$, and every order-three subgroup is conjugate to $\langle r \rangle$, by D.2 or the free profinite product theorem. A lift of the given transposition is $F_2$-conjugate to $s$: a total-group conjugator projects to $C_{S_3}(s) = \langle s \rangle$, and multiplication by $s$ adjusts it into the kernel. Thus there is one kernel-conjugacy class for the order-two lift. It is fixed by every centralizer element, proving solvability of the first equation.

There are exactly two kernel-conjugacy classes of lifts of the specified three-cycle. They are represented by $r$ and $sr^{-1}s^{-1}$. Indeed a total-group conjugator has either an even or an odd image in $S_3$; adjusting by $\langle r \rangle$ or by $s\langle r \rangle$ gives these two representatives. They are distinct because their images in the order-three quotient of $E_{\mathrm{ab}}$ are inverse nonidentity elements. The action of $(\lambda, f)$ preserves the class of $r$ exactly when $\lambda \equiv 1 \pmod 3$, by the computed action on $C_6$. When it does, the same adjustment by $r$ puts its conjugator in the kernel. This is precisely solvability of the second coboundary equation. The argument treats elements, so it retains the label of the three-cycle.

For comparison, Lochak–Schneps, Proposition 3 and Theorem 2 [47], compute the same two nonabelian cohomology sets: one class for $\theta$, two for $\omega$, with the second class selected by $\lambda \equiv -1 \pmod 3$. The first two parts of their proof use only the two-cycle and three-cycle equations, so their computation also applies to $GT_0$. Finally $(-1, 1) \in GT_0$ exchanges the two classes. The homomorphism to $(\mathbb{Z}/3)^\times$ is therefore surjective and the asserted index is exactly two. The residue calculation is independently recorded in the executed program. $\square$

**E.21. [Proved] The arithmetic subgroup and the two fixed points.** The intersection of the usual Galois image $G_{\mathbb{Q}} \to GT_0$ with $GT_0^{(3)}$ is the image of $G_{\mathbb{Q}(\sqrt{-3})}$. The order-three fixed points in $U = \mathbf{P}^1 - \{0, 1, \infty\}$ are $e^{\pm i\pi/3}$, and the order-two lift has one fixed point in $U$. Thus the requirement to fix the chosen order-three section is a proper finite-index restriction, even on the Galois subgroup.

*Manuscript:* 25 bis; 157 verso–160; 312–317.

*Proof.* The multiplier of the Galois action is the cyclotomic character, by its action on the tame inertia at 0. Its reduction modulo 3 is the action on $\mu_3$ and has fixed field $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. E.20 identifies precisely its kernel. The three-cycle $z \mapsto 1/(1-z)$ has fixed-point equation $z^2 - z + 1 = 0$, and the transposition $z \mapsto 1 - z$ has fixed-point equation $z = 1/2$. These points lie in $U$. Complex conjugation exchanges the former pair and fixes the latter point, which also proves properness directly.

On scan 312 the relevant subgroup is obtained by fixing a finite set of critical data; it is therefore of finite index. An assertion that every element of $G_{\mathbb{Q}}$ lifts commuting with a specified one of the order-three lifts would contradict the displayed computation. We record the scan's wording as editorial evidence, not as a settled reading. □

**E.22. [Proved] Normalizer, stabilizer, and the forced lift.** Let $1 \to \pi \to E \xrightarrow{q} \mathcal{G} \to 1$ have centre-free kernel, and let $K \leq E$ be a finite subgroup on which $q$ is injective. Assume $C_\pi(K) = 1$, as holds for a nontrivial rotation stabilizer in the anabelian geometric situation and its completion. Put $G_Q = q(K)$ and let $Q$ denote the $\pi$-conjugacy class of $K$. The restriction

$$q : N_E(K) \longrightarrow \mathcal{G}$$

is injective and has image the stabilizer $\mathcal{G}_Q$ of $Q$. Its inverse on that image is a canonical homomorphism. In a two-group situation $\mathcal{G} = \Gamma G$ with $\Gamma$ centralizing $G$, its restriction gives the forced lift of $\Gamma_Q = \Gamma \cap \mathcal{G}_Q$, and this lift centralizes $K$.

*Manuscript:* 25 bis; 155–157 verso; 307–312.

*Proof.* An element of $N_E(K) \cap \pi$ acts trivially on $q(K)$, and the restriction of $q$ to $K$ is injective. It therefore centralizes $K$, so the kernel is $C_\pi(K) = 1$. A quotient element stabilizes $Q$ precisely when one, and hence every, lift carries $K$ to a $\pi$-conjugate. Multiplying that lift by the appropriate element of $\pi$ produces an element of $N_E(K)$ above it. This proves the image statement and uniqueness of the lift. The inverse is a homomorphism because it is the inverse of an injective homomorphism onto its image. For $\gamma \in \Gamma_Q$, conjugation of its lift on $K$ projects to conjugation by $\gamma$ on $G_Q$, which is trivial. Injectivity on $K$ then makes this conjugation trivial. In the profinite case all these maps are continuous, and the inverse is continuous by compactness of the closed normalizer and the Hausdorff property of the quotient.

For the centralizer, the precise image is $\mathcal{G}_Q \cap C_{\mathcal{G}}(G_Q)$, rather than all of $\mathcal{G}_Q$. When $G_Q$ is nonabelian dihedral, its own elements already distinguish these images. For a reflection in an oval the kernel is $\mathbb{Z}$ or $\widehat{\mathbb{Z}}$ by Part B and D.4, so the stated injectivity is replaced by an exact sequence with that kernel. These computations identify exactly the rigidity used in the manuscript's argument. □

**E.23. [Proved] Explicit discrete centralizers and pointwise kernels.** For the three actions in the question, the full extended geometric groups, their centralizers, their normalizers, and

their pointwise kernels on the exceptional points are as follows. Here $D_{2n}$ has order $2n$.

| $(g,\nu),G$ | $M$ | $C_M(G)$ | $N_M(G)$ | $C_M(G)^\natural$ |
|---|---|---|---|---|
| $(0,3),S_3$ | $S_3\times C_2$ | $C_2$ | $S_3\times C_2$ | 1 |
| $(1,1),C_4$ | $\mathrm{GL}_2(\mathbb{Z})$ | $C_4$ | $D_8$ | $C_2$ |
| $(1,1),C_6$ | $\mathrm{GL}_2(\mathbb{Z})$ | $C_6$ | $D_{12}$ | 1 |

The last column fixes every exceptional point individually, not merely its orbit under $G$.

*Manuscript:* 25 bis; 155–160; 307–317.

*Proof.* For type $(0,3)$, the geometric group is $S_3$ times complex conjugation. The exceptional points are the three transposition fixed points $-1, 1/2, 2$ and the two order-three fixed points of E.21. Complex conjugation fixes the first three and interchanges the last two. The centralizer of $S_3$ is therefore $C_2$ and its pointwise kernel is trivial; its normalizer is the whole product.

Use the matrices

$$R_4=\begin{pmatrix}0&-1\\1&0\end{pmatrix},\qquad R_6=\begin{pmatrix}0&-1\\1&1\end{pmatrix}$$

on $\mathbb{R}^2/\mathbb{Z}^2$, punctured at 0. A rational matrix commuting with either one is $aI+bR_n$. For an integral commuting matrix $a,b\in\mathbb{Z}$. Its determinant is $a^2+b^2$ for $R_4$ and $a^2+ab+b^2$ for $R_6$. Equating this nonnegative norm to 1 gives four and six units respectively, proving the centralizer assertions. Conjugation in the normalizer embeds its quotient by the centralizer in $\mathrm{Aut}(C_n)=C_2$ for $n=4,6$. Reflection in the square or hexagonal lattice realizes inversion, proving the two normalizer assertions.

For $C_4$ the exceptional set outside the puncture is exactly the three nonzero two-torsion points. The matrix $-I$ fixes all three, while $R_4$ interchanges $(1/2,0)$ and $(0,1/2)$. Thus the kernel is $\langle -I\rangle$. For $C_6$ the exceptional set consists of the same three two-torsion points and $(1/3,1/3),(2/3,2/3)$. The three two-torsion points force a power of $R_6$ to be $I$ or $-I$, and the latter exchanges the two order-three points. Hence the kernel is trivial. These lists are exhaustive: each nonidentity power has its fixed points given by $(R_n^j-I)v\in\mathbb{Z}^2$; their union is the displayed finite set. The program enumerates these solutions, the kernels, and the matrix centralizers and normalizers. The norm calculation proves exhaustiveness beyond its bounded integer search. □

**E.24. [Refuted] Recovery of an arbitrary commuting group from its pointwise subgroup.** For an arbitrary specified $\Gamma\le C_{\Gamma'}(G)$, its pointwise subgroup is

$$\Gamma^\natural=\Gamma\cap\Gamma'^\natural,$$

and it need not equal $C_{\Gamma'^\natural}(G)$. Nor is $\Gamma$ determined as a normalizer by $\Gamma^\natural$. For the maximal choice $\Gamma=C_{\Gamma'}(G)$ the proposed equality holds, by the displayed intersection identity.

*Manuscript:* 25 bis; 157 verso; 312.

*Proof.* Both identities follow by testing separately the two defining conditions: membership in $\Gamma$ and pointwise stabilization of each critical datum. To refute recovery, take the first row of E.23. The choices $\Gamma=1$ and $\Gamma=C_2$ have the same pointwise subgroup 1. Its normalizer inside the centralizer $C_2$ is $C_2$, so the procedure cannot recover the first choice. For the proposed equality without the maximal-choice hypothesis, use the second row and take $\Gamma=1$: its pointwise subgroup is 1, whereas the centralizer pointwise kernel is $C_2$. All groups and their kernels are included in the finite check. Thus the failure persists even in the specifically requested discrete cases and has a complete intersection formula as its replacement. □

# 6 F. Statements usable in the later sections

We use $\Gamma_{g,\nu}$ for the extended geometric group permitting puncture permutations and $\Gamma^{!+}_{g,\nu}$ for the pure oriented group, following §19, folio 102, scans 204–205. In each formula below the chosen variant is stated explicitly. The full profinite loop-outer group is $B = \mathrm{Out}_{\mathrm{lac}}(P)$; the closure $H = \overline{M}$ of a chosen geometric subgroup $M$ is a subgroup of $B$. We do not identify that closure with the abstract profinite completion without the congruence assertion required for that identification.

**F.1. [Proved] LM4-F.a: fixed discretizations and prediscretizations.** For $M = \Gamma_{g,\nu}$ and $H = \overline{M} \leq B$, the sets of outer discretizations and prediscretizations are $B/M$ and $B/H$. For a given finite $G \leq B$ their fixed sets are exactly D.16's transporter sets and centralizer-orbit decompositions. In the three types of D.8, every fixed prediscretization has a fixed outer discretization above it. The same lifting conclusion holds for a finite solvable $G$ and genus at most two for the orientation-preserving geometric group covered by Boggi's Theorem C. Over a fixed coset represented by $b$, the fibre is

$$(H/M)^{b^{-1}Gb},$$

with its full labelled-embedding decomposition as in D.16.

*Manuscript:* 28; 178–181; 352–358.

*Proof.* The identifications and the fixed-set computation are proved in D.16, including their inner-automorphism ambiguities. A fixed coset $bH$ has $K = b^{-1}Gb \leq H$. Its fibre consists of $bhM$, $h \in H$; this coset is fixed precisely when $h^{-1}Kh \leq M$, proving the formula. In the three low types D.8 conjugates every finite $K \leq H$ into $M$, giving a point in that fibre. In genus at most two and for solvable $K$, Boggi's Theorem C [9] gives the same conjugation into the oriented geometric group. The congruence subgroup property in these genera identifies its completion with $H$, so the theorem applies to the displayed subgroup. We retain the labelled embeddings in the fibre decomposition because subgroup conjugacy alone does not fix an automorphism of $K$.

The actual based version has $A(P)/J$ in place of $B/M$, as D.16 proves. A chosen genuine lift of $G$ acts on it; an outer subgroup alone acts on its quotient by $P$. These are the exact maps through which the fixed-set assertions of §28, folios 178–181, scans 352–358 are used. For a finite group in the full $B$ not known to stabilize a prediscretization, the same formula still determines the fixed set as a transporter; its nonemptiness is exactly the unresolved recognition question D.12, not an additional conclusion of this lemma. □

**F.2. [Refuted] LM4-F.b, finiteness and equality of the two centralizers.** The centralizer of a finite subgroup $G \leq \Gamma^{+}_{g,\nu}$ need not be finite, either discretely or in the geometric completion. In genus at most two, for finite solvable $G$, the normalizer comparison and the open-kernel argument below give

$$C_{\widehat{\Gamma^{+}_{g,\nu}}}(G) = \overline{C_{\Gamma^{+}_{g,\nu}}(G)}.$$

The corresponding equality with the full loop-outer group in place of the geometric completion is false already for $(g, \nu) = (0, 3)$ and $G = S_3$.

*Manuscript:* 30; 194–196; 383–387.

*Proof.* On a closed genus-two surface let $G$ be generated by the hyperelliptic involution. This involution is central in $\Gamma^{+}_{2}$, so its centralizer is the entire group. A Dehn twist about a nonseparating curve has infinite order: its action on integral homology sends a transverse vector $b$ to $b + a$, and its $n$th power sends it to $b + na$. This also gives infinite order in the

congruence image, since some reduction modulo an integer detects every nonzero $n$. Thus the completed centralizer is infinite as well. For finite solvable $G$, the normalizer comparison, with the source induction and its topology input audited in F.7, gives $N_{\widehat{M}}(G) = \overline{N_M(G)}$ for $M = \Gamma^+_{g,\nu}$. The continuous conjugation homomorphism from this profinite normalizer to the finite group $\mathrm{Aut}(G)$ has open kernel $C_{\widehat{M}}(G)$. Intersecting the dense subgroup $N_M(G)$ with that open kernel gives a dense subgroup of the kernel: every open subset of the kernel is open in the normalizer and meets $N_M(G)$. The intersection is exactly $C_M(G)$, proving the displayed centralizer equality. The map of the completion to the loop-outer group is injective in these genera by the congruence subgroup property. The cited sources state the equality for every finite $G$; F.7 records the precise missing intersection step for their nonsolvable assertion. The passage from any such equality to finiteness would still require the discrete centralizer to be finite, contradicted by the hyperelliptic example.

For the smallest hyperbolic type, $\Gamma^+_{0,3} = S_3$, and $C_{S_3}(S_3) = 1$. Harbater–Schneps identify $C_B(S_3) = GT_0$. This group is infinite: its multiplier contains the cyclotomic image $\widehat{\mathbb{Z}}^\times$ of $G_{\mathbb{Q}}$. These are actual homomorphisms on the same group $\widehat{F}_2$, with the geometric $S_3$ supplied by rational Möbius transformations, so the commutation with the Galois image is the naturality of the fundamental-group action. The two centralizers are therefore 1 and an infinite group. With orientation reversal included the geometric centralizer becomes $C_2$, and the same failure remains.

The statement denoted (c) on scans 383–385, §30, folios 194–195, concerns the centralizer of an arithmetic germ $\Gamma^\natural$. It cannot be replaced merely by a finite $G$: scan 385 itself discusses infinite centralizers of finite groups. The counterexamples above compute the result of the requested replacement explicitly and explain which datum that replacement removes. □

**F.3. [Proved] LM4-F.b, the intrinsic open-image statement.** Let $M \le B$ be an oriented geometric subgroup, $H = \overline{M}$, $N = N_B(H)$, and let $i : G \hookrightarrow H$ be a fixed finite embedding. Conjugation defines an action of $N/H$ on the set of $H$-conjugacy classes of embeddings $G \hookrightarrow H$. The image of the continuous homomorphism

$$C_N(i(G)) \longrightarrow N/H$$

is exactly the stabilizer of $[i]$, and its kernel is $C_H(i(G))$. When the orbit of $[i]$ is finite, this image is open. The orbit is finite for solvable finite $G$ in genus at most two and for every finite $G$ in the three low types of D.8.

*Manuscript:* 30; 195 verso–196; 386–387.

*Proof.* An element $nH$ stabilizes $[i]$ exactly when there is $h \in H$ with

$$ni(g)n^{-1} = hi(g)h^{-1} \quad (g \in G).$$

This is equivalent to $h^{-1}n \in C_N(i(G))$. It proves the image assertion, and the kernel is obtained by intersecting the centralizer with $H$. Because $H$ is compact, the stabilizer is the image of a compact closed transporter and is closed in $N/H$. A closed subgroup of finite index in a profinite group is open: its finitely many closed cosets have open complements, so each coset is open.

For the claimed cases, finite subgroup conjugacy classes in $H$ come from the finite list of geometric classes, by D.8 or Boggi's solvable statement. There are finitely many geometric classes for each fixed finite group: a finite action has an orbifold quotient of bounded signature by the Euler characteristic formula; there are finitely many epimorphisms from each finitely generated orbifold group to the fixed finite group. Passing from subgroup classes to labelled-embedding classes adds at most $|\mathrm{Aut}(G)|$ choices per subgroup class. Thus the orbit is finite and the image is open. F.6 constructs the arithmetic quotient directly from universal monodromy and proves the open-image assertion in that quotient for every finite geometric subgroup and every hyperbolic type. The inclusion of that quotient into $N/H$ is retained there as the exact comparison map. □

**F.4. [Proved] LM4-F.c: the exact canonical lift of a section stabilizer.** Let $\pi$ be a centre-free discrete surface group or its full profinite completion, and put

$$A = \mathrm{Aut}_{\mathrm{lac}}(\pi), \qquad B = \mathrm{Out}_{\mathrm{lac}}(\pi).$$

Let $G \leq B$ be finite and let $s : G \to A$ be a specified lift with $C_\pi(s(G)) = 1$. The natural map

$$C_A(s(G)) \longrightarrow C_B(G)$$

is injective. Its image is the stabilizer of the $\pi$-conjugacy class $[s]$ in the set of lifts of the labelled inclusion $G \hookrightarrow B$. Its inverse is the canonical homomorphism lifting that stabilizer to $A$ and centralizing $s(G)$. In the profinite case it is continuous. For geometrically realized actions with a nontrivial rotation the section-class set is finite, so this stabilizer has finite index and is open in the profinite case.

*Manuscript:* 31; 197 verso–199; 390–393.

*Proof.* The kernel is the inner automorphisms centralizing $s(G)$, identified, by centre-freeness, with $C_\pi(s(G))$. For $b \in C_B(G)$ choose an automorphism representative $v$. The conjugate homomorphism $g \mapsto vs(g)v^{-1}$ still lifts the labelled inclusion, because $b$ centralizes each $g$. It represents $[s]$ exactly when some $p \in \pi$ makes $\mathrm{Inn}(p)^{-1}v$ commute with every $s(g)$. This proves that the image equals the claimed stabilizer. The kernel computation proves uniqueness. The inverse is a homomorphism by uniqueness, since products of commuting lifts still commute with $s(G)$.

For a finitely generated profinite $\pi$, $A$ and $B$ are profinite, and the centralizer is closed. The map is a continuous injection from a compact space into a Hausdorff space, so its inverse on its image is continuous. In the discrete geometric situation Part B proves finiteness of the lift classes; D.3 proves their equality with the profinite classes. The orbit of a class is therefore finite. Its stabilizer is closed, as the image of the compact centralizer, hence open. This proves all topology assertions as well as the group assertion.

For the notation of scans 390–393, §31, folios 197–199, the target $\mathfrak{S}(\pi_0)^G$ is the centralizer of the chosen genuine lift inside the loop automorphism group. The source is the stabilizer of the corresponding section class inside $\mathcal{T}(\pi_0)^G = C_{\mathrm{Out}_{\mathrm{lac}}(\pi_0)}(G)$. A point $P$ determines that section class after a base path; changing the path conjugates the resulting lift by $\pi_0$, exactly as displayed. Thus the canonicity belongs to the chosen section, and the geometric point supplies its conjugacy class. □

**F.5. [Refuted] The unrestricted uniqueness assertion for reflection ovals.** For a reflection fixing an oval, the lifting map in F.4 has kernel $\mathbb{Z}$ in the discrete case and $\widehat{\mathbb{Z}}$ after completion. Consequently it is not injective and its lifts are not unique without further data.

*Manuscript:* 31; 197 verso–199; 390–393; 33; 222 verso–225; 440–445.

*Proof.* Part B and D.4 identify $C_\pi(s(G))$ with the primitive oval subgroup and its closure. Every element of that subgroup induces an inner automorphism centralizing the chosen reflection lift and maps to the identity outer class. The kernel computation in F.4 therefore gives the displayed groups. A nonidentity primitive oval element is an explicit second lift of the identity outer element. The corrected general statement is the exact sequence with this kernel and with image the same section stabilizer, proved by the same representative-and-conjugator calculation. No splitting of that sequence is supplied by the point class alone. □

**F.6. [Proved] LM4-F.b: open arithmetic image for every finite geometric subgroup.** Let $2g-2+\nu>0$, let $M=\Gamma^{+}_{g,\nu}$ permit permutations of the $\nu$ punctures, and retain

$$P=\widehat{\pi}_{g,\nu},\qquad B=\mathrm{Out}_{\mathrm{lac}}(P),\qquad H=\overline{M}\le B.$$

Write $A_{\mathrm{ar}}\le B$ for the image of universal *arithmetic* étale monodromy of the moduli stack of smooth genus-$g$ curves with an unordered reduced divisor of degree $\nu$, over $\mathbb{Q}$. Then $H\triangleleft A_{\mathrm{ar}}$, and put $\Gamma_{\mathrm{ar}}=A_{\mathrm{ar}}/H$. For every finite subgroup $G\le M$, the continuous homomorphism

$$C_{A_{\mathrm{ar}}}(G)\longrightarrow\Gamma_{\mathrm{ar}} \tag{F.6.1}$$

has open image. Its kernel is $C_H(G)$. In the notation $N=N_B(H)$ and $q:N\to N/H$, this is exactly the assertion that

$$q(C_N(G))\cap\Gamma_{\mathrm{ar}}=q(C_{A_{\mathrm{ar}}}(G))\quad\text{is open in }\Gamma_{\mathrm{ar}}. \tag{F.6.2}$$

The same assertion holds with ordered punctures and the pure oriented group throughout. No identification of $H$ with the abstract completion of $M$ is used.

*Manuscript:* 30; 195–196; 385–387.

*Proof.* We first type the arithmetic group and its comparison maps. Denote the indicated moduli stack over $\mathbb{Q}$ by $\mathscr{M}$, and choose a geometric base point and an orientation-preserving topological marking of its complex fiber. Its universal punctured curve supplies the outer monodromy homomorphism

$$R:\Pi_{\mathrm{ar}}=\pi_1^{\mathrm{et}}(\mathscr{M})\longrightarrow B.$$

The geometric comparison and the arithmetic exact sequence are

$$1\longrightarrow\widehat{M}\longrightarrow\Pi_{\mathrm{ar}}\longrightarrow G_{\mathbb{Q}}\longrightarrow 1,\qquad R(\widehat{M})=H. \tag{F.6.3}$$

These are the constructions in [8, §§7.1–7.2, equation (29) and the universal étale monodromy representation]. For the geometric comparison, finite étale covers analytify to finite topological covers; the inverse comparison is compatible on an étale atlas and its double and triple overlaps, so descends to the stack. The marked complex moduli stack has contractible Teichmüller space as its covering, with group $M$, which identifies its topological fundamental group. The universal curve then identifies its monodromy with the completed geometric action on $P$, rather than introducing a separate representation. All these maps preserve peripheral conjugacy classes, since the removed divisor is carried to itself in the universal family. Their base-point ambiguities are inner automorphisms of $P$, already removed in $B$; a change of topological marking conjugates the displayed subgroups by the corresponding element of $M$.

Set $A_{\mathrm{ar}}=R(\Pi_{\mathrm{ar}})$. It is compact and hence closed in $B$. Normality in (F.6.3) gives $H\triangleleft A_{\mathrm{ar}}\le N$. The quotient map induces a continuous surjection

$$\overline{R}:G_{\mathbb{Q}}\longrightarrow\Gamma_{\mathrm{ar}}. \tag{F.6.4}$$

The proof uses this quotient of $G_{\mathbb{Q}}$ as defined. Its kernel and the kernel of $\widehat{M}\to H$ need not be computed.

We next construct a number-field curve with the specified *labelled topological* $G$-action. Finite realization C.5 supplies a compact Riemann surface $X$, a reduced marked divisor $D$, and a faithful holomorphic action of $G$ preserving $D$ whose marked mapping classes are the given inclusion $G\hookrightarrow M$. Choose projective equations for $X$, equations for $D$, and formulas for the finitely many automorphisms and their inverses. Their finitely many coefficients lie in a finitely generated $\mathbb{Q}$-subalgebra $T\subset\mathbb{C}$. Adjoin the coefficients needed for their identities and invert

finitely many nonzero elements. This gives a proper smooth family of curves $\mathcal{X} \to V$, a finite étale reduced divisor $\mathcal{D} \to V$, and the same labelled $G$-action over a nonempty smooth open subscheme $V \subset \operatorname{Spec} T$. Smoothness and disjointness are open conditions. Faithfulness can be kept on this open set as well: for each nonidentity $h \in G$, the equalizer of $h$ and the identity is a proper closed subset of the generic fiber, and this property persists after shrinking. The original inclusion $T \hookrightarrow \mathbb{C}$ defines a point of every such nonempty open, since it has generic underlying point.

Base change to $\overline{\mathbb{Q}}$ and take the connected component whose complex base change contains that point. After a finite extension this component and the family are defined over a number field. It is a smooth connected variety; its complex points form a connected, locally path-connected complex manifold and hence a path-connected space. It contains an $\overline{\mathbb{Q}}$-point, by taking a closed point in any nonempty affine open. Choose a piecewise smooth complex path from the original fiber to this fiber.

For precision, parallel transport along this path preserves the labelled action. A smooth horizontal splitting of the relative tangent sequence can be chosen to agree with the tangent splitting on $\mathcal{D}$: locally the divisor is a disjoint union of sections, and a partition of unity extends their prescribed splittings. Average this splitting over the finite group $G$. Averaging retains its right-inverse property and its prescribed values on the divisor, and makes it $G$-equivariant. The horizontal lift along the compact path has complete transport, because the family is proper and its fibers are compact. Its transport is an orientation-preserving diffeomorphism of pairs commuting with each labelled element of $G$. Thus the endpoint curve has exactly the required topological action. This constructs a deformation and its comparison diffeomorphism; it does not assert that the original complex curve is isomorphic to a curve over $\overline{\mathbb{Q}}$.

All coefficients of the endpoint curve, divisor, and the finitely many labelled automorphisms lie in one finite extension $K/\mathbb{Q}$. Let $x_K : \operatorname{Spec} K \to \mathscr{M}$ be the resulting moduli point. Choose the geometric path to $x_K$ supplied by the transport just constructed and the finite-cover comparison. The morphism $x_K$ then supplies a continuous homomorphism

$$s_K : G_K \longrightarrow \Pi_{\mathrm{ar}}$$

whose composite with $\Pi_{\mathrm{ar}} \to G_{\mathbb{Q}}$ is the inclusion $G_K \hookrightarrow G_{\mathbb{Q}}$. Under the transported marking, $R \circ s_K$ is the Galois outer action on the punctured endpoint curve.

Every $h \in G$ is an automorphism over $K$ of that curve and its marked divisor. Naturality of the fiber functor on finite étale covers gives

$$(R \circ s_K)(\sigma)\, h = h\, (R \circ s_K)(\sigma) \quad \text{in } B \qquad (\sigma \in G_K). \tag{F.6.5}$$

Indeed, pullback by $h$ commutes with scalar conjugation by $\sigma$; choosing paths for the individual base points can alter the induced automorphisms only by inner automorphisms of $P$. Passing to $B$ gives precisely the equality written. The equivariant transport aligns these $h$ with the original labelled inclusion, so (F.6.5) places $R(s_K(G_K))$ in $C_{A_{\mathrm{ar}}}(G)$ itself.

The image of (F.6.1) therefore contains $\overline{R}(G_K)$. The latter has index at most $[K : \mathbb{Q}]$ in $\Gamma_{\mathrm{ar}}$ by the surjectivity of (F.6.4). It is compact and hence closed; a closed finite-index subgroup of a profinite group is open, as proved in F.3. Every subgroup containing it is a union of its open cosets. This proves openness in (F.6.1), with kernel $C_H(G)$ by intersection with $H$.

Finally $q^{-1}(\Gamma_{\mathrm{ar}}) = A_{\mathrm{ar}}$, since $H \subset A_{\mathrm{ar}}$ and $\Gamma_{\mathrm{ar}} = q(A_{\mathrm{ar}})$. An element of $C_N(G)$ whose quotient lies in $\Gamma_{\mathrm{ar}}$ therefore already belongs to $C_{A_{\mathrm{ar}}}(G)$. This proves both inclusions in (F.6.2). Ordering the divisor and using the pure moduli stack changes none of the maps or arguments when $G$ belongs to the pure oriented group.

For §30, folios 195–196, scans 385–387, formula (F.6.2) is the arithmetic open-image assertion with its exact quotient made explicit. F.3 supplies the centralizer image in the entire quotient

$N/H$; F.6 supplies its openness after intersection with the arithmetic image for every finite geometric $G$. The inclusion $\Gamma_{\mathrm{ar}} \hookrightarrow N/H$ is the comparison map between the two statements. Its image has not been identified with all of $N/H$ here, so no larger openness assertion is obtained by suppressing that inclusion. □

**F.7. [Open] The nonsolvable centralizer comparison and its checked finite image.** For an oriented mapping class group $M$ of genus at most two, the normalizer and centralizer comparisons for finite solvable subgroups have the complete proof below. For a geometric $A_5 \leq M = \Gamma^+_{0,\nu}$, the equality

$$C_{\widehat{M}}(A_5) = \overline{C_M(A_5)} \tag{F.7.1}$$

is stated in [9, Theorem C(iii)] and [11, Theorem 5.1(iii)]. The displayed proofs contain an additional closure–intersection equality for which we have not verified a justification. We do not assert that (F.7.1) is false.

There is nevertheless equality of their images in the finite puncture-permutation quotient. Write the invariant puncture set as a union of some of the exceptional orbits of sizes $12, 20, 30$ and $m$ free orbits of size 60. Both images fix the exceptional punctures individually and induce the full group $A_5^m \rtimes S_m$ on the free orbits. In particular, with only exceptional orbits the unresolved part of (F.7.1) lies in the pure profinite mapping class group; its discrete right side is trivial.

F.12–F.14 identify the residual twelve-puncture $A_5$ centralizer with an exact common-involution equalizer in the pure eight-point completion and prove injectivity after the specified ordinary-point forgetting. F.15 proves the full two-point pushing comparison and removes every cocycle in the actual intersection of the two forgetting kernels. Its correcting elements form the explicitly described fixed-subgroup cosets. The original source image has trivial intersection with that kernel; membership of a corrected element in that image remains the full equalizer equation. F.16 constructs a full closed procyclic subgroup of the correcting centralizer that every proper labelled forgetting kills and whose intersection with the source image is trivial. The exact membership test using all proper forgetting images, even with the common-involution centralization equation, therefore has a strictly larger solution group than the original source image. F.17 identifies the sixteen additional punctures required by the literal rational fixed-point drilling and proves its nonfactorization through the original source, using primitive full procyclic winding detectors. F.18 proves that every original residual element has a unique full branch-ordered centralizing lift. Its membership in the rational drilling image is the distinguished fibre of the explicit continuous coset map $\mathfrak{o} : \mathcal{E} \to \mathcal{J}/K^+$. Whether that fibre is the whole residual, and whether the residual is trivial, remain unresolved. F.19 identifies the joint forgetting kernel $D^+$ as the closed ambient normal closure of eight actual cross-collision meridians and proves its exact one-class cocycle comparison with a unique corrector. The canonical correction therefore lies in $D^+$, and the source-membership obstruction has the precise target $D^+/K^+$. The actual drilled image and the full two-track pullback have common explicit splittings over the original rank-eleven source, with kernels $K^+$ and $D^+$ respectively. Equality of those two subgroups and triviality of the original residual remain unresolved. F.20 proves the actual rank-three cross-divisor coset coordinate, its eleven based source actions and all original transports. Exact configuration computations and finite group calculations prove transitivity on the 82 invariant $A_5$ quotients, all $3,321$ distinct paired quotients, and six noncentral quotients of order $60 \cdot 2^{121}$. The specified continuous finite wreath maps consequently have the same full images on $K^+$ and $D^+$, for every coefficient prime. The full source-membership equation remains unresolved. F.21 proves transitivity on all eight full cross-divisor coset spaces and constructs lifts of their entire divisor groups in the actual selected-meridian centralizers. Consequently the closed normal closure of $K^+$ inside $D^+$ is $D^+$, and $K^+$ maps onto the full abelianization and maximal pronilpotent quotient of $D^+$. The exact equality $K^+ = D^+$ and the original residual source-membership equation remain unresolved. F.22 gives the actual two-point kernel as a semidirect product of

two explicit compactly generated profinite kernels, retains its full Artin action, and identifies the interaction kernel killed by both one-point forgettings. The source map has image exactly $K^+$; its joint one-point image is the product of its two actual marginal images. Fullness of those marginals and surjectivity of the displayed source map remain unresolved. F.23 gives an explicit based source-sign symmetry carrying the two actual marginal inclusions isomorphically to one another. It specifies all twenty-seven original source images by exact nine-strand words, with the original basepoint paths and inverse fiber coordinates. For every continuous homomorphism from either named ambient rank-seven group to $S_3$ or $A_4$, the marginal source and marginal target have equal images. The full marginal equalities, the interaction source image and the original residual membership remain unresolved. F.24 proves injectivity of the actual restriction on continuous rank-one crossed homomorphisms and first cohomology for every finite-field character nontrivial on the marginal target. Sixty-four actual source rows have rank seven on four explicit Laurent charts over every characteristic. The signed comparison retains the original source maps. Every scalar affine representation has the same fixed-point set on the marginal source and target, and its two marginal images agree whenever the source image meets the translation subgroup trivially. The remaining nonzero translation intersections, full marginal images and interaction source image remain unresolved. F.25 proves equality of those affine images for every representation whose source translation intersection is linear over the ambient scalar field. This includes every representation with equal ambient and marginal scalar fields, and every representation into $\mathrm{Aff}_1(\mathbb{F}_p)$ for every prime $p$. The chosen scalar lift, its translation conjugation, and the separating coefficient maps are explicit. Unequal scalar fields without that linearity property, the full marginal equalities and the interaction source image remain unresolved. F.26 gives an exact equivalent trace-cocycle kernel criterion for a proper affine marginal image, with the actual conjugation and coefficient homomorphism. Its tensor and Frobenius coefficient maps, including their inverses, identify the remaining rank target as $6m+1$ on $7m$ columns. Complete finite verification and an independently written replay prove equality of the actual marginal images for every scalar field of order nine, even inside a larger original coefficient field. General attainment of the trace rank, the full marginal equalities and the interaction source image remain unresolved. F.27 proves the exact trace rank and affine marginal image equality over every finite coefficient field on the actual two-parameter character locus $(1, s, t, 1, 1, 1, (st)^{-1})$. Its integer Laurent commutators force an essential source line, and actual character-one seed conjugates give an invertible Frobenius Vandermonde comparison. F.28 extends this to the three-parameter locus $(r, s, t, 1, 1, 1, (rst)^{-1})$: an explicit two-dimensional invariant inclusion and four quotient coordinates separate all minimal source submodules across distinct Frobenius factors. The individual restriction theorem then proves summed restriction injective and rank $6m+1$. Both results retain the exact based sign map and include larger original coefficient fields. The general scalar-character locus, full marginal equalities, interaction source image and original residual membership remain unresolved. F.29 proves the larger exact four-parameter locus $(r, s, t, u, 1, 1, (rstu)^{-1})$ over every finite coefficient field. Four actual principal-coordinate charts retain all original source words. Three charts have complete comparisons of their minimal invariant subspaces across Frobenius factors; the exceptional overlap has a forced-lift residual with coordinate $-2$ in its proved odd characteristic. The fourth chart has an essential line detected by actual character-one seed conjugates and an explicit Vandermonde inverse. Summed restriction has rank $6m+1$, and the exact trace and based sign maps give both marginal image equalities on this locus. The general character locus and the original full marginal and interaction comparisons remain unresolved.

*Manuscript:* §30; folios 194–196; scans 383–387; Part D(b) and Part F(b): nonsolvable completion centralizers.

*Source audit and proved comparisons.* The supplied preprint snapshot of [9] has since appeared in *São Paulo Journal of Mathematical Sciences*, volume 20, article 35, published 23 September 2026, DOI `10.1007/s40863-026-00558-y`. The publisher's full text was checked as well: Propo-

sition 2.4 retains the same displayed closure–intersection step, and Lemma 3.2 still invokes that proposition. The audit below applies to both inspected versions; it is a proof-step audit and makes no claim based on the paper's publication status.

For every finite subgroup $J \leq M$, there are canonical continuous maps

$$\widehat{C_M(J)} \longrightarrow \overline{C_M(J)} \hookrightarrow C_{\widehat{M}}(J). \tag{F.7.2}$$

The first is onto by density and compactness. Its injectivity states that the topology induced from $M$ on the discrete centralizer is its full profinite topology. Surjectivity of the second arrow is the ambient centralizer comparison. These are two specific properties of the displayed maps; injectivity of the first arrow does not supply surjectivity of the second.

In [9, Proposition 2.4], choose the displayed generators $x_1, \dots, x_k$ of the finite group, each of prime-power order. The prime-power centralizer theorem does prove

$$C_{\widehat{M}}(J) = \bigcap_i C_{\widehat{M}}(x_i) = \bigcap_i \overline{C_M(x_i)}. \tag{F.7.3}$$

The source then identifies this last intersection with $\overline{\bigcap_i C_M(x_i)}$. The same final equality occurs in the proof of [11, Theorem 5.1(iii)], there indexed by all $x \in J$. Neither displayed argument supplies a closure–intersection hypothesis. This exact step is what is left unverified here. The later virtually compact special application in [11, §6.2] has a separate relatively quasiconvex product-separability input; no such input has been proved here for the punctured-sphere mapping class group.

*A noncircular solvable proof.* We first record the prime-power inputs used in the induction. Goodness and finite virtual cohomological type of mapping class groups of genus at most two, and their finite-index subgroups, permit the direct application of [11, Theorem A]. Thus for a finite $p$-subgroup $A$ we have

$$N_{\widehat{M}}(A) = \overline{N_M(A)}, \qquad C_{\widehat{M}}(A) = \overline{C_M(A)}. \tag{F.7.4}$$

The proof of [9, Theorem A and Corollary B] also supplies the full induced profinite topology on $C_M(A)$ and hence on $N_M(A)$ in this prime-power case. Here is why this use avoids the preceding gap. Its free-action argument is the restriction-injectivity argument on the finite covering surface group. Its ramification-removal argument uses Proposition 2.4 only for the specified group $A$, now a $p$-group, in the good configuration-group extension. For that application the second equality in (F.7.4), directly from [11, Theorem A], replaces Proposition 2.4's intersection argument. Thus the two required inputs are justified without using any nonsolvable assertion. The passage from centralizer to normalizer topology uses its finite-index kernel under conjugation into $\mathrm{Aut}(A)$; this kernel is open in the induced topology, as is detected by any finite quotient in which $A$ injects.

We prove the solvable normalizer comparison, together with its full induced topology, by induction on the order of the finite solvable group, simultaneously for all finite-index subgroups of these mapping class groups. Finite-index passage preserves the comparison: intersect the completed normalizer with the corresponding open subgroup and use density in an open set.

For a nontrivial finite solvable $J$, take a nontrivial $p$-primary subgroup $A$ of the last nonzero term of its derived series. That term is abelian, so its $p$-primary subgroup is characteristic in $J$. Consequently every normalizer of $J$ normalizes $A$. The discrete covering normalizer sequence is

$$1 \longrightarrow A \longrightarrow N_M(A) \longrightarrow Q \longrightarrow 1, \tag{F.7.5}$$

where $Q$ is a finite-index subgroup of the mapping class group of the quotient surface with its branch and puncture decorations. Its genus is again at most two. The exact quotient map is induced by descent of the commuting covering diagram; the decorations record the original

punctures, the removed branch points and the covering subgroup. This is the discrete normalizer sequence in [9, §2.1].

By the prime-power topology input, the completed normalizer in (F.7.4) is $\widehat{N_M(A)}$. Completing (F.7.5) therefore gives its quotient by the finite embedded kernel $A$ as $\widehat{Q}$. The subgroup $J/A \leq Q$ is smaller and solvable. The induction hypothesis applies to its normalizer in $\widehat{Q}$. The inverse image of that normalizer under the quotient in (F.7.5), on either side, is exactly the normalizer of $J$: $J$ is the full inverse image of $J/A$ because it contains $A$. Pulling back the dense discrete normalizer proves

$$N_{\widehat{M}}(J) = \overline{N_M(J)}. \tag{F.7.6}$$

It also proves the topology assertion. Indeed the completion of this inverse image maps to $\widehat{N_M(A)}$; on the quotient by $A$ it is injective by induction, and on $A$ it is the specified inclusion. Its kernel is therefore trivial. This completes the induction.

Conjugation gives a continuous homomorphism

$$N_{\widehat{M}}(J) \longrightarrow \mathrm{Aut}(J)$$

to a finite group. Its kernel is the ambient centralizer and is open. The dense subgroup $N_M(J)$ from (F.7.6) meets that open kernel densely, and its intersection there is exactly $C_M(J)$. This proves

$$C_{\widehat{M}}(J) = \overline{C_M(J)}$$

for every finite solvable $J$, without taking an unjustified intersection of closures.

*The finite image for $A_5$.* Realize the geometric $A_5$ as the icosahedral rotation group. Its exceptional point stabilizers have orders $2, 3, 5$, and the corresponding transitive orbits are $A_5/C_2$, $A_5/C_3$, $A_5/C_5$. Every permutation commuting with this transitive action is right multiplication by an element of $N_{A_5}(C_d)/C_d$. This quotient has order two in each of the three cases; its nonidentity permutation exchanges the two poles of each relevant rotation axis.

An orientation-preserving homeomorphism commuting with an odd-order nonidentity rotation fixes its two poles individually. The local oriented rotation numbers at the two poles are opposite, and an orientation-preserving commuting homeomorphism preserves this local rotation number. This excludes the nonidentity permutation on the 12- and 20-point orbits for the discrete cyclic centralizers. Their completed centralizers have the same finite permutation images by (F.7.4). The full $A_5$ centralizer lies in these cyclic centralizers, so it fixes these exceptional punctures individually.

For an order-two stabilizer choose a tetrahedral subgroup $A_4 \leq A_5$ containing it. An orientation-preserving homeomorphism commuting with $A_4$ fixes every ramification point of the $A_4$ quotient. To see this, its quotient fixes the branch point of order two. It also fixes each of the two branch points of order three, rather than exchanging them, by the preceding local rotation argument. Isotope the quotient map to the identity relative to these three branch points and lift the isotopy. The lift stays in the centralizer; its endpoint is a deck transformation in $Z(A_4) = 1$. The ramification set is finite and invariant along the isotopy, so every one of its points is fixed. The usual discrete covering-centralizer comparison provides a commuting representative for a centralizing mapping class. Since $A_4$ is solvable, the proved centralizer comparison transfers this fixed-puncture property to $C_{\widehat{M}}(A_4)$. It therefore holds for $C_{\widehat{M}}(A_5)$, excluding the last possible nonidentity permutation on the 30-point orbit.

On $m$ free orbits the finite commutant is $A_5^m \rtimes S_m$. Every element of this commutant is already realized by the discrete centralizer. For a direct construction, remove the three branch points from the quotient sphere. Its unramified $A_5$-cover has surjective monodromy onto $A_5$. Move one of the $m$ quotient punctures around a loop, keeping all the others fixed, and lift the isotopy starting at the identity. The lift commutes with $A_5$ and acts on that one free orbit by its specified monodromy, while fixing the other orbits and all ramification points. These loops

realize each factor $A_5$. Half-twists interchanging the quotient punctures realize $S_m$ modulo these factors. Thus the displayed wreath product is the exact finite permutation image on both sides of (F.7.1).

The program `code/part_a/a5_centralizer_permutations.py` constructs $A_5$ as the sixty even permutations of five letters. It checks its order-two and order-three generating pair, the three coset actions, normalizer orders $4, 6, 10$, and commutants of order two. The fixed-pole constraints leave only the identity in each exceptional block; the order-two block uses the six ramification points for the specified $A_4$.

For completeness, the residual intersection question has an exact finite-quotient formulation. Take $u = (1\ 2)(3\ 4)$ and $v = (1\ 3\ 5)$ in $A_5$, which generate it, and put $D_u = C_M(u)$, $D_v = C_M(v)$. For open normal subgroups $V \subset U$ of $M$, let $D_u(V), D_v(V)$ denote their images in $M/V$. Compactness of coset fibres proves

$$\operatorname{im}\bigl(C_{\widehat{M}}(A_5) \to M/U\bigr) = \bigcap_{V \subset U} \operatorname{im}\bigl(D_u(V) \cap D_v(V) \to M/U\bigr). \tag{F.7.7}$$

Indeed choose a lift $t \in \widehat{M}$ of a point in the right side. For each $V \subset U$, the set

$$(\overline{D_u}\,\overline{V}) \cap (\overline{D_v}\,\overline{V}) \cap t\overline{U}$$

is nonempty and compact. These sets are directed under refinement, so their finite intersections are nonempty. Compactness produces a single element in their intersection, which is $\overline{D_u} \cap \overline{D_v} \cap t\overline{U}$. Conversely any such element supplies every finite coordinate. Equations (F.7.3) and (F.7.7) identify the remaining assertion precisely: its right side must equal the image of $C_M(A_5)$ for every $U$. The finite permutation computation proves that equality at the puncture-permutation level. It does not establish it at every refinement. With only exceptional orbits the discrete centralizer is trivial, since the quotient sphere has just the three differently ordered branch points and $Z(A_5) = 1$; the still unproved assertion is then that the pure completed centralizer is trivial.

*The minimal twelve-puncture case.* There is a further exact localization of this remaining group. Write $P = P\Gamma_{0,12}$ and choose $A = A_4 \leq A_5$. The action of $A$ on the twelve vertices is free and transitive. Its quotient has one free puncture and three branch points of orders $2, 3, 3$. The labelled commuting homeomorphism group on the filled sphere is identified, by descent and the unique commuting lift, with the identity homeomorphism component of the quotient fixing those three branch points. It is contractible. Evaluating the free quotient puncture consequently identifies the mapping-class centralizer with the free group $\pi_1(S^2 \setminus \{3 \text{ branch points}\}) = F_2$. Its permutation on the free fibre is the surjective monodromy onto $A_4$. The pure centralizer is therefore the exact kernel

$$1 \longrightarrow C_P(A) = F_{13} \longrightarrow F_2 \longrightarrow A_4 \longrightarrow 1. \tag{F.7.8}$$

Indeed it is the fundamental group of $X = S^2 \setminus R_A$, where the $A_4$ ramification set $R_A$ has $6+4+4 = 14$ points; its rank is also $1 + 12(2-1) = 13$ by Schreier. Label the vertices by $g \in A_4$ using a chosen vertex $z_0$. The inclusion is the configuration map $z \mapsto (gz)_{g \in A_4}$, followed by the quotient by Möbius transformations. The proved solvable comparison and full topology identify its completion with $C_{\widehat{P}}(A)$. In particular the unknown group $C_{\widehat{M}}(A_5)$ lies in this specified embedded $\widehat{F}_{13}$.

Each of the six subgroups $C_5 \leq A_5$ has two fixed vertices and two free five-cycles. Retain either one fixed vertex and one five-cycle, giving twenty-four six-point sets, or retain both fixed vertices and one five-cycle, giving twelve seven-point sets. For each retained set $I$, the continuous forgetting map $r_I$ is defined on $\widehat{P}$ and is equivariant for the specified $C_5$. In either target the quotient has exactly its two branch points and one free puncture. Its discrete $C_5$ centralizer is therefore exactly the deck group $C_5$: the quotient mapping class group fixing these three

decorated points is trivial. Prime-power comparison gives the same completed centralizer. Its intersection with the pure group is trivial, since the deck group permutes the five-cycle. It follows that

$$C_{\widehat{M}}(A_5) \subset C_{\widehat{P}}(A) \cap \bigcap_{I \text{ among these 36 sets}} \ker(r_I). \tag{F.7.9}$$

The six-point maps in (F.7.9) have a proved common kernel even on the discrete $F_{13}$. The antipodal permutation on the twelve vertices commutes with $A_5$. In the regular $A_4$ labelling it is right multiplication by an involution $h \in A_4$. Every retained six-point set contains exactly one member of each antipodal pair. Collisions of two entries $g_i z, g_j z$ at a ramification point with stabilizer $H \leq A_4$ occur precisely when $g_j^{-1} g_i \in H$; these are the right cosets $gH$. Thus there is no collision among the retained entries at either fixed point of $h$. Each six-point configuration map extends across both of these deleted points of $X$. Their nontrivial peripheral meridians in $F_{13}$ map to the identity under all twenty-four maps. Completion retains these meridians as nontrivial primitive procyclic subgroups. Consequently the joint six-point map on the embedded $\widehat{F}_{13}$ is noninjective by these explicit elements.

Every seven-point set retains a collision at each of the fourteen ramification points: seven choices from six pairs force a pair, and seven choices from four triples force a triple collision. The program verifies all twelve sets and the preceding common invisible axis for the six-point sets, using right cosets throughout. It also computes the induced maps on $H_1(F_{13}, \mathbb{F}_{101})$. A meridian at an $A_4$ ramification point maps to the sum of the pair generators within its collision clusters. In the target pure sphere group the pair generators satisfy the vertex-sum and total-sum relations. The two fixed points of each cyclic stabilizer give the same homology column. Each seven-point map and their joint map have rank six, with seven-dimensional kernel; the joint six-point map has rank five. These are exact calculations of the indicated abelian quotients. They neither produce a nontrivial $A_5$ centralizer element nor prove injectivity of the joint seven-point map on $\widehat{F}_{13}$.

F.8 proves that all twelve seven-point kernels in (F.7.9) are the same, both discretely and after completion. It further proves that the discrete kernel contains no nonidentity reducible mapping class and that the completed kernel meets every conjugate of every full source cusp subgroup trivially. The exact four-point characters detecting those cusps and all eighty-four finite subset–inertia checks are supplied there. F.9 subsequently constructs an explicit nontrivial common kernel element with a full procyclic closure, and proves that all twenty-four six-point restrictions factor through these seven-point maps. The thirty-six-fold kernel is therefore nontrivial. The full centralizer comparison and geometric conjugacy remain unproved; no converse to the inclusion (F.7.9) is inferred.

**The exact common-$C_3$ comparison.**

The following maps give an exact model for the residual group in (F.7.1). The fixed intersection displayed below is the remaining unresolved group. All permutations use the five-letter identification of $A_5$ already fixed in F.7–F.10. Put

$$A = \mathrm{Stab}_{A_5}(5), \qquad r = (1\ 2\ 3), \qquad K = \langle r \rangle, \qquad \beta = (1\ 2)(4\ 5).$$

Then $A \cong A_4$, $\beta K \beta^{-1} = K$ by inversion, and $\langle A, \beta \rangle = A_5$.

**The actual quotient homomorphism.** Let $P = P\Gamma_{0,12}$ at the icosahedral configuration. The discrete branched-cover sequence for the $K$ action is

$$1 \longrightarrow K \longrightarrow N(K) \longrightarrow Q \longrightarrow 1,$$

where $Q$ is the finite-index liftable subgroup of the mapping class group of the quotient sphere with two decorated branch points and four free punctures. The centralizer $C_P(K)$ has finite

index in $N(K)$: both the centralizing condition and the pure puncture condition have finite index. Its intersection with $K$ is trivial, since nonidentity deck maps permute the twelve punctures. Its image $J$ in $Q$ is therefore isomorphic to $C_P(K)$ and has finite index in $Q$. Every element of $J$ fixes the four free quotient punctures. It also fixes the two branch points separately: exchanging them conjugates the order-three local deck generator to its inverse, and hence cannot centralize it. Thus

$$C_P(K) \xrightarrow{\sim} J \leq P\Gamma_{0,6}, \qquad [P\Gamma_{0,6} : J] < \infty. \tag{F.7.10}$$

The prime-power comparison and full induced topology proved in F.7 identify $C_{\widehat{P}}(K)$ with $\widehat{C_P(K)}$. Completing the finite-index embedding (F.7.10) identifies it with the specified open subgroup $\widehat{J} \leq \widehat{P\Gamma}_{0,6}$. This gives an injective continuous homomorphism on the actual completed centralizer; no completed intersection identity is used.

By F.7.8 the orbit map identifies $C_P(A)$ with the specified $F_{13} = \pi_1(X)$, where

$$X = \mathbf{P}^1 \setminus R_A, \qquad R_A = \{\infty\} \cup \{z : z(z^4-1)(z^8+14z^4+1) = 0\}.$$

Since $K \leq A$, composition with (F.7.10) is the injection $j : F_{13} \hookrightarrow P\Gamma_{0,6}$ described next. The solvable comparison and full topology for $A$ and $K$ show that its continuous extension

$$\widehat{j} : \widehat{F}_{13} \hookrightarrow \widehat{P\Gamma}_{0,6} \tag{F.7.11}$$

is injective. This is a different, explicitly specified map from the thirty-six forgetful maps whose common kernel was computed in F.9.

**The six labelled quotient points.** In the F.9 tetrahedral coordinate,

$$r(z) = \frac{z+i}{z-i}, \qquad q(z) = z + r(z) + r^2(z) = \frac{z^3+3iz+1-i}{(z-1)(z-i)}.$$

The degree-three quotient $q$ is invariant under $r$. Its branch points upstairs are

$$p_\pm = \frac{(1+i)(1\pm\sqrt{3})}{2},$$

and their images are $3p_\pm$. In the twelve-point labelling of F.8 the four free $K$-orbits, in order, are

$$(0,8,1), \quad (11,3,5), \quad (2,6,9), \quad (7,10,4). \tag{F.7.12}$$

Their representatives have coordinates $z, 1/z, -1/z, -z$. Consequently the unnormalized labelled quotient map is

$$z \longmapsto (3p_+, 3p_-, q(z), q(1/z), q(-1/z), q(-z)). \tag{F.7.13}$$

Its entries are distinct on $X$: any equality between two free quotient points would identify their points in the $A$ orbit, and a collision with a branch point would give a nontrivial stabilizer in $A$.

Put $U(z) = (z-p_+)/(z-p_-)$. Direct substitution gives

$$\frac{q(z)-3p_+}{q(z)-3p_-} = U(z)^3.$$

Normalizing the first, second, and third points in (F.7.13) to $0, \infty, 1$ gives the actual map

$$\Sigma : X \longrightarrow M_{0,6}, \qquad z \longmapsto (s_1(z), s_2(z), s_3(z)),$$

where

$$s_1 = \left(\frac{U(1/z)}{U(z)}\right)^3, \qquad s_2 = \left(\frac{U(-1/z)}{U(z)}\right)^3, \qquad s_3 = \left(\frac{U(-z)}{U(z)}\right)^3. \tag{F.7.14}$$

The homomorphism $j$ in (F.7.11) is precisely $\Sigma_*$, by descent of the same labelled orbit path.

**The second $A_4$ and the exact based involution.** The involution $\beta$ exchanges the two branch points and permutes the four free orbits in (F.7.12) by (0 1)(2 3). On the normalized moduli coordinates its relabelling is therefore

$$B(s_1, s_2, s_3) = (s_1, s_1/s_3, s_1/s_2). \tag{F.7.15}$$

For example, after swapping the branch points the new normalizing coordinate is $u \mapsto s_1/u$, since the first free point has become the old point $s_1$. This derives (F.7.15) from the actual labelled quotient, rather than selecting a conjugate linear action.

Use $z_*$ and the path $\gamma : z_* \to 5 + z_* \to b = 5 + 4i$ from (F.10.12). The point $\Sigma(z_*)$ is fixed by $B$. Let $\tau = \Sigma \circ \gamma$, with the indicated direction. For a chronological loop $\ell$ based at $\Sigma(b)$, the transported involution is exactly

$$\beta_b(\ell) = \tau^{-1} * B(\tau) * B(\ell) * B(\tau)^{-1} * \tau. \tag{F.7.16}$$

The path endpoints in this expression match in order; $B^2 = 1$ and $B\Sigma(z_*) = \Sigma(z_*)$ show directly that $\beta_b^2 = 1$. It induces the conjugation action of the same transported finite mapping class $\beta$ used in F.10 on the injective quotient image (F.7.10).

Since $A$ and $\beta$ generate $A_5$, the exact residual comparison is

$$\widehat{j}(C_{\widehat{P}}(A_5)) = \widehat{j}(\widehat{F}_{13}) \cap \mathrm{Fix}\Big(\beta_b : \widehat{P\Gamma}_{0,6} \longrightarrow \widehat{P\Gamma}_{0,6}\Big). \tag{F.7.17}$$

Indeed centralizing $A_5$ is exactly centralizing $A$ and $\beta$. For elements centralizing $A$, equation (F.7.11) is injective; also $\beta$ normalizes $K$, so both the element and its $\beta$ conjugate lie in the domain of that injection. Equality of their quotient images is therefore equivalent to their equality upstairs.

**Executed exact checks and their scope.** The program `code/part_a/second_a4_c3_attack.py` uses exact arithmetic in $\mathbb{Q}(i, \sqrt{5}, \sqrt{\varphi + 2})$ to check the 60 Möbius matrices, 180 generator products, and 720 vertex-label images. It verifies all four orbits in (F.7.12), all representative functions, the quotient identity, and (F.7.14). It also checks the fixed-locus calculation

$$\Sigma(z) \in \mathrm{Fix}(B) \quad \Longleftrightarrow \quad s_1(z) = s_2(z)s_3(z),$$

whose numerator, up to a nonzero scalar and after cancelling the displayed rational denominators, is

$$z(z^2 + 1)(z^8 + 16z^6 + 46z^4 + 16z^2 + 1). \tag{F.7.18}$$

It verifies exactly that $z_*$ satisfies the relevant quartic factor. Finite intersection of these geometric loci does not by itself calculate the intersection of their completed fundamental groups in (F.7.17).

The auxiliary numerical braid discovery in `work/part_a/f9_c3_johnson_probe.py` currently gives no exclusion at the first nonzero outer Johnson degree. It is explicitly exploratory: the crossings there have not been certified for the actual based map (F.7.16), and a higher-degree comparison must retain that path. No conclusion about (F.7.17) uses that calculation.

**The modular orbifold and the exact $GT_0^1$ image.** There is a further exact description of the ambient peripheral centralizer in the minimal twelve-puncture case. It also gives five specified three-cusp tests for the remaining geometric intersection. The intersection itself remains unresolved.

Let

$$\Delta = \langle a, b \mid a^2 = b^3 = 1 \rangle, \qquad E = \widehat{\Delta} = C_2 \amalg C_3, \qquad d = ab, \quad e = ba. \tag{F.7.19}$$

The isomorphism with the full profinite coproduct follows from the same universal maps on the two finite factors. Use the epimorphisms

| | $a$ | $b$ | kernel |
|---|---|---|---|
| $q_3 : E \to S_3$ | (2 3) | (1 2 3) | $K$ |
| $q_5 : E \to A_5$ | (1 2)(3 4) | (1 3 5) | $H$. |

(F.7.20)

The discrete modular orbifold has one order-two point, one order-three point, and one puncture, whose oriented loop is $d$. In the first quotient $d$ has order two, and in the second it has order five. The corresponding torsion-free covers have the following exact data:

| | degree | cusp width | number of cusps | surface type |
|---|---|---|---|---|
| $K$ | 6 | 2 | 3 | $(0, 3)$ |
| $H$ | 60 | 5 | 12 | $(0, 12)$. |

(F.7.21)

Indeed the quotient maps inject both finite orbifold stabilizers, so the covers are surfaces. Their Euler characteristics are the degree times $-1/6$; the cusp counts are the indices of $\langle q_i(d)\rangle$ in the finite quotient. The resulting genera are zero. Completion of these finite-index inclusions retains their full profinite topology. Thus $K \cong \widehat{F}_2$, $H \cong \widehat{F}_{11}$, and the second extension is the actual centre-free orbifold extension of the icosahedral twelve-puncture action.

Here is the precise identification with the specified action in F.7. Start with that oriented sphere $S$ and its twelve-vertex puncture set $V$. Choose a basepoint $s \in S \setminus V$ outside every nontrivial stabilizer and use its image in the quotient orbifold. Choose ordinary positive orbifold loops $u_2, u_3$ around the order-two and order-three points, with whiskers so that the positive puncture loop is $u_c = (u_2u_3)^{-1}$. Set

$$a = u_3^{-1}u_2u_3, \qquad b = u_3^{-1}; \qquad ab = u_c.$$

Path lifting gives the monodromy map to the actual deck group $A_5$. Its generating pair can be labelled as the pair $q_5(a), q_5(b)$ in (F.7.20), by the exact single-orbit calculation below. This relabels the deck group and does not change its kernel or the given subgroup of the sphere mapping class group. Based loop lifting now identifies the discrete kernel with $\pi_1(S \setminus V, s)$, and its completion identifies $H$ with the same specified profinite surface group. The chosen lift of the cusp whisker selects a vertex $v_0 \in V$ stabilized by $q_5(d)$. The actual identification of the twelve labels is

$$A_5/\langle q_5(d)\rangle \longrightarrow V, \qquad g\langle q_5(d)\rangle \longmapsto gv_0.$$

Under it the cusp $[td^5t^{-1}]_H$ is exactly the puncture $q_5(t)v_0$. In particular the identification of $H$ carries its pure geometric mapping class subgroup to the one used in F.7. Changing the base path changes this identification by an inner surface-group map; changing the chosen lift changes it by the specified deck action. An equivalent choice of the oriented orbifold presentation gives the corresponding based covering isomorphism. These maps conjugate the peripheral outer group and its geometric subgroup together, and carry the displayed $A_5$ to the same geometric action. They therefore carry the intersection below to the identical residual centralizer under that covering identification.

For precision, $x = d^2$ and $y = e^2$ are a full free basis of $K$. Schreier rewriting with transversal $1, a, b, ab, ba, b^{-1}$ gives the following kernel words $t_g s t_{q_3(gs)}^{-1}$:

| $t_g$ | $s = a$ | $s = b$ |
|---|---|---|
| 1 | 1 | 1 |
| $a$ | 1 | 1 |
| $b$ | 1 | 1 |
| $ab$ | $x$ | $y^{-1}$ |
| $ba$ | 1 | $y$ |
| $b^{-1}$ | $x^{-1}$ | 1. |

(F.7.22)

Rewriting $a^2$ and $b^3$ at these six representatives gives only freely trivial words in $x, y$; these twelve relator checks are included in the certificate. The resulting Schreier presentation is exactly $\langle x, y \mid \rangle$, giving the stated basis and then its full profinite basis. Conjugation by $a$ sends $(x, y)$ to $(y, x)$, while conjugation by $b$ sends it to $(y, x^{-1}y^{-1})$. Writing $\omega(x) = y$, $\omega(y) = (xy)^{-1}$, the latter automorphism is exactly $\mathrm{Inn}(y)\omega$. Therefore the outer deck action on $K$ is exactly the geometric $S_3$ used in D.35, with these peripheral generators.

*Characteristic kernels and the finite normalization.* There are six generating order-two/order-three pairs in $S_3$, forming one orbit under its six inner automorphisms. There are 120 such generating pairs in $A_5$, forming one orbit under conjugation by $S_5$. For the latter count, fix $(1\ 2)(3\ 4)$: the generating three-cycles use the fifth letter and one letter from each of its two pairs, giving eight choices. Their products with the involution have order five. The remaining choices generate proper subgroups. The finite certificate below verifies all 300 pairs, as well as the transitive $S_5$ action. Consequently each epimorphism in (F.7.20) has the unique kernel of its indicated quotient type. In particular both $K$ and $H$ are characteristic in $E$.

We also need the exact quotient-automorphism statement. The five $V_4$ subgroups of $A_5$ form an intrinsic set. Its conjugation action is faithful. An automorphism acting trivially on these five subgroups acts trivially on that faithful permutation representation, and hence on $A_5$. Thus $\mathrm{Aut}(A_5)$ embeds in $S_5$; all 120 conjugations by $S_5$ already give automorphisms, so this is the full automorphism group. An odd conjugation exchanges the two $A_5$ conjugacy classes of five-cycles. It follows that an automorphism preserving the conjugacy class of $q_5(d)$ is inner. The certificate verifies the five $V_4$ groups, faithfulness, and both twelve-element five-cycle classes.

*The full peripheral centralizers.* For $T = K, H$, let $\mathcal{P}^1_T$ denote the subgroup of $\mathrm{Out}(T)$ preserving each individual oriented peripheral element up to conjugacy with multiplier one. Put

$$\Omega^1_E = \{[\eta] \in \mathrm{Out}(E) : \eta(d) \sim_E d\}.$$

There are canonical continuous isomorphisms, with the normalizations described below,

$$C_{\mathcal{P}^1_H}(A_5) \xrightarrow{\sim} \Omega^1_E \xrightarrow{\sim} C_{\mathcal{P}^1_K}(S_3) = GT^1_0, \qquad GT^1_0 = \ker(\lambda : GT_0 \to \widehat{\mathbb{Z}}^\times). \tag{F.7.23}$$

We prove the maps rather than identify the two actions only by their surface types.

First, for $m = 2$ and $T = K$, or for $m = 5$ and $T = H$,

$$C_E(d^m) = \overline{\langle d \rangle} \cong \widehat{\mathbb{Z}}. \tag{F.7.24}$$

The element $d^m$ is a primitive cusp of the free group $T$, so $C_T(d^m) = \overline{\langle d^m \rangle}$. This follows, for example, by writing $T$ as the free profinite product of that primitive cyclic factor and its free complement and using factor malnormality as in D.35. The quotient action on the cusp classes is the actual action on $Q/\langle q(d) \rangle$, where $Q = S_3$ or $A_5$. An element centralizing $d^m$ therefore has quotient in $\langle q(d) \rangle$. Multiplying it by a suitable integral power of $d$ puts it in $T$ without changing centralization. This proves the equality in (F.7.24). The procyclic group is a full $\widehat{\mathbb{Z}}$ because $d^m$ already generates such a subgroup: a relation $d^u = 1$ would imply $(d^m)^u = 1$ and hence $u = 0$.

The centralizer $C_E(T)$ is trivial. Its image in $Q$ would act trivially in $\mathrm{Out}(T)$, but that action is faithful on the cusp permutation set. Thus this image is trivial, and the remaining element lies in $Z(T) = 1$. Consequently conjugation embeds $E$ in $\mathrm{Aut}(T)$ as the full inverse image of the specified $Q \leq \mathrm{Out}(T)$.

Let $[\phi] \in C_{\mathcal{P}^1_T}(Q)$. Conjugation by an actual representative $\phi$ on $\mathrm{Aut}(T)$ preserves that inverse image and gives an automorphism $\eta$ of $E$ inducing the identity on $Q$ and restricting to $\phi$ on $T$. Changing $\phi$ by an inner automorphism of $T$ changes $\eta$ by the same inner automorphism of $E$. Also $\eta(d)^m$ is conjugate in $T$ to $d^m$. After that conjugation, (F.7.24) places $\eta(d)$ in $\overline{\langle d \rangle}$; the equation $mu = m$ in $\widehat{\mathbb{Z}}$ forces its exponent to be one. Thus $[\eta] \in \Omega^1_E$.

Conversely, choose $[\eta] \in \Omega^1_E$. Its induced automorphism of $Q$ is inner: for $S_3$ this follows from its full automorphism group; for $A_5$ it follows from preservation of the five-cycle class proved above. If that quotient automorphism is $\mathrm{Inn}(q_0)$, choose $u \in E$ with $q(u) = q_0^{-1}$ and replace $\eta$ by $\eta_0 = \mathrm{Inn}(u)\eta$. The new automorphism induces the identity on $Q$. Different choices of $u$ differ by an element of $T$, so the outer class of its restriction to $T$ is well defined. Changing the representative of $[\eta]$ has the same inner effect after normalization; the centre of $Q$ is trivial, so there is no further quotient ambiguity.

Write $\eta_0(d) = rdr^{-1}$. Because the quotient action is now the identity, $q(r)$ centralizes $q(d)$. In both finite groups that centralizer is exactly $\langle q(d)\rangle$. Hence $r = td^j$ for $t \in T$, and $\eta_0(d) = tdt^{-1}$. For every $s \in E$, the elements $\eta_0(s)$ and $s$ have the same quotient. Therefore the restriction preserves each $T$-cusp class $[sd^m s^{-1}]_T$ with exponent one. It belongs to $\mathcal{P}^1_T$ and centralizes the outer $Q$ action.

These constructions are inverse homomorphisms. For injectivity, a normalized automorphism whose restriction to $T$ is inner can first be adjusted to fix $T$ pointwise. Then $\eta(s)s^{-1} \in C_E(T) = 1$ for every $s \in E$, so it is the identity. In the other direction, an inner automorphism of $E$ arising from a centralizing outer class of $T$ has quotient in $Z(Q) = 1$, hence its conjugator lies in $T$. Continuity follows from conjugation on the specified closed subgroup of $\mathrm{Aut}(T)$ and normalization by the finite quotient. The maps are homeomorphisms because all the groups here are compact Hausdorff. Finally the equality with $GT^1_0$ is precisely the Harbater–Schneps centralizer identification used in D.35, now with the exact basis and outer actions in (F.7.22)[31, Propositions 3–4 and Theorem 2].

Let $\iota : GT^1_0 \hookrightarrow \mathcal{P}^1_H$ denote the inverse of the left-to-right isomorphism in (F.7.23). Genus-zero congruence identifies the completed pure geometric group with its actual image in $\mathcal{P}^1_H$. Thus the remaining group has the exact description

$$C_{\widehat{P\Gamma}_{0,12}}(A_5) = \widehat{P\Gamma}_{0,12} \cap \iota(GT^1_0) \quad \text{inside } \mathcal{P}^1_H. \tag{F.7.25}$$

This proves the relationship with the full $GT_0$ problem; it does not calculate this intersection.

*Explicit substitutions and three-cusp quotient maps.* Write $\gamma = (1, f) \in GT^1_0$ in its normalized representative on $K$, so $x \mapsto x$, $y \mapsto f^{-1}yf$ and $f \in K'$. Its actual extension to (F.7.19) has the particularly simple formulas

$$\eta_f(a) = f^{-1}a, \qquad \eta_f(b) = f^{-1}b, \qquad \eta_f(d) = d. \tag{F.7.26}$$

Indeed the first symmetry is $\theta(f) = f^{-1}$. Direct conjugation on $K$ gives $\gamma\theta = \mathrm{Inn}(f^{-1})\theta\gamma$, so $C_E(K) = 1$ forces the stated image of $a$. Since $\gamma(x) = x$, (F.7.24) with $m = 2$ forces $\eta_f(d) = d$; the displayed image of $b = a^{-1}d$ then follows. These formulas describe the actual automorphism furnished by (F.7.23), including its relations and its invertibility.

The quotient automorphism on $A_5$ fixes $q_5(d)$ and is inner. There is consequently a unique $j \in \{0, 1, 2, 3, 4\}$ for which it is $\mathrm{Inn}(q_5(d)^j)$; equivalently

$$q_5(f) = q_5(e)^j q_5(d)^{-j}.$$

The pure normalized representative on $H$ is

$$\eta^H_f = \mathrm{Inn}(d^{-j})\eta_f\big|_H. \tag{F.7.27}$$

For example, with $c_0 = d^5$ and $c_1 = ad^5a^{-1} = e^5$, put $h_1 = d^{-j}f^{-1}e^j \in H$. Then

$$c_0 \longmapsto c_0, \qquad c_1 \longmapsto h_1 c_1 h_1^{-1}. \tag{F.7.28}$$

This uses the finite quotient to ensure $h_1 \in H$; it is not a formal substitution of a word outside the source of a filling homomorphism.

Choose the cusp labels as the twelve left cosets $A_5/\langle q_5(d)\rangle$, in the breadth-first order specified by the certificate. The corresponding representatives $t_i \in \Delta$ and peripheral words $c_i = t_i d^5 t_i^{-1}$ are recorded there; $t_0 = 1$ and $t_1 = a$. For a three-element label set $J$ define the actual continuous filling homomorphism

$$q_J : H \longrightarrow H_J = H/\langle\langle c_i : i \notin J\rangle\rangle_H \cong \widehat{F}_2. \tag{F.7.29}$$

The isomorphism type follows from filling precisely those nine punctures of the specified sphere; completion preserves the displayed quotient presentation. Since (F.7.27) is pure peripheral, it descends to $H_J$. Thus there is a continuous homomorphism

$$\mathcal{D}_J : GT_0^1 \longrightarrow \mathcal{P}^1_{H_J}, \qquad \mathcal{D}_J(\gamma) = \big[q_J(h) \longmapsto q_J(\eta_f^H(h))\big]. \tag{F.7.30}$$

The brackets on the right specify the outer automorphism induced by the displayed substitution. Any change of representative in (F.7.27) changes it only by an inner automorphism of $H_J$. Formula (F.7.26), the finite determination of $j$, and (F.7.29) define each map without assuming that any two arbitrarily based retained cusp words form a full free basis.

The pure geometric mapping class group of the three-punctured sphere is trivial. The actual forgetting homomorphism on pure mapping classes, its action on the filled fundamental group, and continuity on completions therefore give

$$\iota^{-1}\Big(C_{\widehat{P\Gamma}_{0,12}}(A_5)\Big) \subseteq \bigcap_{|J|=3} \ker \mathcal{D}_J. \tag{F.7.31}$$

For $g \in A_5$, a lift in $E$ induces the actual quotient isomorphism $H_J \to H_{gJ}$ by conjugation; different lifts differ by an inner automorphism. Because $\iota(\gamma)$ centralizes $A_5$ in $\mathrm{Out}(H)$, these isomorphisms intertwine $\mathcal{D}_J(\gamma)$ and $\mathcal{D}_{gJ}(\gamma)$. Their kernels are consequently equal within each actual $A_5$ orbit of three-element subsets.

The 220 triples form exactly five orbits, with the following representatives in the recorded coset labelling:

| $J$ | $\{0,1,2\}$ | $\{0,1,4\}$ | $\{0,1,8\}$ | $\{0,1,10\}$ | $\{0,4,9\}$ |
|---|---|---|---|---|---|
| $\lvert A_5 J\rvert$ | 20 | 60 | 60 | 60 | 20. |

(F.7.32)

The certificate supplies a labelled $A_5$ transporter, as a word in $a, b$, for every triple. Hence the intersection of 220 kernels in (F.7.31) equals the intersection of these five specified kernels. No injectivity or triviality of that intersection has been proved here.

The standard-library executable `code/part_a/modular_a5_gt0_f7.py` has been run. It checks the six and 120 epimorphism pairs and their single postcomposition orbits, both faithful cusp actions, both cyclic cusp centralizers, the five-$V_4$ automorphism argument, the exact twelve Schreier rewritings in (F.7.22), all twelve peripheral words, and all 220 subset transporters in (F.7.32). These finite calculations certify the finite inputs and the displayed words. The centre-free extension, normalization, completion, and filling maps are proved above.

**Exact correspondences for the two orbit-twenty detectors.** The two orbit-twenty tests in (F.7.32) admit the following explicit subgroup and filling description. It gives continuous word substitutions and a common section of the surface-group quotient maps. Faithfulness of the outer-automorphism detectors remains unresolved.

Use the precise $E, q_5, H, K, a, b, d, e$ of (F.7.19)–(F.7.22). Put $T = \langle q_5(b)\rangle$ and $B = q_5^{-1}(T)$. The index-twenty orbifold cover has two order-three points and four width-five cusps, and no order-two points. This is the exact coset action: $a$ has ten two-cycles, $b$ has two fixed cosets and six three-cycles, and $d$ has four five-cycles. Its signature is $(0; 3, 3; 4 \text{ cusps})$.

The following words give a specified full profinite presentation:

$$\begin{aligned} &z = abab^{-1}, \qquad \tau = zaz^{-1},\\ &r = b, \qquad s = \tau b\tau^{-1},\\ &u = (ab^{-1})^4 a, \qquad v = (ab)u(ba),\\ &w = (abab^{-1}a)(ba)^3(b^{-1}aba). \end{aligned} \qquad B = \langle r, s, u, v, w \mid r^3 = s^3 = 1\rangle_{\mathrm{prof}}. \tag{F.7.33}$$

Thus $B = C_3 \amalg C_3 \amalg \widehat{F}_3$ with the displayed inclusion into $E$. The exact Schreier certificate uses twenty right cosets $Tg$ and the prefix-closed breadth-first transversal in $a, b$. Its forty directed generator edges have nineteen tree edges, leaving twenty-one Schreier generators. Rewriting $a^2, b^3$ at the twenty vertices gives forty relators. Sixteen Tietze eliminations, each solving a relator in a generator occurring exactly once, leave the five displayed generators and precisely the two cube relators. The complete transversal, original generator words, substitutions, and eliminations are recorded. Reduction in $C_2 * C_3$ checks the compact words in (F.7.33). These are inverse presentation maps; completion of the finite-index discrete inclusion retains its full profinite topology.

The four based cusp loops and their actual $T$-orbits of $H$ punctures are

| $i$ | $\kappa_i$ in $B$ | $J_i$ |
|---|---|---|
| 0 | $u^{-1}r$ | $\{0,1,2\}$ |
| 1 | $v^{-1}$ | $\{3,4,6\}$ |
| 2 | $wu$ | $\{5,7,8\}$ |
| 3 | $vw^{-1}s$ | $\{9,10,11\}$ |

(F.7.34)

Each is the rewritten word $t_i d^5 t_i^{-1}$ for its recorded transversal representative. The first and fourth triples lie in the $A_5$-orbit of $\{0,1,2\}$; the middle two lie in the orbit of $\{0,4,9\}$.

Fill the other three orbifold cusps and put $B_i = B/\langle\langle \kappa_j : j \neq i\rangle\rangle_B$. To $A = \langle R, S \mid R^3 = S^3 = 1\rangle_{\mathrm{prof}} = C_3 \amalg C_3$ send $r$ to $R$, $s$ to $S$, and use the indicated row:

| $i$ | $u$ | $v$ | $w$ | $\kappa_i$ |
|---|---|---|---|---|
| 0 | $S^{-1}$ | $1$ | $S$ | $SR$ |
| 1 | $R$ | $S^{-1}R^{-1}$ | $R^{-1}$ | $RS$ |
| 2 | $R$ | $1$ | $S$ | $SR$ |
| 3 | $R$ | $1$ | $R^{-1}$ | $RS$ |

(F.7.35)

Substitution kills the three required cusp relators. Conversely those three relators solve for $u, v, w$ as the entries of the row. Inclusion of $r, s$ in the quotient gives the inverse map from $A$. This proves the exact quotient isomorphism and identifies the retained positive cusp as the displayed product without a power ambiguity.

Let $E^+ = \ker(\epsilon : E \to C_2)$, where $\epsilon(a) = 1, \epsilon(b) = 0$. The two-coset Schreier calculation with transversal $1, a$ gives

$$A \xrightarrow{\sim} E^+, \qquad R \longmapsto b, \quad S \longmapsto aba. \tag{F.7.36}$$

Write $p_i : B \to E^+$ for the composite of the row (F.7.35) and (F.7.36). The quotient $B/H = T$ sends

$$(r, s, u, v, w) \longmapsto (\bar{b}, \bar{b}^{-1}, \bar{b}, 1, \bar{b}^{-1}),$$

whereas $E^+/K = C_3$ sends $b$ to its generator and $aba$ to the inverse. Identify these two cyclic groups by their named generators. All four rows intertwine the specified quotient maps, and hence $p_i(H) = K$.

The restriction to $H$ is precisely the filling of the nine punctures outside $J_i$. Indeed the closed $B$-normal closure of the other three cusp words is contained in $H$. Conjugation by $B/H = T$

supplies the three $H$ cusp classes in each omitted orbit, and further $H$ conjugation gives their closed normal closure. Thus

$$\begin{aligned} \langle\langle \kappa_j : j \neq i \rangle\rangle_B &= \langle\langle c_\ell : \ell \notin J_i \rangle\rangle_H, \\ 1 \longrightarrow H_{J_i} \longrightarrow B_i \longrightarrow C_3 \longrightarrow 1, &\qquad H_{J_i} \xrightarrow[p_i]{\sim} K. \end{aligned} \tag{F.7.37}$$

Each omitted cusp has width one for $H \leq B$: $\kappa_j \in H$ and its three lifts have precisely the labels in (F.7.34). No peripheral power is inserted into this normal-closure identity.

There is a common actual section. Define

$$\iota_0 : E \longrightarrow E, \qquad a \longmapsto \tau, \quad b \longmapsto b. \tag{F.7.38}$$

The orders give a homomorphism. On $E^+ = \langle b, aba \rangle$ it maps the two free profinite factors to $r, s$, whose inclusion in $B$ is injective by its retraction onto those factors. Moreover $q_5(\tau)$ is an involution inverting $q_5(b)$, so $q_5\iota_0(E)$ is the order-six normalizer of $T$. Its quotient by $T$ is exactly $\epsilon$. Thus $\ker \iota_0$ is contained in $E^+$, where it is trivial, proving that (F.7.38) is a continuous embedding.

The kernel of $E \to q_5\iota_0(E) \cong S_3$ is the characteristic $K$ proved in (F.7.20). Hence $\iota_0(K) \leq H$. Every $p_i$ is the identity on the displayed two factors after (F.7.36), so

$$p_i\iota_0|_{E^+} = 1_{E^+}, \qquad p_i|_H \circ \iota_0|_K = 1_K. \tag{F.7.39}$$

The program also rewrites $\iota_0(x), \iota_0(y)$ through all four quotient rows and verifies that their images are exactly $x, y$.

For $\gamma = (1, f) \in GT_0^1$, the actual normalized automorphism (F.7.27) extends to an automorphism of $E$ inducing the identity on $A_5$. It preserves $B$ and each filled normal subgroup. Using (F.7.37) for the target identification, the actual detector is

$$\mathcal{D}_{J_i}(\gamma) = \left[ p_i \, \eta_f^H \, \iota_0|_K \right] \quad \text{in } \mathcal{P}_K^1. \tag{F.7.40}$$

Here $\eta_f^H$ is applied to $\iota_0(K) \leq H$. Its preservation of that embedded subgroup is not assumed. The subsequent quotient map is part of the composition. The composition is an automorphism because it is the actual descended automorphism under (F.7.37), evaluated on the section (F.7.39). The two pairs of equal detector kernels are the actual orbit transports of (F.7.31)–(F.7.32).

The four source quotient maps themselves have a nontrivial common kernel:

$$[\kappa_0, \kappa_1] = u^{-1}rv^{-1}r^{-1}uv \neq 1. \tag{F.7.41}$$

Every row kills this word. Its nontriviality in the full completion is detected by $B \to S_3$ sending $r, s, w$ to 1, $u$ to (1 2), and $v$ to (2 3): its image has order three. This source word gives an inner automorphism of $E$, whose class in $\mathrm{Out}(E)$ is the identity. It therefore supplies no nonidentity $GT_0^1$ element in the kernel of (F.7.40). Faithfulness of an outer detector, or of the five-detector family, remains open.

The executed program `code/part_a/modular_c3_correspondence_f7.py` records the full Schreier presentation, its sixteen eliminations, the four three-step filling eliminations, compact word identities, exact finite quotient labels, eight common-section word checks, and the finite detector in (F.7.41).

**An exact normalized detector and its naturality equation.** The first correspondence in (F.7.35) gives a particularly short formula for the orbit-twenty detector. All maps in this paragraph are the based maps already fixed in (F.7.19)–(F.7.40). Thus $x = d^2$, $y = e^2$ is the specified free basis of $K$, and $p_0 : B \to E^+$ is the row retaining $J_0 = \{0, 1, 2\}$.

For $\gamma = (1, f) \in GT_0^1$, let $j \in \{0, \dots, 4\}$ be the actual finite correction in (F.7.27), and put

$$\begin{aligned} W_f &= p_0(d^{-j} f^{-1} e^j) \in K, \\ A_f = \ell_x(W_f), &\qquad B_f = \ell_y(W_f), \\ F_0(f) &= y^{B_f} W_f^{-1} x^{A_f} \in K'. \end{aligned} \tag{F.7.42}$$

Here $\ell_x, \ell_y : K \to \widehat{\mathbb{Z}}$ are the coordinates of the displayed abelianization. The source of $p_0$ in this formula is legitimate: $d^{-j} f^{-1} e^j = h_1$ lies in $H \leq B$ by (F.7.27)–(F.7.28). The exact normalized representative of the detector is

$$\mathcal{D}_{J_0}(\gamma) = \big[(x \mapsto x, \quad y \mapsto F_0(f)^{-1} y F_0(f))\big] \quad \text{in } \mathcal{P}_K^1. \tag{F.7.43}$$

In particular this is a formula in the actual target pure peripheral group; membership of its normalized coordinate in $GT_0^1$ is not used.

To prove the formula, use the two actual $H$ cusp words $c_0 = d^5$ and $c_1 = e^5 = b c_0 b^{-1}$. The row in (F.7.35) and the identification (F.7.36) give

$$p_0(c_0) = (aba)b = x, \qquad p_0(c_1) = bxb^{-1} = y.$$

These images are the previously proved full basis, so (F.7.28) shows that the actual descended automorphism is $\psi_f(x) = x$, $\psi_f(y) = W_f y W_f^{-1}$. This agrees with $p_0 \eta_f^H \iota_0|_K$ because $p_0 \iota_0|_K = 1_K$ and $\psi_f p_0 = p_0 \eta_f^H$ on $H$. Composing $\psi_f$ with $\mathrm{Inn}(x^{-A_f})$ gives the representative in (F.7.43), since $F_0(f)^{-1} y F_0(f) = x^{-A_f} W_f y W_f^{-1} x^{A_f}$. The abelianization of $F_0(f)$ is zero. Uniqueness is the normalized-pair uniqueness: two representatives fixing $x$ differ by conjugation in $\overline{\langle x \rangle}$, and equality of the images of $y$ then leaves a factor in $\overline{\langle y \rangle}$; the two zero abelian coordinates remove both factors.

Consequently the kernel and the equality comparison with the original normalized action have the exact descriptions

$$\begin{aligned} \ker \mathcal{D}_{J_0} &= \{\, (1, f) \in GT_0^1 : W_f = x^{A_f} y^{B_f} \,\}, \\ \{\gamma \in GT_0^1 : \mathcal{D}_{J_0}(\gamma) = \gamma \text{ in } \mathcal{P}_K^1\} &= \{\, (1, f) \in GT_0^1 : y^{B_f} W_f^{-1} x^{A_f} = f \,\}. \end{aligned} \tag{F.7.44}$$

For the first equality one can also argue directly. An inner automorphism fixing $x$ is conjugation by $x^t$. Its equality with $\psi_f$ is exactly $x^{-t} W_f \in C_K(y) = \overline{\langle y \rangle}$, which forces $t = A_f$ and the remaining exponent to be $B_f$. No assertion that the second set in (F.7.44) is all of $GT_0^1$ is made.

The normalized formula is independent of the lift of the finite correction. Replacing $j$ by $j + 5m$, with $m \in \widehat{\mathbb{Z}}$, gives the exact identities

$$h_1 \longmapsto c_0^{-m} h_1 c_1^m, \qquad W_f \longmapsto x^{-m} W_f y^m, \qquad (A_f, B_f) \longmapsto (A_f - m, B_f + m). \tag{F.7.45}$$

Substitution in (F.7.42) leaves $F_0(f)$ unchanged. This also proves explicitly that this normalization absorbs the permissible $H$-inner change.

For completeness the exact naturality defect can be written before any filling. Put $\eta_j = \mathrm{Inn}(d^{-j}) \eta_f$ and

$$L_f = x f^{-1} b, \qquad T_f = L_f f^{-1} a L_f^{-1}.$$

Since $z = xb$, these are $\eta_f(z)$ and $\eta_f(\tau)$, respectively. The actual two basis values are

$$\begin{aligned} \eta_j \iota_0(x) &= d^{-j} (T_f f^{-1} b)^2 d^j, \\ \eta_j \iota_0(y) &= d^{-j} (f^{-1} b T_f)^2 d^j. \end{aligned} \tag{F.7.46}$$

Define two elements of $H$ by

$$\begin{aligned} \Delta_x(f) &= \eta_j \iota_0(x)\, \iota_0(x)^{-1}, \\ \Delta_y(f) &= \eta_j \iota_0(y) \big(\iota_0(f)^{-1} \iota_0(y) \iota_0(f)\big)^{-1}. \end{aligned}$$

For every one of the four exact filling maps, the actual representative of (F.7.40) satisfies

$$\begin{aligned}(p_i\eta_f^H\iota_0)(x) &= p_i(\Delta_x(f))\,x,\\ (p_i\eta_f^H\iota_0)(y) &= p_i(\Delta_y(f))\,f^{-1}yf.\end{aligned} \tag{F.7.47}$$

For $i=0$ the first defect is killed, by the $c_0$ computation above. The second comparison, including its permitted inner normalization, is exactly (F.7.44). A section of the quotient map supplies the evaluations in (F.7.47); it does not supply an additional intertwining identity.

There is also an explicit finite certificate for the size of that section's image. The same $\iota_0$ maps $K$ into $K$, with exact word values

$$\begin{aligned}\iota_0(x) &= xyx^{-1}y^{-1}x^{-1}yx,\\ \iota_0(y) &= yx^{-1}y^{-1}xy^{-1}x^{-1}.\end{aligned} \tag{F.7.48}$$

Reduction in $C_2 * C_3$ verifies both identities. Their abelianization matrix is $\left(\begin{smallmatrix}0 & -1\\ 1 & -1\end{smallmatrix}\right)$, so the induced endomorphism of each free pro-$p$ quotient is an automorphism by its Frattini criterion and the Hopf property. Nevertheless $q_5\iota_0(K)=1$, whereas $q_5(K)=A_5$: the images of $x=d^2$ and $y=e^2$ generate $A_5$, as the finite certificate verifies. Thus $\iota_0|_K$ is an injective, proper full profinite self-embedding with the displayed finite quotient obstruction to surjectivity. Its common-section property is fully compatible with this exact comparison.

The executable `code/part_a/modular_detector_defect_f7.py` checks the two based cusp images, all four common section basis images, the words (F.7.48), their $S_3$ and $A_5$ images, and the normalization, kernel, and lift-independence identities on all 4 373 reduced words of length at most seven. These word checks verify the displayed formulas; they make no faithfulness assertion about (F.7.43) or the full five-detector family.

**One cone conjugator and all five exact filling equations.** The kernel equation (F.7.44) has an exact formulation using a single element of $H$. The full profinite generation assertion needed in this formulation follows from the actual degree-one cusp of a finite orbifold cover.

First, every closed subgroup $L \le E$ satisfying

$$d \in L, \qquad d^6 \in \overline{[L,L]} \tag{F.7.49}$$

is $E$. Indeed a proper such $L$ is contained in a proper open subgroup $U$: separate a point outside $L$ in a finite quotient and take the inverse image of the image of $L$. The subgroup $U\cap\Delta$ has the same finite index, and its full completion is $U$. In the corresponding connected orbifold cover, $d\in U\cap\Delta$ gives a width-one cusp through the base point. A proper cover has at least one more cusp, since the remaining cosets cannot lie in that length-one cycle of $d$. Its cone orders divide two or three. For any prime $p>3$, the cone classes vanish in $H_1(U,\mathbb{F}_p)$; the cusp classes have precisely their total-sum relation, together with independent handle classes. Each cusp class is therefore nonzero. The class of $d^6$ is six times the specified width-one cusp and is nonzero, contradicting (F.7.49).

It follows that every continuous endomorphism $\rho : E\to E$ satisfying $\rho(d)=td^\mu t^{-1}$ for a full unit $\mu\in\widehat{\mathbb{Z}}^\times$ is an automorphism. The compact image $L=t^{-1}\rho(E)t$ contains $d^\mu$ and hence its $\mu^{-1}$ power $d$. Since $E^{\mathrm{ab}}=C_2\times C_3$, the element $d^6$ belongs to the closed commutator subgroup of $E$. Thus $d^{6\mu}\in\overline{[L,L]}$, and its $\mu^{-1}$ power gives (F.7.49). Surjectivity follows, and finite generation of $E$ gives Hopficity. Both the conjugation and the unit-root operation here are performed on the specified image subgroup; no pro-$p$ quotient replaces $E$.

Define the following compact solution space:

$$\mathscr{S} = \{\, g\in H : (dgb^{-1}g^{-1})^2 = 1\,\}, \qquad \eta_g(d)=d,\quad \eta_g(b)=gbg^{-1},\quad \eta_g(a)=dgb^{-1}g^{-1}. \tag{F.7.50}$$

The equation gives the $a^2$ relation, and the $b^3$ relation holds by conjugation. Hence the displayed substitutions define an endomorphism of $E$, and the preceding full boundary test makes it an automorphism. Its action on $A_5$ is the identity because $g \in H$. Consequently its outer class is the element of $GT_0^1$ specified by (F.7.23).

Every element of that $GT_0^1$ has this form. For $\gamma = (1, f)$, use the actual $\eta_j = \mathrm{Inn}(d^{-j})\eta_f$ in (F.7.27). The full free profinite product torsion theorem makes $\eta_j(b)$ conjugate to $b$ or $b^{-1}$; these are the only order-three factor elements in $C_2 \amalg C_3$ [34, Theorem A and Section 3]. Fixing $d$ forces the identity on $E^{\mathrm{ab}} = C_6$, which excludes $b^{-1}$. Choose $u$ with $\eta_j(b) = ubu^{-1}$. Since the quotient action is the identity, $q_5(u) \in C_{A_5}(q_5(b)) = \langle q_5(b) \rangle$. Write $q_5(u) = q_5(b)^r$ and replace $u$ by $g_f = ub^{-r} \in H$. The same source's full free-factor centralizer theorem gives $C_E(b) = \langle b \rangle$ [34, Theorem B and Section 4]. As $H \cap \langle b \rangle = 1$, this $g_f$ is unique. The conjugation map $H \to E$, $g \mapsto gbg^{-1}$, is an embedding of a compact space and thus has a continuous inverse on its image. We have the exact identities

$$\eta_j = \eta_{g_f}, \qquad h_1 = g_f b g_f^{-1} b^{-1}, \qquad f = b d^j g_f b^{-1} g_f^{-1} d^{-j}. \tag{F.7.51}$$

The last formula retains the particular finite $j$ selected for the original normalized pair; it follows from $\eta_f = \mathrm{Inn}(d^j)\eta_j$.

Two solutions $g, h \in \mathscr{S}$ have the same outer class exactly when

$$h = d^{5t} g \quad (t \in \widehat{\mathbb{Z}}). \qquad \mathscr{S}/(g \sim d^{5t} g) \xrightarrow{\sim} GT_0^1. \tag{F.7.52}$$

To prove the assertion, an inner comparison of $\eta_g, \eta_h$ must centralize $d$, so its conjugator lies in $\overline{\langle d \rangle}$ by (F.7.24). Both quotient actions are the identity; since $Z(A_5) = 1$, that conjugator is $d^{5t} \in H$. Comparing the two images of $b$ then gives $h = d^{5t} g$, because the possible remaining factor lies in $H \cap C_E(b) = 1$. Conversely left multiplication by $d^{5t}$ changes $\eta_g$ by precisely $\mathrm{Inn}(d^{5t})$ and preserves (F.7.50). This proves the bijection, and compactness gives the stated homeomorphism of the solution-orbit space with the compact group on the right.

For the first detector the exponent condition sharpens to one procyclic condition:

$$\begin{aligned} \ker \mathcal{D}_{J_0} &= \{\, (1, f) : W_f = x^{A_f} y^{-A_f} \,\} \\ &= \{\, (1, f) : p_0(g_f) \in \overline{\langle x \rangle} \,\}, \\ \mathscr{S} \cap \ker(p_0|_H) &\xrightarrow{\sim} \ker \mathcal{D}_{J_0}. \end{aligned} \tag{F.7.53}$$

For the proof let $\varphi_g$ be the actual automorphism of $E^+$ descended from $\eta_g$ through $p_0$. It sends $b$ to $p_0(g) b p_0(g)^{-1}$, and restricts to the surface automorphism in (F.7.43). When its outer restriction is the identity, that restriction is $\mathrm{Inn}(x^{A_f})$, by (F.7.44). Since $C_{E^+}(K) = 1$, two automorphisms of $E^+$ with this same restriction are equal: their quotient fixes $K$ pointwise, and its difference on each element centralizes $K$. It follows that $\varphi_g = \mathrm{Inn}(x^{A_f})$. Comparing the images of $b$ gives $p_0(g) = x^{A_f}$, since $K \cap C_{E^+}(b) = 1$. Therefore $W_f = x^{A_f} b x^{-A_f} b^{-1} = x^{A_f} y^{-A_f}$. Conversely this expression is exactly the inner kernel equation (F.7.44). Finally (F.7.52) replaces a solution with $p_0(g) = x^t$ uniquely by $d^{-5t} g$ with $p_0(d^{-5t} g) = 1$. This gives the last homeomorphism in (F.7.53). The group law on this slice is explicitly $g \star h = \eta_g(h) g$: here $\eta_g$ descends to the identity through $p_0$, so the product still lies in the slice.

The other four representative fillings can be written in the same actual coordinates. An independent index-sixty Schreier calculation gives $H$ with 61 initial generators and 120 rewritten relators. Fifty single-occurrence Tietze eliminations leave eleven generators and no relators. For each of the five triples in (F.7.32), the nine omitted cusp relators give nine more such eliminations,

leaving a full free basis $P, Q$ with the following values:

| $J$ | $m$ | $c_0$ | $c_m$ | $c_{\text{third}}$ | $P$ in $x, y$ | $Q$ in $x, y$ |
|---|---|---|---|---|---|---|
| $\{0,1,2\}$ | 1 | $QP$ | $Q^{-1}$ | $P^{-1}$ | $yx$ | $y^{-1}$ |
| $\{0,1,4\}$ | 1 | $Q$ | $P^{-1}$ | $Q^{-1}P$ | $y^{-1}$ | $x$ |
| $\{0,1,8\}$ | 1 | $QP$ | $Q^{-1}$ | $P^{-1}$ | $yx$ | $y^{-1}$ |
| $\{0,1,10\}$ | 1 | $Q$ | $P^{-1}$ | $Q^{-1}P$ | $y^{-1}$ | $x$ |
| $\{0,4,9\}$ | 4 | $P$ | $P^{-1}Q$ | $Q^{-1}$ | $x$ | $xy$ |

(F.7.54)

The last two columns are inverse substitutions: they identify $c_0, c_m$ with the specified full basis $x, y$ of $K$. All omitted cusp words go to one. Thus the resulting maps $\pi_J : H \to K$ are precisely the actual fillings (F.7.29) followed by these based quotient isomorphisms. The complete original generators, their words in $a, b$, all relators, all eliminations, and both inverse basis maps are recorded in the certificate. For $J_0$, this is exactly $p_0|_H$, as both identifications send $c_0, c_1$ to $x, y$.

For $g \in \mathscr{S}$ put $h(g) = gbg^{-1}b^{-1} \in H$. Use the original $y = e^2 \in E$ in the following source word:

| $J$ | $V_J(g)$ |
|---|---|
| $\{0,1,2\}, \{0,1,4\}, \{0,1,8\}, \{0,1,10\}$ | $\pi_J(h(g))$ |
| $\{0,4,9\}$ | $\pi_J(h(g)yh(g)^{-1}y^{-1})$. |

(F.7.55)

The whole word in the last row lies in $H$, because $H$ is normal; the map $\pi_J$ is applied to that word and is not evaluated separately on $y$ outside its source. To check the row, the exact label-four whisker is $(ba)^2 = y$, so $c_4 = yc_0y^{-1}$. The substitutions (F.7.50) fix $d$ and $x$, and give $\eta_g(y) = h(g)yh(g)^{-1}$. Consequently the conjugator for $c_4$ is $h(g)yh(g)^{-1}y^{-1}$. For the first four rows the conjugator for $c_1 = e^5$ is $h(g)$, as in (F.7.51). Every descended map therefore fixes $x$ and sends $y$ to $V_J(g)yV_J(g)^{-1}$. If $A_J = \ell_x(V_J(g))$ and $B_J = \ell_y(V_J(g))$, all five exact kernel equations are

$$\begin{aligned} \mathcal{D}_J([\eta_g]) &= [\, x \mapsto x,\ y \mapsto F_J(g)^{-1}yF_J(g)\,], \\ F_J(g) &= y^{B_J}V_J(g)^{-1}x^{A_J}, \\ \mathcal{D}_J([\eta_g]) = 1 \quad &\Longleftrightarrow \quad V_J(g) = x^{A_J}y^{B_J}. \end{aligned} \tag{F.7.56}$$

Here $[\eta_g]$ is read as its exact $GT_0^1$ class under (F.7.23). This supplies the same single full profinite $g$ in every row, together with the quadratic equation (F.7.50).

The four $C_3$-stable triples of (F.7.34) admit one further common word. With the same $\tau$ from (F.7.33), define

$$R(g) = g^{-1}\eta_g(\tau)g\tau^{-1} \in H. \qquad \mathcal{D}_{J_i}([\eta_g]) = 1 \quad \Longleftrightarrow \quad p_i(R(g)) = 1 \quad (0 \le i \le 3). \tag{F.7.57}$$

Indeed $r = b$ is conjugated by $g$, while $s = \tau b\tau^{-1}$ is conjugated by $v(g) = \eta_g(\tau)g\tau^{-1}$. After filling, $r, s$ generate $E^+$. An inner surface action extends uniquely across $E^+$ and its conjugator must be $p_i(g)$, by $C_{E^+}(K) = 1$ and $K \cap C_{E^+}(b) = 1$. Its agreement on $s$ is then exactly $p_i(v(g)) = p_i(g)$, because the other finite factor also has trivial centralizer intersection with $K$. This is (F.7.57), in both directions. Replacing $g$ by $d^{5t}g$ and $\eta_g$ by its specified inner change leaves $R(g)$ unchanged, so this common-word comparison retains the outer normalization.

The actual geometric residual group in (F.7.25) therefore maps into the simultaneous solution set (F.7.50), (F.7.53), and (F.7.56), with the common-word constraints (F.7.57). Triviality of that solution set has not been proved. The nontrivial common source kernel in (F.7.41) supplies no solution to the additional quadratic equation. Conversely the exact section and the primewise images in (F.7.48) supply no step replacing these full quotient equations.

The executed standard-library program `code/part_a/modular_five_fillings_f7.py` records all 95 Tietze eliminations, all five inverse full-basis maps, every original cusp word, the order-three finite centralizer, the five $j$-recovery word identities in (F.7.51), and the exact

label-four conjugator. The orbifold generation test, the full torsion and centralizer inputs, and the simultaneous solution-space identifications are proved above.

This audit concerns the comparison requested for §30, folios 194–196, scans 383–387. F.6's arithmetic open-image proof applies to $A_5$ without using (F.7.1), so that proved arithmetic assertion does not depend on the unresolved intersection. □

**F.8. [Proved] The seven-point kernel and its peripheral restrictions.** In the regular $A = A_4$ labelling of (F.7.8), all twelve seven-point subsets in (F.7.9) form one free orbit for left multiplication by $A$. Consequently their kernels on the specified $F_{13}$ are equal; their kernels on its profinite completion are equal as well. The completed kernel meets every conjugate of every full source cusp subgroup $\widehat{\mathbb{Z}}$ trivially. The discrete kernel contains no nonidentity reducible mapping class under the specified index-twelve inclusion $F_{13} \leq P\Gamma_{0,4}$.

*Manuscript:* §30; folios 194–196; scans 383–387; Part F: nonsolvable finite centralizer comparison.

*Proof.* Here is the actual comparison of maps. Regard the target as $M_{0,I}$ with its seven labels $I \subset A$, and write

$$f_I : X \longrightarrow M_{0,I}, \qquad z \longmapsto [(gz)_{g\in I}].$$

For $h \in A$, let $r_h : M_{0,I} \xrightarrow{\sim} M_{0,hI}$ be the relabelling which assigns to the label $hg$ the former coordinate with label $g$. The configuration defining $f_{hI}(z)$ has coordinate $h(gz)$ at that label. Applying the same Möbius transformation $h^{-1}$ to every coordinate therefore gives the exact identity of maps

$$f_{hI} = r_h \circ f_I.$$

This identity also identifies the base configurations; it induces an isomorphism of the target fundamental groups intertwining the two homomorphisms from $\pi_1(X)$. Functoriality of profinite completion gives the same commuting diagram for their completions. Since $r_h$ and its completion are isomorphisms, the two kernel equalities follow.

The exact label certificate is `code/part_a/seven_point_same_kernel.py`. It constructs the twelve subsets in the action $A_5/C_5$ and gives all twelve left multipliers and the corresponding seven-label permutations. With the certificate's zero-based vertex numbering, one subset is $I = \{0, 1, 2, 3, 4, 7, 10\}$. Its intersections with the three cosets of the normal subgroup $V_4 \leq A_4$ are respectively

| coset | $I$ retained in that coset |
|---|---|
| $\{0, 2, 7, 11\}$ | $\{0, 2, 7\}$ |
| $\{1, 4, 5, 9\}$ | $\{1, 4\}$ |
| $\{3, 6, 8, 10\}$ | $\{3, 10\}$. |

The program checks that none of the twelve subsets contains a full $V_4$ coset. Thus the four-label $V_4$-orbit cross-ratio is not one of the forgetful projections of any of these seven-point maps.

The intersection of the twelve kernels in (F.7.9) is therefore the kernel of any one of these named maps. F.9 below proves it nontrivial by an exact braid certificate. This discrete kernel contains no nonidentity reducible mapping class in the source subgroup of $P\Gamma_{0,4}$. To prove this, use the exact $A_4$ covering

$$X = S^2 \setminus R_A \longrightarrow S^2 \setminus \{b_2, b_3, b_3'\}$$

and its index-twelve injection on fundamental groups in (F.7.8). On the four-punctured quotient sphere, every essential curve separates the moving puncture $q$ and one branch point $b$ from the other two branch points. Let $e = 2$ or $3$ be the order at $b$. The preimage of the disk containing $q, b$ consists of $12/e$ disks, each containing $e$ of the twelve points above $q$. These assertions hold for every such curve: its monodromy subgroup is the conjugate of the inertia subgroup at $b$, so its connected components have degree $e$, and each is a disk branched only at the one point above $b$.

The lift of a nonzero twist power belonging to the index-twelve subgroup is a product of twists about these boundary curves. More precisely, its exponent is divisible by $e$, and each lifted twist has exponent equal to the original exponent divided by $e$. After forgetting five points, at least one of these boundaries remains essential: seven retained points distributed among six pairs or four triples force a disk containing at least two retained points, and its complement contains at least four. The surviving twists all have the same nonzero sign; parallel essential components add their exponents. Independence of twists about pairwise disjoint nonisotopic essential curves makes the resulting mapping class nonidentity. Finally, each nonidentity reducible pure mapping class of a four-punctured sphere is a nonzero power of one such twist: its invariant multicurve has just one component, and cutting along it gives two three-punctured spheres. This proves the asserted exclusion of reducible elements from the discrete kernel.

There is also an injective detector for each full profinite cusp. Let $c$ be a primitive peripheral generator at any one of the fourteen punctures of $X$. Its image in $P\Gamma_{0,4}$ is a conjugate of $T^e$, where $T$ is a twist as above. Thus its lift is the indicated product of twists of exponent one. Choose two retained labels in an essential lifted disk and two outside that disk. Forgetting all other labels gives a specified map to $P\Gamma_{0,4}$. The inside disk contributes one twist $U$. Among the other disks, another twist survives exactly when both outside labels lie in one disk; it then gives the same isotopy class and the same sign. Hence the image of $c$ is $U^d$ with $d \in \{1, 2\}$, up to conjugation.

There is a character $\ell : P\Gamma_{0,4} \to \mathbb{Z}$ with $\ell(U) = 1$. For example the three peripheral twists in the free group $P\Gamma_{0,4} = F_2$ have abelianization vectors $(1, 0), (0, 1), (-1, -1)$, and an essential twist is conjugate to one of these under the pure group. Choose the corresponding integral linear functional. Completing this character and the two forgetting maps gives a continuous homomorphism

$$\widehat{F}_{13} \longrightarrow \widehat{P\Gamma}_{0,7} \longrightarrow \widehat{P\Gamma}_{0,4} \longrightarrow \widehat{\mathbb{Z}}$$

whose restriction to $\overline{\langle c \rangle}$ is $a \mapsto da$. Multiplication by 1 or 2 is injective on $\widehat{\mathbb{Z}}$. The cusp is a full embedded $\widehat{\mathbb{Z}}$ in $\widehat{F}_{13}$, since its primitive generator is a member of a free basis after choosing the omitted sphere peripheral. The same final character detects every conjugate $gc^a g^{-1}$ with $g \in \widehat{F}_{13}$, because its target is abelian. This proves the asserted trivial intersection of the completed kernel with all conjugate cusp groups. The program gives the four retained labels and $d$ in each of the $12 \cdot 7 = 84$ subset–inertia-partition checks; the two fixed points of each inertia subgroup have the same partition.

The exclusion just proved confines nonidentity discrete kernel elements to pseudo-Anosov mapping classes. F.9 supplies an explicit such element by an exact braid certificate. No assertion about a Nielsen–Thurston type for profinite elements is used.

The preceding rank-six calculation is a calculation of its induced abelian map, and does not identify the kernel of the fundamental-group homomorphism. The separate file `code/part_a/seven_point_braid_explore.py` contains a numerical crossing exploration; it is explicitly exploratory and is not used in this map comparison. F.9 replaces the discovery computation by a separate exact rational certificate of a kernel relation. □

**F.9. [Proved] An explicit common kernel element for the seven-point maps.** Every one of the twelve seven-point maps in F.7–F.8 has a nontrivial kernel, both on the specified discrete $F_{13}$ and on its full profinite completion. With the source coordinate, basepoint, and meridians specified below, the same element

$$w = m_{-1} m_i m_1^{-1} m_i^{-1} \tag{F.9.1}$$

lies in all twelve kernels. A source winding character sends $w$ to 1, so its procyclic closure is a full $\widehat{\mathbb{Z}}$ contained in their common completed kernel. The joint kernel of all thirty-six six- and

seven-point maps in (F.7.9) is exactly this same kernel. This refutes injectivity of these particular restriction maps; the residual $A_5$ centralizer comparison in F.7 remains unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Part F, nonsolvable A5 centralizer comparison on the twelve-punctured sphere.

*Proof.* **The specified source and target.** Use the faithful tetrahedral action generated by

$$z \longmapsto -z, \qquad z \longmapsto \frac{z+i}{z-i}.$$

Its fourteen ramification points are infinity and the zeros of

$$D(z) = z(z^4-1)(z^8+14z^4+1).$$

Thus $X = \mathbf{P}^1 \setminus R_{A_4}$ has $\pi_1(X,b) = F_{13}$ for $b = 5+4i$. The seven labels $I = \{0,1,2,3,4,7,10\}$ of F.8, in that order, have the following coordinate functions:

$$(m_0(z),\dots,m_6(z)) = \left(z, \frac{i(z+1)}{z-1}, -\frac{1}{z}, -\frac{z-i}{z+i}, \frac{i(z-1)}{z+1}, -z, \frac{z-i}{z+i}\right). \tag{F.9.2}$$

Here the subscripts $0,\dots,6$ denote positions in the displayed seven-set, not its original labels. The exact finite part of the certificate below verifies all 144 matrix products in this $A_4$, its correspondence with the labels in F.8, and the ramification polynomial $D$ by multiplying the fixed-point polynomials of its seven nontrivial cyclic subgroups.

For $z \in X$, let

$$L_z(t) = \frac{(t-m_0(z))(m_1(z)-m_6(z))}{(t-m_6(z))(m_1(z)-m_0(z))}.$$

The seven orbit points are distinct, so this is the unique Möbius transformation carrying the first, second, and seventh points to $0, 1, \infty$. Define the ordered configuration map

$$\Phi : X \longrightarrow \mathrm{Conf}_6(\mathbf{C}), \qquad z \longmapsto (R_0(z),\dots,R_5(z)), \quad R_j(z) = L_z(m_j(z)). \tag{F.9.3}$$

The rational functions extend across any removable singularities in their displayed expressions; their only poles are among $R_{A_4}$. In particular $R_0 = 0$ and $R_1 = 1$. Appending the seventh point $\infty$ and taking the labelled Möbius quotient gives a map

$$q : \mathrm{Conf}_6(\mathbf{C}) \longrightarrow M_{0,I} \quad \text{with} \quad q \circ \Phi = f_I. \tag{F.9.4}$$

These are actual continuous maps, with the displayed base configurations. The induced map of fundamental groups therefore factors through the pure braid group $P_6 \le B_6$. The inclusion here is induced by the cover from ordered to unordered configurations and is injective by path lifting. We shall prove $\Phi_*(w) = 1$ in $B_6$, which proves $f_{I,*}(w) = 1$ by this factorization.

**The three based source meridians.** Put $r = 27/1000$. For $c = -1, i, 1$, set

$$t_c = (\mathrm{Re}\, c - r) + 4i, \quad b_c = c - r - ri, \quad d_c = c + r - ri, \quad e_c = c + r + ri, \quad a_c = c - r + ri.$$

Let $m_c$ be the loop obtained by traversing the straight polygonal segments with successive vertices

$$b,\ t_c,\ b_c,\ d_c,\ e_c,\ a_c,\ b_c,\ t_c,\ b. \tag{F.9.5}$$

The square is counterclockwise; its stem is followed by its reverse. Each square contains the single ramification point $c$. This also follows directly from the above polynomial: the other eight finite points have real and imaginary coordinates with absolute values $(\sqrt{3}-1)/2$ or $(\sqrt{3}+1)/2$, whereas the remaining finite points are $0, \pm 1, \pm i$. The exact segment check below also verifies

that every stem and every square edge avoids all zeros of $D$. Source loop products in (F.9.1) are chronological concatenation.

**Exact extraction of the three braids.** Multiply every coordinate in (F.9.3) by $\eta = 1 + (173/1000)i$ and project to its real part. This multiplication is an orientation-preserving homeomorphism of the plane. At the basepoint the six coordinate positions, in increasing projected order, are

$$(0, 5, 3, 1, 4, 2).$$

Identify this chamber with the standard real ordered base configuration by a path within the chamber. The chamber of any fixed strict real ordering is convex, so the identification is unique up to homotopy. Read crossings chronologically. Write $j$ for $\sigma_j$, and $-j$ for $\sigma_j^{-1}$; the positive sign means that the strand on the left immediately before the crossing has larger imaginary coordinate. This fixes both the numbering and the crossing convention.

The executed certificate `code/part_a/seven_point_kernel_f9.py` uses only rational and Gaussian-rational arithmetic. For a segment $z = p + (q - p)t$, $0 \le t \le 1$, and each pair of normalized coordinates, write their projected complex difference before taking real parts as $A(t)/B(t)$ with $A, B \in \mathbf{Q}(i)[t]$. It verifies that $A$, $B$, every coordinate denominator, and $D(p + (q - p)t)$ have no complex zero on the real segment: the real and imaginary parts have a greatest common divisor with no real root there. The possible crossings are precisely the roots in $(0, 1)$ of

$$P(t) = \mathrm{Re}(A(t)\overline{B(t)}). \tag{F.9.6}$$

Clearing rational denominators preserves its roots. Sturm isolation over $\mathbf{Q}$ enumerates all of them. Their rational isolating intervals are pairwise disjoint, their multiplicities are one, and $\mathrm{Im}(A\overline{B})$ has a certified constant nonzero sign on each interval. No endpoint is a crossing. The two strands are checked to be adjacent in the current order at every crossing, and the final order is checked against direct exact evaluation at the endpoint.

There are 18 different segments up to reversal, 270 pair checks, and 64 isolated crossings in these segments; the largest crossing-polynomial degree is 12. Reversed segments give the inverse words. Between successive crossings the path lies in a convex ordering chamber. At a crossing only the named adjacent pair changes order, with its certified nonzero imaginary separation. Straightening these intervals and crossing neighborhoods gives the usual elementary braid pieces. Thus the recorded words represent the actual monodromies, not just a linear representation of them. This is also the projection-isotopy argument of Artin's Theorem 3 [2].

The freely reduced words for $m_{-1}, m_i, m_1$ are, respectively, $A, B, C$ below; a line break only separates consecutive parts of the same word:

$$\begin{aligned}
A = (&1, 4, 5, 2, 1, 3, 2, 3, 2, 1, 2, -5, -4, -5,\\
&-5, 4, 5, -2, -1, -2, -3, -2, -3, -1, -2, -5, -4, -1),\\
B = (&1, 4, 5, 2, 3, 1, 4, 2, 3, 4, -3, 1, 2, 3, 4, 5,\\
&-2, 5, 4, 3, 2, 1, -4, -3, -2, -4, -1, -3, -2, -5, -4, -1),\\
C = (&1, 4, 5, 2, 3, 4, 1, 2, 3, 4, 5, -1, 3, 2, -5, -4,\\
&-4, 5, -2, -3, 1, -5, -4, -3, -2, -1, -4, -3, -2, -5, -4, -1).
\end{aligned} \tag{F.9.7}$$

**The braid identity and the source detector.** Use the faithful Artin substitution on $F_6 = \langle X_1, \ldots, X_6 \rangle$:

$$\sigma_j(X_j) = X_j X_{j+1} X_j^{-1}, \qquad \sigma_j(X_{j+1}) = X_j, \qquad \sigma_j(X_k) = X_k \ (k \ne j, j+1). \tag{F.9.8}$$

Its inverse is used for a negative letter. Products of substitutions are composed from the displayed word order; the certificate implements composition by substitution and cancellation of adjacent inverse letters. Artin's Theorem 14, with the inverse generator formula on p. 113,

proves faithfulness [2]. Simultaneously reversing all generator signs is an automorphism of the braid group, and reversing the stacking convention is an anti-automorphism; neither changes the identity test.

For the exact words (F.9.7), substitution gives

$$\mathrm{Artin}(AB) = \mathrm{Artin}(BC), \qquad \mathrm{Artin}(ABC^{-1}B^{-1})(X_j) = X_j \quad (1 \le j \le 6). \tag{F.9.9}$$

This is an integer free-word calculation. The output records all six freely reduced common images for the first equality and all six single-letter images for the second. Faithfulness gives $AB = BC$ in $B_6$. By the certified identification of the three words with the three source monodromies, (F.9.1) lies in $\ker \Phi_*$, and hence in $\ker f_{I,*}$ by (F.9.4).

The map $X \to \mathbf{C}^*$, $z \mapsto z + 1$, gives the integral winding character

$$\ell : \pi_1(X, b) \longrightarrow \mathbb{Z}, \qquad \ell(m_{-1}) = 1, \quad \ell(m_i) = \ell(m_1) = 0.$$

Therefore $\ell(w) = 1$ and $w \neq 1$. Completing this exact character gives $\widehat{\ell} : \widehat{F}_{13} \to \widehat{\mathbb{Z}}$ with $\widehat{\ell}(w) = 1$; this proves nontriviality in the full completion without a detection assumption. The continuous power map $\widehat{\mathbb{Z}} \to \widehat{F}_{13}$, $a \mapsto w^a$, has composite $a \mapsto a$ with $\widehat{\ell}$, so it is injective. Its image lies in the closed completed kernel.

Finally F.8 gives the actual map identities $f_{hI} = r_h \circ f_I$ for the twelve left translates $hI$, with $r_h$ a labelled target isomorphism. They carry this same source word, and all its profinite powers, into the identity in every target.

The twenty-four six-point conditions in (F.7.9) follow from these twelve conditions by specified forgetting maps. For one of the six subgroups $C_5 \le A_5$, let $p_+, p_-$ be its fixed poles and let $O$ be either of its two five-cycles. Put

$$K = O \cup \{p_+, p_-\}, \qquad J_\pm = O \cup \{p_\pm\}.$$

Forgetting the other pole gives the labelled morphism $d_\pm : M_{0,K} \to M_{0,J_\pm}$ and the exact identity $f_{J_\pm} = d_\pm \circ f_K$ on the same based source. The twelve choices of $(C_5, O)$ give all twelve seven-sets, and their two pole choices give all twenty-four six-sets. The separate executed certificate `code/part_a/seven_point_six_factorization_f9.py` lists all twenty-four containments, the omitted pole, and the six positions retained in each seven-tuple. Functoriality of fundamental groups and of full profinite completion proves on both sources that

$$\bigcap_{I \text{ among all } 36} \ker f_{I,*} = \bigcap_{K \text{ among the } 12 \text{ seven-sets}} \ker f_{K,*}. \tag{F.9.10}$$

For the completed equality the starred maps in (F.9.10) mean their completed homomorphisms. Consequently (F.9.1) and its full procyclic closure also satisfy every six-point condition.

The common-kernel injectivity route in F.7 therefore fails. The necessary seven-point restrictions of an element of the residual $A_5$ centralizer still vanish, but the converse implication has not been proved. No centralizer element is constructed here.

The discovery computation used numerical crossings and a finite Burau image to find the short candidate. Neither computation is an input to this proof. The file `code/part_a/seven_point_kernel_f9_output.json` contains the deterministic exact certificate: the coordinate functions, Gaussian-rational path vertices, every crossing polynomial, every rational root interval and sign, the three words, and the Artin identities. The companion stdout file records the successful execution. □

**F.10. [Proved] Discrete procyclic directions in the residual centralizer.** Let $A_5 \le \Gamma^+_{0,12}$ be the geometric icosahedral group acting on its twelve-vertex orbit, and put $P = P\Gamma_{0,12}$. For every nonidentity $x \in P$, its procyclic closure is a full $\widehat{\mathbb{Z}}$, and

$$\overline{\langle x \rangle} \cap C_{\widehat{P}}(A_5) = 1. \tag{F.10.1}$$

In particular, the full procyclic subgroup of the common forgetting kernel constructed in F.9 meets the actual residual $A_5$ centralizer trivially. For that particular word $w$, it already has trivial intersection with the centralizer of the specified involution $\beta = (1\ 2)(4\ 5)$ in $A_5$.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Part F, nonsolvable A5 centralizer comparison in the minimal twelve-puncture case.

*Proof.* **An explicit separating family of integral representations.** We first construct the finite quotients used in the proof. For $n \geq 4$, normalizing the last marked point to infinity and the first two to $0, 1$ identifies

$$M_{0,n} = \mathrm{Conf}_{n-3}(\mathbf{C} \setminus \{0, 1\}).$$

The exact homeomorphism

$$\begin{aligned} \mathrm{Conf}_{n-1}(\mathbf{C}) &\longrightarrow \mathbf{C} \times \mathbf{C}^* \times M_{0,n}, \\ (z_0, z_1, z_2, \ldots, z_{n-2}) &\longmapsto \left( z_0, z_1 - z_0, \left( \frac{z_j - z_0}{z_1 - z_0} \right)_{j=2}^{n-2} \right) \end{aligned} \tag{F.10.2}$$

has inverse $(a, b, (u_j)) \mapsto (a, a + b, (a + bu_j))$. The section fixing the first two factors at $(0, 1)$ therefore gives an injective homomorphism

$$P\Gamma_{0,n} \longrightarrow P_{n-1}, \qquad P_{n-1} \cong \mathbb{Z} \times P\Gamma_{0,n}. \tag{F.10.3}$$

Here $P_{n-1}$ is the ordered plane braid group. All basepoints in (F.10.2) are the displayed normalized configurations. Thus (F.10.3) is the actual direct-factor map, rather than an inference from the ranks of the groups.

Put $r = n - 1$ and $F_r = \langle t_1, \ldots, t_r \rangle$. Artin's faithful braid action gives $P_r \hookrightarrow \mathrm{Aut}(F_r)$ [2, Theorem 14]. We use the homomorphism convention

$$\sigma_i(t_i) = t_i t_{i+1} t_i^{-1}, \qquad \sigma_i(t_{i+1}) = t_i,$$

fixing the other generators, with automorphism products composed in that order. Equivalently, for chronological point-motion concatenation, use the inverse of the right monodromy action on the punctured-plane group. This conversion makes $A(bc) = A(b) \circ A(c)$ and preserves faithfulness. Every pure braid fixes $F_r^{\mathrm{ab}}$.

Let

$$\mu : F_r \longrightarrow \mathbb{Z}\langle\langle X_1, \ldots, X_r \rangle\rangle^*, \qquad \mu(t_i) = 1 + X_i$$

be the integral Magnus expansion, with $(1 + X_i)^{-1} = 1 - X_i + X_i^2 - \cdots$. Here is the injectivity needed below. Write a nonidentity reduced word in maximal syllables as $t_{i_1}^{a_1} \cdots t_{i_s}^{a_s}$, where every $a_j \neq 0$ and successive $i_j$ differ. The coefficient of the monomial $X_{i_1} \cdots X_{i_s}$ is $a_1 \cdots a_s \neq 0$. Indeed that monomial has $s$ successive runs of distinct variables. Omitting a syllable contribution leaves at most $s - 1$ runs, whereas using all syllables and total degree $s$ forces their linear terms. Thus $\mu$ is injective.

For each $d \geq 1$ let

$$T_{r,d} = \mathbb{Z}\langle X_1, \ldots, X_r \rangle / (\text{monomials of degree } > d).$$

An automorphism $\alpha$ of $F_r$ acts on this finite free $\mathbb{Z}$-module by the substitution

$$X_i \longmapsto \mu(\alpha(t_i)) - 1 \pmod{\deg > d}. \tag{F.10.4}$$

The substitution identities for group words show directly that these are homomorphisms of automorphism groups. For an automorphism fixing $F_r^{\mathrm{ab}}$, the right side is $X_i$ plus terms of degree at least two. Order the monomial basis by decreasing degree. The matrix in (F.10.4) is then

upper unitriangular, and all its entries are integers. Combining (F.10.3)–(F.10.4) gives actual homomorphisms

$$\rho_d : P\Gamma_{0,n} \longrightarrow \mathrm{UT}_{N_d}(\mathbb{Z}), \qquad N_d = 1 + r + \cdots + r^d. \tag{F.10.5}$$

This family separates elements. Two different pure mapping classes give different automorphisms of $F_r$ by (F.10.3) and Artin faithfulness. Some generator has different images; injectivity of $\mu$ makes their expansions different in some finite degree. That degree gives a representation (F.10.5) which separates the two classes. In particular, no passage from a discrete lower-central quotient to a completed one is assumed here: the detecting maps are integral matrices and their reductions in finite matrix groups.

**The actual finite-group action on the matrices.** Return to $n = 12$ and put $G = A_5$. Its intersection with $P$ is trivial, as is seen from its faithful puncture permutation. The subgroup $PG$ of $\Gamma^+_{0,12}$ is the actual semidirect product $P \rtimes G$ and has finite index. Thus its profinite completion is the corresponding open subgroup of $\widehat{\Gamma}^+_{0,12}$, with the specified copies of $\widehat{P}$ and $G$.

The discrete centralizer $C_P(G)$ is trivial. Indeed the quotient sphere has exactly the three branch/puncture decorations of orders $2, 3, 5$. The discrete covering normalizer sequence of F.7 has trivial quotient mapping class group, so its centralizer is $Z(A_5) = 1$. For $x \neq 1$ choose $a \in G$ with $axa^{-1} \neq x$, and then choose $d$ in (F.10.5) which separates these two elements. Write $\rho = \rho_d$. On a direct sum of $|G|$ copies of $\mathbb{Z}^{N_d}$, indexed by $g \in G$, define

$$R(h) = \mathrm{diag}_{g\in G}\, \rho(g^{-1}hg), \qquad h \in P. \tag{F.10.6}$$

Let $T_a$ carry the block with index $g$ to the block with index $ag$. Then

$$T_aT_b = T_{ab}, \qquad T_aR(h)T_a^{-1} = R(aha^{-1}). \tag{F.10.7}$$

For example the block with index $g$ on the left of the second identity is $\rho((a^{-1}g)^{-1}h(a^{-1}g))$ $= \rho(g^{-1}aha^{-1}g)$, which is its right-hand block. Consequently

$$ha \longmapsto R(h)T_a \tag{F.10.8}$$

is a homomorphism from the actual group $P \rtimes G$ to $\mathrm{GL}_{|G|N_d}(\mathbb{Z})$. Its reductions modulo every positive integer are finite quotients of this group.

**Every prime coordinate is detected.** Put $W = R(x)$. It is unipotent. By the choice of $d$, $T_aWT_a^{-1} \neq W$, already on the block with index 1. The finite polynomial logarithm

$$L = \log W = \sum_{j\geq 1} \frac{(-1)^{j+1}}{j}(W - I)^j$$

is a rational nilpotent matrix; the sum terminates. The exponential and logarithm polynomials are mutually inverse on nilpotent and unipotent matrices over a field of characteristic zero. Hence

$$T_aLT_a^{-1} - L \neq 0. \tag{F.10.9}$$

In particular this is the same nonzero rational matrix after embedding $\mathbb{Q}$ in $\mathbb{Q}_p$, for every prime $p$.

The reductions of (F.10.8) modulo $p^k$, for all $k$, give a continuous homomorphism from $\widehat{P \rtimes G}$ to $\mathrm{GL}_{|G|N_d}(\mathbb{Z}_p)$. The image of the procyclic power $x^u$, for $u \in \widehat{\mathbb{Z}}$, is

$$\exp(u_pL), \tag{F.10.10}$$

where $u_p$ is its $\mathbb{Z}_p$ coordinate. To check this equality, first take integral $u$; the identity $W^u = \exp(uL)$ follows from the finite polynomial identities. For arbitrary $u_p$, pass to limits of integers in $\mathbb{Z}_p$. The matrix polynomial is continuous, and the powers of $W$ modulo $p^k$ lie in an upper

unitriangular finite $p$-group, so these powers depend only on $u_p$. The limit lies in $\mathrm{GL}(\mathbb{Z}_p)$ because the integral matrices form a closed subset there. This proves (F.10.10) through the specified finite quotients.

Now an element $x^u$ centralizing $G$ commutes with $a$. Apply the continuous representation and then the polynomial logarithm to (F.10.10). It gives

$$u_p(T_a L T_a^{-1} - L) = 0 \quad \text{in } M_{|G|N_d}(\mathbb{Q}_p). \tag{F.10.11}$$

By (F.10.9), $u_p = 0$ for every prime $p$. Thus $u = 0$ in $\widehat{\mathbb{Z}}$ and (F.10.1) follows. The same argument without $T_a$, using any $\rho_d(x) \neq I$, proves that $u \mapsto x^u$ is injective. Its image is exactly the procyclic closure, which is therefore a full $\widehat{\mathbb{Z}}$ as claimed.

**The particular F.9 direction.** The following transport fixes the basepoint convention. Put $\varphi = (1+\sqrt{5})/2$, $\rho = \sqrt{\varphi + 2}$, and $z_* = -i(\rho + \varphi)$. In the tetrahedral coordinate of F.9 the orbit of $z_*$ is the specified icosahedral twelve-point configuration. The extra involution has Möbius matrix

$$\begin{pmatrix} 1 & \varphi - 1 - i\varphi \\ \varphi - 1 + i\varphi & -1 \end{pmatrix}. \tag{F.10.12}$$

This matrix represents (1 2)(4 5) under the fixed five-letter identification of $A_5$, and its twelve-vertex permutation is checked in the certificate below. Use the polygonal source path $\gamma : z_* \longrightarrow 5 + z_* \longrightarrow b = 5 + 4i$. It misses $R_A$: the finite points of $R_A$ have real and imaginary coordinates of absolute value less than $3/2$, the first segment has imaginary part less than $-3$, and the second has real part 5. Transport along the full labelled orbit path $(g\gamma)_{g \in A_4}$ identifies the marked mapping class groups at these two configurations. Equivariant isotopy extension keeps the specified $A_4$ action fixed. The symbol $\beta$ at the F.9 basepoint denotes the transport of (F.10.12) by this path. The exact matrix and vertex-label certificate `code/part_a/second_a4_c3_attack.py` checks all 180 generator products and all 720 vertex images in the number field $\mathbb{Q}(i, \sqrt{5}, \rho)$.

The F.9 source homomorphism is the full twelve-point orbit map of F.7.8, before the stated forgetting. For every source path, its configurations remain invariant under the same fixed Möbius $A_4$. Equivariant isotopy extension along the path, or equivalently the point-pushing lift of the quotient cover in F.7.8, places its monodromy in the actual discrete $C_P(A_4)$. That map identifies the source $F_{13}$ with this discrete centralizer. Thus the word $w$ is a nonidentity element of $C_P(A_4)$, not merely an element whose seven-point images are compatible with $A_4$.

In the fixed five-letter description, $A_4$ fixes the fifth letter and $\beta = (1\ 2)(4\ 5)$ carries it to the subgroup fixing the fourth. These subgroups generate $A_5$; equivalently $\langle A_4, \beta \rangle = A_5$. Therefore $\beta w \beta^{-1} \neq w$, since the opposite equality would put $w$ in the just proved trivial discrete $A_5$ centralizer. We may take $a = \beta$ throughout (F.10.6)–(F.10.11). This proves the last assertion of the claim, on the whole full profinite procyclic closure.

**Executed formula certificate.** The exact-integer program `code/part_a/procyclic_magnus_f10.py` checks the maximal-syllable coefficient on all 484 nonempty reduced words of length at most five in two generators. It verifies the substitution-composition identities and upper unitriangular form for two actual pure Artin automorphisms on the degree-three tensor algebra of rank 40, and checks eleven integral power/logarithm identities. For the actual action of $A_5$ on $\mathrm{Inn}(F_5)$ by permutation of the five free generators, it checks all 180 displayed block-conjugation identities and 60 block-product identities. Finally it checks finite power reductions modulo $p^2$ for $p = 2, 3, 5, 7, 11$. This is a certificate for the algebraic formulas; it does not compute a separating tensor degree for the particular F.9 word. Existence of that degree is proved above by faithful Artin action and the integral coefficient argument.

The conclusion restricts the actual residual centralizer on every displayed discrete procyclic direction. It does not assert that an arbitrary element of that closed residual group lies in one of these subgroups. No such covering assertion is used, and the full equality in F.7 remains unresolved. □

**F.11. [Proved] The rank-six cyclic centralizer and an exact common kernel.** Let $P = P\Gamma_{0,12}$ for the specified geometric icosahedral action in F.7. Fix $D = C_5 \leq A_5$ and $N = N_{A_5}(D) \cong D_{10}$, where $D_{10}$ has order ten. The explicit configuration and marking below give

$$\begin{aligned} C_P(D) &= C_P(N) \cong \pi_1(X_5) = F_6, \\ C_{\widehat{P}}(D) &= C_{\widehat{P}}(N) \cong \widehat{\pi_1(X_5)} = \widehat{F}_6, \qquad X_5 = \mathbf{P}^1 \setminus \{0, \infty, \mu_5\}. \end{aligned} \tag{F.11.1}$$

These are the actual discrete and completed inclusions constructed below. Of the twelve seven-point restrictions in F.7.9, two are trivial on this subgroup; the other ten have exactly the same kernel. Their joint kernel, including all twenty-four six-point restrictions, is the kernel of the specified map

$$f : X_5 \longrightarrow M_{0,7}, \qquad t \longmapsto (1, \zeta^2, \zeta^4, t, t\zeta, t\zeta^4, \infty), \qquad \zeta = e^{2\pi i/5}. \tag{F.11.2}$$

The equalities of kernels hold for the discrete groups and their full profinite completions.

At $b = 2 + 2i$, the precisely based meridians below give

$$w = [m_{\zeta^2},\, m_\zeta m_{\zeta^3} m_0] \in \ker(f_*), \qquad w \neq 1. \tag{F.11.3}$$

Its procyclic closure is a full $\widehat{\mathbb{Z}}$ in this common completed kernel. After the specified basepoint transport and inclusion into $\widehat{P}$, that procyclic subgroup has trivial intersection with $C_{\widehat{P}}(A_5)$, by F.10. Thus the restrictions again have a concrete common kernel, while the residual centralizer in F.7 remains unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Part F, nonsolvable A5 centralizer comparison on the twelve-punctured sphere.

*Proof. The actual action, labels, and basepoint transport.* Use the same five-letter generators and based coset labels as F.7.19–57:

$$a = (1\ 2)(3\ 4), \qquad b = (1\ 3\ 5), \qquad d = ab, \qquad D = \langle d \rangle.$$

The indices $0, \ldots, 11$ refer to the twelve cosets of $D$, in that previously specified order. Put

$$\theta = 2\sin(2\pi/5), \quad \theta^4 - 5\theta^2 + 5 = 0, \quad 19/10 < \theta < 2, \qquad k = 1 - \theta^2, \qquad t_0 = k\zeta^2. \tag{F.11.4}$$

The exact marked configuration is

| $j$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| label of $\zeta^j$ | 1 | 3 | 5 | 6 | 2 |
| label of $t_0\zeta^j$ | 4 | 7 | 9 | 11 | 8 |

$$0 \longleftrightarrow 0, \quad 10 \longleftrightarrow \infty. \tag{F.11.5}$$

Here are actual Möbius transformations realizing the named generators and the normalizer:

$$\begin{aligned} a(z) &= \frac{1 - z}{1 - z/k}, & b(z) &= a(\zeta z), \\ d(z) &= \zeta z, & \tau(z) &= \frac{t_0}{z}, \\ \tau &= (1\ 5)(3\ 4) \in A_5, & \tau d \tau^{-1} &= d^{-1}. \end{aligned} \tag{F.11.6}$$

The finite certificate `code/part_a/c5_centralizer_labels_f11.py` checks these formulas in $\mathbf{Q}(\theta, i)$, the 720 images of the twelve points under the sixty matrices, and all 120 products by the two generators. The sixty distinct matrices therefore form exactly the given $A_5$, with precisely its stated coset action. In particular this supplies an actual equivariant identification of the marked

action, not just an isomorphism of abstract groups. It is the explicit Möbius realization of the based covering choice in F.7; changing that initial marking conjugates all the inclusions and all the named forgetting maps by the same marking isomorphism.

For every $t \in X_5$ replace $t_0$ in (F.11.5) by $t$. This gives a map

$$j : X_5 \longrightarrow M_{0,12}, \qquad t \longmapsto (0, \infty, \mu_5, t\mu_5) \tag{F.11.7}$$

with the individual labels in (F.11.5). The functions $d(z) = \zeta z$ and $\tau_t(z) = t/z$ give a labelled $D_{10}$ action on this entire family: $\tau_t$ exchanges the two poles and sends the first orbit point with index $j$ to the second with index $-j$, and conversely. They satisfy the group relations on every fibre. Monodromy therefore centralizes the actual $D_{10}$ at $t_0$. Equivalently, each of the ten corresponding label permutations fixes the map to labelled moduli (F.11.7), through these specified Möbius transformations, and hence fixes every induced based loop.

The straight segment $\gamma$ from $t_0$ to $b = 2 + 2i$ lies in $X_5$: the exact field calculation gives $\operatorname{Re} t_0 > 2$, so the whole segment has real part at least two. The element of $C_P(N)$ associated to (F.11.3) is explicitly $j_*(\gamma w \gamma^{-1})$. All loop products here mean chronological concatenation. This also fixes the basepoint transport for the seven- and six-point target maps.

*The exact discrete centralizer and its full topology.* The quotient by $d$ is the actual cover $z \mapsto z^5$; its four punctures are $0, \infty, 1, t_0^5$. An equivalent based group description is

$$L = \pi_1(\mathbf{P}^1 \setminus \{0, \infty, 1, t_0^5\}) = F_3, \qquad U = \ker(\chi : L \longrightarrow C_5) = \pi_1(S_{0,12}), \tag{F.11.8}$$

where the positive meridians at $0, \infty$ have $\chi$-values $1, -1$ and the other two have value zero. The inclusion and the quotient map in (F.11.8) are induced by this covering.

We spell out why descent accounts for the entire mapping-class centralizer. Conjugation embeds $L$ in $\operatorname{Aut}(U)$: an element of the free group $L$ centralizing the nonabelian subgroup $U$ is trivial. Its image in $\operatorname{Out}(U)$ is the specified deck group $D$. For an actual automorphism $\alpha$ representing a pure mapping class centralizing $D$, conjugation by $\alpha$ preserves this embedded $L$. Indeed its effect on every deck outer class is the identity, so its effect on the conjugation automorphism of an $l \in L$ differs from that automorphism by conjugation by an element of $U$. The embedding of $L$ makes the resulting element unique. This defines an automorphism $\alpha_L$ of $L$, restricting to $\alpha$ on $U$ and inducing the identity on $C_5$.

The two ramified cusp meridians in $U$ are fifth powers of the corresponding meridians of $L$; the other cusp meridians are their specified unramified lifts. Since $\alpha$ fixes each upstairs peripheral class with exponent one, the uniqueness of roots in a discrete free group shows that $\alpha_L$ fixes each of the four downstairs peripheral classes with exponent one. The discrete Dehn–Nielsen–Baer identification used in A.12 therefore gives a pure mapping class on the four-punctured quotient. Changing $\alpha$ by an inner automorphism of $U$ does not change that downstairs class. The kernel of this descent consists of deck classes, since an inner automorphism of $L$ restricts to precisely such a class on $U$. A deck class fixing the upstairs labels is the identity, because $D$ acts freely on each five-cycle. This proves injectivity of descent on the pure centralizer.

Conversely, every pure class on the quotient preserves $\chi$ and has five commuting lifts. Choose the unique lift fixing the labelled point 1 of the first five-cycle; it then fixes that entire cycle. Identify $P\Gamma_{0,4}$ with $\pi_1(\mathbf{P}^1 \setminus \{0, 1, \infty\}, t_0^5)$ by pushing its fourth point while the first three are fixed, as in A.12. The chosen lift fixes the second five-cycle exactly when this loop lifts to a closed loop under $t \mapsto t^5$. Its image is consequently the kernel of winding about zero modulo five, namely $\pi_1(X_5, t_0)$. The lift is exactly the family (F.11.7). This proves $j_*\pi_1(X_5) = C_P(D)$ and identifies its source as the index-five subgroup $F_6$ of this specified $F_2$. Since its monodromy also centralizes $\tau$, it proves $C_P(D) = C_P(N)$.

There is a particularly direct check of the full induced topology. Retain the eight labels

$$I_8 = \{0, 10\} \cup \{1, 3, 5, 6, 2\} \cup \{4\}. \tag{F.11.9}$$

After this forgetting map, (F.11.7) has the seven fixed points $0, \infty, \mu_5$ and the single moving point $t$. Filling that moving point is precisely the normalization map of A.12. On its point-pushing subgroup the resulting automorphism of $\pi_1(X_5)$ is inner conjugation by the source loop. After completion the exact composite is therefore

$$\widehat{\pi_1(X_5)} \xrightarrow{\widehat{j}_*} \widehat{P} \longrightarrow \widehat{P\Gamma}_{0,8} \longrightarrow \mathrm{Aut}(\widehat{\pi_1(X_5)}), \qquad u \longmapsto \mathrm{Inn}(u). \tag{F.11.10}$$

The final extension is continuous on each finite characteristic quotient. Its displayed restriction is injective because the free profinite group has trivial centre. Thus $\widehat{j}_*$ is injective and its image is exactly the closure of the discrete centralizer. The prime-order centralizer comparison already justified in F.7 identifies that closure with $C_{\widehat{P}}(D)$. Taking closures of $C_P(D) = C_P(N)$ gives both inclusions needed for the second equality in (F.11.1).

*All thirty-six restrictions and the actual normalizer transports.* For $D$ itself the two seven-sets are

$$\begin{aligned} I_A &= \{0, 1, 2, 3, 5, 6, 10\}, \\ I_B &= \{0, 4, 7, 8, 9, 10, 11\}. \end{aligned} \tag{F.11.11}$$

Their maps to labelled moduli are constant: the first configuration is already fixed, and the second becomes fixed under $z \mapsto z/t$. For the other ten sets choose $I = \{1, 2, 4, 5, 7, 8, 10\}$ as representative. In the ordered target labels $(1, 5, 2, 4, 7, 8, 10)$ its map is exactly (F.11.2). The following are the actual $D_{10}$ transports:

| $k$ | $d^k I$ | $d^k \tau I$ |
|---|---|---|
| 0 | $\{1, 2, 4, 5, 7, 8, 10\}$ | $\{0, 1, 2, 3, 4, 7, 11\}$ |
| 1 | $\{1, 3, 4, 6, 7, 9, 10\}$ | $\{0, 1, 3, 5, 7, 8, 9\}$ |
| 2 | $\{2, 3, 5, 7, 9, 10, 11\}$ | $\{0, 3, 4, 5, 6, 9, 11\}$ |
| 3 | $\{1, 5, 6, 8, 9, 10, 11\}$ | $\{0, 2, 5, 6, 7, 8, 11\}$ |
| 4 | $\{2, 3, 4, 6, 8, 10, 11\}$ | $\{0, 1, 2, 4, 6, 8, 9\}$. |

(F.11.12)

The label certificate enumerates the six order-five subgroups, verifies this orbit, and records each transporter as a five-letter permutation, a word in $a, b$, and its permutation of the twelve labels. For $n = d^k \tau^\epsilon$, the actual homeomorphism in (F.11.6) gives the target isomorphism $n_I : P\Gamma_{0,I} \to P\Gamma_{0,nI}$. Naturality of forgetting gives

$$r_{nI} \circ \mathrm{conj}(n) = n_I \circ r_I. \tag{F.11.13}$$

Conjugation by $n$ is the identity on the subgroup (F.11.1), so these ten restrictions have equal kernels there. The same identity extends to the full completions.

Each of the twenty-four six-sets is obtained by removing one of the two poles from its specified seven-set. The certificate lists all twelve pairs of these literal subset inclusions. Forgetting that pole gives the exact factorization of the corresponding six-point map through the seven-point map. Hence adding all twenty-four constraints does not change the joint kernel. Together with (F.11.11)–(F.11.13), this proves the asserted joint-kernel equality for both the discrete and completed groups.

*Precisely based loops and exact monodromy.* Put $r = 31/1000$. For $q \in \{\zeta^2, \zeta, \zeta^3, 0\}$, define $m_q$ by the six successive straight segments through

$$b, \quad q + r + ri, \quad q - r + ri, \quad q - r - ri, \quad q + r - ri, \quad q + r + ri, \quad b. \tag{F.11.14}$$

The square is counterclockwise, with its whisker traversed in reverse at the end. The certificate checks that every segment avoids $0, \mu_5$, and that each square contains exactly its named puncture.

Remove the last, fixed point $\infty$ from (F.11.2). This gives an actual ordered configuration map to $\mathrm{Conf}_6(\mathbf{C})$. Appending $\infty$ and taking labelled Möbius classes recovers $f$. Its loop map

therefore factors through $P_6 \leq B_6$, with the injective inclusion given by the ordered-to-unordered covering. Multiply all six finite coordinates by $1 + (371/1000)i$, and project to their real parts. The increasing base order of the six positions, numbered $0, \ldots, 5$, is $(4, 1, 2, 0, 3, 5)$. Use the convention of F.9: a positive $\sigma_j$ has its left strand above its right strand just before the crossing, and crossings are read chronologically. Identifying the base chamber with the real ordered configuration fixes the braid generators up to its unique chamber isotopy.

The program `code/part_a/c5_seven_kernel_exact.py` uses fractions in the exact field $\mathbf{Q}[\theta]/(\theta^4 - 5\theta^2 + 5)$, with

$$\zeta = \frac{\theta^2 - 3 + i\theta}{2}. \tag{F.11.15}$$

The polynomial is strictly increasing on $(19/10, 2)$ and has opposite endpoint signs. Rational bisection isolates its specified root; exact interval evaluation certifies each nonzero field sign. For every one of the 24 source segments and every one of the 15 coordinate pairs, the projected difference is affine in the segment parameter. Its endpoint signs enumerate all crossings, and its single possible root is computed exactly in the field. Every crossing has nonzero imaginary separation. Distinct crossings have distinct parameters, the strands are adjacent in the current order, and the final ordering agrees with direct endpoint evaluation. The exact source-segment tests also exclude every configuration collision.

The output contains 360 pair checks, 58 simple crossings, and 2530 certified algebraic signs, with a rational isolating interval obtained by 128 bisections. Thus the words describe the actual paths: between crossings the ordering chambers are convex; each crossing has exactly the certified adjacent pair and sign. This is the projection-isotopy argument of Artin's Theorem 3 [2]. Writing signed integers for braid generators, the words for $m_{\zeta^2}$ and for $m_\zeta m_{\zeta^3} m_0$, after free reduction, are respectively

$$\begin{aligned} A &= (-4, -3, -5, -1, -4, -2, -1, -1, 2, 4, 1, 5, 3, 4), \\ B &= (-4, -3, -5, -1, -4, -2, -3, -2, -1, -4, \\ &\qquad -1, -4, -3, -2, -3, 4, -5, -1, 3, 4). \end{aligned} \tag{F.11.16}$$

The complete words for all four individual meridians, of lengths $14, 8, 20, 16$, and every crossing are retained in the output.

Use the faithful Artin substitutions of F.9.8, with the stated inverse substitution for each negative generator. Integer free-word substitution and cancellation give

$$\mathrm{Artin}(AB) = \mathrm{Artin}(BA), \qquad \mathrm{Artin}(ABA^{-1}B^{-1})(X_i) = X_i \quad (1 \leq i \leq 6). \tag{F.11.17}$$

The output records the six common images and the six single-letter final images. Artin's faithfulness theorem, Theorem 14 and formulas (14)–(15) of the original article [2], proves $[A, B] = 1$ in $B_6$. The actual configuration factorization therefore proves $f_*(w) = 1$.

*Nontriviality and the full procyclic subgroup.* Fill all source punctures except $\zeta^2, \zeta, \infty$. This is an actual map to a twice-punctured plane. The loops $m_{\zeta^3}$ and $m_0$ become trivial; the other two loops have winding vectors $(1, 0)$ and $(0, 1)$. In the integral Heisenberg quotient with product

$$(a, b, t)(a', b', t') = (a + a', b + b', t + t' + ab'),$$

their images consequently have the form $(1, 0, s)$ and $(0, 1, t)$. Their commutator is exactly $(0, 0, 1)$, for every $s, t \in \mathbb{Z}$. Thus the image of $w$ is nontrivial. For each prime $p$, reducing these integer matrices modulo every $p^n$ extends this homomorphism continuously to the full source completion and then to $\mathrm{UT}_3(\mathbb{Z}_p)$. The central coordinate of $w^u$ is $u_p$. All prime coordinates are therefore detected, so $u \mapsto w^u$ embeds $\widehat{\mathbb{Z}}$.

The already proved full embedding (F.11.10) preserves this conclusion for $j_*(\gamma w \gamma^{-1})$ in $\widehat{P}$. Its whole procyclic closure lies in all thirty-six kernels by continuity. Finally this is a nontrivial

discrete pure mapping class, so the general procyclic exclusion of F.10 applies and gives the asserted trivial intersection with $C_{\widehat{P}}(A_5)$. No element of that residual centralizer is produced, and no equality between it and this joint kernel is asserted. □

**F.12. [Proved] The common involution quotient and its winding equations.** Use the actual action and source in F.11. Put

$$\tau = (1\ 5)(3\ 4), \qquad \beta = (1\ 3)(4\ 5), \qquad Y = \mathbf{P}^1 \setminus \{0, \infty, \pm\mu_5\}, \qquad t = s^2. \tag{F.12.1}$$

The map $f$ in F.11 induces a split injection on abelianizations, including their full profinite completions. Consequently its completed kernel lies in the closed commutator subgroup of the specified $\widehat{F}_6$.

There is an explicit, fully injective map

$$\widehat{\phi} : \widehat{\pi_1(Y, s_0)} = \widehat{F}_{11} \longrightarrow \widehat{P\Gamma}_{0,8}, \qquad s_0 = i(\theta^2 - 2)\zeta, \quad s_0^2 = t_0, \tag{F.12.2}$$

induced by the configuration

$$\phi(s) = (0, \infty, 1, v_1(s), \dots, v_5(s)), \qquad v_{j+1}(s) = \left(\frac{\zeta^j - s}{\zeta^j + s}\right)^2 \quad (0 \le j \le 4). \tag{F.12.3}$$

The actual quotient action of $\beta$ induces the following based involution on this normalized eight-point moduli space:

$$R(v_1, v_2, v_3, v_4, v_5) = (v_2/v_3,\ v_2,\ v_2/v_1,\ v_2/v_5,\ v_2/v_4). \tag{F.12.4}$$

The residual centralizer in F.7 is exactly the image, under the specified covering inclusion, of the full equalizer

$$\mathcal{E} = \{u \in \widehat{\pi_1(Y, s_0)} : \widehat{\phi}(u) = R_* \widehat{\phi}(u)\}. \tag{F.12.5}$$

In particular this identifies all the groups and maps in the remaining fixed-intersection problem.

Write $e_j^{\pm}$ for the positively oriented source meridian classes at $\pm\zeta^j$. The abelian image of every $u \in \mathcal{E}$ has the form

$$[u] = \sum_{j=0}^{4} a_j (e_j^+ - e_j^-), \qquad a_0 - a_1 + a_2 = 0, \quad -a_1 + a_3 + a_4 = 0 \quad \text{in } \widehat{\mathbb{Z}}. \tag{F.12.6}$$

These are exactly the restrictions of the twenty target winding equalities to that anti-invariant source submodule. Their kernel is the explicitly specified free rank-three $\widehat{\mathbb{Z}}$-module in the proof; no equality with the actual image of $\mathcal{E}$ is asserted.

For the element $w$ in F.11, the lift to this double-cover source has $[w] = e_1^- - e_1^+$. Its first target winding defect is exactly four. Hence its entire procyclic closure meets $\mathcal{E}$ trivially, by this specific abelian quotient computation. Triviality of the full equalizer (F.12.5) remains unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Part F, nonsolvable A5 centralizer comparison on the twelve-punctured sphere.

*Proof. Six exact characters for the seven-point map.* On the target of F.11.2, use the last labelled point as infinity and normalize the first two finite points to $0, 1$. Denote the resulting six finite coordinates by $z_0, \dots, z_5$, so $z_0 = 0, z_1 = 1$. Their pullbacks under $f$ are

$$(z_0, \dots, z_5) = \frac{(1, \zeta^2, \zeta^4, t, t\zeta, t\zeta^4) - 1}{\zeta^2 - 1}. \tag{F.12.7}$$

For distinct $i, j$, the function $z_i - z_j$ is a globally defined nonzero function on this labelled moduli space, so its winding defines a homomorphism to $\mathbb{Z}$. Take the six pairs, in this order,

$$(3,4),\ (3,0),\ (5,0),\ (3,1),\ (5,1),\ (4,0). \tag{F.12.8}$$

The six pullbacks are nonzero constant multiples of

$$t, \quad t-1, \quad t-\zeta, \quad t-\zeta^2, \quad t-\zeta^3, \quad t-\zeta^4. \tag{F.12.9}$$

Thus the induced homomorphism $\ell : P\Gamma_{0,7} \to \mathbb{Z}^6$ satisfies $\ell f_* = \mathrm{ab}$ on $\pi_1(X_5)$, in its ordinary finite-puncture meridian basis. Completion gives the same identity with $\widehat{\mathbb{Z}}^6$. This proves the asserted split injection and the closed-commutator containment. In particular the residual centralizer, which belongs to this kernel by F.7.9 and F.11, lies in the index-two subgroup corresponding to $t = s^2$. The source is exactly $Y$ in (F.12.1); finite-index subgroups have their full induced profinite topology.

*The actual common-involution quotient.* For the family in F.11, the involution $\tau_t(z) = t/z$ has fixed points $s, -s$. On $Y$, label them in that order. The actual quotient map, including these branch labels, is

$$Q_s(z) = \left(\frac{z-s}{z+s}\right)^2. \tag{F.12.10}$$

It sends the two branch points to $0, \infty$. The original pole pair $\{0, \infty\}$ goes to 1. Its other five fibres on the original twelve punctures are

$$\{\zeta^j, s^2\zeta^{-j}\}, \qquad 0 \le j \le 4,$$

and their images are precisely (F.12.3). No two of the eight target points coincide for $s \in Y$. Indeed $v_{j+1}$ is 0 or infinity only at $s = \zeta^j$ or $-\zeta^j$, and equals 1 only at $s = 0, \infty$. For $i \neq j$, the other collision equation is $s^2 = \zeta^{i+j}$, whose two solutions again lie in $\pm\mu_5$.

Here is the full completion comparison for this quotient. The discrete normalizer sequence for $\langle\tau\rangle$ is the specified sequence F.7.5:

$$1 \longrightarrow \langle\tau\rangle \longrightarrow N_M(\langle\tau\rangle) \longrightarrow Q \longrightarrow 1. \tag{F.12.11}$$

Here $Q$ is the group of the quotient sphere with its six punctures and two branch points, with their types retained. It has finite index in the mapping class group with these decorations. The prime-order comparison and the full topology in F.7.4–5 identify its completed source with the actual completed normalizer. Consequently completion of (F.12.11) has exactly the displayed finite deck kernel and has target $\widehat{Q}$. The deck involution permutes the twelve original punctures. Its intersection with the completed pure centralizer is therefore trivial, as detected by the finite permutation quotient. Descent is injective on $C_{\widehat{P}}(\tau)$. The finite-index inclusion of $Q$ supplies its full embedding in the decorated eight-point mapping class completion.

The map from $\widehat{\pi_1(Y)}$ into $\widehat{P}$ is the index-two covering inclusion followed by F.11.10. It is injective and its image centralizes $\tau$. On $Y$ the two branch labels do not exchange, so the quotient map lands in $\widehat{P\Gamma}_{0,8}$. Its composite is exactly $\widehat{\phi}$ in (F.12.2), by (F.12.10). Both components are injective. This proves the full injectivity in (F.12.2), with all branch labels retained.

*The exact second involution and based naturality.* In the twelve-label table of F.11, the six $\tau$-orbits, in the order used for the quotient punctures $1, v_1, \dots, v_5$, are

$$\{0,10\}, \quad \{1,4\}, \quad \{3,8\}, \quad \{5,11\}, \quad \{6,9\}, \quad \{2,7\}. \tag{F.12.12}$$

The element $\beta = (1\ 3)(4\ 5)$ commutes with $\tau$, is outside $D_{10}$, and together with $D_{10}$ generates $A_5$. Its permutation on these six orbits is

$$p = (0\ 2)(1\ 3)(4\ 5). \tag{F.12.13}$$

It exchanges the two branch points. These facts are checked in the same actual Möbius action as F.11, not merely in a separate permutation representation. Specifically the word $b^{-1}abab^{-1}ab$ has the named action of $\beta$, sends $s_0$ to $-s_0$ and $-s_0$ to $s_0$, and in the coordinate $(z-s_0)/(z+s_0)$ has the form $c/z$. The certificate verifies $c^2 = v_2(s_0)$. Thus its actual quotient representative is $V \mapsto v_2(s_0)/V$.

For general normalized quotient coordinates write $v_0 = 1$. After applying the label permutation and renormalizing the branch values to $0, \infty$ and puncture zero to 1, the new coordinate at puncture $i$ is $v_{p(0)}/v_{p(i)}$. This is precisely (F.12.4). At $s_0$ the actual quotient representative just specified proves $R(\phi(s_0)) = \phi(s_0)$. The based action $R_*$ is therefore fixed without an unspecified base path, and agrees with conjugation by that quotient mapping class. The source basepoint comparison with F.11 uses the unique lift, starting at $s_0$, of the already specified straight segment from $t_0$ to $2 + 2i$.

Let $\jmath : \widehat{\pi_1(Y)} \to \widehat{P}$ denote the actual covering inclusion above. For the residual centralizer, (F.12.7)–(F.12.9) show that its elements have unique preimages under $\jmath$. The descent of each such element commutes with the quotient of $\beta$, proving that its preimage lies in $\mathcal{E}$. Conversely, for $u \in \mathcal{E}$, both $\jmath(u)$ and $\beta\jmath(u)\beta^{-1}$ lie in $C_{\widehat{P}}(\tau)$. Their descents agree by (F.12.5). The injectivity proved from (F.12.11) therefore gives $\beta\jmath(u)\beta^{-1} = \jmath(u)$. The element $\jmath(u)$ already centralizes $D_{10}$, so it centralizes their generated group $A_5$. This proves both directions of (F.12.5); the resulting isomorphism is continuous with continuous inverse on these compact subgroups.

*All winding equations on the source homology.* Use the finite source meridians at $0, \zeta^j, -\zeta^j$ as a basis of $H_1(Y; \widehat{\mathbb{Z}})$, omitting the infinity meridian. The map $s \mapsto s^2$ sends the zero meridian to twice the zero meridian downstairs, and sends each $e_j^{\pm}$ to the downstairs meridian at $\zeta^{2j}$. The downstairs abelianization of an element of $\mathcal{E}$ vanishes by the first part of the proof. Since multiplication by two on $\widehat{\mathbb{Z}}$ is injective, its zero coefficient is zero and its two coefficients above each $\zeta^{2j}$ sum to zero. This gives the first expression in (F.12.6).

For clarity the remaining calculation is entirely integral. Take the twenty winding characters on the normalized eight-point space:

$$L_i = \mathrm{wind}(v_i), \quad M_i = \mathrm{wind}(v_i - 1) \quad (1 \le i \le 5), \qquad N_{ij} = \mathrm{wind}(v_i - v_j) \quad (1 \le i < j \le 5). \tag{F.12.14}$$

These give its usual integral first-cohomology basis. One can see the basis directly by forgetting the last variable in turn: the fibre is the plane with the earlier points and $0, 1$ removed, its meridians are detected by the newly added differences, and pure monodromy fixes their abelian classes. The successive ranks are $2, 3, 4, 5, 6$.

The exact functions in (F.12.3) satisfy

$$\begin{aligned} v_{j+1} - 1 &= \frac{-4\zeta^j s}{(\zeta^j + s)^2}, \\ v_{i+1} - v_{j+1} &= \frac{4s(\zeta^i - \zeta^j)(\zeta^{i+j} - s^2)}{(\zeta^i + s)^2(\zeta^j + s)^2} \qquad (i \ne j). \end{aligned} \tag{F.12.15}$$

Their zero and pole orders at the twelve source punctures determine the pullbacks. On an anti-invariant class $\sum a_j(e_j^+ - e_j^-)$ the resulting values are

$$L_{j+1} = 4a_j, \qquad M_{j+1} = 2a_j, \qquad N_{i+1,j+1} = 2(a_i + a_j). \tag{F.12.16}$$

The extra zeros of the last expression, at the two points with $s^2 = \zeta^{i+j}$, cancel on an anti-invariant class.

Put $A_0 = 0$ and $A_{j+1} = a_j$. For $v_0 = 1$, the winding of any difference $v_i - v_j$ is uniformly $2(A_i + A_j)$, and that of $v_i$ is $4A_i$. Under $R$, these functions are transformed using $v_i \mapsto v_2/v_{p(i)}$. The twenty equalities of winding values are consequently equivalent to

$$A_i + A_{p(i)} = A_2 \qquad (0 \le i \le 5). \tag{F.12.17}$$

Here necessity already follows from $M_1$ and $M_4$, whose defects are $2(a_0 - a_1 + a_2)$ and $2(-a_1 + a_3 + a_4)$. Multiplication by two is injective. For sufficiency, substitute (F.12.17) in the winding of $v_2/v_{p(i)}$, $(v_2 - v_{p(i)})/v_{p(i)}$, and $v_2(v_{p(j)} - v_{p(i)})/(v_{p(i)}v_{p(j)})$; these are exactly the three types in (F.12.14). The only two nontrivial relations in (F.12.17) are (F.12.6). Their kernel has the following integral basis, which also gives its full $\widehat{\mathbb{Z}}$-basis:

$$(-1,0,1,0,0), \qquad (1,1,0,1,0), \qquad (1,1,0,0,1). \tag{F.12.18}$$

These equations are necessary for membership in $\mathcal{E}$; their kernel is only the module defined by the displayed winding equations.

*The concrete detector for the element of F.11.* At $t_0$ the chosen root is $s_0 = i(\theta^2 - 2)\zeta$. The lift of the straight path to $b = 2 + 2i$ ends at the negative square root of $b$: throughout that path the real part of $t$ is positive, and $s_0$ is the root with negative real part. The whisker and the small square for $m_{\zeta^2}$ lie in the upper half-plane. Continuation on this sheet therefore lifts that meridian to a positive meridian at $-\zeta$. Its lift on the other sheet is a meridian at $+\zeta$.

The second commutator factor $m_\zeta m_{\zeta^3} m_0$ has winding one about zero, and hence exchanges the two sheets. In the lifted commutator, its path and its reverse cancel in abelianization. The two remaining meridians give exactly

$$[w] = e_1^- - e_1^+. \tag{F.12.19}$$

By (F.12.4), $R^*L_1 = L_2 - L_3$. The winding difference for this class is therefore

$$(L_1 - R^*L_1)\widehat{\phi}(w) = 4(a_0 - a_1 + a_2) = 4. \tag{F.12.20}$$

For a full profinite power $w^u$, the continuous extension of this character has value $4u \in \widehat{\mathbb{Z}}$. It vanishes only for $u = 0$. This is an explicit family of finite cyclic detectors for the procyclic exclusion, and uses the actual common involution quotient.

The executed certificate `code/part_a/c5_common_reflection_f12.py` verifies the common finite labels, the Möbius involution and its branch normalization, the six-character splitting, the twenty-by-twelve integral divisor matrix, the exact involution on the twenty winding characters, and the restriction to the two rows in (F.12.6). It also records the winding defect four. Its larger unreduced matrices have source rank ten and fixed-difference rank six; no rank calculation is used to infer triviality of the full equalizer. □

**F.13. [Proved] The exact homology kernel and the retraction obstruction.** Retain the specified injection $\widehat{\phi}$ of F.12.2. Write $e_0, e_\infty, e_j^+, e_j^-$ for the twelve positive source cusp classes, with the single relation

$$e_0 + e_\infty + \sum_{j=0}^{4}(e_j^+ + e_j^-) = 0. \tag{F.13.1}$$

For $A = \mathbb{Z}$, $\widehat{\mathbb{Z}}$, or $\mathbb{Z}_p$ at any prime $p$, the actual first-homology map induced by $\phi$ has exact kernel

$$\ker \phi_{*,1} = A\,k, \qquad k = e_0 - e_\infty = 2e_0 + \sum_{j=0}^{4}(e_j^+ + e_j^-). \tag{F.13.2}$$

Here the second expression uses the genuine basis omitting $e_\infty$. In particular $k$ is primitive over $\mathbb{Z}$ and has nonzero reduction modulo every prime.

The discrete map, its full profinite completion, and its maps to maximal pro-$p$ completions have no homomorphic left inverse on the stated domains, continuous where profinite groups are involved. The same obstruction holds after every further target-forgetting map. No assertion

about a finite-index target subgroup, the injectivity of a forgotten map, or the full equalizer F.12.5 is made.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), exact residual centralizer and the specified common-involution quotient.

*Proof.* To prove the kernel equality, write an arbitrary class in that basis as

$$z = ce_0 + \sum_{j=0}^{4} (p_j e_j^+ + m_j e_j^-).$$

The twenty target winding characters are the actual integral basis of F.12.14. Their pullbacks from F.12.3 and F.12.15 give

$$\begin{aligned} L_{j+1}(\phi_* z) &= 2(p_j - m_j), \\ M_{j+1}(\phi_* z) &= c - 2m_j, \\ N_{i+1,j+1}(\phi_* z) &= c + p_k + m_k - 2m_i - 2m_j, \quad 2k \equiv i + j \pmod 5 \quad (i \neq j). \end{aligned} \tag{F.13.3}$$

Multiplication by two is injective in each of the specified coefficient rings. Vanishing of the first two rows therefore gives $p_j = m_j = t$ for all $j$ and $c = 2t$. The third row then vanishes identically. This proves (F.13.2), without identifying an actual group kernel with its abelian image.

For completeness, at a specified primary coordinate $p$, the vector $k$ has ten entries equal to one in the displayed $\mathbb{Z}_p$-basis. It is consequently nonzero over $\mathbb{Z}_p$ and modulo $p$. In the full $\widehat{\mathbb{Z}}$-module, multiplying $k$ by the idempotent whose $p$-coordinate is one and whose other primary coordinates are zero gives a nonzero kernel vector supported exactly at that primary coordinate.

There is no homomorphic left inverse of the discrete map $\phi_* : \pi_1(Y) \to P\Gamma_{0,8}$ and no continuous homomorphic left inverse of its full profinite map. Indeed such a left inverse would induce a left inverse on abelianizations, contradicting (F.13.2). More precisely, for every prime $p$ there is no continuous homomorphism

$$r_p : \widehat{P\Gamma}_{0,8} \longrightarrow \pi_1(Y)^{(p)} \quad \text{with} \quad r_p \widehat{\phi} = \pi_p, \tag{F.13.4}$$

where $\pi_p$ is the actual maximal pro-$p$ quotient map. Apply abelianization modulo $p$ to this equality: its left side kills $k \bmod p$, whereas its right side is the identity on the source's $\mathbb{F}_p$-homology. The same argument rules out a left inverse of the induced map between maximal pro-$p$ completions.

Every homomorphism obtained by first applying $\widehat{\phi}$ and then forgetting any target labels also kills $k$ on first homology, by functoriality. Thus none of those composites has a homomorphic left inverse. This does not decide which such composites are injective, nor does it decide the equalizer F.12.5. The obstruction is stated for these displayed domains; passing to a finite-index target subgroup would change its first-homology map and is not covered by this argument.

The separate exact certificate `code/part_a/common_reflection_retraction.py` constructs all twenty divisor rows from (F.13.3), checks the source puncture-sum relation and the primitive kernel vector, computes rank ten over $\mathbb{Q}$, and checks the explicit nonzero kernel vector in the listed finite primary reductions. Its finite reductions verify the displayed formulas; the uniform statement for every prime follows from the coefficient-one argument above. □

**F.14. [Proved] An injective seven-point comparison and its exact remaining defect.** Retain the based map $\phi$ and the involution $R$ of F.12. Label its eight target points

$$0,\ \infty,\ v_0 = 1,\ v_1, \ldots, v_5. \tag{F.14.1}$$

Let $q_{v_0}$ forget the ordinary point $v_0$, retaining both branch labels $0, \infty$. The actual composite

$$\widehat{\psi} = (q_{v_0})_* \widehat{\phi} : \widehat{\pi_1(Y, s_0)} \longrightarrow \widehat{P\Gamma}_{0,7} \tag{F.14.2}$$

is injective. With $v_1$ renormalized to one, its four remaining coordinates and those of the second composite $\psi_R = q_{v_0} R\phi$ are exactly

$$\begin{aligned} \psi(s) &= (v_2/v_1,\ v_3/v_1,\ v_4/v_1,\ v_5/v_1), \\ \psi_R(s) &= (v_3,\ v_3/v_1,\ v_3/v_5,\ v_3/v_4). \end{aligned} \tag{F.14.3}$$

Both maps have the same specified target basepoint. Their second coordinates agree as actual functions.

Forgetting either branch point 0 or $\infty$ from (F.14.1) instead kills each of the two explicitly whiskered source meridians at $s = 0, \infty$ below. Each meridian generates a full closed $\widehat{\mathbb{Z}}$ in the source. Both branch-forgetting composites factor through the specified filling

$$\widehat{\pi_1(Y)} \longrightarrow \pi_1(\widehat{\mathbf{P}^1 \setminus \{\pm\mu_5\}}). \tag{F.14.4}$$

There is also an exact description of the information retained by (F.14.3). Put $P_8 = \widehat{P\Gamma}_{0,8}$, write $K_i$ for its actual kernel of forgetting $v_i$, and define

$$\begin{aligned} \mathcal{E}_7 &= \{u : \widehat{\psi}(u) = \widehat{\psi}_R(u)\}, \\ B &= K_0 \cap K_2, \\ d(u) &= \widehat{\phi}(u)^{-1} R_* \widehat{\phi}(u) \qquad (u \in \mathcal{E}_7). \end{aligned} \tag{F.14.5}$$

Then $d$ takes values in $B$, satisfies

$$\begin{aligned} R_* d(u) &= d(u)^{-1}, \\ d(uv) &= \widehat{\phi}(v)^{-1} d(u) \widehat{\phi}(v) d(v), \\ \mathcal{E} &= \{u \in \mathcal{E}_7 : d(u) = 1\}, \end{aligned} \tag{F.14.6}$$

where $\mathcal{E}$ is the full residual equalizer F.12.5. In the actual point-pushing identification $K_0 = \widehat{F}_6$, $B$ is precisely the closed normal subgroup generated by the meridian about $v_2$. These statements retain both distinct forgetting composites; they do not cancel $q_{v_0}$ from the equalizer.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), residual nonsolvable centralizer and exact quotient comparisons.

*Proof. Filling the two order-five poles.* Let $S_{12}$ be the sphere punctured at the labelled configuration

$$0, \infty, \mu_5, t_0\mu_5,$$

and let $S_{10}$ be obtained by filling the first two punctures. The named rotation $d(z) = \zeta z$ acts on both surfaces. The two fixed poles have become unmarked ramification points on $S_{10}$. After removal of those ramification points, both covering problems are literally the same unramified cover

$$S_{12} \longrightarrow X = \mathbf{P}^1 \setminus \{0, \infty, 1, t_0^5\}, \qquad z \longmapsto z^5. \tag{F.14.7}$$

Its four meridians have the same monodromy values $1, -1, 0, 0$ in $C_5$ as F.11.8.

Write $N_{12}$ and $N_{10}$ for the discrete full normalizers of this specified $C_5$ in the two mapping class groups. The actual covering normalizer sequences of F.7.5 form the commuting diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & C_5 & \longrightarrow & N_{12} & \longrightarrow & Q & \longrightarrow & 1 \\ & & \| & & \downarrow \text{fill} & & \| & & \\ 1 & \longrightarrow & C_5 & \longrightarrow & N_{10} & \longrightarrow & Q & \longrightarrow & 1. \end{array} \tag{F.14.8}$$

Here $Q$ is the same subgroup of the mapping class group of $X$: its elements preserve the covering kernel and the two branch labels as a set, and preserve the two free-puncture labels as a set. The nonzero and zero monodromy values already distinguish these two types. The extra description of a branch point as an original puncture on $S_{12}$ therefore imposes no new permutation condition. On $S_{10}$ those same labels are retained as branch decorations. Descent of a filled representative is the same homeomorphism of $X$, which proves that both right vertical maps are literally the identity after these identifications. The kernel map is the named deck action itself. The exact-sequence argument proves that the middle filling map is an isomorphism.

Both finite groups in (F.14.8) are the same order-five group. The prime-order normalizer comparison and full induced topology established in F.7.4–5 identify their completed normalizers with their actual closures in the ambient completions. Thus completion of the middle isomorphism is the actual injective filling map on the completed normalizer. In particular it is injective on the source subgroup from F.11 and on its index-two subgroup $\widehat{\pi_1(Y)}$. The ambient filling map is defined on the stabilizer of the named pole pair; the normalizer preserves that pair, and the source is pure, so this restriction uses the stated ambient maps.

*The commuting quotient square.* The source still centralizes the actual involution $\tau_t(z) = t/z$ after the poles are filled. On $S_{10}$ this involution permutes the remaining punctures in five pairs. It has the same two unmarked fixed points $s, -s$, labelled over $Y$. Its decorated quotient has these two branch points and five ordinary punctures. The quotient map is still the literal function $Q_s(z)$ in F.12.10. The removed pole pair has image $v_0 = 1$ and every remaining pair has the same image $v_j$.

The completed involution-normalizer quotient again has exactly its named finite deck kernel, by F.7.4–5. That deck involution permutes the ten punctures nontrivially. It has trivial intersection with the completed pure subgroup, detected by the finite permutation map. Hence descent is injective on the pure completed centralizer of $\tau$ in the ten-puncture group. The branch-labelled source lands in the pure seven-puncture target. Write $C_{12}^{\mathrm{br}}$ and $C_{10}^{\mathrm{br}}$ for the subgroups of the two pure completed $\tau$-centralizers whose descents also fix its two branch labels individually. Filling the pole pair preserves this condition. We have the actual commuting square

$$\begin{array}{ccc} \widehat{\pi_1(Y)} & \longrightarrow & C_{12}^{\mathrm{br}} \\ \downarrow\widehat{\psi} & & \downarrow \mathrm{fill}_{\{0,10\}} \\ \widehat{P\Gamma}_{0,7} & \longleftarrow & C_{10}^{\mathrm{br}}. \end{array} \tag{F.14.9}$$

The upper arrow is the F.11–F.12 source inclusion; the lower arrow is the just specified decorated descent. On that source the right arrow is injective by (F.14.8), and the lower arrow is injective on the pure subgroup. Their composite is precisely $(q_{v_0})_*\widehat{\phi}$, by the pointwise formula for $Q_s$. This proves (F.14.2).

*Both seven-point maps and their common coordinate.* After forgetting $v_0$, the three labels $0, \infty, v_1$ give the unique normalization fixing $0, \infty$ and sending $v_1$ to one. This gives the first line of (F.14.3). Substitution of F.12.4 followed by this same normalization gives its second line. Since $R(\phi(s_0)) = \phi(s_0)$ with the exact based representative already proved in F.12, no additional target base path is required. For clarity their common coordinate is

$$\frac{v_3(s)}{v_1(s)} = \left(\frac{(\zeta^2 - s)(1+s)}{(\zeta^2 + s)(1-s)}\right)^2. \tag{F.14.10}$$

Thus forgetting all but its three normalization labels and this coordinate gives equal composites to $M_{0,4}$ as actual maps of spaces. No injectivity of that four-point composite is used.

*The branch-forgetting maps and based meridians.* Forget the branch point 0 instead, retaining $\infty, v_0 = 1, v_1, \ldots, v_5$. Normalize $v_0$ to zero and $v_1$ to one, keeping infinity fixed. Its four coordinates are

$$u_j(s) = \frac{v_{j+1}(s) - 1}{v_1(s) - 1} = \frac{\zeta^j (1+s)^2}{(\zeta^j + s)^2}, \qquad 1 \le j \le 4. \tag{F.14.11}$$

They extend over $s = 0$ and $s = \infty$, with respective values $\zeta^{-j}$ and $\zeta^{j}$. At either end these four values are pairwise distinct and avoid $0, 1, \infty$. Thus this is an extension into the actual seven-point configuration space, not merely a finite limit in its compactification. The other branch-forgetting map has the same property: the actual target coordinate inversion $V \mapsto 1/V$ exchanges the two branch labels and satisfies $1/v_j(s) = v_j(-s)$. Apply the same normalization and extension with $-s$. This proves the space-level factorization through filling the two indicated source punctures and hence (F.14.4) on full completions.

Here are explicit based choices. Put $\epsilon = 1/10$ and use the polygonal whiskers

$$\begin{aligned} \omega_0 &= (s_0,\ -2 + 2i,\ \epsilon + 2i,\ \epsilon), \\ \omega_\infty &= (s_0,\ -2 + 2i,\ 10 + 2i,\ 10). \end{aligned} \tag{F.14.12}$$

From the exact expression $s_0 = \theta - \theta^3/2 + i/2$ and $\theta > 19/10$, its real part is less than $-1$. The first segments therefore lie to the left of the unit circle; the horizontal segments have imaginary part two. The final segment of $\omega_\infty$ lies outside that circle. The final segment of $\omega_0$ has real part $1/10$, different from zero and from every real part of a point of $\pm\mu_5$. Indeed their nonzero real parts have absolute value at least $(\sqrt{5} - 1)/4 > 1/10$. Both whiskers lie in $Y$.

Use the local circles $s = \epsilon e^{2\pi i t}$ and $s = 10e^{-2\pi i t}$, respectively, and return along the reverse whisker. These give positive based source meridians $m_0, m_\infty$. The extended maps in (F.14.11) carry each local circle to a loop filled by its actual disk at the added source point; composing with the image whiskers gives based null-homotopies. The same holds for the other branch-forgetting map. In source abelianization $m_0$ is $e_0$ and $m_\infty$ is $e_\infty = -e_0 - \sum(e_j^+ + e_j^-)$. A finite-puncture winding character takes each of these meridians to 1 or $-1$. Its full continuous extension consequently identifies the closure of each cyclic subgroup with a full $\widehat{\mathbb{Z}}$ and detects every nonzero profinite exponent. Thus these are genuine nontrivial full procyclic subgroups of both branch-forgetting kernels.

*The exact point-pushing defect.* In the target labels of (F.14.1), the permutation underlying $R$ exchanges $v_0$ with $v_2$. Naturality of forgetting with this actual labelled map gives

$$R_*(K_0) = K_2. \tag{F.14.13}$$

For $u \in \mathcal{E}_7$, the defining equality in (F.14.5) says exactly that $d(u) \in K_0$. Since $R_*^2 = 1$, direct multiplication gives $R_* d(u) = d(u)^{-1}$. Equation (F.14.13) then also puts $d(u)$ in $K_2$. The second identity in (F.14.6) follows by expanding the definition on the product $uv$. Both $K_i$ are normal, so the displayed conjugation preserves $B$. The third identity is the defining full equality F.12.5, with both implications now retained inside the explicitly specified seven-point equalizer.

Finally the actual Birman point-pushing diagrams identify $K_0$ with the completion of the fundamental group of the sphere punctured at the other seven labelled points, based at $v_0$. Forgetting $v_2$ restricts on this kernel to the ordinary puncture-filling map $\widehat{F}_6 \to \widehat{F}_5$. The point-pushing subgroup in the seven-point target after forgetting $v_2$ is embedded. The actual completed Birman square is

$$\begin{array}{ccccc} K_0 = \widehat{F}_6 & \longrightarrow & P_8 & \longrightarrow & \widehat{P\Gamma}_{0,7} \\ \downarrow \mathrm{fill}_{v_2} & & \downarrow \mathrm{forget}_{v_2} & & \downarrow \mathrm{forget}_{v_2} \\ K_0' = \widehat{F}_5 & \longrightarrow & \widehat{P\Gamma}_{0,7} & \longrightarrow & \widehat{P\Gamma}_{0,6}. \end{array} \tag{F.14.14}$$

The horizontal quotient maps forget $v_0$; the lower middle group still retains $v_0$. Consequently the kernel of that restriction is exactly $K_0 \cap K_2$. The map of free groups kills the named peripheral meridian about $v_2$ and keeps the other five finite-puncture generators; the same finite presentation and the universal property of completion identify its full kernel with the closed normal subgroup generated by that meridian.

These full Birman maps can also be obtained directly from normalized configuration spaces. Forgetting one variable of an ordered configuration in $\mathbf{C} \setminus \{0, 1\}$ is a fibration with the stated

punctured-plane fibre, and has a continuous section putting the forgotten variable on the positive real axis beyond $2+\sum|z_j|$. The base and fibre are aspherical by induction through these fibrations, giving the actual discrete exact sequences. The split group extension induces the full profinite topology on its finitely generated free kernel: use its characteristic finite-index subgroups and the finite action on each resulting finite quotient in the semidirect product. Completion is then exact, and the forgetting squares are the completions of the displayed topological squares. This justifies the kernel comparison on the full completions used above.

The executed certificate `code/part_a/common_reflection_forgetting_f14.py` checks the two normalized composite formulas, their common coordinate, the two endpoint extensions, the exact source basepoint formula, and the fourteen-by-twenty matrix of the ordinary-point forgetting map on winding characters. The injectivity in (F.14.2) uses the actual normalizer and covering maps, not these homology checks. □

**F.15. [Proved] The two-point kernel and its full involution cocycles.** Retain the actual based configuration of F.12–14 and put $c_0=v_2(s_0)$, $x_0=v_0=1$, $y_0=c_0$. On the sphere with the other six labels retained, set

$$\begin{aligned} M &= \mathbf{P}^1\setminus\{0,\infty,v_1(s_0),v_3(s_0),v_4(s_0),v_5(s_0)\},\\ r(z)&=c_0/z,\qquad X=\mathrm{Conf}_2(M),\qquad \mathcal{R}(x,y)=(r(y),r(x)). \end{aligned} \tag{F.15.1}$$

Thus $\mathcal{R}(x_0,y_0)=(x_0,y_0)$. Write $a_+,a_-$ for the two roots of $z^2=c_0$, which are unmarked points of $M$. Write $\mathsf{r}$ for the actual eight-puncture mapping class of $r$, so conjugation by $\mathsf{r}$ induces the based automorphism $R_*$ of F.12.

There are the following actual completed homomorphisms:

$$\begin{aligned} J&=\pi_1(X,(x_0,y_0)),\\ L_0=\pi_1(M,x_0),&\qquad L_2=\pi_1(M,y_0),\\ 1\longrightarrow B\longrightarrow \widehat{J}&\xrightarrow{p}\widehat{L}_0\times\widehat{L}_2\longrightarrow 1. \end{aligned} \tag{F.15.2}$$

Here $\widehat{J}$ is the kernel in $P_8=\widehat{P\Gamma}_{0,8}$ of forgetting the pair $\{v_0,v_2\}$; $p$ records the two labelled point tracks, and its kernel is exactly $B=K_0\cap K_2$ of F.14. Let $\rho=r_*:\widehat{L}_0\longrightarrow\widehat{L}_2$. The induced involution and its fixed subgroup are

$$R_*(u,v)=(\rho^{-1}(v),\rho(u)),\qquad (\widehat{L}_0\times\widehat{L}_2)^{R_*}=\{(u,\rho(u)):u\in\widehat{L}_0\}. \tag{F.15.3}$$

The actual fixed-configuration map

$$i:M\setminus\{a_+,a_-\}\longrightarrow X,\qquad z\longmapsto(z,r(z)) \tag{F.15.4}$$

induces a full profinite embedding, and

$$\begin{aligned} C_{\widehat{J}}(\mathsf{r})&=\widehat{i}_*\,\pi_1(M\setminus\widehat{\{a_+,a_-\}},x_0)\ \cong\widehat{F}_7,\\ p(C_{\widehat{J}}(\mathsf{r}))&=(\widehat{L}_0\times\widehat{L}_2)^{R_*},\\ C_B(\mathsf{r})&\cong\ker\bigl(\pi_1(M\setminus\widehat{\{a_+,a_-\}},x_0)\xrightarrow{\mathrm{fill}_{a_+,a_-}}\widehat{\pi_1(M,x_0)}\bigr). \end{aligned} \tag{F.15.5}$$

In particular the last kernel is the closed normal subgroup generated by the two based branch meridians in this specified $\widehat{F}_7$.

Every full continuous $R_*$-cocycle in $B$ is a $B$-coboundary. More precisely, there is a homeomorphism of compact spaces

$$\begin{aligned} C_B(\mathsf{r})\backslash B&\xrightarrow{\sim}Z^1(R_*,B):=\{\delta\in B:\delta R_*(\delta)=1\},\\ C_B(\mathsf{r})b&\longmapsto b^{-1}R_*(b). \end{aligned} \tag{F.15.6}$$

Consequently the quotient map $\pi : P_8 \to P_8/B$ gives the exact sequence

$$1 \longrightarrow C_B(\mathsf{r}) \longrightarrow C_{P_8}(\mathsf{r}) \xrightarrow{\pi} (P_8/B)^{R_*} \longrightarrow 1. \tag{F.15.7}$$

For the exact equalizer $\mathcal{E}_7$ of F.14, set

$$\mathcal{B}_u = \{b \in B : b^{-1}R_*(b) = d(u)\}. \tag{F.15.8}$$

This is a nonempty left $C_B(\mathsf{r})$-coset for every $u \in \mathcal{E}_7$. The centralizing representatives of its actual $B$-coset are exactly

$$(\widehat{\phi}(u)B) \cap C_{P_8}(\mathsf{r}) = \widehat{\phi}(u)b^{-1}C_B(\mathsf{r}) \qquad (b \in \mathcal{B}_u). \tag{F.15.9}$$

The source embedding has trivial intersection with $B$. It follows that the intersection of (F.15.9) with $\widehat{\phi}(\widehat{\pi_1(Y)})$ is either empty or the singleton $\{\widehat{\phi}(u)\}$, and is the latter exactly for $u \in \mathcal{E}$. These formulas determine the correction and its ambiguity; triviality of the full residual group $\mathcal{E}$ is still unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), the exact residual centralizer and its common-involution quotient.

*Proof. The labelled configuration and the two tracks.* The actual quotient representative in F.12 is $z \mapsto v_2(s_0)/z$. Its permutation on the eight labels is

$$(0\ \infty)(v_0\ v_2)(v_1\ v_3)(v_4\ v_5). \tag{F.15.10}$$

It has no fixed marked point. Its two fixed points are therefore precisely the unmarked $a_+, a_-$ of (F.15.1). Deleting the invariant pair $\{v_0, v_2\}$ retains an actual homeomorphism of the indicated six-puncture sphere, with no additional base path. Under point pushing, conjugation by its mapping class acts on the ordered two-point fibre by $\mathcal{R}$. This proves both the specified action and (F.15.3).

The discrete Birman fibration identifies $J$ with the kernel of forgetting those two labels. Projecting $X$ to one coordinate has fibre a sphere with seven punctures, and gives a split extension

$$1 \longrightarrow F_6 \longrightarrow J \longrightarrow F_5 \longrightarrow 1. \tag{F.15.11}$$

One continuous section can be seen in a coordinate with one retained puncture at infinity: put the second point on the positive real axis beyond $2 + |x| + \sum_{z \in P_6 \setminus \{\infty\}} |z|$. A path in the fibre adjusts its basepoint to the specified one. The maps in (F.15.11), including that adjustment, remain the ordinary labelled forgetful maps.

The finitely generated free kernel in (F.15.11) has its full induced profinite topology. Indeed, its characteristic finite-index subgroups are cofinal among its finite-index subgroups. For any such subgroup, the quotient of the free kernel is finite and the action of the free quotient on it has finite image; the associated finite semidirect product supplies the required finite quotient of $J$. Completing (F.15.11) is therefore exact. Map this completed sequence into the two successive completed Birman sequences of F.14.14. On its free kernel and its free quotient the maps are the already specified embeddings $\widehat{F}_6$ and $\widehat{F}_5$. An element in the kernel of $\widehat{J} \to P_8$ must first map trivially to $\widehat{F}_5$, then trivially to $\widehat{F}_6$. The map is injective. Conversely, an element in the kernel of forgetting both points can first be matched in the $\widehat{F}_5$ quotient by an element of $\widehat{J}$; the remaining difference lies in the embedded $\widehat{F}_6$. This proves the asserted equality with the actual completed two-point kernel.

Each coordinate projection of $J$ records the track of that named point. The fibre of the first projection maps onto $L_2$ by filling $x_0$, so the product track map is onto $L_0 \times L_2$. The same is

true after completion. The completion of a direct product is the product of the full completions: any homomorphism to a finite group restricts to homomorphisms on its two commuting factors, and their continuous extensions multiply to its unique continuous extension. The kernel of the product track map is the simultaneous kernel of the two point-forgetting restrictions. Inside the embedded $\widehat{F}_6$ it is precisely the kernel of filling the other named point. The last Birman square in F.14 identifies it with $K_0 \cap K_2$, including its closed normal meridian description. This proves (F.15.2) without a closure–intersection assumption.

*The discrete fixed subgroup from the actual quotient cover.* Consider the specified double cover obtained by deleting the two fixed points of $r$:

$$M \setminus \{a_+, a_-\} \longrightarrow (M/\langle r\rangle) \setminus \{\overline{a}_+, \overline{a}_-\}. \tag{F.15.12}$$

The target is a sphere with three ordinary punctures and two branch points removed. Its group is $F_4$. The monodromy takes the two branch meridians to the nonidentity element of $C_2$ and takes the three ordinary puncture meridians to the identity. Its kernel is the actual $\pi_1$ of the source of (F.15.12), of index two and rank seven.

Use the discrete covering-normalizer sequence F.7.5 for $r$ before and after forgetting the pair $\{x_0, y_0\}$. Downstairs this forgets its one ordinary orbit point. The kernel of this downstairs forgetting map is the point-pushing $F_4$ just specified. A point-pushing isotopy lifts from the identity through the double cover, fixing all retained punctures individually. Its endpoint fixes both forgotten labels individually precisely when its monodromy in (F.15.12) is trivial. The finite deck kernel has trivial intersection with the pure upstairs group, because $r$ exchanges the retained punctures in three pairs. Conversely, an element of $J$ that centralizes $r$ descends through this normalizer sequence. After forgetting the orbit point its descent is the identity, since its upstairs filling is the identity and the quotient map has only the displayed deck kernel. It is therefore exactly one of these monodromy-zero point pushes. This proves

$$C_J(\mathsf{r}) = i_*\pi_1(M \setminus \{a_+, a_-\}, x_0) \cong F_7. \tag{F.15.13}$$

The lift of its downstairs point track is the path $(z, r(z))$. Thus the equality uses the actual map (F.15.4), not merely an abstract isomorphism of free groups. Projection to the first track is exactly filling the two fixed points, and projection to the second is its composite with $\rho$.

This $F_7$ also has its full induced profinite topology. The downstairs point-pushing $F_4$ has its full topology in the decorated quotient mapping class group by the completed Birman argument of F.14. Its index-two monodromy kernel inherits its full topology. The pure centralizer is a finite-index subgroup of the normalizer of the specified order-two group, and its descent is injective because the deck element is nonpure. Its image is finite index in the decorated quotient, so this restriction is an isomorphism onto that subgroup. The prime-order normalizer topology in F.7.4–5 consequently transports the full topology of that $F_7$ to its actual subgroup in the eight-puncture mapping class group. It remains full in its subgroup $J$ as well. Hence the completion of (F.15.13) is an embedding of this specified $\widehat{F}_7$ in $\widehat{J}$.

*One discrete conjugacy class of labelled sections.* Let

$$E = J \rtimes \langle \mathsf{r}\rangle \tag{F.15.14}$$

with the actual action already specified. It embeds in the eight-puncture mapping class group: the nontrivial coset is detected by the finite label permutation (F.15.10). Every order-two section in $E$ has that same permutation and fills to the specified mapping class of $r$ on $M$.

We first justify pure conjugacy in the eight-puncture group. The cyclic case of Nielsen realization, in the punctured-surface form of [41, Theorem 4 and Section IV, pp. 255–256], realizes such a section as an orientation-preserving involution of the punctured sphere. It extends over the punctures. With the invariant conformal structure, filling gives a Riemann sphere, and

a nontrivial holomorphic involution has two fixed points. Neither is marked, by (F.15.10). Its quotient and that of $r$ are spheres with two branch points and four specified ordinary orbit points. Choose an orientation-preserving quotient homeomorphism matching those branch points and the four named orbit pairs. Both double covers have monodromy one at the branch points and zero at the ordinary points, so it lifts to an equivariant homeomorphism.

The lift may interchange the two names inside some of the four pairs. Each mismatch can be corrected independently: in the target quotient push that ordinary orbit point once around one branch point, holding the other orbit points and both branch points fixed. Lift the isotopy starting at the identity. The endpoint commutes with $r$, interchanges exactly the two labels in the selected orbit, and fixes the other six labels individually. The covering monodromy proves these exact permutations. Composing the required corrections yields a pure conjugator. Thus every section is $g\mathsf{r}g^{-1}$ with $g \in P\Gamma_{0,8}$.

The actual pure centralizer forgetting map is onto:

$$C_{P\Gamma_{0,8}}(\mathsf{r}) \longrightarrow C_{P\Gamma_{0,6}}(r). \tag{F.15.15}$$

To prove this, take a class on the right. The discrete covering-normalizer map realizes it by a homeomorphism commuting with the actual $r$. Its value at $x_0$ is in the connected surface $M \setminus \{a_+, a_-\}$. Choose a path in that surface from this value to $x_0$. The quotient path moves the one ordinary orbit point while avoiding all branch points and retained punctures. An isotopy moving that point, supported away from those fixed labels, lifts from the identity to an $r$-equivariant isotopy. Its chosen lift carries the image of $x_0$ to $x_0$ and the paired point to $y_0$. Composing its endpoint with the original representative gives a pure eight-label centralizer representative with the same six-label class. This proves (F.15.15).

For a section $s = g\mathsf{r}g^{-1}$ in $E$, its filled class is $r$, so the filled image of $g$ centralizes $r$. Choose $c$ in the left side of (F.15.15) with that same image. Then $h = gc^{-1}$ belongs to $J$ and $s = hrh^{-1}$. Consequently

$$\{\text{order-two sections of } E\}/J \quad \text{has one element.} \tag{F.15.16}$$

*The hypotheses for full profinite section comparison.* We verify them for this particular $E$, rather than assuming that goodness passes to an arbitrary subgroup of a mapping class group. The split sequence (F.15.11) and its exact completion show that $J$ is residually finite: an element with nontrivial image in $F_5$ is detected there, and an element of the free kernel is detected by one of the finite semidirect products used above. The finitely generated group $J$ has cofinal characteristic finite-index subgroups. Taking these to be invariant under the order-two automorphism shows directly that the finite extension $E$ is residually finite and that $\widehat{E} = \widehat{J} \rtimes \langle \mathsf{r} \rangle$.

The free kernel and free quotient in (F.15.11) are good. With a finite coefficient module their cohomology is finite and vanishes above degree one. The discrete and profinite Lyndon–Hochschild–Serre spectral sequences for (F.15.11) therefore have isomorphic second pages: the coefficients $H^b(F_6, A)$ for $b = 0, 1$ are finite, so goodness of $F_5$ applies to them. This proves goodness of $J$ and finiteness of its cohomology with every finite module. The same spectral-sequence comparison for $1 \to J \to E \to C_2 \to 1$ proves these assertions for $E$. The free-by-free group $J$ has a finite two-dimensional classifying complex: take a finite graph for the free kernel over a graph for the free quotient, using the finitely many monodromy homotopy equivalences on its edges. Its universal cover is a tree of contractible covering spaces. In particular $\mathrm{cd}(J) \leq 2$. The resulting finite-type resolutions and the finite-group resolution for $C_2$ show that cohomology of $E$ commutes with filtered unions of coefficient modules. A discrete $\mathbf{F}_2[\![\widehat{E}]\!]$-module is the union of its finite invariant submodules: the span of each finite orbit is finite. Thus the finite-coefficient comparison just proved supplies the comparison for all discrete modules required in [11, Theorem A]. The virtual cohomological dimension and finite-cohomology hypotheses there are now verified as well.

Apply that prime-order theorem to $E \hookrightarrow \widehat{E}$. It identifies the conjugacy classes of its finite 2-subgroups and gives the closure of the centralizer of $\mathsf{r}$. The group $\widehat{J}$ is torsion-free, by its exact extension of the two torsion-free free profinite groups in (F.15.11). Hence every order-two subgroup of $\widehat{E}$ is a section. By (F.15.16) there is one such conjugacy class. A conjugator in $\widehat{E}$ can be replaced by one in $\widehat{J}$: write it as $h\mathsf{r}^{\epsilon}$ and use that $\mathsf{r}^{\epsilon}$ centralizes $\mathsf{r}$. The centralizer comparison, intersected with the open subgroup $\widehat{J}$, gives

$$C_{\widehat{J}}(\mathsf{r}) = \overline{C_J(\mathsf{r})}. \tag{F.15.17}$$

Combining this equality with the exact full-topology statement following (F.15.13) proves the first line of (F.15.5). Its projection is precisely the graph of the full filling epimorphism $\widehat{F}_7 \to \widehat{F}_5$ and its composite with $\rho$. This proves the other two lines, including the named kernel.

*Removing the fixed quotient coordinate.* Take $\delta \in B$ with $\delta R_*(\delta) = 1$. Then $\delta\mathsf{r}$ is an order-two section of $\widehat{E}$. The single section class just proved gives $h \in \widehat{J}$ with

$$\delta = h^{-1}R_*(h). \tag{F.15.18}$$

Its image $p(h)$ is fixed by (F.15.3). By the second line of (F.15.5), choose $c \in C_{\widehat{J}}(\mathsf{r})$ with $p(c) = p(h)$. Then $b = c^{-1}h$ lies in $B$ and direct multiplication yields $b^{-1}R_*(b) = \delta$. This proves surjectivity in (F.15.6). Two such elements $b, b'$ have the same coboundary exactly when $R_*(b'b^{-1}) = b'b^{-1}$, or equivalently $b' \in C_B(\mathsf{r})b$. This proves the asserted fibres. The map is continuous, its source is compact, and the cocycle set is a closed subspace of the profinite group $B$. The bijection is therefore a homeomorphism.

The subgroup $B$ is closed and normal in $P_8$, and $R_*$ preserves it by F.14.13. For any fixed element of $P_8/B$, choose a lift $z \in P_8$. Its defect $z^{-1}R_*(z)$ is an $R_*$-cocycle in $B$. Choose $b$ with the same coboundary by (F.15.6). Then $R_*(zb^{-1}) = zb^{-1}$, giving a centralizing lift with the same quotient image. Its kernel is exactly $C_B(\mathsf{r})$. This proves every map and the surjectivity in (F.15.7).

*The original source and its remaining membership equation.* F.14 places $d(u)$ in the cocycle set of (F.15.6), so (F.15.8) is nonempty and has the specified left coset form. For any one of its elements,

$$R_*(\widehat{\phi}(u)b^{-1}) = \widehat{\phi}(u)d(u)R_*(b)^{-1} = \widehat{\phi}(u)b^{-1}. \tag{F.15.19}$$

Any two centralizing lifts of the same $B$-coset differ on the right by an element of $C_B(\mathsf{r})$, proving (F.15.9). Finally $B \subset K_0$, and F.14 proves that the composite of $\widehat{\phi}$ with forgetting $v_0$ is injective. Thus $\widehat{\phi}(\widehat{\pi_1(Y)}) \cap B = 1$. If two source elements have images in the same $B$-coset, the same injectivity makes them equal. The only possible source representative of (F.15.9) is consequently $\widehat{\phi}(u)$ itself; it centralizes $\mathsf{r}$ precisely for $u \in \mathcal{E}$, by F.12.5. No choice of the correcting element $b$ has been asserted to remain in the original source image. □

**F.16. [Proved] A full procyclic ambiguity invisible to every proper forgetting.** Use $P_8$, the actual involution $\mathsf{r}$, and the source image

$$H = \widehat{\phi}(\widehat{\pi_1(Y, s_0)}) \ \leq P_8$$

of F.12–15. There is an explicitly constructed discrete element $w \in P\Gamma_{0,8}$ such that

$$\begin{aligned} Z_w := \overline{\langle w \rangle} &\cong \widehat{\mathbb{Z}}, \\ Z_w &\subset C_B(\mathsf{r}), \\ q_I(Z_w) &= 1 \quad \text{for every proper labelled forgetting } q_I, \\ Z_w \cap H &= 1. \end{aligned} \tag{F.16.1}$$

Here $q_I$ retains any proper subset $I$ of the eight labels, with the usual induced homomorphism on full pure mapping class completions. In particular all eight individual one-point forgettings kill $Z_w$.

The exact image-membership test ruled out by this construction is the following pullback subgroup:

$$\mathcal{T} = \bigcap_{I \text{ proper}} q_I^{-1}(q_I(H)). \tag{F.16.2}$$

Both inclusions

$$H \subsetneq \mathcal{T}, \qquad H \cap C_{P_8}(\mathsf{r}) \subsetneq \mathcal{T} \cap C_{P_8}(\mathsf{r}) \tag{F.16.3}$$

are strict, witnessed by every nonidentity element of $Z_w$. The same strictness holds when the intersection uses only the eight individual forgettings, or any subfamily of proper forgettings. For each $h \in H$, the elements $hz$ with $z \in Z_w$ have exactly the same images as $h$ under all these maps, and $h$ is their only member in $H$. Within each nonempty correction coset of F.15.9, right multiplication by $Z_w$ likewise preserves centralization and every proper forgetting image. The original source-membership equation and the full residual centralizer remain unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), exact residual centralizer and the common-involution source-membership problem.

*Proof. The actual six-point quotient and its labelled point-pushing group.* Keep the two fixed points $a_+, a_-$ from F.15, with $a_- = -a_+$ and $a_+^2 = c_0$. The specified quotient of $r(z) = c_0/z$ is

$$q(z) = \left(\frac{z - a_+}{z + a_+}\right)^2. \tag{F.16.4}$$

Indeed the coordinate $(z - a_+)/(z + a_+)$ changes sign under $r$. Label the two quotient branch points $A = q(a_+) = 0$, $D = q(a_-) = \infty$. The four ordinary quotient labels and their literal fibres on the eight marked points are

| $i$ | $o_i$ | marked fibre |
|---|---|---|
| 0 | 1 | $\{0, \infty\}$ |
| 1 | $q(1)$ | $\{v_0, v_2\}$ |
| 2 | $q(v_1(s_0))$ | $\{v_1, v_3\}$ |
| 3 | $q(v_4(s_0))$ | $\{v_4, v_5\}$. |

(F.16.5)

These six quotient points are distinct: the four ordinary values are the four different free $r$-orbits, and none is a branch orbit. This is a quotient of the actual configuration, with the same choice of $a_+$ as F.15.

In the pure mapping class group of $(\mathbf{P}^1; A, D, o_0, o_1, o_2, o_3)$, forget the branch label $A$. The actual Birman kernel and point-pushing map are

$$\begin{aligned} L &= \pi_1(\mathbf{P}^1 \setminus \{D, o_0, o_1, o_2, o_3\}, A) \cong F_4, \\ \mathrm{push}_A &: L \hookrightarrow P\Gamma_{0,6}. \end{aligned} \tag{F.16.6}$$

Choose a fixed geometric system of four disjoint whiskers from $A$ to small circles about the four $o_i$, with their interiors avoiding all five other labels. Orient the circles positively. Their based loops $m_0, m_1, m_2, m_3$ are the specified free basis of $L$; the omitted $D$-meridian is the inverse of their ordered product. The based point-pushing map is induced by moving $A$ along these loops and fixing the other five labels. Its inverse on the Birman kernel is the track of $A$ in an isotopy after that label has been forgotten. The aspherical Birman fibration in F.14 makes this inverse independent of the isotopy, and gives both the displayed injection and the full induced profinite topology.

Let $\gamma_i$ be the boundary of a small regular neighbourhood of the $i$th whisker and its two ends. It encloses exactly $A, o_i$. Write $T_i = \mathrm{push}_A(m_i)$. This is the one-turn Dehn twist about $\gamma_i$, with its sign fixed by this point-pushing convention. To see the representative directly, move $A$ once round the small circle at $o_i$ and back along its whisker; outside the regular neighbourhood the endpoint isotopy is the identity, and on its boundary annulus it is precisely that one-turn twist.

*The pure lift, including its annulus degree.* The double-cover monodromy is nontrivial around each of $A, D$, and trivial around every $o_i$. Therefore the inverse image of $\gamma_i$ is one connected annulus covering its downstairs annulus with degree two. The disk it bounds upstairs is the double cover of the disk enclosing $A, o_i$, branched at its one point $A$. Its Euler characteristic is $2 - 1 = 1$, and its boundary is connected, so it is a disk. It contains exactly the two marked points in row $i$ of (F.16.5), together with the unmarked fixed point $a_+$, and no other marked point.

Use angular coordinates in turns on the two annuli, so the covering map is $\theta = 2\vartheta$ modulo one. Choose the radial coordinate $t \in [0, 1]$ according to the preceding twist convention. The square of the downstairs twist and its pure lift are exactly

$$\begin{aligned} T_i^2 &: (\theta, t) \longmapsto (\theta + 2t, t), \\ b_i &: (\vartheta, t) \longmapsto (\vartheta + t, t). \end{aligned} \tag{F.16.7}$$

Both boundary circles are fixed pointwise by the second map. It extends by the identity off the annulus. The deck action there is $\vartheta \mapsto \vartheta + \frac{1}{2}$, which commutes with (F.16.7). Thus $b_i$ is a pure mapping class that centralizes the actual $r$ and descends to $T_i^2$. It is the unique pure lift: the only other lift differs by $r$, whose permutation is the four nontrivial pairs (F.16.5).

The class $b_i$ is the one-turn twist about the boundary of the displayed two-puncture disk. Forgetting either of those two points leaves a disk with one marked point. Its twist is isotopic to the identity relative to that point and all exterior labels: extend the untwisting isotopy across its disk by rotations about that remaining point. Consequently

$$b_i \in C_{P\Gamma_{0,8}}(\mathsf{r}) \ \cap \ \ker q_{p_i} \ \cap \ \ker q_{p_i'}, \qquad \{p_i, p_i'\} = q^{-1}(o_i) \cap \{0, \infty, v_0, \dots, v_5\}. \tag{F.16.8}$$

All kernels here refer to the actual named point-forgetting maps.

*The nontrivial word and all eight forgettings.* Take the commutator convention $[g, h] = ghg^{-1}h^{-1}$ and define

$$\begin{aligned} w &= [[[b_0, b_1], b_2], b_3], \\ \omega &= [[[m_0^2, m_1^2], m_2^2], m_3^2] \in L. \end{aligned} \tag{F.16.9}$$

Every $b_i$ centralizes $\mathsf{r}$, so $w$ does. Forgetting any one of the eight points kills the corresponding $b_i$ by (F.16.8); substituting the identity for any input of the displayed nested commutator gives the identity. This proves all eight individual vanishings. Any proper forgetting factors through one of them, which proves all the asserted vanishings after completion as well. Descent through the actual quotient gives

$$\mathrm{desc}(w) = \mathrm{push}_A(\omega). \tag{F.16.10}$$

In particular this is an identity in the specified Birman kernel, with its based inverse in (F.16.6).

Here is an integral detector for $\omega$. Let $U_5(\widehat{\mathbb{Z}})$ be the group of upper unitriangular five-by-five matrices, and let $E_{ij}$ denote its usual matrix units. The free basis map has a unique continuous extension

$$\widehat{L} \longrightarrow U_5(\widehat{\mathbb{Z}}), \qquad m_i \longmapsto I + E_{i+1,i+2} \quad (0 \le i \le 3). \tag{F.16.11}$$

The four squares go to $I + 2E_{i+1,i+2}$. Successive multiplication in the actual unitriangular group gives

$$\begin{aligned} [I + 2E_{12}, I + 2E_{23}] &= I + 4E_{13}, \\ [I + 4E_{13}, I + 2E_{34}] &= I + 8E_{14}, \\ [I + 8E_{14}, I + 2E_{45}] &= I + 16E_{15}. \end{aligned} \tag{F.16.12}$$

Thus $\omega$ is nontrivial already discretely. For every $z \in \widehat{\mathbb{Z}}$, its full profinite power has image

$$I + 16zE_{15}. \tag{F.16.13}$$

Multiplication by sixteen is injective on $\widehat{\mathbb{Z}}$; this detects every nonzero primary coordinate, including the two-primary coordinate.

*The precise completion maps for this detector.* Let $N_r$ be the discrete normalizer of the specified $\langle r \rangle$ in the eight-puncture mapping class group. Its decorated quotient sequence F.7.5 is

$$1 \longrightarrow \langle r \rangle \longrightarrow N_r \xrightarrow{\text{desc}} Q \longrightarrow 1, \tag{F.16.14}$$

where the two branch labels and four ordinary labels of (F.16.5) are retained. Every pure six-point mapping class preserves the double-cover monodromy: it fixes the peripheral homology classes, and that monodromy is the specified $C_2$-valued homology character. Thus $P\Gamma_{0,6}$ is a finite-index subgroup of this $Q$. The Birman kernel $L$ in (F.16.6) has its full induced topology in $P\Gamma_{0,6}$ by F.14, and hence in $Q$. In particular the actual completed maps give embeddings

$$\widehat{L} \hookrightarrow \widehat{P\Gamma}_{0,6} \hookrightarrow \widehat{Q}. \tag{F.16.15}$$

Set $C_r = C_{P\Gamma_{0,8}}(\mathsf{r})$. It is a finite-index subgroup of $N_r$. Its intersection with the finite deck kernel is trivial, since that nonidentity element permutes the marked points. Consequently descent identifies $C_r$ with its finite-index image $Q_0$ in $Q$. The inverse on $Q_0$ is the unique lift fixing all eight labels individually. The prime-order normalizer and full-topology comparison F.7.4–5 make the completion of this isomorphism the actual continuous descent on the closure of $C_r$ in $P_8$:

$$\overline{C_r} \cong \widehat{C_r} \xrightarrow[\text{desc}]{\sim} \widehat{Q_0} \hookrightarrow \widehat{Q}. \tag{F.16.16}$$

In particular (F.16.10) persists under these maps, and its right side lies in the embedded $\widehat{L}$ of (F.16.15). If $w^z = 1$ in $P_8$, its descent is $\omega^z = 1$ in $\widehat{Q}$. The embedding (F.16.15) then puts this equality in $\widehat{L}$, and (F.16.13) gives $16z = 0$, hence $z = 0$. The usual continuous power map $\widehat{\mathbb{Z}} \to \overline{\langle w \rangle}$ is therefore injective as well as onto. This proves the full procyclic identification in (F.16.1), with the inverse-source and topology comparisons needed to apply the matrix detector.

*The exact obstruction to the proposed membership test.* The eight vanishing maps include $q_{v_0}$ and $q_{v_2}$, so $Z_w \subset B$. Its elements centralize $\mathsf{r}$, proving $Z_w \subset C_B(\mathsf{r})$. F.14 states that $q_{v_0}|_H$ is injective. Since $q_{v_0}(Z_w) = 1$, we obtain $Z_w \cap H = 1$.

For every proper $I$ the identity lies in $q_I(H)$, and $q_I(Z_w) = 1$. Thus $Z_w \subset \mathcal{T}$, while $H \subset \mathcal{T}$ by its definition. Every nonidentity element of $Z_w$ witnesses the first strict inclusion. It also centralizes $\mathsf{r}$ and therefore witnesses the second. The same witnesses work for every subfamily of the displayed maps. For $h \in H$ and $z \in Z_w$, $q_I(hz) = q_I(h)$ for all $I$, whereas $hz \in H$ forces $z \in H \cap Z_w = 1$. Finally $Z_w \subset C_B(\mathsf{r})$ places right multiplication by $Z_w$ inside each actual centralizing correction coset (F.15.9), while all proper forgetting maps remain unchanged. These statements prove precisely (F.16.2)–(F.16.3); they supply no nonidentity element of the original source equalizer.

The executed certificate `code/part_a/common_reflection_forgetting_ambiguity_f16.py` records the four actual orbit pairs, checks the degree-two annulus formulas, all eight individual and all proper word-forgetting consequences, the integer matrix value (F.16.12), and its power formulas in twenty-one finite primary quotients. The geometric lifting and full completion maps are proved above. □

**F.17. [Proved] The actual branch drilling and its nonfactorization.** Retain $Y, s_0, \zeta$ and the labelled map $\phi$ of F.12. Write

$$v_u(s) = \left(\frac{u-s}{u+s}\right)^2 \quad (u \in \mu_5), \qquad a(s) = \frac{\zeta - s}{\zeta + s}, \qquad v_\zeta = a^2. \tag{F.17.1}$$

Thus $v_{\zeta^j} = v_{j+1}$ in the earlier notation. For $u \in \mu_5 \setminus \{\zeta\}$ put

$$A_u = u(2\zeta - u), \qquad B_u = \frac{\zeta u^2}{2u - \zeta}, \qquad \Delta = \{s : s^2 = A_u \text{ or } s^2 = B_u \text{ for one such } u\}. \tag{F.17.2}$$

The set $\Delta$ consists of sixteen distinct points of $Y$. Set

$$Y^\dagger = Y \setminus \Delta, \qquad Y_0 = \mathbf{P}^1 \setminus \{0, \infty, \zeta, -\zeta\}. \tag{F.17.3}$$

There is an actual ten-label configuration map

$$\phi^\dagger(s) = (0, \infty, 1, (v_u(s))_{u \in \mu_5}, a(s), -a(s)) \quad (s \in Y^\dagger). \tag{F.17.4}$$

The two added labels are ordered by these last two formulas. At the basepoint they are the two fixed points of the actual involution in F.12–15. This is a specified ordering; comparison with the other ordering is the literal transposition of these two new labels.

The source fillings and their completed homomorphisms are

$$\widehat{\pi_1(Y^\dagger, s_0)} = \widehat{F}_{27} \xrightarrow{q} \widehat{\pi_1(Y, s_0)} = \widehat{F}_{11} \xrightarrow{q_0} \widehat{\pi_1(Y_0, s_0)} = \widehat{F}_3. \tag{F.17.5}$$

The first fills precisely $\Delta$, and the second fills the eight points $\pm u$ with $u \neq \zeta$. Both maps are onto.

Each based meridian $m_d$ at an added point $d \in \Delta$ is killed by $q$, but $\widehat{\phi^\dagger}_*(m_d)$ generates a full closed copy of $\widehat{\mathbb{Z}}$ in the ten-point pure completion. It is the actual primitive two-point twist at the corresponding collision in (F.17.2), with the base path induced by the specified source whisker. Consequently the particular map $\widehat{\phi^\dagger}_*$ does not factor through $q$.

Forgetting the four labels $v_u$ with $u \neq \zeta$ gives the exact map

$$\chi : Y_0 \longrightarrow \mathcal{M}_{0,6}, \qquad \chi(s) = (0, \infty, 1, a(s)^2, a(s), -a(s)). \tag{F.17.6}$$

Its induced full profinite map is injective and its discrete image has the full induced topology. The retained involution is $V \mapsto a(s)^2/V$. Its actual branched quotient has the four labelled values $0, \infty, 1, t$, where $t = s^2/\zeta^2$. The two maps $Y_0 \to \mathcal{M}_{0,4}$ involved here are specified in the proof: ordinary forgetting has parameter $a^2$, whereas the involution quotient has parameter $t$. Each is a degree-two unramified covering. The first proves the full topology assertion directly.

These statements give an exact additional labelled correspondence for the source-membership problem. They make no injectivity assertion for the whole ten-point map (F.17.4), and do not determine the intersection in F.12.5.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), the exact residual centralizer and its source-membership problem.

*Proof. All collision points of the two new labels.* The rational labels $a, -a$ are the fixed points of $V \mapsto v_\zeta(s)/V$. They are distinct and avoid $0, \infty, 1, v_\zeta$ on $Y$: the excluded cases are $s = 0, \infty, \zeta, -\zeta$. For $u \neq \zeta$, the two exact differences have the factorizations

$$\begin{aligned} v_u - a &= \frac{2s(s^2 - A_u)}{(u+s)^2(\zeta+s)}, \\ v_u + a &= \frac{2(\zeta - 2u)(s^2 - B_u)}{(u+s)^2(\zeta+s)}. \end{aligned} \tag{F.17.7}$$

This proves the stated collision equations, without discarding a denominator or a source puncture.

Here is also a direct separation check. In the specified complex embedding $|u| = |\zeta| = 1$,

$$|A_u| = |2\zeta - u| > 1, \qquad B_u = \frac{1}{\overline{A_u}}, \qquad |B_u| < 1. \tag{F.17.8}$$

Equality in the first inequality would force $u = \zeta$. For two distinct fifth roots $u, w$, $A_u = A_w$ would imply $u + w = 2\zeta$; the equality case in the triangle inequality again forces $u = w = \zeta$. Thus the four $A_u$ are distinct. Their reciprocal conjugates are distinct, and the two sets cannot intersect by (F.17.8). All eight values are nonzero. Their square roots are therefore sixteen distinct points. None has absolute value one or equals $0, \infty$, so none is an old source puncture.

The denominators in (F.17.7) are nonzero at these sixteen points, since the excluded zeros have absolute value one. Every zero displayed there is simple: the $s^2 - A_u$ and $s^2 - B_u$ factors have nonzero derivatives at their roots, and the remaining numerator factors are nonzero. There is only one new collision at each such point. The original eight labels are distinct on $Y$ by F.12. This proves that (F.17.4) is defined precisely on $Y^\dagger$ among the points of $Y$.

The basepoint $s_0$ belongs to $Y^\dagger$. Indeed its actual involution permutes all eight original labels in four nontrivial pairs, by F.15.10. Neither of its two fixed points is one of those labels. The formula in (F.17.1) squares to $v_\zeta(s_0) = v_2(s_0)$, so it selects exactly the required ordered new pair.

*The actual filling maps and local twists.* Choose a fixed system of based geometric whiskers from $s_0$ to small positively oriented circles at the finite punctures of $Y^\dagger$. The circles are disjoint, and the whisker interiors avoid all punctures. The corresponding meridians are a free basis, with the infinity meridian given by their inverse ordered product. Filling the sixteen punctures $\Delta$ sends precisely those meridians to the identity and retains the meridians of $Y$ with their induced whiskers. Filling the eight additional old punctures gives $Y_0$. The same maps after completion are exactly (F.17.5): the finite-target universal property identifies each filling kernel with the closed normal closure of its named meridians. In particular both completed maps are onto, and their composite fills all twenty-four additional points of $Y^\dagger$ relative to $Y_0$.

Let $d^2 = A_u$. In a small disk about $d$, the difference $v_u - a$ has winding one by (F.17.7); every other label difference stays nonzero there. The local monodromy in the ordered configuration is therefore the one-turn pure twist around the two labels $v_u, a$. Concretely, the center and difference of these two coordinates give local configuration coordinates; the center varies in a contractible disk, and the nonzero difference runs once around zero. Extending this representative by the identity outside a small disk containing the two labels gives that twist. Its based conjugating path is the image under $\phi^\dagger$ of the already chosen whisker. For $d^2 = B_u$ the identical argument uses $v_u + a$ and the pair $v_u, -a$. Thus the asserted monodromy includes the actual pair of labels, exponent, and base transport.

For a full profinite detection, normalize the original three labels $0, \infty, 1$ in the ten-point moduli space. Write $Z_u, Z_+, Z_-$ for the coordinates of $v_u, a, -a$. The globally nonzero functions $Z_u - Z_+$ and $Z_u - Z_-$ define their integer winding homomorphisms on its pure mapping class group. Their pullbacks under $\phi^\dagger$ are precisely the two functions in (F.17.7). At the corresponding added meridian the relevant integer winding is one. Completing that character gives the literal equality

$$\widehat{\mathrm{wind}}_{u,\pm}\big(\widehat{\phi^\dagger}_*(m_d^z)\big) = z \quad (z \in \widehat{\mathbb{Z}}). \tag{F.17.9}$$

The continuous power map onto the closure of that twist is injective by (F.17.9), as well as surjective. This proves the full $\widehat{\mathbb{Z}}$ assertion at every primary coordinate. It does not require injectivity of $\widehat{\phi^\dagger}_*$ on the entire source.

Since $q(m_d) = 1$ but its ten-point image is nonidentity, there cannot be a homomorphism $D$ with $Dq = \widehat{\phi^\dagger}_*$. This proves exactly the stated nonfactorization.

*Both diagrams with their literal retained labels.* Let $\mathrm{forget}_{+,-}$ forget the two added labels, and let $\mathrm{forget}_{\mathrm{four}}$ forget precisely the four $v_u$ with $u \neq \zeta$. The based maps on full completions satisfy

$$\begin{aligned} \mathrm{forget}_{+,-}\ \widehat{\phi^\dagger}_* &= \widehat{\phi}_*\, q, \\ \mathrm{forget}_{\mathrm{four}}\ \widehat{\phi^\dagger}_* &= \widehat{\chi}_*\, q_0 q. \end{aligned} \tag{F.17.10}$$

The first target has the original eight labels. The second target has exactly $0, \infty, 1, v_\zeta, a, -a$. Both identities follow from the actual configuration coordinates, with the same basepoint and whiskers. Their common further forgetting retains $0, \infty, 1, v_\zeta$ and has parameter

$$u_0(s) = a(s)^2. \tag{F.17.11}$$

This common four-point target is the ordinary forgetting target, not the branched quotient target described next.

*The six-point double cover and its full topology.* The coordinate $a = (\zeta - s)/(\zeta + s)$ is a Möbius isomorphism from $Y_0$ onto $\mathbf{P}^1 \setminus \{0, \infty, 1, -1\}$. Squaring $a$ is the degree-two unramified covering of $\mathbf{P}^1 \setminus \{0, \infty, 1\}$. It is exactly the ordinary forgetting (F.17.11). The induced fundamental-group image is the index-two kernel of the character that is one on a meridian at zero and zero on a meridian at one, using the $u_0$ coordinate and its induced basepoint.

A finite-index subgroup of a discrete group has its full induced profinite topology: the normal core of a finite-index subgroup of it has finite index in the whole group. Consequently the completed map on this covering is an embedding $\widehat{F}_3 \hookrightarrow \widehat{F}_2$. Since it is the composite of $\widehat{\chi}_*$ with an actual target forgetting homomorphism, $\widehat{\chi}_*$ is injective. The same composite proves the full induced topology of the discrete image of $\chi_*$. This proof asserts full topology for this specified six-point map and uses no inference about the kernel of the ten-point map.

The involution $V \mapsto a^2/V$ fixes the two labels $a, -a$ individually and exchanges the other labels as $(0\ \infty)(1\ a^2)$. Its actual quotient is

$$Q_a(V) = \left(\frac{V - a}{V + a}\right)^2, \qquad \begin{array}{c|c} \text{labelled fibre} & \text{quotient value} \\ \hline a & 0 \\ -a & \infty \\ \{0, \infty\} & 1 \\ \{1, a^2\} & t = s^2/\zeta^2. \end{array} \tag{F.17.12}$$

Direct substitution proves all four entries; in particular $Q_a(1) = ((1 - a)/(1 + a))^2 = s^2/\zeta^2$. Thus the branched-quotient parameter map is the other degree-two unramified covering $Y_0 \to \mathbf{P}^1 \setminus \{0, \infty, 1\}$.

The sheet change $s \mapsto -s$ sends $a$ to $a^{-1}$. The Möbius map $V \mapsto V/a^2$ identifies the two six-point covers over the same $t$ and exchanges exactly the labels $1, a^2$, while preserving the two named branch labels and the labels $0, \infty$. The deck involution exchanges both ordinary pairs. Hence that sheet change cannot be corrected to a pure labelling by the deck involution. Conversely the choices of the two ordinary sheet orderings, modulo the simultaneous deck exchange, are exactly these two possibilities. This proves that $Y_0$ is the actual index-two pure-labelling cover of this four-point quotient family, with all of its labels and the branch order specified.

The executed certificate `code/part_a/common_reflection_drilling_f17.py` checks the eight exact factorizations over $\mathbf{Q}(\zeta)$, all separations of the eight squared collision values, the two quotient and sheet-label formulas, and the coefficient-one meridian detectors in finite primary reductions. It passes 445 exact checks. The topology and full completion comparisons are the explicit covering and winding arguments above. □

**F.18. [Proved] The unique full branch lift and its exact source coset.** Keep the actual eight-point configuration $\mathcal{P}_8$ and involution $\mathsf{r}$ of F.12–17. Put $a_+ = a(s_0)$ and $a_- = -a(s_0)$ with the literal ordering in F.17. Mark these two additional points and retain the same homeomorphism

as $\mathsf{r}^\dagger$. Write

$$\begin{aligned} P_{10}^\dagger &= \widehat{P\Gamma}(\mathbf{P}^1; \mathcal{P}_8, a_+, a_-), \\ P_8 &= \widehat{P\Gamma}(\mathbf{P}^1; \mathcal{P}_8), \\ f : P_{10}^\dagger &\longrightarrow P_8, \qquad \mathcal{J} = \ker f, \\ C^\dagger &= C_{P_{10}^\dagger}(\mathsf{r}^\dagger), \qquad C_8 = C_{P_8}(\mathsf{r}). \end{aligned} \tag{F.18.1}$$

Here $f$ forgets exactly the two named added points. The branch-permutation character $\epsilon : C_8 \twoheadrightarrow C_2$ is defined by the actual decorated quotient of $\mathsf{r}$. Then

$$f : C^\dagger \xrightarrow{\sim} C_8^+ := \ker \epsilon, \qquad C_{\mathcal{J}}(\mathsf{r}^\dagger) = 1 \tag{F.18.2}$$

are an isomorphism of profinite groups and its exact kernel assertion.

For the action $R^\dagger$ on $\mathcal{J}$ by $\mathsf{r}^\dagger$, the full continuous cocycles

$$Z^1(R^\dagger, \mathcal{J}) = \{\delta \in \mathcal{J} : \delta R^\dagger(\delta) = 1\} \tag{F.18.3}$$

have precisely two $\mathcal{J}$-conjugacy classes. Every order-two section has trivial centralizer in $\mathcal{J}$. In particular the map

$$b \longmapsto b^{-1} R^\dagger(b) \tag{F.18.4}$$

is a homeomorphism from $\mathcal{J}$ onto the coboundary class. More generally, for $k \in P_{10}^\dagger$ with $f(k) \in C_8$, the class of $k^{-1}R^\dagger(k)$ is the class labelled $\epsilon(f(k))$.

The original residual group $\mathcal{E}$ of F.12 has a unique full centralizing branch lift

$$\ell : \mathcal{E} \longrightarrow C^\dagger, \qquad f\ell = \widehat{\phi}|_{\mathcal{E}}. \tag{F.18.5}$$

It is continuous and injective. This includes the proof that its branch character is zero; no ordered drilling is assumed in advance.

Here is the exact remaining comparison to the rational drilling of F.17. Put

$$\begin{aligned} L^\dagger &= \pi_1\widehat{(Y^\dagger, s_0)}, \qquad L = \pi_1\widehat{(Y, s_0)}, \\ q : L^\dagger &\twoheadrightarrow L, \qquad \Phi^\dagger = \widehat{\phi^\dagger}_*, \\ K^+ &= \Phi^\dagger(\ker q) = \Phi^\dagger(L^\dagger) \cap \mathcal{J}. \end{aligned} \tag{F.18.6}$$

For $u \in \mathcal{E}$ and any $w \in q^{-1}(u)$, define the actual element

$$b(w) = \ell(u)^{-1}\Phi^\dagger(w) \in \mathcal{J}. \tag{F.18.7}$$

It is the unique solution of

$$b(w)^{-1} R^\dagger(b(w)) = \Phi^\dagger(w)^{-1} R^\dagger(\Phi^\dagger(w)). \tag{F.18.8}$$

The right $K^+$-orbit $b(w)K^+$ is independent of $w$ and defines a continuous map

$$\mathfrak{o} : \mathcal{E} \longrightarrow \mathcal{J}/K^+. \tag{F.18.9}$$

No group structure on this coset space is assumed. Its distinguished fibre is exactly

$$\begin{aligned} \mathfrak{o}^{-1}(K^+) &= \{u \in \mathcal{E} : \ell(u) \in \Phi^\dagger(L^\dagger)\} \\ &= q((\Phi^\dagger)^{-1}(C^\dagger)). \end{aligned} \tag{F.18.10}$$

These are equalities for the specified full maps. Whether this fibre is all of $\mathcal{E}$, and whether $\mathcal{E}$ is trivial, remain unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), full residual centralizer and exact source-membership comparison.

*Proof. The two quotient normalizers and their actual inverse.* On the eight-point sphere the involution is the Möbius map $r(V) = a_+^2/V$, with four nontrivial marked pairs and unmarked fixed points $a_+, a_-$. After marking them, the same map fixes each new label and retains the four old pairs. The decorated quotient before marking has four ordinary orbit labels and two branch points; after marking it has the same four orbit labels and the two branch points ordered individually.

Use the discrete covering-normalizer sequence F.7.5 on both sides of the forgetting map. Its kernel is the same deck group $\langle \mathsf{r} \rangle = C_2$. On either pure centralizer this kernel is trivial, because its nonidentity element permutes the eight old labels. Descent therefore identifies the discrete pure centralizer before drilling with a finite-index subgroup $Q_0$ of the decorated quotient mapping class group. Its image after ordered drilling is exactly the subgroup $Q_0^+$ preserving the two branch labels individually. Indeed, a representative in $Q_0^+$ has its unique lift fixing all old labels, and that lift fixes the two ordered ramification points as well. Conversely every pure lift after drilling has those two properties. The inverse comparison is this same unique pure lift.

The branch permutation on $Q_0$ is onto $C_2$. To see this with the old labels retained, interchange the two branch points along an arc avoiding the four ordinary quotient labels. Lift that quotient homeomorphism. If its endpoint exchanges the two names in some old pair, push that ordinary quotient point once around one branch point and lift the isotopy from the identity. This changes exactly that pair permutation, fixes the branch points individually, and leaves all other pair names fixed. Correcting each mismatch gives a pure upstairs class still exchanging the two unmarked fixed points. Thus $[Q_0 : Q_0^+] = 2$.

The prime-order normalizer and full-topology comparison F.7.4–5 applies to these two actual involutions. In particular both actual pure centralizers are their discrete completions. Their branch-permutation homomorphism is continuous in the induced topology: the full topology admits this finite $C_2$ quotient. Its kernel is open; the completion of $Q_0^+$ is exactly that kernel. Completing the discrete inverse maps above gives the first isomorphism in (F.18.2), for the actual forgetting map $f$. Intersecting its kernel with $\mathcal{J}$ gives the second assertion. No intersection of two unrelated closures is used.

*The actual two-point kernel and its section classes.* Write $S_8 = \mathbf{P}^1 \setminus \mathcal{P}_8$ and

$$J_{\mathrm{d}} = \pi_1(\mathrm{Conf}_2(S_8), (a_+, a_-)). \tag{F.18.11}$$

The action on this ordered configuration fibre is $(z_+, z_-) \mapsto (rz_+, rz_-)$; it does not interchange the two new labels. The two successive Birman sequences, with the completed comparison of F.14–15, identify $\widehat{J_{\mathrm{d}}}$ with the actual subgroup $\mathcal{J}$ in (F.18.1). For completeness, projection to the first point gives the split free-by-free extension

$$1 \longrightarrow F_8 \longrightarrow J_{\mathrm{d}} \longrightarrow F_7 \longrightarrow 1. \tag{F.18.12}$$

A section is obtained by putting the second point far out in a plane coordinate with one original puncture at infinity, and then adjusting the basepoint by a path in the fibre. Characteristic finite-index subgroups of $F_8$ and the finite action of the free quotient on their finite quotients give the full topology on that kernel. Completion is exact. Mapping this sequence into the two completed Birman sequences proves the asserted embedding and equality with the forgetting kernel, just as in F.15.11, with these actual ranks and labels.

Consider the discrete group

$$E_{\mathrm{d}} = J_{\mathrm{d}} \rtimes \langle \mathsf{r}^\dagger \rangle. \tag{F.18.13}$$

It is the actual subgroup of the ten-point mapping class group generated by the two-point kernel and the indicated involution; the nontrivial coset is detected by the old-label permutation. Each order-two section has exactly those four old pairs and fixes the two added labels. Nielsen realization in the punctured form used in F.15 realizes it as a sphere involution. Its two fixed points are precisely the two added marked points.

An ordered quotient homeomorphism matching those two branch points and the four named ordinary orbits lifts to a conjugator to $\mathsf{r}^\dagger$. Correct any old pair-name mismatches by the quotient point pushes just described. This proves that every such section is conjugate to $\mathsf{r}^\dagger$ by a pure ten-point class $g$. The condition that it lies in (F.18.13) is exactly $f(g) \in C_{8,\mathrm{d}}$, the discrete centralizer. The set of its $J_\mathrm{d}$-conjugacy classes is therefore the actual double-coset set

$$J_\mathrm{d}\backslash\{g \in P\Gamma^\dagger_{0,10} : f(g) \in C_{8,\mathrm{d}}\}/C^\dagger_\mathrm{d} \xrightarrow{\sim} C_{8,\mathrm{d}}/C^+_{8,\mathrm{d}} \cong C_2. \tag{F.18.14}$$

The map sends $g$ to the branch permutation of $f(g)$. It is onto because forgetting the two points is onto on pure mapping class groups. If two images have the same branch permutation, their quotient downstairs lifts through $C^\dagger_\mathrm{d}$ by the discrete isomorphism already proved; removing that lift leaves an element of $J_\mathrm{d}$. This proves injectivity. All their centralizers in $J_\mathrm{d}$ are trivial: for the standard section this is the discrete kernel of (F.18.2), and pure conjugation preserves the normal subgroup $J_\mathrm{d}$.

Here are the full completion hypotheses for transporting this calculation. The group in (F.18.12) is good, by the two-row Lyndon–Hochschild–Serre comparison for its free kernel and free quotient, whose finite-coefficient cohomology groups are finite. It has a finite two-dimensional classifying model: use finite graphs for the free kernel and quotient, and mapping cylinders of graph homotopy equivalences representing the finitely many free-group automorphisms. The finite extension (F.18.13) is residually finite as an actual subgroup of the residually finite mapping class group. Its completion is $\mathcal{J} \rtimes \langle \mathsf{r}^\dagger \rangle$, using the finite-index subgroup $J_\mathrm{d}$. The finite-extension cohomology comparison gives goodness and finite virtual cohomological type. The same argument for all discrete two-primary modules follows by taking the filtered union of their finite invariant submodules; the $\mathrm{FP}_\infty$ resolution commutes with these unions. These are precisely the hypotheses used for (F.15.14).

Apply [11, Theorem A] to this actual group and the prime two. It gives the two full conjugacy classes of order-two subgroups and the closure of their centralizers. An $E_\mathrm{d}$-conjugacy between sections is already a $J_\mathrm{d}$-conjugacy: an outside conjugator can be multiplied by the nonidentity element of the section it conjugates to make its quotient label zero, without changing that conjugation. The same argument works after completion. Thus (F.18.14) gives exactly two $\mathcal{J}$-classes. For each discrete section $T$, its centralizer in $E_\mathrm{d}$ is $T$ itself, since its intersection with $J_\mathrm{d}$ is trivial and the quotient is $C_2$. The completed centralizer equality therefore makes the full fixed groups in $\mathcal{J}$ trivial as well. The orbit map from $\mathcal{J}$ to either cocycle class is consequently injective and onto; compactness and the Hausdorff target make it a homeomorphism. This proves (F.18.3)–(F.18.4). For a full $k$ as in the claim with $\epsilon(f(k)) = 0$, let $c \in C^\dagger$ be the unique lift of $f(k)$ under (F.18.2). Then $b = c^{-1}k \in \mathcal{J}$ and $k^{-1}R^\dagger(k) = b^{-1}R^\dagger(b)$. If instead $\epsilon(f(k)) = 1$, a coboundary expression for this element would put $kb^{-1}$ in $C^\dagger$, contradicting (F.18.2) after forgetting. Since there are exactly two full classes, this proves the asserted class label for every full $k$ directly.

*The residual group fixes the two actual stabilizer labels.* We verify this before defining the lift. Let $\jmath : L \to \widehat{P}\Gamma_{0,12}$ be the injective original covering map of F.12. For $u \in \mathcal{E}$, both $\jmath(u)$ and $\beta\jmath(u)\beta^{-1}$ lie in the pure completed centralizer of $\tau$. Their descents agree by the defining equation of $\mathcal{E}$. That descent is injective on the pure centralizer: its only finite kernel is $\langle\tau\rangle$, whose nonidentity element has a nontrivial label permutation. Therefore

$$\beta\jmath(u)\beta^{-1} = \jmath(u). \tag{F.18.15}$$

The element also centralizes $\tau$. This establishes pointwise centralization of $V_4 = \langle\tau, \beta\rangle$; it is stronger than preserving the two lifts $\{\beta, \tau\beta\}$ as an unordered set.

The actual coordinate $W = (z - s_0)/(z + s_0)$ on the original sphere has the formulas

$$\tau(W) = -W, \qquad \beta(W) = \frac{k}{W}, \qquad \tau\beta(W) = -\frac{k}{W}, \qquad k = a(s_0) = \left(1 - \frac{2\theta^2}{5}\right)(1 + 2i). \tag{F.18.16}$$

These follow by conjugating the original Möbius matrices of F.11–12; the exact certificate below checks them in the same coefficient field. The quotient by $\tau$ is $V = W^2$. The fixed points of $\beta$ consequently give $V = k = a_+$, and those of $\tau\beta$ give $V = -k = a_-$. Thus the literal F.17 branch ordering is the ordering of these two distinct stabilizer types.

A discrete class centralizing $V_4$ preserves each stabilizer subgroup individually in the actual quotient cover. Its descent under $\tau$ therefore fixes these two labels individually. The full statement follows from the direct prime-power comparison for the actual two-group $V_4$: its completed centralizer is the closure of its discrete centralizer, with the full topology of F.7.4–5. The branch character is finite and continuous on the target centralizer, so the same zero character holds on that entire closure. Applying this to (F.18.15) proves $\widehat{\phi}(\mathcal{E}) \subset C_8^+$. The inverse in (F.18.2) now defines the continuous injective map $\ell$ in (F.18.5), and proves its uniqueness.

*The exact remaining source-image test.* The square F.17.10 is the literal identity

$$f\Phi^\dagger = \widehat{\phi}\, q. \tag{F.18.17}$$

Since $\widehat{\phi}$ is injective, an element $\Phi^\dagger(w)$ lies in $\mathcal{J}$ precisely when $q(w) = 1$. This proves both inclusions in the equality defining $K^+$ in (F.18.6). It is a closed subgroup, being the compact image of the closed kernel of $q$.

For $q(w) = u \in \mathcal{E}$, (F.18.17) and (F.18.5) put $b(w)$ in $\mathcal{J}$. Because $\ell(u)$ centralizes $\mathsf{r}^\dagger$, direct multiplication gives (F.18.8). Equation (F.18.4) and its trivial fixed kernel prove uniqueness of this solution. Replacing $w$ by $wv$ with $v \in \ker q$ gives the exact formula

$$b(wv) = b(w)\Phi^\dagger(v). \tag{F.18.18}$$

Thus $b(w)K^+$ depends only on $u$. The map on $q^{-1}(\mathcal{E})$ is continuous; the compact surjection to $\mathcal{E}$ is a quotient map. It therefore descends continuously to (F.18.9).

Finally, $\ell(u) = \Phi^\dagger(w)b(w)^{-1}$. The definition and equality in (F.18.6) show that this element belongs to $\Phi^\dagger(L^\dagger)$ precisely when $b(w) \in K^+$. In that case choose $v \in \ker q$ with $\Phi^\dagger(v) = b(w)^{-1}$; then $\Phi^\dagger(wv) = \ell(u)$ centralizes and $q(wv) = u$. Conversely, a source element with centralizing ten-point image fills to $\widehat{\phi}(u)$ with $u \in \mathcal{E}$, and uniqueness in (F.18.2) identifies its image with $\ell(u)$. These are both directions of (F.18.10). In particular the canonical correcting element is unique; its membership in the specified closed subgroup $K^+$ is the remaining literal source test.

The executed certificate `code/part_a/common_reflection_branch_lift_f18.py` checks the original Möbius conjugations, the exact coefficient $k$, the four old marked pairs, and the $V_4$ stabilizer labels. It passes 34 exact checks. The full topological and completion assertions are the arguments above, not finite-quotient lifting assumptions. □

**F.19. [Proved] The two-track kernel and the exact remaining source subgroup.** Use the literal base configuration and ordered branch labels of F.17–18. Partition its ten labels as

$$A = \{0, \infty, 1, v_\zeta\}, \qquad U = \{v_u : u \in \mu_5 \setminus \{\zeta\}\}, \qquad Z = \{a_+, a_-\}. \tag{F.19.1}$$

Let $P_4, P_8, P_6, P_{10}^\dagger$ be the full pure completions on $A, A \cup U, A \cup Z, A \cup U \cup Z$, respectively. The actual forgetting maps form the square

$$\begin{array}{ccc} P_{10}^\dagger & \xrightarrow{\ g\ } & P_6 \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle h} \\ P_8 & \xrightarrow{\ f_0\ } & P_4. \end{array} \qquad \begin{array}{c} \mathcal{Q} = P_8 \times_{P_4} P_6, \\ \pi = (f, g) : P_{10}^\dagger \twoheadrightarrow \mathcal{Q}, \\ D^+ = \ker \pi = \ker f \cap \ker g. \end{array} \tag{F.19.2}$$

In particular the displayed map to this specified profinite fibre product is onto.

For each of the eight pairs $(v_u, a_\pm)$, choose either one of its two collision points in F.17 and retain its actual source whisker. Write $\nu_{u,\pm}$ for the image of that source meridian under $\Phi^\dagger$. Then

$$D^+ = \overline{\langle\langle \nu_{u,\pm} : u \neq \zeta \rangle\rangle}_{P_{10}^\dagger}, \qquad \nu_{u,\pm} \in K^+. \tag{F.19.3}$$

The normal closure is taken in the indicated full ambient group. This does not replace $K^+$ by that normal closure.

All four label sets in (F.19.1) are invariant under the specified involution, and its two new labels are fixed individually. Denote the induced actions by $R$ on each group where no ambiguity arises. With $\mathcal{J} = \ker f$ and $\mathcal{J}_4 = \ker(h : P_6 \to P_4)$, one has

$$\begin{aligned} 1 \longrightarrow D^+ \longrightarrow \mathcal{J} \xrightarrow{g} \mathcal{J}_4 \longrightarrow 1, \\ C_{\mathcal{J}_4}(R) = 1, \qquad H^1(C_2, D^+) = \{1\}. \end{aligned} \tag{F.19.4}$$

Here the last expression denotes the continuous nonabelian cohomology set for this actual action. More explicitly, the map

$$D^+ \longrightarrow \{\delta \in D^+ : \delta R(\delta) = 1\}, \qquad b \longmapsto b^{-1} R(b) \tag{F.19.5}$$

is a homeomorphism. The centralizer map

$$\pi : C^\dagger \xrightarrow{\sim} \mathcal{Q}^R \tag{F.19.6}$$

is an isomorphism of profinite groups.

For the original residual group $\mathcal{E}$, its unique branch lift $\ell$ and the source maps of F.17–18,

$$g\ell(u) = \widehat{\chi}_*(q_0 u), \qquad b(w) = \ell(u)^{-1} \Phi^\dagger(w) \in D^+ \quad (qw = u,\ u \in \mathcal{E}). \tag{F.19.7}$$

Thus $K^+ \subset D^+$, and the exact obstruction map of F.18 has the more precise target

$$\mathfrak{o}_D : \mathcal{E} \longrightarrow D^+/K^+, \qquad u \longmapsto b(w) K^+. \tag{F.19.8}$$

It is continuous and independent of the source lift $w$. Its distinguished fibre is still exactly the fibre (F.18.10).

Here are explicit subgroup maps accompanying that comparison. Set $H^\dagger = \Phi^\dagger(L^\dagger)$ and

$$\begin{aligned} \gamma : L \longrightarrow \mathcal{Q}, \qquad \gamma(u) = (\widehat{\phi}_*(u), \widehat{\chi}_*(q_0 u)), \\ \mathcal{T} = \pi^{-1}(\gamma(L)). \end{aligned} \tag{F.19.9}$$

The map $\gamma$ is an embedding. There is a commutative diagram of exact full profinite sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & K^+ & \longrightarrow & H^\dagger & \xrightarrow{q_H} & L & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & D^+ & \longrightarrow & \mathcal{T} & \xrightarrow{\pi_L} & L & \longrightarrow & 1. \end{array} \tag{F.19.10}$$

Both rows split by the same specified map $j = \Phi^\dagger i$, where $i : L \to L^\dagger$ retains the eleven old free meridians in the source filling. The corresponding inverse coordinates are

$$\begin{aligned} D^+ \rtimes_j L \longrightarrow \mathcal{T}, \qquad (d, u) \longmapsto d\, j(u), \\ \mathcal{T} \longrightarrow D^+ \rtimes_j L, \qquad t \longmapsto (tj(\pi_L t)^{-1}, \pi_L t). \end{aligned} \tag{F.19.11}$$

Restricting these maps gives $K^+ \rtimes_j L \cong H^\dagger$. Moreover

$$\mathcal{T} \cap C^\dagger = \ell(\mathcal{E}). \tag{F.19.12}$$

These are assertions about the actual images and inverse maps; they do not assert $H^\dagger = \mathcal{T}$ or $K^+ = D^+$. The source-membership equation in (F.19.8), and triviality of $\mathcal{E}$, remain unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), the exact residual centralizer and its source-membership comparison.

*Proof. The literal six-point centralizer.* The retained involution on $A \cup Z$ exchanges $(0, \infty)$ and $(1, v_\zeta)$, and fixes $a_+, a_-$. Let $C_6^\dagger$ denote its full pure centralizer, and let $C_4$ denote the full pure centralizer after forgetting $Z$. There is a branch-permutation character on $C_4$; write $C_4^+$ for its kernel. The discrete decorated quotient has two ordinary orbit labels and two branch labels. The deck involution is nonpure before and after adding the branch labels, so descent is injective on both pure centralizers. The actual inverse lift argument of F.18.2 therefore gives

$$h : C_6^\dagger \xrightarrow{\sim} C_4^+. \tag{F.19.13}$$

For this application the prime-order completion and full topology comparison F.7.4–5 is applied to these very four- and six-point actions. The branch permutation is onto: exchange the two quotient branch points, then correct any of the two ordinary pair-name changes by pushing that ordinary quotient point once around a branch point. These corrections preserve the branch permutation. Thus this is the same exact index-two comparison as in F.18, with two old pairs retained.

In fact

$$\widehat{\chi}_*(\pi_1\widehat{(Y_0, s_0)}) = C_6^\dagger. \tag{F.19.14}$$

To check equality with this particular map, descend by the actual quotient $Q_a(V) = ((V - a)/(V + a))^2$. F.17.12 identifies its four ordered quotient values as $0, \infty, 1, t$, $t = s^2/\zeta^2$. There are exactly two choices of pure ordinary sheet names modulo simultaneous deck exchange. The sheet change $s \mapsto -s$, followed by $V \mapsto V/a^2$, exchanges precisely the names in $\{1, a^2\}$; the deck exchanges both ordinary pairs. Hence the degree-two map $Y_0 \to \mathcal{M}_{0,4}$ is exactly the covering belonging to the discrete pure-lift subgroup of this quotient group. Its fundamental group is the entire image of the discrete six-point pure centralizer under descent. The family of involutions on the six labelled points shows that $\chi_*$ lands in that centralizer. Injective descent now proves the discrete equality. The full topology of $\chi_*$, proved using the ordinary $a^2$-cover in F.17, and the prime-order centralizer comparison complete that equality to (F.19.14). Both the quotient parameter $t$ and ordinary forgetting parameter $a^2$ are retained in this argument; no identification of those two maps is used.

*The actual fibre-product map and its completion.* Normalize the three common labels $0, \infty, 1$. The space

$$X = \mathcal{M}(A \cup U) \times_{\mathcal{M}(A)} \mathcal{M}(A \cup Z) \tag{F.19.15}$$

has coordinates $v_\zeta$, four ordered $U$-coordinates and two ordered $Z$-coordinates. The $U$-coordinates are distinct and avoid $A$; the $Z$-coordinates are distinct and avoid $A$. At this stage a $U$-coordinate may equal a $Z$-coordinate. Both projections to the four-point base are the actual configuration forgetting fibrations. The base $\mathcal{M}_{0,4}$ has $\pi_2 = 0$. The homotopy sequences therefore identify $\pi_1(X)$ with the discrete fibre product of the two pure mapping class groups over the four-point group: the kernel on both sides is the product of the two configuration-fibre groups, and the maps on those kernels and on the base group are identities.

Here is also the full-topology comparison for this particular fibre product. The map $\mathcal{M}(A \cup Z) \to \mathcal{M}(A)$ has a continuous section. For the variable coordinate $u = v_\zeta$, place the two new coordinates at $2 + |u|$ and $3 + |u|$. They are distinct and avoid $0, 1, u, \infty$. A path in the connected fibre identifies the resulting basepoint with the specified $(a_+, a_-)$. This gives a based group section. The discrete kernel is

$$J_{4,\mathrm{d}} = \pi_1(\mathrm{Conf}_2(S_{0,4}), (a_+, a_-)), \qquad 1 \longrightarrow F_4 \longrightarrow J_{4,\mathrm{d}} \longrightarrow F_3 \longrightarrow 1. \tag{F.19.16}$$

The latter sequence splits by the same far-point construction in a punctured-plane coordinate. In particular $J_{4,\mathrm{d}}$ is finitely generated.

Consequently the discrete group in (F.19.15) is $J_{4,\mathrm{d}} \rtimes P\Gamma(A \cup U)$, with the action pulled back from the displayed actual section over $A$. For every finite-index subgroup of the kernel, choose a

characteristic finite-index subgroup inside it. The action on this finite quotient has finite image; quotienting the acting group by its action kernel gives a finite semidirect quotient of the whole group. This proves the full induced topology on the kernel. It also extends the action to the completions and proves exactness of the completed split sequence: finite homomorphisms on both factors are separated by these compatible finite quotients. The completed Birman comparison identifies its kernel with the actual $\mathcal{J}_4$ in $P_6$. The same section thus gives

$$\widehat{\pi_1(X)} \cong \mathcal{J}_4 \rtimes P_8 \cong P_8 \times_{P_4} P_6 = \mathcal{Q} \tag{F.19.17}$$

with the actual projection maps. This proves completion for the specified fibre product.

For clarity, surjectivity in (F.19.2) can also be seen directly on those actual full maps. Lift a given $P_8$ component through $f$. Its $P_6$ component differs from the desired one by an element of $\mathcal{J}_4$. The filling map

$$\pi_1 \operatorname{Conf}_2(S_{0,8}) \longrightarrow \pi_1 \operatorname{Conf}_2(S_{0,4})$$

is onto. Apply the two Fadell–Neuwirth sequences: both the one-point quotient $F_7 \to F_3$ and the fibre map $F_8 \to F_4$ are the onto puncture fillings, so a lift on the quotient can be corrected in the fibre. Its completion is onto as well, and these completions are exactly $\mathcal{J} \to \mathcal{J}_4$ by the completed Birman comparisons. Correcting the chosen lift inside $\mathcal{J}$ gives the required pair. This also proves the exact sequence in (F.19.4).

*The eight actual cross-diagonal generators.* The ten-point moduli space is the open subset of $X$ obtained by removing exactly

$$\mathcal{D}_{u,\pm} = \{Z_u = Z_\pm\} \qquad (u \neq \zeta). \tag{F.19.18}$$

These are eight smooth complex hypersurfaces. Each has connected smooth part after removing its intersections with the others: identify the coincident pair, then move the remaining finitely many labelled coordinates along paths avoiding their finite forbidden sets. Different hypersurfaces meet only in complex codimension at least two; their possible intersections represent independent pair collisions.

Every loop in $X$ can be moved off these hypersurfaces. A disk witnessing a nullhomotopy in $X$ can be made transverse to their smooth parts and disjoint from their intersections. Its finitely many intersection points, joined by based arcs in the disk, express its original boundary as a product of conjugates of the corresponding signed normal meridians. Conversely every such meridian dies in $X$. This proves that the discrete kernel is the normal closure of one based meridian for each hypersurface.

Choose its based path to be the actual source whisker from F.17 approaching either of the two simple collision points for that pair. The other labels remain distinct there. The coefficient-one local calculation (F.17.7–9) identifies its normal meridian exactly with $\nu_{u,\pm}$, with that path and a positive one-turn exponent. The finite-target universal property of a quotient now identifies the kernel on full completions with the closed normal closure of these eight elements. Using (F.19.17) gives exactly (F.19.3). Every chosen source meridian is killed by $q$, so every $\nu_{u,\pm}$ belongs to the specified subgroup $K^+$.

There is a useful exact check on what these ordinary winding coordinates detect. The eight global functions $Z_u - Z_\pm$ give a continuous homomorphism

$$\operatorname{wind} : P_{10}^{\dagger} \longrightarrow \widehat{\mathbb{Z}}^{\,8}, \qquad \operatorname{wind}(\nu_{u,\pm}) = e_{u,\pm}. \tag{F.19.19}$$

At either root for a given pair, its own coordinate has winding one and the other seven have winding zero. The selected eight elements already show that the images of both $K^+$ and $D^+$ under this particular homomorphism are all of $\widehat{\mathbb{Z}}^{\,8}$: take the product of their specified full powers to obtain any eight coordinates. No assertion about the whole abelianization of $D^+$ follows from this calculation.

*The two section labels under this filling.* The action on each two-point configuration fibre fixes the two base points individually: $(z_+, z_-) \mapsto (rz_+, rz_-)$. Its fixed subgroup in $\mathcal{J}_4$ is the kernel of (F.19.13), and is therefore trivial. Its continuous section classes have the same two labels as in F.18. Here are the inputs for this smaller actual group. The discrete extension $J_{4,\mathrm{d}} \rtimes C_2$ is the subgroup of the six-point mapping class group generated by the point-pushing kernel and the indicated involution. Each order-two section exchanges the two old pairs and fixes the two new labels. Nielsen realization and the ordered quotient homeomorphism identify it with the standard involution by a pure six-point conjugator. The double-coset comparison of those conjugators is $C_{4,\mathrm{d}}/C^+_{4,\mathrm{d}} = C_2$. The label is the branch permutation of the forgotten conjugator. All their discrete fixed subgroups in $J_{4,\mathrm{d}}$ are trivial, by the discrete version of (F.19.13).

The split free-by-free sequence (F.19.16) gives the actual full topology and the two-row cohomology comparison, just as in F.18.12. Finite graphs for $F_4$ and $F_3$, with mapping cylinders of representatives of their monodromy homotopy equivalences, give a finite classifying complex of dimension two. The extension is residually finite as this actual mapping class subgroup. The finite extension cohomology comparison and finite virtual cohomological-type hypotheses are therefore the same verified ones as in F.18.13–14. The prime-order theorem [11, Theorem A] transports the two conjugacy classes and their centralizers to the full completion. Conjugacy by the extension can be changed to conjugacy by its configuration kernel by removing the quotient-generator power of the conjugator. Thus the full $H^1(C_2, \mathcal{J}_4)$ has exactly these two branch labels.

The filling $\mathcal{J} \to \mathcal{J}_4$ carries the two full classes of F.18 to these two classes with the same labels. At the discrete level this follows by forgetting the other two ordinary quotient orbits: it preserves both named branch points and their permutation. The finite branch characters are continuous by the full-topology comparison, so the identical label statement holds in the completions. The induced map on these two cohomology sets is therefore a bijection.

Let $\delta \in D^+$ satisfy $\delta R(\delta) = 1$. Its image in $\mathcal{J}_4$ is the identity cocycle. The just-proved bijection forces its $\mathcal{J}$-class to be the identity class. By F.18 there is a unique $b \in \mathcal{J}$ with $\delta = b^{-1}R(b)$. Its image $g(b)$ is fixed in $\mathcal{J}_4$, so is the identity. Hence $b \in D^+$. Uniqueness also follows from $C_{D^+}(R) \subset C_{\mathcal{J}}(R) = 1$. The coboundary map is continuous; compactness of $D^+$ and the Hausdorff cocycle space make this continuous bijection a homeomorphism. This proves (F.19.4–5).

For $z \in \mathcal{Q}^R$, choose $k \in \pi^{-1}(z)$. Then $k^{-1}R(k)$ is a cocycle in $D^+$. Its unique corrector $b \in D^+$ makes $kb^{-1}$ fixed, and does not change its image under $\pi$. The kernel of $\pi$ on the fixed subgroup is $C_{D^+}(R) = 1$. This proves (F.19.6), including its actual inverse and continuity.

*The specified source image inside the pullback.* For $u \in \mathcal{E}$, the element $g\ell(u)$ lies in $C_6^\dagger$. Its ordinary four-point image is $f_0\widehat{\phi}_*(u)$. The exact source square (F.17.10–11) identifies that with $h\widehat{\chi}_*(q_0u)$. The latter element also belongs to $C_6^\dagger$ by (F.19.14). Injectivity in (F.19.13) proves the first equality of (F.19.7). For $qw = u$, the same square gives $g\Phi^\dagger(w) = \widehat{\chi}_*(q_0u)$. Thus $g(b(w)) = 1$; its $f$-image is already the identity by F.18. This proves $b(w) \in D^+$. For $qv = 1$, both forgetting images of $\Phi^\dagger(v)$ are the identity, so $K^+ \subset D^+$. The right-coset comparison and its independence are the actual equality $b(wv) = b(w)\Phi^\dagger(v)$. The quotient map $q^{-1}(\mathcal{E}) \to \mathcal{E}$ is a compact surjection, so that equality and continuity give the asserted continuous descended map (F.19.8). The inclusion $D^+/K^+ \to \mathcal{J}/K^+$ is a closed embedding of compact coset spaces. Hence its distinguished fibre is precisely the earlier fibre (F.18.10).

The same source square puts $\gamma(L)$ in $\mathcal{Q}$, and injectivity of $\widehat{\phi}_*$ makes $\gamma$ an embedding. Define $\pi_L$ by the inverse of that embedding after applying $\pi$ to $\mathcal{T}$. It is onto by (F.19.2). For $z = \Phi^\dagger(w) \in H^\dagger$, the equality $\pi(z) = \gamma(qw)$ shows that $q_H(z) = qw$ is independent of $w$. This is a continuous surjection: it is induced by a continuous map on the compact source. Its kernel is exactly $K^+$. These facts prove every map and kernel in (F.19.10).

Use the free based meridians in F.17.5. The filling $q$ kills exactly the sixteen new generators and sends the eleven old finite meridians to a free basis of $L$. Sending that latter basis back

to those eleven specified meridians defines a continuous map $i : L \to L^\dagger$ with $qi = 1$. Thus $j = \Phi^\dagger i$ splits both rows of (F.19.10). Conjugation by $j(L)$ preserves $D^+$, and also preserves $K^+$, the kernel in the upper row. The multiplication and inverse coordinates in (F.19.11) are consequently homomorphisms of the stated semidirect products and are inverse continuous maps. This constructs the common splitting directly from the named free meridians.

Finally an element of $\mathcal{T} \cap C^\dagger$ has projection $\gamma(u)$ for some $u \in L$. Its first component being fixed is exactly the condition $u \in \mathcal{E}$, by F.12. The second component is already fixed by (F.19.14). Equation (F.19.6) gives a unique fixed lift of that pair; (F.19.7) identifies it with $\ell(u)$. Conversely $\ell(u)$ belongs to $\mathcal{T}$ for every $u \in \mathcal{E}$. This proves (F.19.12).

The new rational comparison checked by `code/part_a/common_reflection_cross_kernel_f19.py` is particularly explicit. Writing $w_j = v_{\zeta^j}$, $p = a_+$, $m = a_-$, the normalized action sends

$$(w_0, w_1, w_2, w_3, w_4, p, m) \longmapsto \left(\frac{w_1}{w_2}, w_1, \frac{w_1}{w_0}, \frac{w_1}{w_4}, \frac{w_1}{w_3}, \frac{w_1}{p}, \frac{w_1}{m}\right). \tag{F.19.20}$$

It fixes the retained curve $(w_1, p, m) = (a^2, a, -a)$. For $\sigma = (0\ 2)(3\ 4)$ and $z = p$ or $m$, its cross difference is exactly

$$R(w_j) - R(z) = -\frac{w_1}{w_{\sigma(j)} z}(w_{\sigma(j)} - z). \tag{F.19.21}$$

The coefficient is nonzero in the relevant configuration chart. Thus the eight cross divisors and their labels are preserved in the stated four pairs. The executable checks these identities, the involution identity and the sixteen-column winding matrix from the simple source collisions. It passes 44 exact checks. The fibre-product, completion and section arguments above establish the group assertions; the calculation does not identify $K^+$ with $D^+$. □

**F.20. [Proved] An exact cross-divisor action and its finite wreath images.** Retain the labelled source, the two-track group, and the subgroups $K^+ \le D^+$ of F.17–F.19. Select the cross divisor between the old point $v_1$ and the positive new point $a$. Its full relative path coordinate is $\mathcal{F} = \widehat{F}(x, y, z)$, with the three specified meridians corresponding, in order, to $0, v_\zeta, 1$. The actual source $L = \widehat{\pi_1(Y)} = \widehat{F}_{11}$ has the continuous action

$$A_g(v) = q_g^{-1} \alpha_g(v) p_g \qquad (g \in L,\ v \in \mathcal{F}), \tag{F.20.1}$$

where $\alpha_g, p_g, q_g$ are the literal based braid data specified below. The diagonal-coset map proving this formula, its inverse, and every based source transport are retained in the proof.

Let $\mathscr{Q}$ be the following explicitly enumerated finite quotients of this same $\mathcal{F}$: the 82 source-invariant $A_5$ quotients in the enumeration below, all 3,321 products of two distinct such quotients, and the six mod-two homology-cover quotients

$$Q_\rho = \mathcal{F}/\overline{[N_\rho, N_\rho]\, N_\rho^2}, \qquad N_\rho = \ker(\rho : \mathcal{F} \to A_5), \qquad |Q_\rho| = 60 \cdot 2^{121}, \tag{F.20.2}$$

whose four peripheral images under $\rho$ are all nonidentity. Every action (F.20.1) on every member of $\mathscr{Q}$ is transitive. In each of the six last cases the image of the source filling kernel acts by right translations through a subgroup of index four; the remaining actual source generators connect all four cosets.

For each $Q \in \mathscr{Q}$ and each prime $\ell$ there is an explicitly specified continuous homomorphism

$$\chi_{Q,\ell} : D^+ \longrightarrow \mathbb{F}_\ell[Q] \quad \text{with} \quad \chi_{Q,\ell}(K^+) = \chi_{Q,\ell}(D^+) = \mathbb{F}_\ell[Q]. \tag{F.20.3}$$

It is obtained from the actual one-point forgetting map and the finite wreath assignment in (F.20.15) below. Thus these maps give the same full finite images for the two specified subgroups. This conclusion does not identify $K^+$ with $D^+$ or solve the remaining source-membership equation of F.19.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), the actual residual A5-centralizer and source-membership comparison.

*Proof. The selected two-track space and its actual diagonal.* Normalize the common labels $0, \infty, 1$ as in F.19 and put

$$\begin{gathered} \zeta = e^{2\pi i/5}, \qquad Y = \mathbb{P}^1 \setminus \{0, \infty, \pm\zeta^j : 0 \le j < 5\}, \\ a(s) = \frac{\zeta - s}{\zeta + s}, \quad B(s) = a(s)^2, \quad P(s) = \left(\frac{1-s}{1+s}\right)^2, \quad Q(s) = a(s). \end{gathered} \tag{F.20.4}$$

The letter $Q(s)$ in this display is a point coordinate; a finite quotient $Q \in \mathscr{Q}$ will always be named as such. Forget the other three old labels and the negative new label in the F.19 two-track space $X$. The target is exactly

$$X_1 = \{(B, U, V) : B \neq 0, 1, \quad U, V \notin \{0, 1, B\}\} = \mathcal{M}_{0,5} \times_{\mathcal{M}_{0,4}} \mathcal{M}_{0,5}. \tag{F.20.5}$$

It allows $U = V$; that equality is its selected diagonal. The curve map is literally $s \mapsto (B(s), P(s), Q(s))$. The selected divisor upstairs in $X$ is its inverse image.

The comparison with the original F.17 basepoint is also specified. Write $s_0 = i(\theta^2 - 2)\zeta$, as in F.12, and take the polygonal path with successive vertices

$$s_0, \quad 3s_0, \quad 3s_0 + 6i, \quad 6 + 6i, \quad 2 + 2i. \tag{F.20.4a}$$

It lies in $Y^\dagger$. Indeed $|s_0|^2 = \theta^2 - 1 > 1$. For the four $A_u$ of F.17.2, the imaginary parts of $A_u\overline{s_0^2}$ are respectively $\theta^3/2, -\theta^3/2, (5\theta - 3\theta^3)/2, (-5\theta + 3\theta^3)/2$ in the order $u = 1, \zeta^2, \zeta^3, \zeta^4$; all are nonzero. Thus its first radial segment misses the outer collision roots. The inner collision roots have absolute value less than one. All later segments lie outside the disk of radius two: $3 \operatorname{Re} s_0 < -2$, the middle horizontal-direction segment has imaginary part at least six, and the last segment has both coordinates at least two. Every collision root has absolute value less than two, by $|A_u| < 3$ and $|B_u| < 1$. The exact producer checks these inequalities.

Call this path $\eta$. The based maps are $g \mapsto \eta g \eta^{-1}$ on the source and conjugation by the actual labelled configuration path $\phi^\dagger(\eta)$ on the ten-point target, with the identical forgettings of that path on every smaller target. The composites commute because these are literal paths of the displayed rational labels. They carry the $K^+, D^+$ at $s_* = 2 + 2i$ to the original F.19 subgroups at $s_0$. The homomorphisms in (F.20.3) on those original subgroups include the inverse of this specified basepoint change.

The forgetting projection is a configuration fibration with connected fibers. Over its diagonal the additional three old points and the other new point still vary in their respective connected configuration fibers. Consequently the image of the divisor fundamental group is the full inverse image of the diagonal fundamental group: lift a diagonal loop through this fibration, and correct its endpoint within the connected fiber. The kernel of the projection is represented by loops in that same fiber and already lies in the divisor image.

The same comparison holds on full completions. For clarity, write $P_j = \widehat{\pi_1(\mathcal{M}_{0,j})}$ with the specified label maps. The diagonal inverse image is the fiber product $P_8 \times_{P_5} P_6$; its kernel over $P_8$ is the actual one-point-pushing $\widehat{F}_4$. The discrete sequence has an actual continuous configuration section, obtained by placing the forgotten point on the positive real axis beyond $2 + |B| + |U|$, and then choosing the based fiber path. The characteristic finite kernel quotients used in F.19 give its full induced topology and the explicit completed semidirect-product inverse maps. They identify its completion with this stated fiber product. The identical argument for (F.20.5) gives $\mathcal{Q}_1 = P_5 \times_{P_4} P_5$. Thus the full coset space for the selected divisor in F.19 is carried homeomorphically onto $\mathcal{Q}_1 / \operatorname{diag}(P_5)$. No closure/intersection exchange is used.

*The based source and the two reference transports.* Use source basepoint $s_* = 2+2i$, reference point $R = 8+3i$, and projection $z \mapsto \mathrm{Re}((1+371i/1000)z)$. For each of the eleven centers

$$0,\ 1, \zeta, \zeta^2, \zeta^3, \zeta^4,\ -1, -\zeta, -\zeta^2, -\zeta^3, -\zeta^4 \tag{F.20.6}$$

take the straight whisker from $s_*$ to the northeast corner of the axis-parallel square of half-side $31/1000$ about that center, traverse the square counterclockwise, and return along that same whisker. Call these loops $g_0, \dots, g_{10}$ in the displayed order. The certificate verifies that the squares are pairwise disjoint, each whisker misses all other squares, and the whiskers have distinct directions from their common basepoint. The basepoint lies northeast of every square, so its own whisker first meets that square at its corner. These squares and the embedded star give an actual geometric free basis of $\pi_1(Y)$.

At $s_*$, transport the fourth finite point along the straight segment from $R$ to $P(s_*)$ for the old track and to $Q(s_*)$ for the new track, keeping $0, 1, B(s_*)$ fixed. Both paths avoid those three punctures; their exact cross-product checks are retained. Call these paths $T_o, T_n$. A source loop gives the two based four-strand braids $T_o\, P(g)\, T_o^{-1}$ and $T_n\, Q(g)\, T_n^{-1}$. The notation here retains the actual intermediate three-strand path $B(g)$. At the common reference configuration the projected order is exactly $(0, B(s_*), 1, R)$. Use the same disk basepoint and straighten the four points within their projection-order chamber. A generator $\sigma_j$ is the literal clockwise exchange of adjacent points: the left strand passes above the right strand. For example, after translation and positive scaling, the two paths

$$2t - 1 + 4it(1-t), \qquad -2t + 1 - 4it(1-t) \quad (0 \le t \le 1)$$

have exactly this exchange and the producer's positive crossing. Their squared distance from the midpoint is $((2t-1)^2 - 1/2)^2 + 3/4$, so the pair never collides. These geometric generators give the Artin presentation of the actual braid group. The symbols $x, y, z, d$ below are its ordered free coordinates; the exact identification of $x, y, z$ with the actual point-pushing fibre, including its inverse based loops, is given by (F.20.8)–(F.20.9).

*The Artin coordinate and the diagonal-coset formula.* Use the faithful Artin substitutions

$$\sigma_j : X_j \longmapsto X_j X_{j+1} X_j^{-1}, \quad X_{j+1} \longmapsto X_j, \qquad \sigma_j(X_k) = X_k \ (k \neq j, j+1), \tag{F.20.7}$$

in the word-composition convention of [2]. Path products are read from left to right in time, and the substitution for a displayed word $b_1 \cdots b_r$ is $\phi_{b_1} \circ \cdots \circ \phi_{b_r}$, as in the producer. The displayed substitutions and their inverses satisfy the Artin relations, so the named geometric generators define this homomorphism into the free-group automorphisms. The following fibre-coordinate comparison determines its actual point-pushing action.

For a pure four-strand braid $\beta$, write $\phi_\beta(d) = W_\beta d W_\beta^{-1}$ and let $w_\beta$ be the word obtained by filling $d$ in $W_\beta$. The ambiguity in $W_\beta$ is a power of $d$ and therefore disappears under this filling. Write $\alpha_\beta$ for the filled automorphism of $F(x, y, z)$. Direct substitution gives

$$w_{\beta\gamma} = \alpha_\beta(w_\gamma) w_\beta, \qquad u_\beta = w_\beta^{-1}, \qquad u_{\beta\gamma} = u_\beta \alpha_\beta(u_\gamma). \tag{F.20.8}$$

To identify the actual point-pushing kernel, let $\beta_j = \sigma_3 \cdots \sigma_{j+1} \sigma_j^2 \sigma_{j+1}^{-1} \cdots \sigma_3^{-1}$. The first descending word carries the last strand along the lower-half-plane whisker, the square makes a clockwise lasso about puncture $j$, and the inverse word returns that whisker. Deleting the last strand contracts the old motion within its projection-order chambers. Thus these are the three actual lower-whiskered clockwise lassos, a free basis of the punctured-plane fibre. Direct substitution gives their $u$-images

$$z^{-1}y^{-1}x^{-1}yz, \qquad z^{-1}y^{-1}z, \qquad z^{-1}. \tag{F.20.9}$$

This triangular substitution is its own inverse, by free reduction. The inverse map sends $x, y, z$ to the same three words with $\beta_1, \beta_2, \beta_3$ substituted for $x, y, z$. These are specified conjugates

of the inverse clockwise lassos, so they are actual based puncture meridians. Both composites are the identity on the named free bases. This gives the actual fibre isomorphism and then its full profinite inverse. Adding the last strand gives the standard section on the three-strand braid group, with $u = 1$. The cocycle (F.20.8) therefore makes conjugation by that section exactly the filled Artin substitution, and makes conjugation by a general pure braid $b$ the map $v \mapsto u_b \alpha_b(v) u_b^{-1}$. Its restriction gives the section of $P_5 \to P_4$ used in this coordinate. Filling a puncture kills its lower-whiskered lasso; the same triangular formula commutes with filling the corresponding variable and retaining the order of the others. These are the actual based filling maps in the displayed coordinates.

Here the normalized moduli groups have not been replaced by braid groups without a map. The coordinate isomorphism

$$(z_0, z_1, z_B, z_R) \longmapsto \left(z_0, z_1 - z_0, \ \frac{z_B - z_0}{z_1 - z_0}, \frac{z_R - z_0}{z_1 - z_0}\right) \tag{F.20.10}$$

identifies the ordered configuration space with $\mathbb{C} \times \mathbb{C}^\times \times \mathcal{M}_{0,5}$; its inverse multiplies the last two coordinates by $z_1 - z_0$ and adds $z_0$. The analogous three-point formula has $\mathcal{M}_{0,4}$ as last factor. The original paths keep $z_0 = 0, z_1 = 1$. The braid section commutes with forgetting to these two labelled points, so it restricts to the kernels of the displayed affine-coordinate projections. The section and its retraction extend to full completions and have the same composites.

After the based transports, both factors of $\mathcal{Q}_1$ are this same $P_5$. The mutually inverse coset maps are

$$\begin{aligned} \mathcal{Q}_1/\operatorname{diag}(P_5) &\longrightarrow \mathcal{F}, & (b_o, b_n)\operatorname{diag}(P_5) &\longmapsto b_n b_o^{-1}, \\ \mathcal{F} &\longrightarrow \mathcal{Q}_1/\operatorname{diag}(P_5), & v &\longmapsto (1, v)\operatorname{diag}(P_5). \end{aligned} \tag{F.20.11}$$

The first value belongs to the kernel because both factors have the same base image. Both maps are continuous and their composites are the identity. Left multiplication by the two source braids acts by $v \mapsto b_n v b_o^{-1}$. Substituting $b_o = (p_g^{-1}, \alpha_g)$ and $b_n = (q_g^{-1}, \alpha_g)$ from (F.20.8) proves (F.20.1), including its order of multiplication.

*The complete generator data and exact crossing certificate.* Set $P = xyz$, $k = Pxyx$, and $t = Pxyx^{-1}$. Let $\tau_1, \tau_2$ be the automorphisms (F.20.7) of $F(x, y, z)$ associated with $\sigma_1^2, \sigma_2^2$. The complete source data are

| $i$ | $\alpha_{g_i}$ | $p_{g_i}$ | $q_{g_i}$ |
|---|---|---|---|
| 0 | $\tau_1\tau_2^{-1}\tau_1^{-1}$ | $(xy)x(xy)^{-1}P^{-1}$ | $z^{-1}xy^{-1}x^{-1}$ |
| 1 | 1 | $x^{-2}$ | 1 |
| 2 | $\tau_1^{-2}$ | 1 | $(xy)^{-1}$ |
| 3 | 1 | 1 | 1 |
| 4 | 1 | $ky^{-1}k^{-1}$ | 1 |
| 5 | 1 | 1 | 1 |
| 6 | 1 | $P^2$ | 1 |
| 7 | $(\tau_1\tau_2)^2$ | $tz^{-1}x^{-1}$ | $t$ |
| 8 | 1 | 1 | 1 |
| 9 | 1 | $xy^{-1}x^{-1}$ | 1 |
| 10 | 1 | 1 | 1. |

(F.20.12)

These are values in the specified basis, not independently chosen conjugacy classes.

The executed producer `code/part_a/cross_divisor_monodromy_f20.py` works in $\mathbb{Q}(\theta)$, where $\theta^4 - 5\theta^2 + 5 = 0$ and $19/10 < \theta < 2$, using $\zeta = (\theta^2 - 3 + i\theta)/2$. Every sign is separated by rational interval arithmetic at that particular real root. For each rational configuration segment and each pair of strands it multiplies the projected difference numerator by the conjugate of its denominator. Its real polynomial has exactly the projected crossings; its imaginary

polynomial gives their heights, since the omitted denominator is positive. All possible poles and collisions of either separate track are among (F.20.6): in particular $P(s) = B(s)$ has numerator $4s(1-\zeta)(\zeta - s^2)$, whose remaining roots are $\pm\zeta^3$. The source-basis clearance therefore also checks that these paths stay in the stated configuration spaces.

Exact Bernstein subdivision supplies the root count. On a rational interval, multiplication by the positive binomial coefficients changes its Bernstein sequence to the coefficients of the usual fractional-linear transform for Descartes' rule. Zero sign variation excludes roots and one variation gives exactly one root with multiplicity one. Intervals with larger variation are subdivided. A subdivision point that is an exact root is avoided by choosing another rational fraction; a nonzero polynomial of degree $d$ cannot vanish at $d+1$ such distinct choices. The output retains the complete projected and height polynomials, the terminal root-isolating intervals and their Bernstein signs, and the counts of discarded root-free leaves; the executed code reproduces the subdivisions. The certificate further separates a nonzero derivative and nonzero height on every crossing interval. Intervals for different pairs are disjoint. Their exact order, initial projected order, adjacency at each crossing, and final projected order give the literal braid word.

There are 112 certified forward segments, 672 projected-pair checks, 172 crossings, and 72,380 exact number-field sign calls. The return whisker is the literal inverse path, so its braid is the inverse of its certified forward braid. The two reference transports and all eleven old and new based braids are included. Deleting the fourth strand gives respectively the three nontrivial base braid words $\sigma_1^2\sigma_2^{-2}\sigma_1^{-2}$, $\sigma_1^{-4}$, and $(\sigma_1^2\sigma_2^2)^2$ at indices $0, 2, 7$. The second producer verifies this strand deletion, all entries of (F.20.12), and the inverse substitution (F.20.9) by exact free reduction. Neither producer reads a numerical discovery file.

*The actual full one-point quotient and its finite wreath maps.* Forget the negative new point in F.19's two-point configuration kernel. The two Fadell–Neuwirth rows and their filling maps are

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{F}_8 & \longrightarrow & J & \longrightarrow & \widehat{F}_7 & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \widehat{F}_4 & \longrightarrow & J_4 & \longrightarrow & \mathcal{F} & \longrightarrow & 1. \end{array} \tag{F.20.13}$$

Here $J$ and $J_4$ are the actual eight- and four-puncture two-point groups of F.19, and $D^+ = \ker(J \to J_4)$. Every vertical filling is onto. The full topologies and exact rows are those proved in F.19 by the actual configuration sections and finitely generated free kernels. Put $R_0 = \ker(\widehat{F}_7 \to \mathcal{F})$. The induced map

$$D^+ \longrightarrow R_0 \quad \text{is onto.} \tag{F.20.14}$$

Indeed lift an element of $R_0$ to $J$. Its image in $J_4$ lies in the displayed $\widehat{F}_4$; an element of $\widehat{F}_8$ has its inverse as image. Multiplication by that element gives the required lift in $D^+$ without changing its $\widehat{F}_7$ image.

Choose the geometric free basis of the seven-puncture one-point group consisting of $x, y, z$ and the four old puncture meridians $u_j$. Its filling kills the $u_j$ and retains $x, y, z$. The whisker of the selected $u_1$ is chosen to project to the old reference transport specified above after changing the one-point basepoint by $T_n$; at the original one-point basepoint its projected path is $T_n^{-1}T_o$. The straight $T_n$ also avoids all four other old points, as checked by their nonzero exact cross products. This is an explicit allowable basis adjustment: a difference word $w$ in the retained $x, y, z$ replaces $u_1$ by $wu_1w^{-1}$, fixes the other generators, and has inverse $u_1 \mapsto w^{-1}u_1w$. The difference word is the class of the two stated based paths. Thus the full filling and its splitting are unchanged, with their exact maps retained.

For $\rho : \mathcal{F} \to Q$ in the stated collection and a prime $\ell$, the finite free-group assignment is

$$\begin{aligned} \widehat{F}_7 &\longrightarrow \mathbb{F}_\ell[Q] \rtimes Q, \\ x, y, z &\longmapsto (0, \rho(x)), (0, \rho(y)), (0, \rho(z)), \\ u_1 &\longmapsto (e_1, 1), \qquad u_j \longmapsto (0, 1) \quad (j \neq 1). \end{aligned} \tag{F.20.15}$$

Here $Q$ acts by its actual left regular permutation of the basis $e_a$, $a \in Q$. On $R_0$ it is a homomorphism to the additive module and is onto: conjugating $u_1$ by a retained word with image $a$ gives $e_a$. Composing with (F.20.14) defines $\chi_{Q,\ell}$ and proves its surjectivity on $D^+$. This supplies continuity through an actual finite quotient; it does not complete an infinitely generated discrete meridian module without a topology argument.

The source filling $Y^\dagger \to Y$ kills the sixteen specified positive collision meridians. Its image $K^+$ is their closed normal closure inside the drilled source image, as in F.17–F.19. Forgetting the negative new point kills its eight collision meridians. The assignment (F.20.15) kills the six positive collisions with the other old labels. The remaining two positive meridians have images $e_{a_+}, e_{a_-}$ for the two actual transported relative path classes in $Q$. Their exponent is one, by the simple holomorphic collision and annular calculation of F.17.

Conjugation by a source lift transports these basis elements by exactly (F.20.1). One can see this either by (F.20.11) or directly: a based old meridian is carried to its conjugate by the old track, while the module basepoint is carried by the new track. Their two transports give $q_g^{-1}\alpha_g(v)p_g$ in that order. Different lifts of the same element of $L$ differ by an element of $R_0$, whose conjugation acts trivially on this additive image. Consequently, writing $\mathcal{O}_\pm$ for the two actual source orbits of $a_\pm$, one has the literal finite-image equality

$$\chi_{Q,\ell}(K^+) = \operatorname{span}_{\mathbb{F}_\ell}\{e_a : a \in \mathcal{O}_+ \cup \mathcal{O}_-\}. \tag{F.20.16}$$

The labels $a_\pm$ have not been moved to the identity by an unproved normalization. The transitivity proved next makes their particular values unnecessary for (F.20.3).

*The finite simple quotients and their paired actions.* The executed producer `code/part_a/cross_divisor_finite_f20.py` reads only the preceding exact geometric output. It constructs $A_5$ as the sixty even permutations of five letters, with its exact multiplication and inverse tables. Conjugation by $S_5$ gives all its 120 automorphisms: every automorphism permutes the five Sylow four-groups, and this action is faithful because the resulting faithful $A_5$-action determines each inner automorphism. The conjugacy-class sizes are checked as $1, 12, 12, 15, 20$; no proper nontrivial union including the identity has order dividing 60. This also verifies the simplicity input below.

The program enumerates all $200,160$ generating triples and their $1,668$ automorphism orbits. These are exactly the epimorphism kernels. It evaluates each of the eleven specified $\alpha_{g_i}$ on each triple and finds precisely 82 invariant kernels. For each one it records the unique descending automorphism $\sigma_i$, the two word values $\rho(p_{g_i}), \rho(q_{g_i})$, and the sixty-point permutation

$$v \longmapsto \rho(q_{g_i})^{-1}\sigma_i(v)\rho(p_{g_i}). \tag{F.20.17}$$

Every one of these 82 actions has a single orbit.

For two distinct kernels, the map to $A_5^2$ is onto. Indeed a proper subdirect subgroup of two copies of a nonabelian simple group is the graph of an automorphism: the two projection kernels are normal, and simplicity leaves only the full product or that graph. The graph would identify the two kernels, contrary to their enumeration. The program applies each source generator simultaneously to both coordinates; all $3,321$ such actions have one orbit of size $3,600$. This is a test of the actual joint action, not two separate marginal counts.

*The six noncentral homology-cover refinements.* Exactly six of the invariant triples have $\rho(x), \rho(y), \rho(z), \rho(xyz)$ all nonidentity. For such a triple put $G = A_5$, and consider the connected based Cayley covering graph of the three-petalled rose. Its chain groups and boundary over $\mathbb{F}_2$ are

$$C_1 = \mathbb{F}_2[G]^3, \quad C_0 = \mathbb{F}_2[G], \qquad \partial e_{i,g} = e_{g\rho(x_i)} - e_g. \tag{F.20.18}$$

For a word $w$, let $v(w)$ be its actual oriented edge chain and $a(w) = \rho(w)$. Concatenation gives $v(ww') = v(w) + a(w)v(w')$. Every chain with boundary $e_g - e_1$ is the chain of a word ending at

$g$: subtract one path to $g$ and realize the remaining graph homology class by based loops. Thus (F.20.2) is exactly the finite group

$$\{(v,g) \in \mathbb{F}_2[G]^3 \rtimes G : \partial v = e_g - e_1\}, \quad (v,g)(w,h) = (v + gw, gh). \tag{F.20.19}$$

Its kernel has dimension $180 - 59 = 121$. This description does not assert that this particular extension splits.

Each actual $\alpha_{g_i}$ preserves $N_\rho$ and its mod-two homology kernel. The lifted action is computed by the edge chains of its three actual generator images. If $J_i$ is that semilinear substitution matrix, $\sigma_i$ its permutation of $G$, and $(v_p, a_p), (v_q, a_q)$ the two word values, its affine action is literally

$$(v,a) \longmapsto \left(a_q^{-1}(J_i v - v_q + \sigma_i(a)v_p), a_q^{-1}\sigma_i(a)a_p\right). \tag{F.20.20}$$

The code checks these formulas against direct word evaluation in 660 exact comparisons.

The filling $Y \to \mathbb{P}^1 \setminus \{0, \infty, \pm\zeta\}$ kills exactly $g_1, g_3, g_4, g_5, g_6, g_8, g_9, g_{10}$. Their base automorphisms and new words in (F.20.12) are the identity, so they act by right translations through their old words. Conjugating such a translation by a general source action gives the right translation

$$h \longmapsto p_{g_i}^{-1}\alpha_{g_i}(h)p_{g_i}. \tag{F.20.21}$$

The finite subgroup $H_\rho$ generated by these translations and their indicated conjugates is computed exactly as follows. Choose representatives of its projection into $G$. Schreier products give vectors in $\ker\partial$; their span, closed under the projection's regular action, is precisely its kernel. Add the images of the representatives and this kernel under (F.20.21), and repeat until stable. Each step is finite linear algebra and finite group multiplication. Stability under the given injective maps on a finite subgroup also gives stability under their inverses. This computes the full subgroup generated by the named source filling kernel.

In all six cases its projection is all of $A_5$ and its vector kernel has dimension 119. Hence $[Q_\rho : H_\rho] = 4$. The source action descends to its right cosets, because $A_g(vh) = A_g(v)\,(p_g^{-1}\alpha_g(h)p_g)$. The program reduces those cosets by the specified representatives and row-reduced kernel basis, and finds one orbit of size four in every case. Right translations by $H_\rho$ are themselves source actions, so this proves transitivity on the entire group $Q_\rho$.

These computations prove transitivity for every finite quotient in the statement. Equation (F.20.16) now proves (F.20.3) for every prime $\ell$ and for the actual source collision labels, without requiring a lift of an arbitrary finite candidate. The exact maps and finite tests enlarge the verified winding comparison of F.19. The remaining full coset equation in $D^+/K^+$, and the original residual $A_5$-centralizer, retain their stated scope. □

**F.21. [Proved] Full cross-divisor transitivity and normal generation inside the joint kernel.** Retain the actual groups, source maps, eight selected positive cross-collision meridians, and base whiskers of F.17–F.19. Write

$$G = P_{10}^\dagger, \qquad \varpi : G \twoheadrightarrow \mathcal{Q}, \qquad D^+ = \ker\varpi, \qquad H^\dagger = \Phi^\dagger(L^\dagger), \qquad K^+ = H^\dagger \cap D^+. \tag{F.21.1}$$

For each $u \in \mu_5 \setminus \{\zeta\}$ and $\epsilon \in \{1, -1\}$, let $\mathcal{P}_{u,\epsilon}$ be the full closed image in $\mathcal{Q}$ of its actual cross divisor $v_u = \epsilon a$, based using the source whisker selected for $\nu_{u,\epsilon}$ in F.19. Then the source subgroup $\gamma(L)$ has a transitive left action on the entire compact coset space

$$\mathcal{Q}/\mathcal{P}_{u,\epsilon}. \tag{F.21.2}$$

This holds for each of the eight named divisors, with the original label maps and full profinite completions. The corresponding discrete coset action is also transitive.

The selected meridian has the following actual centralizer-image containment:

$$\varpi\big(C_G(\nu_{u,\epsilon})\big) \supseteq \mathcal{P}_{u,\epsilon}. \tag{F.21.3}$$

Write $D^+_{\mathrm{nil}}$ for the inverse limit of all continuous finite nilpotent quotients of $D^+$. Consequently

$$\begin{gathered} \overline{\langle\langle K^+ \rangle\rangle}_{D^+} = D^+, \\ K^+ \twoheadrightarrow D^+_{\mathrm{ab}} := D^+/\overline{[D^+, D^+]}, \qquad K^+ \twoheadrightarrow D^+_{\mathrm{nil}}. \end{gathered} \tag{F.21.4}$$

In particular every continuous finite nilpotent quotient of $D^+$ has the same image on $K^+$ and on $D^+$. The quotient maps and the normal closure are the displayed actual maps and subgroups. The remaining equality $K^+ = D^+$ and the source-membership equation of F.19 remain unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), the actual residual A5-centralizer and source-membership comparison.

*Proof. Actual source basepoints on each filled cross divisor.* Put

$$\begin{gathered} A(s) = \frac{\zeta - s}{\zeta + s}, \qquad C_u(s) = \frac{u - s}{u + s}, \qquad B = A^2, \qquad U = C_u^2, \qquad V = \epsilon A, \\ k_u = \frac{u - \zeta}{u + \zeta}, \qquad \mathscr{A} = \mathbb{P}^1 \setminus \{0, \infty, 1, -1\}. \end{gathered} \tag{F.21.5}$$

For the selected meridian $\nu_{u,\epsilon}$, let $d$ be its collision point in $Y$. Thus $d^2 = A_u$ for $\epsilon = 1$, and $d^2 = B_u$ for $\epsilon = -1$, in the notation of F.17.2. The sixteen collision points are distinct and lie in $Y$ by F.17. At $d$ exactly this cross collision occurs. The filled map $\gamma : Y \to X$ of F.19.15 is defined at $d$; its selected two-track projection is

$$s \longmapsto \big(A(s)^2, C_u(s)^2, \epsilon A(s)\big). \tag{F.21.6}$$

Its value at $d$ is on the diagonal $U = V$.

Keep the actual source whisker $\omega$ of that meridian, extended to its endpoint $d$. It is a path in $Y$ from the original $s_0$ to $d$, with its open terminal part in $Y^\dagger$. The based source isomorphism is $g \mapsto \omega g \omega^{-1}$ from $\pi_1(Y, d)$ to $\pi_1(Y, s_0)$. On $X$ it is conjugation by the literal path $\gamma(\omega)$. All forgetting projections commute with these maps, because they forget coordinates of this same configuration path. We prove transitivity at the basepoint $d$ and then apply these specified isomorphisms. The cross-divisor subgroup so obtained is exactly the one based by the chosen whisker. No change of the selected meridian or its conjugacy coordinate is required.

*The fourfold square-root cover and its pencil.* The common two-coordinate moduli space is

$$\mathcal{M}_{0,5} = \{(B, U) : B \notin \{0, 1\},\ U \notin \{0, 1, B\}\}.$$

Both coordinates are finite. Its actual connected fourfold covering and an intermediate double covering are

$$\begin{gathered} W = \{(A, C) \in \mathscr{A}^2 : C \neq A,\ C \neq -A\}, \\ W_A = \{(A, U) : A \in \mathscr{A},\ U \notin \{0, 1, A^2\}\}, \\ W \xrightarrow{\rho} W_A \xrightarrow{\beta} \mathcal{M}_{0,5}, \\ \rho(A, C) = (A, C^2), \qquad \beta(A, U) = (A^2, U). \end{gathered} \tag{F.21.7}$$

These maps are coverings because the squared coordinates avoid 0 and $\infty$; their four, respectively two, local square-root choices are distinct. The complements are connected: their projections to $A \in \mathscr{A}$ have connected punctured-plane fibers, and local trivializations arise by moving the finitely many excluded points in disjoint small disks.

There is an everywhere-defined map

$$\kappa : W \longrightarrow \mathscr{B} := \mathbb{P}^1 \setminus \{0, 1, -1\}, \qquad \kappa(A, C) = \frac{C - A}{1 - AC}. \tag{F.21.8}$$

The numerator and denominator cannot vanish together on $W$: that would give $A = C$ and $A^2 = 1$. The values $0, 1, -1$ would force, respectively, $C = A$, $(C - 1)(A + 1) = 0$, or $(C + 1)(1 - A) = 0$. All are excluded in (F.21.7). Let

$$E = \{AC = 1\} \subset W, \qquad W^\circ = W \setminus E, \qquad \mathscr{B}^\circ = \mathscr{B} \setminus \{\infty\}. \tag{F.21.9}$$

For finite $k \in \mathscr{B}^\circ$, the inverse coordinate formula is

$$C = \frac{A + k}{1 + kA}, \qquad A \in \mathbb{P}^1 \setminus \{0, \infty, 1, -1, -k, -1/k, r_+(k), r_-(k)\}, \quad kr^2 + 2r + k = 0. \tag{F.21.10}$$

The eight displayed punctures are distinct for $k \notin \{0, 1, -1, \infty\}$. Indeed the two roots are distinct by $1 - k^2 \neq 0$; neither is 0 or $\infty$; the substitutions $r = 1, -1$ give $2(k + 1), 2(k - 1)$; and the substitutions $r = -k, -1/k$ give $k(k^2 - 1), (k^2 - 1)/k$. The remaining six punctures are pairwise distinct under the same exclusions.

On a small simply connected neighborhood of any such $k$, choose the two distinct analytic roots $r_+, r_-$. All eight puncture paths remain in pairwise disjoint coordinate disks. Within each disk, translate its puncture using a bump function that is one near the path and zero near the disk boundary. These isotopies, with the inverse isotopies, depend continuously on $k$. Their disjoint supports produce an actual local trivialization of the complements in (F.21.10). Consequently $W^\circ \to \mathscr{B}^\circ$ is a locally trivial bundle with the indicated eight-punctured sphere fiber. Its base and fiber are connected, and its fundamental-group sequence is exact.

The old source has constant pencil value:

$$\kappa(A(s), C_u(s)) = k_u. \tag{F.21.11}$$

Here $k_u \notin \{0, 1, -1, \infty\}$ since $u \neq \zeta$, $u \neq -\zeta$, and both $u, \zeta$ are nonzero. The coordinate $s = \zeta(1 - A)/(1 + A)$ identifies its entire fiber with

$$Y_u = \mathbb{P}^1 \setminus \{0, \infty, \zeta, -\zeta, u, -u, t_u, -t_u\}, \qquad t_u^2 = \zeta u. \tag{F.21.12}$$

For $u = \zeta^j$, take $t_u = \zeta^h$ with $h = 3(1 + j) \pmod 5$. The four exponent pairs are

$$(j, h) \in \{(0, 3), (2, 4), (3, 2), (4, 0)\}.$$

The inverse coordinate in (F.21.10) becomes exactly $C_u(s)$. Its exclusions are $s = 0, \infty, \pm\zeta$ from $A \in \mathscr{A}$, $s = \pm u$ from $C_u \in \mathscr{A}$, and $s = \pm t_u$ from $C_u = -A$; $C_u = A$ adds only $0, \infty$. Thus $Y \to Y_u$ fills exactly four of the old punctures, and induces a surjection on fundamental groups. Every loop on $Y_u$ can be moved away from those four points by a small path homotopy. The actual image of $\pi_1(Y, d)$ in $\pi_1(W^\circ)$ is therefore the image of this whole bundle fiber.

*Filling the connected multiplicity-one divisor.* The divisor $E$ is the actual connected smooth curve

$$A \longmapsto (A, A^{-1}), \qquad A \in \mathbb{P}^1 \setminus \{0, \infty, 1, -1, i, -i\}. \tag{F.21.13}$$

The extra exclusions $\pm i$ express $C \neq -A$. At $E$ the local coordinate at the target infinity is

$$\frac{1}{\kappa} = \frac{1 - AC}{C - A}, \qquad \left.\frac{\partial(1/\kappa)}{\partial C}\right|_{C=1/A} = \frac{-A}{A^{-1} - A} \neq 0. \tag{F.21.14}$$

Thus a positive transverse meridian of $E$ maps to a single positive meridian of infinity in the base.

Set $J^\circ = \pi_1(W^\circ, (A(d), C_u(d)))$ and $J = \pi_1(W, (A(d), C_u(d)))$. The inclusion is onto on fundamental groups, and its kernel is the normal closure of the meridians of $E$. To see both statements directly, perturb a loop in the smooth manifold $W$ away from the real-codimension-two submanifold $E$. Perturb a null-homotopy disk relative to its boundary to meet $E$ transversely in finitely many interior points. Delete small disks around those points; the resulting punctured disk expresses the original loop as a product of conjugates of the transverse meridians, with their intersection signs. All positive meridians form one conjugacy class, because $E$ is connected and its complex normal line has its fixed orientation.

Let $N^\circ$ be the fiber image in $J^\circ$ and let $N$ be its image in $J$. The bundle sequence and (F.21.14) give the following quotient map with its specified inverse identification:

$$\begin{aligned} J/N &\cong (J^\circ/N^\circ)/\langle\langle \text{meridian of } \infty \rangle\rangle \\ &\cong \pi_1(\mathscr{B}^\circ, k_u)/\langle\langle \text{meridian of } \infty \rangle\rangle \cong \pi_1(\mathscr{B}, k_u). \end{aligned} \tag{F.21.15}$$

The maps are induced by $\kappa$ and the two actual fillings. Their kernels are the normal subgroups just specified, so the inverse maps are the unique factorizations of these same quotient homomorphisms. In particular

$$N = \operatorname{im}\bigl(\pi_1(Y, d) \to J\bigr) = \ker\bigl(\kappa_* : J \twoheadrightarrow \pi_1(\mathscr{B}, k_u)\bigr). \tag{F.21.16}$$

This is an equality in the discrete fundamental group of the named covering. No assertion about completing this kernel sequence is needed below.

*The new section and the onto pencil map.* Base $W_A$ at $(A(d), C_u(d)^2) = (A(d), \epsilon A(d))$ and put $T = \pi_1(W_A)$. The cover $W \to W_A$ embeds $J$ as a normal subgroup of index two. The map $(A, U) \mapsto A$ has the actual based section

$$\sigma_\epsilon : \mathscr{A} \longrightarrow W_A, \qquad A \longmapsto (A, \epsilon A). \tag{F.21.17}$$

Its image $M = \sigma_{\epsilon,*}\pi_1(\mathscr{A}, A(d))$ maps isomorphically to $\pi_1(\mathscr{A}, A(d))$ under the projection. The kernel of that projection is the actual one-point-pushing group

$$F = \pi_1\bigl(\mathbb{P}^1 \setminus \{0, \infty, 1, A(d)^2\}, \epsilon A(d)\bigr) \cong F_3. \tag{F.21.18}$$

Its injection and kernel description follow from the configuration bundle, because $\pi_2(\mathscr{A}) = 0$.

The pullback of $W \to W_A$ by this very section is the connected double covering

$$\mathscr{R} = \mathbb{P}^1 \setminus \{0, \infty, 1, -1, i, -i\} \longrightarrow \mathscr{A}, \qquad r \longmapsto \epsilon r^2, \tag{F.21.19}$$

and its map into $W$ is $r \mapsto (\epsilon r^2, r)$. Its basepoint is exactly $r_0 = C_u(d)$; the collision equation gives $A(d) = \epsilon r_0^2$. Thus its fundamental-group image is exactly $M \cap J$. It also proves that the projection of $M$ onto $T/J$ is onto. On this pullback the pencil map is literally

$$f_\epsilon(r) = \kappa(\epsilon r^2, r) = \frac{r}{r^2 + \epsilon r + 1}. \tag{F.21.20}$$

At $r_0$ it has value $k_u$, with no intervening basepoint path.

This map induces a surjection $\pi_1(\mathscr{R}, r_0) \to \pi_1(\mathscr{B}, k_u)$. Here is a direct covering proof. Its derivative is $(1 - r^2)/(r^2 + \epsilon r + 1)^2$, and the exact identities

$$\begin{aligned} f_\epsilon(r) - \epsilon/3 &= \frac{-\epsilon(r - \epsilon)^2}{3(r^2 + \epsilon r + 1)}, \\ f_\epsilon(r) + \epsilon &= \frac{\epsilon(r + \epsilon)^2}{r^2 + \epsilon r + 1}, \\ f_\epsilon(r) - \epsilon &= \frac{-\epsilon(r^2 + 1)}{r^2 + \epsilon r + 1} \end{aligned} \tag{F.21.21}$$

identify every removed preimage and both branch values. In particular

$$f_\epsilon : \mathscr{R} \longrightarrow \mathscr{B} \setminus \{\epsilon/3\} \tag{F.21.22}$$

is a connected unramified degree-two covering. The two preimages of 0 are $0, \infty$; the preimages of $\epsilon$ are $i, -i$; the single double preimage of $-\epsilon$ is $-\epsilon$; and the single double preimage of $\epsilon/3$ is $\epsilon$. The poles of $f_\epsilon$ map to the allowed target infinity and are ordinary unramified points. These are precisely the six punctures of $\mathscr{R}$.

Write $G_0 = \pi_1(\mathscr{B} \setminus \{\epsilon/3\}, k_u)$ and let $\chi : G_0 \to C_2$ be the monodromy character of (F.21.22). Its index-two kernel is the image of $\pi_1(\mathscr{R}, r_0)$. A based meridian $c$ of $\epsilon/3$ has $\chi(c) = 1$ by its local double-root expression in (F.21.21). The filling $q : G_0 \twoheadrightarrow \pi_1(\mathscr{B}, k_u)$ kills $c$. For every $g \in G_0$, the element $gc^{-\chi(g)}$ belongs to $\ker \chi$ and has the same image under $q$ as $g$. Hence $q(\ker \chi)$ is the entire target. This proves the asserted surjectivity, using the actual index-two cover and actual filling. The basepoint avoids $\epsilon/3$: its fiber in (F.21.21) consists only of $r = \epsilon$, which is excluded from $\mathscr{R}$.

Combining this surjectivity with (F.21.16) proves the exact subgroup factorizations

$$J = N(M \cap J), \qquad T = JM = NM. \tag{F.21.23}$$

For the first, given $j \in J$, choose $m \in M \cap J$ with $\kappa_*(m) = \kappa_*(j)$ and write $j = (jm^{-1})m$ with $jm^{-1} \in N$. For the second, $M \to T/J$ is onto by the connected cover (F.21.19), and $(M \cap J)M = M$.

*The exact orbit map.* At the diagonal basepoint $d$, the two copies of $\pi_1(\mathcal{M}_{0,5})$ have exactly the same base configuration. Let $P_{5,\mathrm{d}}$ and $P_{4,\mathrm{d}}$ denote these discrete groups with their actual forgetting homomorphism. The mutually inverse diagonal-coset maps are the following, where $[b_o, b_n]$ denotes the indicated right diagonal coset:

$$\begin{gathered} \mathcal{Q}_\mathrm{d} = P_{5,\mathrm{d}} \times_{P_{4,\mathrm{d}}} P_{5,\mathrm{d}}, \qquad \mathcal{Z}_\mathrm{d} = \mathcal{Q}_\mathrm{d} / \operatorname{diag}(P_{5,\mathrm{d}}), \\ \mathcal{Z}_\mathrm{d} \longrightarrow F, \qquad [b_o, b_n] \longmapsto b_n b_o^{-1}, \\ F \longrightarrow \mathcal{Z}_\mathrm{d}, \qquad v \longmapsto [1, v]. \end{gathered} \tag{F.21.24}$$

Their composites are the identity, by right multiplication with $(b_o^{-1}, b_o^{-1})$. The group $F$ here is the same one-point kernel (F.21.18); passage to the cover $W_A$ does not alter its fiber or its inclusion.

For $n \in N$, write $a(n) \in \pi_1(\mathscr{A})$ for its actual $A$-coordinate and let $m(a(n)) \in M$ be its image under (F.21.17). The actual source pair has precisely the image

$$n \longmapsto (n, m(a(n))) \qquad (n \in N). \tag{F.21.25}$$

Indeed its old projection has image $N$ by (F.21.16), and its new projection is $s \mapsto (A(s)^2, \epsilon A(s))$, the displayed section applied to the same $A$-loop. Any element killed by the old projection is also killed by its $A$-projection and hence by the new projection. This proves the equality of the source image with (F.21.25), including its maps.

Left multiplication on (F.21.24) acts on $F$ as

$$v \longmapsto m(a(n))\, v\, n^{-1}. \tag{F.21.26}$$

For every $v \in F$, factor $v^{-1} = nm$ using (F.21.23). Projection to $\pi_1(\mathscr{A})$ gives $a(m) = a(n)^{-1}$. The projection is an isomorphism on $M$, so $m = m(a(n))^{-1}$ and $v = m(a(n))n^{-1}$. This is the image of 1 under (F.21.26). The source action is therefore transitive on the entire discrete coset space.

The connected-fiber forgetting map from the F.19 space $X$ to this selected two-track space carries the selected divisor onto its diagonal. Its divisor image is the full inverse image of the

diagonal group: lift a diagonal loop and correct the endpoint in the connected configuration fiber. Its kernel already comes from that fiber in the divisor. This is the same exact comparison proved in F.20, now with old label $u$ and new label $\epsilon a$. The omitted labels are still three old points and one new point; their distinctness and avoidance conditions give those same connected fibers. It proves the corresponding discrete version of (F.21.2) for each of the four rows of (F.21.12) and both choices of sign.

On full completions, F.19–F.20 identify this precise divisor image with the inverse image of the full diagonal subgroup, and its full coset space with $\widehat{F}_3$. The actual configuration sections and characteristic finite quotients of the finitely generated free kernels used there apply with these same omitted labels. Thus no completion of an unproved intersection is being substituted here. The orbit of a point under the compact group $L = \widehat{\pi_1(Y)}$ is compact and closed in $\widehat{F}_3$. By the discrete transitivity just proved it contains the dense image of $F_3$. It is therefore all of $\widehat{F}_3$. Transport by $\gamma(\omega)$ gives exactly (F.21.2) at the original F.19 basepoint.

*The tubular centralizer lift.* Write $D_i$ for one selected cross divisor in $X$ and $D_i^{\mathrm{reg}}$ for the complement in it of its intersections with the other seven cross divisors. This open submanifold is connected: $D_i$ is connected by the configuration-fiber description above, and paths can be perturbed away from its finite union of complex divisors. The same perturbation makes $\pi_1(D_i^{\mathrm{reg}}) \to \pi_1(D_i)$ onto. The point $\gamma(d)$ belongs to $D_i^{\mathrm{reg}}$ by the distinctness of the sixteen source collision points.

Choose a Hermitian metric on the complex normal line to $D_i^{\mathrm{reg}}$. A sufficiently small positive continuous radius in that line gives a circle bundle lying in the complement of all eight divisors, namely the ten-point moduli space. Such a radius is obtained from a locally finite system of tubular coordinate charts by a positive partition-of-unity minorant of their admissible radii. Over any compact loop the resulting normal circles and their homotopies stay in these tubular charts.

The image of the fiber generator is central in the fundamental group of this circle bundle. Indeed transport around a base loop acts on $\pi_1(S^1) = \mathbb{Z}$ by $+1$, because the normal line is complex oriented. The circle-bundle homotopy sequence is onto on the base fundamental group, since the fiber is connected. Base this construction by the chosen source whisker and its small terminal normal circle. Its fiber generator maps to exactly $\nu_{u,\epsilon}$: F.17.7 gives the simple holomorphic collision, and hence positive exponent one. Filling the divisor takes each of these circle-bundle loops to its base loop in $D_i$. Consequently the discrete centralizer of this very meridian maps onto the entire discrete divisor image in $\pi_1(X)$.

Compose the circle-bundle homomorphism with the full completion $G$. It extends continuously from the full completion of the circle-bundle group. Its compact image centralizes $\nu_{u,\epsilon}$ and its image in $\mathcal{Q}$ contains the closure of the whole discrete divisor image. That closure is exactly $\mathcal{P}_{u,\epsilon}$ by the specified completion comparison above. This proves (F.21.3). Injectivity of the circle-bundle completion is not used.

*Normal generation in the actual kernel.* Let $N_D$ be the closed normal closure of $K^+$ inside $D^+$. Fix one selected meridian $\nu_i$. For $g \in G$, transitivity (F.21.2) factors its image as

$$\varpi(g) = \gamma(l)p, \qquad l \in L, \quad p \in \mathcal{P}_i.$$

Choose $h \in H^\dagger$ over $l$ using the onto map in F.19.10, and choose $c \in C_G(\nu_i)$ with $\varpi(c) = p$ using (F.21.3). Then $d = gc^{-1}h^{-1} \in D^+$ and

$$g = dhc, \qquad g\nu_i g^{-1} = d(h\nu_i h^{-1})d^{-1} \in N_D. \tag{F.21.27}$$

Here $\nu_i \in K^+$ and $K^+ \triangleleft H^\dagger$ by the exact source row F.19.10, so the middle conjugate belongs to $K^+$. F.19.3 states that the closed ambient normal closure in $G$ of these eight $\nu_i$ is $D^+$. Equation (F.21.27) puts every such conjugate in $N_D$, proving $D^+ = N_D$.

In the quotient $D^+_{\mathrm{ab}}$, conjugation by $D^+$ acts trivially. The image of $K^+$ is a closed subgroup, since $K^+$ is compact. The normal-generation equality therefore says that this image is the entire

abelianization. For any continuous finite nilpotent quotient $q : D^+ \twoheadrightarrow P$, put $S = q(K^+)$. The proved equality says that $S$ normally generates $P$. Every maximal proper subgroup $M_0$ of a finite nilpotent group is normal: choose the first term $Z_j(P)$ of its upper central series not contained in $M_0$. Then $Z_{j-1}(P) \subseteq M_0$ and an element of $Z_j(P) \setminus M_0$ normalizes $M_0$, because its commutators with $M_0$ lie in $Z_{j-1}(P)$. The normalizer is strictly larger than the maximal subgroup and hence is all of $P$. A proper $S$ would be contained in such a normal $M_0$; its normal closure would remain there. Therefore $S = P$. Finally the image of compact $K^+$ in $D^+_{\mathrm{nil}}$ is closed and maps onto every defining finite quotient. It is dense by the inverse-limit topology and thus equals $D^+_{\mathrm{nil}}$. This proves all the stated quotient surjections.

The exact symbolic certificate `code/part_a/cross_divisor_geometry_f21.py` checks (F.21.11)–(F.21.12), the coordinate inverse in (F.21.10), the divisor derivative (F.21.14), the section pullbacks, and all branch and puncture identities (F.21.20)–(F.21.21). Its finite checks supplement the bundle, fundamental-group, and compact-orbit arguments written here; those arguments establish the full assertions. □

**F.22. [Proved] The actual two-point kernel, its compact generators, and the remaining source map.** Retain the original F.19 base configuration. Let $p = a(s_0)$, $n = -a(s_0)$, and let $S'$ be the plane with its seven old finite labels removed. Filling the four labels $U = \{v_u : u \neq \zeta\}$ gives the plane $S$ with the three old finite labels $0, 1, v_\zeta$ removed. The actual two-point groups are

$$J = \pi_1(\widehat{\mathrm{Conf}_2(S')}, (p, n)), \qquad J_4 = \pi_1(\widehat{\mathrm{Conf}_2(S)}, (p, n)), \qquad D^+ = \ker(J \to J_4). \tag{F.22.1}$$

There are compatible based point-pushing sections of the two positive-point projections, described below, and geometric free coordinates in which

$$\begin{gathered} V_+ = \widehat{F}(g_1, g_2, g_3, u_1, u_2, u_3, u_4), \qquad Q_+ = \widehat{F}(g_1, g_2, g_3), \\ V = \widehat{F}(X_1, X_2, X_3, U_1, U_2, U_3, U_4, t), \qquad P = \widehat{F}(X_1, X_2, X_3, t), \\ R_+ = \ker(V_+ \to Q_+), \qquad R_1 = \ker(V \to P). \end{gathered} \tag{F.22.2}$$

The displayed fillings kill precisely $u_j$ and $U_j$, respectively. With the actual section $\sigma : V_+ \to J$ and fiber inclusion $\iota : V \to J$, there are mutually inverse continuous maps

$$\begin{aligned} R_1 \rtimes R_+ &\xrightarrow{\Theta} D^+, & (b, g) &\longmapsto \iota(b)\sigma(g), \\ D^+ &\longrightarrow R_1 \rtimes R_+, & d &\longmapsto \big(\iota^{-1}(d\sigma(p_+ d)^{-1}), p_+ d\big). \end{aligned} \tag{F.22.3}$$

Here $p_+$ forgets the negative point; the action in this semidirect product is the actual conjugation by $\sigma$. Its complete free-word recipe is (F.22.11)–(F.22.12).

For a profinite space $T$, write $\mathfrak{F}(T)$ for the free profinite group on continuous maps from $T$, as constructed in the proof. This definition retains the topology on the generator space. Put $I = \{1, 2, 3, 4\}$. Then the exact compact-generator identifications are

$$\begin{aligned} \mathfrak{F}(Q_+ \times I) &\xrightarrow{\sim} R_+, & (q, j) &\longmapsto s_+(q) u_j s_+(q)^{-1}, \\ \mathfrak{F}(P \times I) &\xrightarrow{\sim} R_1, & (v, j) &\longmapsto s_1(v) U_j s_1(v)^{-1}, \end{aligned} \tag{F.22.4}$$

where $s_+$ and $s_1$ retain the displayed free generators.

The other one-point forgetting is the actual $p_- : J \to V_-$, with

$$\begin{gathered} V_- = \widehat{F}(X_1, X_2, X_3, U_1, U_2, U_3, U_4), \qquad Q_- = \widehat{F}(X_1, X_2, X_3), \\ R_- = \ker(V_- \to Q_-), \qquad W = D^+ \cap \ker p_+ \cap \ker p_-. \end{gathered} \tag{F.22.5}$$

One has the exact sequence

$$1 \longrightarrow W \longrightarrow D^+ \xrightarrow{(p_+,p_-)} R_+ \times R_- \longrightarrow 1. \tag{F.22.6}$$

Inside the actual fiber $V$, let $s_-$ retain $Q_-$ and put $\eta(q,e) = s_-(q)es_-(q)^{-1}$ for $e \in \{U_1,U_2,U_3,U_4,t\}$. Then

$$W = \overline{\langle\!\langle [\eta(q,U_j),\eta(q',t)] : q,q' \in Q_-,\ j \in I \rangle\!\rangle}_V. \tag{F.22.7}$$

The literal element $[U_1,t]$ belongs to $W$ and generates a full closed copy of $\widehat{\mathbb{Z}}$.

Finally let $\Delta$ be the sixteen actual source collisions, with the F.19 source meridians and section $j : L \to H^\dagger$. There is a specified continuous map

$$\Lambda : \mathfrak{F}(L \times \Delta) \longrightarrow D^+, \qquad (l,d) \longmapsto j(l)\nu_d j(l)^{-1}, \qquad \operatorname{im}\Lambda = K^+. \tag{F.22.8}$$

Its coordinates in (F.22.3) are given by the actual two projection formulas there, applied to the displayed element. Thus the remaining equality $K^+ = D^+$ is exactly surjectivity of this named continuous homomorphism onto the displayed semidirect product. This statement does not settle that surjectivity or the original residual source-membership equation. Write $\Delta_+$ and $\Delta_-$ for the eight collisions with the positive and negative new point, respectively, and put $\mathcal{K}_\pm = p_\pm(K^+)$. The exact joint source image is their full direct product:

$$1 \longrightarrow K^+ \cap W \longrightarrow K^+ \xrightarrow{(p_+,p_-)} \mathcal{K}_+ \times \mathcal{K}_- \longrightarrow 1. \tag{F.22.8a}$$

Each marginal is the actual image of the corresponding eight-collision source filling kernel, as specified in (F.22.21). Their equality with $R_+$ and $R_-$ remains unresolved.

*Manuscript:* Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part F(b), the original residual A5-centralizer and the exact source-membership problem.

*Proof. Compatible sections at the original basepoint.* All seven old finite points, $p$, and $n$ are distinct by F.17. Choose an embedded finite tree in the plane joining the seven old points and $p$ while avoiding $n$. A sufficiently small regular neighborhood is a closed disk $B$ containing those eight points in its interior and excluding $n$. The inclusions

$$B^\circ \setminus \{\text{seven old points}\} \longrightarrow S', \qquad B^\circ \setminus \{\text{three retained old points}\} \longrightarrow S \tag{F.22.9}$$

induce isomorphisms on fundamental groups based at the original point $p$. Indeed a collar of the disk boundary and the outside annulus retract onto that boundary, so adjoining the outside region changes none of the punctured-disk fundamental-group generators or relations.

On each punctured disk the map $z \mapsto (z,n)$ is an actual based map to its ordered two-point configuration space. Compose its fundamental-group map with the inverse of the relevant isomorphism (F.22.9). This gives sections $\sigma_{\mathrm{d}}$ for $S'$ and $\sigma_{4,\mathrm{d}}$ for $S$. The two disk inclusions and the two maps $z \mapsto (z,n)$ commute with filling the four old punctures. Their inverse group isomorphisms therefore commute too. Hence these very sections form a commuting square with the actual filling $J_{\mathrm{d}} \to J_{4,\mathrm{d}}$. All basepoints here are $(p,n)$; the comparison paths to the F.19 two-point coordinates are the constant paths at that configuration.

The Fadell–Neuwirth sequences have kernels $\pi_1(S' \setminus \{p\},n) = F_8$ and $\pi_1(S \setminus \{p\},n) = F_4$. Their injections follow from $\pi_2(S') = \pi_2(S) = 0$. The full-topology and completed-kernel comparisons are exactly those established in F.19 for these two-point configuration groups. The actual discrete sections extend to full completions, their section identities persist, and their filling square still commutes. Thus

$$J = V \rtimes_\sigma V_+, \qquad J_4 = P \rtimes_{\sigma_4} Q_+, \qquad (b,g) \longmapsto (\overline{b},\overline{g}) \tag{F.22.10}$$

is the actual filling map in these coordinates. The equation $\overline{g} = 1$ puts $g$ in $R_+$, and then $\overline{b} = 1$ puts $b$ in $R_1$. Multiplication and the section inverse give exactly the two maps in (F.22.3).

*Free coordinates and the retained Artin action.* Choose a geometric chart on $B$ in which the old retained points occupy positions $1, 2, 3$, the four other old points occupy positions $4, 5, 6, 7$, and $p$ occupies position 8. The exterior point $n$ is the basepoint for puncture meridians. Such a chart is an orientation-preserving homeomorphism of the disk with the stated finite labelled set: move the points along polygonal arcs avoiding the other points, using successive small disk isotopies. The induced fundamental-group isomorphisms have the inverse chart as their literal inverse. Both filling maps use this same chart and its restriction to the retained labelled set. They therefore commute with the actual coordinate maps, including the basepoints.

Write $X_4 = U_1, \dots, X_7 = U_4, X_8 = t$. Use the literal clockwise exchanges and the Artin substitutions calibrated in F.20.7–9 on $\widehat{F}(X_1, \dots, X_8)$. For the positive point moving once clockwise around old position $i$ along its lower-half-plane whisker, the based geometric free generator $g_i$ has the braid

$$\beta_i = \sigma_7 \cdots \sigma_{i+1} \sigma_i^2 \sigma_{i+1}^{-1} \cdots \sigma_7^{-1}, \qquad 1 \le i \le 7, \qquad u_j = g_{j+3}. \tag{F.22.11}$$

The descending word carries the last strand below the intervening points, its square is the clockwise lasso, and the inverse word returns the same whisker. Deleting that strand gives the trivial old braid by contracting the old motion in its projection-order chambers. These seven lower-whiskered lassos form the actual geometric kernel basis. The section keeps the negative point $n$ fixed outside $B$, so the exact fibre-coordinate comparison below makes its conjugation on the negative-point fiber

$$\alpha_{g_i} = \phi_{\beta_i}, \qquad \phi_{\sigma_k}(X_k) = X_k X_{k+1} X_k^{-1}, \quad \phi_{\sigma_k}(X_{k+1}) = X_k, \quad \phi_{\sigma_k}(X_j) = X_j \ (j \ne k, k+1). \tag{F.22.12}$$

The clockwise braid word is composed in the same chronological and substitution order as F.20, and its inverse word gives its inverse map.

Here is the precise point-pushing identification underlying that conjugation formula. For a pure braid on the eight punctures and the exterior ninth point, let $d$ be the ninth Artin meridian and write $\phi_b(d) = W_b d W_b^{-1}$. Erasing $d$ gives $w_b$; put $u_b = w_b^{-1}$. Exactly as in F.20.8, substitution gives $u_{bc} = u_b \phi_b(u_c)$ after erasing the ninth strand. The section on the first eight strands has $u = 1$. Write $\gamma_i$ for the eight lower-whiskered clockwise point-pushing lassos of the ninth point, given by the same braid pattern with $\sigma_8$ as its first exchange. Their values of $u$ are

$$\tau_i = X_8^{-1} \cdots X_{i+1}^{-1} X_i^{-1} X_{i+1} \cdots X_8, \qquad 1 \le i \le 8. \tag{F.22.13}$$

This triangular substitution is its own inverse: starting with $i = 8$ and substituting downward, each adjacent inverse pair cancels and the image of $\tau_i$ is $X_i$. The actual forward coordinate map is $\gamma_i \mapsto \tau_i(X)$; its inverse is $X_i \mapsto \tau_i(\gamma)$. Both composites are the identity on the named free bases. They therefore give the actual free-group isomorphism and its full profinite inverse between the fiber and these Artin coordinates. The crossed multiplication formula now makes conjugation by the actual section exactly (F.22.12). Erasing any of the four old labels, or erasing $t$, commutes with (F.22.13); the remaining triangular formula has exactly the remaining labels in their retained order. This proves the coordinate form of both fillings.

The plane braid groups in this calculation map to the original normalized moduli groups through the explicit coordinate isomorphism

$$(z_0, z_1, z_2, \dots) \longmapsto \left( z_0, z_1 - z_0, \left( \frac{z_j - z_0}{z_1 - z_0} \right)_{j \ge 2} \right). \tag{F.22.14}$$

Its inverse multiplies the last coordinates by $z_1 - z_0$ and adds $z_0$. The first two old labels are fixed on every loop in $J$, so their two affine coordinates are constant. This is the actual comparison with the normalized F.19 configuration groups.

For completeness, the action extends continuously from $F_7$ to $V_+ = \widehat{F}_7$ on $V = \widehat{F}_8$. Every open subgroup of this finitely generated profinite $V$ contains a characteristic open subgroup: intersect its finitely many subgroups of at most the given index. The action on each resulting finite characteristic quotient has finite image and extends through $V_+$. The compatible inverse automorphisms give the continuous inverse action. This is also the action already supplied by the actual section in (F.22.10).

*A free profinite group on a compact generator space.* For a profinite space $T$, begin with the abstract free group on its underlying set. Take its inverse limit over finite quotients for which the map from $T$ to that finite quotient is continuous; denote the result by $\mathfrak{F}(T)$. Products of finitely many such quotient maps again have continuous generator maps, so these quotients form a directed system. The canonical map $e : T \to \mathfrak{F}(T)$ is continuous and its image generates a dense subgroup. Every continuous map from $T$ to any profinite group $H$ extends uniquely to a continuous homomorphism $\mathfrak{F}(T) \to H$: compose with every finite quotient of $H$, extend on the abstract free generators, and take the compatible inverse limit. This construction proves the precise universal property used here.

Let $Q$ be one of the finitely generated free profinite groups in (F.22.2), and let $I$ be the finite set of the additional generators. Left multiplication on $Q \times I$ induces automorphisms of $\mathfrak{F}(Q \times I)$. This action is jointly continuous. Indeed for any continuous map $Q \times I \to A$ to a finite group, compactness gives an open normal subgroup $N \triangleleft Q$ through which all its finitely many coordinate functions factor. Its translates therefore form a finite set indexed by $Q/N$. Intersect the kernels of the corresponding finite quotient homomorphisms of $\mathfrak{F}$. The intersection is open and $Q$-invariant, and the action on this quotient factors through a finite quotient of $Q$. These invariant open subgroups refine every finite quotient kernel. They prove joint continuity and give the actual profinite semidirect product $\mathfrak{F}(Q \times I) \rtimes Q$.

Let $\widehat{F}(Q; I)$ denote the free profinite group obtained by adjoining these finitely many extra generators $v_j$ to the fixed free basis of $Q$. Its retraction $\rho$ kills the $v_j$, and its section $s$ retains the basis of $Q$. There are inverse continuous homomorphisms

$$\begin{aligned} \widehat{F}(Q; I) &\longrightarrow \mathfrak{F}(Q \times I) \rtimes Q, \\ x &\longmapsto (1, x) \quad (x \text{ in the retained basis}), \qquad v_j \longmapsto (e(1, j), 1), \\ \mathfrak{F}(Q \times I) \rtimes Q &\longrightarrow \widehat{F}(Q; I), \\ e(q, j) &\longmapsto s(q) v_j s(q)^{-1}, \qquad q \longmapsto s(q). \end{aligned} \tag{F.22.15}$$

The second generator map is continuous and $Q$-equivariant, so the universal property and the semidirect-product law give its homomorphism. Both composites fix every retained generator, every $v_j$, and every $e(q, j)$ by the displayed conjugation formula. These sets generate densely, so both composites are the identity. The projections onto $Q$ are precisely $\rho$ and the semidirect-product projection. Their kernels are therefore identified by these inverse maps. Apply this construction first to $(Q_+, u_j)$ and then to $(P, U_j)$ to prove (F.22.4). No full completion of an infinitely generated discrete kernel has been substituted in this argument.

The action in (F.22.3) can now be evaluated on every compact generator, not only on a finite selection. The element $e(q, j)$ of $R_+$ has image $s_+(q) u_j s_+(q)^{-1}$ in $V_+$. Its action is $\alpha_{s_+(q)} \alpha_{u_j} \alpha_{s_+(q)}^{-1}$. For a compact generator $e(v, k)$ of $R_1$, apply that automorphism to $s_1(v) U_k s_1(v)^{-1}$ in $V$, then use the inverse of (F.22.15). The result lies in $R_1$ because the induced action of $R_+$ on $P$ is the identity. The formula is continuous in both $q$ and $v$ by the actual continuous group operations and the proved action continuity.

*The interaction kernel retained by both one-point forgettings.* On the fiber $V$, the negative-point projection $p_-$ fills the positive puncture, so its map $h : V \to V_-$ kills $t$ and retains every other letter. On the section $\sigma(V_+)$ it is the identity element: that section keeps the negative point fixed. Hence

$$(p_+, p_-)(\Theta(b, g)) = (g, h(b)). \tag{F.22.16}$$

The restriction $h : R_1 \to R_-$ is onto. For an element of $R_-$, retain its seven letters in $V$; its old-puncture filling is the identity, so it belongs to $R_1$ and maps back to that element under $h$. Together with the arbitrary $g \in R_+$, this proves surjectivity in (F.22.6), with kernel $W = R_1 \cap \ker h$ inside $V$.

Apply (F.22.15) with retained group $Q_-$ and the five extra letters $U_1, \dots, U_4, t$. The kernel of $V \to Q_-$ is

$$\mathcal{R} = \mathfrak{F}((Q_- \times I) \sqcup (Q_- \times \{t\})) = \mathcal{A} \amalg \mathcal{B}, \tag{F.22.17}$$

where $\mathcal{A} = \mathfrak{F}(Q_- \times I)$, $\mathcal{B} = \mathfrak{F}(Q_- \times \{t\})$, and the coproduct equality follows directly from the continuous-map universal property on the two compact open-and-closed pieces. The two fillings restrict to the retractions $\mathcal{R} \to \mathcal{B}$ and $\mathcal{R} \to \mathcal{A}$. Consequently

$$W = \ker(\mathcal{A} \amalg \mathcal{B} \longrightarrow \mathcal{A} \times \mathcal{B}). \tag{F.22.18}$$

Quotienting $\mathcal{R}$ by the closed normal closure of all the displayed generator commutators makes its two factor images commute. The multiplication map from $\mathcal{A} \times \mathcal{B}$ to this quotient is then a homomorphism. Its composite with the two retractions is the identity, and the reverse composite fixes all generators. These are inverse homomorphisms. Thus their kernel is exactly the stated normal closure. It is invariant under $Q_-$, whose action sends $(q, q')$ to $(aq, aq')$. Its normal closure in $V = \mathcal{R} \rtimes Q_-$ is therefore the same subgroup. This proves (F.22.7), with its actual ambient group.

For every integer $m \geq 2$, send $U_1$ to $I + E_{12}$, send $t$ to $I + E_{23}$, and send the other free generators of $V$ to the identity in the finite group $\mathrm{UT}_3(\mathbb{Z}/m)$. These assignments define a continuous homomorphism. With $[a, b] = aba^{-1}b^{-1}$,

$$[U_1, t] \longmapsto I + E_{13}, \qquad [U_1, t]^z \longmapsto I + (z \bmod m)E_{13} \quad (z \in \widehat{\mathbb{Z}}). \tag{F.22.19}$$

The commutator is killed by both fillings, so it belongs to $W$. The displayed finite images detect every nonzero $z \in \widehat{\mathbb{Z}}$ through some modulus. The exponent map to its closed cyclic subgroup is therefore injective as well as onto, proving the asserted full procyclic subgroup.

*The actual compact-generator source image.* The F.19 source section retains its eleven old free meridians in the rank-twenty-seven source; the sixteen additional free meridians are $m_d$, $d \in \Delta$. Apply (F.22.15) to that very based free system. It identifies

$$\ker(q : L^\dagger \to L) \cong \mathfrak{F}(L \times \Delta), \qquad (l, d) \longmapsto i(l) m_d i(l)^{-1}. \tag{F.22.20}$$

Composition with the actual map $\Phi^\dagger : L^\dagger \to G$ gives exactly (F.22.8), because $\Phi^\dagger i = j$ and $\Phi^\dagger(m_d) = \nu_d$ with the retained source whisker. F.19.10 identifies the image of this source kernel with $K^+$; alternatively its image is precisely the kernel of the induced onto map $H^\dagger \to L$, since the commuting source filling has that kernel. The map in (F.22.8) is continuous and its image is closed, since its domain is compact.

*The two marginal source images and their joint image.* Put $Y^\pm = Y \setminus \Delta_\pm$ and $L_\pm = \pi_1(\widehat{Y^\pm, s_0}) \cong \widehat{F}_{19}$. The source fillings $q_\pm : L_\pm \to L$ retain the eleven old basis elements and kill their eight added meridians. Let $F_+ : G \to P_9^+$ forget the negative point, and let $F_- : G \to P_9^-$ forget the positive point. These are the actual full nine-point forgetting maps; their restrictions to $J$ are $p_+$ and $p_-$ with the displayed one-point kernel targets. The rational source configuration after forgetting the negative point is defined on all of $Y^+$; the only extra singularities of that map are the eight positive collisions. The corresponding statement holds with the signs interchanged. Thus the actual source maps factor as

$$\begin{aligned} F_\pm \Phi^\dagger &= \Phi_\pm r_\pm, \qquad r_\pm : L^\dagger \twoheadrightarrow L_\pm, \\ \mathcal{K}_\pm &= \Phi_\pm(\ker q_\pm) = \mathrm{im}(\mathfrak{F}(L \times \Delta_\pm) \longrightarrow R_\pm). \end{aligned} \tag{F.22.21}$$

Here $r_+$ fills the negative collision points, and $r_-$ fills the positive ones. These equalities retain their based source maps: the signed free basis embeds $L_\pm$ back into $L^\dagger$ by its named old and signed meridians, so $r_\pm(\ker q) = \ker q_\pm$. The last map sends $(l, d)$ to $F_\pm(j(l)\nu_d j(l)^{-1})$.

For $d \in \Delta_+$, forgetting the positive point removes the collision. The remaining configuration map extends across $d$, so the based source meridian has $F_-(\nu_d) = 1$. For $d \in \Delta_-$ one similarly has $F_+(\nu_d) = 1$. For every $l \in L$ these identities give

$$\begin{aligned} F_-(j(l)\nu_d j(l)^{-1}) &= 1 \quad (d \in \Delta_+), \\ F_+(j(l)\nu_d j(l)^{-1}) &= 1 \quad (d \in \Delta_-). \end{aligned} \tag{F.22.22}$$

The maps here are defined on all of $G$; the calculation does not apply $p_\pm$ to an element outside $J$.

Let $K_+^{\mathrm{src}}$ and $K_-^{\mathrm{src}}$ be the compact images of $\mathfrak{F}(L \times \Delta_+)$ and $\mathfrak{F}(L \times \Delta_-)$ in $K^+$ under (F.22.8). Their joint images are $\mathcal{K}_+ \times \{1\}$ and $\{1\} \times \mathcal{K}_-$, respectively, by (F.22.21)–(F.22.22). Given a pair in $\mathcal{K}_+ \times \mathcal{K}_-$, lift its first coordinate to $K_+^{\mathrm{src}}$ and its second to $K_-^{\mathrm{src}}$. Their product in $K^+$ maps to the given pair. This proves (F.22.8a). Its kernel is $K^+ \cap W$ by (F.22.6), giving the exact commuting diagram of actual inclusions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & K^+ \cap W & \longrightarrow & K^+ & \longrightarrow & \mathcal{K}_+ \times \mathcal{K}_- & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & W & \longrightarrow & D^+ & \longrightarrow & R_+ \times R_- & \longrightarrow & 1. \end{array} \tag{F.22.23}$$

The first exact row concerns the proved marginal images, with no enlargement of either marginal.

The point-pushing section $\sigma$ and the source section $j$ have both been retained as their actual maps. To find the coordinates of any generator in (F.22.8), set $z = j(l)\nu_d j(l)^{-1}$, compute the literal one-point forgetting $g = p_+(z)$, and take $b = \iota^{-1}(z\sigma(g)^{-1})$. This is exactly $\Theta^{-1}(z)$, with no identification of the two sections. The first coordinate belongs to $R_1$ and the second to $R_+$ by the proved inverse maps. These formulas identify the remaining source-membership problem in the displayed kernel presentation.

The exact producer `code/part_a/joint_kernel_presentation_f22.py` checks all seven Artin action words and their inverses, their pure conjugators, their boundary product, and their two actual filling squares. It also records the deleted-label conjugators after the old filling, checks that $[U_1, t]$ is killed by both fillings, and checks the central power formula in thirty finite Heisenberg groups. The algebraic formula (F.22.19), proved for every modulus, supplies the full profinite conclusion. The compact-generator construction and all semidirect-product inverse maps are proved above independently of those finite checks. □

**F.23. [Proved] The based source-sign symmetry, complete source braid map, and two exact finite marginal images.** Retain the original groups, source map, basepoints, and two one-point forgettings of F.19–F.22. In particular put

$$\mathcal{K}_\pm = p_\pm(K^+) \le R_\pm \le V_\pm, \qquad V_\pm \cong \widehat{F}_7, \tag{F.23.1}$$

with the actual geometric positive coordinates and Artin negative coordinates of F.22. The source involution $h(s) = \zeta^2/s$, together with the explicit path $\lambda$ below, gives an actual based isomorphism of inclusions

$$\begin{array}{ccc} \mathcal{K}_- & \hookrightarrow & R_- \\ {\scriptstyle \mathcal{B}_+}\downarrow & & \downarrow{\scriptstyle \mathcal{B}_+} \\ \mathcal{K}_+ & \hookrightarrow & R_+. \end{array} \tag{F.23.2}$$

The same map extends to $V_- \xrightarrow{\sim} V_+$. Its formula, inverse, and all filling squares are specified below.

There is a complete exact braid description of the actual homomorphism

$$\Phi^{\dagger} : L^{\dagger} = \pi_1(\widehat{Y^{\dagger}, s_0}) = \widehat{F}_{27} \longrightarrow G. \tag{F.23.3}$$

It uses the twenty-seven specified source lassos, a rational projection perturbation, the original endpoint whiskers, and the original F.20 path $\eta$. The exact arithmetic certificates and twenty-seven full braid words are retained in the accompanying files. The coordinate comparison is the clockwise-exchange and inverse triangular comparison of F.20.7–9 and F.22.11–13.

For each sign, for each of the two groups $H = S_3, A_4$, and for every continuous homomorphism from the indicated full point-pushing group, the following equality holds:

$$\rho(\mathcal{K}_{\pm}) = \rho(R_{\pm}) \qquad (\rho \in \mathrm{Hom}_{\mathrm{cont}}(V_{\pm}, H),\ H \in \{S_3, A_4\}). \tag{F.23.4}$$

This is the exact finite scope of the exhaustive calculations. The original maps $\mathcal{K}_{\pm} \hookrightarrow R_{\pm}$ and $K^{+} \hookrightarrow D^{+}$ remain the specified inclusions; their surjectivity remains unresolved. The joint image and its kernel remain exactly those of F.22.8a.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. The literal source and the sign-changing maps.* Write

$$\begin{gathered} a(s) = \frac{\zeta - s}{\zeta + s}, \qquad v_u(s) = \left(\frac{u - s}{u + s}\right)^2, \qquad u \in \{1, \zeta, \zeta^2, \zeta^3, \zeta^4\}, \\ A_u = u(2\zeta - u), \qquad B_u = \frac{\zeta u^2}{2u - \zeta} \quad (u \neq \zeta), \\ \Delta_+ = \{s : s^2 = A_u,\ u \neq \zeta\}, \qquad \Delta_- = \{s : s^2 = B_u,\ u \neq \zeta\}, \\ Y^{\dagger} = Y \setminus (\Delta_+ \cup \Delta_-). \end{gathered} \tag{F.23.5}$$

The complete finite configuration is

$$\phi^{\dagger}(s) = (0, 1, v_1(s), v_{\zeta}(s), v_{\zeta^2}(s), v_{\zeta^3}(s), v_{\zeta^4}(s), a(s), -a(s)), \tag{F.23.6}$$

with the original additional label at infinity. Its normalization and its induced map to $G$ are those of F.19 and F.22.14. All punctures and their simple collision multiplicities are the actual ones established in F.17.

For $h(s) = \zeta^2/s$ and $u' = \zeta^2/u$, direct substitution gives

$$\begin{gathered} h^2(s) = s, \qquad a(h(s)) = -a(s), \qquad v_u(h(s)) = v_{u'}(s), \\ \frac{\zeta^4}{A_u} = B_{u'}, \qquad \frac{\zeta^4}{B_u} = A_{u'}. \end{gathered} \tag{F.23.7}$$

Thus $h$ is an actual homeomorphism of $Y^{\dagger}$ and $Y$, and it interchanges the two signed sources $Y_{\pm} = Y \setminus \Delta_{\pm}$. Let $\Pi$ permute the old labels by $u \mapsto u'$ and exchange the two new labels. It fixes $0, 1, \infty, v_{\zeta}$. The complete source equality is

$$\phi^{\dagger} h = \Pi \phi^{\dagger}. \tag{F.23.8}$$

This equality commutes with all the original fillings and with the two one-point forgettings, with their signs exchanged.

The based path is explicit. Put $s_1 = h(s_0)$ and use the polygon

$$\lambda = (s_0, s_0 + 6i, s_1 + 6i, s_1). \tag{F.23.9}$$

It lies in $Y^\dagger$. The exact sign-symmetry producer checks that each of its two vertical lines has real coordinate different from every old puncture. For either real coordinate $x$ and every collision square $V = A_u, B_u$, it checks $x \neq 0$ and

$$4x^4 - 4\,\mathrm{Re}(V)x^2 - (\mathrm{Im}\,V)^2 \neq 0. \tag{F.23.10}$$

Indeed a root $x + iy$ of $z^2 = V$ would have $y = \mathrm{Im}(V)/(2x)$ and would make this expression zero. The middle segment has imaginary part greater than two, whereas every old and collision puncture has absolute value less than two. These are exact inequalities in the retained real number field.

With left-to-right path products, the actual based source automorphism and its inverse are

$$\begin{aligned} \mathcal{A}_{\mathrm{src}}(g) &= \lambda h(g)\lambda^{-1}, \\ \mathcal{A}_{\mathrm{src}}^{-1}(g) &= h(\lambda)^{-1}h(g)h(\lambda). \end{aligned} \tag{F.23.11}$$

Both composites cancel to $g$. The square $\mathcal{A}_{\mathrm{src}}^2$ is conjugation with the literal loop $\lambda h(\lambda)$. These maps extend to inverse continuous maps of the full source completions and commute with the original filling $q$.

Put $w = \phi^\dagger(\lambda)$, a path from the original target basepoint $c_0$ to $\Pi(c_0)$. The actual target automorphism is

$$\mathcal{A}_G(z) = w\Pi(z)w^{-1}, \qquad \mathcal{A}_G^{-1}(z) = \Pi^{-1}(w^{-1}zw). \tag{F.23.12}$$

The literal equality (F.23.8) gives $\mathcal{A}_G\Phi^\dagger = \Phi^\dagger\mathcal{A}_{\mathrm{src}}$. It follows on the actual source filling kernel that $\mathcal{A}_G(K^+) = K^+$; the commuting target fillings give $\mathcal{A}_G(D^+) = D^+$ and preserve the corresponding old-point kernel $J$.

Write $F_+$ and $F_-$ for the full $G$ forgettings in F.22. Let $\Pi_+$ be the old-label permutation together with renaming the retained negative new label as positive. Then $F_+\Pi = \Pi_+F_-$. Set $w_+ = F_+(w)$. The two inverse maps are

$$\begin{aligned} \mathcal{B}_+(r) &= w_+\Pi_+(r)w_+^{-1}, \\ \mathcal{B}_+^{-1}(r) &= \Pi_+^{-1}(w_+^{-1}rw_+), \qquad F_+\mathcal{A}_G = \mathcal{B}_+F_-. \end{aligned} \tag{F.23.13}$$

The old-point filling square identifies its restrictions $V_- \to V_+$, and the common-three-point filling square identifies $R_- \to R_+$. These are isomorphisms with the displayed inverse. Finally the equality on $K^+$ and the last formula give $\mathcal{B}_+(\mathcal{K}_-) = \mathcal{K}_+$, proving (F.23.2).

*A full geometric source basis.* Use $s_* = 2 + 2i$ and axis-parallel squares of half-side $h_0 = 10^{-4}$. Their twenty-seven centers are 0, both signs of the five old centers in the first table, and both signs of the eight centers in the second table. Every pair $(r, t)$ below denotes $(r + it)/10^6$.

| old label | $r$ | $t$ |
|---|---|---|
| 1 | 1000000 | 0 |
| $\zeta$ | 309017 | 951057 |
| $\zeta^2$ | −809017 | 587785 |
| $\zeta^3$ | −809017 | −587785 |
| $\zeta^4$ | 309017 | −951057 |

| $u$ | square | $r$ | $t$ |
|---|---|---|---|
| 1 | $A_u$ | 882644 | 1077509 |
| 1 | $B_u$ | 454951 | 555393 |
| $\zeta^2$ | $A_u$ | 80730 | −1390528 |
| $\zeta^2$ | $B_u$ | 41611 | −716735 |
| $\zeta^3$ | $A_u$ | 1260728 | −1131556 |
| $\zeta^3$ | $B_u$ | 439300 | −394290 |
| $\zeta^4$ | $A_u$ | 1685063 | 174411 |
| $\zeta^4$ | $B_u$ | 587159 | 60773 |

(F.23.14)

For each square take the straight whisker from $s_*$ to its northeast corner, traverse its boundary counterclockwise, and return along the same whisker. Call these loops $\gamma_0, \ldots, \gamma_{26}$ in the producer's recorded order: the eleven old punctures first, then the sixteen signed collisions.

Here are the exact checks which make these actual free generators. For every old center the radius-$h_0/2$ disk contains its named old root. For a new center $c$ and its named square $V$, the checks give

$$|c| > \tfrac{1}{2}, \qquad |c^2 - V| < 10^{-5}, \qquad (h_0/2)^2 + 10^{-5} < h_0/2. \tag{F.23.15}$$

On the circle of radius $h_0/2$ about $c$, the linear term $2c(z-c)$ has absolute value greater than $h_0/2$, while $(z-c)^2 + c^2 - V$ has smaller absolute value. Rouché's theorem therefore gives exactly one root of $z^2 - V$ in that disk. The twenty-seven squares are pairwise disjoint by the exact center-distance inequalities. Each whisker stays farther than $\sqrt{2}h_0$ from every other square's center. Its basepoint is northeast of its own square, and its first intersection is that square's northeast corner. All whisker directions are distinct. They consequently form an embedded common-base star. Cutting the punctured plane along this star gives the usual free basis of its fundamental group. Filling the sixteen collision punctures retains precisely the eleven old free generators.

*The exact projection perturbation and crossing certificate.* The field is the same as in F.20:

$$\mathbb{Q}(\theta), \qquad \theta^4 - 5\theta^2 + 5 = 0, \qquad 19/10 < \theta < 2.$$

Real signs are decided by rational isolation of this indicated root and exact field arithmetic. Put $P_0 = 1 + 371i/1000$ and use the literal offsets

$$\delta_j = \frac{10^{-14} j^2}{P_0}, \qquad 0 \le j \le 8, \tag{F.23.16}$$

in the finite-label order of (F.23.6). For each segment and each pair of strands write their original difference as $N/D$. The producer verifies the polynomial inequality, over the entire real parameter interval,

$$|P_0 N|^2 - \left(10^{-14}(i^2 - j^2)\right)^2 |D|^2 > 0. \tag{F.23.17}$$

Every denominator is nonzero on the specified segment. This strict inequality proves that the entire interpolation $z_j \mapsto z_j + t\delta_j$, $0 \le t \le 1$, is collision-free. The polynomial is certified by rational Bernstein subdivision with strictly positive coefficients on every terminal interval. All its coefficients and terminal intervals are retained.

For the perturbed paths, real projected differences are again exact field polynomials after multiplying by positive squared denominators. Bernstein–Descartes subdivision isolates all their zeros, and the certificate checks nonzero height and derivative on each root interval. These intervals are refined until all crossing times are disjoint. At each crossing the two labels are adjacent in the current projection order; the height sign then gives the clockwise Artin letter calibrated in F.20. The final order is checked against the exact endpoint order. The record retains the full polynomials, root intervals, height and derivative signs, subdivision counts, and resulting words. These operations prove the full path word by successive actual projection-chamber comparisons.

The primary run has 137 forward segments, $4,932$ pair checks, and $1,412$ crossings. Its exact sign algorithm made $846,769$ field-sign calls. Return whiskers are the literal reverse paths. The additional chart path below has five segments and ten crossings. Thus the complete comparison uses 142 forward segments, $5,112$ pair checks, and $1,422$ crossings. The initial unperturbed projection had simultaneous events from $0, a, -a$; its rejected run is retained separately. The proved interpolation (F.23.17) is the precise comparison used for the completed run.

*The standard chart and the original endpoint whiskers.* At $s_*$ let $b = \phi^\dagger(s_*)$. Let $r$ have the same old coordinates as $b$ and the two new coordinates $R_p = 8 + 3i$, $R_n = 12 + 6i$. The actual path $T : r \to b$ first moves $R_p$ straight to $a(s_*)$ with $R_n$ fixed, and then moves $R_n$ straight to $-a(s_*)$ with the positive point fixed. Both paths and their entire offset interpolations are certified.

The standard configuration is

$$r_{\mathrm{std}} = (0, 1, 3, 2, 4, 5, 6, 8 + 3i, 12 + 6i). \tag{F.23.18}$$

The path $G : r_{\mathrm{std}} \to r$ moves the five old nonconstant labels one at a time along the specified straight segments, keeping both new points and $0, 1$ fixed. The certified projected order of $r_{\mathrm{std}}$ is the F.22 order $0, 1, v_\zeta, v_1, v_{\zeta^2}, v_{\zeta^3}, v_{\zeta^4}, p, n$. For an explicit straightening, write these standard projected points as $x_j + iy_j$ and take the continuous piecewise linear function $f$ with $f(x_j) = y_j$, affine between adjacent $x_j$ and constant outside them. The oriented plane map

$$x + iy \longmapsto x + i(y - f(x)) \tag{F.23.19}$$

has inverse $x + iy \mapsto x + i(y + f(x))$. Its interpolation using $tf$ changes no projected order. This gives the actual chart and inverse used for the clockwise generators and the lower whiskers; the same chart is restricted for every filling.

At every endpoint $z$ retain the actual path $H_z(t) = (z_j + t\delta_j)_j$. The basepath producer checks the original and shifted orders of all 284 segment endpoints, in $10,224$ exact pair comparisons. Every $H_z$ stays in its original projection-order chamber. Writing a subscript 1 for an offset path, the certified two-parameter homotopies give the following identities in the actual path groupoid:

$$\begin{gathered} \gamma = H_b \gamma_1 H_b^{-1}, \qquad TH_b = H_r T_1, \qquad GH_r = H_{r_{\mathrm{std}}} G_1, \\ GT\gamma T^{-1} G^{-1} = H_{r_{\mathrm{std}}} (G_1 T_1 \gamma_1 T_1^{-1} G_1^{-1}) H_{r_{\mathrm{std}}}^{-1}. \end{gathered} \tag{F.23.20}$$

Thus the computed word is carried to the original standard configuration by its displayed endpoint whisker. The chamber comparison preserves precisely the named clockwise generators. The first two coordinates on an offset loop are the constant values $0, 1 + t\delta_1$. Dividing all coordinates by $1 + t\delta_1$, which is nonzero by the same pairwise certificate, is the literal affine normalization of F.22.14. Consequently (F.23.20) also holds in every original normalized target.

Retain the original F.20.4a path $\eta : s_0 \to s_*$. Its existing exact certificate places it in $Y^\dagger$: the radial first segment misses the old and inner collision roots by its radius, and misses all outer roots by the four nonzero imaginary parts $\mathrm{Im}(A_u \overline{s_0^2})$. The remaining segments lie outside radius two. Let $\mathcal{N}$ be the exact affine normalization of F.22.14. The full original comparison path in the normalized target is

$$\Xi = \mathcal{N}(GT\phi^\dagger(\eta)^{-1}). \tag{F.23.21}$$

All three paths $G, T, \phi^\dagger(\eta)$ keep their first two coordinates literally $0, 1$. Hence the normalization $(z_j - z_0)/(z_1 - z_0)$ leaves every remaining coordinate unchanged at every point of this entire path. Thus $\Xi$ has exactly the original standard and $s_0$ endpoints. Let $b_j$ denote the recorded standard nine-strand braid of $GT\phi^\dagger(\gamma_j)T^{-1}G^{-1}$, interpreted through (F.23.20). Then the complete source formula is

$$\Phi^\dagger(\eta\gamma_j\eta^{-1}) = \Xi^{-1}\mathcal{N}_*(b_j)\Xi, \qquad 0 \le j \le 26. \tag{F.23.22}$$

This follows by cancelling the identical based paths; it keeps the original source basepoint and every original target label.

Extend the finite-point path $\Xi$ by successive supported local disk isotopies. The image of the standard disk containing the first eight points and excluding the ninth specifies the permitted F.22 disk at the original base configuration. The inverse isotopy and (F.23.19) give its inverse chart. Applying the isotopy to a loop gives a square whose two side paths are $\Xi$; that square identifies the induced fiber-coordinate map with the groupoid conjugation in (F.23.22). The original F.22 section still has the constant comparison paths at $(p, n)$. Forgetting labels from the same square proves all its based filling comparisons.

The twenty-seven actual source generators freely generate $\pi_1(Y^\dagger, s_0)$. Their recorded images therefore give its entire homomorphism. The universal full completion maps give the unique

continuous map on $L^\dagger$ with these images. It equals the original completed $\Phi^\dagger$, since the two maps agree on its dense discrete source. This proves the complete-source assertion (F.23.3).

*The two retained source sections and the exact marginal maps.* The eleven old generators give a particular section $i_{\mathrm{geo}} : L \to L^\dagger$ of the original filling $q$. Retain the F.19 section $i$ as well. On the actual target coordinates put

$$c(\ell) = \Phi^\dagger(i_{\mathrm{geo}}(\ell) i(\ell)^{-1}) \in K^+, \qquad j_{\mathrm{geo}}(\ell) = c(\ell) j(\ell). \tag{F.23.23}$$

These formulas follow because both sections have the same $q$-image $\ell$. The exact comparison of the two pullback semidirect coordinates is

$$(d, \ell) \longmapsto (dc(\ell)^{-1}, \ell), \qquad (e, \ell) \longmapsto (ec(\ell), \ell). \tag{F.23.24}$$

Both maps express the same actual element, and their composites are the identity. The original F.19 meridians keep their original whiskers. The twenty-seven new lassos and the old generators define the same actual source filling kernel; restricting the same $\Phi^\dagger$ to that kernel gives exactly $K^+$ in either coordinate system.

Delete strand 9 from each $b_j$ to obtain its positive eight-strand braid, delete strand 8 to obtain its negative one, and delete both to obtain the seven-old-strand braid. The algorithm tracks the literal labels through every crossing. The resulting paths and (F.23.22) are the full $G$ forgetting maps $F_+, F_-$ before their restrictions to $J$. The inverse triangular fiber maps of F.20 and F.22 commute with these fillings: a deleted lower-whiskered clockwise generator becomes trivial, and every retained one becomes the corresponding generator in the retained order. Equivalently, filling the same variables in

$$\tau_i(X) = X_n^{-1} \cdots X_{i+1}^{-1} X_i^{-1} X_{i+1} \cdots X_n \tag{F.23.25}$$

gives exactly the triangular word in that retained order. The additional calibration producer checks the full pure-braid filling identity on every standard pure generator at the ranks and deletion sets used here, as well as the Artin relations and both triangular inverse maps. Thus its finite list of free-word identities proves those homomorphism identities on the entire pure braid groups.

Each of the sixteen collision words has trivial old braid and trivial opposite-sign marginal braid, both by free reduction of the recorded words and by the actual forgetting maps. Its remaining linking vector is supported on its selected old/new pair with value $-1$. This is the counterclockwise simple holomorphic collision in the retained clockwise Artin convention. The full words, including this sign, are retained.

*Exact finite evaluation of the marginal maps.* The following recurrence evaluates each recorded eight-strand word in any finite group, without expanding intermediate free words. Begin with $x_1, \ldots, x_7$, set $x_8 = 1$, and set all conjugator values $c_j = 1$. For a positive letter $\sigma_j$ use

$$(x_j, x_{j+1}, c_j, c_{j+1}) \longmapsto (x_j x_{j+1} x_j^{-1},\ x_j,\ x_j c_{j+1},\ c_j), \tag{F.23.26}$$

and for its inverse use

$$(x_j, x_{j+1}, c_j, c_{j+1}) \longmapsto (x_{j+1},\ x_{j+1}^{-1} x_j x_{j+1},\ c_{j+1},\ x_{j+1}^{-1} c_j). \tag{F.23.27}$$

Induction on the displayed Artin substitutions proves these identities for the images and their labelled conjugators at each prefix. At a pure final word put $u = c_8^{-1}$; the final seven $x_i$ evaluate the filled automorphism $\alpha$. The actual conjugation on the fiber, proved by the cocycle and inverse triangular comparison, evaluates as

$$X_i \longmapsto u x_i u^{-1}. \tag{F.23.28}$$

For the negative marginal these are precisely the Artin coordinates of $V_-$. For the positive marginal, a seven-tuple of values on its actual geometric generators first undergoes $\tau_7$; use those as the initial $x_i$. A collision seed then has value $u$. For a source action apply $\tau_7$ also to the final tuple in (F.23.28). This is exactly the evaluation of the two inverse coordinate maps, since $\tau_7^2 = 1$.

There are eight collision seeds for either sign, and eleven old source automorphisms. They are the computed actual images and actions for $i_{\mathrm{geo}}$. The image of the signed source filling kernel is the closed subgroup generated by all their $L$-conjugates, as follows from the free source basis and its full completed filling. Its image is exactly $\mathcal{K}_\pm$ by F.22.21 and (F.23.23)–(F.23.24).

*The finite candidate reduction with actual peripheral maps.* F.21 and the actual surjections $p_\pm : D^+ \to R_\pm$ give

$$\overline{\langle\langle \mathcal{K}_\pm \rangle\rangle}_{R_\pm} = R_\pm. \tag{F.23.29}$$

Consequently in every finite image its image normally generates the image of $R_\pm$, and has that entire image when the latter is nilpotent, by F.21's finite argument.

The four deleted-puncture coordinates also retain their actual meridians. For a collision over old label $u$, the simple local parameter $a - v_u$, or $-a - v_u$, takes its counterclockwise source circle once around that collision. Transport its whisker by the labelled plane isotopy along the specified source path. The inverse endpoint isotopy carries the local circle to a based peripheral loop in the original seven-punctured plane. Comparing its transported whisker with the specified lower whisker gives the actual conjugating loop between these two peripheral loops. In the positive geometric coordinates this identifies the seed with a conjugate of $u_j^{-1}$. In the negative coordinates (F.23.25) identifies $U_j$ with a conjugate of the inverse clockwise lasso; the seed is consequently a conjugate of $U_j$. These comparisons preserve orientation and the named puncture. In particular, in every finite homomorphism the seed and its named deleted coordinate have exactly the same order.

For $H = S_3$, the proper subgroups are the trivial subgroup, three subgroups $C_2$, and the normal subgroup $C_3$. For $H = A_4$, they are the trivial subgroup, three subgroups $C_2$, four subgroups $C_3$, and the normal subgroup $V_4$. The $S_3$ list follows directly from its three transpositions and two three-cycles; a subgroup with two distinct transpositions, or with both a transposition and a three-cycle, is the full group. The $A_4$ list, its actual element sets and normal closures, is also obtained by the exhaustive finite subgroup closure algorithm in the coefficient certificate. All these proper subgroups are nilpotent. The proper subgroups normally generating $S_3$ are its three $C_2$ subgroups; those normally generating $A_4$ are its four $C_3$ subgroups.

Thus a failure of (F.23.4) requires both $\rho(V_\pm) = H$ and $\rho(R_\pm) = H$. Its marginal image then lies in a proper subgroup $M = C_2$ for $S_3$, or $M = C_3$ for $A_4$. Conjugation in $H$ carries these respective subgroups to one fixed $M$, so all homomorphisms with such a failure occur in the finite enumeration with that fixed subgroup. The peripheral comparison just proved makes all four deleted-coordinate values have order one or two in the first case, and order one or three in the second. At least one is nonidentity, since they normally generate the image of $R_\pm$.

Every seven-tuple in $H^7$ is the values of a unique continuous homomorphism from the actual $\widehat{F}_7$. The complete possible-failure candidate counts are therefore

$$\begin{aligned} S_3 :&\quad 6^3(4^4 - 1) - 3\,2^3(2^4 - 1) = 54,720, \\ A_4 :&\quad 12^3(9^4 - 1) - 4\,3^3(3^4 - 1) = 11,327,040. \end{aligned} \tag{F.23.30}$$

The subtracted families are exactly the tuples contained in a proper subgroup while having a nonidentity deleted coordinate. They are disjoint because that coordinate determines the particular $C_2$ or $C_3$ subgroup.

*The exhaustive source-orbit test and its precise conclusion.* For each candidate tuple evaluate the eight actual collision words by (F.23.26)–(F.23.28), and retain the tuple precisely when all

eight values lie in the fixed $M$. Call the finite set of retained tuples $\mathcal{T}_0$. Each of the eleven actual old source automorphisms acts as a permutation of $\mathrm{Hom}(V_\pm, H)$, by precomposition. It preserves generation of the image. Its actual pure labelled point-pushing action carries each deleted peripheral generator to a conjugate of itself, so it also preserves the candidate order restrictions and the condition that some deleted value is nonidentity. Record all eleven images of every element of $\mathcal{T}_0$. Delete an element when one of these images is outside $\mathcal{T}_0$, and repeat on the retained set.

This computes exactly the largest subset of $\mathcal{T}_0$ invariant under the eleven permutations. Indeed a subset mapped into itself by a permutation of a finite set is also invariant under its inverse. Hence forward invariance under the eleven maps is invariance under their generated group. Their profinite source completion has the same finite orbits: the continuous action on this finite representation set has dense discrete source image and therefore equal finite image. The resulting invariant set consists exactly of the candidates for which every source-conjugate collision seed has image in $M$. Those source-conjugates generate the actual marginal subgroup, so this is the exact remaining failure set.

Both signs were enumerated separately. The complete executed results are

| $H$ | candidates | $\lvert\mathcal{T}_0\rvert$ | first deletion | survivors |
|---|---|---|---|---|
| $S_3$ | $54,720$ | $624$ | $624$ | $0$ |
| $A_4$ | $11,327,040$ | $15,552$ | $15,552$ | $0$ |

(F.23.31)

The table holds for each sign. The outputs retain every element of $\mathcal{T}_0$, its seven coordinates, all eleven image keys and edges, and its deletion layer. Multiplication is literal composition of the enumerated permutations. The $A_4$ run streams the same exhaustive candidate set in finite batches; it does not sample it. The empty invariant sets prove (F.23.4) for every specified continuous homomorphism, including the cases already settled by the finite nilpotent image argument.

*Executed certificate and retained scope.* The complete source geometry is produced by `full_cross_source_geometry_stage.py`; `full_cross_source_basepaths_stage.py` retains the chart and endpoint comparisons; `full_cross_marginal_braids_stage.py` records all full and deleted words. The two finite producers are `marginal_s3_source_stage.py` and `marginal_a4_source_stage.py`. The sign-symmetry producer passes 64 exact checks, and `source_artin_calibration_stage.py` passes $2,473$ exact polynomial and free-word checks. They are retained under `code/part_a/` with their full outputs, stdout, and hashes. Independent isolated execution reproduced all 35 complete-source, symmetry, and finite-scan output files byte for byte; the calibration output was independently reproduced as well.

All original sections, fillings, signs, and based comparison paths enter the displayed maps. The finite claim is the equality of the two images under every map $V_\pm \to S_3$ or $V_\pm \to A_4$. The remaining full marginal inclusion and the interaction subgroup $K^+ \cap W$ of F.22 retain their original source-membership questions. □

**F.24. [Proved] All-field rank-one cocycles and the actual marginal affine fixed points.** Retain the actual continuous inclusions $\mathcal{K}_\pm \le R_\pm \le V_\pm$ and the source maps of F.19–F.23. Let $k$ be a finite field, with its discrete topology, and let $\chi : V_\pm \to k^\times$ be a continuous character whose restriction to $R_\pm$ is nontrivial. Give the additive group of $k$ the left action $g \cdot b = \chi(g)b$, and denote this module by $k_\chi$. The actual restriction maps are injective:

$$\begin{aligned} Z^1_{\mathrm{cont}}(V_\pm, k_\chi) &\longrightarrow Z^1_{\mathrm{cont}}(\mathcal{K}_\pm, k_{\chi|\mathcal{K}_\pm}), \\ H^1_{\mathrm{cont}}(V_\pm, k_\chi) &\longrightarrow H^1_{\mathrm{cont}}(\mathcal{K}_\pm, k_{\chi|\mathcal{K}_\pm}). \end{aligned} \tag{F.24.1}$$

Here a crossed homomorphism satisfies $d(gh) = d(g) + \chi(g)d(h)$, and the principal crossed homomorphisms are $d_b(g) = (\chi(g) - 1)b$, for $b \in k$.

Write $\mathrm{Aff}_1(k) = k \rtimes k^\times$, acting on $k$ by $(b,a)z = b + az$, and let $T_k = \{(b,1) : b \in k\}$ be its translation subgroup. For every continuous homomorphism $\rho : V_\pm \to \mathrm{Aff}_1(k)$ one has

$$\mathrm{Fix}_k\, \rho(\mathcal{K}_\pm) = \mathrm{Fix}_k\, \rho(R_\pm). \tag{F.24.2}$$

For the representations whose scalar character is nontrivial on $R_\pm$, both sides also equal $\mathrm{Fix}_k\, \rho(V_\pm)$. For every representation in the specified class

$$\{\rho \in \mathrm{Hom}_{\mathrm{cont}}(V_\pm, \mathrm{Aff}_1(k)) : \rho(\mathcal{K}_\pm) \cap T_k = \{1\}\},$$

the exact marginal images agree:

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm). \tag{F.24.3}$$

The coefficient calculation proving (F.24.1) has integer Laurent coefficients and proves its rank assertion over every field, including characteristics two and three. Its input is the literal negative marginal braid data of F.23, with the same source lassos and comparison maps. The positive assertion is transported by the explicit F.23.13 isomorphism. The full marginal equalities and the equality $K^+ = D^+$ remain unresolved.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. The retained words and the exact coefficient maps.* We first use the negative coordinates

$$\begin{gathered} V_- = \widehat{F}(X_1, X_2, X_3, X_4, X_5, X_6, X_7), \qquad X_{j+3} = U_j \quad (1 \le j \le 4), \\ R_- = \ker\bigl(V_- \longrightarrow \widehat{F}(X_1, X_2, X_3)\bigr). \end{gathered} \tag{F.24.4}$$

These are the actual F.22.5 filling coordinates. Put $\Lambda = \mathbb{Z}[t_1^{\pm 1}, \dots, t_7^{\pm 1}]$. Let $F_7$ be the discrete free group on these same seven generators, with its dense completion map $F_7 \to V_-$. The universal character and crossed homomorphism are

$$t : F_7 \longrightarrow \Lambda^\times, \quad t(X_i) = t_i, \qquad D : F_7 \longrightarrow \Lambda^7, \quad D(X_i) = e_i, \tag{F.24.5}$$

where $\Lambda^7$ consists of row vectors and $D(vw) = D(v) + t(v)D(w)$. These maps are well defined on reduced words: the inverse formula is $D(v^{-1}) = -t(v)^{-1}D(v)$, which cancels an adjacent letter and its inverse. Equivalently they are the two coordinates of the group homomorphism $F_7 \to \Lambda^7 \rtimes \langle t_1, \dots, t_7 \rangle$ specified on its free generators.

For a character $\chi$ and a continuous crossed homomorphism $d$ as in the statement, let

$$\mathrm{ev}_\chi : \Lambda \longrightarrow k, \qquad t_i \longmapsto \chi(X_i), \qquad v_d = (d(X_1), \dots, d(X_7))^{\mathsf{T}}.$$

This is a ring homomorphism because each $\chi(X_i)$ is nonzero. Induction on the reduced word gives the exact formula

$$d(w) = \mathrm{ev}_\chi(D(w))v_d \qquad (w \in F_7). \tag{F.24.6}$$

All words below are finite words in this dense free group, interpreted in $V_-$ by its stated completion map. In particular, the calculation does not substitute a completion of a different source subgroup.

Let $\nu_0, \dots, \nu_7$ be the eight negative collision words in F.23, in the order

$$(u, \epsilon) = (1,+), (1,-), (\zeta^2,+), (\zeta^2,-), (\zeta^3,+), (\zeta^3,-), (\zeta^4,+), (\zeta^4,-). \tag{F.24.7}$$

The sign here selects the two roots of each negative collision square $B_u$; every word in (F.24.7) belongs to the negative marginal. The original paths $\eta$, $G$, $T$, and $\Xi$ and their affine normalization are exactly those retained in F.23.19–22. Let $B_i : V_- \to V_-$, $0 \le i \le 10$, be conjugation on this

point-pushing group by the image of the $i$th old source generator under the geometric source section $j_{\mathrm{geo}}$ of F.23.23. Thus $B_i$ is a continuous automorphism and

$$B_i(\mathcal{K}_-) = \mathcal{K}_-. \tag{F.24.8}$$

Indeed the source filling kernel is normal in the source; its image is $K^+$; projection to the negative marginal commutes with this conjugation. These are the original maps, with the old and geometric section comparison $j_{\mathrm{geo}}(l) = c(l)j(l)$ from F.23.23–24. The old source indices are 0 for its zero puncture, $1, \dots, 5$ for $+\zeta^j$, and $6, \dots, 10$ for $-\zeta^j$, in each case $0 \le j \le 4$.

Every $B_i$ preserves the abelianization of $V_-$. For the explicit proof in these coordinates, start the decorated Artin recurrence F.23.26–28 with the first seven characters $t_1, \dots, t_7$ and the eighth character one. A crossing exchanges the two characters, and each recorded old braid is pure. Its final conjugation by $u$ does not change them. Hence $t(B_i(X_j)) = t_j$, and

$$A_i = \big(D(B_i(X_j))\big)_{1 \le j \le 7}, \qquad D(B_i(w)) = D(w)A_i. \tag{F.24.9}$$

Both identities follow on generators and then on products and inverses from the crossed-homomorphism identity. Consequently the following sixty-four rows evaluate the crossed homomorphism on actual elements of $\mathcal{K}_-$:

$$\begin{gathered} a_r = D(\nu_r) \quad (0 \le r \le 7), \qquad a_r A_i \quad (0 \le r \le 7, \quad i \in \mathscr{I}), \\ \mathscr{I} = (2, 9, 8, 10, 1, 0, 7). \end{gathered} \tag{F.24.10}$$

Number the rows from zero, first the eight $a_r$, then the eight rows for each index of $\mathscr{I}$ in the displayed order.

*Exact production and rank of the sixty-four rows.* For completeness we give the full arithmetic recipe and the terminal identities of its certificate. Encode an element of the universal affine group by $(a, v)$, where $a \in \mathbb{Z}^7$ is its character exponent and $v \in \Lambda^7$ its crossed-homomorphism row. Then

$$(a, v)(b, w) = (a + b, v + t^a w), \qquad (a, v)^{-1} = (-a, -t^{-a} v). \tag{F.24.11}$$

Initialize the eight Artin images as $(e_j, e_j)$ for $1 \le j \le 7$ and $(0, 0)$ for the erased eighth generator, and initialize all eight conjugators as $(0, 0)$. Apply the exact positive and inverse substitutions F.23.26–27 in this group, using (F.24.11). Put $u = c_8^{-1}$ at the end. The row of $u$ is the seed row; the rows of $u x_j u^{-1}$ form $A_i$ for an old word. This applies the proved decorated recurrence to a specified group, so it evaluates the same actual words as F.23.28. The separate producer records eleven old matrices and eight seed rows. All of them also satisfy

$$D(w)(t_1 - 1, \dots, t_7 - 1)^{\mathsf{T}} = t(w) - 1, \tag{F.24.12}$$

which follows directly from (F.24.5).

Here is the exact row elimination used to check their rank. In chart $j$, work over $\Lambda[(t_j - 1)^{-1}]$. A row normalization multiplies by a Laurent monomial and divides by a common power of $t_j - 1$. A pivot $p$ is accepted only with the exact coefficient identity $p = \varepsilon t^a (t_j - 1)^m$, where $\varepsilon \in \{1, -1\}$ and $m \ge 0$. With coefficient $c$ in its column, replace a later row $r$ by $pr - cr_p$ and normalize again. Each operation has an invertible row multiplier in the displayed ring and preserves the row module there. Every polynomial is stored as its integer coefficients and seven integer exponents; the record retains the input coefficients, shifts, division counts, pivot columns, and resulting rows.

The following are the complete pivot orders, with zero-based row indices and one-based column indices. The last chart first specializes $t_4 = t_5 = t_6 = 1$. Each listed final pivot coefficient

is a signed Laurent monomial, so every pivot remains nonzero in every field on its chart.

| $j$ | pivot rows | pivot columns | row operations |
|---|---|---|---|
| 4 | $11, 8, 0, 6, 4, 1$ | $5, 4, 1, 7, 6, 3$ | 389 |
| 5 | $11, 8, 3, 6, 4, 2, 34$ | $5, 4, 1, 7, 6, 2, 3$ | 444 |
| 6 | $11, 8, 13, 14, 20, 5, 28$ | $5, 4, 6, 7, 2, 1, 3$ | 445 |
| 7 | $0, 2, 4, 14, 39, 6, 46$ | $4, 5, 6, 7, 2, 1, 3$ | 196 |

(F.24.13)

The first chart has one remaining column, column two. After its six displayed pivots, the retained rows numbered five and forty-eight are exactly

$$r_5 = (1 - t_6)e_2,$$
$$r_{48} = \big(t_1t_2t_3t_4t_5t_7(t_6 - 1) - 1\big)e_2.$$

They give the explicit seventh pivot

$$r_{48} + t_1t_2t_3t_4t_5t_7r_5 = -e_2. \tag{F.24.14}$$

Thus all four charts have rank seven. The first three charts cover $t_4 \neq 1$, $t_5 \neq 1$, and $t_6 \neq 1$. The remaining stratum has those three values equal to one and $t_7 \neq 1$, and is covered by the fourth chart. This proves rank seven at every seven-tuple of nonzero field elements for which some $t_j \neq 1$, $4 \leq j \leq 7$. No integer other than 1 or $-1$ is inverted.

The accompanying exact certificate contains every row and operation just described. Its independent verifier imports none of the three producers: it reconstructs the nineteen word evaluations with the group law (F.24.11), reconstructs all sixty-four rows, replays every row operation, proves every division by multiplying back by $t_j - 1$, and checks the triangular pivot orders and (F.24.14). It also replays the longer scalar-ideal certificate retained from the original search. There are 13 982 exact assertions in this independent verification. The manifest fixes the braid input, all producers and outputs, and this verification by size and SHA-256. These finite polynomial identities, together with their invertible operations, establish the asserted rank over all characteristics.

*Restriction, continuity, and the sign comparison.* The restriction $\chi|R_-$ is nontrivial precisely when at least one of $\chi(X_4), \dots, \chi(X_7)$ differs from one: $R_-$ is the closed normal subgroup generated by these four generators, and the target $k^\times$ is abelian. For a crossed homomorphism vanishing on $\mathcal{K}_-$, every row in (F.24.10) annihilates $v_d$ by (F.24.6) and (F.24.8)–(F.24.9). The rank assertion therefore gives $v_d = 0$. The crossed-homomorphism rule then gives zero on the dense group $F_7$; continuity into the finite Hausdorff space $k$ gives zero on $V_-$. This proves injectivity on $Z^1$. For a class restricting to zero in $H^1$, its representative restricts to $(\chi - 1)b$ for some $b \in k$. Subtract that same principal crossed homomorphism on $V_-$ and apply the proved $Z^1$ injectivity. This proves injectivity on $H^1$ with the exact coefficient restriction in (F.24.1).

For the positive sign use the specific continuous isomorphism $\mathcal{B}_+ : V_- \to V_+$ of F.23.13. It carries $R_-$ onto $R_+$ and $\mathcal{K}_-$ onto $\mathcal{K}_+$. Pullback sends $(\chi, d)$ to $(\chi\mathcal{B}_+, d\mathcal{B}_+)$; pullback along $\mathcal{B}_+^{-1}$ is its inverse. It preserves the crossed-homomorphism equations and the principal rows and commutes literally with restriction. Nontriviality on $R_+$ transfers to nontriviality on $R_-$. These maps prove both positive assertions.

*The affine representations and their fixed points.* Write a given representation as $\rho(g) = (d(g), \chi(g))$. Its multiplication law makes $d$ a continuous crossed homomorphism. For a point $z \in k$ put

$$d_z(g) = \rho(g)z - z = d(g) + (\chi(g) - 1)z. \tag{F.24.15}$$

It is another continuous crossed homomorphism for the same character. A point fixed by $\rho(\mathcal{K}_\pm)$ makes $d_z$ vanish there. For $\chi|R_\pm$ nontrivial, (F.24.1) makes $d_z$ zero on $V_\pm$. The inclusions of fixed sets from $\mathcal{K}_\pm \leq R_\pm \leq V_\pm$ now give the three equal fixed sets claimed above.

For $\chi|R_\pm = 1$, the image of $R_\pm$ lies in the abelian group $T_k$. Apply the F.21 normal-generation conclusion through the actual surjection $p_\pm : D^+ \to R_\pm$ of F.22.6 and then through $\rho|R_\pm$. The normal closure of $\rho(\mathcal{K}_\pm)$ in $\rho(R_\pm)$ is $\rho(R_\pm)$; in this abelian group the closure is the subgroup itself. Hence their images agree and their fixed sets agree. This completes (F.24.2).

Finally put $M = \rho(\mathcal{K}_\pm)$ for a representation in the class stated before (F.24.3). Its scalar projection $M \to k^\times$ is injective because its kernel is $M \cap T_k$. Thus $|M|$ divides $|k| - 1$ and is invertible in $k$. The explicit point

$$z_M = |M|^{-1} \sum_{m \in M} m(0) \tag{F.24.16}$$

is fixed by $M$. For $h = (b, a) \in M$, applying its affine formula to this average gives $h(z_M) = |M|^{-1} \sum_{m \in M}(hm)(0) = z_M$, because left multiplication by $h$ permutes $M$. For a nontrivial character on $R_\pm$, (F.24.2) and its stronger assertion put all of $\rho(V_\pm)$ in the stabilizer of $z_M$. The scalar map on that stabilizer has the explicit inverse $a \mapsto ((1 - a)z_M, a)$, so the stabilizer is abelian. F.21 normal generation, transported through the same two actual maps as above, gives (F.24.3). For a trivial character on $R_\pm$, the preceding abelian translation argument already gives (F.24.3).

The remaining finite-affine case consists exactly of representations with $\rho(\mathcal{K}_\pm) \cap T_k$ nontrivial. The argument has not proved equality of the two marginal images in that class. In particular it has not constructed an ambient invariant quotient that kills this translation intersection. The actual inclusions of F.23.1 and the interaction kernel of F.22.6–8 remain the maps whose surjectivity is under investigation. □

**F.25. [Proved] The actual affine marginal images with a common scalar field.** Retain $\mathcal{K}_\pm \le R_\pm \le V_\pm$ and their actual maps from F.19–F.24. Let $k$ be a finite field of characteristic $p$, and let $\rho : V_\pm \to \mathrm{Aff}_1(k)$ be continuous. Write $\chi$ for its scalar character and put

$$\begin{gathered} M = \rho(\mathcal{K}_\pm), \qquad N = \rho(R_\pm), \qquad D = \chi(V_\pm), \qquad C = \chi(R_\pm), \\ U = \{b \in k : (b, 1) \in M\}, \qquad k_0 = \mathbb{F}_p(D), \qquad k_1 = \mathbb{F}_p(C). \end{gathered} \tag{F.25.1}$$

The subgroup $U$ is a vector subspace of $k$ over $k_1$. For the class of these representations in which $U$ is also a vector subspace over $k_0$, the actual finite images agree:

$$M = N. \tag{F.25.2}$$

Consequently the equality holds for every representation in the explicitly specified class

$$\{\rho : \ C = \{1\} \text{ or } k_1 = k_0\}. \tag{F.25.3}$$

In particular, for every prime $p$ and every continuous homomorphism to the full affine group of its prime field,

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm) \qquad (\rho : V_\pm \to \mathrm{Aff}_1(\mathbb{F}_p) \text{ continuous}). \tag{F.25.4}$$

All comparison maps, the choice of an actual scalar lift in $M$, and the coefficient maps used in the proof are specified below. The finite scalar-field class $k_1 \subsetneq k_0$ with $U$ failing to be a $k_0$-subspace remains outside the proved equality; full marginal surjectivity and $K^+ = D^+$ remain unresolved.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof.* Fix either sign throughout. First transport the F.21 normal-generation statement through the actual continuous surjection $p_\pm : D^+ \to R_\pm$ of F.22.6. Its restriction to $K^+$ has image $\mathcal{K}_\pm$. The image of the closed normal closure is the closed normal closure of the image: one inclusion

follows by applying the map to conjugates, and the other by taking the inverse image of a closed normal subgroup containing the image. The image is closed because the source is compact and the target Hausdorff. Thus

$$\overline{\langle\langle \mathcal{K}_\pm \rangle\rangle}_{R_\pm} = R_\pm, \qquad \langle\langle M \rangle\rangle_N = N, \qquad \chi(\mathcal{K}_\pm) = C. \tag{F.25.5}$$

The second equality applies the actual map $\rho|R_\pm$; the third applies its scalar projection to the finite abelian group $k^\times$. When $C = \{1\}$, the group $N$ is a subgroup of the abelian translation group, so (F.25.5) already gives $M = N$.

We now take $C \neq \{1\}$. The set $U$ is an additive subgroup of $k$, hence an $\mathbb{F}_p$-vector subspace. Conjugation of a translation by an actual element of $M$ with scalar $c$ is given by the literal formula

$$(b, c)(u, 1)(b, c)^{-1} = (cu, 1). \tag{F.25.6}$$

The last equality of (F.25.5) supplies such an element for every $c \in C$. Consequently $U$ is stable under every scalar in $C$, hence under the ring $\mathbb{F}_p[C]$. This finite subring of $k$ is a field: multiplication by any nonzero element is an injection of its finite set and therefore is onto, giving its inverse in the ring. Thus $\mathbb{F}_p[C] = k_1$ and $U$ is a $k_1$-subspace, as asserted.

The finite multiplicative subgroup $C$ is cyclic. One direct verification chooses an element of maximal order $n$: the prime-power components of element orders in the abelian group show that every element has order dividing $n$, since otherwise their commuting components would give a larger order. All elements then satisfy $X^n - 1 = 0$ in the field, so $|C| \leq n$. The chosen cyclic subgroup has $n$ elements, proving $|C| = n$. Choose a generator $c \neq 1$ and an actual lift $m = (b, c) \in M$. Define

$$z = \frac{b}{1 - c}, \qquad \tau_z(w) = w + z, \qquad \rho_z(g) = \tau_z^{-1} \rho(g) \tau_z = \big(d(g) + (\chi(g) - 1)z, \chi(g)\big), \tag{F.25.7}$$

where $\rho(g) = (d(g), \chi(g))$. The maps $\tau_z$ and $\tau_z^{-1} = \tau_{-z}$ are explicit inverse affine bijections of $k$, and their conjugations are inverse group automorphisms of $\mathrm{Aff}_1(k)$. The chosen lift now equals $(0, c)$. Conjugating by a translation fixes every element of $T_k$, so the translation intersection remains exactly $U$.

Let $M_z = \rho_z(\mathcal{K}_\pm)$ and $N_z = \rho_z(R_\pm)$. Every element of $M_z$ has scalar $c^j$; multiplication by its element $(0, c^{-j})$ leaves a translation in $U$. Conversely $M_z$ contains all translations in $U$ and the generator $(0, c)$. We have therefore proved the literal equality and the associated inverse coordinates

$$\begin{aligned} M_z &= U \rtimes C, \\ U \rtimes C &\longrightarrow M_z, \qquad (u, a) \longmapsto (u, a), \\ M_z &\longrightarrow U \rtimes C, \quad (b, a) \longmapsto (b, a). \end{aligned} \tag{F.25.8}$$

The group law is $(u, a)(v, a') = (u + av, aa')$, with $U$ stable under $C$ by (F.25.6). Thus the complement used here has been constructed from the actual element $m \in M$.

Now let $U$ be a $k_0$-subspace, as in the first assertion. The field $k$ is a finite-dimensional vector space over $k_0$. Let

$$\mathscr{L}_U = \{\ell \in \mathrm{Hom}_{k_0}(k, k_0) : \ell(U) = 0\}.$$

These exact coefficient maps satisfy

$$U = \bigcap_{\ell \in \mathscr{L}_U} \ker \ell. \tag{F.25.9}$$

For a proof, extend a $k_0$-basis of $U$ to a basis of $k$. Coordinate projections on the additional basis vectors belong to $\mathscr{L}_U$ and detect every nonzero coordinate outside $U$; the reverse inclusion is part of the definition.

Put $d_z(g) = d(g)+(\chi(g)-1)z$, the translation coordinate in (F.25.7). The affine multiplication law makes it a continuous $k$-valued crossed homomorphism. For each $\ell \in \mathscr{L}_U$ the composite

$$\ell d_z : V_\pm \longrightarrow (k_0, +), \qquad (\ell d_z)(gh) = (\ell d_z)(g) + \chi(g)(\ell d_z)(h) \tag{F.25.10}$$

is an actual continuous crossed homomorphism with the character $\chi : V_\pm \to k_0^\times$. Indeed $D \subset k_0^\times$ and $\ell$ is $k_0$-linear, which proves exactly the displayed multiplication identity. All target spaces are finite and discrete, so composition also preserves continuity. Equation (F.25.8) gives $d_z(\mathcal{K}_\pm) \subset U$, and therefore $\ell d_z$ vanishes on $\mathcal{K}_\pm$. The character is nontrivial on $R_\pm$ because $C \neq \{1\}$. Apply the proved F.24.1 restriction injection over the finite field $k_0$. It gives $\ell d_z = 0$ on all of $V_\pm$, for every such $\ell$. Together with (F.25.9) this proves

$$d_z(V_\pm) \subset U, \qquad N_z \subset U \rtimes C = M_z. \tag{F.25.11}$$

The inclusion $\mathcal{K}_\pm \leq R_\pm$ supplies the reverse inclusion. Conjugating back by $\tau_z$ gives the exact equality (F.25.2).

For the class $k_1 = k_0$, the $k_0$-linearity of $U$ has already been proved using (F.25.6). Combining this with the separate case $C = \{1\}$ proves (F.25.3). For $k = \mathbb{F}_p$, both generated subfields are $\mathbb{F}_p$ for every character, so (F.25.4) follows for every prime, including two and three. The proof applies to either of the actual marginals, using its corresponding F.24 injection; that injection already contains the explicit F.23 sign-symmetry comparison.

For the remaining class $k_1 \subsetneq k_0$, (F.25.6) supplies $k_1$-linearity of $U$. The coefficient map (F.25.10) requires $k_0$-linearity. The proof has established the passage precisely for the representations specified in the statement, and supplies no quotient of the actual ambient module for an arbitrary such $U$. □

**F.26. [Proved] The exact trace obstruction and all affine marginal images with scalar field of order nine.** Retain the actual inclusions $\mathcal{K}_\pm \leq R_\pm \leq V_\pm$ of F.19–F.25. Fix either sign, and temporarily write these groups as $K \leq R \leq V$. A proper affine marginal image $\rho(K) \subsetneq \rho(R)$, for a continuous $\rho : V \to \mathrm{Aff}_1(k)$ over a finite field, has an actual continuous affine image of the following form:

$$\begin{aligned} \rho' : V \longrightarrow k_0 \rtimes D, \qquad \rho'(V) = k_0 \rtimes D, \qquad \rho'(R) = k_0 \rtimes C, \\ \rho'(K) \ \leq \ \ker(\mathrm{Tr}_{k_0/k_1} : k_0 \to k_1) \rtimes C, \\ C = \chi(R) = \chi(K) \neq \{1\}, \qquad D = \chi(V), \qquad k_1 = \mathbb{F}_p(C) \subsetneq k_0 = \mathbb{F}_p(D). \end{aligned} \tag{F.26.1}$$

Here $\chi$ is the original scalar character, retained by the explicit conjugation and coefficient map constructed below. In particular the last group in (F.26.1) is a proper subgroup of the displayed $R$ image.

For any finite-field data $k_0, \chi, C, k_1$ as in (F.26.1), including the equality case $k_1 = k_0$, put $q_1 = |k_1|$ and $m = [k_0 : k_1]$. The exact coefficient map is

$$\begin{aligned} \mathscr{T}_\chi : Z^1_{\mathrm{cont}}(V, k_{0,\chi}) &\longrightarrow Z^1_{\mathrm{cont}}(K, k_{1,\chi|K}), \\ d &\longmapsto \mathrm{Tr}_{k_0/k_1}(d|K). \end{aligned} \tag{F.26.2}$$

Its known kernel subspace is

$$\mathscr{P}_\chi = \{\, g \mapsto (\chi(g) - 1)b : b \in k_0, \ \mathrm{Tr}(b) = 0 \,\}. \tag{F.26.3}$$

Every element of $\ker \mathscr{T}_\chi \setminus \mathscr{P}_\chi$ gives a proper actual affine marginal image, and every proper affine marginal image gives such an element for the data in (F.26.1). Thus $\ker \mathscr{T}_\chi = \mathscr{P}_\chi$ for all the stated data is an exact remaining criterion for the full scalar affine image equality. The proof

gives an explicit coefficient isomorphism identifying this criterion with the summed restrictions for the distinct characters $\chi^{q_1^j}$, $0 \le j < m$. Its rank target is $6m+1$ on $7m$ columns; no general attainment of that rank is asserted here.

For the scalar field of order nine that rank is attained. More precisely, for every finite field $k$ and every continuous $\rho : V_\pm \to \mathrm{Aff}_1(k)$ whose scalar values generate a subfield of order nine,

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm). \tag{F.26.4}$$

Consequently this equality holds for every continuous map $V_\pm \to \mathrm{Aff}_1(\mathbb{F}_9)$, including maps whose scalar field is $\mathbb{F}_3$. The finite calculation proving the additional field case exhausts exactly 7 560 character tuples and retains the literal F.23 source automorphisms.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. Reduction through the actual affine image.* Write $H = \rho(V)$, $N = \rho(R)$, $M = \rho(K)$ and let $D, C$ be their scalar images as above. The F.21 normal generation, transported through the actual $D^+ \to R$ of F.22, gives

$$\langle\langle M \rangle\rangle_N = N, \qquad \chi(K) = \chi(R) = C. \tag{F.26.5}$$

These are exactly the maps and equalities proved in F.25.5. A proper marginal requires $C \neq \{1\}$, since $C = \{1\}$ makes $N$ abelian. Put $k_0 = \mathbb{F}_p(D)$ and $k_1 = \mathbb{F}_p(C)$. F.25 excludes equality of these two fields for a proper marginal.

Choose a generator $c$ of $C$ and an actual lift $(b, c) \in M$. Set $z = b/(1-c)$ and conjugate the representation by $\tau_z^{-1}$, where $\tau_z(w) = w + z$. As in F.25.7–8, this gives

$$M_z = U \rtimes C, \qquad U = \{u : (u, 1) \in M\}, \qquad (0, c) \in M_z, \tag{F.26.6}$$

with $U$ unchanged and $k_1$-linear. Let $A_H$ and $A_N$ be the translation subgroups of $H_z$ and $N_z$. Because $N_z$ is normal in $H_z$, conjugation by every actual scalar lift in $H_z$ preserves both subgroups and multiplies translations by its scalar. Thus both are vector subspaces over $\mathbb{F}_p[D] = k_0$. For $a \in A_H$ the exact commutator is

$$[(a, 1), (0, c)] = ((1-c)a, 1) \in N_z. \tag{F.26.7}$$

Since $1 - c \neq 0$ and $A_N$ is $k_0$-linear, this proves $A_H = A_N =: A$. For an arbitrary $(b', d) \in H_z$ the same multiplication gives $[(b', d), (0, c)] = ((1-c)b', 1) \in A$ and hence $b' \in A$. Multiplying by its inverse translation then gives the corresponding pure scalar element. Consequently

$$H_z = A \rtimes D, \qquad N_z = A \rtimes C, \qquad M_z = U \rtimes C, \qquad U \subsetneq A. \tag{F.26.8}$$

Here $A$ is $k_0$-linear and $U$ is $k_1$-linear.

Choose a $k_1$-linear functional $L : k \to k_1$ vanishing on $U$ and nonzero on $A$. To construct one, extend a basis of $U$ by an element of $A \setminus U$, then to a basis of $k$, and give that selected element value one and the other basis vectors value zero.

*The precise trace and functional factorization.* Write $m = [k_0 : k_1]$ and $q_1 = |k_1|$. The trace is

$$\mathrm{Tr}(x) = \sum_{j=0}^{m-1} x^{q_1^j}. \tag{F.26.9}$$

Its $q_1$th power equals itself, using $x^{q_1^m} = x$, so its values lie in $k_1$. It is $k_1$-linear and nonzero: its polynomial has degree $q_1^{m-1} < |k_0|$ and leading coefficient one, and therefore cannot vanish on

all of $k_0$. It is consequently onto $k_1$. Choose $\beta$ with $\mathrm{Tr}(\beta) = 1$. For every $a \neq 0$, $\mathrm{Tr}(a(\beta/a)) = 1$. Thus the pairing $\mathrm{Tr}(ab)$ is nondegenerate, and

$$k_0 \xrightarrow{\sim} \mathrm{Hom}_{k_1}(k_0, k_1), \qquad a \longmapsto (b \longmapsto \mathrm{Tr}(ab)) \tag{F.26.10}$$

is an isomorphism: it is injective and both sides have $k_1$-dimension $m$. This also covers characteristics dividing $m$.

Choose a $k_0$-basis $v_1, \ldots, v_r$ of $k$. For each $i$, (F.26.10) gives a unique $a_i \in k_0$ with $L(tv_i) = \mathrm{Tr}(a_i t)$ for all $t \in k_0$. Define

$$\ell : k \longrightarrow k_0, \qquad \ell\left(\sum_i t_i v_i\right) = \sum_i a_i t_i, \qquad L = \mathrm{Tr}\,\ell. \tag{F.26.11}$$

This is the required $k_0$-linear coefficient map. Its image on $A$ is a nonzero $k_0$-subspace of $k_0$, hence is all of $k_0$; its image on $U$ lies in $\ker \mathrm{Tr}$. The exact homomorphism of finite affine groups is

$$P_\ell : k \rtimes D \longrightarrow k_0 \rtimes D, \qquad (b, d) \longmapsto (\ell(b), d). \tag{F.26.12}$$

The identity $\ell(db) = d\ell(b)$ proves its compatibility with multiplication. All groups here are finite and discrete, so it is continuous. Composing it with the specified conjugation and with $\rho$ gives $\rho'$. Equations (F.26.8)–(F.26.12) prove every image assertion in (F.26.1), with the original scalar character retained.

*The exact trace-cocycle criterion.* For the data in (F.26.2), the restriction of $\chi$ to $K$ lies in $C \subset k_1^\times$. Thus composing a crossed homomorphism with the $k_1$-linear trace on $K$ proves precisely the target cocycle equation in (F.26.2); continuity is preserved. Each principal cocycle in (F.26.3) maps to zero. Because $C \neq \{1\}$, the parameter map $b \mapsto (\chi - 1)b$ is injective, and this subspace has $k_1$-dimension $m - 1$.

The representation $\rho'$ from (F.26.1) has a translation cocycle $d'$ in $\ker \mathscr{T}_\chi$. It is not principal: a principal affine representation fixes a point of the field, whereas $\rho'(V)$ contains every translation, including nonzero translations which move every point. This gives an element outside $\mathscr{P}_\chi$.

Conversely take $d \in \ker \mathscr{T}_\chi \setminus \mathscr{P}_\chi$. The specified homomorphism

$$\rho_d : V \longrightarrow k_0 \rtimes D, \qquad g \longmapsto (d(g), \chi(g)) \tag{F.26.13}$$

is continuous by the cocycle rule. Its translation subgroup is stable under $D$ and hence is a $k_0$-subspace of $k_0$. It cannot be zero. Otherwise the scalar projection would identify $\rho_d(V)$ with the cyclic group $D$. A generator lift $(b, a)$, $a \neq 1$, fixes $b/(1 - a)$, and all its powers fix that same point. Then $d$ is principal; its vanishing trace on $K$, together with $C \neq 1$, makes the parameter trace zero, contrary to the choice of $d$. Thus $\rho_d(V)$ contains all translations of $k_0$. Its $R$ image is normal and has nontrivial scalar image $C$. Formula (F.26.7) therefore forces all translations into that image, giving $\rho_d(R) = k_0 \rtimes C$. Its $K$ image lies in $\ker \mathrm{Tr} \rtimes C$, by (F.26.2), and is proper. This proves both directions of the criterion.

*The coefficient splitting, with its inverse.* The characters $\chi_j = \chi^{q_1^j}$, $0 \leq j < m$, are pairwise distinct: a nonidentity Frobenius power fixing all their scalar values would fix their generated field $k_0$. They all agree on $R$. The exact coefficient isomorphism is

$$S : k_0 \otimes_{k_1} k_0 \xrightarrow{\sim} \prod_{j=0}^{m-1} k_0, \qquad b \otimes a \longmapsto (ab^{q_1^j})_{j=0}^{m-1}. \tag{F.26.14}$$

Choose a multiplicative generator $\gamma$ of $k_0^\times$. Its minimal polynomial $f$ over $k_1$ has degree $m$ and distinct roots $\gamma^{q_1^j}$. Identify the tensor ring with $k_0[X]/f(X)$, with the first $\gamma$ represented by

$X$ and the second factor supplying coefficients. Evaluation at these roots is (F.26.14), and its inverse is

$$(a_j)_j \longmapsto \sum_{j=0}^{m-1} a_j \prod_{r\neq j} \frac{X-\gamma^{q_1^r}}{\gamma^{q_1^j}-\gamma^{q_1^r}} \pmod{f(X)}. \tag{F.26.15}$$

Every denominator is nonzero and the two composites are the identity by evaluation and degree.

Give the first tensor factor the $\chi$ action. Under $S$ its $j$th factor has the $\chi_j$ action. A continuous cocycle on $V=\widehat{F}_7$ is specified by its seven values: any such tuple defines a map from the discrete free group into the finite affine group and extends uniquely to its full completion. Thus (F.26.14) gives the exact comparison

$$\begin{aligned} Z^1_{\mathrm{cont}}(V,k_{0,\chi})\otimes_{k_1} k_0 &\xrightarrow{\sim} \bigoplus_{j=0}^{m-1} Z^1_{\mathrm{cont}}(V,k_{0,\chi_j}),\\ d\otimes a &\longmapsto (ad^{q_1^j})_j. \end{aligned} \tag{F.26.16}$$

Expansion in a $k_1$-basis of $k_0$ likewise identifies the scalar extension of the target of (F.26.2) with $Z^1_{\mathrm{cont}}(K,k_{0,\chi|K})$. Coordinate projections preserve cocycles and continuity, so this target comparison holds for the actual profinite subgroup $K$ as well. On simple tensors, the trace formula proves that $\mathscr{T}_\chi\otimes 1$ is exactly

$$(d_j)_j \longmapsto \sum_j d_j|K. \tag{F.26.17}$$

The scalar extension of $\mathscr{P}_\chi$ is $\{(b_j(\chi_j-1))_j : \sum_j b_j=0\}$, since the trace becomes coordinate summation under $S$. Extension by a field preserves kernels and detects a nonzero vector space by a basis. This proves the claimed equivalent kernel description and the rank target $7m-(m-1)=6m+1$.

*The exhaustive field-of-order-nine character set.* Use the actual negative coordinates of F.24.4 and write $\mathbb{F}_9=\mathbb{F}_3[\iota]/(\iota^2+1)$. Every element is uniquely $a+b\iota$ with $a,b\in\mathbb{F}_3$; the polynomial has no root in $\mathbb{F}_3$, so this quotient is a field. For any scalar field $k_0$ of order nine, choose an element $\iota_0$ of order four in $k_0^\times$. It satisfies $\iota_0^2=-1$. The explicit field map $a+b\iota\mapsto a+b\iota_0$ is an isomorphism: its source is a field, it preserves one, and both fields have nine elements. Its inverse takes the unique two coefficients in this basis. Apply that inverse to both affine coordinates to obtain the corresponding isomorphism of affine groups. It preserves the original source representation and its two subgroup images through this specified coefficient map. Either choice of $\iota_0$ is covered by the complete character enumeration. The field code $a+3b$ in the programs records these two coordinates exactly. Multiplication is

$$(a+b\iota)(c+d\iota)=(ac-bd)+(ad+bc)\iota, \qquad \mathrm{Tr}(a+b\iota)=2a, \qquad \mathrm{Tr}(\iota(a+b\iota))=b. \tag{F.26.18}$$

Thus multiplication of cocycles by $\iota$ is the exact coefficient isomorphism from the imaginary-part kernel computed below to the trace kernel, with inverse multiplication by $-\iota$. It sends real principal parameters to the trace-zero principal parameters $\mathbb{F}_3\iota$.

F.25 covers trivial $C$ and equality of scalar fields. The remaining scalar-field-nine data have $k_1=\mathbb{F}_3$, so $C=\{1,-1\}$. The four deleted character values are in $\{1,-1\}$ with at least one equal to $-1$. Conversely those values give exactly this $C$, because the four deleted generators normally generate $R$ and the character is abelian. The three retained values lie in $(\mathbb{F}_9^\times)^3$ and are not all in $(\mathbb{F}_3^\times)^3$. These conditions give $k_0=\mathbb{F}_9$, because the extension of $\mathbb{F}_3$ has prime degree two. The exhaustive character count is therefore

$$(8^3-2^3)(2^4-1)=7\,560. \tag{F.26.19}$$

The first producer enumerates every such tuple.

For a fixed tuple $t$, write a cocycle-value vector as $v = x + \iota y$, with $x, y \in \mathbb{F}_3^7$, and put $\widetilde{v} = (x, y)^{\mathsf{T}}$. For each of the sixty-four actual F.24 source rows $a(t) = a_{\mathrm{r}} + \iota a_{\mathrm{i}}$ the exact condition that its value have imaginary part zero is

$$(a_{\mathrm{i}}, a_{\mathrm{r}})\widetilde{v} = 0. \tag{F.26.20}$$

The seven-coordinate vector $t - \mathbf{1}$ with real principal parameter gives a nonzero solution. It is nonzero because some deleted value equals $-1$. Every row arises from an actual element of $K$, whose character lies in $\mathbb{F}_3$, so this principal line is contained in the kernel of all the rows and of every further source conjugate.

The first exact row reduction gives the following complete distribution:

| rank on fourteen columns | 11 | 12 | 13 |
|---|---|---|---|
| character tuples | 8 | 396 | 7156. |

(F.26.21)

For the 7 156 rank-thirteen tuples, the kernel is precisely the stated principal line. The other 404 tuples are retained explicitly and passed to the second producer.

*All eleven actual source actions on the remaining tuples.* For each of these 404 tuples, evaluate all eleven matrices $A_i$ from F.24.9 by the same exact Laurent evaluation. Write $A_i = A_{i,\mathrm{r}} + \iota A_{i,\mathrm{i}}$. Its actual action on the fourteen prime-field coordinates is

$$\widetilde{A}_i = \begin{pmatrix} A_{i,\mathrm{r}} & -A_{i,\mathrm{i}} \\ A_{i,\mathrm{i}} & A_{i,\mathrm{r}} \end{pmatrix}. \tag{F.26.22}$$

Start with the eight original negative seed imaginary-part rows. At each round replace their row space $W$ by

$$W + \sum_{i=0}^{10} W\widetilde{A}_i. \tag{F.26.23}$$

Every row lies in the $\mathbb{F}_3$-linear span of the imaginary-part evaluation rows of actual source conjugates of the seeds. The pure source action preserves the character, and its action on cocycles is exactly the precomposition matrix proved in F.24.9. The successive exact ranks are

| stage | rank : number of tuples | | | |
|---|---|---|---|---|
| eight seeds | 5 : 80 | 6 : 284 | 7 : 2 | 8 : 38 |
| one round | 12 : 50 | 13 : 354 | | |
| two rounds | 13 : 404 | | | |
| third round, stable | 13 : 404 | | | |

(F.26.24)

In particular two rounds already supply rank thirteen for every remaining tuple. The third round verifies invariance under all eleven matrices. Their actions are invertible, since they come from the actual source automorphisms. An invariant subspace for an invertible linear map on a finite-dimensional space is also invariant under its inverse. Hence this stable row space accounts for both directions of all source generators.

All arithmetic here is in the explicitly described finite field or in $\mathbb{F}_3$. Every nonzero Laurent variable is evaluated using its powers modulo eight. The field table, all 7 560 character choices, all 64 row evaluations, the 404 retained tuples, the eleven actual source matrices and the complete rank histories are specified by the accompanying producers and their fixed input files. Gaussian elimination uses the exact inverses one and two in $\mathbb{F}_3$. The independent verification imports neither producer. It reconstructs these matrices from the fixed universal source rows and actions, using a separate multiplicative-log Laurent evaluation, and computes the ranks by scalar incremental elimination over $\mathbb{F}_3$. It verifies the entire character set, all principal vectors, all eleven source actions, every closure image, the stable row spaces and all nine coefficient identities

in (F.26.18). The universal source data themselves are the audited literal source evaluations of F.24.

It follows that every continuous cocycle whose imaginary part vanishes on $K$ has values $b(t-\mathbf{1})$ for a unique $b \in \mathbb{F}_3$. Indeed all the actual rows (F.26.20) and their source conjugates must vanish on it, so the proved ranks give precisely that one-dimensional kernel. Conversely those principal cocycles do have imaginary part zero on $K$. Multiplication by $\iota$ in (F.26.18), and its specified inverse, prove $\ker \mathscr{T}_\chi = \mathscr{P}_\chi$ for all 7 560 remaining field-of-order-nine characters.

The criterion proved in (F.26.1)–(F.26.13) now excludes every proper marginal whose scalar field has order nine, even when its original coefficient field $k$ is larger. For a representation into $\mathrm{Aff}_1(\mathbb{F}_9)$ its scalar field is either $\mathbb{F}_3$ or $\mathbb{F}_9$; F.25 handles the former and the present argument handles the latter. Finally, precomposition with the actual continuous isomorphism $\mathcal{B}_+ : V_- \to V_+$ of F.23.13, and its inverse, transports each positive representation to the proved negative case, carrying both $K$ and $R$ exactly as stated there. This proves (F.26.4) for both signs. □

**F.27. [Proved] All finite affine marginal images on an exact two-parameter character locus.** Use the actual negative free basis $X_1, \dots, X_7$ of F.24 and the based comparison $\mathcal{B}_+ : V_- \to V_+$ of F.23.13. Put $Y_i^- = X_i$ and $Y_i^+ = \mathcal{B}_+(X_i)$. For every finite field $k$ and continuous $\rho : V_\pm \to \mathrm{Aff}_1(k)$, let $\chi$ be its scalar character. The character locus

$$\chi(Y_1^\pm) = \chi(Y_4^\pm) = \chi(Y_5^\pm) = \chi(Y_6^\pm) = 1, \qquad \chi(Y_2^\pm)\chi(Y_3^\pm)\chi(Y_7^\pm) = 1 \tag{F.27.1}$$

has the exact image equality

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm). \tag{F.27.2}$$

There is no restriction on the characteristic, extension degree, or size of the original coefficient field.

More precisely, write a negative character in this locus as $(1, s, t, 1, 1, 1, c^{-1})$, where $c = st$, and put $k_0 = \mathbb{F}_p(s,t)$, $k_1 = \mathbb{F}_p(c)$. For $c \neq 1$ and $k_1 \subsetneq k_0$, let $q_1 = |k_1|$, $m = [k_0 : k_1]$ and $\chi_j = \chi^{q_1^j}$, $0 \le j < m$. Writing $K = \mathcal{K}_-$ and $V = V_-$, the actual map

$$\bigoplus_{j=0}^{m-1} H^1_{\mathrm{cont}}(V, k_{0,\chi_j}) \longrightarrow H^1_{\mathrm{cont}}(K, k_{0,\chi|K}), \qquad ([d_j])_j \longmapsto \sum_j [d_j|K] \tag{F.27.3}$$

is injective. At the cocycle level, the kernel of the summed restriction is exactly

$$\left\{ \big(b_j(\chi_j - 1)\big)_j : b_j \in k_0,\ \sum_j b_j = 0 \right\}. \tag{F.27.4}$$

Thus its rank is $6m+1$ on $7m$ cocycle coordinates. The proof uses the literal eleven F.23 source automorphisms and specified source conjugates of one of its eight negative seeds.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. The actual scalar fields and all exceptional cases.* Work first with the negative groups and write $K \le R \le V$. The four deleted generators $X_4, \dots, X_7$ normally generate $R$. The normal generation and actual marginal maps of F.21–F.22, as used in F.25.5 and F.26.5, give

$$\begin{gathered} C = \chi(K) = \chi(R) = \langle c^{-1} \rangle, \qquad D = \chi(V) = \langle s, t \rangle, \\ k_1 = \mathbb{F}_p(C) = \mathbb{F}_p(c), \qquad k_0 = \mathbb{F}_p(D) = k_1(s) = k_1(t), \qquad c = st. \end{gathered} \tag{F.27.5}$$

Indeed the last equalities follow from $t = c/s$ and $s = c/t$. When $c = 1$, the $R$ image consists of translations and is abelian; the actual normal generation therefore gives (F.27.2). For $c \neq 1$ and

$k_0 = k_1$, F.25 gives the same equality. It remains to treat $c \neq 1$ and $k_1 \subsetneq k_0$. Here $s, t \notin k_1$, by (F.27.5), and consequently

$$s\,t\,(s-1)(t-1)(t+1)(c-1) \neq 0. \tag{F.27.6}$$

The factors $s, t$ are nonzero character values. The elements $1, -1$ belong to $k_1$, so neither can equal $s$ or $t$. This includes characteristic two, where $t+1 = t-1$. The case $c = 1$ has already been settled, so every factor in (F.27.6) is justified.

*The exact coefficient map and principal quotient.* All universal word calculations in this proof are performed on the discrete free group on the seven specified generators, with its canonical dense map to $V = \widehat{F}_7$. The coefficient homomorphism from the F.24 Laurent ring is

$$\begin{aligned} \mathbb{Z}[t_1^{\pm 1}, \dots, t_7^{\pm 1}] &\longrightarrow A = \mathbb{Z}[s^{\pm 1}, t^{\pm 1}], \\ (t_1, t_2, t_3, t_4, t_5, t_6, t_7) &\longmapsto (1, s, t, 1, 1, 1, (st)^{-1}). \end{aligned} \tag{F.27.7}$$

Thus a monomial with exponent vector $(a_i)$ maps to $s^{a_2 - a_7} t^{a_3 - a_7}$. The universal derivation on free words has $D(X_i) = e_i$ and $D(uv) = D(u) + \chi(u) D(v)$. Evaluation of its coefficients in a finite field gives the actual cocycle value on that word. Every seven-tuple of generator values defines a homomorphism from the discrete free group to the finite affine group and extends uniquely to its full profinite completion. Consequently the continuous cocycle space is exactly $k_0^7$. No continuous map to an infinite Laurent ring is being used.

Put $\delta = (1, s, t, 1, 1, 1, c^{-1})^{\mathsf{T}} - \mathbf{1}$. Its last entry is $\delta_7 = (1-c)/c \neq 0$. The quotient by principal cocycles has the explicit projection and section

$$\begin{aligned} P(v) &= \left(v_r - \delta_r \delta_7^{-1} v_7\right)_{r=1}^{6}, & S(w) &= (w_1, \dots, w_6, 0)^{\mathsf{T}}, \\ PS &= I_6, & SP(v) &= v - \delta \delta_7^{-1} v_7. \end{aligned} \tag{F.27.8}$$

These formulas identify $Q_\chi = H^1_{\mathrm{cont}}(V, k_{0,\chi})$ with $k_0^6$. Universally they are defined over $A[(1 - st)^{-1}]$; all finite-field evaluations here respect that localization by (F.27.6).

Let $B_l(v) = u_l \alpha_l(v) u_l^{-1}$, $0 \leq l \leq 10$, be the actual old-source automorphisms of F.23. They preserve $K$, including under their inverses, and fix the character because their action on the abelianization of $V$ is the identity. Let $A_l$ be their F.24 precomposition matrices after (F.27.7), so the $i$th row is $D(B_l(X_i))$. Then $A_l \delta = \delta$ and their actual cohomology actions are

$$\overline{A}_l = P A_l S. \tag{F.27.9}$$

The source indices are exactly $0 =$ `old0`, $1, \dots, 5 =$ `old_1_0`, $\dots$, `old_1_4` and $6, \dots, 10 =$ `old_-1_0`, $\dots$, `old_-1_4`.

*A basis with explicit inverse and its actual source action.* For $P_0 = X_1 \cdots X_7$, the character is one, and its derivative row is $(1, 1, s, c, c, c, c)$. Every pure $\alpha_l$ fixes this word. Hence evaluation on $P_0$ induces the cohomology covector $b = (1, 1, s, c, c, c)$, with $b\overline{A}_l = \chi(u_l) b$. For each $r \in \{1, 4, 5, 6\}$, the pure Artin action writes $\alpha_l(X_r) = a_{lr} X_r a_{lr}^{-1}$. Since $\chi(X_r) = 1$, its actual conjugated source action gives $d(B_l(X_r)) = \chi(u_l a_{lr}) d(X_r)$. Thus $e_1^*, e_4^*, e_5^*, e_6^*$ are four more actual quotient covectors. Their common kernel together with $b$ is the line generated by

$$v_0 = (0, 1, -s^{-1}, 0, 0, 0)^{\mathsf{T}}. \tag{F.27.10}$$

Use the basis matrix $J$ with columns

$$v_0, \quad e_2, \quad e_5 - c e_2, \quad e_4 - c e_2, \quad e_1 - e_2, \quad e_6 - c e_2. \tag{F.27.11}$$

Its inverse is the explicit map

$$J^{-1}(w) = (-s w_3,\ b w,\ w_5,\ w_4,\ w_1,\ w_6)^{\mathsf{T}}. \tag{F.27.12}$$

For completeness, $J(z_0,\dots,z_5)^{\mathsf{T}}$ has coordinates

$$\left(z_4,\ z_0+z_1-cz_2-cz_3-z_4-cz_5,\ -s^{-1}z_0,\ z_3,\ z_2,\ z_5\right)^{\mathsf{T}}. \tag{F.27.13}$$

Substitution in either direction gives the identity, so these are actual mutually inverse coefficient maps.

The five displayed quotient covectors give the following form for the source actions in this basis:

$$T_l = J^{-1}\overline{A}_l J = \begin{pmatrix} \nu_l & r_l \\ 0 & \Lambda_l \end{pmatrix}, \qquad \Lambda_l \text{ diagonal}, \qquad (\nu_0,\dots,\nu_{10}) = (1,s,t,1,1,1,s,t,1,1,1). \tag{F.27.14}$$

The last character is computed from the literal source words. More explicitly, for the seven-coordinate lift $\widetilde{v}_0 = S(v_0)$ the verified identities are

$$\begin{gathered} A_l\widetilde{v}_0 = \nu_l\widetilde{v}_0 + \pi_l\delta, \\ \pi_1 = -c, \qquad \pi_2 = \pi_7 = ct, \qquad \pi_6 = s(c-t-1), \qquad \pi_l = 0 \quad (l \notin \{1,2,6,7\}). \end{gathered} \tag{F.27.15}$$

The certificate records every entry of all eleven $A_l$, $\overline{A}_l$ and $T_l$, with the coefficient maps (F.27.7)–(F.27.13).

*An integral determinant forcing the essential line.* Take the additive matrix commutators $E_j = T_1T_j - T_jT_1$ for $j = 2,4,5,6,7$. They kill the first coordinate line and have image in that line, since the lower diagonal blocks in (F.27.14) commute. Their first-row entries on the other five coordinates, in that order, form the exact matrix

$$\begin{gathered} gH, \qquad g = (s-1)(t-1), \\ H = \begin{pmatrix} 0 & 0 & 0 & -(t+1)/t & st \\ 0 & s & -1/t & 0 & 0 \\ 0 & -1/t & 0 & st+1 & 0 \\ (s-1)/(st^2) & s(s-1) & -(s-1)/t & s-1 & 0 \\ (t+1)/(st^2) & 0 & 0 & 0 & st \end{pmatrix}. \end{gathered} \tag{F.27.16}$$

Direct determinant expansion gives the integral Laurent identity

$$\det H = \frac{(s-1)(t-1)(t+1)}{t^4}, \qquad \det(gH) = \frac{(s-1)^6(t-1)^6(t+1)}{t^4}. \tag{F.27.17}$$

All its factors are nonzero after every finite-field evaluation under consideration, by (F.27.6). The identities have integer coefficients and only signed Laurent monomial denominators; no prime is removed from the calculation.

Let $L_\chi = k_0v_0 \le Q_\chi$. It is invariant by (F.27.14). Every nonzero subspace $W \le Q_\chi$ invariant under the eleven actual source operators contains $L_\chi$. Indeed, in the $J$ coordinates the common kernel of the five $E_j$ is precisely the first coordinate line, by (F.27.17). If $W$ is contained in this line, it equals it. Otherwise choose a vector of $W$ with nonzero last five coordinates. Some $E_j$ sends it to a nonzero vector of the first coordinate line; this vector remains in $W$. Since the line has dimension one, it is contained in $W$. This proves the required essential-socle statement by explicit source endomorphisms.

*Actual seed evaluations and all Frobenius factors.* Let $d_0$ be the continuous cocycle with generator values $\widetilde{v}_0$. Take the literal negative seed $a = u_{\texttt{cross_4_-1_1}}$ of index 25. The actual seed calculation gives

$$\chi(a) = c^{-1}, \qquad d_0(a) = c-1. \tag{F.27.18}$$

Both are computed by the same decorated braid words as in F.23; all eight seed rows are audited. Since $a \in K$ and $B_1$ preserves $K$, put

$$\kappa = B_1(a)a^{-1} \in K, \qquad \kappa_r = B_1^r(\kappa) \in K \quad (r \ge 0). \tag{F.27.19}$$

Their characters are one. Using $\pi_1 = -c$ in (F.27.15) gives $d_0(B_1(a)) = (s+1)(c-1)$, hence $d_0(\kappa) = s(c-1)$. For every character-one element $h$, (F.27.15) also gives $d_0(B_1(h)) = s\,d_0(h)$: the principal correction evaluates to zero there. Induction yields

$$d_0(\kappa_r) = s^{r+1}(c-1). \tag{F.27.20}$$

In row coordinates the first two evaluations are $D(a)(A_1 - I_7)$ and $D(a)(A_1 - I_7)A_1$; the independent audit verifies their character-one principal vanishing and the stated values.

Put $s_j = s^{q_1^j}$ and $t_j = t^{q_1^j}$. Since $c \in k_1$, $s_j t_j = c$ for every $j$. The field equality $k_0 = k_1(s)$ proves that $s_0, \dots, s_{m-1}$ are distinct. Indeed an equality would make a nonidentity power of the $k_1$-Frobenius fix $s$, and hence its generated field $k_0$; the powers before $m$ are distinct automorphisms of the field of $q_1^m$ elements.

The coefficient map on cocycles and on cohomology is

$$d \longmapsto d^{q_1^j}, \qquad k_{0,\chi} \longrightarrow k_{0,\chi_j}. \tag{F.27.21}$$

It is additive and semilinear, preserves the cocycle rule, carries principals to principals, and preserves continuity because the field is finite. Its inverse is the power $q_1^{m-j}$, with the identity used at $j = 0$; the two composites are the identity on $k_0$. It commutes with the actual source precomposition operators. Thus the explicit bases, inverse maps, commutator identities and essential lines transport to every $\chi_j$. Write $Q_j = H^1_{\mathrm{cont}}(V, k_{0,\chi_j})$ and $L_j = k_0[d_j] \le Q_j$, where $d_j = d_0^{q_1^j}$. Equation (F.27.20) becomes

$$d_j(\kappa_r) = a_j s_j^r, \qquad a_j = s_j(c-1) \neq 0. \tag{F.27.22}$$

The characters $\chi_j$ all agree on $K$, since $\chi(K) = C \subset k_1^\times$. Evaluation at $\kappa_0, \dots, \kappa_{m-1}$ therefore defines a map on the common target cohomology space: principals vanish on each character-one element. On the summed restrictions of the $m$ essential lines its matrix is

$$(a_j s_j^r)_{0 \le r,j < m}, \qquad \det = \left(\prod_j a_j\right) \prod_{j<l} (s_l - s_j) \neq 0. \tag{F.27.23}$$

The coefficient inverse is explicit. Write

$$\prod_{l \neq j} \frac{X - s_l}{s_j - s_l} = \sum_{r=0}^{m-1} h_{jr} X^r. \quad \text{Then} \quad z_j = a_j^{-1} \sum_{r=0}^{m-1} h_{jr} y_r \tag{F.27.24}$$

is the inverse to $y_r = \sum_j a_j s_j^r z_j$. Evaluation of the displayed interpolation polynomial at every $s_l$ proves both inverse identities. Every denominator in (F.27.24) is nonzero by the proved distinctness and (F.27.22).

*Summed restriction, the trace kernel and the affine image.* The map (F.27.3) commutes with all eleven source operators: on both source and target they are the actual maps $[d] \mapsto [d \circ B_l]$. Consequently its kernel is an invariant subspace of the finite-dimensional direct sum $\bigoplus_j Q_j$. Suppose that kernel were nonzero and choose a nonzero invariant subspace $T$ inside it of smallest dimension. Each coordinate projection $T \to Q_j$ has invariant kernel and image. Minimality makes every nonzero projection injective, with an image having no proper nonzero invariant subspace. The essential-line result puts $L_j$ inside that image; therefore the image equals $L_j$. All projections of $T$ thus lie in their respective $L_j$, and $T \le \bigoplus_j L_j$. But (F.27.23)–(F.27.24) make evaluation after summed restriction injective on this entire direct sum of lines. This contradicts $T$ lying in the kernel and proves (F.27.3).

Zero summed restriction of cocycles now implies that each $d_j$ is principal. Their restrictions have the common character $\chi|K$, so their sum is $(\chi|K - 1)\sum_j b_j$. Since $C \neq 1$, this vanishes

exactly when $\sum_j b_j = 0$. This proves (F.27.4), including both directions. The seven-coordinate description of each cocycle space gives rank $7m - (m-1) = 6m+1$.

For the trace map of F.26.2, a cocycle $d$ in its kernel yields the tuple $(d^{q_1^j})_j$, whose summed restriction is exactly $\mathrm{Tr}_{k_0/k_1}(d|K) = 0$. Thus $d = b(\chi - 1)$ by the result just proved. Trace vanishing on $K$, together with $\chi(K) = C \neq 1$, gives $\mathrm{Tr}(b) = 0$. Conversely these principal cocycles have zero trace restriction. Hence the precise trace kernel is $\mathscr{P}_\chi$ of F.26.3.

Every proper affine marginal image in the current locus would, by the explicit conjugation and coefficient homomorphism of F.26.1–12, give a nonprincipal element of that trace kernel over its scalar field $k_0$. Those maps retain the original scalar character, so all the displayed locus identities and fields remain the same. The proved trace kernel excludes such an element. This establishes (F.27.2) in the negative case, for every original finite coefficient field $k$.

*Certificate and transport.* The symbolic producer applies exactly (F.27.7) to the fixed, previously audited F.24 universal rows. The separate verifier imports neither that producer nor a computer algebra package. It evaluates all eleven actual old-source words and eight actual negative seed words by generic affine multiplication over the integral Laurent ring $A$. It checks the inverse basis maps, the principal-quotient action after clearing $\delta_7$, all source and socle identities, the actual seed evaluations, and (F.27.17) by its 120 signed permutation terms. The verifier performs 699 recorded checks. Thus the determinant certificate is an exact integer identity and applies in every characteristic; its nonzero finite-field factors are separately proved in (F.27.6). The evidence manifest binds both programs, their outputs and stdout, the original braid input and all accepted dependencies.

Finally, a positive representation in (F.27.1) precomposes with the actual continuous isomorphism $\mathcal{B}_+ : V_- \to V_+$ to give exactly the negative character just treated. By F.23.13, this isomorphism and its inverse carry both $\mathcal{K}_-$ and $R_-$ onto their positive counterparts. The equality of the two actual images therefore transports through these specified maps, proving (F.27.2) for both signs. □

**F.28. [Proved] All finite affine marginal images on the three-parameter product-one locus.** Retain the actual bases $Y_i^- = X_i$ and $Y_i^+ = \mathcal{B}_+(X_i)$ of F.27, with $\mathcal{B}_+$ the based comparison of F.23.13. For every finite field $k$ and continuous $\rho : V_\pm \to \mathrm{Aff}_1(k)$ with scalar character $\chi$, the locus

$$\chi(Y_4^\pm) = \chi(Y_5^\pm) = \chi(Y_6^\pm) = 1, \qquad \chi(Y_1^\pm)\chi(Y_2^\pm)\chi(Y_3^\pm)\chi(Y_7^\pm) = 1 \tag{F.28.1}$$

has the exact image equality

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm). \tag{F.28.2}$$

This includes every characteristic, every finite scalar-field extension degree and every larger original coefficient field.

For a negative character $(r, s, t, 1, 1, 1, c^{-1})$, $c = rst \neq 1$, put $k_1 = \mathbb{F}_p(c)$ and $k_0 = \mathbb{F}_p(r, s, t)$. In the case $k_1 \subsetneq k_0$, let $q_1 = |k_1|$, $m = [k_0 : k_1]$ and $\chi_j = \chi^{q_1^j}$, $0 \leq j < m$. The summed continuous $H^1$ restriction for these $m$ characters is injective, and its cocycle kernel is exactly the principal tuples with parameter sum zero. Its rank is therefore $6m+1$ on $7m$ cocycle coordinates.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. The actual fields and the already proved cases.* Use the negative groups $K \leq R \leq V$ first. The deleted generators $X_4, \ldots, X_7$ and the actual normal-generation maps of F.21–F.22 give

$$\begin{aligned} C = \chi(K) = \chi(R) = \langle c^{-1} \rangle, \qquad D = \chi(V) = \langle r, s, t \rangle, \\ k_1 = \mathbb{F}_p(c), \qquad k_0 = \mathbb{F}_p(r, s, t) = k_1(r, t), \qquad c = rst. \end{aligned} \tag{F.28.3}$$

The last equality follows from $s = c/(rt)$; all character values are nonzero. For $c = 1$, the $R$ image is an abelian translation group, and actual normal generation gives (F.28.2). For $c \neq 1$ and $k_0 = k_1$, F.25 gives it. For $r = 1$, the character is exactly $(1, s, t, 1, 1, 1, (st)^{-1})$ in the unchanged actual basis, so F.27 gives both the image equality and the stated summed-restriction result. At the coefficient-ring level this last comparison is the homomorphism $r \mapsto 1$, $s \mapsto s$, $t \mapsto t$, sending $1 - rst$ to $1 - st$ in the corresponding localized Laurent rings. Thus it respects the precise maps used in F.27.

It remains to treat $c \neq 1$, $r \neq 1$ and $k_1 \subsetneq k_0$. These conditions imply only the nonzero factors actually used below: $r, s, t, c, c-1, r-1$. No condition on $s-1$, $t-1$, $r+1$ or the characteristic is imposed.

*The exact invariant inclusion and quotient maps.* The literal coefficient homomorphism from F.24 is

$$\begin{aligned} \mathbb{Z}[t_1^{\pm 1}, \dots, t_7^{\pm 1}] &\longrightarrow A = \mathbb{Z}[r^{\pm 1}, s^{\pm 1}, t^{\pm 1}], \\ (t_1, t_2, t_3, t_4, t_5, t_6, t_7) &\longmapsto (r, s, t, 1, 1, 1, (rst)^{-1}). \end{aligned} \tag{F.28.4}$$

A monomial with exponent vector $(a_i)$ maps to $r^{a_1 - a_7} s^{a_2 - a_7} t^{a_3 - a_7}$. These are identities on the specified discrete free words. Their finite-field cocycle evaluations extend uniquely to $V = \widehat{F}_7$ by the finite affine homomorphisms, exactly as in F.27.7–8.

Let $\delta = (r, s, t, 1, 1, 1, c^{-1})^{\mathsf{T}} - \mathbf{1}$, with $\delta_7 = (1 - c)/c \neq 0$. The actual continuous cohomology space $Q_\chi = H^1_{\mathrm{cont}}(V, k_{0,\chi})$ is $k_0^6$ through the projection and section

$$\begin{aligned} P(v) &= (v_i - \delta_i \delta_7^{-1} v_7)_{i=1}^6, \qquad S(w) = (w_1, \dots, w_6, 0)^{\mathsf{T}}, \\ PS &= I_6, \qquad\qquad\qquad SP(v) = v - \delta \delta_7^{-1} v_7. \end{aligned} \tag{F.28.5}$$

For the eleven actual F.23 source automorphisms $B_l$, $0 \leq l \leq 10$, both directions preserve $K$ and fix the character, as in F.27. Let $A_l$ be their specialized cocycle matrices. Then $A_l \delta = \delta$ and $\overline{A}_l = P A_l S$ are the actual source operators on $Q_\chi$.

The boundary word $P_0 = X_1 \cdots X_7$ has character one and derivative $(1, r, rs, c, c, c, c)$. Pure Artin monodromy fixes it, giving the quotient covector $b = (1, r, rs, c, c, c)$. The character-one meridians $X_4, X_5, X_6$ give the three quotient covectors $e_4^*, e_5^*, e_6^*$, as in F.27: the scalar of their actual conjugator multiplies their cocycle value. The four maps together define

$$\begin{aligned} \pi : Q_\chi &\longrightarrow k_0^4, \qquad \pi(v) = (bv, v_4, v_5, v_6)^{\mathsf{T}}, \\ \iota : k_0^2 &\longrightarrow Q_\chi, \quad \iota(x, y) = (-rx - rsy, x, y, 0, 0, 0)^{\mathsf{T}}. \end{aligned} \tag{F.28.6}$$

Their kernel-image identity is $\ker \pi = \operatorname{im} \iota =: W_\chi$. The coordinate projection $(v_2, v_3)$ is inverse to $\iota$ on this subspace. A coefficient section of $\pi$ is

$$\sigma(z_\partial, z_4, z_5, z_6) = (z_\partial - c(z_4 + z_5 + z_6), 0, 0, z_4, z_5, z_6)^{\mathsf{T}}. \tag{F.28.7}$$

One has $\pi\sigma = I_4$ and $\iota(v_2, v_3) + \sigma\pi(v) = v$. Thus the basis matrix with columns

$$\iota(1, 0), \quad \iota(0, 1), \quad e_1, \quad e_4 - ce_1, \quad e_5 - ce_1, \quad e_6 - ce_1 \tag{F.28.8}$$

has inverse $v \mapsto (v_2, v_3, bv, v_4, v_5, v_6)$. The displayed identities prove both inverse coefficient maps explicitly.

In this basis the actual source matrices have form

$$\begin{pmatrix} G_l & E_l \\ 0 & D_l \end{pmatrix}, \qquad D_l = \operatorname{diag}(\lambda_{\partial,l}, \lambda_{4,l}, \lambda_{5,l}, \lambda_{6,l}). \tag{F.28.9}$$

Equivalently, $\overline{A}_l \iota = \iota G_l$ and $\pi \overline{A}_l = D_l \pi$. All extension matrices $E_l$ are retained in the certificate. These maps give the actual invariant subspace $W_\chi$ and the four actual one-dimensional quotient coordinates.

The literal source-word calculation gives the following comparison table:

| $l$ | $G_l$ | $\lambda_{\partial,l}$ | $\lambda_{4,l}$ | $\lambda_{5,l}$ | $\lambda_{6,l}$ |
|---|---|---|---|---|---|
| 0 | $rI_2$ | 1 | $r$ | $r$ | $r$ |
| 7 | $tI_2$ | $t^{-1}$ | $t$ | $t$ | $ct$ |
| 8 | $I_2$ | 1 | $c$ | 1 | $t^{-1}$ |
| 9 | $I_2$ | 1 | $t^{-1}$ | $c$ | 1 |

$$\lambda_{\partial,2} = rt. \tag{F.28.10}$$

Every source index here is the original index beginning at zero: 0 = `old0`, 2 = `old_1_1`, 7 = `old_-1_1`, 8 = `old_-1_2` and 9 = `old_-1_3`. The independent verifier checks all entries against these exact labelled source words.

There are also explicit actual $K$ evaluations on $W_\chi$. Let $a = u_{\texttt{cross_4_-1_1}}$ be seed 25, with $\chi(a) = c^{-1}$. For $h_2 = B_2(a)a^{-1}$ and $h_7 = B_7(a)a^{-1}$, both characters are one. A representative with generator values $S\iota(x, y)$ satisfies

$$(d(h_2), d(h_7)) = ((1-c)x, s(c-1)y). \tag{F.28.11}$$

Both evaluations therefore descend to cohomology. Their inverse coefficient map is $(u, v) \mapsto ((1-c)^{-1}u, [s(c-1)]^{-1}v)$. The determinant $-s(c-1)^2$ is nonzero in every remaining case. These identities are also verified from the actual seed and source words.

*Frobenius transport and the classification needed for comparison.* For $0 \le j < m$, put $r_j = r^{q_1^j}$, $s_j = s^{q_1^j}$ and $t_j = t^{q_1^j}$. Then $r_j s_j t_j = c$. The semilinear coefficient map

$$d \longmapsto d^{q_1^j}, \qquad k_{0,\chi} \longrightarrow k_{0,\chi_j} \tag{F.28.12}$$

and its inverse, the power $q_1^{m-j}$ with the identity at $j = 0$, preserve cocycles, principals and continuity. They commute with all actual source precomposition maps. Consequently all the preceding inclusions, projections, inverse bases and comparison entries transport to $Q_j = H^1_{\mathrm{cont}}(V, k_{0,\chi_j})$ and its invariant subspace $W_j$ by the substitutions $(r, s, t) \mapsto (r_j, s_j, t_j)$. The characters agree on $K$, since $\chi(K) = C \subset k_1^\times$.

The pairs $(r_j, t_j)$ are distinct. Equality of two pairs would make a nonidentity $k_1$-Frobenius power fix the generators of $k_0 = k_1(r, t)$. It would then fix every element of $k_0$; the nonzero polynomial $X^{q_1^a} - X$, with $0 < a < m$, has degree smaller than $|k_0|$ and cannot vanish on all those elements. Thus

$$(r_j, t_j) = (r_l, t_l) \quad \Longrightarrow \quad j = l. \tag{F.28.13}$$

Also every $r_j \neq 1$, since Frobenius is a field automorphism and $r \neq 1$.

Consider a nonzero invariant subspace $U \le Q_j$ having no proper nonzero invariant subspace. Its intersection with $W_j$ is invariant. If the intersection is nonzero, $U \le W_j$. If it is zero, $\pi_j|U$ is injective. At least one of its four actual coordinate projections is nonzero. That projection has invariant kernel, hence is injective by the defining property of $U$. Thus $U$ is one-dimensional and its source character is one of $\lambda_{\partial,j}, \lambda_{4,j}, \lambda_{5,j}, \lambda_{6,j}$. This argument uses the displayed coordinate maps even when quotient characters repeat; it requires no assumption about distinct quotient characters within one factor.

*Every comparison between different factors.* We now prove that for $j \neq l$ there is no source-intertwining isomorphism between such subspaces of $Q_j$ and $Q_l$. For a source operator acting by scalars $a$ and $b$ on its two sides, an intertwiner $F$ satisfies $(a - b)F = 0$. A nonzero $F$ therefore forces $a = b$. Apply this identity to the exact table (F.28.10).

A boundary-character subspace has source-0 scalar one. Every subspace contained in $W_l$ and every deleted-meridian-character subspace has source-0 scalar $r_l \neq 1$. This excludes every comparison of a boundary character with either of those types.

Two subspaces contained in $W_j$ and $W_l$ would force $r_j = r_l$ and $t_j = t_l$ through sources 0 and 7. This contradicts (F.28.13). For two boundary characters, source 7 forces $t_j = t_l$ and source 2 then forces $r_j = r_l$, giving the same contradiction. For the same deleted type 4 or 5, sources 0 and 7 again determine $(r_j, t_j)$. For type 6 the source-7 value is $ct_j$, and the common nonzero $c$ cancels, with the same result.

For two different deleted types, type 4 versus type 5 is excluded by source 8, whose scalars are $c$ and 1. Type 5 versus type 6 is excluded by source 9, with the same two values. For type 4 in factor $j$ versus type 6 in factor $l$, source 8 would force $t_l = c^{-1}$. Since $c \in k_1$, inverse Frobenius then gives $t = c^{-1}$ and every $t_j = c^{-1}$. Source 9 would consequently compare $c$ with 1, again impossible because $c \neq 1$.

Finally, a subspace of $W_j$ versus type 4 in any factor is excluded by source 8, acting as 1 and $c$. A subspace of $W_j$ versus type 5 is excluded by source 9. For type 6 in factor $l$, source 8 would force $t_l = 1$. Inverse Frobenius gives $t = 1$ and all $t_j = 1$. Source 7 would then compare 1 on $W_j$ with $c$ on type 6, a contradiction. The inverse of an intertwining isomorphism also intertwines, so these comparisons cover both orientations of every pair of types.

*The exact restriction kernel and affine image.* Let

$$\Phi : \bigoplus_{j=0}^{m-1} Q_j \longrightarrow H^1_{\mathrm{cont}}(K, k_{0,\chi|K}), \qquad ([d_j])_j \longmapsto \sum_j [d_j|K]. \tag{F.28.14}$$

This commutes with every actual source operator $[d] \mapsto [d \circ B_l]$. Each individual restriction $Q_j \to H^1(K, k_{0,\chi|K})$ is injective by F.24: its character on $R$ is the same nontrivial $C$, and its coefficient field is finite.

Suppose $\ker \Phi$ were nonzero, and choose a nonzero invariant subspace $T$ inside it with smallest dimension. Each nonzero coordinate projection $p_j|T$ is injective and identifies $T$ with an invariant subspace of $Q_j$ having no proper nonzero invariant subspace. Two nonzero projections would give the explicit source-intertwining isomorphism

$$p_l \circ (p_j|T)^{-1} : p_j(T) \longrightarrow p_l(T). \tag{F.28.15}$$

Its inverse is $p_j \circ (p_l|T)^{-1}$. The comparisons just proved exclude this for $j \neq l$. Thus $T$ has only one nonzero coordinate projection and is contained in one $Q_j$. On that factor $\Phi$ is the injective individual restriction, a final contradiction. This proves the summed $H^1$ restriction injective.

Consequently zero summed restriction of cocycles forces every cocycle to be principal. As all characters agree on $K$ and $C \neq 1$, the exact kernel is

$$\left\{ \left(b_j(\chi_j - 1)\right)_j : b_j \in k_0,\ \sum_j b_j = 0 \right\}. \tag{F.28.16}$$

The converse inclusion follows by direct restriction. Its dimension is $m - 1$ on the $7m$ continuous cocycle coordinates, proving rank $6m + 1$.

A cocycle in the trace kernel of F.26.2 gives the tuple $(d^{q_1^j})_j$ with zero summed restriction. It is therefore principal, and its parameter has trace zero because $C \neq 1$. Conversely every such principal cocycle has zero trace restriction. The precise trace kernel is thus $\mathscr{P}_\chi$ of F.26.3. Every proper affine marginal image would give a nonprincipal trace-kernel element by the actual conjugation and coefficient homomorphism of F.26.1–12. Those maps retain the original scalar character, so they preserve (F.28.1) and the fields (F.28.3). The proved kernel equality excludes such an element. This proves (F.28.2) for the remaining negative case, completing all cases considered above.

*Integral evidence and sign transport.* The symbolic producer applies exactly (F.28.4) to the retained F.24 universal data and records all eleven full matrices, all eight seed rows, all basis

maps and all extension blocks. A separately structured verifier imports neither the producer nor an algebra package. It evaluates all nineteen literal F.23 source and seed braid words by generic affine multiplication over the integral Laurent ring $A$. It checks both basis inverses, the principal-quotient identity after clearing only $\delta_7$, the invariant inclusion, each diagonal quotient coordinate, every entry of (F.28.10), its exact source labels and the actual seed evaluations (F.28.11). It records 781 checks. All coefficient identities have integer coefficients; the only nonmonomial localization is at $1 - rst$, whose nonvanishing was already proved. The evidence manifest binds these programs, outputs and inputs to the exact statement.

For a positive representation, precomposition with the actual continuous isomorphism $\mathcal{B}_+ : V_- \to V_+$ sends the displayed $Y_i^+$ character to precisely the negative character just treated. This map and its inverse carry $\mathcal{K}_-$ and $R_-$ onto their positive counterparts by F.23.13. Equality of the two negative images therefore gives equality of the two original positive images through the specified maps, proving (F.28.2) for both signs. □

**F.29. [Proved] All finite affine marginal images on the four-parameter product-one locus.** Retain the actual bases $Y_i^- = X_i$ and $Y_i^+ = \mathcal{B}_+(X_i)$, where $\mathcal{B}_+$ is the based isomorphism of F.23.13. For every finite field $k$ and continuous $\rho : V_\pm \to \mathrm{Aff}_1(k)$ with scalar character $\chi$, the exact locus

$$\chi(Y_5^\pm) = \chi(Y_6^\pm) = 1, \qquad \prod_{i=1}^{7} \chi(Y_i^\pm) = 1 \tag{F.29.1}$$

has the image equality

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm). \tag{F.29.2}$$

Every characteristic, finite scalar-field extension degree and larger original coefficient field is included.

Write a negative character in this locus as $(r, s, t, u, 1, 1, c^{-1})$, where $c = rstu$. On its nontrivial-scalar part $C = \langle u, c^{-1} \rangle \neq \{1\}$, put $k_1 = \mathbb{F}_p(u, c)$, $k_0 = \mathbb{F}_p(r, s, t, u)$, $q_1 = |k_1|$, $m = [k_0 : k_1]$ and $\chi_j = \chi^{q_1^j}$, $0 \le j < m$. The summed continuous $H^1$ restriction to $K = \mathcal{K}_-$ is injective. At the cocycle level its kernel and rank are exactly

$$\left\{ \left(b_j(\chi_j - 1)\right)_{j=0}^{m-1} : b_j \in k_0,\ \sum_j b_j = 0 \right\}, \qquad \text{rank} = 6m + 1 \quad \text{on } 7m \text{ coordinates.} \tag{F.29.3}$$

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. Actual fields, completed cases and the four remaining charts.* Work first with $K = \mathcal{K}_- \le R = R_- \le V = V_-$. F.21 and the actual marginal surjection of F.22 give $\overline{\langle\langle K \rangle\rangle}_R = R$. Apply this equality to the continuous abelian map $\chi|R$. Normal generation in its abelian image gives $\chi(K) = \chi(R)$. Since the actual deleted generators are $X_4, X_5, X_6, X_7$, their scalars now give

$$\begin{gathered} C = \chi(K) = \chi(R) = \langle u, c^{-1} \rangle, \qquad D = \chi(V) = \langle r, s, t, u \rangle, \\ k_1 = \mathbb{F}_p(C) = \mathbb{F}_p(u, c), \qquad k_0 = \mathbb{F}_p(D) = k_1(r, t), \qquad s = c/(rtu). \end{gathered} \tag{F.29.4}$$

All displayed character values are nonzero. For $C = \{1\}$ the image of $R$ is a translation group, hence abelian. The same normal-generation map gives (F.29.2) directly. This case is precisely $u = c = 1$. For $C \neq \{1\}$ and $k_0 = k_1$, F.25 gives the affine equality, while F.24 gives (F.29.3) with $m = 1$.

For $u = 1$, the original character is exactly $(r, s, t, 1, 1, 1, (rst)^{-1})$ in the unchanged basis. F.28 supplies the affine equality and, on its nontrivial-scalar part, the stated restriction kernel.

The coefficient homomorphism making this comparison is $u \mapsto 1$, fixing $r, s, t$ and sending $c$ to $rst$. It sends each original word evaluation to that of F.28, and sends the principal-pivot denominator $1 - c$ to $1 - rst$. Thus the comparison uses the actual source maps.

We now work with $u \neq 1$, $C \neq \{1\}$ and $k_1 \subsetneq k_0$. The four disjoint coordinate charts are $c \neq 1, r \neq 1$; $c \neq 1, r = 1$; $c = 1, r \neq 1$; and $c = 1, r = 1$. We give the actual principal quotient, invariant maps and source calculations in each chart.

*Coefficient maps, principal quotients and Frobenius.* The first two charts use the coefficient homomorphism

$$\begin{aligned} \mathbb{Z}[t_1^{\pm 1}, \dots, t_7^{\pm 1}] &\longrightarrow A_4 = \mathbb{Z}[r^{\pm 1}, s^{\pm 1}, t^{\pm 1}, u^{\pm 1}], \\ (t_1, t_2, t_3, t_4, t_5, t_6, t_7) &\longmapsto (r, s, t, u, 1, 1, (rstu)^{-1}). \end{aligned} \tag{F.29.5}$$

An exponent vector $(a_i)$ maps to $r^{a_1 - a_7} s^{a_2 - a_7} t^{a_3 - a_7} u^{a_4 - a_7}$. The chart $r = 1$ further applies the homomorphism fixing $s, t, u$ and sending $r$ to 1. All these are derivation identities on the specified discrete free words. Evaluating in a finite field gives the actual cocycle rule $d(ab) = d(a) + \chi(a) d(b)$. Every seven-tuple of generator values defines a homomorphism to the finite affine group and extends uniquely to $V = \widehat{F}_7$. Thus the continuous cocycle space is $k_0^7$; the Laurent calculations are not asserted to define continuous cocycles with infinite coefficient ring.

For the character in a given chart put $\delta = (\chi(X_i) - 1)_{i=1}^7$. Use pivot $p = 7$ when $c \neq 1$, and pivot $p = 4$ when $c = 1, u \neq 1$. Let $I_p$ be the increasing list of the other six indices. Define

$$\begin{aligned} P_p(v) &= (v_i - \delta_i \delta_p^{-1} v_p)_{i \in I_p}, \quad & S_p(w)_p &= 0, \quad S_p(w)_{I_p} = w, \\ P_p S_p &= I_6, & S_p P_p(v) &= v - \delta \delta_p^{-1} v_p. \end{aligned} \tag{F.29.6}$$

Here $\delta_7 = (1 - c)/c \neq 0$ in the first charts; the last charts have $u = h^{-1}$, $h = rst \neq 1$, and $\delta_4 = (1 - h)/h \neq 0$. These two composites identify the actual continuous cohomology quotient $Q_\chi = H^1_{\mathrm{cont}}(V, k_{0,\chi})$ with $k_0^6$ in each displayed coordinate order.

Retain the eleven actual F.23 automorphisms $B_l(v) = u_l \alpha_l(v) u_l^{-1}$, $0 \leq l \leq 10$. They and their inverses preserve $K$ and fix the character. Their cocycle matrices $A_l$ have rows $D(B_l(X_i))$, so $A_l \delta = \delta$. Put $\overline{A}_l = P_p A_l S_p$. For each basis $J$ below the cohomology matrix $T_l = J^{-1} \overline{A}_l J$ is certified by the integral identity

$$\delta_p (A_l S_p J - S_p J T_l) = \delta \left( \mathrm{row}_p (A_l S_p J) \right). \tag{F.29.7}$$

This clears precisely the displayed principal pivot and proves the claimed action after every permitted field evaluation. All source indices in this proof begin at zero: $0 =$ `old0`, $1, \dots, 5$ are `old_1_0`, $\dots$, `old_1_4`, and $6, \dots, 10$ are `old_-1_0`, $\dots$, `old_-1_4`. Their positions in an enumeration beginning at one are $l + 1$; the word labels remain unchanged.

For $P_0 = X_1 \cdots X_7$, the product-one condition makes $d(P_0)$ a covector on cohomology. The pure Artin map fixes $P_0$, and direct conjugation gives $d(B_l(P_0)) = \chi(u_l) d(P_0)$. Also each character-one meridian has $d(B_l(X_i)) = \chi(u_l a_{li}) d(X_i)$, where $\alpha_l(X_i) = a_{li} X_i a_{li}^{-1}$ is its actual pure braid conjugator. These boundary and meridian evaluations are the quotient covectors used below.

Put $r_j = r^{q_1^j}$, $s_j = s^{q_1^j}$, $t_j = t^{q_1^j}$; $u, c \in k_1$ stay fixed. The exact coefficient map

$$d \longmapsto d^{q_1^j}, \qquad k_{0,\chi} \longrightarrow k_{0,\chi_j}, \qquad Q_\chi \longrightarrow Q_j = H^1_{\mathrm{cont}}(V, k_{0,\chi_j}) \tag{F.29.8}$$

is additive and semilinear and preserves cocycles, principals and continuity. Its inverse is the power $q_1^{m-j}$, with the identity at $j = 0$. It commutes with every actual source precomposition. Thus each displayed basis, inclusion, quotient, source identity and inverse map transports by the indicated Frobenius substitutions. All $\chi_j$ agree on $K$, by (F.29.4). Moreover

$$(r_j, t_j) = (r_l, t_l) \quad \Longrightarrow \quad j = l. \tag{F.29.9}$$

Indeed such an equality for distinct indices would make a nonidentity $k_1$-Frobenius power fix the generators of $k_0 = k_1(r,t)$. For $0 < a < m$, the polynomial $X^{q_1^a} - X$ has degree smaller than $|k_0|$, and cannot vanish on the whole field. In either $r = 1$ chart, $k_0 = k_1(t)$, so the $t_j$ are distinct and lie outside $k_1$. In particular $t_j^2 \neq 1$. In either $r \neq 1$ chart every $r_j \neq 1$.

*Chart $c \neq 1$, $r \neq 1$: the three-dimensional core.* Use $p = 7$ and the six coordinates of $X_1, \dots, X_6$. Put $b = (1, r, rs, rst, c, c)$. The boundary and the actual $X_5, X_6$ evaluations give the explicit maps

$$\begin{aligned} \pi(v) &= (bv, v_5, v_6), \\ \iota(x,y,z) &= (-rx - rsy - rstz, x, y, z, 0, 0), \\ \sigma(z_\partial, z_5, z_6) &= (z_\partial - c(z_5 + z_6), 0, 0, 0, z_5, z_6). \end{aligned} \tag{F.29.10}$$

They satisfy $\pi\sigma = I_3$, $\ker \pi = \operatorname{im} \iota =: W$ and $\iota(v_2, v_3, v_4) + \sigma\pi(v) = v$. In particular the basis and its inverse are

$$\begin{aligned} J &= (\iota e_1, \iota e_2, \iota e_3, e_1, e_5 - ce_1, e_6 - ce_1), \\ J^{-1}(v) &= (v_2, v_3, v_4, bv, v_5, v_6). \end{aligned} \tag{F.29.11}$$

Here the $e_i$ inside $\iota$ are the three domain coordinate vectors; the remaining ones are the six quotient coordinate vectors. The displayed decomposition proves both basis composites directly. In this basis every actual source matrix is $\left(\begin{smallmatrix} G_l & E_l \\ 0 & D_l \end{smallmatrix}\right)$, where $G_l$ acts on $W$ and $D_l$ is diagonal on the boundary, $X_5$ and $X_6$ coordinates. Equivalently $\overline{A}_l \iota = \iota G_l$ and $\pi \overline{A}_l = D_l \pi$. The actual source-word table is

| $l$ | $G_l$ | $\lambda_{\partial,l}$ | $\lambda_{5,l}$ | $\lambda_{6,l}$ |
|---|---|---|---|---|
| 0 | $rI_3$ | 1 | $r$ | $r$ |
| 2 | $*$ | $rt$ | $rt/u$ | $rct$ |
| 7 | $tI_3$ | $t^{-1}$ | $t/u$ | $ct$ |
| 8 | $*$ | 1 | 1 | $t^{-1}$ |
| 9 | $*$ | 1 | $c$ | 1 |

(F.29.12)

The stars denote retained full core matrices; the additional exact identities used for them are

$$(G_8 - I_3)^2 (G_8 - (c/u) I_3) = 0, \qquad (G_9 - I_3)^2 (G_9 - (tu)^{-1} I_3) = 0. \tag{F.29.13}$$

These matrix products are integral Laurent identities verified from the literal source words.

We explain the invariant-subspace classification used in this and the next two separating charts. Take a nonzero source-invariant subspace $U \leq Q_j$ with no proper nonzero invariant subspace. Its intersection with the displayed $W_j$ is invariant. A nonzero intersection therefore gives $U \leq W_j$. For zero intersection, $\pi_j | U$ is injective. A nonzero coordinate projection of this map has invariant kernel, so is injective as well. It identifies $U$ with one of the displayed one-dimensional quotient characters. This uses the actual coordinate projections and does not require distinct quotient characters inside a single factor.

For $j \neq l$, compare these subspaces through any source-intertwining isomorphism. Source operators acting by scalars $a, b$ on its two sides give $(a - b)F = 0$, so $F \neq 0$ forces $a = b$. In the present chart source 0 separates a boundary type from every other type, because its scalars are 1 and $r_l \neq 1$. Two core subspaces force $r_j = r_l, t_j = t_l$ through sources $0, 7$, contrary to (F.29.9). Two boundary types force $t_j = t_l$ through source 7 and then $r_j = r_l$ through source 2. For the same type $X_5$ or $X_6$, sources $0, 7$ determine the pair $(r_j, t_j)$ after cancelling the common units $u$ or $c$. The different types $X_5, X_6$ are separated by source 9, with scalars $c \neq 1$ and 1.

A core subspace in factor $j$ isomorphic to type $X_5$ in factor $l$ must be one-dimensional. Source 9 and (F.29.13) force $(c-1)^2(c - (t_j u)^{-1}) = 0$, hence $t_j = (cu)^{-1} \in k_1$. Inverse Frobenius makes $t = (cu)^{-1}$ and all $t_l = t_j$. Source 0 then forces $r_j = r_l$, contradicting (F.29.9). For a core subspace versus type $X_6$, source 8 and (F.29.13) force $t_l^{-1} \in \{1, c/u\}$, so $t_l \in \{1, u/c\} \subset k_1$. Again all $t_j = t_l$, and source 0 contradicts (F.29.9). These comparisons cover every pair of the

four types, including the same-type pairs. The inverse of an intertwining isomorphism covers the opposite orientation.

*Chart $c \neq 1$, $r = 1$: two core coordinates and four quotients.* Here $c = stu$, $k_1 = \mathbb{F}_p(u, c)$ and $k_0 = k_1(t)$. In the same principal quotient $p = 7$, let $b = (1, 1, s, st, c, c)$ and use

$$\begin{gathered}\pi(v) = (bv, v_1, v_5, v_6), \qquad \iota(x, y) = (0, -sx - sty, x, y, 0, 0),\\ \sigma(z_\partial, z_1, z_5, z_6) = (z_1, z_\partial - z_1 - c(z_5 + z_6), 0, 0, z_5, z_6),\\ J = (\iota e_1, \iota e_2, e_2, e_1 - e_2, e_5 - ce_2, e_6 - ce_2),\\ J^{-1}(v) = (v_3, v_4, bv, v_1, v_5, v_6).\end{gathered} \tag{F.29.14}$$

The identities $\pi\sigma = I_4$, $\ker \pi = \operatorname{im} \iota = W$ and $\iota(v_3, v_4) + \sigma\pi(v) = v$ prove the maps and both basis inverses explicitly. All eleven source matrices again have the displayed invariant-core and diagonal-quotient form. Their required entries are

| $l$ | $G_l$ | $\lambda_{\partial,l}$ | $\lambda_{1,l}$ | $\lambda_{5,l}$ | $\lambda_{6,l}$ |
|---|---|---|---|---|---|
| 1 | $*$ | 1 | $u^{-2}$ | 1 | 1 |
| 2 | $tI_2$ | $t$ | $t^{-1}$ | $t/u$ | $ct$ |
| 5 | $I_2$ | 1 | $c^2$ | $t^{-1}$ | $u^{-1}$ |
| 7 | $tI_2$ | $t^{-1}$ | $t$ | $t/u$ | $ct$ |
| 8 | $*$ | 1 | 1 | 1 | $t^{-1}$ |
| 9 | $*$ | 1 | 1 | $c$ | 1 |

(F.29.15)

The smaller core satisfies the exact quadratic identities

$$(G_8 - I_2)(G_8 - (c/u)I_2) = 0, \qquad (G_9 - I_2)(G_9 - (tu)^{-1}I_2) = 0. \tag{F.29.16}$$

They, the basis and every quotient coordinate are independently checked from all eleven literal source words after $r \mapsto 1$.

The five types are now $W, \partial, X_1, X_5, X_6$. Their source-2, source-7 scalar pairs are, respectively, $(t_j, t_j)$, $(t_j, t_j^{-1})$, $(t_j^{-1}, t_j)$, $(t_j/u, t_j/u)$, and $(ct_j, ct_j)$. The same-type comparisons determine $t_j = t_l$, which is impossible for $j \neq l$. A core type versus boundary or $X_1$ would force $t_l^2 = 1$ through these two sources. A boundary or $X_1$ type versus $X_5$ or $X_6$ would likewise force $t_j^2 = 1$ on the first side. These conclusions contradict $t_j \notin k_1$. Types $X_5, X_6$ are separated by source 9. A core type versus $X_5$ would force $c = (t_j u)^{-1}$ by (F.29.16) and source 9, putting $t_j \in k_1$. A core type versus $X_6$ would force $t_l \in \{1, u/c\} \subset k_1$ by source 8. Both are impossible.

The only pair left is boundary in factor $j$ versus $X_1$ in factor $l$. Sources $1, 5$ force $u^2 = c^2 = 1$. Since $c \neq 1$ and $u \neq 1$, the factorization $(a - 1)(a + 1) = a^2 - 1$ in the field gives

$$c = u = -1, \qquad p \neq 2, \qquad s = t^{-1}, \qquad k_1 = \mathbb{F}_p, \qquad t \notin \mathbb{F}_p. \tag{F.29.17}$$

In characteristic two $a^2 = 1$ has only the root 1, so this remaining case is absent.

In (F.29.17) the coordinate order of (F.29.14) is $(x, y, z_\partial, z_1, z_5, z_6)$. Sources $2, 7$ have core $tI_2$ and quotient diagonals $(t, t^{-1}, -t, -t)$ and $(t^{-1}, t, -t, -t)$, respectively. The nonzero factors $t - t^{-1}$ and $2t$ follow from $t \notin \mathbb{F}_p$ and $p \neq 2$. A boundary-character eigenvector must therefore have its $X_1, X_5, X_6$ coordinates zero, by source 2. Its boundary coordinate cannot be zero, because then source 7 would compare $t$ on the core with $t^{-1}$ on that vector. Normalize this nonzero coordinate to one. Source 7 then uniquely forces the first vector in the following exact identities:

$$\begin{gathered}v_\partial = (1 - 2t, 2, 1, 0, 0, 0)^{\mathsf{T}}, \qquad T_7 v_\partial = t^{-1} v_\partial,\\ v_1 = (-t, 0, 0, 1, 0, 0)^{\mathsf{T}}, \qquad T_2 v_1 = t^{-1} v_1,\\ (T_1 - I_6)v_\partial = (T_1 - I_6)v_1 = (3t - 1, -2, 0, 0, 0, 0)^{\mathsf{T}}.\end{gathered} \tag{F.29.18}$$

The boundary scalar at source 1 is 1. Its required eigenvector equation contradicts the nonzero second coordinate $-2$ in (F.29.18). Thus the boundary-character line does not occur as a submodule in this case. For completeness, source 7 first kills the other quotient coordinates of an $X_1$ line; source 2 uniquely forces $v_1$ after its $X_1$ coordinate is normalized. The same residual excludes that line too. The coefficient specialization for these identities is $r \mapsto 1$, $s \mapsto t^{-1}$, $u \mapsto -1$. A monomial $s^a t^b u^d$ maps to $(-1)^d t^{b-a}$, and the pivot becomes $\delta_7 = -2$. The separate literal three-word audit verifies (F.29.7) with this exact pivot for sources $1, 2, 7$, before using its proved odd-characteristic invertibility. The same argument transports by Frobenius to every factor. It excludes the last possible cross-type comparison and completes this chart.

*Chart $c = 1$, $r \neq 1$: the actual pivot $X_4$.* Put $h = rst = u^{-1} \neq 1$. The character is now literally $(r, s, t, h^{-1}, 1, 1, 1)$. Its coefficient map is

$$\begin{aligned} \mathbb{Z}[t_1^{\pm 1}, \dots, t_7^{\pm 1}] &\longrightarrow A_3 = \mathbb{Z}[r^{\pm 1}, s^{\pm 1}, t^{\pm 1}], \\ (t_1, t_2, t_3, t_4, t_5, t_6, t_7) &\longmapsto (r, s, t, (rst)^{-1}, 1, 1, 1). \end{aligned} \tag{F.29.19}$$

The exponent image is $r^{a_1 - a_4} s^{a_2 - a_4} t^{a_3 - a_4}$. This is also the homomorphism $A_4/(rstu - 1) \to A_3$ sending $u$ to $(rst)^{-1}$ and fixing $r, s, t$. The reverse map sends $r, s, t$ to their classes; the relation $rstu = 1$ makes the two composites the identity, including on $u$. The maps are applied to the unlocalized word identities, followed here by localization at $1 - h$ for the actual pivot $X_4$ in (F.29.6). The scalar fields are $k_1 = \mathbb{F}_p(h)$ and $k_0 = k_1(r, t)$ because $s = h/(rt)$.

The six quotient coordinates are now those of $X_1, X_2, X_3, X_5, X_6, X_7$, in that order. Put $b = (1, r, rs, 1, 1, 1)$. The exact maps are

$$\begin{aligned} \pi(v) &= (bv, v_4, v_5, v_6), \\ \iota(x, y) &= (-rx - rsy, x, y, 0, 0, 0), \\ \sigma(z_\partial, z_5, z_6, z_7) &= (z_\partial - z_5 - z_6 - z_7, 0, 0, z_5, z_6, z_7). \end{aligned} \tag{F.29.20}$$

They satisfy $\pi\sigma = I_4$, $\ker \pi = \operatorname{im} \iota = W$ and $\iota(v_2, v_3) + \sigma\pi(v) = v$. Thus the basis maps are explicitly

$$\begin{aligned} J &= (\iota e_1, \iota e_2, e_1, e_4 - e_1, e_5 - e_1, e_6 - e_1), \\ J^{-1}(v) &= (v_2, v_3, bv, v_4, v_5, v_6). \end{aligned} \tag{F.29.21}$$

Every actual source has invariant two-dimensional core and diagonal quotient coordinates $\partial, X_5, X_6, X_7$. The literal table is

| $l$ | $G_l$ | $\lambda_{\partial,l}$ | $\lambda_{5,l}$ | $\lambda_{6,l}$ | $\lambda_{7,l}$ |
|---|---|---|---|---|---|
| 0 | $rI_2$ | 1 | $r$ | $r$ | $r$ |
| 2 | $*$ | $rt$ | $rht$ | $rt$ | $rt$ |
| 7 | $tI_2$ | $t^{-1}$ | $ht$ | $t$ | $t$ |
| 8 | $I_2$ | 1 | 1 | $t^{-1}$ | $h$ |
| 10 | $I_2$ | 1 | $t^{-1}$ | $h$ | 1 |

(F.29.22)

All entries use the original source indices; in particular $10 =$ `old_-1_4`.

The five possible types are $W, \partial, X_5, X_6, X_7$, with the classification proved from the displayed maps above. Source 0 separates boundary from every other type since $r_j \neq 1$. Two core types determine $(r_j, t_j)$ by sources $0, 7$; two boundary types do so by sources $7, 2$. For the same meridian type, sources $0, 7$ determine the same pair, with the common $h$ cancelled for type $X_5$. Every same-type comparison across distinct factors therefore contradicts (F.29.9).

For the different types, core versus $X_7$ is excluded by source 8, with scalars $1, h$. Core versus $X_6$ is excluded by source 10. Core versus $X_5$ in factor $l$ would force $t_l = 1$ by source 10. Inverse Frobenius then gives all $t_j = 1$, and source 7 compares 1 with $h \neq 1$. Types $X_5, X_7$ are excluded by source 8; types $X_6, X_7$ by source 10. For type $X_5$ in factor $j$ versus $X_6$ in factor $l$, source

8 forces $t_l = 1$, hence all $t_j = 1$, and source 10 then compares 1 with $h$. This proves every cross-type comparison, in either orientation. No extra condition on $t-1$ or on the characteristic has entered this chart.

*Chart $c = 1$, $r = 1$: an essential line and actual seed evaluations.* Here the character is $(1, s, t, (st)^{-1}, 1, 1, 1)$, $h = st \neq 1$, $k_1 = \mathbb{F}_p(h)$ and $k_0 = k_1(t) = k_1(s)$. The coefficient map is $r \mapsto 1$ in (F.29.19), sending $h$ to $st$ and retaining the principal pivot $X_4$. Since the field extension is proper, both $s, t \notin k_1$. In particular $s-1, t-1, t+1, h-1$ are nonzero, in every characteristic, as are the units $s, t$.

Keep the six coordinates of the preceding chart and put $b = (1, 1, s, 1, 1, 1)$. The common kernel of the boundary and the four character-one meridian covectors is the line generated by $v_0 = (0, 1, -s^{-1}, 0, 0, 0)$. The following basis and inverse are exact:

$$\begin{aligned} J &= (v_0, e_2, e_5 - e_2, e_4 - e_2, e_1 - e_2, e_6 - e_2), \\ J^{-1}(v) &= (-sv_3, bv, v_5, v_4, v_1, v_6), \\ J(z_0, z_\partial, z_6, z_5, z_1, z_7) &= (z_1, z_0 + z_\partial - z_6 - z_5 - z_1 - z_7, -s^{-1}z_0, z_5, z_6, z_7). \end{aligned} \tag{F.29.23}$$

Substitution proves both composites to be the identity. In these coordinates the actual source matrices and one required lift identity are

$$\begin{gathered} T_l = \begin{pmatrix} \nu_l & a_l \\ 0 & \Lambda_l \end{pmatrix}, \qquad \Lambda_l \text{ diagonal}, \\ (\nu_0, \dots, \nu_{10}) = (1, 1, t, 1, s, 1, 1, t, 1, s, 1), \\ \widetilde{v}_0 = S_4 v_0, \qquad A_2 \widetilde{v}_0 = t\widetilde{v}_0 + st^2\delta. \end{gathered} \tag{F.29.24}$$

All eleven full matrices and the displayed principal correction are independently verified.

For $j = 1, 2, 5, 7, 9$, the additive commutators $T_4T_j - T_jT_4$ kill the first line and have image in that line. Their first rows on the last five coordinates form $gH$, where

$$g = (s-1)(t-1),$$

$$H = \begin{pmatrix} 0 & 0 & 0 & (st+1)(st+s-1) & -t^{-2} \\ 0 & 0 & 1 & -(t+1) & 0 \\ 0 & s/t & -t^{-2} & 0 & 0 \\ (t+1)(st-t+1)/(st^3) & 0 & s & 0 & 0 \\ -(s-1)/(st^2) & 0 & 0 & -(s-1) & 0 \end{pmatrix}. \tag{F.29.25}$$

The independent signed-permutation expansion of the determinant gives

$$\det H = \frac{(s-1)(t-1)(t+1)}{t^6}, \qquad \det(gH) = \frac{(s-1)^6(t-1)^6(t+1)}{t^6} \neq 0. \tag{F.29.26}$$

The factor proof preceding (F.29.23) justifies every nonzero factor, including in characteristic two where $t+1 = t-1$. Every nonzero subspace invariant under the actual eleven sources contains $L = k_0[v_0]$. Indeed, a vector with nonzero last five coordinates is sent by some displayed commutator to a nonzero first-line vector, by (F.29.26). That vector remains in the invariant subspace and spans $L$. A nonzero subspace already contained in that line equals $L$. This proves the essential-line statement directly from the source operators.

Let $d_0$ have generator values $\widetilde{v}_0$. For the actual negative seed $a = u_{\texttt{cross_0_-1_1}}$ of index 13, the word calculation and (F.29.24) give

$$\begin{gathered} \chi(a) = h^{-1}, \qquad d_0(a) = h - 1, \\ \kappa = B_2(a)a^{-1} \in K, \qquad \chi(\kappa) = 1, \qquad d_0(\kappa) = 1 - h. \end{gathered} \tag{F.29.27}$$

In fact $d_0(B_2(a)) = t(h-1) + ht(h^{-1}-1) = 0$. For a character-one word the principal correction in (F.29.24) vanishes. Thus

$$\kappa_n = B_2^n(\kappa) \in K, \qquad \chi(\kappa_n) = 1, \qquad d_0(\kappa_n) = t^n(1-h) \quad (n \geq 0). \tag{F.29.28}$$

The source preserves $K$, and the seed belongs to the original eight negative seeds, so these are actual elements of $K$. Independently, the first two derivative rows $D(a)(A_2 - I_7)$ and $D(a)(A_2 - I_7)A_2$ are checked against (F.29.27)–(F.29.28), including their vanishing on the principal vector.

*The common restriction map and its exact kernel.* Across all four charts consider the actual map

$$\Phi : \bigoplus_{j=0}^{m-1} Q_j \longrightarrow H^1_{\mathrm{cont}}(K, k_{0,\chi|K}), \qquad ([d_j])_j \longmapsto \sum_j [d_j|K]. \tag{F.29.29}$$

It is well-defined because the characters agree on $K$, and it commutes with every source operator $[d] \mapsto [d \circ B_l]$. Each individual restriction is injective by F.24, since $\chi_j(R) = C \neq \{1\}$.

Choose a smallest-dimensional nonzero source-invariant subspace $T \leq \ker \Phi$ under the supposition that the kernel is nonzero. Each nonzero coordinate projection $p_j|T$ is injective, since its kernel is invariant. Its image is a nonzero invariant subspace with no proper nonzero invariant subspace. Two nonzero projections therefore give the exact mutually inverse intertwiners

$$p_l \circ (p_j|T)^{-1} : p_j(T) \longrightarrow p_l(T), \qquad p_j \circ (p_l|T)^{-1} : p_l(T) \longrightarrow p_j(T). \tag{F.29.30}$$

In the first three charts every pair of possible types was excluded above for $j \neq l$. There is therefore only one nonzero projection; $T$ lies in a single factor, contradicting that factor's F.24 restriction injectivity. This proves $\Phi$ injective in those charts.

In the last chart the essential-line argument puts $L_j$ inside each nonzero projection image. Minimality makes that image equal to $L_j$, so $T \leq \bigoplus_j L_j$. Frobenius transport of (F.29.28) gives, on these lines, the evaluation matrix at $\kappa_0, \ldots, \kappa_{m-1}$ and its determinant

$$((1-h)t_j^n)_{0 \leq n, j < m}, \qquad (1-h)^m \prod_{j<l} (t_l - t_j) \neq 0. \tag{F.29.31}$$

All these evaluations descend to the common target cohomology, since the words have character one. The coefficient inverse is explicit: write

$$\prod_{l \neq j} \frac{X - t_l}{t_j - t_l} = \sum_{n=0}^{m-1} h_{jn} X^n, \qquad z_j = (1-h)^{-1} \sum_{n=0}^{m-1} h_{jn} y_n. \tag{F.29.32}$$

It inverts $y_n = \sum_j (1-h) t_j^n z_j$; evaluation of the displayed polynomial at each $t_l$ proves the inverse composites. Every denominator is nonzero by $h \neq 1$ and the proved distinctness of the $t_j$. Thus evaluation after $\Phi$ is injective on $\bigoplus_j L_j$, contradicting the nonzero $T$ in its kernel. This proves the injectivity also in the last chart.

Zero summed restriction of cocycles now forces each cohomology class to be zero, so each cocycle is $b_j(\chi_j - 1)$. Its summed restriction is $(\chi|K - 1)\sum_j b_j$. Since $C \neq \{1\}$, this vanishes exactly for $\sum_j b_j = 0$. The converse follows by direct restriction. The principal vector in every factor is nonzero, and the displayed parameter-sum kernel has dimension $m - 1$. This proves (F.29.3), including the rank $7m - (m-1) = 6m + 1$.

For the exact trace reduction of F.26, the tuple $(d^{q_1^j})_j$ gives

$$\mathrm{Tr}_{k_0/k_1}(d|K) = 0 \quad \Longleftrightarrow \quad d = b(\chi - 1) \text{ with } \mathrm{Tr}_{k_0/k_1}(b) = 0. \tag{F.29.33}$$

The forward implication is the proved cocycle kernel, and the reverse is direct evaluation, using $\chi(K) \subset k_1$. F.26 sends every proper finite affine marginal image through its explicit conjugation

and coefficient homomorphism to a nonprincipal element of this trace kernel over the original scalar field $k_0$. Those maps retain $\chi$, so they retain (F.29.1), $C$, $k_1$ and $k_0$. Equation (F.29.33) excludes that element. This proves (F.29.2) for every remaining negative representation, including those with an original coefficient field larger than $k_0$.

*Integral evidence and the actual sign transport.* The three symbolic producers retain all full source matrices and basis maps for the generic four-parameter chart and the two pivot-$X_4$ charts. Their exact check counts are 754, 691 and 956. The separately structured three-chart verifier imports neither a producer nor an algebra package. It reconstructs all eleven old-source and all eight negative-seed braid words in each of those three integral Laurent rings. Its 3246 checks include (F.29.7), both basis inverses, every invariant inclusion and diagonal quotient, the comparison tables, the integral annihilators, the five printed commutator rows, their 120-term determinant expansion, the source-2 principal correction and the actual seed evaluations. The additional $r = 1, c \neq 1$ verifier uses the same reviewed integer Laurent arithmetic, reconstructs all eleven original words again, and records 996 checks of the new basis, every quotient character and (F.29.16). Root's separate exceptional audit reconstructs sources $1, 2, 7$ directly and checks the exact coefficient map and both residual identities in (F.29.18), with 20 additional recorded certificate assertions.

These are integral coefficient identities. The only nonmonomial principal localizations are $1 - c$ and $1 - h$ in their respective charts; their nonvanishing was proved before each quotient was used. All other determinant and interpolation factors have their explicit nonzero proofs above. The exceptional constant 2 is inverted only after (F.29.17) proves odd characteristic. The evidence manifest binds the source words, programs, outputs, coefficient inputs and completed dependencies to this exact locus and statement.

Finally, a positive representation precomposes with $\mathcal{B}_+ : V_- \to V_+$ to have precisely the negative character just proved in the basis $Y_i^+ = \mathcal{B}_+(X_i)$. This continuous isomorphism and its inverse carry $\mathcal{K}_-, R_-$ onto their positive counterparts by F.23.13. Equality of the two negative images therefore gives equality of the two original positive images through those exact maps. This proves (F.29.2) for both signs. □

**F.30. [Proved] Finite affine product-one images when the first scalar generates the scalar-field extension.** Use the actual bases $Y_i^- = X_i$ and $Y_i^+ = \mathcal{B}_+(X_i)$ under the based isomorphism of F.23.13. For a finite field $k$ and continuous $\rho : V_\pm \to \mathrm{Aff}_1(k)$, let $\chi$ be its scalar character and put $C = \chi(R_\pm)$, $k_1 = \mathbb{F}_p(C)$, $k_0 = \mathbb{F}_p(\chi(V_\pm))$ and $r = \chi(Y_1^\pm)$. The exact locus

$$\prod_{i=1}^{7} \chi(Y_i^\pm) = 1, \qquad k_0 = k_1(r) \tag{F.30.1}$$

has the image equality

$$\rho(\mathcal{K}_\pm) = \rho(R_\pm). \tag{F.30.2}$$

All characteristics, finite extension degrees and larger original coefficient fields are included. On the part $C \neq \{1\}$, put $q_1 = |k_1|$, $m = [k_0 : k_1]$ and $\chi_j = \chi^{q_1^j}$, $0 \leq j < m$. The summed continuous $H^1$ restriction for these $m$ characters is injective. Its cocycle kernel consists exactly of the principal tuples whose parameters sum to zero, and its rank is $6m + 1$ on $7m$ coordinates.

*Manuscript:* Part F root-adjoining intersection question; Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; original F7/F19 residual subgroup-membership reduction.

*Proof. Actual fields and the completed scalar cases.* Use the negative groups $K = \mathcal{K}_- \leq R = R_- \leq V = V_-$ first. Write their product-one character as $(r, s, t, u, v, w, c^{-1})$, where $c = rstuvw$. The normal-generation equality of F.21 maps through the actual marginal surjection of F.22 to

$\overline{\langle\langle K\rangle\rangle}_R = R$. Its image under the continuous abelian scalar map gives $\chi(K) = \chi(R)$. The actual deleted generators $X_4, \dots, X_7$ and (F.30.1) therefore give

$$\begin{aligned} C = \chi(K) = \chi(R) = \langle u, v, w, c^{-1}\rangle, \qquad k_1 = \mathbb{F}_p(u, v, w, c), \\ k_0 = \mathbb{F}_p(r, s, t, u, v, w) = k_1(r). \end{aligned} \tag{F.30.3}$$

All six parameters are nonzero. For $C = \{1\}$, the image of $R$ is an abelian translation group, so the same actual normal-generation map proves (F.30.2). For $C \neq \{1\}$ and $k_0 = k_1$, F.25 gives the image equality. F.24 supplies the individual restriction injection and the stated cocycle kernel with $m = 1$.

We now treat $C \neq \{1\}$ and $k_1 \subsetneq k_0$. Equation (F.30.3) implies $r \notin k_1$, hence $r - 1 \neq 0$. This gives an actual retained-generator principal pivot, independently of the value of $c$.

*Exact coefficient and principal-quotient maps.* The word derivations use the coefficient homomorphism

$$\begin{aligned} \mathbb{Z}[t_1^{\pm 1}, \dots, t_7^{\pm 1}] \longrightarrow A = \mathbb{Z}[r^{\pm 1}, s^{\pm 1}, t^{\pm 1}, u^{\pm 1}, v^{\pm 1}, w^{\pm 1}], \\ (t_1, t_2, t_3, t_4, t_5, t_6, t_7) \longmapsto (r, s, t, u, v, w, (rstuvw)^{-1}). \end{aligned} \tag{F.30.4}$$

Thus an exponent vector $(a_i)$ maps to $r^{a_1 - a_7} s^{a_2 - a_7} t^{a_3 - a_7} u^{a_4 - a_7} v^{a_5 - a_7} w^{a_6 - a_7}$. These are derivations on the specified discrete free words. Their evaluations in a finite field give the actual cocycle rule. Every seven-tuple of generator values defines a homomorphism to the finite affine group and extends uniquely to $V = \widehat{F}_7$. Consequently the continuous cocycle space is exactly $k_0^7$.

Let $\delta = (r, s, t, u, v, w, c^{-1})^{\mathsf{T}} - \mathbf{1}$. Its first coordinate is $\delta_1 = r - 1 \neq 0$. The actual principal quotient uses the maps

$$\begin{aligned} P_1(z) = (z_i - \delta_i (r-1)^{-1} z_1)_{i=2}^7, \qquad S_1(x) = (0, x_1, \dots, x_6)^{\mathsf{T}}, \\ P_1 S_1 = I_6, \qquad S_1 P_1(z) = z - \delta (r-1)^{-1} z_1. \end{aligned} \tag{F.30.5}$$

They identify $Q_\chi = H^1_{\mathrm{cont}}(V, k_{0,\chi})$ with six coordinates corresponding, in order, to the original $X_2, \dots, X_7$. The universal maps are defined over $A[(r-1)^{-1}]$; the proved nonvanishing justifies every finite-field use of that localization.

Let $P_0 = X_1 \cdots X_7$ be the actual boundary word. Its character is one, and its derivative and induced quotient covector are

$$\begin{aligned} D(P_0) = (1, r, rs, rst, rstu, rstuv, c), \\ b = D(P_0) S_1 = (r, rs, rst, rstu, rstuv, c), \\ D(P_0)\delta = 0, \qquad D(P_0) = b P_1. \end{aligned} \tag{F.30.6}$$

The vanishing is the telescoping identity $D(P_0)\delta = \chi(P_0) - 1$; substituting (F.30.5) proves the last displayed equality.

For $\beta = (s, st, stu, stuv, stuvw)$, define

$$\begin{aligned} \pi : Q_\chi \longrightarrow k_0, \quad \pi(z) = bz, \\ \iota : k_0^5 \longrightarrow Q_\chi, \quad \iota(x) = (-\textstyle\sum_{i=1}^5 \beta_i x_i, x_1, \dots, x_5), \\ \sigma : k_0 \longrightarrow Q_\chi, \quad \sigma(a) = r^{-1} a e_1. \end{aligned} \tag{F.30.7}$$

One has $\pi\sigma = I_1$, $\ker \pi = \operatorname{im} \iota =: W_\chi$ and $\iota(z_2, \dots, z_6) + \sigma\pi(z) = z$. In particular the coefficient basis and its inverse are explicitly

$$\begin{aligned} J = (e_2 - s e_1, e_3 - st e_1, e_4 - stu e_1, e_5 - stuv e_1, e_6 - stuvw e_1, r^{-1} e_1), \\ J^{-1}(z) = (z_2, z_3, z_4, z_5, z_6, bz). \end{aligned} \tag{F.30.8}$$

The displayed decomposition proves both basis composites without an assumed invariant complement.

*The actual source operators needed for separation.* Retain the eleven F.23 automorphisms $B_l(a) = u_l\alpha_l(a)u_l^{-1}$, $0 \le l \le 10$. They and their inverses preserve $K$. Their pure Artin action fixes every abelian character, and their cocycle matrices $A_l$ therefore satisfy $A_l\delta = \delta$. Set $\overline{A}_l = P_1A_lS_1$.

Every pure $\alpha_l$ fixes $P_0$. The cocycle formula for conjugation gives $d(B_l(P_0)) = \chi(u_l)d(P_0)$ because $\chi(P_0) = 1$. Combining this with (F.30.6) gives $\pi\overline{A}_l = \chi(u_l)\pi$. Thus $W_\chi$ is invariant under all actual source operators, and $\pi$ is their exact one-dimensional boundary quotient.

Put $\widetilde{\iota} = S_1\iota$. The literal original-word calculation gives the integral identity

$$(r-1)(A_0\widetilde{\iota} - r\widetilde{\iota}) = \delta\ \mathrm{row}_1(A_0\widetilde{\iota}). \tag{F.30.9}$$

Apply $P_1$ and cancel the proved nonzero $r-1$. This gives the actual equality $\overline{A}_0\iota = r\iota$ on the five-dimensional invariant subspace. The same word calculation gives the boundary entries

| $l$ | $\overline{A}_l\|W_\chi$ | $\chi(u_l)$ on $Q_\chi/W_\chi$ |
|---|---|---|
| 0 | $rI_5$ | 1 |
| 2 | $*$ | $rt$ |
| 7 | $*$ | $t^{-1}$ |

. (F.30.10)

The stars stand for the actual core operators; only the displayed entries are used. The indices are the original zero-based $0 =$ `old0`, $2 =$ `old_1_1` and $7 =$ `old_-1_1`, respectively the first, third and eighth source positions. The certificate compares all entries of the three full seven-coordinate source actions with direct evaluations of those labelled decorated Artin words.

*Frobenius transport and all three type comparisons.* For $0 \le j < m$, put $r_j = r^{q_1^j}$, $s_j = s^{q_1^j}$ and $t_j = t^{q_1^j}$. The parameters $u, v, w, c$ are fixed by these powers. The coefficient map

$$d \longmapsto d^{q_1^j}, \qquad k_{0,\chi} \longrightarrow k_{0,\chi_j} \tag{F.30.11}$$

is additive and semilinear. It preserves the cocycle rule, principal cocycles and continuity, and commutes with each actual source precomposition. The inverse is the power $q_1^{m-j}$, with the identity at $j = 0$. It transports every displayed inclusion, quotient, section and inverse basis to $Q_j = H^1_{\mathrm{cont}}(V, k_{0,\chi_j})$ and $W_j$. All characters agree on $K$, since $\chi(K) = C \subset k_1^\times$.

The elements $r_0, \dots, r_{m-1}$ are distinct and each differs from 1. An equality of two would make a nonidentity $k_1$-Frobenius power fix the generator $r$ and hence all of $k_0 = k_1(r)$. For $0 < a < m$, the polynomial $X^{q_1^a} - X$ has degree smaller than $|k_0|$, so cannot vanish on the whole field. Also a conjugate equal to 1 would give $r = 1 \in k_1$, contrary to the proper extension.

Take a nonzero source-invariant subspace $U \le Q_j$ with no proper nonzero invariant subspace. A nonzero invariant intersection with $W_j$ gives $U \le W_j$. For zero intersection, $\pi_j|U$ is injective and identifies $U$ with the one-dimensional boundary character. These are the two possibilities through the exact maps (F.30.7).

For distinct $j, l$, the following comparisons exclude every source-intertwining isomorphism between such subspaces. Two subspaces in $W_j, W_l$ have source-0 scalars $r_j, r_l$; an intertwiner $F$ would satisfy $(r_j - r_l)F = 0$, hence $F = 0$. A core subspace and a boundary subspace have source-0 scalars $r_j$ and 1, and the same argument excludes an isomorphism. For two boundary subspaces, source 7 would force $t_j^{-1} = t_l^{-1}$, so $t_j = t_l$. Source 2 would then force $r_jt_j = r_lt_l$, hence $r_j = r_l$, contrary to the proved distinctness. All character values cancelled here are nonzero. Inverse intertwiners cover both orientations of the mixed comparison. These are all three unordered type comparisons.

*Summed restriction and the finite affine image.* Consider the well-defined actual restriction map

$$\Phi : \bigoplus_{j=0}^{m-1} Q_j \longrightarrow H^1_{\mathrm{cont}}(K, k_{0,\chi|K}), \qquad ([d_j])_j \longmapsto \sum_j [d_j|K]. \tag{F.30.12}$$

It commutes with every actual source operator. Every individual restriction is injective by F.24, because $\chi_j(R) = C \neq \{1\}$. Under the supposition that $\ker \Phi$ is nonzero, choose a nonzero invariant subspace $T$ inside it with smallest dimension. Each nonzero coordinate projection $p_j|T$ is injective, and its image has no proper nonzero invariant subspace. Two nonzero projections give the explicit mutually inverse intertwiners

$$p_l \circ (p_j|T)^{-1} : p_j(T) \longrightarrow p_l(T), \qquad p_j \circ (p_l|T)^{-1} : p_l(T) \longrightarrow p_j(T). \tag{F.30.13}$$

The three comparisons exclude this for $j \neq l$. Hence there is only one nonzero projection, so $T$ lies in one $Q_j$. There $\Phi$ is the injective individual restriction, a contradiction. This proves the summed $H^1$ injection.

Zero summed restriction of cocycles therefore forces each cocycle to be principal. Since all characters agree on $K$ and $C \neq \{1\}$, the exact kernel and its dimension are

$$\left\{ \big(b_j(\chi_j - 1)\big)_j : \sum_j b_j = 0 \right\}, \qquad \dim \ker = m - 1. \tag{F.30.14}$$

The converse inclusion is direct restriction. Each principal vector is nonzero, and each continuous cocycle space has seven generator coordinates. Thus the stated rank is $7m - (m-1) = 6m + 1$.

Apply this result to the tuple $(d^{q_1^j})_j$ of one cocycle in the trace kernel of F.26. The cocycle $d$ is principal, say $d = b(\chi - 1)$. Its Frobenius powers have parameters $b^{q_1^j}$, uniquely because their principal vectors are nonzero. Their parameter sum is exactly the field trace. Consequently

$$\mathrm{Tr}_{k_0/k_1}(d|K) = 0 \quad \Longleftrightarrow \quad d = b(\chi - 1), \qquad \mathrm{Tr}_{k_0/k_1}(b) = 0. \tag{F.30.15}$$

The reverse implication follows directly from $\chi(K) \subset k_1$. The explicit conjugation and coefficient homomorphism of F.26 sends every proper finite affine marginal image to a nonprincipal element of this trace kernel over $k_0$. Those maps preserve the scalar character and both scalar fields. They therefore preserve the product-one condition and $k_0 = k_1(r)$ of (F.30.1). Equation (F.30.15) excludes their resulting nonprincipal element. This proves (F.30.2) for the negative marginal over every original finite coefficient field, including fields larger than $k_0$.

*Integral evidence and actual positive-sign transport.* The coefficient producer applies precisely (F.30.4) to the frozen F.24 universal matrices for the three original sources. The separate verifier imports neither that producer nor an algebra package. It uses the reviewed integer Laurent arithmetic to reconstruct the three actual decorated Artin words and compares every entry of their full seven-coordinate actions with the producer. Its 45 exact checks include both inverse maps in (F.30.8), the boundary kernel and section, the principal-vector identities, (F.30.9) and every boundary value in (F.30.10). The identities have integer coefficients; the only nonmonomial principal localization is $r - 1$, whose nonvanishing was proved from the exact field equality before its use. The evidence manifest binds the statement, programs, original words, coefficient inputs, outputs and completed dependencies.

For a positive representation, precomposition with the actual continuous isomorphism $\mathcal{B}_+ : V_- \to V_+$ sends the values at $Y_i^+$ to precisely the negative coordinates used above. F.23.13 supplies its inverse and carries $\mathcal{K}_-, R_-$ onto their positive counterparts. Thus it preserves the scalar images defining $k_0, k_1$, the first scalar $r$ and all conditions (F.30.1). Equality of the two negative images gives equality of the original two positive images through these specified maps. This proves (F.30.2) for both signs. □

## References


[1] J. M. Aarts and L. G. Oversteegen. Flowbox manifolds. *Transactions of the American Mathematical Society*, 327(1):449–463, 1991. Verified: the author-uploaded original published text was read, including Definition 1.1, Main Theorem 0.5, the construction in Section 4, Remark 4.4, and Definition 5.3 with Theorems 5.6–5.7 on page 460. The regularity condition is the existence of arbitrarily close arc embeddings through all nearby points. The cited Whitney local-section construction was also read in Section 3 of Aarts–Hagopian–Oversteegen, Pacific Journal of Mathematics 150 (1991), 1–12, in the publisher's primary PDF. The auxiliary global flow is not claimed to preserve the original operator action.

[2] Emil Artin. Theory of braids. *Annals of Mathematics, Second Series*, 48(1):101–126, 1947. Verified: the original article scan was opened. The projection-pattern isotopy lemma and Theorem 3 on p.103, Theorem 14 on p.112 (faithful braid substitutions), the generator and inverse formulas (14)–(15) on p.113, and the projection interpretation on pp.125–126 were read. F.9 uses exact rational crossing isolation and the faithful free-group substitution, not faithfulness of a linear braid representation.

[3] Mathieu Baillif and Alexandre Gabard. Manifolds: Hausdorffness versus homogeneity. *Proceedings of the American Mathematical Society*, 136(3):1105–1111, 2008. Verified: Theorem 1.1 on page 1106 and its full proof in Section 4, pages 1109–1110, were read in the published version deposited in the University of Geneva archive. The theorem allows non-Hausdorff manifolds initially and proves that a homogeneous Lindelof manifold is Hausdorff. The proof uses dense Hausdorff open sets, a countable subcover, the Baire property, and homogeneity.

[4] François Béguin, Sylvain Crovisier, and Frédéric Le Roux. Fixed point sets of isotopies on surfaces. *Journal of the European Mathematical Society*, 22(6):1971–2046, 2020. Verified: publisher title, authors, volume, issue, pages, DOI, and publication date; arXiv:1610.00686v2, Proposition 1.5 and Appendix A.1, read directly. Section 23, folios 140–141, scans 279–281 supply the manuscript comparison being addressed. C.66 proves the exact end-extension group topology and parameter-family maps rather than attributing that full conclusion to the source.

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[6] Stephen J. Bigelow and Ryan D. Budney. The mapping class group of a genus two surface is linear. *Algebraic and Geometric Topology*, 1:699–708, 2001. Verified: original complete ten-page article opened. Lemma 2.2, pages 701–702, gives the capped-disc braid central extension for a sphere puncture stabilizer; Propositions 3.2–3.3, pages 703–704, give the genus-two hyperelliptic central extension. The full-twist abelianization is displayed on page 702.

[7] Ilia Binder, Cristobal Rojas, and Michael Yampolsky. Computable Carathéodory theory. *Advances in Mathematics*, 265:280–312, 2014. Verified: publisher title, authors, volume, pages, DOI, and publication date; author research list independently confirms volume and pages. The primary arXiv:1209.6096v2 was read directly: Section 1 states the locally connected Carathéodory theorem, and Section 3.1, printed pages 9–11, gives crosscut chains, Proposition 3.1, and the prime-end compactification theorem. Section 23, folios 140–141, scans 279–281 supply the manuscript comparison being addressed.

[8] Marco Boggi. Automorphisms of profinite mapping class groups, 2026. verified: version 6 of 22 April 2026, the original 53-page PDF and supplied TeX opened; Sections 7.1–7.2

and equation (29) checked. F.6 uses the foundational arithmetic exact sequence, geometric comparison and universal etale monodromy construction, with its own spreading-out and equivariant-transport proof. No assertion from the main automorphism theorems is used.

[9] Marco Boggi. A congruence subgroup property for symmetric mapping class groups. *São Paulo Journal of Mathematical Sciences*, 20:Article 35, 2026. verified: supplied v2 source, original arXiv page, and publisher's full text opened; Theorem C and its induction inspected. F.7 records the unverified closure-intersection step in Proposition 2.4 and supplies the non-circular solvable normalizer/centralizer proof using the direct prime-power input. Published 23 September 2026. Publication status does not supply the missing nonsolvable intersection argument. The reflection argument in D.2 is proved separately.

[10] Marco Boggi and Pavel Zalesskii. A restricted magnus property for profinite surface groups. *Transactions of the American Mathematical Society*, 371:729–753, 2019. Verified: primary author preprint opened; Definition 1.3 and Proposition 3.6 read. Only positive integer exponents are used here, within the stated exponent range. Journal publication data cross-checked against a later primary author bibliography.

[11] Marco Boggi and Pavel Zalesskii. Finite subgroups of the profinite completion of good groups. *Bulletin of the London Mathematical Society*, 57:236–255, 2025. verified: complete supplied source and arXiv HTML opened; Theorems A and 6.3, including conjugacy and normalizer comparison for finite p-subgroups, checked. D.2 uses the geometric specialization for a characteristic prime-order rotation subgroup. F.7 identifies the unverified closure-intersection step in Theorem 5.1(iii); the prime-power comparison and the independent solvable proof do not use that step.

[12] Armand Borel. Sous-groupes compacts maximaux des groupes de Lie. In *Séminaire Bourbaki: années 1948/49–1949/50–1950/51, exposés 1–49*, number 1 in Séminaire Bourbaki, pages 271–279. Société mathématique de France, 1952. Verified: exposé 33, delivered December 1950. The principal theorem and definition of product notation in section 1, page 271, explicitly give a connected maximal compact subgroup K and a multiplication homeomorphism K times H1 times ... times Hs onto the connected Lie group, each Hj a closed one-parameter subgroup isomorphic to R. The displayed statement was visually checked; the reductions for semisimple covers and the general case in sections 3–4, pages 277–278, were read in the original scan.

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[14] Beverly Brechner and Timo Erkama. On topologically and quasiconformally homogeneous continua. *Annales Academiae Scientiarum Fennicae, Series A I Mathematica*, 4(2):207–208, 1979. Verified: both pages of the original 1978/1979 volume were read. Theorem 2 classifies nondegenerate proper compact subcontinua homogeneous with respect to neighborhood extensions as simple closed curves. Only compact components are passed directly to this theorem.

[15] Martin R. Bridson, Marston D. E. Conder, and Alan W. Reid. Determining Fuchsian groups by their finite quotients. *Israel Journal of Mathematics*, 214:1–41, 2016. verified: supplied primary source and original arXiv version opened; Theorem 5.1 and Corollaries 5.2–5.3 checked. The proof below makes the passage from total-group conjugacy to kernel conjugacy explicitly.

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[17] Benjamin Collas, Séverin Philip, and Naganori Yamaguchi. Anabelian geometry for Deligne–Mumford curves, 2026. verified: supplied complete TeX source and primary arXiv record opened; Definition 2.1 and Theorems 1.3 and 2.4 checked. The direction of these assertions, including the prescribed orbifold presentation, is retained in D.22.

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[20] David B. A. Epstein. Curves on 2-manifolds and isotopies. *Acta Mathematica*, 115:83–107, 1966. verified: original full paper opened; Theorem 5.8 for open and closed Mobius bands checked. Theorem 6.3, statement p. 99 and proof pp. 100–101, checked with its proper boundary-pair homotopy hypothesis and explicit preservation of a fixed basepoint. The topological proof paragraph on p. 101 and Appendix Theorem A4 on p. 107 were read directly; original printed pp. 99 and 101 were also rendered and actually viewed for C.91. The annulus boundary interchange is explicitly excluded in C.10.

[21] Edward Fadell and Lee Neuwirth. Configuration spaces. *Mathematica Scandinavica*, 10:111–118, 1962. Verified: publisher's original eight-page scan opened; Theorems 1 and 3 on pages 111–113, the surface homotopy calculations in Section IV, and the aspherical configuration examples on pages 115–116 visually inspected. Publisher metadata confirms 1962, correcting OCR of the scanned year in one search extract.

[22] Benson Farb and Dan Margalit. *A Primer on Mapping Class Groups*, volume 49 of *Princeton Mathematical Series*. Princeton University Press, 2012. Verified: actual 489-page primary book copy opened. Theorems 1.2, 2.5, 4.6 (pp.97–98), the following extended-group and point-push convention remarks, Theorem 8.1 (p.221), Theorem 8.8 (p.234), and the comparison diagram on p.235 read. The lowercase URL path is required. A.12 retains the residual inner ambiguity correctly instead of using the informal uniqueness wording in the last paragraph of p.235.

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[27] Grzegorz Gromadzki and Błażej Szepietowski. On topological type of periodic self-homeomorphisms of closed non-orientable surfaces. *Revista de la Real Academia de Ciencias Exactas, Fisicas y Naturales. Serie A. Matematicas*, 110:303–320, 2016.

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Theorem A and Section 3 on torsion conjugacy into a full profinite free factor, and Theorem B and Section 4 on full free-factor centralizers. The author Luis Ribes's Carleton publication list was independently opened and confirms volume 358 (1985), pages 155–161. The publisher DOI and EuDML retrieval were blocked, so the original article text hosted with the paper on Academia was used.

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[42] Dessislava H. Kochloukova and Pavel A. Zalesskii. Automorphisms of pro-$p$ groups of finite virtual cohomological dimension. *The Quarterly Journal of Mathematics*, 58(1):47–51, 2007. verified: original complete seven-page author manuscript opened; Theorem 2 and Section 1.1 equations (1)–(6), manuscript pages 3–4, checked. The theorem is stated for profinite groups. Publisher metadata verified separately; the citation retains the numbering of the inspected author manuscript.

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[45] Linus Kramer and Raquel Murat García. Fibrations and coset spaces for locally compact groups. *Transformation Groups*, 31:3167–3177, 2026. Verified: published online 27 January 2025. Theorem 3.7 reproduces the free Cartan Lie-action consequence of Palais' theorem with an explicit local trivialization and its inverse; it is the only result used from this paper.

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[49] Armando Martino. A proof that all seifert 3-manifold groups and all virtual surface groups are conjugacy separable. *Journal of Algebra*, 313(2):773–781, 2007. Verified against the publisher metadata and the full author preprint. Theorem 3.7 on printed page 5 states conjugacy separability for every group with a surface subgroup of finite index. Theorem 3.7 is applied to each finite-index subgroup of the explicitly constructed group; the centralizer-closure comparison is proved directly in D.65.

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[59] Ambrus Pál and Gereon Quick. Real projective groups are formal. *Mathematische Annalen*, 392:1833–1876, 2025. Verified: the original open-access publisher article and its metadata were opened. Definitions 1.2 and 6.1, Theorems 1.5 and 7.13, Corollary 4.8, and the stated

real-projective centralizer hypothesis in Theorem 2.1 were read. D.72 supplies its own explicit surjective cochain quasi-isomorphism and recursive defining-system proof; it uses Definition 6.1 to identify its actual target algebra, without asserting real projectivity of the pants extension.

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[63] Eugene Polulyakh. On the theorem converse to Jordan's curve theorem, 2000. Verified: version 1, Theorem 1.1 and Corollary 1.1 on page 7, with the proofs through sections 1.3–2, were read in the primary preprint. A compact connected planar set with exactly two complementary domains is Jordan when its inaccessible subsets from either domain are zero-dimensional. This is applied after compactification with defects confined to one point.

[64] Tibor Radó. Sur la représentation conforme de domaines variables. *Acta litterarum ac scientiarum Regiae Universitatis Hungaricae Francisco-Josephinae, Sectio scientiarum mathematicarum*, 1:180–186, 1923. Verified: original seven-page primary PDF and repository metadata opened; Section 1 theorem on printed page 182 visually checked, and the complete proof on pages 183–186 read. The theorem gives uniform convergence on the closed disk for Riemann maps normalized at the interior origin with positive derivative exactly under Frechet convergence of the enclosing Jordan curves. C85 separately proves its marked-boundary rotation, logarithm, half-plane lift and actual parameter topology comparisons.

[65] Tibor Radó. Über den begriff der riemannschen fläche. *Acta Scientiarum Mathematicarum (Szeged)*, 2:101–121, 1924–1926. Verified: original Section 3, Hilfssatz 2 on page 111 and its full proof through page 114 opened and read; local finiteness is explicit on page 112. The bound-volume title page states 1924–1926; the institutional repository dates the volume and item 1926. The exact bound-volume date is retained here rather than silently selecting a different article-year convention.

[66] Luis Ribes and Pavel Zalesskii. Normalizers in groups and in their profinite completions. *Revista Matemática Iberoamericana*, 30:165–190, 2014. verified: original full publisher PDF opened; Theorem 2.6 and Corollary 2.9 read with the virtually free hypotheses.

[67] Jean-Pierre Serre. Sur la dimension cohomologique des groupes profinis. *Topology*, 3:413–420, 1965. verified: original full paper opened; Section 1, unnumbered theorem and Corollary (2), read. The theorem requires absence of elements of order p in the ambient profinite group.

[68] Jyh-Haur Teh. Topological fundamental groups of locally finite infinite configuration spaces and infinite braids, 2026. Verified: version 1, dated 7 February 2026, opened in the original 43-page PDF. Definition 2.6 on page 5 and equation (2) on page 6 give the vague counting-measure metric and its sum with the ordered coordinate metric. C83 uses only these

topology definitions; none of the preprint's asphericity, infinite normalization, inverse-limit or completeness assertions is imported.

[69] Pekka Tukia. On discrete groups of the unit disk and their isomorphisms. *Annales Academiae Scientiarum Fennicae, Series A I, Mathematica*, (504):1–45, 1972. Verified: the original Academy PDF and its DOI, title, author, year, and series number; the National Library catalogue independently confirms the publication. Section 3.B, Proposition 3.5 and Corollary 3.5.1, printed pages 31–32 (PDF pages 30–31), and their full nested-axis proof were read directly. This section proves extension of the actual inducing homeomorphism for first-kind groups, without finite generation or cocompactness. Section 23, folios 140–141, scans 279–281 supply the manuscript comparison being addressed.

[70] Gareth Wilkes. Profinite rigidity for Seifert fibre spaces. *Geometriae Dedicata*, 188:141–163, 2017. Verified in the publisher's open-access full text: Proposition 5.7 and its proof, with Proposition 2.9; Proposition 6.6 in arXiv v1. The proof applies to every profinite automorphism sending each cone generator to a conjugate of a unit power of itself. The same calculation is Proposition 4.3.6 in the author's 2018 Oxford thesis.

[71] Gareth Wilkes. Relative cohomology theory for profinite groups, 2017. verified: original complete preprint opened; Propositions 1.16, 1.21, 1.26, 5.9 and 5.11 checked. The citation uses this exact preprint numbering.

[72] Gareth Wilkes. Relative cohomology theory for profinite groups. *Journal of Pure and Applied Algebra*, 223(4):1617–1688, 2019. Verified: the cited numbering is that of the 73-page accepted version dated June 19, 2018, deposited in Cambridge Apollo at https://doi.org/10.17863/CAM.37669. Definition 2.1, Propositions 2.3–2.4 and 2.10, Section 3 including Propositions 3.1 and 3.5, Proposition 5.7, and Proposition 6.16 were read in that exact version. Published metadata and DOI were checked against the publisher record and the Cambridge repository. This separate entry does not change the numbering of citations to the earlier preprint.

[73] Gareth Wilkes. Classification of pro-p PD2 pairs and the pro-p curve complex. *Groups, Geometry, and Dynamics*, 14:917–948, 2020. verified: original arXiv v2 PDF opened; Theorem 1.9, Definition 3.2 and Theorem 3.3 read with every presentation parameter and the commutator convention. The finite-character specializations and basis changes are written out in D.19–D.20.

[74] Tatsuhiko Yagasaki. Homotopy types of homeomorphism groups of noncompact 2-manifolds. arXiv:math/0010223, 2000. Verified: supplied primary TeX source opened. Theorem 1.1, the definitions of compact-open homeomorphism groups and fixed compact subpolyhedra, Theorem 2.1 on local extension, and Lemma 2.1 on compact surfaces read. Lemma 2.1 supplies the needed compact marked-surface homotopy triviality; its cited Hamstrom and Scott originals are not independently claimed to have been opened.

[75] Tatsuhiko Yagasaki. Groups of uniform homeomorphisms of covering spaces. *Journal of the Mathematical Society of Japan*, 66(4):1227–1248, 2014. Verified: original publisher PDF and metadata opened; Definition 2.1 on page 1230, all three metric-cover conditions, and Theorem 1.1 and Corollary 1.1 on page 1228 read. C81 proves its metric-cover comparisons directly and retains the compact quotient hypothesis and the need to preserve the named relative subgroup before using the deformation theorem.

[76] Hidehiko Yamabe. On an arcwise connected subgroup of a Lie group. *Osaka Mathematical Journal*, 2(1):13–14, 1950. Verified: the original two-page published proof and the University of Osaka version-of-record metadata were read. The theorem supplies an analytic Lie

subgroup for an arcwise connected subgroup of a Lie group. The present application separately proves closedness and equality with the induced homeomorphism-group topology by an explicit classification of the transitive subgroups.

[77] Naganori Yamaguchi. Center-freeness of finite-step solvable groups arising from anabelian geometry. arXiv:2601.07112v4, 2026. Verified: supplied primary TeX and primary arXiv HTML opened. Proposition 1.3, Theorem 1.5 in finite rank, Corollary 1.6, Proposition 2.6, Theorem 2.9 and Corollary 2.10 checked. A.17 independently proves the precise finite-cover and transfer steps used. A.19 independently proves the completed Magnus embedding it needs and strengthens the closed characteristic-zero case to slimness. No infinite-rank argument from this preprint is used.